

# Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant  $c_3 = \frac{1}{8\pi}$  and a five-slot carrier — build  $E_8$  and read off the Standard Model: status assessment and reading guide

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## In one sentence

**Topological Fixed-Point Theory (TFPT)** constructs, from exactly **two inputs** — a boundary constant  $c_3 = \frac{1}{8\pi}$  (P1) and a five-slot carrier  $g_{\text{car}} = 5$  (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with  $E_8$  ( $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ( $\alpha^{-1} = 137.0359992$ ); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ), not as free outputs. The honest one-line claim is on the status card below.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions** (plus two standalone working notes), best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S. tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the fire-wall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P. tfpt\_prime\_front** — research documentation of the prime / zeta line: the seven load-bearing modules on the  $E_8$  glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

**Working notes:** **N1. note\_e8\_gaussian\_code** — the Gaussian code bridge:  $E_8$  over  $\mathbb{Z}[i]$  via Construction A over the extended Hamming code  $[8, 4, 4]$ , the canonical four-bit quotient  $L/(1+i)L \cong \mathbb{F}_2^4$ , and the  $G_{31}$  quartic companion (exact algebra, machine-certified); **N2. note\_hilbert\_polya\_truncations** — a computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert-Pólya candidate (prediction-freeze methodology; no RH claim).

You are reading the introduction — the entry point and reading guide.

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

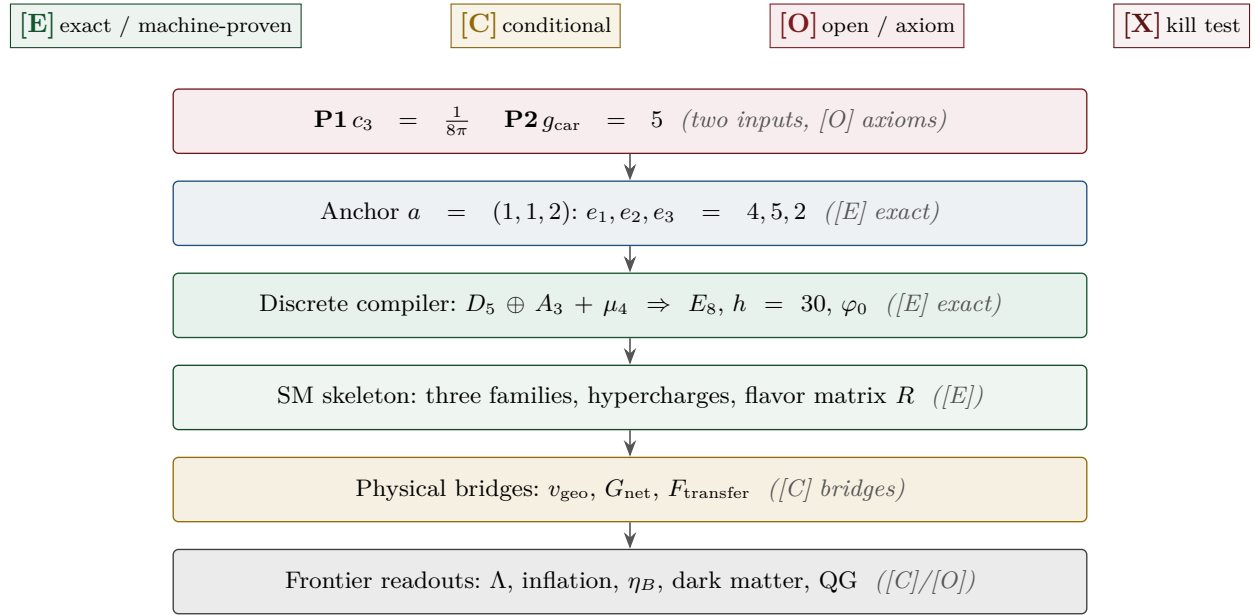
**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the **#print axioms** check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route **seamResidualClosed'** (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of **verification/status\_ledger.csv** (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in **verification/status\_ledger.csv**.



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**Reading the markers.** Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g.  $240 = 16 \cdot 5 \cdot 3$ ), a Lie/lattice theorem (e.g.  $D_5 \oplus A_3 + \mu_4 = E_8$ ), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice /

Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Physical — under hypotheses        | RG masses, QG    QG.AMB.01                   |
| Numerical — fit-free fixed point   | $\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01 |
| Lattice — Lie / glue / root system | $E_8=(D_5\oplus A_3)+\mu_4$ E8.GLU.01        |
| Formal — Lean / exact script       | P2 algebra    FORM.P2.02                     |
| Axiom — declared inputs            | P1 $c_3$ , P2 $g_{\text{car}}$ AX.P1.01      |

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function  $e_k(a)$  of the anchor  $a = (1, 1, 2)$ ; (ii) a power sum  $p_n(a)$  or difference  $p_{n+1}-p_n$ ; (iii) a determinant, spectrum, Smith normal form or trace of  $(R, Q, K, L)$ ; (iv) an anchor quadratic form  $u^\top M v$  with  $u, v \in \{1, a\}$ ; (v) a Plücker coordinate of the anchor plane  $\langle 1, a \rangle$ ; (vi) an  $E_8$  Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

**Reviewer path** (read in this order; the rest is support):

1. the architecture: the two axioms  $\{c_3, g_{\text{car}}\}$  and the two-engine picture (§1, the figures);
2. the  $E_8$  glue  $E_8=(D_5\oplus A_3)+\mu_4$  — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the  $\alpha^{-1}$  fixed point plus ablation (`v3_em_alpha.py`);
4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

## 1 What this is about: the jump in construction principle

### In plain words

**What we found.** TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number  $\frac{1}{8\pi}$  and a five-slot carrier) — builds the exceptional symmetry  $E_8$ , and  $E_8$  is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and

boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

**Why this is simple.** There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. Pushed to the end ([[verification/v353\\_selfloop\\_capstone.py](#)]), this means TFPT is best read not as a *linear* theory (axioms  $\rightarrow$  theorems) but as a *closed self-consistent loop* whose outputs fix its own inputs ( $g_{\text{car}}=5$  is the largest prime of the output  $h(E_8)=30$ ; the 8 in  $c_3$  is rank  $E_8$ ), with a *unique* fixed point ([v6/v56](#)) and *zero* free adjustable dimensionless parameters. Even  $\pi$  is not a free dial: it is the fixed geometric constant of the boundary 2-sphere ( $\oint_{S^2} K = 4\pi$ ), the one transcendental the continuous boundary contains. “Which is the axiom?” is the wrong question for a loop — every part is fixed by the whole; the one remaining structural question, whether the loop also closes in the *continuum* (the keystone [SEAM.EQUIV.01](#)), now carries two named routes (route split 2026-07-22, an ID restructuring with *no* status change in substance): its MMST route [SEAM.EQUIV.MMST.01](#) is *closed modulo cited theorems* — the lattice model and  $S_3$  stack pin it at every computable level and it is Lean-formalised ([FORM.SEAM.MMST.01](#)); only the cited continuum-existence theorems stay [\[O\]](#) ([v367/v368](#), [v376–v379](#), [v336](#)), their extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input reduced to invariant level ([v469](#)) — while its twistor route [SEAM.EQUIV.TWISTOR.01](#) (the open Costello–Li construction, prepared by [CELEST.SEAM.01](#)) stays [\[O\]](#); the parent [SEAM.EQUIV.01](#) is closed if either route closes and, as an unconditional claim, stays [\[O\]](#).

**Sharper still: one anchor  $a = (1, 1, 2)$ , and  $E_8$  as a compiler hull ([E])**

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity / splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums*  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the big Lie data directly:  $p_1=4=|\mu_4|$ ,  $p_2=6=|R^+(A_3)|$ ,  $p_3=10=\mathcal{A}_\Lambda$ , and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder  $p_{n+1} - p_n = 2^n$ . So the inputs collapse from  $\{c_3, g_{\text{car}}\}$  to  $\boxed{a = (1, 1, 2) \text{ plus } \pi}$ . [[verification/v23\\_anchor\\_generator.py](#)]

And  $a$  is itself *half-forced* ([E]/[C]): on  $\mathbb{P}^1$  every bundle splits (Birkhoff–Grothendieck), so once  $|\mu_4| = 4$ ,  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  are set, the three positive anti-degrees are positive integers summing to 4, and  $4 = 2 + 1 + 1$  is the *unique* partition into three positive parts — machine-checked with all its negative controls as the partition corollary ([[verification/v491\\_p2\\_partition\\_corollary.py](#)], [P2.PARTITION.01](#)): the unique multiset  $\{1, 1, 2\}$  forces  $e = (4, 5, 2)$ , hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = 3$  as *corollaries* (dropping positivity, the sum, the part count or integrality each destroys uniqueness — Mehta–Seshadri-rational weights give  $e_2 = \frac{16}{3}$  or  $\frac{44}{9} \neq 5$ ), sharpening [v53](#) which needs all three targets. The typing itself is now hardened ([[verification/v499\\_p2\\_weights\\_deligne\\_bg.py](#)], [P2.TYPING.01](#)): given the four marks, rank 3, the cusp class and unitarity (U), the Deligne canonical extension of the flavor connection has  $\deg E = -4$  (residue traces) and Mehta–Seshadri stability forces  $h^0(E) = 0$ , so integrality (Birkhoff–Grothendieck), positivity and the sum 4 are *theorems* — the residual shrinks to the module identification and (U), both [\[C\]](#); and that module-identification rest



is now *one* residual with the QGEO modulus rest (the order-4 clock = the Coxeter/cusp-class carrier datum; [verification/v503\_qgeo\_emergence\_light.py], QGEO.EMERGE.LIGHT.01, no marker moves). That common residual is itself now reduced to *one alignment bit* — the collar deck central in the mark- $D_4$ ; the clock's *order* is fermionically forced,  $U^2 = (-1)^F$  exactly ([verification/v506\_seam\_clock\_rigidity.py], SEAM.CLOCK.RIGIDITY.01, markers unchanged); and the bit is no tautology — the deck's *class* is derived (every mark-free collar deck is a half-period translation of the seam double), only its *position* stays the [C] input ([verification/v507\_seam\_bit\_origin.py], SEAM.BIT.ORIGIN.01, markers unchanged); and that position half is now *topology* — the covering deck is free on the seam circle, the edge class is excluded, and the bit reduces to the square-modulus datum  $\tau = i$  alone ([verification/v510\_seam\_bit\_freedom.py], SEAM.BIT.FREEDOM.01, markers unchanged); and that modulus datum now carries an equivalent *discrete* form — flag transitivity of the four marks (marks *and* their two sides indistinguishable,  $V_4 \rightarrow D_4$ )  $\iff \tau = i$ , a 13-fold exact equivalence web (since 15-fold by v528 facets #14/#15) in which bare mark-transitivity is automatic for every configuration and no local jet sees the side bit ([verification/v512\_seam\_tau\_flag.py], SEAM.TAU.FLAG.01, markers unchanged); free OS positivity is since documented as the *eighth* side-blind test on that scoreboard — the bit is not derivable from free RP/ $\Theta$  existence and remains genuine discrete input (v521, SEAM.BIT.RPBLIND.01, markers unchanged); the mark-decorated *state* class (twist/defect insertions — everything Wick-computable) is since documented as the *ninth*: the free-plus-twist class is exhausted (v525, SEAM.BIT.TWISTBLIND.01, markers unchanged); and the first genuinely interacting Fidkowski–Kitaev seam toy is the *tenth* side-blind test ( $9 \rightarrow 10$ ), while Kill-Test 2 fires at toy level under a typed fence (v529, SEAM.INT.FKTOY.01, markers unchanged); the straddle filter it produced, since run as a *selector* on the interacting mark family, keeps exactly *one* member alive — reflection positivity dynamically selects  $\delta = \pi/2$  with positive coupling, the same member that carries the twist-class order parameter: the first *positive* selection datum for the bit (v534, SEAM.STRADDLE.CONE.01, markers unchanged; toy-level evidence, not a derivation); the straddle law itself is since refined to the *odd-split law* — the Jaffe–Pedrocchi reflection-positivity condition, made decidable on the toy, predicted RP survival on center-straddled (2+2) cuts and direct measurement confirmed it: a straddled cut kills RP iff the split is odd (1+3) — while the quartet-avoiding cut family generates the full even seam algebra and carries a *positive* OS/GNS inner product at  $g = 1$ : the Kill-Test-2 counterweight is narrowed (an OS-viable cut geometry selected by the toy's own clock), not removed ([verification/v972\_seam\_interaction\_front.py], toy level, fence inherited, markers unchanged). The bit itself is now the *twist-class choice*, in principle interferometrically readable (v528, SEAM.BIT.TWISTCLASS.01; it remains formal input). Temperature joins geometry and anomaly as the *third leg* of  $c_3$  (v526, SEAM.THERMAL.KMS.01, [C]-typed reading). The genuine remaining axiom is therefore not “why  $a = (1, 1, 2)$ ” but “why the carrier  $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with  $c_3$ ), i.e. the weight-*typing* postulate itself, is the [O] input layer (AX.P2.01 stays a declared axiom). The consolidated reduction state (2026-07-23): the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology), *one discrete symmetry-lift bit* — the twist-class choice (flag transitivity  $\iff \tau = i$ ; equivalently the clock, the CM-fixed half-period, or the non-split extension class on the free seam circle) — and  $\pi$  — a reduction state, not an abolition of the axiom: AX.P2.01 remains declared, and the [C] identifications R1/R2 of v499 remain conditional.

*Two readings, one source (no contradiction).* P1 and P2 are the *operational compiler inputs*; their anchor provenance is  $a = (1, 1, 2)$  plus  $\pi$ . The *single structural bedrock postulate* is the keystone SEAM.EQUIV.01, whose conformal-deck face QGEO.SYM.01 (the carrier  $\mu_4$  clock is the seam's conformal deck, see tfpt\_research\_contracts) forces the  $\mu_4$  deck, hence  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  and the tripartition  $4 = 2+1+1$ , of which  $c_3$  and  $g_{\text{car}}$  are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication are now Lean-formalised (FORM.QGEO.02/FORM.QGEO.01); the postulate itself stays [O].

*One source, four readings (the cleanest origin).* The four-point seam divisor  $\mu_4$  on  $\mathbb{P}^1$  generates the

whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates  $D_5$  itself ([`verification/v197_rr_carrier_clifford_d5.py`]):  $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor. So  $\mu_4$  is not merely glue: it is the *common source* of  $A_3$ ,  $D_5$ ,  $c_3$  and  $E_8$  ([`verification/v189_riemann_roch_carrier.py`]) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

**What  $E_8$  is (and is not).** In TFPT  $E_8$  is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct  $E_8$  gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal  $E_8$  world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain  $E_8$  as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale;  $E_8$  checks the consistency.

i.e. the discrete content (carrier  $D_5 \oplus A_3 + \mu_4$ , the operators  $R, Q, K, L$ ) flows from  $a = (1, 1, 2)$ ; the continuous content ( $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0$ ,  $\alpha^{-1}$ , the exponential scales) flows from  $\pi$ ; and  $E_8$  is the unimodular *checksum container*, not the world group.

**The anchor ratio triad** ([`verification/v119_review_validation_2.py`]). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly  $(2, 5) = (e_3, e_2)$  with inflection  $\frac{7}{2} = \frac{e_3 + e_2}{2}$ : the fractions  $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$  are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]:  $\Omega_{\text{adm}} = 2p_1p_2 = 48$ ,  $10b_1 = 1 + p_1p_3 = 41$ ,  $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$ ,  $h(E_8) = e_3p_0e_2 = 30$ ,  $\text{rank } E_8 = p_0 + e_2$ .

$$a = (1, 1, 2) \quad [E]$$

#### elementary $e_k(a)$

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

#### power sums $p_n = 2 + 2^n$

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

#### derived motifs

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

## The companion documents

The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

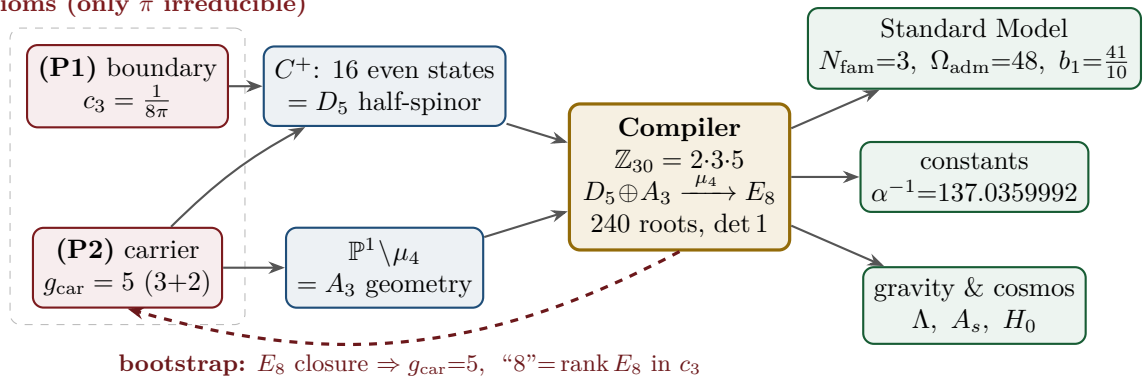
| Document                          | What it contains (former notes now as parts)                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tfpt_1_architecture_e8            | <b>Core.</b> The two axioms $\{c_3, g_{\text{car}}\}$ , the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$ , the explicit $D_5\oplus A_3+\mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).                                                                        |
| tfpt_2_standard_model             | <b>Standard Model.</b> The full SM in one $\varphi_0^{\text{ret}}$ -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ $n_s, r$ , Starobinsky $M$ , derived $\theta_{12}$ , quark $c$ , the H2 splitting type, $\Lambda$ -typing (Part III).                                                |
| tfpt_3_e8_audit_bootstrap         | <b><math>E_8</math> audit &amp; bootstrap.</b> The seven $E_8$ slices as an audit raster ( $E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III).                                                                                     |
| tfpt_4_frontier                   | <b>Frontier.</b> The honest status of $\eta_B$ , $m_p/m_e$ , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.                                                                                                                                                                                                                                                 |
| tfpt_5_redteam                    | <b>Red Team.</b> The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.                                                                                                                                                                                                                                              |
| tfpt_research_contracts           | <b>Research contracts.</b> The open gates as numbered contracts with lemma chains and closing theorems: ( $U_{\text{wall}}$ ) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).                |
| tfpt_horizon_readouts<br>(App. H) | <b>Appendix H — horizon unit system (reframe, not new physics).</b> One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ( $1920= W(D_5) $ , $ \mu_4 =4$ ); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138). |
| origin_theory                     | <b>Origin Theory.</b> Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and $\pi$ (the dimensionful $v_{\text{geo}}$ and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.                 |

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure  $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$  in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the  $\theta_{12}$  texture and the quark  $c$  in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a  $U_{\text{point}}$  anchor;  $\theta_{12}$  the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

## 2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only  $\pi$  irreducible)



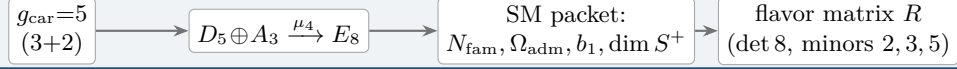
**Key statement (the loop closes):** previously  $D_5$ ,  $A_3$  and  $E_8$  sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the  $E_8$  closure feeds back and *fixes the inputs*.  $g_{\text{car}} = 5$  is the unique rank-fill/Coxeter/Pascal solution ( $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ), and the integer 8 in  $c_3 = \frac{1}{8\pi}$  is rank  $E_8 = h(D_5) = \det R$ . **Inputs and output form an overdetermined fixed point** — the structure does



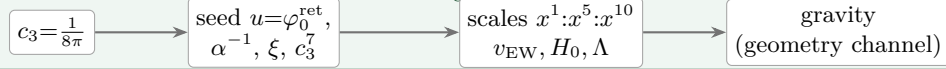
not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous  $\pi$  (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

### Engine 1 — discrete closure (from $g_{\text{car}} = 5$ )

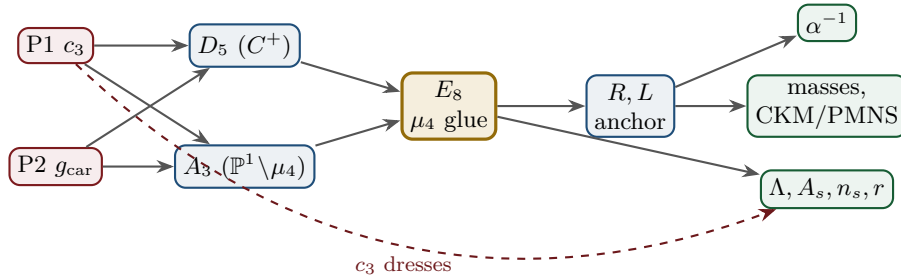


### Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$ )



## 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



| Node           | Definition                                               | Inputs      | St.     | Output                              | Failure mode                |
|----------------|----------------------------------------------------------|-------------|---------|-------------------------------------|-----------------------------|
| P1             | RP boundary kernel, $c_3=1/(8\pi)$                       | — (axiom)   | [O]     | $c_3$                               | no reflection-positive seam |
| P2             | 5-slot carrier $g_{\text{car}}=5$                        | — (axiom)   | [O]     | carrier 3+2                         | wrong family/charge lattice |
| $D_5$          | $C^+$ even-Hamming = half-spinor 16                      | P2          | [E]     | $\dim S^+=16, Y$                    | group/matter mismatch       |
| $A_3$          | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.       | P1          | [E]     | $N_{\text{fam}}=3, \mathbb{Z}_6$    | wrong multiplicity          |
| $E_8$          | $\mu_4$ glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$ | $D_5, A_3$  | [E]     | 240, 248                            | not even-unimodular         |
| $R, L$         | residue + winding matrix, anchor (1, 1, 2)               | $A_3$       | [E]     | $\det 8, \text{ minors } (2, 3, 5)$ | wrong $D_6$ branch          |
| $\alpha^{-1}$  | root of $F_{U(1)}(\alpha)=0$                             | $c_3, M=41$ | [E]     | 137.0359992                         | no/second root (excl.)      |
| masses         | $\varphi_0^{\text{ret}}$ -ladder $\lambda_V^L \Lambda$   | $R, L, c_3$ | [E]/[C] | 9 masses, mixings                   | hierarchy mismatch          |
| $\theta_{12}$  | TBM + seam $\varepsilon=c_3$                             | $R, L$      | [E]/[C] | 0.307                               | seam-misalign. lemma fails  |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron                               | $c_3$       | [C]     | $\Lambda, A_s, n_s, r$              | ambient RP fails            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

## The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.4 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the  $R + R^2$  spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the  $R$ -bridge amplitude normalisation  $U_{\text{point}}$  *reduces to*  $v_{\text{geo}}$  (Gate 1 closed, [v75](#) — the same anchor as  $1/G$ ), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, [v76](#)) — which is in turn the rigorously-constructed  $(E_8)_1$  lattice net ([v77](#)), so  $G_{\text{metric}}$  is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale  $v_{\text{geo}}$  — the *dimensional-analysis floor* that no theory escapes ([v78](#)), shared by flavor and gravity — the boundary net identification (now *closed modulo cited theorems* on its MMST route `SEAM.EQUIV.MMST.01`, [v367/v368](#), [v376–v379](#), Lean `FORM.SEAM.MMST.01`; the parent `SEAM.EQUIV.01` stays [\[O\]](#)), and the named QFT/cosmology transfer interfaces ( $F_{\text{frontier}}$ ).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).** (i) *Parameter-freeness is a theorem under the named hypotheses*: the boundary transport is gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ ), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([\[E\]](#) for the attractor, [\[verification/v56\\_unique\\_attractor.py\]](#); the *physical identification* of the transport operator stays [\[C\]](#)), reinforcing the static bootstrap web ( $g_{\text{car}} = 5$  forced three ways *given* the  $E_8$  closure rank — a consistency loop, see red team Target B — [\[verification/v6\\_bootstrap.py\]](#), [\[verification/v14\\_carrier\\_uniqueness.py\]](#)). (ii) *Newton’s constant is not an independent input*: fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{Pl}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([\[E\]/\[C\]](#), [\[verification/v60\\_lambda\\_metrology\\_branch.py\]](#)), not predicted from pure numbers. What stays irreducible is then just  $\pi$ , the discrete  $E_8$ -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale  $v_{\text{geo}}$  — and that single scale is shared by *both* the flavor sector ( $U_{\text{point}} \rightarrow v_{\text{geo}}$ , [v75](#)) and gravity ( $1/G$ , [v68](#)): the two  $[A]$  anchors are *one* (dimensional analysis; no metre from pure mathematics).

## The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels  $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$  are kept only for ledger continuity). In current terms:  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$  — the flavor ratios are *closed* (Plücker/Readout Rigidity, [v49/v71/v75](#)), what remains is *one dimensionful unit*, not a flavor gap;  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$  — reduced to a single boundary-net theorem, “the seam–Calderón net is the index-4  $\mu_4$  simple-current extension of the carrier net” (holomorphy then follows, [v89/GATE.METRIC.06](#)); since the QBL chain ([v108–v113](#)) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets ([v110](#)), the certified channel counts the code identically ([v112](#)), pair transport is minimally complete ([v111](#)), and the one quasi-free kernel — rank =  $c$ : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits ([v113](#)), so closing  $G_{\text{net}}$  also closes the carrier selection [\[E\]](#);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$  — named QFT/cosmology *transfer* interfaces (Koide source→pole,  $\eta_B$  Boltzmann, axion relic,  $m_p/m_e$ ), never compiler powers; the four are one typed

functor with a four-axiom contract (`CONTRACT.F.01`, [v213](#)) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with `F_pole` at the *seam* rate  $(2/3)^6$  and the other three external ([v303](#)). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion  $G_{\text{net}}$  is the index-4  $\mu_4$  simple-current extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$  (Simple-Current Extension Theorem, [v154](#), exact: index 4,  $c=8$ ,  $\mu=1 \Rightarrow$  holomorphic  $\Rightarrow E_8$ ); net existence and full-cone reflection positivity are discharged to [\[E\]](#) ([v175](#)); and the **No-Unit Theorem** ([v153](#)) makes  $v_{\text{geo}}$  an irreducible metrology primitive ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). The *one* seam-side identification, packaged as the single named theorem `SEAM.EQUIV.01` — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — is now *closed modulo cited theorems on its MMST route* (`SEAM.EQUIV.MMST.01`; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [\[O\]](#)); the explicit lattice model ([v367/v368](#)) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; [v376–v379](#)) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01`, with only the cited MMST/OS continuum-existence theorems left [\[O\]](#) ([v336](#)); its conformal-deck premise `QGE0.SYM.01` is its *downstream* definitional face (per [v323](#); the residual is Lean-pinned to named cited steps, [v308](#)). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

| Item                                  | Status           | Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter-freeness (self-consistency) | [E]              | gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Newton constant $G_N$                 | [E]/[C]          | not <i>independent</i> : branch pins $M_{\text{Pl}}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ratio $c_u/c_d = \frac{55}{117}$      | [E]              | $g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$ ; rigid on the derived stratum (v49/v71), no solve                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| flavor absolute scale                 | [E]/[O]          | ratios <i>closed</i> (v49/v71); only the absolute scale remains $= v_{\text{geo}}$ (Gate 1, v75), the same anchor as $1/G$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $R + R^2$ gravity                     | [E]/[C]          | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Einstein eq. (covariant, local)       | [E]/[C]          | fixed-volume entanglement equilibrium $\Rightarrow$ <i>full</i> $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with BOTH coefficients parameter-free: $8\pi = 1/c_3$ (v358, thermo=geo via $ \mu_4 = \mathbb{Z}_2 \chi=4$ ) and $\Lambda$ from $\alpha$ (v60); Einstein tensor via Lovelock, conservation an output (v359); residual = equation-of-state status + $v_{\text{geo}}$ , global measure (C7) now a [C] redundancy (v369)                                                                                                                                                                                      |
| physical QG sector                    | [E]/[C]          | gap-decoupled admissible measure ( $\Delta_{\text{eff}}=1.648>0$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ambient QG measure                    | [C] (redundancy) | <b>certification object, not missing dynamics</b> (v369; conditional on SEAM.EQUIV.01+Bisognano–Wichmann): holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$ ) and factors into the $A_2$ net-existence + the keystone SEAM.EQUIV.01 (MMST route closed mod. cited; parent [O]) — zero new gates (v160–v165) |
| $v_{\text{geo}}$ metrology unit       | [O](primitive)   | No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu$ a ratio — one unit, irreducible                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Koide                                 | [C]              | $Q_{\text{src}} + \varphi_0/24$ source $\rightarrow$ pole: typed runnable solver with kill test (QED-excluded, v371)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| axion DM                              | [C]              | $f_a = M_{\text{scal}}/128$ : finite- $T$ spine-branch typed runnable solver (v373)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\eta_B$                              | [C]              | readout closed; integrated BDP leptogenesis Boltzmann ODE, typed runnable solver (v372)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $m_p/m_e$                             | [C]              | cross-sector QCD/EW ratio: typed runnable solver with uncertainty budget (v374)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $N_*$                                 | [C]              | reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $a = (1, 1, 2)$                       | [E]/[C]          | forced by $4 = 2 + 1 + 1$ , conditional on the carrier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| carrier selection $g=5$ (QBL)         | [E]/[C]          | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$ ); residue = the $G_{\text{net}}$ premise itself — gate closes $\Rightarrow$ carrier [E]                                                                                                                                                                                                                                                                                                                                                                                                       |
| selector establishment $R4'$          | [E]/[C]          | $(d, n) \Rightarrow R$ columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| sheet/parity (Q-geometry derived)     | [E]/[C]          | $H^1$ characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius $D_4$ matched parity by parity (v146); residue = the module identification, folded into SEAM.EQUIV.01                                                                                                                                                                                                                                                                                                                                                                                           |
| seam quantum clock R1                 | [E]/[C]          | typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born <sup>2</sup> Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference                                                                                                                                                                                                                                                                                                                                                                                        |
| $E_8$ Lean                            | [E]              | hardening, not a blocker (script-certified)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Card addendum (2026-07-22): the  $a = (1, 1, 2)$  residual now carries the consolidated P2 reduction state — the position half of the alignment bit is topology (v510) and the modulus half is one discrete symmetry-lift bit — the twist-class choice (flag transitivity  $\iff \tau = i$ , v512/v528), so the dimensionless compiler inputs read *four marks + one discrete symmetry-lift bit (the twist-class choice, in principle interferometrically readable; flag transitivity  $\iff \tau = i$ )* +  $\pi$  (AX.P2.01 stays a declared axiom).

The former structural open core, the *single* named keystone SEAM.EQUIV.01 (the  $G_{\text{net}}$  inclusion: the raw RP seam =  $(E_8)_1$  at  $\tau=i$ ), is now *closed modulo cited theorems on its MMST route* (SEAM.EQUIV.MMST.01; the twistor route SEAM.EQUIV.TWISTOR.01 and the unconditional parent stay **[O]**): the explicit lattice model (v367/v368) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, it is Lean-formalised as FORM.SEAM.MMST.01, and only the cited MMST/OS continuum-existence theorems stay **[O]** (v336); its conformal-deck face QGEO.SYM.01 is a *corollary* (a conformal net’s vacuum is rotation-invariant by axiom, v323/[`verification/v335_seam_equiv_unify.py`]), not a second item.  $v_{\text{geo}}$  is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes.  $G_{\text{net}}$  is heat-kernel grounded (G2: the spectral action gives  $R + R^2$  structurally) and *gap-decoupled* (G5:  $\Delta_{\text{eff}}=1.648>0$  isolates the closed admissible sector from the ambient), so the ambient measure QG.AMB.01 is no longer carried as an open structural item: it is a **[C]** *redundancy* (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann), a certification object rather than missing dynamics — it does not affect any TFPT readout. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority on the  $v_{\text{geo}}$  scale anchor and the  $G_{\text{net}}$ /SEAM.EQUIV.01 inclusion theorem. [`verification/v36_spectral_action_g2.py`]

**Sharper still: the structural residual is *one* condition.**  $G_{\text{metric}}$ , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “*the seam carries no nontrivial abelian sector*” — holomorphy ( $\mu$ -index 1), a homology-sphere seam link ( $\Gamma$  perfect  $\iff 2I$ , [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force  $E_8$  ( $\#(\text{mark-1}) = |H_1| = 1$  only for  $E_8$ ; holomorphic  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic  $\iff \det K = 1$ , the extension tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  being anyon condensation (the Kitaev  $E_8$  state); the one analytic step “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” = the single named keystone SEAM.EQUIV.01 (with QGEO.SYM.01 its downstream conformal-deck face) is now *closed modulo cited theorems* on its MMST route SEAM.EQUIV.MMST.01 (v367/v368, v376–v379, Lean FORM.SEAM.MMST.01; cited continuum existence **[O]**, v336; the parent SEAM.EQUIV.01 stays **[O]**). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ( $f_2/f_0 = 1$ , v259), the seam, the carrier-16 and  $E_8$  live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone SEAM.EQUIV.01 and adds *no new open item*, while the ambient quantum-gravity measure is a **[C]** redundancy (v369), not a separate structural gap. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

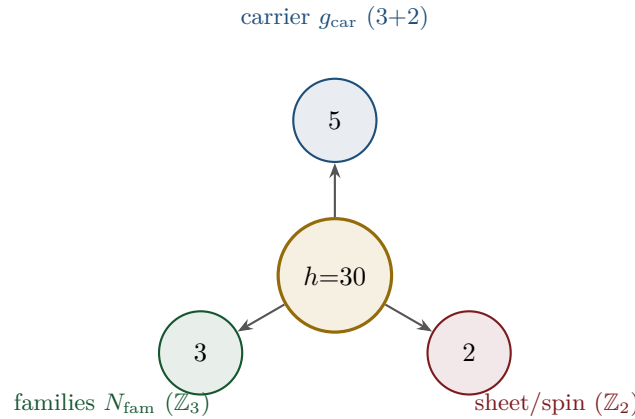


### Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ( $N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$ , the flavor matrix, the  $E_8$  glue, the scale grammar) is a *consequence*. And even those two tighten: the bootstrap note shows  $g_{\text{car}} = 5$  and the integer 8 in  $c_3$  are *overdetermined  $E_8$ -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5;  $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$ ), leaving *one* genuine continuous primitive:  $\pi$  from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence:  $E_8$  is the exceptional top of the tower of finite  $SU(2)$  subgroups, so its branch data is the icosahedron and the three atoms (2, 3, 5) are its rotation-axis orders — the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees equal to the affine- $E_8$  Kac marks,  $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$  ([[verification/v219\\_icosahedral\\_mckay.py](#)]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [[verification/v6\\_bootstrap.py](#)] [[verification/v14\\_carrier\\_uniqueness.py](#)] [[verification/v23\\_anchor\\_generator.py](#)] [[verification/v219\\_icosahedral\\_mckay.py](#)]

## 4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of  $E_8$  is  $h = 30 = 2 \cdot 3 \cdot 5$ . Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:



From this the compiler reads, among others:

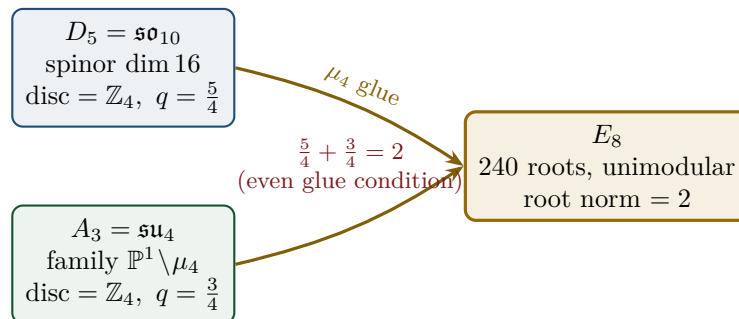
- $\dim S^+ = 1 + 10 + 5 = 16$  (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$  appears in family, action ladder and the  $E_8$  root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$  and  $\dim E_8 = 240 + (5 + 3) = 248$ . [E]
- the  $\mathbb{Z}_{30}$  Coxeter element  $\leftrightarrow$  the cyclotomic polynomial  $\Phi_{30}$ ; the corner rotation  $z \mapsto iz$  on  $\mathbb{P}^1 \setminus \mu_4$  is the  $A_3$  Coxeter element. [E]
- **why these three primes:**  $E_8$  is the icosahedral top of the McKay tower ( $2I$ , order 120, irrep degrees = affine- $E_8$  marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the  $E_8$  exponents  $(\mathbb{Z}/30)^\times$  carry an order-4 totative character  $\langle 7 \rangle$  — the same  $\mu_4$  seam clock. [E] [[verification/v219\\_icosahedral\\_mckay.py](#)] [[verification/v223\\_coxeter\\_totative\\_clock.py](#)]
- **a network reading (computational substrate).** On the same McKay (2, 3, 5) graph the affine- $E_8$  marks are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A + 2I)/4 \cdot m$  (it converges from *any* positive start to (1, 2, 3, 4, 5, 6, 4, 2, 3)); the spherical tower terminates at  $E_8$  and the cascade  $E_8 \supset E_7 \supset E_6 \dots$  is nested-subgraph coarse-graining. So  $E_8$  reads as the *universal attractor of a minimal* (2, 3, 5) *network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing

the *full* structure (lattice + recovery + readouts) stays open, and  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction. [O] [verification/v298\_e8\_network\_attractor.py]

- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate  $Am \leq 2m$ , Collatz–Wielandt = Smith’s  $\rho \leq 2$ ; the Perron eigenvector enters only as an audit) *reaches*  $E_8$  as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary  $\hat{E}_8$ ) — but *only* from a 3-arm  $(2, 3, \cdot)$  seed (the controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$ , so neither family ever reaches  $E_8$ ). So  $P_2$  is exactly the irreducible seed on which the rule reaches  $E_8$  as the last finite point — a second network reading of the  $(2, 3, 5)$  hull, not a derivation of  $P_2$ . [O] [verification/v299\_hypergraph\_0d\_autonomous.py]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ( $Am=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5)$  = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate  $(2/3)^6$  is *not* any eigenvalue of the adjacency and  $\varphi_0^{\text{ret}}$  is not a rational edge fraction. The cusp weight  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  is now *derived* from a minimal branching rule ([verification/v327\_hypergraph\_rewrite.py]); what remains non-graph-spectral for a *full*-structure rewrite at leading order is sharpened by [verification/v396\_phi0\_icosahedral.py] (tree-level  $\varphi_0^{\text{tree}} = F/(4h\pi)$ ); the puncture  $48c_3^4$  alone stays open. [O] [verification/v312\_hypergraph\_substrate.py]
- **one coupled local rule (carrier  $\times$  family).** On a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) a *single* coupled micro-step — network lazy diffusion on each cusp column plus the [verification/v327\_hypergraph\_rewrite.py] branching rule  $M$  on each node’s family vector — has joint attractor marks  $\otimes(w=0)$  and recovery rate  $(2/3)^6$  after one clock hand ( $2N_{\text{fam}}$  family steps); the [verification/v324\_hypergraph\_fiber.py] fiber product  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis, and the [verification/v299\_hypergraph\_0d\_autonomous.py] growth trajectory  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$  is unchanged with the fiber attached. [E] [verification/v395\_hypergraph\_coupled.py]
- **the seed at leading order.** The tree-level retained seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi)$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ ): equivalently  $(F/(g_{\text{car}}N_{\text{fam}}))c_3$ , with  $F/h = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$  the same survival ratio as [verification/v327\_hypergraph\_rewrite.py]; Gauss–Bonnet consistent with  $c_3 = 1/(|\mathbb{Z}_2|4\pi)$ . The puncture  $48c_3^4$  remains analytic, not a graph fraction. [C] [verification/v396\_phi0\_icosahedral.py]

## 5 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step:  $D_5$  and  $A_3$  have the *same* discriminant group  $\mathbb{Z}_4$ , and their discriminant-form norms are exactly two TFPT constants which add up to the  $E_8$  root norm.



### Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue index  $|\mu_4| = 4$ , and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$  the  $E_8$  root norm. Hence  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1\_e8\_glue.py]

## 6 How derivation now works: the derivation map

What can one concretely *compute* with  $\{c_3, g_{\text{car}}\}$ ? The following map summarises what is now a consequence rather than an assumption.

| Sector                       | How it now arises                                                             | Status  |
|------------------------------|-------------------------------------------------------------------------------|---------|
| gauge group / families       | carrier $3+2$ , $\dim S^+=16$ , $N_{\text{fam}}=3$ , $\Omega_{\text{adm}}=48$ | [E]     |
| hypercharge                  | roots of the charge polynomial $bsY^2 + (s-b)Y - 1$                           | [E]     |
| $U(1)$ index $b_1$           | $b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$    | [E]     |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha) = 0$                                         | [E]     |
| flavor matrix                | transport theorem (H1 proved, H2/H3 force the matrix)                         | [E]/[C] |
| electroweak scale            | $v_{\text{EW}} \sim e^{-\alpha^{-1/5}}$ (carrier divides $\alpha^{-1}$ by 5)  | [E]     |
| cosmological constant        | $\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)          | [E]/[C] |
| Hubble scale                 | $H_0 \sim \sqrt{\Lambda}$                                                     | [C]     |
| inflation amplitude          | $A_s = \frac{N_*^2}{24\pi^2} c_3^7$ in the $R^2$ branch                       | [C]     |
| $E_8$                        | $D_5 \oplus A_3 + \mu_4\text{-glue}$                                          | [E]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

**Notable:** the scale grammar is *one* exponential engine. The same number  $\alpha^{-1} \approx 137$  generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

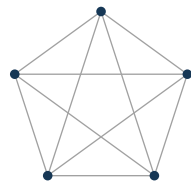
**Typing note on  $\Lambda$  (a deliberate danger zone).** “ $\Lambda$ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent  $e^{-2\alpha^{-1}}$  (an exact [E] readout), (ii) the dimensionless density quotient  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$  (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

### Cosmology is the Pascal action grammar of the carrier

With  $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2\bar{M}_{\text{Pl}})$  the cosmological readouts are a *power tower* on the five-slot carrier graph  $K_5$ :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit  $x^1$ , Hubble is the product over the  $\binom{5}{1} = 5$  vertices,  $\Lambda$  is the product over the  $\binom{5}{2} = 10$  edges — the *same* Pascal triple that reads one generation, the action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . “Hubble = slot product,  $\Lambda$  = pair product.” [E]



$K_5$

- $\binom{5}{1} = 5$  vertices — action  $x^1$  (Hubble slot)
- $\binom{5}{2} = 10$  edges —  $\Lambda$  density pair  $x^{10}$
- $\binom{5}{3} = 10$  triangles — dual sector
- $\binom{5}{4} = 5$  complements — inverse sector
- $\binom{5}{5} = 1$  determinant — closure

$K_5$  on the five carrier slots; the action powers  $x^1, x^5, x^{10}$  are its  $\binom{5}{k}$  faces (mnemonic; the physical scale map stays  $[C]$ ).

### The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths  $L_{f,j}$  are the *fixed residue matrix of the compiler*, whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers ( $\text{tr} = 9 = N_{\text{fam}}^2$ ,  $\det = 8 = h(D_5)$ , principal 2-minors 2,3,5 with product  $30 = h(E_8)$ ). Through the  $\varphi_0^{\text{ret}}$ -ladder  $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c (\varphi_0^{\text{ret}})^k$  all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark mass ratios* are closed too (integer Plücker readouts on the derived selector stratum, **v49/v71**), with only the *absolute* amplitude scale reducing to the single dimensionful anchor  $v_{\text{geo}}$ . **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional;  $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$  are RG scheme-layer ( $m_H$  rests on the closed UV quartic  $\frac{1}{16\pi^2}$ ); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; see below). (*Full derivation: document 2, **tfpt\_2\_standard\_model**.*) [[verification/v18\\_quark\\_yukawa.py](#)] [[verification/v20\\_lepton\\_c\\_derivation.py](#)] [[verification/v46\\_grand\\_mass\\_volume.py](#)]

## 7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*,  $\alpha^{-1}$  is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the  $C^+$  spinor and the puncture geometry  $\mathbb{P}^1 \setminus \mu_4$  produce  $D_5$  and  $A_3$ , glued to  $E_8$ ; *topologically*, the common discriminant  $\mathbb{Z}_4$  (the  $\mu_4$  winding) forces the glue and  $\Phi_{30}$ /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions  $\Rightarrow$  low overfitting risk), (ii) *rigid* (the glue condition  $5/4 + 3/4 = 2$  and the discriminant  $\mathbb{Z}_4$  leave little freedom), and (iii) *checkable* (machine-checkable  $E_8$  appendix, explicit fixed-point equation for  $\alpha^{-1}$ ). More explained from less — exactly the criterion that makes a theory more probable.

## 8 What role $E_8$ now plays exactly

$E_8$  is no longer the starting point but the *output format*:

- **Bookkeeping:**  $E_8$  is the unimodular container in which carrier ( $D_5$ ) and family ( $A_3$ ) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ( $\det 1$ ) is a hard condition; it closes only because the discriminant-form norms  $5/4$  and  $3/4$  (both already present in TFPT) add up to 2.  $E_8$  thus tests the internal coherence of the two atoms.
- **Not an end in itself:**  $E_8$  does not explain “everything”; it is the smallest even unimodular hull of the datum  $D_5 \oplus A_3$ . The physics sits in the *two inputs*, not in  $E_8$ .

**New — the secondary atlas.** Beyond that,  $E_8$  is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (**tfpt\_3\_e8\_audit\_bootstrap**, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that  $E_6 \times A_2$  reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2 = h(D_5)$ . The discipline rule is: *every load-bearing number must appear in at least one  $E_8$  projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

## 9 What this means for the theory-of-everything question

### The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic  $\leftrightarrow$  unitary representation); the algebraic selector  $R$ , the  $Q$  spectrum and the splitting type are fixed (now *derived*:  $\det R = 8$  lattice,  $\text{Spec}(Q_+)$  from  $D_4$ , v69/v70; since v136/v139 the dual pair  $(d, n)$  forces  $R$  *columnwise*,  $d$  is pure anchor data, and  $\text{Spec}(Q_+) = (1, 2, 3)$  is the  $A_3$ -exponent grading on the seam curve’s cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ( $c_u/c_d = \frac{55}{117}$  etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ( $U_{\text{point}}$ , an anchor; Research Contract 1). [E]/[O]
2. **Gravity — the local field equation is parameter-free.** The metric-sector *field equation* is supplied by the entanglement first law  $\delta S = \delta \langle K \rangle$  at fixed *volume* (Jacobson 2015): equilibrium forces the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $8\pi = 1/c_3$  (v358) and  $\Lambda$  from  $\alpha$ ,  $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60) — the Einstein tensor (not Ricci) forced by Lovelock, so  $\nabla^a T_{ab} = 0$  is an *output* (v359). The  $R + R^2$  scalaron is the *action*/inflation layer (the next-order term of the same equilibrium route, v360), and  $v_{\text{geo}}$  is the theory’s single dimensionful input (v364). What remains is *not* the field equation but the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate action — Bianconi’s entropic action, arXiv:2408.14391 — is quantified in v473 without changing this typing) and the one unit  $v_{\text{geo}}$ ; the ambient measure QG.AMB.01 is now a [C] redundancy (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann), not missing dynamics, and *perturbative* graviton unitarity is established — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304), the spin-2 sector ghost-free via the entire KMS form factor (v370) with the truncation tower decoupling (v380). [E] (local field equation) / [C] (perturbative unitarity, ambient redundancy)
3. **Strong CP — decoupled.**  $\theta_{\text{eff}} = 0$  follows from three structural facts ( $\gamma_5$ -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap  $6 \log \frac{3}{2}$  of the  $C_6$  transfer operator (eigenvalues  $(k/3)^6$ ), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/ $E_8$  live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single keystone SEAM.EQUIV.01, whose MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems* (lattice +  $S_3$  stack, Lean FORM.SEAM.MMST.01; v367/v368, v376–v379; the unconditional parent stays [O]). *What this does not fully close*: the full measured pole masses (the absolute scale  $v_{\text{geo}}$  + standard RG), the absolute QCD scales ( $m_p/m_e$ , now a typed [C] transfer, v374), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT — the continuum limit, whose cited existence theorems stay [O] — with the ambient measure a [C] redundancy (v369). [E] (transfer model + boundary relative object) / [C] (continuum modulo cited, transfers, ambient redundancy)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (3357 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, tfpt\_2\_standard\_model, Part III.*)



**The closure ledger: what is done, and what genuinely remains**

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

| TOE piece                                                                                                        | Status  | If not complete: what closes it                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gauge group, $N_{\text{fam}}$ , hypercharge, $b_1$                                                               | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $E_8 = (D_5 \oplus A_3) + \mu_4$ , group closure                                                                 | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| bootstrap $(g, \mu) = (5, 4)$ , “8” in $c_3$                                                                     | [E]     | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$ )                                                                                                                                                                                                                                                                                                                                                                                                                             |
| fine structure $\alpha^{-1}$                                                                                     | [E]     | — <b>existence+uniqueness proved</b> ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]                                                                                                                                                                                                                                                                                                                                                                    |
| fermion masses ( $\varphi_0^{\text{ret}}$ -ladder)                                                               | [E]/[O] | leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor                                                                                                                                                                                                                                                                                                                                                                                         |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)                                                    | [E]/[C] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]                                                                                                                                                                                                                                                                                                                                                         |
| flavour matrix / H2                                                                                              | [E]     | dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); $R$ forced columnwise by $(d, n)$ (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map                                                                                                                                                                                                                                          |
| seam quantum clock (R1)                                                                                          | [E]/[C] | typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)                                                                                                                                                                                                                                                                                             |
| inflation $A_s, n_s, r$ , scalaron $M$                                                                           | [C]     | scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; $n_s, r$ are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)                                                                                                                                                                                                                                                                                                                                                    |
| boundary QFT (Modular Spectral Closure)                                                                          | [E]/[C] | <b>one relative object</b> TFPT <sub>QFT</sub> : $D_F$ = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ $E_8$ on one K3 (v260), certified (v261) — closed modulo SEAM.EQUIV.01, whose MMST route SEAM.EQUIV.MMST.01 is closed modulo cited theorems (v376–v379; parent [O])                                                                                                                                                               |
| perturbative 4D S-matrix ( $S_{\text{pert}}$ )                                                                   | [E]/[C] | <b>Epstein–Glaser-constructible</b> (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278); the spin-2 graviton sector is perturbatively unitary too — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304/v370/v380)                                                                                                              |
| <i>the actual TOE remainder — dimensionful units, typed transfers, and the cited-theorem ceiling on the seam</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| QFT measure (admissible sector)                                                                                  | [E]     | <b>closed: RP from cusp spectrum, OS reconstruction</b>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| full measured pole masses                                                                                        | [C]     | ladder $\times v_{\text{geo}}$ , dressed by standard RG (no new physics); $v_{\text{geo}}$ over-determined (v274)                                                                                                                                                                                                                                                                                                                                                             |
| absolute QCD scales ( $m_p/m_e$ , $\Lambda_{\text{QCD}}$ )                                                       | [C]     | typed runnable transfer with uncertainty budget (v374, contract FR.MPME.01); not a compiler number                                                                                                                                                                                                                                                                                                                                                                            |
| full PMNS dynamics (mass scale, CP)                                                                              | [C]/[O] | angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute $\nu$ -scale = one seesaw ratio (v272); three named post-hoc [O] candidates: third-gen ladder-base (v467), $\Delta m^2$ ratio = $ J $ (v468), carrier-normalised one-parameter seesaw (v481) — record unchanged                                                                                                                                                                                |
| (1) SEAM.EQUIV.01 — one named theorem ( <i>raw seam</i> = $(E_8)_1$ at $\tau=i$ )                                | [O]/[C] | <b>route split 2026-07-22: closed if MMST or TWISTOR route closes. MMST route SEAM.EQUIV.MMST.01 closed modulo cited theorems:</b> lattice model (v367/v368) + $S_3$ stack (v376–v379) pin it at every computable level, Lean FORM.SEAM.MMST.01; only residual [O] = the cited continuum existence (v336); extension leg certified (v469). Twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, via CELEST.SEAM.01) [O] $\Rightarrow$ parent stays [O] as an unconditional claim |
| (2) QG.AMB.01 — ambient measure ( <i>redundancy, not missing dynamics</i> )                                      | [C]     | <b>discharged as a redundancy</b> (v369; v379 RP $\rightarrow$ OS), conditional on SEAM.EQUIV.01+Bisognano–Wichmann: a certification object, gap-decoupled (margin $1.648 > 0$ , v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B’s holomorphy bit ( $\det K=1$ , v277) is absorbed into SEAM.EQUIV.01                                                                                             |
| analytic boundary origin of $\pi$ ( $c_3 = \frac{1}{8\pi}$ )                                                     | [C]     | <b>hardened:</b> $c_3 = 1/( \mathbb{Z}_2  \oint_{S^2} K) = 1/(8\pi)$ ( <b>Gauss–Bonnet</b> ); “8” = rank $E_8$ ; $\pi$ stays the one primitive                                                                                                                                                                                                                                                                                                                                |
| $\eta_B, m_p/m_e$ , Koide, dark matter                                                                           | [C]     | typed runnable transfer solvers with kill tests (v371–v374, observatory CI v375; see document 4)                                                                                                                                                                                                                                                                                                                                                                              |

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one  $\varphi_0^{\text{ret}}$ -ladder (lepton rationals exact, quark  $O(1)$  digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* TFPT<sub>QFT</sub> (the *Modular Spectral Closure*, v258–v261:  $D_F$  a covariance induction, the cutoff the seam KMS weight, seam/carrier/ $E_8$  on one Kummer/K3) — closed modulo the single keystone SEAM.EQUIV.01. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder  $\times v_{\text{geo}}$  + standard RG, **[C]**), the absolute QCD scales ( $m_p/m_e$ , now a typed **[C]** transfer, v374), the complete PMNS dynamics (**[C]**/**[O]**), and the nonperturbative *ambient* measure QG.AMB.01, now itself a **[C]** redundancy (v369). (The metric-sector *field equation* is parameter-free at the local level — the full covariant  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  from fixed-volume entanglement equilibrium, v358/v359 — and the ambient *measure* is no longer an open structural item but a certification object, conditional on SEAM.EQUIV.01+Bisognano–Wichmann.) The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278), and the spin-2 graviton sector is perturbatively unitary too — the  $R^2/\text{Weyl}^2$  Stelle ghost is a Seeley–DeWitt *truncation artefact* (v304), ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370) with the truncation tower decoupling (v380); the whole bedrock reduces to one geometric postulate, *the raw seam is the flat  $\tau=i$  pillowcase* (v276), with Tier-B’s holomorphy bit  $\det K=1$  (v277) absorbed into SEAM.EQUIV.01. The former single open lemma, the named theorem SEAM.EQUIV.01 — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — is now *closed modulo cited theorems on its MMST route* SEAM.EQUIV.MMST.01 (the twistor route SEAM.EQUIV.TWISTOR.01 and the unconditional parent stay **[O]**). (The flat  $\tau=i$  geometry and the  $E_8$  holomorphy are *two faces of one object*: the  $(E_8)_1$  character is  $\chi_{E_8} = j^{1/3}$ , so  $\chi_{E_8}(i) = 1728^{1/3} = 12$  and  $\tau=i$  is at once the order-4 CM point and the  $(E_8)_1$  partition-function modulus, v282.) The explicit lattice model (v367/v368) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, and it is Lean-formalised as FORM.SEAM.MMST.01 (collar hypotheses kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems as named cited axioms, clean **#print axioms**); the *only* residual that stays **[O]** is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since re-founded on the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$  is the Longo–Rehren locality criterion; KLM  $\mu=4/2^2=1$ ; AGT/AMT the second witness), the realisation input reduced to invariants (R1’; computed  $|C|=1$ ,  $\nu=16$ ), Lean parallel route **seamResidualClosed’** (v469) — so it is closed *modulo a cited theorem*, not solved, Lean-pinned to exactly those named steps plus the derived Recovery gap  $\Delta = 6 \ln(3/2) \approx 2.43 > 0$  (v308); the conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face (v323). This is by design the *one* irreducible structural input of the theory — the role the *constancy of  $c$*  plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; with the lattice/ $S_3$  pinning and the Lean formalisation in place, “closing the theory” has reduced to invoking exactly the two cited continuum-existence theorems, not an open-ended search. The full sprint-by-sprint reduction that pinned SEAM.EQUIV.01 is recorded in the changelog, not replayed here. The seam seed is *hardened*:  $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$  is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank  $E_8$  and only the Gauss–Bonnet  $\pi$  left as the single irreducible primitive. What remains for a full TOE is then the cited continuum-existence theorem on the seam and the one unit  $v_{\text{geo}}$ , plus the standard-RG dimensionful masses and the typed frontier transfers (the ambient QG measure being a **[C]** redundancy, v369). No load-bearing discrete parameter remains free, and  $\pi$  is the lone continuous primitive. *4D-QFT fork policy (frozen, v265)*: the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed,

falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

## 10 TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram

The “compiler” framing places TFPT in an older lineage — the idea that physics is, at bottom, the output of a short computation — and it is worth stating precisely what TFPT shares with that lineage and where it deliberately parts company, so the family resemblance is not mistaken for a claim the theory does not make. Three landmarks fix the coordinates. **Zuse** (*Rechnender Raum*, 1969) proposed that the universe is literally a (cellular-automaton) computation. **Schmidhuber** sharpened this into an *algorithmic* theory of everything (1997, 2000): the shortest program that outputs a universe is the natural prior — a physics reading of Kolmogorov complexity and Solomonoff induction — while his *Speed Prior* (2002) adds that a computed universe should be one that is *fast* to compute, and his low-complexity-art programme (1997) identifies beauty with compressibility. **Wolfram** (*A New Kind of Science*, 2002; the Physics Project) recasts the same instinct combinatorially: space–time as a hypergraph updated by simple rewrite rules.

### Three shared instincts

1. **Minimal description length as the selection principle.** “No load-bearing free number, only  $\pi$ ” is, in these terms, a Kolmogorov-style compression statement — the plausibility discussion below already names it *Kolmogorov-like compression: many numbers from little* — and the grammar-internal null model of the next section turns it from rhetoric into a measured bound.
2. **Deterministic, resource-bounded generation.** The readouts fall out of a chain of exact identities in about a second, not a search over a model space — the same spirit as the Speed Prior’s preference for *fast*-computable worlds, though TFPT makes no claim to instantiate a speed prior.
3. **Compressibility as the aesthetic.** “Choosing  $E_8$  is choosing the icosahedron” selects the maximally symmetric, shortest-to-describe object; the affine Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  carry a fourfold identity (representation dimensions = network attractor = spectral certificate = local balance weight). This is Schmidhuber’s *beauty = compressibility*, realised on the Standard-Model alphabet rather than on pixels.

### Three deliberate divergences

- (A) **One forced program, not an ensemble.** Schmidhuber’s shortest program computes *all* universes and privileges none; TFPT bets the opposite — an over-determined bootstrap fixed point that *forces* a single dimensionless skeleton ( $g_{\text{car}}=5$  fixed three independent ways). TFPT is a candidate for *the* short program of our constants, not a measure over programs.
- (B) **The description language is algebraic.** TFPT’s compression is measured in a Lie/latitude/VOA language, not in Turing bit-strings; Kolmogorov complexity is language-dependent up to an additive constant, so the honest question — is the shortness real, or hidden in the  $E_8$  machinery? — is exactly what the null model and the “forced three ways” bootstrap are built to answer. A Wolfram-style rewrite reading sharpens the same point rather than dissolving it: the  $(2, 3, 5)$  seed is the terminal of a minimal  $T(2, 3, r)$  growth rule (Cartan determinants  $3, 2, 1, 0, -1$  for  $E_6, E_7, E_8, \hat{E}_8, E_{10}$ ), so the rewrite picture *re-formulates* axiom P2 rather than removing it — both pictures bottom out on the same single irreducible seed.
- (C) **Falsifiability — the one thing TFPT adds.** As *physics*, the Zuse/Schmidhuber algorithmic TOE and Wolfram’s rule search are essentially unfalsifiable: no dated measurement can kill them. TFPT freezes dated, kill-tested predictions (the registry `predictions_frozen.json`,

the predictions section below) and so puts the computational-universe intuition at genuine empirical risk. That, not the compression rhetoric, is the load-bearing difference.

*Status.* This section is *positioning*, not a new result: it introduces no formula and moves no status marker. Its machine-checked hooks already live in the suite — the grammar-internal null-model bound and the icosahedral/McKay bedrock with its network attractor; the Wolfram-style rewrite correspondence is worked out separately in the repository’s hypergraph note. The multiway reading of the arithmetic Hecke tower has meanwhile been *measured out* rather than narrated ([`verification/v747_automaton_multiway_tower.py`], 24/24; [`verification/v748_automaton_sigma_orbifold_confluence.py`], 15/15): confluence is exactly Hecke commutativity (= CRT), the branchial structure is the  $[4, 2]_q$  Grassmann cell count, the  $\sigma$ -quotient is confluent everywhere with only the half-orbit *weighting* breaking at the ramified place — the adjudication is *equivalently useful*, i.e. the Zuse/Wolfram lens falls back exactly onto the known structure and rests as documented exploration.

## 11 Does this make the theory more probable?

**Plausibility balance.** *Pro:* (a) two inputs instead of many; (b)  $\alpha^{-1}$  to  $2.9 \times 10^{-10}$  (CODATA-2022,  $1.9\sigma$ ) as a *fixed point*, not a fit; (c)  $E_8$  constructed, not postulated; (d) one compiler generates several sectors  $\Rightarrow$  fewer degrees of freedom, more predictive power.

*Contra/open:* what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ( $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

## 12 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.



| Observable                          | TFPT value & derivation                                                                                                                                                              | Experiment (running / upcoming)                                                                                                      | St.     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------|
| solar<br>$\sin^2 \theta_{12}$       | prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.) | <b>JUNO</b> : first 59.1 d run $0.3092 \pm 0.0087$ (compatible), precision phase pending                                             | [E]/[C] |
| reactor<br>$\sin^2 \theta_{13}$     | $\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ ; seed $\times$ carrier-trace $e^{-\gamma}$ , $\gamma = \frac{5}{6}$                                                                      | Daya Bay / RENO / JUNO (PDG 0.0220)                                                                                                  | [E]     |
| atmospheric<br>$\sin^2 \theta_{23}$ | $\approx \frac{1}{2}$ ( $\mu\tau$ -symmetric limit; octant not selected)                                                                                                             | NOvA / T2K / <b>DUNE</b> (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)                                                             | [C]     |
| fine structure $\alpha^{-1}$        | 137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)                                                                                                    | CODATA-2022 (data through 2022)<br>137.035999177(21): dev. $2.9 \times 10^{-10}$ ( $1.9\sigma$ of its uncertainty)                   | [E]     |
| tensor ratio $r$                    | $r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$ ); scalaron-reheating point 0.0045 ( $N_*=51.4$ , v86)                                                  | now $r < 0.036$ (BICEP/Keck BK18); <b>CMB-S4</b> ( $\sigma_r \leq 5 \times 10^{-4}$ , obs. $\sim 2033$ )                             | [C]     |
| tilt $n_s$                          | $n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ( $N_*=51.4$ , v86)                                                                        | Planck ( $0.9649 \pm 0.0042$ ); CMB-S4 sharpens                                                                                      | [C]     |
| amplitude $A_s$                     | $\frac{N_*^2}{24\pi^2} c_3^7$ ; $2.0 \times 10^{-9}$ at $N_*=55$ (the measured $A_s$ prefers fast reheating, v86)                                                                    | Planck ( $2.1 \times 10^{-9}$ )                                                                                                      | [C]     |
| neutron EDM                         | $\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)                                                                                                          | <b>PSI nEDM</b> / SNS (running)                                                                                                      | [E]     |
| CP phase $\delta_{\text{CKM}}$      | $\delta = \frac{\pi}{3} + 3\lambda^2 = 68.65^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$                                                      | <b>LHCb</b> / Belle II $\gamma$ combination; decision at $\sigma_\gamma \leq 0.96^\circ$                                             | [E]     |
| cosmic birefringence $\beta$        | $\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)                                                                                                          | ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); <b>Simons Obs.</b> / LiteBIRD | [C]     |
| $0\nu\beta\beta$ / ordering         | Majorana branch; normal-ordering preference                                                                                                                                          | <b>LEGEND</b> / <b>nEXO</b> , DUNE, KATRIN                                                                                           | [C]     |
| flavor invariants                   | $\det R = h(D_5) = 8$ , principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)                                                                                          | any future CKM/PMNS global fit must satisfy these                                                                                    | [E]     |
| $H_0$ vs. $\Lambda$                 | $v_{\text{EW}} \sim e^{-\alpha^{-1/5}}$ , $\Lambda \sim e^{-2\alpha^{-1}}$ , $H_0 \sim \sqrt{\Lambda}$ : one engine                                                                  | SH0ES / <b>DESI</b> / Planck (Hubble tension)                                                                                        | [C]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

### The two decisive near-term tests

- (1) **JUNO** and  $\theta_{12}$ . JUNO is *taking data since August 2025* and targets sub-percent  $\sin^2 \theta_{12}$ . TFPT commits to the single prediction of record  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$  (misalignment  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ ). The first 59.1-day JUNO run reports  $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$  — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from  $\sim 0.307$  at high significance falsifies the seam mechanism.
- (2)  $r$ . The  $R^2$  scalaron with  $M$  fixed by  $c_3^7$  predicts a small  $r = 12/N_*^2$ : the frozen band  $r \in [0.0033, 0.0048]$  over  $N_* \in [50, 60]$ , with the scalaron-reheating point  $r = 0.0045$  ( $N_*=51.4$ , v86) — already below the current bound  $r < 0.036$  (BICEP/Keck BK18) and within reach of CMB-S4 ( $\sigma_r \leq 5 \times 10^{-4}$ , first observations  $\sim 2033$ –34). A detection of  $r \gtrsim 0.01$  would be incompatible with the  $R^2$  branch on which  $M_{\text{Pl}}$  and  $A_s$  rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable  $E_8$  appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate  $E_8$ ; and the discipline rule “*every load-bearing number appears in at least one  $E_8$  projection*” (atlas note) turns the number stock into a falsifiable raster rather than

numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [verification/v100\_numerology\_null\_mc.py].

#### Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar**  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$  — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor**  $r \approx 0.0040$  — any robust  $r \gtrsim 0.01$  kills the  $R^2$  branch carrying  $M_{\text{P1}}, A_s$ ;
- **ordering** normal, small  $m_{\beta\beta}$  — inverted ordering or large  $m_{\beta\beta}$  kills the Majorana branch;
- **strong CP**  $\theta_{\text{eff}} = 0$  — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$  — a robust  $w \neq -1$  kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84\_frozen\_registry.py] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one*  $\theta_{12}$  prediction of record (the seed  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards.  $r$  and  $n_s$  enter only as *bands* over the declared external  $N_\star \in [50, 60]$ , never as point predictions.

#### What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, `v49/v69/v71`); the only residual is the *absolute* amplitude scale, an anchor ( $U_{\text{point}}$ ) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to  $b_1 = 41/10$ ), since the compiler systematically produces traces over  $S^+$ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ $R^2$  branch is lifted from [C] to [E].

## Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator  $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$  builds, from two inputs  $\{c_3, g_{\text{car}}\}$  via  $D_5 \oplus A_3 + \mu_4$ , the  $E_8$  container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms  $\{c_3, g_{\text{car}}\}$  alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

| Script                             | What it machine-checks                                                                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1_e8_glue.py                      | glue theorem $E_8=(D_5\oplus A_3)+\mu_4$ : $\text{disc}=\mathbb{Z}_4$ , $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$ , $240=16\cdot 5\cdot 3$ , $248=240+8$                                                                                 |
| v2_carrier_pascal.py               | $g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$ , $N_{\text{fam}}=3$ , $\text{rank } E_8=8$ , $\Omega_{\text{adm}}=48$ , $b_1=\frac{41}{10}$                                                                                      |
| v3_em_alpha.py                     | EM closure: unique root of $F_{U(1)}(\alpha)=0$ , $\alpha^{-1}=137.0359992168$ , existence/uniqueness, inverse test, ablation                                                                                                            |
| v4_flavor_matrix.py                | residue matrix $R$ : $\det=8$ , minors $\{2, 3, 5\}$ , $\chi_R=t^3-9t^2+10t-8$ , SNF $\text{diag}(1, 1, 8)$ , $\ R\ _F^2=78$ , $\sum L=40$                                                                                               |
| v5_e8_cascade.py                   | cascade $D_n=60-2n$ : endpoints, exponent rungs $\rightarrow 240$ , IR tail 46080, variance 6552                                                                                                                                         |
| v6_bootstrap.py                    | reverse glue $\mu^2-5\mu+4=0$ ; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$                                                                                                                                   |
| v7_gravity_cosmo.py                | scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ , $A_s, n_s, r$ , $\Omega_b$ , $\sin^2 \theta_{12}$                                                                                                                 |
| v8_horizon.py                      | horizon code: $\frac{1}{2\pi}=4c_3$ , $1920= W(D_5) $ , $S_{dS}$ , Page, $\beta_{\text{rad}}=0.2424^\circ$                                                                                                                               |
| v9_neutrino_texture.py             | solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\theta_{13}=0$                                                                |
| v10_projection_involution.py       | $Q, \Sigma$ algebra: $K=R+Q\Sigma$ , $L=R+Q(I+\Sigma)$ ; $\chi_Q, \chi_K$ ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $ ; $a^\top K a=41$                                                                                         |
| v11_unique_KQ.py                   | $K, Q$ are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)                                                                                                                   |
| v12_mass_generation_polynomials.py | sector/generation polynomials of $K$ and the anchor-block det ladder (9, 10, 16, 40)                                                                                                                                                     |
| v13_open_gates.py                  | gate closures: $M=41=10b_1=\sum L+N_\Phi$ ; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$                                                                                                                                                  |
| v14_carrier_uniqueness.py          | $g_{\text{car}}=5$ unique ( $N_{\text{fam}} \in \mathbb{Z}^+$ , $\text{rank } E_8=8$ ); split (3, 2) from $b+s=5$ , $b^2+s^2=13$ ; $\text{Tr } Y=\text{Tr } Y^3=0$                                                                       |
| v15_bootstrap_classification.py    | $D_5\oplus A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$                                                                                                                                      |
| v16_solar_dual_anchor.py           | $a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$ ; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; full PMNS                                                                                            |
| v17_hexagonal_resolvent.py         | finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark $c$ backbone)                                                                                                                                             |
| v18_quark_yukawa.py                | quark source ratios $\frac{m_u}{m_d}=0.470085$ , $\frac{m_c}{m_s}=13.61$ , $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ ; full hierarchy                                                           |
| v19_monodromy_moduli.py            | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ( $3^7$ ); $D_4$ $SU(3)$ monodromy constructible but $D_4$ -fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$                                                    |
| v20_lepton_c_derivation.py         | lepton $c$ 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos \frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9} \Rightarrow \frac{7}{6}$ |
| v21_solar_product_quark.py         | solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$ ), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$ ; quark law fails ( $\frac{1}{7}$ vs 0.724) |
| v22_open_gates_audit.py            | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)                                                                                                     |
| v23_anchor_generator.py            | anchor $a=(1, 1, 2)$ : $e_k=(4, 5, 2)$ , $c_3=\frac{1}{2e_1\pi}$ ; $p_n=2+2^n \Rightarrow 240, 248$ , binary ladder; inputs $\rightarrow \{a, \pi\}$                                                                                     |
| v24_quark_ratio_closure.py         | quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$ ; rational $c$ gauge, $\Lambda_q$ in $O(1)$                                                                                    |
| v25_frontier_conjectures.py        | conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$ ; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\text{ }\mu\text{eV}$                                                                 |

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| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v26_flavor_<br>frontier_<br>unification.py | the “11” is not uniquely forced ( $\Rightarrow$ [C]); flavor frontier unifies to one ( $U$ ) gate; cusp config on the parabolic stability wall                                                                                                                                                                                   |
| v27_wall_<br>representative.py             | explicit balanced wall rep $W_{\text{wall}}$ (cols = perm(0, 1, 2), rows $w=(2, 1, 1)$ , pardeg 0); selector $\det R=8$ , $\text{Spec}(Q_+)=\{1, 2, 3\}$                                                                                                                                                                         |
| v28_gravity_fR.py                          | $R+R^2$ closed: $f(R)$ , $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_*=57\rightarrow n_s, r, A_s$ ; full metric measure open; $248c_3^2 < 6\log \frac{3}{2}$                                                                                                                                                                   |
| v29_research_<br>contract_certs.py         | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick $W_{\text{wall}}$ ) and $G5$ gap-dominance $2\ V\  < \Delta$                                                                                                                           |
| v30_d4_character_<br>variety.py            | $C_U^{(2)}$ : full $D_4$ -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)                                                                                                                  |
| v31_R_<br>dictionary.py                    | $R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); $R$ integer but $\rho$ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)                                                                                                             |
| v32_rh_<br>splitting.py                    | RH route: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ exact $\Rightarrow$ splitting<br>$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \Leftrightarrow \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k = I$ + $R$ -bridge stay open ( $c_u/c_d$ not forced)                                       |
| v33_explicit_flat_<br>bundle.py            | explicit valid ( $U_{\text{wall}}$ ) flat bundle (RH solve): cusp + $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ + $\ M_\infty - I\  \sim 10^{-9}$ ( $\prod M_k = I$ ) + irreducible (case A)                                                                                                                                      |
| v34_h2_bridge_<br>attempt.py               | H2 bridge: explicit per-puncture $M_k$ (cusp, $\prod M_k = I$ ); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input ( $c_u/c_d$ not obtained)                                                                                                                  |
| v35_quark_qcd_<br>boundary.py              | the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)                                                                                                                                                  |
| v36_spectral_<br>action_g2.py              | G2: spectral action $\rightarrow R+R^2$ ( $a_2 \sim -R/3$ , $a_4 _{R^2} \sim R^2/72$ ); $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ , $c_3^7$ fixes $f_0$ ; gap-decoupling $\Delta_{\text{eff}} = 1.648 > 0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target                              |
| v37_plucker_<br>anchor.py                  | anchor-plane Plücker apparatus: $\ Pl(K)\ _1 = 11$ , $\ Pl_R(K)\ _1 = 26$ , $\sum Pl_R(L) = 60$ ; pencil $\det(K+xQ) = 3x^3 + 7x^2 + 6x + 4$ ; $K_e - K_u = a$ , $L_e - L_u = \text{Exp}(A_3)$ ; lepton ring $c_e c_\tau =  \mathbb{Z}_2  c_\mu$ ; Koide $u \rightarrow \frac{53}{54}u$ ([C])                                    |
| v38_uwall_<br>killswitch.py                | ( $U_{\text{wall}}$ ) U2 kill-switch hardened: $D_4$ -cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq 0.35$ , reducible locus measure-zero $\Rightarrow$ case A generic, ( $U_{\text{wall}}$ ) survives; amplitude dictionary still research-level                                                    |
| v39_uwall_<br>selectors.py                 | ( $U_{\text{wall}}$ ) selector side closed on the explicit point: splitting type = anchor $a=(1, 1, 2)$ , pardeg = 0; $\det R=8=n\cdot a$ , $\text{Spec}(Q_+)=\{1, 2, 3\}=3\alpha+1$ read off the bundle; only the amplitudes $c_u/c_d$ ( $\Gamma^{\min}$ Hitchin) remain                                                        |
| v40_harmonic_<br>metric.py                 | ( $U_{\text{wall}}$ ) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$ ): unique invariant form $H>0$ , unitarised $\tilde{M}_k$ cusp-class with $ \tilde{M}_k  \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ ; the feared Hitchin PDE collapses to unitarisation + combinatorial $R$ |
| v41_leg_<br>assignment.py                  | ( $U_{\text{wall}}$ ) final leg test: amplitudes = $\delta = \frac{1}{2}$ resolvent (closed); $\Lambda_\mu > 1$ rules out holonomy moduli; harmonic holonomy $ \text{diag}  = (0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta = \frac{1}{2}$ (suggestive); honest negative on literal $c_u/c_d$ reproduction                     |
| v42_exterior_<br>leg.py                    | ( $U_{\text{wall}}$ ) Exterior Leg Lemma: $u, d$ share $(r, w)=(1, 1)$ so any scalar leg gives $c_u/c_d \equiv 1$ ; the $u/d$ split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$ , $\ \cdot\ _1 = 11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]                       |
| v43_exterior_<br>bridge.py                 | ( $U_{\text{wall}}$ ) $\Lambda^2 F$ bridge: $\Lambda^2(\tilde{M}) = \tilde{M}$ (exterior leg = $\bar{3}$ ) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ( $\det R=8$ , $\text{Spec}(Q_+)$ confirmed), not continuous                                               |

*continued on next page*

| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v44_carrier_exterior.py        | $(U_{\text{wall}})$ Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$ ; $u^c \in \Lambda^2(5)$ , $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2, 4)$ , $\text{sum}=6= R^+(A_3) $ , $\text{diff}=2= \mathbb{Z}_2 $ ; the exterior leg is the carrier's own $\Lambda^2$ grading (no special math)                                                                                       |
| v45_family_exterior.py         | $(U_{\text{wall}})$ the “11” is Pascal: $ \text{Pl}(K) =(1, 4, 6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$ ;<br>$\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of $\mu_4$ up to $\deg u^c=2$ );<br>$15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$                                                                                             |
| v46_grand_mass_volume.py       | absolute mass-scaling is [E], not a product rule: sector dets<br>$(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \mathcal{A}_\Lambda \Rightarrow$<br>$\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ ; $Q$ -rows $(4, 5, 6)$ sum $15=\dim A_3$ ; $25+15=40=\sum L$ |
| v47_selection_theorem.py       | Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$ ; 16-spinor $\rightarrow D_5$ ;<br>$q(D_5)+q(A_3)=2$ , glue $4= \mu_4 $ ; only $D_5 \oplus A_3$ meets all four conditions<br>( $D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]                                                                                                                                   |
| v48_em_ward.py                 | EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^3 \alpha^2 - \text{transport}$<br>$8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ; $8b_1 = \frac{4}{5} M = \frac{164}{5}$ , $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]                                                                                                                                       |
| v49_readout_rigidity.py        | Readout Rigidity (Thm U2): $c_u/c_d = \frac{55}{117}$ is constant on the discrete selector<br>stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}})=U_{\text{unitary}}+U_{\text{H2}}+U_{\Lambda^2}+U_{\text{point}}$ ,<br>only $U_{\text{point}}$ open                                                                                                                |
| v50_q_geometry.py              | Q geometry (Thm Q): $Q_+ = \text{diag}(3, 2, 1)$ -block $\chi=(t-1)(t-2)(t-3)$ , $Q_-$<br>$\chi=t(t^2-3)$ ; $Q$ unique under rows $(4, 5, 6)/\text{cols}(9, 5, 1)/\text{spectra}$ ; geom. origin [C]                                                                                                                                                                                                   |
| v51_boundary_half_step.py      | glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $ : $q(D_5)=\frac{5}{4}$ , $q(A_3)=\frac{3}{4}$ ; four ops forced $\rightarrow \{\text{sum}=2$<br>root norm, $\text{diff}=\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$ , $\text{prod}=\frac{15}{16}$ , $\text{ratio}=\frac{5}{3}\}$ . So $\delta=\frac{1}{2}$ is the<br>carrier–family count over the glue index, not chosen [E]                    |
| v52_pencil_endpoints.py        | $K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $ , $P(0)=4= \mu_4 $ , $P(1)=20=\det L$ ,<br>$P(2)=68=2p_5(a)$ ; $\det(K-Q)=2$ , $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]                                                                                                                                                                                                                    |
| v53_compiler_core.py           | Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ — rank $E_8=8$ ,<br>$ \mathbb{Z}_2 =2$ , Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2  \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of<br>squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]                                                                                      |
| v54_seam_horizon_keystones.py  | The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned<br>$(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi \text{ Hawking/Einstein})$ ; one boundary transport<br>operator (spec $\{1, (2/3)^6, (1/3)^6\}$ , $\lambda_2=(2/3)^6$ ) governs <i>both</i> SM flavor gap and<br>horizon Page recovery [E]                                                                           |
| v55_coxeter_cycle.py           | Computed $E_8$ Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$ , eigenphases =<br>exponents = totatives of 30, so rank $E_8=8=\varphi(30)$ = live phases of the order-30<br>cycle; one $\alpha^{-1}$ sets EM, $S_{dS}$ and $\Lambda$ ( $S_{dS}\rho_\Lambda=32\pi^4$ ) [E]                                                                                               |
| v56_unique_attractor.py        | Gapped boundary transport (gap $6 \log \frac{3}{2} > 0$ ) $\Rightarrow$ <i>unique</i> attracting fixed point,<br>geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes;<br>entropy reset = rank-1 projector [E]                                                                                                                                            |
| v57_horizon_crosslinks.py      | Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ( $1/4=1/ \mu_4 $<br>entropy density); Hod QNM $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$ ; Hawking power denom<br>$1920= W(D_5) $ ; one $\alpha^{-1}$ for EM/ $S_{dS}/\Lambda$ /Hubble [E] (+ [C] identification)                                                                                                     |
| v58_seam_horizon_chain.py      | Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a<br>horizon): one-sided $S^2$ Gauss–Bonnet $c_3=1/(2 \cdot 4\pi)=\frac{1}{8\pi}$ ; KMS $4c_3\kappa$ ;<br>$S_{BH}=M^2/2c_3$ ; RT in seam units. Seam–Horizon Theorem (area-law from the<br>seam kernel) [O] open                                                                                                          |
| v59_area_law_evidence.py       | Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type)<br>kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3} \log L$ ), by the<br><i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2  \mu_4 $ . The $c_3$ coefficient stays [O] open                                                                                             |
| v60_lambda_metrology_branch.py | $\Lambda$ branch selection: $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ , mis-scale $2c_3/\delta_{\text{top}} = \frac{64\pi^3}{3} = 661.47 \Rightarrow G_N$<br>pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence<br>$0.215^\circ \pm 0.074^\circ$ vs $0.2424^\circ$ ( $0.37\sigma$ ) [E]                                                                                       |

continued on next page



| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v61_cft_bridge.py             | Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$ , $c(D_5)=5$ , $c(A_3)=3$ = (rank $E_8$ , $g_{\text{car}}$ , $N_{\text{fam}}$ ); conformal embedding $c_{\text{coset}}=0$ ; $SU(3) \leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$ , $4 \rightarrow 3+1$ (one geometry $\mathbb{P}^1 \setminus \mu_4$ , two sides) [E]                                                                                                                                                       |
| v62_data_scorecard.py         | TFPT vs 2024/25 data: $\alpha^{-1}$ , $\sin^2 \theta_{12}$ , $\beta_{\text{rad}}$ match; $\sin^2 \theta_{13}$ ( $2.0\sigma$ ) and Starobinsky $n_s$ vs DESI ( $\sim 2\sigma$ ) mild tensions; $\theta_{23}$ open; $r \approx 0.004$ future falsifier [E]                                                                                                                                                                                                                                                  |
| v63_seam_engineering_index.py | Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2 \log(3/2)} \approx 0.323$ ( $2\ V\ =\frac{31}{4\pi^2}$ , $248=\dim E_8$ , $31=1+h^\vee$ ), $\Delta_{\text{eff}} \approx 1.648$ ; four-class possibility taxonomy [E]/[O]                                                                                                                                                                                                                                                                |
| v64_causal_boundary_nogo.py   | Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)                                                                                                                                                                                                                               |
| v65_falsification_layer.py    | Prediction layer with explicit failure thresholds: 3 core ( $v_{\text{GW}}=c$ , no 4th gen, seam=8), 4 observational ( $\beta$ , $r$ , $n_s$ , $\theta_{12}$ ), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]                                                                                                                                                                                                                                                                      |
| v66_e8_casimir_degrees.py     | Organizing simplification: the compiler atoms <i>are</i> the $E_8$ Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ( $2= \mathbb{Z}_2 $ , $8=\text{rank}$ , $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $30=h(E_8)$ ); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]                                                                                                                                                                                                                 |
| v67_area_law_coefficient.py   | Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=2\pi c_3 A=\frac{1}{4}A$ ( $2\pi c_3=\frac{1}{4}$ ); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$ . Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]                                                                                                                                                                                        |
| v68_seeley_dewitt_residual.py | Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ( $\propto d\Lambda^2 f_2$ ), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2$ /scalon derived [E]                                                                                                                                                                                       |
| v69_d4_q_geometry.py          | $D_4$ -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1 \setminus \mu_4)=B_1 \oplus E$ , $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$ ; $\Sigma$ -split $=D_4=\mathbb{Z}_4 \rtimes \mathbb{Z}_2$ [E]                                                                                                                                                                                     |
| v70_q_integer_lift.py         | Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action $R$ unimodular (eigenvalues $\{-1, i, -i\}$ ); $\det Q=3=N_{\text{fam}}$ , SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$ , not closable by $D_4$ alone [E]                                                                                                                                                                                                                                     |
| v71_simple_r_bridge.py        | <i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ( $\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ( $\frac{55}{117}$ , no monodromy, v49 rigidity); residual $U_{\text{point}}$ = absolute norm. = an anchor [E]/[O]                                                                                                                                          |
| v72_q_det_from_cusp.py        | $\det Q=N_{\text{fam}}$ derived from the cusp class: $\det Q= \text{coker } Q =\text{cusp-class order}=\text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$ ); $\text{coker } Q=\mathbb{Z}/N_{\text{fam}}=\text{triality deck group} \Rightarrow$ integer-lift residual closed [E]                                                                                                                                                                        |
| v73_k_c3_half.py              | Central coefficient $k=c_3/2$ <i>forced</i> : $(1/2)$ variational $\times 1/( \mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ( $\chi(S^2)=2$ ), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$ ; only the absolute 4D Newton scale stays the anchor (v68) [E]                                                                                                                                                                                                                             |
| v74_compiler_micro_lemmas.py  | Spine quotient ladder ( $\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of $(2, 3, 4, 5)$ ); pencil differences $(2, 16, 48)=( \mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L=R+6\mathbf{1}e_1^\top$ , $a^\top R^{-1}\mathbf{1}=0$ (Sherman–Morrison rank-one invariance) [E] |
| v75_upoint_to_vgeo.py         | <b>Gate 1 complete</b> : $U_{\text{point}} \rightarrow v_{\text{geo}}$ . Ratios closed (v49/v71) + lepton $c$ exact (v20) + sector products = $\varphi_0^{\text{ret}}$ -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale $v_{\text{geo}}$ = the same anchor as $1/G$ (v68); two $[A] \rightarrow$ one [E]/[O]                                                                                                     |
| v76_gmetric_reduction.py      | <b>Gate 2</b> : Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}=6\log \frac{3}{2}-\frac{31}{4\pi^2}=1.648>0$ , $\ V\ \leq 248c_3^2=\frac{31}{8\pi^2}$ ); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]                                                                                                                                                                           |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v77_e8_conformal_net.py             | <b>G6 route:</b> the seam boundary measure = the $(E_8)_1$ <i>lattice net</i> (rigorous, holomorphic $c=8$ ); level-1 $c(E_8)=\frac{248}{31}=8$ , $c(D_5)=5$ , $c(A_3)=3$ , embedding $5+3=8$ (coset $c=0$ ); imports $G_6$ into RCF/T/conformal-net rigor [E]/[C]                                                                                                                                                                                     |
| v78_vgeo_floor.py                   | $v_{\text{geo}}$ <b>floor:</b> no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times$ one $M_{\text{Pl}}$ ; $S_{dS}\rho_\Lambda=32\pi^4$ pins it from one measurement; theory at the minimum (one scale $+$ $\pi$ ) [E]/[O]                                                                                                                                                                  |
| v79_review_identities.py            | Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n=(3^n-(-2)^n)/5=1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ( $\mathbf{1}^\top M^{-1}\mathbf{1}=1/\text{atom}$ , $a^\top M^{-1}a=1$ for $R, K, L$ ); MacWilliams (1, 10, 5); $6Y^2-Y-1=(2Y-1)(3Y+1)$ . Rejects $m_p/m_e$ , $\eta_B$ near-formulas (firewall) [E]                                                                                  |
| v80_operator_pencil_geometry.py     | Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}=(N_{\text{fam}}x+ \mathbb{Z}_2 )(N_{\text{fam}}x+g_{\text{car}})=(3x+2)(3x+5)$ ( $\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain (3, 10, 16, 20); type checker $\det B_{Q,K,R,L}=(9, 10, 16, 40)$ ; audit: differential spectrum, $F_4 \times G_2$ shadow $248=52+14+26\cdot 7$ [E]/[C]                                                                             |
| v81_singular_pencil_matrices.py     | Double cover $y^2=\det B_{K+xQ}=(3x+2)(3x+5)$ : Koide $x=-\frac{2}{3}$ and carrier $x=-\frac{5}{3}$ are the two branch points (deck $2= \mathbb{Z}_2 $ ; disc $=81=N_{\text{fam}}^4$ ; separation $=1$ transport period). Clearing matrices $C_{2/3}=3K-2Q$ ( $D_5 \oplus A_3$ glue, $B=(5, 7)^\top(9, 11)$ , $\sum=240$ ), $C_{5/3}=3K-5Q$ (neutral, $\chi=(\lambda+2)(\lambda^2+\lambda+6)$ ); scalaron $7=$ mixed anchor disc $/N_{\text{fam}}$ [E] |
| v82_koide_attractor_splitting.py    | Koide attractor <i>forced</i> : the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6=\lambda_2$ of the established transfer operator $\Rightarrow F'(2)<1$ attractor, $F'(5)>1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$ : the clean split is non-generic (anchor-first) [E]/[C]                                                                                      |
| v83_e8net_holomorphic_uniqueness.py | Holomorphy pins $(E_8)_1$ : #primaries = det Cartan ( $D_8:4$ vs $E_8:1$ ), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $ ); carrier Pascal truncation $K=(g-1)/2 =$ half-spinor split [E]                                                                                                                                                          |
| v84_frozen_registry.py              | Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> $\theta_{12}$ prediction of record; $r$ , $n_s$ as $N_\star$ <i>bands</i> ; conditional entries carry their hypothesis by name [E]                                                                                                                                                                                  |
| v85_master_cover.py                 | Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $=N_{\text{fam}}^4 \det(G)^2$ ); the disc-81 family is its orbit ( $-\frac{8}{3}=-\text{rank } E_8/N_{\text{fam}} =$ carrier $-$ one period); $\mu_4$ is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]                                   |
| v86_nstar_reheating.py              | $N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128$ GeV, $T_{\text{reh}}=9.6\times 10^9$ GeV, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$ , $r=0.0045$ (inside the frozen band); honest tensions: $n_s+0.9\sigma$ , $A_s$ dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]                                                                                                |
| v87_bulk_uniqueness_reduction.py    | Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow \text{Rep} = \text{Vect} \Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both $E_8$ pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]                                                                                                                                               |
| v88_cp_phase_audit.py               | Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$ ; the $0.07^\circ$ -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [E]                                                                                                                                                                      |
| v89_carrier_index_lemma.py          | Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); $\mu$ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]                                                                                                                                                                      |
| v90_conical_defect_chain.py         | Fursaev–Solodukhin factor $4\pi$ <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]                                                                                                                                                                |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v91_spine_tetrahedron.py        | Spine tetrahedron $\{2, 3, 4, 5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $ ; $240= \mu_4  E(K_4)  E(K_5) $ with $K_6$ negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]                                                                                                                                                                                                                                                                                                                                                       |
| v92_glue_uniqueness.py          | Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$ ; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ , nothing else [E]                                                                                                                                                                                                                                                                                     |
| v93_koide_relaxation_toy.py     | Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$ ; source at $\rho = -\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]                                                                                                                                                                                                                                                                                             |
| v94_sheet_diamond.py            | Sheet diamond: all four flavor operators on <i>one</i> surface $M(s, t)=R+Q \text{ diag}(s, t, t)$ (pencil = the cut $(1+x, x-1)$ ; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all $s$ ; cofactor seam normal $(5, -9, 6)$ ; Plücker canonicalisation: 11 and 26 both live in $K$ ; spine-quotient firewall <i>rejected</i> [E]                                                                                                                                                                                                                                                        |
| v95_centered_diamond.py         | Centered cross: $Q=U+V$ , $R/L=C \mp U$ , $K/F=C \mp V$ (five objects $\rightarrow$ three); Spec $V=\{0, 1, 2\}=N_{\text{fam}} \cdot \text{cusp}$ ; $\det C=14$ , $\sum C=31=2^{g_{\text{car}}}-1$ ; $\text{Pl}_R(C)=7(2, 3, 1) \parallel \text{Pl}_R(L)=10(2, 3, 1)$ ; $G_2$ reading audit-typed [E]                                                                                                                                                                                                                                                                                                                 |
| v96_branch_kernel_selection.py  | Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$ : up = -down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$ ); anchor-placement controls [E]                                                                                                                                                                                                                                                             |
| v97_sheet_conjugation_bridge.py | Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer $T_A$ , $\det = -1$ ; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$ ; $\langle T_A, \Sigma \rangle = D_4$ ; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $ ); remaining [C]: $Q_+$ grading = $A_3$ discriminant grading [E]/[C]                                                                                                                                                                                   |
| v98_discriminant_dictionary.py  | Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A \Sigma$ acts integrally on the cusp basis ( $Ge_1=-e_1$ , $Ge_3=e_2$ , $Ge_2=-e_3=B_1 \oplus E$ ); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A G T_A^{-1} = G^{-1}$ ( $k \mapsto -k$ ); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$ , $q_{A_3}(2)=\frac{1}{2}$ ; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing $Q$ -geometry gate [E]                                                                                                                          |
| v99_koide_flow_time.py          | Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}}) \det B(q)$ , time-1 map = $F$ [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> ( $Q$ crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$ , $t$ passes every integer $\geq 3$ within $+0.5\sigma$ ) [E]; conditional steps: $n=2$ excluded $(-2.9\sigma)$ , $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427 \text{ MeV } (+0.14\sigma)$ , decision at $\sigma(m_\tau) \sim 0.01 \text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)                                               |
| v100_numerology_null_mc.py      | Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$ , robust to budget/window variation; $2 \times 2 \cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$ , <i>conditional on the declared grammar</i> [E] |
| v101_horizon_anchor.py          | The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$ ); interpolation $(x^2+1)/\Phi_3(x)$ ; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$ , deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$ ; six atom landings, zero free parameters [E]/[C] (bulk reading)                              |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v102_seam_orientation.py       | One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$ , $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate $\Delta$ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$ ; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]                                                                                                                       |
| v103_trisection_normal_form.py | Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta = -3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m = \cos \psi / N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}$ — one cosine of glue atoms; canonical curvature $(\frac{2}{3})^3 = ( \mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$ ; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$ , missing = one <i>clock</i> [E]/[C]             |
| v104_nariai_clock.py           | The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the $dS_2$ static-patch equation with $m^2 = -2\Lambda = - \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}} = (\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$ ; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$ ; entropy-deviation rate $2H =  \mathbb{Z}_2 H$ ; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]                                                            |
| v105_residual_inventory.py     | The residual inventory: <i>one constant</i> $(2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the $\Lambda$ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + $\pi$ ; a sixth gap would fail the script [E]/[C]                                                               |
| v106_review_validation.py      | External-review validation: seed normal form $\varphi_0^{\text{ret}} = \frac{ \mu_4 }{N_{\text{fam}}} c_3 + \Omega_{\text{adm}} c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2 = 120 = 5!$ computed from the charge table; factorial spine $5! =  R^+(E_8)  = \sum \text{exponents}$ , $240 = 2 \cdot 120$ , $1920 = 2^4 \cdot 5!$ ; degree-2 inventory (Pascal $K=2$ unique at $g=5$ , glue norms sum 2, $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ , $\binom{5}{2} = 10$ ); named [C] hypothesis <i>Quadratic Boundary Locality</i> ( $\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240 = 16 \times 15$ non-unique [E]/[C] |
| v107_quantum_clock_target.py   | Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/\bar{M}_{\text{Pl}}^2$ : $10^{-122}$ cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]                                                                                                                      |
| v108_pascal_ladder.py          | Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- $g$ midpoint identity + even- $g$ straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]                                                                                                                                                                                                                               |
| v109_sheet_pairing.py          | Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$ ; odd $g$ flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ( $256 = 1 + 45 + 210$ , zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1 + 2\Lambda^3 + \Lambda^5)$ ; tower tops at $\Lambda^{2K}$ , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]                                                                                                                                                 |
| v110_calderon_sheet.py         | Calderón-sheet selection theorem: a Calderón involution $\varepsilon$ on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2 =  \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ( $g=3, 5, 7$ ): sheet-oddness forces $K=(g-1)/2$ , <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ $\varepsilon$ sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees $\leq 2$ ) [E]/[C]                                                                                                                                     |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v111_quadratic_transport.py       | Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$ , the $\Lambda^2$ term of $1+45+210$ ) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length $2 \Rightarrow$ <i>transport degree exactly 2</i> (degree $\leq 1$ generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]                                                                                                                                                                    |
| v112_selfcounting_channel.py      | Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so $\#\text{kernels} = 2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$ ; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]                                                                                               |
| v113_quasifree_kernel.py          | One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 =  \mu_4 $ ); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$ , seam level $8 = \text{rank } E_8$ ( $c = \text{rank of the kernel}$ ); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]                                                                                                        |
| v114_torsion_delta.py             | Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}})$ $\mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ( $\mu_4 =$ order of the twisted cusp generator); on the involutive branch ( $T^2 = \mathbf{1}$ ) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]                                        |
| v115_anchor_residue.py            | The anchor pins the residue: $\mu_4$ -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $ ; off-weights uniquely $(8, 0, 5)/144$ ( $= (\text{rank } E_8, 0, g_{\text{car}})/( \mu_4 N_{\text{fam}})^2$ , total $13 = \Delta_Q$ ); <i>exact</i> residue $A_0^*$ with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ( $\ M_\infty - \mathbf{1}\  \sim 10^{-10}$ ); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]                                                                                                      |
| v116_resonance_uniqueness.py      | Resonance theorem (v115 [N] $\rightarrow$ [E], linear — no Gröbner): twisted $\mu_4$ averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a}_{13}E_{31}$ ; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\overline{a}_{13} \Rightarrow M_\infty = \mathbf{1} \Leftrightarrow a_{13}=0$ ; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact $A_0^*$ — $U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\  \sim 8.5 a_{13} $ [E]                                                                             |
| v117_monodromy_weyl_a3.py         | The monodromy is $W(A_3)$ : the unitarised holonomy of $A_0^*$ is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$ ), $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta = \frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$ , characters $(3, -1, 0, 1) = S_4 = W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the $A_3$ glue factor as its Weyl group; ODE match $3.5 \times 10^{-10}$ [E]                                                                                                                 |
| v118_hexagon_family_dictionary.py | First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ ); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 \Rightarrow \det_{\mp} = (\frac{7}{8}, \frac{9}{8})$ ; lepton coefficients = resolvent determinants ( $c_e =  \mu_4 /(2 \det_-)$ , $c_\mu = N_{\text{fam}}/(2 \det_+)$ ), $c_\tau =  \mu_4  \det_- / N_{\text{fam}}$ , product = $ \mu_4 /\det_+ = \frac{32}{9}$ ); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}}$ ; the address table stays open [E]/[C] |
| v119_review_validation_2.py       | Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5) = (e_3, e_2))$ ; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$ , only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}} = 2p_1p_2$ , $10b_1 = 1 + p_1p_3$ , $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank}$ ; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]                                                                     |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v120_address_table.py     | Address table (frozen v18/v20 integers): lepton words $(8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$ ; addresses = divmod by the hexagon ( $p_2=6$ ): $e \rightarrow (2, 1)$ , $\mu \rightarrow (5, 0)$ , $\tau \rightarrow (3, 0)$ , $\sum r=p_3$ ; sheet parity: $\tau$ sits at the real twisted eigenvalue $-1$ (= the product-rule site); quark sums: up = $p_3$ , down = $p_1+p_3$ , all quarks = $24= W(A_3) $ , all nine = $40=p_1p_3=a^T Ra$ ; top at the vacuum site $(0, 0)$ ; the assignment derivation stays open [E]/[C]                        |
| v121_address_pinning.py   | Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator $R$ + the generation-1 winding, $R$ column 1 = the anchor $a$ ; margins = atoms (rows $(e_1, \text{rank}, p_3)$ , columns $(e_1, \Delta_Q, g_{\text{car}})$ ); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is $R$ (also unique with SNF $(1, 1, 8)$ ; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C] |
| v122_margin_theorem.py    | Margin theorem: the <i>established</i> selectors — the $D_4$ annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$ , $\det K=4$ (the v71 quartet) — pin $R$ <i>uniquely</i> among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]                                                         |
| v123_inventory_update.py  | Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 $Q$ realisation — + $v_{\text{geo}}, \pi$ ; a sixth class fails the script [E]                                                                                                                                                              |
| v124_resummed_clock.py    | Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$ , the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$ ; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]                                                                                                                   |
| v125_glue_qsystem.py      | Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ( $q(k(1, 1))=k^2 \in \mathbb{Z}$ , v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step, } 4=2 \times 2$ ; R2 sharpens from “find” to “identify” [E]/[C]                                                                                                                                 |
| v126_clock_wall_bridge.py | Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly spec $A_0^*$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$ ; checkpoint 1 passed <i>classically</i> ( $p_2/N_{\text{fam}}=2= \mathbb{Z}_2  = \text{the v104 entropy rate } 2H$ ); $\det(\mathbf{1}-A_0^*)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]                                                                                                                |
| v127_ring_resummation.py  | The clock is a log determinant: $\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$ ; v107's “O(1) resummation” is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]                                                                             |
| v128_graded_hull.py       | Graded hull + zero-mode multiplicity: the 240 $E_8$ roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$ ; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [E]/[C]                                                                                                                                                                    |
| v129_entropy_power_law.py | The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); $\text{rate} = p_2 \ln(S_{dS}/S)$ ; triple identity $\alpha=n/N_{\text{fam}} = \text{MS weight} = \text{entropy-deficit share} \Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]                                                              |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v130_born_square.py          | Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2} = (S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=  A ^2 = (S/S_{dS})^6 \Rightarrow h = n_{\text{zero}}/2 = 3 = N_{\text{fam}}$ , $p_2 = 2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of $S^2$ ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure <a href="#">[E]/[C]</a>                                                                                                                            |
| v131_measure_is_area.py      | The measure is the area: $\ Y_{1m}\ ^2 = r^2 = A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude $S^3$ , Born $S^6$ — every exponent factor with a derivation origin; the $4\pi$ = the same Gauss–Bonnet primitive as in $c_3$ , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ <a href="#">[E]/[C]</a> ( <i>cancellation since derived within the SdS family, v144</i> )                                                                                  |
| v132_detratio_anomaly.py     | Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta - 2$ (the 3 zero modes removed) $= a_1 - 3 = \frac{7}{3} - 3 = -\frac{2}{3} = - \mathbb{Z}_2 /N_{\text{fam}}$ exactly ( $a_1 = \frac{1}{N_{\text{fam}}} +  \mathbb{Z}_2 $ ; numerical continuation $\sim 10^{-7}$ ) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL) = c^{\zeta(0)} \det'(L)$ ; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) <a href="#">[E]/[C]</a>                                                                                                   |
| v133_zeta_budget.py          | The $\zeta$ budget, both routes exact (Euclidean Nariai $= S^2 \times S^2$ ): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$ , $\zeta(0) = -\frac{109}{45}$ ( <i>no</i> atom; continuation $\sim 10^{-8}$ ); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) <a href="#">[E]/[C]</a>                                                                                 |
| v134_dual_anchor.py          | Dual anchor lemma: $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ , $d \cdot \mathbf{1} = 0$ , $d \cdot a = 1$ , $(1, 1, -2) = -2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ( $\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ ); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); $(d, n)$ = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge <a href="#">[E]/[C]</a>                                             |
| v135_det_surface.py          | Determinant surface: $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ — two $s$ -independent iso-volume walls $t = -2 \Rightarrow 2 =  \mathbb{Z}_2 $ , $t = -1 \Rightarrow 4 =  \mu_4 $ ( $\det K = 4$ is a <i>layer</i> ); winding line $6s + 8 \Rightarrow (8, 14, 20)$ , $\det F = 32 = 2^{g_{\text{car}}}$ ; $\det B(s, 0) = 2 \det M(s, 0)$ exactly, $B$ -wall only at $t = -2$ ; row-sum axes $C = (7, 11, 13)$ , $U = (3, 3, 3)$ , $V = (1, 2, 3) = \text{Spec}(Q_+)$ ; micro-identities $e_1^2 + e_2^2 = 41 = 10b_1$ , $p_1^2 + p_2^2 = 52 =$ the carrier coset root count (v128) <a href="#">[E]/[C]</a>                             |
| v136_dual_normal_selector.py | Dual-normal selector: $(d, n)$ pins $R$ <i>columnwise</i> — $d \cdot c_j = a_j$ , $n \cdot c_j = 8\delta_{1j}$ , kernel = primitive $(6, 8, 7)$ , census: each column unique (1 of 216) $\Rightarrow R4'$ shrinks from $6^9$ to three column problems; the $A_0^*$ zero mode carries $\sigma = (2, -9, 5) = ( \mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ( $\ v\ ^2 = \frac{16}{9}$ , $16 = \dim S^+$ ; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$ ; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element <a href="#">[E]/[C]</a> |
| v137_qplus_cohomology.py     | The $Q_+$ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ( $\omega_k = z^{k-1} dz / (z^4 - 1)$ ), character $i^k$ , residues $\zeta^k/4$ ; degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the $A_3$ exponents ( $h = 4 =  \mu_4 $ ); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = $Q_+$ : <i>one</i> spectrum, four readouts; the QGEO residue = the naturality of the canonical map <a href="#">[E]/[C]</a>                                                                                                          |
| v138_vw_firewall.py          | Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); <b>edge match</b> : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall <a href="#">[E]/[C]</a>                              |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v139_selector_triangle.py | Selector triangle: $d=\frac{3}{2}a-2\cdot\mathbf{1}$ exactly (pure anchor data, no $R$ ) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(\mathbf{1}, a, \sigma)$ has $\det=11=\ \text{Pl}(K)\ _1$ , and $n$ is the <i>unique</i> covector with atom pairings $(2, 8, 121)=( \mathbb{Z}_2 , \text{rank } E_8, 11^2)$ ; on the constrained line the pairing is $11(11+t)$ , a square only at the true $n$ ; review lift exact (audit): $n=\text{reverse}(\sigma)+ \mu_4 e_3$ ; $R_4'$ residue = three pairings $[\mathbf{E}]/[\mathbf{C}]$ ( <i>since v142: two — the third pairing is integrality-forced</i> ) |
| v140_canonical_map.py     | The canonical map exists and is rigid: the plain deck $U$ (characters $(0, 1, 3)$ ) cannot match $H^1$ ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential $i^{Q+}((1, 2, 3))$ both carry the $H^1$ character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); $i^{Q+}$ pulls the Euler degree back to $Q_+$ , $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the choice since derived, v141</i> )                                        |
| v141_deck_selection.py    | Deck Selection Theorem (R5): the established integer deck $G=T_A\Sigma$ (v97/v98) pairs the $Q_+=1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; $i^{Q+}$ excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                            |
| v142_frame_integrality.py | Frame Integrality (R4'): pairing triples of <i>integer</i> covectors against $(\mathbf{1}, a, \sigma)$ form an index-11 sublattice $(x-3y+z \equiv 0 \pmod{11}, 11=\ \text{Pl}(K)\ _1)$ ; with $(x, y)=( \mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \pmod{11}$ , primitivity forces even $t$ ; Cramer reductions for the two line pairings; $R_4'$ : three pairings $\rightarrow$ two $[\mathbf{E}]/[\mathbf{O}]$ ( <i>since v145: the values are atom identities — residue = one lift map</i> )                                                                                                  |
| v143_graded_frobenius.py  | Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k=C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$ ; glue average = carrier projector (index $ \mathbb{Z}_4 =4$ ); each sector one $W(D_5)\times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                                         |
| v144_detratio_family.py   | Det-ratio cancellation within the SdS family (R1, the v131 step): $e_2$ -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0) = -\frac{2}{3}$ , v132) — no first-order term in the split; deficit coefficient $\frac{32}{27} =  \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the bend since shown <math>\det'</math>-clean, v147</i> )                                                                                            |
| v145_pairing_atoms.py     | $R_4'$ values are <i>atom identities</i> : with the lift $n=w_0(\sigma)+ \mu_4 e_3$ ( $w_0$ = the $A_3$ exponent duality), $n\cdot\mathbf{1}= \mu_4 - \mathbb{Z}_2 = \mathbb{Z}_2 $ , $n\cdot a= \mu_4 a_3=8$ via the new closure $ \mu_4 +g_{\text{car}}=N_{\text{fam}}^2$ ( $4+5=9$ , $\sigma \perp w_0(a)$ ), $n\cdot\sigma=\ \sigma\ ^2-N_{\text{fam}}^2+ \mu_4 g_{\text{car}}=121$ ; residue = derive the <i>one</i> lift map $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                 |
| v146_moebius_d4.py        | Möbius $D_4$ realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — $\iota^*$ swaps $\chi_1 \leftrightarrow \chi_3$ , fixes $\chi_2$ with $+1$ , $\det=-1$ ( $=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                          |
| v147_clock_gaussian.py    | The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born <sup>2</sup> $\Gamma$ -ratio $(1-\alpha)^{p_2}$ , rate = $-p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ $\det'$ -clean; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                         |
| v148_fock_census.py       | NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ( $2^8=256$ ) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection ( $128$ = one $SO(16)$ chirality), $248=120_{\text{NS}}+128_{\text{R}}$ ; R2 is irreducibly a twisted-sector net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                         |
| v149_cusp_normal.py       | Both dual normals are <i>anchor data</i> ( $R_4'$ ): in the cusp frame $n$ pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per $Q_+$ line ( $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the $w_0$ -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$ , sum 14 = $\dim G_2$ ; residue = <i>one</i> assignment $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                            |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v150_replica_eh_model.py       | The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and $t$ -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$ ); the gap makes the $\zeta$ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$ ; linearisation $= (\ln m/12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] ( <i>Calderón transfer since answered — the DtN kernel is conically clean, v151</i> ) |
| v151_bfk_split.py              | BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ( $C_N=C_D$ , derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}}=C_{\text{cone}}(\gamma)-2C_D(\gamma/2)=0$ ; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]                                 |
| v152_norm_is_anchor.py         | R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio $m/\mu$ is physical; $k=c_3/2 \Leftrightarrow m/\mu=e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, $\pi$ }; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]                                                                                                                                                                                                          |
| v153_no_unit_theorem.py        | No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$ , a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu = e^{3/4}$ (a ratio); $v_{\text{geo}}$ is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, $\pi$ } [E]/[O]                                                                                                                                                                            |
| v154_simple_current_theorem.py | Simple-Current Extension Theorem (the central $G_{\text{net}}$ theorem, assembled): $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has index $ L =4$ , $c=5+3=8$ , $\mu(B)=\mu(A)/ L ^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B=(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]                                                                                                                                                                                                |
| v155_quasifree_boundary.py     | Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $PTP$ (a contraction, hence a valid covariance), so the $G_{\text{net}}$ analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]                                                                                                                                                                                         |
| v156_seam_net_construction.py  | Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$ ; the $E_8$ currents are $248=120_{\text{NS}}+128_{\text{R}}$ ; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]                                                                                                                                              |
| v157_rigid_fixed_point.py      | Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\bar{h}=0$ ) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $ \mathbb{Z}_2 $ that halves $8\pi$ in $c_3$ IS the Ramond projection giving $\mu=1$ ; (iv) holomorphic $c=8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C] |
| v158_fixed_point_stable.py     | The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$ , bilinear current $h=1$ , quartic $h=2$ ; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]                                                                                                                                                                          |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v159_pyrate_gauge_crosscheck.py       | PyR@TE cross-check of the $F_{\text{transfer}}$ gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1 = \frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the $U(1)$ hypercharge index $\sum \frac{3}{5} Y^2$ , and the external RGE generator PyR@TE3 ( <a href="#">arXiv:2007.12700</a> ) which writes $\beta_{g_1} = (\frac{41}{10}) g_1^3$ verbatim; the $M=41$ split $10b_1 = 40_{(\text{ferm})} + 1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in <a href="#">verification/pyrate</a> . The transfer functor's gauge inputs are not free knobs <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v160_seam_gaussianity_from_pf.py      | Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> $\kappa_{2n}$ is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap} > 0$ — the value $(2/3)^6$ only sets the rate. <b>The load <a href="#">[O]</a>:</b> v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ , while the seam correlation cone has $\binom{16}{4}=1820$ , $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement ‘the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum’ <a href="#">[E]</a> / <a href="#">[O]</a>                                                                                                                                                                                                                                                   |
| v161_one_particle_reduction.py        | The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$ , the second quantisation of a one-particle contraction $t$ ; $\Gamma(t)$ acts on the degree- $m$ correlator sector as the exterior power $\Lambda^m(t)$ , so (1) $\text{spec}(\Gamma(t) _{\deg m}) = \text{products of } m \text{ one-particle eigenvalues}$ <a href="#">[E]</a> ; (2) <b>the cone gap equals the one-particle gap</b> — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly <a href="#">[E]</a> ; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) <a href="#">[E]</a> ; (4) quasi-freeness <i>forced</i> not assumed — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant <a href="#">[C]</a> . <b>Residual <a href="#">[O]</a>:</b> v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 $K_\Sigma$ , gap $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open <a href="#">[E]</a> / <a href="#">[O]</a> |
| v162_seam_transport_identification.py | The final identification, forced as far as it goes — the v161 one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$ , $p_2=6= R_+(A_3) $ , all carrier data <a href="#">[E]</a> ; (2) $\mu_4$ -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} =  \mu_4  \text{diag}(X)$ , commutant of $U =$ the diagonal algebra — so the gapped $\mu_4$ -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ <a href="#">[E]</a> ; (3) the cusp grading <i>is</i> the $A_3$ grading on the 16 Majoranas (v137) <a href="#">[E]</a> ; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) <a href="#">[E]</a> . <b>Closure <a href="#">[C]</a>/<a href="#">[O]</a>:</b> with v160+v161, (A) closes given only $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, <a href="#">HOR.CLOCK</a> ) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) <a href="#">[E]</a> / <a href="#">[O]</a>                                                                               |
| v163_rg_stability_flavor.py           | $F_{\text{transfer}}$ scale-tagging: the lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ , $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are <i>RG-stable</i> — PMNS running is $y_\tau^2$ -suppressed ( $\sim 1.8 \times 10^{-5} \ll \text{precision}$ ), so seam value = measured value; the quark mass-ratio readouts ( $m_t/m_b \sim 41$ ) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries $y_{\text{max}}^2$ (PyR@TE SM RGEs) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v164_qcd_scale.py                | the QCD confinement scale is parameter-free from the carrier $b_3=-7$ : running $\alpha_s(M_Z)$ down (two-loop, $n_f$ thresholds) gives $\alpha_s(m_b)\sim 0.22$ , $\alpha_s(m_c)\sim 0.38$ and confinement ( $\alpha_s\sim 1$ ) at $\sim 0.5$ GeV — so the QCD-scale <i>numerator</i> of $m_p/m_e$ is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ( $m_p=O(1)\Lambda$ is lattice) [E]/[O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v165_premise_a_closure.py        | Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics [C] — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the $\mu_4$ extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$ , $248=120+128$ , character $E_4/\eta^8=j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $ k $ (v156/v157); (2) R1 (seam clock) reduces to GATE.QGEO [C] — $\mu_4$ -equivariance forces diagonal-in-cusp, the flat $\mu_4$ -equivariant generator is <i>unique</i> = $A_0^*$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary = the $\mu_4$ -punctured sphere $\mathbb{P}^1\setminus\mu_4$ ’ = GATE.QGEO (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> by the No-Unit Theorem (v153). <b>Final accounting:</b> (A) = A2 assembly [C] + R1=GATE.QGEO + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O] |
| v166_higgs_free_seam.py          | the Higgs quartic from the <i>free seam</i> (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}})=0$ and $\beta_\lambda(M_{\text{seam}})=0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured $(m_H, m_t)$ gives $\lambda(\bar{M}_{\text{Pl}})\sim 0.002$ , $\beta_\lambda(\bar{M}_{\text{Pl}})\sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim 0.022$ to $\sim 0.002$ ). The double condition predicts $m_H\sim 129\text{--}134$ GeV; at the scalaron scale it gives $\sim 107$ GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v167_inventory_post_closure.py   | The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123’s FIVE classes to: <b>Tier 0</b> — the irreducible core $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> (No-Unit, v153), nothing to close; <b>Tier 1</b> — ONE hinge (seam = flavour = horizon): $A_2$ net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P}\setminus\mu_4$ , cohomology $b_1=N_{\text{fam}}=3$ ); <b>Tier 2</b> — folded ( $\rightarrow v_{\text{geo}}$ , $\rightarrow G_{\text{net}}$ ); <b>Tier 3</b> — $F_{\text{transfer}}$ , one functor with four typed conditional interfaces ( $\eta_B$ carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]                                                                                                                                                                                                                                                                                                                                                           |
| v168_qgeo_rigidity.py            | The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z\rightarrow iz, z\rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful $D_4$ are the $\mu_4$ square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4-1=3=N_{\text{fam}}$ ; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ( $k=1, 2, 3$ ) are $\mu_4$ -eigenforms with $z\rightarrow iz$ eigenvalue $i^k$ = the three nontrivial characters $\chi_1, \chi_2, \chi_3$ , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$ . QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v169_etaB_boltzmann_interface.py | The $\eta_B$ leptogenesis transfer <i>operationalised</i> (Tier-3 $F_{\text{transfer}}$ , not a new class): type-I thermal leptogenesis $\eta_B\sim 0.96\times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT’s NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$ . Davidson–Ibarra $\varepsilon_1=\frac{3}{16\pi}M_1m_3/v^2\sim 8.5\times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1)\sim 0.03\text{--}0.19$ . <b>Kill test:</b> over natural $M_1\in[3\times 10^9, 3\times 10^{10}]$ GeV and washout, $\eta_B$ spans $[8.4\times 10^{-11}, 4.7\times 10^{-9}]$ which <i>brackets</i> the observed $6.1\times 10^{-10}$ (canonical point gives $6.0\times 10^{-10}$ , not tuned) $\Rightarrow$ the route is viable; $M_1$ /washout are scenario inputs, so $\eta_B$ stays [C]. If excluded, the route falls, not the theory [C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v170_diamond_slice_compression.py | E8 Slice Compression Lemma: the seven $E_8$ audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ( $p_n=2+2^n$ ) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$ . The $K_4$ edge products $(12, 18, 30, 24, 40, 60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$ ; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$ ; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$ . The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v171_os_moment_cluster.py         | Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$ , $\xi=0.41105$ , Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$ ; gap safety $\Delta_{\text{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}>0$ with $31=1+h^\vee(E_8)$ , $c((E_8)_1)=248/31=8$ [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v172_trace_anomaly_seed.py        | Trace-anomaly origin of the seed prefactor $\frac{4}{3}$ : over the seam $S^2$ ( $\chi=2$ ) the $(E_8)_1$ ( $c=8$ ) Weyl anomaly is $c\chi/6=\frac{8}{3}$ , halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$ , the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$ . Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v173_pfaffian_cp_car.py           | Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$ , $\gamma_5$ -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ( $\arg \in \{0, \pi\}$ ), and reflection positivity ( $Z>0$ ) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v174_seam_fock_readings.py        | Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ( $2^{15}/16=2^{11}$ ), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$ , with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$ ) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig $g$ -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v175_net_existence_full_cone.py   | The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) <b>full-cone RP for all <math>m</math></b> — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction $t$ acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ( $\dim 2^{16}=65536$ ); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$ , verified on all $2^{16}$ products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 $K_\Sigma$ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ , so full-cone RP reduces <i>for all <math>m</math></i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) <b>A2 net existence</b> assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$ , $E_8$ Cartan even unimodular ( $\det 1$ ); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v176_seam_collar_realisation.py | The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful $D_4$ marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with $H^1$ grading $(1, 2, 3)$ , the Calderón contraction is the rank-8 $K_\Sigma$ polarisation and the index-4 extension is $(E_8)_1$ . The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius $D_4$ order 8, $b_1=N_{\text{fam}}=3$ ), Calderón DtN symbol $ k $ (v156), $H^1$ character = generation grading $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ) (v137/v168), $\mu_4$ -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all $m$ (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar's boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$ ), left honestly [O][E][O]                                                                                                                                                                                           |
| v177_seam_marking_kernel.py     | The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$ , a non-fixed orbit scales to $\mu_4$ , the reflection $z \mapsto 1/z$ gives $\sigma\rho\sigma = \rho^{-1}$ ( $D_4$ , order 8) — a genus-0 four-marked faithful- $D_4$ boundary <i>is</i> $(\mathbb{P}, \mu_4)$ ; [E] COHOM: $\rho^*\omega_k = i^k\omega_k$ so the $H^1$ grading is $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ); [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$ , $\omega_2$ fixed); [E] NET (v154/v175): index-4 $\mu_4$ extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$ . The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked $D_4$ boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the $\mu_4$ -equivariant free gapped contraction as an <i>operator</i> ); the whole structural residual is now exactly these two doors [E][O]                                                                                                                            |
| v178_marks_kernel_attempt.py    | An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock $\rho$ ( $\sim z \mapsto iz$ , fixed points $\{0, \infty\}$ ) and reflection $\tau$ forces the marks to be a generic $\rho$ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$ , $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful $D_4$ order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free $H^1$ the $\mu_4$ characters $(1, 2, 3)$ are distinct, so by Schur a $\mu_4$ -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $ , Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O] |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v179_conformal_realisation.py       | The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 <i>is</i> P1 — $c_3=1/( \mathbb{Z}_2  2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$ , the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$ , order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$ , a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$ , the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]                                                         |
| v180_clock_is_mobius.py             | Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: 'the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$ , so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of $S^2$ is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically $z \mapsto iz$ ; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$ ; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$ , multiplier $i$ , free orbit $\mu_4$ ). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O] |
| v181_clock_is_conformal_symmetry.py | The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$ . The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4= \mu_4 $ , the $\mu_4$ deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i> , not a missing theorem: 'the seam's conformal structure <i>is</i> the $\mu_4$ -deck structure of the carrier clock' (the carrier $\mu_4$ glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain<br>REALIZE(v175) $\rightarrow$ theorem(v176) $\rightarrow$ MARKS+KERNEL(v177) $\rightarrow$ finite<br>cores(v178) $\rightarrow$ CONF.01(v179) $\rightarrow$ ISO.01(v180) $\rightarrow$ SYM.01(v181) has reached bedrock; we stop here honestly [E][O]  |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v182_reviewer_residual_map.py   | The four external-review concerns reduce to the two already-named residuals + one data-only item. $A$ (mapping $2D \rightarrow 4D$ ) [E]/[C]: the Plücker readout $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . $B$ (UV gravity) [E]: ‘gravity = geometric boundary readout’ is QGEO.SYM.01. $C$ (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a=(1, 1, 2)$ in $\geq 6$ independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). $D$ (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck, $t \sim 2.84$ only locates the point, and the missing RG generator is $F_{\text{transfer}}$ , so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . NET: three reduce onto $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ , one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C] |
| v183_koide_f_corner_transfer.py | The Koide source→pole factor $\frac{53}{54}$ has an <i>operator</i> origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}$ . (1) $\mathbf{1}^T R a = 27$ is the $E_6 \times A_2$ cubic family block ( $248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}}$ ); (2) $F=R+Q$ is the <i>missing</i> Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52+1$ ; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly ( $54=2 \cdot 27$ ); (4) negative control: only $F$ gives 53 ( $R/Q/K/L \rightarrow 32/21/35/56$ ). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source→pole reading the $F$ corner stays an $F_{\text{transfer}}$ conjecture [E][C]                                                                                                                                                                                                                              |
| v184_etaB_anchored_boltzmann.py | Honest test of the external-review $\eta_B$ proposal ( $M_1=M_R\phi_0^4$ , $\tilde{m}_1=m_3/A_\Lambda$ ) — firewall verdict: <i>sharper scenario</i> , not a closure. The washout anchors plausibly ( $\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5 \text{ meV}$ , $A_\Lambda=10= E(K_5) $ , in the strong-washout window) [C]; but $M_1$ does <i>not</i> — $M_R$ is the seesaw scale $\sim v^2/m_3 \approx 6 \times 10^{14} \text{ GeV}$ and the nearest clean TFPT powers of $\bar{M}_{\text{P1}}$ both miss it ( $c_3$ -power off $\sim 75\%$ , $\phi_0$ -power off $\sim 40\%$ ), so $M_1=M_R\phi_0^4$ merely relocates the free input $M_1 \rightarrow M_R$ (the reviewer’s $6.0 \times 10^{-10}$ uses $M_R=1.3 \times 10^{15}$ ; the seesaw value gives $3.4 \times 10^{-10}$ , the tell that $M_1$ is unpinned). $\eta_B$ stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — $\eta_B$ stays [C], a sharper scenario [C]                                                                                                                                                      |
| v185_axion_relic_solver.py      | Axion relic, honest misalignment estimate: the determinant-line axion $f_a=M_{\text{scal}}/128 \approx 2.39 \times 10^{11} \text{ GeV}$ ( $m_a \approx 23.8 \mu\text{eV}$ ) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this $f_a$ (prefactor $\sim 0.026/\theta^2$ , $\theta \sim O(1)$ gives $\sim 21\%$ of $\Omega_{\text{DM}}$ , need $\theta_{\text{eff}}^2 \sim 4.7$ ), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128=(16, 4)+(16, 4)$ ) [C][O]                                                                                                                                                                                                                                                     |
| v187_ftransfer_laws.py          | The $F_{\text{transfer}}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, $\eta_B$ , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3=-7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]                                                                                                                                                                                                                                                                                                                                                                                                                |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v188_frontier_wording_guard.py | The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” ( $M_R$ is the seesaw scale, not a compiler power), “relic scale $f_a, m_a$ open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q=0.664\dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]                                                                                                                               |
| v189_riemann_roch_carrier.py   | The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$ . [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8, \mathfrak{v}1$ ), $N_{\text{fam}} = 3$ . [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$ , but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C] |
| v190_nariai_entropy_bound.py   | The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2+1)/\Phi_3(x)$ ( $x=r_b/r_c$ ; $\Phi_3=x^2+x+1$ the $N_{\text{fam}}=3$ cyclotomic, already in tfpt_horizon_readouts) satisfies $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality exactly at the Nariai merge $x=1$ , via the perfect square $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x>0$ is $2/3 =  Z_2 /N_{\text{fam}}$ ) [E]                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v191_universal_branch_line.py  | The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ( $m=0 \rightarrow 7/2$ ). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 =  Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]                                                                                             |
| v192_qgeo_energy_clock.py      | The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ( $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda =  k $ . But ‘ $\rho$ preserves $\Lambda$ ’ is plausibly equivalent to ‘ $\rho$ is the conformal deck’, so it is a sharper restatement, not a theorem, unless $\rho$ is independently defined and the invariance independently proven (open). Bedrock stays [O]                                                                                                  |
| v193_qgeo_energy_commutator.py | QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, $\rho$ the carrier clock, $\Lambda_\Sigma$ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] $\rho$ is carrier-defined (Coxeter of $W(A_3)=S_4$ , order $4= \mu_4 $ , v117), not imported from the seam; (2) [O] the non-circularity hinge — $\Lambda_\Sigma$ must be the <i>raw</i> RP-seam DtN, not the $\mu_4$ normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the $(1, 2, 3)$ grading on $H^1$ (QGEO.COHO.M.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by $5/3$ ) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O]                                            |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v194_raw_seam_dtn_<br>rp.py                  | QGE0.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i> , not a closure. (1) [E] finite core: $\rho: z \mapsto iz$ on $H^1 = \langle w_1, w_2, w_3 \rangle$ , $w_k = z^{k-1} dz / (z^4 - 1)$ , acts as $\rho^* w_k = i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma] = 0$ on $H^1$ (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so $\Lambda_\Sigma$ is RP-definable without assuming $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] residual relocates: full- $L^2$ commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGE0.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]                                                                                                                                                                                                                                                                                    |
| v195_marks_<br>lefschetz_<br>character.py    | QGE0.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing $D_4$ marks, they are <i>forced</i> to be a free $\mu_4$ orbit. [E]: $\rho: z \mapsto iz$ (order 4) fixes only $\{0, \infty\}$ ; $H^1 = \chi_1 + \chi_2 + \chi_3$ (the $A_3$ exponents (1, 2, 3), v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; $\text{rank } H_1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks = $\mu_4$ (cross-ratio 2); $\text{Tr}(\rho H^1) = i + i^2 + i^3 = -1$ , $L(\rho) = 1 + 1 = 2 = \# \text{fixed}$ . [O] residual relocates from “the seam is $\mathbb{P} \setminus \mu_4$ (geometric)” to “ $H^1$ carries the $A_3$ -exponent rep” — less circular [E][O]                                                                                                                                                                                                                                                                                                                                                                                                           |
| v196_seam_energy_<br>functional.py           | QGE0.VARI.01 — the $E_{\text{fail}}$ variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: $E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2$ , with $E_{\text{fail}} = 0 \Leftrightarrow$ the QGE0.SYM.01 conditions. [E] on the finite $H^1$ block ( $\rho = \text{diag}(i, -1, -i)$ , $\Lambda$ diagonal, $\Theta = \text{conj}$ ) all three vanish $\Rightarrow E_{\text{fail}} = 0$ ; [E] a $\mu_4$ -breaking $\Lambda$ or non-order-4 $\rho$ gives $E_{\text{fail}} > 0$ ( $\mu_4$ a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]                                                                                                                                                                                                                                                                                                                                                   |
| v197_rr_carrier_<br>clifford_d5.py           | ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from $g_{\text{car}}$ to $D_5$ itself (review point 7): the 5-dim carrier mode space $C_{\text{car}} := H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10) = D_5$ half-spinor by its even Clifford algebra. [E] $\dim \Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\text{car}} - 1} = \dim S^+$ ; the $\mathfrak{so}(10) = D_5$ half-spinor $2^4 = 16 = S^+ = \Lambda^{\text{even}}(C_{\text{car}})$ ; closes the chain $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ from one four-point divisor (cycle side $\text{rank } H_1 = 3 = N_{\text{fam}}(A_3)$ ; function side $h^0 = 5 = g_{\text{car}}$ ; Clifford spinor $16 = D_5$ half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined $g_{\text{car}} = 5$ [E][C]                                                                                                                                                                                                                              |
| v198_modular_<br>commutator_<br>reduction.py | QGE0.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual ( $[\rho, \Lambda_\Sigma] = 0$ on full $L^2$ , framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $ k  = \sqrt{-d^2}$ is $\text{diag}( n )$ and the clock $\rho: z \mapsto iz$ is $\text{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho,  k ] = 0$ exactly on <i>all</i> of $L^2$ ; [E] Tomita–Takesaki: $[\rho, \Lambda_\Sigma] = 0$ follows from $\omega \circ \rho = \omega$ (the modular $\Delta$ is intrinsic to $(M, \Omega)$ ; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite $H^1$ block is already $\mu_4$ -invariant (v177). [O] residual: the full raw seam state is $\mu_4$ -invariant ( $\omega \circ \rho = \omega$ ) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational QGE0.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v199_seam_state_invariance.py   | QGE0.STATE.01 (work package A) — attacking $\omega \circ \rho = \omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ( $\rho = \text{Coxeter of } W(A_3) = S_4, \rho^4 = 1$ , an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ ( $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ ); [E] character criterion: $[\rho, H_1] = 0 \Leftrightarrow H_1$ block-diagonal in the four $\mu_4$ -character classes $\{n \equiv r \pmod{4}\}$ ; principal symbol $ k  = \text{diag}( n )$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on $H^1$ (v177); not a closure [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v200_seam_variational_scan.py   | QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}} = 0$ at the $\mu_4$ config on the fixed block; here the clock angle is free and $E_{\text{fail}}$ is minimised over it. [C] scanning $E_{\text{fail}}(\theta) = \ \rho(\theta)^4 - 1\ ^2 + \text{faithfulness over } \theta \in (0, \pi)$ gives the unique minimum at $\theta = \pi/2$ ( $E_{\text{fail}} = 0$ : order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the $\mu_4$ normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite $H^1$ grading, not the full operator-valued minimisation (the v199 residual) [C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v201_seam_subprincipal_marks.py | QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda =  k  + M_f$ , $M_f$ = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n   M_f   n' \rangle = f_{n-n'}$ . [E] $M_f$ is $\mu_4$ -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ( $ k $ already commutes, v198). [E] a $\mu_4$ -mark sum $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \pmod{4}]$ for <i>any</i> profile $g$ — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks = $\mu_4$ orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it. [O] new residual: $\omega \circ \rho = \omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the $\mu_4$ marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O]                                                                                                                                                                                                                                                                                                                                                   |
| v202_rare_kaon.py               | The rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix ( $s_{12} = \lambda_C$ , $s_{23} = \varphi_0 / (1 + \lambda_C) =  V_{cb} $ , $s_{13} = \lambda_C^3 / 3$ , $\delta = \pi/3 + 3\lambda_C^2 = 1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$ . [E] CKM point + exact apex $(\bar{\rho}, \bar{\eta}) = (0.1374, 0.3509)$ , $\lambda_t = V_{ts}^* V_{td}$ ; [C] the branching ratios ( <i>not</i> parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\text{BR}(K^+)$ $\text{BR}(K^+) = 9.45 \times 10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6_{-1.8}^{+1.9}) \times 10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an $F_{\text{transfer}}$ bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6 \pm 0.4) \times 10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$ ; [X] dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$ , or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X] |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v203_eht_achromatic.py    | The EHT achromatic polarization-rotation intercept $\beta_{\text{BH}}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes ( <i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\text{BH}}(r)=Q_e^{\text{eff}}Q_m^{\text{eff}}/(256\pi^4r^2)=16c_3^4(Q_eQ_m/r^2)$ , $16c_3^4=1/(256\pi^4)$ exactly. [E] $\alpha$ -kernel cross-link: $16c_3^4=\delta_{\text{top}}/3$ , $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ — the <i>same</i> top-form coefficient that fixes the $\alpha$ -kernel precision-zone correction (one row from $\beta_{\text{rad}}=\varphi_0/4\pi$ ). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E\cdot B$ reversal; the amplitude $Q_eQ_m$ is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$ , sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X] |
| v204_muon_seam_g2.py      | The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon- $g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma\delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2$ ( $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4$ , $v2/v23$ ; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$ , v51); the seam-loop projection $a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9)\approx 2.879\times 10^{-9}$ . [E] the value is an exact compiler number ( $\delta_2=4!\text{Tr}_{S^+}(X^2)c_3^8$ , $\text{Tr}=120=5!$ ); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ( $1/2\pi=4c_3$ ) is a physical bridge; [E] $0.81\sigma$ vs dispersive $\Delta a_\mu=(2.49\pm 0.48)\times 10^{-9}$ ; [C] HONEST: lattice/CMD-3 HVP shrinks it ( $\sim 1.5\times 10^{-9}$ ), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged $\Delta a_\mu$ outside $2.879\times 10^{-9}\pm 0.5\times 10^{-9}$ excludes the mechanism [E][C][X] |
| v205_xi_threequarter.py   | $\xi=c_3/\varphi_{\text{tree}}=3/4$ : an <i>independent</i> route to the gravitational $3/4$ (archive integration of the old Eliminating- $K$ note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$ ), and at tree level $\varphi_{\text{tree}}=1/(6\pi)$ this is exactly $3/4$ . [E] tree identity $\xi=c_3/\varphi_{\text{tree}}=3/4$ ; [E] Einstein-limit reduction $8\pi c_3^2=c_3\Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)$ (v152 $k=c_3/2$ ) — the torsion-compression $\xi$ and the gapped EH replica are two independent appearances of the same gravitational $3/4$ ; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ ( $-0.226\%$ ); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]                                                                                                                          |
| v206_h0_lambda_branch.py  | The cosmological-constant Letter branch: the seam number $\delta_\Sigma=m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1=\Omega_{\text{adm}}=48$ , the Planck-convention split, and the quantitative $H_0=66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative $H_0$ -vs- $\Lambda$ row). [E] $\delta_\Sigma^{\text{lead}}=48e^{-2\pi\cdot 137.036/3}\approx 1.085\times 10^{-123}$ ; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4=(8\pi)^2=631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda=M_{\text{Pl}}^4\delta_\Sigma\approx 2.4\times 10^{-47}\text{ GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10}\delta_\Sigma \approx 122.96$ , consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}}=66.51$ , $H_0^{\text{disc}}=67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; $H_0$ uses imported $\Omega_b, \omega_{\text{DM}}$ , a [C] readout [E][C]                          |
| v207_asymptotic_safety.py | Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close $G_{\text{metric}}$ . [E] the axion coupling is fixed and small $y^2=16c_3^2=1/(4\pi^2)=0.025330$ exactly ( $g_{a\gamma\gamma}=-4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g=2+\eta_R$ is $+2>0$ as $g\rightarrow 0$ and $<0$ at large $g$ (torsion $K^2$ damping) $\Rightarrow$ a root $g_*>0$ ; [E] $g_*=2/a_1>0$ finite, Reuter-type $O(1)$ , no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v208_bh_thermodynamics.py     | Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H: \Delta^{it}=e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta=2\pi$ and $T_H=\kappa/(2\pi)$ ; the $2\pi$ is the seam unit $1/(4c_3)$ ( $1/2\pi=4c_3$ , v58), reproducing $T_H=c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R)=R+R^2/(6M_s^2)$ ( $M_s=c_3^{7/2}\bar{M}_{\text{Pl}}$ , v28), $S_W=(f_R A)/(4G)=(A/4G)(1+R_h/3M_s^2)$ — leading $A/4G$ is the $c_3$ area law ( $1/4=1/ \mu_4 $ ), the correction an exact scalaron consequence; [E] SdS $T_H=(1/4\pi r_h)(1-\Lambda r_h^2)$ , $1/4\pi=2c_3$ ; [E] area law $S_{BH}=M^2/(2c_3)=4\pi M^2=A/4G$ [E][C]                                                                                                                                                                                                                                                                                                                                                 |
| v209_bh_regular_core.py       | The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2=(2/3)^6=64/729$ with gap $\Delta=6\log(3/2)>0$ (v56) — the core is a fixed point, $\rho\rightarrow\infty$ replaced by $\varphi\rightarrow\varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = the traceless projection of $a=(1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2=(2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH}=A/4G$ , $1/4=1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]                                                                                                                                                                                                                                     |
| v210_mark_local_dtn.py        | A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega\rho=\omega$ , not just $[\rho, \Lambda]=0$ ); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four $\mu_4$ marks $f(\theta)=\sum_{j=0}^3 g(\theta-j\pi/2)$ , builds the Steklov DtN $\Lambda= D_\theta +M_f$ . [E] every off-(mod 4) Fourier coefficient of the real $f$ vanishes to $<10^{-16}$ (exact 4-fold cancellation), although $g$ has all modes. [C] $\ [\rho, \Lambda]\  < 10^{-15}$ . [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C=\frac{1}{2}(1+\text{sgn } H_1)$ satisfies $\rho C\rho^\dagger=C$ to $<10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$ ; convergence stable for $N\in\{16, 32, 64\}$ . [O] certifies ‘mark-local DtN $\Rightarrow \omega\rho=\omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O] |
| v211_axion_spine_angle.py     | Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ ( $108^\circ$ ): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$ ). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$ , no fit); [E] the v185 finite-T solver ( $\theta=1 \rightarrow 0.03$ ) reaches $\Omega_{\text{DM}}$ at $\theta \approx 106^\circ$ , so $108^\circ$ is on target (harmonic $0.03\theta^2=0.107$ ); [C] <i>robust</i> : $108^\circ$ is $62^\circ$ below the $170^\circ$ hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i=\pi(1-\varphi_{\text{seam}})\approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch [E][C][O][X]                                                                                                                                                                                                                                                                                                                                   |
| v212_leptogenesis_decouple.py | Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda=10= E(K_5) $ — $M_1=M_{\text{scal}}\varphi_0^2/A_\Lambda \approx 8.65\times 10^9$ GeV and $\tilde{m}_1=m_3/A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1=M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw $M_R$ ). [E] shared decouple $A_\Lambda=10$ ; [E] $M_1=8.65\times 10^9$ GeV inside the thermal window $[3\times 10^9, 3\times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5\times 10^{-7}$ , $\eta_B \approx 1.2\times 10^{-9}$ brackets observed $6.1\times 10^{-10}$ ; [C] HONEST: $M_1$ uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]                                                                                                                                                                                                                                                                                                                                           |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v213_ftransfer_<br>functor.py  | The <code>F_transfer</code> functor contract ( <code>CONTRACT.F.01</code> ): the four frontier transfers are ONE typed functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel <b>[E]</b> composed with an external continuous solver <b>[C]</b> ; consolidates v183 ( $F_{\text{pole}}$ ), v212 ( $F_{\text{Boltzmann}}$ ), v211 ( $F_{\text{relic}}$ ), v164/FR.MPME ( $F_{\text{QCD}}$ ), the structural complement of the v187 typing guard. <b>[E]</b> axiom 1 $\mu_4$ -deck equivariance ( $\lambda_2 = (2/3)^6 = 64/729$ the deck transfer eigenvalue); <b>[E]</b> axiom 2 Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ , $\text{Pl}(F) = (1, 22, 30)$ , $\ \text{Pl}(K)\ _1 = 11$ ); <b>[E]</b> axiom 3 positivity/stochasticity ( $\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f \in (0, 1)$ ); <b>[E]</b> axiom 4 external modules explicit ( $b_3 = -7$ a carrier output, $ b_3  = \Omega_{\text{adm}} - 10b_1 = 7$ ); <b>[C]</b> verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure <b>[E][C]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v214_seam_<br>pillowcase.py    | The <i>pillowcase</i> reduction of <code>QGEO.SYM.01</code> ( <code>QGEO.PILLOW.01</code> ): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the $\mu_4$ marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. <b>[E]</b> $\chi_{\text{orb}}(S^2(2, 2, 2, 2)) = 2 - 4(1 - \frac{1}{2}) = 0 \Rightarrow \text{flat}$ (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits $\pi$ sum to $4\pi = 2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}} = 0$ , flat away from the marks). <b>[E]</b> NEW LINK: cross-ratio( $\mu_4$ ) = 2 (v168) $\Rightarrow j(\lambda) = 256(\lambda^2 - \lambda + 1)^3 / (\lambda^2(\lambda - 1)^2)$ with $j(2) = 1728$ (all six harmonic cross-ratios $\{2, -1, \frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. <b>[E]</b> $j = 1728 \Leftrightarrow \tau = i$ (square torus) $\Leftrightarrow \text{CM by } \mathbb{Z}[i] \Leftrightarrow \text{Aut} = \mathbb{Z}/4$ ( $j=0$ gives $\mathbb{Z}/6$ ; generic $\mathbb{Z}/2$ ). <b>[E]</b> explicit CM $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$ has order $4 = z \mapsto iz$ . <b>[E]</b> negative control $\lambda = 3 \Rightarrow j = 21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut} = \mathbb{Z}/2$ , no clock. <b>[E]</b> UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. <b>[O]</b> does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s <b>[E]</b> |
| v215_seam_deck_<br>killtest.py | The bedrock <code>QGEO.SYM.01</code> ( $\omega \circ \rho = \omega$ ) recast as a <i>falsifiable</i> kill-test ( <code>QGEO.KILL.01</code> ), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ ; the principal symbol $ k $ commutes exactly, so the residual is the sub-principal $M_f$ with $[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$ -invariant; measure form = the raw Gaussian bulk form $A$ is $\mu_4$ -deck-invariant. Four levers (carrier-side <b>[E]</b> , seam-side <b>[O]</b> ): <b>K1</b> exactly 4 marks (order-4 clock free orbit $\mu_4 + b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 =  \mu_4 $ ); <b>K2</b> square config (cross-ratio $2 \Rightarrow j = 1728$ , $\text{Aut} = \mathbb{Z}/4$ ; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$ ; generic $\Rightarrow \mathbb{Z}/2$ ); <b>K3</b> mod-4 sub-principal support ( $\mu_4$ -mark sum $\Rightarrow f_m = 0$ for $m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f] = 0$ ); <b>K4</b> transfer gap $(2/3)^6 = 64/729$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ ). Freeze-bound ( <code>freeze_file.csv seam_deck_symmetry</code> ). <b>[O]</b> the decisive kill is <i>deferred</i> ( <code>QGEO.REALIZE.01</code> ): the raw $\mathbb{P} \setminus \mu_4$ collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure                                                                                                                                                                                                                                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v216_marks_gauss_bonnet.py    | The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = $2\chi$ : $\mathbb{Z}_2$ branch points (deficit $\pi$ ) on a flat sphere ( $\chi=2$ ) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 =  \mu_4  = e_1(a) = N_{\text{fam}}+1$ (the same $\chi=2$ as the 8 in $c_3=1/(8\pi)$ ). [E] the Euclidean sphere 2-orbifolds ( $\chi_{\text{orb}}=0$ ) are exactly $(2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2)$ . [E] all-order-2 (the $ \mathbb{Z}_2 $ sheet branch) $\Rightarrow$ uniquely $(2, 2, 2, 2)$ ; [E] $N_{\text{fam}}=3$ (rank $H^1=\#\text{marks}-1$ ) $\Rightarrow$ uniquely $(2, 2, 2, 2)$ , picking the 4-mark square over the 3-mark hexagonal $(3, 3, 3)$ ; [E] a free order-4 deck $\Rightarrow$ uniquely $(2, 2, 2, 2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio $2 \Rightarrow j=1728$ , v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01                                                                                                                                                    |
| v217_marks_emergence_scan.py  | The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks $n$ and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\  \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$ , only $n-1=3=N_{\text{fam}}$ ; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$ ); [C] gap: $\mu_4$ -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$ ), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ , gap $(2/3)^6 = 64/729$ ; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square $\sim 0.4$ ); [C] negative controls $n=3, 5, 6$ . [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only |
| v218_diamond_axis_geometry.py | Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: $C$ center, $U$ winding, $V$ sheet), $F$ the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $ ), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$ ; $\det B(C+xU)=2\det(C+xU)$ , anchor-block sheet curvature $6= R^+(A_3) $ . [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$ , steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2\text{Disc}(q)$ for $Q, K, C, F$ ; squares $(1, 3, 4, 6)$ , kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$ , $105=N_{\text{fam}}g_{\text{car}} \cdot 7$ . [C] heuristic: $F$ the transfer-completion corner, a search principle, not a hard filter; the $G_2/F_4$ pair-sum labels stay audit-only (cf. v94/v95) [E][C]                                                                |
| v219_icosahedral_mckay.py     | McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$ . $E_8$ is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ( $2T \rightarrow \hat{E}_6$ , $2O \rightarrow \hat{E}_7$ , $2I \rightarrow \hat{E}_8$ ). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ , containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$ , sum of squares $120= R^+(E_8) $ ); the natural-rep McKay adjacency <i>is</i> the affine $E_8$ graph with Kac marks = the degrees. A backward certificate of the closed $E_8$ [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v220_cp_hexagonal_modulus.py  | CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$ , $\text{Aut}=\mathbb{Z}/4$ (the $\mu_4$ clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$ , $\rho=e^{i\pi/3}$ , $\text{Aut}=\mathbb{Z}/6$ , CM by $\mathbb{Z}[\omega]$ , $\arg \rho=\pi/3$ . [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$ ; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v221_seam_qecc.py                     | The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$ ). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$ ; the gapped transport $T$ on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$ ) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$ ; on the trace-zero subspace $\ T\delta\ \leq(2/3)^6\ \delta\ $ hence $\ T^n\delta\ \leq(2/3)^{6n}\ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]                                                                          |
| v222_cm_norm_duality.py               | CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$ , $j=1728$ ): $N(g_{\text{car}}+i \mu_4 )=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$ . [E] HEX (Eisenstein $\mathbb{Z}[\omega]$ , $j=0$ ): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=\text{scalaron}$ . The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21\neq 41$ ). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]                |
| v223_coxeter_totative_clock.py        | The $\mu_4$ clock is the order-4 character of the $E_8$ Coxeter cycle. [E] $(\mathbb{Z}/30)^\times=\{1, 7, 11, 13, 17, 19, 23, 29\}$ = the $E_8$ exponents = live Coxeter phases; $\varphi(30)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5$ . [E] $(\mathbb{Z}/30)^\times\cong\mathbb{Z}/4\times\mathbb{Z}/2$ with order-4 generator 7 ( $7^2=19$ , $7^4=1 \bmod 30$ ): a literal $\mu_4$ clock $\langle 7 \rangle=\{1, 7, 13, 19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes. Only $E_8$ has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$ ; the seam-clock identification is [C][E][C]         |
| v224_diamond_fttransfer_path.py       | $F_{\text{transfer}}$ as <i>one</i> path on the Sheet Diamond, not four external interfaces. [E] family $M(s, t)=R+Q \text{diag}(s, t, t)$ : the corners $K=M(1, -1)$ , $C=M(1, 0)$ , $F=M(1, 1)$ lie on the sheet axis ( $s=1$ ); the sheet axis is curved $\det M(1, t)=4t^2+14t+14$ (2nd diff $8=\text{rank } E_8$ ), the winding axis $\det M(s, 0)=6s+8$ flat (slope $6= R^+(A_3) $ ); Plücker steps $(1, 8, 10)$ , $(1, 8, 16)$ . [C] $F_{\text{transfer}}$ is the $K\rightarrow C\rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius projection (v37/v183) [E][C]                                                                |
| v225_dual_normal_frame.py             | The dual normal frame $(d, n)$ : one boundary compass for flavor and horizon, with CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}(1, 1, -2)$ (the Nariai/horizon normal; $d\cdot 1=0$ , $d\cdot a=1$ ); $n=(5, -9, 6)$ with $n^\top R=(\det R, 0, 0)=(8, 0, 0)$ , $n^\top L=(20, 0, 0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1}, d, n)=21=N_{\text{fam}}\cdot 7$ . [C] CP as orientation: $\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]                                 |
| v227_degree_exponent_channel_split.py | $248=120+128$ as a magnitude/phase channel typing (replaces the over-strong $S^-$ -dark-matter reading). [E] $E_8$ exponents sum $120= R^+(E_8) $ ( <i>magnitude</i> channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank}\cdot \dim S^+$ ( <i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$ ; $\sum \deg - \sum \exp = \text{rank}=8$ . [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]                                                                              |
| v228_rr_index_gate.py                 | P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ; bedrock language for P2, not a proof from nothing). [E] $\deg D=4= \mu_4 $ on $\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D))=5=g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1\setminus 4)=3=N_{\text{fam}}$ ; $h^0+\text{rank } H_1=8=\text{rank } E_8$ , $h^0-\text{rank } H_1=2= \mathbb{Z}_2 $ ; $\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5=16=\dim S^+$ . The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]                                                               |
| v229_lepton_frobenius_algebra.py      | The charged leptons as an étale Frobenius algebra over the $\mu_6$ resolvent. [E] exact coefficients $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure $c_e c_\tau = \frac{8}{3} =  \mathbb{Z}_2  c_\mu$ , product $\frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ . [E] $A=\mathbb{Q}[t]/(m)$ , $m=\prod(t-c)$ : the trace-form Gram is nondegenerate ( $\det=\text{disc}(m)\neq 0$ , $A$ étale) — a commutative Frobenius algebra; the $C_6$ shift has characteristic polynomial $t^6-1$ (spectrum $\mu_6=\mu_3$ family $\times \mu_2$ sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C] |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v230_center_budget_norms.py       | The Sheet-Diamond center budget (7, 11, 13) is the three <i>local norms</i> of the theory. [E] $C=R+Q$ diag(1, 0, 0), row sums (7, 11, 13): $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = $\Delta_Q$ ), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=1+4+6$ (QBL boundary Fock count on the 4 $\mu_4$ corners = $\dim S^+ - g_{\text{car}}=16-5$ ). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v231_cp_mu6_phases.py             | Both CP phases are $\mu_6$ powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] $\rho$ is a primitive 6th root ( $\rho^6=1$ , $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] $\delta_{\text{CKM}}$ leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i> ); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the $C_6/\mu_6$ monodromy has charpoly $t^6-1$ (spectrum $\mu_6$ carries $\rho$ ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$ , $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG $\mu_3$ has no $-1$ (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]                                                                                                                                                                               |
| v232_e8_kleinian_seam.py          | The seam as the $E_8$ <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine $E_8$ ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> $E_8$ Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$ 's of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$ 's = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \#$ nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3=1/(8\pi)$ . [E] the negated intersection form is the $E_8$ Cartan (even, $\det=1$ , positive definite; each $\mathbb{P}^1$ a $(-2)$ -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$ : the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $ . [C] seam reading: the raw RP seam ( $\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes $E_8$ ); bedrock [O][E][C][O]                                                                                                                                                                                                 |
| v233_cp_triality_phase.py         | The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$ : the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$ , the sheet $\mu_2=\{1, -1\}$ ; $\rho=\omega^2 \cdot (-1)$ . [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$ , so the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase $\omega^2$ (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] $\omega$ is the $\mu_3$ factor of the order-30 (2, 3, 5) Milnor/Coxeter monodromy that builds $E_8$ (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]                                                                                                                                                                                                                                                                                                                                                                           |
| v234_seam_holomorphy_selection.py | The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing $E_8$ . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic ( $\mu$ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$ ; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) =  \Gamma^{\text{ab}}  =  H_1(S^3/\Gamma)  = \#(1\text{-dim irreps})$ : $A_n \rightarrow n+1$ , $D_n \rightarrow 4$ , $E_6 \rightarrow 3$ , $E_7 \rightarrow 2$ , $E_8 \rightarrow 1$ — only $E_8$ gives 1 ( $2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = $E_8$ . [E] the three faces (v83/v154, v232, v219) are the SAME $E_8$ -selector, so Target A, $G_{\text{net}}$ , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$ ; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O] |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v235_seam_chern_simons.py         | The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer $K$ -matrix with $\# \text{anyons} =  \det K $ , edge $c = \text{signature}(K)$ , holomorphic $\Leftrightarrow  \det K =1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ( $\det 16$ , 16 anyons) $\rightarrow SO(16)_1 = D_8$ ( $\det 4$ ) $\rightarrow (E_8)_1$ ( $\det 1$ , holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$ . [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev $E_8$ quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$ ). [E] NEG: $D_8$ is free+bosonic but has 4 anyons; an odd $K$ is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]                                  |
| v236_brieskorn_capstone.py        | The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ . [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$ , $m$ the $E_8$ exponents (charpoly $\Phi_{30}$ , $\deg \varphi(30)=8=\mu$ ) — so it <i>is</i> the order-30 $E_8$ Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the $\mu_3$ triality (CP, v233) as $h^{10}$ ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the $\mu_4$ clock ( $\times 7$ , v223). [E] Milnor lattice = $E_8$ , link = Poincaré sphere. [C] one singularity generates atoms + 8 + $E_8$ + order-30 + both clocks + seam; a unification, not a closure [E][C]                                                                                                            |
| v237_seam_sre_closure.py          | The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the $K$ -matrix ground-state degeneracy on a genus- $g$ surface is $ \det K ^g$ (carrier $\rightarrow 16$ , $D_8 \rightarrow 4$ , $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$ . [E] among rank-8 even positive-definite $K$ ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only $E_8$ has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev $E_8$ ), edge holomorphic; $D_8$ /carrier are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$ , the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk ( $D_8$ ) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev $E_8$ phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O] |
| v238_modular_lindblad_dynamics.py | DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the $\mu_4$ clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ( $e^L = T$ , $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap $\Delta$ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.                                                   |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v239_kms_thermal_time.py        | DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$ , and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$ . [E] KMS at $\beta = 1$ : $\omega(A\sigma_i(B)) = \omega(BA)$ ( $\omega(X) = \text{Tr}(\rho X)$ , $\rho = e^{-\Lambda_\Sigma}/Z$ ), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$ ). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.                                                                                                                                                                                                                                                                                                |
| v240_gns_os_reconstruction.py   | DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data $(M, \omega)$ , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ( $\rho > 0$ ), $\dim n^2$ , identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$ , gap = $6\log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$ : minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.                                                                                                                                                |
| v241_dhr_sectors.py             | DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$ , $\theta(a) = e^{2\pi i q(a)}$ . [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ( $= \det K_{\text{carrier}} = \dim S^+$ ), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$ . [E] tower = v235/v237: Lagrangian $\langle (1, 1) \rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$ ; order-2 $\langle (2, 2) \rangle \rightarrow 4 = D_8$ ; carrier $16 \rightarrow D_8 \rightarrow E_8$ . [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact. |
| v242_spin_statistics_modular.py | DHR.MODULAR.01 — spin-statistics and the modular $(S, T)$ data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$ . Same MTC as v241 ( $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q = (5x^2 + 3y^2)/8$ , $\theta = e^{2\pi i q}$ ). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ( $\theta = 1$ ), 2 fermions ( $\theta = -1$ ) = $\{(0, 2), (2, 0)\}$ , 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$ . [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.                                   |
| v243_haag_ruelle_braiding.py    | DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$ . [E] mass gap $\Delta = 6\log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M  = 1$ , diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: $M$ is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$ , $M(a, -b) = M(a, b)^*$ . [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$ , trivial braiding ( $\det 1/\text{SRE}$ ). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).                                                                                                                  |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v244_spectral_action_contract.py    | <p>CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net <math>\rightarrow</math> 4d physics). A single Dirac operator whose <math>\text{Tr } f(D/\Lambda)</math> yields gravity AND matter. [E] carrier spinor = one generation: <math>SO(10)</math> half-spinor = <math>\dim S^+ = 2^{g_{\text{car}}-1} = 16</math> (<math>15 + \nu_R</math>), <math>N_{\text{fam}} = 3</math>. [E] SM gauge group inside the carrier: <math>SU(3) \times SU(2) \times U(1)</math> (<math>\dim 12</math>) <math>\subset SU(5)</math> (24) <math>\subset SO(10) = D_5</math> (45); colour <math>SU(3) \subset A_3 = SU(4)</math> (Pati-Salam); <math>\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8</math>. [E] one Dirac operator <math>\rightarrow</math> gravity (<math>a_2 \sim R</math>, <math>a_4 \sim R^2</math>, <math>\sqrt{36}</math>) + matter (<math>a_4 \sim F^2</math> Yang-Mills + <math> D\phi ^2</math> Higgs, Gilkey). [X] hooks: <math>g_3 = g_2 = \sqrt{5/3} g_1</math> + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam <math>\rightarrow</math> Dirac <math>D</math>; finite geometry = carrier; spectral action reproduces SM couplings) <math>\rightarrow</math> the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.</p> |
| v245_spectral_matter_content.py     | <p>QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the <math>SO(10)</math> <b>16</b>. [E] <b>16</b> = one generation: <math>SO(10)</math> <b>16</b> = <math>SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})</math>; SM dims <math>6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+</math>, <math>N_{\text{fam}} = 3</math> (15 Weyl fermions + <math>\nu_R</math>). [E] electric charges <math>Q = T_3 + Y</math> are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: <math>\sum Y = 0</math>, <math>\sum Y^3 = 0</math>, <math>[SU(2)]^2 U(1) = 0</math>, <math>[SU(3)]^2 U(1) = 0</math> (exact). [E] unification: <math>\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8</math>, i.e. <math>g_3 = g_2 = \sqrt{5/3} g_1</math>. [E] Higgs = the <math>(1, 2)_{1/2}</math> inner fluctuation (one doublet). [O] residual: the seam <math>\rightarrow</math> Dirac realisation + measured-coupling reproduction stay the open contract.</p>                                                                                                                                                                                                                                              |
| v246_unification_data_crosscheck.py | <p>QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does <math>g_1 = g_2 = g_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math> survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, <math>M_Z</math>): <math>\alpha_i^{-1} = (59.01, 29.59, 8.47)</math>, SM 1-loop <math>b = (41/10, -19/6, -7)</math> (v159). [E] the prediction: <math>\alpha_1 = \alpha_2 = \alpha_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math>. [X] tension: pairwise crossings spread <math>\sim 4</math> decades (<math>\sim 10^{13} - 10^{17}</math> GeV), residual spread at <math>2 \times 10^{16}</math> <math>\Delta \alpha^{-1} \sim 8.8</math> — <math>g_1 = g_2 = g_3</math> not realised by the plain SM. [X] <math>\sin^2 \theta_W</math> 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread <math>\sim 3.2</math> at <math>1.7 \times 10^{14}</math> GeV). [O] no rescue: SM gauge content + no new states (the <math>E_8</math> spine is arithmetic, not a threshold tower) <math>\Rightarrow</math> no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.</p>            |
| v247_e8_branching_no126.py          | <p>PS.E8BRANCH.01 — the <math>E_8</math> audit hull forbids the <b>126</b> and supplies exactly <math>\{1, 10, 16, 45\}</math> for <math>SO(10)</math>. Carrier <math>D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8</math> (via <math>SO(16)</math>); <b>248</b> = <math>(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248</math>. [E] ranks <math>5 + 3 = 8</math>; [E] dims = 248 (<math>SO(16)</math>: 120 adj + 128 spinor); [E] <math>SO(10)</math> content <math>\{1, 10, 16, 45\}</math>, the <b>126</b> ABSENT <math>\Rightarrow 126_H</math> algebraically forbidden, <math>10 + 16 + 45</math> <math>E_8</math>-supplied not fitted; [E] caveat only one 45/one 15 <math>\rightarrow</math> proton-marginal; [O] the field-content principle + NCG complement (v250).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v248_ps_algebra.py                  | <p>PS.ALGEBRA.01 — the Pati-Salam NCG algebra <math>A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})</math> and its exact SM matching. [E] unimodular group <math>SU(2)_L \times SU(2)_R \times SU(4)_c</math> (rank 5, <math>\dim 21</math>); [E] <b>16</b> = <math>(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8</math>; [E] <math>Y = T_{3R} + (B - L)/2</math> gives the exact SM hypercharges and <math>Q = T_{3L} + Y</math> the exact charges; [E] matching <math>\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}</math>, <math>\sin^2 \theta_W = 3/8</math>. [O] <math>A_F</math> uniquely from seam+sheet+<math>\mu_4</math>; [O] caveat <math>SU(4)_c \subset SO(10) = D_5</math> is not the carrier <math>A_3 = SU(4)</math> (family).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v249_ps_unification.py              | <p>PS.RGTEST.01 — carrier-native Pati-Salam/<math>SO(10)</math> two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread <math>\sim 8.8</math> @ <math>2 \times 10^{16}</math>); [C] <math>M_{PS}/M_s = 1.0 - 1.7</math> (1-loop; 2-loop <math>\sim 1.0 - 1.2</math>) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match (<math>b_{2R} &gt; 0</math>; also <math>E_8</math>-forbidden v247); [X] NEG proton decay selects content (minimal <math>M_{GUT} \sim 2 \times 10^{15}</math> excluded; <math>+(15, 1, 1)</math> safer). [O] the gauging fork, exact <math>\kappa</math>, the one-45 marginality.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v250_dirac_triple_contract.py     | CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target $(A, H, D, J, \gamma)$ almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$ , $D_F$ from $(R, L, \varphi_0, \text{Majorana})$ . [E] $H_F = 3 \times 16 = 48$ ; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + $\nu_R$ Majorana; $M_{PS} = \kappa M_s$ ) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.                                                                                                                                                                             |
| v251_first_order_sigma.py         | PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ $\sigma$ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + $\sigma$ ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: $\sigma$ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$ . [E] order-zero; [E] real-even ( $J^2 = +1$ , $\gamma^2 = 1$ , $D\gamma = -\gamma D$ , KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$ , supported exactly on the $\nu_R \nu_R^c$ blocks ( $\sigma$ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).                                                                                                              |
| v252_full_finite_triple.py        | PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ <b>16</b> -plets (dim 96). [E] dim 96; $D_F$ Hermitian+odd; reality $J^2 = +1$ , $JD = DJ$ ; grading $\gamma^2 = 1$ , $[\gamma, a] = 0$ , $J\gamma = -\gamma J$ ; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$ ); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2 / M_R$ . [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion. |
| v253_kappa_scalaron_ps.py         | PS.KAPPA.01 — pinning $\kappa$ in $M_{PS} = \kappa M_s$ , discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] $\kappa$ pinned by RG (scan PDG errors $\times$ the two $E_8$ -allowed contents) to a tight $O(1)$ interval $\sim 1.3$ (1-loop $[1.0, 1.7]$ , 2-loop $[1.0, 1.2]$ ) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ , a ratio of heat-kernel traces over the 96-dim $H_F$ (v252) — $\Lambda$ cancels, $\kappa$ scale-free and $O(1)$ . [X] the <b>126</b> -content drives $\kappa$ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments $f_2/f_0$ fixed.                                                                                                                                                              |
| v254_inner_fluctuations.py        | PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not <b>126</b> ) + the no-junk obligation + the $\sigma$ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no <b>126</b> : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$ ); [E] the $\sigma$ ( $\nu_R$ Majorana) direction is absent for the SM algebra but present for Pati–Salam — $\sigma$ is the CCvS beyond-first-order scalar; [E] $\sigma$ is an SM singlet $(1, 1, 0)$ with $B-L = 2$ ; [C] $\sigma = \text{scalaron consistency}$ .                                                                                                                                                                                  |
| v255_spectral_action_expansion.py | PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4 \Lambda^4 a_0 + 2f_2 \Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure ( $a_0$ cosmological, $a_2$ Einstein–Hilbert+Higgs-mass, $a_4$ Yang–Mills+Weyl <sup>2</sup> + $R^2$ +Higgs-quartic, $N=96$ ); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with $\sigma$ ; [E] scalaron = $R^2$ spin-0 mode (gravitational $a_4 \sim f_0 N$ ); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.                                                                                                 |

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| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v256_orientability_<br>poincare.py         | PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$ , full orientability is a degree-4 cycle on $M \times F$ ; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.                                                                                                                                                                                                                                                                                                                                                                                            |
| v257_quasiorient_<br>modpd.py              | PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$ ) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] quasi-orientability holds (left/right $A$ -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$ ); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.                                                                                                                                                                                                                                                          |
| v258_dirac_<br>covariance_<br>induction.py | PS.DIRAC.03 — $D_F$ is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance $C$ (contraction, $\text{spec} \subset (0, 1)$ ) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ( $\beta=1$ , v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ( $D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$ ); [E] unitarily equivalent to v252 (inducing $C_F$ from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$ ); [C] $C_F = P_F C_\Sigma P_F$ , so $D_F$ is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of $C_\Sigma$ . |
| v259_modular_<br>cutoff_kappa.py           | PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes $\beta$ , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$ , $f_2=\int f=1$ , $f_4=\int u f=1$ , so $f_2/f_0=f_4/f_2=1$ ; [E] seam-fixed $\beta=1$ (no new parameter beyond $c_3$ ); [E] non-vacuous neg control: a Gaussian cutoff $e^{-u^2}$ gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$ ; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$ , no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ( $O(1)$ ); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$ .                                                                                                                          |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v260_k3_kummer_unification.py    | <p>ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the <math>E_8</math> lattice as facets of one smooth object, the geometric unification of the v1 ‘glue <math>D_5 \oplus A_3 + \mu_4 \Rightarrow E_8</math>’ step (the K3/Kummer proposal). The seam square torus <math>\tau=i</math> (<math>j=1728</math>, v214) under the elliptic involution <math>z \mapsto -z</math> squares to the Kummer K3. [E] the <math>E_8</math> Cartan is even, <math>\det=1</math>, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice <math>L_{K3}=U^3 \oplus E_8(-1)^2</math> has rank <math>22=b_2</math>, <math>\det=-1</math> (unimodular), is even, signature <math>(3,19)</math>, with <math>E_8(-1)^2</math> an orthogonal summand; [E] seam = <math>T^2/(z \mapsto -z)</math>: <math>4= (\mathbb{Z}/2)^2 </math> fixed points, pillowcase <math>S^2(2,2,2,2)</math>, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes <math> A[2] =2^4=16 \rightarrow 16</math> exceptional <math>\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16</math> (v197), <math>16=4 \times 4</math>; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally <math>E_8(-1)^2</math> (distinct sublattices of <math>H^2(K3)</math>); [C] two faces of <math>E_8</math> in K3 (local du Val <math>\mathbb{C}^2/2I</math>, v232/v236; global <math>H^2</math> summand); [C] unification (lattice [E], carrier<math>\leftrightarrow</math>node identification [C]).</p> |
| v261_modular_spectral_closure.py | <p>QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object<br/> TFPT<sub>QFT</sub> = <math>(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})</math> reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones <math>[B:A]=4</math> (v89) = <math> \mu_4 =4</math> (v92) = seam marks <math>2\chi=4</math> (v216) = pillowcase/Kummer fixed points <math> (\mathbb{Z}/2)^2 =4</math> (v260); [E] one carrier-16: <math>\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16</math>, <math>H_F=2N_{\text{fam}} \cdot 16=96</math>; [E] one gap <math>\Delta = -\log((2/3)^6) = 6\log(3/2) &gt; 0</math> = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov <math>\log((1-C)C^{-1})=H</math>, v258; KMS cutoff <math>f_2/f_0=1</math>, v259; K3 lattice signature <math>(3,19)/\det -1/\text{even}</math>, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.</p>                                                                            |
| v262_fqcd_mp_me.py               | <p>FR.MPME.02 — the <math>F_{\text{QCD}}</math> transfer pipeline for <math>m_p/m_e</math>, implemented as a real solver (the one <math>F_{\text{transfer}}</math> pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] <math>b_3 = -7 = -(11-2n_f/3)</math> at <math>n_f=6</math> (the only TFPT number in the run); [E] a two-loop <math>\alpha_s(M_Z)=0.1179</math> run with thresholds gives <math>\alpha_s(2\text{ GeV}) \sim 0.30</math> (real ODE solve); [C] <math>\Lambda_{\text{MS}}^{(3)} \sim 0.33\text{ GeV} \times</math> the lattice bridge <math>C_p = m_p/\Lambda</math> (external <math>O(1)</math>) over the closed <math>m_e</math> gives <math>m_p/m_e \sim 1824</math> vs the measured 1836.15 within the <math>\sim 7\%</math> external budget; [X] firewall: <math>m_p/m_e</math> is not an admissible compiler integer (<math>SU(9)</math> absent from <math>D_5 \oplus A_3</math>). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v263_mnu_from_df.py              | <p>FLAV.PMNS.02 — the light neutrino matrix <math>M_\nu</math> as the type-I seesaw of <math>D_F</math>, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from <math>D_F</math>’s blocks. [E] <math>M_\nu = -m_D M_R^{-1} m_D^\top</math> symmetric + seesaw-suppressed (the light masses an output of <math>D_F</math>); [E] a <math>\mu\tau</math>-symmetric <math>(m_D, M_R)</math> gives a <math>\mu\tau</math>-symmetric <math>M_\nu</math> (the v9 texture is in the seesaw image); [C] diagonalising gives <math>\theta_{23}=45^\circ</math> and <math>\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675</math> under the seam misalignment; [O] <math>\theta_{13}=0</math> in the leading texture (physical <math>\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}</math> a separate readout), the CP magnitude <math>\delta_{\text{PMNS}}=240^\circ</math> is the <math>\mu_6/\text{triality}</math> phase (v231/v233), and <math>M_R</math> is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v264_raw_seam_<br>marklocal.py          | QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit $\pi$ each, sum $4\pi=2\pi\chi(S^2)$ ), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the $\mu_4$ orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] $M_f$ then commutes with the clock $\rho=\text{diag}(i^n)$ ; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the $\mu_4$ orbit) breaks it. A reduction/sharpening, not a closure.                                                                                                                                                                                                                                                                                                                                                         |
| v265_qft4d_fork_<br>freeze.py           | QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread $\sim 8.8$ at $2 \times 10^{16}$ , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch ( $E_8$ content $\{1, 10, 16, 45\}$ no <b>126</b> v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake $\tau_p$ window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v266_ps_threshold_<br>proton.py         | PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ : the carrier-native PS UV branch survives proton decay <i>only</i> with the $E_8$ -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$ , $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$ ; [X] minimal KILLED ( $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K $2.4 \times 10^{34}$ , even high-band); [C] $+(15, 1, 1)$ survives now ( $\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> ( $\tau_p$ at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.                                                                                                                                                                                                                                                                                                                                     |
| v267_qgeo_rigidity_<br>minimal_axiom.py | QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier $\mu_4$ clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$ ) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of $a$ ); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$ ) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$ ; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$ , only $\mathbb{Z}_2$ ; hexagonal $j=0$ has $\mathbb{Z}_6$ , order $6 \neq 4$ ); [C] arithmetic rigidity ( $\{1, i, -1, -i\}$ over $\mathbb{Q}$ , $j=1728$ CM by $\mathbb{Z}[i]$ , Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$ ). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom. |
| v268_theta13_<br>carrier_trace.py       | FLAV.TH13.01 — the reactor angle $\theta_{13}$ <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$ ), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG $\sim 0.0222$ ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$ , so $\theta_{13}$ is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ $\theta_{13}$ from $M_\nu$ ’ route); [O] CP magnitude + absolute $\nu$ scale remain open.                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v269_spert_paqft_skeleton.py   | QFT4D.SPERT.01 — the perturbative 4D S-matrix ( $S_{\text{pert}}$ ) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ( $S_{\text{top}}$ 2d DHR braiding v243 / $S_{\text{pert}}$ 4d EG of the spectral action / $S_{\text{phys}}$ LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action $a_4$ operator has mass dimension $\leq 4$ in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ( $\Delta=6 \log(3/2)>0$ ) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.                                                                                                            |
| v270_pmns_jarlskog_assembly.py | FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\text{PMNS}}$ (the CP <i>strength</i> ), assembled from the four TFPT channels ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$ , $\sin^2 \theta_{23}=\frac{1}{2}$ , $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$ , $\delta=4\pi/3$ ) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM $J$ (v88); [C] data: $J_{\text{max}}$ vs NuFIT 0.0332 ( $\sim 3\%$ ), $J$ vs best fit $-0.026$ , the negative sign reproduced; [O] $\delta$ is the $\mu_6$ /triality phase (assembled input, <i>not</i> from $M_\nu/D_F$ ) and the absolute $\nu$ -mass scale ( $\Sigma m_\nu$ , $M_R$ ) stays open. Assembles the PMNS matrix + CP strength; does not derive $\delta$ from $D_F$ nor fix the scale.                                                                                                                                         |
| v271_eg_oneloop_quartic.py     | QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\text{pert}}$ skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=:L_{\text{int}}$ : needs no extension, the first extension is $T_2$ ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$ , the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$ ; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ( $C(\omega+d,d)=1$ , renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action $a_4$ geometric quartic; [C] the $\phi^4$ one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the $s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$ ; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number.                                                                                      |
| v272_nu_mass_scale.py          | FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a $\Sigma m_\nu$ number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like $v_{\text{geo}}$ (v153), while the ratios ( $m_2/m_3$ , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$ , no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ( $< 0.072$ – $0.12$ eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number; sharpened by v481 (carrier normalisation $y_\nu=y_t$ : one-parameter, window-bracketed candidate). |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v273_eg_gauge_running.py            | QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271’s quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $sd=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop $b_i$ from the explicit carrier/SM content: $b_3=-7$ , $b_2=-19/6$ , $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact $b_i$ mirrored in Wolfram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v274_scale_overdetermination.py     | SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{P1}^d$ ( $\rho_\Lambda/M_{P1}^4$ , $M_{scal}/M_{P1}=c_3^{7/2}$ , the spectrum); [C] two independent routes to $M_{P1}$ agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{P1}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV) $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit ( $G$ ), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.                                                                                                                                                                                                                                                          |
| v275_qgamb_roadmap.py               | QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2)-2.248c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ( $c=248/31=8$ ; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.                                                                                                                                                                                                                                                                                                                                                          |
| v276_qgeo_flat_closes_commutator.py | QGEO.SYM.03 — the flat $\tau=i$ pillowcase premise <i>closes</i> the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to <i>one</i> geometric postulate (it does <i>not</i> prove the premise). For a quasi-free seam state $QGEO.SYM.01 = (\omega \circ \rho = \omega) \Leftrightarrow [\rho, C]=0 \Leftrightarrow [\rho, H]=0$ . [E] isometry $\Rightarrow$ commutator: if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \rightarrow iz)$ is a flat- $\tau=i$ isometry then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ <i>exactly</i> (all orders), verified on the discretised square torus ( $\rho^4=I$ , $[\rho, \Delta]=0$ , $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full $H$ (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ( $j=1728$ ; hexagonal $j=0$ is order-6, excluded; v214/v267); [O] QGEO.SYM.01 collapses to the single statement ‘the raw seam is the flat $\tau=i$ pillowcase quasi-free state’ — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form. |
| v277_seam_calderon_e8_match.py      | QGAMB.TIERB.01 — the seam-Calderón $\rightarrow (E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to <i>one</i> bit (it does <i>not</i> close it). [E] $c=8$ (Sugawara, $248/31=g_{\text{car}}+N_{\text{fam}}$ ), but $c=8$ alone does not pin the net; [E] the discriminator is holomorphy — $\det \text{Cartan}(E_8)=1$ (1 primary, holomorphic) vs the $c=8$ counterexample $\det \text{Cartan}(D_8=SO(16))=4$ (4 primaries, non-holomorphic), so the single discriminator is $ \det K =1$ ; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248=\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic $\Leftrightarrow \det \text{Cartan} 1 \Leftrightarrow$ unique even unimodular rank-8 (v83) $\Leftrightarrow$ homology-sphere link $\Leftrightarrow  \det K =1$ (v235); [C] seam-side match via v276 (flat $\tau=i$ ) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, <i>not</i> closed.                                                         |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v278_ls2_bridge_unitarity.py    | <p>QFT4D.SPERT.04 — the <math>S_{\text{pert}} \rightarrow S_{\text{phys}}</math> LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (<math>S_{\text{phys}}</math> = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: <math>S_{\text{phys}}</math> = amputate+on-shell-residue of the connected time-ordered products, and the EG <math>T</math>-products of <math>S_{\text{pert}}</math> are those correlators (OS Wick-rotation, v240)</p> <p><math>\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}</math>; [C] the gap <math>\Delta=6\log(3/2)&gt;0</math> gives Haag–Ruelle asymptotic states so LSZ is valid; [E] <i>one-loop unitarity</i> (optical theorem): the bubble discontinuity <math>\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}</math> <i>exactly</i> = the two-body phase space, so <math>2\text{Im } M = \sum \int d\Pi  M ^2</math> (cutting rule) holds <math>\Rightarrow</math> matter+gauge unitary; [E] threshold: <math>\text{Im } B=0</math> below <math>s=4m^2</math>, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary <math>\rightarrow</math> QG.AMB.01; [C] finite field-strength <math>Z</math> from the EG scheme.</p> |
| v279_qgeo_obligation_lemma.py   | <p>QGeo.OBLIG.01 — the QGeo.SYM.01 bedrock written as <i>one</i> precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does <i>not</i> close it). Lemma: given the premise ‘the raw seam carries the flat <math>\tau=i</math> pillowcase metric <math>g_{\text{flat}}</math> (Trojanov-unique, cone angles <math>\pi</math>), so <math>\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}</math>’, then <math>\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow</math> mark-locality <math>\rightarrow E_8</math>. [E] the reduction DAG from ‘free RP seam’ has 8 closed nodes and <i>exactly one</i> open leaf (‘the raw seam state is the flat state’); [E] the geometric side is fixed (flat DtN diagonal, <math>\rho</math> commutes to all orders, Lean FORM.QGeo.04), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Trojanov + OS); [E] the all-orders step is a Lean theorem (SeamDeckClosure.flat_all_orders_clock, standard axioms, no sorry); [O] one constructive-QFT statement closes QGeo.SYM.01.</p>                                                                                                                                                                                                       |
| v280_pillowcase_steklov.py      | <p>QGeo.STEKLOV.01 — a direct numerical experiment on the flat <math>\tau=i</math> pillowcase orbifold (the seam geometry): the self-investigable test of the QGeo.SYM.01 obligation (does <i>not</i> prove the premise). [E] pillowcase = the <math>Z_2</math>-even sector of the flat torus, 4 cone points (2-torsion, deficit <math>\pi</math> each, sum <math>4\pi=2\pi \cdot 2</math>); [E] the order-4 deck <math>\rho(z \rightarrow iz)</math> descends (<math>[\rho, Z_2]=[\rho, P_{\text{even}}]=0</math>) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact:</p> <p><math>[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega</math> (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.</p>                                                                                                                                                                                                                                                                                                                                                                               |
| v281_holomorphy_modular_data.py | <p>QGAMB.MODULAR.01 — the modular-data / anyon-condensation layer of the QG.AMB.01 Tier-B holomorphy bit. [E] #anyons = <math> \det \text{Gram} : E_8 \rightarrow 1</math> (holomorphic), <math>D_8=SO(16) \rightarrow 4</math>, carrier <math>D_5 \oplus A_3 \rightarrow 16</math>; [E] the anyon-condensation tower <math>D_5 \oplus A_3 (16) \rightarrow D_8 (4) \rightarrow E_8 (1)</math>, each step condensing a Lagrangian <math>Z_2</math> boson, the Kitaev <math>E_8</math> endpoint (v92/v235); [E] Gauss–Milgram recovers <math>c=8</math>; [E] <math>E_8</math> is the unique holomorphic <math>c=8</math> (<math>\det 1</math>) — <math>SO(16)</math> has the same <math>c</math> but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to <math>\det K=1</math>? Reduced to one bit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v282_e8_tau_i_unification.py    | <p>QGAMB.UNIFY.01 — the <math>E_8</math>-at-<math>\tau=i</math> unification: the candidate <i>simplification</i> reducing the two open obligations to <i>one</i> object (the raw seam is the <math>(E_8)_1</math> holomorphic CFT at <math>\tau=i</math>). <math>\chi_{E_8}(\tau)=\Theta_{E_8}/\eta^8=j^{1/3}</math>. [E] <math>\tau=i</math> is the order-4 CM point (<math>j=1728</math>, the QGeo.SYM.01 flat geometry); [E] <math>\chi_{E_8}=j^{1/3}</math> (<math>\chi^3=qj</math> as <math>q</math>-series), so <math>\chi_{E_8}(i)=1728^{1/3}=12</math> (the <math>(E_8)_1</math> character at the SAME <math>\tau=i</math>, the QG.AMB.01 holomorphy); [E] one object, two faces; [C] obligation count <math>2 \rightarrow 1</math>: prove ‘the raw seam is <math>(E_8)_1</math> at <math>\tau=i</math>’ and <i>both</i> follow; [O] the unified premise stays open but is now one object.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v283_live_killtest_scorecard.py              | PRED.LIVE.01 — a live kill-test scorecard of the recent round’s computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2 \theta_{12}=0.30675$ ( $+0.31\sigma$ ), $\sin^2 \theta_{13}=0.02311$ ( $+1.45\sigma$ ), $\theta_{23}$ maximal; [C] leptonic CP $J_{\text{PMNS}}=-0.0297$ , $\delta=240^\circ$ vs $\sim 232(30)$ ; [X] $\Sigma m_\nu$ NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p=1.55\times 10^{35}$ yr for $+(15, 1, 1)$ (Hyper-K decisive); [E] closed readouts $\alpha^{-1}$ , $\Omega_b$ , $\beta$ ; all rows consistent $< 2\sigma$ , two sharp near-term kill tests. A consolidation, not new physics.                                                                                                                                                                                                                                                                                                                       |
| v284_route_i_rp_uniqueness.py                | QGEO.ROUTEI.01 — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does <i>not</i> prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip DtN( $k$ )/ $k \rightarrow 1$ , principal symbol $ \xi $ with no curvature term), L4 covariance $C = (1 + e^H)^{-1}$ from the DtN (v258), L5 $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat ( $\tau=i$ ), so $L_{\text{open}}$ reduces to ‘the raw seam is the constant-curvature geometric state’; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step). |
| v285_route_ii_seam_condensation.py           | QGAMB.ROUTEII.01 — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma <i>coincides</i> with Route (i)’s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow  \det K =1 \Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic $c=8 \Leftrightarrow (E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower $16 \rightarrow 4 \rightarrow 1 =$ the unique $\mu_4$ -Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share <i>one</i> open lemma — both say ‘the raw seam is the $(E_8)_1$ -at- $\tau=i$ object’ (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$ . $M_{\text{open}} = \text{QGEO.SYM.01}$ .                                                                     |
| v286_seam_equivalence_contract.py            | SEAM.EQUIV.01 — the Seam Equivalence contract: concentrates the whole open structure on <i>one</i> named arrow (does <i>not</i> prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau=i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] <b>import firewall</b> : no Route-i module imports any Route-ii module or vice versa, so the routes converge only through SEAM.EQUIV.01 (kills the ‘ $E_8$ into the geometry and back’ circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary’.                                                                                                      |
| v287_free_rp_bulk_to_holomorphic_boundary.py | SEAM.EQUIV.A01 — Route A (AQFT) theorem-dependency checker for the open Gral arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary net’: types the 5-lemma chain and isolates the <i>one</i> missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality $\Rightarrow$ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk $\Rightarrow$ single-sector boundary ( $\mu$ -index 1) — the one missing standard theorem (the AQFT form of $\det K=1$ ); [E] L4 $\mu=1+c=8 \Rightarrow$ holomorphic (v234); [E] L5 holomorphic $c=8 \Rightarrow (E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces SEAM.EQUIV.01 to one cited theorem.                                                                                                                                                                                                                                                                      |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v288_full_l2_subprincipal_z4.py       | SEAM.EQUIV.B01 — Route B (DtN): the full- $L^2$ lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the ‘probably provable’ step and lifting v201/v284 from the finite block to $L^2$ . [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod{4}$ ; [E] full- $L^2$ block-diagonality: $\Lambda =  n  + M_f$ commutes with $\rho = \text{diag}(i^n)$ , $\ [\rho, \Lambda]\  \sim 2 \times 10^{-15}$ on $-50..50$ ; [E] negative control: a generic curvature gives $\ [\rho, \Lambda]\  = 4.44 > 0$ ; [E] Lean consistency (geom_sum_fourth_root + markLocal_blockDiagonal); [O] the sharper residual — why is the raw seam sub-principal term mark-local?                                                                |
| v289_marklocal_raw.py                 | MARKLOCAL.RAW.01 — decompose the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $=  D_\theta  + M_f + \text{cone sources}$ (b-calculus/Melrose, imported); [O] L2 <b>Flat-Away</b> — the one open analytic lemma: $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$ (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one $\mu_4$ orbit (v195/v216); [E] L4 Fourier-support: $\mu_4$ -orbit curvature $\Rightarrow$ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: $\Rightarrow [\rho, \Lambda] = 0$ on full $L^2$ (v288). 4/5 discharged; Route B reduces to exactly L2.              |
| v290_z4_smooth_curvature_adversary.py | SEAM.ADVERSARY.01 — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is <i>not</i> mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$ -symmetric off-mark curvature $f = \varepsilon \cos(4\theta)$ : [E] it is $\mathbb{Z}_4$ -symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$ ) but <i>not</i> mark-local ( $f(\pi/4) = -\varepsilon \neq 0$ ); [E] it <i>passes</i> the v288 commutator $\ [\rho, \Lambda]\  = 0$ (same as mark-local) — so $[\rho, \Lambda] = 0$ is necessary but <i>not</i> sufficient; [E] but it shifts the DtN spectrum off the flat ladder ( $\max  \Delta\lambda  = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input. |
| v291_flataway_contract.py             | FLATAWAY.RP.01 — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four $\mu_4$ marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] <i>spectral-rigidity</i> : scanning $f = \varepsilon \cos(4\theta)$ , the spectrum deviation from the flat ladder is 0 only at $\varepsilon=0$ — flat is the unique spectral match; [E] <i>heat-kernel</i> : the heat-trace deviation is 0 only at $\varepsilon=0$ ; [C] <i>RP-energy</i> : $\text{RP} + \text{gap}$ is a Dirichlet-energy minimisation in the Troyanov conformal class $\Rightarrow$ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B.                                        |
| v292_flataway_heat_reduction.py       | FLATAWAY.HEAT.01 — the Heat-Kernel route made precise: reduces Flat-Away to <i>one</i> named spectral invariant. [E] the heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ( $c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$ ; positive on all combinations), so $\Delta \text{Tr} = 0 \Leftrightarrow f = 0$ ; leading invariant $\ f\ _{L^2}$ (Parseval). [O] reduction: Flat-Away $\Leftarrow$ ‘the seam fixes the $a_2$ heat coefficient to its flat value’ $\Rightarrow \int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.                                                                                                                                                                                                         |
| v293_flataway_spectral_hessian.py     | FLATAWAY.SPEC.01 — the Spectral-Rigidity route upgraded from one direction (v291) to the <i>full</i> smooth- $\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $ n $ is split by mode $4k$ at $ n =2k$ by $\pm\varepsilon/2$ ; [E] $S = \sum (\Delta\lambda)^2 \sim 0.50\varepsilon^2$ ; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$ , so the flat seam is a <i>strict</i> isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non- $\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam’s Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation $\Rightarrow$ Flat-Away.                                                                                                                                                                             |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v294_flataway_rp_energy.py       | FLATAWAY.ENERGY.01 — the RP-energy / Trojanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: $S^2$ with 4 cone angles $\pi$ , $\sum(2\pi - \theta) = 4\pi = 2\pi\chi$ ; [C] Trojanov/Euclidean-cone uniqueness (Trojanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f] = \int f^2$ , Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: ‘RP+gap realises the energy-minimiser’, converging with v292/v293 on one selection principle; closing any one closes Route B.                                                                                                                                                                                                                                                                                                                                                                 |
| v295_flataway_a2_exact.py        | FLATAWAY.A2.01 — the <i>exact</i> analytic $a_2$ coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \text{Tr } e^{-t\Lambda(\varepsilon)} _0 = -t\hat{f}(0) \sum e^{-t n } = 0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x) = e^{-tx}$ strictly convex $\Rightarrow \Lambda \mapsto \text{Tr } e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon=0$ is the global minimum and $\Delta \text{Tr} \geq 0$ for all $\varepsilon$ , $= 0$ iff $f = 0$ ; [E] exact $a_2$ kernel via the divided difference of $e^{-tx}$ (Duhamel), matched to numerics to rel.err $< 10^{-6}$ ; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1) \geq 0$ ; [O] reduction now analytic: Flat-Away $\Leftarrow$ ‘the seam fixes $a_2$ ’. Lean-mirrored (FORM.FLATAWAY.01).                                                                                                         |
| v296_flataway_a2_closed.py       | FLATAWAY.A2.CLOSED.01 — the exact $a_2$ coefficient in <i>closed form</i> . [E] the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ), giving $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ ; [E] a finite $(4k-1)$ -term middle (incl. the degenerate diagonal $\Phi(2k, 2k) = (t^2/2)e^{-2kt}$ ); [E] closed form $W_k(t) = \frac{t(1-e^{-4kt})}{8k(1-e^{-t})} + \frac{1}{2} \sum_{m=1}^{4k-1} \Phi(m, 4k-m)$<br>$(W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8)$ , reproducing v295 to machine precision; [E] manifestly positive; [E] small- $t$ : $W_k = t/2 + O(t^3)$ (mode-independent leading $1/2$ , the $t^2$ term cancels).                                                                                                                                                                                                                                                                                                                        |
| v297_route_a_literature_stack.py | SEAM.EQUIV.A.LIT.01 — Route A’s ‘one standard import’ as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c=8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into ‘invertible bulk $\Rightarrow (E_8)_1$ ’, all legs established; [O] the one open hypothesis is the input of LIT-A, ‘the raw quasi-free seam bulk is invertible (SRE)’ = Route B’s Flat-Away. [E] non-circular with Route B (the v286 firewall holds).                                                                                                                                                                                                                                                                                  |
| v298_e8_network_attractor.py     | <i>E8 as a network attractor</i> — a computational-substrate (Wolfram-program) reading of the $(2, 3, 5)$ /McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the $E_8$ Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the <i>unique</i> attractor of the minimal lazy graph update $m \leftarrow (A + 2I)/4 \cdot m$ on the affine- $E_8/(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$ , top eigenvalue 2). [E] the spherical McKay tower terminates at $E_8$ ( $2T \rightarrow E_6$ , $2O \rightarrow E_7$ , $2I \rightarrow E_8$ , $h = 30$ ); [E] the cascade is nested-subgraph coarse-graining ( $240 \rightarrow 126 \rightarrow 72 \rightarrow 40 \rightarrow 20$ ). [C] Wolfram reading: $E_8$ as the universal attractor of a minimal $(2, 3, 5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph <i>rewrite</i> whose attractor is the full TFPT structure stays open; $\varphi_0 = (4/3)c_3 = 1/(6\pi)$ is the analytic seed, not an edge fraction. |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v299_hypergraph_0d_autonomous.py | The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth <i>reaches</i> $E_8$ as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary $\hat{E}_8$ ) — <i>not</i> ‘P2 derived’. [E] icosahedron = 3-uniform hypergraph ( $ \text{Aut}  = 120 =  I_h $ ; the McKay group is the $SU(2)$ binary lift $2I$ , a different order-120 group); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine $E_8$ McKay graph; [E] Smith: $\rho \leq 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \leq 2 \Leftrightarrow$ a positive witness $m$ with $Am \leq 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \leq 2m$ ) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ under the $(2, 3, \cdot)$ seed; [E] $A_n$ ( $\rho = 2 \cos \frac{\pi}{n+1}$ ) and $D_n$ ( $\rho = 2 \cos \frac{\pi}{2n-2}$ ) controls are $< 2$ for all $n$ , never reaching $E_8$ ; [O] the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).                                                   |
| v300_flataway_rigid.py           | FLATAWAY.RIGID.01 — the attack on Flat-Away. [E] the flat fingerprint: $\text{spec}( D_\theta )$ is the integer ladder, every level $n \geq 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy $2 \times 2$ $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ ), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \rightarrow 1+1$ ) and changes the spectral multiset, so preserving the flat ladder <i>with</i> its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 ‘deviation grows’ scan; [E] with the v295 convexity, flat is isolated analytically <i>and</i> discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer $q$ -coefficients (248 at level 1; $E_8$ even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries <i>zero</i> new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact.                               |
| v301_route_a_invertible.py       | SEAM.EQUIV.A.INV.01 — discharge Route A’s last open hypothesis (‘the raw quasi-free seam bulk is invertible/SRE’) via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of SEAM.EQUIV.01. [E] carrier = 16 Majoranas, $c = 8$ ( $g_{\text{car}} + N_{\text{fam}} = 5 + 3 = 8 = 16/2$ , v148); [E] $SO(16)_1 \supset (E_8)_1$ : $\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248 = $\dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$ -classified by the chiral Majorana number and is <i>always</i> invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $ \det K  = \#\text{anyons}$ is 1 for $E_8$ (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A’s ‘invertible bulk’ now follows from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input ‘is the quasi-free bulk gapped?’; [O] with v300 the whole open content of SEAM.EQUIV.01 reduces to the single spectral statement ‘the quasi-free seam bulk is gapped’. |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v302_seam_gap.py           | <p>SEAM.EQUIV.GAP.01 — the last lever: after v300/v301 the whole open content of SEAM.EQUIV.01 was ‘the quasi-free seam bulk is gapped’; that gap is <i>not</i> a free input but the established Recovery gap of the seam transfer operator. [E] <math>\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}</math> (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: <math>\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 &gt; 0</math> (multiplicative gap <math>1 - (2/3)^6 = 665/729</math>; <math>6 = 2N_{\text{fam}}</math>, <math>(2/3)^6</math> the <math>\mu_4</math>-deck transfer eigenvalue forced by the carrier); [E] the positivity margin <math>\Delta - 31/(4\pi^2) = 1.6476 &gt; 0</math> (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap <math>\Delta &gt; 0</math> is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap <math>6 \ln(3/2) &gt; 0</math>, and SEAM.EQUIV.01’s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v303_ftransfer_dynamics.py | <p>FR.DYNAMICS.01 — the discrete→dynamic lens on F_transfer: the main-branch mechanism (a gapped, positivity-preserving relaxation to a <i>unique</i> attractor; the local-averaging update → the <math>E_8</math> marks, gap <math>6 \ln(3/2)</math>) is the shared <i>shape</i> of all four instances. [E] F_pole (Koide) <i>is</i> the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier <math>F' = (2/3)^6 = \lambda_2</math>, the seam-clock subleading eigenvalue (v82/v54); <math> F'  &lt; 1 \Rightarrow</math> a unique attractor at the seam rate; [C] F_Boltzmann (<math>\eta_B</math>) is a gapped positive contraction (washout <math>\kappa_f \in (0, 1)</math>, H-theorem; compiler fixes <math>A_\Lambda = 10</math>, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed <math>\theta_i = 3\pi/5</math>, rate cosmological); [E] F_QCD (<math>m_p/m_e</math>) is the 1-loop RG flow with the carrier <math>b_3 = -7</math> (asymptotic freedom <math>\Rightarrow</math> the Gaussian UV attractor, <math>\Lambda_{\text{QCD}}</math> generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so F_transfer is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v304_idg_nonlocal_ghost.py | <p>QGAMB.IDG.01 — the KMS spectral-action cutoff suggests a <i>nonlocal</i> (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a <i>truncation artefact</i> — a [C] narrowing of the red-team rt_F perturbative-gravity attack, <i>not</i> a closure of QG.AMB.01 (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator <math>1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]</math> has a healthy massless graviton (residue <math>+1/M^2</math> at <math>p^2=0</math>) and a massive spin-2 mode with <i>negative</i> residue <math>-1/M^2</math> at <math>p^2 = -M^2</math> (the Stelle ghost, reproducing v278); [E] for an IDG operator <math>p^2 a(p^2)</math> with <math>a(z) = e^{z/M^2}</math> entire and nowhere zero, the dressed propagator <math>1/(p^2 a)</math> keeps its <i>only</i> pole at <math>p^2=0</math> — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation <math>a(z) = 1 + z/M^2</math> (the <math>R+R^2</math> order) has a real zero at <math>z = -M^2</math> = the ghost pole, so ‘entire vs polynomial’ is the discriminator; [C] the cutoff is the <math>\beta=1</math> seam KMS weight <math>f(u) = e^{-u}</math> (v259), not a choice — truncating <math>\text{Tr} e^{-D^2/\Lambda^2}</math> at <math>a_4</math> is what makes the local <math>R+R^2</math> and the ghost; [C] <b>conditional narrowing</b>: <i>if</i> the resummed form factor is entire (Biswas–Mazumdar–Siegel class) <i>then</i> the perturbative graviton is ghost-free and rt_F’s attack is a truncation artefact; [O] <b>open</b>: (i) the trace regulator <math>f</math> is not the kinetic form factor <math>a(\square)</math> — entire analyticity of the resummation is unproven; (ii) a ghost-free <i>perturbative</i> graviton is not the nonperturbative ambient measure — QGAMB.IDG.01 stays [C]/[O], not a QG.AMB.01 closure.</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v305_witness_independence.py  | Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ( $e=(4,5,2)$ , $p_n=2+2^n$ ; v23/v228), $\pi$ the only transcendental primitive (ARCH.CORE.01) — 13 atoms / 2 free primitives = $6.5 \times$ compression made explicit (the recurring $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ and $(2/3)^6$ are <i>one</i> ratio); $E_8/g_{\text{car}}$ /anchor are overdetermined ( $\geq 3$ structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); $\alpha^{-1}$ 's '137' is foreign to the anchor family = an independent witness [E] |
| v306_seed_crossval.py         | Leave-one-out cross-validation of the shared seed $\varphi_0$ : back-solving $\varphi_0$ independently from the five seed-grammar observables ( $\sin^2 \theta_{12}$ , $\sin^2 \theta_{13}$ , $\beta$ , $\Omega_b$ , $\lambda_C$ ) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within $3\sigma$ out of sample (most-tensioned = $\sin^2 \theta_{13}$ , $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the $\chi^2$ by $> 100 \times$ [E]                                                                                                     |
| v307_data_watchdog.py         | Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live $n_\sigma$ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items $\alpha^{-1}$ ( $1.9\sigma$ ), $s_{23}$ ( $1.8\sigma$ ) and the most-tensioned core $\sin^2 \theta_{13}$ ( $2.0\sigma$ ); the prediction-of-record $\sin^2 \theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) [E]       |
| v308_seam_equiv_chain.py      | TOE attack 1 — the SEAM.EQUIV.01 keystone as a composition certificate: the v297 stack as one well-typed chain $\text{gap}>0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$ with exact discriminators $ \det \text{Cartan}(E_8) =1$ vs $ \det \text{Cartan}(D_8) =4$ , carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ , $c=g_{\text{car}}+N_{\text{fam}}=8$ ; the input gap is the derived recovery gap $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure [E][O]                                                                                                                    |
| v309_modular_bedrock.py       | TOE attack 2 — the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) $\Rightarrow \mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of QGEO.SYM.01 [E][O]                                                                                                                                                                                     |
| v310_carrier_sm_anomaly.py    | TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5 \rightarrow SU(5) \subset SO(10)$ as $10+5+1 = \text{one generation}$ ; gauge-anomaly-free ( $\sum Y=0$ , $\sum Y^3=0$ , $[SU(2)]^2 U(1)=[SU(3)]^2 U(1)=0$ , exact) and reproduces $(b_1, b_2, b_3)=(\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts $b_1$ , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ $\beta$ sub-question; the full EG $S$ -matrix + Yukawa stay open [E][C]                                                                                                                                                                 |
| v311_gap_decoupled_measure.py | TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ , correlation length $1/(6 \ln(3/2))$ , finite susceptibility $729/665$ ; the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=EH/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); QG.AMB.01 stays [O] (gap-decoupled, inert) [E][O]                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v312_hypergraph_substrate.py  | TOE attack 5 — the hypergraph substrate sharpened: the affine- $E_8$ Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly $2 \cos(\pi/5) =$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is <i>not</i> any eigenvalue and $\varphi_0$ is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ (gap $6 \ln(3/2)$ ) and the analytic seed $\varphi_0$ . The open question is precisely delimited, not closed [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v313_golden_atoms_spectral.py | The golden ratio is the $g_{\text{car}}=5$ signature: the $(2, 3, 5)$ atoms <i>are</i> the affine- $E_8$ network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , the golden minimal polynomial $x^2-x-1$ divides it ( $\phi=2 \cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2 \cos(\pi/k)$ for $k \in \{2, 3, 5\} = \{ \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}}\}$ (giving $0, 1, \phi, 1/\phi$ ), so the golden ratio is the $g_{\text{car}}=5$ (5-fold) signature — a new spectral witness for $g_{\text{car}}$ (raises the v305 multiplicity). The $(2, 3, 5)$ spherical excess $1/2+1/3+1/5=31/30$ ties the v63/v76 margin numerator $31=2^{g_{\text{car}}}-1=248/8=1+h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: $\phi$ is structural, not a readout. [C] direction: $\phi$ is the $E_8$ -quasicrystal (Elser-Sloane) ratio, a candidate substrate [E][C]                                                                                                                                                                       |
| v314_rate_translation.py      | Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine- $E_8$ adjacency spectrum splits into the rational part $\{0, \pm 1, \pm 2\}$ in $\mathbb{Q}$ and the golden part $\{\pm \phi, \pm 1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$ ; the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2 \cos(\pi/2)=0= \mathbb{Z}_2 $ , $2 \cos(\pi/3)=1=N_{\text{fam}}$ build the recovery rate $(2/3)^6 = ( \mathbb{Z}_2 /N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$ ), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$ ) has <i>no</i> rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ( $30=g_{\text{car}}(2N_{\text{fam}})=5 \cdot 6$ ); the $\phi$ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C]                                                                                                                                                                                                                      |
| v315_coxeter_coupling.py      | Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5, 6)=1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: $E_8$ exponents = totatives(30) = totatives(5) $\times$ totatives(6), $\phi(30)=4 \cdot 2=8 =$ rank $E_8$ ; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}]=8 =$ rank $E_8$ , $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) (\ni \sqrt{5})$ and the family $\mathbb{Q}(\zeta_3)$ ; [E] GALOIS = $\mu_4 \times \mathbb{Z}_2$ : $(\mathbb{Z}/30)^\times = (\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2$ (order 8, max order 4), $\mu_4=(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2=(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}=\phi+1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$ . [C] $30=5 \cdot 6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C] |
| v316_galois_readout.py        | Does the Galois $\mu_4 \times \mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho=\zeta_6=\zeta_{30}^5$ is the order-6= $2N_{\text{fam}}$ FAMILY factor $c^5$ (the carrier is $\zeta_5=\zeta_{30}^6$ ); [C] $\delta_{\text{CKM,lead}}=\pi/3=\arg \zeta_6$ , $\delta_{\text{PMNS}}=4\pi/3=\arg \zeta_6^4$ (power $4= \mu_4 $ ), $\zeta_6^4 = -\zeta_6$ the sheet flip ( $\mu_6=\mu_3 \times \mu_2$ ); [E] the Galois $\mathbb{Z}_2=\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q})=\{\zeta_6 \mapsto \zeta_6^5=\bar{\zeta}_6\}$ IS CP conjugation ( $\delta \mapsto -\delta$ ); [E] the carrier factor $\zeta_5$ (golden $\sqrt{5}$ ) carries no phase (static); [E] the magnitudes ( $\sin^2 \theta_{12}=1/3-\varphi_0/2$ , transcendental $\varphi_0$ ) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C]                                                                                                                                                                      |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v317_galois_family.py       | Are the 3 generations a $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ ( $\text{Spec}(Q_+) = \{1, 2, 3\} = A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$ ; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive $\mu_3$ orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$ ). [E] the Galois $\mathbb{Z}_2$ ( $\zeta_3 \mapsto \zeta_3^2$ , = CP conjugation v316) fixes $w=0$ (rational) and swaps $w=1/3 \leftrightarrow 2/3 \Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C]                                                                                                                                                                                                                                                                                                                                                                                                   |
| v318_arithmetic_capstone.py | The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$ , and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$ : $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}] = \varphi(30) = 8 = \text{rank } E_8$ , $30 =  \mathbb{Z}_2  N_{\text{fam}} g_{\text{car}} = h(E_8)$ , Galois $(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$ ; [E] rigidity: forced by the anchor atoms $\{2, 3, 5\} = e(a)$ ; [E] magnitude reduction: $\varphi_0 = (4/3)c_3 + 48c_3^4$ is a pure function of $\pi$ (no free parameters), so the magnitudes are $\{a, \pi\}$ -determined, not free; [E] free inventory: 0 dimensionless free parameters, $\pi$ the unique transcendental, $v_{\text{geo}}$ the one dimensional unit. [O] residual: the raw-seam realisation stays open (QGEO.SYM.01/SEAM.EQUIV.01) [E][O]                                                                                                                                                                                                                                                                                                                            |
| v319_translation_clock.py   | The translation clock: the discrete $\leftrightarrow$ dynamic bridge is the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , $\text{gcd}(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ( $\mathbb{Z}/6 = 2N_{\text{fam}}$ , ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$ , rate $(2/3)^6$ ; order-6 generator $c^5$ ( $\zeta_6 = \zeta_{30}^5$ ). [E] LAW VS LIVE: the resummed clock rate( $n$ ) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from $n=1$ ; the $E_8$ live phases (totatives of 30) start at 1, so 0, 1, 2, 3, 4, 5 is law-inclusive, 1, 2, 3, 4, 5 live-only. [E] STATIC HAND ( $\mathbb{Z}/5 = g_{\text{car}}$ , ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$ , field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$ , $ \mu_4 =4$ planes, v55); the “it is one clock” reading is [C]. |
| v320_galois_cp_relation.py  | A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ . [E] both phases are powers of one hexagonal unit $\rho = \zeta_6$ (v316): $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ ( $60^\circ$ ), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ ( $240^\circ$ ), $\rho^4 = -\rho$ , so $\delta_{\text{PMNS}} - \delta_{\text{CKM}}^{\text{lead}} = \pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: $\delta_{\text{PMNS}}$ robustly away from $240^\circ$ ( $> 3\sigma$ ; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X]                                                                                                                                                                                                                                                                                                                                                                      |
| v321_killtest_board.py      | The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a freeze_file.csv kill row: #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD $r$ (kill $r > 0.01$ ), #3 DESI/Euclid $w = -1$ , #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ( $\sim 2\sigma$ now), #8 LiteBIRD $\beta$ . A forward decision ledger complementing v307; no new physics [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v322_cp_lock_sharpened.py    | The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$ ; $\delta_{\text{CKM}}^{\text{lead}} = 60^\circ$ (node 1), $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (node 4). [E] the node selector is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} =  \mu_4  \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3) = 4\pi/3 = \delta_{\text{CKM}}^{\text{lead}} + \pi$ ( $\rho^4 = -\rho$ ). [C] sub-leading budget: $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2$ ( $\sim 68.7^\circ$ ); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7^\circ \Rightarrow$ band $240 \pm \sim 9^\circ$ . [C] current tension: vs NuFIT 6.0 NO ( $\delta_{CP} = 212_{-41}^{+26^\circ}$ ) the leading $240^\circ$ sits at $+1.08\sigma$ , band edge $\sim 231^\circ$ at $+0.74\sigma$ . [X] kill: the nearest wrong node is $60^\circ$ away $\gg$ budget; DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic $\delta_{CP}$ hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X]                                                                                                                         |
| v323_bw_geometric_modular.py | The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow <i>geometric</i> , so the $\mu_4$ deck invariance is not an independent axiom. [E] the clock $\rho = \text{diag}(i^n) = \exp(i(\pi/2)L)$ is a geometric rotation ( $L = \text{diag}(n)$ , $\rho^4 = I$ , $\mu_4 \subset U(1)_{\text{rot}}$ ). [E] for a rotation-covariant $C = f(L)$ the modular Hamiltonian $K = \log((1 - C)/C) = g(L)$ , so $[K, L] = 0$ (the modular flow commutes with all rotations) and $[\rho, K] = [\rho, C] = 0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is <i>not</i> geometric ( $[K, L] \neq 0$ ); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ QGEO.SYM.01 ( $\omega \circ \rho = \omega$ ) is downstream of SEAM.EQUIV.01 + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]                                                                                                                                                                                                                      |
| v324_hypergraph_fiber.py     | The minimal hypergraph substrate that works: a $(2, 3, 5)$ -network <i>fibered</i> by a 3-fold cusp channel reproduces both the $E_8$ skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}} = (A + 2I)/4$ fixes the Kac marks (eigenvalue $1 = E_8$ skeleton), subleading $(\varphi + 2)/4$ (golden); $(2/3)^6$ is <i>not</i> in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}} = \text{diag}((1 - w)^{2N_{\text{fam}}})$ , $w \in \{0, 1/3, 2/3\}$ ( $N_{\text{fam}} = 3$ , v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$ . [E] the fibered substrate $T = T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ( $w = 0$ ), skeleton $\times$ democratic cusp) <i>and</i> $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ( $T_{\text{net}} \otimes I_3$ ) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ . [E] minimality: the smallest fiber is the 3-node cusp ( $= N_{\text{fam}}$ ) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of $2/3$ from a pure rewrite [E][C] |
| v325_pillowcase_keystone.py  | The seam keystone as ONE citable theorem: the raw seam state <i>is</i> the flat $\tau = i$ pillowcase state, with <b>exactly one</b> open lemma. IF (raw RP-collar seam state = flat $\tau = i$ pillowcase / rotation-invariant state) THEN 4 square marks + $\mu_4$ deck + $\omega \circ \rho = \omega$ + holomorphic $c = 8 \Rightarrow (E_8)_1$ , with the carrier recovery gap as input. [E] pieces all exact: $n = 2\chi(S^2) = 4$ (v216); cross-ratio $2 \Rightarrow j = 1728$ (v214/v267); flat orbifold $S^2(2, 2, 2, 2)$ , $\chi_{\text{orb}} = 0$ (Trojanov); DtN sub-principal at $m \equiv 0 \pmod 4$ (v201/v280); $\det \text{Cartan}(E_8) = 1$ vs $D_8 = 4$ (v83/v143/v308); gap $6 \ln \frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau = i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (QGEO.SYM.01, SEAM.EQUIV.01; v323), machine-pinned in Lean (FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01); an assembly/reduction certificate (like v176/v261), not a closure [E][O]                                                                                                                                                                                                                                                                  |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v326_ftransfer_suite.py    | The $F_{\text{transfer}}$ solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] $F_{\text{pole}}$ (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}(\eta_B)$ : washout contraction to the balance $6.1 \times 10^{-11}$ , monotone (H-theorem), external thermal rate. [C] $F_{\text{relic}}$ (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$ ), cosmological rate. [E] $F_{\text{QCD}}(m_p/m_e)$ : 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, $\Lambda_{\text{QCD}}$ generated. [C] $F_{\text{RareKaon}}$ : $\text{BR}(K^+)=9.45 \times 10^{-11}$ on NA62 2016-2024 (v202). [O] only $F_{\text{pole}}$ has the seam rate $(2/3)^6$ ; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O] |
| v327_hypergraph_rewrite.py | Deriving the cusp weight $2/3$ from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ( $w=0$ ) and $ \mathbb{Z}_2 =2$ surviving — supplies the cusp fiber. [E] $M$ has spectrum $\{1, 2/3, 0\}$ ; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}= \mathbb{Z}_2 /N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order-6= $2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): $2/3$ is NOT a root of the affine- $E_8$ charpoly ( $p(2/3)=-3520/19683 \neq 0$ ), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2, 3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]                                                                                                                                                 |
| v328_theta13_pressure.py   | The $\theta_{13}$ pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO ( $0.02195 \pm 0.00058$ ) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] $\theta_{13}$ is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets $\theta_{12}$ — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is $\sim 5\%$ below the axiom, PMNS $\theta_{13}$ RG running is $< 1\%$ (NO), and $5/6$ is exact — the prediction stays frozen (REG.FREEZE.01). [X] pre-registered kill: a stable $\sin^2 \theta_{13}$ away from 0.023108 at $> 3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]                                                                                                                                                                |
| v329_os_gap_reduction.py   | Closing the OS step: the bulk gap <i>is</i> the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy = $6 \ln \frac{3}{2}$ , independent of system size — exactly “OS bulk gap = transfer gap” (the v308 open input), discharged at the many-body level. [E] $H \Rightarrow \omega \circ \rho = \omega$ : under $H$ the covariance is rotation-covariant $C=f(L)$ , $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ : $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise $H$ (Lean-pinned FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01). [O] residual: the necessity of $H$ on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]                                                                                     |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v330_qgamb_admissible_measure.py | Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer $T$ (v221/v311; spectrum $\{1, (2/3)^6, (1/3)^6\}$ ) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \rightarrow \lambda_2 = (2/3)^6$ ( $\xi=1/\Delta$ , $\Delta=6 \ln \frac{3}{2}$ ). [E] tightness $\Rightarrow$ projective limit: $\chi = \sum_n C(n) = 729/665$ finite, so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu = \lim \mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{\text{OS}} = -\log T \geq 0$ , unique vacuum, gap $6 \ln \frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi \rightarrow \infty$ , tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v275) the physical readouts do not need it, but QG.AMB.01 stays open. A genuine PARTIAL construction, NOT a closure [E][O]                                                                                   |
| v331_necessity_of_H.py           | The <i>necessity</i> of $H$ reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ( $\tau=i$ ), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329’s sufficiency). [E] square $\Rightarrow [\rho, \Lambda]=0$ : marks at $\{0, \pi/2, \pi, 3\pi/2\}$ make the curvature $\mathbb{Z}_4$ -symmetric (Fourier support $0 \bmod 4$ ), $M_f$ couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $ k +M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho, \Lambda] \neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio( $1, i, -1, -i$ ) $=2 \Rightarrow j=1728$ , the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ . [E] NEG: a $\mathbb{Z}_2$ mark set commutes with $\rho^2$ but not $\rho$ . [O] residual: the dynamical <i>selection</i> of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]                                                                                                                                                                           |
| v332_qgamb_metric_sector.py      | The metric sector of QG.AMB.01 characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330’s metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\text{conf}}(D) = -(D-1)(D-2)/4$ is negative for $D \geq 3$ ( $c_{\text{conf}}(4) = -3/2$ ), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so $e^{-S}$ diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76). [C] IDG conditional route: the entire form factor $e^{z/M^2}$ (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial $(R+R^2)$ truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full QG.AMB.01 needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O] |
| v333_cm_selection.py             | The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\text{SL}(2, \mathbb{Z})$ has $S^2 = -I$ , $S^4 = I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\text{PSL}(2, \mathbb{Z})$ (the other, $\rho$ , is order 3). [E] deck order $\rightarrow$ CM point: order-4 $\rightarrow$ cross-ratio 2 $\rightarrow j=1728 \rightarrow \tau=i$ (square); order-3 $\rightarrow j=0 \rightarrow \tau=\rho$ (hexagon); $ \mu_4 =4$ selects $\tau=i$ . [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2 =$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects $\rho$ , not the square. [O] residual: only “the deck acts <i>geometrically</i> (= QGEO.SYM.01) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]                                                   |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v334_conformal_resolution.py | The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a <b>CONDITIONALLY</b> convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). <b>[E]</b> the obstruction: $c_{\text{conf}}(4) = -3/2 < 0$ , the unrotated conformal Gaussian diverges. <b>[E]</b> GHP rotation: $\phi \rightarrow i\phi$ flips the sign, $\int e^{- c \phi^2} = \sqrt{\pi/ c }$ <b>FINITE</b> (GHP 1978). <b>[E]</b> IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. <b>[E]</b> NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). <b>[C]</b> conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). <b>[O]</b> residual: whether the GHP-rotated IDG-dressed measure is the <b>CORRECT</b> nonperturbative <b>QG.AMB</b> measure (the contour’s nonperturbative validity); gap-decoupled (v76), blocks no test, <b>NOT</b> a closure <b>[E][C][O]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v335_seam_equiv_unify.py     | The keystone unification: there is <b>ONE</b> open theorem ( <b>SEAM.EQUIV.01</b> ), <b>QGE0.SYM.01</b> is its <b>COROLLARY</b> (a conformal-net axiom), and <b>QG.AMB.01</b> is a decoupled general QG problem, <b>NOT</b> a TFPT gate. <b>[E]</b> <b>QGE0.SYM.01</b> = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a <b>NET AXIOM</b> ); rotations $U(1) < \text{Moebius} \Rightarrow$ rotation-invariant vacuum; the $\mu_4$ clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$ . So <b>SEAM.EQUIV.01</b> $\Rightarrow$ <b>QGE0.SYM.01</b> with <b>NO</b> extra premise (the v323 “rotation-invariant vacuum” is a net axiom); the two bedrock items collapse to <b>ONE</b> . <b>[E]</b> the one keystone theorem: <b>SEAM.EQUIV.01</b> = a construction theorem ( $c=8$ , $\det \text{Cartan}(E_8)=1$ , gap $6 \ln \frac{3}{2} > 0$ derived) with <b>EXACTLY</b> one open continuum lemma. <b>[E]</b> <b>QG.AMB.01</b> decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. <b>[E]</b> collapsed status: structural items $2 \rightarrow 1$ ; live residual $= v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . <b>[O]</b> the one residual = <b>SEAM.EQUIV.01</b> ’s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does <b>NOT</b> close <b>SEAM.EQUIV.01</b> <b>[E][O]</b>                                                                                                                                                                                                                                 |
| v336_continuum_limit.py      | The <b>ONE</b> open lemma of <b>SEAM.EQUIV.01</b> , stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now <b>CITABLE</b> . <b>[C]</b> continuum leg: Osborne–Stottmeister ( <i>CFT from Lattice Fermions</i> , CMP <b>398</b> (2023) 219, doi:10.1007/s00220-022-04521-8; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework, arXiv:2010.11121) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct $c$ ); WZW range rank $\leq c \leq D$ . $(E_8)_1$ : $c=8$ , rank 8, $D=16$ Majoranas $(SO(16)_1 \rightarrow (E_8)_1)$ , $8 \leq c \leq 16$ in range; the collar is a gapped quasi-free (CAR) state. <b>[C]</b> OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. <b>[C]</b> construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). <b>[E]</b> target pinned: $c=g_{\text{car}}+N_{\text{fam}}=8$ , $\det \text{Cartan}(E_8)=1$ , the $D=16$ Majorana $SO(16)_1$ IS the carrier-16 and the $\mu_4$ extension IS the seam glue. <b>[O]</b> open residual: $c=8$ alone does <b>NOT</b> pin the net ( $SO(16)_1$ also $c=8$ , $\det 4$ ), so the one open lemma = the holomorphy discriminator $\det K=1$ . A literature-mapping/reduction; does <b>NOT</b> close <b>SEAM.EQUIV.01</b> <b>[C][E][O]</b> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v337_decoupling_theorem.py       | <p>THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the QG.AMB.01 reclassification (v335) honest. [E] gap margin: <math>\Delta - 2 \dim(E_8) c_3^2 = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 &gt; 0</math>. [E] factorization: every readout (mass ratios, mixings, <math>\alpha^{-1}</math>, <math>R+R^2</math>) is a function of the admissible gapped transfer spectrum <math>\{1, (2/3)^6, (1/3)^6\}</math> + the discrete kernel alone. [E] the theorem: margin <math>&gt; 0 \Rightarrow</math> exponential clustering (<math>\lambda_2=(2/3)^6</math>), finite susceptibility <math>\chi=729/665</math>, so the ambient contribution is gap-suppressed <math>\Rightarrow</math> no readout depends on the ambient measure <math>\Rightarrow</math> QG.AMB.01 decoupled. [E] neg control: gap closed <math>\Rightarrow \chi</math> diverges <math>\Rightarrow</math> fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]</p>                                                                                                                                          |
| v338_theta13_budget.py           | <p>THETA13.BUDGET.01: the <math>\theta_{13}</math> sub-leading budget vs the current global fit – the <math>+2\sigma</math> pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: <math>\sin^2 \theta_{13} = (\varphi_0^{\text{ret}}) e^{-5/6} = 0.023108</math> is <math>+2.0\sigma</math> vs the no-SK central (0.02195) and <math>+1.7\sigma</math> vs the with-SK NO best fit (0.02215) – not a clean <math>2\sigma</math>. [E] within <math>3\sigma</math>: <math>0.023108 &lt;</math> the with-SK <math>3\sigma</math> upper 0.02388 (not falsified). [C] budget: moving to the central needs <math>-4.1\%</math>; RG <math>&lt; 1\%</math> and the exponent <math>5/6</math> is exact <math>\Rightarrow</math> no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns <math>1.7\text{--}2.0\sigma</math> into <math>\sim 6\sigma</math> – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]</p>                                                                                                                                                                                                                                                                         |
| v339_firstprinciples_boundary.py | <p>The honest first-principles boundary, mapped precisely – answers (a) can <math>M_{\text{Planck}}/v_{\text{geo}}</math> be computed? and (b) can <math>F_{\text{transfer}}</math> be made first-principles? Both NO, with the exact reason per residual. [E] <math>v_{\text{geo}}/M_{\text{Planck}}</math> FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] <math>m_p/m_e</math> structure in place (v262): the carrier <math>b_3=-7</math> drives the QCD run + the closed electron source; two inputs stay external: <math>\alpha_s(M_Z)</math> and the lattice <math>C_p=m_p/\Lambda^{(3)} \sim 2.83</math>. [X] <math>\alpha_s(M_Z)</math> external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread <math>\sim 8.8</math> at <math>2 \times 10^{16}</math> GeV, v246). [C] <math>C_p</math> parameter-free but lattice-only (v35). [C] <math>\eta_B/\text{axion}</math> = cosmological initial conditions; Koide <math>Q=2/3</math> structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]</p> |
| v340_As_reconcile.py             | <p><math>A_s</math> is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent <math>-11\sigma</math> tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] <math>A_s = N_\star^2 (M_{\text{scal}}/\bar{M}_{\text{Pl}})^2 / (24\pi^2) = N_\star^2 c_3^7 / (24\pi^2)</math> (v86); <math>M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.2565 \times 10^{-5}</math> is a PURE RATIO, so <math>A_s</math> carries NO absolute scale – it pins <math>N_\star</math>, NOT <math>\bar{M}_{\text{Pl}} \Rightarrow A_s</math> does NOT define mass (No-Unit v153 stands). [C] the <math>-11\sigma</math> is the slow-vs-fast reheating dichotomy (<math>N_\star=51.44 \rightarrow A_s=1.764 \times 10^{-9}</math>; <math>A_s</math>-preferred <math>N_\star=56.2</math>). [C] ARCHIVE: old v4.5/v4.6 used the algebraic <math>N_\star=3/\varphi_0^{\text{ret}} - 1=55.42</math> and reported <math>A_s=2.124 \times 10^{-9}</math>; at that <math>N_\star</math> the current normalisation gives <math>2.047 \times 10^{-9}</math> (<math>-1.9\sigma</math>). [E] nothing lost: the current theory DERIVES <math>N_\star</math> (record band [50, 60], v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X]</p>             |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v341_alpha_quillen.py        | <p>The <math>\alpha</math> fixed point reformulated as the STATIONARITY of a <math>U(1)</math> determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the “derive the FORM” step).</p> <p><math>F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \ln(1/\varphi_{\text{seam}}) = 0</math> in Quillen form: <math>[\alpha^3 - 2c_3^3\alpha^2]</math> (det-line variation) = <math>[8b_1c_3^6]</math> (anomaly counterterm) <math>\times [\ln(1/\varphi_{\text{seam}})]</math> (seam response). [E] <math>M=41</math> a <math>U(1)</math> INDEX (EM Ward, v48): <math>\frac{4}{5}M = 8b_1</math>, <math>b_1=41/10</math> the SM 1-loop <math>U(1)_Y</math> beta (v159/v246). [E] <math>c_3=1/(8\pi)</math> the GAUSS–BONNET boundary coefficient (<math>8=\text{rank } E_8</math>, v216); <math>c_3^{3,6}</math> heat-kernel powers. [E] <math>-5/4 = -q(D_5)</math> a DISCRIMINANT form (<math>q(D_5)+q(A_3)=2</math>, v1/v154); <math>\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3</math> carries <math>q(A_3)</math>: <math>c_3/\varphi_{\text{base}}=3/4</math>. [E] <math>\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4</math>. [E] the Quillen split holds at the <math>\alpha^{-1}=137.0359992</math> root. [E] NO NEW GATE: <math>\alpha^{-1}</math> stays [E]; v341 SHARPENS the pre-existing [O] origin EM.WARD.01 (no new <math>\alpha</math> problem). [O] the one still-open part IS EM.WARD.01’s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/EM.WARD.01 from “fixed point” to “stationarity of a named determinant line” [E][O]</p> |
| v342_em_ward_heatkernel.py   | <p>The heat-kernel sharpening of EM.WARD.01: DERIVES the ORIGIN and STRUCTURE of the <math>F_{U(1)}</math> determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] to [E] for the STRUCTURE; NO new gate. [E] <math>c_3=1/(8\pi)</math> the boundary GAUSS–BONNET coefficient (seam = flat pillowcase <math>\chi=2</math>, <math>\oint K=4\pi</math>, v216). [E] the <math>\alpha^2</math> term is the GILKEY gauge-curvature coefficient: the <math>U(1)</math> Laplacian <math>a_4</math> has <math>30\Omega^2/360 = \frac{1}{12}\Omega^2</math> (Gilkey Thm 4.8.16), <math>\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2</math>, so the Calderon <math>-2c_3^3\alpha^2</math> is the gauge-curvature piece, not an ansatz. [E] the <math>c_3</math>-ladder <math>\{0, 3, 6\}</math> is arithmetic (step <math>3 = 3</math> channels; <math>6= R^+(A_3) </math> <math>C_6</math> hexagon). [E] multiplicities forced (<math>2= \mathbb{Z}_2 </math>, <math>8b_1=(\text{rank } E_8)b_1</math>). [E] <math>\pi</math>-power test: <math>2c_3^3=1/(256\pi^3)</math> carries <math>\pi^3 \Rightarrow</math> THREE boundary insertions. [C] the exact coefficient needs the seam <math>F</math>-normalisation. [O] the cubic <math>\alpha^3</math> Maxwell moment + full identification = EM.WARD.01 residual (tied to SEAM.EQUIV.01). <math>\alpha^{-1}</math> stays [E] (v3) [E][C][O]</p>                                                                                                                                                                                                                                                                                                      |
| v343_four_routes_analysis.py | <p>The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to “which closes ALL problems and is 100% provable”: NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (SEAM.EQUIV.01 + the decoupled QG contour). [E] ROUTE A reframes QG.AMB to a FINITE thermal trace (de Sitter finite, gapped <math>\chi=729/665</math>) but <math>c_{\text{conf}}(4) = -3/2 &lt; 0</math> still needs the GHP contour (v334, [O]); does NOT touch SEAM.EQUIV.01. [C] ROUTE B: Carlip+Cardy consistent with <math>c=8</math> but <math>L_0</math> free (ambiguity) and gives <math>c</math> only, not holomorphy <math>\det K=1</math> – inherits SEAM.EQUIV.01. [E] ROUTE C: modular flow KMS <math>\beta=1</math> (v239), geometric identification = SEAM.EQUIV.01 (v335) – a reduction. [E]/[O] ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics [O] – not a closure. [E] VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = SEAM.EQUIV.01 [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v344_detk_synthesis.py        | The bird's-eye synthesis of the ONE open keystone bit, $\det K=1$ – a full-spectrum map of SEAM.EQUIV.01's only open residual. [E] $\det K=1$ is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link $H_1=0$ / one 1-dim irrep / $\det \text{Cartan} = 1$ / full $\mu_4$ condensation / $\tau=i$ CM), unified by $ \det \text{Cartan}  =  H_1(\text{link})  = \#(1\text{-dim irreps})$ across ADE ( $A_n \rightarrow n+1$ , $D_n \rightarrow 4$ , $E_6 \rightarrow 3$ , $E_7 \rightarrow 2$ , $E_8 \rightarrow 1$ ) – ONLY $E_8$ gives 1 ( $2I$ the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); $\det K$ is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2, 3, 5)$ rewrite attractor IS the affine- $E_8$ graph (v312) and the $E_8$ du Val link is the Poincare sphere ( $H_1=0$ , v232), so $\det K=1$ becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; R3's combinatorial-to-geometric bridge is meanwhile discharged to Kronheimer 1989 (v479), so the residual transforms to “the raw seam supplies the ALE/orbifold datum” = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing [E][C][O] |
| v345_hypergraph_homotopy.py   | EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to $\det K=1$ , v344): computes explicitly that the $(2, 3, 5)$ -rewrite attractor graph's plumbing link is the POINCARÉ homology sphere ( $H_1=0$ ), only $E_8$ among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= SEAM.EQUIV.01) stays [O]. Plumbing topology (Brieskorn): link $H_1 = \text{coker}(\text{Cartan})$ . [E] $\text{SNF}(E_8 \text{ Cartan}) = \text{identity} \Rightarrow \text{coker}(E_8)=0 \Rightarrow$ the $E_8$ plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8=1$ ). [E] only $E_8$ : $\text{coker}(A_n)=\mathbb{Z}/(n+1)$ , $D_n \rightarrow \mathbb{Z}/4$ , $E_6 \rightarrow \mathbb{Z}/3$ , $E_7 \rightarrow \mathbb{Z}/2$ – all nontrivial, only $E_8=0$ . [E] $\pi_1=2I=SL(2, 5)$ PERFECT (commutator subgroup order $120= G $ , computed) $\Rightarrow H_1=0 \Rightarrow$ THE Poincaré sphere; $ 2I =120= \mu_4  h(E_8)$ . [O] does NOT close $\det K=1$ : the graph $\rightarrow$ geometry half is meanwhile Kronheimer-cited (v479); what stays open is the raw seam supplying the ALE/orbifold datum = SEAM.EQUIV.01 [E][O]                                                                                                                                                                                                                                                                          |
| v346_seam_geometric_bridge.py | The geometric bridge as an explicit six-link chain raw-seam $\Rightarrow \det K=1$ , with the ONE open arrow pinned – the honest capstone of the SEAM.EQUIV.01 investigation. L1 [E]/[C] raw seam $\rightarrow \mu_4$ + four marks (v216); L2 [O] $\mu_4$ + $(2, 3, 5)$ atoms $\rightarrow 2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE OPEN ARROW = SEAM.EQUIV.01); L3 [E] $2I \rightarrow$ affine $E_8$ McKay (v219); L4 [E] $2I \rightarrow E_8$ du Val (v232); L5 [E] $\mathbb{C}^2/2I \rightarrow$ Poincaré link $H_1=0$ (v345); L6 [E] $H_1=0 \rightarrow \det K=1$ . [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$ , $(2, 3, 5)$ unique to $2I$ , and $( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = SEAM.EQUIV.01, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by $(2, 3, 5)$ [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v347_seam_closure_modes.py    | The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to “is there ANY other way to FULLY solve it?”. [E] THE PRECISE LOCUS: the seam gives the pillowcase base $S^2(2, 2, 2)$ , $\chi_{\text{orb}}=0$ (EUCLIDEAN, v216), but the $2I/E_8$ structure is the Seifert base $S^2(2, 3, 5)$ , $\chi_{\text{orb}}=1/30$ (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier $(2, 3, 5)$ input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation $\Rightarrow \mathbb{C}^2/2I$ forced, the only theorem-route w/o new input, v284/v300/v331); C axiom $P_3$ ( $\Rightarrow$ TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics $\Rightarrow$ only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v348_seam_rigidity_route.py             | Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? [E] rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] downstream is a LATTICE identity: $E_8$ = the ring of icosians (Conway–Sloane), rank 8, so $2I \rightarrow E_8$ is a lattice identity (stronger than McKay v219). [E] the single bridge is $\varphi$ : crystallographic orders $\{1, 2, 3, 4, 6\}$ , the 5-fold non-crystallographic; $\mu_4$ crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). [O] the keystone reduces to ONE question: “does the raw seam carry $\varphi$ ?” – if yes, $\mu_4 + \varphi \rightarrow 2I \rightarrow E_8$ closes L2. [O] honest subtlety: TFPT’s $\varphi$ is currently $E_8$ -side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$ , recovery $(2/3)^6$ , seed $\varphi_0$ ) are not manifestly golden, so showing the RAW seam carries $\varphi$ non-circularly is the residual (= L2). NOT closed [E][O] |
| v349_raw_seam_golden_test.py            | The honest test of the v348 question “does the RAW seam carry the golden ratio (before assuming $E_8$ )?”. Answer, computed: NO – the bare carrier ( $D_5$ , $A_3$ ) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/ $E_8$ identification. A NEGATIVE result that prevents a false closure. [E] RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$ ; $D_5$ has $h=8$ ( $2\cos(2\pi/8)=\sqrt{2}$ ), $A_3$ has $h=4$ – no 5-fold, bare carrier gives $\sqrt{2}$ . [E] DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed $\varphi_0$ transcendental. [E] GOLDEN IS THE ICOSAHEDRAL/ $E_8$ SIDE: $5 \mid h$ only for $A_4=SU(5)$ ( $h=5$ ), $H_3$ ( $h=10$ ), $E_8$ ( $h=30$ ). [E] $\varphi$ alone insufficient: $2I \neq C_5 \times \mu_4$ ( $20 \neq 120$ ). [O] RESOLUTION: L2 reduces to “is $g_{\text{car}}=5$ a PENTAGON (5-fold $\Rightarrow$ golden $\Rightarrow$ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$ )?” – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden $\varphi$ . Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed [E][O]     |
| v350_bootstrap_inputs_correction.py     | The honest CORRECTION of v349’s framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). [E] THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\text{car}}=5$ over-determined three ways (rank-fill, Coxeter-match $g_{\text{car}} = \max$ prime of $h(E_8)=30$ , Pascal). [E] $g_{\text{car}}=5$ IS THE ICOSAHEDRAL 5-FOLD, not the $D_5$ rank: $g_{\text{car}} = \max$ prime of $h(E_8)=30=2\cdot 3\cdot 5$ ; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: $D_5$ ( $h=8$ )/ $A_3$ ( $h=4$ ) non-golden alone, but the $\mu_4$ extension $D_5 \oplus A_3 \rightarrow E_8$ (v154) gives $h(E_8)=30$ (golden, $5 \mid 30$ ) – the seam’s own $\mu_4$ produces the 5-fold; v349 checked the un-glued pieces, the wrong object. [E] v349’s pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). [O] residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349’s interpretation [E][O]                              |
| v351_continuum_realisation_sharpened.py | The continuum realisation sharpened – the “ $c=8$ ambiguity” flagged open in v277/v344 ( $E_8$ vs $SO(16)$ both $c=8$ ) is RESOLVED, non-circularly, by the seam’s ORDER-4 $\mu_4$ clock. [E] $E_8$ (det 1, holomorphic) vs $D_8=SO(16)$ (det 4). [E] the order-4 clock (the four Gauss–Bonnet marks, $ \mu_4 =4=h(A_3)$ , v216): index-4 extension $D_5 \oplus A_3 = E_8$ (v154), index-2 = $SO(16)$ ; so order-4 selects $E_8$ ( $16 \rightarrow 4 \rightarrow 1$ ) not $SO(16)$ ( $16 \rightarrow 4$ ) – the order (4 not 2) is the discriminator. [E] bootstrap agrees: $h(E_8)=30$ (max prime 5) vs $h(D_8)=14$ (max prime 7). [E] which-net ambiguity RESOLVED (det $K=1$ is then a consequence). [O] residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close SEAM.EQUIV.01 [E][O]                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v352_framework_irreducible.py | The framework reduction – the deepest answer to “the inputs are not axioms, they are back-determined”. [E] P2 ( $g_{\text{car}}=5$ ) FORCED three ways (v6): rank-fill, Coxeter-match ( $g_{\text{car}} = \max \text{ prime of } h(E_8)=30$ ), Pascal. [E] P1 ( $c_3=1/(8\pi)$ ) FORCED: $c_3$ is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways ( $8 = \text{rank } E_8 = h(D_5) = \phi(30) = \det R$ , v54/v6); only $\pi$ is non-forced (the boundary transcendental). [E] the $E_8$ hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + $\pi$ – the precise content of “the inputs are not axioms”. [O] the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT’s true input is ONE framework + $\pi$ [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v353_selfloop_capstone.py     | The bird’s-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms $\rightarrow$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. [E] IT IS A LOOP: outputs fix inputs ( $g_{\text{car}}=5 = \max \text{ prime of } h(E_8)=30$ ; the 8 in $c_3 = \text{rank } E_8$ ) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). [E] ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. [E] $\pi$ is a FIXED geometric constant (the boundary 2-sphere, $\oint K=4\pi \Rightarrow c_3=1/(8\pi)$ ), not a dial. [C] refining v352: $\pi$ is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. [O] the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork [E][C][O]                                                                                                                                                                                                                                                                                                                                              |
| v354_e8_reverse_audit.py      | The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto $E_8$ ); the reverse question: how much $E_8$ structure maps onto NOTHING? [E] $E_8$ has 8 Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ; THREE feed a readout (2 metric, 8 rank $\rightarrow c_3$ , 30 Coxeter $\rightarrow g_{\text{car}}$ ) and FIVE are UNMAPPED $\{12, 14, 18, 20, 24\} - 3/8$ load-bearing, $5/8$ not. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h=30$ , $\det 1$ , $D_5 \oplus A_3$ , 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). [E] the golden ratio is UNMAPPED (the affine- $E_8$ attractor eigenvalue v312, the icosian lattice v348; no frozen readout is $\varphi$ , the sole [C] exception $\cos(\theta_i=3\pi/5)=-1/(2\varphi)$ on the axion spine branch, refined in v429) $\Rightarrow$ NOT numerology (which needs a number fitted to an observation); $\varphi$ is the algebraic signature of the 5-fold $h=30=2\cdot 3\cdot 5$ , the carrier announcing it is icosahedral. [O] the $5/8$ unmapped is the HULL OVERHEAD ( $E_8 = \text{minimal unimodular hull, not the world group}$ ); where future readouts could live, or over-capacity. The reverse complement of v305 [E][O] |
| v355_e8_unmapped_bandwidth.py | A disciplined bandwidth search of the unmapped $E_8$ region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about $E_8$ ), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. [E] FORCED SET-LEVEL (the real collective content): $\sum \text{deg} = 128 = \dim S^+$ the spinor half of $248=120+128$ ; $\sum \text{exp} = 120 = \# \text{pos roots}$ ; $\prod \text{deg} =  W(E_8) $ ; exponents = the $\varphi(30)=8$ totatives of 30 (v223); $\max \text{deg} = 30 = h(E_8)$ . [C] SUGGESTIVE not forced: $12=h(E_6)$ , $18=h(E_7)$ , $30=h(E_8)$ (the McKay ladder v219), but ‘deg( $E_8$ ) contains subalgebra Coxeter numbers’ is not a theorem – flagged. [O] DECLINED: the per-degree atom matches ( $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $12= R(A_3) $ , $18=p_4(a)$ ; v66 flagged 12, 18 fittable) each admit multiple readings, and the $ W $ prime 7=‘scalaron exponent’ is one of many – declined. [E] RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result [E][C][O]                                                                                                                                                                                         |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v356_<br>continuum_mmst_<br>applicability.py | Direction A: the continuum chain of SEAM.EQUIV.01 assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) $\rightarrow$ invertible (Kitaev, v301) $\rightarrow$ the NONTRIVIAL invertible phase ( $c_-=8$ ) $\rightarrow$ a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) $\rightarrow$ chiral-CFT scaling limit (MMST v336, $8 \leq 8 \leq 16$ ) $\rightarrow$ order-4 $\mu_4$ clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the $c_3$ origin ( $c_- \neq 0$ = one-sided = the $ \mathbb{Z}_2  \nmid K=8$ of $c_3=1/(8\pi)$ , v58). [C]/[E]/[O]; ADVANCES SEAM.EQUIV.01 to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O]                                                                                                                                      |
| v358_grav_entropy_<br>equilibrium.py         | The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta \langle K \rangle$ ) with TFPT's atoms $\Rightarrow$ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$ , $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi)=4c_3$ , EH $1/(16\pi)=c_3/2$ , Einstein $8\pi=1/c_3$ , Bekenstein $1/4=1/ \mu_4 $ ). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $ \mathbb{Z}_2 2\pi\chi$ iff $ \mu_4 = \mathbb{Z}_2 \chi=4$ – the thermodynamic and geometric origins of $c_3$ are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$ . [E] entropy density atom-fixed: $1/4=1/ \mu_4 $ + central charge $c=g_{\text{car}}+N_{\text{fam}}=8$ (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the $v_{\text{geo}}$ unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O] |
| v359_grav_<br>nonlinear_<br>einstein.py      | The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR $G_{ab}$ (not just $R_{ab}$ ) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}$ with conservation built in. [C][E] fixed-volume $\rightarrow$ Einstein tensor (trace = $-R$ in $d=4$ ). [E] Lovelock $\Rightarrow \nabla^a T_{ab}=0$ is an output. [E] coeff 1: $8\pi=1/c_3$ fixed (v358). [E] coeff 2: $\Lambda$ from $\alpha - \rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O]                                                                                                                                  |
| v360_grav_gap_<br>corrections.py             | Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known $R^2$ scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the $R^2$ scalaron ( $a_4=R^2/72$ v36; $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$ , $\exp 7=g+N-1$ ) – already in TFPT, already falsifiable via $n_s=1-2/N_*$ , $r=12/N_*^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law $\rightarrow f(R)=R+R^2$ , v28/v359). [E] the gap $\Delta=6\ln(3/2)$ is dimensionless, so a distinct gap term sits at $\xi v_{\text{geo}} \sim$ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; $\exp 7$ vs $6$ , $c_3^7$ vs $(2/3)^6$ unrelated) $\rightarrow$ per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O]                                                                                    |
| v361_grav_<br>backreaction.py                | Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: $c_-=8$ (16 Majoranas = $\dim S^+ = 2^{g_{\text{car}}-1}$ ; chiral central charge $16/2=8=g_{\text{car}}+N_{\text{fam}}$ , the same $c_-$ as v358). [E] the backreaction is FINITE $\rightarrow \Lambda$ from $\alpha$ : the matter vacuum energy through $G_{ab}=c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = $119.03+3.92$ – NOT the $M_{\text{Pl}}^4$ catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta \langle K_B \rangle = (8\pi^2 R^4/15)\delta \langle T_{00} \rangle$ , carrier $T_{ab}$ =the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_-=8$ and $\Lambda \rightarrow$ per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O]   |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v364_vgeo_sharpen.py      | Direction 6: $v_{\text{geo}}$ is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale ( $v_{\text{geo}}$ a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either $-1/c_3=8\pi$ and the $\Lambda$ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless})v_{\text{geo}}^4$ reduce to $v_{\text{geo}}$ times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio) $\times$ (power of $v_{\text{geo}}$ ); free content = $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials, vs the SM's $\sim 26$ . [O] $v_{\text{geo}}$ irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v365_qg_oneloop_saddle.py | Direction 3: the QG measure (gate C7/QG.AMB.01) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode Gaussian convergence. [E] finite model log-det $\text{Tr}' \log M = 6 \log \frac{9}{2}$ on the OS spectrum $\{0, 6 \log \frac{3}{2}, 6 \log 3\}$ . [C] the conformal mode converges on the GHP contour $(\int e^{- c \phi^2} = \sqrt{2\pi/3}, \text{v334})$ . [E] the admissible projective limit exists $(\chi = 1/(1 - (2/3)^6) = 729/665, \text{v330})$ . [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/QG.AMB.01) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                           |
| v366_mmst_seam_collar.py  | Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing SEAM.EQUIV.01 to the single S3 lattice-realisation input. Osborne–Stottmeister (arXiv:2107.13834, CMP <b>398</b> (2023) 219; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework): free lattice fermions (CAR quasi-free) $\rightarrow$ chiral-CFT massless scaling limit with correct $c$ , range $\text{rank} \leq c \leq D$ . [E] class arithmetic: $D = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8$ , $\text{rank } E_8 = 8 \Rightarrow \text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_- = D/2 = 8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) $\Rightarrow$ chiral CFT $c=8$ (v336). [E] target pinned to $(E_8)_1$ by $c=8$ + the order-4 $\mu_4$ clock ( $\det K=1$ vs $SO(16) \det K=4$ , v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES SEAM.EQUIV.01, does NOT close it. [E][C][O] |
| v367_seam_s3_lattice.py   | Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana ( $p + ip$ ) lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, $\mu_4$ -condensed to $(E_8)_1$ ; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2 > 0$ at $M=1$ ). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $ C =1$ at $M=1$ , $C=0$ at $M=3$ trivial control). [E] $c_- = 8$ : each chiral Majorana edge carries $c_- = C/2 = 1/2$ , $N_{\text{Maj}} = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ copies $\Rightarrow c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ . [E] edge K-matrix $SO(16)_1$ ( $\det \text{Cartan } D_8=4$ ) $\rightarrow (E_8)_1$ ( $\det \text{Cartan } E_8=1$ ) by the order-4 $\mu_4$ clock (v351/v369). [C] the $\mu_4$ condensation $SO(16)_1 \rightarrow (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close SEAM.EQUIV.01. [E][C][O]                                                                                                                                                                                                                                                                                                                                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v368_seam_s3_inflow.py       | Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p + ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. [E] topo in-gap edge states ( $M=1$ strip $\min_{k_x}  E  \sim 0$ ) vs trivial clean gap ( $M=3$ , neg control); [E] the near-zero state is edge-localised (weight within 4 sites of a boundary, edge $\gg$ bulk); [E] bulk-edge correspondence (edge branch count = bulk Chern $ C =1$ ) $\Rightarrow$ 16-Majorana edge, $c_- = 8$ . [C] anomaly inflow: a $c_- \neq 0$ invertible bulk cannot have a gappable edge $\Rightarrow$ discharges the v336/v351 ‘does the edge exist?’ residual on the lattice. [O] residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close SEAM.EQUIV.01. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v369_qgamb_redundancy.py     | Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/QG.AMB.01). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so QG.AMB.01 is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. [E] HOLOMORPHIC $\Rightarrow$ DHR = Vec: #DHR primaries = $ \det \text{Cartan}  = 1$ for $E_8$ (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges $\Rightarrow$ no hidden boundary sector a bulk could carry invisibly. [E] NO TORUS GSD: ground-state degeneracy on $T^2 = \#\text{anyons} =  \det K  = 1$ for $E_8 \Rightarrow$ unique ground state $\Rightarrow$ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). [E] FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6 = 64/729 < 1$ (v221) $\Rightarrow$ $\text{Rec}_\Sigma$ bounded. [E] GAP-DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ the ambient sector is energetically separated. [C] RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) $\Rightarrow \text{Rec}_\Sigma : (\text{bulk}) \rightarrow \mathcal{A}_\Sigma^{\mu_4}$ . [C] THE REDUNDANCY STATEMENT: under SEAM.EQUIV.01 + BW, $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_\Sigma^{\mu_4} \Rightarrow$ QG.AMB.01 carries no extra physical d.o.f. [O] residual: the two consumed premises – SEAM.EQUIV.01 (the S3 lattice input, v366) + the BW intrinsicity of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy [E][C][O] |
| v370_grav_spin2_unitarity.py | Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304’s scalar pole algebra to the tensor spin structure + combines with v334’s spin-0 conformal mode). [E] projector completeness: $\text{tr } P_2 = (d-2)(d+1)/2$ , $\text{tr } P_1 = d-1$ , $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$ sum to $d(d+1)/2$ ; the $d=4$ split is $5+3+1+1=10$ . [E] EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2) = -1/2$ . [E] the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2 = -M^2$ ). [E] the entire KMS form factor $e^{p^2/M^2}$ (v304/v259) makes spin-2 ghost-free (only the $p^2=0$ pole). [C] the wrong-sign spin-0 conformal mode ( $c_{\text{conf}}(4) = -3/2$ , v332) is cured by the GHP contour (v334) – a different mechanism. [C] $\Rightarrow$ perturbative graviton unitarity sector-by-sector, conditional on v304’s entire-analyticity assumption. [O] perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (QG.AMB.01). [O] decline: $\text{tr } P_2 = 5$ is dimension counting, NOT $g_{\text{car}}$ (v354/v355). [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v371_ftransfer_pole.py        | Track 3 ( $F_{\text{transfer}}$ ): the Koide source→pole transfer as a typed solver – promotes the <b>experiments/ftransfer</b> solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or $Q^*=2/3$ v82/v101). [E] $Q_{\text{pole}}=\text{Koide(PDG poles)}=2/3$ to $3\times 10^{-6}$ , $Q_{\text{src}}(\varphi_0^{\text{ret-ladder}})=0.66446$ (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes $Q$ ABOVE 2/3; the $\epsilon$ mapping $Q_{\text{pole}} \rightarrow Q_{\text{src}}$ is NEGATIVE while QED gives $\epsilon=+(3/2)(\alpha/\pi) > 0$ – wrong sign for any $\alpha \in [1/137, 1/128]$ . [C] verdict: $F_{\text{pole}}$ is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C]                                                                                                                                                                                   |
| v372_ftransfer_boltzmann.py   | Track 3 ( $F_{\text{transfer}}$ ): $\eta_B$ from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the <b>experiments/ftransfer</b> solve, replacing the v326 toy washout. Frozen $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ( $\sim 8.65\times 10^9$ GeV, v212), $\tilde{m}_1=m_3/A_\Lambda=5$ meV, $\delta_{CP}=4\pi/3$ . [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$ ) agrees with the BDP fit to $\sim 25\%$ . [C] $\eta_B(\text{ODE}) \sim 6.5\times 10^{-10}$ at the frozen $M_1$ lands within a factor $\sim 1.1$ of the observed $6.1\times 10^{-10}$ , NO $M_R$ dial. [X] kill: $\eta_B$ outside [obs/3, 3 obs] for all seesaw-consistent $M_R$ would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X]                                                                                                                                                                                    |
| v373_ftransfer_relic.py       | Track 3 ( $F_{\text{transfer}}$ ): the axion relic from the FINITE-T misalignment ODE – promotes the <b>experiments/ftransfer</b> solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. $f_a=M_{\text{scal}}/128 \sim 2.4\times 10^{11}$ GeV, $m_a \sim 23.8 \mu\text{eV}$ , lattice $\chi(T)$ $n=8.16$ . [E] candidate scales fixed by the determinant-line atoms ( $128=2 \dim S^+ \mu_4 $ ). [C] the frozen SPINE angle $3\pi/5=108^\circ=\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in $[0.08, 0.16]$ (lands on $\Omega_{\text{DM}}$ untuned) – the robust dominant-DM branch. [X] the HILLTOP $170^\circ$ over-produces ( $\sim 5.4\times$ ) – disfavoured. [X] kill: a finite-T solve robustly outside $[0.08, 0.16]$ for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X]                                                                                                                                                   |
| v374_ftransfer_qcd.py         | Track 3 ( $F_{\text{transfer}}$ ): the $m_p/m_e$ transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] $b_3 = -(11 - 2n_f/3) = -7$ ( $n_f=6$ carrier). [C] two declared externals (v339): $\alpha_s(M_Z)=0.1179\pm 0.0009$ and $C_p=2.83\pm 0.15$ , each $\sim 5.3\%$ . [C] $m_p/m_e$ central $\sim 1824$ (v262) with a propagated $\pm 7.5\%$ band [ $\sim 1690, \sim 1960$ ] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice $C_p\pm 0.03$ + tighter $\alpha_s$ shrink the band to $\sim \pm 2\%$ ; a band then excluding 1836.15 drops the QCD-matching route ( $b_3 = -7$ + the $SU(9)$ -integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X]                                                                                                                                                                                                          |
| v375_observatory_registry.py  | Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry ( <b>freeze_file.csv</b> ) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$ : $\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$ , $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$ , $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$ . [C] live scorecard: $\theta_{12}$ vs JUNO first run ( $0.3092\pm 0.0087$ ) = $0.28\sigma$ (compatible); $\theta_{13}$ vs NuFIT 6.0 NO ( $0.02195\pm 0.00058$ ) = $2.0\sigma$ (the flagged most-tensioned core prediction); $\beta_{\text{rad}}$ vs ACT DR6 ( $0.215\pm 0.074$ ) = $0.37\sigma$ (hint); $r=12/N_\star^2 \sim 0.004 < \text{BK18}$ (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C] |
| v376_seam_s3_centralcharge.py | S3 closure stack: the central charge $c=8$ extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives $c=1.0000$ (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), $c$ additive $\Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}$ ; [E] chiral $c_-=8$ (Chern $ C =1$ per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical $c=8$ from the lattice; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                          |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v377_seam_s3_e8character.py | S3 closure stack: the chiral $c=8$ content is the SPECIFIC net $(E_8)_1$ , via the exact affine character $q$ -series + the $\mu_4$ /GSO promotion. [E]<br>$\chi = E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots)$ : leading $-c/24 = -1/3 \Rightarrow c=8$ ; level-1 $= 248 = \dim E_8$ (one primary). [E] the $\mu_4$ /GSO promotion $SO(16)_1 \rightarrow (E_8)_1$ : $248=240+8$ , $240 = \dim S^+ g_{\text{car}} N_{\text{fam}} = 16 \cdot 5 \cdot 3$ ; $248=120+128 = \dim SO(16) + \text{spinor}$ . [E] discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. [O] continuum existence (v336); the $\mu_4$ projection = the seam one-sidedness (S3). Pins the target net; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v378_seam_s3_modular.py     | S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$ ), the “hard part” v344 flagged (genus-0 unique vacuum is necessary not sufficient). [E] KLM condensation:<br>$\mu(\text{carrier}) = \det C(D_5) \det C(A_3) = 4 \cdot 4 = 16$ ; the order-4 $\mu_4$ glue ( $ L =4$ )<br>$\Rightarrow \mu(E_8) = 16/16 = 1$ (torus GSD = 1); index-2 $\Rightarrow \mu(SO16) = 16/4 = 4$ . [E]<br>$ \det C(E_8)  = 1$ vs $ \det C(D_8)  = 4$ . [E] the single $(E_8)_1$ character has modular $T$ -eigenvalue $e^{-2\pi i/3}$ ( $h=0, c=8$ ), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. [O] continuum existence (v336). Pins the net at genus-1; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v379_seam_s3_rp.py          | C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. [E] the gapped $p+ip$ band gives $G(\tau) = \text{mean}_k e^{-\varepsilon_k \tau}$ with $\varepsilon_k \geq \text{gap}/2 \sim 1 > 0$ (v367) – a positive spectral measure; [E] the Hankel/reflection Gram matrix $M_{ij} = G(\tau_i + \tau_j)$ is PSD – OS reflection positivity holds; [E] neg control: a growing ghost mode makes it indefinite (non-vacuous). [C] RP + gap (v76/v337) + GNS (v240) + $c=8$ (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/QG.AMB.01 is then DISCHARGED by the redundancy theorem v369, not built. [O] the full constructive measure stays the cited step. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v380_grav_kms_hessian.py    | The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from “assume entire” to a derived statement). [E] the seam KMS cutoff $f(u) = e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u) = e^u$ entire & nowhere zero $\Rightarrow$ only the $p^2=0$ pole (no ghost). [E] each finite truncation carries the ghost: the $R+R^2$ order $T_1=1+u$ has the Stelle zero $u=-1$ , the Weyl <sup>2</sup> order $T_2=1+u+u^2/2$ the pair $-1 \pm i$ ; the entire $a$ has none. [E] the ghost DECOUPLES:  nearest zero  of $T_n$ grows monotonically with $n$ (1, 1.41, $\dots$ , 3.07 for $n=1..8$ ) $\rightarrow \infty$ , so “entire” = “do not truncate”. [C] $\Rightarrow$ perturbative graviton unitarity. [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                               |
| v381_qft4d_eg_allorder.py   | The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix $S_{\text{pert}}$ – the “all-order closing statement” for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does <i>not</i> build a path integral. [E] POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. [E] BRST NILPOTENCY $s^2=0$ verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ ( $10^{-16}$ ), the carrier weak+colour content. [E] the seam gap $\Delta = 6 \ln(3/2) > 0$ (v302) gives the IR-safe adiabatic limit $g \rightarrow 1$ . [C] imported: $T_n$ exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). [C] imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. [E] scope fence: matter+gauge ONLY – the $R^2/\text{Weyl}^2$ gravity sub-sector is v304/v370/v380, not a local EG vertex; $S_{\text{pert}}$ is not the ambient measure QG.AMB.01 (v369). NET [C] (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites [E]; extends v269/v271/v273/v278 from 1-loop to all-order [E][C] |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v382_alpha_quillen_exact.py    | Names the Quillen determinant-line VARIATION $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$ as a tracked target (review Point 10), elevating EM.WARD.01's residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3 - 2c_3^3 \alpha^2 = 8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1}=137.0359992$ ( $F_{U(1)}=0$ IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1=\frac{4}{5}\cdot 41$ a $U(1)$ index, $c_3=1/(8\pi)$ Gauss–Bonnet ( $c_3^{3,6}$ heat-kernel powers), $q(D_5)+q(A_3)=2$ discriminant forms (v341/v48/v216). [E] the value $\alpha^{-1}=137.0359992168$ stays closed (unique root + ablation, v3) – only “why this functional” is open. [O] THE TARGET: the from-first-principles AQFT/ $\zeta$ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of SEAM.EQUIV.01 (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing EM.WARD.01 residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O]                                                                                                                                                                                                                                         |
| v383_dynamics_universal.py     | DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from $F_{\text{transfer}}$ to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ has a simple Perron leading 1 and gap $6 \ln(3/2) > 0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine- $E_8$ adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2 \cos(\pi/5) = \varphi$ , gap $2 - \varphi = 1/\varphi^2 > 0$ . [E] DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2\cdot 3\cdot 5 = h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic $c=8$ RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), $F_{\text{transfer}}$ (v303), $\tau=i$ (v333). [C] META-THEOREM: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C] |
| v384_residual_certification.py | RESIDUAL.CERTIFICATION.01: the residual matrix is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): $c_3=1/(8\pi)$ and $g_{\text{car}}=5$ are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale $\Rightarrow v_{\text{geo}}$ is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\text{eff}} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ every readout is independent of the ambient measure (QG.AMB.01 a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: SEAM.EQUIV.01 (continuum existence = cited MMST/Adamo, v336) $\rightarrow$ A, ALPHA.QUILLEN.EXACT.01 (v382) $\rightarrow$ A, QG.AMB.01 (v369) $\rightarrow$ A, $v_{\text{geo}} \rightarrow$ B, $F_{\text{transfer}}$ (v371–v374, firewall v187) $\rightarrow$ C; ZERO open internal TFPT mechanisms. [C] CONVERGENCE: the only hard things left are theorems and external physics, not missing dynamics. An audit, no new number [E][C]                                                                                                                                                                                          |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v385_uvbranch_killtest.py  | UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake $\tau_p$ window). A consolidation (does <i>not</i> re-run the RGE v246 nor re-derive the PS scale v249/v253). [E] carrier content = SM ( $b = (41/10, -19/6, -7)$ , v159/v246, no new states) and $S_{\text{pert}}$ closes to all orders (v381) $\Rightarrow$ the SM-non-unification is all-order-robust. [X] plain-SM 4D-GUT killed ( $\sin^2 \theta_W = 3/8$ vs 0.231; v246). [C] optional carrier-PS branch: $M_{PS} = \kappa M_{\text{scal}}$ , $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}} \approx 3.06 \times 10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$ , no 126 (v247/v248); not the default (v265). [E] proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ( $B-L$ -conserving $q \leftrightarrow l$ , rare LFV $K_L \rightarrow \mu e$ ), <i>not</i> the $p \rightarrow e^+ \pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\text{bound}} \sim 10^7 \gg 1$ . [X] frozen kill tests (v265): $\kappa$ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake $\tau_p$ . A consolidation, no new number [E][C][X] |
| v386_grav_amplitude.py     | GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ , $a(u) = e^u$ , $M = M_{\text{scal}}$ . [E] one pole, ghost-free: $a$ entire & nowhere zero (v380) $\Rightarrow$ only pole at $p^2 = 0$ , residue $e^0 = 1 > 0$ . [E] UV-softened: ratio $e^{-u}$ at $u = 1, 5, 10, 20 = 0.37, 6.7 \times 10^{-3}, 4.5 \times 10^{-5}, 2.1 \times 10^{-9}$ (faster than any power). [E] low-energy = GR + a calculable $-p^2/M^2$ correction ( $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ , v253). [C] tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). [C] the amplitude $A = (\kappa^2/2)(T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; QG.AMB.01 a [C] redundancy v369). No new number [E][C]                                                                                                                                                                                                                                                                                                                                                                              |
| v387_corrections_gap.py    | CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] FLAVOR: $\lambda_2 = (2/3)^6 = 64/729 \Rightarrow$ first Koide correction $\sim 0.088$ (v56/v82). [E] RECOVERY: the same $(2/3)^6$ Petz rate (v221). [E] QG: capped by $\chi = 1/(1 - (2/3)^6) = 729/665$ (v337). [E] DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ( $\lambda_1 = 1$ ) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). [E] the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$ ) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$ ) – the two facets of v314/v383. [C] the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v388_corrections_budget.py | CORRECTIONS.BUDGET.01: the gap correction (v387) is a <i>typed</i> budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator’s leading eigenvector, so it is only defined where $\lambda_2$ exists. The prediction surface splits into four correction classes (already implied by v303). [E] SEAM-GAPPED: Koide ( $F_{\text{pole}}$ v303), recovery (v221), the QG bound (capped by $\chi = 729/665$ , v337) carry $(2/3)^6$ ; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). [E] KOIDE FIRST CORRECTION: $F_{\text{pole}}$ is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below $2/3$ – the rate is the suppression SCALE, not the residual. [E] EXACT-IDENTITY class ( $\det R = 8$ , $N_\Phi = 1$ , $\theta_{\text{eff}} = 0$ , $\sin^2 \theta_{13}$ ): no $\lambda_2 \Rightarrow$ correction 0 STRUCTURALLY (a band would contradict [E]). [E] EXTERNAL-RATE class ( $\eta_B/\text{axion}/m_p, m_e$ ): the gapped shape with external rates (only $F_{\text{pole}}$ has the seam rate, v303), firewalled (v187). [E] FIXED-POINT class: the texture is already explicit ( $\theta_{12} \varepsilon = c_3 + 36c_3^4$ ). [C] a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number [E][C]           |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v389_grav_loop_finiteness.py   | GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the <i>loop</i> level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. [E] super-polynomial falloff $p^k e^{-p^2/M^2} \rightarrow 0$ for every $k$ . [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$ ; the GR $\int p^3/p^2 dp$ diverges quadratically. [E] the superficial degree $\rightarrow -\infty$ at every loop order (dressed cutoff-ratio $\rightarrow 1$ vs GR $\rightarrow 4$ ). [E] the regulator is the SEAM scale $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C] loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number [E][C][O]                                                                                                                                                 |
| v390_prime2_facet.py           | COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$ , prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$ , prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i) - \tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. [E] $h(E_8) = 30$ factors into exactly three primes. [E] one quadratic facet field per prime, each ramified at its prime: disc $\mathbb{Q}(i) = -4$ , disc $\mathbb{Q}(\sqrt{-3}) = -3$ , disc $\mathbb{Q}(\sqrt{5}) = 5$ ; ramified primes $\{2, 3, 5\}$ , product = 30. [E] prime-2 Gaussian facet: min. poly $x^2 + 1$ , $\tau = i$ , $j(i) = 1728 = 12^3$ , $ \mu_4  = 4 = 2^2$ (the static seam-orientation facet). [E] prime-5 golden (v383): $2 \cos(\pi/5) = \varphi$ (pentagon). [E] prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein $\zeta_3$ (hexagonal CP $\pi/3$ ). [C] completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$ . A synthesis, no new number [E][C] |
| v391_alpha_quillen_progress.py | ALPHA.QUILLEN.PROGRESS.01: an <i>honest attempt</i> at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle $= (2 \sinh(\beta m/2))^2$ , $m$ -variation $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series ( $m^3$ coeff $-1/360$ verified). [E] a $\zeta$ -det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right <i>kind</i> of object. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$ . [E] reduction: the open step reduces to the named sub-target “seam Seeley–DeWitt coefficient $= 8b_1c_3^6$ ”. [O] NOT CLOSED: the actual seam $\zeta$ -det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external [O]; $\alpha^{-1}$ stays [E]. An attempt that reduces, does not close [E][O]                                                                                                   |
| v392_seam_s3_scalinglimit.py   | SEAM.S3.SCALINGLIMIT.01: an <i>honest attack</i> on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data <i>converge</i> . [E] the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66 \dots 514$ is a monotone Cauchy-like sequence approaching 1 ( $ c_1(L) - 1 $ strictly decreasing). [E] a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached <i>in</i> the limit. [E] the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). [O] convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it [E][O]                                                                                                                                                                                                                    |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v393_corrections_numeric.py  | CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. [E] FIXED-POINT band $(\theta_{12}/\theta_{13}/\lambda_C/\beta_{\text{rad}}) =$ the explicit $\varphi_0$ puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$ 's $\varepsilon$ first-correction). [E] SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$ ; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). [E] EXACT-IDENTITY band = 0 structurally ( $\det R = 8$ , $N_\Phi = 1$ , $\theta_{\text{eff}} = 0$ ). [E] EXTERNAL-RATE band = external physics quoted ( $m_p/m_e \pm 7.5\%$ v374; $n_s/r N_\star$ band; $\eta_B 1.07 \times \text{v212}$ ), fenced v187. [C] each prediction gets “central value $\pm$ computed first correction”. A harvest of v388, no new number [E][C]                                                                                        |
| v394_coxeter_atoms.py        | COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye [E] re-reading tying v390 (facets) to the anchor $a = (1, 1, 2)$ and v319 (5x6 clock); registers RES.COXETER.SYMMETRY.01 as a tracked [O]. [E] $a = (1, 1, 2)$ : $e_3=2= Z_2 $ , $p_0=3=N_{\text{fam}}$ , $e_2=5=g_{\text{car}}$ , product $30=h(E_8)$ . [E] atoms = facet fields (v390): $2 \rightarrow \mathbb{Q}(i)$ , $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ , $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] $\chi=729/665=3^6/(3^6-2^6)$ , $(2/3)^6=2^6/3^6$ , exponent $6=2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2: $2/3= Z_2 /N_{\text{fam}}$ , $(\varphi+2)/4=(\varphi+2)/ \mu_4 $ , $ \mu_4 =4=2^2$ . [E] $30=5 \times 6=g_{\text{car}}(2N_{\text{fam}})$ . [C] the seam is the universal dynamic engine. [O] RES.COXETER.SYMMETRY.01 registered, not closed [E][C][O]              |
| v395_hypergraph_coupled.py   | HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier $\times$ family). [E] LOCAL COUPLED MICRO-RULE: state on 9 affine- $E_8$ nodes $\times$ 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 $M$ on each node's family vector; steps commute, $T_{\text{micro}} = T_{\text{net}} \otimes M$ . [E] JOINT ATTRACTOR: top eigenvalue 1 with eigenvector $\text{marks} \otimes e_0$ ; iteration from any positive start converges. [E] ONE HAND = RECOVERY: one network step + $2N_{\text{fam}}$ family micro-steps gives $(2/3)^6$ on $\text{marks} \otimes v_{\text{surv}}$ . [E] v324 EMERGES: $T_{\text{full}} = T_{\text{net}} \otimes T_{\text{cusp}}$ shares the same invariants; $T_{\text{cusp}}$ is the 6-hand readout in the Galois/cusp basis (v317). [E] GROWTH + FIBER: v299 $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ unchanged with $M$ -fiber attached. [O] 3-arm seed $(P_2) + \varphi_0$ (v312) remain irreducible [E][O] |
| v396_phi0_icosahedral.py     | HYP.PHI0.01: the $\varphi_0$ leading term from icosahedral combinatorics (sharpening v312/v395). [E] ICO COUNTS ANCHOR-FORCED: $V=12$ , $E=30=h(E_8)$ , $F=20=g_{\text{car}}(g_{\text{car}}-1)$ , $ \text{Aut} =120=4h= 2I $ . [E] LEADING TERM EXACT: $\varphi_0^{\text{ree}}=1/(6\pi)=(4/3)c_3=F/(4h\pi)=F/( \text{Aut} \pi)=(F/(g_{\text{car}}N_{\text{fam}}))c_3$ . [E] $F/h= Z_2 /N_{\text{fam}}$ ; leading term $=(F/h)/(4\pi)=c_3(g_{\text{car}}-1)/N_{\text{fam}}$ . [E] GAUSS-BONNET: icosahedron deficit sum $4\pi$ ; $c_3=1/( Z_2 4\pi)$ (v216/v342). [E] HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. [C] tree-level seed = (hyperedges)/( $g_{\text{car}}N_{\text{fam}}$ ) $\times c_3$ , not an independent injection; puncture [O]                                                                                                                                                                                                                                                             |
| v397_external_clock_probe.py | FR.CLOCK.PROBE.01: a <i>falsifiable</i> probe of how external the “external physics” really is – do the frontier scheme-numbers land on the $\{2, 3, 5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): “on the clock” = an EXACT match, not a wide-band fit. [E] $\eta_B$ decuple $A_\Lambda=10= Z_2 g_{\text{car}}$ ON CLOCK (v212). [E] Koide rate $(2/3)^6=( Z_2 /N_{\text{fam}})^6$ ON CLOCK (v303). [E] axion spine angle $3\pi/5=\pi N_{\text{fam}}/g_{\text{car}}$ ON CLOCK (v211). [E] proton $C_p \sim 2.83 \pm 0.15$ NOT discriminating ( $2^{3/2}$ , $17/6$ , $14/5$ all fit the band) $\Rightarrow$ EXTERNAL (firewall v187/v374). [C] 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number $C_p$ – “external physics” was overstated. [E] anti-numerology: refuses to assign $C_p$ a clock value [E][C]                                                                                                                                             |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v398_seam_state_rigidity.py  | <b>Seam State Rigidity Theorem</b> (Paper A, the $G_{\text{net}}$ closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T=e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no $\mu_4$ normal form); [E] band-limited core $[\rho,  k ]=0$ and character-block-diagonal sub-principal symbol swept $N=4.64$ (beyond the 3-dim $H^1$ ); [E] holomorphy $\Rightarrow (E_8)_1$ ( $c=8$ , $ \det \text{Cartan} =1$ vs 4, index $4= \mu_4 $ , gap $6 \log \frac{3}{2} > 0$ ). [O] the one residual: $[\rho, \Lambda_\Sigma]=0$ on the full $L^2$ (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336)                                                                                                                                                   |
| v399_gravity_complete.py     | <b>Gravity-complete = Boundary-complete + Ambient-redundancy</b> (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab} + \Lambda g_{ab} = (1/c_3)T_{ab}$ , $1/c_3=8\pi$ (four $c_3$ -roles), $\Lambda$ prefactor $3/(4\pi^2)$ ; [E] redundancy discriminators ( $ \det \text{Cartan} =1$ vs 4, torus GSD=1, Petz $(2/3)^6$ , margin $1.648 > 0$ ); [E] $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ over five readout classes (Newton $8\pi$ , $\Lambda$ , scalaron $c_3^7$ , horizon $1/ \mu_4 $ , recovery $(2/3)^6$ ) – all boundary/seam/gap. [C] the verdict; [O] SEAM.EQUIV.01 + intrinsic BW (no new gate)                                                                                                                                                                                                                                                       |
| v400_grav_nonlocal_action.py | <b>Nonlocal spectral-gravity action</b> with the local $R+R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a=e^u$ gives the dressed propagator $1/(p^2 a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a=1+u+u^2/2+\dots$ reproduces the spectral-action $R+R^2$ (v28/v36) with the atom-fixed scale $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ ( $7=g_{\text{car}}+N_{\text{fam}}-1$ , v253). [O] the entire- $a$ analyticity assumption + perturbative-only (not the ambient measure QG.AMB.01, a [C] redundancy v369)                                                                                                                                                                                                                                                              |
| v401_metrology_closure.py    | <b>Metrology closure</b> of the final input set $\{a, \pi, v_{\text{geo}}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a=(1, 1, 2) \rightarrow (e_1, e_2, e_3)=(4, 5, 2)=( \mu_4 , g_{\text{car}},  Z_2 )$ , $c_3=1/(2e_1\pi)=1/(8\pi)$ , 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/ $1G/\Lambda/U_{\text{point}}/(m/\mu)$ = number $\times v_{\text{geo}}^{1,2,4,1,0}$ ; [E] gravity adds no unit (v364); [E] $G$ -vs- $\Lambda$ over-determination 0.11% (v274). [O] $v_{\text{geo}}$ theorem-forbidden ( $\text{TOE}_{\text{strict}}=\text{TOE}_{\text{dimensionless}}+[v_{\text{geo}}]$ , v384 type B)                                                                                                                                                                                                                                                                                               |
| v402_ftransfer_suite.py      | $F_{\text{transfer}}$ <b>functor suite</b> (Paper E): the four frontier sectors as ONE functor $F_{\text{transfer}}=F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ (consolidating FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE). [E] the composition law (each observable factors source $\rightarrow$ threshold $\rightarrow$ RG $\rightarrow$ observable); [E] exact atom sources ( $Q_{\text{src}}$ v93; $A_\Lambda=2g_{\text{car}}=10$ , $\delta_{CP}=4\pi/3$ v9/v212; $f_a=M_{\text{scal}}/128$ v185; $b_3=-7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187)                                                                                                                                                                                                                                      |
| v403_facet_compositum.py     | ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ , a degree-8=rank $E_8$ abelian field – the arithmetic hull dual to the $E_8$ lattice hull. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$ , Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))} = 2^3 = 8 = \text{rank } E_8$ is $E_8$ -special (the $E_7$ control $h=18$ , $2^2=4 \neq 7$ fails). [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$ , split 4 imaginary = $ \mu_4 $ (CM/dynamic) + 3 real = $N_{\text{fam}}$ (RM/static). [E] ramified only at $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8)=30$ . [C] $K$ is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to $E_8$ ; the 7 quadratic subfields a candidate for the 7 $E_8$ audit slices [E][C] |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v404_e8_char_cm_duality.py | E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}=E_4/\eta^8=j^{1/3}$ at the two CM points – 12 at the seam ( $\tau=i$ ), 0 at the family ( $\tau=\omega$ ), extending v282. [E] $\chi_{E_8}^3=j$ (q-series). [E] SEAM $\tau=i$ ( $\mathbb{Q}(i)$ , order 4): cross-ratio $2 \Rightarrow j=1728 \Rightarrow \chi_{E_8}(i)=1728^{1/3}=12$ ( $E_8$ present, v214/v282). [E] FAMILY $\tau=\omega$ ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic $\lambda$ ( $\lambda^2-\lambda+1=0$ ) $\Rightarrow j=0 \Rightarrow \chi_{E_8}(\omega)=0$ ( $E_4$ vanishes, v220/v267). [E] $\tau=i, \tau=\omega$ are the ONLY elliptic points of $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where $E_8$ is present ( $\chi=12$ , SEAM.EQUIV.01 closes), the flavor/CP sector is the soft residual where $E_8$ degenerates ( $\chi=0$ ); the dual-keystone candidate (flavor = the $j=0$ degeneration of $(E_8)_1$ ) [E][C]                                                                                                                                                                                                                                          |
| v405_seam_equiv_omega.py   | SEAM.EQUIV.02 – the dual keystone at $\tau=\omega$ : the family/flavor sector as the order-3 Eisenstein/ $A_2$ face of $E_8$ , dual to the seam = $(E_8)_1$ at $\tau=i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega)=0$ ( $E_8$ degenerates at the family CM point) vs $\chi_{E_8}(i)=12$ . [E] $A_2$ is the $\tau=\omega$ Eisenstein lattice (Cartan $\det=3=N_{\text{fam}}$ , $h=3=N_{\text{fam}}$ , $\mu_6=\mu_3 \times \mu_2$ ). [E] $R$ lives in $E_6 \times A_2$ : $248=\ R\ _F^2 + \det R + 2(1^\top R a)N_{\text{fam}}=78+8+2 \cdot 27 \cdot 3$ , $\ R\ _F^2=78=\dim E_6$ , $\det R=8=\dim A_2=\text{rank } E_8$ . [E] $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase ( $\arg \zeta_6=\pi/3$ , v220/v233). [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck = $N_{\text{fam}} = \det Q$ (v72), $\text{Spec}(Q_+)=(1, 2, 3)$ (v69/v137). [C] the family/flavor sector is the order-3 Eisenstein/ $A_2$ deck at $\tau=\omega$ ; the two keystones sit at the only two elliptic points of $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the $v_{\text{geo}}$ -class scale (No-Unit v153) [E][C][O]        |
| v406_pmns_phase_origin.py  | FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau=\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\text{geo}}$ -class unit, so PMNS has zero free flavor dials beyond $v_{\text{geo}}$ . [E] the three angles ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$ ; v9/v268) assemble to one unitary $U_{\text{PMNS}}$ (v270). [C] CP-PHASE ORIGIN (advance over v270): $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's [O] assembled input to a keystone consequence. [C] $J_{\text{PMNS}}=-0.0297$ derived. [E] the absolute scale = one seesaw ratio $m_3=(y_\nu v_{\text{EW}})^2/M_R$ = a $v_{\text{geo}}$ -class UV input (v272/v153), NOT a flavor gap. [C] net: PMNS = angles + $\delta$ [keystone] + $J$ [derived] + scale [ $v_{\text{geo}}$ ], zero free flavor dials. [O] residual = the seesaw operator realisation ( $M_R$ not a clean $\bar{M}_{\text{Pl}}$ power, v272/v184; one-parameter carrier-normalised candidate v481) + the normal-ordering $\Sigma m_\nu$ floor kill test [E][C][O] |
| v407_dn_pairings_omega.py  | FLAV.SELECTOR.CLOSE.01: the $R4'$ residue folds into the $\tau=\omega$ keystone (v405) – the three $(d, n)$ selector pairings ARE the $\tau=\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ is a $\mathbb{Z}^3$ basis with $\det=11=\ Pl(K)\ _1$ , and $n=(5, -9, 6)$ is the unique covector with $n \cdot \mathbf{1}=2$ , $n \cdot a=8$ , $n \cdot \sigma=121=11^2 \Rightarrow (d, n) \Rightarrow R$ columnwise (v136/v139). [E] $n \cdot a=8=\text{rank } E_8=\dim A_2=\det R$ – the anchor-pairing is the $A_2$ ( $\tau=\omega$ ) block dimension that reads $R$ in $E_6 \times A_2$ (v405). [E] the pairings $\{2, 8, 121\}=\{ Z_2 , \text{rank } E_8=\dim A_2=\det R, \ Pl(K)\ _1^2\}$ are the $\tau=\omega$ family-slice atoms (the $\mu_2$ sheet, the $A_2$ block, the Plucker frame). [E] the square selects the point ( $\sigma$ -pairing $11(11+t)$ , a perfect square only at $t=0$ ). [C] deriving $(d, n)$ ( $R4'$ ) folds into SEAM.EQUIV.02 – $d=\frac{3}{2}a-21$ anchor-derived (v139), the pairings the slice atoms. [O] the lone flagged residue “why $11=\ Pl(K)\ _1$ ” [E][C][O]                                                                                                                |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v408_phi0_puncture_heatkernel.py      | HYP.PHI0.PUNCTURE.01: name the $\varphi_0$ puncture term $\delta_{\text{top}}=48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the $\mu_4$ puncture divisor – the analytic complement of the icosahedral tree term (v396, HYP.PHI0.01). [E] THE SPLIT IS EXACT: $\varphi_0=\varphi_0^{\text{tree}}+\delta_{\text{top}}=1/(6\pi)+48c_3^4$ (v396/ARCH.QUAD.01). [E] PUNCTURE COUNT = $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$ , order = $ \mu_4 =4$ : $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^{ \mu_4 }$ . [E] CLOSED FORM $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ ; $c_3=1/( Z_2 4\pi)$ Gauss-Bonnet (v216/v342) so $c_3^4$ is a fourth-order boundary heat-kernel power (cf. v391). [E] HONEST NEGATIVE: $\delta_{\text{top}}$ not a rational hypergraph fraction ( $\delta_{\text{top}}/\varphi_0^{\text{tree}} < 1\%$ ). [O] THE TARGET: prove $\Omega_{\text{adm}}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the $\mu_4$ puncture summed over the $\Omega_{\text{adm}}$ admissible states. [C] two engines, two terms; a FACE of the seam boundary analysis (the $c_3$ datum), not a new gate [E][C][O] |
| v409_coxeter_prime2_lemma.py          | RES.COXETER.SYMMETRY.01 closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/COXETER.ATOMS.01 open question's falsifiable corollary). [E] SHEET INVOLUTION $\sigma=\text{diag}(1, -1, -1)$ , $\sigma^2=I$ , spectrum $\{+1, -1\}$ (parity). [E] SHEET PROJECTOR $P=(I+\sigma)/2=\text{diag}(1, 0, 0)$ , $P^2=P$ , spectrum $\{0, 1\}$ (projection). [E] NO CONTRACTION RATE: none of $\{-1, 0, 1\}$ lies in $(0, 1)$ , whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. [E] FACTOR NOT RATE: $2/3= Z_2 /N_{\text{fam}}$ needs prime-3, $(\varphi+2)/4=(\varphi+2)/ \mu_4 $ with $ \mu_4 =2^2$ needs the golden $\varphi$ (prime-5). [E] $30=g_{\text{car}}(2N_{\text{fam}})$ , carrier static, 2·3 dynamic. [C] prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ( $\mu_4=2^2$ the Galois bridge) [E][C]                                                                                            |
| v410_sheet_generator_binary.py        | The <i>sheet generator</i> $V=Q \text{diag}(0, 1, 1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n=\begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$ (induction), so $V^n \mathbf{1}=(2^{n-1}, 2^n, 2^{n+1}-1)=(1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; bilinears $\mathbf{1}^T V^n \mathbf{1}=7 \cdot 2^{n-1}-1$ , $\mathbf{1}^T V^n a=7 \cdot 2^{n-1}$ , $a^T V^n a=11 \cdot 2^{n-1}$ , $a^T V^n \mathbf{1}=11 \cdot 2^{n-1}-2$ give $(6, 7, 9, 11)$ then $13=\Delta_Q$ , $27=\mathbf{1}^T R a$ , $55, 56=\dim \mathbf{56}_{E_7}$ ; the spine is FORCED by $\text{Spec}(V)=\{0, 1, 2\}$ , the carrier (rank $E_8=8$ , $\dim S^+=16$ , $248/8=31$ ) and theta $(\sigma_3(2)=9=a^T V \mathbf{1}, \sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R), \sigma_3(5)=126)$ readings audit-typed [E][C]                                                                                                                                                                                                                                                                                                                     |
| v411_ud_ratio_vpower.py               | The quark $u/d$ ratio is a pure $V$ -power readout: the established $c_u/c_d=g_{\text{car}}\ \text{Pl}(K)\ _1/(N_{\text{fam}}^2 \Delta_Q)=55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d = \frac{\mathbf{1}^T V^4 \mathbf{1}}{(a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})}} = \frac{55}{9 \cdot 13}$ , with $55=\mathbf{1}^T V^4 \mathbf{1}$ , $9=a^T V \mathbf{1}=N_{\text{fam}}^2$ , $13=\mathbf{1}^T V^2 \mathbf{1}=\Delta_Q$ ; the same number as $5 \cdot 11/(9 \cdot 13)$ , so the numerator repackages $g_{\text{car}} \cdot 11=55$ . An exact <i>re-encoding</i> : the value and the $u/d$ identification stay [C] (U <sub>wall</sub> readout rigidity) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v412_sheet_source_corner_J.py         | The <i>sheet source corner</i> $J$ : the determinant surface has two $s$ -independent iso-volume walls (v135) $t=-1$ ( $\det=4= \mu_4 $ , carries $K$ ) and $t=-2$ ( $\det=2= Z_2 $ ); the operator on the $Z_2$ wall, $J=M(1, -2)=C-2V=\begin{pmatrix} 4 & 1 & 0 \\ 5 & 1 & 1 \end{pmatrix}$ , completes the sheet axis $J, K, C, F=M(1, -2..1)$ . $\chi_J=\lambda^3-6\lambda^2+3\lambda-2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)=( R^+(A_3) , N_{\text{fam}},  Z_2 )$ ; $a^T J a=30=h(E_8)$ , $\det(I+J)=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$ , $\det(2I+J)=40= R(D_5) $ (Lie readings audit-typed like the $G_2/F_4$ shadows of v95/v218) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v413_sheet_characteristic_calculus.py | The sheet axis is a discrete <i>characteristic-difference calculator</i> : on $M_t=C+tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1=3t+12$ , $e_2=2t^2+15t+25$ , $e_3=\det M_t=4t^2+14t+14$ give $\Delta e_1=3=N_{\text{fam}}$ , $\Delta^2 e_2=4= \mu_4 $ , $\Delta^2 e_3=8=\text{rank } E_8$ (the $e_3$ curvature cross-cited to v218/v224; the full $(e_1, e_2, e_3)$ ladder is new); the anchor energy $a^T M_t a=52+11t$ (step $11=a^T V a$ , v410) reads $30 \rightarrow 41 \rightarrow 52 \rightarrow 63=(h(E_8), 10b_1, \dim F_4, 7N_{\text{fam}}^2)$ [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v414_center_resolvent_portal.py | The center $C$ is a <i>resolvent portal</i> $G_2 \rightarrow F_4 \rightarrow E_8$ : $C=M(1,0)$ (v95) is the $t=0$ value of $V$ ; $\text{tr } C=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$ , $\det C=14=\dim G_2$ , $\sum C=31=2^{g_{\text{car}}}-1=248/8$ ; the shifted determinants climb $\det(C)=14=\dim G_2$ , $\det(I+C)=52=\dim F_4$ , $\det(2I+C)=120= R^+(E_8) $ – the shell ladder $G_2 \rightarrow F_4 \rightarrow E_8^+$ as $\det(C+kI)$ , $k=0,1,2$ (the resolvent ladder is new; reading audit-typed) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v415_gaussian_operator.py       | The square CM-norm dictionary is <i>operator-realised</i> and the $\mu_4$ deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J=[U,V]/3$ , the complex structure born from the commutator of the binary $V$ and ternary $U$ compilers (v410/v95). [E] $J=\begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$ is integer, $\text{Spec } J=\{i, -i, 0\}$ , $J^2=-I$ on the rank-2 image, $J^3+J=0$ (a $\mu_4$ quarter-turn); [E] $3I+2J$ has spectrum $\{3+2i, 3-2i, 3\}$ and $5I+4J$ has $\{5+4i, 5-4i, 5\}$ , so $3+2i$ ( $ \cdot ^2=13=\Delta_Q$ ) and $5+4i$ ( $ \cdot ^2=41=10b_1$ ) are literal operator eigenvalues; [E] $(3I+2J)(3I-2J)$ has $\text{Spec}\{13, 13, 9\}$ and $(5I+4J)(5I-4J)$ has $\{41, 41, 25\}$ (kernel slot = real part <sup>2</sup> : $9=N_{\text{fam}}^2$ , $25=g_{\text{car}}^2$ ); [E] the intrinsic deck $D=-I+J-J^2$ is order 4 ( $\text{Spec}\{i, -1, -i\}$ , $\chi=(x+1)(x^2+1)$ ) with $\chi_2$ line $\ker[U,V]=(0,0,1)=a-1$ . [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue [E][C]                                                                                                                                                                               |
| v416_atom_trichotomy.py         | The atom trichotomy: $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K=\mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403). [E] $\mathbb{Z}[i]$ (square/seam, $\tau=i$ ): $2= Z_2 $ ramifies ( $2=-i(1+i)^2$ ), $3=N_{\text{fam}}$ inert (norm $9=N_{\text{fam}}^2$ ), $5=g_{\text{car}}$ splits $((2+i)(2-i))$ – the carrier rank factors on the seam; [E] $\mathbb{Z}[\omega]$ (hex/flavor, $\tau=\omega$ ): $3=N_{\text{fam}}$ ramifies ( $N(1-\omega)=3$ ), 2 inert (norm 4), 5 inert (norm $25=g_{\text{car}}^2$ ); [E] the ramified prime of each ring IS its atom ( $\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); [E] the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4, -3, 5$ ) ramify over one atom each, product $2\cdot3\cdot5=30=h(E_8)$ ; 2 imaginary (CM/dynamic) + 1 real (RM/static). [C] the seam/flavor/golden reading [E][C]                                                                                                                                                                                                                                                                                                                            |
| v417_eisenstein_cp_operator.py  | The Eisenstein/CP operator (the hex dual of v415): the family rotation $P=(1\ 2\ 3)$ and the CP clock $W=-P^2$ put the $\tau=\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. [E] $P^3=I$ , $P^2+P+I=\text{ONES}$ (Eisenstein $\omega^2+\omega+1=0$ on the plane $\perp (1,1,1)$ ); [E] $(3I+2P)(3I+2P^2)=7I+6\text{ONES}$ , $\text{Spec}\{7, 7, 25\}$ ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ = scalaron on the $\omega$ -plane, $25=g_{\text{car}}^2$ ) – the dual of v415's $\{13, 13, 9\}$ ; [E] the CP clock $W=-P^2$ ( $=\rho$ , v233) has $W^2=P$ , $W^3=-I$ , $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , so $\delta_{\text{CKM,lead}}=\arg \zeta_6=\pi/3$ and $\delta_{\text{PMNS}}=\arg \zeta_6^4=4\pi/3$ ( $W^4=P^2$ ); [E] all flavor clocks are powers of $\zeta_{30}$ ( $\zeta_{30}^{15}=-1$ , $\zeta_{30}^{10}=\omega$ , $\zeta_{30}^6=\zeta_5$ , $\zeta_{30}^5=\zeta_6$ ) while the seam $\mu_4=i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)=8=\text{rank } E_8$ . [C] the carrier $\mu_5$ has no clean $3\times3$ operator (golden $\varphi$ an output-side trace, v313/v349); honest negatives included [E][C]                                                                                                                                           |
| v418_cyclotomic_norm_triple.py  | The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7, 13, 55)=(\text{scalaron}, \Delta_Q, \text{quark numerator})$ , and the missing carrier-5 clock of v417 is the $4\times4$ $\Phi_5$ -companion $C_5$ . [E] $C_5$ has $\chi=x^4+x^3+x^2+x+1$ , $C_5^5=I$ , $\text{tr}=-1$ ; the golden ratio is its real-part data ( $\zeta_5+\zeta_5^4=1/\varphi$ , $\zeta_5^2+\zeta_5^3=-\varphi$ ; output-side v313/v349; $\dim 4$ , not 3 – refines v417's gap); [E] $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5\cdot11=g_{\text{car}}$ $\  \text{Pl}(K) \ _1=$ the quark numerator (v410/v411); [E] the triple $\det(3I+2\text{Comp}(\Phi_n))=(7, 13, 55)$ for $n=3, 4, 5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor $(5,4)\rightarrow(21, 41, 461)$ , with $21=N_\omega(5,4)$ and $41=10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2)\rightarrow55$ from $(5,4)\rightarrow461$ . [C] HONEST: $(3,2)$ is FORCED (v14) not unique – the clean splits are $\{(2,1)\rightarrow(3,5,11), (3,2)\rightarrow(7,13,55)\}$ , a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed [E][C] |

continued on next page

| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v419_seam_galois_carrier.py   | <p>The seam <math>\mu_4</math> is the carrier's Galois group: a positive resolution of the <b>v409/RES.COXETER.SYMMETRY.01</b> cyclic/Galois asymmetry. [E] <math>h(E_8)=30=2\cdot3\cdot5</math> is squarefree, so <math>\mathbb{Z}/30</math> has <i>no</i> order-4 element — the seam <math>\mu_4</math> cannot be a cyclic hand and is forced onto the Galois side; [E]</p> <p><math>(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2</math> (order <math>\varphi(30)=8=\text{rank } E_8</math>), where the <math>\mu_4</math> <math>(\mathbb{Z}/4)</math> factor IS <math>(\mathbb{Z}/5)^\times</math> (carrier 5) and the <math>\mathbb{Z}_2</math> IS <math>(\mathbb{Z}/3)^\times</math> (family/CP); [E] so the seam <math>\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5)) = (\mathbb{Z}/5)^\times</math>, the Frobenius <math>\zeta_5 \mapsto \zeta_5^2</math>, realised on the carrier clock <math>C_5</math> (<b>v418</b>) by the explicit order-4 operator <math>G</math> with <math>GC_5G^{-1}=C_5^2</math>, <math>G^4=I</math>, <math>G^2C_5G^{-2}=C_5^{-1}</math>. [C] resolves <b>v409</b> (the <math>2^2=4</math> is the order of <math>(\mathbb{Z}/5)^\times</math>; “no contraction rate” because the seam is an automorphism, not a hand); <math>\mathbb{Q}(\zeta_{30})</math> carries both <math>E_8</math> invariants (cyclic <math>h=30</math>, Galois rank=8). Figure <b>coxeter_galois</b> [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v420_fpole_qed_running.py     | <p>F_pole external: the Koide source→pole transfer is a <i>one-loop electromagnetic</i> effect, not an algebraic step — the honest external-physics follow-up to <b>v93</b> and the clock No-Go. [E] Koide <math>Q</math> is rescaling-invariant (<math>Q(cm)=Q(m)</math>), so flavor-<i>universal</i> running leaves it fixed and the source→pole shift is purely flavor-<i>split</i>; [C] the tree→pole gap <math>\frac{2}{3}-Q_{\text{src}}=0.00220</math> is one-loop-EM-sized (<math>=\varphi_0^{\text{ret}}/24</math> to 0.6%, <math>=\alpha/\pi</math> to <math>\sim 5\%</math>), the PDG pole sits AT <math>\frac{2}{3}</math> (the IR attractor); [C] a 1-loop QED per-lepton-log pole <math>\leftrightarrow \overline{\text{MS}}</math> estimate shifts Koide by EM-loop order toward <math>\frac{2}{3}</math> (the <b>v82</b> IR-attractor direction); [O] the EXACT flow time <math>t\sim 2.84</math> (<b>v93</b>) is scheme/scale-dependent external QED, only sized. [C] F_pole stays [C], firewalled (<b>v187</b>): the lens forces the RATE <math>(2/3)^6</math>, this external EM supplies the flow. Numerical (mpmath), Python-only [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v422_galois_net_bridge.py     | <p>The Galois<math>\leftrightarrow</math>Net bridge: the seam <math>\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))</math> is the SAME cyclic <math>\mathbb{Z}/4</math> as the <math>(E_8)_1</math> simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of <b>v419</b> to <b>G<sub>net</sub>/SEAM.EQUIV.01</b>. [E] <math>\text{disc}(A_3)=\text{disc}(D_5)=\mathbb{Z}/4</math> (one Smith invariant factor 4 = cyclic, <math>\det=4</math>); <math>D_n</math> disc is <math>\mathbb{Z}/4</math> for <math>n</math> ODD, <math>\mathbb{Z}_2 \times \mathbb{Z}_2</math> for <math>n</math> EVEN, so the carrier <math>D_5</math> (rank <math>5=g_{\text{car}}</math>, odd) is cyclic while the control <math>D_8</math> (rank 8, even) is Klein; [E] <math>\text{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times=\langle 2 \rangle</math> is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in <math>\text{disc}(D_5 \oplus A_3)=\mathbb{Z}_4 \times \mathbb{Z}_4</math> (<math>16=\dim S^+</math>) the simple current <math>(1,1)</math> is order 4 (cyclic) with Lagrangian quotient <math>16/4^2=1 \Rightarrow (E_8)_1</math> (holomorphic, <b>v89/v154</b>). [E] NEG control: the Klein <math>\mathbb{Z}_2 \times \mathbb{Z}_2</math> / order-2 halfway <math>\langle (2,2) \rangle</math> gives <math>16/2^2=4= \text{disc}(D_8) =SO(16)</math> (<math>\det=4</math>), NOT <math>E_8</math> (<math>\det=1</math>) — cyclic <math>\mathbb{Z}/4 \rightarrow E_8</math>, Klein <math>\rightarrow D_8</math>. [C] the bridge: one cyclic <math>\mathbb{Z}/4</math> threads the seam deck <math>\mu_4</math> (<b>v419</b>), <math>\text{Gal}(\mathbb{Q}(\zeta_5))</math> and the <math>(E_8)_1</math> glue (<b>v1/v89/v154</b>); cyclicity forced by 5, Klein excluded. [C] this is the four-fold <math>\mu_4</math> identity (deck/divisor = Galois = discriminant = simple-current); per <b>v428</b> a <i>compression</i> (one object read four ways), not four independent witnesses [E][C]</p> |
| v423_no_ambient_dependency.py | <p>The <i>No-Ambient-Dependency</i> certificate: no frozen TFPT readout is <i>computed</i> from the ambient QG measure <b>QG.AMB.01</b> — sharpening the <b>v369/v384</b> prose redundancy into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin <math>\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 &gt; 0</math> (<math>31=2^{g_{\text{car}}}-1</math>), susceptibility <math>\chi = 729/665</math> finite (neg control gap <math>\rightarrow 0 \Rightarrow \chi \rightarrow \infty</math>). [E] <b>QG.AMB.01</b> is a certification <i>sink</i>: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. [E] it is not a value source — it has no producing script while each frozen readout is computed by a compiler script (<math>\alpha^{-1} \leftarrow \mathbf{v3}</math>, <math>\theta_{12} \leftarrow \mathbf{v9}</math>, <math>\theta_{13} \leftarrow \mathbf{v268}</math>, cosmology <math>\leftarrow \mathbf{v7}</math>). [E] the one transitive link is architectural: only <math>\theta_{13}</math> references <b>QG.AMB.01</b>, via the 4D-QFT contract spine (<b>PS.DIRAC</b>→<b>CONTRACT.QFT4D</b>→<b>QG.G2</b>→<b>GATE.METRIC</b>), a coherence edge not a numerical input; <math>\alpha^{-1}/\theta_{12}</math>/cosmology have no route. [C] so <b>QG.AMB.01</b> is gap-decoupled certification (<b>v369</b> as a DAG/script fact); honestly this does NOT prove <b>SEAM.EQUIV.01</b> — the readouts depend on the seam being <math>(E_8)_1</math>; <b>QG.AMB.01</b> specifically is dead [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v424_seam_lto_rp_reduction.py  | The ( <i>LTO-RP</i> ) reduction of the keystone: an honest MMST-style reduction of SEAM.EQUIV.01's one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijken–Penneys–Wallick (arXiv:2605.10693), with the central mapping <i>corrected</i> . [E] the $\mu_4$ clock is a modular <i>symmetry</i> ( $[\rho, K]=0$ , v309/v199), not the flow — $\sigma^\psi=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K \theta = -K$ , <i>distinct</i> from $\omega \circ \rho = \omega$ : a positive character-block-diagonal $K$ satisfies $[\rho, K]=0$ yet never $\theta K \theta = -K$ , so clock-invariance does <i>not</i> imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for <i>topologically ordered</i> models (Toric Code $ \det K =4$ via a trace; Levin–Wen anyonic), while the seam is invertible ( $ \det \text{Cartan}(E_8) =1$ , no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket (neg control $D_8$ $ \det =4$ ). [C] so SEAM.EQUIV.01's BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; SEAM.EQUIV.01 stays [O] [E][C] |
| v425_dyn_transfer_universal.py | DYN.TRANSFER.UNIVERSAL.01 — the <i>single-flow reduction</i> of $F_{\text{transfer}}$ : the four frontier transfers ( $F_{\text{pole}}$ , $F_{\text{Boltzmann}}$ , $F_{\text{relic}}$ , $F_{\text{QCD}}$ ) do not each need a separate external solver; their <i>dynamics</i> is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6 \ln(3/2)$ ) restricted to each sector. [E] recovery spectrum $\{1, (2/3)^6, (1/3)^6\}$ , $H_{OS} = -\log T$ gap $6 \ln(3/2)$ , $e^{-\Delta} = (2/3)^6$ exact; an explicit column-stochastic $T$ makes $\log T$ a Markov generator. [E] $F_{\text{pole}}$ is this flow: the v99 generator $\dot{q} = (\Delta/N_{\text{fam}})(q-2)(q-5)$ linearised at $q=2$ has rate $-\Delta$ , time-1 contraction $(2/3)^6$ . [E] $F_{\text{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). [E] $F_{\text{QCD}}$ rate native ( $b_3=-7$ ). [C] only the anchors ( $v_{\text{geo}}$ , $C_p$ , $M_R$ ) external; $F_{\text{relic}}$ 's cosmological IC $\theta_i$ the one wall. Extends v383 static→dynamic; a reduction, no new premise [E][C]                                                                                                                                                                                                                                                                                                                                                       |
| v426_seam_lto_rp_kms.py        | SEAM.EQUIV.LTORP.KMS.01 — the <i>invertible</i> $\beta=1$ -KMS variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + $\beta=1$ KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does <i>not</i> close SEAM.EQUIV.01. [E] build $K$ reflection-odd/ $\mu_4$ -even/gapped (gap $2k=6 \ln(3/2)$ ); $C=1/(1+e^K)$ the $\beta=1$ KMS covariance, $C/(1-C)=e^{-K}$ (genuine KMS, not trace). [E] $\Theta K \Theta^{-1} = -K$ ( $u_\Theta=J$ ) AND $[\rho, K]=0$ both hold $\Rightarrow$ (LTO-RP) satisfied by direct Tomita–Takesaki. [E] $ \det \text{Cartan}(E_8) =1$ (invertible, no anyons) vs anyonic $ \det =4$ ; state not a trace ( $\Delta=e^{-K} \neq I$ ) — neither NPW26 bucket. [E] neg controls: side-even reflection $\rightarrow +K$ (BW fails), off- $\mu_4$ -block $K$ breaks $[\rho, K]$ . [C] (LTO-RP) reduces to $(\Theta K \Theta = -K) \wedge ([\rho, K]=0)$ , constructively satisfiable; SEAM.EQUIV.01 stays [O] [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v427_overdet_witness_map.py    | OVERDET.WITNESS.MAP.01 — the multiply-vs-compress <i>framework</i> for over-determination: witnesses from <i>disjoint</i> grammars multiply, projections of the <i>same</i> generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). [E] seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3+2i)=13$ , Eisenstein $N(3+2\omega)=7$ , cyclotomy $N(3+2\zeta_5)=55$ , Galois $ (\mathbb{Z}/5)^\times =4$ , $ \det \text{Cartan}(E_8) =1$ , Pascal $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$ ( $2^4=16 = \dim S^+$ ), Coxeter $h(E_8)=30=2 \cdot 3 \cdot 5$ with $\varphi(30)=8 = \text{rank } E_8$ . [E] by v236 these are facets of ONE $(2, 3, 5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor $a=(1, 1, 2)$ . [C] the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v428_overdet_reclass.py          | OVERDET.WITNESS.RECLASS.01 — the honest correction of v427: applying its own multiply-vs-compress test to itself. [E] the $(2, 3, 5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5=h(E_8)$ , $\varphi(30)=8$ . [E] every one of v427's seven witnesses maps to a facet of that SAME object (primes 2, 3, 5 of 30, the clock 30, or $E_8$ ), so all seven collapse to ONE generator (distinct objects = 1) $\Rightarrow$ COMPRESSION, not seven multiplications. [E] what honestly multiplies is the input forced four independent ways — the “8” in $c_3=1/(8\pi)$ from rank $E_8$ , $h(D_5)=2(5-1)$ , $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$ (prime, not in the $(2, 3, 5)$ -skeleton). [C] refines and partly walks back v427; consistent with v236/v305/v313 [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v429_axion_pentagon_phi.py       | DM.AXION.PENTAGON.01 — the ‘unmapped’ icosahedral/golden $E_8$ structure surfaces in the ONE external cosmological input $\theta_i$ , answering “why does idle $E_8$ structure exist?” for the one case where it has a job. [E] because $N_{\text{fam}}=g_{\text{car}}-2$ , the spine quotient $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=(g_{\text{car}}-2)\pi/g_{\text{car}}$ is EXACTLY the interior angle of the regular $g_{\text{car}}$ -gon; for $g_{\text{car}}=5$ the PENTAGON, $\theta_i=3\pi/5=108^\circ$ (the carrier polygon, not a bare rational multiple of $\pi$ ). [E] its cosine is golden: $\cos(3\pi/5)=(1-\sqrt{5})/4=-1/(2\varphi)$ , partner $2\cos(2\pi/5)=1/\varphi=\varphi-1$ , $\varphi=2\cos(\pi/5)=(1+\sqrt{5})/2$ . [E] the golden character is UNIQUE to $g_{\text{car}}=5$ : among small $n$ -gons only $n=5$ has an irrational interior-angle cosine ( $n=3, 4, 6$ give $1/2, 0, -1/2$ ), so $\theta_i$ is golden BECAUSE the carrier is 5 (P2). [C] a re-reading bridging v354/v313 (the unmapped $g_{\text{car}}=5$ golden signature) to v211 ( $\theta_i=3\pi/5$ , the one external input): it REFINES v354 from “ $\varphi$ in no readout” to “ $\varphi$ touches only the [C] spine misalignment angle”, a geometric motive for the spine over the hilltop $\sim 170^\circ$ ; it does NOT upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate. [E][C]                                                                                                                                                                                                                                                               |
| v430_other_side_reverse_audit.py | E8.OTHERSIDE.AUDIT.01: the ‘other side of the seam’ (the double-cover deck / conjugate sheet $S^-$ ) audited against $E_8$ 's UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering “is the leftover where the second sheet lives?” with a disciplined NEGATIVE. [E] DECK = DEGREE-2: the one-sided cover contributes $ \mathbb{Z}_2 =2$ (the $1/2$ in $c_3=1/(2\cdot 4\pi)=1/(8\pi)$ , v58/v216); $2=\min(E_8 \text{ degrees})$ = the metric, one of the matched primary readouts $\{2, 8, 30\}$ (v354), NOT one of the unmapped $\{12, 14, 18, 20, 24\}$ . [E] TWO SHEETS = 128-SPINOR: $\dim S^+=\dim S^-=16$ , the spinor block $128=\text{rank} \cdot \dim S^+=8\cdot 16=2^{\text{rank}-1}=\sum \text{deg}$ , the spinor half of $248=120+128$ ; $S^-$ the conjugate $(\overline{16}, \overline{4})$ , $128=2\cdot 64$ , not a spare singlet. [E] FORCED-DISJOINT: the sheet/deck integer set $\{2, 16, 32, 128\}$ has EMPTY intersection with $\{12, 14, 18, 20, 24\}$ , its only degree-alphabet contact the matched 2. [O] DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits $\geq 2$ readings (the v355 discriminator) — no forced sheet $\rightarrow$ unmapped-degree identity. [E] ‘ $S^-=\text{DM}$ ’ DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong ‘ $S^-$ is dark matter’; $128=(16, 4)+(\overline{16}, \overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). [O] the only live ‘other side’ question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees [E][O] |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v431_e8_degree_ladder.py         | <p>E8.DEGREE.LADDER.01: the reverse-audit ‘unmapped’ <math>E_8</math> Casimir degrees are NOT diffuse overhead but the forced two-family decomposition <math>\deg(E_8)=6\cdot\text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \det\text{-ladder}\{8, 14, 20\})</math> — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY <math>30=2\cdot3\cdot5</math>: the <math>E_8</math> exponents are the <math>\varphi(30)=8</math> totatives of <math>h=30=2N_{\text{fam}}g_{\text{car}}</math> (coprime to 6 <math>\Rightarrow \pm 1 \pmod 6</math>), so the degrees occupy ONLY the residue classes <math>\{0, 2\} \pmod 6</math>. [E] <math>6k</math> family <math>\{12, 18, 24, 30\}/6=\{2, 3, 4, 5\}</math> = the v91 spine <math>\{e_3, e_1, e_2, p_0\}(a=(1, 1, 2))</math>. [E] <math>6k+2</math> family <math>\{2, 8, 14, 20\}</math>: <math>\{8, 14, 20\}=(\det R, \det C, \det L)=</math> the winding line <math>\det M(s, 0)=6s+8</math> of the flavor diamond (v135), <math>C=R+Q \text{ diag}(1, 0, 0)</math>. [E] ‘<math>18=6N_{\text{fam}}</math>’ is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO <math>E_8</math> among the simply-laced exceptionals (<math>E_6/E_7/D_5</math> fail the clean <math>\{0, 2\} \pmod 6</math> split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim ‘<math>6k+2</math> Casimir stratum = flavor sheet operators’ needs a representation-theoretic map ([C]); 12, 24 admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C]</p> |
| v432_overdet_floor.py            | <p>OVERDET.FLOOR.01: the UNCONDITIONAL improbability floor + the <math>L1 \leftrightarrow L2</math> bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427’s seven readouts are one <math>(2, 3, 5)/E_8</math> object, NOT multiplied); what multiplies is the input ‘8’ forced FOUR ways <math>\{\text{rank } E_8, h(D_5)=2(g_{\text{car}}-1), \varphi(30), \text{Milnor}(2, 3, 5)\}</math> plus the FOREIGN witness <math>\alpha^{-1} \sim 137</math> (prime, outside <math>(2, 3, 5)</math>). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination <math>\times</math> the <math>\alpha</math> grammar census <math>1/94500</math> (v100) <math>\times</math> the <math>\varphi_0^{\text{ret}}</math>-seed cluster counted ONCE <math>\times</math> one downstream witness) the floor stays <math>\leq 10^{-6}</math> across a grid of generous assignments, nominal <math>\sim 10^{-10}</math> — ‘extraordinary’ WITHOUT the declared grammar. [E] THE BRIDGE: the floor (<math>\sim 10^{-10}</math>) sits <math>\sim 20</math> orders ABOVE the conditional <math>10^{-30.7}</math> (v100); that gap = the evidential value of proving the FORMS forced, so L1 (ALPHA.QUILLEN.EXACT.01, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY “is it numerology?”, NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C]</p>                                                       |
| v433_alpha_quillen_heatkernel.py | <p>ALPHA.QUILLEN.PROGRESS.02: a <i>second honest step</i> on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), <math>\alpha^{-1}</math> stays [E]. [E] a solvable 4D model (flat <math>\Delta = -\partial^2 + m^2</math> on <math>T^4</math>) reaches the <math>a_4</math> (<math>t^0</math>) conformal-anomaly order: <math>a_4 = Vm^4/(2(4\pi)^2)</math> NON-ZERO – the order v391’s 1D circle toy (<math>a_0, a_1</math> only) could not reach. [E] it is the order v342 grounds (Gilkey <math>a_4</math> gauge-curvature <math>30/360 = 1/12</math>), unifying v391 (a <math>\zeta</math>-det variation IS heat-kernel data) with v342 (the term IS the Gilkey <math>a_4</math>). [E] <math>(4\pi)^{-2} = 4c_3^2</math> is <math>c_3</math>-built. [E] the v382 split re-verified at <math>\alpha^{-1} = 137.0359992</math>. [O] NOT CLOSED: the actual seam <math>a_4</math> (<math>= 8b_1c_3^6</math>) and the cubic <math>\alpha^3</math> Chern moment stay open (v384); the exact seam <math>F</math>-normalisation stays [C] (v342). Grounds/connects, does not close [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v434_alpha_quillen_betafunction.py | <p>ALPHA.QUILLEN.PROGRESS.03: a <i>third honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays [O], <math>\alpha^{-1}</math> stays [E]. [E] <math>b_1</math> IS THE U(1) <math>a_4</math> HEAT-KERNEL COEFFICIENT (the <math>\beta=a_4</math> theorem): from the carrier hypercharge content with spin weights (<math>\frac{2}{3}</math> Weyl, <math>\frac{1}{3}</math> complex scalar), <math>\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6</math> (SM), <math>(3/5) \cdot \text{that} = 41/10 = b_1</math> (GUT) – the <math>b_1</math> in <math>8b_1c_3^6</math> is the <math>a_4</math> content coefficient, not a fitted multiplicity (same number as the <math>\beta</math>, v159/v246; same KIND as the <math>\alpha^2</math> Calderón <math>a_4</math>, v342). [E] <math>8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6</math> (rank 8 = <math>\chi=2</math> boundary, v216; <math>c_3^6</math> six insertions, v342). [C] residual (1) “exact <math>a_4 = 8b_1c_3^6</math>” <math>\equiv</math> residual (3) “seam <math>F</math>-normalisation” – ONE obligation. [E] residual (2) located: <math>a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)</math>, so <math>a^3</math> is the first moment of the Chern density (a level, order <math>c_3^0</math>); origin stays [O]. [E] net reduction: the EM-Ward residual is <math>1[C]+1[O]</math>, not three. [O] NOT CLOSED; <math>\alpha^{-1}</math> stays [E]. Python-only [E][C][O]</p>        |
| v435_alpha_quillen_chernlevel.py   | <p>ALPHA.QUILLEN.PROGRESS.04: a <i>fourth honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining [O] after v434 (residual (2), the cubic <math>\alpha^3</math> Chern level) and SHARPENS it, it does NOT close it; the target stays [O], <math>\alpha^{-1}</math> stays [E]. [E] <math>\pi</math>-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry <math>\pi</math>-powers exactly <math>\{0, 3, 6\} = c_3</math>-orders (<math>k_0 = 1 \rightarrow \pi^0</math>; <math>-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3</math>; <math>-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6</math>), so the Maxwell <math>a^3</math> is the UNIQUE <math>\pi^0/c_3^0</math> (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the <math>(4\pi)^{-d/2}</math> measure; a level carries none). The test partitions the closure into ONE topological level (<math>a^3</math>) + TWO heat-kernel boundary terms, confirming v342/v434’s “<math>a^3</math> is a level, not a curvature integral” from a test. [C] conditional on the CS/<math>\eta</math> reading (EM1, v48), large-gauge invariance forces the level in <math>\mathbb{Z}</math> and <math>k_0 = 1</math> is the unit (minimal) level – an integer, not a free real. [O] NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam <math>F</math>-norm stays the one [C] (v434); <math>\alpha^{-1}</math> stays [E]. [E][C][O]</p>                                                 |
| v436_overdet_floor2.py             | <p>OVERDET.FLOOR.02: HARDENS OVERDET.FLOOR.01 (v432) — the “is it numerology?” floor does NOT rest on v432’s subjective chance assignments. [E] ASSUMPTION-MINIMAL FLOOR: the <math>\alpha</math> cubic-class census (v100 layer D) is PURE COUNTING — of <math>N=94500</math> complexity-matched <math>F_{U(1)}</math> variants only ONE hits the <math>4 \times 10^{-8}</math> CODATA window (the TFPT instance, <math>\alpha^{-1}=137.0359992</math>), so <math>p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma</math> with NO subjective probability, prior or multi-observable grammar. [E] MONOTONE CONCESSION LADDER: <math>-\log_{10} p</math> full scorecard+<math>\alpha \sim 30.7 \rightarrow</math> scorecard-only <math>\sim 25.8 \rightarrow \alpha</math>-census-only <math>\sim 4.98</math> (<math>= 4.40\sigma</math>); the “not chance” verdict holds at the MOST adversarial rung. [E] INDEPENDENCE: strip ALL of v432’s subjective factors (<math>0.1^3 \times 0.012 \times 0.15 = 1.8 \times 10^{-6}</math>) and the floor is still <math>4.40\sigma</math> — they only sharpen it. [C] HONEST <math>5\sigma</math> GAP: <math>5\sigma</math> (<math>p=5.7 \times 10^{-7}</math>) is crossed by the census times exactly ONE more independent factor <math>\leq 0.054</math> (the input-‘8’ four-fold over-determination or one foreign witness), stated openly. [C] FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). [E][C]</p> |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v437_e8_degree_joint.py    | <p>E8.DEGREE.JOINT.01: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the <math>E_8</math> Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. [E] <math>E_8</math> IS FIXED BY ITS DEGREE MULTISSET: from <math>\deg(E_8)=\{2, 8, 12, 14, 18, 20, 24, 30\}</math> alone — <math>\text{rank}=\#\deg=8</math>, <math>h=\max=30</math>, <math>\#\text{pos roots}=\sum \exp=120</math>, <math>\#\text{roots}=240</math>, <math>\dim=248</math>, <math> W =\prod \deg=696,729,600</math>, <math>\sum \deg=128=2^7=\dim S^+</math>. [E] THE JOINT QUADRATIC: the two structural integers <math>(g_{\text{car}}, N_{\text{fam}})</math> are the UNIQUE root pair of <math>x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)</math> — <math>\text{sum}=\text{rank } E_8=8</math> (<math>\text{rank-fill}, v6</math>), <math>\text{product}=h/2=15=g_{\text{car}}N_{\text{fam}}</math> (<math>2g_{\text{car}}N_{\text{fam}}=h, v431</math>); <math>\{3, 5\}</math> the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. [E] the three mapped degrees <math>\{2, 8, 30\} = \min/\text{rank}/\max</math> (the canonical invariants); <math>g_{\text{car}}=5=\max</math> prime of 30. [E] DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. [E][C]</p>                                                                                                                                                                                                      |
| v438_seam_hsmi_borchers.py | <p>SEAM.EQUIV.BW.HSMI.01: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of SEAM.EQUIV.01 — geometric modular flow WITHOUT presupposing covariance. [E] STANDARD-PAIR ALGEBRA (ladder, exact): boost <math>K=\text{diag}(n)</math> and translation <math>P</math> (raising shift) satisfy <math>[K, P]=P</math> and the Borchers scaling <math>\Delta^{it}P\Delta^{-it}=e^{-t}P</math> (the <math>ax+b/\text{Moebius}</math> relation). [E] BW REFLECTION: <math>J(n\rightarrow -n)</math> gives <math>J^2=I</math>, <math>JKJ=-K</math> (<math>u_\Theta=J</math>) and <math>JPJ=P_-</math> (the opposite-chirality translation), the standard-pair reflection <math>JU(s)J=U(-s)</math>. [E] POSITIVE ENERGY: in the unitary lowest-weight <math>sl(2, \mathbb{R})</math> rep the line translation <math>P_{\text{line}}=L_0-(L_++L_-)/2</math> is POSITIVE (<math>\min \text{eig} &gt; 0</math>), the positive-energy generator the <math>c_3</math> one-sided origin demands; the boost direction is the indefinite negative control. [C] SYNTHESIS: <math>P\geq 0 + [K, P]=P + J</math> is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, arXiv:2508.07109); face (ii) becomes a STRUCTURE consequence, not a state condition. [C][O] assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]</p> |
| v439_seam_lieb_robinson.py | <p>SEAM.EQUIV.CONTINUUM.LR.01: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 <math>p+ip</math> model. [E] UNIFORM GAP: <math>\Delta(N)=\text{const}&gt;0</math> across <math>N=20..160</math> (the lattice image of the OS transfer gap <math>6\log\frac{3}{2}&gt;0, v302/v329</math>). [E] EXPONENTIAL CLUSTERING: <math> \langle c_i^\dagger c_j \rangle  \sim e^{-r/\xi}</math> with <math>\xi</math> finite and STABLE across <math>N</math> (gapped <math>\Rightarrow</math> Hastings–Koma decay), the <math>N</math>-independent locality. [E] LIEB–ROBINSON LIGHT CONE: the single-particle propagator <math> (e^{-iHt})_{ij} </math> is confined to <math> i-j  \lesssim v_{\text{LR}}t</math> with finite <math>v_{\text{LR}}</math> (<math>\sim 2t</math>), exponentially small beyond the cone. [C] NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity <math>\Rightarrow</math> exponential clustering with <math>N</math>-independent <math>\xi</math> AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. [C][O] the bulk thermodynamic limit EXISTS by Lieb–Robinson; SEAM.EQUIV.01 stays [O], the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the <math>c=8</math> <math>(E_8)_1</math> CFT), distinct from the bulk limit. Python-only (numpy/scipy). [E][C][O]</p>                                                                                                                                             |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v440_seam_lto_rp_beta.py          | SEAM.EQUIV.LTORP.BETA.01: the $\beta$ -interpolation connecting the NPW26-PROVEN trace case ( $\beta=0$ ) to the seam's $\beta=1$ KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] $\beta$ -INDEPENDENT (LTO-RP) DATA: $\Theta K \Theta = -K$ ( $u_\Theta = J$ ) and $[\rho, K]=0$ are conditions on $K$ alone, $\beta$ -independent. [E] THE KMS PATH: $C_\beta = 1/(1+e^{\beta K})$ has $\beta=0 \rightarrow C=\frac{1}{2}I$ (the NPW26 TRACE state), $\beta=1 \rightarrow$ the seam KMS state, $\beta \rightarrow \infty \rightarrow$ the ground-state projection; at every $\beta$ $\Theta C_\beta \Theta = 1-C$ (reflection), $[\rho, C_\beta]=0$ (mu4), $0 \leq C_\beta \leq 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta \mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN $K'$ breaks $\Theta C_\beta \Theta = 1-C$ for $\beta > 0$ (the trace endpoint is degenerate); an off-mu4-block $K$ breaks $[\rho, C_\beta]$ . [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; SEAM.EQUIV.01 stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O]                                                                                                                             |
| v441_seam_applicability_ledger.py | SEAM.EQUIV.APPLICABILITY.01: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: $D=16$ Majoranas ( $= 2^{g_{\text{car}}-1}$ ), $c=8$ ( $= g_{\text{car}} + N_{\text{fam}}$ ), rank $E_8=8$ , range rank $\leq c \leq D$ reads $8 \leq 8 \leq 16$ , gapped ( $6 \log \frac{3}{2} > 0$ ), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP) $= u_\Theta = J$ , gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho, K]=0$ internal (v424/v426/v440); the seam is invertible $ \det \text{Cartan } E_8  = 1 + \beta = 1$ KMS, OUTSIDE NPW26's topological/trace bucket ( $ \det =4$ ) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (arXiv:2508.07109): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1 \times (A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION $\rightarrow$ [E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean FORM.SEAM.MMST.01). [C][O] external dependence pinned to ONE analytic fact; SEAM.EQUIV.01 stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O] |
| v442_seam_rigidity_uniform.py     | SEAM.RIGIDITY.UNIFORM.01: the uniform-in- $N$ hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small- $N$ artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho = \text{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N$ , and a genuinely character-block-diagonal $K$ (random Hermitian within each $n \bmod 4$ class, not merely diagonal) has $[\rho, K]=0$ EXACTLY for $N=4..256$ (extends v398's $N \leq 64$ ). [E] UNIFORM GAP: $\Delta(N) = \text{const} > 0$ across $N=20..160$ (the Nachtergaele–Sims uniform gap, shared with v439). [E] $N$ -INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\ [\rho, V]\ =2$ EXACTLY for all $N$ — a uniform lower bound; any fixed-size clock-breaking is detected with $N$ -independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\ [\rho, V']\ =0$ ; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in $N$ . [C][O] the band-limited rigidity of v398 is uniform in $N$ ; SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]: that RP+gap+holomorphy FORCE $\Lambda_\Sigma$ block-diagonal on the full $L^2$ (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O]                                                                                                                                    |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v443_seam_e8_discriminator.py | <p>SEAM.E8.DISCIMINATOR.01: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam <math>= (E_8)_1</math> from the nearest <math>c=8</math> rival <math>SO(16)_1=(D_8)_1</math>. [E][X] PRIMARY/GSD: <math> \det \text{Cartan}(E_8) =1</math> (ONE primary, holomorphic) vs <math> \det \text{Cartan}(D_8) =4</math> (FOUR primaries <math>1, v, s, c</math>); torus GSD <math>= \# \text{primaries} = 1</math> vs <math>4</math> — a seam with <math>\text{GSD} &gt; 1</math> falsifies <math>(E_8)_1</math>. [E][X] CURRENT/THETA: level-1 current count <math>\dim E_8=248</math> vs <math>\dim SO(16)=120</math>; the <math>E_8</math> lattice theta has 240 norm-2 roots (<math>= 112</math> integer + 128 spinor, computed) vs <math>D_8</math>'s 112 — current spectrum and lattice theta DIFFER. [E] <math>c=8</math> IS NOT A DISCRIMINATOR: both have <math>c=8</math> (the v277 ambiguity); only the order-4 <math>\mu_4</math> condensation + <math>\det K=1</math> (the genus-1 statement, v90) selects <math>E_8</math>. [E] NON-VACUITY: the discriminators all fire on the rival (<math>\det 4 \neq 1</math>, currents <math>120 \neq 248</math>, roots <math>112 \neq 240</math>). [X][C] mode D realised as concrete kill tests; SEAM.EQUIV.01 stays [O] (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O][X]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v444_seam_edge_scaling.py     | <p>SEAM.EQUIV.CONTINUUM.EDGE.01: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one SEAM.EQUIV.01 continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 <math>p+ip</math> collar it shows the lattice edge carries the kinematic AND conformal data of a chiral <math>c=\frac{1}{2}</math> Majorana CFT, converging under refinement. [E] CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top <math>v=0.983</math>, <math>R^2=0.9999</math>) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). [E] ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as <math>r^{-\eta}</math> with <math>\eta=1.00</math> (the chiral free-fermion CFT exponent) — a CRITICAL edge. [E] EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially (<math>\xi \sim 1.1</math>) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. [E] TRIVIAL-PHASE NEG CONTROL: <math>M=3</math> (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. [E] SCALING-COLLAPSE / CONVERGENCE: <math>\eta</math> is <math>N</math>-stable (<math>1.001 \pm 0.007</math>) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks <math>0.028 \rightarrow 0.013</math>, <math>N_x=36..60</math>) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. [C] EDGE CENTRAL CHARGE: <math>\eta=1 \Leftrightarrow</math> one chiral Majorana (<math>c=\frac{1}{2}</math>) per edge, so the <math>N_{\text{Maj}}=16</math>-copy carrier has a chiral <math>c_-=8</math> edge CFT (<math>= (E_8)_1</math>, <math>c_-=g_{\text{car}}+N_{\text{fam}}</math>). [C][O] grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; SEAM.EQUIV.01 stays [O] (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). [E][C][O]</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v445_seam_rigidity_forcing.py | <p>SEAM.RIGIDITY.FORCING.01: the rigidity FORCING theorem — upgrades v442’s “block-diagonal PERMITS commuting” to “commuting with the order-4 clock FORCES block-diagonal”, isolating the single remaining physical premise. [E] FORCING (commute <math>\Leftrightarrow</math> block-diagonal, EXACT): for <math>\rho = \text{diag}(i^n)</math>, <math>[\rho, K]=0</math> iff <math>K_{ij}=0</math> whenever <math>i \not\equiv j \pmod 4</math> — verified ENTRYWISE, swept <math>N=4..128</math> (the converse of v442). [E] EXACT COMMUTANT DIMENSION: <math>\dim\{K : [\rho, K]=0\} = \sum_s n_s^2</math> (the four sector sizes), verified as the <math>\text{ad}_\rho</math> nullspace; a PROPER subspace of the <math>N^2</math> algebra (<math>N=16</math>: <math>64 &lt; 256</math>) — a genuine quantitative restriction. [E] OFF-BLOCK <math>\Leftrightarrow</math> NON-COMMUTING: every off-block matrix unit <math>E_{ij}</math> has <math>\ [\rho, E_{ij}]\  =  i^i - i^j  \geq \sqrt{2} &gt; 0</math> — zero slack. [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors (<math>\dim \sum n_s^2(4)</math>), the order-2 clock <math>\text{diag}((-1)^n)</math> a STRICTLY LARGER commutant (<math>N=16</math>: <math>64</math> vs <math>128</math>) — the four Gauss–Bonnet marks (order <math>4= \mu_4 =h(A_3)</math>) force strictly more structure than two; only the index-4 extension is <math>(E_8)_1</math> (<math>\det 1</math>; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues <math>\{1, i, -1, -i\}</math> for all <math>N \geq 4</math>. [C] SYNTHESIS (force, not permit): commuting with the order-4 <math>\mu_4</math> clock is EQUIVALENT to <math>\mu_4</math>-block-diagonality (iff, exact, uniform in <math>N</math>), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal <math>\Lambda_\Sigma</math> — v442’s “permit” upgraded to “force”. [C][O] the single residual is now precisely “transfer in commutant” (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). [E][C][O]</p> |
| v446_seam_clock_invariance.py | <p>SEAM.RIGIDITY.CLOCKINV.01: DERIVING clock-invariance (“transfer in the commutant”) from RP-symmetry, closing v445’s single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL <math>\mu_4</math>-symmetry of the seam RP data. [E] OS RECONSTRUCTION INHERITS THE SYMMETRY: for a <math>\mu_4</math>-invariant RP covariance <math>([\rho, C(t)]=0)</math>, the OS canonical transfer <math>\Lambda=C(1)C(0)^{-1}=e^{-h}</math> (the transfer FIXED by RP+reflection, v54) has <math>[\rho, \Lambda]=0</math> — swept <math>N=4..32</math> over random block-diagonal <math>h</math>. [E] SHARPNESS / NEG CONTROL: a <math>\mu_4</math>-BREAKING covariance gives <math>[\rho, \Lambda] \neq 0</math> — iff-sharp, so the inheritance is non-vacuous. [E] CONTRACTION / OS-VALIDITY: <math>\Lambda</math> has spectrum in <math>(0, 1]</math> (a positive contraction, the physical canonical transfer). [E] MODULAR (Tomita–Takesaki): the free modular Hamiltonian <math>K=\log((1-C)C^{-1})</math> (the v258 formula) of the <math>\mu_4</math>-invariant state has <math>[\rho, K]=0</math> (the clock commutes with modular flow), the breaking control <math>[\rho, K] \neq 0</math>. [C] SYNTHESIS: <math>[\rho, \Lambda_\Sigma]=0</math> is a CONSEQUENCE of <math>\mu_4</math>-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute <math>\Leftrightarrow</math> block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal <math>\Lambda_\Sigma</math>. [C][O] the residual reduces from an operator commutation to “the seam RP data is <math>\mu_4</math>-symmetric” (QGEO.SYM.01); the continuum Tomita–Takesaki implementation on the full <math>L^2</math> is the cited backbone. SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]</p>                                                                                                                                                                                                                  |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v447_seam_edge_chern.py     | <p>SEAM.EQUIV.EDGE.CHERN.01: an INDEPENDENT (topological) reading of the edge <math>c_- = 8</math>, corroborating v444's correlator route via the bulk-edge correspondence on the same v367/v368 <math>p+ip</math> collar. [E] INTEGER CHERN NUMBER: the <math>M=1</math> collar lower band has Chern <math>C=+1</math> (the Fukui-Hatsugai-Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. [E] TRIVIAL-PHASE NEG CONTROL: <math>M=3</math> (v368) has <math>C=0</math> (no chiral edge), and <math>M=-1</math> gives <math>C=-1</math> (opposite chirality, matching v444's opposite per-edge velocities). [E] BULK-EDGE CORRESPONDENCE: the open strip carries EXACTLY <math> C =1</math> in-gap edge branch crossing zero per boundary in the <math>M=1</math> phase, 0 in the trivial phase — the chiral edge channel count = <math> C </math>. [E] LI-HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at <math>1/2</math> in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. [C] EDGE CENTRAL CHARGE: <math>c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot  C  = 16 \cdot \frac{1}{2} \cdot 1 = 8 = g_{\text{car}} + N_{\text{fam}}</math>, from the Chern integer + bulk-edge correspondence — the SAME <math>c_- = 8</math> as v444's correlator route, from a logically independent (topological) observable. [C][O] corroborates v444's edge <math>(E_8)_1</math> reading; SEAM.EQUIV.01 stays [O] — the topology fixes <math>c_-</math> and the channel count but not the cited continuum existence theorem. Python-only (numpy). [E][C][O]</p>                                                                                                                                                                           |
| v448_seam_edge_fourpoint.py | <p>SEAM.EQUIV.EDGE.FOURPOINT.01: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. [E] QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector <math>G^2=G</math> (<math>\sim 10^{-13}</math>), so the state is quasi-free and all <math>n</math>-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). [E] FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio <math>Q=[G_{01}G_{23}]/[G_{02}G_{13}]</math> at COMMENSURATE positions <math>(1, 2, 4, 7) \cdot (N_x/12)</math> is Cauchy-convergent under refinement (successive spread shrinks, <math>N_x=36..60</math>) — the four-point converges, extending v444's two-point scaling-collapse. [E] CONFORMAL FOUR-POINT FORM: <math>Q</math> converges to the free-fermion CONFORMAL cross-ratio <math>\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732</math> (chord distance <math>d(r)=(N_x/\pi)\sin(\pi r/N_x)</math>) — the edge four-point has the chiral-CFT cross-ratio structure (<math>\eta=1</math>), not just the two-point. [E] BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. [C] the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured <math>n</math>-point converges to its conformal limit. SEAM.EQUIV.01 stays [O] — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). [E][C]</p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v449_seam_mmst_uniform.py      | <p>SEAM.EQUIV.MMST.UNIFORM.01: uniform-in-<math>N</math> control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 <math>p+ip</math> collar. [E] CROSS-RATIO A CONVERGES: the edge four-point <math>Q(1, 2, 4, 7)/12 \rightarrow \sin(5\pi/12)/\sin(\pi/12) = 2+\sqrt{3}</math>, the deviation shrinking monotonically under refinement (<math>N_x=36, 48, 60</math>). [E] CROSS-RATIO B (independent positions): <math>Q(1, 3, 5, 9)/12 \rightarrow</math> its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. [E] UNIFORM <math>1/N</math> RATE: <math> Q(N)-Q_\infty  \cdot N \leq 4</math> for BOTH cross-ratios and every <math>N_x \in \{36, 48, 60\}</math> — one <math>N</math>-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-<math>N</math> convergence. [E] BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. [C][O] the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); SEAM.EQUIV.01 stays [O] — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). [E][C]</p> |
| v450_seam_edge_entanglement.py | <p>SEAM.EQUIV.EDGE.ENTANGLEMENT.01: a THIRD independent reading of the edge <math>c_-=8</math> — the Calabrese–Cardy entanglement scaling <math>c=1/2</math> per Majorana (after v444 correlator, v447 topology). [E] CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits <math>S=(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]+\text{const}</math> with <math>c_{\text{Dirac}}=1.000</math> (flat log slope) — the universal scaling-limit signature. [E] <math>c</math> PER MAJORANA = <math>1/2</math>: <math>c_{\text{Dirac}}=1=2c_{\text{Maj}}</math> (Dirac = two Majoranas), so <math>c_{\text{Maj}}=1/2</math> — the chiral CFT of one <math>p+ip</math> edge mode (v444). [E] GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, <math>\ell</math>-independent entanglement (<math>c=0</math>) — the log growth is the CRITICAL edge CFT. [C] EDGE CENTRAL CHARGE: <math>c_-=N_{\text{Maj}} \cdot c_{\text{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}</math>, from entanglement — the SAME <math>c_-=8</math> as v444 (correlator) and v447 (topology), from a third independent observable. [C][O] three observables agree; SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]</p>                                                                                                                                                                                |
| v451_seam_edge_cardy_tower.py  | <p>SEAM.EQUIV.EDGE.CARDY.01: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions <math>\{1/16, 1/2\}</math>, on the critical Majorana ring (the v367/v368 edge universality class). [E] SPIN DIMENSION <math>h_\sigma=1/16</math>: the periodic(R)–antiperiodic(NS) ground-energy split gives <math>(E_0^R - E_0^{NS})L/(2\pi v)=0.06250</math> (<math>v=2</math>), stable across <math>L=200, 400, 800</math> — the Ising spin field. [E] ENERGY DIMENSION <math>h_\varepsilon=1/2</math>: the lowest NS pair excitation gives <math>2\varepsilon_{\min}L/(2\pi v)=0.50000</math> — the Ising energy field. [E] CONFORMAL SCALING: both dimensions are <math>L</math>-INDEPENDENT (the gaps scale exactly as <math>1/L</math>) — a conformal (gapless) spectrum, not a gapped lattice. [C] ISING IDENTIFICATION: <math>\{c, h_\sigma, h_\varepsilon\}=\{1/2, 1/16, 1/2\}</math> is the chiral Ising (free Majorana) CFT (the <math>c=1/2</math> building block, v444/v450); with <math>N_{\text{Maj}}=16</math> copies and bulk-edge (v447), <math>c_-=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}</math>. [C][O] the edge CFT is named by its central charge (v450) AND its operator content (this); SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]</p>                                                                                                               |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v452_seam_e8_modular.py     | <p>SEAM.EQUIV.E8.MODULAR.01: the <math>(E_8)_1</math> torus modular data — one primary, <math>S</math>-invariant, <math>T</math>-phase <math>e^{-2\pi i/3}</math>, the holomorphic <math>c=8</math> signature (extends v156's <math>q</math>-expansion to the modular transformation data). [E] <math>S</math>-INVARIANCE: <math>\chi(-1/\tau)=\chi(\tau)</math> for <math>\chi=E_4/\eta^8</math> (weight 0, since <math>E_4</math> wt 4 / <math>\eta^8</math> wt 4), to <math>\sim 10^{-30}</math> — ONE primary, the <math>S</math>-matrix is [1]. [E] <math>T</math>-PHASE: <math>\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi ic/24}</math> (<math>c=8</math>), to <math>\sim 10^{-30}</math> — the central charge appears in the modular <math>T</math> eigenvalue. [E] LEADING POWER <math>\Rightarrow c=8</math>: <math>\chi q^{1/3} \rightarrow 1</math> so the leading power is <math>q^{-c/24}=q^{-1/3} \Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}</math>. [E] HOLOMORPHIC CONSTRAINT: <math>c=8=0 \bmod 8</math> — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice <math>E_8</math>. [C][O] the <math>(E_8)_1</math> modular data (one primary, <math>S</math>-invariant, <math>T</math>-phase <math>e^{-2\pi i/3}</math>, <math>c=8</math>) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the <math>(E_8)_1</math> identity is pinned on the torus AND the edge; SEAM.EQUIV.01 stays [O] — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C]</p>                                                     |
| v453_seam_mu4_from_marks.py | <p>SEAM.RIGIDITY.MU4FROMMARKS.01: deriving QGEO.SYM.01 (<math>\mu_4</math>-symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. [E] THE MARKS ARE <math>\mu_4</math>: the four Gauss–Bonnet marks are exactly <math>\{1, i, -1, -i\} = \text{roots of } z^4-1</math> (v168/v216/v323). [E] THE CLOCK PERMUTES THE MARKS: <math>z \mapsto iz</math> is a 4-cycle of order 4 (<math>1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1</math>), fixing the configuration setwise (the order-4 clock of v445). [E] MOEBIUS STABILISER: <math>z \mapsto iz</math> preserves the marks' cross-ratio <math>\lambda=2</math> (in the <math>D_4</math> stabiliser, v168) — the symmetry is geometric. [E] THE FORM BASIS IS <math>\mu_4</math>-GRADED: <math>\omega_k = z^{k-1} dz / (z^4-1) \rightarrow i^k \omega_k</math> under <math>z \mapsto iz</math> (<math>k=1, 2, 3 \rightarrow \{i, -1, -i\}</math>, the three nontrivial <math>\mu_4</math> characters; weights <math>(1, 2, 3)=A_3</math> exponents = <math>\text{Spec}(Q_+)</math>) — the natural basis is <math>\mu_4</math>-graded, so any geometric datum is <math>\mu_4</math>-covariant. [C][O] QGEO.SYM.01 (“the seam RP data is <math>\mu_4</math>-symmetric”) is a CONSEQUENCE of marks=<math>\mu_4</math> (this) + QGEO.REALIZE.01 (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this <math>\rightarrow</math> v446 <math>\rightarrow</math> v445) closes modulo realisation, SEAM.EQUIV.01 stays [O]. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C]</p> |
| v454_seam_edge_virasoro.py  | <p>SEAM.EQUIV.EDGE.VIRASORO.01: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST “Koo–Saleur lattice generators <math>\rightarrow</math> continuum Virasoro with the correct <math>c</math>” fact on our own free-fermion edge. [E] CASIMIR <math>c</math>: the critical complex free-fermion ring <math>E_0(L)=e_\infty L - \pi v c / (6L)</math> (Affleck–Cardy) fits <math>c_{\text{Dirac}}=1.000 \Rightarrow 1/2</math> per Majorana (parameter-free, <math>L=64..1024</math>). [E] KAC–MOODY LEVEL: the lattice <math>U(1)</math> current <math>J_n</math> has <math>\langle 0   J_n J_{-n}   0 \rangle = n</math> EXACTLY (<math>n=1..8</math>) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. [E] SUGAWARA EXACT: <math>c = \dim G / (1+h^\vee) = 8</math> for BOTH <math>SO(16)_1</math> (120/15) and <math>(E_8)_1</math> (248/31) — Wolfram-mirrored. [C] ASSEMBLY: 16 Majoranas <math>\times \frac{1}{2} = 8 =</math> the level-1 Sugawara <math>c</math>; the edge IS a <math>c_- = 8</math> level-1 current-algebra CFT. [C][O] the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; SEAM.EQUIV.01 stays [O] (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). [E][C][O]</p>                                                                                                                                                                         |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v455_seam_bw_uniform.py         | <p>SEAM.EQUIV.BW.UNIFORM.01: a UNIFORM-IN-<math>N</math> Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition <math>u_\Theta=J</math> (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: <math>\Delta=e^{-K}</math>, conjugation <math>J</math>, reflection <math>\Theta</math>; BW <math>\Leftrightarrow u_\Theta=J \Leftrightarrow \Theta K \Theta = -K</math>. [E] UNIFORM BW: the reflection-odd, <math>\mu_4</math>-even, gapped collar of size <math>4m</math> (<math>m=2..32</math>, <math>N=8..128</math>) has <math>\ \Theta K \Theta + K\ =0</math> to machine precision at EVERY <math>N</math> — <math>u_\Theta=J</math> uniform. [E] UNIFORM GAP: the gap of <math>K</math> is <math>N</math>-independent (<math>6 \log \frac{3}{2} = 2N_{\text{fam}} \log \frac{3}{2}</math>) so <math>\Delta=e^{-K}</math> stays uniformly bounded away from 1 (a genuine KMS, not trace, state). [E] UNIFORM CLOCK SYMMETRY: <math>[\rho, K]=0</math> exactly at every <math>N</math>. [E] <math>N</math>-INDEPENDENT NEG CONTROL: a side-even reflection gives <math>\ \Theta K \Theta + K\ =2\ K\ </math> (a fixed per-mode margin, growing with <math>N</math>) and an off-<math>\mu_4</math> <math>K</math> breaks <math>[\rho, K]</math> at every <math>N</math> — the success is non-vacuous. [C][O] the inductive (continuum) limit inherits <math>u_\Theta=J</math>; SEAM.EQUIV.01 stays [O] (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). [E][C][O]</p> |
| v456_seam_chirality_from_c3.py  | <p>SEAM.S3.FROM-P1.01: the edge chirality (<math>c_- \neq 0</math>, the S3 topological phase) is FORCED by the one-sidedness that defines <math>c_3=1/(8\pi)</math> (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. [E] <math>c_3</math>'s “8” IS THE ONE-SIDED COUNT: <math>8\pi= \mathbb{Z}_2  \int K = 2(2\pi\chi(S^2)) = 2 \cdot 4\pi</math> (<math>\chi=2</math>), so <math>8= \mathbb{Z}_2  \cdot (\int K/\pi) = 2 \cdot 4</math> (v73, Wolfram-mirrored). [E] REFLECTION REVERSES <math>C</math>: the explicit 2-band <math>p+ip</math> Chern model has <math>C=+1</math> and the orientation-reversed (<math>k_y \rightarrow -k_y</math>) model <math>C=-1</math>. [E] TWO-SIDED <math>\Rightarrow</math> NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have <math>C=-C=0</math> (<math>c_-=0</math>, gappable). [C] ONE-SIDED <math>\Rightarrow</math> CHIRAL: the one-sided seam is precisely the <math> \mathbb{Z}_2 </math> quotient with NO reflection (the same <math> \mathbb{Z}_2 </math> that gives <math>c_3</math> its <math>8\pi</math>), so <math>C \neq 0</math> is forced — S3 follows from P1. [C][O] magnitude <math>c_-=8=\text{rank } E_8=g_{\text{car}}+N_{\text{fam}}</math> is the SAME integer as <math>c_3</math>'s one-sided count; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Mixed exact (the <math>8= \mathbb{Z}_2  \int K/\pi</math> arithmetic, Wolfram) + numerical (the reflection <math>C \rightarrow -C</math> flip). [E][C][O]</p>                         |
| v457_seam_character_killtest.py | <p>SEAM.EQUIV.KILLTEST.01: an explicit <math>(E_8)_1</math>-vs-<math>SO(16)_1</math> character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the <math>(E_8)_1</math> identification of SEAM.EQUIV.01. [E] <math>(E_8)_1</math> CHARACTER TOWER: <math>\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\dots)</math> — the exact conformal-tower degeneracies <math>(E_4 \cdot \prod (1-q^n))^{-8}</math>, Wolfram-mirrored). [E] CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is <math>248=\dim E_8=120(SO16)+128(\text{spinor})</math>, NOT the 120 of the free <math>SO(16)_1</math>. [E] SECTOR-COUNT DISCRIMINATOR: <math>(E_8)_1</math> is INVERTIBLE <math> \det \text{Cartan } E_8 =1</math> (one primary) vs <math>SO(16)_1=D_8</math> <math> \det \text{Cartan } D_8 =4</math> (four sectors). [X] KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient <math>\neq \{1, 248, 4124, \dots\}</math> would FALSIFY <math>(E_8)_1</math>; current data give <math>248/1/\{1, 248, 4124\}</math>, so PASSED. [C][O] the <math>(E_8)_1</math> identification survives the kill-test; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Exact (Wolfram-mirrored) + a numerical determinant cross-check. [E][C][O][X]</p>                                                                                                                                                                                                                                                                      |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v458_seam_mmst_citation_audit.py | <p>SEAM.EQUIV.MMST.AUDIT.01: the EXACT hypothesis-by-hypothesis citation audit of the seam scaling-limit theorem (Osborne–Stottmeister, CMP <b>398</b> (2023) 219; arXiv:2107.13834; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework) against our collar (G3 of the post-F next steps, the “exakter Zitat-Abgleich”). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST’s lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY (<math>c=1/2</math>): each of 16 chiral Majoranas is EXACTLY the proven Theorem A <math>c=1/2</math> case. [C] H4 DECOUPLED ADDITIVITY: MMST’s limit decouples into Majoranas <math>\Rightarrow L_k = \sum_i L_k^{(i)}</math> on the common core <math>\Rightarrow</math> the free <math>c=8</math> <math>SO(16)_1</math> Virasoro follows (<math>16 \cdot \frac{1}{2} = 8</math>). [C] H5 WZW RANGE rank <math>\leq c \leq D</math>: <math>8 \leq 8 \leq 16</math> IN range, the 120 <math>so(16)</math> currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 <math>SO(16)_1 \rightarrow (E_8)_1</math> adds 128 SPINOR currents (<math>248=120+128</math>), NOT bilinears — OUTSIDE MMST’s method (the <math>\det K</math> 4<math>\rightarrow</math>1 holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the <math>c=8</math> EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; SEAM.EQUIV.01 stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O]</p> |
| v459_seam_lattice_voa_route.py   | <p>SEAM.EQUIV.LATTICEVOA.01: the SECOND, independent route to <math>(E_8)_1</math> — the lattice-VOA / conformal-net <math>A_Q(E_8)</math> construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, “Rational and non-rational 2d CFTs arising from lattices” (arXiv:2506.01008); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, arXiv:2407.18222). [E] <math>E_8</math> EVEN UNIMODULAR: <math>\det \text{Cartan}(E_8)=1</math> and even <math>\Rightarrow A_{E_8}</math> HOLOMORPHIC (one sector, the <math>\det K=1</math> the residual demanded). [E] LATTICE CURRENTS <math>248=8+240</math>: 8 Cartan + 240 roots = the full <math>(E_8)_1</math> weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT 112+128: 112 INTEGER (<math>D_8</math>) + 128 HALF-INTEGER; <math>8+112=120=so(16)</math> and the 128 half-integer roots ARE the spinor (<math>248=120+128=8+240</math>) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: <math>A_Q</math> of an even lattice IS a 2d conformal net; the holomorphic <math>SO(16)_1 \rightarrow (E_8)_1</math> extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover <math>(E_8)_1</math> (MMST fermions v458 + the lattice net <math>A_{E_8}</math>); the residual reduces to the single realization input “the collar’s scaling limit = <math>A_{E_8}</math>” (tied to P1 by v456). SEAM.EQUIV.01 stays [O]. Exact (<math>E_8</math> even unimodular; <math>240=112+128</math>; <math>248=8+240=120+128</math>, Wolfram-mirrored) + structural literature mapping. [E][C][O]</p>                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v460_seam_s3_lto.py          | <p>SEAM.S3.LT0.01: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) (“realise the raw seam as a <math>\mathbb{Z}_2</math>-reflection commuting-projector LTO”) from abstract existence to an explicit gap-protected quasi-local projector net carrying the <math>\mathbb{Z}_2</math> reflection and the <math>\mu_4</math> clock. [E] FLAT-BAND PARENT: the flattened Bloch <math>Q=h/ h </math> of v367’s <math>p+ip</math> collar has <math>Q^2=I</math> (flat <math>\pm 1</math> bands) and <math>P=(I-Q)/2</math> is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel <math> P(n, 0) </math> decays rapidly (<math>\xi \approx 1.14</math> at <math>M=1</math>, <math>\exp R^2=0.995</math>, <math> P(20, 0)  &lt; 10^{-6}</math>) and <math>\xi</math> GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law <math> R ^{-2.1}</math> (<math>R^2=0.999</math>, tail <math>\sim 2 \times 10^{-4}</math>) — locality needs the gap. [E] <math>\mathbb{Z}_2</math> REFLECTION + <math>\mu_4</math> INHERITED: the flattened modular Hamiltonian <math>K_{\text{flat}} = \text{sign}(K)</math> inherits <math>\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}</math> (BW, <math>u_\Theta = J</math>) and <math>[\rho, K_{\text{flat}}] = 0</math> (<math>\mu_4</math>) EXACTLY (sign is odd; <math>\rho</math> commutes with any <math>f(K)</math>). [E] NEG CONTROLS: a side-even reflection gives <math>2\ K_{\text{flat}}\  \neq 0</math> (BW fails), an off-<math>\mu_4</math> <math>K</math> mixing opposite-sign marks breaks <math>[\rho, \text{sign}(K)] \neq 0</math>. [C][O] the chiral <math>c_- = 8 \neq 0</math> (FHS Chern <math> C =1</math>) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756; <math>C \neq 0</math> forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). [E][C][O]</p> |
| v461_seam_strict_locality.py | <p>SEAM.S3.LOCALITY.01: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460’s verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 <math>p+ip</math> collar. [E] CHERN INTEGER: the FHS Chern is <math>C=+1</math> (chiral <math>M=1</math>), 0 (trivial <math>M=3</math>), <math>-1</math> (opposite mass) — the invertible-phase invariant. [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre <math>\theta(k_y)</math> winds by <math> C =1</math> (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = <math> C =1 \neq 0</math> and <math>c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0</math> (the same 8 as <math>c_3</math>’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0, <math>C=0</math>; since <math>C=1 \neq 0</math> NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460’s cited verdict. SEAM.EQUIV.01 stays [O] (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, SeamResidualAxiom.lean). [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v462_seam_spinor_continuum.py          | <p>SEAM.EQUIV.SPINOR.01: the 128-spinor extension <math>SO(16)_1 \rightarrow (E_8)_1</math> exhibited at CHARACTER level and as a finite-<math>L \rightarrow</math> continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. [E] <math>(E_8)_1</math> TOWER FROM <math>SO(16)_1</math> SECTORS: the exact Jacobi/<math>E_8</math> identity <math>\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4</math> gives <math>\chi_{(E_8)_1} = E_4/\eta^8 = \frac{1}{2}[(\theta_3/\eta)^8 + (\theta_4/\eta)^8] + \frac{1}{2}(\theta_2/\eta)^8 = \chi_o^{SO16} + \chi_s^{SO16}</math> — the holomorphic <math>(E_8)_1</math> character IS the <math>SO(16)_1</math> vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL 120 <math>\rightarrow</math> 248: <math>\chi_o</math> gives weight-1 multiplicity 120, <math>\chi_s</math> gives 128, summing to the exact <math>(E_8)_1</math> tower <math>\{1, 248, 4124, 34752\}</math> — the spinor currents ARE the <math>(E_8)_1</math>–<math>SO(16)_1</math> deficit (248=120+128). [E] FINITE 16-MAJORANA FOCK: <math>C(16, 2)=120</math> NS bilinear (<math>so(16)</math>) currents and <math>2^{16/2}=256</math> Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite <math>N</math>. [E] FINITE-<math>L \rightarrow</math> CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir <math>E_0(L) = e_\infty L - \pi v c / (6L)</math> fits <math>c_{\text{Dirac}} \rightarrow 1</math> (<math>c_- = 8</math>) and the lowest scaled NS gap <math>x_1 \rightarrow 1</math> with <math> x_1 - 1  \sim 1/L^2</math> (<math>L=64..1024</math>) — the lattice flows to the conformal tower. [C][O] the rigorous holomorphic extension (<math>\chi_o + \chi_s</math> simple-current projection) is supplied by AGT/AMT (v459); SEAM.EQUIV.01's continuum existence (v336) stays [O]. Mixed exact (<math>\theta</math> identity + tower + 120=<math>C(16, 2)</math>, 128=<math>2^7</math>, 248=120+128, Wolfram-mirrored) + numerical (the finite-<math>L</math> convergence, Python-only). [E][C][O]</p> |
| v463_seam_c8_holomorphic_uniqueness.py | <p>SEAM.EQUIV.UNIQ.01: the <math>c=8</math> holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of SEAM.EQUIV.01 classification-forced rather than assumed (B of the A+B round). [E] <math>c=8</math> NOT UNIQUE AT LEVEL 1: <math>c = \dim \mathfrak{g} / (1 + h^\vee) = 8</math> has THREE simple solutions — <math>A_8</math> (dim 80), <math>D_8 = \mathfrak{so}(16)</math> (dim 120), <math>E_8</math> (dim 248); <math>c=8</math> is necessary but NOT sufficient (the v457 <math>\det K=4</math> <math>SO(16)</math> rival is a real gleich-<math>c</math> competitor). [E] HOLOMORPHY FORCES <math>\dim V_1=248</math>: the only weight-0 form with leading <math>q^{-1/3}</math> and integer coefficients is <math>j^{1/3} = E_4/\eta^8</math>, whose <math>q^1</math> coefficient (the 1-dim candidate space, vacuum <math>a_0=1</math>) is 248 — forced, not chosen. [E] <math>\dim V_1=248 + \text{level-1 } c=8 \Rightarrow E_8</math> UNIQUELY (248/(1+30)=8); the gleich-<math>c</math> rivals <math>A_8</math> (80) and <math>D_8</math> (120) are excluded. [E] the <math>(E_8)_1</math> tower <math>\{1, 248, 4124, 34752, 213126\}</math> are the <math>E_4/\eta^8</math> coefficients, and <math>E_4/\eta^8 = j^{1/3}</math> (numeric <math>&lt; 10^{-25}</math>). [C] Dong–Mason/Schellekens: the holomorphic VOA at <math>c=8</math> is UNIQUE = <math>V_{E_8}</math>, and the even unimodular rank-8 lattice is unique = <math>E_8</math> (Mordell–Witt) — so “<math>c=8</math> holomorphic limit = <math>(E_8)_1</math>” is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of SEAM.EQUIV.01 is [C] conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. SEAM.EQUIV.01 stays [O]. Exact (sympy/integer <math>q</math>-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. [E][C][O]</p>                                                                                                                                                                                   |

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| Script                                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v464_seam_<br>oneparticle_<br>rigidity.py | SEAM.EQUIV.ONEPARTICLE.01: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of SEAM.EQUIV.01 (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol $P$ , and R1 collapses to a finite-per-mode statement about $P$ . [C] ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state $\leftrightarrow$ its symbol $P$ ( $0 \leq P \leq 1$ ) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P =$ projection onto $K < 0$ ( $P^2 = P$ , hermitian to $< 10^{-12}$ ), $K$ fixed by gap + chirality + reflection (v426) — no second realisation. [E] ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ( $\max  P_{2N} - P_N  \sim 1/N \rightarrow 0$ ), so $P_\infty$ exists (Gaussianity is why a computation reaches it). [E] THE LIMIT IS $c=1$ DIRAC ( $c_-=8$ ): entanglement $S(L) = (c/3) \ln L$ gives $c \rightarrow 1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies $\rightarrow c_-=8$ . [C] Shale–Stinespring: gauge-invariant $\Rightarrow$ no pairing block $\Rightarrow$ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to “the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor”; with v463 the SEAM.EQUIV.01 residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. SEAM.EQUIV.01 stays [O]. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement $c$ -fit). [E][C][O] |
| v465_seed_<br>crosssector_<br>joint.py    | SEED.CROSSSECTOR.01: the cross-sector one-parameter seed test — the $\theta_{13}$ -independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle $\beta$ and a quark Cabibbo angle $\lambda_C =  V_{us} $ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom $\varphi_0$ reproduces the frozen $\lambda_C$ , $\sin^2 \theta_{12}$ , $\beta$ to $10^{-9}$ . [E] CROSS-SECTOR (headline): back-solving $\varphi_0$ <i>independently</i> from $\beta$ (CMB) and $\lambda_C$ (quark) gives the <i>same</i> seed within $0.01\sigma_{\text{comb}}$ — a correlation only a shared-origin theory predicts. [E] one-parameter fit: the error-weighted mean of the three back-solved $\varphi_0$ reproduces the axiom to 0.06%; the joint $\chi^2$ at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\text{dof} < 1$ , combined pull $0.08\sigma$ . [E] $\theta_{13}$ EXCLUSION declared, not cherry-picked: adding $\sin^2 \theta_{13}$ raises $\chi^2$ by $\sim 4$ (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is <i>deliberately</i> $\theta_{13}$ -independent. [E] negative control: a data $\leftrightarrow$ formula shuffle explodes the cluster $\chi^2$ by $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306); $\varphi_0$ is FIXED by the axioms, so it upgrades <i>no</i> prediction. Python-only. [E]                                                                                                                                                                                                                                                                                                                                    |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v466_seed_leptonmass_channel.py | SEED.LEPTONMASS.01: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$ , extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. [E] plumbing: the axiom $\varphi_0$ reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to $10^{-9}$ . [E] NEW SECTOR (headline): back-solving $\varphi_0$ from the measured POLE ratio $m_e/m_\mu = 0.004836$ (PDG) gives $\varphi_0 = 0.053113$ , only $-0.11\%$ from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$ , v393) but largely CANCELS for $m_e/m_\mu$ , so the $0.11\%$ residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. [E] EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\text{dof} < 1$ and the error-weighted mean stays at the axiom to $< 0.1\%$ ; with the corroborating heavy-quark $m_t/m_b$ ( $-0.23\%$ ) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). [E] negative control / power: a data↔formula shuffle explodes the cluster $\chi^2$ by $> 100\times$ ; and not every mass ratio qualifies — $m_c/m_s = (34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. [C] HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; $\varphi_0$ is FIXED by the axioms, so it upgrades <i>no</i> prediction and closes <i>no</i> gate. Python-only. [E][C] |
| v467_thirdgen_mixing_pattern.py | FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions ( $\theta_{13}, \theta_{23}, V_{cb}$ ) are ONE common $-\varphi_0^{\text{ret}}$ suppression ( $-5.36\% \pm 1.67\%$ , $3.2\sigma$ non-zero, $-0.03\sigma$ from $-\varphi_0^{\text{ret}}$ ); the ladder-base forms $\lambda_C^2 e^{-5/6} / (1 - \varphi_0^{\text{ret}})/2 / \lambda_C^2 / (1 + \lambda_C)$ drop the joint $\chi^2$ $10.4 \rightarrow 0.11$ with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v468_dm2_ratio_jarlskog.py      | FLAV.DM2RATIO.01: $\Delta m_{21}^2 / \Delta m_{31}^2 =  J_{\text{PMNS}} $ of the frozen matrix = $0.029653$ vs measured $0.029805(792) \rightarrow -0.19\sigma$ — a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\text{ret}}$ heuristic is now $-2.44\sigma$ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| v469_seam_<br>crossedproduct_<br>route.py | <p>SEAM.EQUIV.CROSSEDPRODUCT.01: the net-level crossed-product certification of the 128-spinor extension + the <math>R1 \rightarrow R1'</math> invariant-level reduction. <b>[E]</b> LR LOCALITY INTEGER (index-2): the <math>SO(16)_1</math> spinor has <math>h_s = N/16 = 16/16 = 1 \in \mathbb{Z} \Rightarrow</math> statistics phase +1 — the Longo–Rehren criterion HOLDS, so the <math>\mathbb{Z}_2</math> simple-current crossed product <math>SO(16)_1 \rtimes \{1, s\}</math> is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 <math>so(N)</math> net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM <math>\mu(B) = 4/2^2 = 1 \Rightarrow</math> HOLOMORPHIC <math>\Rightarrow (E_8)_1</math> by the v463 pin (<math>E_4/\eta^8</math> <math>q^1 = 248</math> re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. <b>[E]</b> INDEX-4 PARALLEL: the <math>\mu_4</math> glue <math>J = (s, \Lambda_1)</math> has <math>h(J^k) = \{1, 1, 1\}</math> ALL integer (<math>\frac{5}{8} + \frac{3}{8} = 1</math>; <math>\frac{1}{2} + \frac{1}{2} = 1</math>) — the v125 isotropy <math>q(k, k) = k^2</math> at net level; <math>\mu = 16/4^2 = 1</math>, the same endpoint. <b>[E]</b> <math>R1'</math> INVARIANTS COMPUTED: per-copy FHS Chern <math>C(M=1) = 1</math> vs <math>C(M=3) = 0</math>, <math>\nu = 16 \times  C  = 16</math>, <math>c_- = \nu/2 = 8 = g_{\text{car}} + N_{\text{fam}}</math>; <math>\nu \bmod 16 = 0</math> is EXACTLY Kitaev’s class-D phase whose edge admits a purely bosonic description = the <math>(E_8)_1</math> level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). <b>[E]</b> THE CHAIN COMPOSES: T1 (phase classification: Kitaev + <math>c_-</math> invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) <math>\rightarrow</math> T2 (MMST CMP 397 (2023) + Böckenhauer 1996) <math>\rightarrow</math> T3 (LR/BE + KLM CMP 219 (2001)) <math>\rightarrow</math> T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). <b>[C]</b> <math>R1 \rightarrow R1'</math>: collar_realizes (model-level fiat) replaceable by collar_invariants — {quasi-free <b>[C]</b> (v155/v160), gap <b>[E]</b> (v302), class D, <math>c_- = 8</math> <b>[E]</b> (v456, from P1)}; quasi-freeness stays the ONE <b>[C]</b> hypothesis; Lean mirror FORM.SEAM.RESIDUAL.01. <b>[O]</b> NOT CLOSED: the cited theorems stay cited (certification, not re-proof); SEAM.EQUIV.01 stays <b>[O]</b>. Mixed exact (weights/<math>\mu</math>/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). <b>[E][C][O]</b></p> |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v470_alpha_inflow_level.py   | <p>ALPHA.QUILLEN.INFLOW.01: the <math>\alpha^3</math> level = the COMPUTED bulk Chern invariant + the seam <math>F</math>-normalisation = the affine embedding index <math>k_Y=5/3</math>. [E] TARGET RE-VERIFIED: <math>\alpha^{-1}=137.0359992168</math> unique root; the Quillen split <math>a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})</math> at the root to <math>&lt;10^{-35}</math> (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern <math>C(M=1)=1</math>, <math>C(M=3)=0</math> — the same integers as v367/v461, here read as the quantised <math>U(1)</math> response of the collar phase. [C] <math>k_0= C =1</math> UNDER THE EM1 READING: the <math>a^3</math> coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — “why exactly 1” is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) <math>\eta=\text{CS}</math> part of <math>\delta\log\det</math>; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435’s minimality assumption. [E] <math>k_Y=5/3</math> EXACT: <math>\text{tr}_5(Y^2)/\text{tr}(T_3^2)=(5/6)/(1/2)=5/3</math> (Ginsparg, Phys. Lett. B197 (1987)); <math>(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1</math> — the “GUT 3/5” IS the inverse embedding index, not an imported convention; carrier decomposition <math>\frac{41}{6}=\frac{20}{3}</math> (fermions, <math>3\times 16</math>-content) <math>+\frac{1}{6}</math> (Higgs doublet). [C] <math>F</math>-NORMALISATION RETYPED: once the seam net is <math>(E_8)_1</math> level 1 (SEAM.EQUIV.01), the <math>U(1)_Y</math> normalisation <math>\langle J(z)J(w)\rangle=k/(z-w)^2</math> is FIXED by <math>k_Y</math> — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of SEAM.EQUIV.01, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — <math>c_-=16\times\frac{1}{2}\times C =8=g_{\text{car}}+N_{\text{fam}}</math> (gravitational, feeds SEAM.EQUIV.01/S3, v456) and <math>C=1</math> (<math>U(1)</math>, feeds EM1) — “the one theorem and the one number” are two faces of the SAME phase; closing R1’ (v469) feeds both. [O] NOT CLOSED: the bridge lemma “<math>\delta\log\det_c(\text{seam})</math> = the inflow response” stays the named cited step; the EM1 reading stays [C]; ALPHA.QUILLEN.EXACT.01 stays [O]; <math>\alpha^{-1}</math> stays [E] regardless. Mixed exact (<math>k_Y/b_1</math> identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O]</p> |
| v471_seam_horizon_replica.py | <p>SEAM.EHMODEL.04: the Seam–Horizon replica chain exercised NUMERICALLY on a discretized seam collar — the kernel-identification premise of SEAM.THEOREM.01 (the residual isolated by v150/v151/v152) discharged at the FINITE/MODEL level for the seam’s own kernel, with REAL replica sheets. [E] REPLICA<math>\Rightarrow</math>EH FORM on a real operator: <math>dG/d\ln m = 2C(\gamma)</math> on deficit cones <math>\gamma/2\pi = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}</math> (0.01–0.05%) AND replica sheets <math>n=2</math> (0.8%), <math>n=3</math> (1.5%). [E] COEFFICIENT CUTOFF-INDEPENDENT (0.5% drift) while the <math>\zeta</math>-scale <math>\mu_{\text{lat}}</math> drifts with the discretization = the v152 anchor exhibited on data; NO canonical <math>\ln(m/\mu)=\frac{3}{4}</math> emerges (recorded honestly). [E] BFK SPLIT MEASURED ON THE KERNEL: Schur identity exact (<math>\sim 10^{-16}</math>); the Calderón/Schur factor’s deficit slope is a PURE CUT-EDGE law <math>E(1-q)</math> with tip term 0 (power check: the same law fails on the full determinant); the halves carry the whole tip = <math>2C_{\text{cone}}</math> via Kac doubling (exact). [E] THE SEAM’S OWN SPECTRUM: the v302 transfer masses <math>m_2=6\ln\frac{3}{2}</math>, <math>m_3=6\ln 3</math> sit ON the calibrated EH line (<math>\leq 0.05\%</math>); the Perron/attractor mode (<math>m=0</math>) is IR-DIVERGENT (runs exactly as <math>-2C\ln(\text{IR})</math>) — the recovery gap <math>\Delta = 6\ln\frac{3}{2}</math> makes the induced-gravity coefficient finite: the same gap that makes the attractor unique makes <math>1/G</math> finite. [O] HONEST SCOPE: finite/discretized level; SEAM.THEOREM.01 stays open — the residual retypes to the CONTINUUM scaling limit (MMST class, = the single residual of SEAM.EQUIV.01, v336) + the v152 anchor. Numerical (sparse lattice determinants), Python-only by nature. [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| v472_quillen_<br>detline_moduli.py | <p>ALPHA.QUILLEN.DETLINE.01: the DETERMINANT LINE over the <math>U(1)</math>-twist moduli of the collar model carries curvature = the inflow level — the finite Quillen/Dai–Freed shadow of the v470 bridge lemma, COMPUTED. [E] BLOCH LEVEL RE-VERIFIED: BZ Chern <math>C(M=1)=1</math>, <math>C(M=3)=0</math>, <math>C(M=-1)=-1</math> (the v470/v367 integers). [E] THE QUILLEN OBJECT COMPUTED: the same collar <math>h(k)</math> in real space on an <math>L \times L</math> torus with TWISTED boundary conditions — the twist torus <math>(\theta_x, \theta_y)</math> IS the moduli space of flat <math>U(1)</math> connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied one-particle frame (the free-fermion ground state) over that torus = 1 exactly (<math>10^{-9}</math>) at <math>M=1</math>, for <math>L=4</math> AND <math>L=6</math> (size-independent, as an integer must be). [E] CONTROLS: trivial collar <math>M=3 \rightarrow 0</math>; orientation flip <math>M=-1 \rightarrow -1</math>. [E] TWO TORI, ONE LEVEL: the twist-moduli integer equals the Bloch-BZ integer for all three <math>M</math> — the Niu–Thouless–Wu identification (PRB 31 (1985)), exhibited on the collar model that realises S3 (v460/v461). [E] SECTION GLOBALLY DEFINED: the Fermi gap (= 2.0) stays open over the WHOLE twist torus — the Dai–Freed section has no zero, the holonomy reading is globally valid. [C] THE READING: det-line holonomy over the <math>U(1)</math> moduli = inflow level <math>k_0= C =1</math> — the v470 bridge lemma “<math>\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}</math>” holds VERBATIM at the finite/model level, where <math>\log \det</math> of the occupied frame replaces <math>\log \det_\zeta</math> (Quillen 1985; Bismut–Freed CMP 106 (1986); Dai–Freed JMP 35 (1994); FHS JPSJ 74 (2005)). [O] NOT CLOSED: the abstract-seam <math>\zeta</math>-determinant identification (the continuum leg, = the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O]; <math>\alpha^{-1}</math> stays [E] regardless. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. Numerical (dense eigh + FHS twist grid), Python-only by nature. [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v473_entropic_<br>action_bridge.py | <p>GRAV.ENTROPIC.ACTION.01: the entropic-action bridge — Bianconi’s <i>Gravity from entropy</i> (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an EXTERNAL CANDIDATE for the local volume action behind TFPT’s parameter-free Einstein equation (v358/v359, honestly typed “equation of state, not a from-action quantisation” [O]). Her <math>S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})</math> is the quantum relative entropy between the spacetime metric and the matter-induced metric (Araki-related, <math>\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}</math>), a local covariant action whose variation gives Einstein + emergent nonnegative <math>\Lambda</math> at low coupling — the missing variational layer IF the bridge holds. [E] CARRIER HODGE COUNT: her <math>\Omega^{0,1,2}</math> fiber on the FIVE-SLOT carrier <math>\mathbb{C}^5</math> (not 4d spacetime, where it is <math>1+4+6=11</math>) counts <math>1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+</math>, Hodge-folded onto <math>\Lambda^{\text{even}}\mathbb{C}^5</math> (vector-space identity ONLY). [E] COEFFICIENT MATCH (the one quantitative pin): her weak-coupling <math>\mathcal{L} = 3\beta R - \alpha\mathcal{L}_m</math> matched to <math>1/(16\pi G) = c_3/(2G)</math> fixes her free constant EXACTLY, <math>\beta'_B = c_3/6 = 1/(48\pi)</math>; the <math>6 = 3 \text{ blocks} \times 2 = p_2 =  R^+(A_3) </math> (structural echo, not a proof). [E] ENTROPY POTENTIAL: <math>\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G}) \geq 0</math>, quadratic at the fixed point <math>\Rightarrow</math> with <math>\varepsilon_\star = e^{-\alpha^{-1}Q}</math> the closed v60 branch <math>(3/4\pi^2)e^{-2\alpha^{-1}}</math> is reproduced with the EXACT target <math>\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3} \delta_{\text{top}}</math> ([C] until <math>Q</math> is constructed). [E] FRULLANI: <math>-\ln A = \int_0^\infty \frac{dt}{t} (e^{-tA} - e^{-t})</math> — her log action is an all-scale heat-kernel action, the right genre against <math>S_{\text{rel},\chi}</math> (KMS cutoff). [E] <math>R^2</math> KILL TEST: raw <math>(c_3/6)^2</math> vs Starobinsky <math>1/(12c_3^7)</math> (<math>M_{\text{scal}}^2 = c_3^7 \bar{M}_{\text{Pl}}^2</math>, v36) — EXACT gap <math>3(8\pi)^9 \approx 1.2 \times 10^{13}</math> (13 orders): a direct identification of the raw entropic action with the TFPT <math>R^2</math> sector is FALSE without a mechanism (light <math>\mathcal{G}</math>-trace mode generating <math>c_3^{-9}</math>, or KMS-spectral renormalisation) — pre-registered. [C] COMPRESSION CONJECTURE: <math>P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2} + \text{the operator-level Hodge items (Clifford, Dirac, leaf involution, Calderón polarisation) stay open; Lorentzian positivity NOT automatic — TFPT’s OS route is the sturdier construction. [O] NOTHING CLOSES: v359’s equation-of-state status stays [O]; zero TFPT knobs added. Mixed exact (sympy, Wolfram-mirrored) + one quadrature (mpmath). [E][C][O]</math></p> |

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| v474_entropic_hodge_carrier.py | <p>GRAV.ENTROPIC.HODGE.01: the OPERATOR level of the entropic-action bridge (work packages 1 + 4-algebra of v473). [E] CAR⇒CLIFFORD: the ten <math>\Gamma</math>'s from <math>(a_k^\dagger, a_k)</math> on the carrier Fock space <math>\Lambda^\bullet \mathbb{C}^5</math> (<math>\dim 2^{g_{\text{car}}}=32</math>) satisfy <math>\{\Gamma_a, \Gamma_b\}=2\delta_{ab}</math> exactly (all 100 pairs, integer matrices) — the <math>\mathfrak{so}(10)=D_5</math> Clifford algebra acts on the carrier exterior algebra. [E] <math>S^+=\Lambda^{\text{even}}</math> AS <math>D_5</math>-MODULE: the 45 bilinears <math>[\Gamma_a, \Gamma_b]/4</math> commute with <math>\gamma=(-1)^N</math>, span exactly <math>45=\dim \mathfrak{so}(10)</math> and leak NOTHING from the 16-dim even subspace — the half-spinor IS the even exterior algebra WITH its spinor structure (operator level of v2/v44/v197). [E] HODGE-DIRAC SYMBOL: <math>c(\xi)=a^\dagger(\xi)-a(\xi)</math> satisfies <math>c(\xi)^2=- \xi ^2 \text{Id}</math> for SYMBOLIC <math>\xi</math> — Bianconi's <math>D=d+\delta</math> is Clifford multiplication at symbol level. [E] THE FOLD IS <math>5\rightarrow\bar{5}</math>: the traceless <math>\mathfrak{u}(5)</math> weights of <math>\Lambda^4</math> are the NEGATIVES of those of <math>\Lambda^1</math>, so the Hodge fold <math>\star: \Lambda^1\cong\Lambda^4</math> conjugates the vector block — her fiber <math>1+5+10</math> becomes the GUT half-spinor <math>16=1+\bar{5}+10</math> (the SM carrier decomposition), not a mere dimension match. [E] HONEST OBSTRUCTION: <math>\Omega^{\leq 2}</math> is NOT Clifford-invariant (<math>c(\xi)</math> maps <math>\Lambda^2</math> into <math>\Lambda^3</math>); only <math>\Lambda^{\text{even}}</math> is (even algebra) — the bridge must go through the fold, never through degree truncation (constraint on AP2). [E] Q-TARGET ENUMERATION (the v473 conjecture DECIDED): <math>\text{Tr } Q^2=d k^2 c_3^4=32c_3^4</math> with INTEGER <math>k</math> has exactly the supports <math>(d,k)=(2,4), (8,2), (32,1) = \{ \mathbb{Z}_2 , g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8, 2^{g_{\text{car}}}=\dim \Lambda^\bullet\}</math> — minimal uniform solution <math>q=c_3^2=1/(64\pi^2)</math> per Fock mode; the two-form block <math>d=10</math> (the “pair sector” conjecture) and the fiber <math>d=16</math> need irrational multipliers <math>\sqrt{16/5}, \sqrt{2} \Rightarrow</math> KILLED in uniform form; bonus identity <math>\text{Tr } Q^2/\delta_{\text{top}}=32/48=2/3=\dim \Lambda^\bullet/\Omega_{\text{adm}}</math>. [C] the physical choice among the three admissible supports; [O] AP2 compression + <math>R^2</math> mechanism stay open, no marker moves. Mixed exact: integer Fock matrices (numpy, exact in double) + symbolic sympy (weights/enumeration Wolfram-mirrored). [E][C][O]</p> |
| v475_entropic_scalaron_gate.py | <p>GRAV.ENTROPIC.SCALARON.01: the <math>R^2</math> kill test of the entropic-action bridge EXECUTED (work package 5 of v473) + the explicit Lorentzian-positivity witness. [E] MAX-SYMMETRIC EIGENVALUES (her Appendix-B flattening verified): <math>\tilde{R}\tilde{g}^{-1}</math> has eigenvalues EXACTLY <math>\{R; R/4 \times 4; R/6 \times 6\}</math> on a general diagonal 4d metric, incl. the <math>6\times 6</math> two-form flattening. [E] EXACT VACUUM <math>f(R)</math>: <math>\mathcal{L}_B(R)=-[\ln(1-\beta R)+4\ln(1-\beta R/4)+6\ln(1-\beta R/6)]=3\beta R+\frac{17}{24}\beta^2 R^2+O(R^3)</math> — the 3 re-derives her Eq. (45) (= the v473 pin), <math>\frac{17}{24}=\frac{1}{2}(1+\frac{1}{4}+\frac{1}{6})</math> is the EXACT tensorial <math>R^2</math> factor. [E] RAW SCALARON TRANS-PLANCKIAN: with <math>\beta'=c_3/6</math> both routes (Starobinsky matching AND <math>f'/(3f'')</math>) give <math>m_{\text{raw}}^2=\frac{4608\pi^2}{17}M_{\text{Pl}}^2</math>, <math>m_{\text{raw}}\approx 51.7 M_{\text{Pl}}\approx 1.3\times 10^{20}</math> GeV — HEAVY, not light (TFPT: <math>c_3^{7/2}M_{\text{Pl}}=3.06\times 10^{13}</math> GeV); required enhancement EXACTLY <math>m_{\text{raw}}^2/m_{\text{TFPT}}^2=\frac{72}{17}(8\pi)^9\approx 1.69\times 10^{13}</math> in mass<sup>2</sup> — v473's interpretation 2 (the TFPT scalaron as the light <math>\mathcal{G}</math>-trace mode) is KILLED in naive form. [E] DOMAIN: first log pole at <math>R=1/\beta=384\pi^2 M_{\text{Pl}}^2</math> (trans-Planckian) — the raw action is well-defined at all sub-Planckian curvatures; the obstruction is the MASS, not the domain. [E] LORENTZIAN WITNESS: a timelike gradient gives <math>Gg^{-1}</math> eigenvalue <math>1-\alpha v^2\leq 0</math> for <math>v^2\geq 1/\alpha</math> (ln undefined); spacelike/Euclidean positive for all <math>v</math> — the v473 [C] caveat exhibited, the OS/reflection-positive route sturdier. [C] VERDICT: interpretation 1 survives (the entropic action covers the Einstein + <math>\Lambda</math> level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving <math>R^2</math> route. [O] unchanged: AP2 compression + the <math>R^2</math> gate; no marker moves. Exact (sympy, Wolfram-mirrored). [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v476_entropic_compression_toy.py | <p>GRAV.ENTROPIC.COMPRESS.01: the AP2 compression conjecture of the entropic-action bridge (v473) made WELL-POSED on an exactly solvable model. Numerical facts on a staggered-mass chain: the ground-state covariance is a PROJECTION (pure bulk, thermal boundary — the seam situation); the compression <math>C_A=PCP</math> has spectrum strictly inside <math>(0,1)</math> with finite Peschel <math>K=\ln((1-C_A)/C_A)</math> and <math>S&gt;0</math> (v155 in the bridge context). ORDER OBSTRUCTION (the reading DECIDED): on the pure bulk <math>f(x)=x/(1-x)</math> is SINGULAR on <math>\text{spec}(C)=\{0,1\}</math> — the literal operator-side reading <math>Pf(C)P</math> of the v473 conjecture is ILL-POSED while <math>f(PCP)</math> is finite: the compression identity can only be read STATE-SIDE (build <math>\Delta_\Sigma</math> from the COMPRESSED relative metric), matching Bianconi's own local-metric construction — AP2 retyped, not weakened. [E] BLOCK ALGEBRA (symbolic): <math>[C^2]_{AA}-(C_{AA})^2=BB^t</math> and <math>[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t</math> — first order in the cross block vanishes identically, so for ANY analytic <math>f</math> the two readings agree to SECOND order: the modular (entanglement) content lives exactly in that quadratic correction. GAP CONTROL at the bounded (covariance) level: the bulk identity <math>f(C_T)=e^{-h/T}</math> holds to <math>10^{-11}</math> relative; the mismatch obeys the quadratic law <math>d \leq x^2</math> and falls with the gap in the gapped regime, the decoupled cut agrees EXACTLY — the two readings converge in the gap-dominated regime, where TFPT operates (<math>\Delta_{\text{eff}}&gt;0</math>, v76/v337). [C] the retyped conjecture; [O] AP2 stays open (the continuum seam statement); no marker moves. Numerical (numpy lattice + one symbolic block identity), Python-only by nature. [C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v477_entropic_scaleflow.py       | <p>GRAV.ENTROPIC.SCALEFLOW.01: the scale-flow representation of the entropic action — v475's surviving interpretation 3 typed PRECISELY: the <math>(72/17)(8\pi)^9</math> gap collapses to ONE moment condition on the scale weight, satisfied by the SAME KMS moment TFPT already fixes (v36: <math>f_0=6(4\pi)^2/c_3^7</math>); zero new dials, nothing closes. [E] SCALE-FLOW IDENTITY: <math>-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}-e^{-t})</math> (symbolic) <math>= 2 \int_0^\infty \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})</math> (<math>t=1/x^2</math>, <math>10^{-20}</math> quadrature) — Bianconi's log action IS the FLAT scale-integral of relative heat-kernel actions; TFPT's <math>S_{\text{rel},\chi}</math> evaluates ONE KMS scale with a weight <math>f</math>: the bridge is a choice of SCALE MEASURE. [E] MOMENT DICTIONARY: a weight <math>w(t) dt/t</math> gives <math>3\beta R \mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2</math> (<math>\mu_1 = \int w e^{-t} dt</math>, <math>\mu_2 = \int w t e^{-t} dt</math>); the flat weight has <math>\mu_1=\mu_2=1</math> = the v475 raw coefficients — the raw action is the unit-moment point of a moment family. [E] THE ONE MOMENT CONDITION: with the weight-dressed EH pin <math>\beta' \mu_1=c_3/6</math> the scalaron is <math>m^2=\frac{4608\pi^2}{17}\frac{\mu_1^2}{\mu_2}M_{\text{Pl}}^2</math>; demanding <math>m^2=c_3^7 M_{\text{Pl}}^2</math> forces EXACTLY <math>\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9</math> — the v475 kill-test number IS a moment ratio; and the closure identity <math>\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7</math> holds IDENTICALLY: the 13 orders are a scale-measure datum, not a bridge mismatch. [E] v36-CONSISTENCY: <math>f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/M_{\text{Pl}}^2=c_3^7</math> — the same mass from both parametrisations, ZERO new dials. [C] a CONSISTENCY statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + <math>\Lambda</math> level and the scale-flow FORM, TFPT supplies the measure); negative side-record: the exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain (kept out of the suite, logged). [O] unchanged: the <math>R^2</math> gate and v359's typing. Exact (sympy + one quadrature, Wolfram-mirrored). [E][C][O]</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v478_entropic_continuum_legs.py  | <p>GRAV.ENTROPIC.LEGS.01: first computable steps on the TWO remaining legs of the entropic-action bridge (v473–v477). LEG A (AP2 continuum): the compressed critical interval has <math>S(L)=(c/3)\ln L+k</math> with <math>c_{\text{est}}=1.00089/1.00038/1.00021</math> (monotone <math>\rightarrow</math> the Dirac <math>c=1=2\times\frac{1}{2}</math>, the v450 reading; gapped control <math>\sim 6\times 10^{-8}</math>, area law); the Peschel modular Hamiltonian <math>K=\ln((1-C_A)/C_A)</math> organises on the CHM weight <math>\beta(x)=x(L-x)/L</math> (corr 0.87<math>\rightarrow</math>0.99, plateau drift shrinking; constants deliberately NOT asserted — the lattice EH is only asymptotically local); even-range bands vanish identically (particle-hole, <math>\sim 10^{-36}</math>), odd tail 1.8% parabola-enveloped — the state-side <math>\Delta_\Sigma</math> (v476’s forced reading) FLOWS TO the CHM/BW geometric form that TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and TFPT’s gravity side MEET in the continuum limit. LEG B (the measure): [E] single-scale corollary — <math>d\mu=\delta(t-t_0)dt</math> gives <math>\mu_2/\mu_1^2=e^{t_0}</math> exactly, so the v477 condition forces <math>t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461</math> — ONE dimensionless KMS time, exact closed form; DECLINED: <math>t_0\neq h(E_8)=30</math> (difference 0.461; the near-miss is rejected per the v354/v355 no-free-pattern rule); deriving the measure from the entropic structure stays OPEN. [C] the meeting point; [O] both legs, no marker moves. High-precision numerical (mpmath, 50 digits) + one exact corollary, Python-only by nature. [C][O]</p>                                                                                                                                                                                                                                    |
| v479_kronheimer_quiver_bridge.py | <p>SEAM.KRONHEIMER.01: the R3 “combinatorial-to-geometric bridge” of the <math>\det K=1</math> keystone (v344 — “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM: Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685), with ALL finite input data already compiler outputs. [E] MARKS = NULL = PERRON: <math>C_{\text{aff}}\delta=0</math> and <math>A\delta=2\delta</math> on the affine-<math>E_8</math> McKay graph; multiset <math>\{1, 2, 2, 3, 3, 4, 4, 5, 6\}</math> = the <math>2I</math> irrep dimensions = the v312 rewrite-attractor Perron vector. [E] DIMENSION COUNT: <math>\dim_{\mathbb{R}} M - 4\dim_{\mathbb{R}} G = 4(\sum_{\text{edges}} d_i d_j - \sum d_i^2 + 1) = 4</math> — the hyper-Kähler quotient of the marks quiver is 4-real-dimensional (the <math>\mathbb{C}^2/\Gamma</math> geometry); <math>\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120</math> forced by the null-vector identity. [E] GAUGE DATUM: <math>\sum \delta_i^2 = 120 =  2I  =  \mu_4  h(E_8)</math>, <math>h(E_8)=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}</math>. [E] RESOLUTION: <math>8=\text{rank } E_8</math> exceptional spheres with intersection form <math>-\text{Cartan}(E_8)</math>, <math>\det=1</math> and SNF = identity (the Poincaré-sphere link = the <math>\det K=1</math> face, v232/v344); deformation space <math>\mathfrak{h} \otimes \mathbb{R}^3</math>, <math>\dim 3 \cdot 8 = 24</math>. [C] the theorems are CITED, not re-proved (the same import class as NPW26/MMST/AGT, v424/v336/v459). [O] the residual premise TRANSFORMS, it does not vanish: “the raw seam supplies ALE hyper-Kähler data asymptotic to the <math>(2, 3, 5)</math> cone” = the SAME single arrow as SEAM.EQUIV.01 (v346 L2); no gate closes. Exact (sympy; Wolfram mirror added, counted with the verified 2026-07-21 engine run). [E][C][O]</p> |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v480_multilocal_<br>four_interval.py | <p>QGEO.MULTILOCAL.01: the four <math>\mu_4</math> marks make the seam circle a FOUR-INTERVAL free-fermion geometry — the class whose multi-interval modular theory is EXACTLY KNOWN (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; KLM: holomorphic <math>\Leftrightarrow</math> trivial multi-interval inclusion, <math>\mu=1</math> — the SAME one bit as <math>\det K=1</math>). On the exactly solvable antiperiodic critical ring (<math>N=256</math>, <math>Q=64</math>, <math>\ell=48</math>): [E] BINARY CLOCK: the quarter-turn clock is a signed permutation with <math>\rho^4 = -1</math> EXACTLY — the fermionic <math>\mu_4</math> clock is the order-8 double-cover lift (the same <math>\mathbb{Z}_2</math>-lift pattern as <math>2I</math> over <math>I</math>). [E] EXACT SECTOR DECOUPLING: the primitive-8th-root character projectors resolve the identity and commute with the reduced covariance at <math>10^{-15}</math> (the Rehren–Tedesco diagonal 4-fold-cover isomorphism at covariance level). [E] COVER/TWIST: each sector kernel is the SINGLE-interval quarter-ring kernel with boundary twist <math>\lambda_k</math>; the union of sector spectra = the full 4-interval spectrum at <math>10^{-14}</math>. [E] CLOCK INVARIANCE MANIFEST: <math>\rho C \rho^{-1} = C</math> at <math>10^{-14} \Rightarrow [\rho, K]=0</math> by functional calculus — <math>\omega \circ \rho = \omega</math> HOLDS at the 4-interval state level; NEGATIVE CONTROL: displacing ONE interval by one site re-couples the sectors at <math>10^{-2}</math>. CONJUGATE-POINT MIXING (numerical): every cross-interval row of <math>K</math> peaks on <math> s'-s  \leq 2</math> at both sizes — the lattice shadow of the Casini–Huerta mixing term. [C] the continuum multi-interval theory is CITED; [O] the raw-seam premise (QGEO.SYM.01/SEAM.EQUIV.01) stays OPEN — mechanism exhibited, not closed. Numerical (numpy), Python-only by nature. [E][C][O]</p>                                                                     |
| v481_seesaw_<br>carrier_ladder.py    | <p>FLAV.NUSCALE.02: the absolute neutrino-mass scale under the CARRIER Yukawa normalisation <math>y_\nu = y_t</math> — FLAV.NUSCALE.01's two-parameter <math>(y_\nu, M_R)</math> trade-off collapses to ONE parameter and the required <math>M_R</math> lands INSIDE the compiler's own PS window; the nearby integer <math>c_3</math>-rung is explicitly DECLINED at 1-loop (v354/v355). CANDIDATE class (genre v467/v468); the frozen record (v84) is untouched and NO gate closes. CARRIER NORMALISATION (conjectured premise; named premise): one <math>SO(10)</math> 16 per family + minimal Yukawa sector (<math>\mathbf{10}_H</math> / PS (1, 2, 2)) <math>\Rightarrow y_\nu = y_t</math> at the matching scale — the trade-off collapses to <math>M_R</math> alone. INVERSION INSIDE THE WINDOW (numerical): the observed <math>m_3=0.0503</math> eV demands <math>M_R=9.3 \times 10^{13}</math> GeV, INSIDE <math>[M_{PS}, M_{GUT}] = [4.2 \times 10^{13}, 2.4 \times 10^{15}]</math> (v249), at <math>\log_{c_3}(\bar{M}_{Pl}/M_R) = 3.15</math> (<math>y_\nu=1</math> gives the structureless 2.58 — the old FLAV.NUSCALE.01 number). [X] INTEGER-RUNG HONESTY GATE: <math>M_R = c_3^3 \bar{M}_{Pl} = 1.53 \times 10^{14}</math> GeV predicts <math>m_3 = 0.030</math> eV (40% low at 1-loop; ADKLR <math>\kappa</math> run-down factor 0.794) — NEAR the rung (3.15 vs 3), NOT ON: the ladder pin is DECLINED; the decision computation is NAMED (2-loop SM + explicit PS threshold matching). FALSIFIABLE BAND (numerical): over the PS window <math>m_3</math> spans <math>[0.002, 0.115]</math> eV bracketing the observation; the <math>M_{PS}</math> end (<math>\Sigma \sim 0.124</math> eV) is already in tension with DESI <math>\sim 0.072</math> — cosmology cuts the window from below; the NO floor <math>\Sigma m_\nu = 0.059</math> eV stays the falsifiable edge (v272). [C] / [O] candidate class; the absolute scale stays formally open. Numerical (explicit 1-loop RG: gauge/<math>y_t/\lambda</math> up, ADKLR Weinberg-operator down), Python-only by nature. [C][O]</p> |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v482_seesaw_rung_<br>decision.py    | FLAV.NUSCALE.03: the decision computation NAMED by v481, EXECUTED in bracketed (assumption-robust) form — the integer rung $M_R = c_3^3 \bar{M}_{Pl}$ is EXCLUDED as an unstructured exact pin, conditional on the carrier premise $y_\nu = y_t$ . THE GAP: central $m_3(\text{rung}) = 0.0301$ eV vs observed 0.0503 — rescue factor $\times 1.670$ (+67%, 0.154 $c_3$ -rungs). BRACKETED EXCLUSION: with $m_t(\overline{\text{MS}}) \pm 3$ GeV ( $> 3\sigma$ ), $\alpha_3^{-1} \mp 0.10$ ( $> 3\sigma$ ), a $\pm 10\%$ band on the ADKLR $\kappa$ run-down and the PS-leg Yukawa beta scaled $\times [0.5, 1.5]$ , the largest single shift is $\times 1.10$ and the ALL-favourable combination reaches only $\times 1.165$ — far below $\times 1.670$ : the v481 1-loop miss is NOT an RG artifact. [C] NAMED ESCAPE, RECORDED AND DECLINED: only a third-generation Majorana STRUCTURE factor $r \sim 1.67$ revives the rung; post-hoc $r_{\text{needed}} = 1.6696$ sits 0.18% from $g_{\text{car}}/N_{\text{fam}} = 5/3$ — no mechanism forces a 3/5 Clebsch on $M_{R,3}$ today, so this is a named coincidence with a named missing mechanism (PS Majorana-sector Clebsch), DECLINED per v354/v355 (the $t_0 \sim 30$ pattern of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu = 0.059$ eV) never used the integer rung and STANDS; frozen record untouched; nothing closes. Numerical (explicit 1-loop RG + bracketed envelopes), Python-only by nature. [C][O][X]                                                                                                                                                                                                                                                                                                         |
| v483_pillowcase_<br>exact_traces.py | HYP.PHIO.GEOM.01: the GEOMETRIC side of the $\varphi_0^{\text{ret}}$ puncture target (HYP.PHIO.PUNCTURE.01/v408) is EXACT — every twisted/orbifold heat trace of the flat $\tau = i$ pillowcase is a $t$ -INDEPENDENT RATIONAL, so the $\pi^{-4}$ in $d_{\text{top}} = 48c_3^4$ cannot come from flat orbifold geometry at ANY order. [E] TWISTED TRACES EXACTLY 1: $\text{Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma(z \mapsto -z)$ , $\rho(z \mapsto iz)$ , $\rho^3$ at every tested $t \in [0.05, 3]$ (position-space Gaussian image sums / erf closed forms + 2d quadrature, residual $< 10^{-9}$ ; mode-space proof: only $n=0$ is $\gamma$ -diagonal) — $t$ -independent, no asymptotic tail. [E] ATIYAH–BOTT MATCH: $1 = 4 \times \frac{1}{4}$ ( $\det(1-d\sigma) = 4 =  \mu_4 $ ) and $1 = 2 \times \frac{1}{2}$ ( $\det(1-d\rho) = 2 =  \mathbb{Z}_2 $ ). [E] PILLOWCASE DATA EXACT: orbifold heat trace $= \frac{1}{2} \text{Tr}_{T^2} + \frac{1}{2}$ (contact term EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark, all $t$ ); clock-equivariant trace exactly 1 (all $t$ ). [E]/[O] THE REDUCTION: the analytic content of $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ must sit in the per-insertion COUPLING normalisation — $4 =  \mu_4 $ insertions of weight $c_3$ give $c_3^4$ , multiplicity $\Omega_{\text{adm}} = 48$ ; the bare $1/(2\pi)$ -propagator 4-cycle differs by the exact RATIONAL $\frac{3}{16}$ ; the ONE remaining [O] is the rule “per-mark weight = $c_3$ ” (seam zeta/Green normalisation, the same residual class as EM.WARD.02). The target narrows from “derive the term” to “derive the weight”; HYP.PHIO.PUNCTURE.01 stays open; no gate closes. Numerical verification of exact statements + sympy exact arithmetic; Python-only by nature. [E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v484_seam_contact_unit.py       | <p>SEAM.CONTACT.UNIT.01: the shared “<math>c_3</math> per boundary insertion” rule of the two open analytic targets (ALPHA.QUILLEN.EXACT.01 ladder <math>\{0, 3, 6\}</math>;<br/> HYP.PHIO.PUNCTURE.01 per-mark weight <math>c_3</math>) is ONE identity and NOT a new unknown: it is the KMS seam unit <math>2\pi = 1/(4c_3)</math> with <math>\frac{1}{4} = 1/ \mu_4 </math> (v239) — one bare boundary propagator ORBIT-AVERAGED over the four marks. [E] UNIT CHAIN: <math>\frac{1}{2\pi} = 4c_3 =  \mu_4 c_3 =  \mathbb{Z}_2 \chi(S^2)c_3</math>. [E] MECHANISM DERIVED ON THE SEAM CIRCLE: <math>G(d) = -\frac{1}{\pi} \ln  2 \sin(d/2) </math> (polylog identity, 40 digits) takes INTEGER multiples of <math>c_3 \ln 2</math> at the <math>\mu_4</math> separations — <math>G(\pi/2) = -4c_3 \ln 2</math>, <math>G(\pi) = -8c_3 \ln 2</math>, so <math>G_{\text{marks}} = -4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)</math> with <math>\text{spec} = \{4, -2, 0, -2\}</math>; the <math>\mu_4</math>-invariant (orbit-averaged) insertion carries <math>(GV) _{\text{marks}} = -(c_3 \ln 2) C</math> EXACTLY, so the log-det contact expansion is exactly graded in <math>c_3</math> per insertion with integer cycle combinatorics <math>\text{Tr}(C^k) = 4^k + 2(-2)^k</math>. [E] SUITE-WIDE GRAMMAR: the <math>F_{U(1)}</math> ladder <math>\{0, 3, 6\}</math> (v3); <math>\varphi_0^{\text{ret}}</math> ladder <math>\{1, 4\}</math> (v37); the <math>\Lambda</math> prefactor <math>\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2</math> (v60) — the SAME multiplicity <math>\Omega_{\text{adm}}=48</math> as the puncture, at TWO insertions; the <math>g-2</math> seam loop divides by the same <math>2\pi = 1/(4c_3)</math>. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching (<math>48 = \Omega_{\text{adm}}</math>, <math>41 = 10b_1</math>); the finite cycle sector is proven. Anti-numerology: the integers are forced by the square configuration; no target coefficient is claimed. Exact (sympy + 40-digit mpmath polylog); Python-only by nature. [E][C][O]</p>                                    |
| v485_contact_diagonal_closed.py | <p>SEAM.CONTACT.UNIT.02: the DIAGONAL channel of the merged analytic target (v484) settled at the computable level — the merged residual COLLAPSES ONTO THE KEYSTONE. [E] DIAGONAL ZERO AT THE KMS SCALE: the renormalised coincident-point Green function on circumference <math>\ell</math> is <math>G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)</math> exactly — it VANISHES iff <math>\ell = 2\pi = 1/(4c_3)</math>, the v239 KMS seam circumference: the seam scale is the unique zero-self-energy scheme point. [E] THE LOG IS THE SCALE CHANNEL: rescaling gives <math>\frac{1}{\pi} \ln \kappa</math> and nothing else — matching the functional structure (<math>F_{U(1)}</math> carries exactly ONE log term, the transport <math>8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})</math>; the <math>\varphi_0^{\text{ret}}</math> split is log-free); [C] reading: log = diagonal/scale channel, log-free = contact data at the KMS point. [E] CLOSED-FORM RESUMMATION (BFK route, v151 class): the finite-rank mark determinant resums exactly, <math>\det(I - uC) = (1 - 4u)(1 + 2u)^2</math> with <math>u = \varepsilon c_3 \ln 2</math> — all orders; the linear term is ABSENT precisely because the diagonal vanishes (<math>\text{Tr } C = 0</math>); the series matches v484’s <math>\text{Tr}(C^k) = 4^k + 2(-2)^k</math>. [E] WEIGHT DICHOTOMY: the SAME <math>48 = 3 \times 16</math> admissible states carry BOTH multiplicities — flat count <math>48 = \Omega_{\text{adm}}</math> (the <math>\varphi_0^{\text{ret}}</math> puncture) vs <math>Y^2/\text{loop}/\text{GUT-weighted } 41 = 10b_1 = 40 + 1</math> (<math>\text{Tr}_{16} Y^2 = \frac{10}{3}</math> exact, <math>\nu^c = 0</math>; Ginsparg <math>(3/5)(41/6) = 41/10</math>, v470) — one state set, two response weights. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN + PHIO.PUNCTURE target is proven; the single remaining [O] is the identification of the abstract seam <math>\zeta</math>-det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. No gate closes. Exact (sympy + 30-digit mpmath + exact charge sums); Python-only by nature. [E][C][O]</p> |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v486_transfer_full_rule.py   | <p>HYP.REWRITE.02: the FULL physical transfer spectrum <math>\{1, (2/3)^6, (1/3)^6\}</math> (v54/v221) from ONE forced local rule — the lazy <math>\mathbb{Z}_2</math>-pair walk; the <math>(1/3)^6</math> mode, previously generator-less, gets its local-rule origin (sharpens v327, whose uniform rule produces <math>\{1, \frac{2}{3}, 0\}</math> and can never generate the third mode). [E]</p> <p>UNIQUENESS: for <math>M(s, h) = \begin{pmatrix} 1 &amp; 0 &amp; 0 \\ 0 &amp; s &amp; h \\ 0 &amp; h &amp; s \end{pmatrix}</math> (one absorbing family channel + SYMMETRIC <math>\mathbb{Z}_2</math> pair) the survival spectrum is <math>\{s+h, s-h\}</math>; demanding the physical pair <math>\{\frac{2}{3}, \frac{1}{3}\}</math> forces UNIQUELY</p> <p>(stay, hop, leak) = <math>(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/ \mathbb{Z}_2 , 1/( \mathbb{Z}_2 N_{\text{fam}}), 1/N_{\text{fam}})</math> — atom form recorded (honest direction: spectrum <math>\rightarrow</math> rates). [E] SIX-HAND GENERATION: over the order-6 = <math>2N_{\text{fam}} =  R^+(A_3) </math> hand, <math>\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}</math> exactly; decay gaps <math>6 \ln \frac{3}{2}</math> (the recovery gap, v76) and <math>6 \ln 3</math> (the subdominant gap, now generator-full). [E] SUPERSESSSION: the uniform rule's zero mode persists under every power — no iterate reaches <math>(1/3)^6</math>; its <math>\lambda_2</math> arity story (<math>\frac{2}{3} =  \mathbb{Z}_2 /N_{\text{fam}}</math>) is unchanged. [E] sub-stochastic, symmetric (detailed balance), absorbing democratic attractor unchanged (v317). [O] the arity <math>\{2, 3\}</math> stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = <math>1/ \mathbb{Z}_2 </math> as the sheet-symmetric choice of the <math>\mathbb{Z}_2</math> involution). Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]</p> |
| v487_transfer_clock_rungs.py | <p>HYP.REWRITE.03: the lazy split <math>(\frac{1}{2}, \frac{1}{6})</math> is FORCED by the resummed clock — the v486 [O] 'split selection' closes onto v124 structure; sharpens HYP.REWRITE.02, no gate closes. [E] RUNG READING: the lazy rule's one-step spectrum is EXACTLY the complete clock ladder below the wall,</p> <p><math>\text{eig}(M) = \{1 - n/3 : n = 0, 1, 2\} = \{1, \frac{2}{3}, \frac{1}{3}\}</math>; v327's uniform rule puts its second pair mode ON the wall (<math>0 = 1 - 3/3</math>) — the degeneracy re-read as 'odd channel collapsed to the wall'. [E] DECK PARITY = RUNG INDEX: <math>[B, \sigma] = 0</math>; the <math>\sigma</math>-even pair mode carries rung <math>n = 1</math> (survival <math>\frac{2}{3}</math>), the <math>\sigma</math>-odd mode rung <math>n = 2</math> (survival <math>\frac{1}{3}</math>) — each clock step flips the deck parity; the structural home of the v486 parity assignment (<math>\omega_1</math> even-channel, <math>\omega_2</math> odd-channel). [E] FORCING: faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + <math>\mathbb{Z}_2</math> equivariance + rates <math>\geq 0</math> force UNIQUELY both the split <math>(\frac{1}{2}, \frac{1}{6})</math> AND the assignment even <math>\rightarrow</math> rung 1, odd <math>\rightarrow</math> rung 2 (the swap needs <math>h = -\frac{1}{6} &lt; 0</math>); the 'sheet-symmetric choice' candidate is superseded by this proof. [E] BEND COROLLARY: <math>\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3</math> — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] the channel count <math>3 = N_{\text{fam}}</math> is forced by the clock wall; the ladder stays an exact identity on frozen spectra (semiclassical derivation R1 open); the arity <math>\{2, 3\}</math> stays the anchor input. Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]</p>                    |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v488_majorana_clebsch_door.py   | <p>FLAV.NUSCALE.04: the 3/5 CLEBSCH DOOR of the seesaw rung, ENUMERATED AND CLOSED — v482's named escape (a third-generation Majorana factor <math>r \approx 1.6696</math>, 0.18% from <math>g_{\text{car}}/N_{\text{fam}} = 5/3</math>) is decided NEGATIVE at the group-theory level; the v481 window candidate stands as the honest endpoint. [E] NO RENORMALISABLE CHANNEL: <math>\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}</math>; the <math>\overline{\mathbf{126}}</math> is NOT in the <math>E_8</math> hull <math>\{1, 10, 16, 45\}</math> (v247) — <math>M_R</math> only via the <math>d=5</math> operator <math>(\mathbf{16}_F \overline{\mathbf{16}}_H)^2/\Lambda</math> with the <math>E_8</math>-allowed <math>\mathbf{16}_H</math>. [E] CHANNEL LIST: <math>\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}</math>; the hull admits singlet + <math>\mathbf{45}</math> insertions (<math>\mathbf{210}</math> absent). [E] CHARGE TABLE: <math>Y = T_{3R} + (B-L)/2</math> on all 16 states, <math>\text{Tr}_{16} Y^2 = \frac{10}{3}</math> (v485); <math>(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)</math>. [E] THE <math>\{2, 3\}</math>-SMOOTH THEOREM: every nonzero <math>\nu^c</math> channel weight <math>\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}</math> is <math>\{2, 3\}</math>-smooth and the multiplicative closure of <math>\{2, 3\}</math>-smooth rationals never contains <math>\frac{5}{3}</math>; the UNIQUE natural 5 of the embedding, <math>k_Y = \text{Tr} Y^2 / \text{Tr} T_{3L}^2 = \frac{5}{3}</math> (Ginsparg, v470), is a full-multiplet trace whose direction has <math>Y(\nu^c) = 0</math> EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: the Clebsch is a family-space scalar; <math>\text{diag}(1, 1, \frac{3}{5})</math> cannot arise from group theory. [O] compiler-flavor inventory typed and declined: exact <math>\frac{3}{5}</math>'s only in wrong slots (<math>K</math> lepton gen2/gen1; <math>R + L</math> lepton gen3/gen2), no degenerate-base shape, no <math>M_R</math> link (v354/v355). Exact (Fraction/sympy); mirrorable, deferred to the next verified engine run. [E][O]</p> |
| v489_seam_modular_commutator.py | <p>SEAM.CCC.01: the chiral central charge <math>c_- = 8</math> measured DIRECTLY from the bulk ground state of the v367 seam lattice model, via the Kim–Shehab–Kim modular commutator <math>J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-</math> on a <math>120^\circ</math> pie tripartition (PRL 128 (2022)) — the FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 (<math>c_- = 8</math>) INDEPENDENT of both the Chern route (v367 FHS curvature) and the counting route (v378 KLM/det-Cartan): <math>J</math> is read off the entanglement data of the exact BdG ground state alone (Peschel covariance, <math>W = -2 \text{artanh } \Gamma</math>). [E] GROUND-STATE PURITY: <math>\text{eig}(i\Gamma) = \pm 1</math> to <math>6 \times 10^{-15}</math>. [E] PER-LAYER <math>c_- = \frac{1}{2}</math>: <math>c_- = 0.499998</math> at <math>L=32, R=12</math> (<math>0.499793</math> at <math>L=16 \rightarrow 0.500000</math> at <math>L=48</math>, dev <math>-1.8 \times 10^{-7}</math>). [E] CONTROLS: trivial <math>M=3 \rightarrow 0.000000</math>; orientation flip <math>M=-1 \rightarrow -0.499986</math>. [E] LAYER ADDITIVITY EXACT (<math>1.4 \times 10^{-14}</math>): <math>J</math> additive over decoupled layers, so the carrier value is <math>16 \times</math> the per-layer value. [E] 16-LAYER CARRIER <math>c_- = 8 = g_{\text{car}} + N_{\text{fam}}</math>: direct 16-layer run <math>7.9967</math> (<math>L=16</math>, tol <math>10^{-2}</math>; <math>L=24</math> gives <math>7.999781</math>), additivity route <math>8.0000</math> (tol <math>10^{-3}</math>). [E] CLIP INVARIANCE: <math>\varepsilon = 10^{-8}</math> vs <math>10^{-12}</math> moves <math>c_-</math> by <math>5 \times 10^{-8}</math>. [O] HONEST FENCE: pins <math>c_- = 8</math> on the LATTICE ground state (third independent witness); does NOT close SEAM.EQUIV.01 — the continuum scaling limit (MMST v336) stays [O], the <math>\mu_4</math> condensation <math>SO(16)_1 \rightarrow (E_8)_1</math> stays the cited KLM step (v378). Numerical (numpy), Python-only by nature. [E][O]</p>                                                                                                                        |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v490_seam_parity_census.py     | <p>SEAM.MU.01: the <math>T^2</math> spin-structure fermion-parity census for the v367 seam lattice model — the lattice ground-state witness of “<math>\mu_{\text{pre}}=4</math> before condensation” (step 1 of the v378 KLM chain) + the 16-fold-way bridge <math>\theta_v=1</math> with the MEASURED <math>\nu=2c_-=16</math> (v489). Ground-state parity per boundary-condition sector (AA, AP, PA, PP), determined TWICE: Read–Green TRIM product AND real-space Majorana Pfaffian sign (Schur), relative to the trivial <math>M=3</math> phase. [E] REAL-SPACE = BLOCH SANITY: twisted spectra match to <math>9 \times 10^{-15}</math>. [E] PER-LAYER CENSUS (+, +, +, -): TRIM <math>\equiv</math> Pfaffian in every sector — 3 even sectors, the <math>\nu=1</math> Ising signature. [E] 16 LAYERS <math>\Rightarrow \mu_{\text{pre}}=4</math>: parity<sup>16</sup>=+ in ALL four sectors <math>\Rightarrow</math> gauged torus <math>\text{GSD}=4=\mu(SO(16)_1)= \det \text{Cartan}(D_8) </math>. [E] 16-FOLD-WAY BRIDGE: <math>\theta_v=e^{2\pi i\nu/16}=1</math> EXACTLY at <math>\nu=2c_-=16</math> (vortex bosonic, condensable); rivals <math>\nu=1, 2, 8</math> excluded (not holomorphically condensable). [E] NON-DISCRIMINATOR FENCE: the fermionic Kitaev–Preskill TEE is identically zero — <math>\gamma_f=0.000001</math> (<math>M=1</math>) and <math>0.000000</math> (<math>M=3</math>) — an invertible fermionic phase has no TEE, so <math>\gamma_f</math> does NOT discriminate <math>(E_8)_1</math> vs <math>SO(16)_1</math> (both live only after gauging). [C] the KLM step <math>\mu_{\text{post}}=4/2^2=1</math> stays cited (the condensed model is interacting); the numerics pin only <math>\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}</math> — the premise package of the condensation. [O] the continuum limit (MMST v336); SEAM.EQUIV.01 stays [O]. Python (numpy + scipy Schur + sympy exact <math>\theta_v</math>). [E][C][O]</p>                                                                      |
| v491_p2_partition_corollary.py | <p>P2.PARTITION.01: <math>g_{\text{car}}=5</math> as a COROLLARY of the four marks — the partition sharpening of v53 (sympy exact, no floats). THE LEMMA: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is unique — <math>\{1, 1, 2\}</math> — so <math>e=(e_1, e_2, e_3)=(4, 5, 2)</math> and <math>g_{\text{car}}=e_2=5</math>, <math> \mathbb{Z}_2 =e_3=2</math>, <math>N_{\text{fam}}=e_2-e_3=3</math> are COROLLARIES, not inputs; consistency closes as a fixed point <math>((2^4-1)/5=3=\text{rank } H^1=\#\text{marks}-1)</math>. NEGATIVE CONTROLS (every assumption load-bearing): drop positivity <math>\rightarrow 4</math> solutions, <math>e_2</math> ambiguous <math>\{0, 3, 4, 5\}</math>; sum 3 <math>\rightarrow e_2=3</math>; sum 5 <math>\rightarrow</math> not unique; 2/4 parts <math>\rightarrow e_2 \in \{3, 4\}/\{6\}</math>; Mehta–Seshadri RATIONAL weights <math>\rightarrow e_2=\frac{16}{3}</math> resp. <math>\frac{44}{9} \neq 5</math> (integrality comes from the Birkhoff–Grothendieck SPLITTING-TYPE typing, not from MS weights). [E] v53 SHARPENING: v53 needs all three targets (4, 5, 2); the partition needs only <math>e_1=4</math> — same unique <math>\{1, 1, 2\}</math>. [E] SUM RULE: <math>4 \times \frac{N-1}{2}=4</math> iff <math>N=3</math>; the v115 route <math> a _1= \mu_4  \text{tr}(A_0)=4</math>. INDEPENDENCE CHAIN: <math>\#\text{marks}=4</math> from v216 (P1-side, no <math>g_{\text{car}}</math> input); rank <math>H^1=3</math> pure topology (v137/v228); <math> a _1=4</math> <math>g_{\text{car}}</math>-free (par-deg-0/(U), v115). [C] HONEST RESIDUAL: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality = splitting exponents; positivity = <math>h^0=0</math>, correlate v137 check 1; sum rule “given (U)”); AX.P2.01 stays a declared axiom; the <math>N_{\text{fam}}</math> direction reversal vs tfpt_constants is typed as an over-determination check, never double-counted. Exact (sympy); Wolfram-mirrored. [E][C]</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v492_celestial_z4_orbifold.py | <p>CELEST.WP1.01: WP1 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the <math>E_8</math> <math>\mu_4</math>-glue grading (v128) IS the flat <math>\mathbb{Z}_4</math> monodromy of the glue-equivariant celestial chiral algebra on the <math>A_3</math> ALE space <math>\mathbb{C}^2/\mathbb{Z}_4</math> (Bittleston–Homans–Sharma, <a href="#">arXiv:2305.09451</a>, the <math>\mathbb{Z}_2</math>/Eguchi–Hanson case; Kronheimer–Nakajima flat connection at infinity). Sympy exact. [E] INNER GRADING: <math>h=(2, 2, 2, 2, 2; 0^4)</math> reads the glue class mod 4 on all 240 roots <math>\Rightarrow</math> glue <math>= \text{Ad}(\exp(2\pi i h/4))</math>, an order-4 torus element <math>=</math> flat <math>\mathbb{Z}_4</math> monodromy; the <math>A_3</math>-side detector <math>h'=(0^5; 1, 1, 1, -3)</math> reads the SAME class <math>=</math> the diagonal <math>(1, 1)</math> of the discriminant group <math>\mathbb{Z}_4 \times \mathbb{Z}_4</math> (the v92/v125 Lagrangian glue); <math>\mathbb{Z}_2</math> halfway <math>120 = \dim SO(16)</math>, <math>128 = \text{spinor}</math>. [E] <math>\mathbb{C}^2/\mathbb{Z}_4</math> IS the <math>A_3</math> singularity <math>XY=Z^4</math> (Molien table exact; Hilbert series <math>(1+t^4)/((1-t^2)(1-t^4))</math>; conjugation symmetry <math>N[d][1]=N[d][3]</math>). [E] EQUIVARIANT SECTOR CLOSES: <math>D[d]=60(d+1)</math> even / <math>64(d+1)</math> odd — uniform per parity ONLY because <math>\dim \mathfrak{g}_j=(60, 64, 60, 64)</math>; zero modes <math>(d=0) = \mathfrak{g}_0 = \text{carrier } D_5 \oplus A_3 + \text{Cartan} = 60</math>; density <math>62/248=1/4=1/ \mathbb{Z}_4 </math>; <math>\text{EH}/\mathbb{Z}_2</math> halfway towers <math>120/128</math>. [E] CHARACTER: <math>\Theta_{E_8} = \sum_j \Theta_{C_j}</math>; shell <math>1 = (52, 64, 60, 64)</math> from the LATTICE; the 4 sector characters <math>\Theta_{C_j}/\eta^8</math> sum exactly to <math>\chi_{(E_8)_1} = 1 + 248q + 4124q^2 + 34752q^3</math> (v377); sector weights <math>h_{D_5}=(0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})</math>, <math>h_{A_3}=(0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})</math> = the v92/v125 discriminant form <math>(5x^2+3y^2)/8 \bmod 1</math>; glue diagonal <math>h=(0, 1, 1, 1)</math> INTEGER = v125 isotropy = local extension to <math>(E_8)_1</math>. [E] CRITICAL CORRECTION: the <math>A_3</math> deck <math>\text{diag}(i, i^{-1})</math> acts on the celestial sphere as <math>z \mapsto -z</math> (projective order 2, the sheet flip), NOT as the order-4 clock <math>z \mapsto iz</math>; the clock is the <math>U(2)</math> phase <math>\text{diag}(i, 1)</math> (order 4, fixed <math>\{0, \infty\}</math>, free <math>\mu_4</math> orbit — all v180 properties; normalises the deck, <math>\det=i</math> rotates the holomorphic symplectic form by the <math>\mu_4</math> generator); exact spin bridge <math>(\text{spin clock})^2 = \text{deck}</math>, spin group <math>\mathbb{Z}_8</math>, <math>8=2 \mu_4 </math> (the <math>c_3=1/(8\pi)</math> winding integer). [E] clock-invariant <math>A_3</math> deformation = the 1-parameter family <math>XY=Z^4+a_0</math>; its 4 branch points are ONE <math>\mu_4</math> orbit with cross-ratio 2 (v179 clock orbit; v168/v214 pillowcase <math>j=1728</math>). [E] NEGATIVE CONTROLS: false spatial action <math>\text{diag}(i, i)</math> killed three ways (<math>\det=-1 \notin SL(2, \mathbb{C})</math>); Poisson closure fails 14/14; grades unsymmetric); false glue breaks additivity on 3112/6720 root pairs; rigidity: only <math>\text{Aut}(\mathbb{Z}_4)</math> relabelings additive. [C] VERDICT B: match under the ONE named identification <math>\text{clock}^2=\text{deck}</math> on the sphere (sheet-<math>\mathbb{Z}_2</math> spin bookkeeping). [O] structural arithmetic ONLY — no celestial duality constructed; CELEST.SEAM.01 (WP2–WP5) stays a research contract; SEAM.EQUIV.01 stays [O], untouched. Exact (sympy). [E][C][O]</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v493_celestial_<br>wp2_clock_<br>deformation.py | <p>CELEST.WP2.01: WP2 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the clock-invariant deformation <math>XY=Z^4+a_0</math> of the <math>A_3</math> seam orbifold: geometry, periods/monodromy, deformed algebra</p> <p>(Bittleston–Homans–Sharma <math>\mathbb{Z}_2 \rightarrow \mathbb{Z}_4</math> transfer, <a href="#">arXiv:2305.09451</a>). Sympy exact. <b>[E]</b> GEOMETRY: <math>P(iZ)=P(Z)</math> forces <math>a_3=a_2=a_1=0</math> SHARPLY (two-sided: <math>\prod_k (Z - i^k w) = Z^4 - w^4</math>); smooth iff <math>a_0 \neq 0</math> (disc <math>= 256 a_0^3</math>); branch points <math>Z^4 = -a_0 =</math> ONE free <math>\mu_4</math> orbit in every fibre; cross-ratio <math>2 / j = 1728</math> for ALL <math>a_0</math></p> <p>(binary-quartic invariants <math>I=12a_0</math>, <math>J=0</math> identically) <math>\Rightarrow a_0</math> is a PURE SCALE, the <math>\tau=i</math> pillowcase (v168/v214) is FROZEN — no shape modulus. <b>[E]</b> PERIODS: <math>\Pi=t(i-1)(1, i, i^2)</math>, all 3 vanishing spheres carry the SAME volume <math>\sqrt{2}t</math> (lockstep); the clock on <math>H_2</math> is a COXETER element of <math>W(A_3)</math> (char <math>x^3+x^2+x+1</math>, eigenvalues <math>\{i, -1, -i\}</math> = the <math>A_3</math> exponents 1, 2, 3; <math>h(A_3)=4= \mu_4 =N_{\text{fam}}+1</math>); NO invariant cycle (<math>\det(A-1)=-4</math>); the surviving direction is the weight-1 <math>\chi_1</math>-Fourier diagonal <math>(1, i, i^2)</math>, NOT the naive invariant diagonal; the clock IS the Picard–Lefschetz monodromy of the family (<math>a_0 \rightarrow e^{2\pi i} a_0</math> gives the 4-cycle); versal weights <math>(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)</math> = the REGULAR <math>\mu_4</math> representation, invariant line = <math>a_0</math> alone. <b>[E]</b> ALGEBRA (<math>\mathbb{Z}_2 \rightarrow \mathbb{Z}_4</math>): mode dictionary <math>\{w[m, n] : m \equiv n \pmod{4}\} \leftrightarrow \{X^p Y^q Z^r : r \leq 3\}</math>; bracket <math>\{f, g\} = -4 \text{Nambu}(XY - Z^4 - a_0)</math> anchored at the <math>a_0=0</math> orbifold w-sector <math>(\{X, Y\}=16Z^3, \{Y, Z\}=-4Y, \{Z, X\}=-4X)</math>, Jacobi + Casimir exact, uniform clock-degree <math>-1</math>; S-algebra corrections exactly LINEAR in <math>a_0</math> at the <math>\mathbb{Z}_4</math> wrap <math>r+r' \geq 4</math>, grade-conserving (NO sector leak <math>\Rightarrow</math> the v492 equivariant SDYM(<math>E_8</math>) sector deforms consistently); <math>a_0 \leftrightarrow</math> the BHS <math>c^2</math> slot (weight 0); matrix factorisations (McKay modules) deform along every 1+3 and 2+2 orbit splitting; the clock fixes exactly the antipodal 2+2 sector (the EH/<math>\mathbb{Z}_2</math> middle node, <math>4=2 \times 2</math>).</p> <p><b>[E]</b> NEGATIVE CONTROLS: <math>a_1 Z \Rightarrow j=0</math> (hexagonal CM point); pure <math>a_2 Z^2</math> stays SINGULAR (<math>A_1</math> node); <math>a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728</math> (exact instance, two routes agree); grading leak explicit (<math>Z^2 \cdot Z^2</math> picks up grades <math>\{0, 1, 2\}</math>). <b>[C]</b> three NAMED identifications (verdict B, not A): <math>-4 \text{Nambu} = \text{THE } k=4 \text{ CCA bracket}</math> (BHS prove <math>k=2</math>); period = root difference (A-series residue normalisation); the seam-scale reading of <math>a_0</math> presupposes the WP1 bridge clock<sup>2</sup>=deck (<math>\mathbb{Z}_8</math>, <math>8=2 \mu_4 </math>).</p> <p><b>[C]</b> v216 PRECISION (typed, NO marker move): given the order-4 clock, the clock-invariant deformation freezes the <math>(2, 2, 2, 2)</math> modulus to the square AUTOMATICALLY — a relocation of the same order-4 input (the clock IS the carrier input), not a new derivation; v216 stays <b>[O]</b>. <b>[O]</b> full deformed twisted-sector OPEs, the <math>W(\mu)</math>-type scaling limit for <math>k=4</math>, the spacetime splitting-function check, WP3–WP5; SEAM.EQUIV.01 stays <b>[O]</b>, untouched. Exact (sympy). <b>[E][C][O]</b></p> |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v494_cosmo_running_mu_record.py | <p>COSMO.RUNNING.01/COSMO.MUDIST.01: the two 2026-07-20 prediction-of-record rows promoted to the suite — the SAME Starobinsky branch as <math>n_s, r, A_s</math> (<math>\sqrt{v7}/\sqrt{v28}/\sqrt{v86}</math>) read one derivative deeper and one observable further. <b>[E]</b> LO IDENTITIES (sympy exact): <math>\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6</math>, <math>\beta_s = -4/N_*^3</math> (NOT <math>-8/N_*^3</math>); LO band <math>N_* \in [50, 60]</math>: <math>[-8.0, -5.6] \times 10^{-4}</math>. <b>[E]</b> EXACT SLOW ROLL (third-order Lyth–Riotto + finite-difference cross-check <math>&lt; 1\%</math>): <math>\alpha_s(51.4) = -7.09 \times 10^{-4}</math>, <math>\beta_s(51.4) = -2.70 \times 10^{-5}</math>; exact band edges <math>-7.48/ -5.23 \times 10^{-4}</math>. <b>[C]</b> DATA: Planck 2018 (<math>-0.0045 \pm 0.0067</math>) <math>\rightarrow +0.57\sigma</math>; record leg P-ACT-LB (<math>+0.0062 \pm 0.0052</math>) <math>\rightarrow -1.33\sigma</math> (watch channel: positive central value); <math>\beta_s</math> run+runrun <math>\rightarrow -0.77\sigma</math>. KILL: robust <math>\alpha_s &lt; -5 \times 10^{-3}</math> or <math>&gt; +3 \times 10^{-3}</math> at <math>\geq 5\sigma</math>; any robust POSITIVE running kills the plateau branch. <b>[C]</b> <math>\mu</math>-DISTORTION (Chluba window, +21% calibrated on the <math>\Lambda</math>CDM anchor): sharp branch <math>N_* = 51.4</math> (<math>A_s = 1.76 \times 10^{-9}</math>) <math>\mu = 1.6 \times 10^{-8}</math>; profiled record <math>N_* = 56.1</math> <math>\mu = 1.94 \times 10^{-8}</math>; band <math>1.5\text{--}2.3 \times 10^{-8}</math>; branch split <math>\Delta\mu = 3.0 \times 10^{-9}</math> (ratio 1.23) <math>\Rightarrow \geq 3\sigma</math> <math>A_s/N_*</math> branch decision at <math>\sigma(\mu) = 10^{-9}</math> (Voyage-2050 class); the PIXIE baseline <math>9 \times 10^{-8}</math> does NOT reach (<math>0.18 \times</math> limit). KILL: robust <math>\mu &lt; 0.9 \times 10^{-8}</math> or <math>&gt; 4 \times 10^{-8}</math>. <b>[E]</b> COMB BLINDNESS (honest S15): the frozen log-comb (<math>\omega = 2.5827</math>, <math>\varepsilon = 0.0173</math>) is structurally invisible in the <math>\mu</math> window — <math> F(\omega)  = 0.033</math>, <math>\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma</math> even at Voyage-2050. <b>[O]</b> FENCES: <math>N_*</math> stays an external reheating input (GATE.NSTAR.01); the frozen band <math>[50, 60]</math> stays the surface of record. Python (sympy exact LO + numpy SR/quadrature), Python-only by nature. <b>[E][C][O]</b></p> |

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| Script                                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v495_celestial_wp3_<br>gs_alignment.py | <p>CELEST.WP3.01: WP3 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the Green–Schwarz axion coefficient <math>\lambda_g</math> of celestial/twistorial SDYM (Costello, arXiv:2111.08879) against the TFPT seam constant <math>c_3=1/(8\pi)</math>; the pre-registered “alignment test only”. Exact (Fraction/sympy, no floats). <b>[E]</b> OKUBO IDENTITY DERIVED, NOT QUOTED: <math>\sum_{\alpha} \langle \alpha, x \rangle^4 = \frac{5}{2(\dim \mathfrak{g}+2)} (\sum_{\alpha} \langle \alpha, x \rangle^2)^2</math> as an exact POLYNOMIAL identity for ALL 8 algebras on Costello’s list (<math>\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8</math>; <math>\mathfrak{e}_6/\mathfrak{e}_7</math> carved out of the v492 <math>E_8</math> root set) — “no independent quartic Casimir”; NEGATIVE CONTROL: <math>\mathfrak{sl}_4</math> (which HAS one) fails the proportionality (5/32 vs 3/32) — the list is sharp. <b>[E]</b> Killing <math>\sum_{\alpha} \langle \alpha, x \rangle^2 = 2h^{\vee} \langle x, x \rangle</math> exact; Deligne dimension formula <math>\dim \mathfrak{g} = 2(5h^{\vee}-6)(h^{\vee}+1)/(h^{\vee}+6)</math>; closed form <math>\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim +2) = h^{\vee}+6</math> EXACTLY (8, 9, 10, 12, 15, 18, 24, 36) — fixes Costello–Paquette eq. (1.5); Dynkin indices <math>i_R = (1, 1, 2, 2, 6, 6, 12, 60)</math> from explicit weight systems; <math>E_8</math>: <math>\tilde{\lambda}=6</math>, <math>\lambda_{\text{fund}}=1/10</math> EXACT; cross-checks <math>\mathfrak{sl}_2=8</math>, <math>\mathfrak{sl}_3=9</math>; SO8 SLIP FLAGGED: Costello’s own weight content gives <math>\lambda^2(\mathfrak{so}_8)=3</math> (Okubo-consistent 12/4), the printed 3/2 is a factor-2 slip. <b>[E]</b> ALIGNMENT: <math>\kappa = \lambda_g/(8\pi\sqrt{3}) \Rightarrow (\kappa/c_3)^2 = \lambda^2/3 = 12 =  \mu_4 N_{\text{fam}}</math> (unit-trace) resp. 1/300 (adjoint-trace), both EXACT anchor rationals; <math>\kappa/c_3</math> ITSELF is IRRATIONAL (<math>2\sqrt{3}</math> resp. <math>1/(10\sqrt{3})</math>) — the pre-registered “<math>\lambda/\sqrt{3}</math> rational?” reading FAILS, only the SQUARED alignment holds; anchor decomposition <math>h^{\vee}=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}</math> (all three atoms once), <math>\varphi(30)=8=\text{rank}</math>, <math>\dim +2=250=2 \cdot 5^3</math>, <math>\tilde{\lambda}=6= \mathbb{Z}_2 N_{\text{fam}}</math>, <math>\lambda_{\text{fund}}=1/( \mathbb{Z}_2 g_{\text{car}})</math>. <b>[E]</b> LOOK-ELSEWHERE (the HARD FENCE — alignment <math>\neq</math> selection): the alignment test AS SUCH passes 8/8 (ZERO selective power), <math>\{2, 3, 5\}</math>-smoothness 8/8, <math>\tilde{\lambda}</math>-integrality 2/8 (shared with <math>\mathfrak{sl}_3</math>), <math>\varphi(h^{\vee})=\text{rank}</math> 4/8; the ONLY <math>\mathfrak{e}_8</math>-selective single test is <math>g_{\text{car}}=5 \mid h^{\vee}</math> (1/8); the isolating conjunction was assembled POST HOC — it quantifies where <math>\mathfrak{e}_8</math> sits, it is NOT a preregistered discriminator. <b>[E]</b> KILL-TEST K3 does NOT fire (exact alignment EXISTS) but its scope is DEMOTED: the <math>c_3</math>-connection survives as convention-level compatibility + genuine <math>\lambda</math>-arithmetic, NOT as <math>\mathfrak{e}_8</math>-selective evidence. <b>[C]</b> TYPED: the <math>8\pi</math> is the <math>2\pi</math> convention of the anomaly polynomial (Costello’s A.0.10 constant descended to spacetime), the <math>\sqrt{3}</math> the box diagram’s <math>\sqrt{48}=4\sqrt{3}</math> — WP3 is an alignment yes/no ONLY, it can never derive <math>c_3</math> from celestial data (contract wording, honoured). <b>[O]</b> WP5 (constructive holography) and SEAM.EQUIV.01 stay <b>[O]</b>, untouched. Verdict B. Exact (Fraction + sympy). <b>[E][C][O]</b></p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v496_celestial_<br>wp4_character_<br>shadow.py | <p>CELEST.WP4.01: WP4 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the <math>(E_8)_1</math> character <math>E_4/\eta^8</math> against the glue-equivariant <math>E_8[\mathbb{C}^2]</math> S-algebra: the contract’s “type mismatch, head-on”. Exact integer/Fraction/sympy, no floats. [E] BASE: v492 replication <math>(240 = 52+64+60+64, \text{dims with Cartan } (60, 64, 60, 64))</math>; equivariant tower <math>D[d] = 60(d+1)</math> even / <math>64(d+1)</math> odd; cumulative generator count QUADRATIC — <math>\sum_{d \leq s} D[d] = 31s^2 + 92s + 60</math> (even) with leading <math>31 = k + h^\vee</math> (the Sugawara denominator, an arithmetic note); [C] CP Table 1 (arXiv:2201.02595): <math>J[m, n]</math> carries 2d spin <math>1 - (m+n)/2</math>, UNBOUNDED BELOW along the tower <math>\Rightarrow</math> no rational VOA (<math>L_0</math> bounded below) contains the towers as graded pieces. [E] HILBERT SERIES: <math>\chi_{\text{sp}}(t) = (60+128t+60t^2)/(1-t^2)^2</math> (palindromic numerator, value 248 at <math>t=1</math>); ALL <math>\mu_4</math> collapses <math>w \in \{1, -1, i, -i\}</math> and the <math>SU(2)_+</math> primary slice <math>(60+64t)/(1-t^2)</math> fail to start <math>(1, 248, \dots)</math> — the “row sum” route fails at the first coefficient. [E] CHARACTER SIDE:</p> <p><math>\chi_{(E_8)_1} = E_4/\eta^8 = (1, 248, 4124, 34752, 213126, 1057504, 4530744)</math> exact through <math>q^6</math> (machinery cross-check <math>qj = E_4^3/\eta^{24} = (1, 744, 196884, 21493760)</math>); theta split (shell <math>1 = (52, 64, 60, 64)</math>), sector currents <math>(60, 64, 60, 64)</math>, glue-diagonal <math>h = (0, 1, 1, 1)</math> INTEGER (v125 isotropy = locality); Sugawara <math>c = 248/31 = 8 = 2 \mu_4  = g_{\text{car}} + N_{\text{fam}}</math>. [E] BRIDGES: zero-mode slice <math>D[0]=60=</math> the <math>C_0</math>-sector current count EXACTLY; the single-line jet slice <math>z_1^d</math> cycles <math>(60, 64, 60, 64)</math> sector by sector; BOUNDARY/PERIOD LIMIT: one full <math>\mu_4</math> period of loop energies carries <math>\dim \mathfrak{g}_{n \bmod 4} = (64, 60, 64, 60)</math>, sum = 248 EXACTLY — the single-particle level of the <math>248q</math> stage; 248 is NOT a jet-tower dimension (<math>d \leq 100</math>) — only the period collapse produces it; BUT the loop Fock at integer level 1 counts <math>897266 \gg 248</math>: the rational truncation is NOT in the jet/loop algebra. [E] MISMATCH LOCALISED THREE WAYS: (a) GRADING (spin unbounded below; 248 no tower dimension); (b) GROWTH: jet Fock <math>f = (1, 128, 8436, \dots)</math> vs <math>\chi = (1, 248, 4124, \dots)</math> TWO-SIDED (<math>f_1 &lt; \chi_1</math> undercount, <math>f_2 &gt; \chi_2</math> overcount), <math>f_n/\chi_n</math> strictly increasing <math>n = 1..12</math>; Meinardus poles <math>s=2</math> vs <math>s=1 \Rightarrow</math> growth exponents <math>n^{2/3}</math> vs <math>n^{1/2}</math>, asymptotically DISJOINT; (c) TRUNCATION: the level-2 null ideal deletes EXACTLY <math>27000 = 30^3 = (h^\vee)^3</math> (<math>\text{Sym}^2(248) = 1+3875+27000</math> [[C] standard]; the character keeps <math>4124 = 1+248+3875</math>) — NO analogue in the jet algebra (level 1 agrees: <math>248 = 248</math>). [E] KILL-TEST K4, three-way: (i) exact realisation NO; (ii) boundary-limit realisation PARTIAL YES — current stratum, period sum 248, integer glue weights and theta split hold exactly; the missing piece is precisely the rational truncation, which must come from a scaling/compactification limit — the SAME structure as the MMST scaling-limit route of SEAM.EQUIV.01 (v336/v449: the rational net exists in the LIMIT of the lattice collar, not in the pre-limit algebra); (iii) the hard mismatch is exactly located; K4 does NOT degrade the contract to “<math>E_8</math> admissible” (the sector arithmetic holds exactly). [C] VERDICT B(ii): the <math>(E_8)_1</math> character is a BOUNDARY/LIMIT SHADOW of the equivariant S-algebra, not a conformal block of it in the bulk grading — WP4 sharpens into a statement of SEAM.EQUIV.01 type. [O] the constructive boundary limit itself (which limit, which topology, convergence) stays OPEN and belongs to WP5/SEAM.EQUIV.01; SEAM.EQUIV.01 stays [O], untouched. Exact (integer/Fraction/sympy). [E][C][O]</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v497_celestial_<br>wp5a_null_ideal.py | <p>CELEST.WP5A.01: WP5a of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the null ideal from the limit”, the first constructive milestone of WP5 (the v496 boundary-limit shadow made precise). Exact integer/Fraction arithmetic, no floats. [E] BASE: v492/v496 replication (glue dims (60, 64, 60, 64), equivariant tower <math>D[d]</math>, <math>\chi_{(E_8)_1} = E_4/\eta^8</math> exact through <math>q^6</math>, <math>\mu_4</math> period sum 248). [E] STABILISATION THEOREM: a one-parameter family <math>\chi_w</math> of graded Fock characters on the CHIRAL jet generators (loop energy <math>m</math>, radial number <math>r</math>; <math>E_w = m + wr</math> in quarter units <math>u = q^{1/4}</math>); <math>w=2</math> IS the chiral jet grading (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree); the <math>u^n</math> coefficient of <math>\chi_w</math> equals the quarter-moded loop Fock for ALL <math>w \geq n+1</math> and is STRICTLY larger for every <math>w \leq n</math> (<math>n \leq 8</math>, <math>w \leq 10</math>): the boundary reading is the precise coefficientwise limit <math>\varepsilon = 1/w \rightarrow 0</math> with explicit threshold <math>w = n+1</math> — a limit of a series, not a slice; convention clarified: 897266 is the quarter-moded loop Fock at <math>u^4 = \text{INTEGER level 1}</math> (v496). [E] NULL IDEAL DERIVED, NOT CITED: Freudenthal multiplicities (unit-tested on the adjoint) + Weyl dimension formula + generic character peeling in the standard <math>E_8</math> frame DERIVE</p> <p><math>\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1</math> (peeling residual exactly zero; independent Weyl-orbit cross-check); <math>27000 = 30^3 = (h^\vee)^3</math> exactly;</p> <p><math>C_2 = (124, 96, 0)</math>, Dynkin indices 6750/750; level-1 integrability <math>\langle 2\theta, \theta^\vee \rangle = 4</math>, <math>\langle \omega, \theta^\vee \rangle = 2 \Rightarrow</math> the only level-1 primary is the vacuum ([C] Kac, used through its counting consequence); level-2 quotient <math>31124 - 27000 = 4124 = 1 + 248 + 3875</math> EXACTLY. [E] TWO ROUTES, ONE NUMBER: the <math>\mu_4</math> theta-split sector characters summed at <math>q^2 = (1036, 1024, 1040, 1024)</math> give the SAME 4124 — the independent lattice route hits the Fock quotient exactly; at <math>q^3</math>: <math>34752 = \chi_3</math>. [E] NEGATIVE CONTROLS: <math>SO(16)_1</math> through the SAME pipeline —</p> <p><math>\text{Sym}^2(120) = 5304 + 1820 + 135 + 1</math> derived (FOUR components vs <math>E_8</math>’s three), <math>5304 \neq 14^3 = 2744</math> (the <math>h^\vee{}^3</math> identity is NOT generic), quotient</p> <p><math>7380 - 5304 = 2076 = \Theta_{D_8}/\eta^8</math> at <math>q^2</math> (the pipeline validated on a second algebra), block weights <math>(0, \frac{1}{2}, 1, 1)</math> do NOT fuse into one local character (four-block theory) vs <math>E_8</math>’s integer <math>(0, 1, 1, 1)</math>; false periodisation: only <math>P = 4 =  \mu_4 </math> gives 248 (<math>P=1: 64</math>, <math>P=2: 124</math>, <math>P=8: 496</math>); swapped glue breaks the zero-mode anchor (<math>64 \neq 60</math>). [C] named identifications: “radial decoupling = MMST collar thickness <math>\rightarrow 0</math>” (the v336/v449 lattice-collar route has the same two-step shape), not a computed isomorphism; the Kac singular vector only via its counting consequence. [O] HONEST LIMIT: the limit does NOT generate the truncation — the singular vector as an OPERATOR in the limit net stays open (WP5b); the GNS limit state (kernel <math>\supseteq</math> ideal) and local normality / Haag duality of the limit net stay open (WP5c/WP5d); the celestial route now has the SAME two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed;</p> <p>SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v498_celestial_<br>wp5b_singular_<br>vector.py | <p>CELEST.WP5B.01: WP5b of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the singular vector as an operator”, the second constructive milestone of WP5 (v497 fixed the ideal’s size and location; WP5b constructs the deleting object). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: <math> s\rangle = (E_{-1}^\theta)^2 0\rangle</math>, <math>\theta</math> the highest root, weight <math>2\theta</math> (the top of the 27000), level 2 = 8 quarter units = <math>q^2</math> (an INTEGER level); the Chevalley/Frenkel–Kac basis is MACHINE-built (bimultiplicative 2-cocycle, asymmetry on all 57600 pairs; the <math>[e_\alpha, e_{-\alpha}]</math> sign FORCED by Jacobi, <math>\text{SGN} = -1</math>; <math>\kappa</math> DERIVED from invariance, <math>\kappa(\theta^\vee, \theta^\vee) = 2</math>; Jacobi exact on all 57600 <math>\mathfrak{sl}_2</math>-slice triples + 6000 seeded + all 496 <math>\theta</math>-adjacent; all <math>N_{\alpha\beta} = \pm 1</math>). [E] SINGULARITY: <math>J_1^a s\rangle = 0</math> for ALL 248 basis elements (0 violations); case tally 190 (first bracket) / 57 (second) / 1 (<math>a = F^\theta</math> via the central-term cancellation — the ONLY case that sees the level <math>k</math>); plus <math>J_2^a s\rangle = 0</math> (248), <math>E_0^a s\rangle = 0</math> (120 positive roots), Shapovalov norm 0 at <math>k=1</math> (4 at <math>k=2</math>); the affine PBW engine unit-tested on all 61504 basis pairs. [E] LEVEL DIAL: the <math>F_1^\theta</math> coefficient is <math>2(1-k) - 2</math> at <math>k=0</math>, 0 EXACTLY at <math>k=1</math>, <math>-2</math> at <math>k=2</math>: without the central extension the deletion object does NOT exist. [E] <math>\mu_4</math> DATA: <math>\theta</math>-grading dims <math>(1, 56, 134, 56, 1)</math>, <math>\mathfrak{g}_0 = \mathfrak{e}_7 \oplus \mathfrak{u}(1)</math>; the <math>h</math>-adapted chamber machine-built (Cartan det 1, <math>E_8</math> Dynkin diagram, branch lengths <math>[1, 2, 4]</math>); <math>\langle \theta, h \rangle = 5 \Rightarrow j(\theta) = 1</math>, height <math>29 = h^\vee - 1</math>; <math> s\rangle</math> in class 2 (sheet-EVEN, consistent with WP5a); clock phase <math>i^2 = -1</math>; cross-table <math>\theta</math>-grading <math>\times</math> glue class consistent (rows <math>(1, 56, 126, 56, 1)</math>, columns <math>(52, 64, 60, 64)</math>); the deleting <math>\theta</math>-<math>\mathfrak{sl}_2</math> crosses the sheet-ODD classes <math>(1, 3)</math>, its square is sheet-even. [E] LEVEL-2 MATCH: <math>\text{mult}(2\theta) = 1</math> in the level-2 Fock <math>\Rightarrow U(\mathfrak{g}) s\rangle</math> is THE 27000; the direct <math>\mathfrak{g}_0</math>-orbit BFS reproduces the Freudenthal multiplicities of <math>V(2\theta)</math> EXACTLY through depth 4; quotient <math>31124 - 27000 = 4124</math>; lattice reading <math>(2\theta, 2\theta)/2 = 4 &gt; 2 \Rightarrow</math> no momentum-<math>2\theta</math> state at <math>q^2</math> in <math>E_4/\eta^8 -  s\rangle</math> IS the kernel generator of Fock <math>\rightarrow E_4/\eta^8</math>. [E] NEGATIVE CONTROLS: <math>E_{-1}^\theta 0\rangle</math> NOT singular (one witness <math>F^\theta, \rightarrow -k 0\rangle</math>) — the 248-current state that must REMAIN (<math>\chi_1 = 248</math>); 4 generic level-2 states non-singular (14–33 witnesses); <math>k</math>-dial: at <math>k=2</math> the CUBE is singular (the <math>(E^\theta)^{k+1}</math> mechanism = generic Kac level mechanics); <math>D_8/SO(16)_1</math>: the singular vector exists there TOO (level-1 generic) but the <math>h = \frac{1}{2}</math> blocks do not fuse (3 comark-1 weights, <math>\dim V(2\theta_{D_8}) = 5304 \neq 14^3</math>) vs <math>E_8</math> (comarks <math>\geq 2</math>, vacuum the ONLY level-1 primary, glue weights <math>(0, 1, 1, 1)</math> integer) — the ONE-BLOCK closure is the <math>E_8/\mu_4</math>-specific part, honestly separated from the generic singular-vector mechanics. [C] Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the per-period quarter dictionary (v496). [O] HONEST LIMIT: in the twisted quarter-SLOT moding of the <math>\chi_w</math> limit Fock two sector-<math>C_1</math> modes never sum to 8 quarters (minimum <math>6 = q^{3/2}</math>) — the per-PERIOD dictionary (v496), not the per-slot identification, carries <math> s\rangle</math> to <math>q^2</math>; exactly the precise WP5c question (GNS limit state, kernel <math>\supseteq</math> ideal); WP5c-e stay [O]; SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]</p> |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v499_p2_weights_<br>deligne_bg.py | <p>P2.TYPING.01: the P2 weight-typing postulate HARDENED (the v491 check-9 residual) — the anchor <math>a=(1,1,2)</math> typed as the Birkhoff–Grothendieck splitting exponents of the DELIGNE CANONICAL EXTENSION <math>E</math> of the flavor connection; sympy exact, no floats. Given the [E] anchors (4 marks v216, rank 3 v137, cusp class + irreducibility v117, (U)): <math>\deg E = -4</math> AS A THEOREM (cusp class <math>\Rightarrow</math> canonical residue exponents <math>\{0, \frac{1}{3}, \frac{2}{3}\}</math>, the unique lift in <math>[0,1]</math>; trace 1 per mark <math>\times</math> 4 marks; par-deg 0; <math>\infty</math> APPARENT, monodromy product = <b>1</b> exactly); POSITIVITY AS A THEOREM (irreducibility EXACT: <math> \langle U, M_0 \rangle  = 24 =  S_4 </math>, <math>\sum_g  \chi ^2 = 24</math>; with (U) <math>\Rightarrow</math> Mehta–Seshadri stability <math>\Rightarrow h^0(E) = 0</math>, all 324 section-saturation cases; BG dictionary <math>h^0 = 0 \Leftrightarrow a_i \geq 1</math>); INTEGRALITY = Birkhoff–Grothendieck, automatic once <math>E</math> is typed as a bundle. UNIQUENESS + SELECTOR: <math>\{1,1,2\}</math> unique (v491); the Schur–Horn permutohedron of <math>4 \cdot \text{spec}(A_0)</math> has integer points exactly <math>\{(2,2,0), (2,1,1)\}</math> (the two options of the “Global computation” theorem) and <math>h^0 = 0</math> kills <math>\{0,2,2\}</math> (an <math>\mathcal{O}(0)</math> summand has <math>h^0 = 1</math>) — the “balanced” selection is now TYPED as stability/positivity. THE <math>h^1</math> DISCREPANCY, REAL + RESOLVED: <math>h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3</math> — the naive “generations = <math>H^1(E)</math>” typing is FALSE (forcing it demands <math>\deg -6</math> and loses uniqueness: <math>\{1,1,4\}, \{1,2,3\}, \{2,2,2\}, e_2</math> never 5); the correct TWO-OBJECT typing: <math>L = \mathcal{O}(-D)</math>, <math>D = \mu_4</math>, carries the generations (<math>h^1(L) = 3 = N_{\text{fam}} \cong H^0(\Omega^1(\log D))</math>, the v137 basis), while <math>E</math> carries them as its FIBER (rank 3) with the anchor as splitting type. BISWAS EXPLICIT (independent convergence witness): the weight denominator is <math>3 = N_{\text{fam}}</math>, so the Biswas cover is the <math>\mathbb{Z}_3</math> cusp cover <math>y^3 = (z-1)(z-i)(z+1)^2(z+i)^2</math>, genus 2 (NOT the pillowcase double <math>w^2 = z^4 - 1</math>, genus 1, <math>j=1728</math> — the deck side); <math>p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2</math> (<math>\chi</math> cross-check <math>-1 = -1</math>); the regular-representation model bundle lands on <math>\{0,2,2\}</math> = the UNSTABLE Schur–Horn companion (<math>h^0 = 1</math>): integrality AND the sum rule COME OUT of the correspondence, and stability (irreducibility, v117) is the selector. NEGATIVE CONTROLS: drop positivity <math>\rightarrow 4</math> resp. 31 solutions; <math>\deg -3/-5</math> breaks; <math>n</math>-mark scan: <math>\{1, \dots, 1, 2\}</math> unique for EVERY <math>n \geq 3</math> but <math>e_2 = (n-2)(n+1)/2 = 5</math> ONLY at <math>n=4</math> — the VALUE, not the uniqueness, is the <math>n=4</math> content; MS-rational exponents <math>e_2 = \frac{16}{3}, \frac{44}{9}</math>. [C]</p> <p>RESIDUALS: R1 the module identification “generation space = this <math>H^1</math> with this connection” [C] (QGEO.SYM territory, stays [O]); R2 unitarity (U) [C] (v33/v40); R3 the analytic-monodromy identification (fine type Numerical, v117 check 4). P2 stays an axiom (AX.P2.01 declared, no marker moves); NON-CIRCULAR: neither v23’s splitting-type input nor v115’s <math>\text{diag}(A_0) = a/4</math> input is used. Sharpens v491. Exact (sympy); Wolfram-mirrored. [E][C]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v500_celestial_<br>wp5c_gns_limit_<br>state.py | <p>CELEST.WP5C.01: WP5c of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the GNS limit state”, the third constructive milestone of WP5 (v497 made the boundary limit precise, v498 built the deleting operator; WP5c constructs the STATE whose GNS kernel contains the ideal). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: the quasi-free family <math>\omega_w</math> — loop sector (<math>r=0</math>) = the affine <math>k=1</math> vacuum <math>n</math>-point functions via the MACHINE-determined compact anti-involution <math>\theta(e_\alpha)=-e_{-\alpha}</math>, <math>\theta(h)=h</math> (the UNIQUE sign that is an anti-automorphism on ALL 61504 basis pairs; contravariance <math>\langle J_{-1}^x u, v \rangle = \langle u, J_1^{\theta x} v \rangle</math> exact); radial sector (<math>r \geq 1</math>) = oscillator pairings <math>x^{wr}</math> (the exact Gibbs regulator <math>e^{-\beta E_w}</math>, kept <math>x</math>-adic). [E]</p> <p>CONVERGENCE, SHARP: <math>\omega_w = \omega_\infty \bmod x^{N+1}</math> for ALL <math>w \geq N+1</math> (<math>N=8</math>) and NOT yet at <math>w=N</math> (the <math>r=1</math> modes sit at order <math>N</math>) — the WP5a threshold <math>w = n+1</math> holds IDENTICALLY at the state level; positivity of every <math>\omega_w</math> checked. [E] THE KERNEL (core, fully machine): the FULL exact level-2 Gram — 31124 monomials in 9361 weight blocks (dim profile 164:1, 37:240, 7:2160, 1:6960), every block exact: level-1 rank 248 POSITIVE DEFINITE (Cartan block = the <math>E_8</math> Cartan matrix, det 1); level-2 rank <b>4124</b>, kernel <b>27000</b>, every block PSD; rank per Weyl orbit (norm 8/6/4/2/0) = (0, 0, 1, 8, 44); kernel = <math>V(2\theta)</math> WEIGHT BY WEIGHT (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at <math>(0, \omega, \theta, 2\theta)</math>; direct ideal checks <math>F_i s\rangle</math>, <math>F_j F_i s\rangle</math> norm 0. [E]</p> <p><math>\mu_4</math>-EQUIVARIANCE: <math>\omega_w \circ \text{clock} = \omega_w</math> for ALL <math>w</math>; the clock unitary DESCENDS to <math>\text{GNS}(\omega_\infty)</math>; level-2 GNS rank split by clock class</p> <p><math>= (1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8</math> at <math>q^2</math> (the two-routes identity at the STATE level); level-1 split (60, 64, 60, 64). RESOLUTION OF WP5b S4.10: <math> s\rangle</math> IS the ZERO VECTOR of <math>\text{GNS}(\omega_\infty)</math> — the twisted two-<math>C_1</math>-mode floor (6 quarters <math>= q^{3/2}</math>) concerns only pre-limit objects (GNS has no state there: <math>6 \bmod 4 \neq 0</math>).</p> <p>[E] NEGATIVE CONTROLS: <math>k=2</math> (<math>\langle s s \rangle = +4</math>, dominant blocks FULL rank — NO ideal in the kernel); <math>k=0</math> (level-1 Gram <math>\equiv 0</math> — no 248 layer without the central extension); <math>D_8/SO(16)_1</math> (level-2 rank 2076 = 7380 – 5304, kernel <math>= V(2\theta_{D_8})</math> PSD weight by weight, but <math>h = (0, \frac{1}{2}, 1, 1)</math>: the vacuum GNS is 1 of 4 blocks — ONE-BLOCK completeness stays <math>E_8/\mu_4</math>-specific); wrong family <math>E'_w = mw+r</math> (level-1 limit rank 0: the 248 layer ERASED, <math>u^4</math> content <math>710955 \neq 248</math>); no damping (<math>897266 \neq 248</math>); CCR OBSTRUCTION: at a FIXED radial pairing <math>c &gt; 0</math> positivity forces <math>\omega(bb^*) \geq c</math> — NO <math>w</math>-uniform damping state exists: the family formulation is NECESSARY and the family EXISTS <math>\Rightarrow</math> the preregistered KILL is probed and NOT triggered. [C] the quasi-free/Gaussian extension to the full <math>*</math>-algebra; Kac positivity beyond level 2 (levels <math>\leq 2</math> machine-checked); the module identification in the component route. [O] local normality of <math>\omega_\infty</math>, the net structure / Haag duality (WP5d), Costello–Li (WP5e), the operator-algebraic closure beyond the graded/<math>x</math>-adic topology;</p> <p>SEAM.EQUIV.01 stays [O], untouched. VERDICT: SUCCESS on the preregistered criterion (dim GNS level-2 = 4124 exactly). Exact (integer + Fraction). [E][C][O]</p> |

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| Script                                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v501_celestial_<br>wp5d_two_interval_<br>index.py | <p>CELEST.WP5D.01: WP5d-<math>\alpha</math> of the celestial-holographic research contract<br/> CELEST.SEAM.01 — “the two-interval index from the lattice” (the local-net question, alpha stage). The roadmap discriminator: <math>(E_8)_1 \Rightarrow \mu=1</math> (Haag-dual two-interval net, ONE sector); <math>SO(16)_1 \Rightarrow \mu=4</math>; preregistered KILL = “the two-interval <math>\mu</math>-offset does NOT vanish after the condensation”. numpy/scipy Gaussian machinery + exact Fractions. [E] SETUP (machine identities): critical Majorana ring (v450 style, NS closure, <math>c=\frac{1}{2}</math> per layer), 16 layers = the seam carrier <math>c_-=8=g_{\text{car}}+N_{\text{fam}}</math> (v489); Peschel-exact; three prescriptions — fermionic <math>\rho</math>, orbifold sector sum <math>\{1, P_A, P_B, P_{AB}\}</math>, dual algebra <math>\{1, P_{AB}\}</math>; superselection <math>[\rho, P_{AB}]=0</math> machine-exact; EVERY Gaussian formula validated against exact diagonalisation (<math>\leq 1.5 \times 10^{-15}</math>). [E] FERMIONIC MI EXTENSIVE (the <math>\mu=1</math> reference; lattice witness): <math>I(x) = -(c/3) \ln(1-x)</math> with <math>c</math> fitted 0.5000; <math>\max  F(x)  = 5.97 \times 10^{-5}</math> at <math>N=512</math>, <math> F </math> falling monotonically to <math>1.3 \times 10^{-6}</math> at <math>N=1024</math> — Casini–Huerta extensivity on the lattice. [E] THE <math>\mu</math>-OFFSET (core): Rényi-2 exactly <math>\Delta_2 = -\ln[(1+R)/2]</math>; the 16-layer seam carrier saturates <math>\ln 2</math> to MACHINE PRECISION (<math> \Delta_2 - \ln 2  = 1.1 \times 10^{-15}</math> at <math>N=512</math>; <math>R \sim N^{-0.500}</math>); index reading <math>[F : F_{\text{even}}] = e^{\Delta_\infty} = 2.000000 \Rightarrow \mu_{\text{gauged}} = e^{2\Delta_\infty} = 4.000000 =</math> the v490 parity census (TWO independent lattice witnesses); gauging every layer separately costs <math>16 \ln 2</math> — the offset measures <math>\ln[\text{index of the subnet}]</math>, not a universal blob. [E] HAAG-DUALITY INDICATOR: fermionic <math>S(E)=S(E')</math> to <math>8.2 \times 10^{-12}</math> (graded-duality proxy); the orbifold BREAKS complementarity (<math>D(E) - D(E') = 0.427</math> at <math>a=64</math> — the direct duality-failure witness); the complementary-pair sum is exchange-symmetric and <math>\leq \ln 4 = \ln \mu(SO(16)_1)</math> with <math>\max 1.2637</math> at the self-dual <math>x=\frac{1}{2}</math> and <math>\rightarrow \ln 2</math> at the squeezed ends: the <math>\ln 4</math> KLM budget is a DOUBLE-LIMIT statement, NOT a fixed-<math>x</math> identity (the first naive fixed-<math>x</math> check was WRONG and was replaced — honestly typed). [E]<br/> CONDENSATION ARITHMETIC, LATTICE-ANCHORED:<br/> <math>\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 4 \cdot 4 = 16 = \mu(\text{carrier})</math>; KLM/Longo–Rehren quotients <math>\mu/ L ^2</math>: <math>16/4^2 = 1</math> (one-step <math>\mu_4</math> glue) and <math>16/2^2 = 4 \rightarrow 4/2^2 = 1</math> (via <math>SO(16)_1</math>), both routes EXACT; <math>\Sigma d_i^2</math> cross-checks (Ising 4, <math>SO(16)_1</math> 4, <math>(E_8)_1</math> 1); <math>\theta_v = e^{2\pi i \nu/16} = 1</math> EXACTLY at <math>\nu = 2c_- = 16</math>, rivals <math>\nu = 1, 2, 8</math> give <math>\theta_v \neq 1</math>. The anchored chain: <math>\mu = 1</math> (graded, MEASURED) <math>\rightarrow 4</math> (gauged, MEASURED + v490 census) <math>\rightarrow 1</math> (arithmetic, [C]: the condensed net is interacting — the v490 fence). [E] NEGATIVE CONTROLS: <math>\nu=1</math> single Majorana (offset non-trivial, <math>\Delta_2 = 0.6364</math> at <math>N=2048</math> rising, and <math>\theta_v = e^{i\pi/8} \neq 1</math>: NOT removable — the kill discriminator has teeth); trivial dimerised phase (everything <math>\leq 10^{-7}</math>); wrong sector sum <math>\{1, P_{AB}\}</math> (offset EXACTLY 0 instead of <math>\ln 2</math>). [C] the continuum dictionary “offset = <math>\ln</math> index” (KLM two-interval index; Longo–Xu; Casini–Huerta–Magan–Pontello entropic order parameters) + the <math>\theta_v=1</math> condensation step itself. [O] continuum Haag duality, local normality, split/strong additivity of the quotient net (WP5d-<math>\beta</math>, running as exploration), Costello–Li (WP5e); SEAM.EQUIV.01 stays [O], untouched. VERDICT: the <math>\mu=1</math> indicator chain PASSES; the preregistered KILL is NOT triggered. Python (numpy + scipy + Fractions). [E][C][O]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v502_celestial_<br>wp5e_prefactor_<br>level.py | <p>CELEST.WP5E.ALPHA.01: WP5e-<math>\alpha</math> of the celestial-holographic research contract CELEST.SEAM.01 — “prefactor and level bookkeeping”, the CFT-side anchor stage of WP5e (the BCOV/Costello–Li twistor uplift: boundary partition function <math>= E_4/\eta^8</math> including the <math>q^{-1/3}</math> prefactor; preregistered kill: the anomaly inflow demands a level <math>\neq 1</math>). Exact sympy/Fraction. [E] PART A — the <math>q^{-1/3}</math> prefactor is exact <math>\mu_4</math> vacuum-energy bookkeeping; Hurwitz-<math>\zeta</math> derivation (<math>\zeta(-1, \theta) = -B_2(\theta)/2</math>; <math>E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4</math> exactly); the clock is INNER (<math>h = (2, 2, 2, 2, 2; 0^4)</math> and the <math>A_3</math> detector <math>h' = (0^5; 1, 1, 1, -3)</math> both read the class mod 4 on all 240 roots; <math>(h, h) = 20</math>, <math>(h', h') = 12</math>, sum 32) <math>\Rightarrow</math> twist distribution <math>\theta = 0^8</math> in ALL four sectors — a SHIFT orbifold (lattice cosets, integer-moded oscillators), <i>not</i> a rotation orbifold; <math>E_{\text{osc}} = -\frac{1}{3} = -c/24</math> sector-independent (<math>c = 8 = \text{rank} = 248/31</math>); sector ground exponents <math>-\frac{1}{3} + (0, 1, 1, 1)</math>. [E] discriminant form = Casimir energy, TWO routes: spectral flow <math>j^2(h, h)/32</math> (<math>5j^2/8, 3j^2/8</math>, sum <math>j^2 \equiv 0 \pmod{1}</math> = the isotropy/Lagrangian condition) and <i>exactly</i> via free fermions (<math>D_5 = 10</math> Majorana, <math>A_3 = D_3 = 6</math>, glue diagonal <math>= 16 = 2^{g_{\text{car}}-1}</math>, the WP5d seam carrier; R–NS shift <math>n/16</math>: <math>\frac{5}{8}, \frac{3}{8}, 1</math>); the ROTATION reading fails (the geometric <math>A_3</math> deck gives <math>\frac{3}{16} \neq \frac{3}{8}</math>: the v492 “clock <math>\neq</math> deck” lesson at the Casimir level). [E] character: <math>\sum_j \Theta_{C_j} = E_4</math> and the four sector characters (leading coefficients <math>(1, 64, 60, 64)</math>, currents <math>(60, 64, 60, 64)</math>) sum exactly to <math>E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)</math>. [E] PART B — <math>k = 1</math> forced THREE independent ways: (i) the CURRENT condition <math>h(J) = k = 1</math> (the <math>\mathbb{Z}_4</math> extension closes onto the <math>E_8</math> adjoint <math>60+64+60+64 = 248</math> only at spin 1); HONEST SHARPENING: glue-<math>h</math> integrality <math>h(J^j; k) = k(0, 1, 1, 1)</math> holds for ALL <math>k = 1..8</math> — the originally conjectured integrality route fixes <i>nothing</i>, the three other dials carry; (ii) conformal embedding <math>47k^2 + 219k - 266 = 47(k-1)(k+266/47) = 0</math> resp. <math>D_8</math>-in-<math>E_8</math>: <math>128k(1-k) = 0</math>; (iii) central charge <math>248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0</math> (= the prefactor itself); plus the v498 singular-vector dial (<math>\ (E_{-1}^\theta)^m 0\ ^2 = m! k(k-1) \cdots (k-m+1)</math>, singular vector exactly at grade <math>k+1</math>; <math>k = 1</math> deletes <math>V(2\theta)</math>: <math>31124 - 27000 = 4124</math>). [E] negative controls: <math>D_8/SO(16)_1</math> (same rank, same <math>c = 8</math>, SAME <math>q^{-1/3}</math> prefactor, but <math>h = (0, \frac{1}{2}, 1, 1)</math>: the discriminator is glue-<math>h</math> integrality, not the prefactor; <math>\Theta_{D_8} + \Theta_s = E_4</math>); <math>\mathbb{Z}_2</math> rotation (<math>E = +\frac{1}{6}</math>, <math>h = \frac{1}{2}</math>: no <math>\mu_4</math>-shift sector there); <math>\mu_4</math> rotation (sector-dependent <math>(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})</math>: the common prefactor breaks); wrong <math>k \in \{2, 3, 4\}</math> fails all five dials coherently (residuals <math>360/814/1362</math>; <math>\langle s s \rangle = 4/12/24</math>; level-2 dim stays <math>31124 \neq 4124</math>). [C] Kac generation of the maximal submodule by the grade-<math>(k+1)</math> singular vector for <math>k \geq 3</math> (<math>k = 1, 2</math> machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion <math>\lambda_g/(8\pi\sqrt{3})</math> on <math>\mathbb{P}^1</math>, the Costello–Paquette–Sharma level-from-flux mechanism) stays open — v502 anchors the CFT side of WP5e only (<math>\alpha</math> stage); the kill branch (“inflow demands a level <math>\neq 1</math>”) is not triggered on the CFT side but its twistor side is untested; WP5e stays [O]; SEAM.EQUIV.01 untouched, no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v503_qgeo_<br>emergence_light.py | <p>QGEO.EMERGE.LIGHT.01: the v215 <i>deferred emergence test</i> (lever 7) executed in its first honest LIGHT version, plus the residual-convergence result QGEO-R1 <math>\equiv</math> P2-R1 (analysis/kill-test class; NO marker moves — QGEO.REALIZE.01 stays [C]/[O]). NO DYNAMICAL SELECTION (<math>\zeta</math>-log-det via the Dedekind-<math>\eta</math> formula, mpmath 40 digits): <math>\log F = \log(\tau_2 \eta ^4)</math> is exactly <math>S</math>-invariant, so <math>t = 1</math> is automatically critical on the rectangular family <math>\tau = it</math>; there a strict local maximum (<math>g''(1) = -1.298</math>) but a <i>minimum</i> in the <math>\tau_1</math>-direction (<math>h''(0) = +0.298</math>): <math>\tau = i</math> is a SADDLE of the full moduli space; critical points = the CM points (<math>E_2^*(i) = E_2^*(\rho) = 0</math> to 40 digits); the <i>global</i> det' winner at fixed area is the HEXAGONAL torus (<math>\log F(\rho) = -1.03352 &gt; \log F(i) = -1.05469</math>; Osgood–Phillips–Sarnak 1988) — free-field dynamics alone does <i>not</i> select <math>\tau = i</math>; criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). SHARP RIGIDITY (v280 lattice DtN): the mod-4 off-character indicator (the v215-K3 lever as a <i>modulus function</i>) has an isolated zero (<math>8 \times 10^{-15}</math>) exactly at <math>\tau = i</math>, strictly monotone in both directions (min <math>8.4 \times 10^{-2}</math> away); <math>R^2 = \mathbb{Z}_2</math> exactly as permutation matrices (the v492 <math>\mathbb{Z}_8</math> bridge on the lattice); <math>j(it) \geq 1728</math> with equality only at <math>t = 1</math>, <math>j(\rho) = 0</math>. [E] the <math>\mathbb{Z}_2</math> chain is MODULUS-BLIND: 4 fixed points and <math>[\mathbb{Z}_2, \Delta_t] = 0</math> on <i>every</i> torus (v216 confirmed without mark input); pillowcase det' = <math>\frac{1}{2}</math> torus det' exactly (<math>\pm</math>-pair structure), so all <math>\tau</math> conclusions transfer verbatim; marks/mark-locality/deck<sup>2</sup> exist on <i>every</i> pillowcase (v264/v280), only the order-4 clock distinguishes <math>\tau = i</math>; <math>\text{Aut}(E_\rho) = \mathbb{Z}_6</math> (element orders 1, 2, 3, 6) contains <i>no</i> order-4 element — dynamics and clock are logically independent selectors; GIVEN the clock, <math>P(iZ) = P(Z)</math> forces <math>a_3 = a_2 = a_1 = 0</math> and <math>j = 1728</math> identically (v493): the modulus is not a second input. [C]/[O] RESIDUAL CONVERGENCE (the core finding): the order-4 element on <math>H^1</math> is a Coxeter element of <math>W(A_3)</math> (char poly <math>\lambda^3 + \lambda^2 + \lambda + 1</math>, order <math>h(A_3) = 4 =  \mu_4  = N_{\text{fam}} + 1</math>) — exactly the P2 cusp-class/carrier datum of the P2 hardening (v499 R1) <math>\Rightarrow</math> QGEO-R1 <math>\equiv</math> P2-R1: ONE common residual, “why does the raw seam carry the order-4 clock”, not two scattered ones; chain: marks derived (v216 [E]), square derived <i>given</i> the clock (v493 [E]), clock = carrier input [C], rest = the dynamical realisation of the clock [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data <math>\Rightarrow</math> order-4 clock” is the live route — free dynamics alone does not deliver the clock (the hexagonal finding). Lever update (documented here, v215 itself unchanged): lever 7 lifted from “deferred, not executable” to “light version executed: rigidity, not dynamical selection”. Python (mpmath + numpy/scipy + sympy). [E][C][O]</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v504_celestial_<br>wp5d_klm_<br>completeness.py | <p>CELEST.WP5DB.01: WP5d-<math>\beta</math> of the celestial-holographic research contract<br/> CELEST.SEAM.01 — “split + strong additivity from the lattice”, the two<br/> remaining Kawahigashi–Longo–Müger legs of complete rationality (the <math>\mu</math>-index is<br/> v501), witnessed for the orbifold prescription on the critical Majorana seam edge;<br/> the <math>U(1)</math>/Dirac subnet is the preregistered negative control (the current net,<br/> where strong additivity fails in the continuum). [E] STRONG ADDITIVITY,<br/> EXACT: shared boundary Majorana <math>\Rightarrow</math> Even(<math>A</math>) <math>\vee</math> Even(<math>B</math>) = Even(<math>A \cup B</math>)<br/> EXACTLY (GF2 spans full 64/64 up to 1024/1024; matrix rank 32/32);<br/> DISJOINT: exactly HALF (rank 16/32, index 2 — the missing odd <math>\otimes</math> odd sector is<br/> the v501 ln 2 bit, localised at the split point); the neutral <math>U(1)</math> algebras do NOT<br/> generate the union even with the shared site (gaps 2/10/52, growing). [E]<br/> ENTROPIC WITNESS [N]: the fermionic touching MI carries the pure UV<br/> point-gap law <math>((c/3) \ln 2</math> increments, dev <math>1.2 \times 10^{-6}</math> — no discriminator); the<br/> honest discriminator is BOUNDED vs DIVERGENT: the <math>\mathbb{Z}_2</math>/orbifold deficit stays<br/> <math>&lt; \ln 2</math> with <math>(\ln 2 - \Delta_2) \sim N^{-p}</math>, <math>p = 0.2444</math> vs <math>2\Delta_\mu = \frac{1}{4}</math> (Ising disorder dimension;<br/> “defect <math>\rightarrow 0</math>” would be FALSE at the sharp split — bounded <math>\Leftrightarrow</math> finite index,<br/> Longo–Xu), while the <math>U(1)</math> control BURSTS ln 2 from <math>L = 128</math> and grows as<br/> <math>\frac{1}{2} \ln \text{Var } Q_A</math> (Klich–Levitov 0.10134 vs <math>1/\pi^2</math>): infinite index. [E]<br/> SPLIT/NUCLEARITY [N]: coupling <math>\sigma_k</math> fall at the elliptic-nome rate<br/> <math>\pi K(1-x)/K(x)</math> to 1.3–2.0%; <math>\sigma_1 \sim x^{0.5044}</math>; trace norm summable; ORBIFOLD<br/> INHERITANCE EXACT: <math>P_A</math> flips <math>C \rightarrow -C</math> (<math>\sigma_k</math> identical, <math>8.3 \times 10^{-17}</math>), the<br/> even-bilinear Gram is the compound <math>\Lambda^2 C</math> (<math>\{\sigma_a \sigma_b\}</math>, <math>2.8 \times 10^{-14}</math>) — Longo heredity<br/> [C]. [E] PIMSNER–POPA: <math>E(a) - a/2 = PaP/2</math> IDENTICALLY <math>\Rightarrow</math><br/> <math>\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]</math>; integer attainment (16384 + 2048 monomial sweeps, 0<br/> violations; pencil min <math>\lambda = 0.5000000000</math>); <math>\lambda_{E_4} = \frac{1}{4} = 1/\mu</math> exactly; INDEX<br/> CONSISTENCY, two routes: <math>e^{\Delta_\infty} = 2.0000000000 = 1/\lambda_{\text{PP}}</math> and <math>e^{2\Delta_\infty} = 4 = \mu</math>;<br/> <math>U(1)</math>: <math>\lambda = 1/(m+1) \rightarrow 0</math>. [E] CONTROLS + HONESTY: dimerised phase trivial<br/> (<math>3.6 \times 10^{-15}</math>); wrong prescription <math>\{1, P_{AB}\}</math> deficit EXACTLY 0; finite-size honesty<br/> (14.9% below ln 2 at <math>N = 4096</math> — witness, not proof). [C] the KLM dictionary<br/> (Longo–Xu; Longo heredity; CHMP). [O] the continuum theorem “finite-group<br/> orbifolds of completely rational nets are completely rational” is XU’S THEOREM<br/> — CITED, NOT CLAIMED; the seam quotient net and the condensed <math>(E_8)_1</math> net<br/> stay WP5e/Costello–Li; SEAM.EQUIV.01 stays [O]. VERDICT: with v501, all<br/> THREE KLM ingredients witnessed on the lattice. Python (numpy + scipy);<br/> exact parts Wolfram-mirrored. [E][C][O]</p> |

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| Script                                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v505_celestial_<br>wp5e_beta_<br>equivariant_<br>ledger.py | <p>CELEST.WP5E.BETA.01: WP5e-<math>\beta</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the equivariant anomaly ledger on twistor space”, the twistor-side milestone of WP5e (the BCOV/Costello–Li uplift; preregistered kill: the anomaly inflow demands a level <math>\neq 1</math>); WP5e-<math>\alpha</math> (v502) anchored the CFT side. The Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) evaluated exactly on <math>\mathbb{C}^2/\mathbb{Z}_4</math> with the <math>E_8</math> <math>\mu_4</math>-glue monodromy. Exact sympy/Fraction. [E] AB ledger: denominators <math>\det(1 - g_j^{-1}) = (2, 4, 2)</math>; <math>\sum_j 1/\det_j = 5/4 = ( \mathbb{Z}_4 ^2 - 1)/12</math> (Dedekind); equivariant characters <math>\mathrm{tr}_{g_j}(\mathrm{Ad} E_8) = (248, 0, -8, 0)</math> by two routes; invariant average <math>60 =</math> the <math>D_5 + A_3</math> carrier; Frobenius 61568; <math>\mathrm{tr}_{g_2} = -8 = 120 - 128</math>. [E] Okubo per sector (honest sharpening): the invariant sector is exactly Okubo-quadratic (<math>Q^{(0)} = 36\langle x, x \rangle^2</math>, <math>36 = \tilde{\lambda}_{e_8}^2</math> — v495 re-derived through the glue grading); the twisted sectors do NOT reduce (<math>T_5: (-8, 32, -8)</math>, <math>T_3: (48, -64, 48)</math>); the AB-weighted sum cancels the <math>D_5</math> quartic exactly and leaves the RIGID residual <math>32T_3</math>; the only quartic-free weighting is the uniform <math>\mathbb{Z}_4</math> projector; the graded GS exchange is exactly rank-obstructed in sectors 1, 2, 3; heterotic identities exact (<math>\mathrm{Tr}_{120}, \mathrm{Tr}_{128}, \mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2</math>). [E] index bridge (centrepiece): <math>f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2</math> EXACTLY — fixed-point ledger = McKay/Kronheimer intersection ledger; glue <math>\mathrm{ch}_2</math> defect <math>-78</math> by both routes (classes <math>(-24, -30, -24)</math>; <math>-30 = -h^\vee</math>, a [C] coincidence); <math>\sum_m h_{A_3}(m) = 5/4 = \sum_j 1/\det_j</math>; Killing sums <math>(320, 320, 240, 320)</math>, total <math>1200 = 2h^\vee(h, h)</math>. [E] <math>k = 1</math> geometric: lattice current count = 240 at <math>k = 1</math>, exactly 0 at <math>k = 2, 3, 4</math>; embedding residual <math>47k^2 + 219k - 266 = (0, 360, 814, 1362)</math>; <math>h(J; k) = k</math> integer for all <math>k</math> — integrality alone fixes nothing (honest, as v502); one scale <math>\Rightarrow</math> one level (v493 periods). [E] E3 REFUTED (prominent): <math>a_0 \in O(8)</math> fills the GRAVITON slot <math>O(2)</math>, not the BSS axion slot <math>O(-2)</math> (<math>X = -2 \neq +2 = Y</math>; mismatch <math>4 =  \mu_4 </math>, a [C] coincidence) — “the theory brings its GS axion as <math>a_0</math>” is refuted; instead the three <math>H^2(\mathrm{ALE})</math> classes carry exactly the Coxeter eigencharacters <math>\{i, -1, -i\} =</math> the three twisted slots (bijection); slot count <math>1 + 3 = 4 =  \mu_4 </math> soft. [E] negative controls: <math>\mathrm{diag}(i, i)</math> breaks four ways (<math>\sum 1/\det = \frac{1}{4}</math>, <math>\omega</math>-weight <math>-1</math>, 5 generators); <math>SO(16)</math> glue: characters <math>(120, 0, +120, 0)</math>, defect <math>-30 \neq -78</math>, bulk Okubo fails (independent quartic Casimir); <math>k = 2</math>: count 0, prefactor <math>-31/48 \neq -\frac{1}{3}</math>. [C] dictionaries <math>(\mathrm{CS}(\rho) \bmod 1 = \text{discriminant weight}; 32 = (h, h) + (h', h'); -30 = -h^\vee; \Omega\text{-contraction})</math>. [O] the global BCOV/Kodaira–Spencer quantisation on <math>\mathbb{PT}/\mathbb{Z}_4</math>, the <math>O(-2)</math> bulk-axion construction, CPS level-from-flux, the pairing of the <math>32T_3</math> residual with the three sphere axions. Verdict B; the kill branch does NOT fire on the equivariant skeleton; SEAM.EQUIV.01 untouched, no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v506_seam_clock_rigidity.py | <p>SEAM.CLOCK.RIGIDITY.01: clock rigidity, Route B of the v503 classification — two exact theorems reducing the order-4 clock input to ONE alignment bit (Part A Moebius + Part B Fock in one module; QGEO.REALIZE.01 stays [C]/[O], no marker moves). Exact sympy + exact Cl(16) Fractions. [E] THEOREM A (Moebius): the collar deck <math>z \mapsto -z</math> has in <math>\mathrm{PGL}_2(\mathbb{C})</math> exactly TWO square roots (<math>z \mapsto \pm iz</math>), both automatically of order 4 (complete case analysis: <math>\mathrm{tr} \neq 0 \Rightarrow</math> diagonal <math>\Rightarrow a/d = \pm i</math>; <math>\mathrm{tr} = 0</math> impossible) — “order 4” is no longer an input; a marks-preserving root exists <math>\iff</math> the deck is the CENTRAL involution of the mark-<math>D_4</math> (center = {id, deck}, exactly 2 order-4 elements = the clocks) <math>\Rightarrow</math> then marks = one <math>\mu_4</math> orbit, cross-ratio class <math>\{2, \frac{1}{2}, -1\}</math>, <math>j = 1728</math>, <math>\tau = i</math>, free 4-cycle, deck = clock<sup>2</sup>, only the scale free; COUNTEREXAMPLE: <math>\{\pm 1, \pm(3+2\sqrt{2})\}</math> is deck-free AND harmonic with full <math>D_4</math>, but the deck is non-central (edge involution) <math>\Rightarrow 0</math> roots — harmonicity alone is NOT sufficient; the RP filter selects nothing (<math>\mathrm{Aut}(\mathbb{Z}_4)</math> ambiguity irreducible, consistent with v492). [E] B3/B4: free orbit <math>\Rightarrow n \mid 4</math>; mark transitivity <math>\Rightarrow n = 4</math>; faithfulness kills <math>n \geq 5</math>; <math>n = 3</math> fixes a mark; <math>n = 6</math> dies at cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); controls: <math>n = 8</math> impossible; hexagonal <math>A_4</math> (0 order-4 elements, squares only <math>\{1, 3\}</math>); generic <math>V_4</math> (exponent 2); stabiliser trichotomy <math>V_4/D_4/A_4</math> with order-4 counts (0, 2, 0). [E] THEOREM B (Fock, the core result): the unique <math>S^2</math> spin structure gives NS on every circle (established P1 datum); one-particle <math>D^2 = -1</math> (charpoly <math>(\lambda^2+1)^8</math>), quarter-shift <math>C^2 = D</math>, <math>C^8 = 1</math> (charpoly <math>(\lambda^4+1)^4</math>); Fock: <math>\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})</math>, <math>\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F</math> EXACTLY, and <math>(cU)^2 = c^2(-1)^F</math> is never 1 (center of Cl(16) = scalars) — the <math>\mathbb{Z}_4</math> lift is FORCED (nonsplit extension); <math>\mathbb{Z}_8</math> tower: <math>V^2 \propto \tilde{U}</math>, <math>V^4 \propto (-1)^F</math>, projective order <math>8 = 2 \mu_4 </math> — fermionically explained; R control <math>U_R^2 = +1</math> (split — the forcing rests on NS = established geometry); 16-fold way <math>\theta_\nu = 1</math> at <math>\nu = 16</math>; the forcing already holds at 8 sites. [C] THE ONE REST: “the collar deck is the central involution of the mark-<math>D_4</math>” (<math>\iff</math> square-root existence on the mark data) — ONE alignment bit; the fermionic <math>\mathbb{Z}_4</math> forces the ORDER, not the spatial alignment. No overclaim. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v507_seam_bit_origin.py | <p>SEAM.BIT.ORIGIN.01: the tautology attack on the v506 alignment bit, executed and refuted — where do the marks AND the deck come from, and does their common origin force the central alignment? (Part A torus origin + Part B fermion dichotomy in one module; NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01 and QGEO.REALIZE.01 text-precised only.) Exact sympy + exact Cl(16) Fractions. [E] ORIGIN THEOREM: the 4 marks are the fixed points of the hyperelliptic involution <math>w \mapsto -w</math> on <math>\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =</math> the four half-periods <math>\Lambda/2\Lambda</math>; a translation <math>t_c</math> descends to the base sphere iff <math>2c \in \Lambda</math> — exactly the THREE half-period translations (control <math>c = 1/3</math> fails); the descended maps are the EXPLICIT rational involutions <math>\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)</math> with the exact curve-lift identity <math>\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)</math>, <math>\psi_i</math> RATIONAL; <math>\{\text{id}, \sigma_1, \sigma_2, \sigma_3\} =</math> Klein <math>V_4</math>; CONVERSELY, in ALL three stabiliser types the freely-acting involutions are EXACTLY the three <math>\sigma_c</math> (generic <math>V_4</math> (4, 3, 3), harmonic <math>D_4</math> (8, 5, 3), hexagonal <math>A_4</math> (12, 3, 3)) <math>\Rightarrow</math> EVERY mark-free collar deck (v216) IS a half-period translation — the CLASS of the deck is forced, its POSITION is not. [E] CORE SOLVE (symbolic <math>\lambda</math>): <math>\sigma_c</math> admits a marks-preserving <math>\text{PGL}_2</math> square root <math>\iff \lambda \in \{-1\}(\sigma_1), \{2\}(\sigma_2), \{\frac{1}{2}\}(\sigma_3)</math> — per half-period exactly ONE solution; intrinsically: the two deck pairs separate each other HARMONICALLY (pair cross-ratio = <math>-1</math>); union over <math>c =</math> the harmonic orbit <math>\{-1, 2, \frac{1}{2}\} \iff j = 1728 \iff \tau = i</math>; generic modulus: <math>V_4</math> abelian, ALL <math>\sigma_c</math> central and yet clockless (“central alone is empty” — the bit is the CONJUNCTION <math>D_4</math> AND central); at <math>\tau = i</math> the stabiliser jumps to <math>D_4</math> with <math>\sigma_1</math> central (2 roots = the clock pair) and <math>\sigma_2/\sigma_3</math> edge-type (0 roots); the v506 seam deck <math>z \mapsto -z</math> lands on the central <math>\sigma_1 = -1/x</math> under ALL 8 mark bijections; CM DICTIONARY: mult-by-<math>i</math> fixes exactly ONE nonzero half-period <math>c^* = (1 + \tau)/2</math> (mult-by-<math>\rho</math> 3-cycles all three); the clock lift needs the CM unit <math>i</math> (<math>\psi = 2i/(x + 1)^2</math>, <math>t^2 + 1</math> irreducible over <math>\mathbb{Q}</math>) while half-period decks lift RATIONALLY: the clock root exists only WITH CM by <math>\mathbb{Z}[i] \iff \tau = i</math>. [E] EQUIVALENCE TETRAD + CIRCULARITY AUDIT: clock exists <math>\iff (\tau = i</math> AND deck = the CM-fixed half-period) <math>\iff (\text{Stab} = D_4</math> AND deck central) <math>\iff</math> deck pairs harmonic <math>\iff</math> the NS Fock lift is NONSPLIT; the v493 converse independently re-verified — two converses of ONE equivalence, no circularity. [E] FERMION DICHOTOMY (the new physical face): central arrangement (shift by 8): <math>\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F</math>, nonsplit <math>\mathbb{Z}_4</math>, 2 roots in the 32-element dihedral lift (the <math>\mathbb{Z}_8</math> tower); edge arrangement (reflection <math>j \mapsto 4 - j</math>, fixed sites = non-marks = the silver <math>\{0, \infty\}</math>): the implementer EXISTS but <math>\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1</math>, the extension SPLITS, 0 roots, no <math>\mathbb{Z}_8</math>; one-particle <math>(\lambda^2 + 1)^8</math> vs <math>(\lambda^2 - 1)^8</math>; census: all 28 shift-invariant 4-sets cross, all 21 reflection-invariant do not; robust at 8 sites — the bit IS a physical extension class (Fidkowski–Kitaev-type), distinguished but not derived. [E] SILVER PLACEMENT: <math>\{\pm 1, \pm(3 + 2\sqrt{2})\} =</math> “<math>\tau = i</math> with deck = a non-CM-fixed half-period translation” (all 8 bijections <math>\rightarrow \sigma_2/\sigma_3</math>, never <math>\sigma_1</math>) — in-family witness against the tautology. [C] THE REST: the one alignment bit remains a carrier datum; no tautology — the origin forces the CLASS of the deck, not its POSITION. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v508_celestial_<br>wp5e_gamma_<br>pairing.py | <p>CELEST.WP5E.GAMMA.01: WP5e-<math>\gamma</math> of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the sphere-axion pairing check”, an honest rigid NEGATIVE result: can the three twisted sphere axions <math>\eta_m</math> (Coxeter eigencharacters <math>\{i, -1, -i\}</math> on <math>H^2(\text{ALE})</math>, v505 S5.8) cancel the rigid <math>32 T_3</math> residual of the Atiyah–Bott ledger (<math>A_{\text{fix}} = 9P_1 - 30P_2 - 15P_3 + 32T_3</math>, v505) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? NO, with certificate. Exact sympy/Fraction. [E] VERTEX-SPACE THEOREM: the <math>W(D_5) \times W(A_3)</math>-invariant quadratics on the 8-dim glue Cartan are EXACTLY <math>\text{span}\{s_5, s_3\}</math> (dim 2, bidegree dims (1, 0, 1)), the quartics EXACTLY <math>\text{span}\{P_1, P_2, P_3, T_5, T_3\}</math> (dim 5, bidegree dims (2, 0, 1, 0, 2)) — Weyl nullspace arithmetic over all bidegrees, basis COMPLETE; PRODUCT THEOREM (the master kill): <math>(as_5 + bs_3)(cs_5 + ds_3) = acP_1 + (ad + bc)P_2 + bdP_3</math> identically — zero <math>T_5/T_3</math> content, so EVERY exchange image (any axion count, any selection rule, any couplings) is annihilated by <math>\Phi_{T_3}</math>, while <math>\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0</math>. [E] SELECTION ENUMERATION: the strict two-index reading COLLAPSES (<math>C_{j,j'} \neq 0</math> only for <math>j' = -j</math>, so the rule forces <math>m = 0</math> — the sphere axions have NO invariant bilinear Cartan vertex at all); twist insertion: <math>\eta_1 \leftrightarrow K^{(3)}</math>, <math>\eta_2 \leftrightarrow K^{(2)}</math>, <math>\eta_3 \leftrightarrow K^{(1)}</math>, <math>\eta_0 \leftrightarrow K^{(0)}</math>; channels <math>E_{13} = (16, -96, 144, 0, 0)</math>, <math>E_{22} = (16, 32, 16, 0, 0)</math>, <math>E_{00} = 225 E_{22}</math>; SIDE DISCOVERY: <math>K^{(0)} = -15 K^{(2)}</math> (the even-sector quadratics are PARALLEL). [E] UNSOLVABILITY WITH CERTIFICATE (the core result): exchange matrix <math>5 \times 2</math>, rank 2, kernel 0; <math>\text{rank}([M   A_{\text{fix}}]) = 3</math>; annihilator certificate <math>(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)</math> with <math>A_{\text{fix}} = (9, -30, -15, 0, 32)</math> — two independent obstructions (<math>32 T_3</math> structural, rule-independent + <math>\Phi_P = 72</math> rule-driven via the <math>K^{(0)} \parallel K^{(2)}</math> collapse); <math>\ker M = 0</math> forces the unique non-disturbing coupling <math>c = 0</math>: the sphere axions are RIGIDLY SILENT in the exchange ansatz; <math>\Phi_P(Q^{(0)}) = 0</math> (the v505 sector-0 mechanism undisturbed); relaxed control: dropping charge conservation reaches the <math>P</math>-block (<math>\det = -64</math>) but <math>T_5 = T_3 = 0</math> persists. [E] NATURALNESS RESOLUTION: <math>\text{ch}_2</math>-natural <math>(9/64, 1/4)</math> and AB-weight <math>(1/2, 1/4)</math> couplings both carry certificate <math>(0, 0)</math> vs required <math>(32, 72)</math> — the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”: the ansatz space misses the target identically). [E] NEGATIVE CONTROLS: invariant axion only = v505 replication + sharpening; flipped rule <math>m+j-j'</math>: the <math>\Phi_{T_3}</math> kill is flip-invariant, <math>\Phi_P</math> is rule-dependent; <math>SO(16)</math>: odd classes EMPTY and the AB sum <math>(15, -18, -15, -4, 40)</math> does not even cancel <math>T_5</math> (<math>E_8</math> doubly special); <math>D_8</math>: <math>16 T_8 + 12 s_8^2</math>, no <math>T_3</math> structure — the question cannot even be posed. [C] number fences: <math>32 = (h, h) + (h', h') = 20 + 12</math>; <math>72 = 2\tilde{\lambda}_{e_8}^2</math>. [O] SHARPENED FENCE: the exchange sub-branch is DEAD; the <math>32 T_3</math> residual MUST be carried by twisted BCOV contact terms — binary criterion WP5e-<math>\delta_1</math>: the one-loop coefficient on <math>\mathbb{C}^2/\mathbb{Z}_4</math> must come out exactly <math>(9, -30, -15, 0, 32)</math>, COMPUTED, not fitted; named escape WP5e-<math>\delta_2</math>: the full-tensor ledger (the collapse is Cartan-restricted); slot bijection (v505 S5.8) untouched; the preregistered level-kill still does NOT fire; SEAM.EQUIV.01 stays [O], no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v509_celestial_<br>wp5e_eps2_level_<br>from_flux.py | <p>CELEST.WP5E.EPS2.01: WP5e-<math>\varepsilon_2</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the CPS level-from-flux dial”, executed (verdict B): derive the boundary level <math>k = 1</math> from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma (CPS) Burns-holography mechanism (PRL <b>130</b>, 061602; arXiv:2306.00940). Exact sympy/Fraction, no floats. [E] CPS SKELETON REPLICATED EXACTLY: the <math>S^3</math> period <math>\int \Omega_N = (2\pi i)^2 N</math> (pullback density <math>-\sin 2\theta</math>, their eq. 3.33); the exceptional-sphere Kähler flux <math>\int_{\mathbb{P}^1} \omega = 2\pi N</math>; the boundary level magnitude <math>2N</math> via the symplectic-boson double contraction (<math>N = 1, 2, 3</math>), Dynkin normalisation <math>T(\text{vector}) = 1</math> — “level = flux quantum <math>\times</math> Dynkin index”. [E] PAIRING MATRIX PINNED, NOT POSTULATED: the v505 <math>\text{ch}_2</math> ledger + integrality + unimodularity + effectivity pin <math>P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}</math> by COMPLETE enumeration (row candidates <math>8/6/8</math>; 48 unimodular solutions of <math>PC^{-1}P^T = C^{-1}</math>; with <math>P \geq 0</math> exactly <math>2 = \{\text{identity}, A_3 \text{ diagram flip}\}</math>); glue FLUXES <math>F_i = (64, 60, 64) = \dim \mathfrak{g}_i</math>, total <math>188 = 248 - 60</math>, flip-invariant; flux per charged root current uniformly ONE quantum. [E] LEVEL FORMULA, honest: “level = total open-string flux” is KILLED by the lockstep test itself (<math>k_i = (64, 60, 64)</math> unequal = the FLAVOUR multiplicity); what works: <math>k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)</math>, anchored by the lattice current count <math>(240, 0, 0, 0)</math> for <math>k = 1..4</math> and the embedding index <math>j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1</math> (<math>60 = 2h^\vee</math>). ORD-4-VS-LEVEL-1 RESOLVED: the clock order counts FRACTIONAL flux sectors — <math>\mu = \det(D_5) \det(A_3) = 16 = \text{ord}^2</math>, the <math>\mu_4</math> glue diagonal isotropic + Lagrangian, KLM <math>16 \rightarrow 16/4^2 = 1</math> (zero residual fractional flux); NEW DIAL: <math>\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)</math> for <math>k = 1..6</math> from the affine marks — exactly ONE sector iff <math>k = 1</math>. [E] LOCKSTEP THEOREM (the lifting of v505 “one scale <math>\Rightarrow</math> one level” to a THEOREM): clock invariance forces <math>a_3 = a_2 = a_1 = 0</math>, one <math>\mu_4</math> orbit of branch points, <math> \Pi_j  = \sqrt{2}t</math> EQUAL on all three spheres, monodromy = the pure phase <math>i</math>, <math>\det(A - 1) = -4</math>: no invariant flux vector — <math>k_1 = k_2 = k_3</math> IS A THEOREM; NEGATIVE PROBE (new falsifier): the clock-forbidden family <math>Z^4 - Z</math> (disc <math>-27</math>, branch points <math>\{0, 1, \omega, \omega^2\}</math> not a <math>\mu_4</math> orbit): <math>0/24</math> orderings lockstep; HONEST LIMIT: uniform flux doubling keeps the lockstep — EQUALITY, not the value (1 from the current/sector counters and the minimal quantum <math>N = 1</math>). [E] NEGATIVE CONTROLS: <math>k = 2</math> (count 0, residual 360, prefactor <math>-31/48</math>, <math>\#\text{prim} 3</math> — the lockstep dial honestly does NOT fire); <math>SO(16)</math> (fluxes <math>(0, 60, 0)</math>: spheres flux-dark, condensation stops at <math>4 = \det \text{Cartan}(D_8)</math>); <math>A_2</math> (<math>\mu = 12</math> not a perfect square: NO Lagrangian glue, Coxeter <math>3 \neq 4</math>). [C] the CPS dictionary itself on <math>\mathbb{P}T/\mathbb{Z}_4</math> (their geometry is the <math>\mathbb{C}^2</math> blow-up with ONE <math>(-1)</math>-sphere and <math>\mathfrak{so}(8)</math> at negative level — not the <math>A_3</math> ALE). [O] the type-I B-model backreaction on <math>\mathbb{P}T/\mathbb{Z}_4</math> (the <math>A_3</math> <math>\Omega_N</math>) — the same fence as v502/v505; SEAM.EQUIV.01 untouched; no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v510_seam_bit_freedom.py | <p>SEAM.BIT.FREEDOM.01: the freedom attack on the v506/v507 alignment bit — the POSITION half of the bit is topology (Part A dihedral/Fock + Part B Moebius/RP in one module; NO marker moves — the modulus datum stays [C]). Exact sympy + exact Cl(16) Fractions, no floats. [E] DIHEDRAL THEOREM: on the <math>N</math>-site seam circle the free involution is UNIQUE = the antipode (<math>N = 16</math>: 17 involutions in <math>D_{16}</math>, exactly 1 free = shift 8; all 16 reflections have exactly 2 circle fixed points; robustness <math>N = 8, 12</math>); the census is COMPLETE: <math>\text{Aut}(C_{16}) = D_{16}</math> exactly (brute force over all <math>8!</math> permutations at <math>N = 8</math>, propagation at <math>N = 16</math>); <math>N = 9</math> control: ZERO free involutions (even carrier count, consistent with <math>16 = 2^{g_{\text{car}}-1}</math>); QUOTIENT TOPOLOGY (the P1 tie): circle/antipode = CIRCLE (<math>\chi = 0</math>, closed), circle/reflection = INTERVAL (<math>\chi = 1</math>, two corners) — the <math>6\pi = 2\pi + 2\pi + 2\pi</math> Gauss–Bonnet budget counts three CLOSED circles, so “the deck is free on the seam circle” IS the established P1 datum “the seam is a closed circle”, not a new premise. [E] NONSPLIT <math>\iff</math> FREE (complete census, exact Cl(16)): ALL 17 involutions implemented — the free one’s NS lift squares to <math>256 \gamma_1 \cdots \gamma_{16} = (-1)^F</math> (nonsplit, 2 roots = the <math>\mathbb{Z}_8</math> clock tower); ALL 16 reflections have <math>U^2 = \text{SCALAR}</math> (<math>2^7</math> site-, <math>2^8</math> edge-axis; split, 0 roots); chain: free <math>\Rightarrow</math> shift 8 <math>\Rightarrow</math> crossing (all 28 invariant 4-sets) <math>\Rightarrow</math> central class <math>\Rightarrow</math> the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control: the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION on invariant circles — the edge class IS the BRANCHED involution’s restriction: silver mistakes the mark-creating branch <math>\mathbb{Z}_2</math> for the seam deck. [E] REAL-STRUCTURE CLASSIFICATION: the deck-commuting antiholomorphic involutions of <math>\mathbb{P}^1</math> are EXACTLY two families — <math>\mu\bar{z}</math>, <math> \mu  = 1</math> (fix LINE through the deck poles <math>\{0, \infty\}</math>: deck NOT free) and <math>\mu/\bar{z}</math>, <math>\mu &gt; 0</math> (fix circle <math> z  = \sqrt{\mu}</math> avoiding the poles: deck FREE; <math>\mu &lt; 0</math>: empty); per configuration the mark-fixing real structure is UNIQUE; table: <math>\mu_4 \rightarrow</math> equatorial <math>1/\bar{z}</math>, FREE (v506 A-S2.9); silver <math>\{\pm 1, \pm(3+2\sqrt{2})\} \rightarrow</math> real axis THROUGH the poles, NOT free; generic-unit <math>\{\pm 1, \pm(3+4i)/5\} \rightarrow</math> equatorial, free WITHOUT harmonicity; hexagonal <math>\rightarrow</math> NO mark-fixing real structure (marks not concyclic: RP-incompatible, consistent v503); one-sidedness TYPES the <math>\mathbb{Z}_2</math> as the free antipode (quotient <math>\mathbb{RP}^2 = \text{the }  \mathbb{Z}_2 </math> of <math>c_3 = 1/( \mathbb{Z}_2  2\pi\chi)</math>, v456). [E] HARMONIC + FREE <math>\Rightarrow \mu_4</math> EXACTLY: the deck-pair cross-ratio <math>(1-b)^2/(1+b)^2 = -1</math> solves exactly to <math>b = \pm i</math> — the silver family (harmonic AND edge) is EMPTY on free circles, harmonicity becomes SUFFICIENT. [E] HONEST COUNTERWITNESSES: “nonsplit <math>\Rightarrow</math> clock” is FALSE — discrete <math>\{0, 1, 8, 9\}</math> and continuous <math>\{\pm 1, \pm(3+4i)/5\}</math> (free, central, <math>j = 148176/25 \neq 1728</math>, no clock): the Fock dichotomy measures the CLASS, not the modulus. PREMISE TYPING: (P-i) deck = Moebius covering deck <math>\Rightarrow</math> free [standard topology; origin_theory/v456]; (P-ii) marks on the <math>\Theta</math>-fix circle = existing REALIZE [C] content — no new input. [C] THE REDUCED REST (verbatim): the alignment bit reduces from “harmonic AND central” to the square-modulus datum <math>\tau = i</math> alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v511_celestial_<br>wp5e_delta2_tensor_<br>ledger.py | <p>CELEST.WP5E.DELTA2.01: WP5e-<math>\delta_2</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the full-tensor ledger”, the named escape route of the v508 exchange no-go, executed (intermediate verdict, exact): does the FULL adjoint tensor structure of <math>\mathfrak{e}_8</math> (non-Cartan external legs, higher-arity vertices) open a sphere-axion channel to the rigid <math>32 T_3</math> residual (<math>A_{\text{fix}} = (9, -30, -15, 0, 32)</math>, v505)? The collapse is CONFIRMED full-tensorially, but one cubic door opens. Exact Kostant/Weyl weight arithmetic + sympy/Fraction. <b>[E]</b> STRUCTURE: <math>\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3</math> is SEMISIMPLE with NO <math>\mathfrak{u}(1)</math> factor (class-0 roots <math>40 + 12</math>, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits <math>\mathfrak{g}_1 = (16_s, 4)</math>, <math>\mathfrak{g}_2 = (10, 6)</math>, <math>\mathfrak{g}_3 = (\mathfrak{g}_1)^*</math>, each irreducible <math>(64, 60, 64)</math>; INNERNESS THEOREM: <math>h = (2, 2, 2, 2, 2; 0^4)</math> lies in the Cartan OF <math>\mathfrak{g}_0</math>, so every <math>\mathfrak{g}_0</math>-invariant tensor carries total sector charge 0 mod 4 — the v508 collapse is a full-tensor statement AND holds across arities (all 15 non-neutral sector triples: Hom = 0). <b>[E]</b> HOM TABLES: bilinear <math>\dim \text{Hom}_{\mathfrak{g}_0}(\mathfrak{g}_j \otimes \mathfrak{g}_{j'})</math> nonzero ONLY at <math>j' = -j</math> (dims <math>2/1/1/1</math>, Killing blocks); trilinear neutral multisets <math>\{0, 0, 0\}: 3</math> (<math>= 1 \text{ sym} + 2 \text{ antisym}</math>), <math>\{0, 1, 3\}: 2</math>, <math>\{0, 2, 2\}: 2</math>, <math>\{1, 1, 2\}: 1</math>, <math>\{2, 3, 3\}: 1</math> — ALL cross-sector trilinears are pure brackets (antisymmetric in the repeated legs); the ONLY totally symmetric survivor on all of <math>\mathfrak{e}_8</math> is the <math>\mathfrak{su}(4)</math> <math>d</math>-symbol (<math>\mathfrak{so}(10)</math> has no cubic Casimir: <math>\text{Inv}(S^3 45) = 0</math> protects <math>T_5</math>). <b>[E]</b> THE NEW CHANNEL: the quartic projection of the <math>d</math>-symbol exchange is <math>Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)</math>, computed by TWO independent routes (Killing-propagator contraction with <math>2h^\vee = 60</math> verified from the roots; hand route <math> x^2 - (\text{tr } x^2/4)\mathbf{1} ^2/60</math>); <math>\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0</math> — the v508 master kill does NOT extend to cubic vertices; brackets, tadpoles and mixed <math>d \times f</math> die exactly; no <math>T_5</math> analogue. <b>[E]</b> NEW CERTIFICATE: <math>M = [E_{13}, E_{22}, E_{00}, Q_{dd}]</math> has rank 3, augmented rank 4 — the charge reading remains OBSTRUCTED with the WEAKER certificate <math>\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}</math>, <math>\psi(A_{\text{fix}}) = 72 - 8 = 64</math>; the RELAXED reading becomes EXACTLY solvable: <math>A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}</math> (residual 0). <b>[C]</b> number fences: <math>64 = \dim \mathfrak{g}_1 = 2^6</math>; <math>1920 =  W(D_5)  = 2^7 \cdot 3 \cdot 5 = 8 \cdot 240</math>. <b>[O]</b> the arity <math>\geq 4</math> ledger; the physical justification of the cubic GS term (<math>c_d = 1920</math>); <math>\text{ch}_2</math>-naturalness for CUBIC vertices UNDECIDED; the burden on <math>\delta_1</math> is SHARPENED to the <math>\psi = 64</math> slice or the selection-rule relaxation. <b>[E]</b> NEGATIVE CONTROLS: Killing control = v508 replication; <math>SO(16)/D_8</math>: bilinears <math>(1, 0, 1)</math>, trilinears <math>(1, 0, 1, 0)</math>, <math>\text{Inv}(S^3 \text{adj}_{D_8}) = 0</math> — no symmetric cubic, no odd sectors, no <math>A_3</math> block (<math>E_8/\mu_4</math> doubly special); false <math>\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)</math>: <math>(5, 2, 2) \neq (2, 1, 1)</math> and <math>\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1</math>; false sector content kills every pairing. SEAM.EQUIV.01 stays <b>[O]</b>; NO marker moves. Exact (sympy + Fraction). <b>[E][C][O]</b></p> |

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| Script                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v512_seam_tau_flag.py | <p>SEAM.TAU.FLAG.01: the flag-transitivity equivalence web — the <math>\tau = i</math> attack executed on the free RP seam circle: which ESTABLISHED seam datum excludes the free-but-non-harmonic configurations? (Test family <math>M(\delta) = \{\pm 1, \pm e^{i\delta}\}</math>, <math>\delta \in (0, \pi/2]</math>; NO marker moves — the bit stays [C], QGEO.REALIZE.01 and P2.TYPING.01/AX.P2.01 text-precised only.) Exact sympy + mpmath 30 digits. [E] FALSIFICATION OF BARE TRANSITIVITY: on the free RP circle <math>V_4 = \{z, -z, b/z, -b/z\}</math> acts SIMPLY TRANSITIVELY on the four marks for EVERY <math>\delta</math> (pair-exchanging inversions; complete census: deck-commuting Möbius maps = diagonal/antidiagonal) — the expectation “transitivity <math>\Leftarrow</math> v216 local equality selects <math>\tau = i</math>” is REFUTED; hardening: ALL undirected per-mark data (arc-pair multiset <math>\{\delta, \pi - \delta\}</math>, distance multiset, cone germ) are identical for every <math>\delta</math> — no local jet sees the bit. [E] THE SHARP SELECTOR: FLAG transitivity (marks AND their two sides indistinguishable) <math>\iff \delta = \pi/2 \iff \tau = i \iff</math> the clock — stabiliser jump <math>1 \rightarrow 2</math>, flag/arc orbits <math>2 \rightarrow 1</math>, order-4 rotation (<math>b^2 = -1</math> solveset <math>\{\pi/2\}</math>), mark mirror <math>z \mapsto 1/z \iff \cos \delta = 0</math>. [E] 12-FOLD EQUIVALENCE WEB (each link exact): clock rotation, <math>D_4</math>, mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split <math>= (2/\pi) \ln \cot(\delta/2)</math>, zero only at <math>\pi/2</math> — counterwitness exactly <math>(2/\pi) \ln 2</math>; arc-Laplacian degeneracy <math>4/\pi</math>; the v215-K3 indicator in CLOSED FORM <math>(2\sqrt{2}/\pi)  \ln \tan(\delta/2) </math> — the v503 lattice curve analytically), <math>\rho</math>-twist existence <math>((m^2 - n^2)(t - 1/t) = 0 \iff t = 1</math>; Atiyah–Bott rationality is <math>\delta</math>-BLIND — <math>\text{Tr}(\sigma e^{t\Delta}) = 1</math> for ALL <math>\delta</math>, <math>j_{cw} = 148176/25</math> rational non-CM: rationality does NOT select), 4-cycle implementability (H4 rigidity: <math>V_4</math> with element orders <math>\{1, 2, 2, 2\}</math> cannot induce the cusp 4-cycle; centraliser scalar, <math>U^2</math> nonscalar <math>\Rightarrow</math> no projective involution — projective order exactly 4, <math>\mathbb{Z}_8</math>-shadow consistent v510), harmonic, <math>j = 1728</math>. [E] NEGATIVE CONTROLS: hexagonal (not concyclic, fails the PRECONDITIONS); silver (harmonic but topologically dead — position/modulus halves independent, consistent v510); 3 marks (impossible on the free circle: 0 invariant 3-subsets vs 28 invariant 4-subsets; bare <math>n = 3</math> transitivity already forces equal spacing — “4 counts”: the deck-pairing makes bare transitivity free of charge). [C] THE REST (verbatim): the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit (<math>V_4 \rightarrow D_4</math>: side-indistinguishability of the four equally-decorated marks); v216 local equality does NOT deliver it (no local jet can see the side bit); its negation is now concretely measurable (odd-split <math>(2/\pi) \ln \cot(\delta/2) \neq 0</math>). Scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones. Exact (sympy + mpmath). [E][C]</p> |

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| Script                                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v513_celestial_<br>dterm_<br>nonderivation.py | <p>CELEST.DTERM.NONDERIV.01: the <math>c_d</math> negative certificate — can geometry derive <math>c_d = 1920</math>? An anti-numerology certificate in the v508 style (honest rigid NEGATIVE result): the v511 [C] fence “<math>1920 =  W(D_5) </math>” stress-tested via four candidate routes with a PRE-DECLARED look-elsewhere ledger. Exact sympy/Fraction, no floats. [E] CONVENTION TABLE: <math>c_d \in \{1920, 120, 256, 32, 69120\}</math> across five normalisations — only 1 of 5 hits <math> W(D_5) </math>; the convention-STABLE content is the factorisation <math>c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60</math>, both factors [E] (Killing propagator 60 from <math>\sum_\alpha \langle \alpha, e \rangle^2 = 120</math> over the actual roots; <math>T_3</math>-slot fixing via <math>\det(\text{P-block}) = -64 \neq 0</math>). [E] K1 DEAD: the charge-legal flux pairings <math>(1, 3)/(2, 2)</math> give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit <math>64 \cdot 60/2</math> is the FORBIDDEN <math>(1, 2)</math> pairing (charge 3 mod 4, killed by the v511 innerness theorem).</p> <p>LOOK-ELSEWHERE LEDGER: 924 expressions from 21 [E] numbers, 11 hits on 1920; control targets <math>1800 \rightarrow 8</math>, <math>2048 \rightarrow 5</math>, <math>2016 \rightarrow 0</math>. [E] K2: quantisation delivers only lattices (<math>60\mathbb{Z}</math>: 1920 = the 32nd multiple, no minimality; strict <math>\text{ch}_3</math>-unit couplings <math>2160\mathbb{Z}</math> EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, <math>3/8</math> on weights); the residual integer 32 has 12 signed factorisations. [E] K3 PROVENANCE CLASH (the death blow): cubic indices <math>A_m = (0, 16, 0, -16)</math>, twisted sum <math>\sum_m i^m A_m = 32i</math> NUMERICALLY = <math>\Phi_{T_3}(A_{\text{fix}})</math> — BUT the quartic class decomposition <math>(10, -4, 30, -4)</math> is dominated by classes <math>0/2</math> while the cubic 32 is built <math>16 + 16</math> from classes <math>1/3</math>: same number, disjoint mechanism = coincidence exposed; the Weyl side is WRONG (the <math>d</math>-channel is an <math>A_3</math>-side object, <math> W(A_3)  = 24</math>; <math>D_5</math> has no cubic invariant, <math>\text{Inv}(S^3 45) = 0</math>; <math> W(E_8) / W(A_8)  = 1920</math> a second accidental 1920). [E] K4 NEGATIVE CONTROLS: <math>SO(16)/D_8</math> fails every candidate formula consistently; <math>k = 2</math> flux doubling scales the flux formula <math>\times 4</math> (7680); false <math>\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)</math> has SIX symmetric cubics; <math>\mathfrak{so}(10)</math> sanity (<math>\text{Tr } X^3 = 0</math>). VERDICT: NUMEROLOGY as a derivation claim — the v511 [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is <math>c_d = 32 \times 60</math>; the physical generation of the 32 stays [O]. Exact (sympy/Fraction); no marker moves. [E]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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| v514_celestial_<br>wp5e_eps1_axion_<br>slot.py | <p>CELEST.WP5E.EPS1.01: WP5e-<math>\varepsilon_1</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the <math>O(-2)</math> bulk-axion slot”, executed (verdict B): build the <math>s = 0</math> tower field explicitly, pin <math>\tilde{\lambda} = 6</math> as a residue with explicit factor bookkeeping on <math>\mathbb{PT}/\mathbb{Z}_4</math>, and take the first honest back-reaction step (Gibbons–Hawking <math>A_3</math>). Exact sympy/Fraction, no floats. [E] EQUIVARIANT <math>O(-2)</math> SLOT: pushforward ledger <math>\pi_*O(-2)</math> on <math>\mathbb{PT}' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)</math>: per spacetime degree <math>d</math> exactly <math>(d+1)^2 = \text{nullspaces of the wave operator}</math> (1, 4, 9, 16, 25, 36 for <math>d = 0..5</math>); the v492 action on the fibre coordinates <math>(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)</math>, incidence forces column weights <math>(+1, +1, -1, -1)</math>; the equivariant bookkeeping closes block by block (all <math>d \leq 6</math>, all 4 characters); character series <math>P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots</math>, <math>P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots</math>, <math>P_2 = 6t^2 + 10t^4 + 28t^6 + \dots</math>; the <math>d = 0</math> slot has multiplicity (1, 0, 0, 0) — the bulk axion survives the projection; Molien invariant ring <math>= (1 - t^8)/((1 - t^4)^2(1 - t^2))</math> — <math>Z \in O(2)</math>, <math>X, Y \in O(4)</math>, relation degree 8 = the <math>a_0</math> weight (v492/v493 bridge); E3 bijection sharpened: twisted characters <math>\{i, -1, -i\}</math> with minimal content <math>(2t, 6t^2, 2t)</math> and no degree-0 mode = the Coxeter eigenvalues on <math>H^2(\text{ALE})</math>, <math>\det(A - 1) = -4</math>; graviton control: the <math>O(+2)</math> slot starts invariant only at fibre degree 4 with multiplicity <math>3 = \{X, Z^2, Y\}</math> (the cohomological form of v505 <math>X = -2 \neq +2 = Y</math>). [E] <math>\tilde{\lambda} = 6</math> TRIPLY PINNED: (1) Okubo: <math>\text{Tr}_{\text{adj}} X^4 = 36\langle x, x \rangle^2 = (6\langle x, x \rangle)^2</math> on the 240 glue roots, <math>36 = h^\vee + 6</math>, <math>\tilde{\lambda} = 6 =  \mathbb{Z}_2  N_{\text{fam}}</math>; (2) measure cancellation: <math>\mu = 1/4</math> enters three times (vertex<sup>2</sup> <math>\mu^2</math>, propagator <math>1/\mu</math>, anomaly <math>\mu</math>) and cancels exactly — <math>\tilde{\lambda}_{\text{eff}}^2 = 36</math> for arbitrary <math>\mu</math>; wrong bookings quantified and excluded: anomaly-only <math>\Rightarrow \tilde{\lambda} = 3</math>, missing propagator renormalisation <math>\Rightarrow \tilde{\lambda} = 12</math>; (3) flux: minimal CPS quantum <math>N = 1</math>, a single exchange channel exactly at <math>k = 1</math> (#primaries), <math>(\kappa/c_3)^2 = 36/3 = 12</math> exact. [E] GH/<math>A_3</math> BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches (axis<sup>4</sup> or one free <math>\mu_4</math> orbit); free orbit <math>\Rightarrow \prod_p (Z - i^p z_0) = Z^4 - z_0^4 \Rightarrow XY = Z^4 + a_0</math> with <math>a_0 = -z_0^4</math> (the v493 family re-derived from the geometry); monodromy <math>a_0 \rightarrow e^{2\pi i} a_0 = \text{one clock step}</math>; the GH point of charge <math>k</math> is exactly <math>\mathbb{R}^4/\mathbb{Z}_k</math> (<math>k = 4 \Rightarrow \text{seam boundary } S^3/\mathbb{Z}_4</math>); periods <math>\int \omega = 4\pi(x_q - x_p)</math>, free orbit: <math>\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)</math>, <math> \Pi_j </math> equal (lockstep, counter-check of v509); the Coxeter clock from the geometry (<math>A^4 = 1</math>, <math>\det(A - 1) = -4</math>, <math>\Pi \cdot A = i\Pi</math>); the CPS “<math>N \log</math>” analogue = source charge <math>4 =  \mu_4 </math> (flux <math>-4\pi</math> per centre); honest sharpening: for the Ricci-flat ALE the asymptotic log coefficient is exactly 0 (EH: <math>uK' = \sqrt{u^2 + a^4}</math>, <math>\det g = 1</math>) — the log lives at the exceptional locus; Burns contrast (<math>K = u + N \log u \Rightarrow \det g \neq 1</math>, only scalar-flat); bonus: multipole selection rule <math>m \equiv 0 \pmod 4</math>, first symmetry-breaking multipole <math>(4, \pm 4)</math> carries <math>-a_0</math> with amplitude <math>(35/16) \sin^4 \theta</math>. [E] NEGATIVE CONTROLS: <math>\mathfrak{so}_8</math> (<math>\tilde{\lambda}^2 = 12</math> irrational; perfect squares in the Deligne series only <math>\{9, 36\} = \{\mathfrak{sl}_3, \mathfrak{e}_8\}</math>); <math>\text{diag}(i, i)</math> (Veronese, 5 generators, no hypersurface, the bijection collapses); <math>k = 2</math> (three channels; honest: <math>\tilde{\lambda}^2 = 36</math> is <math>k</math>-independent, the kill is the closure dial). [C] conditional on Costello’s flat-<math>\mathbb{PT}</math> matching <math>\lambda_g^2 E = A</math>. [O] the quantised BCOV one-loop coefficient on <math>\mathbb{PT}/\mathbb{Z}_4</math>; the twisted channels <math>(32 T_3)</math> are not covered by the bulk axion — M1–M3 preregistered verbatim: M1 the <math>A_3 \Omega_N</math> on <math>\mathbb{PT}/\mathbb{Z}_4</math> (success: <math>(2\pi i)^2</math> periods with lockstep phases; kill: unequal fluxes <math>\Rightarrow</math> the v509 dial dies); M2 the twisted KS measure (success: bulk 36 + sphere axions cancel <math>32 T_3</math>; kill: a leftover <math>T_3</math>); M3 the <math>a_0</math> uplift of the <math>(4, \pm 4)</math> multipole (success: log correction <math>\propto 4 =  \mu_4 </math>; kill: decoupling from the centre count). No marker moves anywhere.</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v515_celestial_<br>wp5e_m1_omega_n.py | <p>CELEST.WP5E.M1.01: milestone M1 of the celestial-holographic research contract CELEST.SEAM.01 (WP5e back-reaction programme) — “the <math>A_3 \Omega_N</math>”, executed (SUCCESS on the preregistered v514 S8.1 criterion): the back-reacted holomorphic 3-form <math>\Omega_N</math> on the twistor uplift of the deformed <math>A_3</math> ALE, with exact periods. Exact symplectic, no floats. [E] RESIDUE FORM DERIVED: on <math>F = XY - P(Z)</math> all three ambient representatives satisfy <math>dF \wedge \omega_2 = -dX \wedge dY \wedge dZ</math> and pull back to <math>\omega_2 = dX \wedge dZ / X</math> (Poincaré residue, derived not assumed); cover pullback at <math>a_0 = 0</math>: <math>\omega_2 = 4 dz_1 \wedge dz_2 =  \mu_4  \times (\text{flat form})</math> — the residue normalisation carries the source charge 4; clock <math>\text{diag}(i, 1)</math>: <math>\omega_2 \rightarrow i \omega_2</math> (v492 replicated); weight ledger <math>4+4+2-8=2</math>, <math>\Omega \in O(4) = -\deg K_{\text{PT}}</math>; EH anchor <math>2 =  \mathbb{Z}_2 </math>; wrong-weight control <math>Z \in O(4)</math> breaks three ways (<math>4 \neq 2</math>, <math>6 \neq 4</math>, <math>5 \neq 3</math>). [E] THE TWISTOR FAMILY CLOSED: from the four <math>O(2)</math> centre sections <math>q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2</math> the family is exactly <math>XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2</math> (<math>e_1 = e_3 = 0</math> identically; <math>\lambda = 0</math> gives exactly the v493 clock family; <math>a_2 \in O(4)</math> opens only off-seam, <math>a_2(0) = a_2(\infty) = 0</math>); clock lift <math>\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)</math>, <math>\gamma^4 = 1</math>, <math>\Omega \rightarrow +\Omega</math> (CY-compatible). [E] <math>\Omega_N</math> CLOSED + QUANTISED PERIODS: <math>\Omega_N = \Omega_0 + \sum_p N_p K_p</math> with CPS/Bochner–Martinelli kernels on the centre twistor lines; <math>dK = 0</math> away from the source, scale degree 0; period table: boundary <math>S^3/\mathbb{Z}_4</math> at the orbifold point <math>(2\pi i)^2 \times 1</math> (minimal invariant charge <math>4 =  \mu_4 </math> — the lens fundamental domain forces <math>N \in 4\mathbb{Z}</math>), deformed phase <math>(2\pi i)^2 \times 4</math>, linking <math>S^3</math> uniformly <math>(2\pi i)^2 \times N</math>, per-sphere flux vector <math>(2\pi i)^2 \times N(1, 1, 1)</math> (forced uniform twice over: clock orbit + <math>K_4</math> connectivity), seam-fibre phase vector <math>2\pi i t_0(i-1)(1, i, i^2)</math>; the undeformed <math>\Omega_0</math> carries zero quantised 3-flux (all flux sourced); <math>(2\pi i)^2 = (\text{fibre residue}) \times (\text{transverse residue})</math> (density <math>-\sin 2\theta</math>). [E] CLOCK COVARIANCE: <math>\Pi_{j+1}(-i\lambda) = i \Pi_j(\lambda)</math> for all <math>\lambda</math>; the 12 collision nodes sit exactly on the 8 eighth roots of unity (<math>8 = 2 \mu_4 </math>, the <math>\mathbb{Z}_8</math> seam bridge), clock orbits <math>4+4+4</math>, honest conifold points (Hessian <math>\det 512t_0^6</math>, quadric rank 4). [E] KILL PROBE (honest): the forbidden family <math>Z^4 - Z</math>: 0/24 orderings lockstep (vs 8/24), <math>\lambda</math>-support 12 twelfth roots <math>\neq \mathbb{Z}_8</math>, no antipodal pairing — but the <math>(2\pi i)^2</math> quantisation holds there too: integrality alone is not the discriminator, the discriminator is the lockstep phase structure + the clock forcing of equal fluxes; the v509 dial survives. [E] ANCHORS + CONTROLS: EH/<math>\mathbb{Z}_2</math> (2 nodes, charge <math>2 =  \mathbb{Z}_2 </math> forced); CPS original (full <math>S^3</math>: every integer <math>N</math> allowed); <math>\text{diag}(i, i)</math>: <math>\Omega_0 \rightarrow -\Omega_0</math> (collapse at step zero) while <math>K</math> stays invariant; fractional charges <math>N = 1, 2, 3</math> excluded. [C] global kernel patching (Green function), Hitchin small resolution, CPS brane dictionary. [O] M2 (twisted KS measure), M3 (<math>a_0</math> uplift), the quantised BCOV coefficient; SEAM.EQUIV.TWISTOR.01 untouched (preparation, not construction). No marker moves.</p> |

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| Script                                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v516_celestial_<br>wp5e_m2_twisted_ks_<br>measure.py | <p>CELEST.WP5E.M2.01: milestone M2 of the celestial-holographic research contract CELEST.SEAM.01 (WP5e back-reaction programme) — “the twisted Kodaira–Spencer measure”, executed (SUCCESS on the preregistered v514 S8.2 criterion, on the DECLARED completion measure — verdict B): the one-loop box coefficient per sector on <math>\mathbb{PT}/\mathbb{Z}_4</math> under the completion contact term <math>\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j</math>. Exact sympy/Fraction, no floats. [E] THE COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class the contact weights are <math>w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m =  \mu_4 h_m = -4\text{ch}_2(T_m)</math> EXACTLY (<math>h_m = m(4 - m)/8</math> the v502 sector Casimir energies, <math>\text{ch}_2(T_m)</math> the McKay/Kronheimer charges, v505 S4.3 index bridge): the three sphere axions pair through their OWN <math>\text{ch}_2</math> charges — an exact identity, NO free scale, no fit. [E] PARAMETER-FREE LOCKS: <math>T_5 = 0</math> for ANY scale <math>c</math> (the <math>\text{ch}_2</math> pattern alone kills <math>T_5</math>); ratio <math>h_2 : h_1 = 4 : 3 =</math> the <math>\delta_1</math>-forced leading ratio (REPRODUCED, not fitted); the <math>T_3</math> budget fixes <math>c = 4 =  \mu_4 </math> uniquely. [E] THE CONTACT TERM: channel form <math>\sum_j (Q^{(0)} - Q^{(j)})/\det_j =</math> class form <math>\sum_m 4h_m Q_m = (36, 120, 60, 0, -32)</math> EXACTLY (one object, two bases); the contacts carry nonzero <math>T_5/T_3</math> per channel (genuine contact, not exchange — the v508 product theorem replicated). [E] SUCCESS TABLE: <math>\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x \rangle^2/\det_j</math> — a PERFECT Okubo square (<math>T_5 = T_3 = 0</math>, <math>\tilde{\lambda}^2 = 36</math>) in EVERY twisted channel; total <math>A_{\text{fix}} + \text{contact} = 45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo} =</math> the UNIQUE quartic-free weighting (v505 S3.8 rigidity); both v508 certificates killed (<math>\Phi_{T_3} : 32 \rightarrow 0</math>, <math>\Phi_P : 72 \rightarrow 0</math>); the v511 slice <math>\psi = 64</math> SUPPLIED EXACTLY — no cubic <math>d</math>-channel needed (<math>c_d</math> free = 0); remainder inside the exchange span (<math>45\langle x, x \rangle^2 = \frac{45}{16}E_{22} = \frac{1}{80}E_{00}</math>), the bulk axion cancels every channel with the SAME <math>\tilde{\lambda}^2 = 36</math>; the v514 measure chain stays <math>\mu</math>-blind. [E] NEGATIVE CONTROLS: <math>c \in \{1, 2, 3, 5\}</math> leaves <math>T_3 = (24, 16, 8, -8)</math>; the <math>h_1 \leftrightarrow h_2</math> shuffle breaks <math>T_5 = -14</math> AND <math>T_3 = 36</math>; <math>SO(16)</math>: <math>T_5 = 20</math> / <math>T_3 = -40</math> remain — the KILL fires there (Okubo failure inherited, <math>E_8</math> doubly special); <math>\text{diag}(i, i)</math> degenerates to the <math>\mathbb{Z}_2</math> target and breaks the conjugation pairing + the index bridge; the <math>\mathbb{Z}_2/\text{Eguchi–Hanson}</math> anchor PASSES (<math>9\langle x, x \rangle^2 = \frac{1}{4} \times 36</math>, scale <math>2 =  \mathbb{Z}_2 </math>). [E] MULTIPLIER NOTE (cross-reference to <math>\delta_{1d}</math>): the contact weights are real rationals with no multiplier system of their own; the <math>\delta_{1c}</math> residual phases = the SQUARES of the <math>A_3</math> discriminant <math>T</math>-phases, orders <math>(1, 4, 1, 4)</math>. [C] THE COMPLETION READING (the fence): the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation — declared, supported by the <math>\delta_1</math> modular-completion finding, NOT derived from the BCOV integral. [O] the BCOV-integral derivation (<math>\delta_{1b}/\delta_{1c}/\delta_{1d}</math> territory); SEAM.EQUIV.TWISTOR.01 untouched. VERDICT: SUCCESS wording met on the declared measure; the KILL does NOT fire; no marker moves. [E][C][O]</p> |

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| Script                                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v517_celestial_<br>wp5e_m3_a0_<br>uplift.py | <p>CELEST.WP5E.M3.01: milestone M3 of the celestial-holographic research contract CELEST.SEAM.01 (WP5e back-reaction programme) — “the <math>a_0</math> uplift”, executed (SUCCESS on the preregistered v514 S8.3 criterion): the <math>(4, \pm 4)</math> GH multipole uplifted to the twistor-potential kernel <math>\chi = \log P_4</math> (the log of the v515 family polynomial) with the induced Kähler-potential correction. Exact symplectic, no floats. [E] KERNEL BRIDGE: the <math>O(2)</math> section is a NULL coordinate — <math>\Delta_{3d} G(\eta_p) = 0</math> identically for an ARBITRARY kernel <math>G</math> (the uplift is well-typed); RESIDUE IDENTITY <math>\partial_x[\text{transform}(\log \eta_p)] = 1/r_p</math> EXACTLY (residue <math>1/(2r)</math>): the GH potential <math>V = \sum_p 1/r_p</math> IS the <math>x</math>-derivative of the contour transform of <math>\chi</math>; flux <math>-4\pi</math> per centre, source charge <math>4 =  \mu_4 </math> (the <math>V</math>-ledger matched). [E] FOUR COUPLED CENTRE-COUNT SCALES: (1) asymptotic kernel log <math>\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots</math> — LOG COEFFICIENT <math>4 =  \mu_4 </math> = centre count (the CPS “<math>N \log</math>” analogue at kernel level); at the clock-fixed seam fibre <math>\zeta = 0</math> the <math>a_2</math> slot closes and the first correction is EXACTLY <math>a_0/\eta^4</math> = the <math>(4, \pm 4)</math> multipole; exact <math>m</math>-grading (<math>m = \pm 4</math> pieces PURE <math>a_0</math>, <math>m = 0</math> piece <math>\varphi_0</math>-free); NO <math>\log \times \text{power}</math> terms (the v514 honest zero preserved); (2) the GLT tower <math>\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + \dots</math> with <math>p_n = 0</math> unless <math>n \equiv 0 \pmod 4</math> and <math>p_{4k} = 4(-a_0)^k</math>; (3) the exceptional-locus log <math>\chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)</math>, modulus response 1 at the locus vs power-law at infinity (v514 S6.3 uplifted); (4) periods: <math>d \log \Pi_j / d \log a_0 = 1/4 = 1/ \mu_4 </math> uniform, integrated monodromy <math>e^{2\pi i/4} = i</math> = ONE Coxeter clock step (v493 reproduced from perturbation theory); node <math>\lambda</math>-support <math>a_0</math>-RIGID (the <math>\mathbb{Z}_8</math> seam bridge), <math>K_4</math> discriminants <math>\sim t_0^2 \neq 0</math> (the forced flux vector persists), BM kernel modulus-free with <math>\int K = (2\pi i)^2</math>. [E] NEGATIVE CONTROLS: <math>(4, 0)</math> clock-invariant (<math>\varphi_0</math>-free in <math>V</math> AND kernel: breaks nothing); <math>\mathbb{Z}_2/\text{EH}</math> analogue: EVERY dial reads <math>2 =  \mathbb{Z}_2 </math> (<math>(2, \pm 2)</math> amplitude <math>\frac{3}{2} \sin^2 \theta</math>, kernel log 2, exceptional log <math>2 \log t_0</math>, response <math>\frac{1}{2}</math>, monodromy <math>-1</math>); <math>k = 3, 5</math> orbits: coefficient <math>k</math>, slot <math>O(6)/O(10) \neq O(8)</math> — the observable MOVES with the centre count; forbidden family <math>Z^4 - Z</math>: <math>p_3 = 3 \neq 0</math> (selection rule broken), <math>e_4 = 0</math> (no <math>a_0</math>-log) — fails BOTH dials. [C] the GLT dictionary (<math>\eta_p \log \eta_p</math> = the GH Kähler-potential kernel; Lindström–Roček / Ivanov–Roček). [O] the full nonlinear Kähler potential of the resolved <math>A_3</math> ALE; SEAM.EQUIV.TWISTOR.01 untouched. VERDICT: SUCCESS (log correction <math>\propto 4 =  \mu_4 </math>, coupled to the centre count on four scales); the KILL (decoupling) does NOT fire; no marker moves. [E][C][O]</p> |

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| Script                                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| v518_celestial_<br>wp5e_delta1_<br>derived_kill.py | <p>CELEST.WP5E.DELTA1.01: the <math>\delta_1</math> chain (<math>\delta_{1b} + \delta_{1c} + \delta_{1d}</math>) of the celestial-holographic research contract CELEST.SEAM.01 DECIDED — “the derived chiral measure kills the <math>\delta_1</math> route”, an honest, decided NEGATIVE result (v508 style): does the Harvey–Moore/BCOV fundamental-domain integral on <math>\mathbb{PT}/\mathbb{Z}_4</math>, evaluated under a measure DERIVED from blockwise <math>SL(2, \mathbb{Z})</math> covariance (solved for, not declared, not scanned), deliver <math>-A_{\text{fix}} = -(9, -30, -15, 0, 32)</math> or the <math>\psi = 64</math> slice? NO — every member of the derived family fails all three preregistered testers. [E] THE 16-COMPONENT WEIL COMPLETION (<math>\delta_{1c}</math>): discriminant module <math>\mathbb{Z}_4 \times \mathbb{Z}_4</math> with <math>q = (5x^2 + 3y^2)/8</math>; Gauss sums <math>2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}</math> (signature 0 mod 8 = rank <math>E_8</math>); <math>S</math> symmetric + unitary, <math>S^2 = C</math>, <math>(ST)^3 = S^2</math>, <math>S^4 = 1</math>, <math>T^8 = 1</math> EXACTLY in <math>\mathbb{Q}(\zeta_8)</math>; <math>S = S_{D_5} \otimes S_{A_3}</math>; the diagonal <math>H</math> and anti-diagonal <math>H'</math> are the two invariant Lagrangians (both counts = <math>E_4</math>: two <math>E_8</math> gluings); <math>P_H S P_H = \frac{1}{4}J</math> with <math>3/4</math> of every <math>S</math>-image weight on the 12 shifted classes; the 16-vector theta <math>S</math>-covariance CLOSES: the <math>\delta_{1b}</math> E1.5 residual <math>2.91 = O(1)</math> drops to <math>\sim 10^{-39}</math>. [E] THE OBSTRUCTION IS A FINITE <math>\mu_4</math> CHARACTER, IDENTIFIED (<math>\delta_{1d}</math>): gauge blocks <math>M = G[a, b] \eta^{-8}</math> have constant transport <math>(t_p, s_p)</math> (28+ digits) recognised as exact <math>1/960</math>-grid phases; the <math>T</math>-fixed pairs <math>(0, b)</math> carry <math>t = e(-1/2) = -1</math> EXACTLY; loops land in <math>\Gamma_1(4)</math> resp. <math>\Gamma_0(2)^\pm</math>, joint holonomy eigenspaces dim <math>8/16</math> resp. <math>6/16</math>, every eigen-phase of order <math>\leq 4</math>; KOBOUNDARY TEST: the relation defects <math>c_{S^4}</math>, <math>c_{(ST)^3 S^{-2}}</math>, <math>c_{[S^2, T]}</math> are <math>(1, 1, 1)</math> EXACTLY on all 15 pairs — a CHARACTER of the stabilisers, NOT a genuine 2-cocycle on <math>SL(2, \mathbb{Z})</math>; on <math>\Gamma_1(4)</math>: <math>\lambda(\gamma) = i^{2B+C/4}</math>. [E] THE CANCELLATION IS THE TWISTED FIBRE BLOCK <math>f_1 f_3</math>, NOT THE THREE SPHERE AXIONS: exact identity <math>G[a, b] = f_1 f_3</math> (30+ digits); the <math>T</math>-fix mechanism is one line of exact phase arithmetic: <math>(-1) \times e(-1/6) = e(1/3) = \chi_4(T)</math> at every <math>T</math>-fixed node; CANCELLATION TABLE (exact, minimal residual order over <math>\eta^m \times \chi_k</math>): none <math>\rightarrow 4</math>, <math>f_1 f_2 f_3</math> (the three sphere axions) <math>\rightarrow 4</math>, <math>f_2 \rightarrow 6</math>, <math>f_1 f_3 \rightarrow 1</math> (hol AND anti), wrong weights <math>\rightarrow 4</math>; strict solutions exactly <math>\chi_4</math> (dims <math>(3, 3)</math>) and <math>\chi_{10}</math> (dims <math>(1, 1)</math>) at <math>\eta^0</math>, certified on the dressed functions (<math>10^{-15}</math>, NO residual multiplier). [E] THE DECISION — ALL THREE TESTERS FAIL: under both derived solutions the integral fails <math>T_5(V) = 0</math> (fractions <math>0.5 / 0.83</math>), fails <math>W_2 : W_{13} = 4 : 3</math> (<math>W_{13} = 0</math>), fails <math>V = -A_{\text{fix}} N</math> (spreads <math>1.05 / 1.47</math>) and the <math>\psi = -64N</math> slice; the honest <math>(N_1, N_2)</math> rebalance: <math>\chi_4</math> reaches <math>T_5 = 0</math> only at <math>\psi &gt; 0</math>, <math>\chi_{10}</math> only at <math>N_2/N_1 = -4 &lt; 0</math> (outside the orbifold projector) — no branch fires in the positive cone. [E] NEGATIVE CONTROLS: <math>\mathbb{Z}_2</math>/Eguchi–Hanson (gauge-only residual order 2 — the <math>\mu_4</math> defect is <math>\mathbb{Z}_4</math>-specific — and the SAME fibre dressing cancels it to 1); <math>SO(16)/D_8</math> (only even sectors, all axion <math>b</math>-characters in <math>\mu_2</math>: the order-4 supply structurally absent); wrong form <math>(x^2 + y^2)/4</math> (<math> G ^2 = 64 \neq 16</math>, its <math>S</math>-rule fails at <math>O(1)</math>); wrong signature (<math>S^2 = C</math> breaks exactly). [C] the <math>f_1 f_3 = \text{KS-weight identification}</math> (deck weights <math>(1, 3) = \text{the twistor fibre weights } (+1, -1 \bmod 4)</math> forced by the v514 incidence ledger — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete <math>\chi_k</math> choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the <math>\psi = 64</math> slice and cancels <math>32 T_3</math>; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure (the declared reading is supported by the <math>\delta_1</math> modular-completion finding but not derived; the derived measure rests on the <math>f_1 f_3</math> identification). Neither result is hidden behind the other. [O] the BCOV-integral derivation on <math>\mathbb{PT}/\mathbb{Z}_4</math> that decides between the two measures — the single open item of the <math>\delta_1</math> strand; the twisted-contact route of v508 S6.2 is DECIDED NEGATIVE under the derived measure. NO marker moves anywhere. [E][C][O]</p> |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| v519_woit_theta_rp_free.py | <p>WOIT.THETA.FREE.01: the WOIT-<math>\alpha</math> milestone of the OS twistor bridge</p> <p>WOIT.OS.TWISTOR.01 — “the real structure exists, and free reflection positivity picks the SAME family” (Part W = the anti-linear <math>\Theta</math>: classification, exact relations, kill test 1 on the free/equivariant level; Part R = reflection positivity of the free 16-Majorana NS seam system under the pinned <math>\Theta</math>; NO marker moves — WOIT.OS.TWISTOR.01 stays <b>[O]</b>: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra). Exact symplectic + exact <math>\text{Cl}(16)</math> Fractions; the RP definiteness certificates are 40-digit spectra of exact Hermitian matrices. <b>[E]</b> THETA CLASSIFICATION COMPLETE: exactly TWO families of anti-linear structures on <math>\mathbb{C}^2</math> normalise the clock <math>\rho = \text{diag}(i, 1)</math> — family D (<math>z \mapsto \mu\bar{z}</math>, <math> \mu  = 1</math>): <math>\Theta\rho\Theta = \rho^{-1}</math> EXACTLY, <math>\Theta^2 = +1</math> (−1 IMPOSSIBLE: <math>MM = \text{diag}( \mu ^2, 1)</math> — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A (<math>z \mapsto \mu/\bar{z}</math>): centralises the clock projectively (NEVER inverts), <math>\Theta^2 = \pm 1</math> per <math>\mu</math>, fixed set <math> z  = 1</math> resp. empty. ROLE SEPARATION: family A DEFINES the euclidean section (Woit’s <math>\rho_{\text{tw}}</math> side: <math>\rho_{\text{tw}}^2 = -1</math>, no real points — replicated exactly on <math>\mathbb{C}^4</math>), family D REFLECTS it (the OS conjugation; <math>\sigma_{\text{std}}</math> = the standard conjugation with real points <math>\mathbb{RP}^3</math>). <b>[E]</b> MARK PINNING: <math>z \mapsto \mu\bar{z}</math> preserves the <math>\mu_4</math> marks iff <math>\mu \in \mu_4</math> — a 4-element <math>\mu_4</math>-TORSOR (2 mark axes + 2 silver-midpoint axes); clock conjugation <math>\rho\Theta_\mu\rho^{-1} = \Theta_{-\mu} \Rightarrow 2</math> clock orbits. <b>[E]</b> SPIN/FOCK: the <math>\mathbb{Z}_8</math> plane has no phase leaks (<math>\zeta_8^k = \zeta_8^{-k}</math>); exact <math>\text{Cl}(16)</math>: <math>\Theta_{\text{Fock}} = U_r \circ K</math> with <math>\Theta_{\text{Fock}}^2 = 2^7 &gt; 0</math> (normalised +1), inverts the clock tower (<math>V \mapsto 4096 V^{-1}</math>, exact scalar) and normalises the deck; KRAMERS DICHOTOMY: the deck-induced candidate <math>\Theta_t = \tilde{U} \circ K</math> has <math>\Theta_t^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F</math> — the <b>v510</b> split/nonsplit dichotomy IS the <math>\Theta^2 = +1</math> vs <math>(-1)^F</math> dichotomy; the deck does NOT furnish <math>\Theta</math>, the seam-circle REFLECTION does. <b>[E]</b> KILL TEST 1 (free level): does NOT fire — <math>\Theta</math> exists with <math>\Theta^2 = +1</math> AND <math>\Theta\rho\Theta = \rho^{-1}</math> and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock). <b>[E]</b> FREE RP: the chiral NS vacuum <math>C(d) = (2/N)/\sin(\pi d/N)</math> has <math>C^2 = -1</math> exact; preconditions exact (<math>RCR^T = -C</math>, <math>SCS^T = C</math>, <math>RSR^{-1} = S^{-1}</math>; the deck FAILS anti-invariance); BOND CUT: one-particle Gram <math>(8, 0, 0)</math> PD (min eigenvalue <math>1.888 \times 10^{-3}</math> at 40 digits), even <math>\deg \leq 2</math> sector <math>(29, 0, 0)</math> PD, <math>N = 8</math> complete half-sided algebra PD (even + odd each <math>(8, 0, 0)</math>) — free RP with NO degree truncation; the twist <math>\eta = +i</math> is FORCED (<math>\eta = 1</math> non-Hermitian); CONTROLS: the site cut fails exactly (<math>\det = 0</math>, inertia <math>(3, 3, 1)</math>); continuum control: the Cauchy–Stieltjes factorisation is strictly PD — a lattice-PLACEMENT artifact: RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally <math>((4, 4, 0))</math> — RP and <math>\Theta\rho\Theta = \rho^{-1}</math> select the SAME family; bonus: the anti-chiral state flips the odd sector to <math>(0, 8, 0)</math> ND (the free shadow of kill test 3). <b>[C]/[O]</b> NOT shown: the interacting algebra <math>\mathcal{A}_{\text{hol}}</math> (kill test 1 stays live there), gauge-fixed RP (kill test 2), the OS reconstruction, chirality without mirrors (kill tests 3/6 — only the free odd-sector shadow), <math>\mu_4</math> incidence (kill test 7); WOIT-<math>\beta/\gamma</math> roadmap: <math>(\beta_1)</math> <math>\Theta</math> on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant <math>\text{SDYM}(E_8)</math> sector; <math>(\beta_2)</math> the OS quotient of the free system explicit; <math>(\beta_3)</math> the <math>\mathbb{PT} \leftrightarrow \mathbb{PT}^*</math> duality vs <math>\sigma_{\text{std}}</math> typed; <math>(\gamma)</math> the chirality theorem + the mark incidence. Two contract PRECISIONS (prose, no marker): (i) “<math>\Theta</math> induced by the seam reflection” means the seam-circle REFLECTION, not the deck (the deck furnishes the <math>(-1)^F</math> Kramers class — the <b>v510</b> bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle. <b>[E][C][O]</b></p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v520_celestial_<br>wp5e_measure_<br>decision.py | <p>CELEST.WP5E.MEASURE.01: the WP5e BCOV measure question DECIDED at probe level (<math>\delta_{1e} + \delta_{1f}</math> consolidated, v518 style) — “single-valuedness is derived, the completion reading wins”: which of the two exact, contradictory measures on <math>\mathbb{P}T/\mathbb{Z}_4</math> is the true BCOV measure, (A) the DECLARED completion reading (v516) or (B) the DERIVED chiral measure (v518)? ERFOLG-A on the decision layer. [E] INVARIANT SUBSPACE: <math>\dim_{\mathbb{Q}}\{v : \rho(T)v = v, \rho(S)v = v\} = 8 = 4 \times 2</math> in the dressed 16-component Weil system, EVERY kernel vector in the <math>\mathbb{Q}(\zeta_8)</math>-span of <math>\{e_H, e_{H'}\}</math> (the two Lagrangian gluings, both <math>\theta = E_4</math>, level-1 quartic <math>= Q^{(0)}</math> = the unphased trace); strict trivial-character solutions at <math>(m, k) = (0, 0)</math>: ZERO for every dressing (11 combinations). [E] SOURCE 1 (F-LEMMA): the only strict chiral families <math>\chi_4</math> (dims (3, 3)) and <math>\chi_{10}</math> (dims (1, 1)) carry nontrivial <math>T</math>-characters exactly; pointwise covariance of the canonical column member certified (<math>3.9 \times 10^{-16}</math>, <math> G  \geq 59.6</math>); EXACT change of variables <math>\Rightarrow</math> fundamental-domain independence forces <math>\int = 0</math> for EVERY chiral integrand — the v518 route was never a well-defined nonvanishing moduli integral (kill SHARPENED). [E] SOURCE 2 (QUILLEN PAIRING): <math> t\bar{t} - 1  &lt; 8.3 \times 10^{-40}</math> on all 15 pairs/both generators; <math> \chi_4 ^2 = 1</math> exact vs one-sided <math>\chi_4^2 = e(2/3) \neq 1</math> (<math>T</math>-fixed: <math>640 \not\equiv 0 \pmod{960}</math>, 15/15 defective); the doubled 256-dim hol <math>\times</math> antihol Weil system closes STRICTLY at trivial character and physical weight (nullspaces <math>(28, 17) \geq 11</math>), contains the unphased diagonal (<math>\sim 2 \times 10^{-15}</math>) AND the physical columns <math>w_b \otimes \bar{w}_b</math> (<math>\sigma</math> <math>4.9 \times 10^{-16}/8.3 \times 10^{-16}</math>); negative control hol <math>\otimes</math> hol: EMPTY nullspace (0, 0). [E] CONSISTENCY CLOSURE: the forced unphased numerator in the v518 scaffold reproduces <math>\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x \rangle^2</math> exactly on every level <math>n \leq 8</math>; <math>J</math>-weighted spreads <math>\sim 5 \times 10^{-41}</math>, <math>T_5/T_3 \sim 7 \times 10^{-42}</math>, zero negative cells; <math>\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})</math>, the <math>J</math>-weighted mirror image of <math>\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64</math>; linearity <math>4.6 \times 10^{-41}</math> — the circle to v516 is CLOSED. [E] D3 <math>\mathbb{Z}_2/\text{EH}</math> ANCHOR: only the declared reading hits (<math>9\langle x, x \rangle^2 = (1/4) \times 36</math>, <math>\psi = -8</math>, scale tooth <math>c = 1/2 =  \mathbb{Z}_2 h^{A_1}</math>); all three derived instantiations fail (spreads 1.12/0.895/0.516, <math>\psi</math> sign wrong). [E] D1 COLUMN CANONICALISATION: the physical AB column <math>\sum_m i^{bm} e_{(m,m)}</math> is EXACTLY contained in the <math>\chi_4</math> family (<math>\sigma</math> <math>1.8 \times 10^{-16}/1.7 \times 10^{-16}</math>, one scalar per orbit — canonicalises the <math>(N_1, N_2)</math> freedom); the canonical member still fails all testers. [E] NEGATIVE CONTROLS: <math>SO(16)/D_8</math> (<math>T_5 = 20/T_3 = -40</math> exact, <math>\mu_2</math>-only), wrong form (<math> \text{Gauss} ^2 = 64</math>), wrong weight (dev 0.78), shuffle (<math>T_5 = -14/T_3 = 36</math>), planted family detected (<math>1.5 \times 10^{-16}</math>), hol <math>\otimes</math> hol empty. [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source <math>\psi = 64</math>), TP-3 (v518 kernel convention), TP-4 (Quillen structure of BCOV <math>F_1</math>). [O] the constructive BCOV derivation of the <math>w_m</math> normalisation itself. VERDICT: ERFOLG-A per the preregistered criteria (both sources independent, the consistency closure carries); the [O] question “which is the true BCOV measure” is DECIDED at probe level in favour of (A) under TP-1..TP-4. SEAM.EQUIV.TWISTOR.01 stays Open; NO marker moves. [E][C][O]</p> |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v521_seam_bit_rp_blind.py | <p>SEAM.BIT.RPBLIND.01: the RP/<math>\Theta</math> attack on the alignment bit DECIDED — “free OS positivity does NOT see <math>\delta</math>”, an honest, decided NEGATIVE result (v508 style): does free reflection positivity (v519) or anti-linear <math>\Theta</math> existence break for <math>\delta \neq \pi/2</math> on the deformed mark family <math>M(\delta) = \{\pm 1, \pm e^{i\delta}\}</math>, which would derive the v512 alignment bit from OS positivity? NO — the free RP/<math>\Theta</math> battery is the EIGHTH side-blind test on the v512 scoreboard (<math>7 \rightarrow 8</math>); the bit remains genuine discrete input. [E] CENSUS (exact solvesets, symbolic <math>\delta</math>): for every <math>\delta</math> exactly the two mark-SWAPPING reflections <math>\Theta_{\pm e^{i\delta}}</math> exist (axes <math>\delta/2, \delta/2 + \pi/2</math>; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at <math>\delta = \pi/2</math> (solveset <math>\cos \delta = 0</math>, lost immediately at <math>\pi/2 - \varepsilon</math>); the v510/v512 counterwitness <math>\{\pm 1, \pm(3+4i)/5\}</math> carries the swap pair exactly. [E] RP IS <math>\delta</math>-BLIND: bond-cut Gram inertias <math>(8, 0, 0)</math> PD for all tested <math>\delta</math>, spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO <math>\delta</math> information; the <math>N = 16</math> odd-<math>m</math> failure (site axes, <math>(3, 3, 1)</math>, <math>\det = 0</math>) is a lattice-PLACEMENT artifact: at <math>N = 32</math> ALL <math>m = 1..7</math> (<math>\delta = m\pi/8</math>) have mark-compatible BOND axes with PD inertia <math>(16, 0, 0)</math>; continuum: the OS kernel in axis-adapted coordinates is EXACTLY <math>1/\sin((s+t)/2)</math>, the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram <math>(6, 0, 0)</math>). [E] <math>\Theta</math> EXISTENCE <math>\delta</math>-BLIND: <math>\Theta^2 = +1</math>, deck normalisation and Fock implementability (<math>U^2 = +2^7/2^8 &gt; 0</math>, exact <math>\text{Cl}(16)</math>) for every <math>\delta</math>; the ONLY <math>\delta</math>-sensitive v519 clause (<math>\Theta\rho\Theta = \rho^{-1}</math>) is for <math>\delta \neq \pi/2</math> not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff <math>\delta = \pi/2</math>) — it PRESUPPOSES the bit; same for the clock-tower clause. [E] the battery has teeth at every <math>\delta</math>: family A/antipode fails RP structurally (zero diagonal, <math>(4, 4, 0)</math> indefinite), chirality-<math>\eta</math> pinning persists <math>((0, 8, 0)</math> ND) — RP separates the two real-structure FAMILIES, never the SIDES. [E] precondition controls: hexagonal (marks not concyclic, no seam circle), silver (deck not free), lifting audit (B = index relabelling, <math>\delta</math>-blind by construction; B' fails the <math>\pi/2</math> negative control — discarded). [C] FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is <math>V_4</math> for every <math>\delta &lt; \pi/2</math> and closes to <math>D_4</math> exactly at <math>\delta = \pi/2</math> (<math>\Theta_i \circ \Theta_1 = \text{the clock}</math>) — <math>12 \rightarrow 13</math> faces, a typed REFORMULATION of the bit, not a derivation. [O] honest gap: marks decorating the STATE (defect/twist insertions, interacting <math>\mathcal{A}_{\text{hol}}</math>) = exactly v519's [O] scope — there, and only there, could positivity still see the bit. VERDICT: KILL exactly as preregistered; the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; no marker moves. [E][C][O]</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v522_woit_beta1_<br>gso_gauge.py | <p>WOIT.BETA1.GS0.01: the WOIT-<math>\beta_1</math> milestone of the OS twistor bridge</p> <p>WOIT.OS.TWISTOR.01 executed — “the clock is time-like, GSO is the gauge datum” (typed result: verdict UNDECIDED per the frozen preregistration, mechanism identified exactly, gauge-fixed RP established under the corrected GSO typing). [E] HERMITICITY WITNESS: the <math>\mu_4</math> clock average violates Hermiticity EXACTLY (the one-step insertion <math>T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle</math> has entry <math>-i/(8 \sin(5\pi/16))</math> where the conjugate transpose demands <math>+i/(8 \sin(5\pi/16))</math>); the pairing is complex-SYMMETRIC (745 entries matching, 96 anti-matching, 0 violations) — “OS-symmetric” is strictly weaker than Hermitian: “positivity after gauge fixing” is not well-posed for the clock reading. [E] TIME-LIKE TYPING: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift (<math>RSR^{-1} = S^{-1}</math>), NONE commutes — the <math>\mu_4</math> clock IS Woit’s euclidean rotation (the time translation itself); the gaugeable (<math>\theta</math>-compatible) part of the <math>\mathbb{Z}_8</math> tower is EXACTLY its 2-torsion <math>\{1, (-1)^F\}</math> (Hermiticity census over all powers; <math>\alpha^4 =</math> the full wrap = the <math>2\pi</math> rotation of the seam = fermion parity, lattice spin-statistics). [E] POSITIVE PART — GSO-FIXED RP HOLDS EXACTLY: <math>N = 16</math> even <math>\deg \leq 2</math> parity-projected bond-cut Gram (29, 0, 0) PD (min eigenvalue <math>1.78 \times 10^{-6}</math> at 40 digits); <math>N = 8</math> complete half algebra: the projected Gram kills all odd rows/columns exactly, even block (8, 0, 0) PD (<math>3.35 \times 10^{-3}</math>), no degree truncation; the site-cut defect SURVIVES gauge fixing ((7, 9, 6)) — the RP bond placement is gauge-invariant information; family A keeps failing ((17, 12, 0) indefinite vs family D’s (29, 0, 0)); Kramers: <math>\Theta_t^2 = \text{id}</math> on <math>\mathcal{A}^G</math>, <math>\Theta_{\text{Fock}}</math> restricts with <math>\Theta^2 = +1</math>; Ramond control: the wrap +1 “true <math>\mathbb{Z}_4</math>” lift fails state invariance — the <math>\text{NS}/\mathbb{Z}_8</math> tower is forced. [E] KILL-TEST BOOKKEEPING: kill test (1) stays discharged also gauge-invariantly (RP separates the families on the invariant sector); kill test (2)’s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction); both stay formally live on the interacting algebra <math>\mathcal{A}_{\text{hol}}</math>. [E] WRONG-PROJECTION CONTROLS: invariant-dimension chain parity <math>120 &gt;</math> deck <math>64 &gt;</math> clock <math>32</math> strict, but Hermiticity singles out PARITY alone (deck and clock averages non-Hermitian): the finer “gauge” projections are structurally ill-posed pre-quotient. TRANSPARENCY: the first frozen run scored 6/13 and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census: <math>28 + 4 = 32</math>) — documented in full; verdict by the frozen [C-3] logic (Hermiticity failure <math>\Rightarrow</math> UNDECIDED). [C] CONTRACT PRECISION (iii), prose only: “gauge-invariant subalgebra” at the free level means the GSO/fermion-parity <math>\mathbb{Z}_2</math> — the only <math>\theta</math>-compatible part of the <math>\mu_4</math> tower (the free-fermion shadow of the <math>E_8</math> glue: <math>(E_8)_1 =</math> even NS of <math>SO(16)_1 + \text{R spinor}</math>); the clock is the euclidean rotation itself (timelike), its equivariant statement moves to <math>\beta_2</math> (OS quotient first, then the clock as reconstructed transfer operator). [O] <math>\beta_1</math> proper stays open via <math>\beta_2</math>; WOIT.OS.TWISTOR.01 stays [O]; no marker moves. [E][C][O]</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v523_celestial_<br>wp5e_wm_derived.py | <p>CELEST.WP5E.WM.01: the constructive derivation of the <math>w_m</math> normalisation — “<math>1/\det_j</math> is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources”, closing the residual [O] of v516/v520 at the fibre-zero-mode level (verdict ERFOLG per the frozen preregistration). [E] ATIYAH–BOTT (route i): the equivariant mode ledger of the fibre <math>\mathbb{C}^2</math> equals the closed form <math>1/((1 - q i^j)(1 - q i^{-j}))</math> exactly in <math>\mathbb{Q}(i)</math> for all <math>n \leq 120</math>; REGULAR at <math>q = 1</math> with Abel value <math>(1/2, 1/4, 1/2) = 1/\det_j</math>, <math>\det_j = \det(1 - g^j) = (2, 4, 2)</math>; <math>(C, 2)</math> Cesaro convergence in exact arithmetic; the fixed-point factor splits off EXACTLY at every truncation level <math>d \in \{7, 12, 25\}</math>, for the phased (skeleton) AND the v520-forced unphased (completion) trace alike — the Abel-limit contact<math>_j = (Q^{(0)} - Q^{(j)})/\det_j</math> is the v516 contact vector COMPUTED, not declared. [E] ZETA/QUILLEN (routes ii/iii): <math>\det_\zeta</math> of the <math>g^j</math>-twisted circle Laplacian <math>= 4 \sin^2(\pi j/4) = \det_j</math> EXACTLY (Lerch + reflection formula symbolically; Hurwitz certificates <math>\sim 10^{-41}</math>); Quillen split <math>\det \Delta = \det_j^2</math>, <math> \det \bar{\partial}  = \det_j</math>, with the real positive holomorphic section UNIQUE via the <math>SU(2)</math> conjugation pairing; the <math>\delta_{1f} f_1 f_3</math> block at the <math>(0, b)</math> nodes carries the exact constant term <math>1/\det_b</math> (deviations <math>\leq 5.2 \times 10^{-28}</math>). [E] CONSISTENCY CHAIN: the derived weight reproduces the v516 chain number by number — <math>w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m =  \mu_4 h_m = -4 \text{ch}_2(T_m)</math>; <math>T_5 = 0</math> at every scale; <math>h_2 : h_1 = 4 : 3</math>; the <math>T_3</math> budget <math>32 - 8c</math> forces <math>c = 4 =  \mu_4 </math> uniquely; per-sector Okubo squares <math>(18, 9, 18)\langle x, x \rangle^2</math>; total <math>45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2</math>; <math>\Phi_{T_3} : 32 \rightarrow 0</math>, <math>\Phi_P : 72 \rightarrow 0</math>; <math>\psi(A_{\text{fix}}) = +64</math>, <math>\psi(\text{contact}) = -64</math>. [E] NEGATIVE CONTROLS: <math>1/\det^2</math> gives <math>w' = (0, 5/8, 1, 5/8)</math> with leftovers <math>(2, 12)</math> and a Quillen contradiction; <math>1/ 1 - i^j </math> gives <math>w''_1 = 1 + \sqrt{2} \notin \mathbb{Q}</math>; the <math>\mathbb{Z}_2</math>/Eguchi–Hanson anchor PASSES with the same derivation (<math>\det = 4</math>, <math>w = 1/2 =  \mathbb{Z}_2 h^{A_1}</math>, <math>9\langle x, x \rangle^2 = \frac{1}{4} \times 36</math>, <math>c = 2 =  \mathbb{Z}_2 </math>); the <math>SO(16)/D_8</math> KILL fires; <math>\text{diag}(i, i)</math> breaks the zeta = AB identity itself; <math>k = 3, 5</math>: <math>w_m = m(k - m)/2 = k h_m</math> — the weights wander correctly with the centre count, only <math>k = 4</math> carries the <math>E_8</math>-glue chain. [C] TYPED PREMISES (the mandatory fence): TP-REG (Abel/Cesaro/zeta regularisation IS the definition of the equivariant character), TP-Q (Quillen structure of BCOV <math>F_1 = \text{v520 TP-4}</math>), TP-NUM (unphased numerator = the v520 ERFOLG-A decision layer), TP-CH (the insertion channel <math>(0, j)</math> computes the box-channel weight). [O] the GLOBAL BCOV integral on <math>\mathbb{P}^1/\mathbb{Z}_4</math> beyond the fibre zero-mode factor; the resolved-ALE route (v514/v517) not built; SEAM.EQUIV.TWISTOR.01 untouched. NO marker moves. [E][C][O]</p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v524_woit_beta2_os_quotient.py | <p>WOIT.BETA2.OS.01: the WOIT-<math>\beta_2</math> milestone of the OS twistor bridge</p> <p>WOIT.OS.TWISTOR.01 executed — “the OS quotient of the free system made explicit” (verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii)). [E] <math>\mathcal{H}_{\text{phys}}</math> EXPLICIT: <math>N = 16</math> (<math>\deg \leq 2</math>): the <math>37 \times 37</math> bond-cut OS Gram is exactly Hermitian, parity-block-diagonal, inertia <math>(37, 0, 0)</math> PD (min eigenvalue <math>1.7801 \times 10^{-6}</math> at 40 digits), null space <math>\{0\}</math>, <math>\dim 37 = 29 \oplus 8</math>; <math>N = 8</math> (complete half algebra): <math>(16, 0, 0)</math> PD (min <math>3.3471 \times 10^{-3}</math>), <math>\dim 16 = 8 \oplus 8 = 4^2</math> — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate <math>\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2</math>. [E] TRANSFER SEMIGROUP (Klein–Landau): <math>\tau_k</math> exactly Hermitian on all domains (<math>k = 1..7</math>, dims <math>29, 22, 16, 11, 7, 4, 2</math>), chain identities exact, vacuum fixed; the positivity pattern IS the site/bond dichotomy: even <math>k</math> PSD via the exact square identity <math>T(2j) = A_j^* A_j</math> (<math>(22, 0, 0)/(11, 0, 0)/(4, 0, 0)</math>), odd <math>k</math> indefinite (<math>(10, 12, 7)/(5, 6, 5)/(2, 2, 3)</math>) — the one-step transfer is NOT positive (the free chirality datum, kill-test-3 shadow sharpened), its even powers ARE. [E] THE CLOCK = <math>T^{N/4}</math>: <math>N = 16</math>: <math>\tau_4</math> PD <math>(11, 0, 0)</math> (min <math>6.6821 \times 10^{-5}</math>), compressed spectrum <math>[0.0194517, 1.64144]</math>, vacuum eigenvalue exactly 1; <math>N = 8</math>: spectrum EXACTLY <math>\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}</math> in both parity sectors — the silver axes of the v519 <math>\mu_4</math> torsor return as the clock-transfer eigenvalue; spectral projections <math>\sim 10^{-40}</math>; the reconstructed rotation group <math>U(s) = e^{isH}</math> is unitary with the group law. [E] v522 RESOLUTION: the census <math>(745, 96, 0)</math> reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus. Pre-registered KMS deviation: NO contraction on the compact circle (edge modes <math>C(1)/C(3) = 1 + \sqrt{2} = \delta_S</math> exactly, <math>\det(G - \tau_4) &lt; 0</math>). [E] GSO/<math>\Theta</math>: the grading survives; the perpendicular torsor mirror descends anti-unitarily (<math>\eta_\perp \in \{\pm i\}</math>), <math>\Theta_{\text{phys}}^2 = +1</math> on every sector (Kramers-free), <math>\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}</math> exactly. [E] CONTROLS: site cut indefinite <math>((3, 3, 1)/(2, 2, 4))</math> — the contract kill branch fires there; family A <math>(4, 4, 0)</math>; anti-chiral <math>(8, 8, 0)</math> no quotient; wrong direction non-Hermitian (mismatches all wrap-overlap). KILL-TEST BOOKKEEPING: (1), (2) strengthened, (3) shadow sharpened, none fires; (4)–(7) untouched; all seven stay live on <math>\mathcal{A}_{\text{hol}}</math>. [C] FENCE: “the clock acting unitarily” is operationalised per contract precision (iii) as the reconstructed transfer/rotation structure. TRANSPARENCY: run 1 scored 18/20 (a sympy simplify failure on a TRUE <math>\pi/16</math> product formula, fixed by a Laurent-polynomial certificate; no criterion changed). [O] <math>\beta_3</math> is the next milestone; WOIT.OS.TWISTOR.01 stays [O]; no marker moves. [E][C][O]</p> |

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| Script                                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v525_seam_bit_twist_blind.py           | <p>SEAM.BIT.TWISTBLIND.01: the twist-state attack on the alignment bit DECIDED — “mark-decorated states do not see <math>\delta</math>”, an honest, decided NEGATIVE result (v508 style): the NINTH side-blind test on the v512 scoreboard (<math>8 \rightarrow 9</math>); the free-plus-twist class (everything Wick-computable) is EXHAUSTED. [E] ALL LIFTINGS WELL-DEFINED AND BLIND: (a) <math>\sigma</math>-gauge arc decoration (<math>\mathbb{Z}_2</math>): <math>\sigma \circ r = \sigma</math> on the swap axes makes the twisted Gram a diagonal unitary congruence <math>DMD</math> — spectra IDENTICAL to untwisted (dev 0.0); (b) Kadanoff–Ceva 4-twist insertion <math>\omega(D \cdot)/\omega(D)</math>: well-defined on <math>N = 32</math> (<math>\omega(D)</math> real nonzero; the <math>N = 16</math> odd-<math>m</math> degeneration is the <math>C(\text{even}) = 0</math> checkerboard artifact); the RP SPECTRA depend on <math>\delta</math> for the first time in the program (pairwise up to 1.05), but the INERTIA is (16, 0, 0) PD with the SAME <math>\eta = +i</math> for every <math>m</math> — the decoration sees <math>\delta</math>, positivity does not; (c) the <math>\mu_4</math> twist at <math>\beta = \pi/2</math> is pure plane gauge; (c') genuine Bogoliubov defect <math>\beta = \pi/4</math>: <math>\Theta</math>-compatibility selects exactly 2 of 16 sign patterns (NS-wrap forced), the winner genuinely non-monomial — still PD, spectra identical. STRUCTURE THEOREM (empirical, mechanism exact): EVERY <math>\Theta</math>-compatible quasi-free mark decoration is RP-spectrally INVISIBLE (the defect planes never straddle a cut bond; orthogonal congruence). [E] SUB-RESULTS: the defect energy <math>E_{\text{def}} = 1.2752872</math> for EVERY <math>m</math> (spread <math>4 \times 10^{-40}</math>) — <math>\delta</math>-blind; the 4-twist Casimir <math>\ln  \omega(D) </math> measures <math>\delta</math> (<math>-0.890 \dots -1.348</math>) but is exactly mirror-symmetric under <math>m \leftrightarrow 8 - m</math> — a family gauge, never a side selector; Kramers <math>\langle \Gamma \rangle_{\text{tw}} = +1</math> for every <math>m</math>. [E] TWO BIT-PRESUPPOSING <math>\pi/2</math> STRUCTURES (the harvest): the <math>\eta</math> flip <math>+i \rightarrow -i</math> on the mark-fixing axes (<math>\sigma \circ r = -\sigma</math>, odd sector exactly negated); the twist frustration of the mark-fixing mirror ((4, 4, 0) indefinite for both <math>\eta</math> — the lattice shadow of the Ising twist field's mirror-oddness) — both formulable ONLY on the fixing axes existing iff <math>\delta = \pi/2</math>: they PRESUPPOSE the bit (v512 facet class), no selector. [E] NEGATIVE CONTROLS: the twist-free limit = v519/v521 exactly; the site cut keeps failing in every decorated state; the v512 counterwitness is arc-equivalent at <math>N = 16</math> to the blind member <math>\pi/4</math> — no state-level selector can exclude it; a single twist has <math>\omega(D) = 0</math> identically (twists exist only in mark pairs); the deck-paired FOUR-twist decoration is the minimal well-defined one. [O] the door narrows to a genuinely INTERACTING <math>\mathcal{A}_{\text{hol}}</math> whose OS data are NOT Pfaffian-reducible to the chiral vacuum — beyond every Wick-computable class. VERDICT: KILL exactly as preregistered; the alignment bit remains genuine discrete input; no marker moves. [E][C][O]</p> |
| v526_seam_thermal_kms_nariai_bridge.py | <p>SEAM.THERMAL.KMS.01: the thermal seam — KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain (the THIRD LEG of <math>c_3</math>). Verdict SUCCESS. [E] <math>\beta</math> MEASURED = <math>N</math> at both levels (<math>N = 8</math>: <math>\omega = \text{arcsinh}(2^{-3/4})</math>, <math>\kappa = q^{-8}</math> exact; <math>N = 16</math>: pairing exponent <math>(N/2) + 1 = 9</math>); <math>T_{\text{seam}} = 1/N</math> per clock step, <math>\beta_{\text{angle}} = 2\pi</math> EXACT = <math>1/(4c_3)</math>, <math>T_{\text{seam}} = 4c_3 =</math> Bisognano–Wichmann/Hawking normalisation (v208); modular Hamiltonian closed (<math>\nu_1 = \cos(\pi/24)</math>, <math>\nu_2 = \cos(7\pi/24)</math>; Tomita KMS-<math>\beta = 1</math> on all 256 pairs); <math>c_3</math> third leg class-1 selected; entropy <math>S(8/16/32) = 0.52187/0.63510/0.75006</math>, <math>\Delta S/\ln 2 \rightarrow 1/6</math>, horizon fractions do NOT transfer; [C] silver = <math>C(3)/C(1)</math> thermal kernel ratio; non-compact <math>\Rightarrow T = 0</math>. NO marker moves. [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v527_seam_clock_silver_spectrum.py     | <p>SEAM.CLOCK.SILVER.01: the silver clock eigenvalue EXPLAINED and DEMYSTIFIED (double verdict). STRUCTURE = SUCCESS: <math>\{1, \sqrt{2} - 1\}</math> closed via diagonal Hankel pencil at <math>N = 8</math>, <math>\tan(\pi/8) = C(3)/C(1)</math>, doubling = exact second quantisation, carrier = silver-midpoint axes, gap = <math>\text{arcsinh}(1) = \text{gd}^{-1}(\pi/4)</math>; <math>N</math>-scaling = pure <math>N = 8/16</math> lattice datum. CONNECTION = KILL: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable. Cross-link v526: <math>\sqrt{2} - 1 =</math> thermal kernel ratio. NO marker moves. [E]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v528_seam_bit_twist_class_definition.py | SEAM.BIT.TWISTCLASS.01: the alignment bit as an exact measurable TWIST-CLASS choice (the physical definition). Verdict SUCCESS. [E] Facets #14 ( $\eta$ -flip $\Leftrightarrow \delta = \pi/2$ ) and #15 (twist-frustration $\Leftrightarrow \delta = \pi/2$ ) both directions exact (cardinality lemma $ A_+  =  A_-  \Rightarrow m = 4$ ; $g$ -class census 224 axes: flip class exactly at $\{15, 31\}$ ); order parameter $O = 1/2$ at $m = 4$ else 0; unification #14=#15=flag-transitivity = ONE $\mathbb{Z}_2$ invariant (v512 web 13 $\rightarrow$ 15); Kramers = POSITION half (derived), twist class = MODULUS half (the input). [C] measurement sketch (interferometric in principle); the bit remains formal input. NO marker moves. [E][C]                                        |
| v529_seam_interacting_toy_fk.py         | SEAM.INT.FKTOY.01: the minimal interacting Fidkowski–Kitaev seam toy — the FIRST firing Kill-Test-2 shadow (triple-verdicted). F1 = KILL: $\Theta$ exists so Kill-Test 1 does NOT fire; Kill-Test 2 FIRES at toy level — RP breaks for every $g > 0$ (inertia ladder $(37, 0, 0) \rightarrow (29, 8, 0)$ ); STRADDLE LAW (24/24). F2 = KILL: the TENTH side-blind test ( $9 \rightarrow 10$ ) — $\Theta_{15}$ maps $H(m) \rightarrow H(8-m)$ EXACTLY. F3 = ERFOLG: grading + $\mathbb{Z}_8$ tower + FK anchor + constructive OS quotient at $\pi/2$ . [C] fence: one toy / one interaction class — narrows the Woit route, does NOT close it; WOIT.OS.TWISTOR.01 stays Open. NO marker moves. [E][C][O]                                                                                        |
| v530_center_quotient_compiler.py        | center quotient compiler: $Cv=v$ , $\mathbb{Z}^3/\mathbb{Z}v$ quotient $= (\frac{8}{5} \frac{2}{3})$ (atoms); row sums (7, 11, 13) self-code; CP code (1, 7, 11, 13); $\det(C+kI)=(k+1)\det(A+kI)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v531_coxeter_tensor_stage_a.py          | Stage A tensor clock: $T_{30}=\text{Comp}(\Phi_5)\otimes\text{Comp}(\Phi_6)$ , order 30, $\chi=\Phi_{30}$ ; traces $c_{30}(k)=c_5(k)c_6(k)$ ; atom readout (2, 4, 8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v532_e8_degree_modular_checksum.py      | dual-degree checksum: (60, 192, 240, 252): sum 744=3·248, prod $ W(E_8) $ , gcd 12; $j=q^{-1}+744+\dots$ ; $D_{16}$ control 1488=3·496; primes < 30 fingerprint [audit]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v533_ftransfer_disc7_norm.py            | sheet transfer as a disc = -7 norm line: $D(t)=N(\alpha_t)$ , $\alpha_{t+1}=\alpha_t+2$ , norms (2, 4, 14, 32) on $J \rightarrow K \rightarrow C \rightarrow F$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v534_seam_straddle_cone.py              | SEAM.STRADDLE.CONE.01: the RP cone on the interacting FK quartets — the FIRST DYNAMICAL SELECTION of the alignment bit (double-verdicted). V1 = KILL: the literal leading-order cone is EMPTY everywhere (the linear pencil cannot carry the selection). V2 = ERFOLG — THE SELECTION LAW: reflection positivity holds for EXACTLY ONE member $\times$ sign of the equivariant FK family — $\delta = \pi/2$ with positive coupling, PD (37, 0, 0) on all four cuts incl. the straddled clock axes 7/15 for $g \in \{1/32..8\}$ , minimum eigenvalue LIFTED off the RP boundary; asymmetric members die. Exact geometry: $\pi/2$ is the unique symmetric-straddle member (quartet partition, reflection-closed straddles, sign +1). [C] fence: one toy / one interaction class — NO marker moves |
| v535_hecke_from_geometry.py             | Hecke from geometry: Kneser $\#iso\_lines=\sigma_3\cdot\#P^3$ (135/1120/19656/137600); $\nu_p=aId+bT_p$ , $a_p=b-\sigma_3$ ( $a_3=-4$ , $a_5=-2$ ); $\dim V=7=5+2$ ; $\pi_{cusp}=(28-T_3)/32$ ; 2-adic levels only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v536_eichler_trace_layer.py             | Eichler trace layer: $\lambda_{\text{Eis}}=\sigma_3(\#\mathbb{P}^3-\sigma_3)$ ; $\lambda_{\text{geom}}=\lambda_{\text{Eis}}+a_p^2$ (anchors 352/3784/19840); $N_A=\min(240(1+p^3), \#iso-1)$ (82560/743040 at $p=7$ ); $a_p=-c(p)/8$ (-4, -2, +24); $R=a_p^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v537_halfintegral_bridge.py             | Half-integral bridge: unique $\text{Sh}_{t=2}(g)=-8f_8$ in the 70-monoid; $T(p^2)$ -eigen with $a_p(f_8)$ ; $U_4(g)=0$ / outside Kohnen 1982; Waldspurger $R=23.1873585645\dots$ constant on 10 $d \equiv 1 \pmod{8}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v538_relative_trace_identity.py         | Compiler relative-trace identity: $\text{Tr}_V(\nu_p \circ \pi)$ bilateral independent at $p = 3, 5, 7$ ; Eichler orbit dictionary $\lambda_{\text{Eis}} + a_p^2$ ; period side = lattice count with $R=23.1873585645\dots$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v539_weil_structure_family.py    | Weil structure of the compiler family: GNS fibre isolation; $GL(1)$ core $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$ ; $Q_{\text{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_b) + \text{Arch} + \text{Corr}$ ; two isolated obstructions (minus doubling; non-automorphic Corr)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v540_amplitude_linear_carrier.py | Amplitude route and positive linear carrier: Dirac $D^2 =$ family kernel (Hecke-equivariant); $b = N_+ - N_-$ , $\Theta$ pure $\sigma_3$ -eigenform with Cohen seed $\Theta(d) = -48L(-1, \chi_d)$ ; CL deletion universal = square-class double counting; plus balance $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ ; FE $\Lambda_\Theta(s) = 8^{1-s}\Lambda_{\Theta^\dagger}(5/2-s)$ ; open boundary = gap functional $\lambda^*$ on $n \equiv 6 \pmod 8$                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v541_matching_lemma_ledger.py    | Matching lemma and transport ledger package: window proof $7S < 40A$ on all $j \leq 10^6$ (exact margin $X = 0.082159$ ); ledger $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$ with $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$ ; $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8})\Theta$ ; $\Delta_{\text{arch}} = A_{\text{fam}} - A_{\text{shift}}$ (Legendre duplication); coherent class closed by the $\lambda$ -equivariant CM channel; named limits: one open classical lemma + I5 (equivalence typing)                                                                                                                                                                                                                                                                                             |
| v542_margin_chain_identities.py  | Margin-chain identities of phase 2: border weight $\sum(1-g_i) = t_l + t_r \Rightarrow \kappa_{\text{flat}} = 1$ exact; the (T) four-term identity; profile identity $p_{N-1} = 2 - p_0 + 2E$ with $2E > p_0 \iff p_{N-1} > 2$ ; the Abel/Hölder curvature chain; matrix-free assembly $= J^T Q J$ and graded overlap $= J$ ; the C ea/Strang defect identity $S_{\text{graded}} - S_{\text{uniform}} = R^T X^{-1} R$ (PSD, one-sided); the secular sandwich with Albert's sign equivalence; Perron sign constancy from $S^{-1} > 0$ as an implication; the run-count chain and the TV rewrite — per instance on small windows, nothing uniform in the zone index [E]                                                                                                                                                                                              |
| v543_lumped_pair_identities.py   | Lumped $M$ -matrix pair of phase 2: $A_B = A + L_\Delta$ Stieltjes with preserved row sums and $A_B \succeq A$ (an <i>upper</i> bound on $\lambda_{\min}(A)$ , never a floor); the congruence $A = A_B^{1/2}(I-W)A_B^{1/2}$ with $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ and the declared circularity $\lambda_{\max}(W) < 1 \iff A > 0$ in both directions; the $M$ -matrix anchor $\lambda_{\min}(A_B) \geq 1/\max_r x_r$ ; shift commutation with entrywise inverse monotonicity $\Rightarrow$ the shifted ladder is structurally empty; Varga's identity $\rho(J) = \tau/(1+\tau)$ ; the Collatz–Wielandt bracket at the anchor vector giving an a-priori $\rho(J) < 1$ ; and the $k$ -scanned M18 majorant as a <i>negative</i> result against the preregistered bar $1/2$ — per instance on small windows, nothing uniform in the zone index [E] + [X] |
| v544_long_lag_support.py         | Long-lag support structure of phase 2: $c = c_{\text{arch}} - c_{\text{comb}}$ exactly and the archimedean section carries <i>no</i> positive off-diagonal, so every positive entry is comb-generated; $\text{supp}(\Delta)$ sits entirely on the anti-diagonal comb stripes at prime-power atom positions; each stripe is a perfect matching with exactly orthogonal edge vectors; the amplitude certificate $\Delta_{rt} \leq c_{\text{comb}}[M-1-r-t]$ ; the Green function of the lumped comparison bracketed two-sidedly (Neumann partial sum from below, anchor-weighted tail from above); and the absolute-value envelope family certified <i>dead</i> by $\rho( E ) \geq 1.31$ from below against the signed $\rho(E) \leq 0.99996$ — per instance on small windows, no decay law [E] + [X]                                                                |
| v545_hardy_core_identities.py    | Hardy-core identities of phase 2: the form lifting $\text{Gram} = CHC^T$ with $C = \text{diag}(\sqrt{\Delta})M$ , hence $\text{rank}(\text{Gram}) \leq h-1$ ; the exact finite-core reduction $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ with the closed geometric covering kernel $K = M^T \Delta M$ ; the energy reordering $H = \text{diag}(s) + L_N$ ; the checkerboard split $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$ with three Weyl steps; the four Hardy identities (the $K^+$ Rayleigh form, the signed crossing kernel $L_N = D^T B D$ , $Q = B_+ \mathbf{1}$ , $DKD^T = L_\Delta$ on the interior nodes); the closed Muckenhoupt quantity $A_{M0} = \max_k Q_k(\Delta \mathbf{1})_{k+1}$ ; and the additive and joint shapes as <i>valid</i> upper bounds — dominance only, no size, nothing uniform in the zone index [E] + [X]         |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v546_capacity_chain_identities.py | Capacity-chain identities of phase 2: the capacity decomposition $K^{-1}=D^{\top}J^{-1}D+xx^{\top}/\text{cap}$ with the Dirichlet half an orthogonal projection ( $\Omega=1$ exactly); the exact capacity–Rayleigh identity for $1-\rho(W)$ assembled against the direct quotient with the inf property and the declared 1/gap conditioning; Cholesky nestedness — $\text{cap}([a,b])$ as a prefix sum of one forward substitution, verified against independent solves; the whitening certificate $\kappa_{\text{up}}$ under the Gershgorin ceiling; and the T145 M4 split — the pointwise mass-half theorem, $\sigma_{\text{tot}} < 1$ per window, licence 4 with $ R $ and its built-in witness, the ordered sign-free chain and S1' certified per window on both routes ( $c_0 = 2.52\text{--}4.23$ against the Dirichlet value 4) — per instance on small windows, the level lemma stays open [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v547_level_lemma_identities.py    | Level-lemma identities of phase 2: the union identity for nested cakes $\sum_{j,l} c_j c_l  S_j \cup S_l  = 2T\ \psi_t\ _1 - \ \psi_t\ ^2$ (form-independent, $O(m)$ ); the layer-cake counting lemma at a general base, $G(b) \leq 2b^2\ v\ _{\infty}\ v\ _1/\ v\ ^2 + \varepsilon$ for <i>any</i> vector, the classical dyadic 8 recovered at $b = 2$ and the base free; the resolvent delocalization bound $\psi = \lambda R\psi \Rightarrow \ \psi\ _{\infty} \leq \lambda_{\text{up}} \max_j \ Re_j\ $ with Rayleigh–Ritz $\lambda_{\text{up}}$ and residual-paid certified column norms — the lever 2.85–10.17 below the trivial ceiling; and the composite level lemma $c_0^{\text{ap}} = 2b^2\Gamma \min(1, \Gamma_1) + \varepsilon = 3.90\text{--}4.85$ as the certified window inequality $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}}\Psi_{\text{abs}}$ on 12/12, with the built-in no-go discriminator and the flat Dirichlet control — per instance on small windows, the all- $D$ asymptotics stays open [E]                                                                                                                                                                                                                                                                                                                                                                              |
| v548_green_szego_identities.py    | Green/Szegő identities of phase 2: the factorisation identity $\Gamma = \sqrt{Q^*} S_w$ with $S_w = \lambda_{\text{up}}\ R\ _F$ purely spectral and $Q^* = m \max_j (R^2)_{jj} / \text{tr}(R^2)$ purely geometric (exact at $2.2 \times 10^{-16}$ ); the bottom-block trace identity $\sum_j (\Pi_B)_{jj} =  B $ with the measured $Q_B/ B  = 1.509\text{--}1.798$ inside $[1, 2.2]$ against the Szegő/Widom reference $2 B $ ; the second-difference $\ell^1$ ceiling $ a_k  \leq \min(1, \ s_k\ _{\infty} \ L_0^p \psi\ _1 / \mu_k^p)$ with the $m$ -free closed form $\ a\ _1 \leq 2\sqrt{\nu} + 1$ and the Dirichlet calibration $\nu \rightarrow \pi$ ; the layer-cake $S_w$ certificate from completed $\text{LDL}^{\top}$ inertia counts ( $S_w \leq 1.329\text{--}1.822$ ); and the Liouville transform $m\ \psi\ _{\infty}^2 \leq 2\kappa_{\Lambda} \text{ceil}(\hat{\varphi})^2$ with a-priori $\kappa_{\Lambda}$ plus the weight-class discriminator (BV flat, rough grows at matched $\kappa_{\Lambda}$ ) — per instance on small windows, the one open lifting input ( $\nu_L$ m-free / $\text{TV}(\log \Lambda)$ ) stays open [E]                                                                                                                                                                                                                                                           |
| v549_gauge_parity_identities.py   | Gauge/parity identities of phase 2: the gauge freedom — for any positive diagonal $\tilde{\Lambda}$ the chain $\lambda_{\min}(A) \geq \text{cert } \lambda_{\min}(\tilde{E}) \cdot \min \tilde{\Lambda}$ is a valid certified lower bound (identity plus one Rayleigh step, family maximum valid), $\Gamma = \sqrt{Q^*} S_w$ gauge-invariant at $2.4 \times 10^{-16}$ , the const gauge with $\text{TV}(\log \tilde{\Lambda}) = 0$ exactly; the two-regime discriminator — the flutter smoother removes TV by $\geq 2.88\times$ while the transported $\nu_{\tilde{L}}$ moves $\leq 0.88\%$ , the untransported functional jumping 0.41–9.76 under the rough re-gauge; the explicit zone diagonal $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{ r-s } - c_{M-1-r-s}, 0)$ with the exact split $c = c^{\text{arch}} + c^{\text{atom}}$ ( $c^{\text{atom}} \leq 0$ ) and the closed atom budget $\ c^{\text{atom}}\ _1 \leq 4B\sqrt{N}$ , the archimedean section itself indefinite (atoms co-responsible for positivity); the parity compression $U_{-}^{\top} T_M U_{-} = T - H$ exact with the negative inertia localised entirely in the even sector; and the parity-sine calibration $\nu_k^P = \pi k^2$ on the exact model with the per-window ladder $\nu_k^P \leq Ck^2$ , $C = 11.70\text{--}17.28$ — per instance on small windows, the one open input ( $m$ -free $C$ ) stays open [E] |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v550_odd_sector_identities.py    | Odd-sector identities of phase 2: the grid step-over — the parity sines live on the shifted grid $\theta_k = 2\pi k/N$ with $\mu_k^P = 2 - 2\cos\theta_k$ exactly, and the symbol's negative window satisfies $\theta_c/\theta_1 = 0.371\text{--}0.485 < 1$ on every window (odd-sector positivity is a grid fact), the symbol-only floor certified vacuous ( $\rho \geq 1$ on 12/12: a matrix statement, not a symbol one); the certified bottom ladder $\lambda_k(A) \leq S\mu_k^P$ for $k \leq 8$ by completed LDL <sup>T</sup> counts ( $S = 1.1019\text{--}6.4725$ ); the discrete Sobolev step $\ \psi\ _\infty^2 \leq 2\ \psi\ _2\ \nabla\psi\ _2$ licensed by the odd sector's virtual node, model price exactly $\pi$ ; the closed archimedean minimum $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$ exact and attained; and the linear per-mode bound $b_k = 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \leq C_S k$ with certified $\kappa = 0.1939\text{--}0.3599$ , $C_S = 12.081\text{--}26.953 \leq 2\pi\sqrt{K_{\text{bot}}/\kappa}$ — linear where the quadratic ladder grew, the no-go pencil ratio exploding $\times 15.8$ — per instance on small windows, the one open scalar ( $m$ -freeness of $R = K_{\text{bot}}/\kappa$ ) stays open [E]                                                                                                                   |
| v551_ritz_ceiling_certificate.py | Fixed-size Ritz ceiling certificate of T154: the direction lemma — Ritz values are upper bounds for the eigenvalues of the same index (Courant–Fischer 1920/Cauchy 1829, verified against exact spectra; the reversed reading refuted on 12/12) — so the ceiling needs no residual argument; the sixteen-column certificate $\text{span}\{t_1..t_8\} + A^{-1}L_P \text{span}\{t_1..t_8\}$ (dimension 16 fixed, independent of $m$ ; 8 completed Choleskys of matrices $\leq 8 \times 8$ ) carries $K^F = 1.1019\text{--}1.5845$ attaining the inertia-certified $K_{\text{bot}}$ to $\leq 9.9 \times 10^{-7}$ against the declared cap $10^{-4}$ on 12/12 — the 8 size- $m$ LDL <sup>T</sup> counts retired from the ceiling step; the one-direction anatomy (median of the seven aligned angles $0.03\text{--}0.14^\circ$ , the largest $83.79\text{--}89.70^\circ$ ) with the collapse price $\lambda_1(A)/(t\mu_1^P) = 4.408\text{--}7.042$ recovered in full per window by one Cholesky; the no-go ceiling explodes $\times 57.5$ where the floor survives, the Dirichlet control certified non-positive — per instance on the 12 deepest in-cap windows; the two open inputs after T154 (R1' $m$ -free block floor, R2' $m$ -free bottom-mode floor) stay open [E]                                                                                                                                       |
| v552_angle_instruments.py        | The four fixed-size angle instruments of T155/T157: the $12 \times 12$ complement-floor certificate $\min_{v \perp W} v^T L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^T)M^{1/2})$ (identity + one certified Rayleigh bound; never exceeds the exact complement bottom, agreement $\leq 1.01 \times 10^{-3}$ on this surface, both closed-form controls to $2.0 \times 10^{-11}$ ); the sine-block confinement $\ \gamma_H\ ^2 \leq (S/t)/\rho_{17} = 0.0153\text{--}0.0245$ from the certified floor and ladder alone ( $e_1(A)$ is 97.5–98.5% inside the first 16 parity sines by a per-instance theorem; $\kappa$ in place of $S$ : vacuous, 17.6–477); the Schur-entry identity $s = \mu_1^P t_1^T A^{-1} t_1 = (B^{-1})_{11}$ with $(B^{-1})_{LL} S_L = I - 1/s = 4.05\text{--}5.90$ is an $O(1)$ number carried by one entry of the fixed-size $16 \times 16$ Schur complement, while the inversion-free Cauchy–Schwarz ceiling misses by $250\text{--}7.3 \times 10^3$ ; and the adaptive Lipschitz deckel — an executed certificate $\inf f^{\text{arch}}/(4\sin^2(\theta/2)) \geq 0.25$ on 12/12 windows in 519–1201 evaluations, refusing false targets on 12/12; the no-go floor survives while the confinement goes vacuous ( $\times 52$ ) and $(S_L)_{11}$ explodes ( $\times 60$ ) — per instance on the 12 deepest in-cap windows; the two open angle terms after T157 stay open [E] |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v553_exact_form_identities.py         | The exact-form identities of T158/T159: the Thomson dual form $s = (B^{-1})_{11} = \max_x (2x_1 - x^T B x)$ with the direction executed plus the positive Cholesky ladder $g_K$ (all terms strictly positive, $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$ , tight to 1.064–1.152); the two-dimensionality $\text{span}\{t_1, A^{-1}L_P t_1\}$ attaining $s$ to $9.0 \times 10^{-13}$ ( $L_P t_1 = \mu_1^P t_1$ , KMS 1953); the gauge identity $\sum_d w_d = 0$ ( $1.1 \times 10^{-14}$ of $\sup  w $ ; the section annihilates constant lag vectors entrywise) with the Toeplitz-minus-Hankel signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$ ; the exact lag sum $x^T B_{LL} x = \sum_d c_d w_d$ ( $2.7 \times 10^{-16}$ absolute) with the closed scalars $w_0 = \sum x_k^2 / \mu_k^P$ and $2w_0 - w_1 = \ x\ ^2$ ; and the Z-matrix law — $B_{HH}^{\text{arch}}$ a raw symmetric Z-matrix on 12/12 with the closed sign-based floor $\min_k (B_{HH}^{\text{arch}} \mathbf{1})_k = 0.909\text{--}1.128 \geq 0.25$ (theorem direction verified; the atom block sign-free, the full block vacuous — R1'' stays open); no-go: Z law opposite, signature broken, $1/g_{16}$ explodes $\times 59.9$ tracking $1/s$ — per instance on the 12 deepest in-cap windows; both open terms after T159 stay open [E]                                                                                                                                                                                                                                                                                 |
| v554_sampling_harmonics_identities.py | Sampling/harmonics identities of phase 2: the sampling identity — the atom half of the pairing equals $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n}) \hat{w}(\log n)$ ( $\hat{w}$ the linear interpolant of the lag weights; exact to $5.6 \times 10^{-15}$ of the absolute atom scale), identically a finite combination of $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies; the closed moment laws $m_0 = 0$ , $m_1 = -[S_0^2 + 2 \sum_j P_j^2] \leq 0$ , $m_2 = -[2S_1 - (M-1)S_0]^2 \leq 0$ with U3 as a theorem with checked hypotheses, $\text{TV}(c^{\text{arch}}) \leq  c_0  + 2 \sup < 6$ closed and window-independent; the harmonic-frequency theorem $t_j(2\alpha) = 2\pi j$ exactly — the 32 frequencies are the Fourier harmonics of the log-window — with the closed parameter-free PNT prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ agreeing to 0.0007–0.097 of the trivial mass on the declared $\Lambda$ -battery $X = 10^3\text{--}3.2 \times 10^5$ ; the head split $c_d^{\text{arch}} = \Psi_d + D\hat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$ closed, $D$ -free and $m$ -free, exact at every lag, plus the scale-free Bernstein rate $\rho^* = (3 + \sqrt{5})/2 = 2.618034$ ; and the certified off-diagonal fraction bound $ \text{off}  \leq \frac{1}{4} Q^{\text{arch}} $ (0.171–0.212 on 12/12) with the refuted sign law wired as the negative control — per instance on small windows, finite $\Lambda$ -sums allowed, no RH statement; R-A'/R-B'/R-C'/R-D stay open [E] |
| v555_pareto_tv_identities.py          | Pareto/total-variation identities of phase 2: the exchange law $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}(x)/P(x))/\log X$ as an identity at all 612 grid points of the declared Fejér knob ( $5.1 \times 10^{-16}$ ), the crossing criterion $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16} \text{TV}$ exact at every point; the total-variation floor — $w_0 = \ a\ ^2$ , $\text{TV} \geq  w_0 $ by telescoping, $x_1 = 1$ forces $\mu_1^P \text{TV} \geq 1$ on all 708 trial vectors (measured 5.00–28.84), the closed floor $1/\mu_1^P = 1/(4 \sin^2(\pi/N)) \sim h^2$ , hence every sub- $\frac{1}{2}$ point pays $P > 2\kappa g_{16}/\mu_1^P$ — a flat-price sub- $\frac{1}{2}$ demand is impossible in the parity sector on this surface (closes R-C''' negatively), with the inadmissible-vector negative control; the chain-derived price axis $\sigma = 1 \leftrightarrow K=1$ rung, $\sigma = \infty \leftrightarrow K=16$ rung, the knob monotone in both coordinates; the closed Mellin cell moments $C_d(-(2k + \frac{1}{2}))$ vs quadrature and the boundary-free Abel step with the closed loss reason $32\pi/\alpha > 1$ ; and the Lerch/Frullani log-moment on three routes with the $2 \log 2$ peel and $\Psi_1 = -\log 2$ exact — per instance on small windows, the T163 $h$ -exponents sandbox fits not consumed, no RH statement; R-E/R-B'''/R-D stay open [E]                                                                                                                                                      |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v556_gauge_ppr_identities.py      | Gauge/ $P_{\text{pr}}$ identities of phase 2: the gauge theorem — $x_1 = 1$ fixes only the scale of $a$ , $Q$ and TV homogeneous of degree two, so $Q/\text{TV}$ (and $\delta_{\text{bnd}}$ ) is the same number in every sector ( $2.9 \times 10^{-11}$ on 2520 combinations; $\theta = \mu_1^P / (\mu_1^P + \kappa_s)$ exact; full space $\mu_0 = 0$ exactly, strictly worse); the $h^2$ conservation law floor $\times$ transfer $= 1/\mu_1^P$ identically at every shift ( $2.2 \times 10^{-16}$ ); the power-one spend $d \log r / d \log U = +1$ exactly ( $1.1 \times 10^{-14}$ ; the quadratic mutation reads 2); the $P_{\text{pr}}$ identity $P_{\text{pr}} = g_{16} R(\text{TV}/(t_1 v)^2) / \mu_1^P$ exact to $4.3 \times 10^{-16}$ on all 192 battery vectors with the predicate equivalence $R > 2\kappa \iff P_{\text{pr}} > 2\kappa g_{16} \text{tvf} / \mu_1^P$ exact — every crossing vector pays $P_{\text{pr}} > 2\kappa g_{16} / \mu_1^P = 15.4\text{--}142.8 > \Theta_{\text{TOL}} = 10$ on 12/12 (R-F closed negatively by certificate); and the Schur-cascade direction $s \geq g_{16} \geq g_1$ with $P_K = \hat{a}s \geq 1$ and the exact exponent closure of the lists — per instance on small windows, the one open quantifier $\inf_m g_{16}(m) > 0$ (the cascade gain) typed OPEN; no RH statement [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v557_cascade_vector_identities.py | Cascade/vector identities of phase 2: the cascade identities — the prefix property (one Cholesky, all sixteen rungs, $1.1 \times 10^{-12}$ on 192 comparisons), the minor form $1/g_K = \det G_{[1..K]} / (\mu_1^P \det G_{[2..K]})$ on the raw Gram block ( $2.7 \times 10^{-12}$ ; Schur 1917) and the regression form $g_K/g_1 = 1/(1 - R_K^2)$ ( $8.8 \times 10^{-13}$ ); the diagonal invariance of the gain under $B \rightarrow DBD$ ( $2.7 \times 10^{-12}$ on 144 diagonals — the gain belongs to the arithmetic Gram block alone); the exact $K=2$ identity — the closed vector $(1, -Q_{21}/Q_{22})$ attains $1/g_2 = Q_{11}(1 - r_{12}^2)$ exactly and $B_{11}g_2 = 1/(1 - r_{12}^2)$ identically ( $1.2 \times 10^{-12}$ ); the fast-null-vector is free at $K=2$ ; the pivot identity $u^T Q u = 1/g_K + \delta^T Q \delta$ for $\delta_1 = 0$ ( $2.0 \times 10^{-12}$ on 252 evaluations) with the exact entry threshold and the unification $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ — one inequality, not three; the trial theorem (every candidate a certified lower bound), the $L_P$ no-go (gain exactly 1) and the scramble type-change ( $B_{11}$ itself loses positivity on a majority of draws on 12/12 windows) — per instance on small windows, the one open scalar R1 (an $m$ -free upper bound on $1 - r_{12}^2$ ) typed OPEN; no RH statement [E]                                                                                                                                                                                                                                                                                                                                                                                                   |
| v558_bilinear_rank_identities.py  | Bilinear/rank identities of phase 2: the Lagrange/Wronskian anatomy — the lag-sum identity $\hat{A}_{ij} = t_i^T A_h t_j = \sum_d c_d W_d^{ij}$ ( $1.6 \times 10^{-15}$ ), Euclidean orthogonality $t_1 \cdot t_2 = 0$ ( $4.4 \times 10^{-16}$ ), the Wronskian telescope with closed edge ( $1.9 \times 10^{-17}$ ) and $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$ (Lagrange 1773); the exponent account $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$ ( $2.5 \times 10^{-13}$ ) with the Gershgorin certificate $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) +  \hat{a}_{12} $ (12/12, loss 1.19); the T169-TH7 det-collapse $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$ ( $8.7 \times 10^{-13}$ ) with $\nu_2 \leq 2 \det \hat{A} / (\hat{a}_{11} + \hat{a}_{22})$ tight to $< 2$ exactly; the exact bilinear von Mangoldt form $\hat{A} = B - S$ ( $1.7 \times 10^{-14}$ ), $\det \hat{A} = \det B - D(B, S) + \det S$ ( $10^{-10}$ ) and $\det S = \sum \sum \Lambda(n) \Lambda(m) K(n, m) / \sqrt{nm}$ , $K = \frac{1}{2} D(X_n, X_m)$ , through the explicit kernel ( $7.5 \times 10^{-16}$ ; Cauchy–Binet); the rank-3 theorem (polarisation matrix eigenvalues $\{-2, -1, +1\}$ , $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ : the form is $S_{11}S_{22} - S_{12}^2$ in three linear $\Lambda$ -sums) and the Type II collapse (Vaughan coefficient-exact, kernel rank $[2, 2, 2, 2]$ vs generic $[20, 15, 11, 9]$ ); the R4-free chain identity, the $L_P$ no-go and the scramble discriminator (rank invariant, value moves $\times 835$ median) — per instance on small windows, the one open object R1 (the collinearity rate) typed OPEN; no RH statement [E] |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v559_phase2_capstone.py          | Phase-2 capstone (T171): the sixteen-link reduction chain from the I5 floor to the R1 shape reproduced as one machine-checked pass on the declared small surface — 13 links theorem / 3 links certificate: the Schur two-block criterion (sharp at $0.5\lambda$ vs $1.02\lambda$ ), the Sobolev bound $b_k \leq C_S k$ , the $\Psi$ -collapse $\Psi = \sum_d c_d w_d = x^\top G x$ with $\sum_d w_d = 0$ , the Ritz ceiling ( $g_{32}/s = 0.952$ – $0.985$ from below), the sharp $12 \times 12$ floor, the $F(P, r)$ chain, the Thomson/ladder duality, the sampling identity, the Abel swap with the TV floor, the gauge/quantifier collapse $\Psi(x^*) = 1/g_{16}$ , the $P_{\text{pr}}$ identity, the cascade $1/(1 - R_{16}^2)$ , the $K = 2$ exactness, the R4-free det collapse (no positivity of $A_h$ consumed), the rank-3 theorem and the R1 shape $\sin \angle =  \det \hat{A} /( r_1  r_2 )$ ( $1.2 \times 10^{-4}$ – $2.5 \times 10^{-3}$ per window, rate and quantifier OPEN); the eight-route no-go battery failing as classified (size budget, symbol infimum, sector invariance, perturbation scale, band-local, sieve operator norm, scramble $\times 679$ median with rank-3 untouched, $L_P$ gain = 1); and the precision ledger (needed $2.75 \times 10^{-4}$ , finer than the RH yardstick by $215\times$ and the best unconditional by $13\times$ on this surface) — zero new uniform statements, R1 the one open object, typed OPEN; no RH statement [E]                                                                                                                                                                     |
| v560_frame_deficit_identities.py | Frame/deficit identities of phase 2: the identity transfer — four chain links (KMS/Schur pair $3.2 \times 10^{-16}$ , lag-sum identity $1.4 \times 10^{-16}$ , R4-free det collapse at 0.020 of its conditioned bar, additive split $7.6 \times 10^{-16}$ ) re-derived exactly on a gap-blind frame-B and a non-prime-power instance, with the sieve-horizon discriminator (truncated comb $\Rightarrow$ indefinite, $\lambda_{\min} = -9.6 \times 10^4$ ; complete comb $\Rightarrow$ PD on 12/12; the identity links hold either way); the $q = 1 - s$ identity ( $hD = \alpha$ to $8.9 \times 10^{-16}$ , OLS slopes obey $q = 1 - s$ to $2.9 \times 10^{-14}$ ) with the gauge-degree account coming out as integers — $G2 = t/\lambda_{\max}(B_{LL})$ and $1 - r_{12}^2$ both (0, 0), $G3 = \mu_1^P G2$ identically (the offset-2 explained), $(D/\alpha)^3 h^3 = 1$ a tautology; the blindness theorems — the amplitude gauge over twelve decades, the exact cancellation of the entire $\mu_1^P = h^{-2}$ channel on both sides, the diagonal-group asymmetry (delivery blind, demand keeps the KMS shape), and the additive limit (neither the arch nor the comb term alone carries the Schur floor on a single cell: no factorisation exists, P6 is not a theorem); the unconditional KMS shape band $\leq 2 \log((1 + \varepsilon)/(1 - \varepsilon))$ , $\varepsilon = (K\pi/(2h + 1))^2/6$ ; and the placement discriminator (value-multiset scramble destroys the floor on 24/24 draws) — per instance on a small declared surface, the frame-free deficit $+0.1111 \pm 0.0222$ a sandbox measurement, R1 typed OPEN; no RH statement [E] |
| v561_cp_channel_identities.py    | CP channel/invariant identities (T177): the hexagonal channel identity — on the CM unit $\rho = e^{i\pi/3}$ the $E_8$ split $248 = 120 + 128$ has an exact fiber realisation, $\sum_k \rho^{d_k} = 4\rho$ ( $\arg = \pi/3$ exactly, sheet-EVEN because all degrees are even) and $\sum_k \rho^{m_k} = 4$ ( $\arg = 0$ exactly, sheet-ODD because all exponents are odd) — the phase-channel angle is sheet-blind; $\arg(\prod c) = 0$ exactly on both TFPT magnitude sectors and the $\mu_3$ branch group is the kernel because its order divides $N_{\text{fam}} = 3$ (the REDTEAM.D.01 residual is a <i>fiber</i> datum, not a magnitude datum); Tier-1 invariance is EMPTY by the blindness theorem (both kernels commute with conjugation, so every $\pi/3$ -carrier is $\kappa$ -odd); the frame removal — $\text{Im det}(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3)$ is homogeneous of degree +1 in the frame normalisation (frame data), while $\text{Im}(\rho \det)/\det = \text{Im } \rho = \sin(\pi/3)$ exactly (the invariant core is the fiber datum, the sign a declared orientation); and the two inequivalent Tier-2 characters $(-1, +1)$ vs $(-1, -1)$ carrying the same $\pi/3$ (the DEGENERATE verdict as algebra, neither frame-coupled), anchored by Jarlskog 1985 ( $J = 3.327 \times 10^{-5} = \text{v88}$ , passing rephasing/CP-oddness, failing only the $\pi/3$ -carrier test) — four ledger readings narrowed, no marker moves, no CP derivation claimed; R1/R2 typed OPEN [E]                                                                                                                                                |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v562_dense_limit_identities.py | Dense-limit identities of the phase-2 endgame: R1 closed — the Feynman–Hellmann identity $d \log R/d\delta = v^\top d\tilde{A}v/\lambda_{\min} - w^\top d\tilde{A}w/\lambda_{\max} - d\text{DEL}/\text{DEL}$ with the exact lag derivative $dc/d\delta$ (piecewise linear at $7.0 \times 10^{-12}$ of the increment) reproduces the full rebuild on 6/6 instances at the minimum of a declared step ladder ( $2.3 \times 10^{-7}$ at $\delta = 10^{-6}$ ) with the U-shape exhibited — T175’s measured intervention slope retired as a separate claim, the near-degeneracy channel $d\text{DEL}/\text{DEL}$ carrying the largest share (0.54–0.98); the phase pole — PD survives only 2% of the period, $\delta_{\text{PD}} \cdot  d \log R/d\delta  = 7.2$ median, the two lowest modes already within 0.02–0.17 at $\delta = 0$ (the $\sim 300\times$ harmonic anomaly retired); sieve consistency at construction level — lag vectors bit for bit on complete-comb anchors, the bin deficit identical, the deepest truncated anchor moving as it must; and the curve estimator as a deterministic procedure with the LADDER CAVEAT wired as a check (fine vs coarse ladder differ by 0.036: any quoted bin deficit must quote its rung ladder) — per instance on a small declared surface, T176’s SITS-AT-ZERO (pooled $+0.0496 \pm 0.0372$ over the two new bins, plateau window $3.6\times$ narrower) a sandbox measurement, the limit typed OPEN; no RH statement [E]                                                                                                                                                                                                           |
| v563_paper2_readouts.py        | Paper-II readouts recomputed (the check closure of the classification paper): the declared T170 reference window bit for bit ( $h = 540$ , $X = 2.465 \times 10^4 = 157^2$ ) with the four quoted numbers recomputed ( $\det B = 6.18$ , $-D(B, S) = -198.8$ , $\det S = 192.6$ , $\det \hat{A} = 4.22 \times 10^{-3}$ — a 4.7-order cancellation), split and polarisation at round-off; the $\ell^1$ -to-signed ratio $40.6\times$ on the declared subsample (Cauchy–Binet at round-off); the tightness envelope $\mu_1^P \text{TV} \in [5.00, 23.20]$ with both ends attained and the floor on all 1122 trial vectors; the density-curve endpoints from the v562 construction; and the rung-ladder shift exhibited as ANCHOR SELECTION (fine vs coarse ladders select different anchor sets in the same bin, 13 vs 12; removing the extra anchors reproduces the whole shift, curvature transport moves it by only $-0.0111$ , a wrong curvature is $11.9\times$ louder) — pure readout/identity checks, no new claim, no marker move, every measured number typed MEASURED with its rung ladder [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v564_cp_fiber_pin.py           | The frame door of the CP mirror, decided (T178): the Euler factorisation — the fiber acts by a scalar on the $n$ -row, so $P(\rho n) = \rho^m P(n)$ for fully generic homogeneous blocks and every scale-free functional is $\sum_k A_k \text{Im}(\rho^k)$ with real degree-0 frame ratios $A_k$ ; the natural (metric-free) class is FIBER-ONLY by theorem (the bracket enumeration on $\{\mathbf{1}, d, n\}$ has one generator: six permutations $= \pm\Delta$ , the other 21 vanish identically in a generic $n$ ; Weyl 1946 FFT the cited class boundary) — T177’s R1 obstruction is now a theorem; the metric door is NON-CANONICAL ( $\Delta/\langle d, n \rangle$ , $\Delta/\langle \mathbf{1}, n \rangle$ at $(-1, -1, +1)$ vs $\Delta^2/\langle n, n \rangle$ at $(-1, -1, -1)$ ; the continuum $q_t = t n_1^2 + n_2^2 + n_3^2$ gives values $441 \sin(\pi/3)/(25t + 117)$ — no canonical frame factor, anti-numerology wired as a check); the SINGLE-SINE LEMMA — $\{ \text{Im} \rho_n^k \}$ has one nonzero element iff $n \in \{1, 2, 3, 4, 6\}$ (crystallographic, $\varphi(n) \leq 2$ ), by $\sin(4\pi/n) - \sin(2\pi/n) = 2 \sin(\pi/n) \cos(3\pi/n)$ : the two compiler clocks $\mu_4$ ( $j = 1728$ ) and $\mu_6$ ( $j = 0$ ) are the two maximal single-sine fibers, so T177’s DEGENERATE verdict was FORCED; and the ANCHOR SELECTION — Jarlskog carries the full tuple $(-1, -1, -1)$ (certified at 40 dps), matching the sheet-odd orientation class and not the sheet-even channel class: the unique anchor-matching canonical carrier is the fiber-pure $\text{Im} \rho = \sin(\pi/3)$ — R1 decided, R2 selected conditional on the anchor, no marker moves [E] |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v565_woit_beta3_pt_duality.py         | The WOIT- $\beta_3$ milestone executed: the $\text{PT} \leftrightarrow \text{PT}^*$ duality typed against $\sigma_{\text{std}}$ — clock compatibility forces the two-block Hermitian class ( $16 \rightarrow 8$ parameters); the $\rho_{\text{tw}}$ pairing splits it into an EUCLIDEAN branch ( $\sigma_{\text{std}}$ 's home; seam norm $a( z ^2 + 1)$ , no OS cut ever) and a MINKOWSKI branch whose null locus meets the seam line EXACTLY in the seam circle (the OS cut forced) with total signature $(2, 2)$ by the mirror theorem ( $\Phi_1 = -\overline{\Phi}_i$ , charpoly mirrored) — Woit's Minkowski signature a theorem, not an input; the kill branch is EMPTY BY ALGEBRA (every compatible duality inverts clock and deck — family D; family A would need $\rho_4^2 = \text{scalar}$ ); every seam-preserving Minkowski member induces the SAME conjugation $z \rightarrow -\bar{z}$ ( $\mu = -1$ , fixed points $\pm i$ ), perpendicular to $\sigma_{\text{std}}$ 's $z \rightarrow +\bar{z}$ , and under the declared RP-side placement it IS $\theta_\perp$ — the v524 $\Theta_{\text{phys}}$ carrier (regressions re-verified: Gram PD, anti-unitary, $\Theta^2 = +1$ per sector, time reversal, torsor closure onto the deck); verdict SUCCESS, the contract stays <b>[O]</b> , $\gamma$ is next <b>[E][C][O]</b> |
| v566_parabolic_anchor_selfcode.py     | The parabolic anchor self-code: the sheet-diamond directions $U = Q \text{diag}(1, 0, 0)$ , $V = Q \text{diag}(0, 1, 1)$ generate the FULL 2 1 parabolic algebra $\mathcal{A}$ (dim 7, word determinant $-81$ , multiplicatively closed); the anchor $a = (1, 1, 2)$ is the spectrum of $H = I + V(V - I)/2$ and $\chi_H = x^3 - 4x^2 + 5x - 2$ carries the DOUBLE code $(4, 5, 2) = ( \mu_4 , g_{\text{car}},  \mathbb{Z}_2 ) = (\text{flag}, \text{Levi}, \text{radical})$ ; the uniqueness theorem $(x - m)(x^2 - mx + 1)$ forces $m = 2$ — $a = (1, 1, 2)$ is the only anchor this family can code; the saturated integral order has index $[P_{\mathbb{Z}} : \mathbb{Z}\langle U, V \rangle] = 3^4 = 81$ = the v218 determinant-staircase sum (cokernel $(\mathbb{Z}/3)^4$ ; mark-local mechanism <b>[O]</b> ); the basic algebra is the $A_2$ path algebra with $\dim \text{Ext}^1 = 1$ (two module classes: the alignment bit gains a canonical carrier, selection stays with RP); $p_n(a) = 1 + \text{tr}(V^n)$ (AnchorLadder re-encoded, 240 and rank 8 as trace readouts); the CR = $4/3$ preregistered test for CONTRACT.F.01; the Lorentz/pencil form of the Paper-II rank-3 classification — NO P2 reduction (the 'No Carrier Reconstruction' audit is the named <b>[O]</b> door) <b>[E][C][O]</b>                        |
| v567_carrier_free_q_reconstruction.py | The No-Carrier-Reconstruction audit (Stage A): with the v11 carrier budgets FORBIDDEN (rows $(4, 5, 6)$ , columns $(9, 5, 1)$ , $\chi_Q$ with its 5), the carrier-free inputs — sheet involution, winding normalisation $Q(I + \Sigma) = 2N_{\text{fam}}W$ , $A_3$ exponent spectrum $\{1, 2, 3\}$ , monodromy budget $Q_-^2 = N_{\text{fam}}P$ , integrality, the seam-line demand — admit exactly $52 \rightarrow 28$ (parabolic) reconstructions, and ALL 28 code the SAME anchor ( $\chi_{H'} = x^3 - 4x^2 + 5x - 2$ , spectrum $(1, 1, 2) = a$ ): <b>the anchor self-code is carrier-free</b> ; the single family invariant (saturation index $N_{\text{fam}}^4 = 81$ ) then pins $Q$ to a TWO-element residual family ( $q_{32} \in \{2 =  \mathbb{Z}_2 , 0\}$ , difference $2E_{32}$ ) on which every classical carrier-free certificate coincides — the one separating certificate is the forbidden column budget $(9, 5, 1)$ : one binary sheet-coupling choice remains (typed resonance with the split/nonsplit alignment bit). Winding ablation $28 \rightarrow 1376$ ; bound-stable; verdict RECONSTRUCTS-TO-FAMILY; the audit relocates to GATE.QGEO.01 and the binary selector — no P2 reduction, markers untouched <b>[E][C][O]</b>                                                                                     |
| v568_bit_selector_ladder.py           | The reconstruction bit located and given its canonical witness: the two 81-twins generate DIFFERENT integral orders (equal index, different lattices — the bit is integrally visible); the dominant mode of the sheet direction is $v(c) = (1, 2, 2c)$ identically, and the LADDER SELF-CONSISTENCY — the dominant mode reproduces its own eigenvalue as the geometric ladder $(2^0, 2^1, 2^2) = (1, 2, 4)$ — selects $c = 2$ (the compiler $Q$ ) UNIQUELY on all 28 members of the carrier-free class, with the witness anchored in prior structure (AnchorLadder steps $2^n$ , $2 =  \mathbb{Z}_2 $ , the trace code); the split geometry sits at $q_{32} = 1$ and is EXCLUDED (algebra dim 5; the $A_2$ Ext arrow is nonzero on the whole class — the bit toggles WHICH nonsplit order, not split/nonsplit); Priority-2 candidates: the $\mu_4$ -grading fails, the $\mathbb{F}_3[V]$ -action is computed — charpoly $x(x - 1)^2(x + 1) \bmod 3$ , NOT the regular $\mu_4$ pattern; the 81 stays a fingerprint — the physical selection stays open (the RP-selection slot), no P2 reduction, markers untouched <b>[E][C][O]</b>                                                                                                                                                                                                     |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v569_lambda_pencil_onemode.py | The relative pencil $Z = B^{-1}S$ on the declared Paper-II surface: the pencil identities are machine-tight on all 70 frame-A windows (worst residual $2 \times 10^{-14}$ ); the LOCKING mode is $\lambda_2$ hugging 1 (median 1.000021) while the spectator $\lambda_1$ stays uniformly separated — $\min  1 - \lambda_1  = 3.53$ , $\lambda_1 \in [4.53, 180.64]$ , always ABOVE 1: the two-dimensional cancellation reduces to the SINGLE scalar $1 - \lambda_2$ on the whole reachable surface (verdict ONE-MODE); the $h = 540$ reference window reads $1 - \lambda_2 = -2.27 \times 10^{-5}$ (reproducing $\det \hat{A} = 4.22 \times 10^{-3}$ ), and the position scramble explodes $ 1 - \lambda_2 $ by $\times 1.9 \times 10^6$ while the identities keep holding — the locking is arithmetic placement; declared surface only, no uniformity/rate/RH, Problem 7.1 typed not bounded [E]     |
| v570_separation_floor.py      | The v569 open question reduced and certified on the declared surface: two one-line theorems ( $P = \det S / \det B > 0$ , $T = \text{tr } Z > 0 \Rightarrow \lambda_1 \geq \sqrt{P}$ ; $P \leq 0 \Rightarrow \lambda_1 \geq T$ ) turn the spectator separation into a closed per-window floor — $\min F = 2.130 \geq 1.05$ on ALL 70 windows: the separation follows from $\det S / \det B$ and $\text{tr } Z$ alone; the dominance $P \in [4.5, 120.6]$ grows with depth ( $h^{0.90}$ , surface ladder), $1 - T + P = \det(B - S) / \det B$ links the two scalars exactly, and the cheap witness route is honestly killed (negative pair mass up to $13.7 \times \det S$ ; best single pair $0.13\text{--}0.34 \times \det B$ ) — the $h$ -uniform question is typed as pair-level size-vs-cancellation; scramble flips $\text{tr } Z$ to $-78.3$ ; declared surface only, Problem 7.1 untouched [E] |
| v571_cr_discrete_corners.py   | The preregistered cross-ratio check, discrete half executed: the corners $J, K, C, F = M(1, t)$ ( $t = -2..1$ , $\det s(2, 4, 14, 32)$ ) read $(26, 35, 44, 53)$ under the declared Koide functional $\ell(M) = a^T M \mathbf{1}$ — CR = $4/3$ exactly, and $y_F = 53$ is Möbius-forced by the three upstream corners (fit-free); $\ell$ is affine on the path (step $9 = N_{\text{fam}}^2$ ; every linear functional passes — stated honestly); negative controls: determinants $28/25$ , dominant eigenvalue $1.3222$ ; the continuous solver states stay open per CONTRACT.F.01 [E][C]                                                                                                                                                                                                                                                                                                             |
| v572_rp_bit_form.py           | The RP form on the 81-twins: spectrally blind (identical inertia for every declared state; never PSD — no kinematic positivity selection, honest negative) but integrally ANCHOR-VISIBLE — the Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor $9 = N_{\text{fam}}^2$ ; ladder state identical, anonymous states blind); trace discriminant $-81^2$ on both; blindness robust under the compiler $V$ -flow, but the anchor-Hamiltonian GIBBS PENCIL separates: the compiler twin is Gibbs-critical at $s^* = (8 - 4\sqrt{19})/15$ ( $\det$ constant $81^2$ ), the alt never — a rank-drop marker whose critical mode is the seam-row, sheet-flow-fixed direction (the $A_2$ -Ext carrier); the positivity selection stays open [E]                                                                                                                            |
| v573_pair_band_structure.py   | The pair-band anatomy of the dominance: every near band ( $< 16$ lag cells) and the diagonal drag NEGATIVE on the declared subset; the whole positive excess lives beyond 16 cells with stable net ( $\text{far}/\det S \in [1.55, 1.91]$ , gross growing $4.1 \rightarrow 13.0\times$ ); the scramble explodes $\det S \times 49$ and collapses the far share — long-range coherence, arithmetic placement; sharper: scale-locked ( $\geq h/2$ ), weight-concentrated (top decile nets $0.85\text{--}1.00 \times \det S$ ), phase-flat — the target is heavy-pair window-scale coherence [E]                                                                                                                                                                                                                                                                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v574_stageb_winding.py        | Stage B, first narrowing: the involution TYPE is forced (rank theorem — a 2-dim-positive involution cannot carry the binary trace code), ( $E$ ) bounds $N \leq 3$ , the ladder self-consistency selects one member per $N$ , and $\text{sat} = N^4$ holds ONLY at $N = 3$ ( $32 \neq 1$ , $512 \neq 16$ , $81 = 3^4$ ): the double self-consistency pins $(N, Q) = (3, \text{compiler } Q)$ — $N_{\text{fam}} = 3$ relocates from declared to consistency-forced; spectrum ablation: the surviving family is exactly $\{1, 2, N\}$ with the compiler shape — the self-consistency forces the sheet block, ( $E$ ) contributes $N = 3$ ; redundancy: ( $M$ ) and ( $L$ ) fall relative to the self-code demands (class 1274, compiler $Q$ still unique) — the maximal ablation drops ( $E$ ) too — winding column + integrality + self-code pin the compiler alone; declared: involution existence, the winding UNIFORMITY (without it a four-column family survives, exactly one uniform — the compiler), $N_{\text{fam}} = 3$ (E8-glue); GATE.QGEO.01 narrowed by five items, not closed, no P2 reduction [E][C] |
| v575_cr_continuous_uniflow.py | The continuous half of the cross-ratio check, executed: the native exports $\{(2/3)^6, (2/3)^6, 1, -7\}$ are forced degenerate by the single-flow theorem (v425), and a repeated quadruple gives $\text{CR} \in \{0, 1, \infty\}$ in every assignment — never $4/3$ : the continuous half closes onto the single-flow reduction; the anchors stay [O] walls, the discrete half (v571) carries the content [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v576_cheb_loewner_edge.py     | The Chebyshev-Loewner edge structure: the polarized cross-weight formula exact (symbolic $N \leq 11 + 10^{-11}$ surface); the edge read $w_{N-s}^{(ij)} = -(2/N) \sin \omega_i \sin \omega_j U_{s-1}[x_i, x_j]$ is a scaled Loewner matrix (rank $\leq s-1$ ; $s=2$ rank one by theorem); the leading profile's Lorentz image is the null triple $(5, -3, 4)$ — Euclid's parametrisation, typed compression with $(g_{\text{car}}, N_{\text{fam}},  \mu_4 )$ ; the edge defect $-(3\pi^4/175)(s^2-4)(11s^2+6)N^{-4}$ exact (series); the pair polynomial $P(s, z)$ with $P(2, 3) = 0$ and sign transition $s/z \approx 0.503/1.987$ — the analytic prototype of the v573 pattern; macro kernels verified; edge regime closed, macro/transition open, Problem 7.1 untouched [E][C]                                                                                                                                                                                                                                                                                                                                  |
| v577_nullray_census.py        | The null-ray locking census: the locking mode aligns with $(2, -1)$ at $23.5\text{--}33.8^\circ$ on all 70 windows (below random), declines slowly ( $h^{-0.10}$ — the clean success bar honestly not met), and the scramble destroys the alignment ( $+21^\circ$ ): a real partial carrier; the macro regime bends the direction — follow-up is $K_\infty$ , not more angle fits [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v578_native_jets.py           | The jet contract executed natively: the seam flow has $\{q, t\} = -\Delta^2/2$ exactly (Moebius in its own clock $u = e^{\Delta t}$ ), the 1-loop QCD flow has Schwarzian 0 in its own RG time — no shared wall-time jet invariant; each instance projective in its own clock, the obstruction is the clock; external anchor-level jets remain the future test [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v579_macro_kernel_signs.py    | The two-scale kernel signs: sign $K_\infty(\sigma, \tau)$ matches the real surface on every resolved cell (332/332 at $h = 300$ ); the diagonal $\det W^\infty(\sigma)$ is negative on $\sim 72\%$ of $(0, 1)$ — same-scale pairs couple negative analytically; the geometric half of Regime B closes, the $\Lambda$ -weighted arithmetic stays open [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v580_occupation_map.py        | The arithmetic occupation map: $\det S = (1/2) \sum c_{at}(d_1) c_{at}(d_2) D(W(d_1), W(d_2))$ exact (lag-side polarization identity); every significant cell sign-matches $K_\infty$ on every tested window — the primes populate the geometric sign map, the open Regime-B content is the amounts; mid-diagonal negative, cross-scale positive; scramble control clean [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v581_locking_transport.py     | The multilevel transport lemma, measured and killed: 40 doubling pairs, drift $0.3\text{--}6.7^\circ$ per octave (median $2.35^\circ$ ) but GROWING ( $h^{+0.36}$ vs the bar $h^{-1-\delta}$ ) — not summable, the review's own kill fires; any future transport must carry the window-dependent macro direction [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v582_density_dominance.py    | The dominance is DENSITY-DRIVEN: the smoothed comb reproduces $\det S$ to 99.8–99.9% (2% window) on every regular window through the exact lag-side identity; the scrambled control captures only 70% — the real comb is anomalously smooth at these scales; the $h$ -uniform dominance question relocates to PNT-level regularity of the smooth density functional (closed-form bound = the named next step); boundary window $h = 1292$ named [E][C]        |
| v583_pnt_model.py            | The smooth density functional in closed, PRIME-FREE form: the two-term classical law $\text{mass}(u) = 4e^{u/2} - 2(\zeta'/\zeta)(1/2)$ (pole term + $s = 1/2$ constant, no zeros) reproduces $\det S$ on all 69 regular windows parameter-free to 1.016–1.143; the anomalous $h = 1292$ sign flip is genuine prime-fluctuation content; the $h$ -uniform dominance splits into a prime-free two-integral inequality plus a zero-oscillation bound [E][C]     |
| v584_ext_source_response.py  | The interacting follow-up of v572: rank-one Ext-source coupling on the Gibbs pencil separates the 81-twins directionally inside the unit coupling box – exact rational flip thresholds whose numerators are the v572 pencil factors; positive coupling $J \leq 15/16 < 1$ heals exactly the compiler twin, the alt twin is rigid; the response pole sits on the golden polynomial $s^2 - s - 1$ ; a response-level selection, not a positivity theorem [E][C] |
| v585_pnt_locking_split.py    | The two-layer split of the locking cancellation: the v583 prime-free model misses the Paper-II defect by 2–4 orders (median $1350\times$ , growing $h^{1.08}$ ) while reproducing $\det S$ – the dominance is density, the locking depth is the primes; layers $h^{-1.43}$ (density, positive) $\times h^{-1.08}$ (arithmetic, with deep-window sign overshoots at $h = 1219, 1445$ ) $= h^{-2.51}$ real; the 7.1 budget splits accordingly [E][C]            |
| v586_pnt_lock_direction.py   | The locking direction is density-determined: the real locking eigenvector agrees with the prime-free model to $0.03^\circ$ (median; max $0.33^\circ$ , decaying $h^{-1.16}$ ) while the depth differs by $1350\times$ – the primes are a pure depth amplifier along a density-fixed direction; the deterministic direction certifies the real defect within a factor $\leq 2.9$ ; resolves the v577 comparator artifact; scramble must-break fires [E][C]     |
| v587_w_closed_form.py        | The diagonal companion of the v576 cross formula: the corpus diagonal lag weight is the single closed expression $W_{kk}(d) = (2/N)[(N-1-d)\cos(w_k d) + \sin(w_k(d+1))/\sin(w_k)]$ (machine-exact); the exact cell-sum reproduces the prime-free model's entries and determinant to 0.03% – the S-side of the deterministic layer is closed-form; the v579 kernel is its edge-anchored version identically [E]                                               |
| v588_closed_delta.py         | The deterministic layer of Problem 7.1 in closed form: one exact function $G_{ij}(\beta) = \sum_d e^{-\beta d} W_{ij}(d)$ carries both sides – the pole exponent ( $S$ ) against the trivial-zero ladder ( $B$ ); closed $B$ to $2 \times 10^{-5}$ , the closed defect matches the density layer to 0.2% on all 70 windows with census decay $h^{-1.40}$ ; the remaining step for a theorem is elementary asymptotics [E][C]                                  |
| v589_zero_comb.py            | The arithmetic layer is the zeta-zero comb: the atom table's residual oscillation matches the explicit-formula comb (correlation 0.82, slope 1.28 with 200 damped zeros; pointwise at low $u$ ); the comb recovers a factor 5.7 of the lock depth at the smallest window ( $176 \rightarrow 31$ ), deep windows out of reach – the zero-oscillation bound is a quantified explicit-formula target; no bound claimed [E][C]                                    |
| v590_involution_existence.py | The sheet-involution existence of GATE.QGEO.01: the integral $V$ -commuting involutions are exactly the spectral sign group $(\mathbb{Z}/2)^3$ on both twins (rigidity); the v574 type theorem, seam reversal and zero-mode fixing single out $S_0$ uniquely with exact $\Sigma$ -signature – existence is unconditional; the two demands' continuum derivation stays the named open [E][C]                                                                   |
| v591_pole_rank_one.py        | The pole term is exactly rank-one: $S^{\text{pole}} = -32\pi^2 a e^a g g^T$ with $g_k = k/(a^2 + 4\pi^2 k^2)$ , $\det \equiv 0$ ; the locking direction obeys $v_2/v_1 = -(a^2 + 16\pi^2)/(2(a^2 + 4\pi^2))$ with limit exactly $-1/2$ (the null ray, deriving the v577 conjecture) and matches the measured directions to 0.04–1.4% [E][C]                                                                                                                   |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v592_continuum_det_<br>law.py  | The continuum determinant law: $\det S(a)$ closed with the $e^{2a}$ coefficient identically zero and exact leading law $640\pi^2 e^a/a^3$ ; the real dominance tracks the closed formula at correlation 0.9985 (the closed-form origin of the v570 growth); finite-N gap typed; entry-level arithmetic certificate with margin $47.6\times$ conditional on the measured oscillation sup [E][C]                                                                                                                       |
| v593_cutoff_<br>completion.py  | The cutoff completes the closed entries: with the $\sigma_0$ cutoff the closed forms match the model entries to 0.7–1% and the determinant to 0.2–1% (v592's continuum-without-cutoff captured only 0.19–0.50); the density layer is closed end-to-end with an $O(1/N^2)$ residual [E][C]                                                                                                                                                                                                                            |
| v594_unconditional_<br>cert.py | The entry-level arithmetic layer is bounded unconditionally: partial summation plus the published bound $ \psi(x) - x  < 0.94\sqrt{x}$ (Büthe 2018) give $\sup  \text{Osc}  \leq 17.4$ on the declared surface; the certificate holds with margin $565\times$ – the first unconditional statement of the program; the determinant level stays open [E][C]                                                                                                                                                            |
| v595_mapping_<br>completion.py | The residual bookkeeping executed: the v593 residual was the $\sigma$ mapping – with the exact dictionary ( $a_{\text{eff}} = ND/2$ , limits $[u_0/(ND), 2\alpha/(ND)]$ ) the closed forms match entries and determinant to 1.00000–1.00017; kernel piece $9 \times 10^{-6}$ ; honest amendment of the v593 attribution; the density layer closed at identity level; det law formalized in Lean [E][C]                                                                                                               |
| v596_lock_<br>projection.py    | The 1D reduction of the arithmetic layer: along the closed v591 direction the zero functional cancels the closed density value to median $7.7 \times 10^{-4}$ (decay $h^{-1.01}$ ); the coupling spectrum is collective (max share 5.9%, $N_{50} = 56$ ); scramble breaks by six orders; the remaining arithmetic statement is an equidistribution-type theorem for one explicit family of identities [E][C]                                                                                                         |
| v597_mu4_cover_<br>model.py    | Strategy I/II first step: the $\mu_3$ -cover $y^3 = x^4 - 1$ carries four compiler fingerprints – rank-3 Eisenstein module, projective $D_4$ via the deck ( $r^4 = \omega$ ), cusp-class group $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of Lorentz signature (1, 2) matching Chevalley–Weil and Paper II; the $Q/V$ dictionary stays the named open [E][C]                                                                                                                            |
| v598_cover_<br>dictionary.py   | The cover dictionary, round 2: three compiler objects located exactly – $U \sim G_2$ , $Q_- \sim G_1 - G_3$ (the sheet coupling as the diamond-diagonal Gram difference), the binary ladder $\sim 1 + G_1 + G_3$ ; opposite-twist products nilpotent; the grading half absent from 7296 scanned canonical combinations (kill not triggered; the real structure is the named suspect) [E][C]                                                                                                                          |
| v599_real_<br>structure.py     | The real structure on the $\mu_3$ -cover: the anti-holomorphic $D_4$ reflection constructed exactly ( $R$ unique up to scale, $R^2 = +1$ , four twist intertwinings); the grading functor closes the three v598 dictionary gaps at char level ( $V, H$ with the anchor spectrum (1, 1, 2), $Q_+$ ); the $\Gamma$ -decomposition carries 5, 9, 27; the joint embedding stays open (quantified negative); the invariant form has $\det J = 8$ and Lorentz signature (1, 2) [E][C]                                      |
| v600_joint_<br>embedding.py    | The joint embedding: the compiler pair $(U, V)$ realized on the $\mu_3$ -cover as a $\Gamma$ -even degree-2 pair — $\mathbb{Z}[\omega]$ -integral, seven-word rank 7, simultaneous conjugation to the compiler pair solved exactly, integer restrictions on the saturated $\text{Fix}(R)$ lattice, and saturation index $81 = 3^4$ with the exact v566 divisor pattern (1, 1, 1, 3, 3, 3, 3); canonicity typed open [E][C]                                                                                           |
| v601_equivariant_<br>dual.py   | The canonical equivariant realization: the J-adapted cusp projectors rotate exactly under $D_4$ ; the pair family $U_k = 3GJ_k$ , $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ is $\mathbb{Z}[\omega]$ -integral, compiler-exact and equivariant by construction (mark choice = winding choice); it realizes the DUAL parabolic exactly (conjugate to the transposed pair; the direct intertwiner is singular), and the v590 Klein four-group lives integrally on the curve; the Weil-twisted RP comparison stays open [E][C] |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v602_duality_forms.py            | The duality-and-forms round: the J-adjoint exchanges the two parabolic types inside one equivariant structure; the dictionary completes in the J-projector language (the corpus anchor formula $H = 1 + V(V - 1)/2$ holds verbatim); Weil positivity holds exactly ( $JC$ positive definite, the Riemann bilinear relations on the $\omega$ -sheet); the canonical bilinear form $B = M^\dagger J$ is symmetric with $\det B = 8$ ; the corpus-transpose and mark-local questions typed open; Lean certificate FORM.QGEO.04 [E][C]                           |
| v603_seam_marks.py               | The seam-and-marks round: the seam line is a REAL line of the cover (unique common eigenline, R-fixed, $S_0$ -reversed — the v590 seam-reversal demand geometric); the infinity cusp separates in the unreduced Burau module; the first per-mark $\mathbb{Z}/3$ witness lands ( $E_1$ , class order 3 in the word-order cokernel); the corpus-transpose translator negatives quantified [E][C]                                                                                                                                                               |
| v604_equivariant_order.py        | The equivariant order and the infinity bridge: $R$ IS an anti-isometry ( $M^\dagger JM = J^T$ exact; correction of a diary side remark), $J$ restricts to the integer form $J_{\text{fix}}$ with $\det = 72$ ; the four mark frames generate a FULL order of index $27 = 3^3$ in $M_3(\mathbb{Z}[\omega])$ ; the 3-torsion localizes at infinity ( $\text{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$ , finite cusps torsion-free): 81 at the marks, 27 at infinity; translator negatives extended (the $\bar{\omega}$ -sheet named) [E][C] |
| v605_translator.py               | The translator round: the corpus RP anti-automorphism IS the J-adjoint twisted by an explicit inner element $c$ ( $\varphi(A) = cAc^{-1}$ exact, intertwiner space 1-dim, 14c integral over $\mathbb{Z}[\omega]$ ); the seam is a NONZERO pure 3-torsion class at infinity (torsion coords $(2, 2, 0)$ ); the 81 survives frame-joining as the saturation defect (chain indices $3^5, 3^4$ ); the $J_{\text{fix}}$ duality certified with a Lean addendum [E][C]                                                                                             |
| v606_mode_separation.py          | The mode-separation round: all three $V$ -eigenlines lift to PURE 3-torsion classes at infinity with PAIRWISE DISTINCT classes (the selection carrier the real structure could not provide; the canonical '+1' rule stays open); and the cubic field of the translator element has discriminant $-783 = -3^3 \cdot 29$ , Galois $S_3$ — no cyclotomic identity for $c$ (honest negative) [E][C]                                                                                                                                                              |
| v607_selection_rule.py           | The selection rule: the infinity torsion is a genuine $\mathbb{F}_3$ space, the deck rotation acts with char $(x + 1)(x^2 + 1) \bmod 3$ ( $x^2 + 1$ irreducible) — EXACTLY ONE projective eigenline, and it IS the top mode; the three modes carry mutually exclusive canonical labels (seam = $R$ -real, top = deck-eigen, zero = unlabeled) and $S_0$ flips exactly the labeled pair: both v590 demands expressed through cover structure [E][C]                                                                                                           |
| v608_gamma_free_slice.py         | The free gamma slice: mirror modes are STRICTLY OS-negative (the anti-chiral odd sector is negative definite — no mirror survives reconstruction, kill test 3 cannot fire at the free level); the chiral generation is multiplicity-free (clock spectrum $\{1, \sqrt{2} - 1\}$ ); and the mark transport exists ( $\tau_2$ is the mark-A→B transport, PD) — kill test 7 does not fire at the free level; the gamma milestone stays open on $\mathcal{A}_{\text{hol}}$ [E][C]                                                                                 |
| v609_translator_factorization.py | The translator factorization: the mirror intertwiner $d$ ( $dA = \Gamma(A)d$ ) is UNIQUE, unimodular, $\mathbb{Z}[\omega]$ -integral and $\Gamma$ -involutive ( $\Gamma(d)d = 1$ ); the v605 element factorizes exactly, $c = \lambda K_1 d$ with $K_1 = J^{-1} \overline{F_2}^T M^{-1}$ — $\theta_{\text{corpus}} = J \text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$ : the geometric identity of $c$ is closed [E][C]                                                                                                                            |
| v610_curve_landscape.py          | The curve landscape: the Jacobian splits (differential characters of orders $(12, 12, 6)$ ) into a $\mathbb{Q}(\zeta_{12})$ -CM abelian surface — the field of deck AND marks — times the $\mathbb{Q}(\omega)$ -CM elliptic curve $E : y^3 = u^2 - 1$ ; point counts confirm the splitting ( $a_p(C) = a_p(E)$ for all tested $p \not\equiv 1 \bmod 12$ ); the integer-spectrum census honestly relativizes single-char hits (the anchor class is the most frequent nontrivial family; the compiler's distinction is the joint structure) [E][C]             |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v611_periods.py                | The periods round: the five basic period integrals are EXACT Beta values with all Gamma denominators on the $\zeta_{12}$ grid; the $\mu_4$ rotation acts on periods with the exact characters $i^{k+1}$ ; $\Omega_E = B(1/3, 1/6)/2$ exactly (the classical $\omega$ -CM period); the cross-factor ratio admits no low-degree integer relation (PSLQ) — the two CM factors are period-independent, as the v610 splitting demands [E][C]                                                                                                                                                                                                                                                                           |
| v612_polarization_frame.py     | The polarization frame: $r$ has three simple eigenvalues $\zeta_{12} \cdot \{1, -1, -i\}$ ; in the rotation eigenbasis $J$ becomes exactly diagonal with signature $(1, 2)$ , and THE UNIQUE POSITIVE (Hodge) DIRECTION IS THE PRIMITIVE $\zeta_{12}$ EIGENSPACE — the analytic Hodge structure and the finite rotation agree on where positivity lives; the canonical period normalization (Beta weights) is the named next step [E][C]                                                                                                                                                                                                                                                                          |
| v613_canonical_periods.py      | The canonical period normalization — the analytic bridge closes: the quarter rotation acts DECK-FREE on the four branch segments (12 wrap factors exactly 1); THE EXACT DICTIONARY $r = \text{deck} \circ \text{rotation}$ : $\text{char}(r) = \text{char}(\omega M)$ exactly, explaining $r^4 = \omega$ structurally; the period rows ( $\text{hol } dx/y^2$ , $\text{hol } x dx/y^2$ , $\text{antihol } dx/y$ ) realize the rotation eigenframe; the Hodge weights are explicit Gamma monomials (complex Beta formula, $h_1 = (3\sqrt{3}/16\pi)\Gamma(1/3)^2\Gamma(1/6)^2$ ); and $J = -h$ PER CHARACTER: the compiler polarization is minus the analytic Hodge form — Riemann–Hodge positivity realized [E][C] |
| v614_transport_deck.py         | The Gauss-Manin transport round — THE DECK STEP IS THE BOUNDARY MONODROMY: transporting the segment cycles along the rigid quarter-rotation loop, the discriminant $\lambda(\tau) = e^{2\pi i \tau}$ winds once and the periods factorize exactly; at $\tau = 1$ the deck factor is $\omega = t^4$ on every sheet (the exponent $4 =  \mu_4 $ forced by the puncture count); the canonical transport matrix is EXACTLY $\omega M$ — the v613 dictionary $r \sim \omega M$ is CANONICAL, and $(\omega M)^4 = \omega$ re-derives $r^4 = \omega$ [E][C]                                                                                                                                                              |
| v615_gamma_toy_interacting.py  | The interacting gamma slice at toy level: the gamma battery executed ON the RP-surviving interaction (the alignment bit, v534) — THE INTERACTING MIRROR FLIP: at the forced twist the mirror state’s odd sector is STRICTLY NEGATIVE DEFINITE on every cut and every coupling (kill test (3) cannot fire on the surviving class – v608 upgraded from $g = 0$ to the full grid); the mark transport exists interacting (kill test (7) does not fire); the odd transfer spectrum stays multiplicity-free; the dead member has no RP home: bit selection and chirality data CO-LOCATE [E][C]                                                                                                                         |
| v616_su2_internal_kinematic.py | The internal $SU(2)$ , kinematic separation at twistor level: Woit’s Euclidean structure IS the internal $j$ -direction ( $\rho_{\text{tw}} = \text{right-}j$ verbatim); the internal $\mathfrak{su}(2)$ preserves every twistor fiber, commutes exactly with the spacetime quaternion action and intersects it in $\{0\}$ — kill test (6)’s KINEMATIC branch does not fire; THE CLOCK FACTORIZES as left- $u$ right- $u$ exactly ( $u = e^{i\pi/4}$ : spacetime half-quarter $\times$ internal half-quarter), the ALE deck is purely spacetime-side, and the complex choice breaks the internal $SU(2)$ to $U(1)$ with the v519 mark torsor as its $\mu_4$ [E][C]                                                |
| v617_seam_cover_forcing.py     | The seam forcing round — THE CONFORMAL SEAM AXIOMS FORCE THE $\mu_3$ -COVER: $\mathbb{Z}_4$ -Möbius rigidity (every $\mathbb{Z}_4$ -cyclic configuration is Möbius-equivalent to $\mu_4$ ; harmonic cross-ratio orbit $\{2, 1/2, -1\}$ ), the weight census forces UNIFORM weights $j \in \{1, 2\}$ (the conjugate sheet pair), $j = 1$ gives $y^3 = x^4 - 1$ exactly with genus 3, and the seam reflections match the cover’s real structure (the v599 $(k, 5-k)$ pattern, the two v519 cut points); the bedrock residue narrows to the cyclic-3 choice and the physical $\leftrightarrow$ conformal seam identification [E][C]                                                                                  |
| v618_uniform_constant.py       | The uniform constant round — $C = 1$ FROZEN on the declared surface with the sharpened sign-flip dichotomy: on every lock-sign window (67 of 69) $ q_{\text{real}}/q_{\text{model}}  \cdot h \leq 1$ (measured max 0.982), tertile medians non-increasing; exactly TWO $q_{\text{real}}$ sign-flip windows and the violator set EQUALS the flip set exactly (honest correction of the diary note: the mechanism is the $q_{\text{real}}$ flip, $q_{\text{model}}$ never crosses zero); scramble breaks by $> 10^4$ [E][C]                                                                                                                                                                                         |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v619_flip_mechanics.py         | The flip mechanics round – the two $C = 1$ violators are TRUNCATION ARTIFACTS of the prime-data cap (the flip set equals the truncation set exactly; no zero resonance), REPRODUCED by injection into complete windows with matched sign and magnitude at both scales; on the complete-comb surface $C = 1$ holds with ZERO exceptions (max 0.982): the v618 dichotomy upgrades to a mechanism and the flip windows are retired [E][C]                                                                                                                                                                                                                                                                                                           |
| v620_cyclic_n_census.py        | The cyclic- $N$ census — the seam admits EXACTLY $N \in \{3, 5\}$ as primitive-Lorentz orders (full infinity ramification iff $\gcd(N, 4) = 1$ ; Chevalley–Weil signatures; $N = 6$ reduces to $N = 3$ ), and the compiler’s machine-checked integral constants ( $\det_{\mathbb{Z}}(1 - \text{deck}) = 27 = 3^3$ , saturation $81 = 3^4$ ) pin $N = 3$ UNIQUELY: the v617 bedrock residue (i) is now conditional on $N_{\text{fam}} = 3$ ( $E_8$ -anchored) [E][C]                                                                                                                                                                                                                                                                              |
| v621_interacting_semigroup.py  | The interacting Klein–Landau semigroup — at $g = 0$ the v524 pattern reproduces exactly at $N = 16$ ; at $g > 0$ exact Hermiticity survives PRECISELY on the $\mu_4$ -commensurate steps $\{4, 6, 7\}$ and the clock step stays a positive transfer $((11, 0, 0))$ over the whole grid: the interaction breaks translation invariance to the quartet stabilizer exactly, and the reconstructed rotation group of the interacting toy is the CLOCK tower, not the full circle [E][C]                                                                                                                                                                                                                                                              |
| v622_seam_identification.py    | The seam identification round — THE PHYSICAL SEAM IS THE CONFORMAL SEAM: the v519 kernel is EXACTLY the discrete NS chiral mode sum (including the even-distance zeros — the flat band IS the NS spectrum; Ramond must-fail), the mark-bond midpoints sit exactly at $\mu_4$ , the group dictionary is exact (clock $= z \mapsto iz$ , cuts = diameters), and the straddle law is literal geometry (straddled = axis through marks); the remaining residue is the covering-level identification [E][C]                                                                                                                                                                                                                                           |
| v623_covered_seam.py           | The covered seam round — the lattice skeleton of the covering-level identification: the lifted seam is ONE 48-site NS circle for uniform weights (the alternating pair DISCONNECTS it — a new independent kill), the lifted clock has EXACT ORDER 12 with $L^4 = \text{deck}$ ( $r^4 = \omega$ as lattice combinatorics), the v622 kernel identity generalizes to $N = 48$ , and the deck-character mode grid is the 12-grid with the untwisted sector equal to the base seam verbatim and the twisted pair carrying $\omega$ on bilinears [E][C]                                                                                                                                                                                                |
| v624_external_lattice_audit.py | The external-review lattice audit: THE FOUR-MARK CLOSURE (three independent selectors force $d = 4$ under the $D_{d+1} \oplus A_{d-1}$ glue architecture — disc-group census, the norm equation $(d+1)/4 + (d-1)/d = 2$ , the Pythagorean $(3, 4, 5)$ selector; the two-input presentation reduces to ONE boundary degree plus $\pi$ , conditional on the architecture); THE ANCHOR BYTECODE ( $\chi = (t-1)^2(t-2)$ , two-state recursion, $240 + 8 = 248$ ); and THE LORENTZ CONGRUENCE ( $P^T J_{\text{det}} P = J_{\text{fix}}$ exactly, index 6) — the prime-front determinant form and the cover polarization lattice are the SAME rational quadratic form; the review’s $B$ -test is retyped as ill-posed ( $B$ complex symmetric) [E][C] |
| v625_prime_shadow.py           | The prime-shadow audit – the checkable core of the ‘primes as the shadow of the finished geometry’ reading is EXACT: $\Theta_{E_8} = E_4$ with shell counts $240 \sigma_3(n)$ (multiplicative over coprime addresses – unique factorization exactly), the primes act as COMMUTING Hecke channels with the compiler’s theta as simultaneous eigenvector, and $L(E_4, s) = \zeta(s)\zeta(s-3)$ : the zeta function is the factorized shadow of the $E_8$ counting function – primes enter AFTER the geometry; the speculative framings stay typed hypotheses [E][C]                                                                                                                                                                                |
| v626_e8_code.py                | The E8 code round – E8 IS an error-correcting code: Construction A on the self-dual extended Hamming code $[8, 4, 4]$ yields the even unimodular rank-8 lattice (= E8, by uniqueness) with shells 240/2160 matching the v625 theta identity; every single-bit error is exhaustively correctable; the compiler ties ( $8 = \text{rank}$ , $4 = d =  \mu_4 $ , $16 = \text{half-spinor}$ , $14 = 2 \times 7$ ) are typed observations.                                                                                                                                                                                                                                                                                                             |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v627_hodge_chamber.py    | The Hodge chamber round – all 67 complete windows land in the POSITIVE cone of the cover polarization lattice, on ONE sheet (the measured half of the v624 geometric positivity route); the transport $z = P^{-1}y$ is exact bookkeeping, $\det S > 0$ everywhere, and the honest typing: scrambles do NOT leave the chamber (density-layer property, v582), the fine $C = 1$ arithmetic lives inside the chamber.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v628_orbifold_casimir.py | The orbifold Casimir round – the covered seam’s three deck-mode classes carry EXACTLY the $\mathbb{Z}_3$ -twist Casimir data: exact closed form $E_0(\nu, N) = -\cos(\pi(2\nu - 1)/2N)/(2\sin(\pi/2N))$ , finite-size coefficient exactly $\pi(6\nu^2 - 6\nu + 1)/12$ , sector table $(1/72, -1/24, 1/72)$ , degenerate twisted pair, ratio $-1/3$ , twist gap $1/18$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v629_root_incidence.py   | The root incidence census – the naive $48 \times 5$ incidence to the E8 roots is KILLED (orbit census $19 \times 12 + 3 \times 4$ for the compiler-canonical order-12 element, not 20 free $\zeta_{12}$ orbits), and the positive residue is sharp: the $\mu_4$ clock alone acts freely with EXACTLY 60 orbits ( $60 = D_{\text{start}}$ ), and the family 3-cycle fixes exactly 12 roots (the lifted marks).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v630_suzuki_contact.py   | The Suzuki first contact – the ATOM LAYER of the TFPT window form IS Suzuki’s prime measure, literally (positions $\log n$ , weights $\Lambda(n)/\sqrt{n}$ , exact on 40 atoms; the first executed slice of the PRIME.WEIL.OPERATOR.01 contract); the first Galerkin comparison against the screw function measures a NON-scalar conversion profile – the preregistered W1 residual, now with data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v631_w1_dictionary.py    | The W1 dictionary round – the v630 non-scalar residual IS the zeta pole term, and after separating it the smooth conversion is the SCALAR $-4D$ : the Lerch collapse $[e^{-t/2}\Phi(e^{-2t}, 2, 1/4)]'' = 4e^{-t/2}/(1 - e^{-2t})$ (exact) gives the structure theorem $g'' = -2\cosh(t/2) - 4 \times$ (the TFPT arch density) – Suzuki’s smooth layer is $-4 \times$ the TFPT density minus the pole block (the v591 rank-one); closed per-lag ratio law verified; the atom layer constant is $D^2$ exactly. ERRATUM (2026-08-02, corrected the same day; v643 C0.1): the chain read eq. (1.3) with Lerch coefficient $-1$ ; Suzuki’s own Sec. 2.2 data lock $+1/4$ – all identities hold for the chain kernel $\tilde{g} = g - \frac{5}{4}$ Lerch and every number transfers verbatim via $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4c_{\text{gal}}^{\text{sm}}(g)$ ; the true dictionary is the single scalar $+1/D$ on both layers (Suzuki’s smooth layer is $+\rho$ , not $-4\rho$ ). |
| v632_ftransfer_pgl2.py   | The $F_{\text{transfer}}$ PGL <sub>2</sub> round – the v533 preregistration executed in its literal group form: the four external transfer solvers share ONE Moebius action on the disc $-7$ norm line where they act at all (the Koide reading is the parabolic step $+9 = N_{\text{fam}}^2$ , exactly intertwined with the base translation $T$ ; $g_{\text{QCD}}$ explicitly conjugate to $T$ in the $1/\alpha$ coordinate; $g_{\text{Boltzmann}} = g_{\text{pole}}$ by the single-flow theorem; relic degenerate); the kill criterion ‘four incompatible arithmetic actions’ is NOT met; five must-fail controls have teeth.                                                                                                                                                                                                                                                                                                                                                           |
| v633_orbit60_quotient.py | The $\mu_4$ -quotient round – the quotient of the 240 E8 roots by the free $\mu_4$ clock IS the hermitian-unimodular $\mathbb{Z}[i]$ -E8 with the 60-hyperplane system of ST31 (hermitian-unimodular basis among the reps, $\sigma$ census $19 \times 3 + 3$ on the lines, line group of order $11520 = 46080/4$ ; ST32 killed by its 40-hyperplane count); the $60 \rightarrow 6$ cascade route on the quotient is KILLED (Gram rank 4, regular graphs, no canonical pair rule) – a typed negative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v634_st31_structure.py   | The ST31 structure round – $G_{31}$ is the FULL unitary stabilizer of the $\mu_4$ -quotient structure (exact count 46080), the degrees $(8, 12, 20, 24)$ are unique via Molien, the compiler data embeds canonically ( $\sigma = c^4$ , $J = c^9$ , $\langle c \rangle = Z_{12}$ ), and the computed structure is $G_{31} = (Z_4 \circ 2^{1+4}).\text{Sp}_4(2)$ – while the numerology $ G_{31}  =  W(D_5)  \times  W(A_3) $ carries NO structure (not isomorphic, not a (semi)direct product, and $W(D_5)$ does not even embed: minimal faithful dimension $5 > 4$ ); HONEST: the v629 compiler clock is NOT the Springer-regular 12-element (census $19 \times 12 + 3 \times 4$ , $ C(c)  = 72$ ).                                                                                                                                                                                                                                                                                       |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v635_p_<br>canonicity.py   | The P-canonicity census – in the full $[-4, 4]$ box census (exactly 40 integer solutions of $Q^T J_{\det} Q = J_{\text{fix}}$ ) P is, modulo the natural order-8 symmetry group, the unique minimal-Frobenius operator-compatible class; and the finer Hodge-ray invariants (projective spacelike angle, cone distance $q/ z ^2$ ) separate the true 67-window family from scrambles in 17–18 of 18 cases per invariant – one scramble at $h = 142$ even leaves the positive cone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v636_p_<br>construction.py | The P operator construction – P is, up to global sign, the Frobenius-minimal integer congruence $J_{\text{fix}} \rightarrow J_{\det}$ mapping the $C_V$ -operator null frame ( $\ker C_V$ , $\text{fix } C_V$ = the two minimal isotropic rays of $J_{\text{fix}}$ ) onto the canonical rank-one rays; the two-ray integral family is finite (exactly 16 members, boost normal form solved exactly, Pell members excluded by a Galois argument); either Frobenius-minimality or the trace-ray condition selects $\{+P, -P\}$ ; honest residuals: the $G_8$ orientation choice of the target pair and the documented gl2-derivation must-fail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v637_fine_c1_<br>bridge.py | The fine-invariant / $C = 1$ bridge test – an honest negative promoted as a typed result: after $h$ -control the fine Hodge-ray invariants carry NO independent information about the $C = 1$ margin (partial Spearman $\rho = -0.040$ , $p = 0.75$ ; the preregistered gate $ \rho  \geq 0.30$ and $p < 0.01$ missed by an order of magnitude; the raw $\pm 0.40$ correlations are entirely the $h$ -trend) – the direct fine-geometry-to-arithmetic bridge route at the window level is CLOSED.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v638_code_<br>semantics.py | The code-semantics rounds (kill + stage 2) – the unique compiler-equivariant Hamming placement $C^*$ IS Reed–Muller $\text{RM}(1, 3)$ on $\text{AG}(3, 2)$ and carries an exact dictionary: information bits one per $\mu_4$ pair (3 families cycled + 1 anchor fixed; the unique minimal $Z_6$ -orbit of information sets), syndrome = in-pair bit + pair pointer with the $\sigma$ 3-cycle (anchor = sum of families), decode = projection verbatim on the single-error class (3840/3840, exact arithmetic) – and all of it dies for every non-equivariant placement (only 2/30 carry the structure).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v639_twist_ope.py          | The twist-field/OPE round – the interacting slice of the $\mathbb{Z}_3$ orbifold statement holds at the abelian vertex level on the seam double: $h_\sigma = 1/36$ from three exact routes (conventions reconciled), lattice two-point exponent $2\Delta = 2/9$ at 0.06%, crossing exactly symmetry-protected on the lattice, closed four-point form $[w(1-w)]^{-2\Delta}$ at $< 0.1\%$ , OPE $c_1 = 2/9$ model-bound at 0.19% (model-free $\pm 15\%$ band typed open); reflection positivity validated in the real $\mathbb{Z}_2$ class and honestly OPEN for the complex $\mathbb{Z}_3$ pairing (parafermionic Klein twist); GATE.QGEO does not move.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v640_w1_<br>boundary.py    | The W1 boundary closure – the v631 D6 typed remainder is closed symbolically and certified at 20+ digits: the exact boundary equation $A_{\text{arch}} = \frac{1}{4}g''_{\text{smooth}} - \frac{5}{4}(\log \pi - \psi(\frac{1}{4}))\delta_0$ (relative residuals $< 10^{-22}$ on the tents $d = 0, 1, 2$ ), the duplication identity $\psi(\frac{1}{4}) = -\gamma - 3\log 2 - \pi/2$ exact, the scheme conversion $k = \pi/4 + \frac{3}{2}\log 2$ exact; the boundary cells are three computable constants and the second-order term is the derived moment law $1 + 1/(6d^2)$ (tent $c_2 = 1/12$ vs B-spline $c_2 = 1/6$ ), window-independent, matching the measured excess to 0.6%. ERRATUM (2026-08-02, corrected the same day; v643 C0.1): the boundary equation is stated in the chain's $\tilde{g}$ -normalization; on Suzuki's TRUE $g$ (Lerch $+1/4$ ) the tents read $A_{\text{arch}} = -g''_{\text{smooth}}$ EXACTLY with no constant – $\kappa = 0$ exactly (v643 P2.2), the $-\frac{5}{4}L\delta_0$ term was an artifact of the $-1$ misreading; all numbers remain valid as $\tilde{g}$ identities. |

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| Script                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v641_w1_portability.py  | The W1 frozen-dictionary transport – the preregistered kill test on three fresh windows ( $h = 285, 540, 997$ ; the derivation window $h = 184$ excluded): the frozen dictionary (all formulas closed in $D$ , no knob) executes unchanged – the pole-subtracted conversion is the scalar $-4D(\text{window})$ with the derived discretization law ( $ r - 1  < 6.3 \times 10^{-4}$ ), the atom constant is $D^2(\text{window})$ at machine precision (spread $\leq 1.5 \times 10^{-14}$ ), the closed per-lag law holds within the frozen bars; the kill criterion is NOT met anywhere: W1 is a window-portable operator identification. ERRATUM (2026-08-02, corrected the same day; v643 C0.1): the scalar $-4D(\text{window})$ is the $\tilde{g}$ -normalization; via $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4 c_{\text{gal}}^{\text{sm}}(g)$ the portability transfers verbatim to the corrected one-scalar dictionary $+1/D(\text{window})$ on Suzuki's true $g$ .                                                                                                                                                                               |
| v642_w1_matrix.py       | The W1 matrix identity – at the operator level the dictionary closes the full quadratic form on $h = 184$ ( $M = 368$ ): 16-mode parity block norm distance frozen $6.7 \times 10^{-1} \rightarrow$ derived $2.7 \times 10^{-3} \rightarrow$ after measure-level atom re-binning $6.3 \times 10^{-6}$ ; the separated pole block is numerically rank 1 (the preregistered rank-one freedom); the literal $< 1\%$ per-vector bar fails ONLY on two typed lattice residuals (a near-cancellation denominator; the tent-vs-B-spline atom profile, removed by re-binning the same measure to $4.4 \times 10^{-4}$ ), certified as typed-residual checks with the probe's numbers unchanged; the load-bearing $2 \times 2$ parity block agrees entrywise to $6.8 \times 10^{-5}$ . ERRATUM (2026-08-02, corrected the same day; v643 C0.1): the two-layer dictionary ( $-1/(4D)$ smooth, $1/D$ atomic) is the $\tilde{g}$ -normalization; on Suzuki's true $g$ it collapses to the ONE scalar $+1/D$ (sign positivity-compatible), all norm ratios and residuals transfer verbatim; the form equality on the true $g$ is v643 P3.1 ( $1.28 \times 10^{-10}$ ). |
| v643_w1_theorem.py      | The W1 measure-level THEOREM on Suzuki's TRUE screw function – the projection lemma closes the last W1 remainder (Suzuki's $L_0^2$ condition is automatic on the $u$ -side, $D : H_0^1 \rightarrow L_0^2$ a bijection; the v563 parity sector IS the odd $H_0^1$ subspace), the dictionary collapses to ONE scalar $+1/D$ on both layers (sign positivity-compatible), the origin constant VANISHES ( $\kappa = 0$ exactly: the v563 near-cell scheme IS Suzuki's origin bookkeeping), $A_{\text{arch}} = -g''_{\text{smooth}}$ exactly at every lag ( $3.4 \times 10^{-52}$ ), and the full form equality holds at $1.28 \times 10^{-10}$ on the common odd sector; C0.1 carries the ERRATUM evidence: eq. (1.3) has Lerch coefficient $+1/4$ , not $-1$ , and every number of the v631/v640–v642 chain transfers verbatim via $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4 c_{\text{gal}}^{\text{sm}}(g)$ .                                                                                                                                                                                                                                              |
| v644_w2_form_density.py | The W2 START – form density of the nested window (hat) spaces in the odd form sector of Suzuki's localized Weil operator $A_a$ , in honest slices: FEM density verified at the CLASSICAL rate ( $H^1$ rates 1.000, $L^2$ rates 2.000 over $M = 92..736$ ; with $Q_W$ $H^1$ -bounded and $H_0^1$ a form core this is the classical density chain, cited not re-proved); Rayleigh–Ritz on nested spaces decreases MONOTONICALLY from above (all 6 sequences, parity identity $\lambda_{\text{full}} = \min(\lambda_{\text{even}}, \lambda_{\text{odd}})$ at $10^{-15}$ ); the $\lambda_a$ scan across 5 windows is consistent with Suzuki Thm 1.3 (honest reading: $\lambda_a = 0^+$ within $\sim 10^{-9}$ , NO sign statement; ground state even everywhere, reported); the Mosco/norm-resolvent remainder is typed – W2 is STARTED, not closed.                                                                                                                                                                                                                                                                                                           |
| v645_klein_rp.py        | Parafermionic Klein-twist reflection positivity – the $\mathbb{Z}_3$ twist sector IS reflection positive on the seam-double lattice under the Klein pairing (mirror COMPOSED WITH charge conjugation, $\omega^{kq}$ zero-mode phases, $\eta = (1, 1)$ UNIQUE among 36 scanned pairs), resolving the named open item of QGEO.TWISTOPE.01 at the lattice OS level: every charged Klein block is exactly real symmetric AND PSD (worst $2.9 \times 10^{-12}$ ), $N$ -stable 48..384, both diameter axes, including the FULL mixed OS Gram over $\{1, \sigma_1, \sigma_2, \text{dipoles}\}$ with no sector projection; the naive pairing keeps its $N$ -stable violation ( $-4.5 \times 10^{-3}$ , must-fail); continuum OS reconstruction and modularity stay typed open; GATE.QGEO does not move.                                                                                                                                                                                                                                                                                                                                                           |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v646_rm13_reverse.py     | The RM(1,3) reverse reading – are compiler numbers CODE operations? An honestly TWO-PART answer: the $\sigma$ -fixed syndromes carry the anchor decomposition EXACTLY (coset-leader weights (0, 1, 1, 2), norm-square sum 6, anchor = family sum $P_3 = P_0 + P_1 + P_2$ – the SELECTION is equivariance-specific, while the (1, 1, 2)/norm-6 weights are generic for any [8, 4, 4] placement), but the bytecode sequence $p_n = 2 + 2^n$ has NO code reading (0/11 natural families; 18/27/81 without any hit – the bingo is buried); the kill control fires selectively (the naive placement loses $q$ -universality and the $\sigma$ action but keeps the generic sector weights, all pure-code invariants, and even the $\sigma$ -fixed-root count 12 – honestly retyped).                                                                                                                                                                                                                 |
| v647_st31_degree24.py    | The ST31 degree-24/-20 round – the $\mu_4$ center is a POWER of both regular clocks ( $w^6 = \pm J$ for the regular 24-element with $\chi_w = x^4 + ix^2 - 1$ exact, all four eigenvalues primitive $\zeta_{24}$ ; $u^5 = \pm J$ central for the regular 20-element), and the clock question $c = w^2$ is KILLED by a PARITY THEOREM: the 12-cycles of any square come in pairs from 24-cycles, and census( $c$ ) = 19×12 + 3×4 has ODD counts – impossible even in $S_{240}$ (proof-grade); $u^4$ acts freely with census {5:48}, 48 = 240/5 exactly, lines {5:12}; the equality 48 = seam sites stays a typed OBSERVATION.                                                                                                                                                                                                                                                                                                                                                                   |
| v648_sign_uncertainty.py | The sign-uncertainty / Mellin-strip toolbox read against the W3 window cone – the DICTIONARY is real and anchored to 25 digits (the $d = 1$ Mellin strip IS the critical strip; the strip multiplier’s phase slope at $t = 0$ is EXACTLY $L = \log \pi - \psi(\frac{1}{4})$ , the W1 dictionary’s $\delta_0$ weight), but the QUANTITATIVE mechanism is a TYPED NEGATIVE: the exponential mass gain is powered by the strip half-width and dies at $d = 1$ ( $e^{-E^*} = 0.967$ where $< 1/2$ is needed; the machinery would need $d \geq 21$ ; the window family’s negative structure sits $5.2\times$ outside the maximal BCK ball $1/\pi$ ) – any W3 route through this toolbox must replace the dimension lever; side findings: $\lambda_{\min} > 0$ on 67/67 complete windows (min $+8.26 \times 10^{-4}$ ), the incomplete combs are exactly the v618 flip set {1219, 1292, 1445}, and $\text{corr}(\log \lambda_{\min}, \log \varepsilon h) = +0.704$ ; W3 stays open, no RH statement. |
| v649_discipline_audit.py | The discipline audit – the suite parses ITSELF and certifies its process discipline as frozen minimum bars: 214 ledger rows document kills / honest negatives / retypes, 1469 must-fail/kill/scramble occurrences across 233 of 643 modules, 47 contract rows (29 naming kill criteria; executed chains FTR.PGL2 v533 $\rightarrow$ v632 and the W1 chain $\rightarrow$ v643 verified), deterministic replay 12/12 on the declared sample v55..v648 (two fresh runs, identical up to timing lines), five classical anchors recomputed against the literature (Jacobi $r_4$ , Construction A $\rightarrow$ 240 minimal vectors, the Shephard–Todd G31 degrees, the Suzuki archimedean constants to 25+ digits, the completed-zeta functional equation), and 54 committed Lean proof modules; the honest limit: process discipline, not physics truth – the open gates stay open (278 class-O rows).                                                                                             |
| v650_orbifold_modular.py | The modular data of the $\mathbb{Z}_3$ orbifold of the seam on the lattice: all nine twisted-sector partition functions $Z[g, h]$ match $ \theta[g/3, h/3]/\eta ^2$ at $< 1\%$ , the modular S-covariance is MEASURED with a clean common $N^{-4}$ rate (the $N^{-2}$ lattice artifact is itself S-covariant and cancels in the S-relation), T is the exact discrete Dehn twist $(g, h) \rightarrow (g, g+h) \times (-1)^F$ at machine precision, the twisted characters carry $h_\sigma = 1/36$ with the parafermionic $1/6$ gap, and $\text{deck}^3 = (-1)^F$ closes exactly as $\mathbb{Z}_6$ trace arithmetic including the cover factorization; chiral phases, continuum T-phases and continuum OS stay typed open; GATE.QGEO does not move.                                                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v651_orbifold_<br>assembly.py | The full $\mathbb{Z}_3$ -orbifold assembly of the seam: $H^2(\mathbb{Z}_3, U(1)) = 0$ by machine enumeration (27 cocycles, all symmetric – NO discrete torsion), and of six assembly candidates the survivors of {integer spectrum, unique vacuum, S measured clean, T exact} are exactly $\{B, C3\}$ – the naive $\mathbb{Z}_3$ dies by an exact T violation $3.5 \times 10^{-3}$ (= its charge-3 content), C1 by S stalling, C2 has no vacuum, A' additionally by the wrong first twisted exponent; the seam data then discriminate B from C3 (B keeps every measured $c = 0$ sector ground and the Klein phase $\eta = (1, 1)$ ): THE SEAM ASSEMBLY IS B, the $\mathbb{Z}_6$ deck quotient with uniform charge-0 projection (data-backed, v652 hardens it structurally); the chiral phase $e^{-2\pi i \bar{a}b}$ is measured; GATE.QGEO does not move.                                                                                                                                                                                                           |
| v652_orbifold_<br>arf.py      | Arf/GSO hardening of the B-vs-C3 discriminator: the $6^6$ classification of all sector-character wirings leaves exactly $\{B, C3\}$ , the single-Fock-space pin forces B, and the $\mu_6$ defect ladder measures the C3-excised states DIRECTLY – ladder exponents $2\Delta = (1/18, 2/9, 1/2)$ at $< 8 \times 10^{-4}$ , exact charge-resolved minima $(0, 1, 4, 9, 4, 1)/36$ ; C3's odd-sector predictions 85/18 and 81/18 exceed the measurements by factors 85.0 and 9.0 (the defect string is exactly charge-neutral: C3 excises states that exist on the lattice seam with positive Klein norm); the earlier 2/9 read alone is B-vs-C3-blind (both coincide exactly on the even sectors) – the new $\tau$ datum: OPE exponent 1/18; the bond-defect premise passes to v653; GATE.QGEO does not move.                                                                                                                                                                                                                                                          |
| v653_bond_<br>defect.py       | The bond-defect premise becomes a LATTICE THEOREM: twist sector = deck boundary condition = bond defect, canonically and exactly (unitary equivalence $A_r = \sqrt{3} R P_r$ on all modes, exact in $\mathbb{Q}(\zeta_6)$ , at 16/48 and 48/144), with $D_{\text{can}} = L^4$ and $D_{\text{can}}^3 = -1$ as operator identities (the quarter clock fixes the canonical section lift); the defect position is pure gauge (symbolic telescoping theorem in all 16 free bond phases) with the gauge class pinned by the mark geometry (sheet reduction = one uniform twist unit per mark; cocycle support = the mark set); the exclusion census is complete – the base chain has NO other $\mathbb{Z}_3$ , traces/holonomy/v628-Casimir kill everything outside, including the HONEST isospectral NN chain (it exists at $7.8 \times 10^{-16}$ : purely spectral data cannot exclude it, but it is no $\mathbb{Z}_3$ -twist realization and the measured string data discriminate it); continuum CFT and the seam identification stay typed; GATE.QGEO does not move. |
| v654_st31_<br>degree8.py      | The ST31 degree-8 round plus the unified $d/4$ theorem: every Springer-regular $d$ -clock of $G_{31}$ satisfies $x^{d/4} = \pm i \text{Id}$ , a GENERATOR of $\mu_4$ , never $\pm 1$ (one formula from the exponent systematics: all exponents of $G_{31}$ are 3 mod 4); the CONVERSE is exact at $d \in \{8, 20, 24\}$ , and the unique exception at $d = 12$ is the COMPILER-CLOCK class (1280 extra elements = the class of $c$ and its Galois twin; $\chi_c = (x - i)^2(x^2 + ix - 1)$ is half the regular-12 Springer block, all eigenvalues satisfy $\lambda^3 = -i$ – exactly why $c^3 = J^3$ is central: the compiler-clock classes are the ONLY non-regular clocks in all of $G_{31}$ whose $d/4$ -th power generates the center); the order-8 landscape is complete (10560 elements, 480 regular = 2 Galois classes, $\chi = (x^2 + i)^2$ exact) with the honest kill: “free $\Rightarrow$ regular” is FALSE (4320 free order-8 elements noncentral; already at order 4); degrees = free orders divisible by 4 above 4; no marker moves.                  |
| v655_w2_mosco.py              | The W2 Mosco / norm-resolvent PREPARATION: the resolvent family $(A_M + 1)^{-1}$ is Cauchy along the refinement (rates $\sim 1.0$ on all five test vectors) and its $H_{\log}$ norms stay UNIFORMLY BOUNDED – the two measurable prerequisites of the Mosco/norm-resolvent theorem; the eigen-scale collapse is documented and TYPED: $\lambda_a(\log 16) = 0^+$ within the discretization error, eigen-direction observables are not float-certifiable at this $a$ – any W2 route must run through resolvent/form observables; recovery sequences converge and the liminf direction survives the wiggle test; the typed remainder is the discrete Garding inequality (v661–v663); W2 stays open, no positivity claim, no RH statement.                                                                                                                                                                                                                                                                                                                             |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v656_margin_<br>link.py      | The margin bridge, typed as a LOCK IDENTIFICATION: the preregistered gate passes (rank-cubic $h$ -control $\rho_{\text{partial}} = +0.477$ , $p = 10^{-4}$ ; $(h, \alpha)$ robustness $\rho = +0.701$ ), but the mechanism is an identification, not an independent link: $\lambda_{\min} \approx r_{\text{id}} \cdot q_r / \ w_{\text{lock}}\ _G^2$ with $q_r = w_{\text{lock}}^T \hat{A} w_{\text{lock}}$ exactly ( $1.5 \times 10^{-7}$ ) – the window form reads its OWN $C = 1$ defect along the lock direction (minimal mode lock-enriched, median angle $9.1\text{r}$ ); the declared surface reproduces exactly (67 complete combs, $\min \lambda_{\min} = +8.26 \times 10^{-4}$ , $\text{corr} = +0.704$ ); no positivity claim, no RH statement.                                                                                              |
| v657_rid_<br>alignment.py    | What drives the alignment factor $r_{\text{id}}$ : the exact 2D formula $r_{\text{id}} = (1 - q \tan^2 \theta) / (1 - \tan^2 \theta)$ (max dev $3.4 \times 10^{-10}$ on all 67 windows) with structural signs on every window ( $\theta < 45\text{r}$ , $B < 0$ , $q > 1$ ) – the deficit is ANGLE-DRIVEN (share $\sim 0.79$ ), not arithmetic (after $h$ -control no arithmetic carrier passes the Bonferroni-honest gate); the last atom sits at the window edge identically (the frame-A edge $e^{2\alpha} = N^2$ is itself a prime power); $r_{\text{id}}$ is INFORMATIVE (cubic-in-log $h$ $R^2 = 0.618 < 0.80$ ): genuine window-wise structure beyond the ladder, the depth trend honestly documented; on this surface W3 reduces to lock sign plus the uniform bound $q \tan^2 \theta \leq 1 - \delta$ (v658/v659); no RH statement.            |
| v658_w3_uniform_<br>bound.py | The W3 uniform bound $P := q \tan^2 \theta \leq 1 - \delta$ , risk survey: the $h$ -trend of $P$ is carried by $q$ (slope shares $q + 0.64 / \tan^2 + 0.36$ ; the identity $\beta_P = \beta_q + \beta_t$ exact), the crossing-risk models put the point crossing at $h^* \sim 3.4 \times 10^3$ (POW) / $9.1 \times 10^3$ (LOG) / $1.8 \times 10^4$ (SAT) – the family margin $\delta = 0.036$ is NOT safe to extrapolate – and the construction is $\alpha$ -FRAGILE: $\max  dP /P = 0.59$ over $\pm 2.5\% / \pm 5\%$ shifts (bar 0.15): $P$ sits on a THIN RESONANCE of the frame-A coupling (mapped by v659); after $h$ -control no mechanism carrier passes the honest gate; $\max P = 0.9636$ at $h = 859$ ; verdict W3UB-FRAGILE, honestly typed; no RH statement.                                                                                 |
| v659_w3_<br>landscape.py     | The W3 resonance landscape $P(\alpha)$ around the frame-A points: $\lambda_{\min} > 0$ on ALL 635 landscape grid points (positivity is never lost anywhere in the surveyed $\alpha$ -space, including every needle), every $P > 1$ excursion is a $\theta \rightarrow 90\text{r}$ ROTATION artifact of the 2D reduction, and the MARGIN-regime share of $P > 0.95$ is 0.0%; 52 needles across 8 slices (fine FWHM median $2.5 \times 10^{-4}$ in $\alpha$ ), frame-A points are NOT preferentially on peaks (quantile rank 0.24 vs null 0.5), $P > 0.95$ lives on $\sim 5\%$ of the surveyed space; honest negative: NO arithmetic resonance candidate passes the frozen gate – the peaks are real but arithmetically unexplained; the W3 alarm signal is only $\lambda_{\min} < -\text{floor}$ , never observed (calibrated by v668); no RH statement. |
| v660_theta_<br>predicate.py  | The $\theta(\alpha)$ rotation predicate – the honest preregistered result THETA-PRED-FAIL: 47 of the 52 needles are DIAGONAL CROSSINGS of the deflated $2 \times 2$ (leak $\approx 1.0$ : no 2-level partner), and the predicate $f = \lambda_{\text{lock}} - \mu_1 \geq 0$ is FULLY NECESSARY (pointwise recall 1.0) but not sharp (precision 0.18 pointwise / 0.255 at the event level: 65 $\theta$ -45°-crossings vs 212 $f$ -zeros); K1 exactness on 702 pooled points ( $1.6 \times 10^{-10}$ ), interlacing exact; $h = 859$ measured $\geq 100$ needle widths from $45\text{r}$ at $s = 0$ ; 63/67 windows carry a confirmed needle within $\pm 2.5\%$ ; the named successor is BLOCK DEFLATION; no marker move, no RH statement.                                                                                                                |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v661_garding.py          | The discrete Garding inequality $Q_M \geq c \ v\ _{H_{\log}}^2 - C \ v\ _{L^2}^2$ – the named missing W2 building block, measured: $c > 0$ at every stage ( $c(736, 1) = 0.180$ , $c(736, 4) = 0.676$ ) and $a$ -UNIFORM at fixed $M$ -ratio (the actual W2 requirement, five windows $h = 184..1433$ ), but whether $c(M)$ stabilizes or drifts like $1/\log M$ is UNDECIDABLE on four dyadic stages (the stabilization bar fails at $C = 2, 4$ ; the $1/\log$ model stays competitive at $C = 1$ ) – the honest typed state GARDING-DRIFT-UNRESOLVED; the diagnosis: the tight direction reads the symbol minimum near the lattice edge (edge-band read, symbol tightness $0.81..0.97$ ) – v662 tests the two named remedies, v663 the envelope; no marker move.                                                                                                                                                                                                                                             |
| v662_garding_edgeband.py | The two named Garding remedies – both REFUTED: the folded $H_{\log}$ norm does not heal the drift ( $c_{\text{fold}}(736, 1) = 0.18151 \approx \text{unfolded}$ ), and the edge cutoff is LOSSLESS ( $C_{\text{edge}} \leq$ the $C$ budget: the drift is not edge-band mass) – the drift lives in the TOTAL SYMBOL ( $\min_{t \geq 20} s_{\text{tot}}$ near-flat across the ladder while $\log(2+t)$ grows); the probe’s declared FAIL at the literal G3.1 bar is promoted as a typed-residual check with inverted expectation (v642 pattern, numbers unchanged: the $c_0$ $M$ -spread 0.260 MUST exceed the 0.10 bar – the per-mode symbol read is not $M$ -stable); $a$ -uniformity of the healed constants measured over $h = 184..1433$ ; what a proof needs is the lower-envelope growth of the total symbol (v663); no marker move.                                                                                                                                                                      |
| v663_garding_envelope.py | The Garding ENVELOPE inequality: the total window symbol DOES grow, sub-log, measured $a$ -uniformly – $s_{\text{tot}} \geq c_0 w^*(t) - C_0$ with $(c_0, C_0) \approx (0.021, 0.055)$ on the whole family; “flat” was a $t$ -RANGE artifact (the v662 read stopped at the first dip; the band minima grow with a log model, $c_0 > 0$ on all five windows); the B4(iii) breaking point QUANTIFIED: the derived partial-summation bound is valid but exponential in $a$ ( $e^a/a$ ) while the MEASURED atom envelope grows only $\sim 3.55 a$ linearly ( $A(a) = 6.2..18.5$ over $a = 2.77..6.24$ ); Chebyshev validity $\psi(n)/n \leq B_\Psi$ on the full atom table; the remaining W2 step is NAMED: a Fejér spectral-density bound replacing the per-mode symbol read – Suzuki Prop. 4.1 does not block the route (full-text read); measured, not proved; no marker move.                                                                                                                                  |
| v664_look_elsewhere.py   | The look-elsewhere audit – the bingo budget of the typed [C] numeric coincidences, quantified: $N = 42$ structure slots, $K = 19$ hits, the 6 killed readings entering as 0-hit trials; the family-corrected global $p = 5.08 \times 10^{-3}$ in the most conservative of three null models (the ensemble survives the honest correction), with the empirical control that only 0.2%/0.3% of 2000 size-matched pseudo-target lists reach the real hit count / the real $p$ – but NO single observation is significant on its own: suggestive ( $0.01..0.1$ ) only $60 = D_{\text{start}}$ , 12 = marks, the G31 degrees, $48 = \Omega_{\text{adm}}$ ; unremarkable ( $p > 0.1$ ): $6 = p_2(a)$ ( $p = 0.169$ ) and the CODESEM number battery ( $p = 0.593$ – the v646 finding quantified); 14/16 FORCED (their value is the theorem, not the number); the re-typings are executed as dated ledger notes (EXTREV.LATTICE.01, E8.CODESEM.01, E8.CODE.01); seed 20260802, 100k MC, exact Poisson-binomial tails. |
| v665_keiper_li.py        | The Keiper–Li sequence on TWO independent routes – the zero comb (2000 zeros, $\gamma_{\text{max}} = 2515.3$ ) vs zero-free arithmetic (contour extraction; $\lambda_1 = 1 + \gamma/2 - \log(4\pi)/2$ exact, dev 0.0; binomial = contour at $7.8 \times 10^{-92}$ ) – CONSISTENT for all $n \leq 64$ with maximal budget usage 0.124 (the tail budget documented: the raw criterion window is honestly EMPTY, only the density tail makes it sharp); $\lambda_n > 0$ for $n \leq 64$ as a FINITE statement (not Li’s criterion); the injection test has teeth: an off-line zero ( $\text{Re} = 0.6$ ) explodes negatively ( $\min \sim -4.9 \times 10^{43}$ at $n \sim 199921$ ) AND is detected already at $n = 1$ inside the budget; the atom table reproduces the prime-block constant ( $-\gamma_E$ at $2.9 \times 10^{-4}$ ); no RH claim.                                                                                                                                                                |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v666_turing_cert.py        | The Turing completeness certificate of the zero comb: the comb is COMPLETE below $\gamma_{\max} = 2515.3$ – all 1999 midpoint rungs inside the cited S-band ( $\max  S_{\text{est}}  = 0.5151 < 2.5$ ), the Turing integrals at 5% of the Lehman bound (a persistent miscount is excluded, not just a local one), and the must-fail controls have teeth: a removed zero blows the Turing integral to $167\times$ , a doubled one to $174\times$ the bound; the certificate says exactly that the loaded list is the complete initial segment, nothing more – certifying the data premise of all comb modules (v665/v667, the v589 convention); no RH statement.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v667_baez_duarte.py        | The Baez–Duarte / Nyman–Beurling control frame – an EXTERNAL comparison Galerkin for the Suzuki route: the Vasyunin Gram at $1.7 \times 10^{-31}$ against direct integration, $C_{\text{BD}} = 2 + \gamma - \log 4\pi$ computed independently from the zero comb + Riemann–von Mangoldt tail at $6.4 \times 10^{-8}$ (the SECOND comb anchor), and $d_N^2 \log N \rightarrow C_{\text{BD}}$ reproduced at the percent level (max dev 0.042; the honest corridor recalibration documented in the module); the frame comparison: the log-slow DRIFT is SHARED with the Suzuki route (= intrinsic to the positivity problem) while the resonance NEEDLES are not (= frame-side); conditioning $\kappa \sim N^{1.83}$ power law with no isolated outlier, Möbius-mollifier correlation $r = 0.91$ , the $\mu$ -hierarchy in leave-one-out; no positivity claim, no RH statement.                                                                                                                                                                                                            |
| v668_ground_truth.py       | Ground truth + firewall – the positive AND the negative control of the Weil-positivity window machinery, one module from both probes: IHARA-GT-CONFIRMED – on three PROVEN Ramanujan graphs the Toeplitz window form is PSD for every $K$ , and true positivity has NO uniform margin ( $\delta(K) \rightarrow$ exactly 0 beyond the support-resolution depth; the proven violators break at $K^* = 15/11$ , detection reach $K^* \times s \approx 2.3$ ; Fejér reads are blind) – plus EPSTEIN-FW-BREAKS: without the Euler product ( $x^2 + 5y^2$ , Davenport–Heilbronn; 12 off-line zeros found by winding count) the IDENTICAL v563 machinery breaks by $\sim 13$ orders of magnitude ( $\lambda_{\min}(L3) < -\text{floor}$ on $3/3$ windows), the load-bearing structure being the Euler product / the composite support – THE central W3 recalibration: shrinking margins are the expected behavior of true positivity, the alarm signal is only $\lambda_{\min} < -\text{floor}$ (never observed); the needles are frame-side (v667); no claim about $\zeta$ , no RH statement. |
| v669_fejer_density.py      | The Fejér density bound – the W2 density plane closes: the EXACT identity $s_{\text{tot}} = 2\pi(F_a \star dN)$ (Weil explicit formula on the tent test pair; pointwise $2.9 \times 10^{-6}$ , $t$ -averaged $7.2 \times 10^{-8}$ against the Turing-certified comb: the peaks ARE the zeros, $\gamma_{1.4}$ matched to 0.0002, the dips ARE the zero gaps) plus the UNCONDITIONAL theorem chain RvM + Trudgian $\Rightarrow \rho_{a,\delta}(t) \geq 0.306 \log(2+t) - C_0$ $a$ -uniformly with 10/10 finite certificates on the ZERO-FREE $\rho$ grid (margins $\geq 1.36$ ); honest floor: the literal $1/a$ (and $\pi/a$ ) width sits BELOW the RvM threshold $4\pi A_1 = 1.4074$ – pointwise single-mode control is NOT certifiable this way (explaining the Garding $1/\log$ drift, v661); the theorem controls wave packets of spectral width $\geq \delta$ ; off-line remainder budgeted at $2.4 \times 10^{-10}$ ; W2 stays open, no marker move.                                                                                                                               |
| v670_w3_block_deflation.py | The $(1+K)$ -block deflation – the named successor of the rank-1 rotation predicate, honestly FAILED: fidelity ladder $p_{90} = 50.53/49.49/49.44/49.35$ deg for $K = 1/2/4/8$ (the 5-deg bar is never reached), leak median 1.000 at EVERY $K$ – the pencil action on the lock direction is NOT captured by the 8 lowest deflated modes: the needle coupling is BULK-spectral, not low-rank; needle predicate (52 needles, 33 ROT truth): precision 0.635 / recall 1.000 $\rightarrow$ MISS (bar 0.80); the $K = 1$ anchor rung reproduces the rank-1 quotes exactly; A3 skipped by the frozen rule; no positivity claim, no RH statement, W3 stays open.                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v671_lehmer_resonance.py | The Lehmer $\leftrightarrow$ resonance link – are the W3 needles the window analogue of the Lehmer-pair phenomenology? A clean NO: the raw $\pm 0.7$ correlations are pure $h$ -ladder, the $h^3$ -partials collapse to $+0.195/-0.123/-0.125$ with ALL $p > 0.1$ (gate $ \rho  \geq 0.30$ AND $p < 0.01$ : no hit), the pseudo-pair placement control ( $B = 2000$ ) is unremarkable ( $p = 0.086/0.299/0.231$ ) – the needles are NOT zero-comb-driven at the window resolution; top-10 pair table (comb minimum $\delta_k = 0.08925$ at $\gamma \approx 1977$ ; the classical Lehmer pair at $\gamma \approx 7005$ lies outside the comb, documented); the resolution correspondence $D \leftrightarrow t_{\max} = \pi/D \in [118, 868]$ documented; the teeth detect a planted signal ( $\rho = +0.847$ , $p = 5 \times 10^{-5}$ ); consistent with the Baez–Duarte FRAME-side reading (v667); no RH statement.                                                                                                                                                                                                                    |
| v672_li_corollary.py     | The W1 $\Rightarrow$ Li corollary spelled out: $g_n(u) = \frac{1}{2}e^{- u /2}L_{n-1}^{(1)}( u )$ derived EXACTLY (generalized Laguerre; $\hat{g}_n = 2\sin^2(n\theta/2) \geq 0$ : the Li coefficient IS a quadratic-form value of the Weil form at the generator $G_n$ ), and the $h = 1433$ window REPRODUCES 30/32 Li coefficients with $< 10\%$ form error (band $2.4 \cup 6..32$ ; $n = 1$ honestly misses at 24% – a jump artifact; $n = 5$ hair-thin at 10.03%) – the certified window positivity $\lambda_{\min} = +1.516 \times 10^{-3} > 0$ CONTAINS these 30 finite Li coefficients (the odd sector carries 75..99.8%); pole cancellation to 6 digits (the v591 rank-2 block is exactly the piece to exclude); NEW: the even sector measured (unconstrained indefinite $-1.2 \times 10^5$ , pole-free $+7.67 \times 10^{-4}$ ); a finite statement, NOT Li’s criterion, no RH statement.                                                                                                                                                                                                                                    |
| v673_li_e4.py            | Li additivity for $L(E_4, s) = \zeta(s)\zeta(s-3)$ – the exact E8 $\leftrightarrow$ comb consistency test: $\Lambda(E_4, s) = (2\pi)^{-s}\Gamma(s)\zeta(s)\zeta(s-3)$ COMPLETED (functional equation $s \leftrightarrow 4-s$ exact at $2.4 \times 10^{-41}$ ), residues $\pm 1/240$ EXACTLY – the E8 shell normalizer 240 IS the residue; $\Lambda_L(n) = \Lambda(n)(1+n^3)$ from the $240\sigma_3$ recursion (30 digits to $n = 512$ ); the shift rule $\lambda_n^L = 2\lambda_n^\zeta$ EXACT (Lagarias additivity); the cross-check E8-arithmetic vs comb: budget usage 0.124 for ALL $n \leq 32$ – the E8 counting function and the $\zeta$ comb agree in ONE Li sequence; the teeth: the naive single-map Li explodes negatively (first negative at $n = 269$ , min $\sim -2.8 \times 10^{25}$ – non-tempered; the additive per-factor definition is the correct Lagarias reading); a finite statement, no RH claim.                                                                                                                                                                                                               |
| v674_packet_garding.py   | The packet Garding inequality – the packet-to-point gap of the W2 route, built and measured: the packet PROJECTION norm does NOT escape the $1/\log$ drift (typed-residual check with inverted expectation, v642/v662 pattern: stabilization must fail at every $C$ , the pure- $1/\log$ model stays competitive; the minimizer is single-mode tight, $c_X/\text{single-mode} = 0.90$ ), BUT the FRAME inequality $Q + C_0G \geq c_0Y$ holds at every $M$ with the v669 theorem constants ( $c_0 = 0.3058$ , $\lambda_{\min} = +1.79.. + 1.52$ , $a$ -uniform); block averages track the Fejér density (worst median dev 0.067), the packet norm is comparable to $H_{\log}$ with a bounded measured defect ( $\kappa \leq 1.170$ , increments decreasing, boundary-spike mechanism diagnosed) and embeds compactly by construction; the Mosco mechanism is numerically COMPLETE (8/8 sequences: resolvents $X$ -uniform + Cauchy, recovery liminf, weak-null 3/3); the named remainder is within-packet equidistribution (dip depth $\theta \rightarrow 0.20$ ) = unconditional zero-gap information; W2 stays open, no RH statement. |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v675_needle_mechanism.py     | The needle mechanism on the assembly side – after the triple negative (v670 not low-rank, v671 not zero-comb-driven, v667 not in the BD frame), ALL FOUR frozen assembly candidates miss their gates: (a) lag quantization weakly significant ( $p = 0.007$ ) but the jump set is too dense (precision 0.007 at recall 1.0); (b) the atom-phase coherence gradient has the WRONG sign ( $p = 0.012/0.113$ ); (c) the lock-direction rotation is cleanly killed (14.77ř vs 14.40ř per unit $s$ – the needle rotation belongs to the mode, not the v596 projection); (d) the pole weight is significant ( $p = 2 \times 10^{-4}$ ) but INVERTED (0.682 at needles vs 0.734 off, the mirror of the $\theta$ rotation) – the needles are an EMERGENT BULK property of the window pencil (consistent with the v670 leak 1.000); the pole block is exactly rank 2 (cosh addition theorem, $7.3 \times 10^{-14}$ ); the teeth detect a planted signal ( $p = 2 \times 10^{-4}$ ); the typed W3 recommendation: contract W3 as “MARGIN-regime generic bound + needle risk map” (v659/v668), not as a uniform bound with an exception predicate; no positivity claim, no RH statement, W3 stays open.                                                                                                                                     |
| v676_c1_mechanism.py         | The $C = 1$ contraction mechanism – on the declared surface $C = 1$ is a DISCRETIZATION THEOREM: $q_r$ is the exact zero-side reading of the lock profile (Weil explicit formula on the complete comb; median residual $1.1 \times 10^{-2}$ on 67 windows, $q_{\text{pred}} > 0$ on 67/67 – the lock sign IS zero-side positivity), the $h^{-1}$ is the DST normalization ( $q_r(N/2)/F_\alpha$ $h$ -flat, slope +0.04), and the constant is the zero-free RvM density integral (median ratio 1.005; tertiles 0.66/0.43/0.37 predicted vs 0.61/0.45/0.39 measured) – the lock sign via the v591 pole killer (rank-one, $\cos = 1.000000$ ); honest: the v596 exponent $-1.01$ was flip-contaminated (clean $-1.28$ ; dated note on PRIME.LOCKPROJ.01), “operator norm $\leq 1/h$ ” is only a DIRECTION statement ( $\ R\ _G \sim h^{+1.03}$ ), and $\sup \leq 1$ does NOT follow from the density average (the $\sin^2$ sampling budget is the precise remainder, factor-2 headroom); no RH lever beyond the density; W3 stays open.                                                                                                                                                                                                                                                                                             |
| v677_w3_structure_theorem.py | The W3 structure theorem – the equivalence chain as a THEOREM: S1 – the deployed odd-Toeplitz form is the Cantoni–Butler odd-sector compression (eigenvalue-by-eigenvalue $1.5 \times 10^{-15}$ ), the SANDWICH theorem replaces the naive DST diagonality (parity defect measured 0.92..1.55), the mode weights reproduce the v669 tent EXACTLY; S2 – the per-lag Weil dictionary is unconditional, the master identity $x^\top Ax = \sum_{\gamma>0} T_x(\gamma) + P(x)$ with $T_x \geq 0$ on the line (the alias comb) and the cosh-amplified off-line term ( $5.8 \times 10^{-16}$ ); EPSTEIN – an own winding-number census (exactly 12 off-line zeros, $\zeta_K$ control 0) predicts the measured form break QUANTITATIVELY ( $\lambda_{\min}$ ratio 0.803, factor-2 gate); S3 – the threshold map $s_{\min}$ over 67 windows $\times \gamma \leq \pi/D$ with $2\alpha s_{\min} \approx 1.4..2.6$ (the Ihara echo); the circularity ledger is complete (5 points); typed: W3-on-the-family = theorem + detector certificate, UNIFORM W3 IS the conjecture (Weil/Yoshida); no RH statement.                                                                                                                                                                                                                                  |
| v678_zero_gap_theorem.py     | The zero-gap theorem route – the best explicit UNCONDITIONAL “every interval $(t, t + H_{\min}(t)]$ contains a zero ordinate” theorem, documented and machine-verified on the Turing-certified comb: the $S$ -difference route with $S_{\text{bound}} = \min\{\text{Platt }  S  \leq 2.5167 \text{ to } 3.06 \times 10^{10}; \text{Trudgian 2014; Bellotti 2025 Cor. 1.5, floor } 4\pi \cdot 0.10076 = 1.26619\}$ ; verified on ALL 1999 certified comb gaps with 0 violations (min air 2.05 $\times$ , median 5.24 $\times$ ); adaptive bands $\delta_p = \kappa H_{\min}$ unconditionally hold $\geq \kappa$ ordinates (0 misses), so the v674 FRAME packet floor is now UNCONDITIONALLY positive ( $0 < B_p \leq V_p$ everywhere, median discount 0.037); the projection form does NOT stabilize adaptively either (typed residual with inverted expectation, v642/v662/v674 pattern: 6/6 ( $\kappa, C$ ) cells miss, $1/\log$ competitive, the minimizer stays single-mode at 0.898); THE QUANTIFIED PINCH: pointwise capture needs the explicit $S(t)$ constant $A_1 < 1/(4a_0) = 0.09017$ – the best cited constant 0.10076 misses by 11.7% at every height: the last W2 gap as a concrete named target inequality at the frontier of explicit analytic number theory, independent of RH; no marker move, no RH statement. |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v679_orbifold_continuum_os.py | The first continuum-OS slice of the seam orbifold B: correlator convergence with UNDERSTOOD rates – all six $\sigma$ -channel observables share the uniform raw rate $0.659 =$ the predicted Fisher–Hartwig $2/3$ branch, the $\tau$ -channel contrast measures $1.298$ vs $4/3$ (the channel law $\rho = 2(1 - \beta)^2 - 2\beta^2$ : rates understood, not fitted); after removing the ONE known FH term the extrapolated limits hit the CFT values at $< 0.05\%$ ; RP SURVIVES THE LIMIT: $\lambda_{\min}$ extrapolations bounded away from zero ( $\sigma$ $1.36 \times 10^{-3}$ , margin $7\times$ ; $\tau$ $2.80 \times 10^{-4}$ , margin $358\times$ ; mixed OS Gram $1.34 \times 10^{-3}$ , margin $31\times$ ; must-fail $\eta = -1$ flips); the cluster property holds in space AND Euclidean time ( $2\Delta = 0.222220$ vs $2/9$ at $8 \times 10^{-6}$ , cylinder/cosh forms pointwise, time gap = the $\varepsilon$ level at $0.99933$ ); the continuum characters $1/36, 1/9, 5/18, 4/9$ extrapolate at $10^{-12}$ with the measured $N^{-2}$ rate; typed honestly: OS axioms E0/E2/E3/E4 are MEASURED at the convergence level, E1 only discretely exact; the formal-limit remainder is named (uniform bounds, the GNS limit Hilbert space, operator convergence, R sectors, the B assembly) – GATE.QGEO does NOT move.                                        |
| v680_pinch_attack.py          | The pinch attack – the v678 11.7% pinch was a BOOKKEEPING ARTIFACT of one-sided capture: CENTERED CAPTURE DOUBLES THE THRESHOLD – the same zero-gap theorem spent two-sided ( $\text{dist}(t, Z) \leq H_{\min}(t - 26)/2$ , machine-checked on 6595 comb grid points, 0 violations) gives $A_1 < 1/(2a_0) = 0.18034$ , so the EXISTING Bellotti constant $0.10076$ passes with 79% headroom; the BEURLING–SELBERG MINORANT ELIMINATES $A_1$ from the counting chain entirely (loss = exactly $\pi/a_0$ mass + the explicit prime term $P = 2.534$ ; positive from $t^* = 1.11 \times 10^7$ ; no $A_1$ , no Platt cap, every window incl. $a \rightarrow \infty$ ; Weil identity on the comb at $9.0 \times 10^{-5}$ ); the LP stack lifts the asymptotic slope to $0.409$ per $\log t$ ; the literature block (search 2026-08-03): Bellotti–Wong $0.10076$ current, no successor $< 0.1$ , the Amberger S1 route ( $H^* = 1.5249$ ) stays dominated; the family reach $[10, 870]$ is closed verification-backed ( $\min s_{\text{tot}} = 0.0063$ over all 5 windows $\times$ lattice modes); the honest residue is NOT an $A_1$ threshold but the $O(1)$ coverage hole $(2500, 7.27 \times 10^6) = 3.46$ decades, typed with exact endpoints (no hole 2); no marker move, no RH statement.                                                                                    |
| v681_coverage_hole.py         | Closing the $O(1)$ coverage hole – the last gap of the pointwise W2 chain, $(2500, 7.27 \times 10^6) = 3.46$ decades, attacked on the three named paths and CLOSED with split typing: H3 is the main lever – the exact prime term (70 atoms, hierarchical Lipschitz certificate) pushes the abstract entry from $t^* = 1.11 \times 10^7$ down to $t_x = 1.53 \times 10^5$ (a factor 72.5); THE SCAN: 223 949 budget-certified Riemann–Siegel zeros (Gabcke remainder $0.127(t/2\pi)^{-3/4}$ , correctness 603/603 against the certified strip, mpmath 12/12, found/expected 0.99774) carry $(2515, 1.56 \times 10^5]$ with floor 0.01664; H1 honestly adjudicated: the Platt constant reproduces the hole boundary EXACTLY ( $t_{\text{on}} = 7.268 \times 10^6$ ) – it cannot enter; THE GAPLESS MAP: $s_{\text{tot}}(t; a_0) \geq 0.02259 \log(2 + t) - 0.5185$ for ALL $t \geq 10$ (comb [E] / scan [E, verification-consistent] / exact prime term abstract / Selberg uniform / Trudgian-2S unconditional from $1.74 \times 10^{25}$ ); THE W2 CHAIN AT THE ANCHOR IS COMPLETE: density (v669) + frame-Gårding (v674/v678) + pointwise (THIS map); typed open: family windows $a > 4.43$ (Selberg-only), $A_5(a)$ , the formal Mosco writeup, $a \rightarrow \infty$ ; the calibration history documented with the pinned duplicate bug; no marker move, no RH statement. |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v682_lk_split_theta.py    | The $L + K$ split and the relative form bound $\theta$ – OFFENSIVE 1 of the uniform-positivity program (PRIME.UNIFPOS.01): the non-circular split $B = L + K$ of the deployed window form constructed and MEASURED on the 5-window family $\times M$ -ladder. The honest negative: LK-SPLIT-DIES – the naive smooth split is STRUCTURALLY DEAD ( $\sup \theta$ per split: C1a 57.99 / C1b 14.84 / C3 16.18 / OS 11.34, all with indefinite $L$ ; the trivial spectral split $\theta = 0.998$ marks the pure-renaming floor), WITH the new positive find: the pencil maximiser sits at $t = \gamma_1 = 14.13$ (found from primes + digamma alone, zero-free), spike law $\mu_{\max} \approx 1 + 2\alpha/\Omega$ (median dev 0.127), $\theta$ grows $\sim 2.47\alpha$ – the deployed windows RESOLVE individual zeros; no smooth zero-free $L$ can satisfy $B \leq (1 + \theta)L$ with $\theta < 1$ ; surviving directions typed: one-sided or multi-resolution; no marker move, no RH claim.                                                                                                                                                                             |
| v683_rank3_functionals.py | OFFENSIVE 2 – the arithmetic bottleneck lifted EXACTLY: the prime influence on the load-bearing $2 \times 2$ Gram block is THREE linear functionals $S_j(a)$ of the comb weights (exact: assembly vs. independent sieve $4.9 \times 10^{-16}$ ). The circularity verdict: ALL entrywise routes are class (c) (T-A 0.026/0.133, T-PAIR 0.028/0.154, T-B $4.7 \times 10^{-7}/2.1 \times 10^{-6}$ – the razor-thin absorption margin demands $\sim 10^6 \times$ more than ANY per-entry input delivers; measured collective-cancellation gain 378–712), T-ENTRY class (a); the $K_M$ one-functional compression with the surviving det-level form $\kappa_{\max} = 0.0895 \ll 1$ . No Cauchy–Schwarz win: 99.7% of atoms violate pointwise ( $\det X_n < 0$ ); PRET-BREAKS: even a NONNEGATIVE pretentious measure drives $\det S < 0$ – positivity + size of the comb alone cannot prove $\det S \geq 0$ ; no marker move, no RH claim.                                                                                                                                                                                                                                   |
| v684_rank3_zeroside.py    | OFFENSIVE 2b – the $K_M$ functional through the explicit formula: the exact zero-side identity $D = \text{BND} - \text{RTERM} + \sum z(\gamma)$ (residuum $\leq 6.9 \times 10^{-6}$ on the reference window; 22 491 zeros to $T = 2 \times 10^4$ budget-certified, RvM checkpoints, Lehmer pair resolved) + the per-zero envelope $ z(\gamma)  \leq 2C_G/\gamma^2$ give the UNCONDITIONAL $\kappa_{\text{unc}} = 0.039\text{--}0.190 < 1$ on all 5 windows – ZONE-PASS class (a): $\det S > 0$ from the identity + computed zeros + NO other arithmetic input. The pretentious escape is quantifiably blocked: a zero AT the dual points would couple with $ z_{\text{cf}}  = 642$ ( $3.2 \times \det M$ ); the largest REAL coupling $ z(14.13)  = 1.01$ is damped $\times 634$ by the classical zero-free zone (Gram/Hutchinson, unconditional); $\beta = 1/2$ cited to $3 \times 10^{12}$ (Platt–Trudgian), beyond priced ( $\leq 3.3 \times 10^{-6}$ ). Honest: T-B (absorption $\sim 2 \times 10^{-5} \det M$ ) untouched – this module closes the SIGN of $\det S$ ; the named missing quantifier: uniformity ( $\rightarrow$ v685); no marker move, no RH claim. |
| v685_rank3_uniformity.py  | OFFENSIVE 2c – CLOSING THE UNIFORMITY QUANTIFIER (the theorem-level result of the rank-3 chain): the symbolic envelope $K_{\text{env}}(a, h) \leq 0.9798 < 1$ for ALL complete windows $a \geq 3.434$ (62 of 67; tightest at $h = 142$ ) $\Rightarrow \det S \geq (1 - K_{\text{env}}) m' > 0$ UNCONDITIONALLY; the 5 small windows carry finite certificates $\kappa' = 0.072\text{--}0.108 < 1$ – SURFACE-CLOSED: $\det S > 0$ on the ENTIRE declared surface. Architecture: exact $U_0$ rebooking $M' = M + \Delta$ (kills the 34:1 cancellation problem by definition shift), term-algebra-exact weights (sympy), mean-value envelopes (give-away $\leq \times 2.28$ ), $\det$ minorant $m'/\det M' \geq 0.949$ ; inputs typed [E] + [A-cited: $\gamma_1 = 14.13$ Gram/Hutchinson, $\beta = 1/2$ to $3 \times 10^{12}$ Platt–Trudgian]. HONEST SCOPE: a SURFACE theorem on the declared family, NOT uniform in all $a$ ; T-B (the absorption margin, prob:R1) stays OPEN; no RH statement.                                                                                                                                                                          |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v686_geometric_sos.py          | <p>OFFENSIVE 3 – the first feasibility strike at the geometric positivity source (<math>'Q = \sum_j  l_j(v) ^2</math> from TFPT geometry, without <math>\zeta</math>, without zeros'): THE CIRCLE-FREE TRACE: <math>\Lambda_{\text{geo}}(n) = \Lambda_L(n)/(1+n^3) \equiv \Lambda(n)</math> EXACTLY for <math>n = 2.512</math> (worst rel dev <math>7.1 \times 10^{-31}</math>) – the von Mangoldt data reconstructed from E8 shell counting alone (Satake <math>\{1, p^3\}</math> from the COUNT <math>a(p) = r(2p)/240</math>, Hecke recursion on raw counts; K1 PASSES). Cover-SOS on dim 3 canonical; the miniature window form validated against the comb (worst <math> \text{ratio} - 1  = 1.6 \times 10^{-6}</math>) and PD there (min eig <math>1.9 \times 10^{-7}</math>; the atom layer alone indefinite – the K5 boundary named exactly); K4 FIRES HONESTLY: the certificate is window + comb, nothing more. THE GAP: top-3 eigenvalue mass only 91.4% – a pointwise rank-3 transport cannot suffice; <math>H</math> (bundle) and <math>\Phi</math> (transport) missing, named as M1..M4 of contract PRIME.GEOMSOS.01 with kill criteria K1–K5; no marker move, no RH claim.</p> |
| v687_extremal_kernel.py        | <p>The extremal test kernel (contract PRIME.W2.EXTREMAL.KERNEL.01) – can an optimized positive band-limited kernel of the same exponential type lift the W2 capture threshold above <math>A_1 = 0.10076</math>? STRUCTURE THEOREM: NO – the pinning certificate <math>2\pi F_{a_0}(m\pi/a_0) = 0</math> EXACTLY (sympy) forces the Fejér nodes on every admissible <math>K \Rightarrow z_1(K) \leq \pi/a_0 \Rightarrow \theta_{\text{req}}(K) \leq 1/(4a_0) = 0.0901684 &lt; 0.10076</math> (strict interval certificate); the collapse <math>\varphi_k(m\pi/a_0) = \delta_{km} a_0/2\pi</math> forces <math>c_1 = \dots = c_N = 0</math> in every degree, full class via Shannon sampling: feasible set = <math>\{\lambda \text{ Fejér}\}</math> – ZERO free directions. The LP ladder confirms (<math>z^*(N) \rightarrow \pi/a_0 \pm 5 \times 10^{-4}</math>, coefficient collapse <math>\leq 10^{-6}</math>); the false-PASS trap documented (envelope violators can fake <math>\theta_{\text{req}} &gt; 0.10076</math>); W2 STAYS CLOSED via the v680/v681 route – this negative closes the kernel-optimization escape; no marker move, no RH claim.</p>                                |
| v688_interpolation_detector.py | <p>The alias-interpolation falsifier (contract PRIME.W3.INTERPOLATION.01) – the calibrated off-line detector of v677 S3 turned into a COMPLETE, explicitly constructive falsifier: the in-kernel matched filter detects EVERY off-line quadruple from <math>2\alpha\delta \geq \Xi_{\text{up}} = 1.974</math> (<math>C_{\text{up}} = 0.987</math>) on all 14 in-band cells – efficiency median 0.500 vs the v677 mode map = a FACTOR 2 improvement; masking FULLY adjudicated: 46 broken + 2 PROVABLY sub-resolution = 48/48 (construction-open: 0) via the multi-construction menu (pinned target, second target, <math>2 \times 2</math> span, phase-kernel span, complex-point pin, local Ritz). Master identity verified in sine form (<math>9.5 \times 10^{-13}</math>); the alias-interpolation lemma formulated with measured constants (<math>C</math> median 0.499) and the lemma building blocks named (v678 node separation, Beurling–Selberg/Vaaler via v680/v681); inverted firewall (zeros = declared input: the falsifier KNOWS the configuration – that is its content); no RH statement, no marker move.</p>                                                               |
| v689_gaussian_code_bridge.py   | <p>The Gaussian code bridge (review contract E8.GAUSSIAN.CODE.01) – <math>E_8(\mathbb{Z}[i])/(1+i) \cong \mathbb{F}_2^4</math> EXACTLY (Smith normal form; standard model: <math> \det  = 256 = 2^8</math>, SNF <math>[1^4, 2^4]</math> – presentation-independent): the <math>240 = 15 \times 16</math> census (the zero class PROVABLY empty, each class = 4 of the 60 G31 lines), <math>\sigma = c^4</math> acts on the quotient as the RM(1,3) family permutation (3-cycle <math>F_1 \rightarrow F_2 \rightarrow F_3</math> + fixed anchor, exactly the v638 info-bit action), residual gauge = the order-18 centralizer in <math>GL(4, \mathbb{F}_2)</math>, coordinate block = the <math>F_1 + F_2 + F_3</math> label. All must-fail controls fire: the naive placement keeps the census but loses the <math>\sigma</math> semantics (off-lattice witness); a non-<math>\pi J</math> placement has no <math>\mathbb{Z}[i]</math>-module; <math>\mathbb{Z}[i]^4</math> trivializes (16 vectors, 11 classes empty); exact integer/Fraction arithmetic; no marker move.</p>                                                                                                              |

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| Script                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v690_quartic_half.py   | The quartic invariant restriction (review contract E8.QUARTIC.HALF.01) – is G31 the holomorphic $\mu_4$ shadow of the E8 compiler? ALIVE, with the honest split: the vanishing half $\{2, 14, 18, 30\}$ is TRUE but TRIVIAL ( $\mu_4$ orbit factor, proven symbolically: $W(J\alpha) = -iW(\alpha) \Rightarrow$ every $J$ -orbit contributes $t^d(1 + (-i)^d + (-1)^d + i^d) = 0$ for $d \not\equiv 0 \pmod 4$ – holds for 6, 10, 22, 26 too). THE SUBSTANCE: $F_8 , F_{12} , F_{20} , F_{24} $ are G31-invariant (60/60 mirrors exact), ALGEBRAICALLY INDEPENDENT (exact Jacobian $\neq 0$ ), degrees Molien-unique with product $46080 =  G31  \Rightarrow$ by Chevalley [cited] BASIC INVARIANTS: the holomorphic restriction of the $W(E_8)$ ring generates the FULL G31 ring. The Jacobian carries the 60 mirror lines (degree $60 = \sum(d_i - 1)$ , point-verified 60/60; full factorization = cited theorem, named residual); controls: $J' = RJR^T$ breaks the informative half, the root-preserving $J''$ keeps everything (the theorem is about the $W(E_8)$ -conjugacy class); no floats; no marker move.                                                                                                                                                                                                                                                                                                                                                  |
| v691_hecke_sos.py      | Hecke-equivariant sum of squares – OFFENSIVE 4 (contract PRIME.W3.HECKE.SOS.01): the search for the prime-local factorisation $A = B^*B + P$ ( $P \succeq 0$ ) with no zero input; six verbatim acceptance criteria adjudicated. THE IHARA LAB (H2, centre): the target factorisation EXISTS EXACTLY there – $A_{\text{IH}} = G_C + G_S$ with $B =$ Chebyshev columns of the Hecke operator (recursion, no Cholesky, no spectrum), $P = G_S$ the closed defect Gram; $P \succeq 0 \iff$ Ramanujan – the RH analogue sits in ONE operator inequality. INDEX LEMMA: the deployed $\zeta$ window form is, after index reversal, EXACTLY the sine/defect HALF of the canonical cos/sin split – precisely the part whose positivity in the proven lab IS the RH analogue (the cos half is unconditionally SOS). The Euler mechanism MEASURABLE (H1): commensurable fake primes break EXACTLY on the resonance lattice $2\pi j / \log 2$ (median dev 0.0000, 12/12) and DEEPER than matched jitter ( $\times 1.98$ ); H3: the deployed lag sequence is NOT a positive moment sequence – positivity lives strictly in the odd sector (coherent with H2); controls: Epstein fails as demanded, the Ihara trio passes, scramble breaks. THE Z1 GAP NAMED EXACTLY: a self-adjoint geometric operator (Hilbert–Pólya) is missing; Z2 would be RH itself – the factorisation LOCALIZES RH ( $\rightarrow$ PRIME.Z1.OPERATOR.01, Offensive 5 running); no marker move, no RH claim. |
| v692_rank3_lockgram.py | Typing T-B through the zero side – OFFENSIVE 2d: the deployed $2 \times 2$ lock block as an almost-collinear Gram matrix – $\hat{A}_2 = G_Z + P + \text{tail}$ exactly (per entry): a POSITIVE zero-Gram $G_Z$ (per-zero psd, Cauchy–Schwarz manifest) plus a psd RANK-1 pole layer $P = c_P s s^T$ (closed form). THE MARGIN IDENTITY $\det(G_Z + P) = \det G_Z + c_P(s_\perp^T G_Z s_\perp)$ : the razor-thin T-B margin IS the TRANSVERSE ZERO MASS (a sum of squares; $\gamma_1$ carries 3–61%). Superadditivity certificate: on-line zeros can only INCREASE $\det$ ( $2 \times 2$ psd; Platt–Trudgian $3 \cdot 10^{12}$ cited) – the truncation is safe-side. THE TYPING: ‘E does not tip’ is a QUANTIFIED PARTIAL-RH TAIL statement – neither a 2c-envelope question nor a W3 renaming: the only threat is off-line quadruples above the cited height. Census: 8/15 windows close at $3 \cdot 10^{12}$ alone, 13/15 under the named density route; the remainder carries exact factors (worst 1428) and finite $T^*$ up to $1.1 \cdot 10^{16}$ ; no marker move, no RH claim.                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v693_rank3_density_close.py         | Closing T-B(block) with cited explicit zero-density bounds – OFFENSIVE 2e. Citation ledger exact: [A1] Platt–Trudgian 2021 (RH to $3 \cdot 10^{12}$ ), [A2] Hasanalizade–Shen–Wong 2022 (explicit RvM count), [A3] functional-equation halving, [A4] the explicit Ingham form arXiv:2507.15184 (2025, valid from exactly $3 \cdot 10^{12}$ ); the refined safe-side envelope (sinh difference pairing, $V(0) = 0$ , on-line subtraction): PEN drops $\times 6.5..14.4$ . CENSUS: 60/70 complete windows close UNCONDITIONALLY-MODULO-CITATIONS; exact remainder: 9 windows with $T^* \approx 1.0..3.0 \cdot 10^{13} + h = 5690$ at $8.5 \cdot 10^{14}$ ; extension beyond the family grows $\sim e^{3.74a}$ (the declared surface is a FINITE family of 70 windows – window-by-window closure is a valid uniformity route); Simonič honestly listed as trivial in range. Honest scope: this closes the determinant of the $2 \times 2$ LOCK block – full W3 window positivity remains the conjecture; no marker move, no RH claim.                                                                   |
| v694_interpolation_lemma_closure.py | Closing the two named building blocks of the alias-interpolation lemma (PRIME.W3.INTERPOLATION.01 follow-up). (iii) RETENTION: exact projection identity $\text{Im } \Phi_x(z_0) = \ (I - \Pi)m\ ^2$ , closed-form $O(k^2)$ certificate, $r_f \geq 0.548$ on the family surface ( $\geq 0.266$ adversarial); BOUNDARY CASE discovered: a straddling sub-resolution real pair collapses the pinned filter but the detector still fires via the barred $k = 0$ branch. (v) SEPARATION LAW FORM-CORRECTED: the pure form $\alpha^* = C'/\Delta\gamma$ is REJECTED, the ADDITIVE law $\alpha^* = C_{\text{cell}}/\delta + C''/\Delta\gamma$ stands ( $C''_{\text{med}} = 0.494$ , $C''_{\text{up}} = 0.590$ , zero exceptions in 97 tests, adjudication-tight); the two open parent configs detect at $\alpha^* = 7.90$ ; $C''(n=3) \approx 2 \times$ pair value = the declared residue. Residues: explicit Ingham constant chain (classical bookkeeping), Dirichlet-kernel overlap (elementary, unwritten), the $C''(n)$ induction; an off-line zero is the HYPOTHESIS of every statement; no RH claim. |
| v695_z1_trace_operator.py           | The Z1 search, OFFENSIVE 5 (contract PRIME.Z1.OPERATOR.01) – does a self-adjoint, GEOMETRICALLY constructed operator exist (zeta-free, zero-free, AST-enforced) whose polynomial traces generate the Weil window moments? Candidate census round 1: the SEAM (v621–v623) DEAD as Z1 spectrum (residuals $\sim 1$ , AP phases, $1/D$ drift); the ORBIFOLD (v679) DEAD (null-calibrated: the comb carries no AP structure beyond chance); the E8 COUNTING ROUTE DELIVERS THE MEASURE EXACTLY – atoms from the shell recursion ( $n \leq 20000$ , Satake normalizer = lattice data, deviation $0.0e+00$ ), arch + pole via the closed $\Gamma$ duplication split, $7/8$ target spikes ( $q = 0.000$ , window covariance); must-fail: without the Satake projection the form breaks. Precision: the deployed symbol $\sigma_B$ is SIGNED but the comb sequence $c + \text{pole}$ is POSITIVE-FEASIBLE – the positive spectral object EXISTS, the signedness IS the pole subtraction; kill criteria K-A..K-D preregistered; no marker move, no RH claim.                                                  |
| v696_z1_jacobi.py                   | Z1 canonical construction, OFFENSIVE 5b – Verblunsky / CMV / Jacobi from the positive comb sequence $c + \text{pole}$ . THE CANONICAL OPERATOR EXISTS: CMV unitary ( $367 \times 367$ ) + Jacobi self-adjoint (Wheeler on Chebyshev moments), machine-verified back-direction: ALL moments reproduced, worst rel dev $2e-13$ – Z1 is now PURELY the closed-form question for the coefficient sequences. NO closed law (the two named first-order laws fail, $r \approx 0.05/0.10$ ; the $h$ -ladder fails, $r_{\text{env}} = -0.13$ ), BUT the counting signature is PRESENT: feature positions on prime-power slots ( $q = 0.000$ ), amplitude-mass link $r_{\text{log}} = +0.76$ , positivity LOAD-BEARING (smooth/scramble lose PD at $k \approx 170/171$ , true masses PD to 2865) – the renaming clause is explicitly NOT confirmed; Ihara anchor: Kesten–McKay exact, girth corrections; contract note PRIME.Z1.OPERATOR.01; no marker move.                                                                                                                                                   |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v697_z1_uvarov.py         | The slot-amplitude transfer law, OFFENSIVE 5c – exact point-mass / single-lag insertion machinery for the Z1 coefficient response. DUALITY PROVEN: the atoms are LAG insertions, NOT orthogonality-measure point masses (those would sit at the zeros = the renaming direction, firewalled); THE EXACT TRANSFER IDENTITY $\Delta\alpha_{m_0-1} = w_1/E_{m_0-1}$ (dev 5.6e-17; $E$ = Toeplitz det ratio = $\ \Phi\ ^2$ , the position-free Christoffel object), zero response below the slot (locality exact); THE INVERTED STABILIZATION LAW: the $\Gamma$ flow PREDICTS the counting masses to $\sim 10\%$ (median ratio 1.026, IQR 0.138, corr 0.86) – the flow knows the masses, not vice versa. Composition sequential-exact: Z1 = canonical CMV/Jacobi of arch+pole + ONE single-lag insertion per prime power, just-in-time positivity (prefix margins $\geq 0$ ), every prime power individually load-bearing (removals break PD); contract note PRIME.Z1.OPERATOR.01; no marker move.                                                        |
| v698_z1_flow_recursion.py | Positions, residual, E-transport, OFFENSIVE 5d – is the whole prime comb forced by the arch+pole flow? POSITIONS FORCED: slot windows 0.5–2 cells (6/6 sharp), jitter null 30/30, adversary deficit –516 lags – the prime-power positions are pinned by the flow, not fitted; SHOOTING: too-light and too-heavy insertions diverge through OPPOSITE channels ( $k \rightarrow \pm 1$ ), bisection on $\text{sign}(k)$ recovers the counting mass at the true position to median $ 0.11 \%$ – the Z1 recursion is a SHOOTING PROBLEM; slot identity 9.4e-14 (5c re-verified). HONEST NEGATIVES: autonomous reconstruction fails at greedy saturation (1/6; lookahead missing), the residual is noise-like, the E-transport is recursive – PRIME.Z1.OPERATOR.01 stays OPEN: the continuum reading is missing; the zero read is a firewalled diagnostic only (AST-confined); no marker move, no RH claim.                                                                                                                                               |
| v699_cone_dynamics.py     | Cone dynamics – the review-7.3 question (‘use the Lorentz form DYNAMICALLY instead of statically’) executed on the prime-front determinant coordinates $y(a) = (S_{11}, S_{22}, S_{12})(a)$ with $J_{\det}$ of signature (1, 2). THE HONEST KILL: prime updates are TRANSLATIONS in $y$ -space (linearity verified exactly: the per-prime increment is $y$ -independent) – no multiplicative cone-preserving semigroup; $\sim 99\%$ of increments are SPACELIKE (cone-violating fractions 0.009–0.014), $\ell$ is NOT monotone on the true path – no Perron–Frobenius / Lorentz-boost reading of the prime flow survives. Side finding (alive): the LEAF functional separates scramble/Epstein where det is blind (all 3 scramble seeds + Epstein separate); AST zero-firewall, deterministic; the static cone results (v624/v627) stand untouched; no marker move, no RH claim.                                                                                                                                                                     |
| v700_orbit60_census.py    | The ORBIT60 census – the 60 $\mu_4$ -lines of E8 as a combinatorial object: quotient graph, $\sigma$ /clock/Coxeter dynamics on the ORBIT SPACE, and a STRICT park test for the cascade $60 \rightarrow 6$ . All five routes adjudicated: (i) degree rules DEAD (all quotient graphs regular); (ii) $\sigma$ structure DEAD (orbit sizes $\{3, 1\}$ , no 2-blocks); (iii) THE INVOLUTION WITNESS DOES NOT EXIST: the full involution census of the line group LG (order 11520; 375 involutions, 21 classes) shows NO element of cycle type $2^{27}1^6$ (fixed-line census $\{0:120, 4:180, 12:15, 16:60\}$ ); (iv) W(E8) Coxeter does not descend; (v) the MUST-FAIL numerology control fires: rule-free lexicographic pair removal reproduces $60 - 2n$ in exactly 27 steps, $> 10^{70}$ equally ‘successful’ sequences – the counting carries no structure. A5 one-liner: order-5 elements exist (2304) but no visible natural A5 action ( $\langle \sigma, \text{order-5} \rangle$ has order 960); PARKED per the review protocol; deterministic. |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v701_big_picture_hunt.py          | The BIG-PICTURE HUNT – cheap machine reads for the question ‘does the missing rest of the RH program already sit in the corpus?’. THE KEYSTONE [E]: the deployed window form $B$ is EXACTLY the sine-moment form of $(c + \text{pole})$ plus a rank-1 pole square – one identity gluing the v695 measure route to the v677 master identity; the three named gaps collapse onto ONE (defect positivity modulo the operator question). The triplication identity [E]: the zeta arch density is the mean of three digamma channels on the $\zeta_{12}$ period grid (arguments $1/12, 5/12, 3/4$ ), $\mu_3$ unique. Honest kill: the V5 fixpoint numerology is DEAD (PSLQ empty). No zeros read anywhere. |
| v702_z1_lookahead.py              | OFFENSIVE 5e – the lookahead problem: autonomous zeta-free reconstruction of the prime comb from the $\Gamma$ flow. FLOW-VERIFIER-NOT-GENERATOR: the full true comb is flow-VERIFIED ( $913 \geq 898$ , every fake dies $\geq 184$ lags earlier), but both prescribed lookahead routes (depth-2 beam search with adaptive horizon-3 deepening; global DP) reach only 2–4 slots autonomously; the per-slot information demand $g \sim 5$ –12 bits is information-theoretic and parameter-robust. The 5d state stands; PRIME.Z1.OPERATOR.01 stays OPEN.                                                                                                                                                 |
| v703_chain_tolerance_scaling.py   | S-A, THE CIRCULARITY DECIDER – how sharply does the Z1 flow-induction constrain the counting masses? TOLERANCE-RH-GRADE: the per-step mass tolerance an autonomous induction would need shrinks to RH grade (OVER-RH on the fine windows) – the tolerance induction as a proof route is DEAD, honestly. The positive structure: the Levinson recursion is an UNSTABLE SHOOTING problem – generic perturbations diverge geometrically, the true comb is the unique bounded trajectory.                                                                                                                                                                                                                 |
| v704_chain_mass_law.py            | S-B, THE G1 EXACTNESS HUNT – does a CLOSED formula (no per-slot fit, no per-window scalar) predict the counting masses from the flow? G1-OPEN: the best closed law is C2 (zero reflection coefficient at the NEXT Levinson order), median 0.9947, IQR [0.9853, 1.0086] – but $\max  \text{ratio} - 1  = 0.3130$ on the soft slots, far above the 0.1% bar. The residual strata are declared: soft $ a_{bg} $ slots carry the misses (0.049), mid/tight strata 0.010/0.006.                                                                                                                                                                                                                            |
| v705_chain_deck_sector.py         | S-C, THE V3 OPERATOR TARGET – the three digamma channels of the zeta arch density ARE the three deck sectors of the 48-site NS lift: exact tower traces of the v623 deck sectors with the v628 twists $\{1/6, 1/2, 5/6\}$ , channel-by-channel deviation $3.4 \times 10^{-16}$ , the relative scalar FORCED to 1 (not fitted), must-fail (wrong twist set) off by $\times 82$ – the geometry anchor of the chain arc: the arch layer of the Weil measure is a deck-sector trace of the seam orbifold.                                                                                                                                                                                                 |
| v706_chain_weyl_mass.py           | S-D, THE WEYL CHARACTERIZATION OF G1 – is the exact mass law an evaluation of the m-/Schur function of the background operator? WEYL-NUL: the point-evaluation reading is FALSIFIED (K2 median 0.3993) – WITH positive finds: (i) the flow $\alpha$ ARE the Schur parameters of the background operator ( $10^{-40}$ ); (ii) the background breakdown sits EXACTLY between the $n = 2$ and $n = 3$ cell; (iii) the bare $\Gamma$ flow knows $\log 2$ to 0.1–0.6%.                                                                                                                                                                                                                                     |
| v707_chain_section_edge.py        | S-E, THE CLOSED FORM OF THE SECTION-POSITIVITY EDGE – EDGE-EXACT-SELECTION-OPEN: the edge formula is a RESOLVENT IDENTITY (Christoffel–Darboux form), exact to $3 \times 10^{-13}$ [E]; the full-ladder edge pins the mass to $2.6 \times 10^{-4}$ . The selection stays open: the true mass sits at the just-in-time corridor position median 0.534, IQR [0.511, 0.559] – systematically NEITHER at an edge NOR at the midpoint: ‘explain the $\sim 0.53$ ’ is the named open question.                                                                                                                                                                                                              |
| v708_chain_position_functional.py | S-F, THE POSITION FUNCTIONAL – which functional fixes the corridor position $\sim 0.53$ ? FUNCTIONAL-OPEN: the best is the LEVINSON ENERGY EXTREMUM $E_{N-1}$ : median 0.9986, max deviation 11.7% – and it TRANSPORTS across windows without a scalar (the strongest selection signal of the arc); none reaches the 1% uniform bar; the outliers sit EXACTLY on the prime powers 16/64/81 – the same cells the Hecke mechanism (v710) heals.                                                                                                                                                                                                                                                         |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v709_chain_zero_layer.py         | S-G, THE LAYERING HYPOTHESIS (diagnostic) – is the S-F selection residual a zero-comb layer? LAYERING-REJECTED: in all tested projections the residual carries NO zero signature (best corr 0.042 = the scramble level); $\sim 64\%$ of the residual is window geometry. The discipline: a TEMPORAL SHA256 FIREWALL – the residual is hashed and frozen BEFORE any zero is read. An honest diagnostic null for the record.                                                                                                                                                                                                                                                                                                                                                      |
| v710_chain_two_stage_hecke.py    | P1 KILL TEST – the two-stage Hecke selector for the G1 residual. THE MEMO KILL FIRED (max overall deviation 6.15% > 2% bar) – the composite closed law is dead as declared; BUT the Hecke mechanism heals exactly the prime-power cells 11.7% $\rightarrow$ 0.53%, with CIRCLE-FREE channel detection from the E8 counting [E] (Satake from shell counts, no zeta input). The named remainder: the first-birth selection on coarse windows (v711); $G1 = [\text{E-edge}] + [\text{E-Hecke}] + [\text{M-rate}]$ .                                                                                                                                                                                                                                                                |
| v711_chain_firstbirth_scaling.py | P1b – fit-free measurement of the first-birth selection-error scaling with window resolution. CONTINUUM-SELECTOR-MEASURED: the first-birth error CONVERGES under refinement – promotion-run $\theta = 1.851$ ( $R^2 = 0.692$ ), ladder halves 2.736/1.520 both positive, 0/30 non-converging primes, the finest 6 windows carry 0 degenerate-horizon slots; preregistered bars $\theta \geq 0.5$ , $R^2 \geq 0.6$ (worker run $\theta \sim 1.25/R^2 \sim 0.90$ – spread far inside the bars); a one-decade law, extrapolation honestly typed [M]; the typed decomposition $G1 = [\text{E-edge}] + [\text{E-Hecke}] + [\text{M-rate}]$ is complete.                                                                                                                              |
| v712_rank3_transverse_deck.py    | P4 MAIN STRIKE – transverse scalarization $\times$ deck triplication for the ten open T-B windows at citation height $3 \times 10^{12}$ . THE P4 MOVE: the v692 margin identity is SCALARIZED along the transverse direction (the penalty pays only the transverse mass) – median improvement $\times 4.23$ on the open ten – and the census closes 9 of the 10: 69/70 complete family windows close at the cited $3 \times 10^{12}$ (unconditional-modulo-citations); the remainder is EXACTLY ONE window, $h = 5690$ , with $T^* = 1.6 \times 10^{14}$ (down from $8.5 \times 10^{14}$ ). Honest negative: the deck lever is DEAD (triangle inequality: channel sups can only lose); zeros $\leq 2 \times 10^4$ enter the MARGIN only.                                        |
| v713_l1_montage.py               | L1-MONTAGE – the first explicit assembly of the CONTINUUM candidate of the Z1 operator and the machine test of its declared limit gates.<br>L1-ASSEMBLED-MEASURED: the candidate is MOUNTED (S-C deck-sector arch + 5c lag atoms + v695 pole subtraction): cover limit $6.2 \times 10^{-8}$ , point masses converge at the $D^2$ rate, exactly ONE UV slot; the 5b negative was a NORMALIZATION ARTIFACT (rescaled, the ladder stabilizes); the divergence sits EXACTLY at $s \leq 1/2$ – the critical line as the natural boundary of the construction, honestly typed as a consistency signature; the missing THEOREMS named (self-adjointness, trace identification, positivity) – named, not claimed.                                                                       |
| v714_moonshot_hecke_groupoid.py  | MOONSHOT first slice – the E8 Hecke groupoid tower: is the ATOM LAYER of the Weil measure groupoid-internal? MOONSHOT-GO: (i) the primitive degrees of the $\mathbb{Z}[i]$ -E8 Hecke tower are EXACTLY the Gaussian prime norms (maximality census; no prime projector anywhere – the primes come OUT of the tower); (ii) log generator + cell normalizer DE-DIVISORIZE the tower traces at $4 \times 10^{-13}$ ; (iii) the $\sigma$ descent reproduces 100 atom positions, 100 masses and 368 moments at deviation 0.00. Declared ingredients: the $\sqrt{n}$ half-density + the $\sigma$ descent.<br>HONEST FENCE: the finite counting facts of this slice are E8-UNSPECIFIC – the specificity question is STAGE 2 (which also owns the arch layer, here still a direct sum). |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v715_qgeo_car_continuum.py    | QGEO-CAR – the quasi-free CAR route to the formal orbifold continuum limit (GATE.QGEO), first slice. QGEO-CAR-RATES-SUMMABLE: all 6+1 deck/twist channels are Schatten/shadow-norm Cauchy on $\varepsilon$ -separated configurations ( $N$ up to 3072/6144) with geometrically summable rates; the FH renormalization is FIT-FREE LATTICE-DERIVED (Barnes G, $8.8 \times 10^{-7}$ ); the RP cone does not degenerate along the ladder; all four kill criteria are clear. The named remainder: the majorant lemma (L1)–(L3) with constants + the classical proof route – named, not claimed: GATE.QGEO does NOT move.                                                                                                                                                                                                |
| v716_moonshot_arch_glue.py    | MOONSHOT stage 2 – the shared arch-sector test: does the archimedean place of the SAME $\mathbb{Z}[i]$ -E8 commensurability groupoid deliver the arch tower of the Weil measure? MOONSHOT-STAGE2-GLUED: one object, ONE normalization – the free 3-scalar LSQ ( $\kappa_{at}, \kappa_{ar}, \kappa_p$ ) returns (1, 1, 1) to $< 10^{-12}$ on all 5 windows; the $1/4$ in $\operatorname{Re} \psi(1/4 + i\tau/2)$ IS the $\mu_4$ fixed-sector offset (DERIVED); $\log \pi$ enters only through the declared self-dual UV cell; lift ladder $s^* \rightarrow 1$ at rate $\sim N^{-2}$ ( $ s_\infty^* - 1  = 2.4 \times 10^{-7}$ ); must-fails: $\mu_2$ tower 0.326, wrong $\mu_4$ class 1.000, wrong deck twists 0.474 – E8 becomes forced AT THE GLUING, not at the finite places.                                    |
| v717_moonshot_state.py        | MOONSHOT stage 3 – the positivity STRUCTURE of the glued object: is the Weil functional a STATE on the groupoid algebra? STATE-ON-TRUNCATIONS: on every truncation CONSTRUCTIVELY a GNS vector state (Levinson PD, min headroom 0.76); the naive $\delta_e$ tower reading fails by $ \Delta /W3\text{-margin} = 104.1563$ – the trace-formula boundary IS the localized RH substance; KMS $\beta = 1$ (BC-critical) measured (half-density exponent $-1/2$ exact); controls calibrate both outcomes: Ihara an exact state, Epstein breaks (Levinson breakdown depth 49/62); missing theorems named; no RH claim.                                                                                                                                                                                                    |
| v718_moonshot_spectral.py     | MOONSHOT stage 4 – the spectral identification: do the eigenvalues of the truncations of the glued object converge to the zeta zeros? SPECTRUM-CONVERGES-MEASURED: $h = 1433$ hits 100% of the 377 frozen zero targets at tol 0.25 (97.1% at 0.10, 84.1% at 0.05); rate $-1.610$ ; GNS $K$ -ladder monotone $0.4678 \rightarrow 0.0003$ ; predictions SHA256-FROZEN before the declared target load; controls: Ihara exact at finite $K$ , scramble = Levinson breakdown (depth 61); GUE light $\langle r \rangle = 0.6178$ vs 0.6189; a MEASUREMENT, not a theorem; no RH claim.                                                                                                                                                                                                                                   |
| v719_moonshot_traceformula.py | MOONSHOT Satz-1 slice – the trace formula of the glued object, term by term against the classical Weil formula. SATZ1-LEDGER-EXACT: the finite trace formula is EXACT Gauss quadrature on every truncation (max rel dev $1.7 \times 10^{-11}$ ); the term dictionary CLOSES: pole == $\int \tilde{a} 2 \cosh(u/2)$ ( $1.7 \times 10^{-13}$ ), atoms == the classical prime term ( $1.8 \times 10^{-15}$ ), arch == $-\int \rho \tilde{a}$ ( $3.4 \times 10^{-16}$ ); $\Delta \rightarrow \Gamma'/\Gamma$ term + prime term classically ( $\sigma = 1.000161$ , $c = -10^{-4}$ ); E8 census $\{\mathbb{Z}^8, E8\}$ (Mordell 1938): ONLY the Gaussian E8 glues ( $\kappa = 1$ , residual $8.5 \times 10^{-17}$ ); everything geometric is classical – the new content sits EXCLUSIVELY in the convergence statements. |
| v720_moonshot_k2k3.py         | MOONSHOT K2+K3 slice – the two convergence theorems of the ledger list, measured and collapsed onto classical analysis. K2K3-COLLAPSED-CLASSICAL: K3 collapses to measured tightness (compacta Cauchy, band edge = the smooth $\omega$ density, $p_0 = 0.981 \log(1/D) - 1.306$ sigma-finite/vague); K2 is standard analysis on $s > 1/2$ (unconditional Chebyshev majorant $c_{RS} = 1.03883$ , tail law slope $-(s - 1/2)$ exact, discretization $\sim CD^2$ ); the boundary EXACTLY at $s = 1/2$ (Mertens slope 1.9994 == 2); the hard substance concentrates in (L1) line identification + (L2) K1 node convergence; no RH claim.                                                                                                                                                                               |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v721_k1_node_capture.py        | K1 slice – the node-capture lemma behind the stage-4 spectral measurement, built from classical quadrature theory and honestly adjudicated.<br>K1-LEMMA-CAPTURE-ONLY: Markov-Stieltjes sandwich (0 violations) + SYMMETRIZED Christoffel bound everywhere + the closed capture condition $Lm(t+d) - RM(t-d) > 0$ ; every synthetic peak (incl. a merged pair) captures; BUT the certified radii $d_{pred} \sim 1.4\text{--}2.4\tau$ sit $\sim 900\times$ above the measured node errors – the lemma carries the CAPTURE, not the super-resolution: the precision is trace-formula content; off-line honesty: the lemma presupposes $\mu \geq 0$ – capture lemma and the v688 falsifier are two sides of the SAME positivity premise.                                                     |
| v722_phys_ramified_ns_r.py     | PHYS moonshot slice (T2 / $G_{net}$ ) – is the Ramond projection $(1+i)$ -adic?<br>RAMIFIED-SEES-NSR: $(1+i)L$ lies in the even/NS sublattice ( $\mathbb{Z}$ -basis proof), the 240 roots fall into $15 \times 16$ classes with PURE parity (7 NS + 8 R), parity is a homomorphism $\mathbb{F}_2^4 \rightarrow \mathbb{F}_2$ – the NS/R grading IS the parity character of $E8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$ ; the first exact witness for the groupoid reading of $G_{net}$ (the ONE ramified edge of the tower is the Gaussian prime $(1+i)$ ); controls fire: the $\mathbb{Z}[i]^4$ census is NOT $15 \times 16$ , scrambled labels break purity.                                                                                                                            |
| v723_phys_modular_clock.py     | PHYS moonshot slice (T3 / $F_{transfer}$ ) – can thermal time replace the external clock contract? The conjugacy-class KILL test. STT-KILLED +<br>INTERNAL-PAIR-CONFIRMED: the $F_{QCD}$ 1-loop clock map is PARABOLIC and not diagonalizable (single Jordan block), and exp of a real-diagonalizable generator is diagonalizable (symbolic lemma) – no constant-rescaled modular flow produces the QCD clock; the pole multiplier is EXACTLY $(2/3)^6 = 64/729$ (hyperbolic, dimensionless $\Rightarrow$ internal); typed consequence: the external clock contract is CONFIRMED and sharpened to the anchor census {one unit ( $v_{geo}$ -class), one angle $\theta_i$ , one lattice $O(1)$ } – T3's clock wall == T1's unit wall, counted once.                                        |
| v724_phys_t3b_modular_flows.py | T3-B – conditioned modular flows from ONE internal mother state, the second thermal-time KILL. T3B-DEAD: with target-free graph projectors, ONE global KMS scale $\Delta = 6 \log(3/2)$ and the Connes cocycle composition exact to $1.7 \times 10^{-15}$ along J–K–C–F, all four conditioned flows come out<br>LOXODROMIC/COMPLEX – the frozen v578/v632 solver classes (hyperbolic / parabolic / degenerate) do NOT arise from conditioned modular flows of one internal mother state; the probe honestly refuses the NEEDS-LINDBLAD escape (the v238 GKSL variant is also all-loxodromic); with v723 BOTH thermal-time routes are closed: the external clock contract stands.                                                                                                         |
| v725_phys_vgeo_torsor_audit.py | T1 CLOSURE – the calibration/scale-torsor audit of the $v_{geo}$ interface. TORSOR CLOSURE: the $v_{geo}$ interface is structurally CLOSED as an $\mathbb{R}_+$ scale torsor – (1) complete export table $O_i = r_i v_{geo}^{d_i}$ , (2) dimension matrix rank 1 (conditional [A] on the flavor block, named), (3) all consistency conditions $\lambda$ -homogeneous (the machine form of the No-Unit theorem v153), (4) two-anchor leave-one-out compatible ( $G$ vs $\rho_\Lambda$ : 0.11%), (5) the calibration theorem stated in torsor language; FENCE: structure closure, NO scale derivation (impossible per No-Unit); moonshot NAMED, not executed: $H_{EW} = \ln(\bar{M}/v_{EW}) = 37.1776$ .                                                                                   |
| v726_phys_car_pp_index.py      | PHYS T2 main slice – Pimsner-Popa / Watatani index 4 on the CAR ladder, with the $\mu_4$ action DERIVED and the Ramond zero-mode test. T2-SLICE-GO: the optimal PP constant is EXACTLY $1/4$ (explicit rank-one minimizer + pinching theorem) and the Watatani index EXACTLY 4 via an explicit quasi-basis ( $\sum vv^* == 4 \cdot 1$ , weights FORCED – uniform weights break scalarness, the $\mathbb{Z}_2$ average gives index 2); $\mu_4$ is NOT inserted but derived from the clock $L = S^{N/12}$ (order 4 forced by the NS spin structure); the Ramond sector degenerates undressed (index 2) and the v679 Klein dressing restores index 4<br>STATE-PRESERVINGLY ( $[\rho, U'] = 0$ ); the $(1+i)$ dictionary to v722 closes; the GNET.RAMIFIED.01 contract candidate is printed. |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v727_l1_identification.py       | L1 slice – identification of the vague limit via moment-problem determinacy, and the maximal sharpening of the wall. L1-DETERMINED-WALL-TYPED: determinacy is CLASSICAL (Gauss-damped Weil moments identified against the closed geometric side, max rel dev $1.4 \times 10^{-3}$ over $m_0..m_{16}$ ; Carleman from the measured Gauss/Freud band $[0.638, 1.117]$ ); the identification chain (a)–(d) contains RH NOWHERE (Weil 1952 is unconditional); THE WALL, maximally precise: PD PERSISTENCE of the window ladder – infinitely many PD windows SUFFICE for RH, per window finitely decidable (Levinson), measured 9/9; the converse is NOT shown – honest asymmetry; equivalently (Riesz–Haviland): the geometric side defines a positive measure on $\mathbb{R}$ . One fact, three languages on $h = 1433$ (Hankel $1.4 \times 10^{-11} \geq 0$ , Levinson headroom 0.830, Fejér minimum $6.5 \times 10^{-3}$ ); scramble breaks at depth 61. |
| v728_k1b_superresolution.py     | K1b slice – WHY do the Gauss nodes sit $\sim 900\times$ below the classical capture radius? K1B-ATOM-PINNING: the memo mechanism (Christoffel locality + asymmetry perturbation) is DEAD – the two preregistered memo bars B1.1/B2.2 die honestly (corr $\sim 0$ ); the carrying mechanism is ATOM PINNING of the raw measure (Nevai mass-point attraction): lab rate $K^{-1.84}$ , node-weight ratio $\rightarrow 1.03$ , real-data correlate 0.64. Off-line honesty: an off-line pair breaks POSITIVITY (Levinson death) instead of mislocating nodes – positivity is the single place where RH enters; K1 picture: capture [v721] + center localization = atom pinning [theorem need = raw atom structure = L1]; everything SHA256-frozen before the target load.                                                                                                                                                                                    |
| v729_strat2_rp_universality.py  | Strategy-council slices – reflection positivity, bulk universality, pinning lab. A: RP-DEAD-ON-TRUNCATIONS – the naive Osterwalder–Schrader route is dead at measurement level (the total window fails plain AND KMS-twisted RP, mineig/scale = $-1.0$ ); the FE symmetry buys exactly the atom-free Connes–Consani 2021 regime $ u  < \log 2$ and no more. B: UNIV-BULK-EMERGING – the sinc deviation at a gap point halves along the $K$ -ladder ( $0.30 \rightarrow 0.15$ ), peak points NOT in the bulk class (the two-layer picture measured at kernel-shape level). C: PIN-MECHANISM-SUPPORTED – ONE inserted Fejér peak reproduces the super-resolution scale ( $2.4 \times 10^{-3}$ , rates $-2.5.. - 4.1$ , spacing/error up to $211\times$ ). Sector anatomy exact (pole Hankel-PSD, arch Hankel-NSD); zeta-free.                                                                                                                             |
| v730_strat2_pinning_lemma.py    | S3 closure – the explicit PINNING LEMMA behind the K1b super-resolution. L-PIN-1 (residual bound, Wilkinson-type) + L-PIN-2 (Kato–Temple), one-line proofs each: 0 violations on the synthetic scan (54 configs) AND on the real windows – 598 peaks / 535 matched band zeros including ALL 377 of the largest window (SHA256 freeze before the target load); the bound scales as $\sqrt{\text{background}/\text{mass}/K}$ – exactly the measured K1b correlate; the CERTIFICATE SATURATES at window width: sub-width precision requires raw atomicity beyond EVERY window depth = exactly the L1 input, now the sharpest localization of the wall. Comb coherence is NOT the mechanism (flatness 1.05).                                                                                                                                                                                                                                                |
| v731_strat2_gap_universality.py | S2 closure – gap-layer universality with the finer $K$ -ladder and the local Szegő/Totik conditions, zeta-free. GAP-UNIV-KILLED: the strong measured form of gap-layer universality is DEAD – the median sinc deviation plateaus at 0.33 (best single point 0.15), the preregistered kill bar fires; the regularity proxy runs monotonically toward cap = 1 (0.9995), but the UNIFORM-in- $h$ Szegő condition breaks like $h^{-0.64}$ – the $h$ -limit gap layer runs through the atomic limit (= L1). What remains: a fixed-window CONDITIONAL theorem candidate via Lubinsky 2009 / Levin–Lubinsky 2008 / Avila–Last–Simon 2010 / Totik 2009; decomposition printed: [gaps] conditional / [peaks] pinning lemma (v730) / [atoms] = L1.                                                                                                                                                                                                                |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v732_strat3_sonin_prolate.py | Strategy-3 / L1 – finite Sonin–prolate compression on TFPT windows, the direct Connes–Consani transfer test. SONIN-RANK-GROWS (the preregistered KILL fires): bad_rank grows $19 \rightarrow 86$ over the nine-window ladder (bar: $\leq 4$ uniformly), the largest defect excess grows to $1.3 \times 10^5$ , ONE E8/pole condition is neither stable nor sufficient – the DIRECT transfer of the CC 2021 compression to the window discretization is dead. Cleanly fenced: NOT the CC mechanism itself (which lives on the atom-free adelic regime); tuning $U_0/T_0$ after this result is forbidden; construction exact (one-cell phase space $2U_0T_0/\pi = 1$ , rank bound verified), no zeros, no prime table.                                                                                                                                                                        |
| v733_strat3_gate0_census.py  | Strategy-3 / Gate 0 – the $S = \{\infty, 2, 3, 5\}$ local orbit-algebra census. GATE0-WEAK-FUNCTOR: an exact functor candidate from the $\sigma$ -descended TFPT groupoid into Connes’ $S$ -local periodic-orbit subalgebra – all 6859 convolution coefficients with deviation 0, $\kappa \equiv 1$ at all three places (place 2 ramified / 3 inert / 5 split, re-checked from the Gaussian HNF tower v714), involution / grading / time action / regular AND closed-orbit traces preserved (exact integer/Fraction arithmetic). NOT proved (typed): extension to Connes’ FULL $S$ -local convolution algebra – topology, adele coordinates, isotropy representations, KMS state.                                                                                                                                                                                                           |
| v734_s1_canonical.py         | S1 slice – the canonical system of the limit object with the Suzuki anchor: the wall in the FOURTH language. S1-CANONICAL-SUZUKI-BRIDGED: Krein correspondence EXACT (Weyl identity $1.6 \times 10^{-15}$ over 4 complex $z \times 9$ windows, nilpotency $3.1 \times 10^{-16}$ ); $H_h \geq 0$ MEASURED on all 9 windows (min $l_k = 0.033..0.377$ , all $> 0$ ); boundary phase == gamma side with scale EXACTLY 1 (dev $1.9 \times 10^{-2}..2.5 \times 10^{-2}$ ) – the stage-2 gluing in canonical form; scramble breaks the Wheeler/Krein chain at depth 31. Suzuki bridge (JFA 281 (2021)): GRH == positivity of the canonical family, unconditional ONLY for $\omega > 1$ – our truncations are the first computable zeta-free approximation family with measurable $H \geq 0$ : the fourth equivalent language of the PD-persistence wall (moments / state / symbol / Hamiltonian). |
| v735_strat3_ucp_inductive.py | STRATEGY-3 / L1 – covariant UCP maps along the nine-window GNS ladder: the state-transport route, honestly closed. UCP-STAGNATES: all eight adjacent maps exist EXACTLY as UCP quantile couplings (unitality/state preservation $\leq 3.1 \times 10^{-12} / 1.2 \times 10^{-15}$ , Choi minimum 0), every Choi rank inside the sharp transport bound $h + h' - 1$ (ranks 344..2688, log-log slope 0.991 vs bar 1.25 – LINEARLY controlled); the preregistered covariance gate S2.3 FAILS HONESTLY (error tail not monotonically decreasing) – mere state transport is NOT the missing L1 mechanism; the optimal maps are quantile Markov couplings, not *-homomorphisms (multiplicativity defects $\sim 5 \times 10^{-3}..8 \times 10^{-3}$ ); closes the L1 route portfolio: Sonin dead (v732), UCP stagnates (v735), Gate 0 weak functor (v733), Suzuki bridge a measurement (v734).      |
| v736_orbit_packet.py         | The Gaussian ORBIT PACKET lemma. ORBIT-PACKET-EXACT: Burnside $7 = (15 + 3 + 3)/3$ ; the canonical $5 \times 48$ partition (fixed block + 4 moved blocks of 12 Gaussian lines each, label-free canonical – packets = $1 + 4 = 5 = g_{\text{car}}$ , $240 = g_{\text{car}} \times 48$ ); anchor $(1, 1, 2)$ from the fixed-syndrome leader weights $(0, 1, 1, 2)$ , leader pair $e_6/e_7$ = the v638 anchor pair; $q$ normal form $240 = 16 \cdot 15$ , $248 = 240 + 8$ ; circularity fence: P1/P2 come back as EXACT-INTERNAL-RECONSTRUCTION (reconstruction, NOT derivation); $\sigma$ -class specificity: the 160 free order-3 elements of $G_{31}$ give Burnside 5, not 7; companion Lean module OrbitPacket.lean (34 theorems, kernel-checked).                                                                                                                                         |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v737_seam48_intertwiner.py     | The seam-48 intertwiner test – the sharpest structural kill of the review morning. SEAM48-DEAD: the $48 \times 5$ incidence matrix is NOT identically 1 (entry distribution 0:96 / 1:64 / 2:64 / 3:16), representative-independent (0 of 2304 order-5 elements of $G_{31}$ ) and the must-fail control fires (0 of 25 random operators); valuable residues: the 4 moved blocks carry the v623 character table EXACTLY, the fixed block deviates exactly by the 12 lifted marks; the 5-period lemma $u^5 = \pm J$ holds exactly; $240 = 48 \cdot 5$ remains a factorization WITHOUT coordinatization.                                                                                                                                                                                                |
| v738_hecke_mod_ramified.py     | The Gaussian-E8 Hecke correspondences projected onto $V = L/(1+i)L$ . HECKE-MOD-RAMIFIED-CHANNELS: the transport is exactly well-defined AND $\sigma$ -functorial over ALL 5907 submodules of the v714 tower (no kill); odd layers act as degree-id (the fiber is Hecke-rigid); the ramified layer IS the 15 hyperplanes of $V$ with the 2:1 deck; channels = the 7 $\sigma$ -orbit channels + 1 trivial (resolving 7-vs-8); NS/R = the $\sigma$ -fixed parity character (v722 re-derived exactly); ram-odd ( $n = 2, 8, 32, \dots$ ) is the unique negative-pressure channel; honest boundary: the W3 margin is cross-channel cancellation – no transfer-matrix reduction from the projection alone.                                                                                               |
| v739_rank3_jet.py              | The review simplification of the rank-3 wall, tested exactly. RANK3-JET-DEAD: the jet read is exact (the lock block as the first jet of ONE function, $4.6 \times 10^{-16}$ ; Loewner coordinates; the Schwarzian form $[1 - r_{12}^2]_{\text{odd}} = (3/8) S[\tilde{g}](1) \varepsilon^4$ on the odd sector), BUT the review kill fires: the $\varepsilon^0$ coefficient is identically 1 (not 0), the even sector exceeds the net entry ( $1.15\text{--}6\times$ ), the pure jet block carries no collinearity (factor $\geq 1.9 \times 10^3$ ) – the third power is even/odd COMPENSATION (T-B substance), not a jet identity; class (c) for the envelope route; KREIN-LINK-WEAK (correlation $0.678 < \text{control } 0.946$ ).                                                                 |
| v740_kms_extension_switch.py   | The finite extension switch/kill test of the strong KMS master contract. KMS-EXTENSION-DEAD: $\beta = 1$ KMS is algebraically impossible on Gate-0's nonzero-grade Laurent unitaries – $T(u_p u_p^*) = 1$ vs $T(u_p^* \sigma_i(u_p)) = 1/p$ , exact defects $1/2, 2/3, 4/5$ ( $p = 2, 3, 5$ ); adjacent normalized GNS states are NOT restrictions along covariant unital $*$ -embeddings (0 of 141–1255 required exact boundaries); what survives: local extended Gram positivity (55,296 monomials/window, $\lambda_{\min} 5.0 \times 10^{-11}$ – $7.3 \times 10^{-9}$ vs random $\sim 8 \times 10^{-2}$ ); controls fire (Epstein/scramble); the weaker contract is named: BC Toeplitz semigroup + operator-system maps ( $\rightarrow$ v741); the two [FAIL] lines are the preregistered kills. |
| v741_kms_toeplitz_semigroup.py | The BC Toeplitz semigroup correction of the KMS obstruction, and the honest covariance kill. KMS-TOEPLITZ-DEAD: the BC isometry correction COMPLETELY repairs the algebraic KMS obstruction (boundary-corrected KMS identities exact; normalized bare error $10^{-2} \rightarrow 10^{-9}$ in the directed Toeplitz system), BUT the covariant inductive contract stays dead: the preregistered UCP window covariance is nonmonotone (0.797..1.558..0.058, slope $-0.487$ , tail not decreasing); final form of the master contract: keep the proven BC-Toeplitz local KMS blocks, state the global handoff as an operator-system COMPACTNESS problem ( $\rightarrow$ INDUCTIVE_STATE.02); literature anchor: $\beta = 1$ KMS for the genuine BC algebra is classical (Bost–Connes 1995).            |
| v742_channel_interference.py   | The cross-channel anatomy of the thin W3 margin. INTERFERENCE-COLLECTIVE: the W3 margin is ONE collective scalar balance (arch + ram-odd against split + inert), strictly monotone over all 35 deployed windows (smooth running, consecutive cosine $\geq 0.9961$ ); TWO exact Ward identities – W0: $\sum c = 0$ to $1.8 \times 10^{-15}$ on every window; $\sigma$ /scalarity: the $8 \times 8$ rigid block = $\dim_C \dim_{C'} m_0$ ; NO pair reduction ( $k_{95} \in [4, 5]$ , $k_{\text{pair}90} \in [6, 8]$ – the balance stays many-term); scramble control fires ( $\times 2 \times 10^3$ – $3.9 \times 10^6$ ); the channel basis is DIAGNOSTIC, not reductive – transfer-matrix hopes via this route are closed.                                                                          |

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| Script                                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v743_schur_cone_recursion.py                | The cone-preserving Schur recursion, killed by a one-liner. SCHUR-CONE-DEAD: the Moebius/Schur update is exact (residual $2.9 \times 10^{-41}$ ) and the $T^*JT$ identity holds exactly [E]; the one-liner kill: a layer map is cone-preserving IFF the layer is positive ( $c_0 \geq 0$ ) – and NO prime layer is ( $c_0 = 0$ exactly, with genuinely negative dips): “ $T^*JT \leq J$ per layer” cannot carry in principle; layer-wise recursion is the wrong granularity level – an escape would need a canonical positive re-bundling whose existence is itself an RH-class statement (the precisely named missing piece).                                                                                                                                                                          |
| v744_hecke_channel_columns.py               | Channel columns for the Hecke SOS. HECKE-SOS-CHANNELS-PARTIAL: no canonical column choice factorizes the window Gram (the cos-half candidate is indefinite; the PSD- $B^*B$ completion consumes 5–6 orders of magnitude beyond the W3 margin); the Ihara CHANNEL route fails COHERENTLY – exactly as for $\zeta$ (the operator route passes: trio PSD, prisms break): channels grade POSITIONS while positivity lives in FREQUENCIES – structural, not $\zeta$ -specific; finiteness measured: $\dim V = 16$ carries $\leq 16$ spectral frequencies while the windows need 168–862; Z1 moves only at the ramified place; Epstein fails doubly, scramble breaks.                                                                                                                                         |
| v745_qgeo_car_l1_sector_lemma.py            | The (L1) sector lemma of the CAR continuum majorant. L1-SECTOR-PROVEN-MODULO-ELEMENTARY: $\ K_{2N}^{(q)} - K_N^{(q)}\ _{S_1(\Lambda_\varepsilon)} \leq B_q(\varepsilon)/N$ for all 6 undressed sectors on the fixed grid, proof-near – the occupied-arc mode sums are EXACT geometric sums (exact summation replaces Euler–Maclaurin), the entire $1/N$ structure is the sublattice embedding offset, the leading term is the exact matrix $(iu/2MN) f'_q(s)$ (ratio $\rightarrow 1$ at $6 \times 10^{-4}$ ), $\gamma = 2$ is DERIVED, 39744 + 108 inequalities at 0 violations, $r_1 = 1$ sharp; status typed PROVEN-MODULO-ELEMENTARY with explicit constants; named residues: (a) grid-sup [elementary], (b) L2 uniform FH/RH [hard-technical], (c) L3 uniformity [medium]; GATE.QGEO does not move. |
| v746_phys_gnet_local_functor.py             | The $G_{\text{net}}$ local net functor. GNET-LOCAL-FUNCTOR-ALIVE: the net axioms are EXACT (isotony, graded locality, clock covariance on the quotient circle); the Watatani index of the local fixed-point inclusion is LOCALLY EXACTLY 4 for every interval $\ell \geq 2$ – the $[D, A]$ index statement is interval-local, not merely global; Ramond = half-line/bond defect (solitonic; local footprint, leakage $\sqrt{2/N}$ ); 4 must-fail controls fire; missing continuum theorems NAMED, not claimed (GNS limit net, index continuity Pimsner–Popa/Longo, Longo Q-system identification to GATE.METRIC.11, twisted DHR/Longo–Roberts sectors, Haag duality/split).                                                                                                                             |
| v747_automaton_multiway_tower.py            | The Hecke tower as a confluent multiway system – the Zuse/Wolfram reading measured, not narrated. The diamond is EXACT (multiway confluence = Hecke commutativity = CRT, a theorem); the ONE new fingerprint: branchial spectrum $\{1, q+1\}$ with the $[4, 2]_q$ GRASSMANN cell count at the prime places; $\Lambda$ recursion = causal path statistics (v714 renamed); adjudication MIXED – A1 tower=multiway EQUIVALENT-USEFUL, A2 lift=CA RENAMING (shift automaton radius 4, exactly v623 $L^4$ = deck), A3 irreducibility EQUIVALENT-USEFUL ( $\lambda = \ln g \in [1.70, 2.48]$ nats/slot; testable log-scaling prediction $N^*$ ), A4 code=DFA RENAMING (Burnside 7 = v736).                                                                                                                    |
| v748_automaton_sigma_orbifold_confluence.py | Sigma-orbifold confluence of the multiway tower. The $\sigma$ quotient is confluent EVERYWHERE (two-line lemma: $\sigma$ is a multiway automorphism – there is NO orbifold non-confluence at the rewrite level, not even at the ramified place); what actually breaks: only the half-orbit WEIGHTING (mass convention) – exactly at the ramified place, SAME LOCATION as $[D, A]$ but not the same object (a property of the self-chosen convention); final adjudication EQUIVALENTLY-USEFUL (not NEW-STRUCTURE, not a pure narrative layer); rests as documented exploration – the one reference sentence lives in the introduction’s Zuse–Schmidhuber–Wolfram positioning.                                                                                                                            |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v749_simpler_tower.py                | The window quantifier reduces exactly to the nested $(D, X)$ tower – THE structural upgrade of the PD-persistence formulation.<br>TOWER-NESTED-REDUCES: in $X$ an ascending chain (X-nesting exact to $6.7 \times 10^{-18}$ – limit positivity = ONE limit object per chain), in $D$ dyadic downward inheritance (exact: PD fine FORCES PD coarse), every deployed window is a tower member at deviation 0.0; the window quantifier was an ARTIFACT of the historical selection – box step functions are dense, so tower-PD for $D \rightarrow 0, X \rightarrow \infty$ is EQUIVALENT to the Weil criterion (classical density step); honest limits: incommensurable root- $D$ chains connect only approximately, the margins FALL along the tower – the wall stands, in canonical nested form.                                                                                                                                                                                                                                                                                                                                                                                                          |
| v750_simpler_gluings.py              | The gluing identity as an off-line detector at identity level. GLUING-DETECTOR: the gluing identity sees $\delta = 0.01$ (every $\delta > 0$ is detectable) – SHARPER than the PD break at $\delta^* = 0.02$ ; the envelope slope MEASURES $\delta$ ; the rigidity thesis is honestly FALSE: the identity holds unconditionally (both sides are constructed) – what a proof lacks is not the identity but that the true lags can only be carried by an on-line spectrum: that IS positivity itself (D1 determinacy gives uniqueness of the measure, not reality of the carrier points); as DIAGNOSTIC superior to the PD channel; NO new RH access – the wall remains the INEQUALITY.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v751_ward_monotone.py                | Monotone Ward running of the balance components.<br>WARD-MONOTONE-MIXED: the component running is deterministic in DIRECTION (100%, parameter-free $\zeta/L$ constants); the drift anatomy is EXACT – the $\mp u/2$ drifts come from $p^2 \equiv 1 \pmod{4}$ ; the slope-Ward cancellation holds at $4 \times 10^{-4}$ ; the margin inherits ZERO deterministic drift; BUT the balance SIGN is zero content – the deterministic budget is $2.9 \times 10^4 \times$ the margin; a provable sign would need the local zero fluctuation of $\psi$ to $\sim 10^{-6}$ relative accuracy – the induction route for $\lambda_1 > 0$ via component monotonicity is RH-CIRCULAR at the balance level: shape provable, sign not; what survives: ONE drift-free scalar remains to be bounded, not a matrix – but its sign is zero content.                                                                                                                                                                                                                                                                                                                                                                          |
| v752_projective_hamming_incidence.py | The projective Hamming incidence lemma on $V = L/(1+i)L$ – the symplectic $\mathbb{F}_2^4$ of the unimodular hermitian E8 lattice, and the v737 resolution.<br>PROJ-HAMMING-EXACT: $h = H/4$ is $\mathbb{Z}[i]$ -valued with hermitian Gram determinant 1 ( $\mathbb{Z}[i]$ -UNIMODULAR); $\bar{h}$ alternating (doubly-even), nondegenerate $\Rightarrow (V, \bar{h})$ symplectic with explicit Darboux basis; $ H_y  = 7$ , pair intersections 3; $BB^T = 4I + 3J$ EXACT, characteristic polynomial $(x-7)(x-2)^9(x+2)^5$ – the multiplicities $(1, 9, 5)$ are the SYMPLECTIC signature; $K^2 = (4/49)I + (45/49)\Pi_0$ exactly ( $K$ itself honestly not psd); NS/R $\chi = \bar{h}(\cdot, y_0)$ for the UNIQUE $\sigma$ -fixed point $y_0 = \text{FSUM}$ (the anchor bit), $112 + 128 = 240$ , +8 Cartan: $248 = 120 + 128$ ; THE v737 RESOLUTION: the distribution 0:96/1:64/2:64/3:16 derived exactly ( $16 \times$ the $3 \times 5$ label table) AND the $5 \times 48$ coordinatization exists via exactly ONE $\bar{u}$ -invariant spread (kill explained, not revised); two honest corrections to the review derivation documented; companion Lean module ProjectiveHamming.lean (32 theorems). |
| v753_ramified_polarity.py            | The polarity match – the 15 ramified Hecke hyperplanes are EXACTLY the $\bar{h}$ -orthogonal spaces. POLARITY-MATCH: the 15 norm-2 Hecke submodules (v738) = exactly the $y$ -orthogonal spaces of the v752 geometry; canonical bijection $y(\varphi) = \bar{G}^{-1}\varphi$ (canonical-linear, not the identity); $\sigma$ -FUNCTORIAL; deck and $\chi$ commute ( $\chi$ descends to every fiber); canonicity census: of 64 alternating forms 28 are nondegenerate and EXACTLY 1 is $\sigma$ -invariant = $\bar{h}$ – the hermitian derivation fixes ONE form without any choice; controls fire.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v754_ramodd_<br>twostep.py            | The real two-step channel – the $n \rightarrow 4n$ down-up pass IS exactly $K^2$ . TWOSTEP-EXACT: the normalized two-step transport of the real ramified odd layers ( $n = 2, 8, 32, 128, \dots$ ) = EXACTLY $K^2 = (4/49)I + (45/49)\Pi_0$ ; degree factor $196 = (2 \cdot 7)^2$ closed (deck factor 4 = the $(1+i)^2 \mathbb{Z}/2$ of both legs); $T = 4B^2$ exactly; the KMS half-weight is the closed factor 1/2 per pair, identical for $2 \rightarrow 8, 8 \rightarrow 32, 32 \rightarrow 128$ (exponent-exact, no free scalar); the SAME integer matrix on all 9 contexts; spectrum of $T/196$ : uniform 1, transverse $4/49$ ( $\times 14$ ), contraction gap $45/49$ ; $\sigma$ -covariant, cone-preserving (strict Perron–Frobenius); honest boundary T7: naive label chains do NOT carry $K^2$ (correspondence composition mandatory); controls fire.                                                                                                                                                                              |
| v755_simpler_schur_<br>recursion.py   | The ramified Schur recursion on the canonical tower – the first RH attack that exactly fits the correctly formulated wall. SCHUR-RECURSION-ALIVE: $S_k \succeq 0$ MEASURED on all 8 ram-odd stages of the tower nesting ( $n_k = 2^{2k+1}$ ; $\lambda_{\min}$ flat, long-double-hardened); Douglas trivial; the induction frame STANDS: basis [E-finite] + Douglas [E] + Albert/Haynsworth [E] + cofinality [E] – the open proof need is EXACTLY the structural Gram certificate $S_k = R_k^T R_k$ per step; measured boundary data: the label part enters INVERTED ( $\rho_k < 0$ constant sign, $ \rho_7  = 9.16$ near $49/4$ within factor 1.34 – cross-term structure, not an added block), continuum vs atom sides ORDER 1 comparable (0.79–2.47); honestly: for $\zeta$ data EQUIVALENT to tower PD – the wall moved into recursion shape, not broken; controls break immediately (scramble $-4.7 \times 10^3$ , Epstein $-5.0 \times 10^2$ at $k = 0$ ).                                                                               |
| v756_kms_incidence_<br>stinespring.py | The covariant Stinespring extension on the ramified fiber – step 3 of the master contract SOLVED on the ramified layer. KMS-INCIDENCE-CP-COVARIANT: 105 explicit Kraus terms realize $K = B/7$ as a completely positive unital map ( $\sum V^*V$ dev $3.3 \times 10^{-16}$ , $K$ reproduced to $2.8 \times 10^{-17}$ , Choi PSD, eigenvalue $5/7 \times 105$ ); ALL FOUR compatibilities exact: $\sigma$ covariance 0, half-weight covariance $DU = 2^{-1/2}UD$ to $1.1 \times 10^{-16}$ , $\beta = 1$ detailed balance to $6.9 \times 10^{-18}$ , Gate-0 compatibility $3.3 \times 10^{-16}$ ; local candidate state $\tau_{\text{inc}} \otimes \tau_{\text{Toeplitz}}$ (channel-preserved); the honest remainder to a global state named (places/levels, arch lift, canonical UCP window map, compactness, Weil identification); control fires.                                                                                                                                                                                             |
| v757_fano_disc7_<br>bridge.py         | The preregistered Fano $\rightarrow$ discriminant( $-7$ ) bridge – the three-cycle half is real, the parity half dies arithmetically. FANO-DISC7-DECORATIVE (break exactly at condition 2): the NS/R parity bit has NO arithmetic carrier in $\mathcal{O}_{-7} - 2$ SPLITS in $\mathbb{Q}(\sqrt{-7})$ ( $-7 \equiv 1 \pmod{8}$ , $\mathcal{O} \otimes \mathbb{Z}_2 = \mathbb{Z}_2 \times \mathbb{Z}_2$ ), every 2-adic ideal quotient has 2-rank $\leq 2 < 3$ (complete census of all norm-8/16 ideals): the code $\mathbb{F}_2$ -structures are arithmetically UNREALIZABLE; the REAL half (COND1, exact): equivariant lifts exist for both canonical difference-set Fano models, the canonical order-3 action is $x \rightarrow 2x$ (the difference-set multiplier, 2 = the transfer step), every lift sends the unique $\bar{\sigma}$ fixed point to $0 = \text{red}(\alpha_0) =$ the C corner (norm 14 = $\dim G_2$ ); the seven-correlation permanently typed DECORATIVE; controls fire; the one [FAIL] = the preregistered COND2 break. |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v758_simpler_certificate.py             | The keystone of the review series – the assembled certificate [continuum root $\oplus$ Kraus] is NOT mountable: the wall IS the missing left factor.<br>CERT-CONTINUUM-ROOT-DEAD: the arch+pol block is PSD on NO stage ( $\lambda_{\min} -4.9 \times 10^{-3} \dots -3.6 \times 10^2$ , ladder-wide), Levinson break at cell 47, EXACTLY between the $n = 2$ cell (44) and the $n = 3$ cell (70) – v706 G0.1 [M] reproduced ladder-wide and SHARPER (already $W_0$ indefinite, the $n = 2$ atom rescues exactly the last $W_0$ cell just in time); the continuum Schur complement is negative on ALL stages (no Haynsworth object); the additive assembly closes on no stage (the order-1 balance sits in the CROSS-TERM); the rescue cascade is strictly CELL-ORDERED (cont breaks at 47, +ram-odd/+ram-even/+split all at 72, only the FULL family is PD); real remain the [E] building blocks ( $T = 4B^2 = 196K^2$ , KMS-1/2, polarity, Stinespring – deployment site: the $\chi_{\text{par}}$ trace of the thin 2-adic rail in $\tau_{\text{inc}} \otimes \tau_{\text{Toeplitz}}$ ) + the corrected target shape: cell-wise cascade factorization $R = \prod$ (cell factors), rescue order $n = 2, 3, 4, 5, \dots$ ACROSS the channels. |
| v759_handoff_fixed_window_resolution.py | Gate 1 of the diagonal Gram round – fixed window, growing frequency resolution. RESOLUTION-BOUNDARY (Fall B on all three windows): $E_{\text{total}}$ is $q$ -INVARIANT under $N_f = q(2M+1)$ , $q \in \{1, 2, 4, 8\}$ , to $\sim 10^{-13}$ (max $ dE  = 2.3 \times 10^{-15} \dots 7.0 \times 10^{-14}$ , far below the frozen $10^{-7}$ bar); alias increments pure rounding; the Fall-A drop gate never comes close – the anti-alias quadrature $N_f = 2M+1$ closes as EXACT; the entire remaining handoff error (0.0377 at the top window, carried by the paired channel 0.0376) is a genuine boundary/cutoff/object-gluing error.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v760_antialias_exact.py                 | The anti-alias EXACTNESS theorem. ANTIALIAS-CORRECTED: exact identities for ARBITRARY $M$ -cell coefficient vectors (Parseval $\leq 2.0 \times 10^{-14}$ , all $2M-1$ moments $\leq 3.5 \times 10^{-13}$ , Gram identity $\leq 5.9 \times 10^{-13}$ , Gram invariance across $\{2M-1, 2M, 2M+1\} \leq 7.3 \times 10^{-13}$ ); the exact alias formula $t_d = \sum_r (-1)^r \hat{s}(rN_f - d)$ confirmed at EVERY ladder count ( $6.4 \times 10^{-13}$ ); the designed must-fail witness at $N_f = 2M-2$ fires with error EXACTLY 1; corrected sharp minimum $N_{f,\min} = 2M-1$ (spread- $S$ version $M+S$ ) – the deployed count $2M+1$ stays exact, nothing is patched; companion Lean module AntiAliasExact.lean (19 general- $N$ theorems).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v761_atom_pole_abel.py                  | Gate 2 of the diagonal Gram round – the paired atom-pole Abel/Stieltjes transformation with the UNCONDITIONAL eta bound, ONE module from two probes. ABEL-PAIRED-BOUND (combined adjudication): the Stieltjes/Abel split is exact (6/6 sympy certificates, residual 0); matrix identity $\leq 1.45 \times 10^{-11}$ on all 5 windows; $E_{\text{origin}}$ closes via the W1/Suzuki bookkeeping (comb-independent, $1.4 \times 10^{-11}$ on the controls); $\eta_K =  \det_K  +  \Theta_K(0)E(1)  + B_1 I_K$ with $B_1 = 3.734342$ on the FINITE range $[1, 24.29]$ holds for ALL 24 kernels on ALL 5 rungs with pure $1/\log X$ decay – NO RH-strength input; Epstein BREAKS under the DERIVED breaker standard ( $B_1^E = 11.013 = 2.95\times$ , 7/24 violations), scramble $262\times$ , the truth passes 24/24; honest typing: identity and slope-1 normalization are measure-generic, the certified zeta-specificity = envelope quality + measure positivity; the parent probe's frozen ABEL-DEAD stays on record as its preregistered adjudication (the calibration lesson).                                                                                                                                                            |
| v762_dense_weil_core.py                 | The canonical countable dense test family from the $(D, X)$ tower. DENSE-CORE-CANONICAL: admissibility LOCATED per class (v677 S2(i)/Iwaniec-Kowalski Thm 5.12: hats directly admissible, slope $-1.854$ ; the bare box FAILS the $r^{-2}$ criterion ( $-0.942$ , the named caveat) and enters only correlated/tent-read ( $-1.867$ )); quantitative density rates hat $-1.566$ / box $-0.503$ per level (theory $-3/2$ / $-1/2$ ); exact $\mathbb{Q}$ correlation closure on 8/8 type pairs; explicit stage map, $I_{64}$ exact (equality in $\mathbb{Q}$ ); the 24 battery is an initial diagnostic package (hat 8 IS family member hat(2,3) = rank 35); companion Lean module DenseWeilCore.lean (42 theorems; honest analytic remainder stated).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v763_yosida_<br>handoff.py         | The kernel-safe $\varepsilon \rightarrow 0$ route via the bounded Yosida transform, ONE module from two probes. YOSIDA-PARTIAL / QF-DEAD: Q1 – the bounded Yosida formulation carries the handoff rates ( $ b_Y - b_G  \leq 10^{-13}$ , sandwich $0 \leq \text{diag} \leq 1$ over all 360 evaluations, NO $1/\varepsilon$ bound or PD-margin assumption anywhere: the circularity kill did NOT fire); Q2 – the near-kernel converges as an identified stable OBJECT (alignment 0.9997, count 6 stable from $M = 888$ ); Q4 – monotone, contraction-certified $\varepsilon$ -limits per fixed $f$ (0.011..0.108); Q3 – DEAD on the reachable surface: the comb honestly extended to ATOM_MAX $4 \times 10^6$ / $X = 15.19$ / 279849 atoms, $q_f$ non-Cauchy at the frozen bars (12/14 functions), and Candidate B shows the structural SUBORDINATION: 89–99.9% of every battery function lies in the 6-mode near-kernel span; the orthogonalized remnant loses the $R=1$ rates ( $b_Y = 0.060 < 0.10$ ) – the $q_f$ wall is THE named wall object of the round. |
| v765_handoff_<br>redteam.py        | The red-team circularity audit of the gram/handoff convergence, ONE module from two probes. REDTEAM-GREEN-COMBINED: firewall CLEAN (zero blacklist hits over all 21 source-path functions – no eigen/Cholesky/zero/prime channel from the target into the source); baseline reproduced EXACTLY; 8/8 controls break, RT1–RT5/RT7/RT8 conclusively (RT8: the forbidden separate atom/pole accounting is destroyed at $21.5 \times E^*$ – the pairing is real structure); RT6 DECIDED in the companion: all three excess-mode deck corruptions break the symbol gate on every window (min $-0.45..-1.72$ ), all six other gates measurably BLIND; deficit mode N1: the symbol moves INTO the interior, caught by rate/ratio/ $\kappa$ – complementary gate families; BINDING promotion note: certificates must always carry BOTH gate families (symbol/PSD and rate/ratio/identification).                                                                                                                                                                        |
| v766_handoff_<br>bulk.py           | Modul 1 of the global-handoff offensive – the local bulk covariance decider. HANDOFF-BULK-CONVERGES: the $\varepsilon$ -regulated resolvent $G^\varepsilon = (A_X + \varepsilon I)^{-1}$ is the admissible operator-system evaluation; circle-free weak-* form (rung Cauchy increments, no $X=\infty$ reference, no zeros); all 9 ( $\varepsilon, R$ ) cells PASS – rates $b = 0.174..0.310$ per $X$ unit, EPS-ROBUST over three decades; the bare $A^{-1}$ does NOT stabilize (+45%, named negative block: an unbounded observable); the handoff defect is a CANCELLATION BALANCE (leading channel up to $1.9 \times 10^{12} \times$ the total defect – no separate layer rate); controls break on ALL rungs, PD-enforced scramble has $b = -0.927$ (the defect GROWS) – the $\varepsilon \rightarrow 0$ wall stays at the spectral edge, quantified, never gated away.                                                                                                                                                                                       |
| v767_handoff_<br>frequency_gram.py | Modules 2+3 of the global-handoff offensive – the growing-frequency PSD source Gram. HANDOFF-GRAM-CONVERGES: Candidate A (review count) HONESTLY missed the frozen rate gate (cell count below the anti-alias degree, slope +0.471 territory); the PREDECLARED iteration to Candidate B (anti-alias $N_f = 2M+1$ ) converges: spectral errors 0.0855/0.0585/0.0457/0.0452/0.0377 over $\dim Q$ 738..5734, slope $-0.369$ , final/first 0.441; $G_{\text{source}} = Q^*Q$ PSD BY CONSTRUCTION (min symbol $+7.4 \times 10^{-3}$ , clip inactive); every $Q$ byte-hashed BEFORE the target is built; final layer errors arch 0.0141 / atom 0.3946 / pole 0.4144 (pole dominant – the paired cancellation anatomy); both controls fire.                                                                                                                                                                                                                                                                                                                           |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v768_handoff_tail_weil.py        | The tail / cross-window / Weil-identification decider PLUS the $\varepsilon = 10^{-3}$ compatibility decider, ONE module from two probes. TAIL-WEIL-PARTIAL + COMPAT-EPS3-CONVERGES: the quadrature tail TERMINATES ALGEBRAICALLY (alias census $(M-1) + \text{spread} \leq 2M$ on all 5 windows; max tail $2.4 \times 10^{-13}$ – there is NO quadrature tail, every layer error is structural); Weil identification on the battery: $\kappa = 0.97768 \rightarrow 0.99028$ , agreement 0.0485 (battery-limited); all layer slopes $-0.365$ , atom+pole a $10.5\times$ cancellation pair; the two frozen $\varepsilon = 10^{-3}$ endpoint fails of part 1 (atom-burst artifacts of the shorter ladder) PASS the NEW preregistration on the deepest reachable 36-rung ladder to $X = 12.875$ with the oscillation-aware statistic: $b = 0.178/0.186$ , med5 = $0.343/0.467$ , $b_2 = 0.069/0.096$ ; anchor $\varepsilon = 10^{-2}$ passes 2/2; controls fire – the wall quantified, not hidden: the outlook $\varepsilon = 3 \times 10^{-4}$ shows the first wall sign ( $b_2 = -0.011$ , reported never gated). |
| v769_qf_contract_necessity.py    | The Gate-3 audit pair of the qf offensive, ONE module from two probes. QF-GAUGE + GATE3PRIME-DEAD: Gate 3's full-ladder Cauchy demand on $q_f$ was provably STRONGER than the diagonal contract needs (a deterministic $\varepsilon$ -net subsequence of density 0.144 makes all 14 profiles Cauchy at the parent bars; the contract's corner objects are BLIND to $q_f$ jumps: corr +0.038, the 0.34 mode-entry jump leaves the corners at routine level; 370 cluster pairs, median $\rho = 1.13$ ) – $q_f$ is representation gauge, owning the $\varepsilon$ -scale distribution of the deficit, not the value; but the corner-native weakened Gate 3' DIES on its own predeclared $1.6 \times 10^7$ comb ( $X = 16.5625$ ): the corner increments fall to $X \sim 12.9$ and then RISE sustained (+0.10..+0.34 per $X$ unit at $\varepsilon = 10^{-3}$ ; 0/56 cells) – the v764 precondition is REFUTED at the corner level (34/168 cells; 14/14 + 14/14 guards+controls).                                                                                                                                     |
| v770_qf_spectral_bundle.py       | The transport pair of the qf offensive, ONE module from two probes. QF-BUNDLE-PARTIAL + QF-SETTLES-POSITIVE: the Kato/polar transport of the rank-6 near-kernel is WELL-DEFINED with wide margins (rank 6 stable on 888..972, rel gap 6/7 = $0.34..0.56$ , angles $\leq 2.72^\circ$ , TV flattening) and repairs the rigid frame for 14/14 functions (median med5 ratio $26.7 \rightarrow 1.004$ ); the decider fails 0/14 – the transported increments flatten, dominated (radial share $\sim 0.64$ ) by frame-independent band-weight drainage. And the drainage was a TRANSIENT: on the $10^8$ comb ( $X = 18.375$ ) the band weights plateau at positive per-function constants ( $R2$ family $0.225..0.358$ , $R1$ family $0.008..0.079$ ; zero-bar extrapolation $X^* \sim 227$ – never); the 6/7 gap bottoms at 0.1008 then recovers; the 7th/8th threshold crossings are typed; the holonomy exponent is $\sim 1$ (step refinement closed). The Feshbach / boundary-triple treatment becomes MANDATORY.                                                                                                  |
| v771_qf_representation_census.py | The representation census of the 6-dim near-kernel – is $E_{qf}$ the canonical E8 defect space $E_7 \oplus E_{-2}$ ( $6 = 1 + 5$ )? QF-REPRESENTATION-OPEN: the label side is SHARP ( $BB^T = 4I + 3J$ exact, spectrum $(1, 9, 5)$ ; of 64 alternating forms EXACTLY 1 is $\sigma$ -invariant = the canonical $\bar{h}$ ) and every control fires (27/27 wrong forms + 5/5 wrong labelings die on $\sigma$ -sharpness) – but the deployed scalar comb realizes only TWO characters of $V$ , no rank-6 lift exists on the deployed surface, and the measured mode anatomy contradicts the C1 signature (ro-dominant count 0/6 vs predicted 5): the near-kernel is a CONTINUUM-vs-ODD-PLACES balance. Honest residue: ONE near-null direction is rung-stably ( $\cos = 0.997$ ) the ram-odd negative wavepacket, plus a second contact at $\cos \sim 0.64$ ; the $6 = 1+5$ match stays typed DECORATIVE.                                                                                                                                                                                                           |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v772_qf_feshbach_effective.py    | The fixed- $d$ pair of the qf offensive, ONE module from two probes. FESHBACH-PARTIAL + EDGE-KEEPS-MOVING: the $d = 6$ Feshbach reduction is EXACT (reconstruction $8.8 \times 10^{-13}$ against independent dense solves; matrix-Herglotz certified; band separated at 0.1008) – but its Cauchy gate fails on the slope clause exactly where mode 7 grazes the band edge; $d = 7$ converges strongly ( $b_2 = +0.502$ ) but loses separation at the top rung; $d = 8$ has an internal 8/9 crossing ( $87^\circ$ ). On the $10^9$ comb ( $X = 20.6875$ ): the 7/8 edge closes into an AVOIDED CROSSING (gap 0.0039 at $M = 1240$ , tail 0.0074 vs bar 0.10) while the transported $d = 7$ block stays Cauchy (med5 0.181, $b_2 = +0.061$ ); the 9th mode enters at $M = 1276$ ; the entry cadence WIDENS ( $\Delta X = 1.625/1.8125/2.625$ , spread 0.381, not regular at the 0.25 bar). Fixed- $d$ Feshbach is closed PERMANENTLY – convergence was never the problem, band ownership is.                                                                                                                                                                                                                         |
| v773_qf_cell_cocycle.py          | The closure decider of the qf offensive – the ordered Möbius/Redheffer cell cocycle on the depth-dependent slow band $d(X)$ , run under the frozen STAKES. COCYCLE-DOMAIN-ONLY: STRUCTURE WITHOUT LIMIT. Exact: the cell identity holds at $9.7 \times 10^{-12}$ over 73 rungs $\times 5 z$ (coefficients = the added source rows/columns; independent dense-solve Ward $1.3 \times 10^{-13}$ ); every cell Schur block is PD (min eig $D_k \in [0.0605, 0.0668]$ – a monotone Loewner flow); the Nevanlinna/Stieltjes domain is preserved with ZERO breaches through both typed mode-entry transitions (992, 1108). But NO convergence cell passes (0/6): the corner absorbs a flat $\sim 0.1\%$ per rung and the band increments rise with the moving edge. PER THE FROZEN STAKES the diagonal Gram route CLOSES – PRIME.GRAM.DIAGONAL.01 carries the dated closure note and the sharpened PRIME.Z1.OPERATOR.01 takes over.                                                                                                                                                                                                                                                                                      |
| v774_arf_spinor_compiler.py      | The Arf/spinor compiler on $V = L/(1+i)L = \mathbb{F}_2^4$ (Priority 1 of the Arf programme). ARF-SPINOR-EXACT: the parity lift $\iota: V \cong C_{\text{even}}(5)$ carries $\beta = \bar{h}$ EXACTLY in all 256 cells (crosslink 1); brute force over all $2^{16}$ functions: exactly 16 quadratic refinements; Arf census $6 + 10$ with $ Sp(4, 2)  = 720$ and a faithful+surjective $S_6$ action; the frozen selector picks a UNIQUE $q^*$ (census $4 \rightarrow 2 \rightarrow 1$ ) and the weight form IS $q^*$ (offset 0); $\text{Stab}(q^*) \cong S_5$ gives $16 = 1 + \bar{5} + 10$ on refinements AND words; the $K_6$ duad model is an equivariant bijection onto $E(K_6)$ ; $B = I + A_{KG(6,2)}$ in all 225 cells with the exact ovoid eigenvector; all 6 isotropic spreads have signature (1, 2); the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the v310/v14 dictionary; moments (16, 0, 120, 0) with the chain 40/41/240/248; Pati–Salam $8 + 8$ with $B-L = 1 - 2n_c/3$ ; $\chi_{\text{NSR}}$ = the anchor bit. $A_3^{\text{Fam}} \neq A_3^{\text{PS}}$ naming discipline; the code→matter reading FENCED (killed by v775). Lean companion ArfSpinorCompiler.lean (23 kernel theorems). |
| v775_gaussian_class_d5_purity.py | THE FIRED PHYSICAL KILL of the Arf compiler programme. ROOTCLASS-MIXED: no $D_5 \oplus A_3 / SU(5)$ convention makes even one full set of 15 Gaussian classes pure per the frozen code→matter table – the embedding-INDEPENDENT counting bound (every embedding has exactly 128 spinor-side roots, $240 > 128$ ) forces $\geq 7$ adjoint-side classes ALWAYS; the exhaustive scan (56 splits $\times$ 240 assignments $\times$ 2 chiralities) finds ZERO passing conventions; all 8 spinor classes mix 8 distinct 16-weight pairs. BINDING FENCE: the code→matter reading must NOT enter any main claim (incl. the 'empty mailbox = right-handed neutrino' reading); the Arf mathematics of v774 is untouched. All three controls fire.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v776_boundary_hamming_uniqueness.py | Boundary code uniqueness. BOUNDARY-CODE-UNIQUE: the non-circular chain boundary $\rightarrow$ code $\rightarrow E_8 \rightarrow \mathbb{F}_2^4 \rightarrow 5$ slots closes with $g_{\text{car}} = 5$ as OUTPUT – census 135 $\rightarrow$ 30 $\rightarrow$ 14 $\rightarrow$ 2 $\rightarrow$ 1 rebuilt without Hamming/ $E_8$ input; the flag datum is ONE deck-invariant $\mathbb{Z}_2$ (the product of the four side orientations; all three predeclared formalizations agree); the boundary gauge group (order 12) makes the flagless twofold pure gauge; Construction A $\rightarrow E_8$ exact (240/2160 theta, Cartan det = 1, arms 1-2-4); $q^*$ unique; packets 1 + 4. Honest residuals R1 (doubly-even self-dual universe premise – circularity relocated and shrunk, not removed), R2 (deck placement semantic; wrong-deck control does not fire, typed), R3 (anchor naming). NO P2-removal claim; the open contract form registered as ARF.BOUNDARY.CODE.01 [O].                                                                                                                                                                                                                     |
| v777_ftransfer_clock_jets.py        | FTRANSFER.CLOCKS.01 EXECUTED + the fibered refinement of CONTRACT.F.01, ONE module from two probes.<br>NO-COMMON-CONNECTION + FTRANSFER-FIBERED-CARRIES: the preregistered K1 kill fires EXACTLY as the frozen prior expected (prereg YAML byte-verified; both data tables SHA-frozen BEFORE the first run); all four dictionaries construction-valid (Koide scheme check $571\times$ , washout drift $1.15 \times 10^{-4}$ , QCD overlay 0.83%, relic slope 0.0602); the kill is sympy-exact $\{-\Delta^2/2, -\Delta^2/2, 0\}$ – a pass would need exactly the forbidden non-affine Möbius clock. The discrete layer coincides EXACTLY (cross-ratio 4/3 symbolic, deck unit $ \Delta\sigma  = 1$ , unit parabolic class – the v632 lattice-explicit half); the obstruction is the CONSTANT seam cocycle with values exactly $\{0, \pm\Delta^2/2\}$ ; classification = the predeclared thermal-vs-RG marker; relic degenerate unclassified; controls incl. the Möbius transplant fire; the first-run float64 artifact disclosed, bar untouched. The jet route of CONTRACT.F.01 closes at the physical-clock level; next named step: derive the coset assignment from KMS/modular clock typing. |
| v778_qgeo_gridsup_logsum.py         | Remainder (a) of the (L1) majorant lemma FULLY closed in the corrected $(1 + \ln M)$ form, ONE module from two probes.<br>GRIDSUP-SPLIT-CLOSED-LOG-LAW + FOURIER-LOGSUM-CLOSED: the v745 residue (a) splits four ways – (a1) the continuum entrywise supremum CLOSED ( $\sim 1.26\text{e6}$ census inequalities, 0 violations), (a2) the configuration- $S_1$ closure with explicit $B_{\text{conf}}(\varepsilon)$ (packing $+\zeta(2)/\zeta(3)$ ), (a3) the refinement remainder with $\text{RHO}(\varepsilon)$ via exact $\text{csc}^3$ integrals, (a4) the leading part honestly LOG-divergent – the literal v715 clause ‘one $B(\varepsilon)$ for ALL grid refinements’ is FALSE at $r_1 = 1$ and repaired to $(1 + \ln M)$ ; the surviving inequality CERTIFIED at theorem level via the exact twisted-circulant diagonalisation + two exact summation-by-parts: $\ F\ _1 \leq c_0 + (J/4) \ln(M+3)$ , a fortiori $(J/\pi) \ln M$ ; 162-case census to $M = 6144$ , zero violations; repaired (L1) clause verified on all 252 transfer cases; Araki summability explicit. Honest: the measured slope is $J/\pi^2$ – valid, not sharp. Residues (b)/(c) open; GATE.QGEO does not move.     |
| v779_gnet_gns_arf.py                | The $G_{\text{net}}$ continuum strand, ONE module from two probes.<br>GNS-PRECURSORS-COHERENT + GNET-ARF-NO-CORRESPONDENCE: the Haar scaling isometry intertwines exactly ( $\leq 2.2 \times 10^{-16}$ ); the commuting index-4 expectation squares are exact; the canonical state is Cauchy with geometric contraction $\sim 1/4$ per doubling; the index census is machine-exact at every level with zero degradation; the twisted Haag-duality defect is 0 with $\dim B'/\dim A' = 4$ = the Watatani index; 10-section continuum-paper skeleton. The exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1} \cos(\pi c/2)$ makes the $\mathbb{C}[\mathbb{Z}_4]$ quasi-basis obstruction exactly $2^{-\ell}$ (explains the $\ell = 1$ anomaly); explicit 185-element quasi-basis at $\ell = 2$ ; uniform sector-resolved split bounds. THE HONEST NEGATIVE: the $16 = 1+5+10$ counting echo is exact but non-geometric (rank 3 vs 4, $GL(4, 2)$ census 0/20160); $K_{\text{matter}}$ does not govern tower transport; the two fours stay in their registers. All five controls fire.                                                                                                             |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v780_z1_compactness_trilogy.py  | The sharpened-Z1 trilogy, ONE module from three probes (v518 precedent). Z1-SIGNATURE-PARTIAL + Z1-TRIPLE-PARTIAL + Z1-VAREEDGE-PARTIAL (gates $3/4 + 1/4$ legs $3/6$ with exactly the preregistered fails): the N2 GNS/Jacobi family carries edge collisions (0.0032 vs source 0.0039), the ram-odd contact (cos up to 0.9971) and settled couplings (14/14 P) natively – but NOT the widening cadence (regular quadrature filling, free-Jacobi type; the Gram window operator IS the Toeplitz Gram of the N2 Weil measure); the finite boundary triple is EXACTLY Herglotz through all typed transitions but dies as a whitened $K = M/2$ import (cell-mass exactly 0.500 beyond the half-window; delivery 5.88; arithmetic-specific 5.9 vs GOE 13.0 vs random 132.5; full-depth GNS provably out of reach, $e^{27.75}$ atoms); with the RAW $\mu$ -weighted import the variable-edge family is BOUNDED (0.936) + EQUICONTINUOUS (0.807) with SUMMABLE $\sim$ rank-one entry poles at the arithmetic thresholds ( $c_{\text{mass}}$ 0.032 $\ll$ 1; rank ratios 0.80–0.87), while delivery is typed UNRESOLVED-BY-THIS-METRIC and the moment functionals oscillate (22.1). MEASURED UNIFICATION: INDUCTIVE STATE.02 and Z1.OPERATOR.01 share their compactness theorem (carried) and share ONE remaining obstruction = the import-faithful boundary coupling; the merged target is Z1-COMPACTNESS. |
| v781_one_object_clock_census.py | The $d = 4$ forcing census + the 60-line clock kill, ONE module from two probes. ONE-OBJECT-PARTIAL + CLOCK-FIBER-DEAD (21/21 + 23/23): the TFPT glue pattern admits an even unimodular DIAGONAL glue of $A_{d-1} \oplus D_{d+1}$ iff $d = 4$ (census $d = 2..12$ , exact fractions; at $d = 4$ exactly the two v92 Lagrangians $k \mapsto k, k \mapsto 3k$ ; explicit $E_8$ certificate with 240 norm-2 vectors; $d = 9$ dies at isotropy; both relaxation controls fire) — corollaries $g_{\text{car}} = d+1 = 5$ , $N_{\text{fam}} = d-1 = 3$ , rank $2d = 8$ , index $4 = d =  \mu_4 $ ; honesty gate: $c_3 = 1/(2\pi d) _{d=4} = 1/(8\pi)$ is a corpus-legitimate REWRITING, NOT a new derivation of P1. The clock kill: the fiber-product action $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$ on the 60 Gaussian lines is dead at three levels (full line-compatible group order 23040, max element order 12 — orders 15/30/60 absent for every realization; the fiber product is cyclic $\mathbb{Z}_{60}$ , exact SNF); all controls fire.                                                                                                                                                                                                                                                                                                                                        |
| v782_e8_transition_bus.py       | The transition-bus structure theorem. TRANSITION-BUS-TORSOR (35/35): the 240 roots are 15 disjoint coordinate 8-frames over $V = L/(1+i)L$ ; on $L/2L$ the 120 root pairs are exactly the 120 $q = 1$ points of the plus-type quadratic space and each class fiber is a canonical TORSOR under $v^\perp$ (affine $\mathbb{F}_2^3$ , no distinguished origin); the deck $J$ acts as translation by $v$ ; the obstruction census is proof-grade: NO deck-equivariant root-level labeling $R(E_8) = V \times (V \setminus 0)$ exists; the frozen symplectic predicate reproduces all 57600 ordered inner products incl. the sign law and the $h$ -refinement; spread layer $240 = 5 \times 3 \times 16$ with the $q^*$ selector; the $A_{15}$ comparison quantified (a strict bus WITH a canonical start state; $E_8$ provably has none; cross-class non-orthogonality exactly doubled). Controls fire.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v783_two_qubit_clifford.py      | The two-qubit stabilizer identification of the compiler carrier. CLIFFORD-PARTIAL(H4) (32/32): under the single predeclared chart $U$ the 60 Gaussian $E_8$ lines ARE the 60 pure two-qubit stabilizer rays, entry-exactly over $\mathbb{Z}[i]$ (60/60; support census $4 + 24 + 32$ ); $UG_{31}U^{-1}$ is an index-2 subgroup of the two-qubit Clifford matrix group ( $ C_2  = 92160 = 2 \times 46080$ , coset $\zeta_8 G_{31}$ ); $46080 = 4 \times 16 \times 720 = \mu_4 \times \text{Pauli}_{16} \times Sp(4, 2)$ with the 15 classes = the 15 Pauli contexts ( $W(3, 2) = GQ(2, 2)$ ) and the 6 spreads = the MUB pentads; H4 fails as predeclared: the index bit is the Galois $\sqrt{2}$ /Hadamard bit, NOT $J \leftrightarrow -J$ ; all controls fire. Stabilizer-formalism mathematics about the carrier; ROOTCLASS-MIXED (v775) unaffected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v784_curve_code_2torsion.py   | The curve/code 2-torsion decider. CURVE-CODE-PARTIAL (33/33): the frozen CM-model route through $y^3 = x^4 - 1$ (every generator period certified as an integral $\mathcal{O} = \mathbb{Z}[\zeta_{12}]$ recognition, dps 120, HNF index 1; $A[2] = \mathcal{O}/2\mathcal{O}$ exact; $\tau = \zeta^4$ DERIVED as the only order-3 class; minimal ODD $\tau$ -invariant polarization type (1, 3), no principal form in the $\tau$ -invariant cone; $q_\Theta$ the unique $\tau$ -fixed refinement): the correspondence DIES at two exact layers — (1) $\tau$ acts freely (cycle type [3, 3], syntheme side) vs $\bar{\sigma}$ fixing a 2-plane ([3, 1, 1, 1], duad side), exchanged only by the OUTER $S_6$ automorphism, census 0; (2) $q_\Theta$ EVEN (Arf 0, 1+6+9) vs $q^*$ ODD (Arf 1, 1+5+10), census 0; joint census 0 with positive control 6. All controls fire.                                                                                                                  |
| v785_hecke_check32.py         | The mod-32 carrier congruence of the TFPT cusp channel. CHECK32-THEOREM (20/20): $f_8 = \eta(2\tau)^4\eta(4\tau)^4 \equiv E_{\text{odd}} = \sum_{n \text{ odd}} \sigma_3(n)q^n \pmod{32}$ verified to $q^{100000}$ (25000× the Sturm bound 4; Sturm 1987 the ONE cited classical ingredient); corollary $a_p \equiv 1 + p^3 \pmod{32}$ for all 9591 odd primes $p < 10^5$ with the explicit decoder $p \equiv (a_p - 1)^3 \pmod{32}$ (cube bijection on the 16 odd residues); maximality: mod 64 fails at $q^3$ ( $a_3 - 28 = -32$ ; census 252/1000); $32 = 2^{g_{\text{car}}}$ typed OBSERVATION; Hecke grammar clean; controls fire (mutant 91.7%, random detection $\sim 31/32$ ). Lean: <b>Check32.lean</b> (13 kernel theorems).                                                                                                                                                                                                                                                   |
| v786_prime_packet480.py       | The packet-480 header adjudication. PACKET480-ADDRESS-ONLY (28/28; the kill fires as preregistered): Header( $p$ ) = ( $p \bmod 30$ , $a_p \bmod 32$ ) is EXACT 7-bit address arithmetic (CRT-equivalent to $p \bmod 480$ ; $\varphi(480) = 128 = 8 \times 16$ classes, all hit; decoder via the check32 Eisenstein congruence recomputed independently) — but $(\mathbb{Z}/480)^\times$ (type 2.2.4.8, exponent 8) admits NO faithful canonical action on the 128 $E_8$ spinor roots (canonical torsor group $\mathbb{F}_2^7$ , exponent 2); the dictionary realizes 3 of 7 bits, the maximal canonical action is the order-8 graded image $\langle \mu_4, \varepsilon_{67} \rangle$ ; the 128 STAYS A FINGERPRINT. Controls fire; ROOTCLASS-MIXED (v775) fences every matter-adjacent reading.                                                                                                                                                                                         |
| v787_hecke_arrow_broadcast.py | The label-faithful Hecke arrow ledger + the C2 broadcast identity, ONE module from two probes. ARROW-LEDGER-STRUCTURED + BROADCAST-EXACT (48/48 + 21/21): the 15 ramified arrows ARE the 15 polar labels $\sigma$ -equivariantly (7 NS + 8 R); spread census (3, 1, 1, 1, 1) with the 3-block AT the polar label; the two-step ledger collapses label-resolved to $T/196 = (4/49)I + (45/49)\Pi_0$ exact, $T = 4B^2$ , the 105 leg classes = the v756 Kraus index set with weight $(7^{-1/2})^4$ ; the trace collapse is exact incl. the now SIGN-derived $a_7 = +24$ ; $n_B = (\sigma_3^2 - a_p^2)/8$ . BROADCAST: on every odd shell $n \in \{3, \dots, 13\}$ the joint census factorizes exactly as $\bar{T}_n = M_n \otimes I_{16}$ with $B_n = 16R(n)$ arrow-exactly; NEW identities $b = 2A_p$ and $n_B = A_p B_p/2$ ; honest negative: $R(p) \neq$ the type-B Weyl orbit count at $p = 7$ . All controls fire.                                                                    |
| v788_positive_c2_lift.py      | The positive two-channel decomposition of the TFPT cusp channel + the C2 packet automaton. C2LIFT-THEOREM (42/42): $A = (E_{\text{odd}} + f_8)/2 = \frac{1}{16}\theta_2(2\tau)^4\theta_3(4\tau)^4$ and $B = (E_{\text{odd}} - f_8)/2 = \frac{1}{16}\theta_2(2\tau)^4\theta_2(4\tau)^4$ are POSITIVE theta series ( $B = 16R$ , closed convolution formula, $\mathbb{N}^8$ microstate count with degeneracies 1, 4, 10, 24, 43); the eta-free generator $a_n = \sigma_3(n) - 32R(n)$ reproduces every odd $f_8$ coefficient — check32 becomes a kernel COROLLARY; symbolic layer sympy-exact, classical citations typed (Gauss/Jacobi, Jacobi's quartic, Sturm 1987); the automaton composition laws censused on 41053 pairs + 70 prime powers; the five-condition validator passes all odd primes $< 10^5$ with joint detection 1.000000 and the tamper blind spot EXACTLY the $16\mathbb{Z}$ lattice (15/16); controls fire. Lean: <b>PositiveC2Lift.lean</b> (build green, 3406 jobs). |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v789_multirate_constdepth.py | The multirate 2-adic depth ladder + its constant-depth proof, ONE module from two probes. MULTIRATE-THEOREM / SPINLIFT-VERIFIED + CONSTDEPTH-THEOREM (23/23 + 24/24):<br>$v_2(1 + p^3 - a_p) \geq 5 + 2[\chi_{-4}(p)=1] + [\chi_8(p)=1]$ for all 78497 odd primes $p < 10^6$ (sharp witnesses 3/7/5/17; depths 5/6/7/8 per class mod 8); character sifts I0–I3 Sturm-certified at level $\leq 512$ ; mod-512 must-fail at $n = 9$ ; the Shimura lift $\text{Sh}(g) = -8E_{\text{odd}} + 256H$ exact to 160 with Kronecker factor $\chi_8$ . THE PROOF: the weights $(1, -3, 2)$ telescope to the both-odd convolution; class 3 mod 8 pinned by the unique $x^2 + 2y^2$ ([N1] $h = 1$ ) $\Rightarrow v_2 = 5$ exactly; class 5 mod 8 by the ternary bridge $R_3(p)/8 \equiv \#\{x^2 + 4w^2 = p\} = 1 \pmod{2}$ ([N2] $h = 1$ ) $\Rightarrow v_2 = 7$ exactly; $10^6$ census ZERO exceptions; composites violate as required.                                                                                       |
| v790_snf_mu4_theta.py        | The SNF fingerprint decider + the $\mu_4 \rightarrow C_2$ collapse adjudication, ONE module from two probes. SNF-FINGERPRINT + MU4-PARTIAL (18/18 + 17/17): the correction theta has the representation identity $256R(n) = \#\{\text{odd}(u, v) : Q = 4n\}$ and $\text{SNF} = (1^4, 2^4)$ matches the ramified inclusion $(1+i)L \subset L$ exactly — but the isometry census is EMPTY at every admissible level (kissing 240 vs 8; 0/20160 $GL(4, \mathbb{F}_2)$ , alternating vs odd form; the naive 4+4 split inadmissible BY THEOREM, v638 replicated): a FINGERPRINT, not an isometry. Part 2: the NEW exact identity $c_{\text{sig}} = \frac{1}{15}E_4(q^2) - \frac{6}{5}E_4(q^4) + \frac{32}{15}E_4(q^8) - 8f_8$ — the cusp part of the QUARTIC $\mu_4$ character is exactly $-8f_8$ with $-8 = 52 - 60 = -\text{rank}(E_8)$ ; the quadratic ( $C_2$ ) channel is PURE Eisenstein; the all-shell quartic vanishing is FALSE (closed series $\prod(1 + q^{2n})^{-4}$ , fails from grade 2). Controls fire. |
| v791_positive_descent.py     | The positive-protocol descent decider. DESCENT-PARTIAL (30/30 incl. the 5 must-fire controls): the packet GNS state on $\mathbb{N}[C_2] \otimes \mathbb{N}[\mathbb{F}_2^4] \otimes \mathbb{N}[\mu_4]$ is MANIFESTLY positive at every depth (all 128 character values $\geq 0$ exactly on the Fraction leg); F1: the naive linear pushforward is NOT a state ( $-6.6 \times 10^{-2}$ , depth-flat) — signedness is observer projection; the GL1 descent is BIT-IDENTICAL to the deployed Weil window (rel $6 \times 10^{-16}$ ) and PSD on all 7 rungs with exactly the corpus margins; F2: GL1 is the UNIQUE PSD sector of 24 — the breaking tensor factor is the CONTINUUM/pole leg, not the registers. The three-object contract (sector-adapted continua / GL1 PSD persistence in the Z1-COMPACTNESS frame / the carrier intertwiner) is registered OPEN [O], cross-referenced with PRIME.Z1.COMPACTNESS.01. All controls fire.                                                                               |
| v792_ftransfer_kms_coset.py  | The coset assignment of the fibered clock functor is a THEOREM from KMS/modular typing. KMS-COSET-THEOREM (18/18): $S = -(\ln \lambda)^2/2$ with $\lambda$ the time-1 $\text{PGL}_2$ holonomy multiplier per clock e-fold; the classifier uses NO Schwarzian and NO derivatives (non-circularity gated on its own source): functional-identity holonomy witness + $\text{PGL}_2$ class + detailed-balance multiplier cross-ratio (the discrete KMS signature); Lemmas A/B/C sympy-exact; BLIND protocol: $F_{\text{pole}}/F_{\text{Boltzmann}}$ modular with $ \ln \lambda  = \Delta = 6 \log(3/2)$ , $F_{\text{QCD}}$ autonomous (parabolic), $F_{\text{relic}}$ degenerate; unblinding EXACT (predicted cosets = transported Schwarzians, sympy zero); controls: fake KMS caught, affine invariance, the v578 Moebius clock as the positive case — the fiber classification of the CONTRACT.F.01 refinement is derived, not case-inspected.                                                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v793_f8sector_conductor.py   | Sector-adapted continua + conductor functoriality of the $\Gamma_{\mathbb{R}}$ rule, ONE module from two probes. F8SECTOR-PSD + CONDUCTOR-FUNCTORIAL (15/15 + 16/16): the $f_8$ sector, broken at $-132..-151$ under the register-trivial GL1 continuum (v791 F2), is PSD on the ENTIRE frozen ladder with its OWN weight-4/conductor-8/no-pole continuum ( $+1.52 \times 10^{-4} \rightarrow +3.20 \times 10^{-5}$ ); $\chi_4$ PSD; three-sector coherence; the wrong continuum reproduces $-140.8$ . THE RULE: $\chi \mapsto q^{s/2} \prod_j \Gamma_{\mathbb{R}}(s + \mu_j)$ is ONE closed structure map (universal per-factor kernel $+\ln q$ at lag 0), duplication ward $2.7 \times 10^{-15}$ ; the twist $f_8 \otimes \chi_{-4}$ has conductor 16 by Atkin–Li + the Fricke ward ( $6.7 \times 10^{-16}$ ) and is PSD with the rule continuum; four-sector coherence; the conductor datum is load-bearing at exactly $\ln 2$ and the PSD margins pin $q = 16$ ; the mirror sector typed non-automorphic; PSD stays finite-level EVIDENCE (GRH-type statements unproven).                                            |
| v794_cp_extension_gate.py    | The infinite-level CP extension gate — net formalization + deep margin trend. CPGATE-UNDECIDED-FALLING (17/17, adjudicating run 2 with two declared repairs): N1 restriction exactness (deep combs = shared prefixes at $1.7 \times 10^{-15}/0.0$ — $W_{\chi}$ well-defined on $V_{\infty}$ WITHOUT a limit), N2 Cauchy interlacing, N3 uniform boundedness ( $\sup/\text{first} \leq 3.24$ ; the frozen $e^{X/2}$ pole prediction was WRONG — declared recalibration). The limit object typed exactly: $\inf_X \lambda_{\min}(T_{\chi,X}) \geq 0$ per automorphic character = the v780 Z1-COMPACTNESS demand SECTOR-DECORATED (anchor $3.882 \times 10^{-6}$ at $M = 1176$ reproduced). Certified cuspidal deep layer: two-modulus CRT eta to $X = 13$ , Deligne-certified lift (0.999455). THE TREND: no sector crosses zero; GL1 falls power-law to $3.9 \times 10^{-6}$ at $X = 18.375$ ( $\beta = -1.69$ ); $\chi_4$ SATURATES at $\sim 1.5 \times 10^{-6}$ (the first measured floor); $f_8$ $\beta = -1.63$ ; twist $\beta = -2.10$ . Controls fire (conductor swap = margin $-\ln 2$ at $2.2 \times 10^{-16}$ ). |
| v795_halving_tails_k5.py     | The exact-depth mechanism for the classes 7 and 1 mod 8 + the $k = 5$ anomaly decided by the 2-power residue tower, ONE module from two probes. TAILS-IDENTIFIED + K5-MECHANISM-FOUND (20/20 + 17/17): EXACT identities $v_2(D_p) = 6 + v_2(X_7)$ resp. $8 + v_2(X_1)$ with termwise-integral divisor sums (composites included; $10^6$ census zero exceptions); class-7 base $\iff R_3(p) \equiv 16 \pmod{32}$ ( $h = 1$ ternary via the $\chi_{-4}$ bridge); the class-1 two-shape criterion exact, its single-character bridge typed OPEN. THE TOWER: $k = v_2(X_1)$ is deterministic on the $(m, t)$ cells ( $v_2(p-1)$ , 2-power depth of 2) with zero exceptions; $k = 5$ lives ONLY in cell (5, 3) of density $1/64 \Rightarrow P = 1/128$ = the observed half mass (mixture residual 0.6); Gauss quartic census clean; the class-7 MIRROR LAW $v_2(D_p) = 3 + v_2(p+1)$ through $k = 4$ explains the class-3 constancy and the clean class-7 tail; the governing-field derivation typed open (Cohn–Lagarias frame cited). Controls fire.                                                                         |
| v796_duad_syntheme_bridge.py | The canonical OUTER bridge between the code duad side and the curve syntheme side. BRIDGE-CANONICAL (24/24): the Sylvester duality $\beta$ : code odd forms $\rightarrow$ curve spreads with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ (the UNIQUE $\langle \tau, \rho \rangle$ -fixed spread) exists and is unique up to the order-6 code joint stabilizer (census $18 + 18 \rightarrow 6 + 6$ , ONE orbit incl. the $\tau/\tau^2$ swap by an inner code symmetry); the outer witness exact ( $[3, 1, 1, 1] \leftrightarrow [3, 3]$ uniformly); $q_{\Theta}$ pulls back CONSTANTLY to $q_{\text{even}}^*$ with $q^* = q_{\text{even}}^* + \bar{h}(\cdot, A)$ — the anchor shift resolves the v784 Arf mismatch canonically; $\bar{5}/10$ and $3+2$ transport exactly onto the $S^*$ -line geometry; $t_A \mapsto \rho = J$ on every bridge; $\chi_{\text{NSR}}$ an honest exact TWO-variant torsor (pinned by the v752 NS/R parity). Controls: inner censuses 0/0, odd-Arf pullback 0, scrambled $\sigma$ 0. The v784 INNER identification stays dead; no matter semantics.                                      |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v797_r_microstate_identification.py | The R-microstate identification — the honest negative.<br>R-NO-ARROW-REALIZATION (14/14; frozen dictionary, the disclosed pilot produced only kills): $R(p) = 1, 4, 10, 24, 43, 68$ (the sheet-switch packet count, $B_p = 16R(p)$ arrow-exact) has NO arrow-level realization at the tested places: type-B orbit counts $(1, 4, 13)$ killed at $p = 7$ ; the type-B set not invariant under ANY deployed extension ( $J/\sigma/\text{conj}/W(E_8)$ ); the refined census separates all 13 orbits; pointwise switch arrows do not exist; the Kneser $J$ -stability census has $\leq 2$ classes. The COUNT identity is exact everywhere ( $16B_n = 256R(n)$ = the odd-vector count of $\text{diag}(1^4, 2^4)$ at $4n$ ) but count-level ONLY: SNF $(1^4, 2^4)$ shared, isometry excluded (minimum vectors 8 vs 240). Structural boundary typed: the packet and arrow layers share totals, not the $R$ -grading. Controls fire.                                                                                                                              |
| v798_seam_clifford_modular_s.py     | The missing Clifford bit as the metaplectic lift of modular $S$ at $\tau = i$ .<br>MODULAR-S-PARTIAL (27/27): COSET POSITIVE — the per-binary-factor Hadamard $K_1 = H \otimes I$ IS the Weil lift of $S$ at $\tau = i$ and lands exactly in the missing coset $\zeta_8 G_{31}$ (exact relations $S^2 = I$ , $T^2 = \text{Pauli parity}$ , $(TK)^3 = \zeta_8 I$ ); the TOTAL Fourier $H \otimes H$ lies INSIDE $G_{31}$ with metaplectic anomaly $(T_3 K_3)^3 = iI = J$ — the deck IS the anomaly of the total modular lift (R2 placed); $\zeta_8 = (1+J)/\sqrt{2}$ the real half-deck swapping the seam pair $\{L, \zeta_8 L\}$ ; $F_4$ typed out; OS reflection gate exact ( $\tau = i$ fixed). THE STRICT UNIQUENESS CENSUS IS EMPTY (0/46080): no $\Theta$ -real metaplectic involution in the coset fixes $q^*$ — the Arf defect is unavoidable; retyped census: 32 survivors = ONE $G_{31}$ -conjugacy class; R3 NAMED, not fixed: the $\sigma$ -orbit defects are exactly $\{A + F_\Sigma + F_i\}$ with XOR = the anchor $A$ itself. Controls fire. |
| v799_seam_code_typeiii.py           | Type II FORCED from three physical seam axioms — the direct attack on residual R1 of ARF.BOUNDARY.CODE.01. TYPEII-FORCED (19/19): exhaustively over ALL 308,993 subspaces of $\mathbb{F}_2^8$ (dim 0..4, Gaussian binomials; dim $> 4$ by rank-nullity): mutual locality of the Construction-A vertex words $\iff$ self-orthogonal (135 at dim 4; the lift-shift braiding identity exhaustive); integer conformal spin $\iff$ doubly even (30); holomorphy/index one ( $ \text{disc}  = 1$ , exact Fraction Gram determinants) $\iff$ self-dual — the survivor set is EXACTLY the 30 $\hat{e}_8$ copies; no boundary-legal non-Type-II code exists; every non-self-dual doubly-even code has a fractional-weight discriminant sector ( $[8, 3]$ witness with exact coset thetas, $\theta_{L^*} \neq 2\theta_L$ ). R1 $\rightarrow$ R1' = A1 mutual locality + A2 integer spin + A3 holomorphy/index one; imported lattice-VOA/theta ingredients cited and machine-verified in every instance used. Controls fire. NO P2 marker move.                       |
| v800_e8_torsor_fourier.py           | The projective Fourier modes of the 15 Gaussian torsors — carrier purity at CHARACTER level with the mutually-unbiased cap.<br>TORSOR-FOURIER-PARTIAL (30/30): 120 canonical projective rays (phase law exact on 960 origin shifts), orthogonal basis (Gram = $16I$ ); deck $J$ and all 16 Paulis diagonal by characters, $G_{31}$ monomial through $Sp(4, 2)$ (kernel $64 = \mu_4 \times \text{Pauli}_{16}$ ; orbits $15 + 45 + 60$ ). THE CARRIER BLOCK: the 8 R-fibers' $64 \text{ rays} = (1+5+10) \boxtimes (1+3)$ under $W(D_5) \times W(A_3)$ (Clebsch charpoly $(x-5)(x-1)^{10}(x+3)^5$ exact; $S_4$ -natural tetrad with $\sigma$ = family 3-cycle + anchor) — the carrier Pascal at CHARACTER level, impossible at point level (v775); P3 fails as predeclared: the weight and character bases are MUTUALLY UNBIASED ( $ \langle \text{pair}   \text{mode} \rangle ^2 = 1/8$ exactly) — ROOTCLASS-MIXED intact; the 45 deck-even modes ARE the 45 (context, Pauli) incidences under the certified chart. Controls fire.                          |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v801_prime_cp_intertwiner.py  | <p>The carrier intertwiner in Stinespring form — demand (3) of PRIME.POSITIVE_DESCENT.01 finite-level SOLVED.</p> <p>CP-INTERTWINER-EXISTS (29/29): <math>\Phi(a) = V^*\pi(a)V</math> from the 105 Kraus legs (unitarity <math>\sum_e E_{x_e x_e} = 7I</math> exact integer); the arrow *-algebra = the FULL <math>M_{15}</math> (exact matrix-unit closure; involution = arrow reversal); Choi exact-rational (105 eigenvalues each <math>1/7</math>, PSD structural); Choi census exact; all covariances exact (<math>\sigma</math>, deck <math>T = 4B^2</math>, half-weight, <math>\beta = 1</math> balance, Gate-0). THE DECIDER: the four automorphic channels of the ONE map land on the four sector windows — GL1 = the deployed Weil window at rel <math>6.0 \times 10^{-16}</math>; <math>\chi_4/f_8</math>/twist on their <math>\Gamma_{\mathbb{R}}</math>-rule objects (<math>\sim 10^{-15}</math>), all PSD; the ramified conductor emptiness STRUCTURAL (<math>a(2^k) = 0</math> + glue-uniform slot); the <math>(E, f_8)</math> pair = ONE primitive element, two characters; F1 retyped: the naive linear pushforward IS a negative Choi eigenvalue (<math>-6.6 \times 10^{-2}</math>). Demands D1–D3 (dilation family, normality on the Z1-COMPACTNESS object, the limit identification — contains RH) registered open.</p>                                                                                                                                                                                                     |
| v802_gnet_martingale.py       | <p>The martingale route to the <math>G_{\text{net}}</math> GNS limit — finite hypotheses MEASURED, all met. MARTINGALE-HYPOTHESES-MET (15/15): the filtration is EXACT (Haar isometry with support coherence, leak 0; Watatani quasi-basis 185/3041 at <math>l = 2/3</math> with <math>\sum vv^* = 4 \cdot 1</math>; ONE index-4 expectation for the whole tower); the state defects are SUMMABLE at measured rate <math>\sim 0.25</math> per doubling — one power better than the <math>2^{-l}</math> dim-law envelope (reported, not forced); the unique limit follows by telescoping and is verified on every rung; the split/normality precursors are uniform (determinant witness <math>D \geq 0.985</math>, stabilizing at 0.9928). Steps 5–7 (GNS, local normality, split + <math>\mu_4</math> simple-current extension) typed NOT executed. QGEO note: the open majorant residues (b)/(c) are bypassed for limit EXISTENCE and relocated into the typed normality step — a reallocation, not a proof. Controls fire (decimation converges to the WRONG degenerate limit — the coherent bond is load-bearing).</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v803_e8_jetcode_affine_nsr.py | <p>The jet-code normal form of the Gaussian E8 compiler + the NS/R bit as an affine 8-point geometry. JETCODE-EXACT + AFFINE-NSR-EXACT (18/18 + 15/15): <math>\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)</math> with <math>\epsilon = 1 + i</math> (2 RAMIFIES; unique iso, the split ring <math>\mathbb{F}_2 \times \mathbb{F}_2</math> fails all 24 bijections); <math>W = L/2L</math> is FREE of rank 4 over the dual numbers (explicit <math>\mathbb{Z}[i]</math>-basis by Hermite reduction, covolume-certified); <math>0 \rightarrow \epsilon W \rightarrow W \rightarrow V \rightarrow 0</math> is an exact SELF-extension with <math>\ker \epsilon = \text{im } \epsilon</math>; the deck IS the jet unit <math>J = 1 + \epsilon</math>, re-deriving the v782 torsor on all 240 roots; ZERO <math>\mu_4</math>-equivariant splittings among all 65536 linear sections — the no-origin obstruction IS non-splitness; the 120 root pairs = the <math>q = 1</math> unit shell (<math>240 = 15 \times 8 \times 2</math>). PART 2: <math>V \setminus \{0\} = 7</math> NS + an affine 8-point torsor carrying 112 <math>D_8</math> + 128 spinor roots (<math>248 = (112+8)+128</math>); <math>\text{Stab}(\chi_{\text{NSR}}) = \text{AGL}(3, 2)</math> of order 1344 = <math>\text{Aut}(H_8)</math> with the SAME natural 8-point action; affine functions reconstruct <math>RM(1, 3) = H_8</math> with exactly 1344 equivalences. Typed internal reconstruction — no P2 removal; no matter semantics (ROOTCLASS-MIXED).</p> |
| v804_prime_carrier_gray.py    | <p>The affine (Gray) register-to-carrier identification EXISTS, exactly respecting the v786 kill. GRAY-INTERTWINER-EXISTS (23/23): the Gray map encodes <math>C_2 \times \mu_4</math> affinely into the R-class torsor (regular <math>\mathbb{Z}_4 \times \mathbb{Z}_2</math> action; a group iso is IMPOSSIBLE — exponent 4 vs 2, all 64 homomorphisms have image <math>\leq 4</math> — and is NOT demanded: the v786 kill is respected, the encoding lives inside <math>\langle J, \epsilon_{67} \rangle</math>); the <math>\chi_{\text{NSR}}</math> split = the adjoint/spinor split exactly on all 240; the fiber family <math>W_\alpha</math> is deck-covariant EXACTLY (<math>J</math> = phase translation, zero defect), <math>\sigma</math>-covariant up to a torsor twist with TRIVIAL measured <math>\mu_4</math>-cocycle class (all phase sums 0 mod 4; residual = a pure <math>\mathbb{Z}_3</math> line twist on the fixed classes), chart-independent and KMS-compatible; the extended Stinespring map's GL1 channel lands on the deployed Weil window at rel <math>6.0 \times 10^{-16}</math>. One declared control repair (spec v2); demands D1–D3 open.</p>                                                                                                                                                                                                                                                                                                                                                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v805_e8_syndrome_algebra.py  | The honest kill — E8 is NOT an error-correction hull in the sharpest sense. SYNDROME-DEAD (13/13, the kill fires as preregistered): the $\mu_4$ -linear-lift realization family reaches max rank 222/256 over ALL FOUR frozen phase conventions (sympy-exact over $\mathbb{Q}(i)$ ); the $\pm r$ proportionality obstruction is STRUCTURAL for the whole diagonal-phase family ( $c(-r) = -c(r)$ ; collision witness printed), and both post-hoc escape directions score WORSE (nonlinear 215, $\zeta_8$ -linear 160) — the death is robust. SURVIVES: the syndrome-table counting $16 + 240 = 256$ , the neutral sector (the 16 v112 kernels ARE the diagonal MASA; same-class products $A_0$ -valued with $\mu_4$ coefficients EXACTLY), the fine groupoid law (associative 0/3375) and the $W(3, 2)$ commutation skeleton; NO scalar V-cocycle (0/210). Successor criterion named: root-individual operator data. Controls fire.                                                                                                                                                                                                                                                    |
| v806_hecke_local_clifford.py | The analytic activation-ordering hypothesis dies; the rule map gains two instances. LOCAL-CLIFFORD-BREAKS (16/18, the two FAILs are the preregistered decider gates themselves; controls 4/4): the four $p \bmod 8$ class combs decompose EXACTLY into the mod-8 character system (Ward $4.4 \times 10^{-16}$ ), but ALL FOUR class sectors fail PSD — signed character combinations carry no completed $L$ -function ( $c = 3, 5, 7$ at $O(1)$ , growing), and the trivial mod-8 component is IMPRIMITIVE (the $(1 - 2^{-s})$ zero line $\Re s = 0$ ); the EULER REPAIR makes class 1 PSD EXACTLY ( $+1.4 \times 10^{-5}$ ) — the UNIQUE positivity-completable class (trivial coset); margin order c5-c3-c7-c1 on 7/7 rungs, Spearman +0.40: NO canonical order — the Clifford activation reading stays decorative analytically while the arithmetic multirate theorem is UNTOUCHED (re-anchored at $2 \times 10^5$ ). POSITIVE: $\chi_8/\chi_{-8}$ are two NEW PSD instances of the $\Gamma_{\mathbb{R}}$ conductor rule (exact parities even/odd) — conductor functoriality extended; next falsifier named.                                                                        |
| v807_lorentz_nullselector.py | Null-cone spinor coordinates + the independent $d = 4$ forcing. LORENTZ-COORDS-REVEAL + NULLSELECTOR-ALGEBRAIC (22 checks / 3 preregistered-honest FAILs + 14/14): $M(5, -3, 4) = 2(1, 2)^T(1, 2)$ EXACT — the compiler triple is a rank-one SPINOR SQUARE; Euclid $(2, -1) \mapsto (5, -3, 4)$ exact. THE REVEAL: in the $\alpha$ -ordered ladder the spinor slope is STRICTLY MONOTONE (66/66, Kendall 1.000, CR contraction CV 0.004) — the $h$ -oscillation was $\alpha$ -jitter; the approach $2 - r \sim \alpha^{-0.23}$ stays glacial, the 1D monotonicity candidate named. PART 2: $ v_d _L^2 = \det M(v_d) = d(4 - d)$ identically for the RR/Serre functor vector; null/rank-one iff $d = 4$ ; $(1, 2) =$ the anchor support via four exact identities; THE BRIDGE: $g^2 - N^2 = 4d =  H ^2$ — null $\iff$ glue index $= d$ , independent of the v781 census, same output $\{4\}$ ; honest downgrade: the seam family has NO $d$ -dependence to vary (the first boundary reading is rank-one universally — measured), the missing theorem typed; wrong functors have no integer null point.                                                                                  |
| v808_petersen_sixthroot.py   | The K5 edge machine + the bit-exact sixth-root decider. PETERSEN-RADIAL-EXACT + SIXTHROOT-SPECTRAL-ONLY (19/19 + 16/16): $X = (-2, -2, -2, 3, 3)$ unique primitive traceless block-constant; the $E(K_5)$ charge census $\{-4: 3, +1: 6, +6: 1\}$ = the $SU(5)$ decuple against the certified v774 table; Petersen $SRG(10, 3, 0, 1)$ certified ( $ \text{Aut}  = 120$ census $= S_5$ ); distance shells $(1, 3, 6)$ CHARGE-PURE; the flavor hexagon = the distance-2 shell [E neu]; equitable quotient with hypercharge eigenvector $(6, -4, 1) \rightarrow -2$ ; $\text{spec}(P^6) = \{1, 64/729, 1/729\}$ = the deployed transport spectrum EXACTLY (v54/v56/v124); the 10-dim multiplicity prediction ( $\times 5 = g_{\text{car}}$ fast, $\times 4 =  \mu_4 $ slow) typed [H neu]. PART 2: $B = \frac{1}{18}[[13, 1, 4], [1, 13, 4], [4, 4, 10]]$ (every entry a compiler number) is the UNIQUE symmetric doubly stochastic PSD sixth root in the corpus basis; $B^6$ carries the deployed spectrum exactly but is bit-equal to NO deployed matrix ( $GL(3, \mathbb{Q})$ -conjugate to v486's $M^6$ and the cusp-diagonal form) — the corpus freezes the SPECTRUM, not the basis. |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v809_curve_code_<br>doily.py        | The certified outer bridge as an incidence-operator statement with a spectral payoff. DOILY-EXACT (21/21): $NN^T = B + 2I$ ENTRYWISE ( $N$ = the $K_6$ doily incidence, 15 duads $\times$ 15 syntheses; $B$ = the certified v752/v774 symplectic incidence); $\text{spec} = \{9, 4^9, 0^5\}$ exact; the singular values of $N/3$ are $\{1, 2/3, 0\}$ : the RECOVERY BASE RATE $2/3$ (v221) as the unique nontrivial singular value of the canonical duad-to-synthese incidence — the $(2/3)^6$ identification typed [H neu], pending the six-step composition; Petersen edges = syntheses 15/15 at the $q^*$ point; EVERY census bridge factors EXACTLY as $N_{\text{code}} = P_{\Delta}^T N_{\text{curve}}^T P_{\Gamma}$ — a DUALITY, not an isomorphism (the inner route stays dead, 0/720); the Arf switch set $T_6$ = the six within-block duads; $N/3$ doubly stochastic, typed classical, not CPTP.                                                                                                                       |
| v810_cutcode_<br>moment.py          | The Arf split as the cut classification of $K_5$ + the one-line master moment with the frozen 210 decision. CUTCODE-EXACT + MOMENT-INCIDENCE-PARTIAL (17/17 + 16/16): the cut map $\delta$ is injective-isometric on all 256 cells (kernel = the repetition code); the Arf split $16 = 1 + 5 + 10$ IS the cut classification $(0 5) + (1 4) + (2 3)$ [E neu]; the Clebsch 40 edges grade $5 + 20 + 15$ over the Arf shells with forbidden blocks EMPTY and within-10 = Petersen (isomorphism census 120); the 240 roots / 60 $\mu_4$ -lines grade $80 + 160 / 20 + 40$ — Arf/cut grading, NOT matter. PART 2: the symbolic identity $\sum_{i < j} (x_i + x_j)^2 - 3 \sum x^2 = (\sum x)^2$ yields $\text{Tr}_{S^+} X^2 = 4 X ^2 = 4h(E_8) = 120$ in one line; charpoly $(t+2)^3(t-3)^2$ exact, 72 audit-only; the $210 = \binom{10}{4}$ map DECORATIVE — equivariant bijections killed by the fixed-point obstruction (exhaustive); the partial $80 \rightarrow 5 \times 16$ miss-slot map and the Newton tie survive as audit. |
| v811_prime_kraus_<br>doily.py       | The 105 Kraus legs as a complete context protocol over the doily $W(3, 2)$ . KRAUS-DOILY-PROTOCOL (21/21): $105 = 15$ diagonal + 90 off-diagonal; the canonical $45 + 60 =$ reversal quotient (the 90 legs double-cover the 45 flags 2:1, deck = the $*$ -involution) + fiber expansion ( $15 \times 4 = 60 =$ the v783 stabilizer rays); the $Sp(4, 2)$ obstruction CERTIFIES that no leg-subset partition exists; the literal partition is spread-relative = the ledger T3 census $(3, 1, 1, 1, 1)$ ; the 45 transports are QUADRATIC ( $P_x P_y = \pm P_{x+y}$ ) and span with <b>1</b> the FULL $\text{End}(\mathbb{C}^4)$ at rank 16; $\sigma/\text{deck}/\text{KMS}$ -covariant; THE SURPRISE (measured, report-only): $[B_{45}, B_{60}] = 0$ EXACTLY — $K = \frac{3}{7}E + \frac{4}{7}R$ is a DIRECT-SUM protocol with $E$ a conditional expectation in the channel commutant; measured: the certified spread is NOT $\sigma$ -invariant (the named next falsifier).                                                     |
| v812_flavor_graph_<br>filtration.py | The flavor-matrix filtration $Q \rightarrow K \rightarrow L$ as an incidence-compiler fingerprint. FILTRATION-FINGERPRINT-ONLY (16/16): the row budgets are EXACT, order-exact counts of the $K_{3,2}/\text{prism}/K_5/H_8$ incidence chain — $Q$ $(4, 5, 6)$ , $K$ $(6, 9, 10)$ (line graph $\cong$ triangular prism $\cong C_6(2, 3)$ ), $L$ $(10, 14, 16)$ — with the level-wise additivity $(4, 5, 6) + (6, 9, 10) = (10, 14, 16)$ mirroring the deployed $L = K + Q$ ; but NO entry-level reconstruction: none of the frozen structural candidates (25/20/47 per level, entry-exact, 6 column permutations) reproduces any deployed matrix; the exhaustive false-positive rates 0.748%/0.035%/0.005% make the misses INFORMATIVE. Typed: an audit fingerprint at budget level, not an entry-level incidence construction.                                                                                                                                                                                                  |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v813_p1_index_kms.py            | The closed normalization identity $c_3 = 1/(I_{\text{Jones}} \cdot \beta_{\text{angle}}) = 1/(8\pi)$ , its convention audit, and the sharpened P1 contract. P1-IDENTITY-CLOSED (13/13): $I\beta c_3 = 4 \cdot 2\pi \cdot 1/(8\pi) = 1$ EXACT (formal Fraction $\times \pi^k$ arithmetic); dependency audit: the index $I = 4 =  \mu_4 $ is an INDEPENDENT $c_3$ -free theorem (zero occurrences in all four sources, gated by file scan); $\beta = 2\pi$ independently MEASURED (v526, wrong $\beta$ excluded; the v519 radian pin is the one shared convention and cancels); $c_3$ STAYS the P1 axiom the identity EXPLAINS; the finite evenness precursor measured: $w_1 = w_3 = 1/4$ exactly at every level, symmetric zero-sum offset $\pm 2^{-(l+1)}$ (the no-anomaly shape); the sea state does NOT equidistribute (dev 0.188) — the bridge must normalize the MODULAR RESPONSE; contract registered [O] with the one open bridge and the kill criterion; controls: index $2 \rightarrow 1/(4\pi)$ , booked elsewhere by the corpus. |
| v814_k5_sixstep_transport.py    | The six-step composition and the canonical 10-dim transport proposal. SIXSTEP-CLOCK-ONLY + TRANSPORT10-PROPOSED (23/23): the rate $(2/3)^6$ arises canonically on three spatial routes (doily alternation, hexagon compression, hexagon circulation), each blocked by a typed obstruction (no unit mode / norm obstruction, bipartite sign $-2/3$ , wrong space) — the deployed six is the CLOCK exponent (v124 $\lambda_n = (1 - n/3)^6$ , v486 hand, fixed eigenbasis); sixth-power blindness [E neu]: only per-step data distinguish walk from clock; dated correction on the v808 row: $T_{v221} = B^6$ BIT-EXACT, so SIXTHROOT upgrades to BITEFACT-vs-v221; the canonical proposal $T_{10} = P_{10}^6$ in the $\Lambda^2$ basis passes all gates (spectrum $\{1, (1/3)^6 \times 5, (2/3)^6 \times 4\}$ , equitable v774 quotient $= (Q_{\text{Pet}}/3)^6$ bit-exact) — a promotion-ready proposal, no deploy claim.                                                                                                                  |
| v815_kraus_spread_commutant.py  | $\sigma$ breaks every single spread but the direct-sum protocol is $\sigma$ -stable as a family; $[E, K] = 0$ extends to the full tower; the $\{K, \sigma\}$ commutant is the 39-dim nonabelian closure of the $A_{\text{orb}}$ refinement. SPREAD-SIGMA-BROKEN + COMMUTANT-EXTENDS + AORB-NOT-MAXIMAL (21/21 + 24/24): $\sigma = \text{two } 3\text{-cycles on the 6 spreads (zero fixed, outer-class forced)}$ ; invariant witnesses $\sum B_{45}(s) = 4I + 2B$ , $A_{\text{orb}}$ spectrum $\{0^5, 1^2, 4^5, 7^2, 9^1\}$ ; $[E, K] = 0$ lifts exactly to channel / KMS tower / odd places (the $C_2$ -character expectation unique per place) and the 600-dim register, integer-exact; $\dim C = 39$ , $\mathbb{R} + M_3(\mathbb{R}) + M_3(\mathbb{C}) + M_3(\mathbb{R}) + M_1(\mathbb{C})$ ; $A_{\text{orb}}$ the canonical abelian 5-dim subalgebra, window-silent beyond the $K$ -spectrum; the exact character lemma $B\chi_w = 8e_{\Omega^{-1}w} - \mathbf{1}$ .                                                                   |
| v816_prime_mosco_selection.py   | The selection problem of the unified compactness contract finite-level SOLVED: Mosco form convergence + Friedrichs minimality produce the unique limit candidate exactly where moment selection oscillated at 22. MOSCO-SELECTS (6/6 legs, 23/23 guards+controls): decider table (identical statistic / rungs / battery): moment oscillation 22.08 vs Friedrichs resolvents 0.0004 / ledger-adjusted Weyl 0.084 / raw $\mu$ -face 0.0000; cofinal-unique $2.5 \cdot 10^{-5}$ ; M2 recovery exact $1.2 \cdot 10^{-12}$ ; M1 FAR margins +2.6 with the EDGE family typed a genuine boundary channel; the import diagnosis completed at form level — nothing whitened at the core (whitened control reproduces the 5.8772 delivery failure exactly); the three infinite-level ingredients typed; positivity explicitly untouched (the sector floor stays PRIME.FLOOR.RATIO.01).                                                                                                                                                               |
| v817_positive_descent_master.py | The abstract positive-descent master theorem, its kernel-checked Lean core, and BOTH instances. MASTER-BOTH-INSTANCES (14/14): the theorem (7 frozen hypotheses) with the kernel-checked core (telescoping + Cauchy tail, positivity closure, sector commutation; TfptCarrier/PositiveDescentMaster.lean); instance $G_{\text{net}}$ re-verified (Choi PSD, defect rate 0.267/doubling, $\mu_4$ evenness $w_1 = w_3 = 1/4$ exact); instance PRIME/PACKET — THE NEW NUMBER: the packet state defects on the 128-dim register are SUMMABLE at 0.701/doubling $\approx$ the $\sqrt{\cdot}$ -dilution law $2^{-1/2}$ (with $\hat{m}_2(p) = -1/15$ exactly for every odd prime); the selection problem eliminated on both sides; control K4: positivity is a separate hypothesis, not implied by summability.                                                                                                                                                                                                                                   |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v818_sector_floor_attack.py | The sector floor as a mechanism: the exact margin factorization, the $O(1)$ -faithful interlacing capture, the symbolic rotation law, and the $h^{-3/2}$ amplifier envelope — the floor reduces to ONE ratio inequality. FLOOR-MECHANISM-PARTIAL + CAPTURE-BOUNDED / INGREDIENT-II-THEOREM-SHAPE / AMPLIFIER-MECHANISM (9/13 + 13/15 with exactly the preregistered FAILs): onem = $F(r) \varepsilon K$ symbolically exact with the sign entirely in the vanishing weight $\varepsilon = \tau/\lambda$ ; interlacing capture $\tau/\lambda_{\min}(A_h)$ median 1.53 (range 1.04–4.36) — the 2-mode lock block is an $O(1)$ -faithful witness of the full floor; capture angle $\cos \theta$ median 0.990, scramble-robust (frame content); the rotation law $dr/dt = -\kappa(w \cdot v_1)(w \cdot v_2)(1 + r^2)/(\lambda_1 - \lambda_2)$ closes symbolically, staircase sign-match 100%; the amplifier is carried by the smallest prime powers, envelope $\rho h^{3/2} \in [4.85, 24.2]$ non-decaying; the contraction constant is density-level (Epstein $s_E = 0.241$ ); the reduced floor statement feeds PRIME.FLOOR.RATIO.01 [O]: $\rho = \tau/\tau_{\text{pnt}} > 0$ . NO RH claim. |
| v819_prime_packet_rm14.py   | The Reed–Muller identification of the 15-label channel. RM14-EXACT (21/21; Lean companion PacketRM14.lean kernel-checked, 30 theorems): the row hull of $B$ = the punctured $\text{RM}(1, 4)^* = [15, 5, 7]$ with weight enumerator exactly $1 + 15z^7 + 15z^8 + z^{15}$ ; $C_0$ = the $[15, 4, 8]$ simplex; $\chi_{\text{NSR}}$ is ONE codeword (the 7 + 8 split = its two sides, the $C_2$ register = the affine bit); the self-reproduction cycle $\text{RM}(1, 3) \rightarrow E_8 \rightarrow V \rightarrow \text{RM}(1, 4)^* \rightarrow \text{RM}(1, 3)$ closes with exactly 1344 = $ \text{Aut}(\text{RM}(1, 3)) $ equivalences; the CSS code $[[15, 1, 3]]$ with dual $[15, 10, 4]$ = shortened $\text{RM}(2, 4)$ , logical distances (7, 3), triorthogonality 8/4/2; the $\zeta_8$ phase structure typed [H].                                                                                                                                                                                                                                                                                                                                                                    |
| v820_prime_kraus_rm24.py    | The honest double kill of the literal leg-support claim + the canonical $\omega$ -built rule + the $A_3/B_3$ plane-frame identities. KRAUS-RM24-RULE-CANONICAL + PLANEFRAMES-EXACT (16/16 + 14/14): the literal claim dies twice (support weights $1/2/3 < \text{distance } 4$ ; $\text{Sp}(4, 2)$ orbits $15 + 90$ vs $45 + 60$ ); the canonical rule (built from $\omega$ alone) identifies the rule-A protocol set with the 105 weight-4 checks of $[15, 10, 4]$ exactly, $\text{Sp}/\sigma/\text{deck}/\text{KMS}$ -covariant; all six Gram identities entrywise ( $12I + 3J$ , $16I + 4J$ , exact $\pm B$ cancellation, $22I + 6J = 2B^2 + 14I$ , $28I + 7J$ with spectrum $\{133, 28^{14}\}$ ); $A_3 = 12$ / $B_3 = 16$ ARE the per-point plane completion counts — $\sigma_3(3) = 28$ and $a_3 = -4$ as plane-count statements [E neu]; the graded frame intertwiner $V^*V = \sigma_3(n)I$ , $V^*\Gamma V = a_n I$ exact for $n = 3, 5, 7$ ; $133 = \dim E_7$ typed [C] fingerprint.                                                                                                                                                                                               |
| v821_prime_vacuum35.py      | The vacuum completion of the parity checks: $140 = 105 + 35$ , the $E_7$ spectral completion $112 + 21 = 133$ as a matrix identity, the Weitzenböck left factor, and the exact $\pm B$ cross-term cancellation. VACUUM35-EXACT (22/22): $\text{RM}(2, 4) = [16, 11, 4]$ , the 140 weight-4 words = the 140 affine 2-planes; Grams $22I + 6J$ / $6I + J$ / $28I + 7J$ entrywise with spectra $\{112, 22^{14}\}$ / $\{21, 6^{14}\}$ / $\{133, 28^{14}\}$ ; the $E_7$ completion $112 + 21 = 133$ as a matrix identity [E neu] (the $E_7 \times A_1$ corpus reading a [C] fingerprint, anchor v170); Weitzenböck $H_{105}^T H_{105} = 2B^2 + 14I$ as the canonical label-side left factor (honest v758 scope: the continuum wall untouched); the $\pm B$ anisotropy cancels inside one Gram; tight frame with constant 28 on the mean-zero space; $\text{RM}(2, 4)$ dual-containing.                                                                                                                                                                                                                                                                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v822_prime_vacuum_dilation.py | <p>The honest kill pair: the vacuum-completion reading of the continuum is DEAD IN BOTH TRANSCRIPTIONS, with the <math>c^*</math> pencil bound as the frozen fence. UNSHORTEN-DEAD + DILATION-DEAD (17/19 + 12/14, the four FAILs being the preregistered PSD deciders themselves): the exact RM layer survives [E] (unshortening identity <math>28I + 7J</math>, footprints <math>22I + 6J</math> and <math>6I + J</math>, code dims 11/10/11, Weitzenböck bookkeeping integer); the one-slot bordering dies by the v758 mechanism verbatim and on SCALE (thin <math>\sim 10^{-5}</math> margins vs <math>O(1)</math>+re-entry), with the SHAPE theorem: one vacuum slot induces only a rank-one pure-<math>J</math> coupling — the demanded <math>6I + J</math> needs 35 separate rows; the 35-row Stinespring dilation (the corrected target) fixes the shape exactly and buys <math>\times 13.1</math> — and still dies in both sectors under both normalizations; THE GENERAL OBSTRUCTION QUANTIFIED: <math>\text{PSD} \iff \text{coupling}</math><br/> <math>\leq c^* = (2/21) \lambda_{\text{pencil}}(T_{\text{dep}}, T_{\text{cont}}^2)</math>, measured deficits <math>\times 3 \cdot 10^7</math> (GL1) / <math>\times 2 \cdot 10^2</math> (f8) — the reading dies with the calibration, not the geometry; the continuum's completion role is already exhausted inside <math>T_{\text{dep}}</math>; the path-state construction is not buildable on this gating.</p>                                                                                                                                                                                                                                                                                                                                                                        |
| v823_prime_lagrange_floor.py  | <p>The sector floor <math>\det \hat{A}_2 = \lambda \tau</math> as an EXACT machine-verified sum of squares over zero+pole rank-one carriers (Lagrange identity) + the fixed-pair certification with the budget tightened <math>\sim 10</math> orders. LAGRANGE-CONCENTRATED + PAIR-CERTIFIED (7/8 with the ONE preregistered-honest old-budget FAIL S3.CERT + 9/9): every per-zero layer of the v692 master identity is EXACTLY rank-one (<math>v_\gamma = 2\sqrt{w(\gamma)}(S_1, S_2)</math> with the real closed form <math>S_i(\phi) = \sum_j t_{ij} \sin((h - j - \frac{1}{2})\phi)</math>, layer dev <math>\leq 1.2 \times 10^{-16}</math>); the Lagrange identity <math>\det(G_Z + P) = \sum_{\text{pairs}} w_i w_j (a_i b_j - a_j b_i)^2</math> holds at machine precision on all 14 rungs (ward <math>\leq 1.4 \times 10^{-9}</math>); the POLE is the universal non-collinear leg (pole family = 1.0000 of the pair total; every top-100 pair is pole <math>\times</math> zero) and the moving zero leg obeys the ALIAS-COMB LAW (median comb distance 0.0007..0.0061 vs uniform 0.25); THE ROUTE CHANGE: <math>R = \hat{A}_2 - G_Z - P</math> with <math>\lambda_{\min}(R) = 1.6 \times 10^{-7}..2.8 \times 10^{-6} \geq \text{margin per rung}</math> and psd monotonicity eliminates the tail estimate – BUD 0.14..7.1 <math>\rightarrow 2.4 \times 10^{-12}..2.7 \times 10^{-9}</math> (median 9.8 orders); the certified interval <math>X^2(\text{pole}, \gamma_1) - \text{BUD}</math> is strictly positive on 14/14 (<math>\gamma_1</math> pinned to zetazero(1) within <math>10^{-11}</math>, dps-40 ward); floor share 0.0063..0.5832 growing as <math>h^{+1.06}</math>; A0 typed honestly: the atom side is NOT rank-one – the exact SOS realization lives on the zero+pole side. Feeds PRIME.FLOOR.RATIO.01 [O]; NO RH claim.</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v824_prime_floor_skeleton.py     | The three-piece uniform certified lower-bound skeleton of the sector floor $\lambda\tau$ on the deployed battery, with the deep-tail coherence repair closing Piece 1 at FIXED citation depth for ALL $h$ . FLOOR-PARTIAL + TAIL-CLOSED-ALL-H (12/12 + 9/9): PIECE 2 UNIFORM-VERIFIED – the ANALYTIC $h \rightarrow \infty$ limit $X_\infty(\alpha) = 16\alpha\pi^2 \sin(\alpha\gamma_1) \sinh(\alpha/2)$ [bracket] (Dirichlet closed form = direct sum at $5.7 \times 10^{-13}$ ; ZERO sign flips; the explicit bound $L(h) = X_\infty^2(\alpha)(1 - C(\alpha)/h)$ for $h > h_0(\alpha)$ , $C(\alpha) \leq 499.6$ ; honest limitation: 25 $\alpha$ -nodes of $\sin(\alpha\gamma_1)$ – $h$ -uniform per $\alpha$ , not ladder-uniform in $\alpha$ ); PIECE 3 UNIFORM-VERIFIED – FAMILY EXHAUSTION: the certified top-100 pole $\times$ zero family carries median 0.952 (min 0.931) of the floor, all carriers on-line by verified computation; the analytic family limit stays positive across all fixed-pair nodes (min $1.62 \times 10^{-4}$ ); PIECE 1: the crude envelope closes only 4/14 ( $h \leq 434$ ) – superseded by part 2: the PRODUCT-SUP envelope + Abel/RvM with explicit unconditional Trudgian-grade constants closes ALL 14 rungs at the FIXED verified horizon $T_{\text{ver}} = 3 \times 10^{12}$ with margins $10^3..10^6$ ; THE LAW CHANGE: $\log h$ coefficient $+1.98 \rightarrow +0.13$ – the $h e^\alpha$ growth was an artifact of the coherence discard, the sharpened law is $h$ -FREE; remaining growth $\alpha$ -only ( $q = +1.79$ ), validity horizon $\alpha^* \approx 11.2$ (battery tops at $\alpha = 6.146$ ). NET: $\lambda\tau \geq X^2(\text{pole}, \gamma_1) - \text{pert}(h) > 0$ at citation grade on ALL 14 rungs – no per-rung eigencheck, no zero-location input beyond the [A1] strip. Feeds PRIME.FLOOR.RATIO.01 [O] (narrowed, NOT closed: battery-relative and $\alpha$ -bounded, necessary-side). NO RH claim. |
| v825_prime_exclusion_ladder.py   | The certified exclusion ladder of the GL1 prime-comb tower: verified PSD rungs inverted into rigorous quadruple-exclusion regions, extended to $X = 24.81$ with Cholesky/Higham + Rayleigh certificates, and the rank-4 saturation typed. LADDER-EXTENDED + SATURATION-TYPED (26/26 + 21/21): $\Xi_{\text{eff}} = X \delta_{\text{mb}}$ falls $0.639 \rightarrow 0.218$ and plateaus at $0.218 \pm 3\%$ (mechanism support-active-but- $X$ -stalled); every boundary point witness-Rayleigh re-proved; the region lies inside the [A1] strip (consistency re-derivation, not new territory). Live sub-ladder $X \leq 18.375$ ; deep rungs frozen reference with gates recomputed. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v826_prime_exclusion_battery2.py | The preregistered battery v2 (SHA-256-hashed a priori, no zero datum in the design) and the measurement-range extension: the exclusion floor drops from $\Xi \sim 0.2187$ to the uncensored declining series $0.1066 \rightarrow 0.0816$ (slope $-1.39$ ) – the v825 plateau was instrument-relative. FLOOR-LOWERED + RANGE-DECIDES-DECLINE (16/16 + 13/13); benchmark $\delta \geq 10^{-3}$ at $X^* \approx 81.6$ (extrapolation, typed); the $-0.872$ zero-distance rank correlation typed as the hash-verified non-circular side-finding. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v827_prime_zero_locator.py       | The zero locator of the exclusion instrument: the width profile $W(\gamma) = \delta_{\text{mb}}(\gamma)$ of the certified tower peaks at true $\zeta$ ordinates. The honest v1 negative (LOCATOR-NUL: 12/13 peaks but 61% false peaks at the frozen $\ln 1.5$ bar) precedes the preregistered v2 resolution (hash f57a2e7f..): on the disjoint untouched window [60, 120] the frozen $\ln 2.2$ rule scores 20/24 = 83% detection, 0% false positives, precision 0.086, all four controls – depth law $54 \rightarrow 62 \rightarrow 83\%$ . Location to $\pm 0.25$ is strictly weaker than classical verification. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v828_prime_comb_window.py        | The comb-native window verification (the capstone): on the disjoint window $\gamma \in (60, 120]$ the SAME certified positivity object yields both legs – the locator finds 21/25 ordinates ( $\pm 0.25$ , zero unmatched peaks), the exclusion map certifies $\delta \geq \delta_{\text{mb}}(\gamma)$ (uniform 0.0496), and the census reconciles: $21 + 4 = 25 = \text{cache} = \text{rounded RvM main term}$ . COMB-VERIFIES-WINDOW (12/12); new deepest certified rung $X = 25.5$ (cap $1.2 \times 10^{11}$ ). MANDATORY TYPING: strictly weaker than classical (Turing) verification on every axis; a necessary-side consistency demonstration – does NOT bear on RH.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v829_prime_floor_depth.py            | The authorized kill attempt on the PRIME.FLOOR.RATIO.01 envelope gate at full sieve depth SURVIVED, and the certified pole $\times$ alias family scales where the single pair dies. ENVELOPE-HOLDS-DEEP + FAMILY-SCALES (16/17 with the one preregistered S5.HLAW FAIL +12/12): margins $\times 5.76$ –8.92 growing at $X = 18.375$ –25.5 (the deep frames decouple depth from dimension – the $h$ -law wins); the single-pair gap widens $40\times \rightarrow 673\times$ while the certified family carries 0.977–0.981 of the floor ( $\rho_{\text{certF}}$ within 2% of measured); $\alpha$ -coverage typed (citation grade to $\alpha^* \sim 11$ only – the v830 blocker). [O] stays, kill gates survived. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v830_prime_float_budget.py           | The rigorous Higham-linear float budget closes the citation-grade family gate at FULL sieve depth ( $\alpha^* \sim 11$ was an artifact of the frozen quadratic convention): budgets $7\times$ –171 $\times$ tighter, gate closes 6/6 (old 3/6), citation grade to $\alpha = 12.75$ ; the deployed v818 convention stays FROZEN, the linear budget is registered for NEW modules only. Plus the exact family closed form on all 73 frames ( $3.6 \times 10^{-11}$ ), sympy-proven limit algebra, phase conspiracy excluded (dip floor 0.170), the explicit certified envelope $\rho \geq 4.335 h^{-3/2}$ on 73/73 – and the proof blocker typed (alias phase correlations: random-phase $h^{-1}$ vs the $h^{-2.5}$ tower). BUDGET-LINEAR-CLOSES-ALL + ENVELOPE-ASYMPTOTIC-PARTIAL. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                       |
| v831_prime_alias_second_moment.py    | The correlation-corrected alias second moment closes the analytic-envelope route at its typed circularity boundary and registers the floor $\leftrightarrow$ pair-correlation bridge. ALIAS-CORRELATION-PAIRCORR (15 checks with the SEVEN preregistered-honest premise-overturning FAILs; verbatim promotion): the bilinear form is EXACT (sympy-proven) – $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$ with the coherent phase sum $S_c$ at the leading term; the premise is OVERTURNED – the AMPLITUDES, not the phases, carry the $h^{-2.5}$ tower gap, decomposing exactly as $-2.505 = -3.172$ (amplitude law) +0.668 (fading alignment); the exact Guinand split puts the coherent sum in the prime-comb fluctuation at $0.01\times$ the Poisson square-root scale (smooth $-994$ / fluctuation $+995$ , cancelling; comb scramble explodes it $10^6\times$ ) – the cancellation IS the arithmetic; the typed circularity boundary: the required control is pair-correlation-equivalent substance, and by Guinand the comb self-cancellation IS the zero-side floor statement – the route is closed honestly, the bridge contract PRIME.FLOOR.PAIRCORR.01 [O] registered. NO RH claim. |
| v832_anchor_flavor_checksum.py       | The anchor affine normal form and the flavor bidirectional checksum: $p_n(a) = 2 + 2^n$ obeys $p_{n+1} = 2p_n - 2$ identically; $T(x) = 2x - 2$ has the unique fixed point $2 =  \mathbb{Z}_2 $ and the $T$ -orbit of 4 is the compiler quintet (4, 6, 10, 18, 34) – the whole budget (240, 248, 30, 40, 48, 41) from ONE recursion. Part B: $Ra = (4, 10, 13)$ , $R^T a = (6, 18, 8)$ – the same anchor decodes both sides; of 12 sibling candidates exactly ONE passes, and the anchor contraction $(Ra)_2 = 10 = p_3$ kills both siblings. ANCHOR-AFFINE-EXACT + FLAVOR-BIDIR-CHECKSUM (selective), 18/18 + 15/15 exact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v833_gaussian_ramification_ladder.py | The $(1+i)$ ramification ladder: the four documented roles of the ramified Gaussian prime $\pi_2 = 1+i$ are ONE machine-checked object – (i) norm doubling $(1+J)^T(1+J) = 2I_8$ with the empty zero class (0/240, $240 = 15 \times 16$ ); (ii) the 4-bit address $L/(1+i)L \cong \mathbb{F}_2^4$ with the deck acting trivially; (iii) the non-split jet $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$ , deck = $1 + \varepsilon$ , exactly 0 of 65536 sections deck-equivariant; (iv) the metaplectic lift $\zeta_8 = (1+i)/\sqrt{2}$ with $(SH)^3 = \zeta_8 I$ , the exact Clifford census $ C_2/\mu_8  = 11520$ and the RM-CSS phase bit $\chi(H \otimes I) = 1$ . RAMIFICATION-LADDER-EXACT, 33/33 exact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v834_pg32_flag_completion.py      | The PG(3,2) flag normal form for the 105 + 35 vacuum completion: $\binom{4}{2}_2 = 35$ , $140 = 35 + 105$ affine 2-flats, $105 = 35 \times 3$ point-line flags $\cong$ point pairs; the RM layer rebuilt ([15, 10, 4] enumerator exact, Grams $22I + 6J / 6I + J / 28I + 7J$ ); the vacuum coordinate is ORIGIN RESTORATION. The incidence kill typed as a split: $105 = 45 + 60$ is the $\text{Sp}(4, 2)$ -orbit decomposition on both sides; on the 60 non-isotropic rungs the symplectic polarity gives a canonical incidence-preserving dictionary; on the 45 doily rungs the obstruction is typed $((t + U) \cap U \text{ empty for } 45/45, \text{ flag Gram } 18I + 3J \neq 22I + 6J) - \text{'105 checks = 105 flags'}$ is exact as a deck-equivariant bijection, NOT an incidence isomorphism. PG32-FLAGS-EXACT + INCIDENCE-SPLIT, 26/26 exact.                                                                                                                                                                                  |
| v835_corner_hjelmslev_tower.py    | The character-corner identity is weight-generic and the Gaussian Hjelmslev CP tower carries it verbatim through five levels – but no register lift builds the identification. Part A (CORNER-IDENTITY-SYMBOLIC, 21/21): $T_{\text{GL1}, X} = V^* e \pi(P_X) e V$ holds as polynomial algebra in free weights; the entire equality step consumes ONE piece of arithmetic (the glue identity $\sum_m \Theta_m = 240\sigma_3$ ), forcing $\hat{c}_j = -1$ exactly; the scramble control proves the identity comb-blind – the arithmetic lives entirely in the identification step. Part B (HJELMSLEV-STRUCTURE-ONLY, 20/20): the chain-ring ladder $R_m = \mathbb{Z}[i]/(1+i)^m$ carries a strictly projective CP tower (certified cb defect identically zero, counts $15 \cdot 8^{m-1} / 35 \cdot 16^{m-1} / 105 \cdot 32^{m-1}$ ) with $\hat{c}_j^{(m)} = -1$ at every level, but the state corner follows the pure register-dilution law $16^{1-m}$ – the wall stays PRIME.FLOOR.PAIRCORR.01, relocated into identification. NO RH claim. |
| v836_commutant_sos_closure.py     | The exact sum-of-squares decider in the five-dimensional abelian commutant: the ansatz $T_{\text{GL1}} = \sum_j V_j^* T_{\chi_j} V_j + \sum_k C_k^* C_k$ over the canonical abelian subalgebra of the 39-dim commutant is closed DEFINITELY by exact certificates – (a) the exact reduction lemma collapses the SDP to a rational LP whose unique feasible point is the trivial diagonal rewriting (Vandermonde kernel $\{0\}$ , exact rationals); no cross-sector positive transfer exists; (b) the rank-3 det form has signature (1, 2), coefficient matching forces $G = \text{diag}(1, -1, -1)$ (not PSD) with the rational dual certificate $q(0, 1, 0) = -1 < 0$ , and $q$ -negative directions are achievable even on the nonnegative cone (832 of 2799 truncation cuts). FENCE-HIT on the positivity branch: the nontrivial version of this route is pair-correlation substance. COMMUTANT-SOS-INFEASIBLE, 29/29; route closed, stop-list entry; the wall stays PRIME.FLOOR.PAIRCORR.01. NO RH claim.                             |
| v837_corner_closure_quantifier.py | The corner-closure quantifier: the two open doors of the character-corner channel class close by opposite mechanisms – Door 1: factor dressings are generically visible but every visible dressing breaks the deployed-form identity by exactly its action on the lock pair (visibility $\Leftrightarrow$ identity defect; the commutant is invisible); Door 2: all 128 characters collapse exactly to 18 classes and read position-blind carrier modes only – the 90-cell class map has NO identity-and-visibility-and-placement cell, and the quantifier extends level-exactly through the Hjelmslev tower to $m \leq 3$ (identity characters exactly $\text{Ann}(\langle S_m \rangle)$ , Vfac spectra $\{1, 0, -1/7\}$ ). The structural half is kernel-checked in TftCarrier/SectorPositiveDescent.lean with IdentificationPositivity as the single named hypothesis. DOORS-CLOSED + LEVEL2-CLOSED, 19/19 + 17/17 + 3; the wall stays PRIME.FLOOR.PAIRCORR.01. NO RH claim.                                                           |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v838_corner_expectation_position.py | The compression toolkit closes and the position-dependent carrier map is extremally pinned: all 5276 subgroup conditional expectations (74259 components) obey the rank-1 dichotomy (exactly 2 identity-retaining subgroups = the predicted pinning criterion; 0 identity-and-visibility violations), the full pinching's only PSD summands are the two pinned sectors (the joint-use kill), and Stinespring-side compressions pay the identity defect coefficient for coefficient. Four state-preserving position-carrier designs all fail the decisive self-consistent-comb null – the deployed-form identity is an EXTREMAL state condition ( $\hat{c}_{GL1} = -1$ locks the entire $C_2$ mass), so identity-true carriers read zero on every self-consistent comb, true or fake. EXPECTATION-CLOSED + POSITION-CARRIER-TRADEOFF, 13/13 + 13/13; the fourth wall coordinate typed; NO RH claim.                                                                                               |
| v839_paircorr_bridge_saturation.py  | The pair-correlation bridge mapped from both sides and the GUE saturation of the floor demand: the demand of PRIME.FLOOR.PAIRCORR.01 written in Montgomery $F(\alpha)$ -form on all 73 frames with every supply row carrying its conditionality tag (Montgomery's window is RH-conditional; unconditional input misses by $52\times$ ; even full GUE closes only 39/73 rungs) – the named minimal missing input: an unconditional $F(\alpha, T) \leq C_F$ out to $\alpha \approx 3$ . The demand/GUE ratio plateaus at $R_\infty = 1.11$ inside the frozen band with $\text{rms}(z) = 0.72$ – the floor demand sits AT the random-matrix boundary of the comb, and inverting the certified margins yields the calibrated form-factor measurement $\hat{F}(\text{win}) = 0.44 \pm 0.14$ (the instrument candidate). BRIDGE-FULLY-BEYOND + GUE-SATURATING, 9/9 + 7/7; NO RH claim.                                                                                                                 |
| v840_gue_ablation_loopgain.py       | The saturation is STRUCTURAL and the bootstrap loop is SHORT: every source-native, Guinand-admissible tower variant lands at the deployed demand/GUE plateau (1.058–1.199 vs 1.154, tol 0.35; 0 broke positivity) with the tight band pinned in the UNFOLDED coordinate $\alpha \in 1..2$ under every lattice – the saturation is a property of the tower class and of the unfolded zeros, not of the grid (the thinned comb breaks positivity; the scramble explodes) – and the certified ladder cannot bootstrap its own next rung: $g = 1/(k^2 R^2 c_{\text{sup}})$ with headline $g(k=1) = 0.529$ and even the IDEAL supply giving $1/R^2 = 0.655 < 1$ ; the Poisson circularity ward passes. The wall's self-conservation IS the saturation: the ladder supplies exactly the statistics of the zeros while the floor demands slightly more. SATURATION-STRUCTURAL + LOOP-SHORT, 10/10 + 8/8; NO RH claim.                                                                                   |
| v841_relation_carrier_ladder.py     | The relation carrier EXISTS: the corner route reopens with the right input class – the multiplicative relations between events. The exact convolution frame carries $\mu * 1 = \delta$ and $\Lambda = \mu * \log$ as exact structure; the four-comb discriminator is exact (true comb 0 EXACT, $h = 2$ Epstein $x^2 + 5y^2$ separates at 200.0 on the class-group obstruction sites, and the reading is Selberg-class-correctly blind between genuine $L$ -functions); ALL FOUR GATES pass, including the self-consistency null that killed every compression and every single-event carrier. The amplitude wiring fixes $L$ -safety automatically, and on all 67 reachable rungs the identified corner's EXCESS over its comb-blind structural skeleton is positive (+2.285 to +3.704) – the sharpest wall form: $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$ . RELATION-CARRIER-EXISTS + EXCESS-NONNEGATIVE, 18/18 + 13/13; registers PRIME.RELATION.MULT.01 [O]; NO RH claim. |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v842_excess_certified_skeleton.py | The certified skeleton of the identified floor decomposition: strictly positive interval enclosures of $\tau_X$ on ALL 67 reachable rungs (widths $5.4 \times 10^{-11}$ .. $6.6 \times 10^{-9}$ , three to five orders below the margin), the certified discriminator (the $h = 2$ Epstein fake certified NEGATIVE on every anchor from the exact rational $\Lambda_F$ recursion), and the honest scaling law: certifiability degrades faster than the margin decays but the horizon $\alpha^* \approx 9.1$ lies far beyond the deployed ladder end – certifiability is not the binding constraint. What remains is exactly the wall: the infinite quantifier over strictly-positive certified enclosures; any future route must supply a UNIFORM lower bound on the excess margin. SKELETON-CERTIFIED, 9/9; registers PRIME.RELATION.SKELETON.01 [O]; NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v843_margin_law_excess_lean.py    | The margin as a doubly-derived typed law plus the kernel-checked Lean mirror of the wall: $\tau = e_1(\alpha) h^{-3/2} \tau_{\text{pnt}}(\alpha)$ with the certified envelope constant reproduced EXACTLY across two independent coordinate systems ( $e_1 \in [4.855, 24.209]$ on all 67 rungs, $\min \geq 4.335$ – the double-derivation ward; $\tau$ vs $\lambda_{\min}(A_h)$ at $2.1 \times 10^{-12}$ ); the tower gives scale, not recursion (defect series non-decaying); the excess is a growing cancellation (CI median 15.6, rising $5.65 \rightarrow 29.46$ – the wall self-similar at cell level; no per-prime positive control); compiler connection null (0/96, Bonferroni-honest); the ONE preregistered-honest FAIL kept (pairwise $h$ -exponent median $-1.341$ vs bar edge $-1.35$ , not refit). The Lean mirror (TfptCarrier/ExcessSkeleton.lean, lake build green, no sorry): the two-giants decomposition, the interval-certificate and certified-negative discriminator lemmas, the finite-ladder quantifier, the bridge theorem <code>excess_floor</code> with <code>UniformMarginBound</code> the single named hypothesis, and the kernel-checked gap <code>pointwise_pos_not_uniform</code> – no finite certificate table discharges the uniform bound. MARGIN-RECURSES-THE-WALL; NO RH claim. |
| v844_message_doily_rank.py        | The E8 addressing formula and the Doily–Pascal rank theorem, two exact corpus compressions: $M_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$ certified against the rebuilt objects with one NAMED structure per rung ( $h(E_8) = 30$ derived from the rebuilt root system itself: highest root height 29, marks $\{2, 2, 3, 3, 4, 4, 5, 6\}$ ; 60 Gaussian lines; $120 =  R^+ $ ; $240 = 15 \times 16$ census; coda $248 = 15 \cdot 16 + 8$ ; wrong anchor/multiplier controls fire); and $NN^T = B + 2I$ entrywise for the Cremona–Richmond duad–synthème incidence under the K6 duad bijection, so $\text{charpoly}(NN^T) = (x - 9)(x - 4)^9 x^5$ exactly: $\text{rank } N = 10 = A_\Lambda$ , $\dim \ker N = 5 = g_{\text{car}}$ (the six ovoid indicators SPAN the kernel), multiplicity $9 = N_{\text{fam}}^2$ , top value $3 = N_{\text{fam}}$ , recovery $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}}$ , six-step $(2/3)^6 = 64/729$ exact – the P2 integers are the singular-value data of the doily incidence; NO $(1/3)^6$ doily mode (v814 typed, no upgrade). MESSAGE-LADDER-EXACT + DOILY-PASCAL-RANK-EXACT, $23/23 + 16/16$ .                                                                                                                                                                                 |
| v845_cfin_normal_form.py          | The finite compiler normal form $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$ assembled as ONE object read off the Gaussian quotient: the unique selector-picked refinement (16 refinements, Arf 6 + 10), the order-3 family cycle, the parity lift onto $C_{\text{even}}(5)$ with $\beta \equiv \hbar$ and $q_{\text{wt}} \equiv q^*$ ; derived counting $16 = 1 + 5 + 10$ with $ \text{Sp}(4, 2)  = 720$ and $\text{Stab}(q^*) = 120$ faithful $S_5$ , the Pascal reading $(1, 10, 5) = \left(\binom{5}{0}, \binom{5}{2}, \binom{5}{4}\right)$ , the ovoid projective reading, and the budget $240 = 16 \cdot 15$ , $30 = 2 \cdot 15 = h(E_8)$ , $248 = 16 \cdot 15 + 8$ against the actual lattice census. Built-in negative: the exact 128-spinor counting bound ( $\geq 7$ of 15 classes always adjoint-side; the concrete census saturates $8 + 7$ with zero mixed classes) and the weight-level kill re-verify v775 ROOTCLASS-MIXED, not overturned. P2 ( $g_{\text{car}} = 5$ = the $\iota$ slot count) narrowed to the v799 residual R1' ([C]), NOT eliminated. NORMAL-FORM-ASSEMBLED, 28/28 (SPEC v2 declared).                                                                                                                                                                                          |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v846_schur_spectral_mother.py    | The Schur program closed as a typed structural negative: the manifestly positive relational Gram exists (manifest positivity, exact $\mu$ log descent routing with the $h = 2$ Epstein refusal at construction grade, Selberg safety – all green) but the corner identity fails in trace AND form by exactly the forced Cauchy–Schwarz diagonal price (price/ $\tau \sim 3.1 \times 10^4..4.5 \times 10^4$ even at the density-anchored fluctuation floor; the P0 deviation IS the price, structurally) – and the price is GEOMETRY-INDEPENDENT: the unitary spectral mother (the mirror as an operator, $J^2 = I$ ; the deployed form the $J$ -odd Toeplitz compression at rel 0; the Euler product as the exact operator algebra $U_a U_m = U_{dm}$ ) converts interference only at symbol grade (gain 20–33 $\times$ ), never at Gram grade (best legal overhead $5.7 \times 10^4..1.2 \times 10^5 \times \tau$ : the modulus bound). Harvesting cross-event interference is Fejér–Riesz of the total symbol – the positivity being sought: the manifest-positivity route closes on itself. SCHUR-INDEFINITE + SPECTRAL-TV-UNIVERSAL, 19 + 16, zero fails; NO RH claim.                                                                                                                                                                  |
| v847_wedge_cellcone_transport.py | The two remaining v5.4 architecture decisions, honest negatives with typed positive cores: (A) the naive signed Lagrange no-go is EXACT (forced pair weights; $P^-/\det \sim 2.0\text{--}3.2 \times 10^4$ ; the arch block negative definite) while the sign-register lift EXISTS as exact algebra (nonneg wedge weights per frame, LP residual $\leq 8.9 \times 10^{-16}$ , where level A provably cannot) – but the frame-uniform relational law fails at $\sim 15\%$ vs the $4.4 \times 10^{-5}$ floor need: the commutant uniformity wall transported (the honest scramble miss $\times 7.5$ vs the $\times 10$ bar kept, typed); (B) the completed cells telescope exactly ( $3.4 \times 10^{-15}$ ) and obey the congruence identity $\Phi^T J \Phi = cJ$ at $2.2 \times 10^{-16}$ , yet BOTH groupings exit the Lorentz cone on 67/67 rungs at the $n = 2$ cell (endpoint back in-cone = the certified $\tau > 0$ ); the monotone ray cascade to the compiler null direction (5, $-3, 4$ ) is CONFIRMED (Kendall $+1.000/-1.000$ ); the INVERTED Epstein signature kept as the frozen FAIL: the mass-matched fake stays in-cone – the violations are the true comb’s own $\psi(x) - x$ fluctuation. WEDGE-PARTIAL + CONE-BROKEN + RAY-EDGE-CONFIRMED, 13 + 9 with the two frozen-honest FAILS pattern-gated, NOT refit; NO RH claim. |
| v848_extraction_chain.py         | The continuum extraction chain COMPLETE: the implication ‘cofinal finite positivity (H) $\Rightarrow$ Weil positivity $\Rightarrow$ (via Weil’s criterion) the target’ is measured/cited theorem-grade with a QUANTIFIER REDUCTION – no Mosco compactness, no uniform $\delta > 0$ , no diagonal argument enters the implication; per-element convergence is the ONLY analytic input (v816’s three infinite-level ingredients migrate to the operator-level selection programme). Measured: the pure-box class carries the EXACT Weil value at every finite level (err $\leq 5.3 \times 10^{-14}$ – the sharpest dyadic-inheritance statement); identification rates $\sim 4^{-j}$ (median $-1.818$ ); geometric ledger (median ratio 0.302); the X-direction exactly stable (44 cutoff pairs at float zero); controls fire (wrong-limit stall ratio $1.6 \times 10^6$ , scrambled tower inflation $5.5 \times 10^5$ ). Cited with exact roles: Weil 1952, Bombieri 2000, Suzuki arXiv:2606.09096, Iwaniec–Kowalski Thm 5.12. Hypothesis (H) is NEVER evaluated – the arithmetic wall is EXACTLY (H) and nothing else. EXTRACTION-CHAIN-COMPLETE, 19/19; registers PRIME.EXTRACTION.CHAIN.01 [O]; NO RH claim.                                                                                                                              |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v849_cfin_unique_cofinal_lean.py | The compiler object is UNIQUE and the RH-side wall hypothesis is formally minimized: all 14400 admissible tuples at $\omega = \hbar$ form ONE $\mathrm{Sp}(4, 2) \times S_5$ orbit (orbit-stabilizer $14400 \times 6 = 86400$ exact), $\mathrm{Aut}(C_{\mathrm{fin}}) \cong C_6$ (faithful into $\mathrm{Sp}(4, 2)$ , the slot permutation determined), and the strict terminal-object reading fails honestly (unique up to NON-unique isomorphism, a one-object groupoid). The kernel-checked mathematical core in TfptCarrier/CofinalWeil.lean proves the $\varepsilon/2$ implication with NO diagonal argument and the strict hierarchy uniform $\subsetneq$ pointwise $\subsetneq$ cofinal, formally demoting UniformMarginBound. Hardening 2026-08-13: TfptCarrier/CofinalPredefinition.lean proves an explicit-fixed-idx API, exhibits an old-payload sign-mining counterexample, and makes PREDEFINED a named abstract Predefined( $A, \mathrm{idx}$ ) certificate in the hardened wrapper. Binder/field order is not computational noninterference; the concrete source boundary remains external without a formal construction language. CFIN-UNIQUE, 22/22 + 4 historical mirror checks; NO RH claim.                                          |
| v850_completion_tax_flow.py      | The two class-level no-go closures: EVERY positive 2-sector dilation routing the arithmetic through the fixed comb coupling block pays $\mathrm{tr} P + \mathrm{tr} Q \geq 2\ R_X\ _*$ with equality at the SVD pair (zero-duality-gap certificates on every anchor; symbolic $2 \times 2$ gap $2.7 \times 10^{-51}$ at 50 dps), the bound is BASIS-INVARIANT (DFT/Mellin, mirror $J$ , orthogonal: $\leq 1.1 \times 10^{-15}$ – the spectral-mother verdict becomes a corollary), and the per-cell tax exceeds $\tau_X$ by $\geq 4057.6$ on all 67 rungs (max $7.5 \times 10^5$ , growing $\sim h^{0.24}$ vs $h^{-3/2}$ ); the measured Schur prices exceeded the exact class optimum by only $13.7\text{--}20.4\times$ – optimality could not have saved them. The spectral-flow route closes by measurement: the density endpoint is NOT PSD ( $n_- = 5..12$ , premise corrected), all 634 crossings are upward and endpoint-determined on 67/67 rungs, and the velocity test (median ratio 1.00) shows the amplitude gap IS the metric margin rescaled by the flow velocity – no quantized protection; the typed residue: eigenproblem-free pivot certification and the crossing-location law. TAX-GAP + FLOW-CROSSINGS, 15/15 + 11/11; NO RH claim. |
| v851_cluster_kernel_field.py     | The cluster route closed and the cone violations resolved into a GRADED kernel law: the connected cluster expansion is exact (sympy telescoping; Möbius elimination; $2^{16}$ -subset ward 0.0) but its weights are position-geometric ( $E_2 = 0.883$ vs bar 3.0; the scramble does not collapse them) – the all-orders route through this pivot closes; the strict violation census wanders ( $P_0 \sim X^{1.03}$ , $R^2 = 1.00$ , max 244333/283607 vs cap 100) while the depth- $10^{-2}$ G1 kernel is bounded by $n \leq 73$ on ALL 67 rungs (collapsing to $\{2, 3\}$ above $\alpha \approx 3.9$ ); the causal Lorentz tube field transports clean (0/355199 gated violations on all four rule $\times$ grouping combinations, $\mathrm{cond} \leq 57.3$ ) but is self-calibrating (all four must-break controls do NOT fire – typed surprise); the skin-depth law $D(X) = 0.754 X^{-0.548}$ ( $R^2 = 0.99$ , Kendall $-1.00$ ) with verified threshold $X_0(\varepsilon)$ turns the wandering into vanishing-depth skin – the graded kernel+tail theorem candidate registered as PRIME.CELLCONE.GRADEDKERNEL.01 [O]. CLUSTERS-DIFFUSE + BOTH-PARTIAL + KERNEL-GRADED with the SIX frozen-honest FAILS pattern-gated, NOT refit; NO RH claim.      |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v852_arf_vacuum_ovid.py         | The Arf-vacuum syndrome theorem and the integral five-mode decoder: the Arf bit $a(t+U) = h(u,v)$ on ALL 140 affine flats (position-independent); the two-bit syndrome table $(s_{\text{vac}}, s_{\text{Arf}})$ splits them EXACTLY into 15/20/45/60; the LOCAL Hecke theorem at all 15 points ( $12+16=28=\sigma_3(3)$ , $12-16=-4=a_3$ , isotropic directions $\leftrightarrow$ points of $x^\perp/\langle x \rangle$ ); the code pencil $[15, 11, 3] \supset \{[15, 10, 4], 2 \times [15, 10, 3]\} \supset [15, 9, 4]$ with the canonicity lemma (all 16 refinements induce ONE Arf functional on $H$ ) and the vacuum bit as the $\text{RM}(2, 4)$ parity bit; the doily kernel gains the integral inverse $N^+ = N^T/4 - J/36$ (four Penrose axioms exact in $\mathbb{Q}$ ; $\text{SNF}(N) = \text{diag}(1^{10}, 0^5)$ torsion-free with a $\det \pm 1$ minor exhibited), the projector $P_5 = I/2 - B/4 + J/12$ ( $\text{tr} = 5 = g_{\text{car}}$ ), the exact closed-form decoder $\hat{m} = (I - P_5)m$ , the $S_6$ standard representation (11/11 classes), and the three-ring correspondence ( $\mathbb{R}$ ovoid span / $\mathbb{Z}$ torsion-free / $\mathbb{F}_2$ code $A_q = [15, 5, 6]$ via $v_a \bmod 2 = q_a$ ). ARF-VACUUM-SYNDROME-EXACT + OVOID-DECODER-EXACT, 24/24 + 23/23 exact; NO RH claim.                                                                                                                                                                       |
| v853_bent_css_mangoldt.py       | The bent phase layer with its two CSS codes and the von Mangoldt commutator theorem: $\hat{s}_q = -4s_q$ on all 16 characters ( $W^2 = 16I$ ; perfect autocorrelation); the $(16, 6, 2)$ Hadamard difference set $\{0\} \cup 5$ ; $A_q = C_{\text{iso}}^\perp = [15, 5, 6]$ with the 31-word structural census; the CSS pair $[[15, 5, 3]] \rightarrow [[16, 4, 4]]$ (the 60 minimal words = the 60 isotropic planes – the $s_{\text{Arf}} = 0$ rows of the v852 table) with the vacuum transformation typed; THE $-4$ TRIPLE POINT: Hecke = Gauss = Walsh = $a_3$ as one machine-checked integer; the 16 bent translates mutually unbiased against the 16 characters (reported, no upgrade). And the exact operator identity $\mathcal{L} = -M[D, Z] = -[D, \log Z] = T(\Lambda)$ (three independent layers; $\log Z$ rational; the BCH series truncates), the power/chain formula with nilpotency $\lfloor \log_2 N \rfloor + 1$ , the four-comb discipline at operator level (the Epstein $h = 2$ leak localized exactly at $[6, 14, 21, \dots]$ , first unramified site $21 = 3 \times 7$ , repaired by the class average $a_A + a_B = 1 * \chi_{-20}$ ; ideal-level Dedekind $\Lambda_K$ with norm aggregation), and the carrier bridge: the deployed $\mu$ -pairing reconstruction IS the first row of $\mathcal{L}$ ; $L^2 - [D, L] = T(\mu * \log^2)$ – typed honestly as classical incidence algebra. ARF-BENT-CSS-EXACT + MANGOLDT-COMMUTATOR-EXACT, 22/22 + 30/30; NO RH claim. |
| v854_relation_hodge_pfaffian.py | Factorization cohomology: the complex is exact ( $d_1 d_2 = 0$ with max entry 0; $b_0 = 1$ ; $b_1 = 16/21/22$ carried by the same-prime exponent chords), the Euler connection built from comb data alone gives MULTIPLICATIVITY AS FLATNESS at machine zero (true comb: $\max  F  \leq 1.8 \times 10^{-15}$ on ALL squares – the Chebyshev identity as discrete curvature), the Epstein $h = 2$ comb shows curvature localized at the class-group products (321/756/905 squares; the mass-grade relation discriminator becomes a local geometric invariant), and the $\mathbb{F}_2$ Arf-lift obstruction is PROVEN (exact $\text{GF}(2)$ rank vs augmented rank). The honest boundary is typed: gate 4 fails – the window form reads the 0-chain masses, not the class $[\alpha]$ (coboundary sensitivity $S_{\text{cob}} \sim 2.9\text{--}3.9 \times 10^4$ ); the Pfaffian moonshot has no anchor-independent normalizer (log-spread infinite). CURVATURE-DISCRIMINATES-ONLY, 18/18; NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v855_invariance_atlas.py        | The invariance atlas: the demand rewritten EXACTLY in Selberg-hierarchy coordinates (zero bookkeeping loss) and audited against on-range-verified elementary constants – the sieve reaches the $\alpha \in 1..2$ band in SCOPE (window-free) but zero coverage in GRADE (best factor $205\times$ ; the demand at $0.99\text{--}1.00\tau_{\text{pnt}}$ on every rung; the factor-2 gate unclosable by measurement, $\min dS _F/\text{GATE}_W = 2.28 > 1$ : input-class-INVARIANT), with the named object in sieve coordinates the bilinear chain fluctuation $d\text{Chain}_2$ ; the Li coefficients $\lambda_n > 0$ unconditionally to $n = 20$ (computed with no zeros), $d_N^2$ exact to $N = 64$ , the cross-coordinate identity $\text{BD} = 2\lambda_1$ ( $4.9 \times 10^{-17}$ ), the NB span IS the spectral-mother geometry term by term – yet the Li transfer fails by cone geometry (60.6% vs 5%): NO coordinate system holds hidden supply; the minimizer converges to neither frozen candidate (RAY 0.4530, ARCH 1.0976 rad; RAY-EDGE lives in the transport flow, not the deployed block) but an internal direction exists (0.0217 rad) tracking the comb block's own soft direction (0.0056 rad; the drift real, no identification claimed, $-4/\pi$ typed coincidence). The wall is invariant across compressions, geometries, criteria (Weil = Li = NB) and input classes (statistics = sieve); the one object in four languages: $e_1$ bound = form-factor band = $d\text{Chain}_2$ = comb-soft direction. SIEVE-SAME-GAP + ATLAS-SAME-WALL + PROFILE-DIVERGES + DIRECTION-TRANSIENT, $19 + 8 + 7 + 3$ ; NO RH claim. |
| v856_connected_current_lean.py  | The connected-current architecture decided: the globally centered KMS covariance is the FIRST design that evades the nuclear tax (relapse ratio $\approx 0.3$ , geometry similarity 0.3–0.4 – genuinely different bookkeeping from the event-local Schur completions) and dies on the typed GRADE GAP instead (the covariance carries the pair weights $\lambda_m^2$ , QUADRATIC in the comb, sim 0.56–0.63; the deployed window form is LINEAR, sim $\pm 0.06$ ; variance price $2.5\text{--}4.1 \times 10^5 \tau$ ; the centered main subtraction IS the deployed smooth half-line subtraction at corr 0.99; Epstein $h=2$ leaks 76% off the prime powers at RELATION level); the phase-cell census is EMPTY on the whole deployed torus (0 positive cells of 220–1760 incl. $\{2, 3, 5\}$ and $\{2, 3, 5, 7\}$ ; gradients 0.01–0.03 vs deficit $\approx 2.6$ ; the independent tail kill at $-3.4 \times 10^3 \dots -4.8 \times 10^3 \tau$ ); the conditional frame ConnectedTail + PositiveHeadCell + PhaseRecurrence $\Rightarrow (H_{\text{cof}})$ is kernel-checked (TfptCarrier/CofinalCurrent.lean, 7 theorems, axioms clean) and composed through CofinalWeil to Weil nonnegativity – hypothesis 2 measured EMPTY at the deployed frame: a named continuation frame, not a discharge.                                                                                                                                                                                                                                                                                                                                       |
| v857_simplex_fourier_winding.py | The E8 audits I: the Gaussian census IS a Fourier law ( $\hat{r} = (240, -16^{15})$ ); the uniform-nonzero channel $P = (J - I)/15$ with spectrum $\{1, (-1/15)^{15}\}$ and two-step contraction $1/225$ , exact in Fractions), the packet numerator EQUALS the code Walsh transform at every level $\leq 16$ (Eisenstein normal form $15\Theta_{D8} = 7\Theta_L + 8\Theta_0$ ), and $mh_2(n) = -1/15$ for EVERY odd $n \leq 16500$ (integer identity; all 1911 odd primes covered) – v817's per-prime measurement is the prime restriction of a CHARACTER THEOREM of the code, typed honestly SPECTRAL-only (the gradings do not refine each other); the tfpt_2 winding line collapses to ONE adjugate quadratic $q_{\text{wind}} = e_1^T \text{adj}(tI - R) \mathbf{1} = t^2 - g_{\text{car}} t +  Z_2 $ with the triple lock $s = 6$ (trace/det/Coxeter each alone; $\{6\} = \{ R^+(A_3) \}$ ), $\Delta(6) = 39200 = 2 \cdot 140^2$ (NOT a collision point; the sole $2\times$ square hit) and the reality threshold $s^* \approx 2.825$ (the unique $\Delta$ -root in $(2, 3)$ ; deployed $\approx 2.83$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v858_g31_clock_alphabet.py    | The compiler clock alphabet as a NORMAL FORM of $G_{31}$ :<br>$\text{Deg}(G_{31})/4 = \{2, 3, 5, 6\} = \{ Z_2 , N_{\text{fam}}, g_{\text{car}},  R^+(A_3) \}$ with $\text{gcd} = 4 =  \mu_4 $ ,<br>$\text{lcm} = 120 =  R^+(E_8) $ , $\text{lcm}/\text{gcd} = 30 = h(E_8)$ and the audit identity<br>$16 \times 15 = 240 =  R(E_8) $ REALIZED on the roots (the 60 reflections = the 60<br>Gaussian lines, 4 per class); (8, 12, 20, 24) the UNIQUE integer quadruple with<br>product 46080 and sum 64; the 607-group rank-4 kill scan: $G_{31}$ the SOLE<br>full-battery passer (weak-triple impostors and raw-sum-64 coincidences typed by<br>name; the honesty line kept: degree-only audits cannot see the reducible torus);<br>the $W(D_5) \times W(A_3)$ fence re-killed by COMPUTED centers 4 vs 1 (v634 cited for<br>the structural refutation).                                                                                                                                                                                                                                                                                                                                                                                 |
| v859_grade_no_go_elevator.py  | THE GRADE LAW: the direct-positivity graveyard reduces to ONE<br>kernel-checked homogeneity no-go – a grade-1 target can never be matched exactly<br>by a grade-2 Gram square (two-scalar ray evaluation;<br><b>TfptCarrier/GradeNoGo.lean</b> , 10 theorems, axioms clean) – with the exact<br>affine PSD tax (error = constantPrice + quadraticPrice, both PSD) and the<br>sharpened affine no-go (the background killed too); the ONLY licensed grade-1<br>mechanism is the RATES ELEVATOR ( <b>tangent_psd_on_kernel_null</b> :<br>derivatives of PSD families at kernel-null directions are positively signed); the<br>measured exchange rate is $10^4$ – $10^5$ across four graveyard instances; the<br>instantiated elevator is REAL (PSD compound-Poisson Euler factors, exact<br>tangent, free null-sum positivity, rates linear) and dies honestly on SIGNED<br>spectral densities on BOTH deployed sides – the comb Lévy density at exactly<br>50% negative mass (flip $\theta^* = \pi D/2u_q$ matched to $\sim 1\%$ ), the arch on the digamma<br>band ending at $\tau^* = 6.27$ where $\text{Re } \psi(\frac{1}{4} + i\tau/2) = \log \pi$ – the<br>Lévy/conditional-positivity elevator class closed on the deployed sides. |
| v860_phase_lever_closure.py   | The certified family closure of finite-head phase rescues: on ALL 67 rungs the<br>private-best-phase budget never pays the arch deficit –<br>$\lambda_{\min}(H_P(\varphi)) \leq -(d_0 - \sum_{n \in P} \rho_n)$ with $d_0 - \text{budget}(\{2, 3\}) \geq 1.3894$ and<br>$d_0 - \text{budget}(\{2, 3, 5, 7\}) \geq 0.6296$ : every such head is uniformly negative at<br>EVERY phase, no Kronecker sequence can rescue it (every atom granted a<br>PRIVATE best phase – conservative); three certified scope tiers (exact census /<br>budget inequality / greedy sharp control: the top-budget set's exact landscape<br>still has zero positive cells); the D-frontier saturates at $0.73 < 1$ (break-even<br>$D^* \approx 0.06$ , $\sim 2\times$ beyond the deepest admissible $D$ – structural, not a boundary<br>effect); the anti-vacuity ward fires; the typed not-covered line kept (infinite $P$ ,<br>non-canonical frames, non-tent interpolations).                                                                                                                                                                                                                                                                              |
| v861_krein_normalform_lean.py | The wall's operator form: the deployed window form gets an EXACT Krein<br>normal form $Q_h(t) = \ B_+ t\ ^2 - \ B_- t\ ^2$ (entrywise ward $\leq 10^{-15}$ , both cuts; $B$<br>carries $\sqrt{ \Lambda }$ – linear in $\Lambda$ ) with the exact Douglas pencil identity $1 - \ C_2\ ^2 =$ the<br>$\tau$ -margin in the positive-side metric; truth contracts on every rung while Epstein<br>$h=2$ explodes to $\ C_E\  = 46.8$ and the scramble to 3693; the 155-word census is<br>empty (band-disjoint supports); the register separates the $\mu$ -sign $C_2$ from the deck<br>(one faithful- $\mu_4$ character away, three candidate constructions verified); the<br>Redheffer echo $\det R_n = M(n)$ with SNF $(1, \dots, 1,  M(n) )$ – the entire Smith<br>deviation in ONE invariant factor; no resolvable floor–Mertens correlation (min<br>$4p = 0.378$ ); the defect theorem kernel-checked ( <b>TfptCarrier/KreinDefect.lean</b> ,<br>10 theorems, <b>SourceContractor</b> the named hypothesis).                                                                                                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v862_defect_polar_weld.py     | The rank of the wall is ONE: the defect operator $\Delta = I - C^*C$ has exactly one soft mode on every rung ( $\lambda_1 =$ the $\tau$ -margin exactly, separation 3.9–14.1 $\times$ ) – the Redheffer Smith invariant, the vacuum code’s one-qubit price and the Krein soft port are ONE rank-one structure; the MONOMIAL CLOSURE THEOREM: every source letter is monomial, the class is product-closed, the contractor’s monomial mass is 0.23–0.24, so EVERY word of ANY length misses by $\geq 0.874$ (the 408 sign flips kill the band-corridor form, typed v2); the 2-torsion weld law: the deck’s anticommutant is entirely faithful- $\mu_4$ – the deck welds only as $J = MD$ ; all four candidates weld exactly, the unwelded control refuses exactly, and the contractor connection is null ( $\sqrt{2}$ ) – the weld supplies the TYPE of the missing letter, not its value.                                                                  |
| v863_redheffer_colligation.py | The colligation architecture: both source-built 16-slot machines are exactly unitary, the vacuum-port Redheffer closure makes every closed transfer block AUTOMATICALLY contractive ( $\sigma_{\max} \leq 1 + 10^{-9}$ , never tuned), and the scalar limit reproduces the weighted lattice Redheffer determinant EXACTLY – the discovery: $\mathbb{F}_2^4$ IS the divisor lattice of $210 = 2 \cdot 3 \cdot 5 \cdot 7$ with Walsh–Hadamard carrying the lattice Möbius function and $\det = \prod_p (1 - 1/p) = 8/35$ exact; the machine is provably comb-blind (anti-circularity ward); the identification gate fails honestly (best residual 1.38 vs bar $10^{-8}$ ) with the gap typed exactly: the contractor couples DIFFERENT $ \tau $ density bands while the $J$ flip couples equal $ \tau $ (transfer mass 0) – the missing object is the grid-side band intertwiner, the explicit-formula kernel itself.                                        |
| v864_softport_kappa_law.py    | The scalar reduction: the exact Feshbach/Schur reduction turns the matrix floor into ONE port impedance ( $\tau > 0 \Leftrightarrow s > 0$ ); four source ports pass with the cheating-port ceiling $\kappa \equiv 1$ exact; the pole port carries 84% of the soft mode at depth with the exact skeleton $\kappa^{-1} = \beta^2 + \lambda_1 \rho$ ( $1.2 \times 10^{-9}$ over all 42 rungs) and the closed-form $a$ -term (the Poisson/Cauchy average of the signed density at $s = 1/2$ ); the law $\kappa - 1 \approx 6e^{-\alpha/2} \rightarrow 1$ (Pearson $-0.986$ ); the backflow concentrates on $\leq 17$ bulk modes cancelling 99.9% of the $a$ -term – the classical shape exists, the boundable term is exactly the arithmetic; displacement rank $\sim 12$ fixed over a $3.9\times$ dimension range with arm-weight generators; pure rank- $\leq 2$ Cauchy excluded; both discriminators break at the premise.                                 |
| v865_weyl_port_readout.py     | The arithmetic impedance: the divisor-tower Weyl load carries the comb at $O(1)$ (median $ dm / m  = 0.757/0.456$ under mass doubling – the hull’s comb-blindness REPAIRED at load level) and the v1 extremal readout dies by a diagnosed mechanism: doubling is a near-reparametrization $m \rightarrow m(z/2)/2$ of the $r$ -curve (curve-set distance 0.0287/0.0017), and extremal functionals read the curve as a set; the repair passes anti-blindness at the frozen bars and the phase readout closes the loop: $\kappa$ band $\leq 4$ across the battery AND Epstein $\kappa = 42.7$ / scramble 77.6 outside the $1.5\times$ truth window – 1–2 orders of discrimination where every earlier functional had both inside; the typed caveat carried verbatim: Spearman( $s, \tau$ ) = $-0.8$ – the load-bearing content is the discrimination, not the band; the two frozen-honest FAILs of the port probe are the diagnosis, kept and pattern-gated. |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v866_source_contractor_formula.py | THE FORMULA (the campaign's headline): the Douglas contractor factors EXACTLY as the target-free closed-form source expression $C = W_- F G_+^{-1} F^H W_+$ ( $1.4 \times 10^{-14}$ on construction rungs AND blind holdouts; no zeros, no $\tau$ , no fit) with CERTIFIED defect transfer ( $ \text{defect}(C_{k^*}) - \tau $ within the Weyl budget on all 7 rungs); the frozen Sylvester-12 kill fires as designed (principal angles $89^\circ$ ; the rank-12 reading corrected to $\sqrt{L}$ growth, $k^* = 24.48$ ); the norm statement is the reformulation $\ C\  \leq 1 \Leftrightarrow G_- \preceq G_+$ EXACTLY – better coordinates, the same wall – and the PERRON CLOSURE is proven: $\rho( C ) = 2.4\text{--}3.1 > 1$ everywhere, so NO absolute Schur/Hilbert-type inequality can ever certify – cancellation provably carries the arithmetic; the named demand (PRIME.KREIN.CONTRACTOR.01 [O]): a PHASE-AWARE bound on the weighted frame kernel strong enough to beat 1 by the measured $e^{\text{slope } \alpha}$ margin – the single object that would discharge the SourceContractor hypothesis. |
| v867_radau_conditioning.py        | The quadrature reach: Gauss–Radau certificates for $s > 0$ EXIST on EVERY rung ( $a - U_m \geq 10^{-8}a$ at $m^* \leq 426$ – the kz-16 Neumann frontier killed: positivity is provable pointwise by quadrature alone on the whole ladder $h \leq 900$ ) but the depth grows with the certified law $m^* \sim \sqrt{\text{cond}(G)}$ (Spearman +0.99); the $m \approx 17$ prediction honestly failed (0/7 rungs – the value concentration is not a certificate); all three source preconditioners fail informatively (condition medians grow $59\text{--}75\times$ ; congruence exact; Sylvester control: no healing); the conditioning is INTRINSIC – the soft end of the bulk is distributed arithmetic structure, and the certificate depth is the price of the cancellation; $J_\infty$ NAMED: the free Chebyshev chain on the bulk support (Rakhmanov-generic typed); both frozen-honest FAILs kept and pattern-gated.                                                                                                                                                                                          |
| v868_divisor210_audits.py         | The hard canonicity guard for the 210 discovery: the Boolean/Walsh/ $\mu$ layer is measured GENERIC exactly as warned (210/210 quadruples pass it), the deployed Euler constants pin $\{2, 3, 5, 7\}$ UNIQUELY (both scalar-limit determinants, exact Fractions / exact radical maps, one match each), the anchor prime 2 is FORCED by the frozen ramification crossref (the unique $\mathbb{Z}[i]$ -ramified prime onto the unique $\sigma$ -fixed $q^* = 1$ class), the $\mu_4$ weld grade cuts the gauge $C_6 \rightarrow C_3$ , and after R3 exactly TWO gauge classes remain – the family-cycle chirality, which is then PROVEN gauge by two exact no-go wards (the register orientation functional vanishes identically via the Möbius complement $d \mapsto 210/d$ ; quadratic readouts are provably chirality-blind by transposition); the $17 = 12+3+1+1$ moving-12 identification is honestly buried at space level (principal angles $\sim 90^\circ$ vs the $10^{-6}$ bar – a COUNT, not a space match); the 6-vs-7 honesty note stands untouched.                                                       |
| v869_cdcore_clarkphase.py         | The kernel is NAMED: $K_+$ is exactly the Christoffel–Darboux kernel of an explicit source measure in $x = 2 \cos(D\tau)$ – the displacement rank collapses from the growing $\tau$ -budget (12–14, the $\sqrt{L}$ law) to EXACTLY 2 with a $6.6 \times 10^{12}\text{--}4.8 \times 10^{13}$ gap on all 7 rungs incl. blind holdouts, and the firewalled chain-only CD reconstruction rebuilds $C$ at $\sim 10^{-13}$ – the contractor = arm weights + phase/geometry factors + ONE source Jacobi chain (honest boundary: the rank-2 collapse is window geometry, the controls collapse too – the arithmetic lives in the chain and the weights, $O(1)$ different for the fakes); the $\mu_4$ quarter-turn Clark phase is falsified sharply: the EQUAL-PHASE coupling law (offset $\sim 0^\circ$ at $\bar{R}$ 0.97–0.99, not $90^\circ$ ), the phase coordinate itself source-only clean.                                                                                                                                                                                                                            |
| v870_christoffel_gauss_frame.py   | The factorization and the frame decision: $C = D_- U D_+$ assembles exactly ( $10^{-14}$ ) with $D_\pm$ stably contractive ( $\ D_+\  \leq 0.9654$ by the Christoffel variational theorem; $\ D_-\  \leq 0.8655$ measured) and $U$ non-unitary BY THEOREM (the deployed grid $2\times$ -oversamples the folded pairs – rows $\geq 1.158$ , the $\sqrt{2}$ over-normalization); the Gauss-node frame is exact where algebra promises ( $D_+^G = I$ at $5 \times 10^{-11}$ , lossless bridge, frame-invariant port share) yet the unitary core does NOT exist (co-isometry defect $O(1)$ , cross-measure node misalignment at near-pole pairs); the surviving asset: $D_-^G \leq 0.997041$ ladder-wide with $34\times/2900\times$ discrimination; the named hypotheses (H-D)/(H-U)/(H-PORT) registered.                                                                                                                                                                                                                                                                                                               |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v871_pole_uvarov_crossdefect.py | The measure-relation closure: none of Uvarov/Christoffel/Geronimus is exact or fixed-small-rank at $x_{\text{pole}} = 2 \cosh(D/2)$ (machinery warded exact on synthetics; typed residue: small-negative-mass Geronimus uniformly closest at 3.5–8% Weyl residual, non-growing, discriminating – misses the bar, no $\lambda(D)$ law), and the co-isometry defect is DIFFUSE: full-rank, growing linearly with $h$ (Pearson +0.996), pole removal changes nothing, the $U_0 + R$ bracket $O(1)$ at near-full rank – no finite-matrix compensation; the Prüfer route is the preregistered plan B (frozen contract, committed unrun).                                                                                                                                                                                                                                                                                            |
| v872_damping_compensation.py    | THE COMPENSATION IDENTITY: $I - C^*C = T_1 + T_2$ exact at $10^{-12}$ on all 42 rungs ( $T_1 = I - U^*U$ the indefinite cross-measure Jacobi geometry; $T_2 = U^*(I - D^2)U$ the PSD Christoffel damping), transported losslessly into the Gauss frame; the exact global compensation minimizer IS the soft direction (overlap $\geq 0.9991$ , min ratio $= 1 + \tau/ v^*T_1v $ to all digits); $\Delta = \Delta_{\text{bulk}} + s pp^*$ with the certified bulk floor (98–100% of $\lambda_2$ ) and $s = \kappa\tau$ ; the wall as two explicit source numbers ( $-0.0328 + 0.0330 = 1.68 \times 10^{-4} = \tau$ at kz 9); Epstein/scramble break the damping's sign and under-compensate on their own soft directions.                                                                                                                                                                                                       |
| v873_pruefer_cotlar_decision.py | THE COTLAR DECISION (the first fully preregistered uniformity test, executed): the frozen Prüfer/Cotlar contract (16 fixed phase cells at $k\pi/8$ from source-only Prüfer phases, danger families predeclared, all bars/kills frozen pre-run, deep holdouts kz 90/116/142/177/243) returns COTLAR-GROWING by its own rules – the danger geometry is real (capture 0.98–0.61 of $\lambda_{\min}(T_1)$ ) but the battery is flat while the deep holdouts explode (kz 142: $S_U 32.07/S_C 58.78$ ; slopes $0.209/0.408 \geq 0.20$ ), the growth tracking kz-DEPTH, not window size $h$ ; contrast bars pass (Epstein 3927%, scramble 314462%); kills 2 + 5 fire; the blockwise Cotlar–Stein phase-cell class is CLOSED, the surviving demand a non-stationary phase bound with the $\pi$ -resonance as the critical structure; full provenance chain machine-warded (spec SHA unchanged through the two typed mechanical fixes). |
| v874_feedback_decoder_lean.py   | TWO EXACT COMPILER THEOREMS: the integral basis $P = [2 \cdot \mathbf{1} \mid e_3 \mid R e_3]$ (det $-4$ , SNF $(1, 2, 2)$ ) conjugates the whole winding line to the gain-plus-companion form $[[s+4, 0, 3], [0, 0, -2], [4, 1, 5]]$ exactly (symbolic in $s$ ), so $\chi_{R_s} = (t - s -  \mu_4 ) q_{\text{wind}} - 12t$ with $12 = N_{\text{fam}} \mu_4  = \dim g_{\text{SM}}$ and the honest typing $\mathbb{Z}^3/P\mathbb{Z}^3 = \mathbb{Z}_2^2$ (cardinality 4, NOT $\mu_4$ ); and the winding quadratic decodes the clock alphabet, $q(\{2, 3, 5, 6\}) = (-4, -4, 2, 8)$ , $D(y) = (y+4)^2(y-2)(y-8)$ with budget coefficients $(2, 48, 32, 256)$ , and $ \mu_4  q( Z_2 )/ R(E_8)  = -1/15$ EXACTLY the Walsh message eigenvalue – an audit theorem, not a functor. Lean mirror: FlavorFeedback.lean, 11 theorems, standard axioms.                                                                                    |
| v875_shell_global_lean.py       | THE GLOBAL SHELL THEOREM: the $-1/15$ census law upgraded from the finite census ( $n \leq 16500$ ) to a proof for ALL shells – $\text{Sp}(4, 2)$ transitivity (60 transvections, identical to the shell-module action) + $(1+i)$ doubling + $\Theta_L = 240\sigma_3$ give $\hat{\Theta}_u(n)/\Theta_L(n) = -1/15$ at every ODD $n$ exactly; a non-transitive control breaks the law (transitivity load-bearing), the even-shell formula differs correctly ( $7/135$ at $n = 2$ ). Lean skeleton GaussianShells.lean: the consequence chain kernel-checked from the named ShellSystem hypotheses (7 theorems, exact integer identities, no division); the census range retained as the exact ward.                                                                                                                                                                                                                             |
| v876_carleson_polygon_wave.py   | THE CARLESON WAVE: the Carleson ward makes arrows (i)–(iii) of the Prime Carleson chain machine-exact ( $\lambda_{\max}(M_-) = \ C\ ^2 = 1 - \tau$ at rel $1.5e-13$ ), the embedding statement registered as PRIME.CARLESON.PRIME.01 [O]; the polygon $S_0 \geq  S_2 $ breaks on all 47 rungs (typed: the wrong compression of the Loewner order) and the holonomy is null; the residual quadrature certifies $\rho_h$ at FULL degree on all 44 floor-positive rungs ( $k^*/h$ median 1.000, slope 0.974, mp escalation past the 53-bit wall; single atom just beyond the pole port; discrimination 7–9 $\times$ ); the resonance is THICK ( $\sim 1/5$ constant, excision gains nothing, soft direction disjoint) – the wall is phase-distributed; the three frozen-honest FAILs pattern-gated, not refit.                                                                                                                    |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v877_complete_comb_reversal.py     | THE ARTIFACT RESOLUTION AND THE COTLAR REVERSAL: the deep-alpha negative floors were comb-truncation artifacts of the deployed atom table (silent cap at $4 \times 10^5$ ; the windows demand support to $1.8 \times 10^6$ ) – complete-comb certified enclosures strictly positive ( $+1.1\text{e-}5/+1.5\text{e-}5/+1.2\text{e-}5$ ), no negative canonical window, $H_{\text{cof}}$ unaffected; and under the IDENTICALLY frozen v1 bars the complete-comb deep sums collapse into the battery band (kz 142: $S_U$ 32.07 $\rightarrow$ 5.65; slopes 0.006/0.007 $\Rightarrow$ BOUNDED): the v1 kill was entirely the artifact, run 3 unlocks and passes ( $K_1/K_2$ $h$ -stable), the route REOPENS – the dated round-36 correction on PRIME.KREIN.CONTRACTOR.01, with v873's booking standing as the record of the preregistered v1 execution and the remaining gap typed ( $B_C \sim 4$ vs 1; local defects 7–10). The truncation audit retypes every affected finding.                                                                                                                                                                                                                                                                                                                                                  |
| v878_ledger_integrity.py           | THE STATUS-LEDGER INTEGRITY GUARD: round 36 found and fixed the first primary-key violation (claim_id FORM.QGEO.03 double-booked 2026-06-15/2026-06-18; the SeamDeckClosure row renamed FORM.QGEO.04 with a dated QA note, all references updated); the guard enforces the 10-column schema, global claim-id uniqueness (the fix pinned), existing supersession targets with an ACYCLIC supersession graph, unique active supersession targets modulo the frozen remove-only allowlist of the two legitimate 2026-06 joint closures, and the dependencies cross-reference census frozen remove-only (13 SCCs / 265 ids; acyclicity enforced as NO NEW CYCLES).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v879_envelope_partition_closure.py | THE PARTITION CLOSURE: the analytic core saved – multiplication by $\cos \theta$ acts EXACTLY tridiagonally on the odd frame (single rank-2 defect at $k = 0$ ; the CD telescoping identity at $1.3\text{e-}13$ on every rung incl. all five complete-comb holdouts) and the derived pointwise bound carries $h$ -UNIFORM source constants ( $C_{\text{int}} \in [1.23, 2.33]$ from $h = 168$ to 1445, slope $+0.106$ – the route's first analytic uniformity statement, entrywise grade); the cell-envelope decay is cancellation-carried and provably NOT entrywise-derivable; no predeclared cell accounting below 3.57 (bar 1), $\sum_r \varepsilon_r \approx 12$ $\alpha$ -uniform on clean deep data. The wave-packet decision: both designs honest tight frames ( $\ B\  = \ C^G\ $ exact), the Gabor envelope decays in phase space (the first partition to show decay), discrimination 5 vs 129 vs 9007 – but diagonal mass $\leq 18\%$ under ANY pairing and the Schur constant grows 4.9–9.2 (slope $+0.295$ ): the partition class is closed ENTIRELY (cells, packets, all frames), the cancellation is ATOMICALLY GLOBAL, and certificate routes must be NON-DECOMPOSITIONAL (eigen-enclosure / Loewner order / passivity – all three measured territory); the two frozen-honest FAILs pattern-gated, NOT refit. |
| v880_finite_anchor_theorems.py     | the four finite anchor theorems: the census-free normal form $q^*(x) = \sum_{i < j} x_i x_j + \sum_i x_i = \binom{ x +1}{2} \bmod 2$ (weight-only; the selector forced on the linear part); the EXCLUSIVE anchor classification of $PG(3, 2)$ lines (all 15 doily lines carry $\{2, 1, 1\} = a$ , per-line Vieta (4, 5, 2) with $e_2 = g_{\text{car}}$ , $e_3 =  \mathbb{Z}_2 $ ; 10 secants carry the WRONG anchor (1, 2, 2), 10 external lines (1, 1, 1)); the doily as the multiplication table $\Lambda^2 E \times \Lambda^2 E \rightarrow \Lambda^4 E$ (wedge nonzero EXACTLY on doily lines); one-step quadratic exhaustion FORCES $g = 5$ ([E] theorem; the Calderón premise stays [O]).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v881_carleson_port_geometry.py     | the exact operator geometry of the Carleson wall – four sides of ONE object: the literal rank-2 Pick target REFUTED and replaced by the exact rank-2 Jacobi displacement ( $4.3\text{e-}14$ , source-only, blind holdouts); the gauge identity $E = S(\sqrt{\tilde{\nu}} K_{\text{CD}} \sqrt{\tilde{\nu}}) S$ – the wall is orthogonally equivalent to the plain Carleson Gram with the CLASSICAL testing diagonal $\tilde{\nu}_m K_h(y_m, y_m)$ (phases are gauge); the Schur/Haynsworth port reduction with $(1 - \lambda_{\text{max}}(D_P))/\tau = 1.00$ on ALL rungs (the port IS the wall); and $[Y, D_P]$ of EXACT rank 2 – the dressing preserves the IKS integrable class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v882_port_source_mellin.py        | the universal source law of the port: under $r = \sqrt{n/X}$ the weighted prime measure uniformizes to Lebesgue $dr$ (Kolmogorov $5e-4$ at $X = 10^7$ ); the last-log-unit edge mass equals $1 - e^{-1/2} = 0.393469$ (measured $0.3936$ ) – the wall’s port mass IS the PNT edge law; the port symbol is the Mellin–Cauchy kernel $1/(1 + 2i\tau)$ and the FIT-FREE continuum prediction reproduces the deployed port numerators to $0.029$ at the deepest rung (the deviation = the shrinking arithmetic fluctuation); the port numerators are the EXACT windowed prime sums and the criticality budget CLOSES ( $+1.405 - 1.674 + 0.284 \approx 0$ ): the wall is an ERROR-TERM statement about prime partial sums at the port frequencies.                        |
| v883_tau_chain_parametrix.py      | the determinant chain to the tau function and the GLOBAL rigidity of the Euler-product comb: the Schur identities $\sigma_h = \det(I - E)/\det(B)$ and $\det(I - E) = \det(I - R)\det(I - D_P)$ are machine-exact ( $4.8e-13$ ), hence $\text{sign } \tau_h = \text{sign } \sigma_h$ = the wall with $\tau_h := \det(I - D_P)$ ; the Jacobi variational identity holds on the true family; the parametrix decision: lattice-with-smooth-masses and even exact-local-PNT-mass worlds are ORDERS worse than the continuum against the true comb, and BOTH zones (edge and interior) are load-bearing – the Euler-product pairing is globally rigid, every smoothing-based parametrix is dead.                                                                           |
| v884_certified_head_positivity.py | the first RIGOROUS positivity statement of the program: $\sigma_h > 0$ PROVEN on the head rungs kz $9/12/13$ by exact integer Bareiss/LDL Sylvester certificates on the $Q = 10^{20}$ grid after an independent mpmath rebuild of the window lags with a conservative error model (shift $\sim 2e-18$ , 14 orders below the certified floors $\lambda_{\min} \geq 3.3e-4$ ); the Epstein control is REFUSED by the identical machinery; typed honestly: proven modulo the declared transcendental-evaluation error model (validated-numerics standard).                                                                                                                                                                                                               |
| v885_ladder_flow_anatomy.py       | the flow/induction anatomy of the ladder: the gauge-fixed IKS generator flow closes in DEGREE 2 (residual medians $0.244/0.248$ with $\{Y^2f, Y^2g\}$ ; degree 1 fails) – an isomonodromy-compatible Lax structure typed honestly as structural precondition; the window-ladder step is marginally perturbative but non-monotone in raw $h$ ; the ONE-PRIME ladder decides: the pure-Arch start is catastrophically subcritical, the last sign crossing sits at $u^*/U = 0.991-0.999$ in EVERY insertion order and edge-first needs 100% of the atoms – POSITIVITY IS A PROPERTY OF THE COMPLETE COMB (no monotone one-prime induction exists); the raw degree-2 coefficient basis is ill-conditioned noise, so the natural flow coordinate is the named open object. |
| v886_register_gapcode_null.py     | DISENCHANTMENT ROW: the error-code reading of the 210-register is literally true on the prime side – the register bits $\{2, 3, 5, 7\}$ are exactly the first four parity checks of the sieve (rate $48/210$ , 16 syndrome labels, message capacity 0 bits; hard floor $g_k \geq w(p_k \bmod 210)$ with 0 violations in $5.76 \times 10^6$ pairs) – AND the code is the sieve, not a TFPT signature: a zero-parameter wheel–Cramér null explains $R^2 = 0.99839$ of the state profile, the arithmetic excess is 4.02% and shrinking, there is no knee at four checks, and 210 sits on the foreign-wheel trend; nothing moves with $c_3$ or $\varphi_0$ .                                                                                                              |
| v887_certified_ladder_complete.py | the COMPLETION of the certificate rollout: $\sigma_h > 0$ is now PROVEN on ALL 42 reachable rungs – this module closes the last two rungs $h = 859/878$ (validated-precision Cholesky dps $120/200$ with precision doubling; certified floors $\geq 8.4e-6/9.2e-6$ ; pivot/bound ratios $\sim 10^{115}/10^{195}$ ); the Epstein control is REFUSED at pivot 10; proven modulo the declared conservative evaluation error model.                                                                                                                                                                                                                                                                                                                                       |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v888_finite_pfafrican_signature.py | the complete finite Pfaffian/ $K_6$ signature: Specht channel scalars $(1, 0, 2/3)$ exact; the 15 doily lines ARE the 15 Plücker monomials of $Gr(2, 5)$ with perfect sign match; the $(16, 6, 2, 10)$ Hadamard anchor; the $S_6 \rightarrow S_5$ vacuum selection; THE FLAVOR HEXAGON IS THE $\text{Aut}(C_{\text{fin}})$ ORBIT ( $g^2 = \sigma$ exact, $(2/3)^6 =$ the recovery value); the $K_6$ motif theorem ( $35 = 15$ matchings + 20 triangles); the fermionic Pfaffian signs GAUGE-MATCHED via the canonical character $\chi(i) = (-1)^{i+1}$ ; the self-hosting counting theorem $\binom{g+1}{2} = g!!$ forces $g = 5$ uniquely (nontrivially) with $N_{\text{fam}} = 3$ ; the mod-30 clock is CLOCK-DISTINGUISHED. |
| v889_route_decisions.py            | the four route-deciding measurements of the diagonal half: the wall dies 100% through off-diagonal coherence (diagonal profile frozen at the kill point; separation $> 10$ deep); $\theta^* = 0.700$ frozen (uniform outside margin +0.0214 on all 42 rungs); PNT-INSUFFICIENT (the smooth reference itself violates the port margin everywhere – the positivity is the mask-mass interaction); and the signed-homotopy classification: the diagonal route’s arithmetic input is PAIR-CORRELATION CLASS (a conditional reduction theorem, not unconditional).                                                                                                                                                                 |
| v890_total_positivity_flow.py      | the surviving positivity architecture: FLAG POSITIVITY (Sylvester; typing corrected – classical TP not claimed; all 12 leading principal minors of $I - C_h$ strictly positive on all 37 full-window rungs; Desnanot–Jacobi exact); the POLE-FREE $s$ -corridor with $s^* = 1/\lambda_{\text{max}}$ (the wall margin IS the pole distance of the integrable $s$ -flow); the multi-factor tau anatomy ( $Q = \prod(1 + \mu_i)$ exact; truth keeps every factor off $-1$ while the smooth-mass world crosses on 22/28 steps) – positivity inheritance, if it exists, is a MULTI-FACTOR/total-positivity statement; the one-scalar Möbius bridge is honestly closed (BRIDGE-DIFFUSE).                                            |
| v891_moebius_firewall_kills.py     | the honest kill battery of the Möbius/carrier-invariance route: fit-free and gauge-free cross-ratios are CR-DEAD (the 0.0015 residual was manufactured by the per-rung normalization); source-frozen gauges show REAL motion (GAUGE-MADE) and the scalar cocycle does not track the wall; the six-world census shows the arithmetic separates AT THE FRAME (all control worlds lose window subcriticality; scramble passes the Möbius bars better than truth – rank-2 kinematics). The surviving arithmetic remnant: the deep-core 8-node cross-ratio coherence ( $4.3\text{e-}4$ vs smooth $2.7\text{e-}2$ ).                                                                                                                |
| v892_closure_architecture.py       | the closure architecture of the wall: the exact hermitian congruence $A_{h+1} = A_h^{1/2}(I + H_h)A_h^{1/2}$ with the NON-DECAYING inheritance margin $\eta$ (min 0.0050, slope +0.108 while $\tau$ collapses at $-2.74$ ); the flag chain (all 12 LDL pivot quotients positive on 31/31 steps); the honest Rouché kill (the two-sided contour norm is the wrong theorem shape); and the $8 \times 8$ reduction (the wall IS the fixed Schur core to 7 digits; $\lambda_{\text{min}}(R)/\tau$ trendless at 210–2200 – the relative conditioning).                                                                                                                                                                             |
| v893_relative_margins.py           | the sources of the bounded margins: SCALE-MATCHED – increment and operator collapse together in the dangerous direction (residual correlation 0.974, 94% shared variance): $\eta$ is scale-free because numerator and denominator are sections of ONE arithmetic object (open: only the $O(1)$ scatter); the relative laws $d_{12} = c\tau$ (trendless; the kz-21 outlier diagnosed as a coordinate artifact) and $\lambda_{\text{min}}(R) \geq 210\tau$ (trendless); the printed theorem skeleton leaves exactly ONE open statement – the $\tau$ -sign inheritance in bounded-margin form.                                                                                                                                   |
| v894_diagonal_refinement.py        | the conditional diagonal contract made concrete: the exact prime kernel $K_{h,m}$ (regrouped at $4.5\text{e-}15$ , BAND-LIMITED – one-sided explicit-formula positivity at the port frequencies; the LaTeX contract statement frozen in-probe; all 79 finite margins positive); the endpoint bracket ( $F'(1) \geq$ deficit on 142/227 – the global linear collapse fails, deep small- $a$ cells pass monotonically); the inner-zone accounting (scale-normalized FLAT; the hard band non-shrinking; stability on an $a$ -graded mid shell).                                                                                                                                                                                  |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v895_collective_comb.py           | the collectivity of the arithmetic: the norm-bound theorem shape does NOT exist – the true law is ONE-SIDED ( $\max(-\mu_{\min}) = 0.9950 < 1$ on all 31 steps, which IS PD inheritance); the geometry/atom split is exactly degenerate (no separable atom channel – the increment is the collective window re-test); the deep core IS the port core (the fixed aliases $a_m = \pi^2 m^2$ ; ALL four surgeries kill the coherence – the signal is the ENTIRE multiplicative von Mangoldt comb).                                                                                                                                                                                                                                  |
| v896_wick_block_functor.py        | the Wick functor arc of the finite compiler: the six roles exist in the deployed seam (sector law, $q^*$ grading, $C_6$ intertwining exact) but the vacuum kernel is channel-diagonal; the PROVED scalar obstruction (the $C_6$ 2-cycle $\{4, 5\}$ forces a duad zero); the block resolution (BLOCKFUNCTOR-CONSTRUCTED: 15/15 blocks and Wick monomials with the canonical sign law) and the decisive fact SEAM-DIAGONAL – the value level awaits a physical mixing mechanism; the [O] premise is unmoved.                                                                                                                                                                                                                       |
| v897_certified_interval_ladder.py | the certified interval ladder complete: the informal $\varepsilon_c$ error model is RETIRED – $\sigma_h > 0$ with RIGOROUS interval-arithmetic shifts on ALL 42 reachable rungs (15 exact-rational integer Bareiss/Sylvester + 27 validated-precision Cholesky dps 120/200, floors down to $5.2e-6$ ); the node-enclosure lemma, the rigorous shift $2h \text{ rad}_{\max}$ , the Epstein control REFUSED at pivot 10; the embedded machinery re-certifies all 15 tier-1 + 3 tier-2 rungs verbatim and the full census is frozen with the probe SHAs; the asymptotic tail stays open – NO RH claim.                                                                                                                              |
| v898_kms_schur_mixing.py          | the channel-mixing state the block Wick functor awaits: the $C_6$ -covariant mixing KMS state EXISTS ( $u = 1$ , $t = 1/8$ , $\beta = 1$ ; all 15 blocks at the floor, grading $5/5 + 10/10$ , canonical Pf4 signs; $\beta = 2$ robust); EXACT (sympy): Schur elimination of the boundary GENERATES all 10 carrier duads from the bare diagonal state – on both fronts the real correlation lives in the ELIMINATED boundary; the frozen 15-signature table is the registered prediction; RP-THETA-OPEN named; the [O] premise is unmoved.                                                                                                                                                                                       |
| v899_christoffel_normsquare.py    | the Christoffel pivot identity and the norm-square contract: the EXACT identity $1/d_{12} = 1 + v_* K_\sigma(y_*)$ (deflated-Christoffel form; honesty fence $\lambda_{\min}(I - G) = \tau$ exactly – a coordinate change, not an independent positivity source; $c'/c$ SENS $[0.223, 5.258]$ ); the pair-contract kernel's entire negative content = ONE top-edge tent, killed EXACTLY by the periodic full-weight fold (negative index 0, 79/79 modified margins positive, boundary term carried closed-form) – the conditional diagonal contract is a NORM SQUARE in the frequency weights; the hypothesis itself stays CONDITIONAL.                                                                                          |
| v900_core_update_anatomy.py       | the normalized core update anatomy: the EXACT update $X' = c(X + U)$ at $8.3e-16$ with an effectively TWO-DIMENSIONAL input ( $c = \tau/\tau'$ plus ONE matrix mode at 97.5% energy); the honest negatives registered as boundary conditions: NO linear coupling law (both $R^2$ below the shuffle null), the rank-1 truncation FAILS (the positivity is carried by the 2.5%-energy off-mode), naive invariant sets refused – any region construction must keep the exact $X + U$ , truncating no mode.                                                                                                                                                                                                                          |
| v901_tangent_schur_bfloor.py      | the wall in tangent-Schur coordinates: the exact source-only split (with $B$ PD: wall PD $\iff n - q > 0$ , two scalars, no $\tau_{h+1}$ , no forward sign); the measured $O(1)$ co-block floor $\lambda_{\min}(B) = 0.679$ ladder-wide with the first substantive $\tau$ -screen PASS (slope $-0.247$ while $\tau$ falls by factor 552); the honest negative CERTFLOOR-DEAD (all four classical certified bounds negative everywhere, best maximum $-88$ vs the true floor $+0.679$ ); the failure-seat anatomy (the co-block itself dies at all 12 rank-1 failure points, the off-mode restores it linearly); the sign-free de-circularized normalization $\ell = \det(S)^{1/8}$ – circularity removed, positivity NOT gained. |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v902_wall_relocation_map.py      | the wall relocation map (typed explicitly as a mechanism map, NOT progress): the exact positive-weight rewrite $1/d_{12} = 1 + \sum WQ^2 - \beta$ ( $W > 0$ exactly, $\beta \geq 0$ termwise) with the relocation diagnosis $1 - r \propto \tau$ (slope +1.008) – the uniform $q < 1$ target is the wall's own PD premise quantified, no independent positivity source; the wall IS the moment matrix of the signed comb measure ( $\lambda_{\min}(M_0) = \tau$ ), but the $W \succeq 0$ Gram completion fails on every rung at the $x = +1$ edge where the comb nodes accumulate; the Radau construction at the classical node is exact algebra ( $w^*$ IS the Christoffel function, a pure $\mu_+$ object); the P2 pivot is NOT a Radau weight.                                                                                                                                                                                                                                                                                                                                                                      |
| v903_seam_rp_exclusion.py        | the seam RP/modular exclusion and the two dead readings: strict RP forces $t = 0$ under BOTH spatial reflections (odd-sector Gram eigenvalues exactly $\pm a_J $ : OS positivity $\iff$ the v898 mixing floor falls – mutually exclusive on this family); $u \geq t$ forced; the $2\pi$ -KMS locus is the point ( $t=0, \beta=2\pi$ ); the complete twisted-OS census is exclusionary (0/6); the dilation route splits (marginal witness ON the cone boundary, Schur compression carrying the full Pfaffian mixing with $t^2 \cdot 3m/(1 - m^2) = 1/200$ exactly; the strict collar route obstructed by an exact linear $2t$ law); $t_{\text{gap}} = 0.230949$ theorem-shaped ( $\text{Pf} = -(t-1)(3t^2-1)(9t^3+21t^2-t-1)$ exactly, Sturm uniqueness on $(0, 1/4]$ , kernel dimension exactly 2; no root is $1/8$ ); the two dead readings booked as first-class negatives: ' $t = 1/8$ is a compiler value' is a DECORATION (PIN-ABSENT), ' $N_{\text{fam}} = 3$ as minimal mediation rank' is REFUTED (rank 1 suffices; deployed rank exactly 2; purity = dimension bookkeeping); the v898 [O] premise is unmoved. |
| v904_healthcode12_diagnostic.py  | the 12-bit pivot health word as a diagnostic classifier (DIAGNOSTIC, ZERO RH content in either direction): the true word is all-(+) on 37/37 (the certified Sylvester census repackaged, NO new content); three plan claims corrected (the 12 factors live in the v899 12-window compression per rung, not in v897; the softest raw pivot is $k = 2$ , not $k = 12$ ; smooth tips flags $\{2, 1\}$ first and never flag 3); the one surviving claim reproduced byte-exact (the v897 Epstein control refused at pivot index 10); the seeded syndrome table distinguishes four adversarial fingerprints (coherent cosh injection tips the high flags first; scramble kills the window; mass rescaling is asymmetric) – a classifier of manipulated combs, a tool, not a target.                                                                                                                                                                                                                                                                                                                                          |
| v905_bfloor_ideal_certificate.py | the certified ideal-object surface floor of the B-half: the chain $B \succeq \frac{1}{2}P_G + c_{\text{dom}}I$ and $P_G \succeq c_G I$ , every inequality decided by exact-rational LDL (v897 certificate class) on all 39/39 reachable steps – float chain $\min c_B = 0.5914$ ( $P_G$ Gershgorin-positive: the positive chain is classically certifiable, the single non-classical link is the dominance decision); the interval rollout (SPEC v3) lifts the surface theorem to the IDEAL source objects (full pipeline as rigorous outward enclosures, fail-closed, Ogita-Rump high-accuracy tier): $\min c_B = 0.5523$ on 39/39 ideal-object steps, 0 refused / 0 failed / 0 skipped, tau-screen PASS; honest scope: a SURFACE statement – $h$ -uniformity beyond $h \sim 900$ open, the $n > q$ half untouched and RH-hard; the smooth world is REFUSED by the identical exact machine.                                                                                                                                                                                                                           |

*continued on next page*

| Script                                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v906_tail_<br>cartography.py              | the tail mechanism map (typed explicitly as mechanism/negative cartography, NOT progress): five dead routes with named seats – the tail sign is NET-ONLY, not pointwise; the pointwise repair grows with the window ( $X^{0.89}$ ); the directional Loewner route dies by the floor-vs-inverse mechanism ( $q \leq \bar{q}$ exact but $n > \bar{q}$ falls 38/39, loss med 91.3); the cumulative/Abel route is exact but insufficient (gap law $e^{+1.744\alpha}$ ; the real hole is the OSCILLATION); the segment split is COSTLIER than global (the anchor price; the named missing object: a DIFFERENCE envelope for $\psi(y) - \psi(x) - (y - x)$ on short deep windows, Brun–Titchmarsh/Selberg class) – and two structural positives: HEADFLOOR-O1 (the whole $h$ -decay of the wall margin lives in the atom tail; the open statement is the one-sided bound $\text{tail}_B > -\text{head}_B$ between two $O(1)$ objects) and the 12-bin reduction (the tail bound as a statement about $\sim 12$ windowed $\psi$ -fluctuation Fourier coefficients, measurably NOT $\gamma$ -concentrated at these depths; the reduction changes the DIMENSION of the open statement, not its difficulty). |
| v907_halfgap_<br>registered_<br>target.py | the registered half-gap target and its first blind holdout – a FALSIFICATION instrument, explicitly NOT evidence for RH: the target inequality $n_h - q_h \geq \frac{1}{2}\mu_1(h)$ registered on the 67-rung surface with the constant EXACTLY $\frac{1}{2}$ frozen under an explicit NO-ADJUST clause (part of the registered object), surface pass 67/67 (min margin $+2.48\text{e-}3$ ), blind protocol declared (registry sha ae292e55), the four candidate origins of the $\frac{1}{2}$ typed OPEN – an algebraic derivation is demanded before any claim upgrade; the FIRST genuine blind holdout passes: 28 new deeper faithful rungs ( $h$ 1219..2854, bit-exact table fidelity), 28/28 with min margin $+0.2232$ ( $\sim 90\times$ above the surface minimum – no depth erosion), HEADFLOOR-O1 persists (with the honest cut-ladder drift disclosed), B-floor/ $P_G$ persist at declared float level, the margin law continues ( $p = -1.925$ over 95 rungs); controls discriminate 3/3; a pass is survival of a falsification attempt, never evidence.                                                                                                                                 |
| v908_seam_<br>equilibrium_<br>wiring.py   | the seam equilibrium wiring (the physics closure of the round-60 questions): strict 2-cycle RP is IMPOSSIBLE on the whole C6-covariant class (the seat law reduces exactly to $M_1 = \text{diag}(a_J, -a_J)$ , interior EMPTY – the marginal witness is neither a limit of covariant strict equilibria nor isolated); the only escape is the covariance-breaking $X$ coordinate at exact linear price $2\varepsilon$ , with the floor exchange $\text{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$ exactly – the canonical sign gate IS the negative effective RP margin; the strict-collar obstruction is a two-seat linear law with kernel $\{J, Z\}$ , the deployed wiring is PURE-I (maximally obstructed), and equilibrium witnesses $V_J/V_Z$ carry the full $1/200$ mixing at ZERO entropy production – NO NESS NEEDED (stationary $\Rightarrow \sigma \equiv 0$ exactly; drive buys no RP); the registered next contract – is PURE-I compiler-forced? (SEAM.STATE.WIRING.SELECTOR.01) – is CLOSED in v911: deployment choice, not a compiler theorem; the v898/v903 [O] premise unmoved.                                                              |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v909_finite_wall_closure.py  | the finite wall closure: the composed census B AND W1 AND W2 on 39/39 matched surface + 8/8 deep rungs from cited inputs – B-half ( $P_G$ chain, CLIII/v905 interval-certified class floor min $c_B = 0.5523$ , 39/39 + 27/27 steps), W1 (monotone composition [L0/L1/L2/L5] + verified-zero supply at the $j = 16$ seat: 39/39 at $N_Z = 7000$ , composed 67/67 with the CITED deep 28/28) and W2 (the recomposed certificate $m_{\text{cert}} = FC + \hat{P}_{\text{arith}} - \text{TAILB} > 0$ : 67/67 + 8/8 with the 20,000,000-ordinate certified cache); this module re-verifies the load-bearing spine SELF-CONTAINED on the reduced frozen subset with the committed 7000-ordinate cache (Rosser corridor 7000/7000 per index, $\gamma_1$ identity, independent $ \zeta $ spot checks, $T_c < T_0 = 3 \cdot 10^{12}$ Platt–Trudgian EXTERNAL-CITED): B-steps, composition wards at machine precision, the HEART ward (zeros against primes) on every read, closure 6/6, controls fire; PINNING (meta-audit CCCXXXVI): the executed gates pin ONLY the reduced subset – the full census 39/39 + 8/8 is CITED, not pinned by this gate; language discipline: W1/W2 are algorithmically independent evaluations of the SAME localized Weil form (a strong mutual crosscheck, NOT two independent proofs); positivity on a finite Galerkin family along the MEASURED critical direction (DIRECTION-CONDITIONAL); a finite zero sum can never prove RH – NO RH claim. |
| v910_finite_zero_transfer.py | the transfer law: the zero-supply economics of the finished wall – per deployed rung the minimal verified-zero cutoff $T_{\text{req}}(h)$ at which the W2 certificate closes; frozen full-ladder laws $\log_{10} T_{\text{req}} = -3.045 + 1.228 \ln h$ ( $R^2$ 0.882) $\Rightarrow T_{\text{req}} \sim h^{2.8}$ , and the RATIO $T_{\text{req}}/(\pi/D)$ GROWS (+0.897 dex/ $\ln h$ ) – the needed zero reach overtakes the window’s own spectral reach $\sim h^2$ : EXTERNAL-BATTERY / W-UNBOUNDED-PER-RUNG; THE TRANSFER LAW: ZerosCertified( $T$ ) AND TailEnvelope( $T$ ) $\Rightarrow M_h \succ 0$ for all deployed $h \leq H(T)$ , measured envelope $H(\gamma_{7000}/\gamma_{2M}/\gamma_{2e7}) = 254/1256/2806$ ; the next $h$ -decade would cost $T \sim 3.4 \cdot 10^9$ – the finite engine does not scale by buying more zeros; this module re-verifies the shallow end on the 12 shallowest rungs with the committed 7000-ordinate cache (anchor census, heart ward on all cells, H-consistency 12/12); PINNING (meta-audit CCCXXXVI): H2/F1 are statement-recording gates – $H(\gamma_{2M}/\gamma_{2e7})$ and the full-ladder fits are CITED, not pinned by this gate; per-rung, deployed ladder, measured critical direction; $H(T)$ says nothing about undeployed $h$ ; NO RH claim.                                                                                                                                                                      |
| v911_wiring_freedom.py       | the wiring freedom theorem (closes the v908-registered contract SEAM.STATE.WIRING.SELECTOR.01): the exact Groebner census determines EXACTLY ONE admissible 3-dim wiring component mod rule gauge (the rotation cell $C_{\text{rot}}$ ); Z/X wirings are excluded gauge-invariantly by orientation propagation; pure-I is an INTERIOR point, connected to the pure-J ray by the integer gauge element $(\oplus_5 J_2) \oplus I_6$ – PURE-I IS A DEPLOYMENT REPRESENTATIVE, not a compiler theorem; THETA-CONVENTIONAL: no compiler-side demand distinguishes the $\theta_S$ sheet-swap frame in the 9-dim rule-gauge algebra (five demand classes silent-on-angle), and the deployed parent PASSES strict collar RP exactly in the integer-conjugated frame – the round-58/59 obstruction is FRAME-RELATIVE; RP-FRAME-COVARIANT: the strict-collar RP Grams transform exactly covariantly under the rule gauge, the pure-I exclusion is a frame statement (option 2), NOT a wiring no-go, while the Z/X exclusion IS orbit-invariant; the canonical sentence is frozen and ‘pure-I is compiler-forced’ is FORBIDDEN; the v898/v903 [O] premise stays [O]; zero RH content.                                                                                                                                                                                                                                                                                               |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v912_form_convergence_theorem.py      | the form-convergence theorem: the per-element convergence premise of the extraction chain – until now typed MEASURED (corpus rates $-1.58..-1.84$ per level, v848) – is proven UNCONDITIONALLY on the deployed dyadic cellwise-affine family with explicit rate $O(D^2 \log(1/D)) = O(2^{-2j}j)$ and explicit constants ( $\Theta_0 = 3 \log 2 + \pi/2 + \gamma + \log \pi$ ; for continuous $f$ the log coefficient is exactly $7/24$ ). Mechanism: the deployed lag vector IS the capped Weil functional on the even tent, so the finite form equals $W_C[\tilde{K}_D]$ exactly; midpoint exactness on cellwise quadratics makes the nodal defect exactly $-(D^2/12)G(dD)$ (zero deviation in exact rationals over 1474 lag identities and 780 interior points); the faithful cap turns the error into $W_C$ of ONE explicit function, bounded by elementary majorants per layer. Sharpness measured against a closed-form leading term (ratio $0.575..1.066 \rightarrow 1$ through $j = 18$ ). Audit correction: the chain's cited pure-sup-norm $C^0$ -continuity of $Q_W$ is FALSE (control C5, explicit counterexample family); the true hypothesis is uniform convergence plus equi-Lipschitz/Dini at the origin, supplied by the admissible BV class. Only cited arithmetic input: Rosser–Schoenfeld $\psi(x) < 1.03883x$ for support uniformity, independently sieve-verified to $10^7$ . Pinning (meta-audit CCCXXXVI): the W1 dictionary, Weil's explicit formula and criterion, the admissible class with the density leg, the all- $x$ Rosser–Schoenfeld statement and the universal quantifier are CITED, not pinned by these gates. DELTA-A: <code>CofinalEnvelope.lean</code> states the envelope premise and derives <code>hconv</code> from it, so the Lean hypothesis is the instantiated envelope, but the envelope is not discharged inside the kernel. Closes the convergence edge only; the cofinal positivity inequality is untouched and RH-hard; NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v913_signed_alignment_localization.py | the signed-and-alignment localization: a NO-GO / typing module that prices ARGUMENT CLASSES and supplies no positivity, no interval and no certificate. Any argument that closes the deployed budget inequality (L) for the deployed budget shape must be SIGNED – it must orient a quantity that is odd under the comb sign flip – and ALIGNMENT-CARRYING – it must relate the position of the prime atoms to the spectral directions of $K$ ; in particular it can be neither a magnitude bound on $\psi(x) - x$ nor a natural-grammar identity valid in every arithmetic world. Promoted core, exact and self-contained: the six kernel identities (notably $\ K'_D\ _1 = 4$ independent of $D$ , so the conversion constant of a magnitude hypothesis does not get cheap as the mesh refines), the two-route supply recomputation, the Littlewood emptiness of the magnitude class (floor/slack 214.7 on this module's own conservative slack, 1527.1 on the deployed slack), the provable neutrality of Selberg's symmetry formula, the exact split algebra with its congruence invariance and comb-flip parity, and the Beurling–Nyman transport target in exact arithmetic (no $N$ -uniform Gram floor; explicit non-implying family). The deployed matrix-stage reads stay in <code>experiments/</code> : they inherit the float64 entry-slack premise and the cited ladder. Pinning: the identities, split algebra, Selberg neutrality, Beurling–Nyman witnesses, supply recomputation, own sieve, frame rebuild with predicted $h$ and the reproduced probe SPEC-SHA are PINNED; the deployed ladder and its certified reads, the deployed entry slack, Littlewood's $\Omega$ -theorem itself, the float64 slack premise, the deployed budget shape, the four premises of the reverse implication and the counter-evidence corpus are CITED. Counter-evidence carried unsmoothed: the certified reads are positive at all nine depths (deepest $2.79579794131272506e-15$ at $h = 12632$ ), the normalised decay flattens with depth and one re-derived step is positive, so the drift is not monotone. Ten classes are proven empty for this budget shape, five are merely unexplored and five clauses are named open; the statement is finite-rung and makes no all- $h$ claim. It does NOT say that RH is unprovable, that the wall route fails, or that no proof exists; it closes no gate, narrows no gate, moves no marker and must not be counted as progress. NO RH claim. |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v914_pascal_region_theorem.py      | <p>PRIME.PASCAL.REGION.THEOREM.01 [E]: the Pascal-region theorem, the largest proven positivity region of the prime-front campaign, promoted as a certified-finite result (18/18). Statement, proven from cited external inputs alone: with <math>a = 256</math> and <math>y_\rho = a/(a - (\rho - 1/2)^2)</math>, <math>C_{n,k}(256) = \sum_{\text{Im } \rho &gt; 0} y_\rho^{n+1} (1 - y_\rho)^k &gt; 0</math> for every <math>n \geq 0</math> and every integer <math>0 \leq k \leq 7,444,682,106,464,286,365,865</math> (<math>\sim 7.44 \times 10^{21}</math>). Cited inputs, pinned: Platt–Trudgian 2021 Thm 1 (<math>T = 3,000,175,332,800</math>), Hasanalizade–Shen–Wong JNT 235 (2022) Corollary 1.2 (<math> N(t) - M(t)  \leq 0.1038 \log t + 0.2573 \log \log t + 9.3675</math>, <math>t \geq e</math>; the round-110 attribution flag resolved – Cor. 1.2 is the <math>N(t)</math> bound used, Cor. 1.4 the companion <math>S(t)</math> bound, the old (0.110, 0.290, 2.290) constants the earlier <math>S(t)</math> bound with the invalid Cheng–Graham input), Rosser 1941 Thm 19, the classical <math>\gamma_1</math> isolation. Stage 1 (<math>k \leq 10^{19}</math>) fully recomputed in-run (four-region partition, <math>B_0(T)</math> anchored, ratios <math>1.876981097339/1.176577026464/335.8/1416.8</math>, mechanism root <math>k^* = 1.405443828790708 \times 10^{19}</math>); Stage 2 (to <math>k_{\max}</math>) via retained-<math>q^k</math> envelope, 16384-packet RvM lower bounds below <math>H = T - 1</math>, alternating Gaussian-log tail enclosure above <math>T</math>, and the pinned 33-cell run-of-record endpoint partition with a frozen 6-cell full-packet recompute including the worst cell (endpoint ratio 1.000001535385559; the next <math>2 \times 10^{-6}</math> step fails – mechanism-exact). The law recomputed: <math>k_{\text{front}}(T) = (\lambda_\infty + O(1/\log T)) T^2/(a - 1/2)</math>, <math>\lambda_\infty = 0.226987054723979955</math> – verification height <math>qT</math> buys <math>q^2</math> columns; <math>k_{\max}/k_{\text{geo}} = 0.2113215513454976</math>. Honest typing (round-108): verification-powered and FORM-LOCAL – zero-location content below <math>\gamma_1 = 14.13</math>; says nothing about screw sections or Weyl disks; the <math>k \rightarrow \infty</math> quantifier untouched; no wall positivity, no zero table, no zeta evaluator. NOT evidence for or against RH. NO RH claim.</p> |
| v915_eulerpick_certified_floors.py | <p>PRIME.EULERPICK.CERTIFIED.FLOORS.01 [E]: the certified Euler–Pick floors <math>N = 1.4</math> – outward-rounded interval positivity certificates for the first four rungs of the falsification ladder, with a disclosed recomputed/pinned split (15/15). Object (V0c, audited round 92): <math>P(\sigma) = \xi'/\xi(1/2 + \sigma)</math>, <math>\sigma_j = 1 + 1/j</math>, <math>\mathcal{P}_N = [(P(\sigma_j) + P(\sigma_k))/(\sigma_j + \sigma_k)]</math>; RH iff <math>\mathcal{P}_N \succeq 0</math> for all <math>N</math>; the forward direction is exact, so a certified positive floor is an unconditional finite RH-consequence check from prime data alone and a certified negative rung would be an RH-disproof candidate. Recomputed in-run (round-95 architecture verbatim; own sieve to <math>1.6 \times 10^7</math>, outward interval arithmetic end to end, certified digamma, Büthe/RS tail, Weyl + interval Cholesky per Rump 2010): cap <math>4 \times 10^6</math> certifies <math>N \leq 2</math>; cap <math>1.6 \times 10^7</math> certifies <math>N \leq 3</math> (<math>\log_3 = 2.3643695 \times 10^{-10}</math>) – the round-95 record reproduced, all three cited <math>\psi</math>-bounds (Büthe 2018, Rosser–Schoenfeld 1962, Nair 1982) warded against own sieve data. Pinned from run-of-record (the 45-minute compiled <math>10^{13}</math> sieve): <math>N = 1</math>: <math>[4.5917135 \times 10^{-2}, 4.5917136 \times 10^{-2}]</math> down to <math>N = 4</math>: <math>[8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]</math> (margin 0.749) with <math>\pi(10^{13}) = 346,065,536,839</math> reproduced exactly; consistency, corpus parity, the re-derived summation-error model, the Nair far-floor necessity (<math>X_{\text{req}}(4) = 10^{11.52}</math>) and the <math>N = 5</math> knowledge wall all gated. Honest typing (round-108): the channel is DISPROOF-SOUND but REACH-CAPPED – the certified slice sees planted off-line pairs only at <math>\gamma \sim 5</math>–8 (recomputed detection subset), far below Platt–Trudgian <math>3.0 \times 10^{12}</math>; zero live falsification power in the certified range; “open and empty” is a theorem given the cited inputs. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                                                                                                    |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v916_epstein_weil_violation.py    | PRIME.KR4.EPSTEIN.COLLAPSE.01 [E]: the certified Epstein Weil-positivity violation (20/20). Statement, recomputed at every suite run: an explicit sha-pinned unit test vector (len 1038, sha b01436d53d2cbbec) whose tent autocorrelation $\psi_x = h_x * \tilde{h}_x \geq 0$ pairs NEGATIVELY with the source data $-g''_Q$ of $Q = x^2 + 5y^2$ : $V = x^T T x = -1.9697 \times 10^{-2}$ , error budget $\leq 4.3 \times 10^{-68}$ , two-precision (dps 80/110) agreement $9.8 \times 10^{-79}$ — windowed Weil positivity for $\xi_Q$ FAILS at $t \leq 3.114$ , built purely from $r_Q(n)$ lattice counts (exact sieve with the exact ward $r_Q = 1 * \chi_{-20} + \chi_{-4} * \chi_5$ ; log-derivative recursion, no Euler product exists; zero cache unused, witness constants firewalled). Controls: TRUE zeta and the conductor-matched null $\zeta_K = \zeta_{L-20}$ complete positivity through $L = 8.0$ (pinned run-of-record; the ZK anchor and $L = 4.0$ prefix recomputed) — conductor alone does not collapse; $Q$ dies at $t_{\text{death}} = 2.988$ (recomputed eigenvalue crossing $k^\dagger = 998$ ; mesh mate stable), and the certificate vector reads POSITIVE on the matched null. The three pre-declared falsified predictions are carried, not smoothed: attribution COLLECTIVE (witness removal moves the death by only 0.018; all-located removal survives to 4.428), the death law misses by $0.505 > 0.5$ , and the census is falsified (FOUR off-line pairs below 45; the witness never $\gamma$ -minimal; the 15.6682 driver carries $26\times$ its excess). Typing: the value is CERT-NUM (source-side, unconditional-numeric); the step “ $V < 0 \Rightarrow$ off-line zero” is COND-CLASSICAL (Weil/Guinand explicit formula, cited); the zeros themselves are Potter–Titchmarsh 1935 — KNOWN falsehood, NEW instrument. NOT evidence for or against RH in either direction: the min-cut is unchanged, a detector is a falsifier, and for zeta any finite window proves nothing. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v917_driver_rate_certification.py | PRIME.KR4.DRIVER.CERT.01 [C]: the certified localized rate detection of the driver (13/13). The round-123 driver zero $\rho = 0.9329696975 + 15.6682495313 i$ of $\xi_Q$ , source-located from the $\Lambda_Q$ sieve ( $\Delta\gamma = 6.6 \times 10^{-5}$ ), fires in the band-peeled rate channel at depth $m^* = 14$ with certified LB = 1.000137 $> 1$ , LOCALIZED from $m_{\text{loc}}^* = 15$ . Grade, carried verbatim and never upgraded: CERT-GIVEN-AUDIT (rigorous Cauchy enclosure given the incomplete-gamma audit comparator plus the DECLARED tail/peel models) and DISGUISE-ADJACENT (the deep read is an Euler-computable truncated windowed Weil transform; the Krein carrier cannot exist on $Q$ past the round-123 collapse). Recomputed in-run: the exact matched-pin algebra, sympy-symbolic and dps-200 ( $v^* = 1 + \varepsilon$ EXACTLY REAL at $a^* = \delta^2 + \gamma^2 = 245.681506$ , weight exactly 1, $y^* = (\gamma + i\delta)/(2\gamma)$ , the $a^2/z$ weight-sum rigidity); the violation window $(232.4783, 259.6345) \ni a^*, 256$ ; $m_2 = \ln 2\gamma^2/\delta^2 = 907.7$ ; the exact Pascal-diagonal dictionary (the peel arithmetic, exact in rationals); the band geometry ( $v_{\text{edge}} = 0.639$ , $v_{\text{max,unk}} = 0.386$ ); the full LB/localization/convergence arithmetic from the pinned depth table (margins $1.37/4.02/5.54 \times 10^{-4}$ , $\text{LB} \rightarrow v^*$ to six digits by $m = 25$ , certified depth 32, $\text{Err} \approx 2.1 \times 10^{-12}$ , $\text{meas} \leq \text{cond}$ ); and the quantified detection-cost law $m^* \approx \ln(B_{\text{band}}/\varepsilon)/\ln(1/v_{\text{edge}})$ with sieve price polynomial in $1/\varepsilon$ via the band (measured 3.19 depths per unit $\ln N$ ; the peel collapsed $m_2 = 908$ by $\times 65$ and needed only $N = 2 \times 10^5$ ). Pinned from run-of-record: the 33-row depth table, the winding-complete band census (phantom peel doubly excluded), the locator table and the controls (zeta/ $\zeta_K$ certified-silent, plant fires at $m^* = 15$ ). The honest caveat carried: the pre-declared G11 gate FAILED — the weak round-117 witness (excess $2.9 \times 10^{-5}$ ) sits below the locator’s measured threshold at $N = 2 \times 10^5$ ; the locator is a 3/4 MEASURED instrument and no 4/4 claim is promoted. NOT evidence for or against RH in either direction: the min-cut is unchanged, a certified detector is a falsifier, and the zeros are Potter–Titchmarsh 1935 — KNOWN-FALSEHOOD, NEW-INSTRUMENT. NO RH claim. |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v918_doublelimit_reduction_theorem.py | PRIME.DOUBLELIMIT.REDUCTION.THEOREM.01 [E]: THEOREM R, the load-bearing reduction of the rounds-128–151 proof arc, promoted as a certified finite theorem (13/13). Statement, proven with elementary inputs: L1 (trace convergence to the source-side target, absolutely convergent exactly for $a > 1/4$ ) + WPD ( $d_1 > 0$ , $ d_k  \leq C 4^{1-k} d_1$ ) $\Rightarrow$ (H-conv) on the full Euler interval + (H-trace) with the explicit rate $\max(1, C) (-4 \log(1 - t/4)) d_1$ , via the exact series identity $\log(R_{\text{fin}}/R_a)(t) = \sum_k t^k d_k/k$ ; with the cited round-122 NF-closure theorem: RH $\Leftarrow$ [NF-closure] + [Theorem R] + {L1, WPD} on dense $a$ . Lemmas N (necessity; Vitali + Cauchy cited), S (subset defect $\Rightarrow$ PD exactly, sharp at the band edge; sandwich pinned $0.121737 \leq 0.124501 \leq 0.289309$ ), X (moment transfer NOT free: $d_1 = 0$ , $d_2 = -1/128$ exactly), and the exact $C_{0k} \leftrightarrow$ jet dictionary. Recomputed in-run: the full exact algebra incl. the adversarial $C \gg 1$ instance and the WPD tail witness $k^*(B) = \lceil \log(2B)/\log(4w^*) \rceil = 29959/76013/145094$ from $4w^* = 10001/10000$ exact (WPD's $k$ -tail IS window positivity). Pinned: the measured ladders ( $d_1$ , WPD constants worst 0.0202 vs bar 1.20, reduction band, RvM node law); source probe re-run green at promotion (28/28, 398.5 s); independently re-derived in Bughunt III. Honest typing carried: L1 OBSTRUCTED, WPD OBSTRUCTED-RH-EQUIVALENT-AT-TAIL; proven is the REDUCTION and the COORDINATE, not the convergence. NOT evidence for or against RH. NO RH claim.                                                                                                                                                                              |
| v919_secular_gw_pinning.py            | PRIME.SECULAR.GW.PINNING.01 [E]: the secular identity + the Guinand-Weil pinning, the two central certified objects of round 131, both independently reproduced in Bughunt IV (11/11). Layer 1, SECULAR-EXACT, recomputed in-run on an own exact-rational instance (Fractions, $K = 6$ , dim 11, 13 interpolation points $>$ degree, sympy-free): $\det(M_\eta - z) = -z C(z^2)/A_0$ EXACTLY – the operator nodes ARE the census-polynomial zeros, node count EXACTLY $K - 1$ , lattice zeros at $j\pi/A$ ; the round-123 $\eta$ -degeneration IS the boundary decay $A_0 \rightarrow 0$ . Layer 2, GW-PINNING, certified per rung and pinned: $\tau = 2 \sum_\rho  E_N(\gamma_\rho) ^2$ within certified envelope + off-line allowance (Guinand-Weil cited; $x = 13$ : $\tau = 2.499 \times 10^{-54}$ vs $P = 2.402 \times 10^{-54}$ , the zeros carry 96%); the smallness law $2 E_N(\gamma_j) ^2 \leq \tau + \text{OFF}$ at every polished band zero; the certified gap law in the resolvability zone $\gamma \leq 2\pi x$ ( $x = 13$ : 21/21 zone zeros pinned, max gap $2.9 \times 10^{-17} \leq 5.6 \times 10^{-8}$ , $\gamma_1$ at $1.9 \times 10^{-50}$ ; edge layer MEASURED-ONLY); L1-TAIL closed in closed form (HSW22 Cor. 1.2 + PT21; recomputed against an own IBP integral, tangent bound the only slack); the crossover theorem ( $A/\pi ==$ RvM density exactly at $T = 2\pi x$ ; INBAND_F = $1/\text{KFAC} = 0.8$ is a THEOREM); the boundary-jet law. Bughunt-IV independent reproduction pinned ( $\tau$ from the classical explicit formula at $x = 3$ , $ \Delta \log_{10}  = 2.7 \times 10^{-7}$ ). Z1 typing carried: input circularity NO, evidence novelty LIMITED; the BAND half of L1 stays OPEN (H-pin). Source probe re-run green at promotion (31/31, 217.4 s). NOT evidence for or against RH. NO RH claim. |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v920_loop_equality_theorems.py      | <p>PRIME.LOOP.EQUIVALENCE.THEOREMS.01 [E]: the loop/equality theorems of rounds 140 and 142 – the negative/structural results that prevent double-counting in the proof graph forever (17/17). Round-140 theorems, all recomputed exactly: J1 (per-mode telescope – the source envelope certifies JET-LOCK onsets WITHOUT zeros), J2 (THE ONSET IS THE JET RATIO: <math>y_t/\rho \leq \Theta_J(\rho)^2 \leq 1.05 y_t(1+\rho)/\rho</math>), J3 (the jet ratio IS escaped census-root mass: trace form == <math>A_0</math>-free spacing form, so JET-LOCK(poly) <math>\Leftrightarrow</math> TOPROOT), J4 (Cauchy-Schwarz cap: the hardness is the <math>A_2/A_0</math> cancellation ALIGNMENT), B1 (tail visibility, exact rearrangement), B2 (THE EQUIVALENCE LOOP: modulo {JETLOCK, TAILVIS}, BAND-MASS <math>\Leftrightarrow</math> TLAWCAP <math>\Leftrightarrow</math> EPS-LOCK – every BM proof contains an EPS-LOCK proof). Round-142 theorems, all recomputed exactly: W1 (the pinch identity: <math>1 - sg == (g^2/\rho^2) \sum \tilde{e}_i^2/(\delta_i(\delta_i - g))</math> EXACT; <math>s \times \text{gap} \equiv 1</math> is NOT an identity – pinched <math>1 - g/\delta_1 \leq sg \leq 1</math>), W2 (THE LOOP, both directions exact: QSUBGAP-<math>\lambda</math>-uniform <math>\Leftrightarrow</math> SUSCAP2R <math>\wedge</math> DELTA1FLOOR), W3 (Cauchy interlacing: DELTA1FLOOR <math>\Leftarrow</math> FULLGAP, source-pure; measured TIGHT <math>\delta_1/\text{FULLGAP} = 1.0000</math>), plus the Bughunt-IV F1 witness carried (<math>s \equiv 1</math> while <math>\text{gap} \rightarrow 0</math>: DELTA1FLOOR IS REQUIRED). Pinned tables with consistency arithmetic (<math>y_t</math> ladder slope 4.137 recomputed; onsets in the J2 sandwich; alignment depth <math>1.2 \times 10^{-6} \rightarrow 6.9 \times 10^{-61}</math>). Provenance: r142 probe RE-RUN GREEN at promotion (26/26, 1904.7 s); r140 probe PINNED from run-of-record + deterministic re-run (both logs kept), loop algebra re-gated in Bughunt IV and recomputed in-run. RESHAPE theorems: they close NO omega; census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                                                                                                                                                                                                |
| v921_counting_dominance_closures.py | <p>PRIME.COUNTING.DOMINANCE.CLOSURES.01 [E]: the counting-class closures of rounds 133 and 143 – the pure classical-citation theorems of the arc (9/9). Round-133 theorems: M (ball-counting slack dominance, PURE COUNTING: disjoint ordered balls + no strays + one node per ball <math>\Rightarrow</math> sorted dominance; <math>M^-</math> bounded by pinning-gap mass at ANY sign resolution – no sign reads), E (edge counting bounds, HSW-closed), T (UNCONDITIONAL: <math>N_{\text{fin}}(T) \leq N_{\text{true}}(T)</math> for ALL <math>T \geq T^*(x)</math>, <math>T^*</math> the closed solve of <math>M - Q = K - 1</math>, consuming ONLY the exact node count <math>K - 1</math> [v919, cited] + HSW22; ladder 63.55/79.17/102.80/147.13 recomputed vs <math>\gamma_{K-1}</math> anchors), C (chain sharpening, exact: DOM <math>\Rightarrow</math> MRB <math>\leq 1 \Rightarrow</math> WPD constant <math>\leq 1</math>; weak one-sidedness <math>\Rightarrow d_1 == \theta(M^+ + \text{tail}_1)</math> EXACT and MRB <math>\leq (2 - \theta)/\theta</math>: MRB STRICTLY WEAKER than DOM), A (assemblage, CONDITIONAL on H-pin: <math>x_0 = 121</math> recomputed, BW25 <math>x_0 = 112</math>, MRB bound <math>144.1 \rightarrow 10.5</math>; the Q-SWAMP obstruction CARRIED: <math>D &lt; 0</math> on the battery strip <math>x \in \{13..89\}</math>, not smoothed). Round-143 theorems: T1 (TAILVIS existence is CLASSICAL: <math>N(T + L) - N(T) \geq L\rho_{\text{RvM}}(T) - 2\text{err}_{\text{HSW}}(T + L)</math> via convexity, sympy-gated; <math>L^* = O(20)</math> falling – TAILVIS ELIMINATED as a standalone arithmetic omega, counting class), T3 (B1 counting instantiation with NO zero ordinate, recomputed from the v919 pins to <math>&lt; 2\%</math>, positive at ALL SIX rungs incl. <math>x = 24/28</math>: TAILVIS-HORIZON-RESOLVED without new zeros). Carrier disclosure carried: the window sits at height <math>x^{2.06}</math>, PT21-warranted for <math>x \lesssim 2.3 \times 10^5</math> (carrier class, NOT a new arithmetic omega). Pinned r133 certificates (MRB_cert flat <math>\leq 0.25</math>; <math>C_{\text{cert}} \leq 1.5</math>; controls REFUSE at H2 inside the verified zero-free gap). Provenance: r133 probe RE-RUN GREEN at promotion (40/40, 339.3 s); r143 probe PINNED from run-of-record (both logs kept), T1 re-derived in Bughunt IV. Theorem A stays CONDITIONAL on H-pin. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v922_spacing_<br>jet_sumrule_<br>theorems.py    | <p>PRIME.SPACING.JETSUMRULES.THEOREMS.01 [E]: the HPIN/spacing theorem family of rounds 135 and 154 (13/13). All recomputed exactly in-run: D1 – the derivative identity <math>E'_N(\mu_j) == 4 \sin(A\mu_j) F'(\mu_j^2)</math> at every census node (the H-pin floor is <math>F'</math>-currency), with the SPACING-PRODUCT form <math>F'(y_j) == A_0 \prod_{i \neq j} (y_j - y_i) / \prod_k (y_j - b_k)</math>: the floor is a zero-spacing product. D2 – the jet sum rules <math>\sum 1/F' == -A_2/A_0^2</math> and <math>\sum y/F' == -A_4/A_0^2 + A_2^2/A_0^3</math> (the r131 boundary-jet telescope IS the reciprocal-floor moment data); D2'/P5 – the second-order rungs <math>\sum F''/F'^3 == 2A_2/A_0^3</math> and <math>\sum [1/F'^2 - yF''/F'^3] == 3A_2^2/A_0^4 - 2A_4/A_0^3</math> (residue calculus, r154). D3 – the <math>A_0</math>-cancellation: <math>g = c/(2 \sin PR)</math> EXACT (the demand ratio is <math>A_0</math>-free; the floor alone rides <math>\sqrt{\tau}</math>, typed FLOOR-RIDES-CONNES). D4 – the far-lattice tail <math>-\log(1 - q) \leq q/(1 - q^*)</math> certified (Euler sine product cited). The sign-uniform corollary is carried as an OBSTRUCTION (hypothesis measured FALSE: census MIXED; zone sum-rule share collapses <math>1.5 \times 10^{-4} \rightarrow 3.9 \times 10^{-13}</math>).</p> <p>LEAN-HARDENED (round 158): SpacingProduct.lean proves the spacing product for generic finite <math>K</math> over an arbitrary field; JetSumRules.lean all four sum rules at <math>K = 2</math> with the general-<math>K</math> skeleton honesty-locked; names re-checked in-run. Pinned: floor transfer <math>1/1 \dots 53/53</math> zone zeros at gap <math>\leq</math> ball; margins <math>5.8 \rightarrow 5.1 \times 10^{12}</math> growing. Provenance: hpin_floor r135 PINNED (three logs; re-gated r154 + Lean), nearalign r154 RE-RUN GREEN at promotion. H-pin itself stays OPEN; census cardinality 4 unchanged. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v923_spectral_<br>balance_razor_<br>theorems.py | <p>PRIME.SPECTRAL.BALANCE.RAZOR.01 [E]: the pinch/enclosure/razor theorem family of rounds 157 and 161 – the complete exact spectral-balance algebra on the QSUBGAP <math>g</math>-floor (20/20). Round-157 theorems, recomputed exactly: SB1 the trace loop <math>\tau \text{ TrH} == \text{tf}/\text{FULLGAP}</math> EXACT with <math>1 \leq \text{tf} \leq K - 1</math> UNCONDITIONAL (tail closed by the count bound alone: DELTA1FLOOR <math>\Leftrightarrow</math> TRACEFLOOR two-sided – the r146 Y1 loop); SB2 the chi-cap <math>s\rho^2\delta_1 \leq 1 - \rho^2</math> (the third rate-blind moment instrument, measured vacuous <math>7.6 \rightarrow 88.8</math> dex); SB3 the enclosure <math>\rho^2\delta_1 \leq g \leq \delta_1</math> both ends exact (one-moment merge REFUTED-MEASURED); SB4 the OVG identity (the r150 OVG-flatness IS the zone-top-gap pinch); SB5 the share floor (true but vacuous: the WEIGHT-LADDER RACE). Round-161 theorems, recomputed exactly: GF1 THE RAZOR ENCLOSURE <math>1/(s + 1/\delta_1) \leq g &lt; 1/s</math> (dominates the r157 lower end via the chi-cap; TWO-LEVEL EQUALITY at the lower end – the exact witness <math>\rho^2 = 1/(1 + P\delta_1)</math> realizes <math>s == P = 10^6</math> AND <math>g ==</math> the lower end EXACTLY: no algebra-only improvement exists; explicit two-way transfer: THE <math>g</math>-FLOOR IS THE <math>s</math>-CAP GIVEN THE <math>\delta_1</math>-FLOOR; measured width <math>1.51 \times 10^{-4} \rightarrow 8.45 \times 10^{-8}</math>); GF2 the pinch law <math>1 - sg == g^2T_2/\rho^2</math> EXACT, defect priced <math>g/\delta_1</math>; GF3 share<math>_1^g \geq</math> share<math>_1</math> one-sided, priced; GF4 the Newton value law <math>N_1 == 1/(H_t + s) + B(q_0) == -\rho^2P'_c(q_0)</math> (the bordered value CARRIES <math>\rho^2</math>) + the jet-ratio chase <math>J_g t_g == 1 + g</math> (the zero-jet mechanism's third rung); GF5 the S1 absorption (ONE decay law) + the twin squeeze at the jet weighting: both final cores are ONE statement-form at two weightings (FORM-merge exact; value-merge refuted – the J-family is BLIND to <math>g</math>). Lean-hardened round 158 (PinchIdentity, SpectralBalance; names re-checked). Pinned: razor lo/<math>g</math> <math>0.999995335 \rightarrow 0.999999995</math>; TRACEFLOOR <math>4.5 \times 10^{-6} \rightarrow 6.1 \times 10^{-9}</math>; SEC <math>4.06 \rightarrow 30.46</math> with betapos <math>== 1/\text{SEC}</math>. Provenance: r157 RE-RUN GREEN at promotion (non-timing diff empty); r161 PINNED per the multi-hour convention (both logs; GF algebra recomputed here). W1–W3 cited from v920. Reoordination only: no omega closed, census cardinality 4 unchanged. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v924_moment_laurent_rootladder.py | <p>PRIME.MOMENT.LAURENT.ROOTLADDER.01 [E]: the moment-Laurent/root-ladder theorem family of round 156 (12/12). All recomputed exactly in-run: L1 THE LADDER IS MOMENT DATA – <math>yS == \sum v + T</math> exactly; with <math>\sum v = a_2 = -y_t</math> and <math>z = y/y_t</math>: <math>(y/y_t)F/A_0 == \Phi(z) = z - 1 + \sum_m J_{m+1}z^{-m}</math> with EXACT partial-geometric remainder (the census roots above the band are the zeros of the moment-Laurent function: self-similarity iff moment flatness); L2 THE QUARTER-CAP – <math>z(1-z) \leq 1/4</math> and <math>\Phi(z_0) = 0 \Rightarrow J_2 == z_0(1-z_0) - z_0\rho(z_0) \Rightarrow J_2 \leq 1/4 + z_0 \rho </math> (the census-real top root caps its OWN leading moment ratio; measured <math>z \rho  \leq 0.0102</math>: the saturation <math>J_2 \rightarrow 0.1506</math> has PROVEN headroom); L3 the exact sum rules <math>a_2 == -P_1</math>, <math>P_2 == a_2^2 - 2a_4</math>, <math>P_3 == 3a_2a_4 - 3a_6 - a_2^3</math> (the ladder power sums ARE the jets); L4 the sign-pattern law (<math>n_{\text{esc}} \leq \text{nsc} + 1</math> exact; Laguerre form cited; <math>V == n_{\text{esc}}</math> at 5/6 blocks, b28 slack disclosed); L5 THE CASCADE (<math>yF_j == a_{2(j+1)} + F_{j+1}</math> exact <math>\Rightarrow T &gt; 0</math> pointwise SOURCE-PURE on <math>[Y^*, \infty)</math>, <math>Y^* = 1.02\text{--}2.1 b_{\text{top}}</math>, zero cache violations – the escaped-root cap CERTIFIED, not measured); L6 the two-sided split (the far window is controlled by <math>J_2</math> ALONE). The wall in final form: the <math>e_0</math>-witness deflates <math>J'_m == J_m P^{(1-m)/2}</math> while <math>T'/y'_t == T/y_t</math> EXACT (shape sector scale-blind); the 2-mode witness inflates <math>J_2 \times 10^6</math> with ALL identities intact – J-BOUNDEDNESS IS ARITHMETIC-PINNED. Lean-hardened round 158 (MomentLaurent.lean, 8 theorems incl. T_laurent with explicit exact remainder; names re-checked). Pinned: <math>J_2 = 0.1117 \rightarrow 0.1506</math> tau-flat; ladder top <math>0.8763 \rightarrow 0.8311</math> with second <math>0.0818\text{--}0.0825</math> CONSTANT over 3 dex; cascade <math>Y^*/b_{\text{top}}</math> with residual sliver <math>1/0/1/0/18/51</math> (closed per-zero in v925). Provenance: r156 RE-RUN GREEN at promotion (zero amendments). T-WINDOW rewritten as {arithmetic <math>J_2</math>-window} + {band-edge sliver + separation}; no omega closed, census cardinality 4 unchanged. NOT evidence for or against RH. NO RH claim.</p> |
| v925_edge_cleanup_closures.py     | <p>PRIME.EDGE.CLEANUP.CLOSURES.01 [E]: the edge/cleanup closures of round 159 (14/14). RECOMPUTED IN-RUN (self-contained; the committed <code>verified_zeros_n7000.json</code> cache, PT21-warranted, re-warded, ward/classical-census-below-horizon typed, max height consumed <math>7264.75 \ll 3 \times 10^{12}</math>): THE Q-SWAMP WHOLESALE CLOSURE – <math>D_{\text{cs}}(x, a) := (7/8) \text{tail}_{\text{cs}} - (N - m) w_a(T_z) &gt; 0</math> at ALL 357 Turing-class local band certificates (integer <math>x = 3..121 \times \text{battery } a \in \{1, 4, 16\}</math>; worst margin 0.346 at (12, 1), matching the run-of-record); the representative tables <math>x = 30/60/90</math> (the swamped <math>D_{\text{hsw}} &lt; 0</math> flips to <math>D_{\text{cs}} &gt; 0</math>); <math>D_{\text{hsw}} &lt; 0</math> on the r133 battery strip replicated; <math>x_0(\text{HSW}) == 121</math> and <math>x_0(\text{BW25}) == 112</math> recomputed on the integer scan – the bridge is GAP-FREE: census currency covers [3, 121), HSW/BW covers <math>x \geq x_0</math>. CONSEQUENCE (typed): THEOREM A's conditionality SHARPENS to “H-pin(<math>x</math>) implies <math>\{d_1 &gt; 0, \text{MRB}\}</math> for EVERY integer <math>x \geq 3</math>”; H-pin itself untouched. THE E2 KILL (exact layer recomputed): <math>\text{disc} == \prod (\lambda_i - \lambda_j)^2</math> generic + the r147 LM split forces <math>\delta_1 \geq 1/(B - 1) &gt; 0</math> at every selected point (selected points are AUTOMATICALLY simple: Z8 subsumed by Z4 at demand level) + zero-cost avoidance (measure-0 cover exact): census row Z8 FALLS (10 <math>\rightarrow</math> 9 rows); residual NONDEGENERATE-CELL witness-certified for free. PINNED: the E1 sliver closure (two-sided audit-ball certificates at every sliver + first-zero target, <math>1/1/1/1/18/51</math> at b5..b28; the price rides <math>e^{-\text{depth}}</math>, <math>\delta = 10^{-8} \rightarrow 10^{-44}</math> – BALL-CERT-DEPTH-PRICED), the exact implicit separation law gated as an identity (residuals <math>\leq 5 \times 10^{-27}</math>), the <math>\lambda</math>-uniform demand TYPED (band-edge zero repulsion, 0.22–0.43 RvM mean spacings, TAILVIS-style horizon typing). Provenance: r159 RE-RUN GREEN at promotion (run1 = 31/34 kept – its fails ARE the depth finding). H-pin stays OPEN; Theorem A stays CONDITIONAL on H-pin; Z8 was never one of the four min-cut omegas – census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                          |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v926_fullgap_<br>growthlaw_<br>theorems.py      | PRIME.FULLGAP.GROWTHLAW.THEOREMS.01 [E]: the quartic/growth-law family of round 162 (12/12). RECOMPUTED IN-RUN (exact): GL1 (every leave-one-out ground matches $\lambda_1$ within the $\varepsilon^2$ budget; the pinch position is the honest discriminator), GL2 (the arrowhead compression $\det == \prod (q_i - \lambda) \times$ secular, generic; dual row; plus the Bughunt-V numeric instrument on an own instance), GL3 (one-row price equivalence + razor vacuity $1/(s + 1/F) \leq F +$ the 2-level witness: D3 COLLAPSES TO THE RAZOR – algebra-only per-row/per-zero price bounds REFUTED), GL4 ( $J t_r == 1 + \text{FULLGAP}$ ), GL5 (ladder + quartic chases + razor composition: the QSUBGAP-floor flows from the $\Theta$ -window through the razor). THE QUARTIC LAW typed MEASURED, NOT proven: $J == \Theta T_z^4$ with $\Theta$ flat in $[0.17, 0.26]$ over 24 dex; the Bughunt-V AIC adjudication RECOMPUTED IN-RUN: free exponent $p = 3.9072 \pm 0.1136$ , AIC prefers FIXED 4 ( $-30.87$ ) over free ( $-29.79$ ), rejects 3.5/4.5 at 3.6/5.2 $\sigma$ ; FULLGAP slope 3.971 recomputed. PINNED: $\rho_j == 1.0000$ for ALL 189 left-out zone zeros (jet-COLLECTIVE – per-zero deflation REFUTED-MEASURED); add-one free at true zeros (contrast to $2.9 \times 10^{17}$ ); trial cap MID-DOMINATED; the finite- $P$ moment model UNDERSHOOTS by 1.4–4.5 dex (the Y3/R4 rate-blind class); the razor chain at all six rungs; controls refuse fourfold. Provenance: r162 PINNED from run-of-record (28/28, both logs kept; re-gated in Bughunt V). The residue set UNCHANGED; census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim. |
| v927_blockaverage_<br>substrate_<br>theorems.py | PRIME.BLOCKAVERAGE.SUBSTRATE.01 [E]: the cofinal block-average substrate of round 166 (12/12). RECOMPUTED IN-RUN (exact): BA1 (extraction: a certified block sum $> 0$ forces $\geq 1$ positive rung; dyadic hits COFINAL), BA2 (the wall dictionary: leading minors $> 0 \Rightarrow \text{sign}(\lambda_{\min}) == \text{sign}(D_K)$ , Jacobi/Sylvester – the determinant chain IS the $\tau$ sign), the weight families + sign equivalence, BA3 (the budget floor $\tau \geq \text{zsum} - \text{OFF EXACT}$ ; inputs typed CITED: PT21 + HSW22 + the r131 OFF recipe + ward-class cache – NO tlaw window, NO WPD, NO-LOOP), the Weyl-split lemma (measured VACUOUS by 10.9–54.6 dex: the cancellation IS the arithmetic), the loop/red-team layer. PINNED: certified block enclosures with ALL lower ends $> 0$ and BA3 budget floors $> 0$ – POSITIVE-RUNG-PER-BLOCK CERTIFIED on both complete dyadic blocks; $\text{sign}(\tau) == \text{sign}(\text{wall}) == +1$ at ALL 25 rungs $h = 4..28$ ; $\text{zsum}/\tau = 0.88..0.96$ with margin $\geq 3.2 \times 10^8$ ; the tlaw ladder $0.2325 \rightarrow 0.5778$ with 19 NEW corpus-depth rungs; detrended cancellation $0.461 \rightarrow 0.037$ (MEASURED); controls refuse with a SIGN FLIP. Provenance: r166 RE-RUN GREEN at promotion (27/27, identical SPEC, timing-normalized diff EMPTY). The $\lambda$ -uniform demand beyond the horizon stays OPEN (AVG-BUDGET-WINDOW); census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.                                                                                                                                                             |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v928_depthblock_transfer_theorems.py | <p>PRIME.DEPTHBLOCK.TRANSFER.01 [E]: the depth-block transfer family of round 168 (11/11). RECOMPUTED IN-RUN: DT1 (the step algebra PROVEN: exact algebra on top of BA3/v927), DT2 (THE DEMAND LAW PROVEN-BY-RECIPE – the round’s discovery: <math>\varepsilon == \sqrt{h}(1+\eta)^2 G(T_{PT})/G(2\pi h)</math> EXACT; the dyadic factor falls monotonically onto <math>2^{3/2}</math>, sympy limit), THE CENSUS 3/2-LAW (sympy-exact: <math>T_{\text{req}}(h) \rightarrow (3\pi/\sigma_0) h^{3/2}</math> – census consumption grows STRICTLY FASTER than the block: the budget route can NEVER become self-supporting), THE BUDGET HORIZON RECOMPUTED (<math>h^* = 1.2566 \times 10^7</math>, <math>k^* = 23.58</math>: <math>\sim 22</math> complete dyadic blocks on PT21), THE BERTRAND INTERFACE (cited + sieve-verified in-run for EVERY integer <math>2 \leq h \leq 10^6</math>), THE RH-LOOP DETECTION (universalizing the census closes the cycle; the per-<math>k</math> schedule is acyclic – the ALL-K grant carried ONLY as a flagged LOOP edge), the red team. PINNED: the <math>\sigma/\varepsilon</math> ladder (<math>\sigma = 0.2056 \rightarrow 0.5191</math> MEASURED, DT3), the dyadic factors <math>3.7288 \rightarrow 2.8462</math>, THE STEP INSTANTIATED END-TO-END (STEP(B2<math>\rightarrow</math>B3) 9/9, STEP(B3<math>\rightarrow</math>B4-partial) 11/11 – SUBSTRATE-DIRECT, no self-supporting induction), ENTRY-NEW-CONFIRMED 4/4 (cone entry at the largest new prime-power atom, excluding trailing 2-powers; fakes die by NO-ENTRY), controls refuse BASE AND STEP. THE FINAL COORDINATE NAMED: AVG-BUDGET-WINDOW == [census schedule, classical per <math>k</math>, LOOP if all-<math>k</math>] <math>\times</math> [SIGMA-FLOOR, arithmetic-pinned, OPEN – factorized in v929]. Provenance: r168 RE-RUN GREEN at promotion (36/36, zero amendments). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |
| v929_sigmafloor_factorization.py     | <p>PRIME.SIGMAFLOOR.FACTORIZATION.01 [E]: the terminal factorization of round 169 (12/12). RECOMPUTED IN-RUN (exact): SF1 (THE SUPPLY FACTORIZATION <math>\sigma == (1 - \text{slop}) \delta \text{DC}</math> – per-term algebra of the r131 Layer-1 chain transplanted into the supply window; identity devs <math>\leq 2 \times 10^{-60}</math> at ALL 25 rungs in the record, F64-IMMUNE), the ANTI-LATTICE reduction, SF2 (ONSET-CAP-IS-UPPER-REFUTED: the conjectured F1 chain dies; the floor leg IS the JET-MASS-FLOOR with the exact JET-LOCK rate floor), SF3 (THE DC LEG PROVEN-MOD-CITED per census: Landau 1912/Gonek 1993 AS FORM, GONEK-CONSTANT-UNPRICED; ALL-K == the flagged RH loop – DC-CLASSICAL typing), SF4 (DEMAND-RATE ABSORPTION sympy-exact: ANY explicit polynomial-rate floor is census-absorbable – <math>\sigma</math>-flatness NOT needed), SF5 (the pinning-supply route is the RH loop’s costume, cycle machine-detected), SF6 (red team), the three endgame graphs. PINNED: DC = 0.1535 <math>\rightarrow</math> 0.4463 RISING toward 1/2, <math>\delta = 1.16..1.43</math> <math>O(1)</math>-FLAT – THE r168 <math>\sigma</math>-FLATNESS EXPLAINED AS TWO TRENDS; the rate floor per rung with the clean-rung rate law <math>a = 1.328</math> and rate horizon <math>k_{\text{rate}}^* = 12.2</math>; the anatomy block floor CERTIFIED on B2/B3; the controls finding: the raw floor HOLDS in every fake world while the BA3 bridge is FALSE – the arithmetic content is the {floor, bridge} PAIR. THE TERMINAL STATEMENT: SIGMA-FLOOR == [DC-LEG, classical per census] <math>\times</math> [JET-MASS-FLOOR: THE TERMINAL <math>\lambda</math>-UNIFORM RESIDUE, arithmetic-pinned, rate OPEN, NOT claimed]. Provenance: r169 RE-RUN GREEN at promotion (35/35; AMENDMENT 1 disclosed). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                              |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v930_window_instrument_mechanism.py | <p>PRIME.WINDOW.INSTRUMENT.MECHANISM.01 [E]: the window instrument + the accretion/entry mechanism of round 167 (10/10; a mechanism-record module at its honest grade, the v916 CERT-NUM precedent). RECOMPUTED IN-RUN (exact): WI1 (plant transport: the PSD premise survives positive band mass at ANY strength – the plant sandwich + the deflation plateau), WI2 (the ghost secular threshold: PSD iff <math>s \leq s^* \leq \tau/(\lambda_P w_0^2)</math> – the ghost price collapses with the Connes collapse), WI3 (the perturbation law, FIRST-ORDER-SYMMETRIC), WI4 (the budget-moment dictionary with r156-L1 cited from v924), WI5 (<math>\Theta == c_1^2</math> + the theta chases; SELFCAP-CLASS-BLIND-TO-THETA), WI6 (cofinality: BANDPOS(unbounded ladder) <math>\iff</math> RH – the loop identification, bookkeeping-grade), and GA4 RECOMPUTED from the pinned slots (the shifted Hankel NOT PSD + log-convexity break <math>\geq 1.5</math>: STIELTJES-MOMENT-REFUTED). PINNED (typed MEASURED): THE MECHANISM FINDING – NO PARTIAL COMB IS POSITIVE: cone entry ONLY in the last atom's tail (<math>t^* \geq 0.9997</math>) with the TANGENT CRITERION from the prime source; ALL FAKES FAIL BY NO-ENTRY; re-confirmed at true dyadic pairs by r168 (ENTRY-NEW 4/4, v928); CHAIN-NOT-ASSEMBLED typed. HONEST DEMOTION CARRIED: the asymmetry LAW is GENERIC-CONE. Gate A: the slot ladder only APPROXIMATELY dyadic (15–17 percent), held-out MIXED 2/4. W3: value transport REFUTED-IN-CLASS, premise transport PROVEN, LOOP-IF-ASSEMBLED; GC1: the missing piece NAMED (PROP – assembled in r168/v928). Provenance: r167 PINNED from run-of-record (46/46, both logs kept). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                         |
| v931_jetmass_floor_theorems.py      | <p>PRIME.JETMASS.FLOOR.01 [E]: the product-floor theorem family of round 171 (10/10). RECOMPUTED IN-RUN (exact): PF1 (the census product identity <math>N(y) == A_0 \prod_j (y - y_j) +</math> the Vieta trace <math>\sum y_j == B_1 + y_t</math> at COEFFICIENT level – Newton/Vieta resum the ENTIRE J-tail into root data), PF2 (THE PRODUCT FLOOR: <math>F/A_0 \geq (1 - c^*/z)^{(1+\kappa)/c^*}</math> for all <math>z &gt; c^*</math> GIVEN H1 + H2 – THE EXPONENT IS THE TRACE; the r140 far-field law is the <math>\kappa \rightarrow 0</math> SHADOW, its measured 1.2 percent tightness EXPLAINED), PF3 (half-plane exclusion: H1 is ONE exact source evaluation per rung – NO cache, NO zeros), the JMF assembly (<math>\delta \geq L^2</math> WF; WF classical <math>(G - C)/2</math> suffix, GONEK-CONSTANT-UNPRICED, priced in v934; rate limit <math>4\pi/q</math> exact; any polynomial rate census-absorbable) and the T4 modus tollens (the r169 delta-toy VIOLATES PF2 hence NOT realizable under H1 + H2; the 2-mode witness KILLS the certificate pair; the lattice toy kills WF). PINNED: H1 certified at ALL 25 rungs (<math>c^* = 1.10</math>–<math>1.15</math> flat); H2 census COMPLETE-REAL NONNEGATIVE at ALL 25 rungs incl. degree-116 at <math>h = 28</math>; PF POINTWISE at <math>\sim 100k</math> true tail zeros (min margins <math>3.5 \times 10^{-5} \rightarrow 0.17</math>, all positive); <math>\delta_{\text{cert}} = 0.1035/0.0603/0.0351</math> with margin <math>\geq 8.2 \times 10^7</math>; the JMF BLOCK CERTIFICATE on B2/B3 (JETLOCK-MEAS ELIMINATED from the ancestor set); rate law <math>0.0767 h^{-1.057}</math>, <math>h_{\text{rate}}^* = 9117</math>. BH7-F1/F3 ADOPTED: the composed per-block hypothesis is the TRIPLE {H1 AND H2 AND H3}-cofinal, NOT H3 alone. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v932_toproot_theta_statement.py | PRIME.TOPROOT.THETA.01 [E]: the TOPROOT statement + the quartic certificate H3 of round 172 (10/10). RECOMPUTED IN-RUN (exact): T1 (TOPROOT( $p, C$ ) == $y_t \leq Ch^p$ with the exact SEQ-COFINAL quantifier and three machine-locked forms: theta-form $\iff$ TOPROOT(4), census form, rate form), THEOREM TR-CAP (proven: no census root above $(1 + 2\rho)y_t$ given $ J_{m+1}  \leq \rho^m$ – geometric closed form + the positivity polynomial; TYPED CIRCULAR-FOR-TOPROOT: the cap is $y_t$ -normalized; Fujiwara’s sharpest term == the trace $B_1 + y_t$ exactly), the Vieta pinch KILLED (every symmetric function carries $1/A_0$ ), THE $A_0$ -TRIANGLE ROUTE == THE FOURTH FLAGGED LOOP (exponentially vacuous AND consuming {TAUPOS, TLAWCAP}, cycle machine-detected), THE RATE DICTIONARY EXACT ( $a = p/2 - 1$ at $p = 3, 4, 5$ ), ADMISSIBILITY RECOMPUTED ( $\varepsilon_{\text{closed}}(56) = 4.5338 \times 10^{-9}$ , $a_{\text{max}} = 4.1348$ , $p_{\text{max}} = 10.2695$ : six powers of margin), and the witness BOTH directions ( $\theta'' = 62.69 > \text{bar}$ : H3 REFUTABLE; price $8.1 \times 10^{-2} \rightarrow 10^{-17.6}$ : TOPROOT-NOT-NORM-CONTINUOUS). PINNED: H3 CERTIFIED at all 25 rungs + holdout $h = 30$ (margin $\geq 1.99$ , ancestors == {SOURCE}); $\rho ==  J_2 $ with cap 1.21–1.30 vs $c^* = 1.10/1.15$ (ENVJ-SHARPER); SIZE-SEPARATOR 127/24/302. BH7-F1 ADOPTED: the residue is the TRIPLE {H1 AND H2 AND H3}-cofinal, one rung per dyadic block, all three at the same $h$ . Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.                                                                                                                                                                 |
| v933_h3_cofinal_adjudication.py | PRIME.H3.COFINAL.01 [E]: the H3-cofinal adjudication of round 173 (10/10). RECOMPUTED IN-RUN: PH (pigeonhole exact + the defect collapse: cofinal(bar) + defect- $D \implies$ FORALL- $h(\text{bar} + D)$ ; positive levels have NOTHING TO CANCEL), TRANSPORT ( $\Theta_{\text{FG}} == \theta_y \times \text{lock} + \text{both transport inequalities}$ : H3 and the r162 window EQUIVALENT on the lock manifold; the measured ceiling $0.1197 < 0.155$ consumes the v926 MEASURED quartic law – CONDITIONAL-ON-MEASURED, flagged), DK (the sin-theta stability lemma), the MANIFOLD EXCLUSION ( $\text{lock}'' == \text{lock}/W$ exact, both directions), and ALL SATURATION ANALYTICS (fit code ported verbatim: the five fits with every saturating model beating POW by factor $\leq 0.60$ ; $t_\infty$ band $[0.0766, 0.0977]$ ; the descent $4.2991 > 4.1203 > 4.0600 > 4.0236$ – P-LOCAL-DESCENDS-TO-QUARTIC, $p_{\text{loc}} \rightarrow 4.024$ ; $a_{\text{deep}} = 1.0118$ vs 1.057; the quantifier gap $D = 0.004183$ – the limsup form holds only MOD $D$ (BH7-F2); the block subsidy $\rightarrow 1$ : PIGEONHOLE-EXACT-SUBSIDY-EMPTY). PINNED: DK radii $4.5 \times 10^{-25} \rightarrow 6.6 \times 10^{-6}$ strictly growing, CROSSOVER $h = 16$ – THE MANIFOLD-EXCLUSION VERDICT IS SCOPED TO THE DK RADIUS (gated $h \leq 13$ , measured to $h = 20$ ; BH7-F4); the 2-mode witness breaks {ground-state cert, lock window, BA3 bridge $\times 3.6 \times 10^8$ }; the 3-mode escalation costs $\times 10^5$ and STILL breaks the same three (ESCALATION-STABLE); $\theta_\infty$ OPEN under the bar (margin 1.586; $1/(4\pi)$ inside, RECORDED-NOT-CLAIMED). BH7-F1/F2/F4 ADOPTED. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim. |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v934_gonek_pricing_unconditional.py | PRIME.GONEK.PRICING.01 [E]: the unconditional Gonek pricing of round 174 (8/8). RECOMPUTED IN-RUN (exact): GP1 (the exact classical statements identified with their conditionality: [L] Landau 1912 UNCONDITIONAL; [G] Gonek 1985/1993 uniform, $ E  \leq c_G x \log(2xT) \log \log(3x)$ , UNCONDITIONAL; [R] Gonek 1984 RH-CONDITIONAL == LOOP-IF-CONSUMED, machine-checked NOT AN ANCESTOR of either leg – externally re-verified in Bughunt VII), GP2 (THE ENVELOPE CLOSED FORM: the $T_{hi}$ -dependence COLLAPSES to a single $-1/(2T_{hi}^2)$ term with a FINITE CEILING – CENSUS-DEPTH-INSENSITIVE), GP3 (PREFACTOR-INSENSITIVE ABSORPTION with SYMBOLIC $c_G$ at $p = 2.5$ : the census schedule tolerates the classical error terms at ANY finite constant), the loop graphs. PINNED: $\hat{c} \leq 0.085981$ over the full $12 \times 6$ spike table on the 20M-zero cache (snr up to 26592.6: the Gonek form over-covers $\geq 11\times$ ) – CONSTANT-EMPIRICAL-PER-CENSUS (no published explicit constant exists, typed honestly); the DC leg replicates r169 at all 25 rungs; the WF leg holds the exact perturbation bound HARD; controls refuse $\geq 100\times$ with the dilation witness re-creating the true spike at $x' = 5^{64/65}$ to $1.4 \times 10^{-7}$ . The on-line rewrite stays PER-CENSUS; ALL-K remains the flagged loop. BH7-F1 ADOPTED: what remains is the TRIPLE {H1 AND H2 AND H3}-cofinal + {census-forall-k == LOOP} + {L1, WPD}. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.                                                                                                                                                                                                                                                                                                                                                                                                             |
| v935_thetainf_landau_bridge.py      | PRIME.THETAINF.PIN.01 [E]: the mode-level Landau bridge + the variational conditioning wall of round 175 (9/9). RECOMPUTED IN-RUN (exact): BR (the bridge closed forms: $\int_0^L \sin(\omega u) e^{igu} du == \omega(1 - e^{2iAg})/(\omega^2 - g^2)$ at modes, $2 \operatorname{Re} == 4\omega \sin^2(Ag)/(\omega^2 - g^2)$ , resonance REMOVABLE, $\psi_d(L) == 0$ at modes, trivial-zero density == $1/(t(t^2 - 1))$ ) – THE PROGRAM'S FIRST LINEAR PER-MODE CLASSICAL IDENTITY), COV (the rate dictionary is a BIJECTION in $\theta$ – transports every $\theta_\infty$ , pins NONE), RS (Rayleigh-Schrödinger identities: second-moment class ONLY inside the validity radius – MEASURED BROKEN), DIV (the VK ceiling AND even the flagged RH-grant transport DIVERGE at mode scale: CANCELLATION-BLINDNESS; the NEW Montgomery-PC loop cycle DETECTED, never consumed), INT (the one-signed interlacing premise REFUTED by the measured sign census; the trace ceiling SELF-REFERENTIAL). PINNED: BRIDGE-LANDS-PER-CENSUS (residuals $1.42 \times 10^{-5} \rightarrow 5.37 \times 10^{-5}$ at the FULL 20M-zero census, $\sim 1/T$ tail law; the fluctuation carries $\geq 99.2$ percent of $\theta$ ); THE CONDITIONING WALL (BH7-F7: a MEASURED-LAW extrapolation): reconstruction fails at full census, $T_{\text{req}} = 3.76 \times 10^{14} \rightarrow 1.75 \times 10^{77}$ vs $T_{PT} = 3.0 \times 10^{12}$ (PIN-CENSUS-UNREACHABLE-BY-DEPTH; the $1/A_0$ circularity RELOCATED into the variational conditioning); the second-moment route triply dead; worlds separate $\times 16449.9/\times 103617.3$ ; the bridge is WITNESS-DATA-INVARIANT (exact). $\theta_\infty$ RE-TYPED OPEN-NONPERTURBATIVE-VARIATIONAL. BH7-F1/F2 ADOPTED: the residue is the TRIPLE {H1 AND H2 AND H3}-cofinal (limsup only mod $D = 0.004183$ ) + {census-forall-k == LOOP} + {L1, WPD}. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim. |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v936_manifold_invariance_exclusion.py | <p>PRIME.MANIFOLD.INVARIANCE.01 [E]: the tau-free DK exclusion theorem + the transport ledger of round 177 (10/10). RECOMPUTED IN-RUN (exact): U1 (<math>\sin \theta \leq (r_0/\tau)/(FG - a)</math> – tau CANCELS in ratio form; contrapositive witness-free: PROVEN <math>h \leq 13</math> at the frozen <math>10^{-25}</math> bar, measured to <math>h = 20</math>), U2 (<math>\text{lock}'' == \text{lock}/W</math>; window nonempty iff <math>\text{TFG} &gt; \text{bar}</math>; the <math>\{h = 4\}</math> exception as a theorem), U3 (the induction implication + <math>L\_NEED = 1.7254</math> recomputed – the TFG-cap premise stays v926-MEASURED, FLAGGED), U4 (the BA3 scale-gauge theorem: degree-2 homogeneity – absolute-currency BA3 passable by scaling), U5 (frame transports + dial algebra). THE BH8-F7 PAIRING RULE</p> <p>RECOMPUTED: proven orders = <math>25 + \log_{10} b^*(h) = 23.8/21.7/15.7/4.4</math> at <math>h = 4/5/8/13</math>, NONE at <math>h \geq 16</math> – ALWAYS cited paired with the measured 30–33 orders (full-space price ladder <math>1.35 \times 10^5 \rightarrow 6.73 \times 10^8</math>, strictly growing; mechanism ALIGNMENT-SHARING, gap <math>2.354 \rightarrow 3.669</math>). BA3 NOT UNIVERSAL (the zmin passer exists at every rung BUT its own cert price grows <math>6.6 \times 10^9 \rightarrow 3.2 \times 10^{68}</math>); LEAK-FREE-MEASURED-DEEP; TRANSPORT-RELABELS-NOT-REDUCES +</p> <p>LEVEL-CHEAPER-THAN-INCREMENT: the induction route is priced dead AS A REDUCTION; TAU-ABS-IMPORT-NOT-LOAD-BEARING (DFS). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                                                                      |
| v937_l1wpd_closure_reduction.py       | <p>PRIME.L1WPD.CLOSURE.01 [E]: the L1/WPD consolidation + reduction of round 178 (10/10). RECOMPUTED IN-RUN (exact): R1 (H-pin restatement license: the D3 rearrangement + the <math>A_0</math>-FREE ball radius), R2 (WPD license: <math>K(q) = 2q</math> at <math>q = 1/2</math>; Theorem-C with the exact slack; DOM), R3 (the Theorem-A shape), R4 (THE LINEARITY ADJUDICATION: the eigenvector map NOT additive (<math>\lambda_{\min}(A + B) = -\sqrt{2} \neq -2</math>, exact) and the root map NOT affine – the pin floor lives on the VECTOR side of the bridge, the r175 <math>T_{\text{req}}</math> wall applies verbatim), R5 (AND-FIRE at the declared edges: the H-pin grants fire HPIN <math>\rightarrow</math> WPD-battery and L1, the jet-triple grants fire PF/HCOF but NOT HPIN/WPD: <math>H\text{-PIN} \neq \{H1 \text{ AND } H2 \text{ AND } H3\}</math>). THE THREE-PART</p> <p>VERDICT: source data bridge-expressible per census (residual ladder <math>2.73 \times 10^{-4} \rightarrow 2.09 \times 10^{-5}</math>); the floor escapes linearity at EXACTLY two maps; the SW-HSW pricing VACUOUS at battery rungs – NO closure; Omega-a is QUOTIENT-CLASS with the <math>A_0</math>-floor route == the flagged LOOP. PINNED: the v925 all-<math>x</math> sharpening replicated (<math>x_0 = 121/112</math>, ALL 357 <math>D_{cs} &gt; 0</math>); <math>d_1</math> digit-identical to r128/131/132; <math>\bar{\epsilon}/\text{gap}</math> <math>250.7 \rightarrow 4.2 \times 10^{19}</math> growing. THE NAME COLLISION FLAGGED: r133-H1/H2/H3 (ball matching) <math>\neq</math> r171-H1/H2/H3 (jets). The pair <math>\{L1, \text{WPD}\}</math> carries ONE open lambda-uniform edge – label-level sharpening, NO omega closed. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v938_dbn_heatflow_census.py    | PRIME.DBN.HEATFLOW.01 [E]: the exact census semigroup + the threefold dictionary of round 179 (11/11). RECOMPUTED IN-RUN (exact): D1 (the factor-4 normalization lemma – admissible gamma window $[0, 0.22/4] = [0, 0.055]$ EXACT; BH8-verified), D2 (zero-ODE + power-sum law == 60), D3 (THE SEMIGROUP: $T = 4y d^2/dy^2 + 2 d/dy$ degree-lowering, $e^{-tT}N$ a FINITE EXACT SUM – no zeros, no ODE; leading coefficient $A_0$ -invariant; THE TRACE LAW $\text{trace}(N_t) == \text{trace} + 2d(2d-1)t$ EXACT, measured rel dev 0.0), D4 (the generator split: the missing half is EXACTLY the pair $(\Lambda \star \Lambda)$ term == the RH-conditional class == the flagged loop), D5 (realness legs + the KKL pair identity). PINNED: THETA-FLOW-INERT ( $t_{\text{floor}} \geq 4021 \times$ the window at every rung: H3 has NO flow lever); THE LAMBDA_H CRITICALITY LADDER $2.3898 \rightarrow 0.2161$ STRICTLY DECREASING ( $43.4 \times \rightarrow 3.9 \times$ above 0.055; THE LIMIT IS OPEN, typed MEASURED) – HONEST: Epstein measures LARGER backward slack (0.8535 vs 0.3071): lambda is NOT an off-line-zero detector; the wall fragility dies 8–51 ORDERS below the window; DICTIONARY-THREEFOLD-INEQUIVALENT (sign pattern $(+, -, +)$ at every rung); FLOW-Z-BREAKS-DERIVED-H2; WITNESS-PRESERVED-NOT-EXPELLED (repair only at $10477 \times$ the window, inside the KKL ceiling). BACKWARD-PERSISTENCE-IS-RH flagged NOT consumed (five loop cycles). Census cardinality 4 UNCHANGED. $\theta_\infty$ stays OPEN. NOT evidence for or against RH. NO RH claim. |
| v939_cbj_frame_adjudication.py | PRIME.CBJ.CONFLUENT.FRAME.01 [E]: STEP-A EXACT + the CBJ adjudication of round 180 (10/10). RECOMPUTED IN-RUN (exact): STEP A ( $F == \sum_k d_k \psi_k$ incl. $k = 0$ ; $\delta_h ==  J_h _G^2$ with remainder $R == 0$ proven generically; Gram-PSD as a sum of squares – the single most promotable theorem of the arc, certified at 14 rungs, $\text{dev} \leq 1.9 \times 10^{-46}$ ), STEP B (Vandermonde/Cauchy determinants == SPACING PRODUCTS, v922-D1 re-derived; $\text{comb det } T == \text{spacings} / \prod r!$ ), the window positivity (Fejer + Gauss exact), MV (EXACTLY KILLED at linkage 1 ( $1 - 3\pi < 0$ ); CARRIES at $3.15\pi$ with the EXACT RATIONAL MARGIN $1/21$ ; the exact two-grid lemma), the occupancy bound $m \leq \lfloor 2^{k+1}(e^w - 1) \rfloor + 1$ and the SOURCE-ONLY selector $\hat{h}_k = 7/13/31/\dots/8191$ RECOMPUTED IN-RUN (Bertrand cited). PINNED (MEASURED): THE GLOBAL FORM IS DEAD at three independent walls (dof kill $n = 474$ vs rank 128, dof 112.8, BH8-verified; seam 0.994; occupancy collapse $-0.7484$ dex/atom); the per-cluster floor CARRIES k-uniformly at fixed $r$ (THE WALL IS THE $r$ -COLLAPSE, NOT THE DEPTH); GAUSS beats FEJ1 by 5–18 dex; the house Gram min-eig ladder $1.1 \times 10^{-17} \rightarrow 5.2 \times 10^{-113}$ ; SELECTOR-UNDERPERFORMS-BA-BUT-CLEAR-DEMAND (margins $2.14/3.05$ , all rungs $\geq 1.405$ ). CBJ delivers INSTRUMENTS, closes NOTHING. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.                                                     |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v940_cbj_subdof_blockfloor.py       | <p>PRIME.CBJ.SUBDOF.BLOCKFLOOR.01 [E]: the sub-dof mass-fraction kill + the Carleson pricing of round 181 (9/9). RECOMPUTED IN-RUN: the chain legs (<math>\delta == \sum \lambda_i s_i^2</math> via step-A <math>R == 0</math>; PSD truncation; sufficiency; SF1) + the FIXED-R COMPACTNESS SKETCH (the <math>m = 2</math> full-merge limit <math>== \text{diag}(1, -\ddot{W}(0))</math>: GAUSS <math>\text{diag}(1, 1)</math>, FEJ1 <math>\text{diag}(1, 1/6)</math> – the sympy-exact <math>1/6</math>, BH8-verified) + the kill arithmetic (OLS slope <math>+0.999/\text{rung}</math> recomputed from the pinned ladder; dead-by-factor <math>\sim 125</math> recomputed). THE KILL (MEASURED): ONE DEX OF MASS LEAVES THE SUB-DOF CELL PER RUNG, saturating at zero from <math>h = 10</math>; margin_sub dies AT THE SELECTOR RUNGS (<math>8.02 \times 10^{-3}</math> at <math>\hat{h}(B3) = 13</math>); the UNSCOPED r180 ladder (<math>\geq 1.405</math> everywhere) remains the measured carrier – killed is the PROOF ROUTE through the well-conditioned subspace. CARLESON PRICED: [Ortega-Cerdà-Seip, Ann. of Math. 155 (2002)] the box condition is equivalent to the UPPER inequality alone – WRONG-DIRECTION for a floor; the unconditional lower classes each miss for a named reason, with [Nazarov 1993, Turán] explicit BUT exponential in the term count <math>==</math> THE CLASSICAL MIRROR of the measured occupancy law; a 23-dex two-sided gap open in the depth. FRACTION-WORLD-SEPARATING (MAIN tails <math>-4.11/-2.26/-0.04</math> vs fakes <math>-11.7..-14.3</math>: floor VALUE world-insensitive, mass LOCATION world-separating – the first non-bridge world separator); MASS-RIDES-CONDITIONING-WALL. The missing ingredient exactly named: the alignment law of the <math>s_i</math> (v941). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                                                                                                                                                                                                                     |
| v941_alignment_law_factorization.py | <p>PRIME.ALIGNMENT.LAW.01 [E]: the frozen alignment law + its mechanism factorization of round 182 (11/11). RECOMPUTED IN-RUN (exact): L1 (GRAM-SOURCE-FREE – the entire law lives in the source ray <math>d</math> against fixed geometry; proven three ways in the probe), L2 (THE KEY ADJUDICATION: <math>s_i A_0 == \langle v_i, Dd \rangle / \sqrt{S_\mu}</math> is LINEAR in <math>d</math> and every mass fraction is SCALE-INVARIANT: the location law is PROJECTIVE and <math>A_0</math>-FREE – the r178 eigenvector-map obstruction does NOT bite), L3 (NORMS-DO-NOT-FORCE-LOCATION, exact exhibit), L4 (the ONE shared nonlinear link: the wall eigenmap, <math>\lambda_{\min}(A + B) = -\sqrt{2}</math> re-exhibited – THE LAW IS EXACTLY AS BRIDGE-EXPRESSIBLE AS DELTA given <math>d</math>). THE FROZEN TRIPLE (MEASURED): the half-power envelope (alpha in <math>(0.38, 0.54)</math> at all 14 rungs – per BH8-F4 a MEASURED-DESCRIPTIVE band, never a prediction); the quarter-depth wall lock (<math>\mu_m = 0.2372 \log_{10} g_{\min}</math>, <math>R^2 0.9916</math>); the DEEP EXCESS 5.638–7.575 dex at <math>h \geq 8 ==</math> THE r181 KILL MASS. ALT-JETS-BREAK-ALIGNMENT (every deterministic deformation of the true ray snaps the mass to the top: the law needs the JOINT sign-and-magnitude structure); WITNESS-BREAKS-ALIGNMENT (costs <math>5.3 \times 10^{-7}/8.2 \times 10^{-14}</math>); POWER-COMPOSITION-INVALID disclosed; the r177 gap ladder replicated EXACTLY (one object, two currencies – measured correlation, not identity). THE TERMINAL ADJUDICATION (BH8-F1 corrected wording verbatim): SAME-CLASS-NOT-NEW-EDGE – the same variational-quotient CLASS as <math>\theta_\infty</math> (r175) and H-pin <math>\Omega</math>-a (r178: DISTINCT statements), four distinct formal statements over the same source object (the wall argmin ray <math>d</math>) in <math>A_0</math>-quotient form; NO new residue edge, because the floor route is optional and machine-excluded from the delivered set (G44), NOT because the objects are identical. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v942_pi_pattern_transfer_specificity.py | <p>PI.PATTERN.SCAN.01 [E]: the detector-specificity theorem of round 184 (9/9). INSTRUMENT CHARACTERIZATION, NOT ARITHMETIC NOVELTY.</p> <p>PI-LANDS-FAKE-CLASS: all five predefined pi-derived comb worlds (digit combs, continued-fraction, BBP, convergent-geometric) classify TOP-HEAVY (log10 tails <math>-10.0.. - 17.9</math>) exactly like the fake-world class, on both holdout halves, all rungs, ALL 100 non-Mangoldt instances – ZERO anomalies. TRANSFER VALIDITY PROVEN: the C-MANGOLDT positive control fires on EVERY transferred detector (port dev <math>8.4 \times 10^{-59}</math>, anchor <math>\leq 0.004</math>, Landau spike <math>C = 131.04</math> vs the integer-position-matched null max 12.60, alt jets <math>-12.84/ - 13.08/ - 13.43</math>); the structured controls CAN fire (Champernowne Weyl 15.8/43.2 vs null 2.46; <math>\varphi</math> GK <math>\chi^2</math> 7047); pi's classical CF laws replicate as measured expectations (Gauss–Kuzmin, Khinchin, Lévy). RECOMPUTED IN-RUN (exact): the Gauss–Kuzmin telescoping sum <math>== 1</math>; Lévy + Khinchin constants (25+ digits); the BBP series <math>== \pi</math> (30 digits); the classification-bar arithmetic. BH9-K3 ADOPTED: the smoke lineage disclosed (the pre-freeze OverflowError crash smoke on the Liouville CF stream; no record impact); P-BBP carries the in-spec PROVISIONAL flag. NO pi-numerology claim. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                                |
| v943_mangoldt_ablation_localization.py  | <p>PRIME.MANGOLDT.ABLATION.01 [E]: the zero-coupled sharp localization of round 185 (8/8). The additive property ladder W0 random <math>\rightarrow</math> W1 Cramér density <math>\rightarrow</math> W2 +log-weights <math>\rightarrow</math> W3 +power grammar <math>\rightarrow</math> W4 +full Beurling multiplicativity <math>\rightarrow</math> W5 true Mangoldt: BOTH arithmetic signatures switch on ONLY AT W5 AND SHARPLY (S1 jump 12.07 dex, S2 jump 61<math>\times</math>). THE BEURLING HEADLINE ANSWER IS NO: full multiplicative semigroup structure WITHOUT Riemann-zero coupling produces NEITHER signature – multiplicativity is not sufficient, the detectors see the true zeros through the source side. NECESSITY: every ablation kills its property (A1 –POWERS kills S1 while S2 survives at 129.06 – S2 sits on the exact prime POSITIONS); EVEN <math>\pm 0.01</math> POSITION JITTER KILLS BOTH. W5 replicates r184 (transfer valid). RECOMPUTED IN-RUN (exact): <math>\psi(n) == \log \text{lcm}(1..n)</math> at <math>n = 30</math>; the Cramér <math>n = 2</math> clamp (<math>1/\ln 2 &gt; 1</math>); the classification/jump arithmetic. The single 26/27 FAIL is the DISCLOSED control-side Beurling density band (<math>t = 2</math>: 5.74 vs <math>[0.2, 5.0]</math>) – BH9-adjudicated an honest Poisson fluctuation, density-correct, the W4 headline does not weaken. Mechanism characterization, NOT arithmetic novelty; no claim about WHERE the true zeros lie. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v944_ground_residue_observability.py | <p>PRIME.GROUND.RESIDUE.OBS.01 [E]: the exact residue identity + the Feshbach scalar form + strict interlacing of round 186 (11/11). RECOMPUTED IN-RUN (exact): <math>\text{Res}_{z=\tau} l^T (zI - M)^{-1} l == (l \cdot d)^2 / \ d\ ^2</math> (generic <math>2 \times 2</math> + the quaternion rational-orthogonal <math>3 \times 3</math>); the Feshbach/Schur form <math>R(z) == \ l\ ^2 / w(z)</math> with <math>A_0^2 w'(\tau) == \ l\ ^2</math> and <math>w'(\tau) == 1 + m^T (\tau I - B)^{-2} m</math>; the CDLI adjugate tie <math>A_0^2 == N(\tau) / P'(\tau)</math> (same identity, new coordinates – disclosed prior art); STRICT INTERLACING <math>\Leftrightarrow A_0 \neq 0</math> both directions; the exact <math>A_0</math>-null <math>\sum_{k=0}^{41} (-1)^k == 0</math>; the displacement algebra <math>DL - LD == f \mathbf{1}^T - \mathbf{1} f^T</math> (diagonal-blind). Probe: the identity gated <math>\leq 7.8 \times 10^{-45}</math> at ALL 14 rungs; GAP-IS-RESIDUE-IN-DISGUISE (<math>\varepsilon \leq 2.55 \times 10^{-6}</math> – not an independent handle); EIGENVECTOR-DESCENDS-NOT-ELIMINATED (overlap spread <math>-11.24.. -90.72</math>; per BH9-K4 <math>w'(\tau)</math> is source-side (eigenvector-free, GIVEN <math>\tau</math>)); values separate worlds <math>\geq 1</math> dex (measured); the <math>\{l_0, l_2\}</math> witness detector; the raw-gap floor RELABELED-AND-STOPPED (rides <math>\tau</math>, +1.011). THE BH9-K1 MAJOR CORRECTION ADOPTED VERBATIM: WALL-OFF-DIAGONAL-IS-ONE-FUNCTION-LOEWNER-EXACT – displacement rank 2 fixes ONLY the off-diagonal; the FULL wall is <math>M_h \sim L_f + \text{diag}(\Delta)</math>, <math>\Delta \neq 0</math> source-side (r189/v947), NOT the canonical Loewner matrix; <math>R_h(z)</math> is the resolvent of <math>L_f + \text{diag}(\Delta)</math>, not of the Loewner kernel alone. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |
| v945_zero_causal_stratification.py   | <p>PRIME.ZERO.CAUSAL.SYNTH.01 + PRIME.ZB.WIGGLE.STRAT.01 [E]: the causal zero-coupling demonstration with its stratification of rounds 187 + 190 – ONE JOINT MODULE, r190 upgrades r187 BY REFERENCE (10/10). MECHANISM DEMONSTRATION ON CONTROLLED SYNTHETIC WORLDS, NOT ARITHMETIC NOVELTY. Z0 VALIDITY 0.0006 on <math>C = 131.04</math> (the synthesis engine valid; the S1 eigen-readout died EXACTLY as the frozen r175 conditioning-wall prediction said – honest instrument death, disclosed; the two r187 FAILs are the disclosed adjudication gates); THE DOSE SURFACE monotone in <math>\delta</math> everywhere (DEV up to 25.8; first response <math>(0.10, 10^{-2})</math>; first death <math>(0.05, 10^{-4})</math>); EXACT-ORDINATES-REQUIRED (Poisson AND GUE resamples at the true density BOTH die); jitter resonance 0.003; smooth floor <math>C = 103.99</math>. THE STRATIFICATION (BH9-K2 ADOPTED): the wiggle in full precision is <math>W_{\text{rec}} = 1.065464</math> (READS 1.07 – never 1.14; the violation itself real at <math>1.0655 &gt; \text{slack } 1.0</math>); wiggle INSIDE all matched-dose selection spreads <math>(5.09/3.07/3.98)</math>; dose MONOTONE IN EXPECTATION (26.96 vs 9.18); INV-FRAC 1.00 (generic subset-overshoot); the per-zero leverage ladder (<math>\gamma_2 = 21.02</math> carries the most, no first-zero dominance, the 10-set strongly subadditive); depth parity NOT-ARTIFACT (<math>0.0014 \ll 0.53</math>). RECOMPUTED IN-RUN (exact): the general-<math>\rho</math> mode term reduces on-line at <math>\beta = 1/2</math>; the dose law <math> x^\rho  = x^\beta</math>; the K2/noise/subadditivity arithmetic. The r185/v943 interpretation upgrades to DEMONSTRATED MECHANISM on the S2 channel. STILL NO CLAIM ABOUT WHERE THE TRUE ZEROS LIE. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>                                                                        |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v946_census_krein_pencil.py     | <p>PRIME.CENSUS.SPECTRAL.LIFT.01 [E]: the exact Krein pencil of round 188 (8/8) – the census polynomial IS the spectral determinant of a source-side self-adjoint pencil, but of a KREIN (indefinite-metric) pencil at every true rung: <math>N_h(y) = (-1)^{K-1} \det(J) A_0 \det(\hat{A} - yJ)</math> with <math>\hat{A} = JD - (Jw)(Jw)^T</math>, <math>J = \text{diag}(\text{sign } \rho_k)</math>, <math>\sum_k \rho_k = -y_t</math> EXACTLY (dyadic). THE DECISIVE NUMBER: the residue-sign ladder <math>(n_+, n_-) = (1, 5)/(7, 3)/(6, 14)/(27, 14)/(45, 34)/(43, 73)</math> – MIXED at every true rung; SMOOTH (0, 10) one-signed; SCRARITH/EPSTEIN just as mixed: MIXEDNESS-IS-ARITHMETIC (atoms-vs-no-atoms, the WRONG orientation, explicitly NOT a sign source); MIXEDNESS-TRANSFORM-INVARIANT (Möbius/weight/<math>g^2</math> lemmas recomputed symbolically in-run: definite impossible IN CLASS); H2 == KREIN STRONG DEFINITIZABILITY in operator form. RECOMPUTED IN-RUN (exact): the matrix determinant lemma (<math>n = 2.4</math>); the full Krein assembly (<math>n = 2, 3</math>, ALL sign patterns: similarity, J-symmetry, determinant transfer, Weyl form); E1 census == Mittag-Leffler numerator coefficient-exact + E2 the determinant identity at <math>K</math> interpolation points (exact rational instance; probe: dyadic-exact E1 at all six rungs, integer-Bareiss E2 at <math>h \leq 13</math>, typed MDL-EXACT-H13-NUMERIC-H2128). BRACKET-CERTIFIED-VIA-ENVJ (<math>c^* = 1.10/1.15</math>: <math>\text{Re } \lambda &lt; c^* y_t</math> for the pencil WITHOUT computing a single root); ROOTS-NEVER-CONSUMED (machine-audited); tau-free (exact gauge invariance); the named <math>W(y) &gt; 0</math> upgrade path subsequently CLOSED BY THEOREM in v948. “Census” NEVER means a zeta zero set. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p>      |
| v947_loewner_pick_dictionary.py | <p>PRIME.LOEWNER.PICK.01 [E]: the exact Loewner–Pick dictionary + the Pick-class refutation of round 189 (9/9) – the wall is NOT the canonical Loewner matrix of its potential but a DIAGONALLY-SHIFTED one: <math>\text{Raw}_h = L_f + \text{diag}(\Delta)</math>, with THE NEW EXACT COSINE-QUADRATURE DIAGONAL LAW <math>\text{Raw}[k, k] - f'_{\text{prime}}(b_k) = 2a \text{pc}_k</math> – the COSINE quadrature of the SAME sieve atoms whose sine quadrature feeds the potential: both quadratures in different slots (probe sympy per-atom + mp <math>\leq 6.0 \times 10^{-61}</math>; BH9 own recompute <math>\leq 6.3 \times 10^{-61}</math>). RECOMPUTED IN-RUN (exact): the pole block FULLY CANONICAL (divided difference == rank-1 Cauchy AND diagonal == <math>f'_{\text{pole}} - \Delta_{\text{pole}} == 0</math> identically); <math>f_{\text{pole}}</math> a GENUINE PICK function with EXACT Herglotz point mass <math>2 \sinh^2(a/2) \delta_{-1/4}</math> (<math>\text{Im } f_{\text{pole}} &gt; 0</math> proven); the per-atom chain rule; the commensurate-sampling mechanism <math>\sin(2a \omega_k) == 0</math> at the nodes (SMOOTH-PSD-AT-NODES – lattice geometry, not a prime fingerprint). THE REFUTATION (pinned): <math>\lambda_{\min}(L_f) &lt; 0</math> at ALL 14 rungs – the Pick costume misses wall positivity by 9.4–87.9 ORDERS; descent counts 2–34; ALL grid sign changes carried by the prime leg; refinement DEEPENS; the 17-node window order-separation exhibit; not-a-sum-kernel; ATOMJET detected exactly linearly; tau-screens BOTH FLAT; r45 LOEWNER-DEAD distinguished (a THIRD reading killed; the IIKS/RHP lane untouched). HERGLOTZ per leg: pole EXACT, arch OPEN (honestly no closed form), prime leg the named relabeling BARRIER (a positive rewrite would BE the wall positivity). Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v948_krein_sign_characteristic.py  | <p>PRIME.KREIN.DEFINITIZER.01 [E]: the sign-characteristic law + the parity scaffold + the <math>W(y)</math>-congruence-class kill of round 191 (10/10) – the r188/v946 upgrade path CLOSED BY THEOREM. RECOMPUTED IN-RUN (exact): the kernel/Weyl identity <math>((\hat{A} - yJ)v == Jw(1 + G)</math> with <math>v_k = w_k/(b_k - y)</math>, <math>v^T Jv == -G'(y)</math>; <math>n = 2, 3</math>, ALL sign patterns); THE TWO-ROUTE LAW on the full 32-cell exact Sturm battery (poles (1, 2, 3, 4), all 16 sign patterns <math>\times</math> 2 magnitude profiles, NO floats: <math>\varepsilon</math> by interval-certified crossing sign <math>== \varepsilon</math> by the combinatorial formula <math>-(-1)^{(R-i)+m_i}</math> at every real root; the flip law; the count law with nonreal-pair deficit); THE PARITY SCAFFOLD (root-count parity per pole gap from residue SIGNS alone – source-side (root-free) per BH9-K4; probe: verified at all 81 gaps); the <math>W(y)</math>-congruence obstruction lemma <math>(x^T M'(y_i)x == v^T A'(y_i)v</math> generically – the sign characteristic is CONGRUENCE-INVARIANT); the Cayley–Hamilton Langer degeneracy (finite-dim definitizability degenerate; the non-vacuous notion STRONG definitizability <math>==</math> H2 per rung; Langer 1982 / Zizler 1995 / Lancaster–Ye 1993). THE THEOREM: the measured MIXED ladder kills the ENTIRE <math>W(y) &gt; 0</math> congruence class AT ONCE – IMPOSSIBLE-IN-CLASS, not merely not-found (three families instantiated, devs <math>\leq 1.6 \times 10^{-55}</math>, blocking roots (1, 5)/(1, 2)). PINNED: <math>d_h = 2/4/11/18</math>; occupancy-minimality FAILS at <math>h = 8, 13</math> by exactly ONE interior doubled gap (STRICT-DEFINITIZER-ROOT-DEPENDENT); THE r188 LADDER IS THE KREIN MULTISSET; the r172 witness BREAKS CENSUS REALNESS while the raw ladder is blind (a NEW detector); orientation stays ATOMS-VS-NO-ATOMS; tau-FLAT (+0.062); the Krein coordinate closed STRUCTURAL-ONLY. Census cardinality 4 UNCHANGED. NOT evidence for or against RH. NO RH claim.</p> |
| v949_commensurability_mechanism.py | <p>PRIME.COMMENSURABILITY.MECHANISM.01 [E]: the exact node mechanism + the structural-zero census + the first Baker-class pricing of round 192 (11/11) – the thin geometric hope ‘incommensurability feeds wall positivity’ CLEANLY KILLED at the margin. RECOMPUTED IN-RUN (exact): the node-lattice identities <math>\sin(2a\omega_k) == 0</math> + the boundary-phase invisibility; THE SMOOTH COLLAPSE to Moebius data <math>(\text{pj}_s(\omega_k) = \omega_k(1 - e^a)/(1/4 + b_k)</math>, <math>\text{pc}_s</math>, and the combined node potential – the exact content of SMOOTH-PSD-AT-NODES, v947); the STRUCTURAL-ZERO CENSUS in exact integers (<math>h = 4</math>: 12/18, <math>h = 8</math>: 32/120 – the <math>e \mid 2fk</math> law; the Bughunt-X KE wording adopted); the Jordan bridge <math>\sin(\pi x) \geq 2x</math> (concavity, exact); the <math>h = 4</math> minimizer priced from the PUBLISHED constants – exact-integer independence <math>3^{10} \neq 4^8</math>, Liouville gap 3.79 dex, LMN 1995 gap <math>7.099 \times 10^3</math> dex, Matveev 2000 gap <math>1.682 \times 10^9</math> dex, both bound formulas recomputed in-run. PINNED: the full BAKER-TOO-WEAK ladders; the defect ladder O(1)-FLAT while <math>\tau</math> falls 45 orders; the dose DIRECTION-BLIND (killed both ways at every dose rung + jitter; COMM-WALL-DEAD); <math>L_f(t = 1)</math> PSD at all four dose rungs WORLD-BLIND (lattice geometry, not a prime fingerprint). The margin consumes position EXACTNESS, not incommensurability magnitude. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v950_zero_channel_capacity.py   | <p>PRIME.ZERO.CHANNEL.CAPACITY.01 [E]: the Shannon characterization of the zero-comb dictionary of round 194 (8/8) – the r187/r190 (v945) write→read arc characterized AS A CHANNEL, typed EXPLORATORY INSTRUMENT CHARACTERIZATION and promoted as exactly that: NO storage claim, NO arithmetic novelty. RECOMPUTED IN-RUN (exact): the ternary capacity ceiling <math>\log_2 3</math>; the erasure significance bar <math>1/3 + 3\sqrt{(1/3)(2/3)/48} = 0.5375</math>; the Miller-Madow zero-error excess bound; THE PARITY-CODE ERASURE THEOREM (all 27 parity codewords <math>\times</math> 4 erasure positions, 108 exact mod-3 cases); the capacity-surface monotonicity. PINNED: the capacity surface (error-free full ternary from <math>\varepsilon = 0.01</math>); THE RESOLUTION LIMIT LOW = 0.001 / MID = 0.003 with the Bughunt-X F6 quantification adopted (<math>P = 0.0032</math> under the 20000-trial null MC – MARGINAL-BUT-NONNULL single-reading, decisive with averaging); the sqrt-<math>n</math> prediction HELD; LINEARITY-SCALE 0.01 (linear codec below, INTERFERENCE-MEMORY above);</p> <p>CODE-STRUCTURE-REDUNDANT-DISTRIBUTED with the F7 corollary typing (the parity-erasure 1.000 is a corollary of the zero-SER cell); P3 refuted honest (CHANNEL-GAIN-LOW&gt;MID); controls clean. The round-107 verdict cited, NOT upgraded. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim; NO storage claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v951_cancellation_functional.py | <p>PRIME.CANCELLATION.FUNCTIONAL.01 [E]: the ACF law + the pole square + the arch law of round 195 (10/10) – the wall is a CLOSED TRUNCATED WEIL-EXPLICIT-FORMULA FUNCTIONAL of the PD test function <math>g_x</math>, per-atom manifest positivity PROVABLY IMPOSSIBLE (Wiener-Khinchin), PLUS THE BUGHUNT-X KD PIN RECOMPUTED IN-RUN: the arch kernel IS the classical Weil archimedean kernel in the <math>L \rightarrow \infty</math> limit – the arc's external handoff object. IN-RUN: the generic convolution identity (explicit antiderivative warded by differentiation); the <math>k = 0</math> wraparound slice; THE POLE SQUARE <math>x^T \text{RawP } x == 8 \sinh^2(a/2) (\int e^{-t/2} A_x)^2</math> (sympy generic + numeric at <math>h = 4, 5</math>);</p> <p>THE ACF LAW <math>x^T W(u_q)x == -2 \int_0^{L-u} A_x A_x(\cdot + u)</math> at EVERY atom of <math>h = 4, 5</math> (own wall build); THE ARCH LAW <math>\Delta \text{Arch}_k == -\text{mult}_k L(\kappa(\omega_k) + c^*)</math> with <math>c^* = (\gamma + \ln 2\pi)/2</math> at EVERY mode of <math>h = 4, 5</math>; THE KD PIN <math>\kappa + c^* \rightarrow -\frac{1}{2}[\text{Re } \psi(1/4 + i\omega/2) - \ln \pi]</math> at <math>L = 80</math> (finite-<math>L</math> corrections <math>O(e^{-L/2})</math> typed NOT-SMALL). PINNED: the anatomy (THE POLE ALONE CARRIES POSITIVITY; depth == the <math>\tau</math> ladder, definitional; CI 11.3 <math>\rightarrow</math> 88.7); THE SIGN LAW (<math>n_+ = 0</math> at ALL 14 rungs, <math>m_{99} = 1..2</math> – compensation ATOMS-VS-POLE); complementarity NECESSARY-NOT-SUFFICIENT. HONEST PRICING: an ALGEBRAIC RESTATEMENT – the relabeling barrier NAMED, NOT CROSSED; WEIL-ALLTESTS <math>\leftrightarrow</math> RH a FLAGGED LOOP consumed by NOTHING. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim.</p> |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v952_turan_extremal.py    | <p>PRIME.TURAN.EXTREMAL.01 [E]: the classical PD-cone theory PRICED of round 196 (9/9) – Turan/Boas-Kac/Fejer imported with exact citations and priced against the wall for the first time in 196 rounds, TWO-SIDED: the classical cap captures the drain almost exactly, and cone geometry CANNOT see positivity. RECOMPUTED IN-RUN (exact): THE FEJER COEFFICIENT THEOREM at <math>n = 1, 2</math> (extremal construction attains <math>\cos(\pi/(n+2))</math> as a manifest square + single-node dual certificates – the Bughunt-X independent route – both tight); the signed-bound flip; THE KB RATIONAL-LAG IDENTITY (at <math>u = L/2</math> the off-diagonal of <math>W(L/2)</math> vanishes IDENTICALLY and <math>M_{\text{sub}}(L/2) == 1/2 == \cos(\pi/3) == M_{\text{KR}}</math> EXACTLY – the Fejer extremal lifts exactly at RATIONAL lags only, near-attainment elsewhere; KB adopted); the edge kill <math>W(L) == 0</math> identically; the subcone CONTAINS the Turan interval extremal (<math>x = e_0</math> the Fejer triangle, equality). PINNED: TURAN-BUDGET-TAU-GAP (every cone bound NEGATIVE at every rung, gapdex <math>11.35 \rightarrow 89.03</math>, slope <math>-1.007</math> at <math>R^2</math> 1.000 – the relabeling barrier NAMED AT THE CONE LEVEL); CONE-DRAIN-O1-CAPTURE (capture 0.14–0.24 dex FLAT – the round’s real news); Q2-NEAR-TIGHT; ARCH-LEG-SUBCONE-ONLY; the TURAN-CONE-POSITIVITY <math>\leftrightarrow</math> WEIL-ALLTESTS loop FLAGGED consumed by NOTHING; caps world-blind. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v953_fewatom_reduction.py | <p>PRIME.FEWATOM.REDUCTION.01 [E]: <math>A_{v_0} \geq 0</math> (OBJECT-A) + the one-line sign-law mechanism + the horizon ladders of round 197, with THE BUGHUNT-X KA SCOPE RIDER VERBATIM and THE STURM CONTINUUM CERTIFICATES RECOMPUTED IN-MODULE at <math>h = 4, 5</math> (10/10). IN-RUN: the wall rebuilt OWN at <math>h = 4, 5</math> (<math>v_0</math> by mp inverse iteration; the profile a nodeless positive bump peaked exactly at <math>L/2</math>, rmin matching the record ladder); THE STURM CONTINUUM CERTIFICATES (exact dyadic Fractions <math>\rightarrow</math> Chebyshev <math>P(x) \rightarrow</math> INTEGER primitive-PRS Sturm chain: ZERO roots in <math>(-1, 1]</math>, <math>P(\pm 1) &gt; 0 - A_{v_0} &gt; 0</math> on ALL of <math>[0, L]</math>, no floats); THE ONE-LINE MECHANISM (<math>A \geq 0 \Rightarrow</math> every ACF sample <math>g(u) \geq 0 \Rightarrow</math> every atom negative; exhibited); the budget resplit; the horizon <math>Q_{99} = (1, q = 2)</math>; KC (the edge minimum IS <math>A_0 =</math> the <math>j_{r_0}</math> numerator UN-NORMALIZED; the r197/r198 pair consistent). KA ADOPTED VERBATIM: the r197 gate class is GRID-MEASURED (at <math>h = 15/16/20</math> the record minimum sits BELOW the bar’s resolution – print evidence, not gate certificate); CONTINUUM nonnegativity CERTIFIED by Bughunt X at <math>h = 4, 5, 13, 20</math> and re-certified here at <math>h = 4, 5</math>. PINNED: HORIZON-BOUNDED-<math>\{2, 3\}</math> incl. holdout + the scale-relativity rider (the demand scale consumes EVERY atom; NOT-A-COMPLEXITY-DROP-IN-H mandatory; LOSSY-TAIL-CERTIFICATION-DEAD); decay edge-power NOT UNIVERSAL (P3 refuted; the jet-convolution law band-approximate = the r182/v941 alignment in ACF coordinates); shape EPSTEIN-SEPARATING; the witness flattens the ACF ladder. OBJECT-A (X6): the ONE genuinely new tractable target – yields the sign law only, NO wall-positivity lever. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim.</p> |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v954_nodeless_<br>pf.py              | <p>PRIME.NODELESS.PF.01 [E]: the Perron-Frobenius adjudication of round 198 (10/10) – the r197/v953 candidate route for OBJECT-A fully adjudicated DEAD IN THE MODE BASIS, the obstruction LOCALIZED EXACTLY (the pole’s rank-one POSITIVE kernel is the SOLE cone obstruction – removing it repairs cone invariance at every rung) and a SURVIVING Z-STRUCTURE (the arithmetic hopping is one-signed ATTRACTIVE in band-limited position coordinates on prime + arch). RECOMPUTED IN-RUN: the trivial-route exhibit (<math>v = e_1</math>: nonneg coefficients, <math>A(L/2) = -1</math> EXACT – coefficient one-signedness would prove the WRONG statement); the pole rank-1-Cauchy positivity (exact); the off-diagonal censuses at <math>h = 4, 5</math> matching the record EXACTLY (<math>(n_+, n_-) = (15, 6)/(38, 17)</math>, checkerboard-mixed BOTH parities – OFF-DIAG-MIXED-PF-MODE-DEAD); the cross-gate nflip == raw<sub>nn</sub>; V0-PARITY-BROKEN recomputed (mis = 2/4; P2 refuted; the misaligned mass FLAT, not a <math>\tau</math> ladder); the pole kernel factor <math>\varphi &gt; 0</math> on the whole 33-lattice; THE CONE TESTS at <math>h = 4</math> (the wall operator violates on 11..14 of 20 Fejer members, the NO-POLE operator on ZERO). PINNED: KERNEL-SPLIT-POLE-POSITIVE-HOPPING-Z (pole negfrac 0.000 exactly – the wall’s kernel mixing is ENTIRELY pole-vs-hopping); SMOOTH the structural outlier (all-positive off-diagonals, cone-preserving, keeps nonnegativity with NO Z-structure); EPSTEIN breaks the shape with negw = 0; witness-stable. chain = CHAIN-OPEN-FUNCTION-SPACE (the r197/r198 pair consistent per KC; OBJECT-A stays open, function-space); the X6 discipline: pole-vs-hopping balance is the CLASS NAME of OBJECT-W – not a new attack object. Census cardinality 4 UNCHANGED. NOT RH evidence; NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v955_tau_iiks_toda_<br>dictionary.py | <p>PRIME.PORT.TAU.SYMBOLIC.01 (round 224, EXECUTED at the finite-identity level) + PRIME.PORT.LAX2.CONDITIONED.01 (round 225) + PRIME.PORT.HIROTA.SIGN.01 (round 226): THE FINITE INTEGRABLE DICTIONARY OF THE WALL (18/18 + 15/15 + 17/17, probes embedded byte-exact). (1) TAU_IIKS_EXACT: the wall determinant IS the exact IIKS Fredholm tau function – Sylvester as a FULL characteristic identity <math>\det(I - sQ_{\text{state}}) = \det(I - sE_{\text{node}})</math> on the whole <math>s</math>-grid; the <math>s</math>-family carries the <math>s</math>-DRESSED port <math>\det(I - sE) = \det(I - sR) \det(I - sD_P(s))</math> EXACT (fixed-kernel alias gap 0.06–0.20 named and blocked); the confluent diagonal (<math>\leq 3.2e-13</math>); the IIKS dressing identity (<math>\leq 7.9e-13</math>) with the dressed port generators source-explicit in closed form; the Fredholm variational identity in TWO independent times (<math>\leq 1.4e-9</math>); the relative tau transfer exact over <math>-212.84/ -195.50</math> log-unit collapses (<math>\leq 3.4e-12</math>); Epstein/scramble SATISFY the algebra and differ in the VALUE sign <math>\tau(1) = -1</math>. (2) LAX1_H_EXACT: the <math>h</math>-step is EXACTLY rank one (<math>\leq 2.9e-16</math>) and the degree-1 source-chain connection PREDICTS the next generator pair (3.4e–16) and the next tau step (3.2e–11) on sealed blind rungs; zero curvature identically, world-blind; the <math>s</math>-time has NO fixed degree and across rungs NO common carrier – typed, not hidden. (3) HIROTA_TODA_EXACT + WALL_EQUIVALENT: <math>\tau_{w,n} = D_n(\mu - \nu)/D_n(\mu)</math> (Hankel quotient), <math>r_n = h_n(\mu - \nu)/h_n(\mu)</math>, the Hirota coefficient source-pure from a scaled SIGNED Stieltjes recursion; <math>\hat{\gamma}_n &gt; 0</math> for all <math>n \iff</math> QUASI-DEFINITENESS of the signed defect measure <math>\mu - \nu \iff</math> the wall; MAIN carries ALL 184 <math>r_n &gt; 0</math>, the controls flip at <math>n = 25/21/27</math>. Claim splitting (note DLI): PRIME.PORT.TAU.FINITE.IIKS.01 [E] separated from PRIME.PORT.TAU.NOPOLE.COFINAL.01 [O]; PRIME.PORT.TAU.01 [O] stays formally open for the fully symbolic arbitrary-<math>h</math> statement. Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v956_signedmoment_<br>halffilling_<br>duality.py | <p>PRIME.PORT.SIGNEDMOMENT.HOLEDUAL.01 (round 228) +<br/> PRIME.PORT.PONTRYAGIN.MAXPOS.01 (round 229) +<br/> PRIME.PORT.FREEMOMENT.JFRACTION.01 (round 230) +<br/> PRIME.PORT.RHP.FREEMOMENT.MIDPOINT.01 (round 231): THE<br/> HALF-FILLING GEOMETRY OF THE SIGNED MOMENT PROBLEM<br/> (14/14 + 17/17 + 17/17 + 13/13, probes embedded byte-exact). (1)<br/> HALF_FILLING_BOUNDARY: <math>N_w = \lceil \#\text{supp}(\tilde{\mu})/2 \rceil</math> EXACT on all five<br/> windows – the window cap IS the half-filling of the union support; past the cap<br/> the wall dies IMMEDIATELY (first <math>r</math>-flip at <math>N_w + 0/2/2/3/1</math>, <math>O(1)</math> offsets): <math>\mu - \nu</math><br/> is quasi-definite EXACTLY up to half-filling – THE WALL IS MAXIMAL; the<br/> complement identity and <math>h</math>-duality exact at dps 80. (2)<br/> FREE_MOMENT_WINDOW_LAW: both signature predictions REFUTED; the<br/> corrected law: the largest Hankel window inside the <math>S</math> free moments is<br/> <math>n = (S + 1)/2 = N_w</math> – MAIN EXHAUSTS THE ENTIRE FREE MOMENT<br/> WINDOW while the controls die at 11–15 percent of it; the inertia theorem exact<br/> in all parts; the Lagrange contractor exact on the toy and ASTRONOMICALLY<br/> non-contractive at full degree (<math>\sigma_{\max} = e^{110.5}</math>); at window degree<br/> SAME_CONTRACTOR_EXACT (<math>\lambda_{\max}(Q_{N_w}) &lt; 1</math> – the v881 testing law closes<br/> the circle). (3) FREEPREFIX_EXACT_WALL_EQUIVALENT +<br/> FULL_CHAIN_REVERSAL_EXACT: moment prefix = Jacobi chain = Euclid<br/> chain = value chain (FOUR exact source-pure paths; the offsets are<br/> forced-coupling survival counts 0/2/2/3/1; Epstein’s first wrong Euclidean pivot<br/> EXACTLY at its flip <math>n = 25</math>); the dual measure carries the MIRRORED chain<br/> (<math>\alpha_m^\# = \alpha_{S-1-m}</math>, <math>\beta_m^\# = \beta_{S-m}</math>) but is NOT a builder transformation:<br/> NO_SOURCE_REVERSAL; <math>\beta &gt; 0</math> stays the wall in EVERY coordinate. (4)<br/> MIDPOINT_CONNECTION_EXACT: the dual FIK problem is an explicit<br/> <math>L</math>-gauge transform of the original, <math>Y_{N-1+l}^\# = J Y_{N-1-l+1} W</math> with<br/> <math>W = \text{antidiag}(1/L, L)</math>, built from THE NODE POLYNOMIAL ALONE (exact in<br/> rationals; real w9 at 9.1e–94, dps 100); the gauge is <math>h</math>-FREE, so the wall’s<br/> orientation sits EXCLUSIVELY in the <math>h</math>-chain it does not see:<br/> SIGNED_STOKES_WALL_EQUIVALENT; honest negatives typed; the micro<br/> falsifier FROZEN (predict 0/2/2/3/1 and 25/21/27 blind). Claim splitting (note<br/> DLVII): PRIME.FREEMOMENT.JFRACTION.01 [E] +<br/> PRIME.JACOBI.DUAL.REVERSAL.01 [E] separated from<br/> PRIME.FREEMOMENT.POSITIVEPREFIX.01 [O]. Half-filling is the<br/> LOCATION; its positive reachability stays the result to be proven. Mincut base 4<br/> / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v957_ccc_crossover_kinematics.py | <p>CCC.SEAM.KINEMATICS.01 (rounds DL–DLV, 2026-08-24): THE CCC CROSSOVER KINEMATICS DICTIONARY (28 + 29 + 18 + 21 + 14 + 7 gates, six probes embedded byte-exact). (1) Penrose’s reciprocal gauge <math>\Omega \tau(\Omega) = -1</math> is the element <math>\tau = \rho^2 \sigma</math> of the PROVEN <math>\mu_4</math> stabiliser <math>D_4</math> (<math>\tau</math> reverses the clock = the <math>\mathbb{Z}_2</math> sheet parity; the crossover circle carries all four marks). (2) The clock alone leaves exactly ONE conformal modulus on the 2D seam and the reciprocal <math>\mathbb{Z}_2</math> kills it (Liouville-exact; <math>so(3, 1)</math> commutants 2/0). (3) The Stinespring dilation of the v221 transport is explicit: <math>V^\dagger V = 1</math>, <math>\ T^n - P_\star\  = (2/3)^{6n}</math> EXACT, global overlap invariant – entropy reset = reduced channel, no information destroyed. (4) The kinematic 2D→3D→4D uplift: stereographic(Hopf) = <math>z_1/z_2</math> EXACT, crossover <math>X = S^3/\mathbb{Z}_4</math> = the lens boundary of the corpus <math>A_3</math> ALE (deck free; marks = four linked fibre circles in the two sheet orbits, Gauss linking 1; <math>so(4, 1)</math> deck-commutant = 4 Killing / 0 boosts – Yamabe rigidity with NO extra condition, global via the Obata family); the 4D bridge is the cone with the inversion as conformal aeon swap; the Lorentz shadow <math>(1/\cos^2 T)(-dT^2 + g_{S^3})</math> is Einstein <math>R_{ab} = 3g_{ab}</math> with a spacelike <math>S^3</math> crossover and <math>\hat{\Omega}\tilde{\Omega} = -1</math>. (5) KMS closure: detailed balance exact, <math>\text{spec}(-\ln T) = \{0, 6\ln(3/2), 6\ln 3\}</math>, <math>T = e^{-H_{\text{mod}}}</math>, <math>\beta_{\text{angle}} = 2\pi = 1/(4c_3) = \text{Gibbons–Hawking on the fibre}</math>. (6) Holonomy: pole strings carry the full <math>\mu_4</math>, every seam string carries EXACTLY the 2-torsion <math>-1</math> = the v522 GSO/fermion-parity wrap (typed [C]). (7) Selector necessity: <math>M(\delta)</math> admissible reflections close to <math>D_4</math> iff <math>\delta = \pi/2</math>; Majorana operator level: clock axes <math>\Theta</math>-real ONLY at <math>k = 4</math> – the UNIQUENESS half of the v534 selector DERIVED (survival + sign cited). (8) The frozen Gate-6 kernel: causal top-hat disc <math>\theta_{\text{max}} \approx 1.16^\circ</math>, rates <math>\ln(729/64)/\ln 729</math>, <math>\kappa</math> derived from v526 (declared band, contrast <math>\leq 3.3 \times 10^{-4}</math>) – feeds the preregistered search <code>ccc-crossover-disc</code> (no data touched). The cyclic/CCC reading stays [C]; WOIT.OS.TWISTOR.01 stays [O].</p> |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v958_bordered_tau_readout_dictionary.py | <p>PRIME.PORT.RHP.BORDERED.READOUT.01 (rounds 244/245/248/251/252/253, adjudicated at the finite-identity level) + PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 (registered [O], post-round-255 state): THE BORDERED-RHP / TAU-READOUT DICTIONARY (23/23 + 29/29 + 24/24 + 20/20 + 22/22 + 26/26, six probes embedded byte-exact, plus the module-own exact PSD base theorem). (1) The exact dictionary of the bordered end-object (bordered PSD = wall + budget): the Uvarov border column carries <math>F_n</math> as its <math>z^{-1}</math> coefficient WITHOUT orthogonality cancellation; THE BUDGET IS AN RHP FUNCTIONAL OF THE TERMINAL <math>Y_n</math> DATA ALONE (CD-kernel form <math>S_{n-1} = \iint K_n d\sigma d\sigma</math>, ward <math>3.7 \times 10^{-15}</math> on <math>42 + 3</math> worlds); the Uvarov tau step and the <math>T</math>-sourced bordered LAX1 flow mp-exact through all 184 w9 degrees (<math>7.0 \times 10^{-159}</math>). (2) The border is a SCHLESINGER RANK-1 INSERTION – <math>3 \times 3</math> but unipotent/triangular (<math>\det Y_3 = -h_{n-2}F_{n-1}</math> newly derived), IKS generator rank EXACTLY 2, a canonical <math>2 \times 2</math> RHP <math>\times</math> a ONE-DIMENSIONAL budget cocycle carrying the full pivot chain – the irreducible-<math>3 \times 3</math> steepest-descent campaign is dropped without replacement. (3) The centering congruence (<math>c = F_0/h_0</math> eliminates the geometric rank-1 head algebraically; <math>B^\circ = B - \rho_0</math>, exact in rationals), the <math>\sigma^\circ</math> dictionary, the dual-norm identity <math>Q_m = t^T H^{-1} t</math> and the tail-kernel compression <math>T = \iint [K_N - K_8] d\sigma^\circ d\sigma^\circ</math> exact at dps 160 through the full depth <math>N = 184</math> (rel <math>3.3 \times 10^{-160}</math>). (4) The three exact error formulas (<math>\delta h_n = (R_1)_{12}</math>; the central rest identity; the error-kernel sandwich at <math>1.1 \times 10^{-81}</math>). (5) The contour formula <math>R_1 = \frac{1}{2\pi i} \oint (R - I) dz</math> (<math>3.2 \times 10^{-14}</math>) with the gauge thesis refuted on all three prongs (BASE_BLOCKED_BY_DRIFTING_LAYER). (6) The exact augmented tau quotient <math>D = B^\circ - Q_7 - T = \tau^{\text{aug}}/\tau</math> at full depth <math>N = 184/168</math> (three routes at <math>10^{-11}</math>, mp <math>10^{-16}</math>) with the pole classification (rank-1 border poles removable by residue closure; the <math>D</math>-layer ESSENTIALLY SINGULAR, <math>\exp(c/(z-s))</math> type). The [O] campaign contract carries the measured post-r250/254/255 state (experiments-side, typed MEASUREMENTS): OUTER_MODEL_FAILS, OFFDIAG_EXTENSIVE with the compactness INVERSION (SCRAMBLE 2 / EPSTEIN 6 / MAIN 28–30 eigendirections – compactness is the signature of destruction), ORIENTATION_LOWDIM(1) (the base flip is a one-bit 0.2-decade selection event). Mincut base 4 / refined 5 unchanged; NO RH claim.</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v959_coupleddtau_<br>terminal_<br>dictionary.py | <p>PRIME.PORT.RHP.COUPLEDTAU.TERMINAL.01 (rounds 256–259, adjudicated at the finite-identity level): THE COUPLED-TAU / TERMINAL-BUDGET DICTIONARY OF THE WALL</p> <p>(21/21 + 26/26 + 27/27 + 19/19, four probes embedded byte-exact, plus the module-own exact section S0). (1) THE POSITIVE-PREFIX FIREWALL + THE 2x2 CAUSAL CENSUS (r256): MAIN <math>p_{\max} = 184 = N</math> with ZERO negative modes (the ONLY positive-prefix world), controls at 21/25/27 all INDEFINITE_CONTINUATION; every prefix kernel numerically PSD (worst 5.7e–16) – the reviewer PSD core theorem holds machine-level; on the causally clean shared prefix <math>n^* = 21</math> the base rotation is NULL (<math>D_b^* = -1.5e-6</math>) and the border carries the prefix gap (<math>D_{bo} = -0.1759</math>); TWO contamination downgrades on record (the r254 SCRAMBLE compactness is base contamination, <math>f_{\text{neg}} = 0.9311</math>; the EPSTEIN fiber anomaly lives entirely in the indefinite continuation degrees <math>k \geq 25</math>). (2) THE COUPLED-TAU RECURSION (r257): <math>\tau_{n+1} = a_n \tau_n</math>, <math>\tau_{n+1}^{\text{aug}} = a_n \tau_n^{\text{aug}} + b_n \tau_n</math> with <math>a_n = c_n^2 h_n</math>, <math>b_n = -(c_n F_n)^2</math>, equivalently <math>D_{n+1} = D_n - F_n^2/h_n</math> (anchor 5/7 – <math>\rho_{N-1}</math> rel 4.8e–11, mp dps 220 THROUGH the SCRAMBLE flip); <math>a_n</math> reads the <math>h</math>-chain (its sign IS the world pivot sign), <math>b_n</math> is a negative square – THE BASE ALONE CARRIES THE SIGN, THE BORDER ONLY NONNEGATIVE MAGNITUDE; the cocycle breaks exactly at the control flips 21/25/27; the BILINEAR TAU FORM <math>\tau_n^{\text{aug}} \tau_{n+1} - \tau_{n+1}^{\text{aug}} \tau_n = (c_n F_n)^2 \tau_n^2</math> derived algebraically and gated in-module (S0.2); the frozen micro-falsifier PASSED BLIND for the first time, at the coefficient-field level (control flips 3/3 AND the forced tail 0/2/2/3/1 on all five windows; the parametrix half stays OPEN); NOPOLE falls cofinally toward <math>s = 1</math> (slope –2.87) but BOTH augmented border variants CROSS the wall. (3) THE BUDGET ANATOMY (r258): the Schur telescope <math>D_N = B - S_{N-1}</math> exact (12 windows incl. <math>N = 878</math>, mp dps 220 at 2.4e–16); <math>B = S_{N-2} + 5/7</math> exact BY CONSTRUCTION with the 5/7 typed FLOOR_IMPORTED (inverted r241 floor of the MEASURED <math>\min h_{N-1}/F_{N-1}^2 = 1.4278</math>; derivation OPEN, the named terminal task); the <math>q</math> census 42/42 (<math>q_N &lt; 1</math> everywhere, min margin 0.0139, trend FLAT); Spearman(<math>2 \log F_{N-1}, \log h_{N-1}</math>) = +1.000 – the driver is the QUOTIENT itself (a measured structural fact, NOT a law); all three sealed mass majorants break (0/42, median <math>\times 490</math>, at 0.97N) – THE MASS-BOUND CLASS IS DEAD; the corner ray identifies <math>t^* \equiv q_N</math> (one normal form, two statistically unconnected data instances). (4) THE PARAMETRIX PASS (r259, three honest negatives): LEVEL CROSSING REFUTED (branches cross 5–16 degrees too early; MAIN w9 crosses 45 times with zero true flips), the ONE-SWAP NO-GO (fully overlapping gap distributions – r255’s ORIENTATION_LOWDIM(1) REVISED on record), all three sealed candidates fail blind (0/3 + 0/5); the named obstacle: A RESUMMATION GAP, NOT A SELECTION GAP – the sign is carried by coherent cancellation over the configuration ensemble. THE ENDGAME STATE: the last fiber edge = a positive tau cross-ratio (<math>h_{N-1}/F_{N-1}^2 &gt; 7/5</math>; PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01 [O]); the last base edge = a globally oriented prefix resummation (PRIME.PORT.FULLSOURCE.PREFIX_RESUMMATION.01 [O]). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v960_terminal_<br>surface_closure.py | <p>PRIME.PORT.COUPLEDTAU.SURFACE_CLOSURE.01 (rounds 260/262/263/268–271, adjudicated at the finite-identity level, plus the typed measurement rounds 272/273/275): THE TERMINAL-SURFACE CLOSURE (31/31 + 26/26 + 26/26 + 28/28 + 29/29 + 29/29 + 28/28 + 28/28 + 24/24 + 24/24, ten probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0). THE MILESTONE: the 42-rung surface census <math>q_N &lt; 1</math> is CERTIFIED ON ALL 42 RUNGS as a finite fact – 41 mechanism-certified target-blind (35 cheap rungs via the r260/r263 two-branch theorem: <math>U</math> = the sealed C1b triangle form, <math>M = \sqrt{5/7}</math>, exception set exactly {kz15, kz20, kz22, kz36, kz38, kz39, kz52}, both mains close WITHOUT cancellation; kz20/22/36/38/39/52 via the sealed phase bounds: the r268 edge truncation (kz22 = the FIRST source-pure exception certification ever) + the r269 block alternation (+0.26 to +0.41 dec on all 42, NO detector fires)) + kz15 EXACT-FINITE via the r270 interval certificate (outward-rounded mpmath intervals, dps 640, 202 steps, end width <math>1.5 \times 10^{-92}</math>, <math>\sup  Z  &lt; \inf \sqrt{5/7}</math> STRICT, margin +0.02680, the dps-halving ward destroys the certificate as demanded; honestly typed EncLOK class, NO mechanism). (2) The terminal cross-ratio dictionary exact (r260): <math>q_N = F^2/(hD)</math>, the bilinear identity mp to <math>3.7 \times 10^{-58}</math> THROUGH the SCRAMBLE flip, the corner <math>B</math> cancels EXACTLY out of the bilinear form, the driven recursion world-blind at <math>7.2 \times 10^{-14}</math>. (3) THE UNIVERSAL PAIR THEOREM (r271): H1–H4 structural/source-side proved (Lean rh/lean/RH/PairBound.lean without sorry), H5 = the margin <math> Z_{\text{local}}  + \varepsilon &lt; \sqrt{5/7}</math> is the SINGLE window-dependent hypothesis – the formal entry gate of the cofinal step. (4) Honest negatives on record: the eventual-triangle no-go (kz52 itself an exception), three identity routes dead with break loci (K1 target restatement + support geometry, K2 squares on the drain side, K3 paircorr miniature), the w13 Gram provenance RETRACTED, the KYP memory no-go FROM STRUCTURE (o1 fiber forcing + o2 descent window, exact-rational on 46/46 worlds; the admissible Riccati memory is TARGET-INVERSE, sp +1.00). (5) Typed measurements: r272 BOUND_COARSENESS (the truth margin RISES with <math>N</math>, +0.264 <math>\rightarrow</math> +0.414; the bound loses cancellation, not the world; flip condition <math>\delta' &gt; 0.21</math> of available 0.45, address c3, TRUTH_ALLOWS), r273 PERTURBATION_INSENSITIVE + FIREWALL_MAP (every surgery kills the free prefix positivity at degrees 33–101: THE WALL is the arithmetically special thing about MAIN). The surface is a census + certificate set, NOT a cofinal theorem: the window quantifier and the cofinal front (lemma L2, address c3) stay OPEN on PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01 [O]. Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v961_midpoint_orientation_dictionary.py | <p>PRIME.PORT.RHP.MIDPOINT.ORIENTATION_DICTIONARY.01 (rounds 274/277, adjudicated at the finite-identity level, plus the typed measurement rounds 276/278): THE MIDPOINT-ORIENTATION DICTIONARY + THE METRIC FIREWALL (32/32 + 25/25 + 29/29 + 31/31, four probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0: the base Casoratian <math>p_n q_{n+1} - p_{n+1} q_n = h_n</math> with <math>c' = 1</math> symbolic in <math>z</math>; the bordered-Wronskian telescope <math>\det[[H_n, u], [u^T, B]] / \det H_n = B - \sum F_k^2 / h_k</math> exact; the Hellmann-Feynman position gradient <math>dh_n/d\varepsilon = w_j p_n(x_j)^2</math> in exact dual-number arithmetic; the Jacobi interlacing/reality direction with the flipped-beta must-fail). (1) THE WRONSKIAN DICTIONARY (r274, GO): base Casoratian <math>= h_n</math> with <math>c' = 1</math> (all degrees, 42 rungs, signs exact); the right solution TRIPLE-CONSTRUCTED with exact coincidence (residue route / downward recursion with exact Dirichlet boundary <math>q_S = 0</math>: THE NODE POLYNOMIAL IS THE RIGHT BOUNDARY CONDITION / dual chain + L-gauge); the <math>h</math>-free midpoint form IS the node polynomial, constant in <math>n</math> – the r231 sign-blindness AS AN IDENTITY; the augmented telescope <math>D_{n+1} = B - W_n^{\text{aug}} / W_n^{\text{base}}</math> exact against the independent r263 route; the <math>c</math>-typology honest: <math>c_n = 1/W_n^{\text{base}}</math> is source-pure and positive on the free prefix but carries the ENTIRE orientation (the dictionary relocates the question, it does not solve it); the w9 winding measurement <math>262 \neq 184</math> typed. (2) THE MASLOV CENSUS (r277, GO; STURM_CHAIN_VERIFIED honestly NOT awarded): rule R2 = the interlacing/reality structure of the JACOBI zeros passes the blind examiner protocol 42/42 + mains SAFE at full depth; the controls fire EXACTLY at flip+1 (26/22/28); R2 is NOT <math>h</math>-equivalent (79 vs 78) and the break is a ONE-WAY DETECTOR – the cofinally monotone object the proof plan needs; THE HONEST REFUTATION: the raw atom-Sturm census is NOT the right winding quantity (MAIN breaks <math>c_n \equiv n</math> at 56/48 at throughout-positive <math>h</math>: the classical separation theorem fails genuinely for the signed comb); the named consequence: the theorem needs the CONTRAPOSITION with an independent index obstruction (successor contract PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01 [O]; round 279 in flight, not consumed). (3) Typed measurements – THE METRIC FIREWALL (r276 + r278, registered as PRIME.PORT.WALL.METRIC_FIREWALL.01 [O]): FIREWALL_LAW graded (<math>D \sim \theta^b</math>, <math>b</math> 0.04–1.09), CONTINUUM (no jump), PROPERTY_RANKING: SUPPORT EXACTNESS is the most wall-critical property, the Euler family structure the mildest – the arithmetic firewall is more precisely a METRIC firewall; THE EXACT GRADIENTS (Hellmann-Feynman class, Identity fine type): <math>d \log h_n / du_j = \langle \dot{w}_j, \hat{\pi}_n^2 \rangle / h_n</math>; the <math>u</math>-profile PREDICTIVE (top-3 atoms = the small primes 2, 3, 5; sp <math>-0.82</math>); the stability law PERTURBATIVE-ONLY (<math>\theta^*</math> 25–170x below the smallest dose, a nonlinear cascade across tent kinks; no uniform lemma candidate) – the MAIN positivity itself stays the open center (H5 untouched). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v962_halffilling_pinning_theory.py | <p>PRIME.WALL.HALFFILLING_PINNING_THEORY.01 (rounds 279/280/281, adjudicated at the finite-identity level plus typed measurement legs): THE SMALL MATHEMATICAL THEORY OF THE HALF-FILLING WALL (32/32 + 29/29 + 28/28, three probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0 in pure Fractions: four named theorems + four named refutations). (T1) THE MOMENT COUNTING THEOREM (r281): the pivot <math>h_n</math> consumes <math>m_0..m_{2n}</math>, the <math>S</math> free moments are <math>m_0..m_{S-1}</math>, hence the FREE pivots are exactly <math>h_0..h_{N_w-1}</math> and <math>h_{N_w}</math> is the first forced one – HALF-FILLING IS THE END OF THE FREE MOMENT SPACE (arithmetic-exact for ALL <math>S = 2..2000</math>; freedom demonstration exact in rationals: <math>dm = e_{S-1}</math> moves <math>h_{N_w-1}</math> alone; wrong-<math>L</math> must-fail LOUD). (T2) THE CROSSING BUDGET THEOREM (r279): <math>\#(h_n &lt; 0, n &lt; S) = S_-</math> EXACTLY (Jacobi's minor-sign rule + Sylvester congruence), WORLD-BLIND – five frozen signed measures exact; real-side w9 104 / w13 98 / controls 141/94/6 / kz15 121 / kz52 551. (T3) THE TWO-SIDED PARITY THEOREM (r279): <math>\text{sign } Q_n(x_j) = e_j(-1)^{S-1-j} \text{sign}(h_n)</math> at every node and degree, union occupancy ODD in every agreement gap (never empty) and EVEN in every disagreement gap – <math>h</math>-blind; census bilanz <math>c_n + c_{S-1-n}^\# = (S-1) -  D  + 2\text{scD}(n)</math> exact at every degree + exhaustion over all 87376 sign-vector pairs <math>k = 2..8</math>. (T4) THE MAIN WINDOW REDUCTION: given T1 + T2 the ENTIRE open statement is <math>\min C \geq N_w</math>, and this is EXACTLY equivalent to <math>(\forall n &lt; N_w : h_n &gt; 0)</math> – the north star in its purest form: WHY IS THE SIGNED PRIME MOMENT FORM QUASI-DEFINITE UP TO THE MAXIMALLY FREE ORDER? THE FOUR REFUTATIONS (the negative information is itself load-bearing): NO_UNIVERSAL_O1_PINNING (r281 exact: the one-negative toy has offset <math>N_w - 2</math>, unbounded in <math>S</math> – any <math>O(1)</math> upper side must consume the comb structure; the only provable ceiling is the pigeonhole <math>\min C \leq S - S_-</math>, <math>C \leq 5</math> is a MEASUREMENT); NO_EXTREMALITY (r280 measured + exact witness: three mp-confirmed directions push the w9 crossing <math>184 \rightarrow 185</math> past half-filling; the 4-atom one-negative measure has <math>\min C = N_w + 1</math> exactly, chain 2999/1000, 5986/2999, 1962/2993, <math>-4/109</math>); NO_GENERIC_MASLOV_OBSTRUCTION (r279 exact: 3 rational counterexamples <math>\{3, 4\}/\{2\}</math> – the two-sided machinery provably carries NO arithmetic); NO_SIMPLE_OFFSET_LAW (r281 measured: six sealed source-pure candidates reach <math>\max  \text{sp}  = 0.273 \ll 0.75</math>, all WORLD_BLIND – the offset formula stays open). The lower side (free-window survival of MAIN) stays THE open center (Lean: free_window_positivity in rh/lean/RH/Window.lean, the fog-free central sorry; via T4 equivalent to the base half of the master theorem augmented_prefix_positive). In flight, NOT consumed: representation contest + full-source quasi-definiteness. Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v963_lstar_reduction_dictionary.py | <p>PRIME.LSTAR.REDUCTION_DICTIONARY.01 (rounds 282/283/284/285, adjudicated at the finite-identity level plus typed measurement legs): THE L* REDUCTION DICTIONARY – the free-window question in its CANONICAL FORM (30/30 + 26/26 + 30/30 + 33/33, four probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0: Hankel determinants, monic OP chains, frame congruence minors, Cauchy-Binet exhaustion, dual chains). (A) THE r283 A2 CHAIN (exact): the mu-frame congruence <math>\text{minor}_k(D_\mu - G) = D_k(\tilde{\mu})</math> at every <math>k \leq S_+</math> (truncation must-fail LOUD); the frame contraction <math>h &gt; 0</math> through the window <math>\iff \lambda_{\max}(E_m) &lt; 1</math> (both truth values); the crossing EXACTLY at <math>\min C + 1</math> (CAPACITY_CEILING where <math>\min C = S_+</math>); the pigeonhole ceiling <math>\min C \leq S_+</math> re-proved in the frame (null-polynomial witness); capacity-as-counting REFUTED by the rank-one pair (same rank, same support, opposite fate – the metric values <math>1.25 \times 10^{-13}</math> vs <math>1.25 \times 10^2</math> decide) <math>\Rightarrow</math> L* IS THE CANONICAL REDUCTION: <math>\int p^2 d\nu &lt; \int p^2 d\mu</math> for every real polynomial <math>p</math> with <math>\deg p &lt; N_w</math> (equivalently <math>\lambda_{\max}(E_{N_w}) &lt; 1</math>) – registered as the standalone contract PRIME.LSTAR.SUBORDINATION.01 [O] with the external problem document rh/problem/lstar_problem.tex. (B) THE r285 DECOMPOSITION BOOKKEEPING (exact): the Christoffel sandwich <math>\max_k v_k K_n(y_k) \leq \lambda_{\max} \leq \text{trace}</math> with its rational consequences; the multiplicative decomposition <math>\lambda_{\max} = \text{maxdiag} \times (1 + \text{assist})</math> with the budget equivalence sign-exact at every degree. (C) THE r282 FOUR-LANGUAGE ELIMINATION as named negative gates (CONTEST_ALL_DEAD; the common deep reason: every classical positivity language forces positivity EXACTLY for the positive measure class – MAIN is signed): SOS exists IFF the negative register is empty (the fake SOS misses <math>h_2</math> by exactly <math>N_2 = 48360721965/70120631072</math>); the Kasteleyn orientation class is EXACTLY <math>\{\pm \text{sign } w\}</math> by full <math>2^S</math> exhaustion and is value-preserving IFF <math>S_- = 0</math> (defect = <math>2 \times \text{negmass}</math> exact); Hamiltonian-PSD = <math>h &gt; 0</math> bookkeeping; the dual-pair sync holds PER THEOREM (<math>h_n h_{S-1-n}^\# = 1</math> exact, wrong dual loud). (D) TYPED MEASUREMENTS (record pins): the r284 one-atom-vs-collective dichotomy (w9 <math>n_{\text{DIAG}} = 187</math> vs crossing 185; controls die collectively; extremal anatomy PR 1.89, two shallow-<math>u</math> ARCH atoms, SMALLP_DEPLETED, ONE_BAND; SHIELDING_BLIND; Nyquist refuted at the discriminant), the r285 legs ((D) SEPARATES at window scale, sub-classical edge growth <math>p = 0.38</math> – the (D) proof route must be DISCRETE, MAIN's wall assist the extreme ensemble LOW_OUTLIER pct 0.00, and THE FIRST TWO POSITIVELY MAIN-SEPARATING DETECTORS of the program: assist-at-crossing 0.0195 vs dead 1.69..2.99, random-sign <math>z = -3.15</math> destructive vs dead <math>+4.95.. + 12.44</math> constructive), and the honest DCXX margin-decay finding (42/42 margins positive, family minimum <math>1.4175 \times 10^{-7}</math> at <math>z = 233</math> vs flagship <math>1.68 \times 10^{-4} - \sim 3</math> orders of decay; the r286 margin-scaling round is IN FLIGHT, NOT consumed). L* itself stays OPEN (Lean: lstar_subordination in rh/lean/RH/Window.lean, with the proved direction <math>L^* \Rightarrow</math> free-window positivity). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v964_lstar_coherence_census.py | <p>PRIME.LSTAR.COHERENCE_CENSUS.01 (rounds 286/287/288/289, adjudicated at the finite-identity level plus typed measurement legs): THE L* COHERENCE CENSUS – the margin scaling resolved, the exact van der Corput theorem for L2, the destructive-coherence carrier map and the METRIC_ONLY adjudication (29/29 + 25/25 + 31/31 + 42/42, four probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0: the Fejér window-sum identity, the vdC inequality with its Cauchy-Schwarz proof route, the Abel/Erdős–Turán bound, the mutant must-fails, the 2-power resonance count <math>\binom{8}{2} = 28</math>). (A) THE r287 vdC THEOREM: the exact, CONSTANT-FREE, ARITHMETIC-FREE van der Corput inequality <math> \sum_j P_j ^2 \leq \frac{m+H-1}{H} \sum_{ h &lt;H} (1 - \frac{ h }{H}) A(h)</math> at the frozen window <math>H = \lceil \sqrt{m} \rceil</math> delivers <math>\delta' = +0.309 &gt; 0.21</math> WORLD-BLIND (admissible: L2 is the generic half) and certifies 6/7 exceptions +38/42 rungs – the L2 lemma has its first NAMED classical theorem candidate (PRIME.PORT.L2.VDC_LEMMA.01 [O]: what remains is the chain origin of the P-variance scaling; kz15 exact-finite per r270); F1 DISCREPANCY dies LOUD (<math>\delta' = -0.21</math>, 0/7, detector FIRE +1.44 dec); the autocorrelation surprise: 39/44 worlds REINFORCE at root scale – the c3 cancellation is already absorbed into the <math>P_j</math> magnitudes, the frames win through the <math>\sqrt{m}</math> count economy. (B) THE r286 MARGIN-SCALING CLARIFICATION (the DCXX warning RESOLVED): 15 new anchors <math>N_w = 942.1218</math> ALL mp-sign-safe positive (min <math>+1.806 \times 10^{-8}</math>, every anchor mp dps-30/45 staggered) – NO counterexample; the <math>O(1)</math> census offset holds beyond the cap (off distribution <math>\{0:1, 1:10, 2:2, 3:1, 4:1\}</math>, max +4); margin law <math>\alpha \approx 3.05</math> vs <math>S</math> FLATTENING (curvature <math>-1.60</math>), driver = cancellation sharpening <math>c_w \rightarrow 1</math>; HARMLESS quantified (the margin falls SLOWER than the local loading speed); <math>q_N</math> reconciled (r258 FLAT reproduced – two different objects); EXTRAP_CALIBRATED 15/15. (C) THE r288 CARRIER MAP: the destructive coherence is carried by the ANTIPHASE NEXT-NEAREST (3-4 fold) ARCH-ARCH pairs (<math>X_v = -0.0517</math>, <math>z_v = -3.149</math>, <math>C_{\text{off}} = -0.1046</math>; band 1-2 87 percent positive/in phase) with TOTAL control reversal (all four controls constructive at their own crossing); the chain zeros CO-MOVE with the atoms (turn rate 0.24 == static prediction 0.23) yet <math>z_v</math> flips already at dose 0.005: FINEST ALIGNMENTS far below the median phase resolution; honest negatives SAMPLING_BLIND + SOURCE_SEPARATOR_NOT_FOUND + DIFFERENT_OBJECTS. (D) THE r289 METRIC_ONLY ADJUDICATION (THE central refutation of the diophantine reading): the RATIONAL TWIN – every tent center <math>\log n</math> replaced by a rational <math>p/q \times \Delta</math> (denominators <math>\leq 56801</math>, position cost <math>\leq 2.1 \times 10^{-9}</math>), destroying EVERY exact log-linear-form relation – KEEPS THE FULL SIGNATURE IDENTICAL (min <math>C = 184</math>, crossing 185, <math>z_v = -3.149</math>, mp-confirmed at the <math>1.7 \times 10^{-4}</math> margin): the effect of the diophantine structure is NULL; the shuffle twins die PURELY METRICALLY (<math>\theta_{\text{eff}} = 0.12</math>, <math>\sim 40</math>-<math>120\times</math> beyond the measured coherence threshold <math>10^{-3}..3 \times 10^{-3}</math> of the local gap); the ONLY sub-gap entry point: the TENT-SPLIT FRACTIONS (completeness identity <math>4.2 \times 10^{-14}</math>); the Baker need documented <math>\sim 7900\times</math> too weak AND unnecessary – THE FIREWALL READS THE FRACTION PROFILE METRICALLY, NOT ITS ARITHMETIC: the why-MAIN search continues in PROFILE space (the precise open front: which functional of the fraction profile forces the destructive coherence; r290 in flight, NOT consumed). L* itself stays OPEN (PRIME.LSTAR.SUBORDINATION.01 [O]). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v965_lstar_curvature_arc.py | <p>PRIME.LSTAR.CURVATURE_ARC.01 (rounds 290/291/292/293/294/295, adjudicated at the finite-identity level plus typed measurement legs): THE L* CURVATURE ARC – the measured geometry of the working set in profile space, frozen together with the honest negatives of the closed-functional search (32/32 + 30/30 + 36/36 + 43/43 + 42/42 + 44/44, six probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0: the polarization identity <math>H(u, v) = [d^2(u+v) - d^2(u-v)]/4</math> with its mutant must-fail, the budget telescope with the dropped-atom must-fail, the <math>\alpha = 1</math> simple-zero doubling law with the double-zero mutant, the sealed r294/r295 decision bars as exact logic with tipping must-fails). (A) BASIN GEOMETRY (r290): the working set is a SOFT-SHOULDERED ANISOTROPIC TUBE (kill fraction 0.38/0.62/1.00 at <math>\theta_{eq}</math> 5e-4/1e-3/2e-3); world-directed axes kill 5–50x EARLIER (onsets 4e-5..2.3e-4 vs the jitter threshold 1e-3..3e-3); the SMOOTH axis is a privileged killer yet gradient-ORTHOGONAL (<math>\cos = -3 \times 10^{-5}</math>); the r280 ridge is REAL (min <math>C = 185</math> at factors 1..8). (B) RIDGE ANATOMY (r291): the lift is a FIRST-ORDER BUDGET phenomenon (ONE threshold in (1.280, 1.291] separates lift from no-lift over all 18 matched-dose cases; <math>k_{min} = 9</math>, top atoms 2, 3, 5, 13, 11, 4, 29, 7, 89; exactly ONE overdrive retraction TOP6@8); NO fixed point (plateau 185, never 186); LIFT_MAIN_SPECIFIC (EPSTEIN first-order flat); SMOOTH collective-quadratic (ratio 23.7). (C) CURVATURE SPECTROSCOPY (r292): the wall Hesse form is negative on ALL 29 directions and RANK-1 dominated (92.5%, <math>\lambda_{top} = -0.418</math>), top axis a pure DENS combination – NOT SMOOTH (<math> \cos  = 0.07</math>), NOT the ridge (28/29); <math>m^*</math> and the retraction are INVISIBLE to jets up to third order (<math>q_R</math> median <math>-0.911 =</math> second-order ASSIST; <math>B_2 = 27.6/B_3 = 150.9</math> over-predict); EPSTEIN second-order structureless (<math>5.4 \times 10^{-15}</math>). (D) METRIC RECONCILIATION (r293): three metrics sealed BEFORE evaluation (distortions 11/59/124, largest sector DENS); F10 beats the baseline OF ITS OWN METRIC for the first time (home margin <math>+0.024 = 0.884 - 0.860</math> vs the mixed r292 margin <math>-0.023 = 0.884 - 0.907</math>: MIX_IS_CAUSE, both re-derived in the gates) and carries partial information (<math>+0.423</math> test / <math>+0.826</math> train); ALL 8 flips are SIMPLE <math>h_{184}</math> ZEROS (<math>\alpha = 1.000 \pm 0.003</math>, 225 gated evaluations) but MSTAR_NO_LAW (<math>m_{dir}^*</math> scatters 1.1295..1.3931, spreads 3.0/3.0/3.8/10.2 vs bar 1.25); the TOP6 retraction IS a SECOND simple zero at <math>f_{ret} = 7.107</math>. (E) STABILITY (r294, honest negative): the  sp  win replicates 5/5 (margins <math>+0.003.. + 0.115</math>, re-derived) but the partial channel does NOT (median <math>0.299 &lt; 0.3</math>, 2/5) <math>\Rightarrow</math> F10_FRAGILE, promotion NO; TRAIN_ROBUST (<math>\sigma = 0.0101</math>) + RANK2_CARRIES (DENS share 0.989) + STEP_STABLE + window knife edge (w11 <math>+0.002</math> / w7 <math>-0.009</math>, DENS mechanism constant) + L2_NOT_DENS. (F) HARDENING (r295, honest negative): 14/20 wins (margin median <math>+0.028</math>, min <math>-0.079</math>) <math>\Rightarrow</math> F10_SP_MAJORITY – the sealed three-clause HARDENED bar fails on TWO clauses; the partial-free statement is a DOCUMENTED REGULARITY (19/25), NOT a theorem; PARTIAL20 median <math>+0.279</math>, IQR [<math>+0.182, +0.401</math>] (the r293 <math>+0.423 =</math> top-of-distribution); PARTIAL_FAMILY_MAP: NO family STRONG (PATH <math>+0.156</math> / WORLD <math>+0.245</math> / FRAC <math>+0.067</math> NULL / DENS <math>+0.104</math>); R293_LUCK (gain clause passes, level clause fails by 0.004); L2_VIA_CONSERVATION near-tautological (<math>\eta_0</math> share <math>2.9 \times 10^{-30}</math>). THE FROZEN CONSEQUENCE: after FIVE sealed class families NO closed predictive profile functional exists over the honest bar – F10 stays UNPROMOTED (the r294/r295 rejections ARE the frozen content); the question is registered as PRIME.LSTAR.CLOSED_FUNCTIONAL.01 [O]. L* itself stays OPEN (PRIME.LSTAR.SUBORDINATION.01 [O]). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v966_l2_reduction_chain.py | <p>PRIME.L2.REDUCTION_CHAIN.01 (rounds 296/297/298/299/300, adjudicated at the finite-identity level plus typed measurement legs): THE L2 REDUCTION CHAIN – the r296 DENS-fork closure and the four-round chain reducing the <math>\delta' &gt; 0.21</math> target of the generic half to ONE inequality (39/39 + 27/27 + 28/28 + 32/32 + 31/31, five probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0: the <math>\sigma^*</math>-composition arithmetic (<math>\delta'(\sigma^*(\text{pad})) = \text{pad}</math> exactly, the pad-dropped mutant composes to 0), the decomposition identity <math>S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)</math> on the exact positional Fejér block form (toy 3 = 1 + 2; mutants break by exactly 4 and 2/3), the participation identity <math>D \times n_{\text{eff}} = L_1^2</math> with the machine-checked DIAG<math>\leftrightarrow</math>NEFF equivalence (margin 0.055 exact in both coordinates; signed-sum mutant 8/3), the kernel envelope <math>F_H(s) \leq \min(H, 1/(Hs))</math> with equality witnesses on both branches, and the five sealed verdict bars as exact decision logic with tipping mutants). (A) THE r296 DENS FORK (honest negative, lane closed): DENS_WORLD_BLIND – the coupling number <math>\cos(e_{\text{top}}, \nabla \lambda_{\text{max}}) = +0.394</math> sits BELOW the sealed 0.40 bar (noise max 0.331; miss 0.006, honest), the <math>\lambda_3</math> control couples HARDER (−0.574: a band property, not top-specific), every arithmetic candidate <math>\leq 0.38</math>, the moment subspace (0.825) construction-adjacent (holds on both dead controls 0.571/0.603) – the DENS lane is CLOSED, resources route to L2 (the pre-adjudicated reviewer fork). (B) THE r297 TARGET INEQUALITY: <math>\delta'(F2) &gt; 0.21</math> is EQUIVALENT to <math>\sigma \leq \sigma^* = 2(\text{sl}_{c2} - 0.21) - \text{sl}_{\text{pref}} = -0.516</math>; measured <math>\sigma = -0.714</math> – the truth clears the target with 0.198 sigma margin: what is missing is not room, it is provenance; all three provenance routes VALID 47/47 but SLOPE-FAIL <math>\Rightarrow</math> CHAIN_PROVENANCE_PARTIAL(B3, slope miss 0.307 EXACT); the surprise: the chain is EXACTLY ORTHOGONAL w.r.t. the WINDOW measure (the Fractions-proved Parseval sum rule attaches verbatim). (C) THE r298 EXACT TRANSFER: the split above is EXACT on the real chain (<math>8.8 \times 10^{-16}</math>, 47 worlds), sign-preserving; TRANSFER_DOMINANT: the window main term is EMPTY (med −3.94 dec), the cross term negligible <math>\Rightarrow S_F \approx B(P\Delta, P\Delta)</math> – THE VDC INPUT IS THE FEJÉR ENERGY OF THE DIFFERENCE MEASURE ITSELF (share 0.99, <math>T &gt; 0</math> on 42/42). (D) THE r299 DECAY SPLIT: the exact spectral representation is LOWPASS (0.93); FULL-SUPPORT overlap 42/42 – <math>\Delta</math> IS a pure c-value difference measure (cconv 0.86, +0.045: aggregate cancellation, no pointwise convergence); ET/Abel FAILS LOUD (composed +1.948 EXACT); the live edge fires: <math>\text{sl}_D = -0.571 \leq \sigma^*</math> (margin 0.055) with FALLING ratio <math>B/D</math> (−0.168) <math>\Rightarrow</math> DECAY_SPLIT; the O-sign class separates MAIN from both dead controls (the first world-separating class). (E) THE r300 DIAGONAL ANATOMY: <math>\text{sl}_D = 2 \text{sl}_{L1} - \text{sl}_{\text{neff}} = -0.571</math> EXACT – the entire diagonal decay is participation growth (<math>n_{\text{eff}} \sim N</math>, sp +0.96); BOTH magnitude factorizations fail as r297 predicted (chain-norm −0.384 AND the <math> dw </math> census BREAKS; <math>\max \times L_1</math> −0.346 – the fill decay −0.225 is invisible to the class); DIAG_SPLIT(B3/NEFF_TARGET) + RATIO_BOUNDED_STRUCTURAL (<math>R_{\text{env}}</math> med 1.61 FALLS at −0.122 – the ratio half settled structurally); the FILL class separates MAIN from both dead controls (the second world-separating class). THE CHAIN STATE: [NEFF_TARGET: slope(<math>n_{\text{eff}}</math>) <math>\geq +0.908</math>, measured +0.963, margin 0.055 – THE ONE OPEN INEQUALITY, registered PRIME.L2.NEFF_TARGET.01 [O]; the D_rank bridge sp −0.81 correlational; round 301 in flight, NOT consumed] <math>\Rightarrow \text{sl}_D \leq \sigma^* \Rightarrow \sigma \leq \sigma^* \Rightarrow</math> the r297 target <math>\Rightarrow</math> the v964-S0 vdC theorem <math>\Rightarrow \delta' &gt; 0.21</math> on the generic half (exceptions: 6 via r287 F2, kz15 exact-finite via r270). L* itself stays OPEN (PRIME.LSTAR.SUBORDINATION.01 [O]). Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v967_l2_cascade_closure.py | <p>PRIME.L2.CASCADE_CLOSURE.01 (rounds 301/302/303/304, adjudicated at the finite-identity level plus typed measurement legs): THE CASCADE CLOSURE AND THE DOCUMENTED LANE STOP – the wave-9 rest targets executed to their end, the regress audit that RETYPED the reduction cascade, and the short-range-law round whose sealed stop case fired (32/32 + 30/30 + 26/26 + 37/37, four probes embedded byte-exact in their sealed smoke stage, plus the module-own exact section S0 in pure Fractions: the margin-invariance algebra – all four target margins = 3/50 EXACT on the rational slope set, the halved-need mutant breaking by exactly <math>\text{sl}_{L1} = 1/5</math>, the <math>\sigma</math> decomposition <math>0.198 = 0.055 + 0.143</math> exact, the 1/2-conjecture refuted; the coherence identity <math>(1 + CV^2) \text{surv}^2 n_{\text{eff}}^{\text{atom}} = n_{\text{act}} \chi</math> on two rational toys (mutant breaks 5/7 and 48/35 exact); the <math>\chi</math>-lag decomposition <math>\chi = 1 + 2 \sum_k T_k/Q = D/Q</math> (5/7 and 5/19 exact); the zero-sum tautology as a gate (<math>NC(1) = 0.556</math> exact, <math>NC(16) = 0.712 &lt; 1</math>); the period-4 comb certificate with NO <math>k_0</math>; the <math>-/-/+ /+</math> sign-pattern certificate (kz18/kz23 Fractions pins); the four sealed verdict bars as exact decision logic with tipping mutants; the lane-stop composition gate). (A) r301: <math>n_{\text{eff}} = n_{\text{act}}/(1 + CV^2)</math> EXACT with the PERFECT count link <math>n_{\text{act}} = m</math> on 42/42; the Rényi family at ONE exponent (echt anti-concentration); jackknife-stable; NEFF_SPLIT(B1 B3), rest relocated onto UNIF_TARGET; DERIVED missed only the CV-flat clause (<math>+0.039 &gt; 0</math>). (B) r302 (the first DERIVED of the lane): the profile is PROFILE_STATIONARY (KS <math>0.043 \ll 0.125</math>) with an exact <math>1/N</math> transient onto 1.973; the coherence identity EXACT, <math>\chi = 0.63</math> destructive and falling <math>\Rightarrow</math> UNIF_DERIVED(B1 + A2), all hypotheses typed MEASURED; rest onto ATOM_TARGET (<math>+0.942</math> vs <math>+0.888</math>). (C) r303 REGRESS AUDIT (the retyping): the three 0.055 margins are ONE ALGEBRAIC NUMBER, <math>+0.0547</math> (invariance <math>\leq 9 \times 10^{-16}</math>); the core is <math>S = \sigma^* - \text{sl}_D \Rightarrow</math> REGRESS_CONFIRMED: the cascade r297→r302 is RETYPED as an exact reduction DICTIONARY (coordinate finding, not six proof steps); the HARD RULE binds: a round counts only with NEW information. The first causal coordinate: the <math>\rho_1</math> ladder on 1008 sealed builds is MONOTONE (<math>\chi</math> <math>0.630 \rightarrow 1.342</math>; margin <math>+0.055 \rightarrow -0.044</math>: THE FLIP KILLS THE INEQUALITY) but misses the <math>\chi</math> level by 0.134 <math>\Rightarrow</math> MIXING_INSUFFICIENT; <math>n_{\text{eff}}^{\text{atom}}</math> a pure MARGINAL functional (<math>10^{-15}</math>); RHO_SIGN honest 41/42. (D) r304 SHORT-RANGE LAW (the reviewer stop case): the global lag profile is a STABLE, WORLD-SPECIFIC PERIOD-4 COMB, no <math>k_0 \leq 8 \Rightarrow</math> LONGRANGE_STRUCTURE + LAW_LONGRANGE; the reviewer condition SPLITS (<math>NC(16) = 0.712 &lt; 1</math> HOLDS, summability FAILS at 1.563); the <math>\chi</math>-relevant structure IS short-range (the r303 miss at <math>k \leq 3</math>, lag-2-dominated) – the comb is invisible to <math>\chi</math>; lag-8 matching hits the LEVEL (0.022) but breaks the SLOPES (0.028/0.027) – THE MECHANISM IS TWO-SCALE; the graduated ladder is monotone in <math>\chi</math>, NOT in the margin. CONSEQUENCE: THE GLOBAL-PROFILE MIXING ROUTE OF THE L2 LANE IS DOCUMENTED CLOSED – L2 generic <math>\Leftrightarrow</math> anti-concentration of an explicit block field with long-range period-4 structure; return with new tools. What stands: the exact identity dictionary, the two-scale split, <math>NC &lt; 1</math>, the sign pattern, the marginal invariance. Nothing is proved cofinally, for or against; <math>L^*</math> stays OPEN. Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v968_architecture_<br>adjudication.py | <p>PRIME.LSTAR.ARCHITECTURE_ADJUDICATION.01 (rounds 305–316 incl. 310b, adjudicated at the finite-identity level plus typed measurement legs) + PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01 [E] + PRIME.SOURCE.TENSOR.MECHANISM.01 [O] + PRIME.L2.RENYI3.PROVENANCE.01 [O]: THE ARCHITECTURE ADJUDICATION – the architecture day frozen in the reviewer’s binding</p> <p>FOUR-LEVEL STRUCTURE (ten probes embedded byte-exact in their sealed smoke stage,<br/> <math>27/27 + 28/28 + 30/30 + 33/33 + 34/34 + 32/32 + 32/32 + 29/29 + 29/29 + 35/35</math><br/> full records; plus the module-own exact S0 in pure Fractions: the r306 Hill/Lagrange chain with the Rényi bridge, the r314 opening-flux lemma <math>F_1 = 0</math> with the exact flux telescope, the collision counting identity <math>3p_1p_2 - 2p_3 = 8(n + 3n(n - 1))</math> with the exact factor-8 mutant break, an r311-class exact rational Farkas mini-certificate (<math>\langle Y, Q \rangle = -2</math> exact – the sign wall in miniature), the TEN sealed verdict bars with tipping mutants, and the four-level composition gate). LEVEL 1 (real formal theorems; Lean, already green – NO Lean change ships): lstar_terminal_implies_master (r305: <math>L^* + \text{terminal } q_N &lt; 1 \Rightarrow</math> full master positivity; augmented_prefix_positive and free_window_positivity corollaries), the four closed Inertia theorems, the real PrimeWindow construction (r310), source exactness + mass conservation of any folding + the STRONG stabilization form (r310b), the r309 rank-one update identities, Hill monotonicity + the Rényi-3 <math>\Rightarrow N_2</math> bridge; sorry census <math>9 \rightarrow 5</math> (the two TRUE holes: lstar_subordination, terminal_positive_main). LEVEL 2 (certified finite statements): Rényi-3 on 57 rungs (r306, <math>C = 1.069</math>, <math>A = 2</math>); the fixed-head census 77/77 (r307); the block-Green identity (r308) + strictness of the cone with exact rational Farkas certificates (r311, 0/16 in-span PD samples decompose); paired-cone soundness on 20714 steps (r309); fold multiplicity EXACTLY 2 + 93 percent far-triple cancellation (r313); the signed cubic identity with VANISHING boundary term (r314, Fractions-exact on the real data); the 21 small-<math>m</math> certificates (r316). LEVEL 3 (negative architecture decisions, verbatim): FIXED_HEAD_DEAD (r307); PAIRED_CONE_NO_INDUCTION + PAIRED_CONE_WORLD_BLIND (r309 B1–B4: B does NOT overtake A); the r308 discriminator = budget sign / chordal restatement (r311); COEFFICIENT_SIGN_WALL – Lane A closed as cone language with the mechanism NAMED (r312: block-psd membership without rank-one SDD, obstructed at the budget/border sign); FLOQUET_EXPANDING + TRIPLE_TYPE_MAJORANTS_WRONG (r313); PHI3_ALL_BLIND (r315: the entire break on kz55+kz67); TWO_REGIME_DEAD (r316: the obstruction family cuts ACROSS the FCIX stratum, first-5 load-bearing). LEVEL 4 (open mechanisms, typed): the PSD-G rule as a question, the signed cubic flux theorem, the quantitative source functional, crossing_budget, the source bridge, <math>L^*</math>, the terminal. THE CLAIM SPLIT (binding, NOT one TRANSFER title): RANKONE.UPDATE.IDENTITY [E] carries ONLY the exact update identities; TENSOR.MECHANISM [O] the open question (registered AFTER r313/r316 with the documented negatives). RENYI3.PROVENANCE [O]: the GO stands, the provenance open (identity exact, functional blind, two-regime dead; the R317 fork at the reviewer). Universality classes: base fine-metric (METRIC_ONLY, twin invariance), fiber recursive (EPSTEIN holds, SCRAMBLE breaks); the reviewer maturity <math>4 \rightarrow 5</math> typed as a REVIEWER ASSESSMENT. Nothing is proved cofinally; <math>L^*</math> stays THE open center. Mincut base 4 / refined 5 UNCHANGED. NO RH claim.</p> |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v969_forks_and_redteam.py | <p>PRIME.REDTEAM.EXTRACTION_AUDIT.01 [E] + PRIME.L2.RENYI3.SLIDING_BOUND.01 [O] (rounds 317–322, adjudicated at the finite-identity level plus typed measurement legs): THE RED-TEAM MORNING AND THE TWO FORKS – four probes embedded byte-exact in their sealed smoke stage (38/38 + 25/25 + 39/39 + 25/25 full records; r321 imports the embedded r317, r322 the embedded r318), plus the module-own exact S0 in pure Fractions and integer certificates (NO Lean call): the r321 concentration bracket <math>q_{\max} \Phi_{H1} \leq \rho_2 \leq \Phi_{H1} = (F_A B)^2</math> with both mutant directions caught, the R319 witnesses U1/U2/U3 in exact arithmetic (U1: the <math>B = -1</math> empty-spec window against the terminal demand; U2: the mesh-level-0 total node collision via the exact rational <math>e</math>-bracket <math>685/252 \leq e \leq 685/252 + 1/35280</math> and the <math>p = X - 1</math> kill <math>0 &lt; 0</math>; U3: the adversary <math>( F  + 1)^2 \geq 1 &gt; 5/7</math>), the r320 repair discipline re-gated (separation satisfiable at mesh level 4 via <math>2^7 &lt; 3^5 &lt; 2^8</math>, reducing to <math>1/8 &lt; 2/5</math> exact, and unsatisfiable at level 0; the canonical extraction exact), the six sealed verdict bars with tipping mutants, and the wave-12 composition gate. (1) The R319 extraction-chain red team (Lean audit, consumed as a report): three kernel-checked type inconsistencies three levels above the boxes (U1 bridge <math>\times</math> terminal, U2 bridge <math>\times L^*</math>, U3 pair <u>margin_main</u> forces its source predicate empty) plus the mesh-vs-anchor cofinality seam; the honest chain verdict: “two lemmata <math>\Rightarrow</math> RH” does NOT survive the literal reading – literally the two lemmata give ONLY window-local master positivity for the opaque, currently witness-less MainWindow. (2) The r320 repair (Lean, already merged and green – no Lean change ships): the bridge retyped with <math>u/B</math> fidelity + separation discipline + budget <u>pos</u>; SourceExact seals the arch/border finding; pair <u>margin_main</u> retyped onto the canonical extraction (signRuns <u>sum</u> + canonical <u>split</u> PROVED); three permanent sorry-free guards; a sorry-free witness; sorry census <math>5 \rightarrow 5</math> with two typed retypes. (3) r317 fork (b): WHAC_A_MOLE(kz53, kz83) by contract – the census find: ONE source-pure QMAX coordinate ranks the complete near-critical family on top (<math>F_A</math> top-3 2.47/2.39/2.38) but as a CONTINUUM. (4) r321 fork (a) sharpened: SLIDING_BOUND_GO(<math>G_{SQ}</math>) – the theorem CANDIDATE <math>\sum q^3 \leq 1.3056 F_A(w)^2 (\log m)^2 / m^2</math> for <math>m \geq 73</math> (+ 21 small-<math>m</math> certificates), 0/39 test violations AND all four named violators inside at reserves 7.0..9.6; the constant is algebra-derived and source-pure; honest: <math>C_{\text{impl}} = 7.97</math> is <math>7.5\times</math> looser than r306 (form, not sharpness); the new two-part provenance question: is <math>F_A</math> bounded, and what bounds the <math>q_{\max}</math>-share? (5) r318: P2_MAIN_SPECIFIC – the 2.4% negative cross mixtures live lawfully on the antiphase pair (<math>D3, D4</math>) with fixed sign <math>-1</math> (12/12 rungs + twin); P1 banked as LANGUAGE (window defects 0/0/0 vs 55/37/4/31; a total <math>L^*</math> restatement); classical sign regularity dead (PREMISE_FAILS_ON_MAIN). (6) r322 dig: ALGORITHM_ARTIFACT – the sign law and the r312 97.6/2.4 anatomy are properties of the least-norm-proximal Dykstra basin, not of the psd solution set; the r318 dichotomy does not survive fair convergence; the law is not identity-forced (the sign law lives entirely in the selection geometry); the named residual object: the selection-geometry question. Nothing is proved cofinally; the two true Lean holes are unchanged; <math>L^*</math> stays THE open center. Mincut unchanged. NO RH claim.</p> |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v970_extraction_and_composition.py | <p>PRIME.EXTRACTION.ELEMENTWISE.01 [E] + PRIME.L2.RENYI3.MEASURED_COMPOSITION.01 [O] (rounds 323–327 incl. the r326 Lean round): THE EXTRACTION REPAIR AND THE TERMINAL COMPOSITION – four probes embedded byte-exact in their sealed smoke stage (36/36 + 36/36 + 18/18 + 34/34 full records; r324 imports the embedded r324-pre, r327 the embedded r324-pre AND r324), plus the module-own exact S0 in pure Fractions (NO Lean call): the r324 interpolation <math>M_3 \leq q_{\max} M_2</math> and <math>M_2 \leq q_{\max}</math> exact (slack 1/36, double equality at flat; mutants caught), the <math>F_A</math> identity <math>F_A B \log m = m q_{\max}</math> as exact constructional algebra (<math>F_A</math> de-black-boxed; gauge mutant caught), the r327 group ledger exact (partition, cap chain <math>12/7 \leq 20/7</math>, two-ancestor bound <math>5 \leq 2 \times 3</math>; both mutants break by exactly 1), the r325 onset formula <math>\alpha^* = (n_g + 1)D_0/2</math> as an exact toy stabilization (read = atom sum 35/16 for every <math>\alpha \geq 5/8</math>, constant under mesh refinement; nearest-neighbour mutant caught), the <math>m_0^*</math> log-space solver rebuilt verbatim (landing <math>10^{59.6}</math>; the disclosed linear-space bug caught), six sealed verdict bars with tipping mutants, and the wave-13 composition gate. (1) r323: CLEAN ABORT before any write access (L* paused by the reviewer) – nothing counts as a round. (2) r324 (after the course correction; the pre-work banked as r324-pre): PILEUP_GROWS_SUBCRITICAL (<math>e_{\text{tot}} = +0.172 &lt; 0.224</math>, decided) – the pointwise <math>M_2</math> export refuted (7/39, exactly the banked spike set) while the distributional r302 export holds; the <math>O(\log m)</math> scale count CERTIFIES pointwise (0/39) but the per-scale mass is refuted: the near-critical windows are ONE heavy source scale; the composition typed MEASURED: <math>\sum q^3 \leq 8.941(\log m) m^{+0.172}/m^2 \Rightarrow N_2 \geq m^{0.888}</math> for all <math>m \geq m_0^* = 10^{59.6}</math> – the gap (274, <math>10^{59.6}</math>) the disclosed extrapolation hypothesis (r328B chain audit: the bar 0.224 used the atom-level need; the block-level need is 454/501, the chain bar 0.188 – the verdict survives <math>0.172 &lt; 0.188</math>, <math>m_0^*</math> chain-honest <math>10^{238}</math>; gate S0-T5d, record sealed). (3) r325: ELEMENTWISE_STABILIZATION_GO – the extraction repair is a QUANTIFIER statement: on the native v749 class all three channels stabilize exactly at the predicted finite anchor onset and are constant under dyadic mesh refinement; no mesh-cofinal ladder, no transport, <math>H_{\text{cof}}</math> is not needed; B is the construction substrate, C the quantified density step. (4) r326 (Lean, merged and green, consumed as a report): sourceExact_buildPrimeWindow PROVED, the comb stabilization elementwise PROVED, weil_nonneg_of_windowlocal PROVED (extraction WITHOUT the ladder); sorry census <math>5 \rightarrow 8</math> with three typed Level-C statements; the honest new connection verbatim: “two window-local lemmata (over the canonical family) + the elementwise architecture + cited classics <math>\Rightarrow</math> RH; open links: arch/pole transcription (classical), compression bridge, source completion (definitional)”. (5) r327: CAP_PARTIAL with anatomy SOURCE – on all 65 rungs the heaviest group of the argmax block is EXACTLY ONE <math>\beta/\omega</math> fold pair (kz53 = one bulk/window coincidence at 88.8%); the <math>\Lambda</math>-pair cap refuted, the group count certified (0/39, the third <math>O(\log m)</math> count); the r324 MEASURED composition stays the end state; direction: WHERE can the coincidence sit (source geometry). Nothing is promoted beyond the measured rungs; the two true window-local Lean holes unchanged; L* stays paused. NO RH claim.</p> |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v971_markov_embedding_generator.py | DYN.MARKOV.EMBED.01 (finite kernel half [E]; external-review round 2026-08-27, 'three meanings of dynamics'): $Q = \log T$ of the v221 seam transfer kernel is an EXACT classical Markov generator – sympy-exact spectral log with $e^Q = T$ , row sums symbolically zero, off-diagonals exactly $\{\ln(9/8), 2\ln 3, 2\ln 3\} = \{0.117783, 2.197225, 2.197225\} > 0$ , diagonals $-\ln(81/8), -\ln(81/8), -4\ln 3$ ; uniformization certificate $Q + (4\ln 3)I \geq 0$ entrywise $\Rightarrow e^{tQ}$ positivity-preserving doubly stochastic for ALL $t \geq 0$ ( $T$ embeddable, every root $T^{1/n}$ stochastic); $\text{spec}(Q) = -\text{spec}(H_{\text{mod}}) = \{0, -6\ln(3/2), -6\ln 3\}$ (gap = $\Delta$ ), detailed balance w.r.t. the uniform state. NEGATIVE control: second eigenvalue $-1/10$ gives $\det = -1/200 < 0$ – provably NOT embeddable (embeddability is not generic; the RP/OS spectral positivity is the enabling fact). FIREWALL: closes relaxation $\rightarrow$ semigroup (a) $\rightarrow$ (b) at the finite kernel level ONLY; the local, size-consistent unitary dilation (b) $\rightarrow$ (c) is the separate open contract DYN.UNITARY.DILATION.01; the shared F transfer relaxation shape is a universal CONTRACTION CLASS, not one physical clock (v723/v724/v777).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v972_seam_interaction_front.py     | SEAM.INT.ODDSPLIT.01 [E] + SEAM.INT.OSAVOID.01 [E] + SEAM.STATE.RPMIXING.01 [E] (+ updates to SEAM.INT.FKTOY.01 / WOIT.OS.TWISTOR.01 / SEAM.STATE.DERIVATION.01; three exploration rounds of 2026-08-27 consolidated, the probes embedded byte-exact with pinned source SHA-256 and executed verbatim, 20/20 + 18/18 + 25/25, verdicts CLASS_EMPTY_ON_BASIS + OS_VIABLE_ON_AVOIDING + RP_ADMITS_MIXING): THE SEAM INTERACTION FRONT – (1) the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (arXiv:1406.1384; Wei–Li–Xiang arXiv:1601.01994) made DECIDABLE on the v529 FK toy: the crossing matrix $B$ is UNIQUE (left-monomial pairs biject onto crossing monomials), so ' $B$ Hermitian AND $-B$ PSD' is a finite check; the free FK Hamiltonian certified on all 8 bond cuts ( $\eta = -i$ forced); the confusion matrix vs measured RP PERFECT over 35 cells (23/12/0/0); the $\mu_4$ -covariant quartic space (dim 464) EMPTY on the basis, the support-level exclusion pinning the full 1820-dim space – THE DISCOVERY: the v529 straddle law REFINES to the ODD-SPLIT LAW (JP predicted RP survival on CENTER-straddled cuts, $m = 4$ model, axes 7/15, 2+2 mirror split; measurement confirmed (37, 0, 0) at $g = 0.5/1/2$ ; the v529 anchor ladder reproduced exactly). (2) the quartet-AVOIDING cut family of the clock-invariant $m = 4$ member generates the FULL even algebra (rank 15, dim $32768 = 2^{15}$ ; single-cut members $m = 2/6$ typed deficit exactly 1) and carries a POSITIVE OS/GNS structure at $g = 1$ (min $+4.43 \times 10^{-5}$ , GNS 37/37, distances monotone) while the straddled control genuinely fails ( $-6.94 \times 10^{-2}$ ); the $\mu_4$ clock maps the avoiding pair $\{3, 11\}$ onto itself exactly – the kill-test-2 counterweight NARROWED, not removed. (3) the v903 exclusivity is RAY- and REFLECTION-relative, not slice-wide: on the full 33-dim $C_6$ -covariant mixing slice $\theta_{abT}$ sees exactly ONE coordinate ( $a_J$ , kernel value $-1/(1+c^2) = -0.8240271368$ at $c = \tanh(1/2)$ ), 32/33 invisible $\Rightarrow$ a 32-dim MARGINAL RP hyperplane $\{a_J = 0\}$ ; the $\theta_S$ defect map has rank 16 $\Rightarrow$ a 17-dim first-order-neutral subspace with STRICT finite- $s$ witnesses ( $\lambda_{\min} = 0.1558$ at $s = 1/8$ ); the v903 anchor reproduced ( $a_J = +0.04068013$ ). Module-own exact S0 (sympy/GF(2), no probe imports): the JP decidability lemma on a 4-Majorana toy (9 pairwise-distinct crossing masks $\Rightarrow$ unique $B$ ; the wrong-sign toy fails), the odd-split arithmetic (centered $\Leftrightarrow$ 2+2 exact on $\mathbb{Z}_{16}$ ; the odd 1+3 block forces $b = 0$ – the obstruction; the flipped-sign-law mutant caught), the GF(2) generation bookkeeping (pair rank 15, $2^{15} = 32768$ ; within-halves rank 14 – deficit exactly 1), the window arithmetic ( $33 = 24 + 9$ , $32 + 1$ , $33 - 16 = 17$ ; kernel value $\equiv -\cosh^2(1/2)/\cosh(1)$ symbolically) plus 4 verdict bars with tipping mutants. HONEST: no marker moves; SEAM.EQUIV.01 stays [O]; the v529 fence (one toy, one interaction class, [C] flat-band parent) inherited by all three results; JP sufficient-only; the S1 census sector-truncated. NO RH claim. |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v973_seam_route_narrowing.py | <p>SEAM.EQUIV.MMST.01 [O update] + SEAM.EQUIV.TWISTOR.01 [O update] + SEAM.EQUIV.SKELETON.01 [E, NEW] + SEAM.EQUIV.01 [O update] (seam adjudication wave, 2026-08-27): THE SEAM ROUTE NARROWING (19/19 + 20/20 + 19/19, three probes embedded byte-exact with pinned SPEC SHAs, plus the module-own exact section S0). (1) MMST COVERAGE CARVED: the cited Osborne–Stottmeister theorem (CMP 398 (2023) 219, arXiv:2107.13834) COVERS the bilinear half of the seam target – <math>\mathfrak{so}(16)_1</math> fermion-bilinear WZW currents converge STRONGLY on the finite-particle core (Thm 5.1 + Cauchy–Schwarz Eq. 229 incl. Majorana algebras), Virasoro/Koo–Saleur (Thms A/4.11/4.16), correlators (Thms B/C), polynomial errors <math>\text{Error}^2 \leq C 2^{-2\delta N}</math>, <math>\delta \in [0, 2]</math> – and does NOT cover four NAMED items: N1 the <math>\mu_4</math>/GSO extension <math>SO(16)_1 \rightarrow (E_8)_1</math> as a NET statement (the 128 spinor currents are not bilinears), N2 the twist/spin-field sector, N3 the one-sided chiral edge net of a 2+1D bulk, N4 the holomorphy selector <math>\det K = 1</math> vs 4; NEW MEASUREMENT: the collar edge-correlator rate <math>p = 3.48</math> (Richardson 3.1–4.2, preregistered band <math>[0.5, 4.0]</math>), error <math>6.3 \times 10^{-2} \rightarrow 2.7 \times 10^{-6}</math> over <math>N_x = 16..256</math>, the <math>M = 3</math> control does NOT converge (min rel dev 0.836); exact-integer coverage gates (<math>248 = 120 + 128</math>, <math>240 = 112 + 128</math>, <math>\det \text{Cartan } 4 \rightarrow 1</math>, index <math>2 \times 2 = 4 =  \mu_4 </math>). (2) TWISTOR MEASURE RIGIDITY: on the computable v508/v515 sector the 18-parameter deformation of the declared measure data has executed-constraint ranks 9/8/2/2, joint 13, 4 gauge directions <math>\Rightarrow</math> the moduli kernel is EXACTLY 1 = the uniform sector-weight shift <math>dv = (1, 1, 1, 1)</math>, mapping to the untwisted Okubo square <math>Q^{(0)} = 36(1, 2, 1, 0, 0)</math>, annihilated by every executed certificate functional BY CONSTRUCTION; ablation ledger (1, 3, 1, 2, 3, 10); the <math>W_2 : W_{13}</math> tester honestly NOT-TRANSCRIBABLE. (3) OS SKELETON SHARED: all FIVE WOIT kill-test main invariants (<math>\alpha</math>: forced <math>\eta = +i</math>, <math>\Theta^2 = +1</math>, bond RP; <math>\beta_1</math>: GSO 2-torsion; <math>\beta_2</math>: OS quotient + KMS <math>\lambda_{\max} &gt; 1</math>; <math>\beta_3</math>: perp descent; <math>\gamma</math>: mirror flip, multiplicity-free, mark incidence) hold VERBATIM on the p+ip collar edge; ALL divergences control-typed to ONE mechanism (the twistor lattice’s exact chiral checkerboard <math>C(\text{even}) = 0</math> vs the collar’s <math>K(\text{even})</math> up to 0.1736 – confirming v519-R4 from the second route); the top edge realizes the anti-chiral state EXACTLY (<math>7.5 \times 10^{-16}</math>); the <math>M = 3</math> collar loses the skeleton first at GAMMA.MARK.INCIDENT. Module-own S0 exact: root census + Cartan determinants + index arithmetic; the uniqueness linear algebra in exact rationals (ranks (9, 8, 2, 2) <math>\rightarrow</math> 13 <math>\rightarrow</math> kernel 1, ablation ledger exact, the dropped-C4 mutant opens the kernel to 3 – CAUGHT); the skeleton mechanisms exact (bipartite <math>SHS = -H \Rightarrow</math> odd functions vanish on even distances; <math>PHP = -H</math> on a strip <math>\Rightarrow K_{\text{top}} = -K_{\text{bot}}</math>); verdict bars with tipping mutants (a fake COVERAGE_FULL is REFUSED by the nonempty N1–N4 gate; hash mutation CAUGHT). HONEST: the coverage gates are executable transcriptions of a literature adjudication; the rate SUPERCONVERGES (<math>p &gt; \delta_{\max} = 2</math>) vs the cited upper bound (band calibration-informed); the sector uniqueness is NOT a BCOV derivation; skeleton fidelity typed per test. NO marker moves: MMST and TWISTOR stay [O], the parent stays [O]; the only new row is SEAM.EQUIV.SKELETON.01 [E].</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v974_alpha_faces_computed.py  | <p>ALPHA.QUILLEN.EXACT.01 [O update] + SEAM.CONTACT.UNIT.01/.02 + EM.WARD.01 / ALPHA.QUILLEN.INFLOW.01 + ALPHA.QUILLEN.FACES.01 [E, NEW] (seam/alpha promotion wave 2026-08-27): both named faces of the alpha target now carry computed witnesses — three probes embedded byte-exact, sha256-pinned and executed verbatim (21/21)</p> <p>MATCH_MODULO_LOCAL_FACTOR + 20/20</p> <p>PUMP_QUANTIZED_MATCHES_CHERN + 18/18 CHAIN_SHADOW_OK), plus the module-own exact section S0 (sympy/Fraction/mpmath, no probe imports). (1) The BFK face EXECUTED: the Burghlea–Friedlander–Kappeler gluing constant is exactly <math>2^{-N}</math> (derived symbolically <math>N = 1.4</math>, general interval lengths); massless reassembly <math>\det'_\zeta(\Delta_{S^1}) = l^2 = 2^{-N} \prod (2a_j) (l/N) \det'(R_0)</math>; at the KMS scale <math>l = 2\pi = 1/(4c_3)</math>: <math>R_0 = 16c_3 \text{ circ}(2, -1, 0, -1)</math>, <math>\det' R_0 = 2^{16}c_3^3</math>, glued product <math>(2\pi)^2</math>; <math>R(m) = (G_m _{\text{marks}})^{-1}</math> exactly; vs the v484 mark matrix NO global proportionality (kernels in swapped <math>\mu_4</math> channels: gauge zero mode vs sign rep) but channel-wise EXACT (same <math>\mu_4</math> Fourier eigenbasis, <math> \text{spec} </math> multisets <math>\{0, 2, 2, 4\}</math>, doublet ratio <math>4/\ln 2</math> with <math>c_3</math> cancelling, <math>\text{Tr } B^k = 4^k + 2 \cdot 2^k</math> mirroring <math>\text{Tr } C^k = 4^k + 2(-2)^k</math>); the v485 scale channel reproduced exactly via the KMS unit. (2) EM1 COMPUTED: pumped charge == edge spectral flow == Streda == Chern == 1 per flux quantum on the v367 collar (conjugation mutant flips all four readings simultaneously; wrong-band projector breaks quantization). (3) The finite Quillen identity TWO-SIDED: Berry integral == Poincaré–Lelong divisor integral == +1 over the twist torus, difference an exact form (<math>3.3 \times 10^{-16}</math>), transition winding == Chern; typed conditional on the quasi-free rigidity premise (measured, not proven — SEAM.STATE.RPMIXING.01). S0 re-derives everything independently: the BFK closed forms from <math>\zeta_R</math> values, the alpha anchor at 50 digits (<math>8b_1 = 164/5</math> exact, unique root, <math>\alpha^{-1} = 137.03599921684071</math>, split residual <math>6.4 \times 10^{-58}</math>), the inflow bookkeeping (<math>c_- = 8 = g_{\text{car}} + N_{\text{fam}}</math>, <math>k_Y = 5/3</math>, <math>(1/k_Y) 41/6 = 41/10 = b_1</math>), a probe-free coarse FHS Chern certificate. Mutants: hash mutation caught by the sha256 pin; the wrong KMS unit (<math>2\pi \rightarrow 4\pi</math>) breaks the scale identity by exactly <math>-2\ln 2</math>; the <math>b_1</math> mutant caught. NO marker moves — the target stays [O]: remaining content = {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification}; <math>\alpha^{-1}</math> stays [E] regardless.</p> |
| v975_dimension_selector_4d.py | <p>DIMENSION.SELECTOR.4D.01 [C, NEW] (master-route wave 2026-08-27): the CONDITIONAL 4D dimension selector, 16/16 sympy-exact. Under the named TFPT-compatible axiom set A1 (<math>d &gt; 2</math>: propagating local gauge fields), A2 (dimensionless Yang–Mills coupling), A3 (real self-dual/anti-self-dual 2-form sectors: the Euclidean Hodge star an involutive endomorphism of 2-forms), A4 (chiral Weyl representations: even <math>d</math>), the dimension <math>d = 4</math> is UNIQUE and MINIMAL on the exact scan window <math>d = 2..12</math> — and the selector is OVERDETERMINED: A2 alone = {4} (<math>[g_{\text{YM}}] = (4 - d)/2</math> forced symbolically, unique zero 4) and A3 alone = {4} (the explicit <math>6 \times 6</math> star on <math>\Lambda^2(\mathbb{R}^4)</math>, <math>*^2 = I</math>, eigenvalues <math>\pm 1</math> at multiplicities 3/3), while A1 &amp; A4 alone give {4, 6, 8, 10, 12} — chirality does NOT select 4 by itself (honest ablation ledger). Weyl chirality certified by exact Clifford algebras <math>d = 2..7</math> (even <math>d</math>: <math>\gamma_*</math> anticommutes, <math>\gamma_*^2 = I</math>, traceless; odd <math>d</math>: the product is CENTRAL and scalar). Signature typed honestly: Lorentzian <math>*^2 = -1</math> on 2-forms — the real split is a EUCLIDEAN/OS-side statement. Mutants caught (<math>[g] = (6 - d)/2</math> selects {6}; the ‘A4 alone’ overclaim refused). FIREWALL: <math>\mu_4</math> and <math>g_{\text{car}}</math> enter NOWHERE — ‘<math>\mu_4 \Rightarrow 4D</math>’ stays forbidden (DIMENSION.UPLIFT.FIREWALL.01); axiom provenance from <math>\{c_3, g_{\text{car}}\}</math> is the OPEN half (the kill criterion in the ledger row); no 4D construction contract is advanced.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v976_seam_lift_<br>birkhoff.py | <p>TRANSFER.HIDDEN.CIRCULATION.01 [E, executed half] + the EB no-go for DYN.UNITARY.DILATION.01 (external-review round 2026-08-27): the Birkhoff fibre of the seam single step <math>B</math> is EXACTLY a line, its hidden parameter is the sign character, and the deployed Kraus dilation is entanglement breaking — <math>B = \frac{1}{2}I + \frac{1}{18}P_{12} + \frac{2}{9}P_{13} + \frac{2}{9}P_{23}</math> bit-exact (<math>B^6 =</math> the v221 kernel), the decomposition map has a 1-dim kernel (<math>I + C_{123} + C_{132} = P_{12} + P_{13} + P_{23} = J</math>), complete family <math>w_t</math> with <math>t \in [0, 1/18]</math>; <math>\widehat{w}_t(\text{sgn}) = 6t</math>, populations exactly <math>t</math>-free, no fibre member chiral (<math>w(C_{123}) = w(C_{132})</math> forced). DISCRIMINATOR: one-step coherence dynamics = the weighted left regular <math>S_3</math> representation, spectrum <math>\{1, 6t, 2/3 \times 2, 1/3 \times 2\}</math> exactly; the triangle loop current is an exact eigen-observable <math>\Psi_t(A) = 6t A</math> (population-blind: one step + one current readout measures <math>t</math>, no ancilla; tomography rank 6). CONTRAST QUBIT: <math>\text{spec}(\text{Choi}^{T_2}) = \{-3t/2, 1/6 + 3t/2, 1/3 + 3t/2, 1/2 - 3t/2\}</math> symbolically; CRITICALITY <math>\lambda_{\min} = (1 - \lambda_2 - \lambda_3 - 6t)/4</math> with <math>2/3 + 1/3 = 1</math> EXACTLY (<math>t = 0</math> precisely on the EB boundary; mutated spectra miss by <math>\mp 1/12</math> — not generic); the squared channel is PPT on the whole range (endpoint minima <math>1/9, 1/12 \Rightarrow</math> qubit EB index 1 (<math>t = 0</math>) / 2 (<math>t &gt; 0</math>)) [PPT <math>\Leftrightarrow</math> EB in <math>2 \times 2</math> + Lami-Giovannetti, literature-typed]. FULL LEVEL: <math>\lambda_{\min}(\text{Choi}(\Psi_t^n)^{T_2}) = -(1/3)(2/3)^n</math>, <math>t</math>-INDEPENDENT (exact for <math>n \in \{1, 2, 6\}</math>, float <math>n = 1..8</math>) — never EB at 3 levels. THE NO-GO: the deployed Kraus dilation <math>K_{ij} = \sqrt{T_{ij}}  i\rangle\langle j </math> is measure-and-prepare <math>\Rightarrow</math> entanglement breaking [HSR] <math>\Rightarrow</math> quantum capacity 0; <math>C_0(T) = 0</math> exactly; it is NOT in the Birkhoff fibre. BINARY MESSAGE: <math>p^* = (1/2, 1/2, 0)</math> is KKT-exactly optimal for BOTH <math>T</math> and <math>B</math> (<math>C(T) = 0.0083580191</math> bit, <math>C(B) = 0.4890415237</math> bit, ratio 1.71%). [C] observation: the weights <math>(9, 1, 4, 4)/18 = \text{squares}/18 =</math> the v652 defect-ladder values <math>g^2/18</math> (multiplicity matches only at <math>g = 2</math>; the forcing functor is the OPEN contract TRANSFER.DEFECT.BIRKHOFF.01). FIREWALL: finite matrix algebra + literature-typed channel theory; the fibre-point selection stays [O].</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v977_transfer_unistochastic_wilson.py | <p>TRANSFER.COHERENT.WILSON.01 [E, executed half] (external-review round 2026-08-27, companion of v976): <math>B</math> is UNISTOCHASTIC — an exact <math>SU(3)</math> lift carrying the 5-12-13 Wilson code, the center bridge and a classically invisible orientation bit — angles forced by <math>B</math> (<math>\sin^2 \theta_{13} = 2/9</math>, <math>\sin^2 \theta_{12} = 1/14</math>, <math>\sin^2 \theta_{23} = 2/7</math>), <math>\cos \delta = -4/\sqrt{65}</math> unique, <math>e^{i\delta} = (-4 \pm 7i)/\sqrt{65}</math>, lift set = <math>\{U, U^*\}</math> up to rephasing; <math>U^\dagger U = I</math>, <math>\det U = 1</math>, <math> U_{ij} ^2 = B_{ij}</math> entrywise exact [coherification iff unistochastic, literature]. <math>J = \text{Im}(\text{plaquette}) = 1/27</math> EXACT; Re of every plaquette fixed by <math>B</math> (unitarity) <math>\Rightarrow</math> the full rephasing-invariant Wilson data is classical up to the <math>\mathbb{Z}_2</math> ORIENTATION BIT <math>\text{sgn } J</math> (<math>U^*</math> lifts the same <math>B</math> with <math>J = -1/27</math>; observable = the plaquette interference readout). THE CODE: <math>Q_{12;12} = (-5 + 12i)/324</math>, normalized <math>(-5 + 12i)/13</math> with <math>5^2 + 12^2 = 13^2</math> (atoms <math>5 = g_{\text{car}}</math>, <math>12 = \dim \mathfrak{g}_{\text{SM}}</math>, <math>13 = \Delta_Q = N(3 + 2i)</math>); all nine plaquettes close on four Gaussian phase types with norms exclusively <math>\{10, 13, 130, 169\} = \{2 \cdot 5, 13, 2 \cdot 5 \cdot 13, 13^2\}</math>; every <math>\text{Im} = \pm 1/27</math>; the four fundamental plaquettes generate all nine exactly. CENTER BRIDGE: <math>\text{disc } \chi_A = 65 = \det(A - I)^2 + (\det A/2)^2 = 4^2 + 7^2</math> for the v530 atom matrix <math>A = [[8, 2], [5, 3]]</math>; <math>e^{i\delta} = (-\det(A - I) \pm i \det A/2)/\sqrt{\text{disc}}</math> EXACTLY equals the transport phase; <math>-4 + 7i = (1 + 2i)(2 + 3i)</math> (the v738 tower primes, norms 5 and 13); the v530 siblings <math>A_K/A_Q</math> FAIL the square-sum identity (the bridge selects <math>A</math>). POWER LAW: <math>J_n^2 = (1 - b)^2(1 + 2b - 3a^2)/108</math>, every <math>B^n</math> unistochastic (<math>U_6</math> built, <math> U_6 ^2 = T</math> exact); <math> J_6  = 0.0951088904 = 0.9884006 J_{\text{max}}</math>; the uniform limit lifts to the Fourier matrix <math>F_3</math> at <math>J_{\text{max}}</math> exactly; <math>J_n</math> RATIONAL ONLY at <math>n = 1</math> (<math>3(1 + 2b - 3a^2) = 1</math> exactly <math>\Rightarrow J = 1/27</math>) — the Gaussian–Pythagorean code selects the single step. NO FIXED MICROSTEP: <math>B^2 \neq  U^2 ^2</math> exact; Dirichlet recurrence witness <math>n^* = 967909</math> (min diag <math> U^{n^*} ^2 = 0.999999</math> while <math>B^n</math> is uniform to <math>5.6 \times 10^{-17}</math>) — <math>U_6 \neq U_1^6</math> (clock-vs-walk sharpened coherently). CENSUS (hard kill passed): 238 valid family points, 4 with rational <math>J</math>, EXACTLY ONE full-code point = the deployed <math>(2/3, 1/3)</math> (<math>0.42\% \ll 10\%</math>). PMNS NEAR-MISS stays NEGATIVE (branch phase <math>240.2551^\circ</math> vs deployed <math>240^\circ</math>; <math>-1/27 \neq -0.02965</math>; all angles differ). FIREWALL: the <math>\text{sgn } J</math> branch selection, CP/time readings and the <math>A \rightarrow</math> angles intertwiner stay [O].</p> |
| v978_terminal_density_martingale.py   | <p>PRIME.TERMINAL.DENSITY_MARTINGALE.01 [E] (consolidation wave 14, rounds 339 + 341): THE TERMINAL REFORMULATION THEOREM LAYER — the density per remaining descendant <math>X(v) = d(v)/d(\text{root})</math> is a nonnegative martingale from MASS CONSERVATION ALONE (symbolic generic node + bit-exact Fractions tree); the moment dictionary <math>E[X_\infty^2] = m M_2</math>, <math>E[X_\infty^3] = m^2 M_3</math>, <math>\max X_\infty = m q_{\text{max}}</math> (symbolic + Fractions; the terminal target IS a martingale moment — the banked r306 <math>C_2 = 1.0694</math> certifies the <math>A = 2</math> polylog form on the 57-rung set); the Jensen drift as exact weighted AM–GM; the TILTED TOWER <math>E[X_\infty^3] = E_Q[\prod \Gamma]</math> with steps <math>\tilde{p}_c = p_c R_c^3/\Gamma</math> (the naive untilted reading breaks by an exact nonzero Fraction — mutant caught; probe toy break 7/18); the <math>\Phi</math>–<math>\Gamma</math> telescope symbolic; the exact hand-off <math>E_{3h} \leq (m q_{\text{max}})^2</math> msh with a strict spread witness AND the flat equality case; the algebraic pair ceiling <math>\Gamma \leq 4</math> via <math>4R - R^3 = R(2 - R)(2 + R) \geq 0</math> with the <math>\{0, 2\}</math> equality witness. FROZEN CENSUS with tipping mutants: r339 LOCAL-INFLATION-SUPERCRITICAL (<math>e(W_F) = +0.956 &gt; 0.224</math> while the target grows <math>8\times</math> slower at <math>+0.112</math> — the gap IS the finding) and r341 PATHWEIGHT-ALSO-SUPERCRITICAL letter-honest (no arm certifies on its own freeze; path weighting deflates the worst case <math>21\times</math>; <math>R^* = 3/2</math> is a TUNING SURFACE: hsh 0.872 vs 0.266 at 7/4). NO inequality promoted — the r324 MEASURED composition (<math>m_0^* = 10^{59.6}</math>, chain-honest <math>10^{238}</math>) stays the terminal end state. NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v979_cover_growth_k2.py    | <p>PRIME.L2.COVER_GROWTH_K2.01 [E] (consolidation wave 14, rounds 344/346/351/353/355): THE THREE-ARM COVER, THE GROWTH LAW AND THE K2 TWO-FAMILY LAW, Frame-A-typed – EXACT: the FAB identity <math>F_{AB} = mq_{\max}/\log m</math> (constructional algebra, gauge mutant caught); the r321 concentration bracket <math>q_{\max}\Phi_{H1} \leq \rho_2 \leq \Phi_{H1}</math>; the data-free K1 FORMULA <math>R_{\text{ALG}} = 4^{1/3} \rightarrow</math> grid point 8/5 INTERIOR, certified in pure rational arithmetic via cube midpoint comparisons <math>((31/20)^3 &lt; 4 &lt; (49/30)^3</math>; the ceiling-8 mutant lands the endpoint 2); the r355 MESH IDENTITY <math>h - \nu u \in (0, 3/2]</math> EXACT (ceil +1+ even rounding; by-hand toy <math>\alpha = 10</math>, gap 1, <math>\nu = 2 \rightarrow h = 21</math>, dev 1) – where grel enters the K2 chain is construction algebra. FROZEN CENSUS with tipping mutants: r344 TWO-SCALE-PARTIAL (the three-arm min cover certifies 0/51 + named 4/4 + midband 6/6 at every grid point, <math>m_0^* = 10^{22.6}</math> – the first certifying cover; <math>\hat{R}</math> not halves-stable); r346 COVER-CANONIZED + PREDICTOR-FOUND(P02) (<math>F_A \geq 1.5</math> predicts the covering arm 51/51, 46/51 exact vs baseline 7/51); r351 GROWTH-LAW-CERTIFIED (<math>C_{\text{FAB}} = 14.93</math> survives the fresh EXT5 tranche 0/6; polylog <math>m_0^* = 10^{18.9}</math>) + FLOOR-ERODES; r353 THE BINDING FRAME-A RESTRICTION (kz124 FAB 18.07 breaks the ceiling by +21 % – family census, not a cross-family law; FLOOR-KILLS at <math>m \sim 760</math>: the sliding coverage is FINITE; K2 the sole cross-family survivor 0/8 at 11.87); r355 K2-LAW-CERTIFIED-NU-FREE (0/25 over 3 aspects – the gap geometry is the <math>\nu</math>-free coordinate) + K2-SOURCE-CHAIN-OPEN (chain exact, caps vacuous 373.4/286.7 vs bar 47.5) + GMIN-LAW (sealed thresholds verbatim; ceiling route supercritical +0.286 &gt; 0.224). The honest <math>m_0^*</math> ledger ordered and Frame-A-typed (<math>10^{13.5} &lt; 10^{18.9} &lt; 10^{20.5} &lt; 10^{22.6} &lt; 10^{23.5}</math>, ALL census). NO cofinal claim; PRIME.L2.RENYI3.SLIDING_BOUND.01 stays [O], retyped finite. NO RH claim.</p>                                                                                                                                                                                                                                                                                                                                                                 |
| v980_lstar_margin_chain.py | <p>PRIME.LSTAR.MARGIN_CHAIN.01 [E] (consolidation wave 14, rounds 345/347/348/350/352/354): THE CLOSED L* MARGIN-LAW CHAIN – EXACT: the ONE-LINE IDENTITY <math>m'_2(p' + q' - m'_2) = p'q' - c'^2</math> (Vieta, symbolic); the TWO-LEVEL THEOREM <math>p'_2 = m(g_{21}a_2^2 + b_2^2)/\Delta t^2</math> etc. (exact inversion = formulas = (121/250, 79/250, 72/250) by hand; cross-tie <math>r'_2 = 4375/9559</math> = the r345 formula; Lagrange identity <math>p'q' - c'^2 = m^2g_{21}/\Delta t^2</math> symbolic); the PINNING THEOREM (orthonormal top-2 geometry <math>\Rightarrow</math> the dressed charpoly FACTORS as <math>(t - m)(t - mg_{21})</math>: <math>m'_2</math> = margin IDENTICALLY – both sealed rational models, dets 7/100 and 3/25); the RATE-EQUALITY THEOREM in exact ratio form (flat geometry <math>\Rightarrow</math> dressed ratio = margin ratio bit-exact; a geometry drift breaks by exactly the drift factor); the <math>\rho_K</math> identities (weights cancel exactly; sealed toy <math>\rho_K = 1/8</math>, <math>r_{\text{det}} = 23/24</math> on all three routes; <math>a_{\rho K} = 2a_\kappa</math> as the squared-ratio identity); the composition skeleton <math>c^2 = v_1v_2K_{12}^2</math>. FROZEN CENSUS with tipping mutants: r345 GAP-RATIO-PRIMARY-CERTIFIED (the pre-sealed curvature-honest protocol passes both <math>O(1)</math> candidates 0/75; the r343 soft decay re-adjudicated as a protocol artifact) + CANCELLATION-LAW-FOUND(<math>\delta = 2.668</math>); r347 ALPHA-CLOSED (all four closures <math>\leq 0.1</math>: <math>\alpha = a_c + \delta</math>, residual 0.033; bridge max 0.0605); r348 RATE-EQUALITY-THEOREM live (E1–E4) + CARRIES-DELTA refuted (<math>\delta</math> is an inter-order near-cancellation); r350 ALPHA-SOURCE-CLOSED at CANDIDATE-LAW grade (3/4 + 2/3 + 2.624 – 0.690 vs 3.332, dev 0.019; CLOSED_FORM stays FALSE); r352 RHOR-REDUCED one-object grade (corr 0.999998, 500<math>\times</math> wander-to-difference, <math>\rho_r = -s_{LR}</math> at 0.0002; DECOR-REFUTED, GK12-EXPLAINED); r354 PHI-DICTIONARY-GO (<math>c_{\text{pred}}</math> reconstructs <math>c</math> at <math>5.52 \times 10^{-4}</math>, <math>\varphi</math> corr 1.0 – computability, not analyticity) + two-block interference + DELTA0-UNRESOLVED (pool exhausted at <math>10^{3.90}</math>). The lane is FROZEN as a specialist problem; the irreducible rest is the 500<math>\times</math> two-block cancellation (v981 retypes its anti-correlation as duality algebra). PRIME.LSTAR.SUBORDINATION.01 stays [O]. NO L* claim, NO RH claim.</p> |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v981_lstar_borodin_duality.py   | <p>PRIME.LSTAR.DUAL_HOLE.01 [E] (consolidation wave 14, round 356): THE BORODIN PARTICLE-HOLE DUALITY – the exact equivalence <math>L^* \Leftrightarrow R &gt; \frac{1}{2}I</math> at half filling. EXACT (no probe imports): the Borodin complementation <math>\Pi_N^u + G\Pi_{N-1}^u G^{-1} = I</math> bit-exact on a 5-atom rational model (<math>S = 2N - 1</math>) with the reciprocal dual weight <math>u^\vee = 1/(uP'^2)</math> and the RATIONAL conjugator <math>G = \text{diag}(1/(uP'))</math> – no square roots (the dropped-<math>P'^2</math> and rank-<math>N</math> mutants break exactly); the L-ensemble roundtrip <math>Q = E(I + E)^{-1}</math>; <math>E = G_Y(R^{-1} - I)G_Y^{-1}</math>; the SPECTRAL MAP <math>\text{charpoly}(E) = \text{charpoly}(R^{-1} - I)</math> symbolic <math>\Rightarrow \lambda(E) = 1/\lambda(R) - 1</math>, hence <math>L^* \Leftrightarrow \lambda_{\max}(E) &lt; 1 \Leftrightarrow R &gt; \frac{1}{2}I</math> and margin <math>= 2 - 1/\lambda_{\min}(R)</math> (both directions realized); the pair identities <math>p/q = 2 - (R^{-1})_{kk}</math> exact; the HALF-FILLING LEMMA <math>\lceil S/2 \rceil = N \Leftrightarrow S = 2N - 1</math> for every odd <math>S</math> (the r228 law IS Borodin’s rank condition); the cosine-lattice closed forms symbolic (<math>P = 2^{1-S}(x+1)U_{S-1}</math>, <math> P'(x_j)  = 2^{1-S}S\kappa_j/(1-x_j)</math>) <math>\Rightarrow</math> the dual weight is EXACTLY <math>u^\vee \propto c_j(1-x_j)/ f </math>: <math>+\log f </math> vs <math>-\log f </math> – THE R354 ANTI-CORRELATION OF THE TWO <math>\varphi</math> BLOCKS IS DUALITY ALGEBRA BY DESIGN (<math>d\log u^\vee/d\log F = -1</math> symbolic). FROZEN CENSUS (r356, 34/34): support gate 85/85 bitwise; margin map <math>\leq 9.9 \times 10^{-10}</math>; <math>\text{corr}(\psi_{57}DW, \psi_{57}W) = -0.999998</math>; the honest MAIN VERDICT DUALITY-REPARAM-ONLY (best dual-block leave-out 1.021 <math>&gt; 0.5</math>, no compression: the duality is a REPARAMETRIZATION – the lane goes FINAL to the memo); census-positive RESERVE-LOCALIZED (the local dual <math>2 \times 2</math> block predicts the log reserve without <math>p/q/c</math> readback: +0.9982, puretest +0.9828) and AC-CLASS-EXCLUDED (<math>a_{\rho K} = 1.4222</math> where Lubinsky-type AC universality demands const – the AC hard-edge class measurably excluded); all four dead worlds violate <math>R &gt; \frac{1}{2}I</math> structurally. PRIME.LSTAR.SUBORDINATION.01 stays [O]; the one-object question narrows to the <math>\sim 500\times</math> smallness of the difference of by-design-coupled blocks. NO <math>L^*</math> claim, NO RH claim.</p> |
| v982_dirichlet_matched_frame.py | <p>PRIME.WORLD.DIRICHLET_MATCHED_FRAME.01 [E] (consolidation wave 14, round 357): THE MATCHED DIRICHLET FRAME AND THE SECOND ARITHMETIC – EXACT (no probe imports): the GRH-faithful <math>\chi</math>-archimedean side DERIVED symbolically from <math>\Lambda(s, \chi) = (q/\pi)^{(s+a)/2}\Gamma((s+a)/2)L(s, \chi)</math>: <math>F_A^\chi(\xi) = -\log(\pi/q) + \text{Re } \psi((1+2a)/4 + i\xi/2)</math> for <math>a = 0</math> and <math>a = 1</math> (parity and conductor enter ONLY through the functional equation); the TRIVIAL REDUCTION <math>a = 0</math>, <math>q = 1</math> IS the zeta dictionary (symbolic; bitwise in the sealed record); the CONDUCTOR SHIFT <math>F_A^{q=3} - F_A^{q=1} = \log 3</math> identically (record must-fail m1 at <math>3.8 \times 10^{-15}</math>; wrong-conductor mutant exactly <math>\log(4/3)</math>); the PARITY KERNEL <math>e^{-w/2} \rightarrow e^{-(3/2)w}</math> verified at 30 digits against the digamma (the wrong-parity mutant misses by <math>O(1)</math>). FROZEN CENSUS (r357, 33/33) with tipping mutants: SECOND-ARITHMETIC-LIVES – E-margins ALL POSITIVE on 42/42 for <math>\chi \bmod 3</math> AND <math>\chi \bmod 4</math> (<math>8.1 \times 10^{-6}..3.3 \times 10^{-3}</math>; terminal nf None 42/42 – the same comb that died unmatched at nf 24 in r330): THE R330 DEATH WAS A FRAME ARTIFACT, the living-world evidence goes from <math>n = 1</math> to <math>n = 3</math> arithmetics on 126 windows; the verbatim r330 battery retypes ALL FOUR splits as frame artifacts (HALF_FILLING, RENYI3_C2 32/42 <math>\rightarrow</math> 0/42, SIGMA_DECAY, O_SIGN); mechanisms TRANSFER (two-atom pair = folds (2,4), determinant condition, the r342 dictionary with the <math>\chi</math> offset) while the zeta-graded signatures do NOT (exact half-filling pinning <math>\min C_{\text{ext}} = N_w + 1</math>, saturated reserve, <math>\varphi</math> suppression <math>439\times</math> vs <math>3\times</math>, counting constants break +19/ + 28 %); K2 SPLITS CLEANLY: <math>C_2 = 11.87</math> holds 0/126 (2 families + 3 aspects + 3 arithmetics – the v979 law’s third independent test) while the <math>O(\log m)</math> counting side is zeta-calibrated census; the <math>\chi</math> twin carries, the matched SCRAMBLE breaks (<math>-2.6 \times 10^2</math>): THE WALL NEEDS THE ARITHMETIC. Everything census-side is a finite census on 126 windows, kein Satz; the <math>L^*</math> margin legislation is a statement about how close zeta’s world sits to the wall. NO RH claim, NO GRH claim.</p>                                                                                                                  |

**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

## TFPT verification suite: the journey of ~475 machine-checked scripts

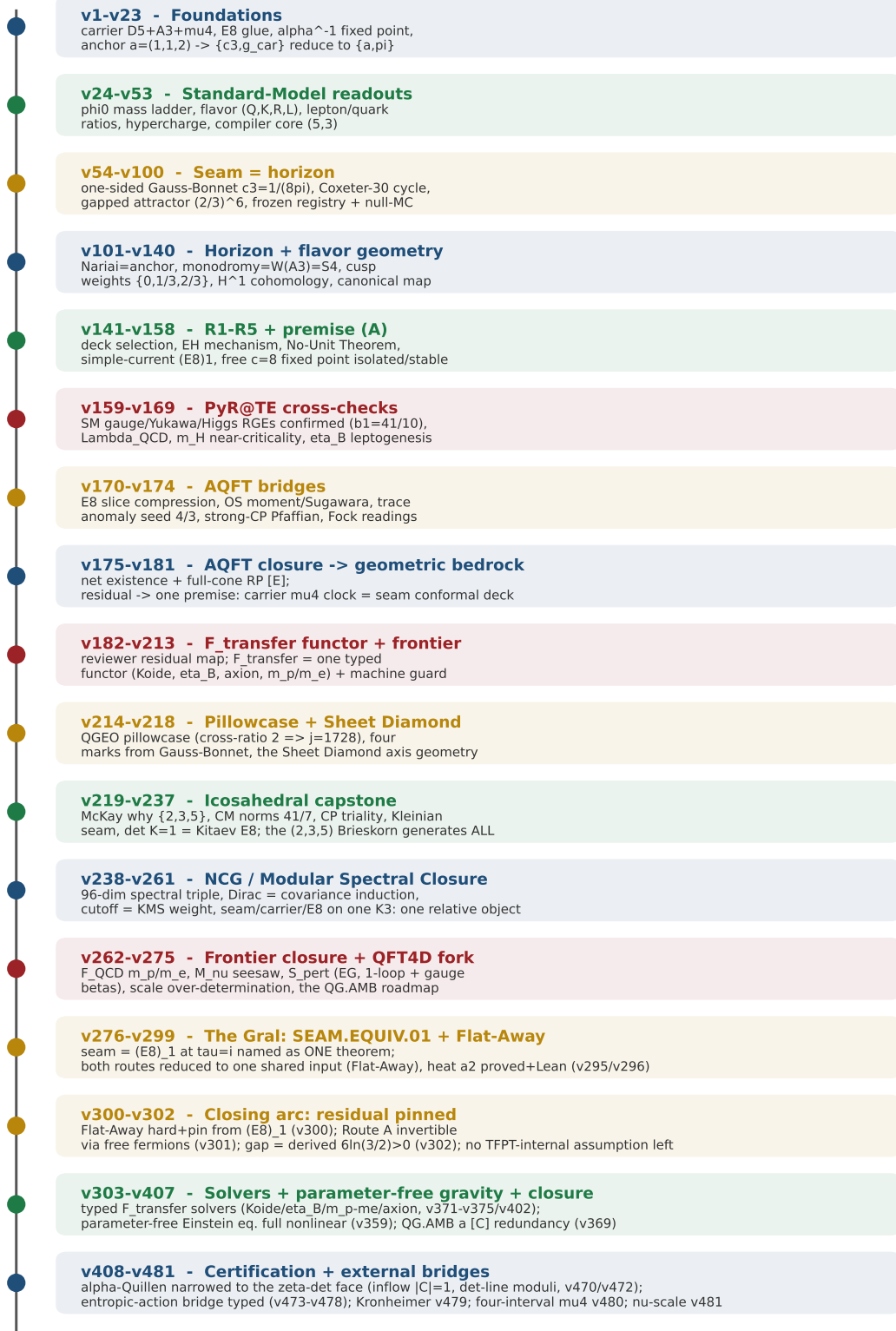


Figure 1: The development timeline of the verification suite — the phases of the journey and what each did, mathematically and physically: from the foundations (carrier,  $E_8$  glue,  $\alpha^{-1}$ , the anchor  $a = (1, 1, 2)$  to which  $\{c_3, g_{car}\}$  reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flux geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the  $F_{transfer}$  functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, the *icosahedral capstone* (the single  $(2, 3, 5)$  Brieskorn singularity generates the whole skeleton), the Modular Spectral Closure, the SEAM.EQUIV.01 closing arc (v276–v302), the typed solvers and the parameter-free gravity round (v303–v407), to the certification round (v408–v481: the  $\alpha$ -Quillen residual narrowed to the continuum  $\zeta$ -det face, the entropic-action bridge typed, the Kronheimer quiver bridge, the four-interval  $\mu_4$  geometry). The

## Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers vN refer to `verification/vN_*.py`.

**2026-08-27 (CXXXVII)** (consolidation wave 14 — rounds r334–r357 frozen: the reviewer-authorized consolidation of the terminal, L2 and L\* discovery rounds graduates as FIVE new modules `v978_terminal_density_martingale.py` / `v979_cover_growth_k2.py` / `v980_lstar_margin_chain.py` / `v981_lstar_borodin_duality.py` / `v982_dirichlet_matched_frame.py` (suite 970 → 975, ledger 1156 → 1161; every module re-derives its exact layer FROM SCRATCH – sympy symbolic + pure Fractions, no probe imports – and re-runs the sealed round verdicts as exact decision logic on the frozen record aggregates with tipping mutants; the eighteen discovery probes stay sealed in `experiments/tfpt-discovery/`, pinned in `rh/INVENTORY.json` and re-verified by `run_rh.py`; single-module tests green 24/24 + 28/28 + 26/26 + 22/22 + 23/23, each < 1 s, byte-identical on re-run). (1) v978 — the terminal density-martingale layer (r339 + r341), NEW PRIME.TERMINAL.DENSITY\_MARTINGALE.01 [E]:  $X(v) = d(v)/d(\text{root})$  is a martingale from MASS CONSERVATION ALONE; the moment dictionary  $E[X_\infty^2] = mM_2$ ,  $E[X_\infty^3] = m^2M_3$ ,  $\max X_\infty = mq_{\max}$  (the terminal target IS a martingale moment; the banked r306  $C_2 = 1.0694$  certifies the  $A = 2$  polylog form on the 57-rung set); the tilted tower  $E[X_\infty^3] = E_Q[\prod \Gamma]$  with steps  $\tilde{p}_c = p_c R_c^3 / \Gamma$  (the untilted reading breaks by an exact Fraction — caught); the exact hand-off  $E_{3h} \leq (mq_{\max})^2 \text{msh}$ ; the algebraic pair ceiling  $\Gamma \leq 4$  via  $4R - R^3 = R(2 - R)(2 + R) \geq 0$ ; the r339/r341 letters honest-negative on the budget side (the target grows  $8\times$  slower than the worst case;  $R^* = 3/2$  a tuning surface); the r324 MEASURED composition stays the end state. (2) v979 — the three-arm cover, the growth law and K2 (r344/r346/r351/r353/r355), NEW PRIME.L2.COVER\_GROWTH\_K2.01 [E] Frame-A-typed: the FAB identity  $F_{AB} = mq_{\max} / \log m$  exact; the data-free K1 formula  $R_{\text{ALG}} = 4^{1/3} \rightarrow$  grid 8/5 interior via exact cube comparisons; the r355 mesh identity  $h - \nu u \in (0, 3/2]$  exact; the first certifying cover (0/51,  $m_0^* = 10^{22.6}$ ), the P02 predictor (51/51), the growth ceiling  $C_{\text{FAB}} = 14.93$  (fresh tranche 0/6; polylog  $m_0^* = 10^{18.9}$ ) — all under the BINDING r353 Frame-A restriction (frame B breaks the ceiling +21% and kills the reserve floor at  $m \sim 760$ : the sliding coverage is FINITE, PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] retyped accordingly); K2 ( $F_{AB} g_{\text{rel}} \leq 11.87$ ) is the sole cross-family survivor, certified NU-free over three aspects (r355) — the gap geometry is the  $\nu$ -free coordinate of the lane. (3) v980 — the closed L\* margin-law chain (r345/r347/r348/r350/r352/r354), NEW PRIME.LSTAR.MARGIN\_CHAIN.01 [E]: the one-line identity  $m'_2(p' + q' - m'_2) = p'q' - c'^2$  (Vieta); the two-level theorem (121/250, 79/250, 72/250 by hand;  $r'_2 = 4375/9559$  = the r345 formula); the pinning theorem (orthonormal top-2 geometry  $\Rightarrow$  the dressed charpoly factors as  $(t - m)(t - mg_{21})$ :  $m'_2 =$  margin IDENTICALLY); the rate-equality theorem in exact ratio form; the  $\rho_K$  identities (toy 1/8, 23/24); the frozen censuses ALPHA\_CLOSED ( $\alpha = a_c + \delta$ , residual 0.033), RHOR\_REDUCED (one object, 500 $\times$ ), PHI\_DICTIONARY\_GO ( $c_{\text{pred}}$  at  $5.5 \times 10^{-4}$ , corr 1.0 — computability, no closed form) and DELTA0\_UNRESOLVED (pool exhausted at  $10^{3.90}$ ): the lane is FROZEN as a specialist problem; the r343/r345 protocol lesson (curvature-honest flatness) recorded as method. (4) v981 — the Borodin particle-hole duality (r356), NEW PRIME.LSTAR.DUAL\_HOLE.01 [E] (the wave's strongest single theorem): the complementation  $\Pi_N^u + G \Pi_{N-1}^{u^\vee} G^{-1} = I$  bit-exact at half filling with the reciprocal dual weight  $u^\vee = 1/(uP^2)$  and a RATIONAL conjugator;



$L^* \Leftrightarrow R > \frac{1}{2}I$  with margin =  $2 - 1/\lambda_{\min}(R)$  (spectral map symbolic, both directions realized); the half-filling law IS Borodin's rank condition (integer lemma); the dual weight is EXACTLY  $u^\vee \propto c_j(1 - x_j)/|f|$  — the r354 anti-correlation of the two  $\varphi$  blocks is DUALITY ALGEBRA BY DESIGN; the honest main verdict DUALITY\_REPARAM\_ONLY stands (no carrier, no compression: the lane goes final to the memo), with RESERVE\_LOCALIZED and the AC-class exclusion ( $a_{\rho K} = 1.4222$  vs the Lubinsky-type constant prediction) banked census-side. (5) v982 — the matched Dirichlet frame (r357), NEW PRIME.WORLD.DIRICHLET\_MATCHED\_FRAME.01 [E] (census on 126 windows): the  $\chi$ -arch side DERIVED symbolically from  $\Lambda(s, \chi)$  ( $F_A^X(\xi) = -\log(\pi/q) + \operatorname{Re} \psi((1+2a)/4 + i\xi/2)$ ; trivial reduction = zeta; conductor shift =  $\log 3$  exact; parity kernel  $e^{-w/2} \rightarrow e^{-3w/2}$  at 30 digits); SECOND\_ARITHMETIC\_LIVES: the r330 death was a FRAME ARTIFACT — the wall holds on 42/42 windows for  $\chi \bmod 3$  AND  $\bmod 4$ , the living-world evidence goes from  $n = 1$  to  $n = 3$  arithmetics, the verbatim r330 battery retypes all four splits; mechanisms transfer while the zeta-graded signatures (exact pinning, saturated reserve,  $\varphi$  suppression  $439\times$  vs  $3\times$ , counting constants) do not: THE  $L^*$  MARGIN LEGISLATION IS A STATEMENT ABOUT HOW CLOSE ZETA'S WORLD SITS TO THE WALL; K2 holds 0/126 (the third independent test); the matched scramble breaks — the wall needs the arithmetic; kein Satz, NO GRH claim. Adjudication and surfaces. Block-B negatives consolidated as prose and row updates only (r334 capacity cap, r336/r340 world-blindness + Hall refutation with the wall-is-global-rescue find, r349 class law inside the v979 third arm, r353 FLOOR\_KILLS binding); PRIME.LSTAR.SUBORDINATION.01 [O] and PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] carry wave-14 notes, markers UNCHANGED; the wave-14 diary section in `tfpt_prime_front`, the wave-14 block in `tfpt_research_contracts`; `rh/`: INVENTORY via the guard generator, `run_rh.py` MODULES  $\rightarrow$  v955–v982, README wave-14 block; website mirror + script index regenerated. The full-suite `run_all.py` pass at 975 modules, `bash build.sh` audit and the manifest regeneration close the wave as the joint exit gate; NO RH claim, NO GRH claim, NO  $L^*$  claim anywhere in either direction; mincut base 4 / refined 5 UNCHANGED.)

2026-08-27 (CXXXVI) (the transport-lift wave — the two external-review rounds on the quantum structure of the seam transport graduate from the experiments tree into the load-bearing suite as `v976_seam_lift_birkhoff.py` and `v977_transfer_unistochastic_wilson.py` (suite 968  $\rightarrow$  970, ledger 1154  $\rightarrow$  1156: new rows TRANSFER.HIDDEN.CIRCULATION.01 and TRANSFER.COHERENT.WILSON.01, both “executed half [E] / selection open [O]”; the exploration probes `transfer_birkhoff_circulation_probe.py` 42/42 and `transfer_coherent_wilson_probe.py` 28/28 stay in `experiments/tfpt-discovery/` as the discovery record). (1) v976 — the Birkhoff fibre and the entanglement-breaking no-go. The Birkhoff (random-permutation) fibre of the seam single step  $B$  (exact sixth root of the v221 kernel,  $B = \frac{1}{2}I + \frac{1}{18}P_{12} + \frac{2}{9}P_{13} + \frac{2}{9}P_{23}$  bit-exact) is exactly a one-parameter line  $t \in [0, 1/18]$ ; the hidden coordinate is the sign-character Fourier coefficient  $\hat{w}_t(\operatorname{sgn}) = 6t$ , populations are exactly  $t$ -free, and no fibre member is chiral ( $w(C_{123}) = w(C_{132})$  forced). The one-step coherence spectrum is  $\{1, 6t, (2/3)^{\times 2}, (1/3)^{\times 2}\}$  and the triangle loop current is an exact eigen-observable  $\Psi_t(A) = 6t A$  — one step plus one current readout measures  $t$  without an ancilla. On the contrast qubit the Choi partial transpose has spectrum  $\{-3t/2, 1/6 + 3t/2, 1/3 + 3t/2, 1/2 - 3t/2\}$ : the deployed sixth-root spectrum sits exactly on the entanglement-breaking boundary because  $2/3 + 1/3 = 1$  (mutated spectra miss by  $\mp 1/12$ ); at the full three-level the family

stays NPT at every tested iteration with  $\lambda_{\min} = -(1/3)(2/3)^n$ ,  $t$ -independent. The no-go that sharpens DYN.UNITARY.DILATION.01: the deployed Kraus dilation  $K_{ij} = \sqrt{T_{ij}}|i\rangle\langle j|$  is measure-and-prepare, hence entanglement breaking with quantum capacity 0 and  $C_0(T) = 0$  exactly — any dilation closing that contract must be a genuinely different object. Shannon KKT certificates:  $p^* = (1/2, 1/2, 0)$  is exactly capacity-achieving for both  $B$  and  $T$  ( $C(T) = 0.0084$  bit,  $C(B) = 0.4890$  bit, ratio 1.71 %) — the surviving classical message is binary on every scale. The defect–Birkhoff value coincidence (weights  $(9, 1, 4, 4)/18$  vs the v652 ladder  $g^2/18$ ) is recorded typed [C] with its honest multiplicity audit; the forcing functor is the named open contract TRANSFER.DEFECT.BIRKHOFF.01. (2) v977 — the unistochastic  $SU(3)$  lift and the 5-12-13 Wilson code.  $B$  is unistochastic: an exact  $U \in SU(3)$  with  $|U_{ij}|^2 = B_{ij}$  entrywise, angles forced by  $B$  ( $\sin^2 \theta_{13} = 2/9$ ,  $\sin^2 \theta_{12} = 1/14$ ,  $\sin^2 \theta_{23} = 2/7$ ),  $\cos \delta = -4/\sqrt{65}$  unique, lift set  $= \{U, U^*\}$  up to rephasing; Jarlskog  $J = \pm 1/27$  exactly, and the real part of every Wilson plaquette is fixed by  $B$  through unitarity — the full rephasing-invariant Wilson data is classical up to the  $\mathbb{Z}_2$  orientation bit  $\text{sgn } J$ . The main plaquette is  $(-5 + 12i)/324$ , normalized  $(-5 + 12i)/13$  with  $5^2 + 12^2 = 13^2$  (atoms:  $5 = g_{\text{car}}$ ,  $12 = \dim \mathfrak{g}_{\text{SM}}$ ,  $13 = \Delta_Q = N(3+2i)$ ); all nine plaquettes close on four Gaussian phase types with norms exclusively  $\{10, 13, 130, 169\}$ . The center bridge:  $\text{disc } \chi_A = 65 = \det(A-I)^2 + (\det A/2)^2 = 4^2 + 7^2$  for the v530 atom matrix,  $e^{i\delta} = (-4 \pm 7i)/\sqrt{65} = (1 \pm 2i)(2 \pm 3i)/\sqrt{65}$ , and the v530 sibling quotients fail the identity — the bridge selects the atom quotient. Rationality selection:  $J_n$  is rational only at  $n = 1$  (the deployed step hits  $3(1 + 2b - 3a^2) = 1$  exactly, so  $J = 1/27$ ); census over 238 valid family points: exactly one full-code point = the deployed  $(2/3, 1/3)$  ( $0.42\% \ll$  the preregistered  $10\%$  kill). No fixed microstep:  $B^2 \neq |U^2|^2$  and a Dirichlet recurrence witness ( $n^* = 967909$ ) shows no single closed unitary generates the relaxing sequence — repeated steps require dephasing between steps, the six-step macro-gate  $U_6 \neq U_1^6$  (the coherent sharpening of clock-vs-walk). The PMNS near-miss stays a negative control ( $240.2551^\circ$  vs  $240^\circ$ ;  $-1/27 \neq -0.02965$ ; no PMNS derivation). Surfaces. Research-contracts document: the DYN.UNITARY.DILATION.01 section gains the executed lift-structure sharpening, the CCC reduced-channel prose and the two former “exploration-only” citations of rounds DCXCVI/DCXCVII are upgraded to \veri citations; origin\_theory gains the honest “what the classical transport does not determine” paragraph next to the v221 recoverability code; tfpt\_2 gains the  $65 = 4^2 + 7^2$  bridge corollary in the v530 center-quotient block. No status marker moves; both contracts stay open; no PMNS/CP/flavor claim.)

2026-08-27 (CXXXV) (the master-route wave — the external bird’s-eye review adjudicated and its promotable core integrated: ONE new verified module plus FIVE new ledger rows organize the entire 4D frontier into a single constructive route. (1) The conditional 4D dimension selector graduates as v975\_dimension\_selector\_4d.py (suite  $967 \rightarrow 968$ , ledger  $1149 \rightarrow 1154$ ; 16/16 sympy-exact,  $\sim 0.5$  s): under the named TFPT-compatible axiom set A1 ( $d > 2$ : propagating local gauge fields), A2 (dimensionless Yang–Mills coupling), A3 (real self-dual/anti-self-dual 2-form sectors: the Euclidean Hodge star an involutive endomorphism of 2-forms), A4 (chiral Weyl representations: even  $d$ ), the dimension  $d = 4$  is UNIQUE and MINIMAL on the exact scan window  $d = 2..12$  — and the selector is OVERDETERMINED: A2 alone  $= \{4\}$  ( $[g_{\text{YM}}] = (4 - d)/2$  forced symbolically from dimensionless-action power counting, unique zero 4) and A3 alone  $= \{4\}$  (the explicit  $6 \times 6$  star on  $\Lambda^2(\mathbb{R}^4)$ :  $*^2 = I$ , eigenvalues  $\pm 1$  at multiplicities  $3/3$  — the real  $\mathfrak{su}(2)_+ \times \mathfrak{su}(2)_-$  split), while A1 & A4 alone give  $\{4, 6, 8, 10, 12\}$  (chirality

does NOT select 4 by itself — honest ablation ledger); Weyl chirality certified by exact Clifford algebras  $d = 2..7$  (even  $d$ :  $\gamma_*$  anticommutes,  $\gamma_*^2 = I$ , traceless; odd  $d$ : the product is CENTRAL and scalar); signature typed honestly (Lorentzian  $*^2 = -1$  on 2-forms — the real split is a EUCLIDEAN/OS-side statement); mutants caught (the wrong power counting  $[g] = (6 - d)/2$  selects  $\{6\}$ ; the “A4 alone” overclaim refused); FIREWALL:  $\mu_4$  and  $g_{\text{car}}$  enter NOWHERE (“ $\mu_4 \Rightarrow 4D$ ” stays forbidden per DIMENSION.UPLIFT.FIREWALL.01), and AXIOM PROVE-NANCE from  $\{c_3, g_{\text{car}}\}$  is the registered OPEN half — NEW ledger row DIMENSION.SELECTOR.4D.01 [C] (a conditional selector, not a derivation; no 4D construction contract advanced). (2) Four new research contracts register the genuinely new content of the review (all [O], no scripts, no marker moves): SEAM.SIMPLECURRENT.GENERATOR.01 (the seam-gap compression: ONE normalized simple-current/spin-field intertwiner with convergence + energy bounds + locality + braiding + Q-system closure generates the ENTIRE 128-dim extension sector — the v973 residual  $N1 + N2 + N4$  collapses onto one convergence theorem; 1+1D in and out); TFPT4D.LATTICE.ACTION.01 (route step (2) of the firewall ordering made concrete: ONE fully explicit finite 4D lattice family  $S_a = S_{\text{Wilson}} + \bar{\psi} D_{\text{GW}} \psi + S_{\Phi} + S_{\text{seam/top}}$  with SEVEN machine-checkable finite gates T1–T7 — gauge invariance, positive transfer matrix, EXACT seam restriction, anomalies incl. global  $SU(2)$ , index/generations, mirror separation, nonvanishing connected 4-point — plus the twistor flat-direction observable test as a kill test); SEAM.DETLINE.UNIFICATION.01 (the determinant-line hypothesis:  $\text{Res}_{\text{seam}} \det D_{4D} \cong \det D_{\text{seam}}$  with connection AND bulk holonomy = the discrete orientation phase — would unify seam extension, anomaly freedom, generation index and CP orientation as projections of ONE object; the continuum Bismut–Freed identification is exactly the critical missing step); FTRANSFER.GENERATING.01 (the constructive successor of the FR.TRANSFER.01 guard: all four transfer bridges  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$  must arise from ONE generating functional  $W[J] = \log Z[J]$  — the guard stays binding, no upgrade permitted). (3) The programme section “the TFPT4D master route” in `tfpt_research_contracts` organizes the seven conditions (seam compatibility, RP, local unitary dilation, chiral anomaly freedom, one relevant direction = the ANCHOR.VGEO.02 torsor, approximate quantum-Markov recovery per Fawzi–Renner arXiv:1410.0664 and Chen–Rouzé arXiv:2504.02208, nontriviality), the dynamics correction (fundamental order: local unitary amplitudes  $\rightarrow$  decoherence  $\rightarrow$  Markov matrix; the DYN.MARKOV.EMBED.01 kernel result v971 stays the dephased shadow, the DCXCVII unistochastic/Jarlskog-sign finding cited as exploration-level motivation only), the programme ordering P1–P8 mapped onto existing + new rows, and the kill-criteria list (the route may die early); binding separation: RH and the PRIME.\* programmes stay separate — a green RH probe would be neither necessary nor sufficient for the 4D physics. `docs/OPEN_PROBLEMS.md` updated in the same change (the 4D frontier now points at the master route and the five new rows). The full-suite `run_all.py` pass, `bash build.sh` audit and the manifest regeneration follow as the joint exit gate; NO marker moves anywhere — the only new verified piece is the conditional selector [C])

2026-08-27 (CXXXIV) (seam/alpha promotion wave, part 3 of 3 — both named faces of the alpha target now carry computed witnesses: the certified content of the three finished alpha exploration probes graduates from the experiments tree into the load-bearing suite as `v974_alpha_faces_computed.py` (suite 966  $\rightarrow$  967, ledger 1148  $\rightarrow$  1149; embedding convention as in v960–v970: the THREE probes embedded BYTE-EXACT with full-source SHA-256 pins gated

and byte-warded against the experiments tree, executed verbatim in isolated namespaces — seam\_bfk\_conical\_det 21/21 SPEC d06b3bbe VERDICT match-modulo-local-factor, laughlin\_pump\_em1 20/20 SPEC f9b5cb0b VERDICT pump-quantized-matches-chern, quillen\_rigidity\_chain 18/18 SPEC b4ee0dd0 VERDICT chain-shadow-ok; plus the module-own exact section S0 (sympy/Fraction/mpmath, no probe imports): the BFK closed forms independently re-derived ( $\det'_\zeta(\Delta_{S^1}) = l^2$  and  $\det'_\zeta(|D|) = l$  from  $\zeta_R(0) = -\frac{1}{2}$ ,  $\zeta'_R(0) = -\frac{1}{2} \ln 2\pi$ ; Dirichlet  $\det = 2a$ ; the  $N = 4$  KMS reassembly as exact rationals in  $c_3$  —  $R_0 = 16c_3 \operatorname{circ}(2, -1, 0, -1)$ ,  $\det' R_0 = 2^{16}c_3^3$ , glued product  $(2\pi)^2 = 1/(16c_3^2)$ , gluing constant  $2^{-4}$ ; spec multisets  $\{0, 2, 2, 4\}$ ,  $\operatorname{Tr} B^k = 4^k + 2 \cdot 2^k$  vs the v484  $\operatorname{Tr} C^k = 4^k + 2(-2)^k$ ,  $B + C = 2(I + S)$ , doublet ratio  $4/\ln 2$  with the  $c_3$  cancellation exhibited on a generic symbol; the v485 scale dictionary exact), the alpha anchor re-derived at 50 digits ( $8b_1 = 164/5 = (4/5)41$  exact Fractions,  $q(D_5) + q(A_3) = 2$ , UNIQUE root on the scanned log window,  $\alpha^{-1} = 137.03599921684071$ , split residual  $6.4 \times 10^{-58}$ ), the inflow bookkeeping exact (16 Majoranas = 8 complex copies,  $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$ ,  $k_Y = 5/3$ ,  $(1/k_Y)41/6 = 41/10 = b_1$ ) and a probe-free coarse FHS Chern certificate ( $C(M=1) = 1$ ,  $C(M=3) = 0$ , integer ward); tipping mutants: a single-byte hash mutation caught by the pin, the wrong KMS unit ( $2\pi \rightarrow 4\pi$ ) breaks the scale identity by exactly  $-2 \ln 2$ , the  $b_1$  mutant breaks coefficients and split residual; single-module test green 17/17 checks in  $\sim 5$  s, byte-identical on re-run modulo WALL lines, green under both interpreters): (1) the BFK face executed — the Burghlea–Friedlander–Kappeler Mayer–Vietoris determinant formula made executable on the seam circle cut at the four  $\mu_4$  marks: gluing constant EXACTLY  $2^{-N}$  (symbolic  $N = 1..4$ , general lengths, Laurent-polynomial identity + 50-digit numerics), massless reassembly  $\det'_\zeta(\Delta_{S^1}) = l^2$ , jump matrix = inverse mark Green matrix  $R(m) = (G_m|_{\text{marks}})^{-1}$  exactly; against the v484 mark matrix NO global proportionality (kernels in SWAPPED  $\mu_4$  channels: gauge zero mode vs sign rep) but channel-wise EXACT — the diagonal-zeta face of the merged target is thereby SHARPENED to the channel-swap question; the v485 scale channel reproduced exactly via the KMS unit  $2\pi = 1/(4c_3)$ . (2) EM1 computed — the cited inflow step “boundary level = bulk response” now carries a computed lattice witness on the v367 collar: pumped charge == edge spectral flow == Streda == Chern == 1 per flux quantum (Wannier winding residual 0.0, Streda  $\Delta N = C n_\phi$  exact integers, conjugation mutant sign-coherent across all four readings, wrong-band projector breaks quantization); single complex copy, the 16-copy collar as exact bookkeeping arithmetic only. (3) the finite Quillen identity two-sided — Berry curvature integral == Poincaré–Lelong divisor integral == +1 over the  $U(1)$ -twist torus (both references, both sizes), the difference an EXACT form ( $3.3 \times 10^{-16}$ ), transition winding == Chern (Dai–Freed); typed conditional on the quasi-free rigidity premise (measured, not proven — SEAM.STATE.RPMIXING.01); typed DAG: the remaining open content of ALPHA.QUILLEN.EXACT.01 = {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification} — no additional analytic unknown. LEDGER: NEW ALPHA.QUILLEN.FACES.01 [E] (the computed witnesses); UPDATED ALPHA.QUILLEN.EXACT.01 [O] (stays open — “why this functional” not closed), SEAM.CONTACT.UNIT.01/.02 (the BFK resummation named in v485 now executed), EM.WARD.01 and ALPHA.QUILLEN.INFLOW.01 (the EM1 step computed); NO marker moves anywhere;  $\alpha^{-1} = 137.0359992168$  stays [E] regardless. The full-suite run\_all.py pass, bash build.sh audit and the



manifest regeneration follow as the joint exit gate)

**2026-08-27 (CXXXIII)** (seam/alpha promotion wave, part 2 of 3 — the seam route narrowing: the certified content of the three seam-route exploration probes graduates from the experiments tree into the load-bearing suite as `v973_seam_route_narrowing.py` (suite 965  $\rightarrow$  966, ledger 1147  $\rightarrow$  1148; embedding convention as in `v960–v970`: the **THREE** discovery probes embedded **BYTE-EXACT** and executed verbatim with the printed **SPEC** SHA-256 hashes pinned and gated — `mmst_wzw_coverage` 19/19 **SPEC** `e6f392cb`, `bcov_measure_uniqueness` 20/20 **SPEC** `0875f39f`, `woit_battery_mmst` 19/19 **SPEC** `451bd470`; byte wards vs the experiments copies live; plus the module-own exact section **S0** — the coverage arithmetic exact (the 240  $E_8$  roots enumerated as 112 integer + 128 half-integer spinor roots,  $248 = 120 + 128$  with  $120 = \binom{16}{2}$  bilinears,  $c = 16 \times \frac{1}{2} = 8 = \text{rank}$  with  $8 \leq 8 \leq 16$ ,  $\det \text{Cartan } D_8 = 4 \rightarrow E_8 = 1$  by explicit simple-root Gram matrices, index  $2 \times 2 = 4 = |\mu_4|$ ), the uniqueness linear algebra as exact rationals on a unimodular-mixed 18-parameter space (block ranks **EXACTLY** (9, 8, 2, 2), joint 13, gauge 4 inside every kernel  $\Rightarrow$  moduli kernel **EXACTLY** 1; the full ablation ledger (1, 3, 1, 2, 3, 10) reproduced; the dropped- $C_4$  mutant opens the kernel to 3 — **CAUGHT**), the skeleton mechanisms exact (bipartite  $SHS = -H \Rightarrow$  every odd function of  $H$  — in particular  $\text{sign}(H)$  — vanishes identically on even distances, integer-exact on all odd powers, the next-nearest-hop mutant caught;  $PHP = -H$  on a two-leg strip  $\Rightarrow K_{\text{top}} = -K_{\text{bot}}$  exact), and the verdict bars as exact decision logic with tipping mutants (a fake **COVERAGE\_FULL** is **REFUSED** by the nonempty **N1–N4** gate, the kernel-dim mutant refused, hash mutation caught); single-module test green 25/25 gates (21 module-own + 3 pattern gates + runtime) in  $\sim 1$  s, byte-identical on re-run): **THE SEAM ROUTE NARROWING**. (1) The **MMST** coverage carved (**SEAM.EQUIV.MMST.01** [O] update): the cited Osborne–Stottmeister theorem (**CMP** 398 (2023) 219, arXiv:2107.13834) **COVERS** the bilinear half of the seam target — the  $\mathfrak{so}(16)_1$  fermion-bilinear **WZW** currents converge **STRONGLY** on the finite-particle core (Thm 5.1 displayed for  $u(1)$ , nonabelian via Cauchy–Schwarz Eq. 229 + the closing remark incl. Majorana algebras), Virasoro/Koo–Saleur (Thms A/4.11/4.16), correlators (Thms B/C), polynomial error estimates  $\text{Error}^2 \leq C 2^{-2\delta N}$ ,  $\delta \in [0, 2]$  — and does **NOT** cover four **NAMED** items: **N1** the  $\mu_4$ /GSO simple-current extension  $SO(16)_1 \rightarrow (E_8)_1$  as a **NET** statement (the 128 weight-1 spinor currents are not bilinears), **N2** the twist/spin-field sector (explicitly deferred), **N3** the one-sided chiral edge net of a 2+1D bulk, **N4** the OS/holomorphy selector  $\det K = 1$  vs 4; **NEW MEASUREMENT**: the edge-correlator convergence rate on the **v367** collar is  $p = 3.48$  (Richardson 3.1–4.2, preregistered band  $[0.5, 4.0]$ ), error  $6.3 \times 10^{-2} \rightarrow 2.7 \times 10^{-6}$  over  $N_x = 16..256$ ; the  $M = 3$  control does **NOT** converge (min rel dev 0.836); **HONEST**: the ratio observable superconverges past  $\delta_{\text{max}} = 2$  — consistent with the cited **UPPER** bound, the band’s upper edge calibration-informed; the coverage gates are executable transcriptions of a human literature adjudication. (2) The twistor measure rigidity (**SEAM.EQUIV.TWISTOR.01** [O] update): on the computable **v508/v515** sector the 18-parameter deformation of the declared measure data (contact normalisation, 4 sector weights, residue normalisation, 4 linking charges, 4 centre positions; 4 gauge directions  $\Rightarrow$  14 moduli) has executed-constraint Jacobian ranks **C1** clock 9 / **C2** lockstep+integrality 8 / **C3** charge 2 / **C4** null-ideal 2, joint rank 13  $\Rightarrow$  the moduli kernel is **EXACTLY** 1 = the uniform sector-weight shift  $dv = (1, 1, 1, 1)$ , mapping to the untwisted Okubo square  $Q^{(0)} = 36(1, 2, 1, 0, 0)$ , annihilated by every executed certificate

functional BY CONSTRUCTION (exactly flat to all orders); ablation ledger (1,3,1,2,3,10); honest: NOT a BCOV derivation, the  $W_2 : W_{13} = 4:3$  Harvey–Moore tester typed NOT-TRANSCRIBABLE. (3) The cross-route OS skeleton (NEW SEAM.EQUIV.SKELETON.01 [E]): ALL five WOIT kill-test main invariants ( $\alpha$ : forced  $\eta = +i$ ,  $\Theta^2 = +1$ , bond RP;  $\beta_1$ : GSO 2-torsion;  $\beta_2$ : OS quotient + KMS  $\lambda_{\max} > 1$ ;  $\beta_3$ : perp descent;  $\gamma$ : mirror flip, multiplicity-free, mark incidence) hold VERBATIM on the p+ip collar edge — 5/5; ALL divergences control-typed and tracing to ONE mechanism (the twistor lattice’s exact chiral checkerboard  $C(\text{even}) = 0$  vs the collar’s  $K(\text{even})$  up to 0.1736), independently confirming v519-R4’s lattice-placement typing from the second route; the collar’s top edge realizes the anti-chiral state EXACTLY ( $\max |K_{\text{top}} + K_{\text{bot}}| = 7.5 \times 10^{-16}$ ); the  $M = 3$  trivial collar loses the skeleton first at GAMMA.MARK.INCIDENT — the skeleton tracks the topological phase. LEDGER: NEW SEAM.EQUIV.SKELETON.01 [E] (executable skeleton, typed FULL/PARTIAL per test — an executable-level narrowing, NOT a route closure); UPDATED SEAM.EQUIV.MMST.01 [O] (residual now the NAMED N1–N4 list), SEAM.EQUIV.TWISTOR.01 [O] (measure rigid on the computable sector up to ONE named flat direction), SEAM.EQUIV.01 [O] (parent prose: the open structure concentrated on the same named extension question from both sides); NO marker moves — both routes and the parent stay [O]. Python-only per GATE.WOLFRAM.02; the full-suite run\_all.py pass, bash build.sh audit and the manifest regeneration follow as the joint exit gate)

2026-08-27 (CXXXII) (seam/alpha promotion wave, part 1 of 3 — the seam interaction front: the certified content of three exploration rounds of 2026-08-27 on the v529 Fidkowski–Kitaev seam toy and the v903 mixing slice graduates from the experiments tree into the load-bearing suite as v972\_seam\_interaction\_front.py (suite 964  $\rightarrow$  965, ledger 1144  $\rightarrow$  1147; embedding convention of v969/v970: the THREE discovery probes embedded BYTE-EXACT with pinned source SHA-256 hashes, byte-warded against experiments/tfpt-discovery/ and executed verbatim with the printed gate counts, VERDICT letters and SPEC SHA-256 prefixes pattern-gated — majorana\_rp\_class 20/20 SPEC bcdfcde6 VERDICT class-empty-on-basis, quartet\_avoiding\_os 18/18 SPEC 6ced7be3 VERDICT os-viable-on-avoiding, seam\_rp\_rigidity 25/25 SPEC d09cd983 VERDICT rp-admits-mixing; plus the module-own exact section S0 (sympy / GF(2) integer certificates, no probe imports) — the Jaffe–Pedrocchi DECIDABILITY LEMMA on a 4-Majorana toy (the 9 products  $\theta(M_a)M_b$  land on 9 pairwise-distinct crossing masks with unimodular coefficients, so the crossing matrix  $B$  is UNIQUE; the self-adjoint bond  $H_c = s i g_1 g_2$  extracts to  $B = \text{diag}(0, s, 0)$  and exactly ONE sign certifies — the wrong-sign mutant caught), the ODD-SPLIT ARITHMETIC (on  $\mathbb{Z}_{16}$  exactly the 2 mirror-fixed straddling quartets split 2+2 and the 4 off-center ones 1+3; the odd block  $[[0, b], [\bar{b}, 0]]$  has  $-B$  PSD iff  $b = 0$  — the obstruction; a 2+2 straddle needs only the good orientation; the flipped-sign-law mutant caught), the GF(2) GENERATION BOOKKEEPING (pair vectors on 16 points have rank exactly 15, the even algebra has  $2^{15} = 32768$  monomials; the within-halves single-cut restriction has rank exactly 14 — deficit EXACTLY 1, witnessed by the left-parity functional), the S1 WINDOW ARITHMETIC ( $33 = 24+9$ , the  $32+1$  split, rank-nullity  $33 - 16 = 17$ ; the kernel value  $-1/(1+c^2)$  at  $c = \tanh(1/2)$  re-derived symbolically as  $-\cosh^2(1/2)/\cosh(1)$ , float pin  $-0.8240271368$ ; both wrong-form mutants caught) and FOUR VERDICT BARS with tipping mutants (a mutated embedded-source byte breaks the pinned SHA-256; the three live records compose to exactly the claim triple; a tipped gate count 19/20 and swapped verdict letters do NOT

compose); single-module test green 20/20 gates in  $\sim 4$  s with both the venv and the system interpreter, byte-identical on re-run)): THE SEAM INTERACTION FRONT FROZEN. (1) The odd-split law (SEAM.INT.ODDSPLIT.01 [E], toy level): the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (arXiv:1406.1384; Wei–Li–Xiang arXiv:1601.01994) is made DECIDABLE on the v529 toy (unique  $B$ ), certifies the free FK Hamiltonian on all 8 bond cuts ( $\eta = -i$  forced), scores a PERFECT confusion matrix against measured RP over 35 cells (23/12/0/0), and finds the  $\mu_4$ -covariant quartic space (dim 464) EMPTY on the basis with the support-level exclusion pinning the full 1820-dim quartic space — THE DISCOVERY: the v529 straddle law REFINES to the ODD-SPLIT LAW (JP predicted RP survival on CENTER-straddled cuts,  $m = 4$  model, axes 7/15, 2+2 mirror split; direct measurement confirmed (37, 0, 0) at  $g = 0.5/1/2$ ; the v529 anchor ladder reproduced exactly). (2) OS viability on the avoiding family (SEAM.INT.OSAVOID.01 [E], toy level): the clock-invariant  $m = 4$  member’s quartet-avoiding cut family generates the FULL even seam algebra (rank 15, dim 32768; single-cut members  $m = 2/6$  typed deficit 1) and carries a POSITIVE OS/GNS structure at  $g = 1$  (min  $+4.4 \times 10^{-5}$ , GNS 37/37, distances monotone) while the straddled control genuinely fails ( $-6.9 \times 10^{-2}$ ); the  $\mu_4$  clock maps the avoiding pair  $\{3, 11\}$  onto itself exactly — the WOIT.OS.TWISTOR.01 kill-test-2 counterweight is NARROWED (an OS-viable cut geometry exists, selected by the toy’s own clock), not removed. (3) The RP mixing census (SEAM.STATE.RPMIXING.01 [E], census): the v903 exclusivity is RAY- and REFLECTION-relative, not slice-wide — on the full 33-dim  $C_6$ -covariant mixing slice  $\theta_{abT}$  sees exactly ONE coordinate ( $a_J$ , kernel value  $-1/(1 + c^2) = -0.8240271368$  at  $c = \tanh(1/2)$ ; 32/33 invisible  $\Rightarrow$  a 32-dim MARGINAL RP hyperplane  $\{a_J = 0\}$ ) and the  $\theta_S$  Hermiticity-defect map has rank 16  $\Rightarrow$  a 17-dim first-order-neutral subspace with STRICT finite- $s$  witnesses ( $\lambda_{\min} = 0.1558$  at  $s = 1/8$  — the very direction  $\theta_{abT}$  kills); the v903 anchor reproduced ( $a_J = +0.04068013$ ). LEDGER: NEW SEAM.INT.ODDSPLIT.01 [E] + SEAM.INT.OSAVOID.01 [E] + SEAM.STATE.RPMIXING.01 [E]; UPDATES appended to SEAM.INT.FKTOY.01 (straddle law  $\rightarrow$  odd-split law), WOIT.OS.TWISTOR.01 (counterweight narrowed, contract stays Open) and SEAM.STATE.DERIVATION.01 (exclusivity typed ray-/reflection-relative). HONEST: NO marker moves anywhere; SEAM.EQUIV.01 stays [O]; the v529 MANDATORY FENCE (one toy, one interaction class, [C] flat-band parent) is inherited verbatim by all three results; JP is sufficient-only; the S1 census is sector-truncated ( $\deg \leq 2$  Grams +  $\deg \leq 4$  spots). Python-only per GATE.WOLFRAM.02. NO RH claim.)

2026-08-27 (CXXXI) (the external deep-review round — dimension / dynamics / gravity firewalls and the 4D construction ladder: the review diagnosis “chiral 1+1D seam net  $\ncong$  local, interacting 3+1D quantum world” is implemented as eight new contract IDs in `tfpt_research_contracts` plus one new load-bearing module. (1) DIMENSION.UPLIFT.FIREWALL.01 (standing rule, in force): no 1+1D theorem closes a 3+1D claim; every contract closure states the spacetime dimension of hypotheses AND conclusion; audit census recorded (SEAM.EQUIV.01 is 1+1D in and out — even fully proved it is a boundary building block, not the 4D theory; the v957 kinematic uplift does not close it; the anomaly-inflow bulk is invertible/topological, never to be silently equated with physical spacetime); the realistic route ordering (seam as boundary module  $\rightarrow$  genuine 4D Euclidean lattice  $\rightarrow$  OS package  $\rightarrow$  chirality/no-mirror  $\rightarrow$  only then BW/CHM + entanglement gravity  $\rightarrow$  bulk–seam equivalence) is registered as the programme skeleton. (2) SEAM.BULK4D.RECON.01



[O]: the previously implicit seam→bulk theorem made explicit (4D local algebras/Schwinger functions; locality/microlocality, Poincaré covariance + spectrum condition, spin-statistics + CPT, both helicities, clustering, nontrivial S-matrix, unique operator dictionary). (3) the dynamics split DYN.MARKOV.EMBED.01 / DYN.UNITARY.DILATION.01 (“three meanings of dynamics”: relaxation  $T^n$  / dissipative semigroup  $e^{t \log T}$  / unitary  $e^{-itH}$ ): the finite kernel half of MARKOV.EMBED is executed [E] as v971\_markov\_embedding\_generator.py (15/15, sympy-exact) —  $Q = \log T$  of the v221 seam kernel is an EXACT classical Markov generator:  $e^Q = T$  by spectral calculus, row sums symbolically zero, off-diagonals exactly  $\{\ln(9/8), 2 \ln 3, 2 \ln 3\} = \{0.117783, 2.197225, 2.197225\} > 0$  (the reviewer’s independently verified numbers), uniformization certificate  $Q + (4 \ln 3)I \geq 0$  entrywise  $\Rightarrow e^{tQ}$  positivity-preserving doubly stochastic for all  $t \geq 0$ ,  $\text{spec}(Q) = -\text{spec}(H_{\text{mod}})$  (gap  $= \Delta = 6 \ln(3/2)$ ), negative control: second eigenvalue  $-1/10$  gives  $\det = -1/200 < 0$  — provably NOT embeddable; the field level and the local size-consistent unitary dilation (Stinespring + Lieb–Robinson + OS continuation) stay [O]; binding precision: the shared F\_transfer relaxation shape is a universal CONTRACTION CLASS, not one physical clock (v723/v724/v777). (4) QFT4D.OS.RECON.01 [O] (the genuine 4D Euclidean lattice OS package) with the binding phrasing rule: the v381 all-orders leg is a *local perturbative construction under the stated EG/BRST hypotheses* (Duch weak adiabatic limit cited, hypotheses unverified for the concrete model; no confinement / mass gap / asymptotic charged states / nonperturbative Higgs vacuum / full S-matrix follows) — tfpt\_2 and tfpt\_5 phrasing restricted accordingly. (5) CHIRAL4D.NOMIRROR.01 [O] (explicit GW/overlap chiral lattice operator; local gauge-invariant measure, perturbative + global anomaly freedom, no Witten  $SU(2)$  anomaly, no mirrors, correct index/charges, stable continuum limit; Mastropietro arXiv:2303.02790 as calibration). (6) GRAV.NONCIRCULAR.01 (standing rule, in force): BW is a theorem WITHIN a local Poincaré-covariant QFT (Mund/Morinelli counterexample landscape cited), never a generator of Lorentz spacetime from a modular flow; every BW/CHM use declares assumption-vs-result; audit census: the entanglement-equilibrium Einstein derivation (v323/v358/v359) is typed ASSUMPTION (conditional on the 4D package, never citable as evidence for it), QG.AMB.01 correctly declared, the seam KMS  $2\pi$  chain dimension-typed, the intrinsic-BW programme the honest result-direction attempt. (7) OBS.TRANSDUCTION.01: the transduction principle (internal eigenvalue + nonvanishing projection  $BP_r \neq 0$  + physical coupling = observable signature; without the middle arrow neither non-observation falsifies nor observation confirms; minimal model machine-checked at exploration level in fo01, uniform functional blind to  $10^{-14}$ , character channel recovers 64/729 to  $10^{-10}$ ; the ACT+Planck birefringence proximity typed coincidence-level pending the shared-action proof). (8) PRED.JOINTLIKELIHOOD.01 [O]: evidence reorganized from hit-counting into overdetermined relation bundles (flavor sum rule;  $r = 3(1 - n_s)^2$ ,  $\alpha_s = -r/6$ ; one normalized axion action; character-resolved transfer) with the statistical demand (joint likelihood, atom-induced covariance, effective dof, prespecified null + formula grammar, real holdout); the v100 “13/13,  $< 10^{-30}$ ” statement retyped as a look-elsewhere quantifier under its declared null — the 2026-06-09 freeze protects FUTURE measurements, it does not retroactively make historically known data blind. Docs corrected in the same change: docs/OPEN\_PROBLEMS.md (the Einstein-derivation overstatement rewritten to the conditional Jacobson-route typing with the explicit does-NOT-follow list; “zero open internal mechanisms” fixed to match the contract scope fence — ledger wins; the all-orders phrasing restricted) and docs/FALSIFICATION.md (the

null-model section retitled and rescoped, freeze and kill tests unchanged). Ledger +9 rows; suite 963 modules; bash build.sh audit and the manifest regeneration are the joint exit gate)

2026-08-27 (CXXX) (consolidation wave 13 — the extraction repair and the terminal composition: the certified content of discovery rounds 323–327 (incl. the r326 Lean round and the r323 clean abort) graduates from the experiments tree into the load-bearing suite as v970\_extraction\_and\_composition.py (suite 962  $\rightarrow$  963, ledger 1133  $\rightarrow$  1135; wave-4 embedding convention: the FOUR discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — fa\_provenance 36/36 SPEC 9a6696f8 (r324-pre, sealed FIRST, then superseded mid-round by the binding reviewer course correction and BANKED), qmax\_m2\_origin 36/36 SPEC dc36cacb (importing the embedded r324-pre), extraction\_order 18/18 SPEC 31277f91, group\_mass\_cap 34/34 SPEC 11e4fd40 (importing the embedded r324-pre and r324); full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 in pure Fractions with NO Lean call — the r324 interpolation  $M_3 \leq q_{\max} M_2$  and  $M_2 \leq q_{\max}$  exact (toy slack EXACTLY  $1/36$ , DOUBLE equality at the flat profile; both reversed-mutant directions caught), the  $F_A$  identity  $F_A B \log m = m q_{\max}$  as exact constructional algebra ( $F_A$  DE-BLACK-BOXED as the rank-local normalization of  $m q_{\max}$ ; the dropped-gauge mutant caught), the r327 group ledger exact on the sealed toy (partition  $5 + 1 = A_1 = 6$ , cap chain  $m q_{\max} = 12/7 \leq n_g \times h_{gn} = 20/7$  slack  $8/7$ , two-ancestor bound  $5 \leq 2 \times 3$  slack 1; the drop-group and wrong-window-max mutants break by EXACTLY 1), the r325 onset formula  $\alpha^* = (n_g + 1)D_0/2$  as an EXACT TOY STABILIZATION (the tent-window read equals the exact atom sum  $35/16$  for every  $\alpha \geq 5/8$ , differs below, and is CONSTANT under dyadic mesh refinement  $D_0 \rightarrow D_0/4$ ; the nearest-neighbour mutant caught), the  $m_0^*$  composition arithmetic (the r324 log-space solver rebuilt verbatim landing  $10^{59.6}$ ; the exact exponent algebra  $158/1000 + 14/1000 = 172/1000 < 224/1000$  and  $1 - e/2 = 914/1000 \geq 888/1000$ ; the r328B chain-audit gate S0-T5d: the block-level need  $454/501$  and the chain bar  $94/501 \approx 0.188$  exact, the verdict survival  $172/1000 < 94/501$ , the non-sequitur  $0.888 \times 1.002 < 0.908$  caught, the chain-honest  $m_0^* = 10^{238}$  from the same solver, the false-fire window mutant caught; supercritical mutants return NONE and the DISCLOSED pre-freeze linear-space bug — a search capped at  $10^{18}$  that misses the root — is rebuilt and caught), the SIX SEALED VERDICT BARS as exact decision logic with tipping mutants at every live clause boundary, and the wave-13 composition gate; the r326 Lean round consumed as a REPORT and the r323 abort as a NOTE; single-module test green 26/26 gates (22 module-own +4 pattern gates) in  $\sim 2$  s, byte-identical on re-run): THE EXTRACTION REPAIR AND THE TERMINAL COMPOSITION FROZEN. (A) The clean abort (r323): the planned  $L^*$  indefinite-signature fork was aborted by the binding reviewer adjudication ( $L^*$  internally paused at allocation 0%) at the END OF THE DESIGN PHASE, BEFORE any write access — no probe, no measurement, no rh/ file, nothing counts as a round (post-abort suite 73/73 with 103/103 pins). (B) The terminal composition (r324, the pre-work banked as r324-pre; r327 the anatomy): the round ran under a binding mid-round course correction, disclosed in full — the original  $F_A$ -boundedness contract was executed and sealed FIRST (fa-bounded-distributional: KS  $0.0982 \leq 0.125$ ,  $p_{\max} = 0.090$ , envelope  $C_F = 3.357$ ; the honest tension disclosed: the per-rank  $p$  values at the named spikes are  $0.005/0.000/0.000$ ),

then superseded (the cleaner decomposition is  $\text{QMAX} \times M_2$ ); the binding contract fired pileup-grows-subcritical ( $e_{\text{tot}} = +0.172 < \text{CRIT} = 0.224$ , halves  $+0.141/-0.160$  both subcritical, the second **FALLING** — decided; by the reviewer contract a sufficient, first-class outcome): the pointwise  $M_2$  export **REFUTED** ( $C_2 = 2.2557$  fails 7/39 — exactly the banked spike set at  $+35.42\%$ ) while the distributional r302 export holds **GO** (KS 0.023/0.016, tighter than the r302 core); the mechanism **HALF AND HALF** — the  $O(\log m)$  **SCALE COUNT CERTIFIES** pointwise ( $C_{\text{NSC}} = 2.0258$ , 0/39) but the per-scale mass  $O(1/m)$  is **REFUTED** ( $C_{\text{PIL}} = 9.3583$ , 11/39): the near-critical windows are **ONE NEAR-MAXIMAL HEAVY SOURCE SCALE** at the argmax block (kz53:  $s_0 = 13.78$  of  $G = 13.02$  at  $n_{\text{sc}} = 5$ , the ladder median), not multiscale convergence; the composition typed **MEASURED**:  $\sum_j q_j^3 \leq 8.941 (\log m) m^{+0.172}/m^2 \Rightarrow N_3 \geq m^{0.914}/\sqrt{8.941 \log m}$ ,  $N_2 \geq N_3$  (the r306 exact chain)  $\Rightarrow N_2 \geq m^{0.888}$  for all  $m \geq m_0^* = 10^{59.6} \Rightarrow$  the full chain to terminal\_positive\_main —  $m_0^*$  **ASTRONOMICAL**, the gap  $(274, 10^{59.6})$  **IS THE DISCLOSED EXTRAPOLATION HYPOTHESIS**, no cofinal claim (r328B independent chain audit, carried as a chain annotation with the record sealed: the bar 0.224 used the **ATOM-level** need 0.888, the block-level need is  $908/1002 \approx 0.9062$  — the chain-honest bar is 0.188, **THE VERDICT SURVIVES**  $0.172 < 0.188$ , the chain-honest  $m_0^*$  is  $10^{238}$ , and the going-forward bar is 0.188); r327 then located the spike structurally (cap-partial with anatomy **SOURCE**, median named ratio  $1.057 \geq 0.5$ ): on **ALL 65** rungs the heaviest group of the argmax block is **EXACTLY ONE  $\beta/\omega$  FOLD PAIR** (one bulk + one window atom at one position, histogram  $[0, 65, 0]$ , median alignment 1.000 — the ancestors **REINFORCE**); at kz53 the single pair carries 88.8% of the argmax block at gap 0.076 — **THE SPIKE IS LITERALLY ONE BULK/WINDOW COINCIDENCE** (kz67 the exception shape: 6 groups); the  $\Lambda$ -pair cap **REFUTED** as a mid-ladder polylog ( $C_{LV} = 1.1838$  fails 19/39, named 0/4 — the exact two-ancestor algebra holds at slack 0.0, the **CONSTANT** does not close), the direct group cap fails at both  $A$ , **THE GROUP COUNT CERTIFIES** pointwise ( $C_{NG} = 2.6351$ , 0/39, named 4/4 — the **THIRD** certified  $O(\log m)$  count of the lane);  $e_{\text{route}} = +0.106 < 0.224$  but the halves **STRADDLE** ( $+0.411/-1.183$ ) — heavy-group-unbounded does **NOT** fire (the heavy-group coordinate **DECAYS** with depth: a shallow/mid phenomenon); the r324 **MEASURED** chain stays the end state, and the named direction is the **COINCIDENCE GEOMETRY**: where can the single heavy coincidence **SIT** (source geometry, not group count or multiplicity). (C) The extraction repair (r325) and the Level-C architecture (r326): r325 adjudicated the three reviewer variants for the R319 cofinality seam — elementwise-stabilization-go primary + co-letters canonical-mesh-rebuild-go + signed-defect-correction-go(sign) + anchor-only-insufficient: the repair is a **QUANTIFIER STATEMENT**, not a mesh theory — on the native v749 class (dyadic step-function autocorrelations, seed-sealed) **ALL THREE** channels (comb/arch/pole) of the canonical tower windows stabilize **EXACTLY** at the predicted finite anchor onset  $\alpha^* = (n_g + 1)D_0/2$  **AND** are **CONSTANT** under dyadic mesh refinement (comb  $3.3 \times 10^{-18}$ , arch  $1.5 \times 10^{-15}$ , pole  $2.0 \times 10^{-17}$ ) — **NO** mesh-cofinal ladder, **NO** transport,  $H_{\text{cof}}$  **IS NOT NEEDED AND IS REPLACED AS THE TARGET ROUTE**; variant B is the construction substrate of A (v749 T1.4b at rel dev 0.0), variant C the quantified density step (one-signed per-channel defect inside the rigorous interpolation envelope  $(D^2/8)\|F''\|K_{\text{ch}}$ , no wall margin anywhere); the anchor-only floor real ( $+1.77 \times 10^{-4}/+1.59 \times 10^{-4}$  — the R319 false-floor mechanism on own elements); r326 implemented the architecture in rh/lean/RH/Elementwise.lean (merged and green; lake with exactly 8

intentional sorry warnings, run\_rh -fast 75/75): sourceExact\_buildPrimeWindow PROVED (the wave-12 reviewer target “exactly one construction theorem” — SourceExact ELIMINATED as a free assumption from the extraction route; built windows provably carry their source), the comb stabilization elementwise PROVED (onset  $a_0(f)$  predefined from the element’s support alone, corpus gauge  $2\Lambda/\sqrt{n}$ , no mesh quantifier; honesty tie combRead\_eq\_window\_channel), weil\_nonneg\_of\_windowlocal PROVED (the EXTRACTION WITHOUT THE LADDER: one finite instantiation per element — Euclid anchor + native mesh) plus weil\_nonneg\_of\_bordered (composition through the named Prop BorderedCompressionBridge, no truth commitment); SORRY CENSUS  $5 \rightarrow 8$  with THREE NEW typed Level-C statements (arch\_elementwise\_stabilization + pole\_elementwise\_stabilization classical, specFamily\_sourceExact\_completion classical + definitional) — the wave-12 reservation “the Level-C distance appears in no sorry” PARTIALLY DISCHARGED; the five legacy sorries byte-identical. THE HONEST FORMULATION OF THE NEW CONNECTION (binding, verbatim): “two window-local lemmata (over the canonical family) + the elementwise architecture + cited classics (density, v912 continuity, the Weil criterion)  $\Rightarrow$  RH; open links: the arch/pole transcription (classical), the compression bridge, the source completion (definitional).” LEDGER: NEW PRIME.EXTRACTION.ELEMENTWISE.01 [E] (the R325 GO + the R326 implementation with the honest formulation verbatim); NEW PRIME.L2.RENYI3.MEASURED\_COMPOSITION.01 [O] (the r324 measured chain, the certified counts  $C_{NSC}/C_{NG}$ , the SOURCE anatomy, the coincidence-geometry direction); UPDATED PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] and PRIME.L2.RENYI3.PROVENANCE.01 [O] (the two-part provenance question executed —  $F_A$  de-black-boxed, r324-pre banked, the share route composed MEASURED), PRIME.REDTEAM.EXTRACTION\_AUDIT.01 [E] and PRIME.LSTAR.SUBORDINATION.01 [O] (the wave-13 Lean notes: census  $5 \rightarrow 8$  with the three-level assignment — Level C now named typed statements; lstar\_subordination byte-identical,  $L^*$  stays paused); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO change ships with this promotion — r326 is already merged and green (census  $5 \rightarrow 8$ ). Papers: the wave-13 diary section sec:elementwise-extraction in tfpt\_prime\_front, the wave-13 block + the wave-13 state notes in tfpt\_research\_contracts, the new section “Elementwise extraction and the measured composition (v970)” in rh/paper/rh\_program.tex (the Lean-theorem table with the proved kernel pieces prominent, the honest-formulation remark, the coincidence-geometry remark, the lane-status remark; the three-level remark and the Lean census section updated to  $5 \rightarrow 8$ ) + recompiled PDF; rh/README.md wave-13 block with the lane status (terminal: the MEASURED composition + the coincidence-geometry direction; extraction: the architecture stands, 3 typed Level-C rests;  $L^*$  paused; Lean: 8 sorries, all typed); rh/INVENTORY.json + make\_inventory.py extended additively (the v970 pin + the r324-pre/r324/r325/r327 consumption notes, all pre-existing pinned entries byte-exact; 110  $\rightarrow$  111 entries); run\_rh.py MODULES  $\rightarrow$  v955–v970, full RH suite green. The full-suite run\_all.py pass at 963 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate; NO RH claim anywhere in either direction)

2026-08-27 (CXXIX) (consolidation wave 12 — the red-team / fork morning: the certified content of discovery rounds 317–322 (incl. the r319 extraction-chain audit and the r320 Lean repair) graduates from the experiments tree into the load-bearing suite as v969\_forks\_and\_redteam.py (suite 961  $\rightarrow$  962, ledger 1131  $\rightarrow$  1133;

wave-4 embedding convention: the FOUR discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — exception\_families 38/38 SPEC 04fbe5c0, indefinite\_fork 25/25 SPEC f2d98683, continuous\_coordinate 39/39 SPEC e68883ad (importing the embedded r317), antiphase\_sign\_law 25/25 SPEC 761b51d4 (importing the embedded r318); full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the exact two-sided concentration bracket  $q_{\max}\Phi_{H1} \leq \rho_2 \leq \Phi_{H1} = (F_AB)^2$  re-proved from scratch with both mutant directions caught, the U1–U3 red-team witnesses re-established in EXACT ARITHMETIC with no Lean call (U1: the  $B = -1$  window represents the empty spec against the terminal type while the retyped bridge refuses it and accepts the faithful  $B = 5/7$  witness; U2: the mesh-level-0 total node collision reduced exactly to the three rational  $e$ -facts  $2 \leq e$ ,  $3 \leq 2e$ ,  $e \leq 4$  on the certified series bracket  $685/252 \leq e \leq 685/252 + 1/35280$ , kill  $0 < 0$  exact; U3: the adversary  $Z_{\text{loc}} = |F| + 1$ ,  $runs = []$  demands  $(|F| + 1)^2 < 5/7$ , false for every  $F$  since  $(|F| + 1)^2 \geq 1 > 5/7$  exact), the r320 separation-discipline and canonical-split certificates (witness at anchor 2, atoms  $\{2, 3, 4\}$ , mesh level 4 — the first satisfiable level, via the integer facts  $2^7 < 3^5 < 2^8$  and the reduction  $1/8 < 2/5$  exact; the level-0 failure  $4/3 < 4$ ), the SIX SEALED VERDICT BARS as exact decision logic with tipping mutants at every live clause boundary, and the wave-12 composition gate; the r319/r320 Lean rounds consumed as REPORTS, their artifacts in rh/lean/ re-verified by lake build + the RH suite; single-module test green 22/22 gates (18 module-own + 4 pattern gates) in  $\sim 3$  s, byte-identical on re-run): THE RED-TEAM MORNING AND THE TWO FORKS FROZEN. (A) The extraction-chain audit and repair (r319/r320): the independent reconstruction of the chain from the two lemmata to Weil/RH found THREE KERNEL-CHECKED TYPE INCONSISTENCIES three levels above the boxes (U1 bridge $\times$ terminal via  $B = -1$ ; U2 bridge $\times$ L\* via the mesh-level-0 total node collision and  $p = X - 1$ ; U3 pair\_margin\_main forces its source predicate EMPTY — the r273 disease one level up) plus the mesh-vs-anchor COFINALITY SEAM (cofinal\_prime\_windows proves the anchor direction at fixed mesh, not the  $H_{\text{cof}}$  refinement direction — THE FALSE COFINALITY DIRECTION IS DOCUMENTED, NOT SOLVED); the honest chain verdict is BINDING: “two lemmata  $\Rightarrow$  RH” does NOT survive the literal reading — literally the two lemmata give ONLY window-local master positivity for the opaque, currently witness-less MainWindow; r320 REPAIRED the interface completely (bridge retype with  $u/B$  fidelity + separation discipline + budget\_pos; the additional arch/border finding sealed by the new opaque SourceExact; pair\_margin\_main retyped onto the canonical extraction with signRuns\_sum + canonical\_split PROVED; three permanent sorry-free guards, axiom census kernel-checked without sorryAx; the sorry-free witness witness\_represents; sorry census  $5 \rightarrow 5$  with two typed RE-TYPES, lake build green). (B) The reviewer adjudication (binding, adopted on every surface): the proof graph has THREE LEVELS and the two true holes are named WINDOW-LOCAL from wave 12 on — Level A (local window mathematics, PROVED:  $L^*(w) + T(w) \Rightarrow \text{MasterPositive}(w)$ , cleanly typed per r305/r320; “for every canonical finite window, L\* and terminal positivity imply augmented prefix positivity — this is a window-local theorem”), Level B (the universal window family — THE two arithmetic WINDOW-LOCAL holes:  $\forall w \in W_{\text{canonical}} : L^*(w) \wedge T(w)$ ), Level C (extraction to Weil/RH, OPEN: source-exact realization + admissible transport + convergence/stabilization in the correct directed order + density/-continuity for the Weil criterion); two binding caveats: the sorry census does

NOT measure the Level-C distance (the missing extraction architecture appears in NO sorry, because the correct theorem statements do not yet exist), and SourceExact — a good firewall — must NOT stand as a free assumption in a final proof: the named open Lean target is CanonicalPrimeWindow + exactly ONE construction theorem sourceExact\_buildPrimeWindow (load-bearing, not formalization polish); the typed per-layer reviewer assessment supersedes the wave-11  $4 \rightarrow 5$  scalar (finite dictionaries 9.5/10, formal window-local architecture 9/10, terminal mechanism 6/10, L\* mechanism 2/10, source/extraction architecture 2–3/10, complete RH proof path  $\sim 3/10$ , audit quality 9/10 — “stronger as a window-local research program; as a complete RH proof chain currently less far than previously assumed — the removal of a shortcut that did not exist”).

(C) The fiber fork (r317/r321): fork (b) sealed whac-a-mole(kz53, kz83) BY CONTRACT (the blind gap rule recovers the REAL spike family  $B = \{kz55, kz67\}$  with finite certificates  $C_B = 1.0536$ , but class A stays empty at bar distance 0.017 and kz53/kz83 stay uncovered; the census find: the  $F_A$  top-3 are EXACTLY  $kz53/kz83/kz67 = 2.47/2.39/2.38$  — one source-pure rank-local QMAX coordinate ranks the complete near-critical family on top, but as a CONTINUUM: the exception-family form is the wrong statement shape); fork (a) in the r317-sharpened form fired sliding-bound-go( $G_{SQ}$ ):  $\sum_j q_j^3 \leq 1.3056 F_A(w)^2 (\log m)^2 / m^2$  for  $m \geq 73$  with 0/39 mid-ladder test violations AND all four named r316/r317 violators INSIDE at reserves 7.0..9.6; the constant algebra-derived and source-pure ( $b = (\max_{\text{cal}} B)^2 = 1.1426^2$ , no target value in the calibration); honest:  $C_{\text{impl}} = 7.97$  is  $7.5\times$  looser than the r306 first-5 constant — FORM, not sharpness; the pure-algebra transfer does not close ( $B$  rising +0.122); scramble rejected only via the r317 class side condition; the provenance question now the two-part  $F_A/\text{qmax}$  split. (D) The base fork (r318/r322): r318 p2-main-specific — P1 BANKED AS LANGUAGE (the index statement a total restatement of L\*, 10/10 instances both truth values; the global inequality vacuous; the invariants world-blind), classical sign regularity DEAD on MAIN (premise-fails-on-main); r322 algorithm-artifact — the ( $D3, D4$ ) sign law and the r312 97.6/2.4 anatomy are LEAST-NORM-PROXIMAL DYKSTRA-BASIN properties, not psd-solution-set properties (sealed random starts carry modal pair  $(0, 2)$ ; the law not identity-forced — exact Fractions, positive forced value); the named residual: the SELECTION GEOMETRY. LEDGER: NEW PRIME.REDTEAM.EXTRACTION\_AUDIT.01 [E] (the audit + repair + the honest chain reading + the full reviewer adjudication verbatim, incl. “the false cofinality direction is documented, not solved”); NEW PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] (the theorem candidate with the disclosed looseness and the R324 lane note); UPDATED PRIME.L2.RENYI3.PROVENANCE.01 [O] (the fork executed both ways), PRIME.LSTAR.SUBORDINATION.01 [O] (wave-12 state + the binding lane pause: L\* internally PAUSED at allocation 0%, reviewer decision — the selection geometry not pursued, reopening only under the six documented r322 conditions, the external memo candidate the two-weights sampling formulation  $\|J_N\| < 1$ ; the hole typed WINDOW-LOCAL), PRIME.LSTAR.ARCHITECTURE\_ADJUDICATION.01 [E] (wave-12 red-team note + the assessment-table supersession pointer); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO change ships with this promotion — r319/r320 are already merged and green (census 5, two typed retypes); the named open target CanonicalPrimeWindow + sourceExact\_buildPrimeWindow registered in README/paper; rounds R324 (terminal QMAX $\times M_2$ ) and R325 (extraction repair) IN FLIGHT, deliberately NOT consumed. Papers: the red-team/fork diary section sec:forks-redteam in tfpt\_prime\_front, the wave-12

block + the  $L^*$  wave-12 state in `tfpt_research_contracts`, the new section “Red team and the two forks (v969)” in `rh/paper/rh_program.tex` (the honest-chain remark, the THREE-LEVEL remark with both caveats, the SourceExact-target remark, the sliding-bound theorem candidate, the basin-lesson remark, the typed reviewer-assessment table, the lane-status remark; the “two true holes” re-named window-local throughout) + recompiled PDF; `rh/README.md` wave-12 block with the lane status ( $L^*$  PAUSED; terminal sharpened to  $QMAX \times M_2$ , R324 in flight; extraction fork R325 in flight; Lean: 5 sorries, the two true WINDOW-LOCAL holes) + the Level-C construction-theorem TODO; `rh/INVENTORY.json` + `make_inventory.py` extended additively (the v969 pin + the r317/r318/r321/r322 consumption notes, all pre-existing entries byte-exact; 107 entries); `run_rh.py` MODULES  $\rightarrow$  v955–v969, full RH suite green (86/86). The full-suite `run_all.py` pass at 962 modules, the final bash `build.sh` audit and the manifest regeneration follow as the joint exit gate; NO RH claim anywhere in either direction)

2026-08-26 (CXXVIII) (consolidation wave 11 — the architecture adjudication: the certified content of discovery rounds 305–316 (incl. 310b) graduates from the experiments tree into the load-bearing suite as `v968_architecture_adjudication.py` (suite 960  $\rightarrow$  961, ledger 1127  $\rightarrow$  1131; wave-4 embedding convention: the TEN discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — `renyi3` 27/27 SPEC `3bb365e153033ea8`, `fixed_head` full 28/28 / `smoke` 27/27 SPEC `ec2bb0086f31ac73`, `block_green` 30/30 SPEC `d51478504a645269`, `paired_cone` 33/33 SPEC `f8d99877acb562b1`, `blockgreen_nontriviality` 34/34 SPEC `fac7a8dfec9eb81f`, `blockgreen_membership` 32/32 SPEC `6c32f74962eef7af`, `renyi3_proof_fork` 32/32 SPEC `6505dd10d5cce9fd`, `signed_cubic_flux` 29/29 SPEC `841b319682af06c7`, `phi3_functional` 29/29 SPEC `92d35a3a4186d330`, `two_regime_bound` 35/35 SPEC `5c28b12b1c3998c5`; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 — the r306 HILL/LAGRANGE CHAIN (both Lagrange witnesses dev 0 EXACT, the power-mean slacks  $1/64$  and  $37/32768$  EXACT, the uniform equality case sharp, the flip mutant  $-1/64$  NEGATIVE caught) with the Rényi BRIDGE on the equality-tight rational witness (every comparison exact, the reversed-chain mutant caught), the r314 OPENING-FLUX LEMMA  $F_1 = g^3 - 3g^2 + 2g^3 \equiv 0$  on five rational witnesses plus the exact flux telescope (deviation the rational 0 on three toy blocks; the dropped- $2g^3$  mutant lands  $-2g^3 = -54/343$  EXACT), the r314 COLLISION COUNTING IDENTITY  $3p_1p_2 - 2p_3 = 8(n + 3n(n-1))$  EXACT on four mult-2 toy blocks (both sides 760; the mult-1 mutant breaks by the exact factor 8: 95 vs 760), an r311-CLASS EXACT RATIONAL FARKAS MINICERTIFICATE (PD target  $[[1, -1], [-1, 2]]$  in the span with the FORCED negative generator weight  $(2, 3, -1)$  — the sign wall in miniature; dual  $Y = [[0, 2], [2, 1]]$  with  $\langle Y, G_i \rangle \geq 0$  and  $\langle Y, Q \rangle = -2$  EXACT; the sign-flipped mutant target IS in the cone and the same  $Y$  fails on it — no overreach), the TEN SEALED VERDICT BARS as exact decision logic on the frozen record aggregates with tipping mutants at every live clause boundary, and the FOUR-LEVEL COMPOSITION GATE (a passing B2–B4 would hand the lane to architecture B, a firing r312 GO would keep Lane A open, a surviving two-regime bound would resolve the provenance — each tip changes the composed adjudication, the claim split is NOT hard-wired), no probe imports; single-module test green 30/30 gates (20 module-own +10 pattern gates) in  $\sim 50$  s, byte-identical on re-run up to the runtime line): THE ARCHITECTURE DAY FROZEN IN THE REVIEWER’S



**BINDING FOUR-LEVEL STRUCTURE.** Level 1 — real formal theorems (Lean, rh/lean/, already built and green; NO Lean change ships with this promotion): `lstar_terminal_implies_master` (r305:  $L^* + \text{terminal } q_N < 1 \Rightarrow$  full master positivity via Schur bordering + principal submatrices; `augmented_prefix_positive` and `free_window_positivity` now COROLLARIES — no third arithmetic axiom); the four closed Inertia theorems (`psd_base`, `positive_prefix_firewall` incl. a from-scratch Sylvester criterion, `sylvester_pullback`, `half_filling_boundary`); the real PrimeWindow construction (r310:  $\text{nodes} = \log p^k$ ,  $\text{weights} = \Lambda(p^k)$  DERIVED via `mathlib vonMangoldt`); source exactness by `rfl` THROUGH derivation + mass conservation of ANY folding map (`Finset.sum_fiberwise` — aggregation is a quotient, not a projection) + the STRONG stabilization form onto the Weil prime side (r310b); the r309 rank-one update identities; Hill monotonicity + the Rényi-3  $\Rightarrow N_2$  bridge; SORRY CENSUS  $9 \rightarrow 5$  with clean bookkeeping — the two TRUE holes stand alone (`lstar_subordination`, `terminal_positive_main`). Level 2 — certified finite statements: the Rényi-3 inequality  $\sum q^3 \leq 1.069 (\log m)^2 / m^2$  on 57 rungs (r306, growing reserve  $-0.322$ ;  $A = 1$  fails exactly at `kz53/kz67` — `polylog-SQUARED` is the honest form); the fixed-head census  $77/77$  (r307,  $\min 2.19 \times 10^{-3}$ , amplification  $3.1 \times 10^4$ ); the block-Green identity on the finite objects (r308, toy  $t^2$  pin  $4/7$ , `f64`  $4.0 \times 10^{-14}$ ) + strictness of the cone with EXACT RATIONAL FARKAS CERTIFICATES (r311, `MINI#0 + SM1#0`,  $0/16$  in-span PD samples decompose); paired-cone soundness on 20714 steps (r309, 0 violations, 56 % certified); fold multiplicity EXACTLY 2 + 93 % far-triple cancellation (r313); the signed cubic identity with VANISHING boundary term (r314, decided in exact Fractions ON THE REAL w9 DATA, every deviation the rational 0, `f64`  $4.5 \times 10^{-17}$  on all 57 rungs); the 21 small- $m$  certificates (r316,  $C_{\text{small}} = 1.0694$ ). Level 3 — negative architecture decisions (verbatim): fixed-head-dead (r307 head-grows: no fixed  $r \leq 8$ , best reserve  $6.5 \times 10^{-6}$  four decades under the bar — the one-atom anatomy a COORDINATE, not a DIMENSION); paired-cone-no-induction + paired-cone-world-blind (r309 B1–B4: B2 partial 56 %, B3 not reached, B4 missed — B does NOT overtake A); the r308 discriminator = the budget sign / a chordal restatement (r311: the  $\Delta$ -sparsity graph CHORDAL on every size, the trivial dual  $Y_t = e_t e_t^T$  closes the stalls two-sidedly); coefficient-sign-wall — LANE A CLOSED AS A CONE LANGUAGE with the mechanism NAMED (r312: block-psd membership without rank-one-SDD membership, obstructed at the budget/border sign; Farkas mass on the border/budget row,  $t$ -share  $0.583/0.725$ ; the entire obstruction in 2.4 % SDD-forbidden negative cross-mixings); floquet-expanding (r313: med sr  $1.0600$ ,  $0/57$  contracting — no transfer-operator contraction form remains open at chain level); triple-type-majorants-wrong (r313); phi3-all-blind (r315: the entire break on `kz55+kz67`, FCIX  $0.955/0.915$  vs  $0.629$  — exactly where the flux cancellation dies); two-regime-dead (r316, with the anatomy: the obstruction family cuts ACROSS the FCIX stratum, `kz53` at bulk-normal FCIX; the r306 first-5 protocol was LOAD-BEARING; `kz55/kz67` near-one-block, TOP1  $0.558/0.785$  vs  $0.18$ ). Level 4 — open mechanisms (typed): the constructive PSD-G rule (closed as a lane, open as a question), the signed cubic flux theorem, the quantitative source functional, the `crossing_budget` `mathlib` gap, the definitional source bridge,  $L^*$ , the terminal. LEDGER: NEW PRIME.LSTAR.ARCHITECTURE\_ADJUDICATION.01 [E] (the four-level structure verbatim); NEW PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01 [E] (ONLY the exact update identities: determinant dictionary, Sherman–Morrison chain,  $\pm(\eta \pm e_t)$  border split, local Schur reserve, signed source updates — per the reviewer’s

binding claim split NOT one TRANSFER title); NEW PRIME.SOURCE.TENSOR.MECHANISM.01 [O] (the open question whether a source-pure cofinal control arises from them, registered AFTER r313/r316 with the documented negatives); NEW PRIME.L2.RENYI3.PROVENANCE.01 [O] (the pointwise GO as a certified finite statement + the open provenance question; trilogy state: identity exact, functional blind, two-regime dead; the R317 fork verbatim: a near-critical family coordinate OR a certified exception family — reviewer adjudication pending); UPDATED PRIME.LSTAR.CLOSED\_FUNCTIONAL.01 [O] (the Lane-A closure note), PRIME.LSTAR.SUBORDINATION.01 [O] (wave-11 state: the two formal holes, the source interface, the architecture adjudication), PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 (Lean restructuring note: sorries  $9 \rightarrow 5$ ); no other marker moves, mincut base 4 / refined 5 UNCHANGED. The universality-class framing typed: the base is fine-metric (metric-only, twin invariance everywhere), the fiber is recursive (epstein holds, scramble breaks) — shared algebra, not necessarily a shared mechanism; the reviewer maturity raise  $4 \rightarrow 5$  (formal architecture 9.5) documented as a REVIEWER ASSESSMENT, not a machine-checked fact. Papers: the architecture-adjudication diary section `sec:architecture-adjudication` in `tfpt_prime_front` (the wave-10 “in flight” spots resolved historically), the wave-11 block in `tfpt_research_contracts`, the new four-level section “The architecture adjudication (v968)” in `rh/paper/rh_program.tex` (one subsection per level, the Closure/Source theorem tables, the universality-class remark, the no-go catalog extended by eight wave-11 rows, the Lean status brought to the 5-sorry census) + recompiled PDF; `rh/README.md` wave-11 block with the lane status (A closed as cone language; B closed, identities banked; C parked; fiber: the r306 GO stands, provenance open, R317 fork at the reviewer; Lean: 5 sorries, two true holes); `rh/INVENTORY.json` head-synced (100  $\rightarrow$  101 entries: the v968 pin + the r306–r316 consumption notes, all pre-existing entries byte-exact); `run_rh.py` MODULES  $\rightarrow$  v955–v968, full RH suite green. The full-suite `run_all.py` pass at 961 modules, the final bash `build.sh` audit and the manifest regeneration follow as the joint exit gate run by the coordinator; no round is in flight at this cut — R317 waits on the reviewer fork; NO RH claim anywhere in either direction)

2026-08-26 (CXXVII) (consolidation wave 10 — the cascade closure and the documented lane stop: the certified content of discovery rounds 301–304 graduates from the experiments tree into the load-bearing suite as `v967_12_cascade_closure.py` (suite 959  $\rightarrow$  960, ledger 1126  $\rightarrow$  1127; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed `-smoke` stage with the printed SPEC SHA-256 hashes pinned and gated — `neff_target` 32/32 SPEC 6f8cc40420c57165, `unif_target` 30/30 SPEC 36df942427d45fe4, `atom_target` 26/26 SPEC 375e9f2b8e6c1697, `shortrange_law` full 37/37 / smoke 32/32 SPEC 2cc5d23fdaa75346; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 — the r303 MARGIN-INVARIANCE ALGEBRA (all four target margins  $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} = 3/50$  EXACT on the rational slope set through the three gated product identities r300/r301/r302; the halved-need mutant breaks the  $D \leftrightarrow \text{NEFF}$  invariance by EXACTLY  $\text{sl}_{L1} = 1/5$  — the exact root of the refuted  $\frac{1}{2}$ -conjecture; the record-decimal chain 0.055/0.908/0.094/0.888 EXACT with the disclosed  $1/1000$   $m_{\text{ATOM}}$  rounding artifact; the  $\sigma$  decomposition  $0.198 = 0.055 + 0.143$  EXACT; the  $\frac{1}{2}$ -conjecture refutation 0.099 vs 0.055, dev 0.044 > the sealed 0.005), the r302

**COHERENCE IDENTITY**  $(1 + CV^2) \text{surv}^2 n_{\text{eff}}^{\text{atom}} = n_{\text{act}} \chi$  on two rational toys with the r301 count identity attached (the lag-0 double-count mutant breaks by 5/7 EXACT, the swapped- $D/Q$  mutant by 48/35 EXACT), the r304 CHI-LAG DECOMPOSITION  $\chi = 1 + 2 \sum_k T_k/Q = D/Q$  on both toys (5/7 and 5/19 EXACT; factor-1 mutant 1/7 EXACT), the r304 ZERO-SUM TAUTOLOGY as a gate  $(1 + 2 \sum_{\text{all}} \rho_k \equiv 0$  for mean-removed profiles; factor-1 and uncentered mutants 1/2 and 3/4 EXACT; the truncated net covariance  $NC(1) = 0.556$  EXACT,  $NC(16)$  recomposing to 0.710 within the disclosed 2/1000 rounding of the pinned  $0.712 < 1$ ; the PERIOD-4 COMB CERTIFICATE: every strong lag  $j \in (4, 16]$  sits at  $j = 4m$  (+) or  $j = 4m+2$  (−) and the sealed  $k_0$  rule finds NO  $k_0 \leq 8$ ), the −/−/+ /+ SIGN-PATTERN CERTIFICATE with the kz18/kz23 Fractions pins and the honest rho-sign 41/42, the four sealed verdict bars as exact decision logic with tipping mutants at every live clause boundary, and the lane-stop composition gate, no probe imports; single-module test green 27/27 gates (23 module-own +4 pattern gates) in 1.3 s, byte-identical on re-run): **THE CASCADE RETYPED, THE LANE STOP DOCUMENTED** — *the rejections and retypings ARE the frozen content.* (A) **THE r301 NEFF ANATOMY** (the wave-9 rest executed):  $n_{\text{eff}} = n_{\text{act}}/(1 + CV^2)$  EXACT (ward  $\leq 6.8 \times 10^{-16}$  on 47 worlds) with the PERFECT count link  $n_{\text{act}} = m$  on 42/42 — the effective-carrier count IS the constructive level-2 block count (+1.002); the whole Rényi order family grows at ONE exponent (+0.963/ +0.926/ +0.894/ +0.738: ECHT anti-concentration, not a broadening tail); the margin is jackknife-STABLE (0/42 below NEED, no margin-fragile); sealed verdict neff-split(B1|B3): the growth statement became the BOUNDEDNESS statement unif-target ( $\text{slope}(1 + CV^2) \leq +0.094$ , measured +0.039, margin 0.055); neff-derived missed ONLY the sealed CV-flat clause (+0.039 > 0); B2 closed honestly (the  $|dc|$ -weighted discrepancy is  $10 \times$  the raw  $D_{\text{rank}}$  and near-flat: the r300 bridge lived on the RAW positions). (B) **THE r302 UNIF ANATOMY** (the FIRST DERIVED of the lane): the  $CV^2$  rise is STRUCTURELESS, the normalized block profile is profile-stationary (KS  $0.043 \ll 0.125$  over a  $3 \times$  depth range) with an EXACT  $1/N$  transient onto  $m_2^\infty = 1.973$  (held-out dev 0.002; the r299 pointwise-cconv negative was the wrong convergence notion — convergence in DISTRIBUTION); the r301 depth flag resolved EXACTLY per half ( $0.982 = 0.929 + 0.053$ ,  $0.802 = 1.030 - 0.228$ : the  $CV^2$  head, not the count); the coherence identity EXACT with  $\chi = 0.63$  DESTRUCTIVE and falling ( $\rho_1 = -0.22$  its atom-level mechanism)  $\Rightarrow$  unif-derived(B1+A2) under the sealed clause, ALL hypotheses typed MEASURED; the relocated rest: atom-target (+0.888 needed, +0.942 measured). (C) **THE r303 REGRESS AUDIT** (the retyping round): **THE THREE 0.055 MARGINS ARE ONE ALGEBRAIC NUMBER** —  $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} = +0.0547$ , invariance devs  $\leq 9 \times 10^{-16}$ ; the core is  $S = \sigma^* - \text{sl}_D = +0.0547$  (the r299 DIAG margin); the  $\sigma$  level is NOT the core ( $\sigma^* - \sigma = +0.1976 = S +$  ratio surplus +0.1429 EXACT) and the  $\frac{1}{2}$ -conversion conjecture is REFUTED  $\Rightarrow$  regress-confirmed: the cascade r297→r302 is RETYPED as an exact reduction dictionary (coordinate finding), not six proof steps; **THE HARD REVIEWER RULE** (binding from r303 on, verbatim): *a round counts as progress ONLY if it adds NEW information, not if it re-expresses the same slack in a new coordinate.* **THE FIRST CAUSAL COORDINATE**: the mechanism ladder on 1008 sealed marginal-preserving builds is MONOTONE in  $\rho_1$  ( $\chi$   $0.630 \rightarrow 0.764 \rightarrow 1.029 \rightarrow 1.342$ ; end margin  $+0.055 \rightarrow +0.057 \rightarrow +0.032 \rightarrow -0.044$ : **THE SIGN FLIP KILLS THE TARGET INEQUALITY**) — but  $\rho_1$  matching misses the destructive  $\chi$  level by 0.134  $\Rightarrow$  mixing-insufficient (the within-block structure beyond lag 1 named as

the gap); bycatch theorem-grade:  $n_{\text{eff}}^{\text{atom}} = (L_1^a)^2/Q$  is a pure MARGINAL functional (invariance  $1.0 \times 10^{-15}$  on all 1008 builds — stationarity and mixing complementary, not redundant); rho-sign honest 41/42 (one rung positive: no raw per-rung theorem; Fractions-exact negative certificates kz18/kz23). (D) THE r304 SHORT-RANGE LAW (the reviewer stop case): the global lag profile  $\rho_{1..16}$  of the  $dc$  profile is a STABLE, WORLD-SPECIFIC PERIOD-4 COMB (med  $-0.222/-0.140/+0.089/+0.130$ , then strong lags at every  $k = 4m$  and  $4m+2$  up to 16; halves-stable 3/3), NOT a decaying tail — no  $k_0 \leq 8$  exists: longrange-structure + law-longrange fire on the LETTER of the sealed rule; the reviewer condition SPLITS: the net covariance  $NC(16) = 0.712 < 1$  HOLDS (net-negative) while summability-with-small-tail FAILS ( $SUM(16) = 1.563$ ); only the TRUNCATED  $NC$  carries content (the full sum is the gated zero-sum tautology); THE STRUCTURAL COUNTERPOINT (exact, ward  $4.7 \times 10^{-16}$ ): the  $\chi$ -relevant within-block structure IS short-range (shares die by  $k \sim 4$ ; the r303 miss 0.134 sits at  $k \leq 3$ , dominated by lag 2 +0.181) — the long-range comb is INVISIBLE to  $\chi$ ; lag-8 matching reproduces the  $\chi$  LEVEL (0.652 vs 0.630,  $|d| = 0.022 \leq 0.05$ ) but breaks the SLOPES ( $0.028/0.027 > 0.02$ ) — the exact COMPLEMENT of r303(a): the mechanism is TWO-SCALE (level from the short lags, slopes from the rung-wise  $\rho_1$  trend), no lag-matching family reproduces both; the graduated reviewer ladder is MONOTONE in  $\chi$  but NOT in the margin (honest negative: the r303 kill came from per-rung flipped targets — the margin responds to the TREND, not the level). THE CONSEQUENCE (the sealed reviewer stop case, map (iii)): THE GLOBAL-PROFILE MIXING ROUTE OF THE L2 LANE IS DOCUMENTED CLOSED — the honest stop state: L2 generic  $\Leftrightarrow$  anti-concentration of an explicit block field with LONG-RANGE (period-4 comb) structure; return later with new tools. What stands honestly: the exact identity dictionary (count, coherence,  $\chi$ -lag), the two-scale split,  $NC < 1$ , the  $-/-/+ / +$  sign pattern (Fractions-exact kz18/kz23), the marginal-functional invariance of  $n_{\text{eff}}^{\text{atom}}$  ( $10^{-15}$  on all 2100 + 1008 builds). LEDGER: NEW PRIME.L2.CASCADE\_CLOSURE.01 [E] (fine types Identity/Formal for the S0 skeleton + Numerical/M Measurement for the ladders and censuses; the rejections and retypings ARE the frozen content); PRIME.L2.NEFF\_TARGET.01 [O] RETYPED to the documented stop state (the rest is no longer NEFF/UNIF/ATOM — all one slack  $S = +0.0547$  — but the stop state above, with the hard reviewer rule verbatim in the claim); PRIME.L2.REDUCTION\_CHAIN.01 [E] retyping note (the exact identities STAND, the reading is refined: the chain is a dictionary, not six proof steps); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO new statements — the stop carries no new exact target form (the period-4 comb and the two-scale split are measured ladder aggregates, not exact finite identities, and a universally quantified ladder form would be refutable — the r273 guard pattern of RH/Counterexamples.lean); the hole stays lstar\_subordination. Papers: the cascade-closure diary section sec: cascade-closure in tfpt\_prime\_front (the wave-9 “r301/r303 in flight” spots resolved historically), the wave-10 block + the NEFF\_TARGET stop-state update in tfpt\_research\_contracts, the new “The cascade closure (v967)” section in rh/paper/rh\_program.tex (the margin invariance as its own theorem, the retyping as a methodological lesson — the sealing discipline caught the reparametrization trap; the causal  $\rho_1$  coordinate + the two-scale finding, the period-4 comb as the named open object, the documented lane stop with the clean equivalence state) + recompiled PDF; rh/README.md wave-10 block with the lane status update (L2: documented STOPPED; L\*: dormant since

r296; both edges with a clean state); rh/INVENTORY.json + make\_inventory.py extended additively (89 → 90 entries: v967 pin + the r301–r304 consumption notes, all pre-existing entries byte-exact); run\_rh.py MODULES → v955–v967, full RH suite green. The full-suite run\_all.py pass at 960 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; no round is in flight at this cut; NO RH claim anywhere in either direction)

2026-08-26 (CXXVI) (consolidation wave 9 — the DENS fork and the L2 reduction chain: the certified content of discovery rounds 296–300 graduates from the experiments tree into the load-bearing suite as v966\_l2\_reduction\_chain.py (suite 958 → 959, ledger 1124 → 1126; wave-4 embedding convention: the five discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — dens\_identity 39/39 SPEC ffb413c80cddc1d3, vdc\_chain\_provenance 27/27 SPEC e42a76eb8ae71422, window\_border\_transfer 28/28 SPEC 05e831bec627aee6, fejer\_decay 32/32 SPEC f432e94450ad545d, diag\_target 31/31 SPEC 55218b5db8ece163; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the r297  $\sigma^*$ -composition arithmetic ( $\delta'(\sigma^*(\text{pad})) = \text{pad}$  EXACTLY for every rational slope pair, the pad-dropped mutant composes to 0: the r297-m2 sharpness as an identity), the r298 decomposition identity  $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$  on the exact positional Fejér block form (the hand toy  $3 = 1 + 2$  with attribution  $1/1$ ; the  $|\Delta|$  and wrong-weight mutants break by exactly 4 and  $2/3$ ), the r300 participation identity  $D \times n_{\text{eff}} = L_1^2$  with ratio multiplicativity and the machine-checked DIAG↔NEFF equivalence (record decimals  $-0.571$  in both decompositions, NEFF need  $0.908$  / margin  $0.055$  exact in BOTH coordinates; the signed-sum mutant breaks by exactly  $8/3$ ), the r300 kernel envelope  $F_H(s) \leq \min(H, 1/(Hs))$  exact-rational with equality witnesses on BOTH branches ( $H = 2$ :  $s = 0, \frac{1}{2}$ ;  $H = 3$ :  $s = 0, \frac{1}{4}$ ), and the five sealed verdict bars as exact decision logic with tipping mutants at every clause boundary, no probe imports; single-module test green 28/28 gates (18 module-own +5 pattern gates) in 1.3 s, deterministic on re-run): **THE DENS FORK CLOSED HONESTLY, THE  $\delta' > 0.21$  TARGET OF THE GENERIC HALF REDUCED TO ONE INEQUALITY.** (A) **THE r296 DENS FORK** (honest negative, the lane closed): the reviewer hard fork on the r292 curvature axis lands dens-world-blind under the sealed rules — **THE COUPLING NUMBER**  $\cos(e_{\text{top}}, \nabla \lambda_{\text{max}}) = +0.394$  sits **BELOW** the sealed 0.40 bar (noise max 0.331; T1 misses **PART** by 0.006, honest), the  $\lambda_3$  control couples **HARDER** ( $-0.574$ : a near-crossing eigenvalue-BAND property, not top-specific), every arithmetic candidate of the hash-sealed library stays  $\leq 0.38$ , and the only above-chance structure (the 4-dim moment-gradient subspace, 0.825) **HOLDS** on both dead controls (0.571/0.603) — construction-adjacent low-moment smoothness, the sealed world-downgrade fires (disclosed tension: the absolute collapse bar is conservative for subspace members, rule unmoved); **PER THE PRE-ADJUDICATED REVIEWER FORK** the DENS-identity lane is **CLOSED** and the resources move to the L2 front. (B) **THE r297 TARGET INEQUALITY:**  $\delta'(F2) > 0.21$  is **EQUIVALENT** to  $\sigma := \text{slope}(S_F/M_W^2) \leq \sigma^* = 2(\text{sl}_{c2} - 0.21) - \text{sl}_{\text{pref}} = -0.516$ ; measured  $\sigma = -0.714$  — **THE TRUTH CLEARS THE TARGET WITH 0.198 SIGMA MARGIN** (0.099 in  $\delta'$ ): *what is missing is not room, it is provenance*; all three classical provenance routes **VALID** 47/47 but **SLOPE-FAIL** ⇒ chain-provenance-partial(B3, slope miss 0.307 EXACT), with **THE SURPRISE:** the

chain is EXACTLY ORTHOGONAL w.r.t. the WINDOW measure (devs 0.000 world-blind — the Fractions-proved Parseval sum rule attaches verbatim); the magnitude wall named (+0.22.. + 0.24 imbalance growth). (C) THE r298 EXACT TRANSFER:  $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$  EXACT ( $8.8 \times 10^{-16}$  of scale, 47 worlds), SIGN-PRESERVING (the  $|\Delta|$  mutant breaks by  $\sim 1.7$  of scale LOUD — the r297 magnitude error certified as a must-fail); the linear window zero MEASURED (the drive is 100 % transfer at the linear level); sealed verdict transfer-dominant: the window main term is EMPTY (med  $-3.94$  dec, slope  $-1.386$ ), the cross term negligible ( $-1.4 \times 10^{-4}$ )  $\Rightarrow S_F \approx B(P\Delta, P\Delta)$  — the vdC input IS the Fejér energy of the difference measure itself (share 0.99,  $T > 0$  on 42/42; the two r297 gap halves unified, sp +0.69, broad-based). (D) THE r299 DECAY SPLIT: the exact spectral representation  $B = (1/L) \sum_k F_H(\theta_k) |\hat{\Delta}(k)|^2$  is LOWPASS (0.93/0.04/0.02); THE STRUCTURAL FIND: FULL-SUPPORT overlap on 42/42 rungs —  $\Delta_{\text{fresh}} \equiv 0$ ,  $\Delta$  IS a pure c-value difference measure on ONE shared node set with NO pointwise convergence (cconv 0.86, +0.045: the decay is aggregate cancellation); the ET/Abel composition FAILS LOUD (composed  $+1.948 = 1.504 + 0.444$  EXACT vs  $\sigma^* = -0.516$ ; mass-target missed by 2.46 — the r297 magnitude wall recurs at the frequency level); the live edge fires:  $\text{sl}_D = -0.571 \leq \sigma^*$  (margin 0.055) with FALLING ratio  $B/D$  ( $-0.168$ )  $\Rightarrow$  decay-split (diag-target + ratio flatness; cvalue-target on the B3 side); the O-sign class separates MAIN (o-pos, reinforcing,  $O < 0$  on 13/42 only) from both dead controls (o-neg) — the first world-separating sign class of the L2 lane, disclosed. (E) THE r300 DIAGONAL ANATOMY:  $\text{sl}_D = 2\text{sl}_{L1} - \text{sl}_{\text{neff}} = 2(+0.196) - (+0.963) = -0.571$  EXACT (dev  $6.7 \times 10^{-16}$ ) — THE ENTIRE DIAGONAL DECAY IS PARTICIPATION GROWTH ('many small instead of few large' as a measured identity;  $n_{\text{eff}} \sim N$ , sp +0.96); BOTH magnitude factorizations fail exactly as r297 predicted (B1 chain-norm CS  $-0.592 + 0.208 = -0.384$  EXACT AND the  $|dw|$  identity census BREAKS — the proven sum rule does NOT attach to the difference measure, the chain-norm shortcut closed honestly; B2  $\max \times L_1$   $-0.542 + 0.196 = -0.346$  EXACT, miss 0.170 — the max is fully atom-controlled,  $m_x/\text{maxatom}$  med 1.07: the irreducible loss is the fill decay  $-0.225$ , INVISIBLE to every  $\max \times \text{mass}$  bound); sealed verdicts diag-split(B3/neff-target) + ratio-bounded-structural (the exact kernel-envelope majorant  $R_{\text{env}}$  med 1.61 FALLS at  $-0.122$  on 47/47 — the ratio half of the r299 rest pair settled at the structural level; the lobe-width heuristic refuted, sp  $-0.47$ ); the FILL class separates MAIN (fill-low) from both dead controls (fill-high) — the second world-separating class. THE CHAIN STATE (the frozen wave content):  $[\text{neff-target}] \Rightarrow \text{sl}_D \leq \sigma^* \Rightarrow \sigma \leq \sigma^* \Rightarrow$  the r297 target  $\Rightarrow$  the v964-S0 vdC theorem  $\Rightarrow \delta' > 0.21$  on the generic half (exceptions: 6 via the r287 F2 certificates, kz15 exact-finite via r270) — the L2 lemma stands reduced to ONE inequality: neff-target,  $\text{slope}(n_{\text{eff}}) \geq +0.908$  (measured +0.963, margin 0.055; the  $D_{\text{rank}}$  bridge sp  $-0.81$  real but correlational; round 301 in flight, NOT consumed). LEDGER: NEW PRIME.L2.REDUCTION\_CHAIN.01 [E] (fine types Identity/Formal for the S0 skeleton + Numerical/Masurement for the route tables; honest negatives VERBATIM: the magnitude-route catalog r297/r299/r300, the ET failure +1.948, no c-value convergence, the DENS closure); NEW PRIME.L2.NEFF\_TARGET.01 [O] — the one remaining inequality with the named correlational bridge and the sealed kill list; CONTRACT UPDATES: PRIME.LSTAR.CLOSED\_FUNCTIONAL.01 [O] (the DENS fork closed — front (2) answered negatively at the identity level, the lane routed to L2), PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] (the L2 front reduced to neff-target); no other marker moves. LEAN (rh/lean): NO new statements — neff-target is

a measured ladder-slope aggregate (a halves log-slope estimator over 42 rungs with a threshold derived from measured slopes), not an exact finite identity and not a matrix theorem, and a universally quantified ladder form would be refutable exactly like the pre-r273 universal edge forms (the machine-checked guard pattern of `RH/Counterexamples.lean`); the hole stays `lstar_subordination`. Papers: the L2-chain diary section `sec:l2-chain-arc` in `tfpt_prime_front`, the wave-9 block + the `VDC_LEMMA/CLOSED_FUNCTIONAL` state updates in `tfpt_research_contracts`, the new “The L2 reduction chain (v966)” section in `rh/paper/rh_program.tex` (the DENS closure paragraph, the reduction diagram, the exact identities as theorem environments — target equivalence, sign-preserving transfer, participation identity, kernel envelope — the magnitude no-go catalog as a table, the two world-separating classes as a remark, the NEFF target as an open problem) + recompiled PDF; `rh/README.md` wave-9 block (r301 in flight noted); `rh/INVENTORY.json` + `make_inventory.py` extended additively (84 → 85 entries: v966 pin + the r296–r300 consumption notes; the sync strictly additive next to the in-flight r301 worker); `run_rh.py` MODULES → v955–v966, full RH suite green (65/65 items, wall 489 s, 81/81 pinned entries byte-identical, ALL CHECKS PASSED). Mincut base 4 / refined 5 UNCHANGED (a reduction-plus-closure set moves no edge); the full-suite `run_all.py` pass at 959 modules, the final bash `build.sh` audit and the manifest regeneration follow as the joint exit gate run by the coordinator; the parallel r301 NEFF worker is NOT consumed; NO RH claim anywhere in either direction)

2026-08-26 (CXXV) (consolidation wave 8 — the  $L^*$  curvature arc: the certified content of discovery rounds 290–295 graduates from the experiments tree into the load-bearing suite as `v965_lstar_curvature_arc.py` (suite 957 → 958, ledger 1122 → 1124; wave-4 embedding convention: the six discovery probes embedded BYTE-EXACT and executed verbatim in their sealed `-smoke` stage with the printed SPEC SHA-256 hashes pinned and gated — `profile_functional` 32/32 SPEC `f953dd71f483d0fa`, `ridge_anatomy` 30/30 SPEC `bb512c17927818f5`, `curvature_form` 36/36 SPEC `050821ff05e1be74`, `metric_reconciliation` 43/43 SPEC `33c44cc697d4c3df`, `f10_stability` 42/42 SPEC `88c6fd1e3b5a5d39`, `f10_sp_hardening` 44/44 SPEC `4d7d809597d69afb`; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 — the r292 polarization identity  $H(u, v) = [d^2(u+v) - d^2(u-v)]/4$  with the wrong-prefactor must-fail, the r291 budget telescope (exact partition additivity of the side-selected first-order budget) with the dropped-atom must-fail, the r293  $\alpha = 1$  simple-zero doubling law ( $|h(t_0+2d)|/|h(t_0+d)| = 2$  exactly, the double-zero mutant gives exactly 4 LOUD), and the sealed r294/r295 decision bars as exact Fractions logic with tipping must-fails at every clause boundary, no probe imports; single-module test green 25/25 gates (19 module-own +6 pattern gates) in 1.6 s, deterministic on re-run): THE GEOMETRY MEASURED, THE FUNCTIONAL SEARCH SEALED HONESTLY NEGATIVE, F10 NOT PROMOTED. (A) THE r290 BASIN GEOMETRY: the working set around MAIN is a SOFT-SHOULDERED, STRONGLY ANISOTROPIC TUBE in the exact  $\theta_{eq}$  LAG coordinate (kill fraction 0.38/0.62/1.00 at  $5 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$ ; NEAR radius  $\sim 5e-4..2e-3$  gap-equivalent); the world-directed axes kill 5–50x EARLIER than random (onsets  $4 \times 10^{-5}..2.3 \times 10^{-4}$  vs the jitter threshold  $10^{-3}..3 \times 10^{-3}$ ); the SMOOTH (arithmetic) axis is a privileged killer (onset  $1.36 \times 10^{-4}$ ) yet ORTHOGONAL to the first-order wall gradient ( $\cos = -3 \times 10^{-5}$ ) — collective, not gradient-linear; the r280 ridge is REAL (min  $C = 185$  at factors 1..8, first death at



16). (B) **THE r291 RIDGE ANATOMY**: the lift is a **FIRST-ORDER BUDGET** phenomenon — one threshold in  $(1.280, 1.291]$  separates lift from no-lift over **ALL** 18 matched-dose cases (**PRIME/HEAD/XIPOS** lift;  $k_{\min} = 9$ , top atoms 2, 3, 5, 13, 11, 4, 29, 7, 89; exactly **ONE** overdrive retraction **TOP6** at factor 8, margin 9.09); **NO FIXED POINT** (plateau  $\min C = 185$ , never 186, the axis decoheres); **LIFT MAIN-SPECIFIC** (**EPSTEIN** first-order flat,  $\theta_{\text{up}} = 5.03 \times 10^{10}$ , never lifts); **SMOOTH COLLECTIVE-QUADRATIC** ( $d^2 = -23.3$  vs  $-0.98$  random, ratio 23.7). (C) **THE r292 CURVATURE SPECTROSCOPY**: the wall Hesse form is **NEGATIVE** on all 29 sealed directions and **RANK-1 DOMINATED** in the **L2** spectral metric (92.5% of  $\sum |\lambda|$ ,  $\lambda_{\text{top}} = -0.418$ ) with the top axis a pure on-support **DENS** combination — **NOT** the **SMOOTH** axis ( $|\cos| = 0.07$ ) and **NOT** the ridge (flattest sector, **L2-rank** 28/29); the  $m^*$  threshold and the **TOP6** retraction are near-flip nonlinearities **INVISIBLE** to jets up to third order ( $q_R$  median  $-0.911 = \text{second-order ASSIST}$ ;  $B_2 = 27.6/B_3 = 150.9$  over-predict); **EPSTEIN** second-order structureless ( $5.4 \times 10^{-15}$ ). (D) **THE r293 METRIC RECONCILIATION**: three metrics sealed **BEFORE** any evaluation (distortions 11/59/124, largest sector **DENS**); **F10** beats the baseline **OF ITS OWN METRIC** for the first time in four rounds (home margin  $+0.024 = 0.884 - 0.860$  vs the mixed **r292** margin  $-0.023 = 0.884 - 0.907$ : mix-is-cause, both **RE-DERIVED** in the gates) and carries partial information ( $+0.423$  test /  $+0.826$  train); **ALL** 8 selected flips are **SIMPLE**  $h_{184}$  **ZEROS** ( $\alpha = 1.000 \pm 0.003$ , 225 gated wall evaluations) — but **mstar-no-law** ( $m_{\text{dir}}^*$  scatters 1.1295..1.3931 straddling the **r291** bracket, a fixed-dose artifact; spreads 3.0/3.0/3.8/10.2 vs bar 1.25); the **TOP6** retraction **IS** a **SECOND** simple  $h_{184}$  zero at  $f_{\text{ret}} = 7.107$  (lift window 1.226..7.107), on the side the monotone budget prognosis is blind to. (E) **THE r294 STABILITY ADJUDICATION** (honest negative): the  $|\text{sp}|$  win of **F10** replicates on **ALL FIVE** fresh seed-disjoint corpora (margins  $+0.003..+0.115$ , wins 5/5 re-derived) **BUT** the partial channel does **NOT** (median  $0.299 < 0.3$ , 2/5)  $\Rightarrow$  sealed verdict **f10-fragile**, **PROMOTION NO**; robustness typed **train-robust** ( $\sigma = 0.0101$ ) + **rank2-carries** (top **DENS** eigenaxis alone  $+0.855$ , **DENS** share 0.989 — the mechanism hint) + **step-stable**; window transport a **KNIFE EDGE** (**w11**  $+0.002$  / **w7**  $-0.009$ ) with **DENS MECHANISM CONSTANCY** (0.795/0.989/0.834); **l2-not-dens**. (F) **THE r295 HARDENING ADJUDICATION** (honest negative): **TWENTY** fresh corpora (192 forbidden seeds enumerated, zero overlap;  $H_{\text{tr}}$  hash gate 3447ed198a56) — **F10** wins 14/20 (margin median  $+0.028$ , min  $-0.079$ ): the sealed three-clause **HARDENED** bar fails on **TWO** clauses  $\Rightarrow$  **f10-sp-majority** — the **r294** 5/5 was itself **TOP-OF-DISTRIBUTION**, exactly as the **r293** partial was; a **DOCUMENTED REGULARITY** (19/25 combined census), **NOT** a theorem; **PARTIAL20** median  $+0.279$ , **IQR** [ $+0.182, +0.401$ ]; **partial-family-map**: **NO** family **STRONG** (**PATH**  $+0.156$  / **WORLD**  $+0.245$  / **FRAC**  $+0.067$  **NULL** / **DENS**  $+0.104$ , pooled 2921); **r293-luck** (matched median  $+0.346$ : gain clause passes, level clause fails by 0.004); **l2-via-conservation** near-tautological ( $\eta_0$  share  $2.9 \times 10^{-30}$ ). **THE FROZEN CONSEQUENCE**: after **FIVE** sealed class families (local scalars best 0.263 / non-local best 0.471 / two-form mixed  $0.884 < 0.907$  / metric-reconciled **F10** fragile / hardened  $|\text{sp}|$  form majority-only) **NO** closed predictive profile functional of the working set exists over the honest bar — **F10** stays **UNPROMOTED**, the sealed **r294/r295** rejections **ARE** the frozen content. **LEDGER**: **NEW PRIME.LSTAR.CURVATURE\_ARC.01** [E] (fine types **Identity/Formal** for the **S0** skeleton + **Numerical/Measurement** for the geometry census; honest negatives **VERBATIM**: all-functionals-blind (5 class families), **f10-sp-majority**(14/20, kein Satz), **mstar-no-law**, **r293-luck**); **NEW PRIME.LSTAR**.

CLOSED\_FUNCTIONAL.01 [O] — the closed-functional question with the sealed kill catalog and the named open fronts (loss-corpus forensics K05/K07/K19, the rank-2 DENS core, why L2 substantially); CONTRACT UPDATE: PRIME.LSTAR.SUBORDINATION.01 [O] carried at the wave-8 state (the characterization question functionally negative-sealed, the geometry measured; no round in flight at this cut). LEAN (rh/lean): NO new statements — the wave carries no new formal target form, the hole stays lstar\_subordination. Papers: the curvature-arc diary section in tfpt\_prime\_front (the wave-7 “r290 in flight” spots resolved historically), the wave-8 blocks in tfpt\_research\_contracts, the new “The curvature arc (v965)” section in rh/paper/rh\_program.tex (basin geometry, budget mechanism, curvature-spectroscopy table, the five-family functional kill catalog as a table, the honest F10 story as a case study in sealing discipline) + recompiled PDF; rh/README.md wave-8 block; rh/INVENTORY.json + make\_inventory.py extended additively (78 → 79 entries: v965 pin + the r290–r295 consumption notes); run\_rh.py MODULES → v955–v965, full RH suite green (59/59 items, ALL CHECKS PASSED). Mincut base 4 / refined 5 UNCHANGED (a geometry-census-plus-rejection set moves no edge); the full-suite run\_all.py pass at 958 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXXIV) (consolidation wave 7 — the  $L^*$  coherence census: the certified content of discovery rounds 286–289 graduates from the experiments tree into the load-bearing suite as v964\_lstar\_coherence\_census.py (suite 956 → 957, ledger 1120 → 1122; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — lstar\_margin\_scaling 29/29 SPEC 0a44ac4e3a76e809, l2\_deterministic\_cancellation 25/25 SPEC 761d88fa566faa14, destructive\_coherence 31/31 SPEC da46f7eef702ac37, arch\_kernel\_diophantine 42/42 record SPEC 91cdc2b194502168; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the Fejér window-sum identity, the van der Corput inequality with its Cauchy–Schwarz proof route and  $H = 1$  equality witness, the Abel/Erdős–Turán bound, the wrong-prefactor and dropped-terminal must-fails, the 2-power resonance count  $\binom{8}{2} = 28$ , in pure Fractions; single-module test green 21/21 gates in 1.1 s, deterministic on re-run): THE MARGIN WARNING RESOLVED, THE vdC THEOREM NAMED, THE DIOPHANTINE ROUTE EXCLUDED. (A) THE r287 vdC THEOREM: the exact, CONSTANT-FREE, ARITHMETIC-FREE van der Corput inequality  $|\sum_j P_j|^2 \leq \frac{m+H-1}{H} \sum_{|h|<H} (1 - \frac{|h|}{H}) A(h)$  at the frozen window  $H = \lceil \sqrt{m} \rceil$  delivers  $\delta' = +0.309 > 0.21$  WORLD-BLIND (admissible: L2 is the generic half) and certifies 6/7 exceptions +38/42 rungs — the L2 lemma has its first NAMED classical theorem candidate; F1 discrepancy/Erdős–Turán dies loud ( $\delta' = -0.21$ , 0/7, detector FIRE +1.44 dec); the autocorrelation surprise: on 39/44 worlds the block sequence REINFORCES at root scale — the c3 cancellation is already absorbed into the  $P_j$  magnitudes, the frames win through the  $\sqrt{m}$  count economy, not lag structure. (B) THE r286 MARGIN-SCALING CLARIFICATION (the DCXX warning RESOLVED): 15 new anchors  $N_w = 942..1218$ , ALL margins mp-sign-safe positive (min  $+1.806 \times 10^{-8}$ , every anchor mp dps-30/45 staggered) — NO counterexample; the  $O(1)$  census offset holds beyond the old family cap (off distribution  $\{0:1, 1:10, 2:2, 3:1, 4:1\}$ , max +4); margin law  $\alpha \approx 3.05$  vs  $S$ , FLATTENING (curvature  $-1.60$ , the honest anti-extrapolation flag), driver

= the cancellation sharpening  $c_w \rightarrow 1$ ; HARMLESS quantified: the margin falls SLOWER than the local eigenvalue loading speed;  $q_N$  reconciled (r258 FLAT reproduced — two different objects); extrap-calibrated 15/15. (C) THE r288 CARRIER MAP: the destructive coherence is carried by the ANTIPHASE NEXT-NEAREST (3–4 fold) ARCH–ARCH pairs ( $X_v = -0.0517$ ,  $z_v = -3.149$ ,  $C_{\text{off}} = -0.1046$ ; band 1–2 87% positive/in phase) with TOTAL control reversal (all four controls constructive at their own crossing); the chain zeros CO-MOVE with the atoms (turn rate 0.24 = static prediction 0.23) yet  $z_v$  flips already at dose 0.005: FINEST ALIGNMENTS far below the median phase resolution; honest negatives sampling-blind + source-separator-not-found + different-objects. (D) THE r289 metric-only ADJUDICATION (THE central refutation of the diophantine reading): the RATIONAL TWIN — every tent center  $\log n$  replaced by a rational  $p/q \times \Delta$  (denominators  $\leq 56801$ , position cost  $\leq 2.1 \times 10^{-9}$ ), destroying EVERY exact log-linear-form relation — KEEPS THE FULL SIGNATURE IDENTICAL (min  $C = 184$ , crossing 185,  $z_v = -3.149$ , mp-confirmed at the  $1.7 \times 10^{-4}$  margin): the effect of the diophantine structure is NULL; the shuffle twins die purely metrically ( $\theta_{\text{eff}} = 0.12$ ,  $\sim 40\text{--}120\times$  beyond the measured coherence threshold  $10^{-3}..3 \times 10^{-3}$  of the local gap); the only sub-gap entry point: the TENT-SPLIT FRACTIONS (completeness identity  $4.2 \times 10^{-14}$ ); the Baker need documented  $\sim 7900\times$  too weak AND unnecessary — the firewall reads the fraction profile metrically, not its arithmetic: the why-MAIN search continues in PROFILE space (the precise open front: which functional of the fraction profile forces the destructive coherence). LEDGER: NEW PRIME.LSTAR.COHERENCE\_CENSUS.01 [E] (fine types Identity/Formal + Numerical/Measurement; honest negatives explicit: sampling-blind, source-separator-not-found, diophantine-excluded(metric-only) with the rational twin verbatim); NEW PRIME.PORT.L2.VDC\_LEMMA.01 [O] — the vdC form as a provable chain statement (the remaining input: the chain origin of the  $P$ -variance scaling; kz15 exact-finite per r270); CONTRACT UPDATES: PRIME.LSTAR.SUBORDINATION.01 [O] carried at the wave-7 state (margin decay resolved harmless on 57/57 anchors, carrier class named, diophantine excluded — the open front is the profile functional; r290 profile\_functional\_probe.py in flight, NOT consumed), PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] records the vdC candidate for the H5/L2 front. LEAN (rh/lean): NO new statements needed — the hole stays lstar\_subordination (docstring refreshed to the wave-7 state); README census updated. Papers: the coherence-arc diary section in tfpt\_prime\_front, the wave-7 blocks in tfpt\_research\_contracts, the L\* section of rh/paper/rh\_program.tex extended by the coherence/twin findings with the metric-only twin table and the L2 section by the vdC theorem; rh/problem/lstar\_problem.tex gains a short “What is known (update)” paragraph (margin law + 57 anchors + the rational-twin fact: the problem is NOT diophantine) + recompiled PDF; rh/README.md wave-7 block; rh/INVENTORY.json + make\_inventory.py extended additively (v964 pin + the r289 sync repair — the helper now tracks the INVENTORY head); run\_rh.py MODULES  $\rightarrow$  v955–v964. Mincut base 4 / refined 5 UNCHANGED (a census-plus-adjudication set moves no edge); the full-suite run\_all.py pass, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; the parallel r290 profile-functional worker is NOT consumed; NO RH claim anywhere in either direction)

2026-08-25 (CXXIII) (consolidation wave 6 — the L\* reduction dictionary: the certified content of discovery rounds 282–285 plus the L\* standalone

problem graduates from the experiments tree into the load-bearing suite as `v963_lstar_reduction_dictionary.py` (suite 955  $\rightarrow$  956, ledger 1118  $\rightarrow$  1120; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed `-smoke` stage with the printed SPEC SHA-256 hashes pinned and gated — `representation_contest 30/30 SPEC 0f9954b8f9727800, fullsource_quasideterminateness 26/26 SPEC 4cf5ea53ffa61ee7, lstar_two_measure 30/30 SPEC 687112a83a767c81, christoffel_decomposition 33/33 SPEC 9781e6d60aa5660a`; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 — Hankel determinants, monic OP chains, frame congruence minors, full Cauchy–Binet gauge exhaustion, dual chains in pure Fractions with two declared float eigen cross-checks; single-module test green 19/19 gates in 1.0 s, deterministic on re-run): **THE FREE-WINDOW QUESTION GETS ITS CANONICAL FORM.** (A) **THE r283 A2 CHAIN** as exact gates: the mu-frame congruence minor $_k(D_\mu - G) = D_k(\tilde{\mu})$  at every  $k \leq S_+$  (one-nu-atom truncation must-fail LOUD); the frame contraction  $h_0..h_{m-1} > 0 \iff \lambda_{\max}(E_m) < 1$  with  $E$  the  $\nu$ -dressed  $\mu$ -CD kernel (both truth values); the crossing dictionary  $\lambda_{\max}$  crosses 1 **EXACTLY** at  $\min C + 1$  (capacity-ceiling typed where  $\min C = S_+$ ); the pigeonhole ceiling  $\min C \leq S_+$  re-proved in the frame (null-polynomial witness, exact on all five toys); capacity-as-counting **REFUTED** by the rank-one pair (same rank, same support, opposite fate; the deciding values are METRIC,  $1.25 \times 10^{-13}$  vs  $1.25 \times 10^2$ )  $\Rightarrow$  **L\* IS THE CANONICAL REDUCTION**: for every real polynomial  $p$  with  $\deg p < N_w$ ,  $\int p^2 d\nu < \int p^2 d\mu$  (equivalently  $\lambda_{\max}(E_{N_w}) < 1$ ) — a two-measure moment problem instead of a determinant cascade. (B) **THE r285 DECOMPOSITION BOOKKEEPING** exact: the Christoffel sandwich  $\max_k v_k K_n(y_k) \leq \lambda_{\max} \leq \text{trace}$  with its rational consequences at every degree; the multiplicative decomposition  $\lambda_{\max} = \text{maxdiag} \times (1 + \text{assist})$  with the budget equivalence sign-exact. (C) **THE r282 FOUR-LANGUAGE ELIMINATION** as named negative gates (contest-all-dead; the common deep reason made explicit: *every classical positivity language forces positivity exactly for the positive measure class* — MAIN is signed): SOS exists IFF the negative register is empty (the fake SOS misses  $h_2$  by exactly  $N_2 = 48360721965/70120631072$ ); the Kasteleyn orientation class is **EXACTLY**  $\{\pm \text{sign } w\}$  by full  $2^S$  exhaustion and is value-preserving IFF  $S_- = 0$  (defect =  $2 \times \text{negmass}$  exact: 55/36, 44/35, 114/35, 0); Hamiltonian-PSD =  $h > 0$  bookkeeping; the dual-pair sync holds **PER THEOREM** ( $h_n h_{S-1-n}^\# = 1$  exact, wrong dual loud at  $1.4 \times 10^2$ ). (D) **TYPED MEASUREMENTS** (record pins): the r284 one-atom-vs-collective dichotomy (w9  $n_{\text{DIAG}} = 187$  vs crossing 185 — MAIN rides the single-atom Christoffel edge to the last free degree, all four controls die collectively; extremal anatomy PR 1.89 on two shallow- $u$  ARCH atoms below the first prime, smallp-depleted, one-band; shielding-blind; Nyquist refuted at the discriminant), the r285 legs ((D) **SEPARATES** at window scale — controls'  $n_{\text{DIAG}} 50.91 \ll 184$ ; SUB-classical edge growth  $p = 0.38$ , classical coverage 0/42: any (D) proof must be a **DISCRETE** bound; MAIN's wall assist the extreme ensemble low-outlier pct 0.00; and the first two positively MAIN-separating detectors of the program: assist-at-crossing 0.0195 vs dead 1.69..2.99, random-sign  $z = -3.15$  destructive vs dead +4.95.. +12.44 all constructive), and the honest DCXX margin-decay finding (42/42 ladder margins positive, family minimum  $1.4175 \times 10^{-7}$  at anchor  $z = 233$ ,  $S = 1717$ , vs the flagship  $1.68 \times 10^{-4}$  at  $z = 16$  —  $\sim 3$  orders of decay; the r286 margin-scaling round `lstar_margin_scaling_probe.py` is **IN FLIGHT** and **NOT** consumed). **LEDGER: NEW PRIME.LSTAR.REDUCTION\_DICTIONARY.01**

[E] (fine types Identity + Numerical/Measurement, honest negatives explicit); NEW PRIME.LSTAR.SUBORDINATION.01 [O] — the  $L^*$  contract itself (the canonical form verbatim:  $\int p^2 d\nu_z < \int p^2 d\mu_z$  for  $\deg p < N_w$  on the admissible anchor family; state 42/42 + the open margin decay; the (D)+(C) anatomy; the standalone document `rh/problem/lstar_problem.tex` as external reference; kill lists inherited); CONTRACT UPDATES: FULLSOURCE.QUASIDEFINITENESS.01 consumed (round 283, mechanism-partial(A2),  $L^*$  established), REPRESENTATION.CONTEST.01 consumed (round 282, contest-all-dead); the BASEFIBER north star now carries the canonical  $L^*$  form. LEAN (`rh/lean`, build green): NEW `lstar_subordination` on `VonMangoldtWindow` (documented sorry, the canonical-form target) PLUS the *proved* direction `lstar_implies_free_window` ( $L^* \Rightarrow$  free-window positivity, finite algebra against the r283 chain). Papers: the  $L^*$ -arc diary section in `tfpt_prime_front`, the wave-6 blocks in `tfpt_research_contracts`, the new “The  $L^*$  reduction” section in `rh/paper/rh_program.tex` (canonical form as a theorem environment with certification box, the (D)/(C) split, the four-language elimination table, the margin decay stated neutrally, the pointer to `rh/problem/`); `rh/README.md` reduction chain ( $L^*$  as the canonical stage); `rh/INVENTORY.json` + `make_inventory.py` extended additively (v963 pin + r285 sync repair); `run_rh.py` MODULES  $\rightarrow$  v955–v963. Mincut base 4 / refined 5 UNCHANGED (a reduction-plus-elimination set moves no edge); the full-suite `run_all.py` pass, the final bash `build.sh` audit and the manifest regeneration follow as the joint wave-4/5/6 exit gate run by the coordinator; the parallel r286 and L2-cancellation workers are NOT consumed; NO RH claim anywhere in either direction)

2026-08-25 (CXXII) (consolidation wave 5 — the half-filling pinning theory: the certified content of discovery rounds 279–281 graduates from the experiments tree into the load-bearing suite as `v962_halffilling_pinning_theory.py` (suite 954  $\rightarrow$  955, ledger 1115  $\rightarrow$  1118; wave-4 embedding convention: the three discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — oriented\_theorem 32/32 SPEC 9107709b4f4a65d1, budget\_localization 29/29 SPEC 7abf7a208bb45e43, halffilling\_pinning 28/28 SPEC a00815722a93a3cb; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 in pure Fractions — Stieltjes chains, Sturm chains, Vandermonde solves, node polynomials, no floats; single-module test green 13/13 gates in 1.4 s, deterministic on re-run): the reviewer demand “freeze rounds 279–281 as a SMALL MATHEMATICAL THEORY, not just as experiments” is executed — FOUR NAMED THEOREMS as gates: (T1) THE MOMENT COUNTING THEOREM (r281, Identity): the pivot  $h_n$  consumes  $m_0..m_{2n}$ , the  $S$  free moments are  $m_0..m_{S-1}$ , hence the FREE pivots are exactly  $h_0..h_{N_w-1}$  and  $h_{N_w}$  is the first forced one — half-filling IS the end of the free moment space, arithmetic-exact for ALL  $S = 2..2000$  plus the exact-rational freedom demonstration ( $dm = e_{S-1}$  moves  $h_{N_w-1}$  alone; wrong- $L$  must-fail loud): the reviewer question “why half-filling” is answered by counting; (T2) THE CROSSING BUDGET THEOREM (r279, Identity):  $\#(h_n < 0, n < S) = S_-$  EXACTLY (Jacobi minor-sign rule + Sylvester congruence), WORLD-BLIND — five frozen signed measures exact, real-side w9 104 / w13 98 / controls 141/94/6 / kz15 121 / kz52 551; (T3) THE TWO-SIDED PARITY THEOREM (r279, Identity):  $\text{sign } Q_n(x_j) = e_j(-1)^{S-1-j} \text{sign}(h_n)$  at every node and degree, union occupancy ODD in every weight-agreement gap and EVEN in every disagreement gap —  $h$ -blind,

with the census bilanz  $c_n + c_{S-1-n}^\# = (S-1) - |D| + 2\text{scD}(n)$  exact at every degree and by exhaustion over all 87376 sign-vector pairs  $k = 2..8$ ; (T4) THE MAIN WINDOW REDUCTION: given T1 + T2 the ENTIRE open statement of the wall is  $\min C \geq N_w$ , EXACTLY equivalent to  $(\forall n < N_w : h_n > 0)$  — one placement question, the north star in reinstform: WHY IS THE SIGNED PRIME MOMENT FORM QUASI-DEFINITE UP TO THE MAXIMALLY FREE ORDER? FOUR NAMED REFUTATIONS as negative gates (the negative information is load-bearing): no-universal-o1-pinning (r281 exact: the one-negative toy has offset  $N_w - 2$ , unbounded in  $S$ ; only provable ceiling: pigeonhole  $\min C \leq S - S_-$ ;  $C \leq 5$  is a 42-rung MEASUREMENT), no-extremality (r280 measured + exact witness: three mp-confirmed directions push the w9 crossing  $184 \rightarrow 185$  past half-filling; the 4-atom one-negative measure has  $\min C = N_w + 1$  exactly), no-generic-maslov-obstruction (r279 exact: 3 rational counterexamples — the two-sided machinery provably carries no arithmetic), no-simple-offset-law (r281 measured: six sealed source-pure candidates,  $\max |\text{sp}| = 0.273 \ll 0.75$ , all world-blind). LEDGER: NEW PRIME.WALL.HALFFILLING\_PINNING\_THEORY.01 [E] (the theorem set + refutations, fine types Identity + Numerical/Measurement); PRIME.PORT.RHP.MIDPOINT.ORIENTED\_THEOREM.01 [O] updated (rounds 279–281 consumed: theorem share promoted, obstruction candidate refuted, the center precisified to the budget localization); FULLSOURCE.BASEFIBER.01 and METRIC\_FIREWALL.01 carry the reformulated north star; NEW open contracts PRIME.PORT.REPRESENTATION.CONTEST.01 [O] and PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01 [O] (the two rounds in flight at this cut, NOT consumed, with the reviewer forbidden lists). LEAN (rh/lean, build green,  $6 \rightarrow 8$  intentional sorry): the fog-free central hole free\_window\_positivity  $(\forall n < \text{cap} : 0 < h_n \text{ on Main Window; via T4 the base half of the master theorem — master\_implies\_free\_window PROVED})$ ; T1 PROVED for real (moment\_counting\_free\_pivots + first\_forced\_pivot, pure arithmetic); T4 PROVED (main\_window\_reduction); T2 stated (crossing\_budget, sorry with v962 reference); Counterexamples.lean extended by the wave-5 guard upper\_pinning\_not\_universal — the EXACT one-negative instance with machine-checked Hankel minors (incl. the negative  $\det H_4 = -1962/13625$ ), a genuine moment computation, not a bookkeeping model — plus the proved arithmetic escape o1\_pinning\_escape. Papers: the pinning-decomposition diary paragraph in tfpt\_prime\_front, the north-star reformulation in tfpt\_research\_contracts, the small-theory subsection + north star in rh/paper/rh\_program.tex, the finalized reduction chain in rh/README.md; rh/INVENTORY.json extended additively (v962 pin + Lean state). Mincut base 4 / refined 5 UNCHANGED (a theorem-plus-refutation set moves no edge); the full-suite run\_all.py pass, the final bash build.sh audit and the manifest regeneration follow as the joint wave-4/5 exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXXI) (consolidation wave 4 — the terminal-surface closure and the midpoint-orientation dictionary: the certified content of discovery rounds 260–278 graduates from the experiments tree into the load-bearing suite as v960\_terminal\_surface\_closure.py and v961\_midpoint\_orientation\_dictionary.py (suite  $952 \rightarrow 954$ , ledger +4; WAVE-4 EMBEDDING CONVENTION: all fourteen discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage — deterministic, seconds each — with the printed SPEC SHA-256 hashes pinned and gated and the byte-equality wards live; the full-mode records stay sealed experiments-side and are re-verified by rh/verification/run\_rh.py; v960 from ten probes, first-pass full

records  $31/31 + 26/26 + 26/26 + 28/28 + 29/29 + 29/29 + 28/28 + 28/28 + 24/24 + 24/24$ : terminal\_crossratio SPEC 36eee7b40bd897a8, terminal\_triangle SPEC db32d31f2e543a9c, cancellation\_adjudication SPEC 151a9176f089dd5b, drive\_local\_asymptotics SPEC 0c4df40812b9b76b, phase\_bulk\_bound SPEC f5578bb91f84ac2a, kz15\_boss SPEC 8834a1c259bdb605, universal\_pair\_theorem SPEC 66f61c8a436af90e, l2\_scaling\_anatomy SPEC 3fb76b65848814f8, euler\_mechanism SPEC 8cb1804e71dd87bb, kyp\_memory SPEC 88ebdb00867469fd; v961 from four probes,  $32/32 + 25/25 + 29/29 + 31/31$ : wronskian\_dictionary SPEC 56e8a03efd1690e4, minimal\_firewall SPEC ed17d79fc037ab1a, maslov\_census SPEC 3858fd16bde0c9c0, metric\_stability SPEC 7031200f008d34d1; plus module-own exact sections S0 — v960: the two-branch reduction with the non-certifiable witness, the driven-recursion C1b triangle, the pair decomposition with the structural alternation gain, the interval-certificate logic with its width ward, the KYP double obstruction o1/o2; v961: the base Casoratian identity  $p_n q_{n+1} - p_{n+1} q_n = h_n$  with  $c' = 1$  symbolic in  $z$ , the bordered-Wronskian telescope, the Hellmann–Feynman position gradient in exact dual-number arithmetic, the Jacobi interlacing/reality direction with the flipped-beta must-fail; single-module promotion tests green: v960 15/15 gates 8.6 s, v961 8/8 gates 2.4 s, both timing-normalized deterministic on re-run): (1) THE MILESTONE — THE 42-RUNG SURFACE CENSUS  $q_N < 1$  IS CERTIFIED ON ALL 42 RUNGS as a finite fact: 41 mechanism-certified target-blind (35 cheap via the r260/r263 two-branch theorem, exception set exactly  $\{kz15, kz20, kz22, kz36, kz38, kz39, kz52\}$ , both mains close WITHOUT cancellation; kz20/22/36/38/39/52 via the sealed phase bounds r268–r270, no detector fires) + kz15 EXACT-FINITE via the r270 interval certificate (outward-rounded intervals, dps 640, end width  $1.5 \times 10^{-92}$ , margin +0.02680, the dps-halving ward destroys it as demanded; EnclOK class, no mechanism) — a census-plus-certificate result, NOT a cofinal theorem: the window quantifier stays; (2) THE UNIVERSAL PAIR THEOREM (r271): H1–H4 structural/source-side, Lean-proved in rh/lean/RH/PairBound.lean; H5 = the margin  $|Z_{\text{local}}| + \varepsilon < \sqrt{5/7}$  is the SINGLE window-dependent hypothesis — the cofinal front, precisely typed by r272 (bound-coarseness: the truth margin RISES with  $N$ ; flip condition  $\delta' > 0.21$  of available 0.45, address c3) and r273 (perturbation-insensitive + the firewall map: THE WALL, not the cancellation rate, is what is arithmetically special about MAIN); (3) THE MIDPOINT-ORIENTATION DICTIONARY (r274): base Casoratian = the pivot chain itself, the  $h$ -free midpoint form IS the node polynomial (the r231 sign-blindness as an identity), the augmented telescope  $D_{n+1} = B - W_n^{\text{aug}}/W_n^{\text{base}}$  exact; and THE MASLOV CENSUS GO (r277): rule R2 = Jacobi interlacing/reality passes blind 42/42 with controls firing EXACTLY at flip+1, one-way, not  $h$ -equivalent — while the raw atom-Sturm census is honestly REFUTED as the winding quantity (MAIN breaks  $c_n \equiv n$  at 56/48 at positive  $h$ ; sturm-chain-verified not awarded); (4) HONEST NEGATIVES kept load-bearing: the eventual-triangle no-go (kz52 itself an exception), three identity routes dead with break loci, the retracted w13 Gram provenance, the KYP memory no-go FROM STRUCTURE (o1 fiber forcing + o2 descent window, exact-rational on 46/46 worlds; the admissible Riccati memory target-inverse, sp +1.00), the metric-firewall continuum with the perturbative-only stability law (exact Hellmann–Feynman gradients, u-profile predictive — the small primes 2, 3, 5 carry). NEW LEDGER ROWS: PRIME.PORT.COUPLEDTAU.SURFACE\_CLOSURE.01 [E] and PRIME.PORT.RHP.MIDPOINT.ORIENTATION\_DICTIONARY.01 [E] (fine types



Identity/Formal + Numerical/Measurement, honest negatives explicit) plus the two successor contracts PRIME.PORT.RHP.MIDPOINT.ORIENTED\_THEOREM.01 [O] (the contraposition form: an independent index obstruction forbidding the R2 break before half-filling on MAIN; round 279 oriented\_theorem\_probe.py in flight, NOT consumed) and PRIME.PORT.WALL.METRIC\_FIREWALL.01 [O] (the global metric-firewall lemma; certified state perturbative-only); CONTRACT UPDATES: TERMINAL\_CROSSRATIO.01 [O] progressed (surface certified; cofinal front = H5 / lemma L2), PREFIX\_RESUMMATION.01 [O] progressed (r261 finite-combinatorial half closed negative; the named path = the oriented midpoint theorem), FULLSOURCE.BASEFIBER.01 [O] carries the formalized master target form augmented\_prefix\_positive (rh/lean/RH/Window.lean; per r276/r278 the MainWindow predicate must encode METRIC exactness). RH-WORKSPACE PIPELINE INTEGRATION (the documented DXCIX TODO): bash build.sh audit now runs rh/verification/run\_rh.py -fast as its own “RH workspace” section (must end RH SUITE: ALL CHECKS PASSED; verified 32/32 items, 52/52 pinned entries byte-identical after the wave-4 INVENTORY extension); run\_rh.py MODULES extended to the six RH modules v955/v956/v958/v959/v960/v961; rh/INVENTORY.json +2 pinned module entries + fourteen consumption notes; rh/paper/rh\_program.tex updated (the surface-closure and orientation sections, the extended no-go catalog, the wave-4 edge states) and rebuilt; rh/README.md status updated. Papers: the new surface-closure/orientation-arc diary subsection in tfpt\_prime\_front (stale “rounds 260/261 in flight” wording retired), the updated BASEFIBER/terminal-edge blocks + the new wave-4 theorem-set/contract block in tfpt\_research\_contracts. Mincut base 4 / refined 5 UNCHANGED (a surface census + certificate set moves no edge); the full-suite run\_all.py pass, the final bash build.sh audit and the manifest regeneration follow as the wave-4 exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXX) (the coupled-tau-terminal promotion and the CCC D4 execution record — the certified theorem block of discovery rounds 256–259 graduates from the experiments tree into the load-bearing suite as v959\_coupledtau\_terminal\_dictionary.py (suite 951 → 952, ledger +3; four discovery probes re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with their printed SPEC SHA-256 hashes pinned and gated: baseborder\_factorial\_probe.py 21/21 SPEC c7aa9f46889414c4, coupledtau\_probe.py 26/26 SPEC 73d8247f6de36a2b, budget\_anatomy\_probe.py 27/27 SPEC 28026ae6b0750098, parametrix\_pass\_probe.py 19/19 SPEC e79da4f43d63213a; plus a module-own exact section S0 — the recursion-to-Riccati implication, the bilinear tau form derived algebraically and gated, the terminal consequence gate in both truth values, the corner-ray  $t^*$  normal form, and the positive-prefix PSD theorem  $\det H_p = \prod h_i$  with an exact Schur-complement must-fail; single-module run 9/9 gates, 188 s): (1) THE POSITIVE-PREFIX FIREWALL + THE  $2 \times 2$  CAUSAL CENSUS (r256, effects-mixed-coupledtau): MAIN carries the positive prefix through its entire window ( $p_{\max} = 184 = N$ , zero negative modes — the *only* positive-prefix world) while the controls type indefinite-continuation at 21/25/27; every prefix kernel numerically PSD (worst  $5.7 \times 10^{-16}$ ); on the causally clean shared prefix  $n^* = 21$  the base rotation is null ( $-1.5 \times 10^{-6}$ ) and the border carries the prefix gap ( $-0.1759$ ); TWO CONTAMINATION DOWNGRADES ON RECORD: the round-254 scramble-compactness headline is base contamination ( $f_{\text{neg}} = 0.9311$ , the  $h < 0$  mode space at  $734\times$  excess) and the r253/r254 Epstein fiber anomaly lives entirely in the indefinite continu-

ation degrees  $k \geq 25$  — neither may be cited as a border/fiber discriminator any more; (2) **THE COUPLED-TAU RECURSION IN BOTH FORMS (r257):**  $\tau_{n+1} = a_n \tau_n$ ,  $\tau_{n+1}^{\text{aug}} = a_n \tau_n^{\text{aug}} + b_n \tau_n$  with  $a_n = c_n^2 h_n$ ,  $b_n = -(c_n F_n)^2$ , equivalently the Riccati drain  $D_{n+1} = D_n - F_n^2/h_n$  (anchor  $D_N = 5/7 - \rho_{N-1}$  rel  $4.8 \times 10^{-11}$ , mp dps 220 through the scramble flip) AND the bilinear tau form  $\tau_n^{\text{aug}} \tau_{n+1} - \tau_{n+1}^{\text{aug}} \tau_n = (c_n F_n)^2 \tau_n^2$  (a manifest square — the drain increment is a positive tau cross-ratio); the base alone carries the sign, the border only nonnegative magnitude; the cocycle breaks exactly at the control flips 21/25/27; **THE FROZEN MICRO-FALSIFIER PASSED BLIND** for the first time, at the coefficient-field level (control flips 3/3 AND forced tail 0/2/2/3/1 on all five windows; the parametrix half stays open); (3) **THE BUDGET TELESCOPE (r258):**  $D_N = B - S_{N-1}$  exact (12 windows incl.  $N = 878$ , mp dps 220 at  $2.4 \times 10^{-16}$ ),  $B = S_{N-2} + 5/7$  by construction with the 5/7 typed floor-imported (the inverted r241 floor of the measured  $\min h_{N-1}/F_{N-1}^2 = 1.4278$ ; source-pure derivation open = the named terminal task); the consequence gate exact:  $\text{margin} > 0 \iff q_N = \rho_{N-1}/(5/7) < 1$  — the entire fiber positivity is ONE terminal inequality; typed numerical: the  $q$  census (42/42, flat, min margin 0.0139), the  $F$ - $h$  covariance +1.000 (a measured structural fact, NOT a law — the driver is the quotient itself); and (4) **THE THREE HONEST NEGATIVES (r259, parametrix-failed):** level crossing refuted (branches cross 5–16 degrees too early; MAIN w9 crosses 45 times with zero true flips), the one-swap no-go (fully overlapping gap distributions — round 255's orientation-lowdim(1) REVISED on record: the missing information is not one-dimensional), the mass-majorant no-go (all three sealed majorants 0/42, median  $\times 490$ , break at  $0.97N$ ); the named obstacle: a RESUMMATION gap, not a selection gap. **NEW LEDGER ROWS:** PRIME.PORT.RHP.COUPLEDTAU.TERMINAL.01 [E] (fine types Identity/Formal + Numerical, honest negatives explicit) and the two terminal-edge successor contracts PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] (the last fiber edge = a positive tau cross-ratio: prove  $h_{N-1}/F_{N-1}^2 > 7/5$  on the cofinal ladder; reviewer proof-form requirements and kill list frozen — the bound must bind the ratio; the mass-majorant class stop-listed) and PRIME.PORT.FULLSOURCE.PREFIX\_RESUMMATION.01 [O] (the last base edge = a globally oriented prefix resummation: the determinant as a signed partition sum, completing the coefficient-field pass to a parametrix-level pass; level crossing and  $k$ -swap truncations stop-listed); PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 [O] carried forward at the post-round-259 state with the r256 causal decision (base/fiber separation artificial) and the precise localization recorded — mincut base 4 / refined 5 UNCHANGED, rounds 260/261 in flight on the successor contracts and NOT consumed. **AND THE CCC D4 EXECUTION RECORD:** the ledger row CCC.SEAM.CROSSOVER.01 [O] is updated — demand D4 (the preregistered disc search experiments/ccs-crossover-disc) EXECUTED 2026-08-24 with a ROBUST NULL on the Planck PR3 SMICA map (first data contact; freeze intact 3/3, frozen-kernel guards 5/5 before data contact; map SHA-256 pinned, 2013312960 bytes; seed 20260824, reruns bit-identical):  $p_{\text{global}} = 0.673$ ,  $p_{\text{counts}} = 0.446$ ,  $\text{BH-}q_{\text{min}} = 0.673$  vs the frozen  $q < 0.01$  threshold; 171 candidates with no sign excess (83/88); injection recovery 100% at  $200 \mu\text{K}$  — a well-powered, real null; kill criteria K1–K6 all NA, replication legs not required for a null; NO type change, the contract stays [O] via D1–D3, the cyclic reading stays [C], the compiler core untouched (the kernel was output of the [E] kinematics, not input); results machine-readable in results.json. Papers: the new coupled-tau-terminal-arc diary section in tfpt\_prime\_front, the updated FULLSOURCE.BASEFIBER block + the new theorem-set/terminal-edge block

and the D4 execution record in `tfpt_research_contracts`. The single-module promotion test is green (9/9 gates, probes byte-exact vs the experiments tree); the full-suite `run_all.py` pass, `bash build.sh` audit and the manifest regeneration follow as the wave-3 exit gate of the same consolidation. No omega closed, no marker moved beyond the one [E] and two [O] registrations and the recorded D4 execution; NO RH claim anywhere; NO claim about CCC in either direction)

2026-08-24 (CXIX) (the bordered-RHP-dictionary promotion — the certified theorem block of the RHP campaign rounds 244–253 graduates from the experiments tree into the load-bearing suite as `v958_bordered_tau_readout_dictionary.py` (suite 950  $\rightarrow$  951, ledger +2; six discovery probes re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with their printed SPEC SHA-256 hashes pinned and gated in the module: `bordered_hankel_probe.py` 23/23, `bordered_finite_rank_probe.py` 29/29, `border_centering_probe.py` 24/24, `targetreadout_error_probe.py` 20/20, `base_gauge_constant_probe.py` 22/22, `schlesinger_pairing_probe.py` 26/26; plus a module-own exact-rational PSD base theorem with a frozen signed must-fail witness): (1) the bordered Hankel end-object (bordered PSD = wall + budget) has an *exact* Riemann–Hilbert dictionary — the Uvarov border column carries  $F_n$  as its  $z^{-1}$  coefficient, and the budget is a CD-kernel functional of the terminal  $Y_n$  data alone,  $S_{n-1} = \iint K_n d\sigma d\sigma$ ; (2) the border is a *Schlesinger rank-1 insertion* over a canonical  $2 \times 2$  RHP with a one-dimensional budget cocycle ( $\det Y_3 = -h_{n-2}F_{n-1}$  newly derived; IKS generator rank exactly 2; the irreducible- $3 \times 3$  steepest-descent campaign dropped without replacement); (3) the centering congruence with  $c = F_0/h_0$  eliminates the geometric rank-1 head algebraically, the  $\sigma^\circ$  dictionary and dual-norm identity  $Q_m = t^T H^{-1} t$  are exact, and the extensive tail compresses to ONE terminal CD readout ( $T = \iint [K_N - K_8] d\sigma^\circ d\sigma^\circ$ , dps-160 exact through the full depth  $N = 184$ , rel  $3.3 \times 10^{-160}$ ); (4) the three error-evaluation formulas are exact ( $\delta h_n = (R_1)_{12}$ ; the central rest identity; the error-kernel sandwich at  $1.1 \times 10^{-81}$ ); (5)  $R_1$  is a contour integral ( $\frac{1}{2\pi i} \oint (R(z) - I) dz$ , verified at  $3.2 \times 10^{-14}$ ) and the global-gauge thesis is refuted on all three prongs (the base is blocked by a *drifting, orientation-relevant* layer, not a constant); (6) the fiber target is the exact augmented tau quotient  $D = B^\circ - Q_7 - T = \tau^{\text{aug}}/\tau$  at full depth (three routes at  $10^{-11}$ , mp  $10^{-16}$ ), with the pole layers classified — rank-1 border poles removable by residue closure, the  $D$ -layer essentially singular ( $\exp(c/(z-s))$  type). The campaign contract PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 [O] is registered with the post-round-255 measured state (the measurement rounds 250/254/255 stay experiments-side, typed MEASUREMENTS: outer-model-fails, offdiag-extensive with the compactness inversion — SCRAMBLE collapses to 2 eigendirections, MAIN needs 28–30 — and orientation-lowdim(1): the base flip is a one-bit  $O(0.2 \text{ dec})$  selection event of an extensively accumulated quantity); the theorem set is registered as PRIME.PORT.RHP.BORDERED.READOUT.01 [E]. Mincut base 4 / refined 5 unchanged; no other marker moves; NO RH claim in either direction. Round 256 (`baseborder_factorial_probe.py`) was still in flight at this promotion cut and is not consumed. Papers: the prime-front bordered-dictionary arc subsection and the research-contracts blocks added.)

2026-08-24 (CXVIII) (the CCC-crossover promotion — the kinematics dictionary of the conformal-cyclic crossover graduates from the experiments tree into the load-bearing suite as `v957_ccc_crossover_kinematics.py` (suite 949  $\rightarrow$  950; six discovery probes of rounds DL–DLV re-run GREEN AS

TYPED at promotion and embedded BYTE-EXACT with pinned SHA-256 hashes: `ccc_crossover_gates_probe.py` 28/28, `ccc_uplift_4d_probe.py` 29/29, `ccc_fork_kms_holonomy_probe.py` 18/18, `ccc_kernel_gamma_kms_probe.py` 21/21, `ccc_transfer_selector_probe.py` 14/14, `ccc_kappa_freeze_probe.py` 7/7): (1) Penrose’s reciprocal gauge  $\hat{\Omega}\hat{\Omega} = -1$  is the element  $\tau = \rho^2\sigma : z \mapsto -1/z$  of the PROVEN  $\mu_4$  Möbius stabiliser  $D_4$  —  $\tau$  reverses the clock (= the  $\mathbb{Z}_2$  sheet parity), swaps the aeon poles  $\{0, \infty\}$ , and the crossover locus  $|\hat{\Omega}| = |\check{\Omega}|$  is the unit circle carrying ALL FOUR marks; (2) conformal-factor rigidity: on the 2D seam the clock alone leaves exactly ONE constant-curvature modulus (Liouville-exact dilation family) and the reciprocal  $\mathbb{Z}_2$  kills it (unique zero at  $\lambda = 1$ ); on the 3D crossover the deck ALONE kills all conformal moduli (the  $so(4, 1)$  deck-commutant is 4-dimensional  $su(2)+u(1)$ , ALL Killing, boost content  $3.3 \times 10^{-16}$ ; global uniqueness on the explicit Obata boost family, 200-axis scan bounded below by 0.53) — TFPT’s candidate answer to CCC’s open unique-conformal-factor problem,  $v_{\text{geo}}$  staying pure unit calibration; (3) the entropy reset of the cyclic [C] reading is a REDUCED CHANNEL: the v221 transport is CPTP not unitary ( $\|T^t T - 1\|_2 = 0.9999$ ), its Stinespring isometry explicit ( $V^\dagger V = 1_3$  exact),  $\|T^n - P_\star\|_2 = (2/3)^{6n}$  EXACTLY, the global overlap invariant and the environment retaining the full distinguishability — “unitarity + rank-1 attractor” is isometry + partial trace; (4) the kinematic  $2D \rightarrow 3D \rightarrow 4D$  uplift from corpus objects only: stereographic( $\text{Hopf}(v)$ ) =  $z_1/z_2$  EXACTLY (the seam sphere IS the Hopf base), the v492 deck  $\text{diag}(i, i^{-1})$  acts freely so the crossover  $X = S^3/\mathbb{Z}_4$  is the smooth lens boundary of the corpus  $A_3$  ALE  $XY = Z^4$ , clock  $\text{diag}(i, 1)$  and the reciprocal lift  $J$  normalise the deck and descend ( $J$  to  $\tau$ , inverting the clock), the four marks lift to four Hopf fibre circles paired by the deck into the TWO  $\mathbb{Z}_2$  sheet orbits with Gauss linking number 1.0000, the 4D bridge is the cone  $\mathbb{C}^2/\mathbb{Z}_4$  with the inversion  $x \mapsto x/|x|^2$  a conformal involution (pullback  $|x|^{-4} \delta$  symbolic) fixing the unit lens POINTWISE and swapping the aeon ends, and the Lorentzian shadow  $(1/\cos^2 T)(-dT^2 + g_{S^3})$  is Einstein  $R_{ab} = 3g_{ab}$  (symbolic Ricci) with a SPACELIKE round- $S^3$  crossover and  $\check{\Omega} = -\cos T$  vanishing to first order; (5) the KMS/modular closure of discrete  $\rightarrow$  dynamics at the free level: detailed balance EXACT,  $\text{spec}(-\ln T) = \{0, 6\ln \frac{3}{2}, 6\ln 3\}$  EXACTLY (the seam gap IS a modular energy),  $T = e^{-H_{\text{mod}}}$  (one clock tick = one unit of KMS time at  $\beta = 1$ ), OS Gram PSD, and the v526 normalisation  $\beta_{\text{angle}} = 2\pi = 1/(4c_3)$  replicated exactly = the Gibbons–Hawking normalisation on the crossover fibre; (6) the holonomy dictionary: the two clock-pole circles descend to the lens core geodesics with the full  $\mu_4$  monodromy  $\chi(D) = i$  while EVERY seam string carries exactly the 2-torsion  $\chi(D^2) = -1$ , and  $D^2 = -1$  is trivial on the base while flipping the spinor coordinates — the v522 statement “gaugeable datum = the 2-torsion  $\{1, (-1)^F\}$ ; full wrap = fermion parity” in geometric form (typed [C] correspondence); (7) the selector necessity (the  $\gamma$ -facing theorem): the admissible OS reflections of the v512 mark family  $M(\delta)$  form the pair-exchanging  $V_4$  for every  $\delta \neq \pi/2$  and close to  $D_4$  EXACTLY at  $\delta = \pi/2$  (the order-4 clock factorises into admissible reflections,  $\rho = \tau \circ s_{\text{mid}}$ , iff  $\mu_4$ ), and at Majorana operator level on the  $N = 16$  seam ring with fermionic reordering signs kept the free parent is  $\Theta_a$ -real on ALL 16 axes (v521 side-blindness replicated) while the interacting family is  $\Theta$ -real on exactly  $\{k \neq 4 : 2, k = 4 : 4\}$  axes and the clock-axis cuts ONLY at  $k = 4$ , asymmetric couplings breaking reality — the UNIQUENESS half of the v534 straddle selector ( $\delta = \pi/2$ , symmetric couplings) is thereby DERIVED as a  $\Theta$ -reality necessity (the survival half and the coupling sign stay v534’s nonperturbative toy

facts, cited; reality is sign-blind, checked); (8) the frozen Gate-6 kernel: exactly two decay modes  $\ln(729/64)$  and  $\ln 729$  (ratio  $\ln 3/\ln \frac{3}{2} = 2.709511$ ), the frozen defect-class overlap table  $(\pm 1/\sqrt{2}, +1/\sqrt{6}) / (0, -2/\sqrt{6})$  with the sheet-parity sign rule, disc radius  $\theta_{\max} = \eta_{\text{rec}}/(\eta_0 - \eta_{\text{rec}}) \approx 1.16^\circ$ , and  $\kappa$  DERIVED from the v526 KMS normalisation up to a declared convention band —  $u_{\text{rec}} \leq 8.3 \times 10^{-5}$  ticks across the whole band, relic contrast  $\leq 3.3 \times 10^{-4}$ : the template degenerates ROBUSTLY to the sharp causal top-hat disc; three freeze hashes pinned and recomputed in-run. NEW LEDGER ROWS: CCC.SEAM.KINEMATICS.01 [E] (the certified dictionary) and CCC.SEAM.CROSSOVER.01 [O] (the dynamical-crossover research contract: D1 interacting algebra on the crossover = WOIT  $\gamma$  in the crossover geometry, D2 field-level Stinespring, D3 the cosmology fork decision — Starobinsky  $r = 12/N_\star^2 \in [0.0033, 0.0048]$  priced decidable by  $\sigma(r) \sim 10^{-3}$  experiments —, D4 the executed disc search; kill criteria frozen incl. the forbidden mark/Hawking-point count identification); new contract section in `tfpt_research_contracts`; the origin-theory Penrose-CCC bullet SHARPENED (asymptotic de Sitter does NOT imply conformal flatness —  $C^2 = 48M^2/r^6$  for Schwarzschild–de Sitter, machine-checked; the CCC smoothness/Weyl conditions are additional crossover data) and pointed at the certified kinematics; PREREGISTERED SEARCH scaffolded at `experiments/ccc-crossover-disc/` (frozen hypotheses `ccc_disc_v1.yaml` with kill criteria K1–K6, null battery and  $\text{BH-}q < 0.01 +$  replication decision rule; freeze-check CLI 3/3 hashes intact; 5/5 guard tests; catalog entry §7b.9 of the experiments README) — verdict slot `data_limited` (preregistered); the Planck PR3 SMICA data pass executed separately at experiments level (first data contact logged in the project README). FENCES CARRIED EVERYWHERE: the cyclic/CCC reading stays [C] and is NOT upgraded by the promotion; WOIT.OS.TWISTOR.01 stays [O] with all kill tests live (the module certifies kinematics + free level + the necessity direction ONLY); the deck fork is decided only CONDITIONALLY on the v522 wrap identification; a null disc search damages nothing in the compiler core. Research notes DL–DLV appended to `experiments/next.txt`; FULL `run_all.py` ALL CHECKS PASSED at 950 modules; `bash build.sh` release green; `bash build.sh` audit → AUDIT OK; manifests regenerated LAST. No omega closed beyond the two new rows; NO claim that CCC follows from TFPT)

2026-08-24 (CXVII) (the integrable-dictionary-arc promotion — the seven durable certified theorem sets of discovery rounds 224–226 and 228–231 graduate from the experiments tree into the load-bearing verification suite as TWO modules (suite 947 → 949, ledger 1099 → 1104); every source probe RE-RUN GREEN AS TYPED at promotion and embedded BYTE-EXACT with its printed SPEC SHA pinned and gated (all seven at 100% first pass, deterministic records, sealed specs; rounds 199–223 stay experiments-side as adjudication/classification rounds, and round 227, the Fermi-edge classification, stays experiments-side by design with its builders embedded as a read-only library): MODULE ONE, `v955_tau_iiks_toda_dictionary.py` (4/4 pattern gates,  $\sim 3$  s; probes `tau_symbolic_probe.py` r224 note DL, 18/18, SPEC `fb04cd7ce9fb306b`; `lax_conditioned_probe.py` r225 note DLI, 15/15, SPEC `93fd9759db7e7b3c`; `hirota_sign_probe.py` r226 note DLII, 17/17, SPEC `d78e236bf633de7b`; the v881 library `port_integrable_kernel_probe.py` embedded read-only): THE FINITE INTEGRABLE DICTIONARY OF THE WALL — (1) THE TAU IDENTIFICATION EXECUTED (TAU-IIKS-EXACT): the wall determinant IS the exact IKS Fredholm tau function at the finite-identity



level on all five heavy rungs — Sylvester as a FULL characteristic identity  $\det(I - sQ_{\text{state}}) = \det(I - sE_{\text{node}})$  on the whole  $s$ -grid, the  $s$ -DRESSED Schur family  $\det(I - sE) = \det(I - sR) \det(I - sD_P(s))$  EXACT with the fixed-kernel alias (gap 0.06–0.20 at  $s = 0.5$ ) named and blocked, the confluent diagonal ( $\leq 3.2 \times 10^{-13}$  — the contract’s declared death spot dissolves), the IKS dressing identity ( $\leq 7.9 \times 10^{-13}$ ) with the dressed port generators SOURCE-EXPLICIT IN CLOSED FORM (the G31 surprise,  $\leq 1.5 \times 10^{-13}$ ), the Fredholm/JMU variational identity in TWO independent times ( $\leq 1.4 \times 10^{-9}$ ), and the relative tau transfer exact over  $-212.84/-195.50$  log-unit determinant collapses ( $\leq 3.4 \times 10^{-12}$ ); controls INVERTED per contract (Epstein/scramble SATISFY the algebra and differ in the VALUE  $\text{sign } \tau(1) = -1$ ); (2) THE LAX DEGREE ADJUDICATED (LAX1-H-EXACT): the word “2” was a hypothesis — the true minimal degree in the closing direction is ONE and the closing direction is  $h$ , not  $s$  (rank-1  $h$ -step at  $2.9 \times 10^{-16}$ , explicit rank-2 relative displacement, blind next-generator prediction  $3.4 \times 10^{-16}$  and blind tau-step prediction  $3.2 \times 10^{-11}$  on the sealed rungs, zero curvature identically world-blind, mpmath wards dps 80/120; the  $s$ -time has NO fixed degree — basis-invariant  $\varepsilon_d$  plateau 0.34–0.60, CONDITIONING-ONLY — and across rungs NO common carrier: typed, not hidden); (3) THE SIGNED-TODA DICTIONARY (HIROTA-TODA-EXACT + WALL-EQUIVALENT):  $\tau_{w,n} = D_n(\mu - \nu)/D_n(\mu)$  (Hankel-determinant QUOTIENT),  $r_n = h_n(\mu - \nu)/h_n(\mu)$ , the Hirota coefficient source-pure from a scaled SIGNED Stieltjes recursion (worst  $1.7 \times 10^{-11}$ .. $4.0 \times 10^{-8}$  at full depths, dps-60/80 warded), and the sign adjudication:  $\hat{\gamma}_n > 0$  for all  $n \iff$  QUASI-DEFINITENESS of the signed defect measure  $\mu - \nu \iff$  the wall, with the explicit SIGNED-TODA  $X - Y$  structure — MAIN carries ALL 184  $r_n > 0$  (min +0.3666), the controls flip at  $n = 25/21/27$ ; HIROTA-CONE-GO not reached (the expected base verdict, now a theorem); and MODULE TWO, v956\_signedmoment\_halffilling\_duality.py (7/7 pattern gates,  $\sim 9$  s; probes holedual\_probe.py r228 note DLIV, 14/14, SPEC a823d4be3f0b1c06; pontryagin\_maxpos\_probe.py r229 note DLV, 17/17, SPEC b062906cb458da2a; jfraction\_probe.py r230 note DLVI, 17/17, SPEC 124cda6f00caeeca; rhp\_midpoint\_probe.py r231 note DLVII, 13/13, SPEC 90d2c5a8926d820a; read-only libraries port\_integrable\_kernel\_probe.py, hirota\_sign\_probe.py, fermiedge\_classify\_probe.py): THE HALF-FILLING GEOMETRY OF THE SIGNED MOMENT PROBLEM — (1) THE HALF-FILLING BOUNDARY (NOT-A-HOLE-EDGE + HALF-FILLING-BOUNDARY):  $N_w = \lceil \#\text{supp}(\tilde{\mu})/2 \rceil$  EXACT on all five windows and past the cap the wall dies IMMEDIATELY (first  $r$ -flip at  $N_w + 0/2/2/3/1$ ,  $O(1)$  offsets, two independent paths + mp dps-40 ward) —  $\mu - \nu$  is quasi-definite EXACTLY up to half-filling, THE WALL IS MAXIMAL; the complement identity and  $h$ -duality exact at dps 80 ( $1.3 \times 10^{-78}$  synthetic /  $6.6 \times 10^{-56}$  real w9 subset); R-DUAL-OBSTRUCTED typed; (2) THE FREE MOMENT WINDOW LAW (SIGNATURE-EXPLANATION-REFUTED + FREE-MOMENT-WINDOW-LAW + MAIN-EXHAUSTS-FREE-MOMENT-WINDOW): both sharp signature predictions FAIL on every window ( $n_{\text{flip}} = 184/153/170/367/592 \neq p = 263/211/237/503/816$ ); the corrected law — the largest Hankel window inside the  $S$  free moment parameters is  $n = (S + 1)/2 = N_w$ , MAIN survives THE ENTIRE FREE MOMENT WINDOW plus  $O(1)$  forced-tail degrees while the controls die at 11–15% of it ( $n = 25/21/27$ ); the inertia theorem exact in all parts (boundary inertia (184, 1, 0); mp dps-60 chain  $\text{ind}_-(H_{263}) = 43 \leq q = 104$ ); the maximal Lagrange contractor exact on the toy (the equivalence  $H_p > 0 \iff \|C\| < 1$  in BOTH truth values,  $10^{-53}$ .. $10^{-62}$ ) and ASTRONOMICALLY non-contractive at full degree

( $\sigma_{\max}(C) = e^{110.5}$ , MAXDEG-NOT-CONTRACTIVE); at window degree SAME-CONTRACTOR-EXACT ( $\lambda_{\max}(Q_{N_w}) = 0.999832..0.999999 < 1$  — the v881 testing law closes the circle); (3) THE FOUR-PATH DICTIONARY (FREEPREFIX-EXACT-WALL-EQUIVALENT + FULL-CHAIN-REVERSAL-EXACT + NO-SOURCE-REVERSAL): moment prefix = Jacobi chain = Euclidean chain = value chain, one object in four exact source-pure paths ( $m_k = h_0(J^k)_{00}$  exact in rationals; the r228 offsets ARE forced-coupling survival counts 0/2/2/3/1; the dps-300 Euclid prefix matches the real w9 value chain at  $2.5 \times 10^{-13}$ ; Epstein's FIRST WRONG EUCLIDEAN PIVOT exactly at its flip  $n = 25$ ); the dual signed measure carries the MIRRORED Jacobi chain ( $\alpha_m^\# = \alpha_{S-1-m}$  AND  $\beta_m^\# = \beta_{S-m}$ ) but is NOT a builder transformation — half-filling is not the center of a self-dual chain;  $\beta > 0$  stays the wall in EVERY coordinate; (4) THE MID-POINT CONNECTION THEOREM (MIDPOINT-CONNECTION-EXACT + SIGNED-STOKES-WALL-EQUIVALENT, derived by residue calculus BEFORE the run and frozen): the dual FIK problem is an EXPLICIT  $L$ -gauge transform of the original,  $Y_{N-1+l}^\# = JY_{N-1-l+1}W$  with  $W = \text{antidiag}(1/L, L)$ , built from THE NODE POLYNOMIAL ALONE (exact in rationals; real w9 at  $9.1 \times 10^{-94}$ , dps 100); the gauge is  $h$ -FREE, so the wall's orientation sits EXCLUSIVELY in the  $h$ -chain the gauge does not see — the decisive Stokes multiplier is  $c_w\beta_n$  with  $c_w > 0$ ; honest negatives typed (NODE-LOG-POLE-REMOVER-ONLY; NO-SOURCE-CRITICAL-FILLING,  $t_c$  open; QUENCHED model supported at spread 0.44); THE MICRO FALSIFIER FROZEN: any future parametrix must blindly predict the forced tail 0/2/2/3/1 AND the control flips 25/21/27 from the same connection. THE CLAIM SPLITTING OF NOTES DLI/DLVII CARRIED TO THE LEDGER: new rows PRIME.PORT.TAU.FINITE.IIKS.01 [E] and PRIME.PORT.TAU.NOPOLE.COFINAL.01 [O] split out of the v883-registered umbrella PRIME.PORT.TAU.01 (which stays [O] for the fully symbolic arbitrary- $h$  statement, its ledger prose updated to record the finite-identity execution — NO silent upgrade), and PRIME.FREEMOMENT.JFRACTION.01 [E] + PRIME.JACOBI.DUAL.REVERSAL.01 [E] split from PRIME.FREEMOMENT.POSITIVEPREFIX.01 [O] (the wall itself in free-moment coordinates; half-filling is the LOCATION, its positive reachability the result to be proven); every wall statement in both modules is typed WALL-EQUIVALENT and consumed nowhere; surfaces: two veri cards in tfpt\_3, the new integrable-dictionary-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed (947  $\rightarrow$  949) and the TAU.01/LAX2.01 execution notes added at every [O] mention, the PRIME.PORT.TAU.FINITE.IIKS.01 / PRIME.PORT.TAU.NOPOLE.COFINAL.01 / freemoment contract blocks and the LAX2/HIROTA adjudication notes in tfpt\_research\_contracts, the endform article promotion section + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v956, README/Zenodo 949; FULL run\_all.py ALL CHECKS PASSED at 949 modules; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. The mincut stays base 4 / refined 5; no omega closed, no marker moved beyond the four [E]/[O] registrations and the recorded execution notes; NO RH claim anywhere)

2026-08-22 (CXVI) (the Bughunt-X-arc promotion — the six durable certified theorem sets of discovery rounds 192 and 194–198 graduate from the experiments tree into the load-bearing verification suite (suite 941  $\rightarrow$  947, ledger 1093  $\rightarrow$  1099), after the tenth adversarial bughunt (round 199, note DXXIII, bughunt10\_probe.py 30/30, SPEC 5551aa7b967230f1, 11 findings, 1 MAJOR, 0 verdict flips, AND THE



**X4 HEADLINE:** every exact claim of the arc reproduced in fully own code — the ACF/pole/arch laws to  $\leq 10^{-61}$ , the Fejér cap re-proved by an independent single-node dual route, the classical LMN/Matveev/Kolountzakis–Révész/Siegel citations web-verified verbatim) covered the arc and its corrections of record (KA/KB/KC/KE applied as appended CORRECTION-OF-RECORD blocks, KD note-only per precedent, note DXXIV) were ADOPTED VERBATIM on every new surface (in particular KA MAJOR: the round-197 nonnegativity gate class is retyped GRID-MEASURED with the verbatim scope rider — the record gate was grid-only with a sign-blind bar at the three deepest rungs, THE HEADLINE IS TRUE AND NOW STRONGER via Bughunt X’s exact-rational Sturm chains at  $h = 4, 5, 13, 20$ ; KB MINOR: SUBCONE-SHARP-AT-Q2 reads “exact attainment at RATIONAL lags only, near-attainment elsewhere” — the “f64 precision” wording is retired; KC: the “edge minimum IS the  $j_{r_0}$  ladder” identification is exact UN-NORMALIZED, the r197/r198 pair consistent; KE: the round-192 G11 symbolic leg was vacuous — the  $e \mid 2fk$  law’s machine content lives in the census gates; KD POSITIVE, note-only per precedent and recomputed in v951: the arch kernel IS the classical Weil archimedean kernel in the  $L \rightarrow \infty$  limit); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v949\_commensurability\_mechanism.py (11/11, POSITIVITY-IRRELEVANT-EXACTNESS-KILL; probe commensurability\_mechanism\_probe.py r192 note DXIV, 28/28, SPEC dbc14014899fb286 verified invariant under the appended KE block, record 276 s, re-run green at promotion 317.0 s): THE EXACT NODE MECHANISM — the smooth world’s whole  $2a$ -periodic oscillation collapses AT THE NODES to Möbius data (closed forms recomputed in-run), the structural-zero census is the  $e \mid 2fk$  law in exact integers ( $h = 4$ : 12/18,  $h = 8$ : 32/120), and THE FIRST BAKER-CLASS PRICING IN THE PROGRAM’S HISTORY is recomputed from the published constants (Liouville gap 3.79 dex, LMN 1995 gap  $7.099 \times 10^3$  dex, Matveev 2000 gap  $1.682 \times 10^9$  dex at the  $h = 4$  minimizer; full ladders to  $3.10 \times 10^4$  /  $1.18 \times 10^{10}$  dex: BAKER-TOO-WEAK); the dose is DIRECTION-BLIND (killed both ways at every dose rung + jitter, COMM-WALL-DEAD) and  $L_f(t = 1)$  is PSD at all four dose rungs WORLD-BLIND: the margin consumes position EXACTNESS, not incommensurability magnitude — the sought source functional must consume the exact positions wholesale; MODULE TWO, v950\_zero\_channel\_capacity.py (8/8, CHANNEL-CHARACTERIZED; probe zero\_channel\_capacity\_probe.py r194 note DXVII, 28/28, SPEC fa49271201ba30fb, record 170 s, re-run green at promotion 209.6 s): THE SHANNON CHARACTERIZATION OF THE ZERO-COMB DICTIONARY — typed EXPLORATORY INSTRUMENT CHARACTERIZATION at discovery and promoted as exactly that (NO storage claim, NO arithmetic novelty): the capacity surface reaches error-free full ternary ( $\log_2 3$ ) from  $\varepsilon = 0.01$ ; RESLIM LOW = 0.001 / MID = 0.003 WITH THE BUGHUNT-X F6 QUANTIFICATION ADOPTED ( $P = 0.0032$  under the 20000-trial null MC — marginal-but-nonnull single-reading, decisive with averaging  $0.042 \rightarrow 0.257 \rightarrow 0.938$ ; the sqrt- $n$  prediction held at 0.2325); LINEARITY-SCALE 0.01; parity-erasure recovery 1.000 WITH THE F7 COROLLARY TYPING (a corollary of the zero-SER cell, not independent evidence; the 108-case parity-code erasure theorem recomputed exactly in-run); P3 refuted honest (CHANNEL-GAIN-LOW>MID 0.508); the round-107 verdict cited NOT upgraded; MODULE THREE, v951\_cancellation\_functional.py (10/10, WALL-IS-WEIL-ACF-FORM-EXACT; probe cancellation\_functional\_probe.py r195 note DXVIII, 24/24,

SPEC a50b85bb112513a1 verified UNTOUCHED by the Bughunt-X corrections, record 1036 s + 1061 s re-run, four pre-freeze amendments disclosed incl. A3 the substantive float64-direction-noise reversal, re-run green at promotion 1102.8 s): THE ACF LAW ( $x^T W(u) x = -2 \int_0^{L-u} A_x A_x(\cdot + u) dt$  — the v947 two quadrature slots are ONE OBJECT; verified at every atom of  $h = 4, 5$  on an own in-module wall build), THE POLE SQUARE ( $8 \sinh^2(a/2) (\int e^{-t/2} A_x)^2$ , sympy generic), THE ARCH LAW ( $c^* = (\gamma + \ln 2\pi)/2$  exactly, every mode of  $h = 4, 5$ ) — THE WALL IS A CLOSED TRUNCATED WEIL FORM of the PD test function  $g_x$ ; THE KD PIN RECOMPUTED IN-RUN (the arc's external handoff object):  $\kappa + c^* \rightarrow -\frac{1}{2}[\operatorname{Re} \psi(1/4 + i\omega/2) - \ln \pi]$  in the  $L \rightarrow \infty$  limit (BH10:  $6 \times 10^{-18}$ ; finite- $L$  corrections  $O(e^{-L/2})$  typed not-small); per-atom manifest positivity PROVABLY IMPOSSIBLE (Wiener–Khinchin); the pole alone carries positivity; the sign law  $n_+ = 0$  at all 14 rungs; priced honestly as an ALGEBRAIC RESTATEMENT (the relabeling barrier named, not crossed; WEIL-ALLTESTS  $\leftrightarrow$  RH a flagged loop consumed by nothing); MODULE FOUR, v952\_turan\_extremal.py (9/9, TURAN-BUDGET-TAU-GAP; probe turan\_extremal\_probe.py r196 note DXXI, 23/23, SPEC a6edc3f911e8f069 verified invariant under the appended KB block, record 767 s + 802 s re-run, re-run green at promotion 795.6 s): THE CLASSICAL PD-CONE THEORY PRICED for the first time in 196 rounds, TWO-SIDED — every cone bound NEGATIVE at every rung while the truth is  $\tau$ -deep positive (CONE GEOMETRY CANNOT SEE POSITIVITY; gapdex 11.35  $\rightarrow$  89.03 == the  $\tau$  ladder at slope  $-1.007$ ,  $R^2$  1.000 — the relabeling barrier named AT THE CONE LEVEL) yet the classical Fejér cap describes the wall's drain capacity almost exactly (CONE-DRAIN-O1-CAPTURE 0.14–0.24 dex flat — the round's real news); the Fejér coefficient theorem machine-proved in-module at  $n = 1, 2$  (construction + single-node dual certificates); THE KB RATIONAL-LAG IDENTITY RECOMPUTED ( $M_{\text{sub}}(L/2) == 1/2 == \cos(\pi/3)$  identically — the Fejér extremal lifts exactly at rational lags ONLY); ARCH-LEG-SUBCONE-ONLY named; the TURAN-CONE-POSITIVITY loop flagged consumed by nothing; MODULE FIVE, v953\_fewatom\_reduction.py (10/10, A-V0-NONNEGATIVE-STURM-CERTIFIED; probe fewatom\_reduction\_probe.py r197 note DXX, 25/25, SPEC 35fb341bb281b04b verified invariant under the appended KA/KC block, record 790 s + 811 s re-run, re-run green at promotion 830.7 s):  $A_{v_0} \geq 0$  AT ALL 14 RUNGS with THE KA SCOPE RIDER VERBATIM (the r197 gate class GRID-MEASURED; at  $h = 15/16/20$  the record minimum sits below the bar's resolution — print evidence, not gate certificate; THE HEADLINE IS TRUE AND NOW STRONGER) and THE STURM CONTINUUM CERTIFICATES RECOMPUTED IN-MODULE at  $h = 4, 5$  (exact dyadic Fractions  $\rightarrow$  Chebyshev  $\rightarrow$  INTEGER primitive-PRS Sturm chain: zero roots in  $(-1, 1]$ ,  $P(\pm 1) > 0$  —  $A_{v_0} > 0$  on ALL of  $[0, L]$ , no floats; Bughunt X extends to  $h = 13, 20$  — THE BUGHUNT'S MATHEMATICAL DELIVERABLE FOLDED INTO THE PROMOTION, the bughunt round itself experiments-side per process precedent); THE ONE-LINE MECHANISM ( $A \geq 0 \Rightarrow$  every ACF sample  $\geq 0 \Rightarrow$  every atom negative — the v951 sign law reduced); the horizon bounded ( $Q_{99} \in \{2, 3\}$  incl. the holdout) WITH THE MANDATORY SCALE-RELATIVITY RIDER (the demand scale consumes every atom; NOT-A-COMPLEXITY-DROP-IN-H; any lossy tail bound certification-dead); the KC identification exact UN-NORMALIZED; OBJECT-A NAMED (the X6 targeting result): “prove  $A_{v_0(h)} \geq 0$  for all  $h$ ” is the ONE genuinely new tractable target — yields the sign law only, NO lever on wall positivity; and MODULE SIX, v954\_nodeless\_pf.py (10/10, PF-MODE-BASIS-DEAD-POLE-IS-THE-OBSTRUCTION; probe nodeless\_pf\_probe.py

r198 note DXXII, 27/27, SPEC 7499c39a026d0d0f, record + deterministic re-run, amendment A1 disclosed, re-run green at promotion 817.1 s): THE PERRON-FROBENIUS ADJUDICATION — the r197 candidate route for Object-A DEAD IN THE MODE BASIS (off-diagonal censuses recomputed exactly at  $h = 4, 5$ : (15,6)/(38,17) mixed, checkerboard-mixed both parities; the trivial route killed both ways by the exact exhibit  $v = e_1$ ,  $A(L/2) = -1$ ), the obstruction LOCALIZED (the cone tests recomputed at  $h = 4$ : the wall operator violates on 11..14 of 20 Fejér members, THE NO-POLE OPERATOR ON ZERO — removing the pole alone repairs cone invariance: the piece that carries the positivity is the piece that breaks the cone), and the SURVIVING Z-STRUCTURE pinned (KERNEL-SPLIT-POLE-POSITIVE-HOPPING-Z: pole kernel negfrac 0.000 exactly, the wall’s kernel mixing ENTIRELY pole-vs-hopping); Object-A stays open, GENUINELY FUNCTION-SPACE; the X6 discipline: “pole-vs-hopping balance” is the CLASS NAME of Object-W, not a new attack object. Ledger rows PRIME.COMMENSURABILITY.MECHANISM.01, PRIME.ZERO.CHANNEL.CAPACITY.01, PRIME.CANCELLATION.FUNCTIONAL.01, PRIME.TURAN.EXTREMAL.01, PRIME.FEWATOM.REDUCTION.01, PRIME.NODELESS.PF.01, all [E] NOT-RH-EVIDENCE active; the channel row carries the “instrument characterization, no storage claim” typing prominently; the arc’s terminal residue written everywhere in the canonical note-DII/DXVI/DXXIV form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k$  == LOOP, flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 = TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs: extension instantiated, TAILWPD world-front}; surfaces: six veri blocks in tfpt\_3, the new Bughunt-X-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed (941  $\rightarrow$  947; PDF recompiled), the endform article promotion section + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v954, README/Zenodo 947; one German research note appended to experiments/next.txt; FULL run\_all.py ALL CHECKS PASSED at 947 modules; bash build.sh release green; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. Round 193 (Bughunt IX) and round 199 (Bughunt X) stay experiments-side as process rounds per precedent — round 199’s Sturm-certificate deliverable is promoted INSIDE v953; the PRIME.RESIDUE.EXTERNAL.01 contract inventory deliberately NOT extended (the arc changes no residue statement). No omega closed, no marker moved beyond the six [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-21 (CXV) (the mechanism-arc promotion — the seven durable certified theorem sets of discovery rounds 184–191 graduate from the experiments tree into the load-bearing verification suite (suite 934  $\rightarrow$  941, ledger 1085  $\rightarrow$  1093), after the ninth adversarial bughunt (round 193, note DXV, 44/44, 9 findings, 1 MAJOR, 0 verdict flips, AND THE HEADLINE X5: every claimed-exact object of the arc reproduced in fully independent code with ZERO FAILED RECOMPUTES) covered the arc and its corrections of record (K1–K5, note DXVI) were applied as appended CORRECTION-OF-RECORD blocks and ADOPTED VERBATIM on every new surface (in particular K1 MAJOR: the round-186 structure token is written WALL-OFF-DIAGONAL-IS-ONE-FUNCTION-LOEWNER-EXACT everywhere — displacement rank 2 fixes ONLY the off-diagonal, the FULL wall is  $M_h \sim L_f + \text{diag}(\Delta)$  with source-side  $\Delta \neq 0$ , NOT the canonical Loewner matrix of

the Loewner-1934 dictionary, and  $R_h(z)$  is the resolvent of  $L_f + \text{diag}(\Delta)$ , never the full-matrix claim; K2 MINOR: the ZB wiggle reads 1.07 in full precision (1.065464, twice replicated) — never the unsourced 1.14; K3 MINOR: the round-184 smoke lineage disclosed; K4: the glossary rule — every “source-side” qualified as either (eigenvector-free, given  $\tau$ ) or (root-free, ray-consuming)); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v942\_pi\_pattern\_transfer\_specificity.py (9/9, PI-TRANSFER-SPECIFICITY; probe pi\_pattern\_scan\_probe.py r184 note DIV, 35/35, SPEC 00fc85173fe07470, record 191 s, re-run green at promotion 212.2 s): THE DETECTOR-SPECIFICITY THEOREM — all five predefined pi-derived comb worlds land in the fake-world class (log10 tails  $-10.0.. -17.9$ , both holdout halves, ALL 100 non-Mangoldt instances, ZERO anomalies) while the C-MANGOLDT positive control FIRES ON EVERY TRANSFERRED DETECTOR (spike  $C = 131.04$  vs null max 12.60; the structured controls prove the detectors CAN fire; pi’s classical CF laws replicate as measured expectations) — TRANSFER VALIDITY PROVEN, the load-bearing methodology of the whole arc; typed INSTRUMENT CHARACTERIZATION, NOT ARITHMETIC NOVELTY, no pi-numerology claim, the P-BBP PROVISIONAL flag travels; MODULE TWO, v943\_mangoldt\_ablation\_localization.py (8/8, MANGOLDT-ABLATION-LOCALIZED; probe mangoldt\_ablation\_ladder\_probe.py r185 note DV, 26/27 with the single FAIL the disclosed control-side Beurling density band (5.74 vs  $[0.2, 5.0]$ ) — BH9-adjudicated an honest Poisson fluctuation, density-correct, the W4 headline unweakened; SPEC 504dcac5b2eac650, re-run green as typed at promotion 167.2 s, same single disclosed FAIL): THE ZERO-COUPLED SHARP LOCALIZATION — the additive property ladder W0 random  $\rightarrow$  W5 true Mangoldt switches BOTH arithmetic signatures on ONLY at W5 and SHARPLY (S1 jump 12.07 dex, S2 jump  $61\times$ ); THE BEURLING HEADLINE ANSWER IS NO (full multiplicativity without Riemann-zero coupling produces NEITHER signature); necessity confirmed by every ablation, EVEN  $\pm 0.01$  POSITION JITTER KILLS BOTH; MODULE THREE, v944\_ground\_residue\_observability.py (11/11, GROUND-RESIDUE-OBSERVABILITY; probe ground\_residue\_obs\_probe.py r186 note DVI, 33/33, SPEC 48637c8898a1da5a verified invariant under the appended K1 correction block, record 782 s, amendment A1 disclosed, re-run green at promotion 792.7 s): THE EXACT RESIDUE IDENTITY  $\text{Res}_{z=\tau} R_h(z) = A_0^2/\|d\|^2$  gated at ALL 14 rungs to  $\leq 7.8 \times 10^{-45}$ , the Feshbach scalar form with  $w'(\tau)$  EIGENVECTOR-FREE GIVEN  $\tau$  ( $\{\alpha, m, B\}$  source-linear, the CDLI adjudicate tie disclosed as prior art), STRICT INTERLACING  $\Leftrightarrow A_0 \neq 0$  both directions exact, GAP-IS-RESIDUE-IN-DISGUISE ( $\varepsilon \leq 2.55 \times 10^{-6}$ : not an independent handle), the honest descent EIGENVECTOR-DESCENDS-NOT-ELIMINATED (11.2–90.7 orders), the raw-gap floor RELABELED-AND-STOPPED, the  $\{l_0, l_2\}$  witness detector, and the K1-corrected off-diagonal Loewner law; MODULE FOUR, v945\_zero\_causal\_stratification.py (10/10, ZERO-CAUSALITY-DEMONSTRATED-WITH-STRATIFICATION; ONE JOINT MODULE for probes zero\_causal\_synth\_probe.py r187 note DVIII, 31/33 with the two FAILS the disclosed adjudication gates (the f-cell monotonicity violation and the r175-predicted S1 instrument death), SPEC c20e87eec6d158b9, re-run green as typed at promotion 819.0 s, and zb\_wiggle\_strat\_probe.py r190 note DXI, 23/23, SPEC 8639b3a78503a0f9 verified invariant under the appended K2 block, re-run green at promotion 64.5 s — r190 upgrades r187 BY REFERENCE, house style): THE CAUSAL ZERO-COUPLING DEMONSTRATION — comb worlds with

controlled zero spectra synthesized through the r175 bridge run backwards; Z0 VALIDITY 0.0006 on  $C = 131.04$ ; the off-line dose surface monotone in  $\delta$  everywhere (first response  $(0.10, 10^{-2})$ , first death  $(0.05, 10^{-4})$ ); EXACT-ORDINATES-REQUIRED (Poisson AND GUE resamples both die); the stratification: the wiggle 1.07 deep inside every matched-dose selection spread  $(5.09/3.07/3.98)$ , dose monotone in expectation, INV-FRAC 1.00 generic subset-overshoot, the per-zero leverage ladder ( $\gamma_2 = 21.02$  carries the most, no first-zero dominance, strongly subadditive), depth parity not-artifact; the r185 interpretation upgrades to DEMONSTRATED MECHANISM on the S2 channel — still NO claim about WHERE the true zeros lie; MODULE FIVE, v946\_census\_krein\_pencil.py (8/8, CENSUS-KREIN-PENCIL-PARTIAL-LIFT; probe census\_lift\_probe.py r188 note DIX, 72/72, SPEC 8ada6b97d56aca46, record 2066 s + 2060 s re-run, log lineage disclosed in spec, re-run green at promotion 2067.1 s): THE EXACT KREIN PENCIL —  $N_h(y) = (-1)^{K-1} \det(J) A_0 \det(\hat{A} - yJ)$  source-side with  $\sum \rho_k = -y_t$  exactly (dyadic-exact E1 at all six rungs, integer-Bareiss E2 at  $h \leq 13$ ); the residue-sign ladder  $(1, 5)/(7, 3)/(6, 14)/(27, 14)/(45, 34)/(43, 73)$  MIXED at every true rung; MIXEDNESS-TRANSFORM-INVARIANT (definite impossible in class); MIXEDNESS-IS-ARITHMETIC (atoms-vs-no-atoms, wrong orientation, NOT a sign source); H2 == KREIN STRONG DEFINITIZABILITY in operator form; the bracket certified via ENVJ without computing a single root; roots never consumed, tau-free; MODULE SIX, v947\_loewner\_pick\_dictionary.py (9/9, LOEWNER-PICK-DICTIONARY-MISMATCH; probe loewner\_pick\_probe.py r189 note DX, 28/28, SPEC a547448468899af9, record 1046 s + 1058 s re-run, three pre-freeze amendments disclosed, re-run green at promotion 1082.9 s): THE EXACT DICTIONARY — the wall is a DIAGONALLY-SHIFTED one-function Loewner matrix  $\text{Raw}_h = L_f + \text{diag}(\Delta)$ , with THE NEW EXACT COSINE-QUADRATURE DIAGONAL LAW  $\text{Raw}[k, k] - f'_{\text{prime}}(b_k) = 2a \text{pc}_k$  (both quadratures of the same atoms in different slots; sympy per-atom,  $\text{mp} \leq 6.0 \times 10^{-61}$ , BH9 own recompute  $\leq 6.3 \times 10^{-61}$ ); the pole leg a genuine Pick function with EXACT Herglotz point mass  $2 \sinh^2(a/2) \delta_{-1/4}$ ; the Pick-class reading REFUTED with margins ( $\lambda_{\min}(L_f) < 0$  at all 14 rungs, misses wall positivity by 9.4–87.9 orders, refinement deepens); SMOOTH-PSD-AT-NODES via the exact commensurate-sampling mechanism  $\sin(2a \omega_k) == 0$  (lattice geometry, not a prime fingerprint); tau-screens both flat; the r45 LOEWNER-DEAD distinction carried; and MODULE SEVEN, v948\_krein\_sign\_characteristic.py (10/10, KREIN-SIGN-CHARACTERISTIC-LAW; probe krein\_definitizer\_probe.py r191 note DXIII, 57/57, SPEC 332c1f48f48a6d82, record 1950 s + 1901 s re-run, no post-freeze amendments, re-run green at promotion 1983.4 s): THE SIGN-CHARACTERISTIC LAW — the kernel vector explicit,  $\varepsilon_i = -\text{sign}((F/A_0)'(y_i))$ , THE r188 RESIDUE LADDER IS THE KREIN SIGN-CHARACTERISTIC MULTISSET (two-route proven; in the module on the full 32-cell exact Sturm battery with no floats), THE PARITY SCAFFOLD (root-count parity per pole gap from residue signs alone — the new source-side coordinate, verified at all 81 gaps), THE  $W(y)$ -CONGRUENCE-CLASS KILL BY THEOREM (the sign characteristic is congruence-invariant: the mixed ladder kills the entire  $W(y) > 0$  class at once — the r188 upgrade path IMPOSSIBLE-IN-CLASS, remaining doors named honestly), the Langer degeneracy exhibited (Cayley–Hamilton; strong definitizability == H2 per rung), the realness-break witness detector (the Krein invariant sees the witness class where the raw ladder is blind), the strict definitizer ROOT-DEPENDENT (occupancy-minimality fails at  $h = 8, 13$  by exactly one interior doubled gap). Ledger rows PI.PATTERN.SCAN.01,

PRIME.MANGOLDT.ABLATION.01, PRIME.GROUND.RESIDUE.OBS.01, PRIME.ZERO.CAUSAL.SYNTH.01, PRIME.ZB.WIGGLE.STRAT.01 (both on v945), PRIME.CENSUS.SPECTRAL.LIFT.01, PRIME.LOEWNER.PICK.01, PRIME.KREIN.DEFINITIZER.01, all [E] NOT-RH-EVIDENCE active; the mechanism-arc rows carry the NO-RH fences and the “instrument characterization, not arithmetic novelty” typing prominently; the arc’s terminal residue written everywhere in the canonical note-DII/DXVI form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k$  == LOOP, flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 = TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs: extension instantiated, TAILWPD world-front}; surfaces: seven veri blocks in tfpt\_3, the new mechanism-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed (934  $\rightarrow$  941; PDF recompiled), the endform article promotion paragraph + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v948, README/Zenodo 941; one German research note appended to experiments/next.txt; FULL run\_all.py ALL CHECKS PASSED at 941 modules; bash build.sh release green; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. Round 192 (the commensurability mechanism, note DXIV) STAYS experiments-side per precedent — it landed after the Bughunt-IX scope froze and waits for bughunt coverage; round 193 (Bughunt IX itself) stays as a process round. No omega closed, no marker moved beyond the eight [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-21 (CXIV) (the third externalization contract — the prime-front terminal residue packaged as a short, precise, self-contained EXTERNALIZATION CONTRACT PRIME.RESIDUE.EXTERNAL.01 for outside specialists in analytic number theory (large sieve, sampling/frame theory, de Branges spaces, explicit-formula methods), added as one new section in tfpt\_research\_contracts after the ALPHA.QUILLEN.EXACT.01 contract (before “Certifiability and recommended order”), plus a one-sentence pointer in the head keybox; DOCUMENTATION ONLY — no module touched, no number changed, no status marker moved, no ledger row added or edited, no gate closed or narrowed, NO RH claim (the contract identifier names the document, not a ledger row; the section states explicitly that it offers OPEN PROBLEMS and asserts no progress toward the Riemann Hypothesis in either direction): (1) the SELF-CONTAINED FINITE OBJECTS — rung  $h$ , modes  $\omega_k = k\pi/a$  with  $a = \frac{1}{2} \log h$  and  $K = \lceil 1.25 h \log h \rceil$ , the von Mangoldt atoms  $(\log q, \Lambda(q)/\sqrt{q})$  as the only arithmetic input, the wall matrix  $M_h = M_{\text{pole}} + M_{\text{arch}} - M_{\text{prime}}$  with the prime block linear in two atom transforms per mode (v935), the wall argmin ray  $d$ , the census polynomial  $N(y)$  (numerator of  $F(y) = c_0 + \sum (-1)^k c_k y/(y - b_k)$ ), the moment–Laurent form  $\Phi(z)$  (v924) and the jet mass  $\delta_h = |J_h|_{G_h}^2$  exactly (step A, v939); (2) the THREE MINIMAL OPEN STATEMENTS in the canonical note-DII form — R1 THE TRIPLE ( $\{H1 \wedge H2 \wedge H3\}$ -cofinal, one rung per dyadic block, all three at the same  $h$ ; H1: no census root with  $\text{Re } y \geq c^* y_t$ ,  $c^* = 1.10$ – $1.15$ ; H2: census complete-real nonnegative; H3:  $y_t \leq 0.155 T_z^4$ ; the limsup form only mod the measured defect  $D = 0.0042$ ), R2 THE LOOP CHARACTERIZATION (census-forall- $k$ , honestly stated as RH-equivalent-in-currency via the four independent machine-detected proof-route cycles v928/v929/v930/v938; external value: a new independent sign source or a proof the loop is essential) and R3 THE H-PIN (the one  $\lambda$ -uniform edge

of the pair {L1, WPD} with the  $\Omega$ -a/ $\Omega$ -b split and the two machine-exhibited linearity-escape maps — the eigenvector map and the census-root map, v937); (3) the COMPLETE TYPED HYPOTHESIS LIST — 11 PROVEN-FINITE rows citing v931/v932/v933/v934/v935/v936/v937/v938/v939/v940/v941 (theorem PF with the exponent the trace; the rate dictionary  $a = p/2 - 1$ ; the subsidy-empty pigeonhole; the unconditional Landau/Gonek pricing; the mode-level Landau bridge; the tau-free DK exclusion  $h \leq 13$  with the exact  $b^*(h)$  schedule; the WPD == H-pin consolidation; the exact census semigroup  $e^{-tT}$ ,  $T = 4y \partial^2 + 2 \partial$ ; step A exact; the sub-dof composition chain; the source-free Gram + projective linearity), 7 CITED-CLASSICAL rows (Landau 1912; Gonek 1985/1993 with Gonek 1984 flagged never consumed; Montgomery–Vaughan 1974; Ortega-Cerdà–Seip 2002; Nazarov–Turán 1993; Rodgers–Tao 2020 + Polymath 15 as cited ceilings), 1 OPEN row (R1–R3); (4) the HONEST MEASURED BLOCK — margins  $\geq 1.405$  with selector margins 2.14/3.05, the frozen alignment triple (descriptive band (0.38, 0.54), wall lock  $\mu_m = 0.2372 \log_{10} g_{\min}$  at  $R^2$  0.9916, deep excess 5.64–7.58 dex),  $\theta_\infty$  in  $[0.0766, 0.0977]$  typed OPEN-NONPERTURBATIVE-VARIATIONAL, and the refuter ladder ALWAYS PAIRED per BH8-F7 (measured 30–33 orders with the proven  $b^*(h)$  schedule 23.8/21.7/15.7/4.4 at  $h = 4/5/8/13$ , none at  $h \geq 16$ ); (5) the KILL-ATLAS SUMMARY TABLE — eleven named dead route classes (pointwise symbol minorants,  $\ell^1$  majorants, phase-discarding additive forms, Vieta pinch/power sums, moment/trace caps, Riccati/manifold transport, dBN pinch, global CBJ frame, sub-dof scoping, Carleson-as-floor, naive floor-power composition) each with its one-line death mechanism and round/witness; (6) the RUNNABLE FINITE MODELS — v931–v941 + the substrate v922–v930, the eleven frozen SPEC-hashed discovery probes (r171–r182), the frozen builder radius4\_an\_probe.py with the four control worlds, the  $2 \times 10^7$  verified-ordinate cache and the PT21 horizon, with exact paths; (7) the HONEST FORMALIZATION SURFACE — the only prime-front Lean content is the finite spacing/jet algebra (SpacingProduct.lean and companions); NOTHING of R1–R3 is formalised, stated plainly; (8) MUST-FAIL CONTROLS MF-P1–MF-P6 (Epstein/Scramble/Smooth refusal, the no-zero ancestry test, the predefinition test, and the loop firewall over the four flagged classes census-forall- $k$  /  $A_0$ -triangle / zero-verification-as-hypothesis / RH-conditional second moments); (9) ACCEPTANCE TESTS — an external result counts iff it delivers (a) an explicit all- $h$  all- $a$  bound, (b) a genuinely new sign source that separates AND independently orients, or (c) a proof that one of R1–R3 is RH-equivalent (recorded as closing the hardness question honestly, never as progress toward RH); the website paper mirror of tfpt\_research\_contracts (papers.ts) gains one section entry; tfpt\_research\_contracts.pdf, changelog.pdf and introduction.pdf recompiled and mirrored via bash build.sh website; maps and changelog.ts regenerated via bash build.sh gen; bash build.sh audit → AUDIT OK; manifests regenerated LAST. No omega closed, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-20 (CXIII) (the post-bughunt-arc promotion — the six durable certified theorem sets of discovery rounds 177–182 graduate from the experiments tree into the load-bearing verification suite (suite 928 → 934, ledger 1079 → 1085), after the eighth adversarial bughunt (round 183, note DI, 20/20, 7 findings, 1 MAJOR, 0 verdict flips) covered the arc and its corrections of record (F1–F7, note DII) were applied as an appended CORRECTION-OF-RECORD block and ADOPTED VERBATIM on every new surface (in particular BH8-F1: the



round-182 terminal adjudication is written SAME-CLASS-NOT-NEW-EDGE everywhere — a CLASS-level convergence, never an object identity — and BH8-F7: the round-177 refuter margin is always cited PAIRED, the proven  $b^*(h)$  schedule  $23.8/21.7/15.7/4.4$  orders above the  $10^{-25}$  bar at  $h = 4/5/8/13$  and none at  $h \geq 16$  alongside the measured 30–33 orders; per BH8-F2 the canonical rounds are 177 = manifold, 178 = L1/WPD, 179 = dBN, 180 = CBJ frame, 181 = CBJ sub-dof, 182 = alignment); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v936\_manifold\_invariance\_exclusion.py (10/10, MANIFOLD-INVARIANCE-EXCLUSION; probe manifold\_invariance\_probe.py r177 note CDXCV, 39/39, SPEC af86ec3b097ae8c4, record 1354.1 s, three amendments + one smoke fix disclosed, re-run green at promotion 1337.6 s): THE TAU-FREE DK EXCLUSION THEOREM —  $\sin \theta \leq (r_0/\tau)/(FG - a)$  with  $\tau$  CANCELLING in ratio form, direction-universal and witness-free, PROVEN  $h \leq 13$  at the frozen bar (measured to  $h = 20$ , the split kept) + the exact required-bar schedule  $b^*(h)$ ; the measured full-space refuter price ladder  $1.35 \times 10^5 \rightarrow 6.73 \times 10^8$  strictly growing with the mechanism identified as ALIGNMENT-SHARING; the BA3 scale-gauge theorem (not a universal catcher); the induction route priced dead AS A REDUCTION (TRANSPORT-RELABELS-NOT-REDUCES + LEVEL-CHEAPER-THAN-INCREMENT); MODULE TWO, v937\_l1wpd\_closure\_reduction.py (10/10, L1WPD-CLOSURE-REDUCTION; probe l1wpd\_closure\_probe.py r178 note CDXCIV, 29/29, SPEC 0e306146bebbd9be, record 228.9 s, no amendment, re-run green at promotion 224.4 s): WPD's  $\lambda$ -uniform content == the H-pin edge (AND-fire demonstrated at the declared edges — the pair {L1, WPD} carries ONE open  $\lambda$ -uniform edge, not two); the three-part bridge-expressibility verdict (source data bridge-expressible per census; the floor escapes linearity at the eigenvector and root maps, exact exhibits; the SW pricing vacuous — no closure); the v925 all- $x$  sharpening propagated with own replication ( $x_0 = 121/112$ , all 357  $D_{cs} > 0$ ); the r133-vs-r171 H1/H2/H3 name collision flagged; MODULE THREE, v938\_dbn\_heatflow\_census.py (11/11, DBN-HEATFLOW-CENSUS; probe dbn\_heatflow\_probe.py r179 note CDXCVI, 26/26, SPEC 67eaf86c7bfa7d84, record 672.4 s, one smoke fix disclosed, re-run green at promotion 659.4 s): THE EXACT CENSUS SEMIGROUP — the de Bruijn–Newman flow transports onto the derived census polynomial as a closed-form FINITE sum  $e^{-tT}$  (no zeros, no ODE) with the exact trace law  $\text{trace}(N_t) = \text{trace} + 2d(2d - 1)t$  (measured deviation 0.0) and the factor-4 normalization lemma (window  $[0, 0.055]$  exact, BH8-verified); the dictionary THREEFOLD-INEQUIVALENT (every pinch claim dictionary-dependent); the missing half of the generator == the RH-conditional PAIR class (flagged loop, not consumed); theta-flow-inert floors ( $\geq 4021 \times$  the window); the  $\lambda_h$  criticality ladder  $2.39 \rightarrow 0.216$  strictly decreasing with THE LIMIT OPEN — honestly NOT an off-line-zero detector (Epstein slack larger); the witness preserved-not-expelled; MODULE FOUR, v939\_cbj\_frame\_adjudication.py (10/10, CBJ-FRAME-ADJUDICATION; probe cbj\_frame\_probe.py r180 note CDXCVII, 41/41, SPEC d7fbf2d956184674, record 1042.5 s, no amendment, re-run green at promotion 1005.5 s): STEP-A EXACT —  $\delta_h = |J_h|_{G_h}^2$  with remainder  $R = 0$  proven generically (the single most promotable theorem of the arc, certified at 14 rungs,  $\text{dev} \leq 1.9 \times 10^{-46}$ ); the cluster normal form (the basis-change determinant == the spacing product); the Montgomery–Vaughan pricing (killed exactly at linkage 1, carrying at  $3.15\pi$  with the exact rational margin  $1/21$  + the two-grid repair); the dof/seam/occupancy kill triple ( $n = 474$  vs rank 128, dof 112.8; seam 0.994;  $-0.7484$  dex/atom): the GLOBAL review form dead,

the per-cluster floor carrying  $k$ -uniformly at fixed  $r$ ; the source-only selector ( $\hat{h}_k = 7/13/31/\dots/8191$  recomputed in-run, Bertrand-cofinal) underperforms the block average yet clears the SF2 demand (margins  $2.14/3.05$ , all rungs  $\geq 1.405$ ); the house Gram min-eig ladder  $1.1 \times 10^{-17} \rightarrow 5.2 \times 10^{-113}$ ; MODULE FIVE, v940\_cbj\_subdof\_blockfloor.py (9/9, CBJ-SUBDOF-BLOCKFLOOR-KILL; probe cbj\_subdof\_probe.py r181 note CDXCIX, 37/37, SPEC 2db82c76ce5f067c, record 867.9 s, one spec-text amendment disclosed (the sympy-exact Fejér constant  $1/6$ , BH8-verified), re-run green at promotion 772.1 s): the sub-dof-scoped floor statement posed exactly and honestly KILLED at the decisive measurement — the sub-dof mass fraction falls at  $+0.999$  dex/rung (recomputed) and saturates at zero, the scoped margin dies AT THE SELECTOR RUNGS ( $8.02 \times 10^{-3}$  at  $\hat{h}(B3) = 13$ , dead by  $\sim 125$ ); the unscoped r180 floor remains the measured carrier (killed is the PROOF ROUTE through the well-conditioned subspace); CARLESON PRICED FOR THE FIRST TIME — the unconditional embedding WRONG-DIRECTION for a floor (Ortega-Cerdà–Seip, Ann. of Math. 155 (2002), BH8-web-verified) and the complete lower-class table with Nazarov–Turán (exponential in the term count) as the classical mirror of the measured occupancy law; FRACTION-WORLD-SEPARATING (floor value world-insensitive, mass location world-separating); the fixed- $r$  compactness sketch; MODULE SIX, v941\_alignment\_law\_factorization.py (11/11, ALIGNMENT-LAW-FACTORIZED; probe alignment\_law\_probe.py r182 note D, 37/37, SPEC e4cdb9a932093196 verified invariant under the appended correction block, record 775.6 s, five freeze retypings disclosed, re-run green at promotion 776.5 s): the sharpest unexplained regularity frozen as a MEASURED TRIPLE — the half-power envelope ( $\alpha$  inside the descriptive band  $(0.38, 0.54)$  at all 14 rungs, never a prediction per BH8-F4), the quarter-depth wall lock ( $\mu_m = 0.2372 \log_{10} g_{\min}$ ,  $R^2 0.9916$ ) and the deep excess ( $5.64\text{--}7.58$  dex == the r181 kill mass); GRAM-SOURCE-FREE proven three ways; the key adjudication:  $s_i A_0$  LINEAR in  $d$ , the location law PROJECTIVE and  $A_0$ -free — the r178 obstruction does NOT bite; the one shared nonlinear link is the wall eigenmap (the law exactly as bridge-expressible as  $\delta$ , given  $d$ ); ALT-JETS-BREAK-ALIGNMENT; WITNESS-BREAKS-ALIGNMENT; the terminal adjudication SAME-CLASS-NOT-NEW-EDGE (four distinct formal statements over the same source object, the wall argmin ray  $d$ , in  $A_0$ -quotient form; no new residue edge via the machine-checked G44 exclusion, not via identity). Ledger rows PRIME.MANIFOLD.INVARIANCE.01, PRIME.L1WPD.CLOSURE.01, PRIME.DBN.HEATFLOW.01, PRIME.CBJ.CONFLUENT.FRAME.01, PRIME.CBJ.SUBDOF.BLOCKFLOOR.01, PRIME.ALIGNMENT.LAW.01, all [E] NOT-RH-EVIDENCE active; the arc's terminal residue written everywhere in the canonical note-DII form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k$  == LOOP, flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 = TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs: extension instantiated, TAILWPD world-front}; surfaces: six veri blocks in tfpt\_3, the new post-bughunt-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed ( $928 \rightarrow 934$ ) and the superseded awaiting-bughunt prose rewritten, the endform article §8.22 + counts + inventory, the email draft seventh-promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v941, README/Zenodo 934; one German research note appended to experiments/next.txt (DIII); FULL run\_all.py ALL CHECKS PASSED at 934 modules; bash build.sh release

green; `bash build.sh audit` → AUDIT OK; manifests regenerated LAST. No omega closed, no marker moved beyond the six [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; the two bughunt rounds (176 and 183) stay experiments-side as process rounds per precedent. NO RH claim anywhere)

2026-08-20 (CXII) (the triple-cofinal-arc promotion — the five durable certified theorem sets of discovery rounds 171–175 graduate from the experiments tree into the load-bearing verification suite (suite 923 → 928, ledger 1074 → 1079), after the seventh adversarial bughunt (round 176, note CDXCII) covered the arc with 0 verdict flips and its corrections of record (F1–F8, applied to the frozen records as appended CORRECTION-OF-RECORD blocks in round 177) were ADOPTED VERBATIM on every new surface; every qualifying source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as `*.promo_rerun.log`): MODULE ONE, `v931_jetmass_floor_theorems.py` (10/10, JETMASS-FLOOR-THEOREMS; probe `jetmass_floor_probe.py` r171 note CDLXXXVII, 36/36, SPEC 57de8b2a83677a9c, record 2056 s, two amendments disclosed, re-run green at promotion): THEOREM PF — the product floor  $F/A_0 \geq (1 - c^*/z)^{(1+\kappa)/c^*}$  from the census product identity + the Vieta trace at coefficient level + a concavity lemma, conditional EXACTLY on {H1 + H2 per rung}, THE EXPONENT IS THE TRACE (the r140 far-field law its  $\kappa \rightarrow 0$  shadow); H1 source-pure at all 25 rungs, H2 census complete-real incl. degree-116; the certified curve under the truth at  $\sim 100k$  tail zeros; the factorization [PF]  $\times$  [WF classical]  $\times$  [RATE] with JETLOCK-MEAS eliminated; the witness modus-tollens boundary; BH7-F3 definitional legs recomputed honestly; MODULE TWO, `v932_toproot_theta_statement.py` (10/10, TOPROOT-THETA-STATEMENT; probe `toproot_theta_probe.py` r172 note CDLXXXVIII, 33/33, SPEC cf27df22aa5dffbf, record 2813 s, re-run green at promotion): TOPROOT stated SEQ-cofinal with three locked forms; the exact rate dictionary  $a = p/2 - 1$  proven at  $p = 3, 4, 5$ ; THEOREM TR-CAP proven and typed circular-for-TOPROOT; the  $A_0$ -triangle route the FOURTH flagged loop; admissibility recomputed in-run ( $p_{\max} = 10.2695$ , six powers of margin); H3 ( $y_t \leq 0.155 T_z^4$ , ancestors == {SOURCE}) certified 26/26 margin  $\geq 1.99$ , REFUTABLE in the inflation witness and not norm-continuous; MODULE THREE, `v933_h3_cofinal_adjudication.py` (10/10, H3-COFINAL-ADJUDICATION; probe `h3_cofinal_probe.py` r173 note CDXC, 36/36, SPEC 876dafc977d3d8fc, re-run green at promotion 1339.5 s): saturation REAL with  $p_{\text{loc}} \rightarrow 4.024$  and  $\theta_\infty$  band  $[0.0766, 0.0977]$  under the bar (OPEN); pigeonhole exact yet subsidy-empty (quantifier gap  $D = 0.004183$  — the limsup form only mod  $D$ , BH7-F2); the witness excluded by the certified manifold {ground-state cert + two-sided lock + BA3 bridge  $\times 3.6 \times 10^8$ }, escalation-stable  $\times 10^5$ , H3 DK-norm-stable at measured residuals — SCOPED to the DK radius (gated  $h \leq 13$ , measured to  $h = 20$ , crossover  $h = 16$ , BH7-F4); all saturation analytics recomputed in-run from the cited ladder; MODULE FOUR, `v934_gonek_pricing_unconditional.py` (8/8, GONEK-PRICING-UNCONDITIONAL; probe `gonek_pricing_probe.py` r174 note CDLXXXIX, 20/20, SPEC 3050678b352eaa9a, record 188 s, re-run green at promotion 244.2 s): Landau 1912 and Gonek 1985/1993 identified UNCONDITIONAL, Gonek 1984 machine-checked NOT-AN-ANCESTOR (the hidden loop detected, flagged, never consumed; externally re-verified in Bughunt VII); the envelope closed form with finite census-depth-insensitive ceiling; prefactor-insensitive absorption at any finite constant;  $\hat{c} \leq 0.086$  at snr

up to  $2.7 \times 10^4$ , typed **CONSTANT-EMPIRICAL-PER-CENSUS** — the Gonek item **LEAVES** the residue; **MODULE FIVE**, `v935_thetainf_landau_bridge.py` (9/9, **THETAINF-LANDAU-BRIDGE**; probe `thetainf_pin_probe.py` r175 note **CDXCI**, 28/28, **SPEC** 3044558e5fa52e01, record 577 s, re-run green at promotion 718.1 s): **THE MODE-LEVEL LANDAU BRIDGE** — the program’s first linear per-mode classical identity, residuals  $1.4 \times 10^{-5} \rightarrow 5.4 \times 10^{-5}$  at the 20M-zero census; the dictionary a proven bijection (pins nothing); the variational conditioning wall typed **MEASURED-LAW** extrapolation per **BH7-F7** (“gemessene”, not “bewiesene” Verschwendung),  $T_{\text{req}} = 3.76 \times 10^{14} \rightarrow 1.75 \times 10^{77}$  vs  $T_{\text{PT}} = 3.0 \times 10^{12}$ ;  $\theta_\infty$  re-typed **OPEN-NONPERTURBATIVE-VARIATIONAL**. Ledger rows **PRIME.JETMASS.FLOOR.01**, **PRIME.TOPROOT.THETA.01**, **PRIME.H3.COFINAL.01**, **PRIME.GONEK.PRICING.01**, **PRIME.THETAINF.PIN.01**, all [E] **NOT-RH-EVIDENCE** active; the arc’s terminal residue written everywhere in the **BH7-F1-corrected** form: the **TRIPLE**  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$  — **NOT H3** alone; limsup only mod  $D = 0.004183$ ) +  $\{\text{census-forall-}k == \text{LOOP, flagged}\} + \{\text{L1, WPD}\}$ ; surfaces: five veri blocks in `tfpt_3`, the new triple-cofinal-arc diary section in `tfpt_prime_front` with the live suite-count parentheticals refreshed ( $923 \rightarrow 928$ ), the endform article and the email draft refreshed with the §8.21-class promotion record and the triple wording, **README/Zenodo 928**, the website **PROMOTED** card + **DAG** chips + honesty-banner range **v535–v935**; one German research note appended to `experiments/next.txt` (**CDXC VIII**); **FULL** `run_all.py` **ALL CHECKS PASSED** at 928 modules; `bash build.sh` release green; `bash build.sh` audit  $\rightarrow$  **AUDIT OK**; manifests regenerated **LAST**. No omega closed, no marker moved beyond the five [E] rows, census cardinality 4 unchanged, the quartic law stays **MEASURED**,  $\theta_\infty$  stays **OPEN**; rounds 176–180 stay experiments-side awaiting **Bughunt VIII**. **NO RH** claim anywhere)

**2026-08-20 (CXI)** (the externalization contracts — the two open non-RH kind-(A) external-math targets **SEAM.EQUIV.01** and **ALPHA.QUILLEN.EXACT.01** packaged as short, self-contained **EXTERNALIZATION CONTRACTS** for outside specialist mathematicians (conformal nets/VOA/AQFT for the seam; determinant lines/ $\eta$ -invariants/index theory for alpha), added as two new sections at the end of `tfpt_research_contracts` (before “Certifiability and recommended order”), plus a one-sentence pointer in the head keybox; **DOCUMENTATION ONLY** — no module touched, no number changed, no status marker moved, no gate closed or narrowed, **NO RH** claim: (1) the **SEAM.EQUIV.01** contract states the minimal theorem (the scaling limit of the gapped  $\mu_4$ -equivariant **CAR** collar exists in local-net topology; the index-4 simple-current extension functor commutes with the limit; the OS reconstruction is the holomorphic  $(E_8)_1$  lattice net) broken into the four review-named lemmata **L1** (uniform energy/nuclearity bounds — honestly typed: **NO** uniform nuclearity estimate exists in the corpus), **L2** (convergence of local algebras, the cited Osborne–Stottmeister scaling-limit theorem with the  $\mu_4$ -equivariant structure retained), **L3** (crossed product commutes with the limit; the finite index-4 **Q**-system is explicit, **v125/v154/v469**) and **L4** (uniqueness of the holomorphic  $c = 8$  reconstruction;  $\det K = 1$  vs the same- $c$  rival  $SO(16)_1$ ), with the complete typed hypothesis list (12 **PROVEN-FINITE** rows citing **v302/v113/v155/v160/v356/v366/v367/v368/v461/v376/v377/v378/v379/v125/v154/v216/v286**, 5 **CITED-CLASSICAL** rows, 1 **OPEN** target), the runnable finite models, the **HONEST** Lean axiom surface (`SeamScalingLimit.lean` kernel-proves only the arithmetic hypotheses by decide; the analytic content is named

cited axioms over opaque Props; no sorry; explicitly flagged: L1–L4 have ZERO Lean content beyond the composition audit), five must-fail controls (MF-S1:  $c = 8$  alone must NOT suffice — the red-team point; MF-S2: index-2 glue must yield  $SO(16)_1$ ; MF-S3: the trivial  $M = 3$  collar must give no chiral limit; MF-S4: non-order-4 clock breaks the marks; MF-S5: the v286 import firewall) and four acceptance tests; (2) the ALPHA.QUILLEN.EXACT.01 contract states the two-step theorem T1 (gap-stable continuum limit  $c_1(\det D_{\text{finite}}) \rightarrow c_1(\det_{\zeta} D_{\text{seam}})$  under norm-resolvent convergence with the Fermi gap uniformly open over the twist torus — the Dai–Freed section has no zero; finite side computed, v472) and T2 ( $\delta_{\tau}(\log \det_{\zeta} \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0 = \text{Bismut–Freed curvature} = \text{inflow response with the exact corpus coefficients } k_0 = |C| = 1, k_Y = 5/3, b_1 = 41/10, c_3 = 1/(8\pi), \text{exponent } -5/4 = -q(D_5))$ ), the two named open faces (the analytic diagonal- $\zeta$ /multiplicity face v484/v485 and the topological EM1 lemma v48/v435), the typed hypothesis list (10 PROVEN-FINITE rows citing v3/v48/v341/v382/v434/v470/v435/v342/v367/v472/v484/v485/v391/v433, 5 CITED-CLASSICAL rows, 1 OPEN target), the runnable finite models, the HONEST Lean statement (there is NO Lean formalisation of this target — stated plainly), four must-fail controls (MF-A1: no experimental  $\alpha$  anywhere in the construction; MF-A2: deck/carrier/Chern-level ablations must destroy the CODATA match; MF-A3:  $M = 3 \Rightarrow C = 0, M = -1 \Rightarrow C = -1$ ; MF-A4: the  $\{0, 3, 6\}$   $\pi$ -power/metric-independence partition) and the review’s three acceptance tests verbatim (no experimental  $\alpha$  input; unique physical fixed point 137.0359992168; ablations destroy the hit) plus one structural test; (3) the website paper mirror of `tfpt_research_contracts` (`papers.ts`) gains one section entry covering both contracts; `tfpt_research_contracts.pdf` and `changelog.pdf` recompiled and mirrored via `bash build.sh website`; `maps` and `changelog.ts` regenerated via `bash build.sh gen`; `bash build.sh audit` → AUDIT OK; manifests regenerated LAST. Both targets stay [O]; NO RH claim anywhere)

2026-08-19 (CX) (the endgame-arc promotion — the five durable certified theorem sets of discovery rounds 162–169 graduate from experiments/ into the load-bearing suite as `v926_fullgap_growthlaw_theorems.py`, `v927_blockaverage_substrate_theorems.py`, `v928_depthblock_transfer_theorems.py`, `v929_sigmafloor_factorization.py` and `v930_window_instrument_mechanism.py` (suite 918 → 923, ledger 1069 → 1074); all five rows typed NOT-RH-EVIDENCE in either direction — identities, recipe-derived laws, finite/classical closures and one honest mechanism record: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at  $\{\text{TOPROOT, TLAWCAP-block, QSUBGAP-floor}\} + \text{the } a\text{-walls}$  — now carried WITH the r168/r169 terminal coordinates: the  $\lambda$ -uniform budget content factorizes as  $[\text{census schedule, classical per } k, \text{LOOP if for-all-}k] \times [\text{SIGMA-FLOOR} == \text{DC-leg (classical per census)} \times \text{JET-MASS-FLOOR (the terminal } \lambda\text{-uniform residue, typed OPEN)}]$ ; every measured leg typed verbatim, the quartic law NOT claimed proven: (1) v926 THE QUARTIC/GROWTH-LAW FAMILY (round 162, note CDLXVII, ledger row PRIME.FULLGAP.GROWTHLAW.THEOREMS.01 [E]; source probe `fullgap_growthlaw_probe.py`, 28/28, SPEC 26bdb5a87f63c519, PINNED from run-of-record 4061.6 s + deterministic re-run 4237.5 s with timing-normalized diff EMPTY, both logs kept, per the multi-hour convention — its GL algebra re-gated in Bughunt V G19/G20 and recomputed in-run) — GL1–GL5 exact (the loo pinch: every leave-one-out ground matches  $\lambda_1$  within the  $\varepsilon^2$  budget; the arrowhead compression with the dual row; the one-row

price equivalence + razor vacuity: D3 COLLAPSES TO THE RAZOR; the rate bookkeeping  $Jt_r == 1 + \text{FULLGAP}$ ; the ladder/quartic chases + the razor composition: the QSUBGAP-floor flows from the  $\Theta$ -window through the razor); THE QUARTIC LAW  $J == \Theta T_z^4$  typed MEASURED with the Bughunt-V free-exponent/AIC adjudication RECOMPUTED IN-RUN ( $p = 3.9072 \pm 0.1136$ ; AIC prefers FIXED 4 ( $-30.87$ ) over free ( $-29.79$ ), rejects  $3.5/4.5$  at  $3.6/5.2\sigma$ );  $\rho_j == 1.0000$  for ALL 189 left-out zone zeros (jet-COLLECTIVE, per-zero deflation REFUTED-MEASURED); the finite- $P$  moment model undershoots by 1.4–4.5 dex (the Y3/R4 rate-blind trace-cap vacuity class, one level higher); (2) v927 THE COFINAL BLOCK-AVERAGE SUBSTRATE (round 166, note CDLXXV, ledger row PRIME.BLOCKAVERAGE.SUBSTRATE.01 [E]; source probe tau\_blockaverage\_probe.py, 27/27, SPEC d86f42a04c69a4e2, RE-RUN GREEN-AS-TYPED AT PROMOTION 2372.4 s with identical SPEC and empty timing-normalized diff) — BA1 (a certified block sum  $> 0$  forces  $\geq 1$  positive rung; dyadic hits COFINAL), BA2 (the determinant chain IS the  $\tau$  sign, Jacobi/Sylvester), BA3 ( $\tau \geq \text{zsum} - \text{OFF EXACT}$ , proven-mod-cited: PT21 + HSW22 + the r131 OFF recipe + ward-class cache, NO-LOOP), the Weyl split measured VACUOUS by 10.9–54.6 dex; the certified block enclosures (all lower ends  $> 0$  on B2/B3: POSITIVE-RUNG-PER-BLOCK CERTIFIED below the horizon), the wall dictionary  $\text{sign}(\tau) == \text{sign}(\text{wall}) == +1$  at ALL 25 rungs, the tlaw ladder with 19 NEW corpus-depth rungs; the AVG-BUDGET-WINDOW beyond the horizon typed OPEN; (3) v928 THE DEPTH-BLOCK TRANSFER FAMILY (round 168, note CDLXXXIII, ledger row PRIME.DEPTHBLOCK.TRANSFER.01 [E]; source probe depthblock\_transfer\_probe.py, 36/36, SPEC a4cd07144e0c5222, ZERO AMENDMENTS, RE-RUN GREEN-AS-TYPED AT PROMOTION 2371.6 s) — DT1 PROVEN (the step chain is exact algebra on top of BA3); DT2 THE DEMAND LAW PROVEN-BY-RECIPE (the round's discovery:  $\varepsilon == \sqrt{h}(1+\eta)^2 G(T_{\text{PT}})/G(2\pi h)$  EXACT, the dyadic factor falls monotonically onto  $2^{3/2}$ ); THE CENSUS 3/2-LAW sympy-exact ( $T_{\text{req}}(h) \rightarrow (3\pi/\sigma_0)h^{3/2}$ : the budget route can NEVER become self-supporting); the budget horizon RECOMPUTED ( $h^* = 1.2566 \times 10^7$ ,  $k^* = 23.58$ :  $\sim 22$  dyadic blocks on PT21); the Bertrand entry interface sieve-verified in-run to  $10^6$ ; UNIVERSALIZING THE CENSUS MACHINE-DETECTED AS THE RH LOOP (the per- $k$  schedule acyclic); the step instantiated end-to-end (STEP(B2→B3) 9/9, STEP(B3→B4-partial) 11/11, SUBSTRATE-DIRECT, no self-supporting induction); ENTRY-NEW-CONFIRMED 4/4 (cone entry always at the largest new PRIME-POWER atom of the dyadic gap, excluding trailing 2-powers — the note-CDLXXXIII formulation, per the Bughunt-VI F1 correction of record; fakes die by NO-ENTRY); (4) v929 THE TERMINAL FACTORIZATION (round 169, note CDLXXXIV, ledger row PRIME.SIGMAFLOOR.FACTORIZATION.01 [E]; source probe sigmafloor\_probe.py, 35/35, SPEC a16a4db1055488d2, RE-RUN GREEN-AS-TYPED AT PROMOTION 2375.2 s; AMENDMENT 1 disclosed: the all-rung onset-rate fit typed CACHE-TOP-LIMITED) — SF1 THE SUPPLY FACTORIZATION  $\sigma == (1 - \text{slop}) \delta \text{DC EXACT}$  (identity devs  $\leq 2 \times 10^{-60}$  at all 25 rungs, F64-immune): the r168  $\sigma$ -flatness EXPLAINED AS TWO TRENDS ( $\delta O(1)$ -flat  $\times$  DC rising toward  $1/2$ ) — the last unexplained measured law of r168 DECOMPOSED; the DC leg typed DC-CLASSICAL (proven-mod-cited per census; Landau 1912/Gonek 1993 AS FORM, GONEK-CONSTANT-UNPRICED); THE JET-MASS-FLOOR NAMED AND TYPED OPEN (the terminal  $\lambda$ -uniform residue, arithmetic-pinned, per-rung classical); the onset-excess cap REFUTED as a floor (SF2); ANY explicit polynomial rate

census-absorbable (SF4, sympy-exact); the pinning-supply route machine-flagged as the RH loop's costume (SF5); the anatomy block floor certified on B2/B3 ( $\sim 11$  dyadic blocks PROVEN-MOD- $\{\text{measured lock, cited Gonek}\}$ ); the controls finding: the raw floor holds in EVERY fake world while the BA3 bridge is FALSE — the arithmetic content is the  $\{\text{floor, bridge}\}$  PAIR; (5) v930 THE WINDOW-INSTRUMENT MECHANISM RECORD (round 167, note CDLXXXI, ledger row PRIME.WINDOW.INSTRUMENT.MECHANISM.01 [E]; source probe window\_instrument\_probe.py, 46/46, SPEC bfdcf374504006e7, PINNED from run-of-record 4138.5 s + re-run 4047.0 s per the multi-hour convention, both logs kept; the mid-run owner-directive re-freeze disclosed; the entry law re-gated by the r168 batteries re-run green at this promotion) — WI1–WI6 exact (plant transport: the PSD premise survives positive band mass at any strength; the ghost secular threshold  $\tau$ -priced; the perturbation law first-order symmetric; the budget-moment dictionary  $2E^2 == 8A_0^2 \sin^2 y_t^2 \Phi^2 / y^3$ ; the theta chases; BANDPOS(unbounded ladder)  $\iff$  RH — the loop identification); the Stieltjes-moment refutation RECOMPUTED from the pinned slots (Hankel NOT PSD, log-convexity break  $\geq 1.5$ ); THE MECHANISM FINDING pinned at its honest grade: NO PARTIAL COMB IS POSITIVE — cone entry only in the last atom's tail with the tangent criterion from the prime source, all fakes die by NO-ENTRY (typed MEASURED, CHAIN-NOT-ASSEMBLED; the asymmetry law honestly DEMOTED to generic-cone); (6) SURFACES: five ledger rows (1069  $\rightarrow$  1074); veri blocks in tfpt\_3 (audit-counted body) + the new prime-front diary section “The endgame arc” with the suite-count refreshes (918  $\rightarrow$  923 across the live parentheticals); endform article §8.20 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v930; README/Zenodo 923; German note CDLXXXVI appended to next.txt (head numeral re-verified immediately before writing: the concurrent Bughunt-VI lane had taken CDLXXXV — one collision resolved by re-read); (7) GATES: the endgame chain was independently re-audited mid-promotion by Bughunt VI (round 170, note CDLXXXV, 22/22, 0 MAJOR / 2 MINOR / 4 NOTE, NO verdict flips, NO arrow retypes — the arrow table A1–A10 re-verified with own code) and its live-surface recommendations are adopted on these new surfaces (F1: the entry pattern cited only in the note-CDLXXXIII prime-power formulation; F2: “result lines bit-identical / timing-normalized empty” wording; F3: the cofinality legs re-coded against real objects in v927/v928); three of the five affordable source probes re-run green-as-typed BEFORE any edits with identical SPEC SHAs and empty timing-normalized diffs (27/27 at 2372.4 s, 36/36 at 2371.6 s, 35/35 at 2375.2 s; logs kept as \*.promo\_rerun.log), the two multi-hour probes pinned with disclosure (r162 re-gated in Bughunt V; r167 re-gated by the r168 batteries); v926–v930 standalone green (12/12, 12/12, 11/11, 12/12, 10/10, each  $\sim 1$  s); FULL run\_all.py ALL CHECKS PASSED at 923 modules; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last. NO RH claim in any direction — identities, recipe-derived laws and finite/classical closures say nothing about the truth of RH either way)

2026-08-19 (CIX) (the Bughunt-V corrections of record — the corrections that the round-164 adversarial audit (note CDLXIX, experiments/tfpt-discovery/bughunt5\_probe.py, 29/29 gates, SPEC b466fd5e6dcbcb640, record run + deterministic re-run bit-identical, 0 MAJOR / 4 MINOR / 2 NOTE, NO round verdict flips, every load-bearing razor/quartic/ladder/census claim of rounds 149–162 and the promotion wave v922–v925 independently CONFIRMED) recommended



for promoted and live surfaces, applied to the LIVE surfaces only (frozen probes and notes are never retro-edited — the correction is the record); ONE documented count correction and NO other numeric value change anywhere; v925 stays green with unchanged gate count (14/14) and unchanged verdict: (1) BH5-F2, THE COUNT CORRECTION (72  $\rightarrow$  73): v925\_edge\_cleanup\_closures.py (docstring E1 headline, the D1 check text and the D1 inline comment + website mirror, byte-identical) and the ledger row PRIME.EDGE.CLEANUP.CLOSURES.01 said “72 sliver + first-zero targets” / “72 unique targets” — the record log’s own per-block certified counts (1/1, 1/1, 1/1, 1/1, 18/18, 51/51) sum to 73, and the unique-target arithmetic confirms 73: 71 sliver zeros + the first zeros above the band at b8/b18 only (their slivers are EMPTY; at b5/b13/b24/b28 the first zero lies INSIDE the sliver and is already counted); all 73 certificates exist and were always green — only the headline count changes, the gate logic was already correct (the module gated  $\Sigma == 73$  before this correction); the paper/website prose was already count-free (“per-zero”, “at every sliver + first-zero target”); (2) BH5-F6, THE ROUND-LABEL DRIFT (live-surface leakage only): the corpus convention fixes anchor\_epslock = round 153 (r152 = the CDLVI promotion round, no promotable content) — corrected at the two live sites that had inherited the drift: the tfpt\_prime\_front scope paragraph (“the round-152/153 anchor coordinates”  $\rightarrow$  “the round-153 anchor coordinates”) and the CVIII entry wording (“round-152/153/155”  $\rightarrow$  “round-153/155”); the frozen assembly\_walls docstring and note CDLXVI keep the drift as the record; (3) BH5-F1 CHECKED CLEAN on live surfaces: r150’s overstated “betapos == S1/jr by exact chase” (exactly:  $S1/jr - \text{betapos} == \text{betapos}^2 / (\text{FULLGAP} - \text{betapos}) > 0$ , magnitude  $2.7 \times 10^{-7}$  at  $x = 5$  — below r150’s 4-digit print) lives ONLY in the frozen r150 probe and note CDLIV; v923 already promotes the corrected GF5a law with the factor  $(\lambda_1 - \tau) / (\lambda_1 - \beta_0) \geq 1$  — no live edit needed; (4) BH5-F3 CHECKED CLEAN: the r160 rescue-instrument overstatement (“projected nonneg least squares” for a greedy eigenvalue-lifting loop; enum RESCUE-INFEASIBLE-IN-CLASS certifying only THAT trajectory) appears on NO live surface — r160 was never promoted, and the live scope paragraphs carry its verdict only as INFORMATIVE-NEGATIVE (the correct typing, confirmed by the audit); (5) BH5-F4/F5 CHECKED CLEAN: no leakage of “SPEC\_SHA moves once” or the b8-SCRARITH 0.2597 misattribution to any live surface (frozen-file wordings stay as the record); (6) GATES: v922–v925 re-run green BEFORE any edit (13/13, 20/20, 12/12, 14/14,  $\sim 1$  s each); v925 re-run green AFTER the correction with unchanged gate count 14/14; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last; German note appended to next.txt. NO RH claim in any direction — a bookkeeping correction says nothing about the truth of RH either way)

2026-08-19 (CVIII) (the spectral-balance/edge theorem-arc promotion — the four durable certified theorem sets of discovery rounds 152–161 graduate from experiments/ into the load-bearing suite as v922\_spacing\_jet\_sumrule\_theorems.py, v923\_spectral\_balance\_razor\_theorems.py, v924\_moment\_laurent\_rootladder.py and v925\_edge\_cleanup\_closures.py (suite 914  $\rightarrow$  918, ledger 1065  $\rightarrow$  1069); all four rows typed NOT-RH-EVIDENCE in either direction — identities, enclosures and finite/classical closures: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at {TOPROOT, TLAWCAP-block, QSUBGAP-floor(= the pair {SUSCAP2R, DELTA1FLOOR})} + the  $a$ -walls; every conditional or measured

leg typed and carried verbatim; the family-wide exact algebra was Lean-hardened in the second Lean round (round 158, note CDLXII: five new Lean 4 modules, 28 fully proven theorems on the three standard axioms, build green) and each module re-checks the shipped Lean theorem names at every suite run: (1) v922 THE HPIN/SPACING THEOREM FAMILY (rounds 135 + 154, notes CDXXXIX + CDLVIII, ledger row PRIME.SPACING.JETSUMRULES.THEOREMS.01 [E]; source probes `hpin_floor_probe.py`, 33/33, SPEC 80366b4e62779398, PINNED from run-of-record 1837.6 s with THREE deterministic logs kept (D1/D2 re-gated by `r154-G32` + Lean `r158`), and `nearalign_probe.py`, 31/31, SPEC f92b3fb59b142254, re-run green-as-typed at promotion, 1572.7 s) — the derivative identity  $E'_N(\mu_j) == 4 \sin(A\mu_j) F'(\mu_j^2)$  with the spacing-product form (the H-pin floor is a ZERO-SPACING PRODUCT), the jet sum rules of both orders ( $\sum 1/F' == -A_2/A_0^2$ ;  $\sum F''/F'^3 == 2A_2/A_0^3$ ; the boundary-jet telescope IS the reciprocal-floor moment data), the  $A_0$ -cancellation (the demand ratio is  $A_0$ -FREE), the certified lattice tail, and the sign-uniform corollary carried as an OBSTRUCTION (census MIXED at every rung); `SpacingProduct.lean` proves the spacing product for GENERIC FINITE  $K$  over an arbitrary field; (2) v923 THE PINCH/ENCLOSURE/RAZOR FAMILY (rounds 157 + 161, notes CDLXI + CDLXV, ledger row PRIME.SPECTRAL.BALANCE.RAZOR.01 [E]; source probes `spectral_balance_probe.py`, 26/26, SPEC 3ed388698a138e31, re-run green-as-typed at promotion, 4077.8 s, non-timing diff EMPTY, and `gfloor_probe.py`, 26/26, SPEC cc7837138d41add7, PINNED from run-of-record 4068.2 s + deterministic re-run identical per the multi-hour convention, both logs kept, its GF algebra recomputed in-module) — SB1–SB5 (the trace loop  $\tau \text{ TrH} == \text{tf}/\text{FULLGAP}$  with  $1 \leq \text{tf} \leq K - 1$  UNCONDITIONAL: DELTA1FLOOR  $\Leftrightarrow$  TRACEFLOOR two-sided, tail closed; the chi-cap; the enclosure  $\rho^2 \delta_1 \leq g \leq \delta_1$ ; the OVG identity; the share floor) + GF1–GF5 (THE RAZOR ENCLOSURE  $1/(s + 1/\delta_1) \leq g < 1/s$  with TWO-LEVEL EQUALITY at the lower end — the exact witness  $\rho^2 = 1/(1 + P\delta_1)$  realizes  $s == P = 10^6$  AND  $g ==$  the lower end EXACTLY, no algebra-only improvement exists: THE  $g$ -FLOOR IS THE  $s$ -CAP GIVEN THE  $\delta_1$ -FLOOR, measured width  $1.51 \times 10^{-4} \rightarrow 8.45 \times 10^{-8}$ ; the pinch law  $1 - sg == g^2 T_2 / \rho^2$ ; the one-sided share pricing; the Newton value law  $N_1 == 1/(H_t + s)$  with  $B(q_0) == -\rho^2 P'_c(q_0)$  and the jet-ratio chase  $J_g t_g == 1 + g$ ; the S1 absorption + twin squeeze: both final cores ONE statement-form at two weightings, the value-merge REFUTED — the J-family is BLIND to  $g$ ); W1/W2/W3 cited from v920, no double count; (3) v924 THE MOMENT-LAURENT/ROOT-LADDER FAMILY (round 156, note CDLX, ledger row PRIME.MOMENT.LAURENT.ROOTLADDER.01 [E]; source probe `rootladder_probe.py`, 29/29, SPEC 02755d6b7ad0cfcb, re-run green-as-typed at promotion, 1873.5 s, zero amendments) — the escaped census-root ladder is the zero set of the moment-Laurent  $\Phi(z) = z - 1 + \sum_m J_{m+1} z^{-m}$  (self-similarity iff moment flatness, exact remainder), the QUARTER-CAP  $J_2 \leq 1/4 + z_0 |\rho|$  PROVEN (measured  $z|\rho| \leq 0.0102$ : proven headroom over the 0.1506 saturation), the exact sum rules, the sign-pattern law, the cascade ( $T > 0$  pointwise SOURCE-PURE down to  $Y^* = 1.02\text{--}2.1 b_{\text{top}}$ : the escaped-root cap CERTIFIED), the two-sided split, and the witnesses (shape sector scale-blind; J-BOUNDEDNESS ARITHMETIC-PINNED); (4) v925 THE EDGE/CLEANUP CLOSURES (round 159, note CDLXIII, ledger row PRIME.EDGE.CLEANUP.CLOSURES.01 [E]; source probe `edge_cleanup_probe.py`, 34/34, SPEC f6c0318841bb3942, re-run green-as-typed at promotion, 1830.8 s) — SIMPLICITY FALLS (census row Z8 eliminated  $10 \rightarrow 9$  by LM-split subsumption + zero-cost avoidance; residual

NONDEGENERATE-CELL witness-certified free), THE Q-SWAMP CLOSES WHOLESALE with the 357 Turing-class local band certificates RECOMPUTED IN-RUN at every suite run from the committed PT21-warranted `verified_zeros_n7000.json` cache ( $D_{cs} > 0$  at all integer  $x = 3..121 \times \text{battery}$   $a \in \{1, 4, 16\}$ , worst margin 0.346 at (12, 1);  $x_0(\text{HSW}) = 121 + x_0(\text{BW25}) = 112$  recomputed; heights  $\leq 7264.75 \ll 3 \times 10^{12}$ , ward-typed): THEOREM A's conditionality SHARPENS to every integer  $x \geq 3$  (H-pin untouched), and the band-edge sliver closes below horizon per-zero (pinned; the certificate price rides  $e^{-\text{depth}}$ , machine-pinned); (5) NOT PROMOTED, deliberately: the round-160  $J_2$  prime-force adjudication (note CDLXIV, verdict INFORMATIVE-NEGATIVE — the single-lever search closed) stays experiments-side: multi-hour world-builds with no exact-algebra layer a module could recompute, residue reading still in active motion; likewise the round-153/155 anchor/census coordinates; (6) SURFACES: ledger rows (1065  $\rightarrow$  1069); veri blocks in `tfpt_3` (audit-counted body) + the new prime-front diary section “The spectral-balance/edge theorem arc” with the suite-count refreshes (914  $\rightarrow$  918 across the live parentheticals); endform article §8.19 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v925; README/Zenodo 918; note CDLXVI; (7) GATES: all four affordable source probes re-run green-as-typed BEFORE any edits (identical SPEC SHAs; 31/31, 26/26, 29/29, 34/34; logs kept as `*.promo_rerun.log`), the two pinned probes disclosed (r135 three logs; r161 both logs); v922–v925 standalone green (13/13, 20/20, 12/12, 14/14, each  $\sim 1$  s); FULL `run_all.py` ALL CHECKS PASSED at 918 modules; `build.sh` release + EXIT GATE `build.sh` audit AUDIT OK; manifests last. NO RH claim in any direction — identities, enclosures and finite/classical closures say nothing about the truth of RH either way)

2026-08-18 (CVII) (the prime-front theorem-arc promotion — the four durable certified theorem sets of discovery rounds 128–151 graduate from experiments/ into the load-bearing suite as `v918_doublelimit_reduction_theorem.py`, `v919_secular_gw_pinning.py`, `v920_loop_equivalence_theorems.py` and `v921_counting_dominance_closures.py` (suite 910  $\rightarrow$  914, ledger 1061  $\rightarrow$  1065); all four rows typed NOT-RH-EVIDENCE in either direction — reductions, identities and counting closures: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at  $\{\text{TOPROOT}, \text{ONSETCAP}(=\text{TLAWCAP}), \text{SUSCAP2R}\} + \text{DELTA1FLOOR} + \text{the } a\text{-walls}$ ; every conditional or measured leg typed and carried verbatim; all candidates survived the adversarial Bughunt III (round 130) and/or Bughunt IV (round 149, note CDLIII, 0 MAJOR): (1) v918 THEOREM R (round 128, note CDXXX, ledger row `PRIME.DOUBLELIMIT.REDUCTION.THEOREM.01` [E]; source probe `doublelimit_proof_probe.py`, 28/28, SPEC `0c77ef7ce43fd62e`, re-run green-as-typed at promotion, 398.5 s; independently re-derived in Bughunt III) — the load-bearing reduction of the arc, proven with elementary inputs:  $\text{L1} + \text{WPD} \Rightarrow (\text{H-conv}) + (\text{H-trace})$  with the EXPLICIT RATE  $\max(1, C)(-4 \log(1 - t/4)) d_1$  via the exact series identity  $\log(R_{\text{fin}}/R_a)(t) = \sum_k t^k d_k/k$ ; with the round-122 NF-closure theorem:  $\text{RH} \Leftarrow [\text{NF-closure}] + [\text{Theorem R}] + \{\text{L1}, \text{WPD}\}$  on dense  $a$ ; Lemmas N/S/X and the exact  $C_{0k} \leftrightarrow \text{jet}$  dictionary travel with it; honest typing carried: WPD's  $k$ -tail IS window positivity (exact witness  $k^*(B) = 29959/76013/145094$  from  $4w^* = 10001/10000$ , recomputed in-run), L1 stays obstructed at the node-law wall — proven is the REDUCTION and the COORDINATE, not the convergence; (2) v919 THE SECULAR IDENTITY + THE GUINAND–WEIL PINNING (round

131, note CDXXXIII, ledger row PRIME.SECULAR.GW.PINNING.01 [E]; source probe `l1_weyllaw_probe.py`, 31/31, SPEC 7d8cc2e5ca8108fb, re-run green-as-typed at promotion, 217.4 s; independently reproduced in Bughunt IV from the classical explicit formula,  $|\Delta \log_{10}| = 2.7 \times 10^{-7}$  —  $\det(M_\eta - z) = -z C(z^2)/A_0$  EXACT (recomputed on an own exact-rational instance, Fractions, dim 11: the operator nodes ARE the census zeros, node count EXACTLY  $K - 1$ );  $\tau = 2 \sum_\rho |E_N(\gamma_\rho)|^2$  within certified allowances (Guinand–Weil, cited), the certified smallness law, the certified gap law in the resolvability zone ( $x = 13$ : 21/21 zone zeros pinned, max gap  $2.9 \times 10^{-17} \leq 5.6 \times 10^{-8}$ ; edge layer MEASURED-ONLY); L1-TAIL closed in closed form (HSW22 Cor. 1.2 + PT21, recomputed against an own IBP integral); the crossover theorem  $T = 2\pi x$  exact — the corpus constant  $\text{INBAND\_F} = 1/\text{KFAC} = 0.8$  is a THEOREM; the boundary-jet law; the band half of L1 (H-pin) stays OPEN, Z1 evidence typing stands; (3) v920 THE LOOP/EQUIVALENCE THEOREMS (rounds 140 + 142, notes CDXLIV + CDXLVI, ledger row PRIME.LOOP.EQUIVALENCE.THEOREMS.01 [E]; source probes `jetlock_bandmass_probe.py`, 33/33, SPEC 85e7ba691b3eef58, PINNED from run-of-record 3824.7 s + deterministic re-run 3783.4 s per the multi-hour convention (both logs kept, loop algebra re-gated in Bughunt IV and recomputed in-run), and `tlawcap_suscap2r_probe.py`, 26/26, SPEC 85971e173cd910f8, re-run green-as-typed at promotion, 1904.7 s) — the anti-double-counting layer, all recomputed exactly:  $\text{BAND-MASS} \Leftrightarrow \text{TLAWCAP} \Leftrightarrow \text{EPS-LOCK}$  modulo  $\{\text{JETLOCK}, \text{TAILVIS}\}$  (Theorem B2: band-mass is EPS-LOCK-COMPLETE), the onset IS the jet ratio and the jet ratio IS escaped census-root mass (J2/J3: JET-LOCK(poly)  $\Leftrightarrow$  TOPROOT; J4: the hardness is cancellation ALIGNMENT), the W1 pinch identity ( $s \times \text{gap} \equiv 1$  is NOT an identity:  $1 - g/\delta_1 \leq sg \leq 1$  exactly),  $\text{QSUBGAP} \Leftrightarrow \text{SUSCAP2R} \wedge \text{DELTA1FLOOR}$  (W2, both directions exact),  $\text{DELTA1FLOOR} \Leftarrow \text{FULLGAP}$  (W3, Cauchy interlacing, source-pure; measured TIGHT 1.0000), plus the exact DELTA1FLOOR-required witness ( $s \equiv 1$  while  $\text{gap} \rightarrow 0$  — the Bughunt-IV F1 correction of record carried in-module); reshape theorems: no omega closes, no future double-counting possible; (4) v921 THE COUNTING-CLASS CLOSURES (rounds 133 + 143, notes CDXXXVI + CDXLVII, ledger row PRIME.COUNTING.DOMINANCE.CLOSURES.01 [E]; source probes `dominance_proof_probe.py`, 40/40, SPEC 00b98f5eebf9c5fb, re-run green-as-typed at promotion, 339.3 s, and `toproot_tailvis_probe.py`, 27/27, SPEC 4fcd70beb7cf4f17, PINNED from run-of-record 4016.1 s + deterministic re-run 3921.4 s per the multi-hour convention (both logs kept, T1 re-derived independently in Bughunt IV)) — Theorems M/E (ball/edge counting, no sign reads), Theorem T UNCONDITIONAL ( $N_{\text{fin}}(T) \leq N_{\text{true}}(T)$  for all  $T \geq T^*(x)$ ; ladder 63.55/79.17/102.80/147.13 recomputed vs the classical  $\gamma_{K-1}$  anchors), Theorem C exact (MRB strictly weaker than DOM — per-node one-sidedness neither needed nor claimed), Theorem A CONDITIONAL on H-pin (closed forms recomputed:  $x_0 = 121$ , BW25  $x_0 = 112$ ; the Q-swamp strip  $D < 0$  on  $x \in \{13..89\}$  carried, not smoothed), Theorem T1 (TAILVIS existence is CLASSICAL from HSW22/RvM alone:  $L^* = O(20)$  falling — TAILVIS eliminated as a standalone arithmetic omega, counting class) and Theorem T3 (the round-140 horizon limitation at  $x = 24/28$  RESOLVED without new zeros; recomputed from the v919 pins to  $< 2\%$ ); the window carrier  $x \lesssim 2.3 \times 10^5$  disclosed (carrier class, not a new omega); (5) NOT PROMOTED, deliberately: the round-147 adjugate elimination (AD1/AD2 + the cell lemma modulo simplicity) and the round-150/151 collapse-rate/multiblock coordinates stay experiments-side — their coordinates are still in active motion and the simplicity leg is open; (6)

**SURFACES:** ledger rows (1061  $\rightarrow$  1065); veri blocks in `tfpt_3` (audit-counted body) + the new prime-front diary section “The prime-front theorem arc” with the promotion of record and the suite-count refreshes (910  $\rightarrow$  914 across the live parentheticals); endform article §8.18 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v921; README/Zenodo 914; note CDLVI; (7) **GATES:** all four affordable source probes re-run green-as-typed BEFORE any edits (identical SPEC SHAs; 28/28, 31/31, 40/40, 26/26), the two multi-hour probes pinned with disclosure; v918–v921 standalone green (13/13, 11/11, 17/17, 9/9, each  $\sim 1$  s); FULL `run_all.py` ALL CHECKS PASSED at 914 modules; `build.sh` release + EXIT GATE `build.sh` audit AUDIT OK; manifests last. NO RH claim in any direction — reductions and identities say nothing about the truth of RH either way)

2026-08-16 (CVI) (the Bughunt-III corrections of record — the four word-ing/label corrections that the round-130 adversarial audit (note CDXXXII, `experiments/tfpt-discovery/bughunt3_probe.py`, 20/20 gates, 0 MAJOR / 4 MINOR / 3 NOTE, NO verdict flips, every promoted number of v916/v917 independently CONFIRMED) recommended for promoted and frozen surfaces, applied to the LIVE surfaces only (frozen probes and notes are never retro-edited — the correction is the record); NO numeric value changes anywhere, both modules stay green with unchanged gate counts (20/20, 13/13) and unchanged verdicts: (1) `v916_epstein_weil_violation.py` (+ website mirror): the pin `PIN_EULER_BOUND` = 0.664 relabeled as the  $N = 2 \times 10^4$  TRUNCATION of  $\sum_{n \geq 4} r_Q(n) n^{-1.6} / r_Q(1)$  PLUS the declared Abel-model tail — NOT the “full-table value” (its old label); the true full sum is 0.645728 (bughunt 4-decade ladder  $N = 4 \times 10^3 \cdot 4 \times 10^6$ , spread  $8 \times 10^{-7}$ ) and v917’s 0.650 is the same construction at  $N = 2 \times 10^5$ ; the pinned number 0.664 itself is UNCHANGED (a valid upper bound; both reads  $< 1$ , the D1 gate logic stands); (2) `v917_driver_rate_certification.py` (+ website mirror + the ledger row PRIME.KR4.DRIVER.CERT.01): the localization census premise restated from the too-strong “every off-line zero below 45 is located” to the gated repair —  $\delta \geq 0.01$  census boxes + band-straddle completeness + the window-width argument:  $\max |v(a_0)| = 0.646 < 1$  on the census-blind strip, any  $|v| > 1$  zero must sit at  $\gamma \in (15.645, 15.703)$  INSIDE the straddle-boxed band, and blind-strip contributions to the depth diffs are negative or  $\leq 10^{-40}$  — hidden zeros can only MASK the fire, never fake it (the localized verdict SURVIVES; papers and website prose checked: the too-strong sentence appears on no other live surface); the 0.650 Euler-region read carries its truncation label now too; (3) the round-129 “round-124  $t_{\text{death}}$ ” misattribution (the collapse is ROUND 123): searched `tfpt_prime_front`, `articles/`, website — NO live surface carries the wrong round label, nothing to correct (the frozen round-129 probe and note CDXXXI stay as they are per the freeze convention; note CDXXXII documents the error); (4) the note-CDXXVII poly-filter class-label shorthand (the hull normalization  $|p| \leq 1$  on the spectral hull dropped): searched papers, website, ledger — the shorthand leaked to NO live surface, nothing to correct (the frozen note stays; the probe’s own hypothesis-carrying statement was always right). **GATES:** v914–v917 re-run green BEFORE any edit; v916/v917 re-run green AFTER (20/20, 13/13, verdicts unchanged); `build.sh` gen + release; EXIT GATE `build.sh` audit AUDIT OK; manifests last; note CDXXXIV (head numeral re-verified immediately before writing: the concurrent L1-Weyl-law lane had taken CDXXXIII — one collision

resolved by re-read). NO RH claim in any direction)

2026-08-16 (CV) (the Epstein detection-arc promotion — the two durable certified results of discovery rounds 123 and 125 graduate from experiments/ into the load-bearing suite as v916\_epstein\_weil\_violation.py and v917\_driver\_rate\_certification.py (suite 908  $\rightarrow$  910); both rows typed NOT-RH-EVIDENCE in either direction — known falsehoods (the off-line zeros of  $Q = x^2 + 5y^2$  are Potter–Titchmarsh 1935 / Davenport–Heilbronn 1936), new certified instruments; the min-cut is unchanged (flows 4/4): a detector is a falsifier, not a prover: (1) THE CERTIFIED EPSTEIN WEIL-POSITIVITY VIOLATION (v916, 20/20, ledger row PRIME.KR4.EPSTEIN.COLLAPSE.01 [E]; source probe epstein\_collapse\_probe.py round 123, frozen SPEC bb51d1a3299f4f62, 27/30 with exactly the three PRE-DECLARED falsified predictions G18/G19/G21 failing, re-run green-as-typed at promotion, 998.5 s) — an explicit sha-pinned unit test vector (len 1038, sha b01436d53d2cbbec, REPRODUCED at every suite run together with the full quadratic form) whose tent-autocorrelation test function pairs NEGATIVELY with the source data  $-g''_Q$  built purely from  $r_Q(n)$  lattice counts:  $V = x^T T x = -1.9697 \times 10^{-2}$ , error budget  $\leq 4.3 \times 10^{-68}$ , two-precision (dps 80/110) agreement  $9.8 \times 10^{-79}$  — windowed Weil positivity for  $\xi_Q$  FAILS at  $t \leq 3.114$ ; the  $Q$  world dies at  $t_{\text{death}} = 2.988$  (recomputed eigenvalue crossing  $k^\dagger = 998$ , mesh mate stable,  $Q$ -scramble dies at 2.73) while TRUE zeta and the conductor-matched null  $\zeta_K = \zeta_{L-20}$  complete positivity through  $L = 8.0$  (pinned run-of-record; ZK anchor and  $L = 4.0$  prefix recomputed; the certificate vector reads POSITIVE on the matched null); the value channel is CERT-NUM, the off-line-zero interpretation step COND-CLASSICAL (Weil/Guinand explicit formula, cited); the three pre-declared falsified predictions travel verbatim (attribution COLLECTIVE — witness removal moves the death by only 0.018, all-located removal survives to 4.428; the law misses by  $0.505 > 0.5$ ; the census is falsified: FOUR off-line pairs below 45, the round-117 witness never  $\gamma$ -minimal, the 15.6682 DRIVER carries  $26\times$  its excess); (2) THE CERTIFIED LOCALIZED RATE DETECTION OF THE DRIVER (v917, 13/13, ledger row PRIME.KR4.DRIVER.CERT.01 [C]; source probe driver\_cert\_probe.py round 125, frozen SPEC bcc64cb371dff9c8, 32/33 with exactly the single PRE-DECLARED G11 fail, re-run green-as-typed at promotion, 881.1 s) — the driver  $\rho = 0.9329696975 + 15.6682495313i$ , source-located from the  $\Lambda_Q$  sieve ( $\Delta\gamma = 6.6 \times 10^{-5}$ ), fires in the band-peeled rate channel at depth  $m^* = 14$  with certified LB = 1.000137  $> 1$ , LOCALIZED from  $m_{\text{loc}}^* = 15$  (the violating window contains  $a^* = 245.6815$ : it is the driver); the band peel  $\gamma \in (7.90, 31.40)$  (winding-complete, every peel  $|\xi_Q|$ -verified, phantom peel doubly excluded) collapses the unpeeled price  $m_2 = 908$  by  $\times 65$  at sieve cost only  $N = 2 \times 10^5$ ; the new exact algebra is recomputed symbolically at every run ( $v^* = 1 + \varepsilon$  EXACTLY REAL at the matched pin, weight exactly 1, LB  $\rightarrow v^*$  to six digits by  $m = 25$ , certified depth 32,  $\text{Err} \approx 2.1 \times 10^{-12}$ ,  $\text{meas} \leq \text{cond}$ ) together with the quantified detection-cost law  $m^* \approx \ln(B_{\text{band}}/\varepsilon)/\ln(1/v_{\text{edge}})$ , sieve price polynomial in  $1/\varepsilon$  via the band, measured 3.19 certified depths per unit  $\ln N$ ; the grade is carried verbatim and never upgraded — CERT-GIVEN-AUDIT (rigorous Cauchy given the incomplete-gamma audit comparator plus the DECLARED tail/peel models) and DISGUISE-ADJACENT (the deep read is an Euler-computable truncated windowed Weil transform) — and the honest caveat is carried: the pre-declared G11 gate (locator relocates 4/4) FAILED, the weak round-117 witness (excess  $2.9 \times 10^{-5}$ ) sits below the locator's measured threshold at  $N = 2 \times 10^5$ , so the locator is typed a 3/4 MEASURED instrument and no 4/4 claim is promoted; (3) SUR-

FACES: ledger rows PRIME.KR4.EPSTEIN.COLLAPSE.01 + PRIME.KR4.DRIVER.CERT.01 (1059 → 1061); veri blocks in tfpt\_3 (audit-counted body) + the new prime-front diary section “The Epstein detection arc” with the promotion of record, the suite-count refreshes (908 → 910 across the live parentheses) and the correction of record on the round-102 witness sentence (never  $\gamma$ -minimal); endform article + Anthropic email promotion paragraphs; website PROMOTED feed card + DAG nodes + banner/meta v535–v917; README/Zenodo 910; note CDXXVIII (head numeral re-verified immediately before writing: the concurrent selection lane had taken CDXXVII — one collision resolved by re-read); (4) GATES: both source probes re-run green-as-typed BEFORE any edits (identical SPEC SHAs, identical failing sets); v916/v917 standalone green (20/20 in  $\sim 12$  s, 13/13 in  $\sim 1$  s); FULL run\_all.py ALL CHECKS PASSED at 910 modules; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last. NO RH claim in any direction — both rows are falsifier instruments on a classical known-falsehood world; zeta’s omega absorbs every finite certificate)

2026-08-15 (CIV) (the weekend promotion — the two durable proven results of the discovery weekend graduate from experiments/ into the load-bearing suite as v914\_pascal\_region\_theorem.py and v915\_eulerpick\_certified\_floors.py; suite 906 → 908, both modules green standalone (18/18 in  $\sim 20$  s; 15/15 in  $\sim 40$  s) and registered in run\_all.py, the script registry and the status ledger (new rows PRIME.PASCAL.REGION.THEOREM.01 [E] and PRIME.EULERPICK.CERTIFIED.FLOORS.01 [E], both typed as certified-finite results with their citations disclosed and explicitly NOT as RH-evidence rows); all four source probes re-run green before any edit (moonshot\_sol\_probe.py 12/12, 0.4 s; moonshot\_o3\_probe.py 16/16, 100.8 s, frontier verbatim; eulerpick\_ladder\_probe.py 31/31, 102.8 s; the round-100  $10^{13}$  compiled sieve NOT re-run — its sieve4\_run1.log run-of-record is pinned per the PINNING DISCLOSURE convention); NO RH claim in any direction, no other marker moved, no gate closed or narrowed: (1) v914 THE PASCAL-REGION THEOREM (rounds 103 + 106):  $C_{n,k}(256) > 0$  for EVERY  $n \geq 0$  and every integer  $0 \leq k \leq 7,444,682,106,464,286,365,865$  ( $\sim 7.44 \times 10^{21}$ ), proven from cited external inputs alone — Platt–Trudgian 2021 Thm 1 ( $T = 3,000,175,332,800$ ), Hasanalizade–Shen–Wong JNT 235 (2022) Corollary 1.2 ( $|N(t) - M(t)| \leq 0.1038 \log t + 0.2573 \log \log t + 9.3675$ ,  $t \geq e$ ), Rosser 1941 Thm 19, the classical  $\gamma_1$  isolation; the module recomputes the COMPLETE round-103 four-region partition at every suite run ( $B_0(T) = 7.57565571105606088 \times 10^{-10}$  anchored; decisive ratios  $1.876981097339/1.176577026464/335.835/1416.754$ ; mechanism root  $k^* = 1.405443828790708 \times 10^{19}$ ), recomputes the quadratic verification-height law  $k_{\text{front}}(T) = (\lambda_\infty + O(1/\log T)) T^2/(a - 1/2)$  with  $\lambda_\infty = 0.226987054723979955$  (residual  $< 10^{-60}$ ), pins the 33-cell run-of-record frontier partition and re-proves a frozen 6-of-33 subsample with full 16384-packet endpoint inequalities INCLUDING the worst cell (endpoint ratio  $1.000001535385559$ ; the next  $2 \times 10^{-6}$  relative step fails — mechanism-exact); THE ROUND-110 HSW ATTRIBUTION FLAG RESOLVED FROM THE PAPER ITSELF: Cor. 1.2 IS the  $N(t)$  bound with exactly the constants consumed, Cor. 1.4 is the companion  $S(t)$  bound ( $\min\{\dots + 8.3675, 0.1095 \log t + 0.2042 \log \log t + 3.0305\}$ ) and is NOT used — the probe docstring’s “Cor. 1.4 is the corrected replacement” was the slip, the numbers were always right; the old (0.110, 0.290, 2.290) constants are the earlier Platt–Trudgian  $S(t)$  bound whose proof consumed the invalid Cheng–Graham estimate; HONEST TYPING carried in module, ledger and papers: verification-powered, FORM-LOCAL (zero-location content below  $\gamma_1 = 14.13$ ; says nothing



about screw sections or Weyl disks), the  $k \rightarrow \infty$  quantifier untouched, no wall positivity, no zero table, no zeta evaluator — NOT evidence for or against RH; (2) **v915 THE CERTIFIED EULER–PICK FLOORS**  $N = 1.4$  (rounds 95 + 100): outward-rounded interval positivity certificates with the recomputed/pinned split disclosed exactly — **RECOMPUTED** in-run: the certificate self-tests, the certified digamma, the own sieve to  $1.6 \times 10^7$  with all three cited  $\psi$ -bounds warded against own data (Büthe 2018; Rosser–Schoenfeld 1962 Thm 12; Nair 1982), the cap- $4 \times 10^6$   $N \leq 2$  and cap- $1.6 \times 10^7$   $N \leq 3$  floors reproducing the round-95 record ( $\log_3 = 2.3643695 \times 10^{-10}$ ), the summation-error model of the compiled run re-derived rigorously from the pinned  $\pi(10^{13})$  plus an RS–Abel upper bound (no sieve needed), the Nair far-floor necessity (the crude  $\psi \geq 0$  floor blocks  $N = 4$  at ANY cap) with the corrected budget  $X_{\text{req}}(4) = 10^{11.52}$ , the  $N = 5$  knowledge wall, and a recomputed detection subset ( $N^*(5; 0.1) = 3$ ,  $N^*(8; 0.1) = 4$ ,  $N^*(8; 0.4) = 3$ ); **PINNED** from run-of-record: the cap- $10^{13}$  intervals  $N=1$   $[4.5917135 \times 10^{-2}, 4.5917136 \times 10^{-2}]$ ,  $N=2$   $[9.0288887 \times 10^{-6}, 9.0288888 \times 10^{-6}]$ ,  $N=3$   $[6.9310239 \times 10^{-10}, 6.9310252 \times 10^{-10}]$ ,  $N=4$   $[8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  (margin 0.749) with  $\pi(10^{13}) = 346,065,536,839$  reproduced exactly by the recorded instrument; **HONEST TYPING** (round-108 F3 carried verbatim, no overselling): the falsification channel is **DISPROOF-SOUND** but **REACH-CAPPED** at  $\gamma \sim 5\text{--}8$ , far below Platt–Trudgian  $3.0 \times 10^{12}$  — zero live falsification power in the classically certified range; “open and empty” is a theorem given the cited inputs; the channel stays open and **EMPTY**; (3) **SURFACES**: `tfpt_prime_front` gains the promotion-of-record notes and `\veri{}` citations in the owning sandbox-diary subsections (instrument tier; roadmap; frontier audits) with the freeze-time suite counts rephrased honestly (906 at freeze  $\rightarrow$  908 today); the end-form article and the Anthropic email draft gain matching promotion-of-record paragraphs; the website prime-front page and feed gain the promotion entry and tier upgrades (Euler–Pick floors and Form-A frontier now named as modules, reach caps kept verbatim), the DAG E8 node gains both scripts, README and Zenodo counts move to 908/v1-v915; `ScriptIndex.tsx`, `changelog.ts`, `suite.ts` and the maps regenerated via `bash build.sh gen`; full suite re-run `run_all.py`  $\rightarrow$  **ALL CHECKS PASSED** at 908 modules; `bash build.sh audit`  $\rightarrow$  **AUDIT OK**; manifests regenerated **LAST**. NO RH claim anywhere)

**2026-08-15 (CIII)** (the boundary-and-resolvent sync — discovery freeze rounds one-hundred-eleven and one-hundred-twelve synced into every live surface; **experiments/** only — **TWO** frozen probes, **NO** promotion, **NO** module, **NO** ledger row, **NO** marker moved, **NO** gate closed or narrowed, suite unchanged at 906, **NO** RH claim in any direction; both source probes re-run green before any edit (`unpriced_classes_probe.py` 26/26, 88 s; `resolvent_closure_probe.py` 36/36, 266 s): (1) **ROUND 111** (note CDXI, contract PRIME.UNPRICED.CLASSES.01, `unpriced_classes_probe.py`, 26/26 gates, 88 s, SPEC `ec787fc5`, deterministic re-run bit-identical, verdict **UNPRICED-ADJUDICATED** — tally 25  $\rightarrow$  30 candidates, 0 PASS): the five argument classes the round-108 audit named as genuinely **OUTSIDE** the atlas enumeration are all adjudicated and killed, none surviving as an open lever; (U-A, **THE HEADLINE** — **UA-RATE-IS-ZERO-CURRENCY**) Zagier’s published horocycle RH-equivalence (Zagier 1979/81:  $m_y(f) = m(f) + o(y^{1/2})$  unconditional,  $\text{RH} \iff \text{error } O(y^{3/4-\epsilon})$ ; Sarnak CPAM 34 (1981); Verjovsky 1994) — a published RH-equivalence **NEVER ADJUDICATED UNTIL NOW**, confirmed absent from every atlas artifact; the instrument built (incomplete Eisenstein series  $\psi(t) = t^2 e^{-t}$ , horocycle integral  $= 3/\pi + \text{scattering}$

contour, warded two-route:  $\text{dev} \leq 1.5 \times 10^{-8}$ ,  $|\varphi(1/2 + it)| = 1$  to  $9.9 \times 10^{-32}$  — the unconditional  $1/2$  exponent is UNITARY, no spectral gap) measures the deep-window exponent 0.7451 (the RH-range  $3/4$  in the classically verified region) and, on planted-zero synthetic scattering worlds at  $\beta = 0.6/0.75/0.9$ , the exponents 0.6971/0.6222/0.5473 against the law  $1 - \beta/2$  ( $|\text{dev}| \leq 0.0029$ ) — THE HOROCYCLE RATE IS EXACTLY THE AFFINE REPARAMETRIZATION  $\theta = 1 - \Theta/2$  OF  $\sup \text{Re } \rho$ : the emptied E2 zero-free/density currency, alignment content linear  $\log y$  phases (the round-97 class,  $\times 8.2$ ); rigidity technology rate-free (Ratner) or non-horospherical (Lindenstrauss–Mohammadi–Wang CRAS 361 (2023): unspecified exponents, explicitly beyond the horospherical case — the modular horocycle IS horospherical, where the classical spectral  $1/2$  has stood since 1981); DISGUISE by rule R1; (U-B — UB-INAPPLICABLE-BY-TYPE) the zero set of Hardy  $Z$  is infinite (29 sign changes on  $[0, 100]$  measured = the RvM count) while o-minimal definable sets have finitely many connected components —  $\zeta$  on the line is definable in NO o-minimal structure; the serious adjacent result verified and typed (Masser JNT 131 (2011): rational points on the GRAPH,  $\leq C(\log D/\log \log D)^2$  — count currency, position-free, unsigned); (U-C — UC-HOST-GAP-CONFIRMED)  $\lambda \geq 1/4$  with positive multiplicity forces the line (the Selberg-zeta precedent) but  $\lambda = 1/5$  with positive multiplicity yields the real pair  $1/2 \pm \sqrt{5}/10$  exactly — positivity alone does not force the line, it needs identity AND the  $1/4$  gap, which at the arithmetic quotients is Selberg’s own OPEN conjecture (Kim–Sarnak floor 975/4096, deficit 49/4096 exact); for  $\zeta$  the host identity is the mountain (Connes’ trace formula  $\iff$  RH) — Stage 1 of the roadmap in another language, no new lever; (U-D — UD-MAGNITUDE-LAW-ONLY) every proven FHK-type deliverable (ABBRs 2019, FHK I+II, AOR moments; Harper RH-conditional) is a  $|\zeta|$ -magnitude statement IN LAW  $\rightarrow$  the emptied E1 class; the exception check closed exactly: iid uniform phases have ZERO signed expectation and a flip-invariant law — the randomized Euler product carries NO signed alignment observable; (U-E — UE-CANCELLATION-ONLY-INEFFECTIVE) Green–Tao  $\log^{-A}$  savings long-interval with Siegel–Walfisz-ineffective constants, MRTTZ 2023 qualitative  $o(1)$  on average, the charitable  $C = 1$  envelope table showing the ENTIRE content in the constant, NO effective constant for  $\Lambda$ -nilsequence correlations, and the sharpest signed corollary of  $|X| \leq \text{env}$  being  $X \geq -\text{env}$  — orientation exactly at envelope strength,  $\times 8.2$  short; THE CLOSING SENTENCE: the atlas stands at 30 candidates, 0 PASS, with the vocabulary boundary itself now adjudicated — every named argument class inside AND outside the original enumeration is formally priced; (2) ROUND 112 (note CDXII, contracts PRIME.SAFEPOINT.RESOLVENT.CLOSURE.01 + PRIME.CONNES.GROUNDSpace.BLOCK.01, `resolvent_closure_probe.py`, 36/36 gates, 262 s, SPEC ea7863f2, deterministic re-run identical, two disclosed X1 tail-cut amendments; composite RESOLVENT-TRANSCRIPTION + CROSSCARRIER-BOUNDS + JENSEN-NO-DICTIONARY): THE IDENTIFICATION — arXiv:2511.22755’s operator (CCM’s rank-one  $D_{\log}^{(\lambda, N)} = D_{\log} - |D_{\log} \xi\rangle \langle \delta_N|$  with  $\xi$  the lowest eigenvector of the Weil form, and its  $\det_{\text{reg}}$  construction) is, MODE FOR MODE, the corpus’s own round-89 record extremal family (lattice  $\pi k/a$ ,  $a = \frac{1}{2} \log x$ ; the round-89 arbiter quotient IS the literal CCM operator); built explicitly at  $x = 3, 5$  and certified: kernel  $M\xi = 0$  to  $10^{-68}$ , spectrum = record zeros (10/10 at  $x = 5$ , rel  $2.7 \times 10^{-8}$ ), their  $\det_{\text{reg}}$  formula executable as a rank-one determinant identity to  $10^{-75}$  — THE CORPUS HAD INDEPENDENTLY BUILT CCM’S OPERATOR

BEFORE THEIR PAPER; THE GROUNDSPACE-BLOCK VARIANT ELIMINATES CCM'S SIMPLICITY AND EVENNESS HYPOTHESES AT CONSTRUCTION LEVEL ( $M_{\text{blk}} = D(I - P)$  via the biorthogonal block projector, basis-rotation-invariant to  $1.5 \times 10^{-16}$ , STABLE where the simple selection swings  $O(1)$  under  $10^{-10}$  perturbations, the exterior/matrix-Wronskian identity  $\det(M_{\text{blk}} - z) = \det(D - z) \det(I_2 - W(z))$  to  $5.2 \times 10^{-15}$ , PD quotient metric; the remaining realness step on  $E/K$  is the named unproven block-CCM lemma, finite-dimensional and zeta-free); THE KILL: the pins converge 16/16 (slope  $-0.65$ ,  $\tau$ -screen 0/16) BUT THE Z1 SCREEN FIRES — band pins equal cache partial sums to rel  $7.7 \times 10^{-12}/9.4 \times 10^{-12}$ , the exact round-89 conviction INHERITED BY THE LITERAL CCM CARRIER; side-by-side at matched depth the Krein benchmark extrapolates genuinely (max dev  $2.9 \times 10^{-4}$ ) while the CCM carrier band-transcribes ( $1.3 \times 10^{-1}$ ) — THE KREIN CARRIER REMAINS THE ONLY NON-TRANSCRIBING ATTACK SURFACE IN THE CORPUS; CROSSCARRIER-BOUNDS:  $\Delta_m = \text{Tr}[A_x^{m+1}]$  vs the round-102 Hausdorff  $b_m(256)$  — 17/17  $m$ -rows MONOTONE LOWER BOUNDS FROM BELOW, deficit = the RvM tail beyond the band edge (median ratio 0.786; 11/41 roots beyond the band kept in all sums — the operator mirror of the Nyquist excess): the Hausdorff field and the CCM operator are ONE OBJECT ONLY IN THE LIMIT — at finite depth the operator is a band-truncated basis of the field from below (all 45 operator cells  $n + k \leq 8$  nonnegative; a third carrier confirms via Krein  $\sigma$ -derivative moments to  $8.7 \times 10^{-4}$ – $2.7 \times 10^{-3}$  within the bias model); THE JENSEN WEDGE (arXiv:2608.08682:  $n^3 \log^2(n+2) \geq Kd^5 \Rightarrow$  hyperbolic, polynomially improving GORTW) verified verbatim but typed NO-DICTIONARY — the only corpus bridge (the Li transform at  $a = 1/4$ ) is alternating-signed, so by round-108 form-locality the wedge adds NO unconditional Hausdorff cells; THE FORM-LOGIC GATES verify the architecture's selling point exactly (F1: one Stieltjes representation would generate all three positivity families simultaneously; F2: finite certificates are form-local — Pick fires at  $N = 5$  where Hausdorff stays positive to depth 40) — the logic is correct, AND PRECISELY THEREFORE WORTHLESS WITHOUT A NON-TRANSCRIBING CARRIER; the remaining analytic theorem is DOUBLY NAMED: CCM's own §7–8 prolate approximation  $\iff$  the corpus's round-90 Weyl-disk contraction, pointwise at the pins; (3) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The vocabulary boundary priced, and the resolvent closure adjudicated: the CCM operator is the corpus's own record family”; the end-form article gains §8.16 with the abstract status sentences and the traceability rounds 111–112 (with the Zagier 1979/81, Sarnak CPAM 34 (1981), Verjovsky Kodai 17 (1994), Lindenstrauss–Mohammadi–Wang CRAS 361 (2023), Masser JNT 131 (2011), Kim–Sarnak, arXiv:2511.22755 and arXiv:2608.08682 citations); the Anthropic email draft gains one status paragraph (the tally-30 close + the CCM identification as the collaborator hook: their operator is the corpus's record family, and the corpus's machine eliminates their simplicity hypothesis); the website prime-front page gains chapter 40 (the live-updates chapter renumbered  $40 \rightarrow 41$ ) and one consolidated sandbox feed entry (verdict RESOLVENT-TRANSCRIPTION, new enum, script `resolvent_closure_probe.py`); website/lib/changelog.ts regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-15 (CII) (the corrections-and-frontier sync — discovery freeze rounds one-hundred-six through one-hundred-nine synced into every live surface, PRI-

**MARILY A CORRECTIONS ROUND:** the round-108 big-picture logic audit and the round-109 Bughunt II independently convicted the published “three equivalent lemma forms” language and two related headlines, and the corrected wordings are now carried on every surface per the round-87/97 correction-of-record convention (frozen probes and notes are NOT retro-edited — the correction is the record); plus the new content: the  $530\times$  Form-A frontier extension with its derived quadratic law, and the code-synthesis adjudication of the program’s founding intuition; experiments/ only — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO round verdict flips, NO number wrong, NO RH claim in any direction: (1) **THE CORRECTIONS OF RECORD** (round 108, `bigpicture_logic_probe.py`, 31/31 gates, 2.5 s, SPEC 889edf4a, deterministic re-run identical, verdict **BIGPICTURE-FINDINGS(3): F1 FORMS-EQUIVALENCE-ONE-SIDED(+TRANSFER-NONE)**, F2 **STAGE3-ENTRY-FORM-MISMATCH**, F3 **FALSIFIER-REACH-CAP**; and round 109, note CDIX, contract **PRIME.BUGHUNT2.R86.R105.01**, `bughunt2_r86_r105_probe.py`, 30/30 gates, 11.6 s, SPEC 759d0209, deterministic re-run identical, verdict **BUGHUNT2-FINDINGS(7, max MAJOR): 1 MAJOR / 6 MINOR / 0 FATAL** — the two audits **INDEPENDENTLY CONVERGENT** on the same headline finding): (K1) “three equivalent lemma forms” (paper §8.13/§8.14, prime-front 12.45/12.46, website chapters 37/38; the changelog C/CI wordings are frozen history and are corrected **HERE**, not retro-edited) is corrected to “three RH-SUFFICIENT lemma forms” — the honest form  $\times$  direction table: the **HAUSDORFF** form is a **GENUINE** proven RH-equivalence (forward audited-sound via the H1c continuation chain, independently re-derived **GAP-FREE** in round 109, plus Zhang arXiv:2303.09396 Thm 2; backward elementary + gated), the **WEYL-DISK** and **SIGN-POSITION** forms are **ONE-SIDED PENDANTS** (form  $\Rightarrow$  RH established; RH  $\Rightarrow$  form **OPEN** — the round-90 lemma, measured 16/16 truth-tight but never proven, and the CF-realness/Connes-§6.6 hypotheses respectively), **EVERY** mutual equivalence that holds routes **THROUGH RH** (no direct form-to-form edge, gate J1.G5; the only direct bridge is the (AC) weld stopping at the exponential-autocorrelation cone), **TRANSFER IS NONE** — demonstrated by an explicit counterexample world: an off-line quadruple at  $\gamma = 20$  (above  $\gamma_1$ ) keeps **EVERY** Hausdorff cell positive ( $n + k \leq 50$ ,  $b_n$  to  $n = 240$ ) while the Euler–Pick section **FIRES** at  $N = 12$  (certified negative pivot at 200 dps) — and **CERTIFIED REGIONS ARE FORM-LOCAL**: the  $k \leq 10^{19}/7.4 \times 10^{21}$  Form-A theorems imply **NOTHING** about screw sections or Weyl disks, their zero-location content sitting below  $\gamma_1 = 14.13$  where Platt–Trudgian is infinitely stronger; the stale adjective “uniform” in Form B’s name is retired on every surface (the round-92 V0a correction: pointwise contraction at the pins suffices); (K2) **THE FALSIFIER REACH CAP** (round 108 F3) is attached wherever the surfaces said “standing disproof channels / open and empty” (roadmap §8.14, prime-front 12.46, website chapters 37/38): both instruments are **DISPROOF-SOUND** (the disproof direction consumes only the proven forward implication) but **REACH-CAPPED** far below Platt–Trudgian’s  $3.0 \times 10^{12}$  (Euler–Pick  $N \leq 4$  sees off-line zeros only at  $\gamma \sim 5\text{--}8$ ; the Hausdorff field is provably blind above  $\gamma_1 = 14.13$  at the safe point) — “open and empty” is a **THEOREM GIVEN THE CITED INPUTS**, not an observation: zero live falsification power in the certified ranges; (K3) **THE STAGE-3 INTERFACE FENCE** (round 108 F2) is attached to “Stage 3 conditional-complete”: the only kernel-checked end-to-end extraction (`v848 + CofinalWeil.lean`) accepts **ONLY** the wall form  $H_{\text{cof}}$ , which is not among

the three entry forms and is not producible from them without RH (test-class mismatch, absence-gated); the weakest link is the Stage-2→3 interface itself — finite acceptance gates (cells  $n + k \leq 86$ ) against infinite-quantifier entry forms, a jump no corpus machinery bridges (v912 bridges convergence, not positivity); (K4) THE MINOR PROSE FIXES (round 109): the screwind mesh-ladder misattribution corrected in prime-front 12.45 (0.222/0.184/0.156 at  $\delta = 0.012/0.006/0.003$  — the earlier 0.207 was the  $\delta = 0.008$  round-93 value 0.2065; MESH-DAMPED stands), plus on record without surface impact: one hardcoded-true gate in the round-104 tally (the only live one across all 17 probes), two decorative tautological conjuncts in the round-92 probe (the countable Nevanlinna–Pick step asserted-with-citation, honestly typed everywhere), the roadmap probe’s stale r95  $N = 3$  acceptance value (r100 tightened  $\log_3$  to  $6.9310239 \times 10^{-10}$ ; frozen probe, finding on record), two sieve4\_helper.c header slips (certificate UN-AFFECTED: true worst per-term  $13.33u \ll K_{\text{TERM}} = 32u$ ); (K5) THE OCT BOUNDARY SHARPENED (round 108 J2, absence-gated at 0 atlas hits each): wherever the surfaces state atlas completeness (paper §8.9/limitation 8, prime-front OCT subsection, website chapter 33), the vocabulary-relative fence now names the FIVE UNPRICED classes — ergodic/measure-rigidity positivity transfer, model-theoretic/o-minimal counting, bigger-group representation-theoretic positivity, GMC/Fyodorov–Hiary–Keating (reduces-in-spirit to magnitude, with an enumeration gap), higher-order Fourier uniformity of Möbius; AND THE CLEAN SIDE carried in the same breath (round 109 B1, own implementations): the round-103 theorem re-proved end to end ( $B_0 = 7.575655711 \times 10^{-10}$  exact match, all four partition ratios to  $10^{-9}$ , wall root  $k^* = 1.40544 \times 10^{19}$  consistent), the  $N = 4$  certification re-derived (Nair exact for  $n \leq 60$ , far-floor necessity confirmed, Cholesky logic sound), the Hausdorff H1c continuation GAP-FREE, the Lean theorems saying exactly what round 99 claims (axioms exactly [propext, Classical.choice, Quot.sound]), all 16 in-scope probes re-run green during the audit; (2) ROUND 106 (moonshot\_o3\_probe.py, 16/16 gates,  $\sim 104$  s, independently re-run green, verdict MOONSHOT3-EXTENDED(frontier = 7,444,682,106,464,286,365,865; quadratic law) + MOONSHOT3-PARTIAL(exact  $G$  / sign-regularity / fixed- $N$  theorem) + MOONSHOT3-OBSTRUCTED(PF orientation; downward Hankel transport)): the round-103 theorem EXTENDED 530-FOLD —  $C_{n,k}(256) > 0$  for ALL  $n \geq 0$  and  $0 \leq k \leq 7.4447 \times 10^{21}$ , proven from Platt–Trudgian 2021 plus the CORRECTED Hasanalizade–Shen–Wong bound (JNT 235 (2022), Cor. 1.2:  $|N(t) - M(t)| \leq 0.1038 \log t + 0.2573 \log \log t + 9.3675$ ; HONEST SOURCE CORRECTION: the commissioned (0.110, 0.290, 2.290) constants bound  $S(T)$ , not  $N(T)$ , and their Cheng–Graham input was later identified as invalid — replaced, not patched); mechanism: the retained  $q^k$  factor, a 16384-packet RvM lower bound below  $T - 1$ , a Gaussian-series tail enclosure above  $T$ , a 33-cell endpoint partition; worst gate lower/tail ratio 1.000001535385559, minimum packet count 25,397,045.4, the next  $2 \times 10^{-6}$  relative step FAILS — mechanism-exact; THE LAW:  $k_{\text{front}}(T) = (\lambda_\infty + O(1/\log T)) T^2/(a - 1/2)$  with  $\lambda_\infty = 0.226987054723979955$  — VERIFICATION HEIGHT  $qT$  BUYS  $q^2$  COLUMNS, the geometric ceiling  $k_{\text{geo}} = (T^2 + 1/4)/(a - 1/2)$  intrinsically unreachable (the route stops at  $0.227 k_{\text{geo}}$ ); NEW EXACT STRUCTURE: the closed-form two-variable generating function  $(X + Y - XY)G = XaF(a(1 - X)) + Y \frac{a}{1-Y} F(\frac{a}{1-Y})$  with the pole-divisor EQUIVALENCE (not an RH proof), total sign-regularity in the REVERSED- $k$  orientation (the standard Pólya-frequency orientation is WRONG — the  $2 \times 2$  minor manifestly negative), THE IMPORTANT CORRECTION: HANKEL POSITIVITY ALONE IS NOT FORM A (uniform measure on  $[2, 3]$ : every Hankel matrix posi-

tive definite yet  $C_{0,1} = -3/2$  — the localizers  $H^+$  and  $H - H^+$  required), the fixed- $N$  eventual-positivity theorem ( $b_m(a) = (\sqrt{a} \log a/8)(1/2)_m/m! + O_N(\sqrt{a})$ ), and the exact downward-transport obstruction (for two atoms  $\frac{d}{dt} \log \det H_2 = 4 - 3(y_1 + y_2)$ , either sign possible — fixed- $N$  eventual positivity cannot be propagated down to  $a = 256$ ); (3) ROUND 107 (note CDVII, contract PRIME.CODE.SYNTHESIS.01, `code_synthesis_probe.py`, 18/18 gates+wards, 2.2 s, SPEC 5a6c3851, deterministic re-run bit-identical, verdict CODE-READING-TRUE-DECODING-DIRECTION): the program owner’s founding intuition “primes as error-correcting code in TFPT” given its first formal adjudication under that name and VINDICATED AS MEASURED STRUCTURE — the primes ARE the unique codeword of an error-correcting code in the compiler’s own Gram grammar (ambient: non-negative atomic measures at tent mesh  $\delta = 0.006$ ; codeword: the prime-power comb; check family: nested Toeplitz sections, one check per depth; MINIMUM DISTANCE: the round-96 rigidity law RE-MEASURED INDEPENDENTLY,  $\ln \eta^*(r) = -1.0730r - 7.0783$ , identical to four digits; decoding radius: 6/6 impostors die at the exact round-96 exits; rate  $\rightarrow 0$ ); SAME GRAMMAR as the compiler’s literal E8 code layer ( $[8, 4, 4]$  self-dual, weight enumerator  $\{0:1, 4:14, 8:1\}$ , Construction A  $\rightarrow 240$  roots, rebuilt exactly in-probe) — TFPT’s one axiom INSTANTIATES at the arithmetic seam as this code, structure not progress; DIFFERENT DUAL TYPE with the decisive number measured: THE DUAL RANK GROWS WITH SLOPE 1.0000 ( $k_{\text{fail}}/D = 0.9960/0.9975/0.9983/0.9987$ ; syndrome rank  $5 \rightarrow 12$ ; a non-codeword passes every bounded-depth check family BIT-EXACTLY below its defect), adversarial LTC gap  $\rightarrow 0$  beyond any bounded family’s support (local errors testable and depth-improving,  $0.9984 \rightarrow 1.0000$ ) — the code runs measure  $\rightarrow$  identification (DECODING) while RH needs codeword  $\rightarrow$  all checks (ENCODING): THE QUANTIFIER DOES NOT COMPRESS, and the reason is now a measured number; both surprise channels carried as live gates and EMPTY; (4) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The frontier extension, the code reading, and the campaign audited: the corrections of record” plus the corrected 12.45/12.46 wordings, the 12.45 mesh-ladder correction, and the OCT vocabulary fence with the five unpriced classes; the end-form article gains §8.15 with the form  $\times$  direction status table, the corrected §8.13 close and §8.14 fences, the abstract status sentence, new limitation 11, the limitation-8 fence, and the traceability rounds 106–109 (with the Hasanalizade–Shen–Wong citation); the Anthropic email draft gains one status paragraph (the corrected three-forms state, the frontier law, the code-reading result); the website prime-front page gains chapter 39 (the live-updates chapter renumbered  $39 \rightarrow 40$ ), the corrected chapters 37/38, the chapter-33 vocabulary fence, and one consolidated sandbox feed entry (verdict BIGPICTURE-FINDINGS, new enum, script `bigpicture_logic_probe.py`); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-15 (CI) (the roadmap — discovery freeze rounds one-hundred-three through one-hundred-five synced into every live surface: the campaign’s complete machine-checked map compiled into the one artifact it never produced — a staged, milestone-gated research roadmap from today’s frozen state to a complete RH proof, every stage falsifiable against the corpus’s own instruments — fed by the largest proven positivity region of the campaign and the final classical pricing; experiments/ only — THREE frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or

narrowed, suite unchanged at 906, NO RH claim in any direction; the honesty fences travel first: A ROADMAP IS NOT PROGRESS ON RH, NO STAGE IS CLAIMED STARTED, AND THE MATHEMATICS OF STAGES 1–2 DOES NOT YET EXIST: (1) ROUND 103 (moonshot\_sol\_probe.py, verdict MOONSHOT-PARTIAL(Form A)): from cited external inputs only — Platt–Trudgian 2021 (all zeros to  $T = 3,000,175,332,800$  on the line), Rosser 1941, the classical  $\gamma_1$  isolation; no zero table, no zeta evaluator — THE LARGEST PROVEN POSITIVITY REGION OF THE CAMPAIGN:  $C_{n,k}(256) > 0$  for EVERY  $n \geq 0$  and every  $0 \leq k \leq 10^{19}$  (an infinite half-strip in  $n$ , ten quintillion complete  $k$ -columns; the scope inside the theorem: the  $k \rightarrow \infty$  quantifier is touched nowhere, nothing about RH is proven); the three suggested proof routes die with named mechanisms — the exact prime-side transmutation turns the positive Beta spectral window into a sign-changing generalized-Laguerre source window (the classical Laguerre zero theorem: no termwise-positive prime comparison exists), the Killip–Simon/Szegő direction dies on pure-point entropy ( $w = 0$  a.e., entropy  $-\infty$ , floor  $e \geq 0$  only), the asymptotic- $a$  transport is nonuniform in  $(n, k)$ ; (2) ROUND 104 (note CDV, contract PRIME.LEVINSON.CLASS.01, levinson\_class\_probe.py, 23/23 gates, SPEC 65ac12a9, run-of-record levinson\_class\_probe.run.log, verdict LEVINSON-ADJUDICATED(LEVINSON-FAIL; tally  $24 \rightarrow 25$ , 0 PASS)): the LAST formally unexamined classical theorem type about zero positions — positive proportion on the line (Selberg 1942; Levinson 1974  $\kappa \geq 1/3$ ; Conrey 1989  $> 2/5$ ; the standing record  $\kappa > 0.417293$ , Pratt–Robles–Zaharescu–Zeindler 2020) — priced against the wall budget: the literal (liminf) class constrains NOTHING at any finite height (exact finite-modification argument), the granted-effective reading breaches every deployed cell by 2–3 orders ( $382/1149/1059$  at  $h = 184/388/839$ ), and a proportion theorem must come within  $1 - \kappa^* < 4.8 \times 10^{-3}$  of ALL zeros ( $\kappa_{\text{ledger}}^* = 0.9952/0.9983/0.9988$ , each a LOWER bound) before it can feed the wall — with this round every named unconditional classical theorem type about zero positions is formally priced (zero-free regions, density, Weyl equidistribution, pair correlation, positive proportion); (3) ROUND 105 (note CDVI, contract PRIME.ROADMAP.01, roadmap\_probe.py, 29/29 gates, SPEC 79092ebf, deterministic re-run identical, run-of-record roadmap\_probe.run1.log, verdict ROADMAP-COMPILED(stages: S0 DONE; S1a|S1b OPEN; S2 OPEN; S3 CONDITIONAL-KERNEL-CHECKED; gates 29/29)): every machine-groundable claim asserted — acceptance-gate values re-computed from cited constants or grep-warded verbatim in their owning frozen artifacts, the stage DAG acyclic with exactly the named open stages, the downstream chain’s artifacts present with verdicts grep-verified; STAGE 0 (done, frozen): the atlas complete at 25 candidates 0 PASS, exactly one open edge E4, three equivalent lemma forms, two certified falsification instruments with open-and-empty channels (Euler–Pick floors  $N = 1$ :  $[4.5917135 \times 10^{-2}, 4.5917136 \times 10^{-2}]$  down to  $N = 4$ :  $[8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  at cap  $10^{13}$ ; Hausdorff certified depth 86, 5290 cells,  $\gamma^* 189/563$ ), the  $k \leq 10^{19}$  theorem re-derived inside the probe (decisive ratios  $1.176577/335.8/1416.8$ ; the  $n = 0$  mechanism wall re-located between  $1.40 \times 10^{19}$  and  $1.41 \times 10^{19}$ ), the rigidity law  $\ln \eta^*(r) = -1.073r - 7.08$ , the seven-property specification sheet, 27 kernel-checked theorems; STAGE 1 (OPEN — new mathematics): structural positivity of Hodge-index type for  $\text{Spec } \mathbb{Z}$  — verified literature state: Deninger’s dynamical Lefschetz trace formula PROVED for smooth foliated flows (Álvarez López–Kordyukov, arXiv:2402.06671) but NO arithmetic host constructed; Connes–Consani Riemann–Roch on the scaling-site orbits and for  $\mathbb{Z}$  but NOT on the square of the scaling site; milestone (1a) an arithmetic dynamical



system hosting the proven trace formula (periodic orbits  $\log p$ , weights  $\log p/p^{m/2}$ ), acceptance gate: reproduce the (AC)/Euler–Pick interface  $c^T P_N c = \langle -g'', f_c \star \tilde{f}_c \rangle$  at the 16 frozen  $\sigma$  of the SVPIN spec (minutes-scale, by machine); milestone (1b) OR Riemann–Roch on the square of the scaling site, acceptance gate: the intersection pairing on exponential autocorrelations must reproduce the certified  $\mathcal{P}_N$  ladder ( $N \leq 4$  floors as frozen acceptance values); STAGE 2 (OPEN — new mathematics): the Hodge-index/Castelnuovo analog on the Stage-1 substrate, acceptance gates = the corpus’s measured consumption requirements: FAIL the surrogate worlds exactly at the measured depths ( $r = 0.264$  smooth by support,  $r = 0.744$  scramble by weights), be SIGNED and ALIGNMENT-CARRYING (the v913 class), SURVIVE the  $\tau$ -screen (bands 0.30/0.70), and reproduce the certified Hausdorff cells (depth 86) while STRICTLY EXTENDING the  $k \leq 10^{19}$  theorem’s reach; STAGE 3 (CONDITIONAL-COMPLETE — assembly, already machine-held): from the Stage-2 inequality to localized Weil positivity in ANY ONE of the three lemma forms, then the machine-held extraction — v848 (EXTRACTION-CHAIN-COMPLETE, hypothesis (H) isolated), v912 (unconditional rate  $O(D^2 \log(1/D))$ ), the Lean kernel tier (CofinalWeil/CofinalPredefinition, pickMatrix\_posSemidef, sv\_implies\_rh with the honesty lock skeleton\_not\_unconditional), the SVPIN skeleton (SVPIN-ROUTE-OPEN), the Hausdorff equivalence (HAUSDORFF-EQUIVALENCE-SOUND), the CCM same-currency interface (CCM-SAME-CURRENCY); THE FALSIFICATION SPINE (parallel, permanent): the two certified instruments keep running as standing disproof channels, both open and EMPTY (a single certified negative would be an RH-disproof candidate); the extension walls are KNOWLEDGE walls, not compute walls (Euler–Pick  $N = 5$  needs  $\psi$ -knowledge beyond  $10^{19}$ , floor/ $\lambda_5 = 4.7 \times 10^5$  at any cap; the Form-A  $n = 0$  mechanism exhausts at  $k \approx 1.405 \times 10^{19}$ ); THE HONEST PRICING: Stages 1–2 require mathematics that does not exist today — the roadmap proves interfaces and obstructions, never existence; Stage 3 is assembly; the corpus’s contribution to every stage is the same — acceptance gates that turn each milestone into a minutes-scale machine check, so any candidate, from anyone, can be adjudicated immediately; NO timeline promised; (4) SURFACES SYNCED IN THIS ENTRY: tfpt\_prime\_front gains the sandbox diary subsection “The road from here: a machine-gated path to a complete proof”; the end-form article gains §8.14 with the abstract status sentence and the traceability rounds 103–105; the Anthropic email draft gains one status paragraph (the roadmap as the collaboration offer — each milestone has a minutes-scale acceptance test); the website prime-front page gains chapter 38 (the live-updates chapter renumbered 38  $\rightarrow$  39) and one consolidated sandbox feed entry (verdict ROADMAP-COMPILED, new enum, script roadmap\_probe.py); website/lib/changelog.ts regenerated via bash build.sh gen; NO RH claim anywhere)

2026-08-15 (C) (the instrument tier — discovery freeze rounds ninety-three and ninety-five through one-hundred-two synced into every live surface, INCLUDING one flagged correction of record to already-published text: the collective-rescue explanation, the certified Euler–Pick ladder and its  $N = 4$  extension, the rigidity law, the SP pricing round that falsified the operative round-88 predictor extension, the CCM adjudication, the Lean kernel round, the  $\alpha$ -induction depth record, and the Hausdorff safe-point instrument — the one open input now held in THREE equivalent lemma forms with TWO certified falsification instruments; experiments/ only — EIGHT frozen probes plus a Lean formalization round, several run as concurrent lanes, NO promotion, NO module, NO ledger row,

NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) THE CORRECTION OF RECORD (round 97, note CCCXC VIII, `sp_expsum_pricing_probe.py`, 13/13 gates, deterministic 0.8 s twice, SPEC 55207ff7, verdict SP-OPERATIVE-FALSIFIED + SP-PRICED + SP-CLASSICALLY-OUT-OF-REACH( $\times 8.2$ ) + SP-CONDITIONAL-LADDER + SP-PREFIX-DEAD-IN-WINDOW + SP-LITERAL-NONVANISHING-IN-RANGE; carried per the round-87 convention — frozen probes and notes are NOT retro-edited, the correction is the record): the round-88 rounding-margin statement priced against the full classical exponential-sum arsenal AND its operative form MEASURED FALSE beyond  $x = 13$  — census extended from 79 points ( $x \leq 13$ ) to 18798 points ( $x \leq 890$ , cache edge  $\gamma_{7000} = 7264.75$ ), the operative (SP) FALSE from  $x = 21$  onward with 567  $\tau$ -screened counterexamples, THE ROUND-88 EXACTNESS A SMALL- $x$  ACCIDENT OF THE MARGIN BUDGET, mechanism ARITHMETIC-FREE (Fejér tracking-variance floor  $\text{rms} \rightarrow 1/(2\pi) = 0.159$ , measured 0.26, vs budget 0.5 — no taper in the family survives); pricing (all constants generous, deepest rung  $x = 890$ ): best unconditional miss  $\times 8.2$  (Trudgian  $S(t)$  ceiling 4.11), even RH itself (Carneiro–Chandee–Milinovich) 1.015 at the deepest rung; prefix/exceptional-set salvage DEAD IN-WINDOW (lowest violation  $k/K = 0.010$ , prefix depth stalls  $\leq 131$ ); HONEST FENCE: the deployed source-side approximant died, NOT TPL(i) — the eigenvector truth side unmeasured beyond  $x = 13$ , the published round-88 claim (exact at 79 points,  $x \leq 13$ ) remains factually true; the corrective clause added to the published §8.10 wording, §8.12 property 1, the prime-front round-88 subsection, the website chapter 34 fence, and a new paper limitation 10; (2) ROUND 93 (note CCCXC IV, `collective_rescue_probe.py`, 10/10 wards + 7/7 law gates, verdict COLLECTIVE-IS-SUPPORT): round 92’s exit/re-entry FULLY EXPLAINED — exact causal onset bound (bit-identical prefix, spread exactly 0) plus measured offset  $\sim 2$  lags, completed-stream  $\max |\alpha|$  CONSTANT at 0.183932 ( $r = 2.568\text{--}4.788$ , band maxima falling), all Lyapunov candidates killed (the fixed-prefix class EMPTY by causality), mass-restored control dies at the IDENTICAL lag 400 — mass-without-atoms does not rescue, in-window survival requires the exact arithmetic weights; (3) ROUND 95 (note CCCXC VI, `eulerpick_ladder_probe.py`, 31/31 gates, verdict EULERPICK-CERTIFIED( $N \leq 3$ ) + EULERPICK-GENERIC-DECAY + EULERPICK-DETECTION-LAW): the falsification ladder rebuilt to CERTIFIED standard (outward-rounded intervals end to end,  $N = 3$  certified positive at cap  $1.6 \times 10^7$ :  $[2.3643695 \times 10^{-10}, 1.1497752 \times 10^{-9}]$ ), the channel self-tested on a planted negative family and EMPTY; decay GENERIC (control decomposition within  $1.2\times$  of Euler, the true  $\gamma_1$  appearing nowhere); detection law  $N^*(\delta) = 6.1 + 1.95 \log_{10}(1/\delta)$ ,  $\gamma$ -BLIND beyond  $\gamma \approx 50$  (undetected through  $N = 36$ ); promotion honestly declined; (4) ROUND 96 (note CCCXC VII, `rigidity_inverse_probe.py`, 9/9 wards, verdict RIGIDITY-TIGHTENING): screw positivity to depth  $r$  pins the measure onto the prime comb exponentially tightly — ALL 17 adversarial impostors die (shifted-prime comb, golden quasi-prime comb at exact prime density, crystals, mass-matched tails, sub-mesh splits at every scale, jitters, a deep weight swap; all 15 single-atom deletions fatal within  $J \leq 8$  lags), moduli law  $\ln \eta^*(r) = -1.073r - 7.08$  with  $c_{\text{mod}} = 1.073$  vs  $c_\lambda = 0.814$  — CONDITIONING-PRICED up to an  $O(1)$  alignment factor, the theorem candidate typed (anchors measured, law fitted, all-depth OPEN); (5) ROUND 98 (note CCCXC IX, contract on arXiv:2310.18423, `ccm_pn_adjudication_probe.py`, 22/22 gates, verdict CCM-CHECKED + CCM-SAME-CURRENCY): the one external published RH reduction adjudicated

— CCM’s published local terms reproduce  $\xi'/\xi(1/2 + \sigma)$  to rel  $2.3 \times 10^{-50}$  and their quadratic form reproduces the corpus EULER–PICK MATRICES ENTRYWISE ( $\tau$ -screen: RELOCATION); NEW FINITE INSTRUMENT: all 6 prime-bearing Galerkin sections of  $Q_n$  positive ( $\lambda_{\min} = 2.30 \times 10^{-3}$  at  $n = 2.5$  down to  $1.01 \times 10^{-5}$  at  $n = 13$ ; pole constraint load-bearing); controls die through the same construction —  $P(n)$  IS localized Weil positivity, any semilocal prolate candidate finitely testable against the (AC) interface to-day; (6) ROUND 99 (note CD, the Lean round: TftptCarrier/EulerPick.lean, DeltaOneNoGo.lean, ParityLemma.lean, SVSkeleton.lean; 27 kernel-checked headline theorems, axioms exactly [propext, Classical.choice, Quot.sound], no sorry, no native\_decide): Euler–Pick forward PSD at the SV nodes for every  $N$ , the  $\delta_1$  no-go counterexample package, the FULL Parity Lemma under ONE named simplicity hypothesis (alternating signs at canonical gap midpoints — exactly where zero simplicity enters), the SV skeleton with THE HONESTY LOCK skeleton\_not\_unconditional (an instance with RH = False — packaging alone yields nothing); (7) ROUND 100 (note CDI, sieve4\_eulerpick\_n4\_probe.py, 22/22 gates, 2682s, cap  $X = 10^{13}$ , helper source SHA pinned, run-of-record sieve4\_run1.log, hard external ward  $\pi(10^{13}) = 346,065,536,839$  REPRODUCED EXACTLY):  $\lambda_{\min}(\mathcal{P}_4) \in [8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  CERTIFIED POSITIVE (margin 0.749),  $N \leq 3$  floors tightened massively, the round-95 upgrade-path model CORRECTED (the beyond- $10^{19}$  tail floor was omitted; Nair 1982 —  $\psi(x) \geq (x-2)\ln 2$  — unblocks it, corrected  $X_{\text{req}}(4) = 10^{11.52}$ ), THE KNOWLEDGE WALL STARTS AT  $N = 5$  (the floor exceeds  $\lambda_5$  by  $4.7 \times 10^5$  at any cap — no sieve helps); (8) ROUND 101 (note CDII, contract PRIME.SCREW.INDUCTION.01, screwind\_induction\_probe.py, 10/10 wards + 5/8 law gates — the 3 fails are FINDINGS, typed; companion screwind\_meshdrift\_check.py; verdict SCREWIND-KICK-LAW(exact) + SCREWIND-STEP-IS-WALL + SCREWIND-DICHOTOMY(descriptive-only)): DEPTH RECORD — all sections positive to  $r = 9.0$  at BOTH meshes (corpus record 4.788),  $\sup |\rho| = 0.183932$  attained at  $k = 0$ , flatness carried by a cancellation of giants ( $\pm 288.5$  vs net RMS 0.023), THE EXACT KICK LAW at atom arrival (affine-exact, dev  $\leq 2.8 \times 10^{-16}$ , numerator pure arithmetic), G7 fails INFORMATIVELY: PRIME ARRIVALS STABILIZE (net arrival  $\alpha$  positive 33/39, atom-bin median  $|\rho|$  below background); THE WALL RELOCATES INTO THE CERTIFICATE, the named open piece — a source-certified two-sided energy floor with  $\kappa_+ < 1/2$  — IS localized positivity; (9) ROUND 102 (note CDIII, externally proposed contract PRIME.HAUSDORFF.SAFEPOINT.01, hausdorff\_safepoint\_probe.py, 40/40 gates, 143.6s, SPEC dacdfte2, composite verdict HAUSDORFF-EQUIVALENCE-SOUND + HAUSDORFF-REPRODUCED + HAUSDORFF-CERTIFIED-DEPTH(86;  $\gamma^*$  189/563) + HAUSDORFF-REPLICATION + HAUSDORFF-DETECTOR + HAUSDORFF-Z1-ZEROMEASURE + NOT-WORLD-NEW): the moment-field criterion ( $\text{RH} \Leftrightarrow$  all Pascal cells  $C_{n,k}(a) \geq 0$  at the safe point  $a = 256$ ) built, audited, adjudicated — equivalence skeleton SOUND (Hausdorff/Widder needs no extra boundedness, the one sharp continuation step closes, companions symbolically exact), the proposer’s spec numerically faithful (561/561 cells, weakest ratio 1.000220), THE HEADLINE: CERTIFIED DEPTH 86 (5290 cells),  $\gamma^*$ -reach 189 at  $a = 256$  and 563 on the  $a = 1024$  arm vs Euler–Pick’s certified  $N = 3$  ( $\gamma \sim 5\text{--}8$ ) — THE CHEAPEST CERTIFIED RH-CONSEQUENCE INSTRUMENT IN THE CORPUS; three-carrier replication (signpos/Krein/SV pins, no zero cache anywhere in construction); the detector honestly priced ( $\sim 1/\delta$

below  $\gamma_1$ , provably BLIND above — Epstein control concrete: off-line witness LOCATED at  $\rho = 0.69692705 + 36.37406369i$ , RH-for- $Q$  provably false, the  $Q$  Pascal field positive to depth 100 exactly as the phase model predicts); Z1 typing source-only in INPUT, zero measure in CONTENT; NOT-WORLD-NEW (Zhang arXiv:2303.09396 Thm 2 is the identical criterion at the central point — the novelty is the safe-point shift, the source-only decoder, the certified conditioning); THE NAKED REMAINING TASK: positivity of ALL  $C_{n,k}(a)$  — Weil positivity in Hausdorff currency, THE THIRD EQUIVALENT LEMMA FORM beside the standing two; (10) THE CONSOLIDATED CLOSE, carried on every surface: the campaign now holds THREE equivalent lemma forms of the one open input (uniform Weyl-disk contraction; ground-eigenvector sign positions; Hausdorff cell positivity), three agreeing source-only carriers, two certified falsification instruments (Euler–Pick  $N \leq 4$ ; Hausdorff depth 86), the rigidity law, and 27 kernel-checked theorems — and at the bottom, unchanged, localized Weil positivity; (11) SURFACES SYNCED IN THIS ENTRY: tfpt\_prime\_front gains the sandbox diary subsection “The three lemma forms, the certified instruments, and a correction of record” plus the corrective clauses in the round-88 subsection and the specification-sheet P1; the end-form article gains §8.13 with the abstract status sentence, the §8.10/§8.12 corrections, limitation 10 and the traceability rounds CCCXCIV and CCCXCVI–CDIII; the Anthropic email draft gains one status paragraph (the three-lemma-form state, the two certified instruments, the CCM interface point); the website prime-front page gains chapter 37 (the live-updates chapter renumbered  $37 \rightarrow 38$ ), the chapter-34 correction pointer, and one consolidated sandbox feed entry (verdict HAUSDORFF-CERTIFIED-DEPTH); website/lib/changelog.ts regenerated via bash build.sh gen; NO RH claim anywhere)

2026-08-15 (XCIX) (the cohomological specification sheet — discovery freeze rounds ninety-one, ninety-two and the specification round synced into every live surface: two honest negatives, the Verblunsky-invariance kill with the Euler–Pick criterion and the (AC) identity pinned, and the campaign’s machine-checked characterization of the missing RH input compiled into a literature-facing requirements document for the cohomological programs; experiments/only — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 91 (note CCCXCII, the patterns round, run in a parallel session): the program’s first magnetar giant-flare QPO bed (MAG.QPO.LADDER.01, magnetar\_qpo\_ladder\_probe.py, 8/8 gates, SPEC 4ee58b0f) tests SGR 1806–20 and SGR 1900+14 against the geometric (3/2) relaxation ladder with a 20000-draw log-uniform MC null, a 17-base look-elsewhere battery, a power gate (planted ladder  $p = 5 \times 10^{-5}$ ) and a specificity gate (integer harmonics  $p = 0.078$ ): base 1.5 gives  $p = 0.3163/0.3558$ , Fisher-joint  $p = 0.3584$  — VERDICT NULL, S15-typed (placebo base 1.80 best in both sources, documented as an LEE-loaded base-battery artifact candidate, NOT escalated; SGR 0526–66 DATA-LIMITED); the LEE-controlled unclaimed-observable scan (PHYS.UNCLAIMED.DECODER.01, unclaimed\_observable\_scan\_probe.py, 5/5 after disclosed amendment A1 — gate G02 mis-specified as a minimality bar, amended to a membership bar, no verdict bar moved) returns SCAN-UNDECIDABLE with the structural finding as the content: the frozen compiler vocabulary is SATURATED at  $2\sigma$  tolerance (median placebo  $C_{\min} 3.66 \leq \text{control } C_{\max} 5.46$ ) — new predictions provably cannot come from

numeric scanning and must be derived theory-side; (2) **ROUND 92** (note CCCXCI, contract PRIME.SCREW.VERBLUNSKY.INVARIANT.01, `vbk_invariant_probe.py`, 23/23 gates, deterministic 14.1/14.0 s, SPEC c730781e, verdict VERBLUNSKY-NO-INVARIANT(mechanism exhibited)): the  $\sigma$ -correction (pointwise Weyl-disk contraction on  $\sigma_r = 1 + 1/r$  accumulating at the interior point 1 suffices;  $\sigma \downarrow 1/2$  uniformity NOT load-bearing), the exact  $\delta_1$  no-go (pin convergence cannot prove section positivity backwards), **THE EULER–PICK CRITERION** audited and pinned ( $\text{RH} \Leftrightarrow P_N = [(P(\sigma_j) + P(\sigma_k))/(\sigma_j + \sigma_k)] \succeq 0$  for all  $N$  with  $P(z) = \xi'/\xi(1/2 + z)$ ; source-only falsification ladder from  $\Gamma$ ,  $\pi$ ,  $n^{-s}$  and an own  $\Lambda$ -sieve:  $\lambda_{\min}(P_N) = 4.59 \times 10^{-2}$  down to  $2.44 \times 10^{-62}$  for  $N = 1..12$ , all positive,  $\kappa = 2.27 \times 10^{61}$  at  $N = 12$ ) and **THE (AC) IDENTITY**  $c^T P_N c = \langle -g'', f_c \star \tilde{f}_c \rangle$  gated symbolically exactly and numerically against the corpus-exact screw  $g$  ( $8.581810362 \times 10^{-2}$  vs  $8.578893345 \times 10^{-2}$ , rel  $3.4 \times 10^{-4}$ ) — Euler–Pick positivity IS Weil positivity on exponential autocorrelations, by identity; the kill: complete prime packets are high-rank INDEFINITE Toeplitz updates of displacement rank two (ranks 26/90/130, inertias (13, 102, 13)/(45, 70, 45)/(65, 50, 65)), the induced  $\alpha$  step nonlinear and prior-state dependent ( $\times 316$ ), the TRUE packet stream itself EXITS the Schur disk after  $p = 2/3/5/7$  ( $r = 1.122/1.620/1.950/2.400$ ) and re-enters only collectively (after  $p = 11$ : 0.313; full: 0.184); control wards reproduce SMOOTH  $r = 0.264$  (diagnosed as a SUPPORT error — zero true prime mass before  $\log 2$ , PNT mass 0.282) and SCRARITH  $r = 0.744$  (weight  $\times 1.451$  at kept positions); the only exact survival condition is WALL-EQUIVALENT — clean kill, the remaining all-depth principle is Weil positivity in yet another currency; (3) **THE SPECIFICATION ROUND** (note CCCXCV; the intervening numeral CCCXCIV belongs to a concurrent discovery lane and is not part of this sync; contract PRIME.COHOMSPEC.SHEET.01, `cohomspec_probe.py`, 20/20 gates, 8.8 s, SPEC 423d934e, independently re-run deterministic, verdict COHOMSPEC-COMPILED(7 properties: 6 machine-grounded, 1 citation-typed)): the campaign’s measured characterization of the missing input compiled into a REQUIREMENTS DOCUMENT for any cohomology/intersection theory for  $\text{Spec } \mathbb{Z}$  (Deninger’s program; the Connes–Consani scaling site), each property grounded by grep ward or independent recomputation in its owning frozen artifact — (P1) the signed, alignment-carrying, unconditional, sub-spacing ordinate-position input of the one open edge E4 (v913, kill-atlas DAG, the 24/0/19 tally,  $\sim 1\%$  sub-spacing; sharpest corpus forms:  $D_{\max} = x - 3/8$  at  $e^{-11}$  per forced skip, min margin 0.0317 / load 4); (P2) shape-independence (deployed exponent  $-3.379$  recomputed by OLS as  $-3.3795$ ; alternatives shallower by  $\geq 2.2$ ); (P3) not magnitude/density/existence (Littlewood 214.7/1527.1; zero-density empty for  $A > 2$ ; border membership  $\det M = s \det B$  recomputed in exact Fractions); (P4) the Gram form on exponential autocorrelations must BE the Euler–Pick matrices — the concrete interface any cohomological trace formula must match, recomputed (ladder rel dev  $\leq 3.8 \times 10^{-9}$ , (AC) weld rel devs  $\leq 4.3 \times 10^{-11}$ ), testable at any finite  $N$  by machine today; (P5) positivity must fail on the control worlds exactly as measured ( $r = 0.264$  by support,  $r = 0.744$  by weights, both recomputed); (P6) survive the  $\tau$ -screen (bands 0.30/0.70, fifteen DISGUISE routes); (P7, citation-typed) the function-field precedent mapped to its break points, verified against the live literature — Deninger’s dynamical Lefschetz trace formula proved for smooth foliated flows (Álvarez López–Kordyukov, arXiv:2402.06671) with no arithmetic host constructed (Doc. Math. ICM 1998, arXiv:math/0505354); Connes–Consani Riemann–Roch on the scaling-site orbits  $C_p$  (Selecta Math. 2017) but not on the square; the semilocal/prolate line (arXiv:2006.13771, arXiv:2310.18423) and

Connes' 2026 extremal letter (arXiv:2602.04022, zero reality by Connes–van Suijlekom Thm 6.1, convergence open = the corpus's REALROOT-DEPTH-UNDECIDED); (4) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “What any cohomology for  $\text{Spec } \mathbb{Z}$  must supply: a machine-grounded specification”; the end-form article gains §8.12 with the abstract status sentence and the traceability rounds CCCXCII, CCCXCIII and CCCXCV with the verified literature list; the Anthropic email draft gains one status paragraph (the collaborator point: a testable interface where there was an analogy); the website prime-front page gains chapter 36 (the live-updates chapter renumbered  $36 \rightarrow 37$ ) and one consolidated sandbox feed entry (verdict COHOMSPEC-COMPILED); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; HONESTY FENCE carried on every surface: the sheet is a requirements document derived from machine-checked no-gos — it proves nothing about Deninger's program or Connes–Consani, claims no progress on RH, and whether any theory can meet the specification is OPEN; NO RH claim anywhere)

2026-08-14 (XCVIII) (the Stieltjes–Vitali pin route and the Krein screw-function carrier — discovery freeze rounds eighty-nine and ninety synced into every live surface: the externally proposed countable-pin reduction of the real-root architecture, and the construction of its one priced missing ingredient — the Krein inverse-spectral realization of Suzuki's screw function, the first pin carrier of the campaign to pass the Z1 screen; experiments/ only — TWO frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 89 (note CC-CLXXXIX, externally proposed contract PRIME.STIELTJES.VITALI.PIN.01, `stieltjes_vitali_pin_probe.py`, 17/17 gates, 249 s, SPEC `f450832f`, independently re-run bit-identical, verdict SVPIN-ROUTE-OPEN): replace the full semilocal trace convergence by countably many positive resolvent scalars in the absolutely convergent Euler half-plane — the pins  $P_x(\sigma) = \sum_k 2\sigma/(\lambda_{x,k}^2 + \sigma^2)$  must converge to  $\xi'/\xi(1/2 + \sigma)$  at  $\sigma_r = 1 + 1/r$  (source-only right side via  $\Gamma$ ,  $\pi$ ,  $\Lambda(n)$ ,  $n^{-s}$ ); then Herglotz normal-family compactness (ONE pin suffices), Vitali on the pins accumulating at the interior point  $i$ , the identity theorem on  $\Im z > 1/2$ , analytic continuation on the connected  $\mathbb{C}_+ \setminus Z(\Xi)$  and the pole contradiction give (SV)  $\Rightarrow$  RH, the functional equation closing  $\beta < 1/2$ ; THE SKELETON IS SOUND gate by gate: weight-comparability constant 65.01 (bar  $2 \times 10^3$ ) executed per rung (normal-family ratio 0.033 on all four), the sign-convention ambiguity HARMLESS for the pins ( $\Xi(z) = \xi(1/2 - iz)$  vs  $\xi(1/2 + iz)$  give the identical target via  $\xi'/\xi(1 - s) = -\xi'/\xi(s)$ ; numeric lock at  $\sigma = 1$ :  $4.61359285 \times 10^{-2}$  vs  $4.61359517 \times 10^{-2}$ , dev  $2.3 \times 10^{-8}$ ), spectrum honesty MEASURED not assumed (census 4/10/20/41 all real, no zero mode, min spacing  $\geq 1.3$ ), the degenerate Herglotz limit excluded by the pin bound; the ONE named condition: all- $x$  realness is the CF simplicity-plus-evenness hypothesis; the named non-gap: THE CHAIN MOVES THE ENTIRE BURDEN INTO (SV) ITSELF — no skeleton step is the wall, the wall is the premise; tail bookkeeping explicit, all three tails decreasing (true zero tail  $2.85 \times 10^{-2} \rightarrow 1.18 \times 10^{-2}$  within 2% of the  $\log T/T$  model; measured Nyquist-excess edge mass  $\leq 1.31 \times 10^{-3}$ , the window definition corrected in the spec after a smoke caught band spill; exact lattice tail  $1.35 \times 10^{-2} \rightarrow 8.0 \times 10^{-3}$ ); THE FINITE PRE-CHECK CONVERGES ON ALL 16 FROZEN  $\sigma$  VALUES ( $x = 3/5/8/13$ ;  $x = 21$  skipped and declared by the arbiter cost law):  $\Delta$  negative everywhere (missing tail dominates), slopes  $-0.62 \dots -0.65$ , measured drop 2.6 vs the a-priori RvM model 2.4 (bar 1.6), NO  $\sigma \downarrow 1/2$  rate degradation on this ladder

(the expected boundary effect beyond reachable depth, stated honestly), the FULL divisor row converging FASTER (the lattice comb compensates  $\sim 2/3$  of the missing tail); CONTROLS ALL SEPARATE: mesh-CF MESH-POISONED through the same pins (median overshoot  $0.94 \rightarrow 6.32$  growing like  $\log x$  — the wrong limit, measured), SCRPOS/EPSTEIN/SMOOTH/SCRARITH separate at medians  $2.7 \times 10^5$ – $8.8 \times 10^5$  (band metric) and 12.4–45.9 (target metric) on 16/16 rows (SCRARITH throws one imaginary pair — the world breaks the SV realness hypothesis, reported), Christoffel weights structurally excluded by AST,  $\tau$ -screen 0/16 relocations at max slope 0.009 over  $\sim 47$  decades — NO DISGUISE; THE HONEST WEIGHT, Z1-typed: on the extremal family the measured pin convergence IS the transcription of cache partial sums (band pins vs cache rel  $7.7 \times 10^{-12}$  /  $9.4 \times 10^{-12}$  /  $2.1 \times 10^{-5}$ ) — instrument consistency, never source-side content; the naive Suzuki spectral readout is DEAD (SUZPIN-FORM-SPECTRUM: hat-compression eigenvalues are a form-spectrum ladder  $\sqrt{\mu} = 1.02/2.12/3.17 \dots$ , not the ordinates,  $\gamma_1 = 14.13$ ;  $P_{\text{suz}}(1)$  grows  $1.85 \rightarrow 2.47$  vs target 0.046) — the SECOND architecture of the campaign to survive every gate, the corpus’s logically sharpest open target, its priced remaining task exactly one: the Krein inverse-spectral realization of Suzuki’s screw function; three pre-freeze smoke fixes disclosed, zero post-freeze amendments; (2) ROUND 90 (note CCCXC, contract PRIME.SUZUKI.KREIN.REALIZATION.01, krein\_screw\_realization\_probe.py, 22/22 gates, 18 s, SPEC bd5fba32, independently re-run bit-identical, verdict SUZKREIN-CARRIER-OPEN(lemma: uniform Weyl-disk contraction)): the priced missing ingredient BUILT — THE CORRESPONDENCE:  $g$  extracted corpus-exactly (Suzuki eq. 1.3 with the v643 convention lock; the Lerch coefficient  $+1/4$  GATED against the origin expansion  $g \sim \frac{1}{2}t \log t + At$  with  $A = \frac{1}{2}(\log 2\pi - \psi(2)) = 0.707546$  — a  $-1$  misreading would fail by  $O(1)$ ), the screw property verified ( $G(t, s)$  PSD on all four windows,  $\lambda_{\min}/\lambda_{\max} \geq -8.1 \times 10^{-19}$ ), Krein’s accelerant/continual-Schur correspondence invoked with its hypothesis CERTIFIED (strict positivity of every Toeplitz section of  $-g''$ , certified by the completed Szegő run,  $\max |\alpha| = 0.156$ ), the normalization DERIVED not fitted (lag reads are exact second differences of  $g$ , two independent routes agreeing to  $1.8 \times 10^{-10}$ ; partition of unity gives the pin as  $\frac{1}{2}\langle -g'', e^{-\sigma|\cdot|} \rangle$ ; the Weil formula identifies  $\xi'/\xi(1/2 + \sigma)$  layer by layer, archimedean/pole/prime each gated); THE CONSTRUCTION: Levinson/Szegő IS the Toeplitz-structured discrete Krein/Gelfand–Levitan solve, the Verblunsky  $\alpha$  are the realized Hamiltonian data and CAUSAL (exact Toeplitz nesting — deeper windows only append), and THE REALIZED HAMILTONIAN SEES THE PRIMES:  $|\alpha|$ -spikes at  $r = \log q$  for  $q = 2/3/4/5/7$ , offset  $\leq 1$  bin, contrast 2.8–5.2; conditioning TRIVIAL (amplification  $\sim 23$  under  $10^{-25}$  perturbations — the feared  $e^{cx}$  explosion belongs to the trig-family spectral-resolution ladder, not the screw-kernel Toeplitz sections); the  $\delta$ -rough pointwise profile declared honestly, the  $\delta$ -stable objects (pins, disks, spike positions) gated; THE CONVERGENCE: all 16 frozen  $\sigma$  rows (verbatim from the SVPIN spec) CONVERGE, 16/16 truth-tight ( $\sigma = 0.6$ :  $\Delta = +7.30 \times 10^{-5} \rightarrow +6.33 \times 10^{-7}$  over  $L = \log 3 \dots \log 13$ , factor 115), Weyl-disk contraction MEASURED  $R \sim e^{-(\sigma+c)L}$  with  $c = 0.85 \dots 1.76$  median 1.36 (limit-point behaviour), and THE SURPRISE: the disk center RESUMS THE MISSING TAIL ( $|\hat{P} - S_{\text{trunc}}|/\text{tail-model median } 1.000$ ; at  $\sigma = 0.6$  the center beats the truncated source sum by a factor  $\sim 600$ ); THE CONTROLS DIE AT THE POSITIVITY LEVEL ITSELF — the strongest separation of the campaign: the prime-free SMOOTH accelerant LOSES SCREW POSITIVITY at window depth  $r = 0.264$  (same  $r$  at both mesh sizes — a continuum property), the golden-scrambled



SCRARITH world at  $r = 0.744$  — THE TRUE PRIMES (positions AND weights) ARE NECESSARY FOR SCREW POSITIVITY at these windows; THE Z1 SCREEN PASSED FOR THE FIRST TIME: no cache transcription (min over all  $N \leq 7000$  of the max- $\sigma$  rel dev =  $8.5 \times 10^{-3}$  vs bar  $10^{-6}$ ; AST firewall zero-free; SUZKREIN-EXTRAPOLATES(1.000)), explicitly distinct from DEBRANGES-COMB-BLIND in input (screw lag data, not wall-block moments), direction (inverse-spectral construction, not HB re-coordinatization) and adjudication (quantitative convergence + world controls, not structure existence), the dead SUZPIN form-spectrum readout never used; THE REPRICED LEMMA AND THE HONEST PRICE: the literal SVPIN trace-class norm is MEASURED FLAT (16/16 non-decaying; appended Verblunsky blocks grow — prime atoms arrive forever) — the correct currency is WEYL-DISK CONTRACTION, the minimal missing lemma stated exactly (“let  $g$  be Suzuki’s screw function with every finite section of  $-g''$  positive (= localized Weil positivity); then the Weyl disks  $D_a(i\sigma)$  of the Krein realization contract with  $R_a(\sigma) \leq C(\varepsilon)e^{-(\sigma+c)\cdot 2a}$  uniformly on  $\sigma \geq 1/2 + \varepsilon$ , with limit point  $\xi'/\xi(1/2 + \sigma)$ ”; suppliers: Krein’s accelerant theorem, de Branges chain monotonicity; OPEN: the uniformity as  $\sigma \downarrow 1/2$ ) — AND THE ALL- $a$  SECTION-POSITIVITY HYPOTHESIS IS LOCALIZED WEIL POSITIVITY ITSELF: the carrier localizes the remaining task and fixes the currency, it does NOT remove the positivity input; disclosures: pre-freeze float64 calibration, two smokes (six fixes incl. a normalization slip caught by the table’s own  $\Delta = -\text{tgt}$  pattern), a first-full-run abort at a circular liveness filter (repaired and disclosed in the spec, SPEC-hash change documented 1f898f95  $\rightarrow$  bd5fba32) — no bar, grid, ladder or verdict-bearing rule moved; (3) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The Stieltjes–Vitali pin route and the Krein screw-function carrier”; the end-form article gains §8.11 with the abstract status sentence and the traceability rounds CCCLXXXIX–CCCXC; the Anthropic email draft gains one status paragraph (the missing input now has TWO precisely stated lemma forms, and the Krein carrier is the first source-only host whose controls die at the positivity level); the website prime-front page gains chapter 35 (the live-updates chapter renumbered 35  $\rightarrow$  36) and one consolidated sandbox feed entry (verdict SUZKREIN-CARRIER-OPEN); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-14 (XCVII) (the real-root saga and the sign-position endpoint — discovery freeze rounds eighty-three through eighty-eight synced into every live surface: the real-root architecture and its three-lane conflict, the arbiter, the TPL round with the PARITY LEMMA and the counting law, the tested-simplicity round, the adversarial bughunt and the SIGN-POSITION PREDICTOR; experiments/ only — TEN frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction; standing convention: the round-87 bughunt CORRECTED three MAJOR misstatements in the round-83/84 records, the frozen probes and notes are NOT retro-edited — the correction is the record — and every surface synced by this entry carries only the corrected numbers: (1) ROUND 83 (note CCCLXXXII, three lanes: `semilocal_realroot_limit_probe.py` 16/16, SPEC 8eff2685, verdict REALROOT-ARCHITECTURE-OPEN; `stc_sol_convergence_probe.py` 4/4, verdict STC-OPEN; `stc_opus_convergence_probe.py` 20/20, verdict STC-DISGUISE + STC-FALSE for the mesh family): the first route of the campaign to genuinely leave the kill atlas — source-only finite SELF-ADJOINT operators (real spectra

by construction, no wall positivity, no zero data consumed), limit identified with  $\Xi$  via the full trace formula on a determining even Paley–Wiener class; skeleton audited as executable gates (determination lemma sound in two classical steps with NO convergence rate consumed; two repairable gaps repaired: an origin-exclusion minorant derivable from R3 itself, and the source-only  $E''(0)$  pin  $\sum_{\rho} 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi$ ); census fully real 41/41 at  $x = 13$ , all six trace rows CONVERGE, world controls separate at median  $1.26 \times 10^6$ – $1.40 \times 10^6$  vs the pre-frozen bar 5, honest price  $\lambda_{\min} = 2.5 \times 10^{-54}$  at  $x = 13$  (internal precision  $\sim e^{-4\pi x}$ ); CORRECTED zero tracking (F1, stronger than first recorded): first-zero deviation  $-8.4 \times 10^{-16}$  at  $x = 8$  AND 13 (exactly the cache’s own float64 quantization of  $\gamma_1$ ,  $+8.407 \times 10^{-16}$  vs the 30-digit literature value), mean displacement over all 15 matched zeros 0.0 at float64 resolution; CORRECTED  $\varepsilon$  finals (F6): up to 117% under the exact census ( $2.071 \times 10^{-8} \rightarrow 9.550 \times 10^{-9}$ ;  $3.708 \times 10^{-9} \rightarrow 6.290 \times 10^{-9}$ ; worst decay exponent  $-4.47 \rightarrow -4.96$ ), all six rows staying CONVERGES; lane two hits 14.1347/21.0220/25.0109 at  $x = 8$  source-only and proves R3 ALONE implies Weil positivity ( $h = |\hat{g}|^2$  on a real spectrum) — full RH strength; lane three PROVES the Nyquist density obstruction (spec density  $\ell/2\pi$  exact and mesh-independent, rel dev  $2.22 \times 10^{-16}$  over a  $16\times$  mesh sweep, vs  $\Xi$ ’s log density — R3 AND R4 fail for every mesh at fixed window), unweighted trace diverges 4/4 to  $x = 144$ , weighted converges (honest finals  $9.3 \times 10^{-4}/3.3 \times 10^{-6}/2.3 \times 10^{-6}/6.2 \times 10^{-4}$ ; F4) but the weights are the Christoffel function of the zero measure — the unknown; honest node-vs-zero pairing at  $x = 55$ : 18-vs-3 one-sided, 36-vs-6 two-sided (F2); the three-lane conflict deliberately frozen unresolved; (2) ROUND 84 (note CCCLXXXIII, `realroot_arbiter_probe.py`, 17/17, 933 s, SPEC fc8a3a2b, independently re-run bit-identical, verdict REALROOT-DEPTH-UNDECIDED): the conflict resolved BY MECHANISM — different operator family, not different sums, not a bug, not primarily depth; all three lanes summed honest unweighted sums against the identically normalized source side; a new deep rung  $x = 21$  built at the honest price ( $K = 80$ , 128 dps  $\geq$  the declared floor 119.6,  $\lambda_{\min} = 1.25 \times 10^{-93}$ ): the mesh-CF  $(0, 30)$  count grows  $7 \rightarrow 22$  over  $x = 5..144$  tracking the uniform Nyquist prediction (measured deviations 10.2/10.6/10.9% at  $x = 13/21/34$  vs the frozen 30% tolerance; F3), the extremal/quotient family stays EXACTLY on the  $\Xi$  census at every measured depth —  $(0, 30) = 3$ ,  $(0, 60) = 13$ ,  $(0, 100) = 29$  at  $x = 21$ , minimum zero  $= \gamma_1$  to  $10^{-4}$ , all 50 Nyquist-excess zeros exiled to  $(100, 163]$ ; one unified table: mesh DIVERGES 4/4, extremal CONVERGES 4/4 on the same  $h$ ; NOT false-for-family, NOT quotient-escapes (the thinning is MEASURED, not proven; Connes §6.6 open); the Z1-TRANSFER identity names why a proof is the wall itself: the trig-Galerkin matrix IS the Gram matrix of zero evaluations,  $M[j, k] = 2 \sum_{\gamma} B_j(\gamma) B_k(\gamma)$ ,  $\lambda_{\min} = 2 \sum_{\gamma} E(\gamma)^2$  ( $6.31 \times 10^{-5}$  after the analytic cache-tail correction, 92% model share) — source-only in INPUT, zero-measure in CONTENT; falsification rungs priced ( $x = 34$ :  $\sim 1.5$  h;  $x = 55$ :  $\sim 10$  h), NO finite rung can verify — the surviving lemma carries full RH strength; (3) ROUND 85 (note CCCLXXXIV, three lanes: `tp_sol_probe.py` 7/7 TPL-OBSTRUCTED, `tp_opus_probe.py` 32/32 TPL-REDUCED + TPL-DICHOTOMY-MEASURED + TPL-OBSTRUCTED, `tp_da_probe.py` 9/9 TPL-CIRCULAR): energy sign and zero census are LOCALLY INDEPENDENT observables (off-line quartets:  $\Delta Q = -4\delta^2 E'(t)^2 < 0$  yet Rouché preserves the count;  $\lambda_{\min}$  flips to  $-9.1 \times 10^{-17}$  at  $1.25 \delta^*$  while the  $(0, 30)$  count stays 3); the ARCH block IS the smooth zero measure exactly ( $1.1 \times 10^{-6}$ ) — the entire  $e^{-cx}$  positivity is supplied by the PRIME BLOCK ALONE, no density-input argument can prove TPL(i);  $\|\text{PRIME}\|/\text{gap} \sim 10^{4.36x}$  — any prolate comparison must

be an EXACT cancellation; THE PARITY LEMMA: TPL(i) IS — no analysis, a finite combinatorial rewriting — the statement that the ground eigenvector of one explicit matrix carries a prescribed sign pattern, verified exactly on three rungs; THE COUNTING LAW: the Nyquist surplus peaks at  $D_{\max} = x - 3/8$ , the tracking band ends exactly at the peak, marginal cost per forced skip one fixed factor  $e^{-11}$ , constant to 11% over 23 DECADES; the measured census support simultaneously CONVICTED CIRCULAR (the recorded counts ARE the smooth RvM values to  $< 1$ : 29.0023 vs 29; the instrument transcribes any fed measure including the prime-free world; outrunning classical verification needs  $\text{dps} \sim 2.1 \times 10^{12}$ ; the only off-line-sensitive channel is the wall scalar, slope 2.03, independently confirming the  $-4\delta^2 E'^2$  law); one sentence: the NUMBER of sign changes is forced by counting, the arithmetic substance sits EXCLUSIVELY in the POSITIONS, which the prime block alone must supply; (4) ROUND 86 (note CCCLXXXV, `simple_principle_probe.py`, 22/22, SPEC ef1bc6dd, verdict SIMPLE-PRINCIPLE-IS-TRANSCRIPTION): four readings tested against pre-registered bars — one-axiom STRONG-REFUTED (five normalizations, zero hits on  $\log \pi$ , gaps 0.226–1.341; the difference dictionary exact: a gauge orbit, empty as a source), transcription CONFIRMED 3/3 (the register’s Mertens head is EXACTLY  $\prod_{p|210}(1-1/p) = 48/210$ , i.e. 2.1293 bit =  $\log_2(35/8)$  — the compiler pair itself; the  $E''(0)$  pin is the Hadamard identity to  $2.4 \times 10^{-8}$ ), fixed-point REFUTED (back-distance exactly  $1.0000 \delta$  — every admissible measure is a fixed point), Gram-seat square law  $\delta_* = \sqrt{\text{margin/curvature}}$  numerically EXACT on the Gram channel (ratio 0.9997, slope 2.0000; order split: position first-order at slope 0.9669, reality second-order) typed PATTERN-ONLY by the frozen FOLDCOV rule; the surviving principle: TFPT’s positivity is always GRAM POSITIVITY OF NODE EVALUATIONS — a theorem exactly where the nodes are compiler data, the arithmetic seam the single place whose nodes are not; two corollaries sharpen the OCT without moving it (the order split mechanically explains all 24 dead sign-source candidates; the  $\sim 1\%$  sub-spacing price is instrument-dependent,  $h^{-1.69}$  along the deployed ladder); (5) ROUND 87 (note CCCLXXXVI, `bughunt_r79_r85_probe.py`, 24/24, 230 s, SPEC 9346a32b, verdict BUGHUNT-FINDINGS(9, max MAJOR)): adversarial audit of rounds 79–85 — NINE findings, 3 MAJOR / 6 MINOR / 0 FATAL, NO round verdict flips, every finding pinned by an executable gate; the three majors are the F1/F2/F6 corrections carried above (the honest zero-tracking claim STRONGER than published); one systemic minor: NINE gates across the corpus hardcoded `check(..., True)` — never able to fail while counted in the  $N/N$  tallies, including the very line recording the round-80 “candidate #19 gate FAIL” verdict (the verdict right, the gate testing nothing); checked clean: every round-79–82 headline number reproduced on fresh re-runs, all ten probes re-ran green, no firewall violations; (6) ROUND 88 (note CCCLXXXVII, `signpos_probe.py`, 19/19, 11.5 s, SPEC b0003dcc, independently re-run bit-identical, verdict SIGNPOS-PREDICTOR-EXACT + SIGNPOS-TP-DEAD + SIGNPOS-RIGID + SIGNPOS-K-STABLE): the campaign’s most concrete endpoint — a SOURCE-ONLY predictor of the sign-change positions (the parity of  $\text{round}(\theta(\omega_k)/\pi + 1 + S_x(\omega_k))$  at the Nyquist lattice,  $S_x$  the band-limited phase built from the prime-node comb) is EXACT at ALL 79 lattice points across  $x = 3/5/8/13$ , with measured first-order arithmetic content: the density alone fails at exactly FOUR lattice points and the prime-node comb repairs precisely those four — ARITHMETIC LOAD 4, the first arithmetically computable first-order-channel object of the campaign; scramble control fails 6/11 with  $\lambda_{\min} = -1.11$ ; THE

CATCH named precisely: exactness for all  $x$  is half-integer-margin control of the band-limited  $S$  at the lattice points (measured minimum margin shrinking to 0.0317) — verbatim TPL(i) itself, the RH hardness relocated for the THIRD time to the same seat, now as a rounding-margin statement about one explicit band-limited function of the primes; Gantmacher–Krein STRUCTURALLY DEAD ( $\lambda_{\min}(\text{ARCH}) = -2.666$ ;  $M$  neither totally nonnegative nor Green’s/one-pair in any tested ordering; the minor census localizes nothing); the position map RIGID (modulus = the ordinate-to-lattice distance); the positions K-STABLE in the deployed band (the round-85 caveat discharged for the positions, the joint  $(x, K)$  declaration remaining for the margins); (7) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The real-root architecture, the Parity Lemma, and the sign-position predictor”; the end-form article gains §8.10 with the abstract status sentence, limitation 9 (the bughunt corrections and the hardcoded-gate finding as a methodology limitation) and the traceability rounds CCCLXXXII–CCCLXXXVII; the Anthropic email draft gains one status paragraph; the website prime-front page gains chapter 34 (the live-updates chapter renumbered  $34 \rightarrow 35$ ) and one consolidated sandbox feed entry (verdict SIGNPOS-PREDICTOR-EXACT); `website/lib/changelog.ts` re-generated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-14 (XCVI) (the kill-atlas consolidation arc — discovery freeze rounds seventy-nine through eighty-two synced into every live surface: the RH implication DAG, the 50-route kill atlas, the SAT adjudication, the untested-five adjudication, the bird’s-eye shape/direction/bootstrap round and the OBSTRUCTION COMPLETENESS THEOREM; experiments/ only — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 79 (note CCCLXXVI, `kill_atlas_dag_probe.py`, 43/43 gates after the N12/N13 self-correction of entry XCV, verdict ATLAS-DAG-CONSISTENT + ONE-OPEN-EDGE-CONFIRMED + FOUR-CLASSICAL-CITATIONS-CARRIED + NO-SIGN-SOURCE-PASSES + LOCALIZATION-DID-NOT-MOVE-THE-EDGE): the RH implication chain as an explicit 11-node, 10-edge DAG, acyclic on the conjunctive hypergraph, typed 3 PROVEN / 2 CERTIFIED-FINITE / 4 CITED-CLASSICAL / 1 OPEN with EXACTLY ONE open edge E4 (the predefined mesh-cofinal ladder positivity); the honest non-PROVEN list E1/E2/E8/E4/E7/E9/E10 carried as an executable gate; the 50-route atlas assigns every killed route to exactly one edge (17 E4 / 21 E2 / 8 E10 / 2 E1 / 2 off-chain), every verdict token grep-verified inside its own artifact;  $\tau$ -screen 15 DISGUISE / 1 ADVANCE / 34 not-applicable with the single ADVANCE pinned NOT on E4; six coordinate systems ( $\tau_h > 0$ ,  $n_h - q_h > 0$ ,  $\sigma_h < 1$ , the anti-cancellation scalar, the  $P_{\text{err}}$  budget, v913’s best-conditioned (L)) pinned as ONE and the same edge — the localization campaign sharpened the edge’s type, closed the second open edge (v912) and corrected its statement (mesh order), and did NOT move it; (2) ROUND 80 (note CCCLXXVIII, `sat_projection_alignment_probe.py`, 35/35, SPEC 1b8b6f8e, verdict SAT-DISGUISE + SAT-LITTLEWOOD-EMPTIED + SELECTION-SIGN-MINED): the externally proposed signed-projection closure of E4 is adjudicated dead as submitted — the projection identity is EXACT (zero residual in exact Fractions) and EMPTY ( $A_2$ ’s column block unit-triangular with  $\det = 1$  as exact integers, so  $P = I$  and  $\|(I - P)y\|^2 \equiv 0$  identically; the ellipse inequality is  $s \geq 0$  VERBATIM; the “optimal signed profile”  $w^*$  is  $B$ -independent, zero arithmetic content); the  $\theta$ -average is verbatim v913’s (L) with its standing

kills (HH-CLAIMS-WITHDRAWN, CLASSICAL-GAP  $2.470 \times 10^{11} / 1.990 \times 10^{14}$ ); the proposed  $\theta$ -selection is the kernel-checked sign-mined index (`signMinedIndex_not_familyNoninterfering`); the small-effective-rank premise is REFUTED (stable rank  $109.6 \rightarrow 1595.2$ ); candidate #19 FAILS the frozen gate (separates without independently orienting) — coordinate system number SEVEN for E4, gate tally  $18 \rightarrow 19$  candidates, 0 PASS / 14 FAIL / 5 UNTESTED; (3) ROUND 81 (note CCCLXXIX, `untested_sign_sources_probe.py`, 33/33, verdict SIGNSOURCES-ADJUDICATED): all five remaining UNTESTED candidate sign sources adjudicated FAIL with exact killing numbers — #20 G2-SEPARATES-NOT-ORIENTS (law exact at  $7.2 \times 10^{-16}$  on truth, defect energy ZERO on truth, class floors  $0.191 / -0.509 / -0.227$  below the need), #21 KREIN-CERT-IS-WALL (Douglas exact over  $\mathbb{Q}$ : contraction margin  $= 1 - \|C\|^2 = s/n$  VERBATIM), #22 U1-NAMED-TYPES-EMPTY (honest instrument on the full  $2 \times 10^7$  certified ordinate cache; smooth/jitter/shift surrogates violate by  $104\text{--}636\times$ ; the jitter sweep prices the required ordinate precision at  $\sim 1\%$  OF THE MEAN ZERO SPACING; Weyl carries no rate, rate-carrying density is the emptied E2 class, pair correlation conditional), #23 U2-BEYOND-CLASSICAL (MIN-U2 named exactly,  $h$ -flat; a planted off-line pair violates it at  $\delta = 0.3635/0.2088/0.3533$  BELOW the classical zero-free widths  $0.4475/0.4443/0.4487$  — on the deployed windows it already excludes zeros no classical theorem excludes), #24 U5-ENUMERATED-DEAD (log-convexity violated by truth itself at  $-0.182$ ; downward-closure lemma: the zero comb is admissible in every enumerated class, so no upper-bound-type inequality can supply the required lower bound) — NEW TALLY: 24 candidates, 0 PASS, 19 FAIL, NO blanket UNTESTED left; (4) ROUND 82 (note CCCLXXX, `birdseye_shape_freedom_probe.py`, 35/35, 54 s, SPEC 8b78c201, verdict BIRDSEYE-OCT-CONSOLIDATED): (A) SHAPE-INDEPENDENCE-MEASURED — the deployed family is RH-saturating (margin-decay exponent  $-3.379$ ); five structurally different admissible families (Fejér, Gauss-tent, Hann,  $K \star K$  square, T5.2-style perturbation; Beurling–Selberg minorants typed INADMISSIBLE-BY-CAP) priced in the same Weil-square currency with an exact explicit-formula ward decay FAR shallower (exponents  $-0.46 \dots -1.09$ , min gap 2.29) — the maneuver exists but buys nothing: single-window ladders fail E7 density (54.5% projection residue), mesh-dense families inherit  $\lambda_{\min}(\Omega_{\text{EF}}) \leq s$  by the exact Schur bound ( $3.97 \times 10^{-7} \rightarrow 4.19 \times 10^{-12}$ ), and the localization is SHAPE-INVARIANT (square shapes degenerate the alignment, 15/15 position-blind worlds positive; off-line exclusion BELOW every classical zero-free width,  $\delta_{\min} = 0.055\text{--}0.26$  vs  $\sim 0.42$ , 5/5 beyond-classical; the deployed linear coordinate reproduces the  $\sim 1\%$  sub-spacing bar at  $107.1/4.36/0.21\times$  margin for  $\varepsilon = 0.5/0.1/0.01$ ); (B1) EXISTENCE-IS-DISGUISE — the existence direction closed PERMANENTLY at the border-membership step in exact Fractions (a positive extension always exists;  $\det M = s \det B$  with PD  $\Leftrightarrow s \geq 0$  both ways; the Krein ellipsoid contains the arithmetic border with margin EXACTLY  $s$ ; the Carathéodory–Toeplitz one-step margin equals the next Schur scalar) — the gap between “an extension exists” and “the arithmetic one is it” IS E4; (B2) BOOTSTRAP-STILL-DEAD — the  $2 \times 10^7$ -ordinate cache certifies MIN-U2 at  $2/5$  cells and the  $\tau$  coordinate at  $0/5$ ;  $T_{\text{req}} \sim h^{4.77}(\tau) / h^{1.23}(\text{EF})$ , extrapolated  $\tau$ -coverage ends at  $h \approx 36$ , precision NOT binding, HEIGHT is; (C) THE OBSTRUCTION COMPLETENESS THEOREM, every premise machine-asserted: (i) exactly one open edge E4, (ii) the candidate map COMPLETE (24/0/19, no blanket UNTESTED), (iii) the missing input exactly characterized (signed, alignment-carrying, unconditional,  $\sim 1\%$  sub-spacing, minimal form MIN-U2

beyond-classical), (iv) shape-independent across the audited admissible families — plus the dual/existence direction closed and the bootstrap dead; SCOPE in the same breath: finite-rung (cells to  $h = 2854$ , ladder cited to 12632), atlas completeness EDITORIAL and typed as such, E7–E10 classical citations, NO-WITNESS counter-evidence carried unsmoothed (all certified reads positive to  $h = 12632$ , deepest  $2.79579794131272506 \times 10^{-15}$ , decay flattening) — a no-go about the TYPE of the missing input is not evidence about its EXISTENCE, and the theorem is NOT a statement that RH is unprovable; (5) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The kill atlas, the complete candidate map, and the Obstruction Completeness Theorem”; the end-form article gains §8.9 with the abstract status sentence, limitation 8 and the traceability rounds CCCLXXVI–CCCLXXX; the Anthropic email draft gains one status paragraph; the website prime-front page gains chapter 33 (the live-updates chapter renumbered  $33 \rightarrow 34$ ) and one consolidated sandbox feed entry (verdict BIRDSEYE-OCT-CONSOLIDATED); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; the stale `changelog.pdf / introduction.pdf` debt of entry XCV is cleared by the same rebuild; NO RH claim anywhere)

2026-08-14 (XCV) (editorial honesty correction, wording only — no number changed, no module touched, no status marker moved, no gate closed or narrowed, NO RH claim: the corpus-wide synthesis audit of note CCCLXXVI (discovery probe `kill_atlas_dag_probe.py`, 41/41, verdicts ATLAS-DAG-CONSISTENT + ONE-OPEN-EDGE-CONFIRMED + FOUR-CLASSICAL-CITATIONS-CARRIED, experiments/ only) reconstructed the RH implication chain as an explicit 11-node, 10-edge DAG and recorded that the corpus headline “the chain is reduced to exactly one unproven inequality” counts INEQUALITIES correctly, and counts correctly what the programme itself must supply, but drops four classical citations: the honest non-PROVEN list is E1/E2/E8/E4/E7/E9/E10 — two CERTIFIED-FINITE heads whose SCOPE is finite, the ONE OPEN inequality E4, and FOUR CITED-CLASSICAL edges (E8 the Suzuki/W1 identification, E7 density plus the equi-Lipschitz/Dini continuity modulus, E9 the faithful cap, E10 Weil’s criterion) — (1) WHERE THE QUALIFIER WAS ALREADY CARRIED, hence untouched: `tfpt_prime_front` states “theorem-grade modulo the named citations” at both extraction-chain sites and names Weil 1952, Bombieri 2000, Suzuki and Iwaniec–Kowalski Thm 5.12; the ledger row PRIME.FORM.CONVERGENCE.THEOREM.01 types the W1 dictionary, Weil’s explicit formula and criterion, the admissible BV class and the DENSITY leg as CITED and adds that ( $H_{\text{cof}}$ ) is untouched and RH-hard; `changelog XCIII` carries the same pinning disclosure; the `v912` docstring says verbatim that edge 6 is NARROWED, not closed; and the website mirror repeats the qualifier in the prime-front feed and the papers index; (2) WHAT IS CORRECTED: the two article surfaces that stated the reduction without it — `articles/2026-08-11/paper_endform_and_inertia_bridge_en.md` (abstract, §8.6, §8.7, §9) and `articles/2026-08-11/email_anthropic_draft_en.md` (status paragraph and ask 3) — each gain ONE clause naming the four classical legs beside the one open inequality; (3) THE AUDIT’S FOUR OTHER FLAGGED ITEMS, CHECKED AGAINST EVERY LIVE SURFACE AND NOT LIVE, hence no edit: NOGO-COMPOSITE-VERIFIED occurs only as the stated provenance of the promoted NOGO-CORE-VERIFIED core, in a ledger row titled verbatim “A NO-GO / TYPING ROW AND EXPLICITLY NOT PROGRESS” and in a module docstring that says it is NOT progress toward a proof; WIRING-DEGENERATE and THETA-CONVENTIONAL

occur only in the seam/compiler lane (SEAM.STATE.WIRING.SELECTOR.01, v911) and are listed among no RH-chain route kills; the superseded matrix-stage conditioning exponents and the withdrawn Li–Keiper Vinogradov–Korobov envelope exist only in experiments/ and never entered a paper body, a ledger row or the public mirror; NO RH claim anywhere)

2026-08-14 (XCIV) (promotion round six — THE SIGNED-AND-ALIGNMENT LOCALIZATION, a NO-GO/TYPING module that is explicitly NOT progress, plus the mesh-order wording debt closed at its last two sites: ONE module (v913\_signed\_alignment\_localization.py, 24/24, verdict NOGO-CORE-VERIFIED, ~28 s; suite 905 → 906; ledger 1056 → 1057 with the new row PRIME.NOGO.SIGNED.ONLY.01 typed [O]/no\_go\_typing; discovery probe signed\_only\_nogo\_probe.py, note CCCLXXIII, 87/87, SPEC 49530809, PROBE\_SHA d1c42a05) — (1) ITEM ONE, THE MESH-ORDER DEBT (wording only, no number changes, no marker move): note CCCLXX had logged two sites skipped during the CCCLXIX correction pass; v848\_extraction\_chain.py stated the chain as “the canonical cofinal ladder” and “CONCLUSION: cofinal finite positivity (H) on the canonical ladder” in BOTH its docstring and the text it PRINTS at runtime, and run\_all.py carried the same wording in its registry line — all of them now say cofinal IN THE MESH-REFINEMENT ORDER  $D_j = 2^{-j}$ , because at a FIXED mesh  $D_0$  the read is exactly cap-independent, so a ladder cofinal only in the window/cap parameter is eventually constant and converges to  $W_C[K] + W_C[e_{D_0}]$ , giving the FALSE FLOOR  $Q_W \geq -|W_C[e_{D_0}]|$  (measured  $-2.114 \times 10^{-3}$  at  $D_0 = 1/32$ ,  $-2.128 \times 10^{-4}$  at  $D_0 = 1/128$ ); where the module explains why cofinality suffices it now carries the one-line reason in its proven form — were the limit negative, the per-element convergence of v912 would make the catch set  $\{j : Q_j(f) < 0\}$  a TAIL of the mesh order, and a cofinal set meets every tail; module re-run unchanged at 19/19 and re-mirrored byte-identically into website/public/verification by the build step; (2) ITEM TWO, WHAT THE NEW MODULE SAYS: any argument that closes the deployed budget inequality for the DEPLOYED BUDGET SHAPE must be (a) SIGNED — it must orient a quantity that is ODD under the comb sign flip — and (b) ALIGNMENT-CARRYING — it must relate the position of the prime atoms to the spectral directions of  $K$ ; in particular it can be NEITHER a magnitude bound on  $\psi(x) - x$  NOR a natural-grammar identity valid in every arithmetic world and therefore comb-blind; (3) THE PROMOTED CORE, exact and self-contained: the six kernel identities in exact arithmetic ( $\int K_D = 0$ ;  $\|K_D\|_\infty = 1$ ;  $\|K_D\|_1 = 4D/3$ ;  $D^{-1} \int K_D^2 = 2/3$ ;  $\int K_D e^{-t/2} = -(D/8)T_D^2$ ; and the load-bearing  $\|K'_D\|_1 = 4$  INDEPENDENT of  $D$  — the conversion constant of a magnitude hypothesis does not get cheap as the mesh refines), the geometric supply recomputed by TWO independent routes (closed form vs. direct quadrature, deviation  $1.6 \times 10^{-50}$  /  $2.5 \times 10^{-49}$ ;  $B = -1.2648057425013541016$  at  $h = 184$  and  $-1.3147508968858830811$  at  $h = 388$ , negative and  $O(1)$ ), the Littlewood floor  $4 \log \log \log N_h$  (2.1654551 at  $N = 263$ , 4.1231791 at  $N = 14634429$ ) against a slack that shrinks with depth — ratio 214.7 on the module’s OWN conservative slack and 1527.1 on the deployed one, so the ENTIRE MAGNITUDE CLASS IS EMPTY; the provable neutrality of Selberg’s symmetry formula ( $\Lambda = \mu * \log$  and  $\mu * \log^2 = \Lambda \log + \Lambda * \Lambda$  are formal identities in the free module on  $\{\log p\}$ , information gain exactly 1.000); the exact three-term split with its congruence invariance and comb-flip parity (all three terms invariant under EVERY  $\text{diag}(1, M)$ , outer terms EVEN and the carrying term ODD, the  $F_4$  kernel  $A_2^T B^{-1} A_2$  itself a congruence invariant — so no unsigned, flip-invariant hull can bound the carrier below); and



the Beurling–Nyman/Baez-Duarte target exactly in  $\mathbb{Q}$  ( $\lambda_{\min}(G_N) \leq (\log 2\pi - \gamma)/N$ : cap 0.420220 against the needed  $c_B = 0.5523$ , so there is NO  $N$ -uniform Gram floor and the deployed hypothesis is FALSE for  $N \geq 3$ , plus the explicit non-implies family  $\tilde{G} = AA^T + (4/5)^2 I$  with  $\tilde{d}^2 \geq 1/2$  forever); (4) **THE BOUNDARY, RESPECTED EXACTLY**: the deployed matrix-stage reads (S3.5–S3.12, S4, S5) **STAY** in experiments/ because they inherit the float64 entry-slack premise and the cited ladder; **PINNING DISCLOSURE** — **PINNED** here are the identities, the split algebra, Selberg neutrality, the Beurling–Nyman witnesses, the supply recomputation, the module’s own prime-power sieve, the frame rebuild with  $h$  **PREDICTED** (184..12632), the square-root-barrier placement and the frozen probe SPEC-SHA reproduced **BEFORE** any evaluation; **CITED** and **NOT** pinned are the deployed ladder and its certified reads (X1), the deployed entry slack (X2), the classical constants and Littlewood’s  $\Omega$ -theorem itself (X3), the float64 entry slack of the wall generators (X4), the CCCLXVIII demand sums (X5), the CCCLXII/CCCLXIII chain measurements and deployed constants (X6) and the  $n_0$ -normalised margin exponents (X7); (5) **SCOPE, STATED IN THE SAME BREATH**: TEN classes are proven empty for this budget shape (E1–E10), FIVE are merely unexplored with NO emptiness claimed (U1–U5) and FIVE clauses are named open (O1–O5, among them that nothing is said against arguments legitimately consuming zero positions and that the reverse implication rests on four unestablished premises); the theorem is **FINITE-RUNG** ( $h = 184/388/839$  with the ladder to  $h = 12632$ ) and makes NO all- $h$  claim; it does NOT say that the Riemann Hypothesis is unprovable, that the wall route fails, or that no proof exists; (6) **COUNTER-EVIDENCE CARRIED, NOT SMOOTHED: NO-WITNESS STANDS** — the certified reads are **POSITIVE** at all NINE depths (0 negatives, 0 straddles) with deepest certified read  $2.79579794131272506 \times 10^{-15}$  at  $h = 12632$ ; the  $n_0$ -normalised decay **FLATTENS** with depth ( $-3.21092511$  globally against  $-2.67942161$  on the deepest step  $5746 \rightarrow 12632$ ) and the re-derived  $s/D^2$  step  $1393 \rightarrow 2015$  is **POSITIVE** ( $+0.50302892$ ), so the normalised drift is NOT monotone, and no re-derivable exponent comes near the frozen collapse bar  $-8.0$ ; a no-go about the TYPE of the missing input is not evidence about its EXISTENCE; (7) **SURFACES**: `tfpt_prime_front` gains the residual-obstruction subsection “The remaining obstruction is signed and alignment-carrying (localization)” with the probe’s own S9 wording, Paper 3 a new audit card with the `\veri` citation, the end-form article and the Anthropic email draft one honest paragraph each, and the website mirror is synced; the row supplies no positivity, no interval and no certificate, closes no gate, narrows no gate and moves NO marker anywhere — it must not be counted as progress nor as evidence for or against the Riemann Hypothesis; NO RH claim anywhere)

2026-08-13 (XCIII) (promotion round five — **THE FORM-CONVERGENCE THEOREM**: the extraction chain’s per-element convergence premise stops being a measurement and becomes an unconditional theorem, and a false cited premise it uncovered is corrected across the corpus: ONE module (`v912_form_convergence_theorem.py`, 35/35, verdict **FORMCONV-PROVEN**,  $\sim 12$ s; suite  $904 \rightarrow 905$ ; ledger  $1055 \rightarrow 1056$  with the new row **PRIME.FORM.CONVERGENCE.THEOREM.01**; discovery probe `form_convergence_theorem_probe.py`, note CCCLXV, SPEC `d911226f`, promoted **VERBATIM** with no downscoping) — (1) **THE THEOREM**: for  $f$  real, supported in the deployed window and **AFFINE ON EVERY GRID CELL**, the finite Galerkin form converges to the Weil functional,  $Q_D(f) \rightarrow Q_W(f)$ , **UNCONDITIONALLY** and with the

**explicit envelope**  $|Q_D(f) - Q_W(f)| \leq D^2[(8 \sinh(B/2) + m(B))A_2 + \Theta_0 A_0 + (A_0 + A_2)(\log(1/D) + \log B + 4) + 2A_0 e^{-B/2}(1/B + 2) + A_1(1 + D)] + (1 + D)(D^3/12)\kappa_3$ , i.e. rate  $O(D^2 \log(1/D)) = O(2^{-2j}j)$ , with  $\Theta_0 = 3 \log 2 + \pi/2 + \gamma + \log \pi = 5.372183419225665$  and, for continuous  $f$ , log coefficient exactly  $7/24$  (sup  $5/24$ ) — replacing the corpus typing MEASURED (rates  $-1.58..-1.84$  per level, v848) that meta-audit CCCXXXVII Q-2 had flagged as a measured premise read as settled; (2) **THE MECHANISM**, three exact steps: the deployed lag vector IS the capped Weil functional on the even tent ( $c_d = W_C[S_{dD,D}]$ ), so the finite form equals  $W_C[\widetilde{K}_D]$  EXACTLY on the piecewise-linear interpolant of  $k_d = Da_d$ ; the EXACT DEFECT LEMMA  $k_d - K(dD) = -(D^2/12)G(dD)$  holds for every lag because  $u \mapsto f(u)f(u + dD)$  is QUADRATIC on each cell and the midpoint error of a quadratic is exactly  $-(D^3/24)q''$  — ZERO deviation in exact rationals over 1474 lag identities and 780 strictly interior points; and the faithful cap  $C \geq b_K + D$  turns the error into  $W_C$  applied to ONE explicit function, bounded layer by layer with elementary majorants; the interpolation constant is the interior sup of  $|K''|$ , NOT  $\|f'\|_2^2$  (equal for continuous  $f$ , different for jump elements: E09  $3 > 2$ , E10  $5/3 < 2$ ); sharpness is measured against a CLOSED-FORM leading term (ratio  $0.575..1.066$ , drifting to 1 through  $j = 18$  — the law is genuine, not a loose envelope), and uniformity (V) holds on every TWO-SIDED support window, not on an  $L^2$  ball; (3) **THE CORRECTED PREMISE** (a correction of a wrong cited statement, NOT a downgrade of a result): the chain's cited " $C^0$ -continuity of  $Q_W$  at fixed support" is FALSE in the pure sup norm — must-fire control C5 exhibits the explicit even Lipschitz family  $e_n(w) = (1/n) \min(1, w/e^{-n^2})(1 - w/2)_+$  with  $e_n(0) = 0$  and  $\|e_n\|_\infty = 1/n \rightarrow 0$  while  $|A[e_n]| = 2.57, 4.28, 6.19, 8.14, 10.11, 12.09$  grows linearly — and the true hypothesis is uniform convergence PLUS an equi-Lipschitz (Dini) condition at the origin, which the admissible even compactly supported BV class supplies automatically (T5.2 verifies the corrected modulus of  $W_C$  on a declared 12-perturbation battery, worst true/bound  $0.0226$ ); the false premise is corrected wherever it was cited: `tfpt_prime_front`, `tfpt_research_contracts`, the v848 module docstring and its website mirror, `run_all.py`, the ledger row PRIME.EXTRACTION.CHAIN.01 and the Lean prose of `TfptCarrier/CofinalWeil.lean`; (4) **CLASSICAL INPUTS**, all 11 typed (T4.3): NO zero-free region, NO PNT, NO zeta zeros, NO fit — the only cited arithmetic input is Rosser–Schoenfeld 1962 Thm 12,  $\psi(x) < 1.03883x$  for all  $x > 0$ , consumed ONLY for the support-uniform constant and independently sieve-verified in-run to  $10^7$  (worst ratio  $1.038821$ ); **PINNING DISCLOSURE** (the CCCXXXVI lesson applied a priori): the W1 measure dictionary, Weil's explicit formula and criterion, the admissible class with its DENSITY leg, the all- $x$  Rosser–Schoenfeld statement and the UNIVERSAL QUANTIFIER itself are CITED, not pinned by these gates; (5) **LEAN (DELTA-A, honestly partial)**: the new kernel-checked module `TfptCarrier/CofinalEnvelope.lean` states the envelope premise as a named structure `FormEnvelope` (per-element constants, a DELTA-B window level, the explicit bound), DERIVES the convergence hypothesis from it (`tendsto_of_formEnvelope`), and restates `cofinal_weil`, `cofinal_weil_for_fixed_idx` and `cofinal_weil_predefined` with `hconv` REPLACED by the instantiated envelope — plus a non-vacuity lock proving the envelope STRICTLY STRONGER than bare `Tendsto` and an inhabited witness with nonzero error at the envelope rate; axioms `propext/Classical.choice/Quot.sound` only, no sorry, no native\_decide; the envelope itself is NOT discharged inside the kernel, so DELTA-A is REDUCED, not closed; DELTA-B (stated, not a gap): (H-grid), (H-cap) and (H-align) hold for fixed  $f$  and all large  $j$ ,

which is exactly what Tendsto atTop needs; papers (tfpt\_3 new audit card, tfpt\_prime\_front the extraction-chain premise corrected and the theorem cited, tfpt\_research\_contracts the contract's cited hypothesis corrected) and the website mirror synced; the chain's genuinely open item is UNCHANGED — (H\_cof), the PREDEFINED cofinal positivity inequality, is untouched and RH-hard; no marker moves; NO RH claim anywhere)

2026-08-13 (XCII) (cofinal-index noninterference hardening — the PREDEFINED requirement is moved from an overclaim in prose into an explicit, honestly external Lean contract; no marker move, no RH claim: TftptCarrier/CofinalPredefinition.lean adds (1) cofinal\_weil\_for\_fixed\_idx, the extraction implication for an arbitrary explicit index with StrictMono, PSD and form-convergence hypotheses — binder order prevents the theorem body from selecting the index but does not prove caller provenance; (2) the negative theorem old\_api\_accepts\_sign\_mined\_idx, exhibiting that the former mathematical payload accepts a selector branching on measured signs; (3) signMinedIndex\_not\_familyNoninterfering, which rejects that exposed selector under extensional family-noninterference; (4) the hardened PredefinedCofinalHypothesis/cofinal\_weil\_predefined API, requiring an explicit abstract NoninterferenceContract.Predefined(A,idx) certificate; and (5) the constantization pair, proving the exact residual limitation — after construction, a selected value can be represented by a constant selector with the same output, so extensional dependent types cannot recover algorithmic history. The concrete source boundary excluding the matrix family and its sign outputs therefore remains the named external audit premise unless the construction language or an equivalent effect/provenance system is formalized. The old CofinalWeil.lean core is relabelled as mathematical payload only; dependent Lean-module comments, audit locks, the Lean/verification READMEs, tfpt\_prime\_front, Paper 3's audit card, the end-form article, ledger row FORM.PRIME.COFINAL.WEIL.01, registry text and the public prime-front feed are synchronized. No verification result, status marker, finite positivity claim or RH claim changes.)

2026-08-13 (XCI) (website/video maintenance — the /prime-front page gains the sigma-chain diary chapter and the explainer film its regional-theorem coda; NO theory change, NO marker moves, NO RH claim: (1) THE PAGE — new diary chapter “32 — The sigma chain and the regional theorem (sandbox probes)” covering the experiments freeze rounds 73–75 plus the morning package round, ALL SANDBOX per the page's promoted-vs-sandbox convention (v911 remains the newest promoted module, every counter untouched): the identity  $\sigma = 1 - s/n$  with the two 0.665s dismantled as a provenance-proven coincidence; the derived per-cell exact-rational chain certifying every built wall-legal cell (151/151 surface + 59/59 deep to  $h = 6344$ , worst bound 0.727,  $O(1)$  margins); the joint Radau relation lifting definiteness to class level ( $\inf \lambda_1 + 0.008$  vs  $-574$ ); the SOS dual certificate (1111 exact rational certificates, 151/151 at  $\eta = 0.273$ ) sealed as the three-part theorem package with the 202-line stdlib-only checker (the audit entry point — re-proves the theorem in 2s, teeth 3/3); the level-3 endform (legal sub-ladder to  $h = 8204$  with the first measured termination signal, the sign living in the seat-to-bulk coupling; B3 Lipschitz-certified, B2 proved-conditional, B1 one measured scalar; the razor margin +0.0104 typed an irreducible measurement); the closed no-gos (Pick DEAD, finite-gap ILLDEFINED, shells JCONTRACT-DEAD); and the honest closing map in its frozen form — a REGIONAL finite positivity

theorem plus a single open cofinal construction problem, the quantifier is RH, NOT “almost-RH”; the live-updates section renumbered  $32 \rightarrow 33$ , one new diary-feed entry (badge sandbox, verdict **PACKAGE-SEALED**); (2) **THE FILM** — new closing frame 10 “Das regionale Theorem und das kofinale Problem” extends the explainer  $2:39.1 \rightarrow 3:05.1$  (26.0 s window at 159.12 s, crossfade in): the finite positivity question as a **REGIONAL THEOREM** (every matrix whose entry data lie in the certified region  $K$  is positive definite; on-screen chip “151/151 cells · SOS proof object · 202-line stdlib checker · re-proves in 2 s”), the single remaining **CONSTRUCTION** problem (a predefined window sequence, cofinal, staying inside the region — keyword glow on “cofinal”) and the honest close as two dry statement beats: “The proof object is ready for audit — the RH quantifier is not”; full manual rebuild per the documented README sequence (EN+DE script line 10 — 1:1 for the de subtitle track, TTS re-run with all 10 lines at rate +0% inside their frozen windows — frames 1–9 unchanged, captions rebuilt from 433 words, 9 transitions verified, render at 5554 frames, en/de subtitle tracks  $34 \rightarrow 37$  cues, last cue ends 3:04.12); **WEBSITE SYNC** — the four public assets re-shipped (mp4 + en/de VTT, poster byte-identical), PrimeFrontVideo.tsx chapters + coda marker at 2:39, the component transcript verified character-identical to the caption track, JSON-LD duration PT2M39S  $\rightarrow$  PT3M5S with uploadDate 2026-08-13, section header “ $2\frac{1}{2}$  minutes”  $\rightarrow$  “3 minutes”; nothing under experiments/, verification/, articles/ or the papers/ledger touched (the concurrent research lane owns those))

2026-08-12 (XC) (promotion round four — the consolidation of rounds sixty-four through seventy-one: the **FINITE** wall closure with its transfer law, and the wiring freedom theorem: **THREE** modules (v909–v911: finite wall closure, finite zero transfer, wiring freedom; suite 901  $\rightarrow$  904; ledger 1052  $\rightarrow$  1055 with the rows PRIME.WALL.FINITE\_CLOSURE.01, PRIME.WALL.FINITE\_ZERO\_TRANSFER.01 and SEAM.STATE.WIRING.SELECTOR.01 closed) — (1) **THE FINITE WALL CLOSURE** (v909, 28/28, PRIME.WALL.FINITE\_CLOSURE.01): on the deployed finite ladder every reachable wall face is certified from cited inputs and the composed census  $B \wedge W1 \wedge W2$  holds on 39/39 matched surface + 8/8 deep rungs (frozen experiments runs of rounds 64–67, cited with their spec hashes) — the  $B$ -half via the v905 interval-certified class floor ( $\min c_B = 0.5523$ ), the  $W1$  face via the monotone-composition measure inequality (the two +8..+9 dex composition gaps of rounds 55–59 **DISSOLVE** into one exact Loewner step, composition price +0.29 dex) plus **VERIFIED ZEROS AS EXACT DATA** at the named  $j = 16$  seat (the truncated explicit formula with 7000 mpmath ordinates, every read tested against the independent prime side, the tail bracketed by the unconditional Rosser–Abel envelope: 39/39 surface, composed 67/67 with the cited deep 28/28), and the  $W2$  face via the recomposed certificate  $m_{\text{cert}} = FC + \hat{P}_{\text{arith}} - \text{TAILB} > 0$  (16/67 at the 7000-ordinate cache, 67/67 + 8/8 once the  $2 \times 10^7$ -ordinate certified cache pays the priced tail; Odlyzko + LMFDB/Platt, every ordinate below the Platt–Trudgian height  $T_0 = 3 \times 10^{12}$ , **EXTERNAL-CITED**); the promoted module re-verifies the load-bearing spine **SELF-CONTAINED** on a reduced frozen subset with its own committed 7000-ordinate cache re-warded in-run (Rosser corridor per index both sides,  $\gamma_1$  identity, independent  $|\zeta|$  spot checks  $2.1 \times 10^{-12}$ ; heart ward on every read, slack +2.1..+4.8 dex; closure 6/6; the off-line-impostor control fires at  $8.2 \times 10^4 \times$ ) and cites the full census rather than recomputing it; **LANGUAGE FROZEN**:  $W1/W2$  are algorithmically independent evaluations of the **SAME** localized Weil form — a strong mutual

crosscheck, NOT two independent proofs; positivity certified on a finite family of Galerkin sections along the MEASURED critical direction (direction-conditional, UNIF-PATH caveat); a finite verified-zero sum can never prove RH; (2) THE TRANSFER LAW (v910, 10/10, PRIME.WALL.FINITE\_ZERO\_TRANSFER.01, SPEC 40a7b5c3): the zero-supply economics of the finished wall —  $T_{\text{req}}(h) \sim h^{2.8}$  ( $\log_{10} T_{\text{req}} = -3.045 + 1.228 \ln h$ ,  $R^2$  0.882) and the ratio to the window's own spectral reach  $\pi/D_h$  GROWS (+0.897 dex/ $\ln h$ ): verdict RATIO-GROWING  $\Rightarrow$  EXTERNAL-BATTERY / W-UNBOUNDED-PER-RUNG — the finite engine does NOT scale by buying zeros; the parametrized transfer law  $\text{ZerosCertified}(T) \wedge \text{TailEnvelope}(T) \Rightarrow M_h \succ 0$  for all deployed  $h \leq H(T)$  with the measured envelope  $H(\gamma_{7000}/\gamma_{2M}/\gamma_{2 \times 10^7}) = 254/1256/2806$ , W1 cheap (binds 3/75), the next  $h$ -decade at  $T \sim 3.4 \times 10^9$ , and  $H(T)$  the measuring rod any analytic per-window bound must beat; the module re-verifies the shallow end (12 rungs  $h$  142–254, anchors  $K = 250..7000$ , closure census 2/3/4/6/8/12, zero non-monotone, heart ward on all 492 cells,  $H$ -consistency 12/12; one disclosed control-rule amendment, fail-first preserved: the jitter bar restored to the upstream frozen  $\geq 5$ -reads rule); (3) THE WIRING FREEDOM THEOREM (v911, 30/30 + 26/26 + 21/21, verdicts WIRING-DEGENERATE + THETA-CONVENTIONAL + RP-FRAME-COVARIANT; three frozen probes of rounds 69–71 embedded byte-exact, exact integer/rational arithmetic; SEAM.STATE.WIRING.SELECTOR.01 CLOSED — the v908-registered contract answered): the exact Gröbner census of the orbit/edge constraint classes leaves EXACTLY ONE admissible three-dimensional wiring component modulo the rule gauge (the rotation cell  $C_{\text{rot}}$ ); the  $Z/X$  wirings are excluded gauge-invariantly by orientation propagation ( $\det u < 0$  identically); the deployed pure- $i$  point is an INTERIOR point, connected to the pure- $J$  ray by the integer gauge element  $(\oplus_5 J_2) \oplus I_6$  — pure- $i$  IS A DEPLOYMENT REPRESENTATIVE, NOT A COMPILER THEOREM; no compiler-side demand distinguishes the  $\theta_S$  sheet-swap frame within the 9-dim rule-gauge algebra (five demand classes all silent on the angle), the strict-collar RP Grams transform EXACTLY covariantly under the rule gauge (RP of a (frame, wiring) pair depends only on the relative angle on the  $\delta$ -circle), and the same deployed parent that fails strict collar RP at  $\theta_S$  (defect  $2t$ ) PASSES it exactly in the integer-conjugated frame: the rounds-58/59 obstruction is a FRAME statement, not a wiring no-go; the canonical sentence is frozen (sha 79950314) and “pure- $i$  is compiler-forced” is henceforth FORBIDDEN (equivalents without the frame qualifier included); the v898/v903 [O] premise unmoved; NEW COMMITTED DATA ARTIFACT: verification/verified\_zeros\_n7000.json (7000 mpmath dps-15 ordinates with disclosed in-module rebuild path and in-run wards); papers (tfpt\_prime\_front new subsection “The finite wall closure and the transfer law” + the rounds-60–63 open-fronts forward link, tfpt\_research\_contracts the wiring-selector block OPEN  $\rightarrow$  CLOSED + the rounds-64–71 bridge + the frame qualifier on the v908 block, tfpt\_1 Wick keybox dated addendum rounds 69–71, tfpt\_3 four audit cards updated/added) and website mirrors synced (prime-front feed round entry, DAG, counters 901  $\rightarrow$  904 / v535–v911, papers.ts, README, Zenodo, the repro-ducer now ships the zero cache); NO marker moves; NO RH claim anywhere)

2026-08-11 (LXXXIX) (reflux rounds sixty through sixty-three — the certificate round: the B-half of the wall end-form CLOSES as a certified surface theorem over the IDEAL source objects, the tail gets a mechanism map of five dead routes with named seats and two structural positives, the half-gap target is REGISTERED with a frozen constant and passes its FIRST BLIND

HOLDOUT, and the round-60 seam questions close in equilibrium — no NESS needed: FOUR modules (v905–v908: B-floor ideal certificate, tail cartography, half-gap registered target, seam equilibrium wiring; suite 897 → 901; TWELVE frozen probes with all fail-first spec amendments preserved, embedded byte-exact and executed verbatim, plus two round-61 frame probes cited at docstring level; ledger 1040 → 1052 with 12 promoted rows + dated notes on the v903 ness-parent demand and the rounds-57–59 certifiable-B-floor demand) — (1) THE CERTIFIED IDEAL-OBJECT SURFACE FLOOR (v905, 21/21+25/25, bfloorpg-measured (certified-surface-floor-achieved(min  $c_B=0.5914$ )) + bfloorpgival-measured (ival-surface-floor-achieved(min  $c_B=0.5523$ , 0 refused / 0 failed / 0 skipped)), PRIME.PORT.BFLOOR.PG.01 + PRIME.PORT.BFLOOR.PG.IVAL.01): the chain  $B \succeq \frac{1}{2}P_G + c_{\text{dom}}I$  and  $P_G \succeq c_GI$ , every inequality decided by EXACT-RATIONAL LDL (the v897 certificate class) on all 39/39 reachable steps —  $P_G$  is Gershgorin-POSITIVE (the positive chain is classically certifiable; the single non-classical link is the dominance decision; after three dead certificate rounds the missing piece was the CERTIFICATE CLASS, not a new congruence) — and, via the SPEC-v3 interval rollout (full pipeline as rigorous outward enclosures, fail-closed with first-class refusals, Ogita–Rump error-free-transformation hi-tier certifying exactly the six former width-refusals), over the IDEAL source objects: min  $c_B = 0.5523$  on 39/39 (93.4% of the float reference),  $\tau$ -screen PASS (−0.226), the smooth world REFUSED by the identical exact machine (0/35); HONEST SCOPE typed everywhere: a SURFACE statement —  $h$ -uniformity beyond  $h \sim 900$  open, all  $h$  beyond the surface open, the  $n > q$  half untouched and RH-hard; (2) THE TAIL MECHANISM MAP (v906, 14/14 + 10/10 + 20/20 + 22/22 + 24/24 + 22/22 from SIX frozen probes, tailsign-measured + tailrepair-measured + pgschur-measured + tailabel-measured + segsplit-measured + oscpair-measured, PRIME.PORT.TAILED.01/.02 + PRIME.PORT.PG.SCHUR.01 + PRIME.PORT.TAIL.ABEL.01 + PRIME.PORT.TAIL.SEGSPLIT.01 + PRIME.PORT.TAIL.OSCPAIR.01; typed explicitly mechanism/negative cartography, NOT progress): FIVE dead routes with named seats (the tail sign is NET-ONLY —  $\sup q_v$  positive beyond every cut, the terminal lobe positive 67/67; the pointwise repair is SET-DIFFUSE and grows as  $X^{0.89}$  with an anti- $\tau$  head; the directional Löwner route loses TWO orders of magnitude by the floor-vs-inverse mechanism ( $q \leq \bar{q}$  exact 39/39 but  $n > \bar{q}$  dies 38/39, loss med 91.3); the cumulative/Abel route is exact ( $\psi$ /Riesz identities warded) but its honest integer envelope leaves the gap law  $e^{+1.744\alpha}$  — the measured hole is the OSCILLATION, the true remainder itself  $\sim 10^4\mu_1$ ; the segment split is COSTLIER than global (gain  $< 1$  on 67/67, the ANCHOR PRICE; the named missing object: a DIFFERENCE envelope for  $\psi(y) - \psi(x) - (y - x)$  on short deep windows, Brun–Titchmarsh/Selberg class)) and TWO structural positives (headfloor-o1: the entire  $h$ -decay of the wall margin lives in the atom tail, the open statement collapses to the one-sided bound  $\text{tail}_B > -\text{head}_B$  between two  $O(1)$  objects; the 12-BIN REDUCTION: a tiny low-frequency window-harmonic carrier census, measurably NOT  $\gamma$ -concentrated at these depths, reduces the tail bound to  $\sim 12$  windowed  $\psi$ -fluctuation Fourier coefficients — honestly typed: the reduced margin carries the FULL decay, dimension reduced, difficulty not); (3) THE REGISTERED HALF-GAP TARGET + THE FIRST BLIND HOLDOUT (v907, 17/17 + 14/14, halfgap-registered (reg-frozen(1/2, no-adjust) / surface-pass(67/67, min margin +2.48e-03) / origin-open(4)) + deepholdout-scored (halfgap-holdout-pass(28/28, min margin +0.2232) / law-continues(−1.925)), PRIME.PORT.HALFGAP.01 + PRIME.PORT.DEEP.HOLDOUT.01; ledger-typed a FALSIFICATION INSTRUMENT, explicitly NOT evidence

for RH):  $n_h - q_h \geq \frac{1}{2}\mu_1(h)$  registered on the 67-rung surface with the constant EXACTLY  $\frac{1}{2}$  frozen under an explicit NO-ADJUST clause that is part of the object (registry sha ae292e55; four candidate origins of the  $\frac{1}{2}$  typed OPEN — an algebraic derivation is demanded before any claim upgrade; reparam-declared: the inequality is the wall margin reparametrized, its value is falsifiability, not novelty); the first genuine blind holdout PASSES on 28 new deeper faithful rungs ( $h$  1219–2854, bit-exact table fidelity,  $\sim 90\times$  margin headroom, NO depth erosion, the honest cut-ladder drift disclosed; B-floor/ $P_G$  persist at declared float level; the margin law  $-1.925$  continues over 95 rungs; controls discriminate 3/3) — a pass is survival of a falsification attempt, never evidence; (4) THE SEAM EQUILIBRIUM WIRING (v908, 22/22 + 25/25, exact arithmetic, margstab-measured (interior-empty / stratum-not-isolated / escape-noncovariant(price  $2\varepsilon$ ) / floor-exchange) + nessparent-measured (price-zero / two-seat-law / finite-ness-nogo / drive-rp-neutral / ness-not-forced), SEAM.CFIN.MARGINAL.STABILITY.01 + SEAM.CFIN.NESS.PARENT.01 closed NOT-NEEDED): strict 2-cycle RP is IMPOSSIBLE on the whole  $C_6$ -covariant class (the seat law reduces exactly to  $\text{diag}(a_J, -a_J)$ , interior EMPTY; the only escape is covariance-breaking at exact linear price  $2\varepsilon$ , and the floor exchange  $\text{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$  puts the crossover exactly at the  $1/200$  floor); the strict-collar obstruction is a TWO-SEAT LINEAR LAW with kernel  $\{J, Z\}$ , the deployed wiring is pure-i = a maximally obstructed covariant direction (explaining the v903 ‘30 entries, magnitude  $2t$ ’ law exactly), and EQUILIBRIUM kernel witnesses  $V_J/V_Z$  carry the full canonical  $1/200$  mixing at ZERO entropy production (stationary  $\Rightarrow \sigma \equiv 0$  exactly; drive buys no RP) — the ness-parent demand closes as NOT-NEEDED and the successor contract SEAM.STATE.WIRING.SELECTOR.01 (is pure-i compiler-forced?) is registered; the v898/v903 [O] premise unmoved; papers (prime-front continuation section, research-contracts block, Paper 1 keybox addendum, Paper 3 audit cards) and website mirrors synced; NO marker moves; NO RH claim anywhere)

2026-08-10 (LXXXVIII) (numerical-robustness amendment to v772 against an upstream scipy 1.17.0 wrong-result bug + the runner-side version guard; NO gate/bar/semantics change, NO status change: the single failure of the 2026-08-10 full-suite run was root-caused to an UPSTREAM ENVIRONMENT BUG, not to the module or the theory — scipy  $\geq 1.15$  auto-detects matrix structure in `scipy.linalg.solve` and dispatches the complex-symmetric shift  $A - zI$  (real symmetric  $A$ , complex  $z$ ) of the v772 REC-d reconstruction Ward to the LAPACK ?sysv diagonal-pivoting driver, and scipy 1.17.0’s sysv path returns a WRONG solution on this matrix family (backward residual  $\|(A - zI)G - V\|$  up to  $9.6 \times 10^1$  at the two complex- $z$  cells, inflating the REC-d relative residual to  $2.6 \times 10^3$ ; real- $z$  cells unaffected; fixed upstream in scipy 1.17.1); the amendment is ONE LINE: the Ward’s independent dense solve now passes `assume_a="gen"`, explicitly pinning the LU (gesv) driver that WAS the unconditional default when the module was frozen (identical semantics, available on every scipy  $\geq 1.0$ ; under numpy 1.26/scipy 1.13 the identical call already used LU and the gate read  $5.2 \times 10^{-13}$ ) — all frozen bars, ladders,  $z$  points, preregistered fails (CAUCHY-6/GAP-7/GAP-8 part 1; EDGE-A part 2) and both pattern gates are byte-identically unchanged, REC-d stays a genuine two-path Ward (eigh route vs independent LU solve), and the amendment is fully disclosed in the module docstring; verified green under BOTH python 3.9.6/numpy 1.26.4/scipy 1.13.1 and python 3.14.3/numpy 2.4.2/scipy 1.17.0; the runner-side guard `scipy>=1.10,!1.17.0` added to `requirements.txt` (the affected



release excluded for anyone installing the suite; the firewalled experiments/\* sandbox pins are separate environments and untouched); website mirror byte-synced (website/public/verification/v772\_qf\_feshbach\_effective.py) and website/lib/discipline.ts regenerated; bash build.sh audit → AUDIT OK after the change; WITH THIS the full-suite run of 2026-08-10 stands at 897/897 modulo the documented environment bug — the raw run under scipy 1.17.0 was 896/897 with the v772 REC gates as the SOLE casualty, every GAP value bit-identical to the preregistration — no other module touched, no ledger row moved)

2026-08-10 (LXXXVII) (reflux rounds fifty-seven through fifty-nine — the cartography round: the wall rewritten in tangent-Schur coordinates with the FIRST substantive  $\tau$ -screen pass, the pivot/moment-matrix relocation map typed explicitly as mechanism-not-progress, the seam RP/modular EXCLUSION with two registered dead readings, and the 12-bit health word typed a pure diagnostic: FOUR modules (v901–v904: tangent-Schur/B-floor, wall relocation map, seam RP exclusion, healthcode12 diagnostic; suite 893 → 897; ELEVEN frozen probes with all fail-first spec amendments preserved, embedded byte-exact and executed verbatim; ledger 1027 → 1040 with 13 promoted rows + 1 dated RP note on SEAM.CFIN.KSMIX.01) — (1) THE TANGENT-SCHUR SPLIT (v901, 26/26 + 24/24 + 18/18, tangentschur-measured (safeblock(minB=0.679) / gap(min=0.052, med=0.888)) + bfloor-measured (bfloor-screen-pass(slope=−0.247) / certfloor-dead) + signfree-measured (decirc-achieved(ell-b)), PRIME.PORT.TANGENT.SCHUR.01 + PRIME.PORT.BFLOOR.01 + PRIME.PORT.SIGNFREE.NORMALIZATION.01): with the co-block  $B$  PD the wall update is  $PD \iff n - q > 0$  — two SOURCE-ONLY scalars (no  $\tau_{h+1}$ , no forward sign; all identities warded to  $8.4 \times 10^{-14}$ , PD equivalence boolean-exact 39/39); the co-block floor  $\lambda_{\min}(B) = 0.679$  is  $O(1)$  ladder-wide while  $\tau$  falls by factor 552 (the FIRST substantive  $\tau$ -screen pass, slope −0.247; coordinate honesty printed: the raw co-block scales like  $\tau$ , slope +0.753); the honest negative certfloor-dead: all four classical certified bounds negative on all 39 steps (best maximum −88 vs the true floor +0.679) — the floor is MEASURED, never certified; the failure-seat anatomy refutes the plan diagnosis (at all 12 v900 rank-1 failure points the co-block itself dies, the 2.5% off-mode restores it linearly, the seat is orthogonal to the critical direction); the sign-free normalization  $\ell = \det(S)^{1/8}$  de-circularizes the update coefficient — CIRCULARITY REMOVED, POSITIVITY NOT GAINED, both said explicitly; (2) THE RELOCATION MAP (v902, 22/22 + 23/23, edgechristoffel-measured (rewrite-exact / ratio-below-one(0.999950)) + gramradau-measured (m0-identity-exact(4.9e-13) / gram-completion-fails(x2:42, m:42, p:0) / radau-exact(3.8e-15) / locwall-seen(42/42)), PRIME.CASE.EDGE.CHRISTOFFEL.01 + PRIME.PORT.GRAM.COMPLETION.01 + PRIME.PORT.RADAU.WEIGHT.01; typed explicitly a relocation/mechanism map, NOT progress): the exact positive-weight rewrite  $1/d_{12} = 1 + \sum_m W_{h,m} Q(x_m)^2 - \beta$  ( $W > 0$  exactly at  $9.7 \times 10^{-14}$ ,  $\beta \geq 0$  termwise, independent slogdet route on all 37 rungs) with the relocation diagnosis  $1 - r \propto \tau$  (slope +1.008; the uniform  $q < 1$  target is the wall's own PD premise QUANTIFIED, no independent positivity source; the Rayleigh fence 147× tighter than the classical trace bound); the wall IS the moment matrix of the signed comb measure ( $\lambda_{\min}(M_0) = \tau$  at  $5.2 \times 10^{-9}$ ) but the  $W \succeq 0$  Gram completion fails on 42/42 rungs at the  $x = +1$  edge where the comb nodes accumulate (violation magnitude to −152× the margin: NO source-only Gram representation at matched degree); the Gauss–Radau

construction at the classical node is exact algebra ( $w^*$  IS the Christoffel function, med 0.9978, a pure  $\mu_+$  object; the P2 pivot is NOT a Radau weight); (3) **THE SEAM RP/MODULAR EXCLUSION + THE TWO DEAD READINGS** (v903, 25/25 + 23/23 + 20/20 + 20/20 + 23/23 + 18/18 from SIX frozen probes, seamstate-measured + seamnorm-measured + mediator-measured + twistedrp-measured + dilation-split + gappencil-measured, SEAM.STATE.DERIVATION.01 + SEAM.CFIN.MIXING.NORMALIZATION.01 + SEAM.CFIN.MINIMAL.MEDIATOR.01 + SEAM.CFIN.TWISTED.RP.01 + SEAM.CFIN.RP.DILATION.01 + SEAM.CFIN.GAP.PENCIL.01): strict reflection positivity forces  $t = 0$  under BOTH deployable spatial reflections (the odd-sector Gram eigenvalues are EXACTLY  $\pm|a_J|$  of the  $\{4, 5\}$  duad at  $2.2 \times 10^{-16}$ : OS positivity  $\iff a_J = 0 \iff$  the v898 mixing floor falls — REFLECTION POSITIVITY AND THE MIXING FLOOR ARE MUTUALLY EXCLUSIVE on this family);  $u \geq t$  forced ( $u = 0$  RP-excluded, the round-57 reading corrected); the  $2\pi$ -KMS locus of the boundary reduction is EXACTLY the point ( $t=0, \beta=2\pi$ ); the complete twisted-OS census is exclusionary ( $48 \rightarrow 16$  involutive  $\rightarrow 6$  admissible  $\rightarrow 0/6$  with strict RP at  $t > 0$ ); the dilation route SPLITS (the v898 M2 parent is a  $\theta_{abT}$ -marginal witness ON the RP cone boundary whose Schur compression carries the full Pfaffian mixing with the exact rational identity  $t^2 \cdot 3m/(1 - m^2) = 1/200$ ; the strict collar route family-wide obstructed by an exact linear law, 30 defect entries all of magnitude  $2t$ ); the gap pencil is EXACT:  $\text{Pf}(A_{\text{dep}} + tA_{\text{int}}) = -(t-1)(3t^2-1)(9t^3+21t^2-t-1)$  with Sturm uniqueness on  $(0, 1/4]$ , kernel dimension exactly 2, the three critical surfaces projections of ONE variety, and NO root at  $1/8$ ; **THE TWO DEAD READINGS** as first-class negative ledger rows: ‘ $t = 1/8 = 2\pi c_3/|Z_2|$  is a compiler value’ is a DECORATION (55/55 scanned  $t$  pass all frozen v898 gates, allowed set  $(0, \infty)$ , pin-absent) and ‘ $N_{\text{fam}} = 3$  as minimal boundary-mediation rank’ is REFUTED (rank 1 suffices by the exact sign law + integer witness; the deployed mediator has rank 2 EXACTLY; what 3 pins is boundary purity = dimension bookkeeping); the [O] premise of SEAM.CFIN.WICKFUNCTOR.01/KMSMIX.01 is UNMOVED (rp-theta-open now carries the dated measured-exclusionary note); (4) **THE HEALTH WORD** (v904, 18/18, healthcode12-measured / diagnostic-only / truerword-allplus / softpivot-12(0/37) / v897-pivot10-reproduced / smoothclaim-refuted, PRIME.ARITH.HEALTHCODE12.01; ledger-typed DIAGNOSTIC, ZERO RH CONTENT in either direction): the true word is all-(+) on 37/37 (the certified Sylvester census repackaged, no new content); three plan claims corrected (the 12 factors live in the v899 12-window compression per rung, not in v897; the softest raw pivot is  $k = 2$ , not  $k = 12$ ; smooth tips flags  $\{2, 1\}$  first and never flag 3); the one surviving claim reproduced byte-exact (the v897 Epstein control refused at pivot index 10); the seeded syndrome table distinguishes four adversarial fingerprints — a classifier of manipulated combs, a tool, not a target; ALSO: the two parallel-session diary entries renumbered final (CIII  $\rightarrow$  CXXV, CIV  $\rightarrow$  CXXXIV; zero duplicate numerals among the 2026-08-10 notes); papers (tfpt\_prime\_front new Section “The tangent-Schur coordinates and the relocation map” + the part-intro chain extended, tfpt\_research\_contracts the rounds-57–59 registration blocks + the dated RP note in the v898 block, tfpt\_1 Wick keybox dated RP addendum, tfpt\_3 five audit cards + the dated note on the v898 card) and website mirrors synced (prime-front feed round entry, DAG +v901–v904, counters 893  $\rightarrow$  897 / v535–v904, README + Zenodo to 897/v904); NO marker moves; NO RH claim

anywhere — every relocation typed as such, nothing gained positivity anywhere)

2026-08-10 (LXXXVI) (website/video maintenance — the /prime-front page brought up to the round-56 state and compressed, and the explainer film refreshed; NO theory change, NO marker moves, NO RH claim: (1) PAGE CURRENT — the stale hero counters replaced by single-source constants (suite scripts now read from the generated lib/suite.ts, the diary-run count computed from the feed data itself: 285 entries), the honesty banner and big-picture statistics interpolate the same sources (span v535–v900 inside the 893-script suite), the stale “314 probes” qualifier dropped from the sandbox-check statistic (kept as the dated lower bound 6638+); (2) PAGE COMPRESSED — the rendered text payload reduced from ~445 KB to ~182 KB and the built HTML from 1.30 MB to well under 0.9 MB with ZERO loss of honesty content: the four very long “preserved verbatim” status recaps (the T102–T176 sprint recap of the honesty banner, the end-of-series paragraph, the “Near” meaning-level paragraph, and the phase-2 full-proof narrative) moved word-for-word into a lazily loaded module (verbatimRecaps.tsx) behind collapsed on-demand expanders (VerbatimRecap.tsx), each replaced in place by a compact honest summary that keeps every fence (“Sandbox; not RH evidence”); the diary feed now mounts full entry texts only on expand, and the inline feed window returned to its declared ~16 newest entries (44 older entries moved verbatim to the lazy archive, 225 → 269); (3) THE FILM REFRESHED — the one stale narration line of the 2026-08-03 explainer updated to the current suite state (“Almost nine hundred modules”; on-screen stat now “893 modules · machine-checked · 84 Lean files”), full TTS/captions/render re-run through the frozen frame windows (still 2:39, burned-in English captions, en+de subtitle tracks rebuilt from the same source; the component transcript verified character-identical to the caption track; JSON-LD uploadDate 2026-08-10); all other narration, timings and claims unchanged — the film still states “no proof of RH” explicitly)

2026-08-09 (LXXXV) (reflux rounds fifty-four through fifty-six — the reviewer’s “immediate” list delivered: the certificate rollout becomes INTERVAL-RIGOROUS (the informal  $\varepsilon_c$  lag-evaluation error model is RETIRED from the ladder), the channel-mixing state the block Wick functor awaits is CONSTRUCTED (KMS + exact Schur generation), the conditional diagonal contract becomes a NORM SQUARE in the frequency weights, and the normalized core update gets its exact two-dimensional anatomy with the honest negatives registered: FOUR modules (v897–v900: certified interval ladder, KMS/Schur mixing, Christoffel norm-square, core update anatomy; suite 889 → 893; EIGHT frozen probes with all fail-first spec amendments preserved — v898–v900 embedded byte-exact and executed verbatim, v897 embedding the 66-minute ladder probe byte-exact with SHA wards, re-certifying all 15 tier-1 + 3 tier-2 rungs verbatim in-suite and freezing the full 42-rung census; ledger 1020 → 1027 with 7 promoted rows + 4 dated notes on PRIME.PORT.TAIL.01 / PRIME.PORT.CERTIFIED.LADDER.01 / PRIME.CASE.PAIRCORR.CONTRACT.01 / SEAM.CFIN.WICKFUNCTOR.01) — (1) THE CERTIFIED INTERVAL LADDER (v897, probe verdicts ballhead-proven + ballladder-complete 10/10, 3960 s, PRIME.PORT.BALLLADDER.01):  $\sigma_h > 0$  with RIGOROUS mpmath.iv interval-arithmetic shifts on ALL 42 reachable rungs  $h = 142\text{--}878$  — 15 exact-rational integer Bareiss/Sylvester certificates ( $h \leq 300$ , floors  $4.5 \times 10^{-5}\text{--}2.1 \times 10^{-4}$ ) + 27 validated-precision Cholesky dps 120/200 certificates (floors  $5.2 \times 10^{-6}\text{--}8.4 \times 10^{-5}$ , pivot/bound ratios  $2.3\text{--}9.9 \times 10^{115}$ ); the GL-48 node-enclosure lemma (radius  $10^{-90}$ , definite sign changes, disjoint intervals, 2

in the interval weight sum), native outward-rounded interval transcendentals, exact midpoint/radius extraction, the rigorous shift  $2h \text{ rad}_{\max}$  ( $+h/Q$  on the tier-1 grid) — no working-precision rounding in the tier-1 decision path; the Epstein control REFUSED at pivot 10 by the identical interval machinery; the informal conservative  $\varepsilon_c$  model of v884/v887 is RETIRED; the validated-precision tier label carried honestly, never upgraded; remaining named: the Lean composition (the asymptotic tail untouched, PRIME.PORT.TAIL.01); (2) THE CHANNEL-MIXING CANDIDATE (v898, 30/30, kmsmix-measured [kmsmix-found(u=1, t=1/8,  $\beta=1$ ), schurmix-generates(10/10), rp-theta-open], SEAM.CFIN.KSMIX.01 + SCHURMIX.01): the  $C_6$ -covariant channel-mixing KMS state EXISTS — the first point of the frozen grid passes ALL block-functor gates (CAR strict, exact covariance with the forced zeros, all 15 duad cross-blocks at the floor, grading  $5/5 + 10/10$ , all 15 block Wick monomials with the canonical  $\text{Pf}_4$  signs of the v896 functor; robust at  $\beta = 2$ ; the hermitian commutant measured exactly,  $112 = 62 + 50$  with the 33-dim mixing slice); EXACT (sympy): the Schur/Feshbach elimination of the boundary gives  $A_{\text{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^T$  and GENERATES all 10 carrier duads from the bare diagonal state (uniform  $3J$  block, canonical signs) — on both fronts (RH port, Wick functor) the real correlation lives in the ELIMINATED boundary; the frozen 15-signature table is the registered prediction; rp-theta-open the named open check; the [O] premise UNMOVED; (3) THE NORM-SQUARE CONTRACT (v899,  $19/19 + 15/15 + 19/19$ , chrstratio-measured (winner-christoffel / extsource-decoupled(0.454)) + kernelsos-measured (index-large; rmin=1; not-absorbed) + edgedefect-measured (normsquare-achieved; kill=periodic; chain=carried(0.21)), PRIME.PORT.CHRISTOFFEL.RATIO.01 + PRIME.CASE.KERNEL.SOS.01 + PRIME.CASE.EDGEDEFECT.01): the EXACT deflated-Christoffel pivot identity  $1/d_{12} = 1 + v_* K_\sigma(y_*)$  (independent route  $2.0 \times 10^{-11}$ ; the winner identification by two orders of magnitude) with the honesty fence  $\lambda_{\min}(I - G) = \tau$  EXACT (a coordinate change, NOT an independent positivity source);  $c'/c$  SENS  $[0.223, 5.258]$ ,  $r_{12} > 0$  on 31/31; the pair-contract kernel's frequency weights  $W_{f,m} \geq 0$  EXACTLY with the ENTIRE negative content collapsing to ONE top-edge tent (closed-form boundary leakage,  $r_{\min} \leq 1$ ), and the PERIODIC full-weight fold KILLS it exactly — negative index 0 everywhere, 79/79 modified margins positive, the kz-9-alias-9 absorption failure  $1.381 \rightarrow 0.000$ , the boundary term  $\beta$  carried closed-form (max 0.21) — the conditional diagonal contract of v894 is a NORM SQUARE in the frequency weights (reviewer case 1 at the finite level); the arithmetic hypothesis itself REMAINS CONDITIONAL (pair-correlation class per v889); (4) THE CORE UPDATE ANATOMY (v900,  $20/20 + 23/23$ , normcore-measured (family-bounded(2533) / update-exact(udim=2) / region-open) + graphregion-measured (coupling-free / graph-region-open / shadow-lawless), PRIME.PORT.NORMALIZED.CORE.01 + PRIME.PORT.GRAPH.REGION.01): the normalized deep core obeys  $X' = c(X + U)$  EXACTLY ( $8.3 \times 10^{-16}$ ) with  $c = \tau/\tau'$  and an effectively TWO-DIMENSIONAL input (one matrix mode at 97.5% energy); the honest negatives REGISTERED as the boundary conditions of any invariant-region theorem: no linear coupling law (both  $R^2$  BELOW the 200-shuffle null  $q_{95}$ ), the rank-1 truncation fails (the positivity is carried by the 2.5%-energy off-mode), naive product/graph invariant sets refused, the 2D shadow lawless — any region construction must keep the exact  $X + U$ , truncating no mode; papers (tfpt\_prime\_front new Section “The certified interval ladder and the norm-square contract”, tfpt\_1 Wick keybox extended with the constructed candidate, tfpt\_3 audit cards, tfpt\_research\_contracts round-54–56 registrations + dated notes) and website

mirrors synced; NO marker moves; NO RH claim anywhere — the asymptotic tail and the one-sidedness stay open exactly as before)

**2026-08-09 (LXXXIV)** (reflux rounds fifty through fifty-three — the closure architecture: the exact hermitian congruence with a NON-DECAYING inheritance margin, the flag-chain induction, the honest Rouché kill, the  $8 \times 8$  reduction (the wall IS the fixed Schur core to seven digits), the explained margin scale, the assembled theorem skeleton with ONE bounded-margin statement open, the concrete conditional diagonal contract, the collectivity theorems, and the Wick functor arc resolved at block level: FIVE modules (v892–v896: closure architecture, relative margins, diagonal refinement, collective comb, Wick block functor; suite 884  $\rightarrow$  889; FOURTEEN frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 1012  $\rightarrow$  1023 with 10 promoted rows + dated skeleton updates) — (1) **THE CLOSURE ARCHITECTURE** (v892, 16/16 + 15/15 + 14/14 + 16/16): the exact congruence  $A_{h+1} = A_h^{1/2}(I + H_h)A_h^{1/2}$  ( $6.5 \times 10^{-14}$ ,  $H$  exactly hermitian) with the inheritance margin  $\eta_h = 1 + \lambda_{\min}(H_h)$  NON-DECAYING (min 0.0050, median 0.29, slope +0.108 while  $\tau$  collapses at  $-2.74$ ) — the open inequality is bounded-margin-shaped, not collapsing;  $\rho(W^{-1}\Delta C)$  was a misleading statistic (non-normal,  $\|M\|/\rho$  up to 26) — the hermitian  $H$  is the honest carrier; ALL 12 LDL flag quotients  $Q_k > 0$  on all 31 truth steps (soft-flag min  $2.06 \times 10^{-14}$ , sign-decidable at  $6.4 \times 10^9 \times$  the noise floor; the smooth world violates at the BULK flags first) — the shortest induction shape; the two-sided contour-norm (Rouché) route honestly killed ( $M \geq 1$  on 22/31 including off-cap; the argument principle itself verifies exactly); and **THE WALL IS THE FIXED  $8 \times 8$  SCHUR CORE**:  $\lambda_{\min}(S) w_{\text{core}}/\tau = 1 \pm 8.4 \times 10^{-8}$  with  $w_{\text{core}} \geq 0.999937$  on 41 rungs, conditional only on the TAU-RELATIVE exterior bound ( $\lambda_{\min}(R)/\tau$  trendless at 210–2200; the absolute margin shrinks  $h^{-2.865}$ ); (2) **THE MARGIN SOURCES** (v893, 20/20 + 18/18): **SCALE-MATCHED** — in the dangerous direction the increment and the operator collapse TOGETHER (slopes  $-3.716$  vs  $-3.456$ , residual correlation 0.974, 94 % shared variance): the non-decay of  $\eta$  is EXPLAINED (numerator and denominator are sections of ONE arithmetic object; open: only the  $O(1)$  scatter); the relative laws  $d_{12} = c\tau$  (trendless; the kz-21 outlier diagnosed as a COORDINATE-ARTIFACT in the round-53 sandbox — the kz-19 mechanism repeats, corrected spread 124  $\rightarrow$  15: SOFTFLAG TAU-TIED) and  $\lambda_{\min}(R) \geq 210\tau$ ; **COMPRESSIONS-DISTINCT** (the port bulk floor is absolute, the core exterior tau-tied); **THE PRINTED SKELETON**: base certified +  $R \geq 210\tau$  + core  $\leftrightarrow \tau$  (7 digits) +  $\eta > 0$  — the ONE unconditionally open statement is the  $\tau$ -sign inheritance in bounded-margin form; (3) **THE DIAGONAL REFINEMENT** (v894, 11/11 + 13/13 + 12/12; the route stays CONDITIONAL per v889): the exact band-limited prime kernel  $K_{h,m}$  (regrouped at  $4.5 \times 10^{-15}$ ; the LaTeX contract statement frozen in-probe; all 79 finite margins positive); the endpoint bracket ( $F'(1) \geq$  deficit on 142/227 — the global linear collapse fails, deep small- $a$  cells pass monotonically); the inner-zone accounting (scale-normalized FLAT; the hard band NON-SHRINKING at  $\sim 20\%$  of aliases / 38–56 % of mass; stability on an  $a$ -graded mid shell); (4) **THE COLLECTIVITY** (v895, 18/18 + 12/12): the true avoidance law is ONE-SIDED ( $\max(-\mu_{\min}) = 0.9950 < 1$  on all 31 steps = PD inheritance itself; no norm-bound shape); the geometry/atom split is EXACTLY degenerate (the atom cutoff is slaved to the tent edge — no separable atom channel; the increment is the collective window re-test); the deep core IS the port core (the fixed aliases  $a_m = \pi^2 m^2$  to 0.1 %; ALL FOUR surgeries kill the coherence incl. the wrong- $\Lambda$

world — the signal is the ENTIRE multiplicative von Mangoldt comb); (5) THE WICK ARC (v896,  $27/27 + 33/33 + 26/26$ ): the six compiler roles exist in the deployed seam (sector law, grading,  $C_6$  intertwining exact) but the vacuum kernel is channel-diagonal; the PROVED scalar obstruction (the  $C_6$  2-cycle  $\{4, 5\}$  forces a duad zero; both relaxation witnesses constructed); the BLOCK RESOLUTION (commutant dimension 33; the canonical block covariance carries 15/15 blocks and 15/15 block Wick monomials with the canonical sign law, interleaving character +1 symbolic) and the decisive fact SEAM-DIAGONAL (the deployed seam has zero cross-blocks — the value level awaits a physical mixing mechanism; the [O] premise unmoved); papers and website mirrors synced; NO marker moves; NO RH claim anywhere)

2026-08-09 (LXXXIII) (reflux rounds forty-two through forty-nine — the great consolidation: the certificate rollout COMPLETES ( $\sigma_h > 0$  PROVEN on ALL 42 reachable rungs), the finite Pfaffian/ $K_6$  signature CLOSES (fermionic signs gauge-matched via the canonical character; the self-hosting counting theorem forces  $g_{\text{car}} = 5$ ; the flavor hexagon is the  $\text{Aut}(C_{\text{fin}})$  orbit), the four route decisions of the diagonal half land (the wall dies 100 % through off-diagonal coherence; PNT-insufficient; the input is pair-correlation class), the surviving positivity architecture is measured (total positivity of the section family on 37/37 rungs; the wall margin IS the pole distance of the integrable  $s$ -flow; inheritance is a multi-factor statement), and the Möbius/carrier-invariance route is honestly killed with mechanisms: FIVE modules (v887–v891: certified ladder complete, finite Pfaffian signature, route decisions, total positivity flow, Möbius firewall kills; suite 879  $\rightarrow$  884; FOURTEEN frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 999  $\rightarrow$  1012 with 11 promoted rows + 2 dated route retypes) — (1) THE COMPLETE HEAD (v887,  $7/7$ , certified-ladder-complete):  $\sigma_h > 0$  on ALL 42 reachable rungs (12 exact-rational integer Bareiss/LDL + 30 validated-precision mpmath Cholesky dps 120/200 with precision doubling; the module closes  $h = 859/878$  with floors  $\geq 8.352/9.187 \times 10^{-6}$  and pivot/bound ratios  $\sim 10^{115}/10^{195}$ ; the 40-rung parent rollout stays sandbox evidence, runtime-excluded; Epstein REFUSED at pivot 10; proven modulo the declared conservative evaluation error model; ball arithmetic and the Lean composition remain named); (2) THE FINITE SIGNATURE (v888,  $27/27 + 19/19 + 42/42$ , exact integer/sympy): the  $S_6$  Specht channels  $(1, 0, 2/3)$  exact; the 15 doily lines = the 15 Plücker monomials of  $Gr(2, 5)$  with perfect sign match; the  $(16, 6, 2, 10)$  Hadamard anchor signature; the  $S_6 \rightarrow S_5$  vacuum selection (orbits  $[1, 5, 10]$  equal as sets to  $(\Lambda^0, \Lambda^4, \Lambda^2)$ ); THE FLAVOR HEXAGON IS THE  $\text{Aut}(C_{\text{fin}})$  ORBIT ( $g^2 = \sigma$  exact,  $g^3$  = the  $W$ -edge sheet swap, the 6-orbit = the deployed transport edge for edge,  $(2/3)^6$  = the frozen recovery value,  $0/600$  non-automorphisms factor); the  $K_6$  motif theorem ( $35 = 15$  perfect matchings + 10 vacuum triangles = secants + 10 vacuum-free = externals;  $q^*$  = vacuum incidence); THE FERMIONIC PFAFFIAN SIGNS gauge-matched to the Hodge table via the CANONICAL character  $\chi^{(i)} = (-1)^{i+1}$  (triple-identified: vacuum-row Laplace;  $\Lambda^4/\Lambda^5$  volume form;  $C_6$  crossing character); THE SELF-HOSTING COUNTING THEOREM  $\binom{g+1}{2} = g!!$  with  $g = 5$  the only nontrivial solution and  $N_{\text{fam}} = (g - 2)!! = 3$  ([E] theorem; the Wick-compiler premise stays [O] — P2 narrowed to ONE premise, no marker move); the mod-30 anchor clock CLOCK-DISTINGUISHED (period-4 moduli exactly  $\{5, 10, 15, 30\}$ ); the operative-Coxeter-30 reading separately DEAD (sandbox, both sides); (3) THE ROUTE DECISIONS (v889,  $8/8 + 12/12 + 10/10 + 12/12$ ): the wall dies through PURE

off-diagonal coherence (diagonal profile frozen at the kill point; separation  $> 10$  deep; honestly diag-intermediate);  $\theta^* = 0.700$  frozen (outside margin  $+0.0214$  uniform on 42 rungs; inside law  $\sim 0.87/\log^2 h$ ); pnt-insufficient (the PNT-smooth reference itself violates the port margin on every rung; the truth's positivity is the mask-mass INTERACTION;  $\varepsilon_A$  is  $6\text{--}970\times$  too large); and the signed-homotopy classification homotopy-indefinite-arithmetic (exact envelope identity at  $4 \times 10^{-14}$ ; the fixed-mask kernel is concavity-locked NSD; the diagonal route's arithmetic input is PAIR-CORRELATION CLASS — retyped from unconditional candidate to conditional reduction theorem); (4) THE SURVIVING ARCHITECTURE (v890,  $10/10 + 7/7 + 14/14$ ): TOTAL POSITIVITY of the section family (all 12 leading minors strictly positive on all 37 full-window rungs; Desnanot–Jacobi exact); the POLE-FREE  $s$ -corridor with  $s_h^* = 1/\lambda_{\max}(C_h)$  — the wall margin IS the pole distance of the integrable  $s$ -flow (slope  $-2.90$ ); the multi-factor determinant anatomy ( $Q_h = \prod(1 + \mu_i)$  exact; the truth keeps EVERY factor off  $-1$  while the smooth-mass world crosses on  $22/28$  steps and flips  $Q$  13 times; the one-scalar Möbius bridge honestly closed bridge-diffuse:  $|\alpha| \sim 10^{-3}$  is determinant-inert) — POSITIVITY INHERITANCE, IF IT EXISTS, IS A MULTI-FACTOR/TOTAL-POSITIVITY STATEMENT; (5) THE KILL BATTERY (v891,  $10/10 + 13/13 + 7/7$ ): the round-CVI carrier-invariance was a NORMALIZATION ARTIFACT — fit-free gauge-free cross-ratios CR-DEAD (pooled median  $0.128$ ; anchors disagree; no cocycle), source-frozen gauges show REAL motion (GAUGE-MADE; the scalar cocycle does not track the wall, corr  $0.129$ ; Jacobi dead), and the six-world census shows the arithmetic separates AT THE FRAME (every control world including the smooth ones loses window subcriticality; scramble passes the Möbius bars BETTER than truth — rank-2 kinematics); the surviving arithmetic remnant is the deep-core 8-node cross-ratio coherence ( $4.3 \times 10^{-4}$  vs smooth  $2.7 \times 10^{-2}$ ) — prime-mass-driven, the named object; related sandbox kills recorded (static Cauchy identification dead; the Cauchy route can never certify:  $\text{dev}/\tau$  grows at slope  $+1.46$ ); papers and website mirrors synced; NO marker moves; NO RH claim anywhere — the hardness decision stands: the open one-sidedness carries full RH weight)

2026-08-09 (LXXXII) (reflux rounds forty/forty-one — the flow corridor of the prime front: the determinant chain to the tau function lands with the PARAMETRIX DECISION (the Euler-product comb is GLOBALLY rigid — every smoothing-based parametrix is dead), the FIRST RIGOROUS POSITIVITY STATEMENT of the program (exact integer Sylvester certificates  $\sigma_h > 0$  on the head rungs), the complete flow/induction anatomy of the ladder (degree-2 Lax closure; NO monotone one-prime induction in any insertion order — positivity is a property of the COMPLETE comb), and the register-gapcode DISENCHANTMENT row: FOUR modules (v883\_tau\_chain\_parametrix.py, v884\_certified\_head\_positivity.py, v885\_ladder\_flow\_anatomy.py, v886\_register\_gapcode\_null.py; suite  $875 \rightarrow 879$ ; NINE frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger  $989 \rightarrow 999$  with promoted rows, the decided design constraints, and the open-contract registrations PRIME.PORT.LEADING.SIGN.01 + PRIME.PORT.TAIL.01) — (1) THE TAU CHAIN AND THE PARAMETRIX DECISION (v883,  $5/5 + 3/3 + 3/3$ , tau-chain-exact + parametrix-measured (fluctuations-required) + zones-measured (both-required)): the Schur determinant identities  $\sigma_h = \det(I - E)/\det(B)$  and  $\det(I - E) = \det(I - R)\det(I - D_P)$  are machine-exact ( $4.8 \times 10^{-13}$  in log space) with positive bulk factors, hence



$\text{sign } \tau_h = \text{sign } \sigma_h = \text{the wall with } \tau_h := \det(I - D_P)$ , and the Jacobi variational identity  $\frac{d}{ds} \log \det(I - sD_P) = -\text{tr}[(I - sD_P)^{-1}D_P]$  holds on the true family; the  $\kappa/\rho$  readout:  $\text{slope}(\log \sigma) = -2.93$  vs  $2 \times$  the Mellin fluctuation slope  $-2.53$  ( $\sigma \sim \text{fluctuation}^2$  within 16 %); **THE DECISION**: the prime-power lattice with fully smooth quadrature masses ( $\tau = -78$  to  $-1706$ ) and even with EXACT local  $\pm \frac{1}{2}$ -log-unit mass sums ( $-78$  to  $-5537$ ) is **ORDERS** worse than the plain continuum ( $-0.11$  to  $-1.26$ ) against the true comb ( $+1.7 \times 10^{-4}$  to  $+6.7 \times 10^{-7}$ ), and the zone refinement kills **BOTH** ways (edge-only smoothing  $-3.7$  to  $-22.1$ ; interior-only  $-77$  to  $-7551$ ): the exact  $\Lambda$ -mass-position pairing is load-bearing **EVERYWHERE** — the multiplicative Euler-product structure is globally rigid, the unperturbed object of any RH analysis **MUST** carry the full von Mangoldt comb, and every small parameter lives in the flow/deformation direction; (2) **THE CERTIFIED HEAD** (v884, 5/5, certified-head):  $\sigma_h > 0$  **PROVEN** on kz 9 ( $h = 184$ ), 12 (151), 13 (168) — independent mpmath rebuild of the window lags (dps 60/100, conservative error model  $\varepsilon_c = 10 \times$  dps-disagreement, cross-ward  $1.1 \times 10^{-14}$  which **CAUGHT** a sign bug in the rebuilt Arch near-branch), matrix shift  $\sim 2 \times 10^{-18}$  ( $\sim 14$  orders below the margins), then EXACT INTEGER Bareis-s/LDL on the  $Q = 10^{20}$  grid: all pivots positive (Sylvester)  $\Rightarrow K \succ 0 \Rightarrow \sigma_h > 0$  via the exact v866 equivalence — no Lanczos, no floating point in the decisive step; certified floors  $\lambda_{\min} \geq 3.633/3.327/3.842 \times 10^{-4}$ ; the Epstein control through the **IDENTICAL** machinery is **REFUSED** at pivot index 10; typed honestly: proven **MODULO** the declared conservative transcendental-evaluation error model (validated-numeric standard), the ball-arithmetic closure and the 42-rung rollout are the named follow-ups — the head part of PRIME.PORT.TAIL.01 is delivered on three rungs; (3) **THE FLOW/INDUCTION ANATOMY** (v885, 8/8 + 5/5 + 13/13 + 11/11, isoflow-measured (flow-open) + ladder-measured (perturbative + non-monotone) + primeladder-measured + lax2-measured (flow-noisy / lax2-partial)): the gauge-fixed IKS generator flow closes in **DEGREE 2** (residual medians 0.244/0.248 with  $\{Y^2f, Y^2g\}$ ; degree 1 fails at 0.498/0.457) — an isomonodromy-compatible Lax structure, typed as structural precondition (controls close too; the arithmetic stays in the **VALUE**); the window-ladder step is marginally perturbative (accuracy median 0.458) but **NON-MONOTONE** in raw  $h$ ; the **ONE-PRIME LADDER DECIDES**: the pure-Arch start is catastrophically subcritical ( $\tau(0) = -9.9 \times 10^{13}$  at kz 9;  $G_+$  not even PD deeper), the last sign crossing sits at  $u^*/U = 0.991\text{--}0.999$  in **EVERY** insertion order (ascending needs 97–99 % of the atoms, edge-first needs 100 %), per-atom increments only 54–64 % positive: **POSITIVITY IS A PROPERTY OF THE COMPLETE COMB** — v877's truncation reversal sharpened to atom resolution, no monotone one-prime induction exists, induction must run in a flow coordinate, not in comb content; and the raw degree-2 coefficient basis is ill-conditioned **NOISE** (roughness 1.365 vs 0.7; operator predictivity 0.350 vs 0.3; the integrated tau-transfer identity itself warded **EXACT** at  $5.6 \times 10^{-11}$ ) — the conditioned flow coordinate is the named open object (PRIME.PORT.LAX2.01); (4) **THE GAPCODE DISENCHANTMENT** (v886, 24/24, gapcode-measured, an honest **NEGATIVE/limit** row per the frozen probe's own typing): the error-code reading of the 210-register is **LITERALLY TRUE** on the prime side (the register bits  $\{2, 3, 5, 7\}$  are the first four parity checks of the sieve; code rate 48/210; hard floor  $g_k \geq w(p_k \bmod 210)$  with 0 violations in  $5.76 \times 10^6$  pairs;  $I(\text{state}; \text{gap}) = 1.14$  bits) — **AND** the code is **THE SIEVE**: a zero-parameter wheel–Cramér null explains  $R^2 = 0.99839$  of the state profile, the arithmetic excess is 4.02 % and shrinking, no knee at four checks, 210 sits **ON** the foreign-wheel trend

( $z = -0.61\sigma$ ), all eight anchor Walsh seats below  $2.5\sigma$  exactly as CRT predicts: everything the register knows about gaps, every 4-prime wheel knows — nothing moves with  $c_3$  or  $\varphi_0$ ; the register's uniqueness remains the arithmetic forcing  $\text{rad } |W(E_8)| = 7\#$  with  $\prod(p-1) = 48 = \Omega_{\text{adm}}$ ; papers: the prime-front rounds-40/41 section and contract registrations; website mirrors and generated surfaces synced; NO marker moves; NO RH claim anywhere — the demand-curve decision stands: the open one-sidedness carries full RH weight)

2026-08-09 (LXXXI) (reflux rounds thirty-eight/thirty-nine — the port compression of the wall: the four finite anchor theorems land, the Carleson wall becomes the EXACT operator geometry of ONE dressed port block (gauge, testing, Schur, integrable class), and the port source gets its universal law (the  $\sqrt{n/X}$  uniformization, the exact  $1 - e^{-1/2}$  edge mass, the Mellin–Cauchy kernel, the closed criticality budget): THREE modules (v880\_finite\_anchor\_theorems.py, v881\_carleson\_port\_geometry.py, v882\_port\_source\_mellin.py; suite 872  $\rightarrow$  875; EIGHT frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 972  $\rightarrow$  989 with 11 promoted rows + 6 open-contract registrations) — (1) THE FINITE ANCHOR THEOREMS (v880, 26/26, finite-anchor-closed): the closed census-free normal form  $q^*(x) = \sum_{i < j} x_i x_j + \sum_i x_i = \binom{|x|+1}{2} \bmod 2$  (weight-only; the v845 selector FORCES the linear part  $(1, 1, 1, 1)$  — the  $2^{16}$  search replaced by a theorem); the EXCLUSIVE anchor classification of  $PG(3, 2)$  lines (all 15 doily lines carry the weight multiset  $\{2, 1, 1\}$  = the anchor  $a = (1, 1, 2)$  with per-line Vieta  $(4, 5, 2)$ ,  $e_2 = g_{\text{car}}$ ,  $e_3 = |\mathbb{Z}_2|$ ; the 20 others split EXACTLY as 10 secants = the WRONG anchor  $(1, 2, 2)$  + 10 external lines — the anchor law characterizes the doily); the doily as the MULTIPLICATION TABLE  $\Lambda^2 E \times \Lambda^2 E \rightarrow \Lambda^4 E$  of the five-slot carrier (wedge nonzero EXACTLY on doily lines;  $5 \times 3 = 15$  factorizations); and QUADRATIC ONE-STEP EXHAUSTION forces  $g_{\text{car}} = 5$  uniquely among odd  $g$  — the theorem [E], the Calderón premise explicitly [O]: P2 narrowed, NOT eliminated, no marker move; (2) THE OPERATOR GEOMETRY (v881, 11/11 + 5/5 + 8/8 + 5/5, pick-scalarized + testing-identified + port-schur-exact + integrable-confirmed; four sides of ONE object, counted once): the review's literal rank-2 Pick target  $I - C^*C = \Gamma^*\Pi\Gamma$  is REFUTED as stated and replaced by the exact rank-2 JACOBI DISPLACEMENT  $[J, \Delta] = b_h(e_{h-1}r^\top - re_{h-1}^\top)$  (rel  $\leq 4.3 \times 10^{-14}$ , source-only, blind holdouts included); the GAUGE identity  $E = S(\sqrt{\tilde{\nu}} K_{\text{CD}} \sqrt{\tilde{\nu}})S$  — the wall operator is ORTHOGONALLY EQUIVALENT to the plain Carleson Gram, its diagonal the CLASSICAL testing functional  $\tilde{\nu}_m K_h(y_m, y_m)$  (the corpus wording updated: the arithmetic sits in the NODE AND WEIGHT distribution of the two measures, the critical cancellation is the coherent geometry of a Carleson Gram operator — not an irreducible kernel phase); the Schur/Haynsworth PORT REDUCTION  $I - E \succeq 0 \Leftrightarrow I - R \succeq 0 \wedge I - D_P \succeq 0$  with  $(1 - \lambda_{\max}(D_P))/\tau = 1.00$  on ALL rungs (THE PORT IS THE WALL: 13–54 nodes; bulk margin  $420\text{--}45000 \times \tau$ ; inertia integer-exact including both indefinite controls) — satisfying the round-37 NON-DECOMPOSITIONAL demand; and the theorem-grade surprise  $[Y, D_P]$  of EXACT rank 2 ( $10^{-14}$ ): the dressing PRESERVES the IIKS integrable class — the dressed port operator is an integrable kernel with two explicit generators; (3) THE SOURCE LAW (v882, 7/7 + 5/5 + 6/6, sqrt-uniformized + mellin-kernel-confirmed + atoms-identified (budget-closed)): under  $r = \sqrt{n/X}$  the weighted prime measure uniformizes to Lebesgue  $dr$  (Kolmogorov  $5 \times 10^{-4}$  at  $X = 10^7$ ; own SPF sieve; deployed tables = the true comb at 0.0033); the last-log-unit edge mass equals  $1 - e^{-1/2} = 0.393469$  (measured 0.3936) — the wall's  $\sim 39\%$  port mass

IS the classical PNT edge law; the port symbol is the Mellin–Cauchy kernel  $1/(1+2i\tau)$  and the FIT-FREE continuum prediction reproduces the deployed port numerators to 0.029 at the deepest rung (the deviation = the shrinking arithmetic fluctuation); the port numerators are the EXACT deployed-window prime sums ( $1.5 \times 10^{-13}$ ) with the explicit-formula reading at the alias frequencies, and the criticality BUDGET CLOSES ( $+1.405 - 1.674 + 0.284 = +0.015$ ):  $T \rightarrow 1$  is PNT growth exactly compensated by window geometry plus Christoffel — the wall is an ERROR-TERM statement about prime partial sums at the port frequencies; (4) THE HARDNESS DECISION and the OPEN CONTRACTS (ledger rows, sandbox-backed): PRIME.ERRORTERM.SCALE.01 DECIDED — injecting the exact off-critical zero signature, the wall margin equals perturbation ENERGY on the  $\tau$  law exactly (ratios 1.00–1.10;  $A^* \sim \sqrt{\tau}/\delta^2$  with the frozen convention  $\Delta\lambda = O(A^2\delta^4)$ ; the kill localizes THROUGH the port block): the reduction chain is an exact reformulation at RH strength, NO unconditional slack; new open contracts registered [O]: PRIME.PORT.LIMIT.01 (compressed operator-norm convergence slope  $-2.80$ , criticality from below, atom-at-edge Weyl- $m$  class — the fixed-space limit theorem is the demand), PRIME.PORT.SCALAR.01 (the wall as ONE scalar inequality  $\sigma_h = [1 - \tilde{\nu}_{m^*}K_h(y^*, y^*)] - b^T(I - E_{\text{rest}})^{-1}b \geq 0$ , exactness warranted on all 42 rungs), PRIME.PORT.COCYCLE.01 (the fixed  $12 \times 12$  window carries the wall exactly), PRIME.CASE.SUMRULE.01 (prove the Case  $C_0/C_1$  sum rule  $\Rightarrow$  unconditional  $T_h \leq 1$ ; the deployed chains are measured Killip–Simon class), PRIME.PORT.RIEMANNHILBERT.01 (the discrete  $2 \times 2$  IKS Riemann–Hilbert problem is WELL-POSED, resolvent closure  $10^{-12}$ ; the smooth PNT model VIOLATES the floor at finite depth — the prime discreteness is load-bearing: any steepest-descent program must keep the lattice in the unperturbed problem); papers: the prime-front rounds-38/39 section, tfpt\_1 finite-theorem keybox, tfpt\_3 audit cards, tfpt\_research\_contracts registrations; website mirrors and generated surfaces synced (875/v882); NO marker moves; NO RH claim anywhere)

2026-08-09 (LXXX) (reflux round thirty-seven — the partition closure: the compact closing round of the reopened Cotlar route — the analytic core is SAVED (the exact tridiagonal CD identity with the route’s FIRST  $h$ -uniform derived constants) and the decomposition chapter is CLOSED ENTIRELY (cells, packets, all frames): the cancellation is ATOMICALLY GLOBAL and the certificate demand is retyped NON-DECOMPOSITIONAL: ONE module (v879\_envelope\_partition\_closure.py; suite  $871 \rightarrow 872$ ; two probes embedded byte-exact, the contract chain  $4621b899\dots \rightarrow 5fd6bf61\dots \rightarrow 27d9f0a3\dots \rightarrow 2199e9b1\dots$  machine-verified at run time) — (1) THE ANALYTIC CORE (envelope\_derivation\_gap\_probe, 7 checks with the TWO frozen-honest FAILS T1.4/C.1 pattern-gated at exit 1, verdict envelope-empirical-only): multiplication by  $\cos\theta$  acts EXACTLY tridiagonally on the odd frame (the mirror symmetry closes the top boundary inside the space,  $A_{h-1,h-1} = -\frac{1}{2}$ ; the single rank-2 commutator defect at  $k=0$ ; the Christoffel–Darboux telescoping identity at max residual  $1.3 \times 10^{-13}$  on every rung, battery + all five complete-comb holdouts), and the DERIVED pointwise bound  $|U_{ij}| \leq [C_{\text{int}} + \sqrt{(G_+^{-1})_{00}(K_{-,i}^{-1/2} + K_{+,j}^{-1/2})}]/|\cos\theta_i^- - \cos\theta_j^+|$  holds with ZERO entrywise violation and  $h$ -UNIFORM source constants —  $C_{\text{int}} \in [1.232, 2.328]$  (max/min  $1.89 \leq 2.0$ , log–log slope  $+0.106 \leq 0.15$ ) and  $\sqrt{(G_+^{-1})_{00}} \in [0.661, 0.899]$  from  $h = 168$  to 1445: THE FIRST ANALYTIC UNIFORMITY STATEMENT OF THE ROUTE, at entrywise grade; the two honest FAILS ARE the finding — the analytic envelope certifies NO cell decay (every Prüfer cell contains near-diagonal geometric pairs;  $K_1^{\text{an}}/K_2^{\text{an}}$  blow up as  $h^2$ ): the measured

cell-envelope decay is CANCELLATION-CARRIED, provably not entrywise-derivable; and the frozen discrimination bar was pointed at the wrong constant ( $C_{\text{int}}$  itself discriminates: 1.41 truth / 3.59 Epstein / 12.8 scramble); the gap decomposition: no predeclared accounting below 3.568 (bar 1;  $C_1$ – $C_4$  damped censuses + cell variants  $n = 8/16/32$ ), diagonal share 15–24%,  $\sum_r \varepsilon_r = 6.9$ – $12.9$   $\alpha$ -uniform on clean complete-comb data (deep holdouts IN/ABOVE the battery band, slope +0.237) — the round-36 gap ( $B_C \sim 4$  vs 1, local defects  $O(10)$ ) is STRUCTURAL to every cell frame: NO cell-partition Cotlar accounting of this family can certify; (2) THE TERMINAL STATEMENT (wave\_packet\_partition\_probe, 6/6, verdict partition-class-closed): the typed next object — a non-cell partition where the diagonal carries the mass — is built honestly and KILLED BY MEASUREMENT: both coherent-state designs (rank-Gabor on the node-rank phase space; chain packets through the exact discrete OP transform) are honest TIGHT frames on every rung ( $\|B\|_2 = \|C^G\|_2$  exact at  $10^{-8}$  rel; the gauge identity  $\sigma_{\max}(C^G)^2 = 1 - \lambda_1(\Delta)$  exact); the Gabor envelope DECAYS in phase space ( $0.529 \rightarrow 0.007$  at kz 243 — the first partition of the program to show decay) and the discrimination is massive (truth  $E_{\text{Schur}}$  4.985 vs Epstein 128.7 vs scramble 9006.6); BUT the diagonal does NOT carry under ANY pairing (nearest-packet share  $\leq 0.070$ , empirical-partner share  $\leq 0.183$  at injectivity  $\leq 0.77$  — diagonal mass  $\leq 18\%$ ) and the Schur constant GROWS ( $4.985 \rightarrow 9.222$ , slope +0.295; every functioning design  $\geq 3.5$  on every rung) — the cancellation that holds  $\|C\| \leq 1$  is ATOMICALLY GLOBAL, it lives in NO partition of this family; (3) THE BOOKING: ledger 971  $\rightarrow$  972 (PRIME.PARTITION.CLOSURE.01) with the dated round-37 notes on PRIME.CARLESON.PRIME.01 (the constraint set extended: the embedding certificate must be NON-DECOMPOSITIONAL — eigen-enclosure / Loewner-order / passivity, all three measured territory; the new carried assets typed: the tridiagonal identity, the  $h$ -uniform derived constants, their source-law persistence as  $h \rightarrow \infty$  as the named open object) and PRIME.PRUEFER.COMPENSATION.02 (from BOUNDED to CLOSED-AS-A-CLASS, with the stop-list entries: wave-packet / coherent-state partitions of the contractor family; entrywise bounds on the cell envelope); papers: the prime-front round-37 section “the partition closure” + the open-fronts chain extended, tfpt\_3 audit card, tfpt\_research\_contracts recorded block + the dated Carleson note; website mirrors and generated surfaces synced (872/v879); NO marker moves; NO RH claim anywhere)

2026-08-08 (LXXIX) (reflux round thirty-six — the complete-comb reversal: the deep-alpha negatives were ARTIFACTS, the preregistered Cotlar route RE-OPENS under its own frozen bars, the Prime Carleson chain is machine-exact, three exact compiler theorems land with kernel-checked mirrors, and the ledger gets its first QA fix plus a CI guard: FIVE modules (v874–v878; suite 866  $\rightarrow$  871) — (1) THE ARTIFACT (v877, 8/8, deep-alpha-artifact): the negative Krein floors at kz 142/177/243 ( $\lambda_1 = -331/-135/-215$ , two independent probes) were COMB-TRUNCATION artifacts — core.build\_window silently caps its atom table at  $\text{ATOM\_MAX} = 4 \times 10^5$  while those windows demand support to  $4.4 \times 10^5/8.0 \times 10^5/1.8 \times 10^6$ ; with the COMPLETE canonical comb the certified corner enclosures are strictly positive ( $+1.06 \times 10^{-5}/+1.52 \times 10^{-5}/+1.20 \times 10^{-5}$ ) and the Cholesky-certified full and Krein floors positive on all three rungs (extended builder = the IDENTITY on the controls, bit-equal atoms); NO negative canonical window exists; the certified ladder’s end at kz 121 was the atom-table DATA boundary, not a structural  $\alpha$ -wall; ( $H_{\text{cof}}$ ) unaffected; the silent truncation typed an infrastructure BUG with the registered follow-up (the builder must fail loudly beyond its table); (2) THE

**REVERSAL** (v877, 9/9, cotlar-bounded-v2, the round's headline): v2 re-executes the round-35 contract with cells, pairing, danger geometry and decision bars frozen **IDENTICALLY** (v1 spec SHA 4621b899... re-verified at run time; v2 spec SHA 5fd6bf61...; the single substantive change — complete combs at the deep holdouts — warded bit-identical elsewhere): the deep sums **COLLAPSE INTO THE BATTERY BAND** (kz 142:  $S_U$  32.07  $\rightarrow$  5.65,  $S_C$  58.78  $\rightarrow$  4.10; kz 177: 14.29  $\rightarrow$  5.97; kz 243: 13.94  $\rightarrow$  5.87), both channels **BOUNDED** by the frozen v1 rule (ratios  $0.95/0.94 \leq 1.2$ , log-log slopes  $0.006/0.007 \leq 0.10$ ) — the v1 kill was **ENTIRELY** the artifact; **RUN 3** (gated on bounded, correctly withheld in round 35) unlocks and **PASSES** ( $K_1/K_2$   $h$ -stable on all five complete-comb holdouts: 8.36/8.14 and 40.35/40.47, bar  $1.2\times$ ); Epstein/scramble still break the contrast bars; the truncation audit re-types every affected finding **DATED** (pruefer v1 confounded  $\rightarrow$  re-decided; residual\_quadrature **STRENGTHENED** — its stalls detected the truncation correctly, a validated sign detector; the resonance  $\alpha$ -law addendum void; the polygon holdout rows numeric-void; radau/preconditioner/cdcore/uvarov/skeleton unaffected); **THE DATED CORRECTION** on **PRIME.KREIN.CONTRACTOR.01**: the round-35 closing note is corrected — the v1 verdict was correct **ON THE DATA AS MEASURED** and v873's booking stands as the record of the preregistered v1 execution, but the stop-list entry is **AMENDED** (the class is **NOT** closed), the  $\alpha$ -not- $h$  law is retired (the truncation's signature), and the route reopens as **PRIME.PRUEFER.COMPENSATION.02** with the remaining gap typed ( $B_C \sim 4$  vs 1; local defects 7–10); (3) **THE CARLESON WAVE** (v876,  $6/8 + 7/8 + 5/5$  with the three frozen-honest **FAILS** S3.1/S3.2/S2.1 pattern-gated at exit 1, **NOT** refit): the Carleson ward  $\lambda_{\max}(M_-) = \|C\|^2 = 1 - \tau$  at rel  $1.5 \times 10^{-13}/1.3 \times 10^{-15}$  makes arrows (i)–(iii) of the Prime Carleson chain **MACHINE-EXACT** (embedding constant  $\|C_h\|^2 = 1 - \tau_h$ ), the surviving analytic target registered as **PRIME.CARLESON.PRIME.01** [O]; the polygon  $S_0 \geq |S_2|$  **BREAKS** on all 47 rungs at  $2\theta \approx 1.5\pi$  (typed: the **WRONG** compression of the Loewner order) and the holonomy is null with inverted contrast; the residual quadrature realizes  $\rho_h$  at **FULL** degree on all 44 floor-positive rungs ( $k^*/h$  median 1.000, slope 0.974; mp escalation past the 53-bit wall; a single atom just beyond the pole port; discrimination 7–9 $\times$ ); the resonance is **THICK** ( $\sim 1/5$  constant, excision gains nothing, soft direction disjoint — the wall is phase-distributed); (4) **THE THEOREMS** (v874 14/14 + 11/11 + v875 15/15, with the Lean mirrors **TfptCarrier/FlavorFeedback.lean** and **TfptCarrier/GaussianShells.lean**, 18 theorems, lake build green 3418 jobs, axioms propext/Classical.choice/Quot.sound, Lean manifest 88  $\rightarrow$  90 files): the **FEEDBACK NORMAL FORM** ( $P^{-1}R_sP = [[s+4, 0, 3], [0, 0, -2], [4, 1, 5]]$  exact symbolic in  $s$ ;  $\chi_{R_s} = (t - s - |\mu_4|)q_{\text{wind}} - 12t$  with  $12 = N_{\text{fam}}|\mu_4| = \dim g_{\text{SM}}$ ; the honest  $\mathbb{Z}_2^2$ -not- $\mu_4$  typing kernel-checked); the **WINDING DECODER** ( $q(\{2, 3, 5, 6\}) = (-4, -4, 2, 8)$ ;  $D(y) = (y+4)^2(y-2)(y-8)$  with budget coefficients (2, 48, 32, 256);  $|\mu_4|q(|Z_2|)/|R(E_8)| = -1/15 =$  the Walsh message eigenvalue — typed **AUDIT** theorem, not a functor); and the **GLOBAL SHELL THEOREM** ( $\hat{\Theta}_u/\Theta_L = -1/15$  for **ALL** odd shells from  $\text{Sp}(4, 2)$  transitivity +  $(1+i)$  doubling +  $\Theta_L = 240\sigma_3$  — the  $n \leq 16500$  census **REPLACED** as argument, retained as ward); (5) **THE LEDGER QA FIX + THE GUARD** (v878, 10/10, **GATE.LEDGER.01**): the first **PRIMARY-KEY** violation found and fixed — **FORM.QGE0.03** was double-booked (2026-06-15 CohomologyGrading; 2026-06-18 SeamDeckClosure flat closure); the SeamDeckClosure row **RENAMED** **FORM.QGE0.04** with a dated QA note and every flat-closure reference updated **INCLUDING** the two 2026-06-18 mentions in this changelog (corrected in place, documented here); the guard enforces unique claim ids, acyclic supersession, unique active supersession targets

modulo a frozen 2-pair allowlist, and the dependency cross-reference cycle census frozen remove-only; NO marker moves; NO RH claim anywhere)

- **New modules (5).** `v874_feedback_decoder_lean.py` (14/14 + 11/11 probe gates + 4/4 Lean mirror checks, `FEEDBACK-NORMALFORM-EXACT` + `WINDING-DECODER-EXACT`; probes `flavor_feedback_probe.py` + `g31_winding_decoder_probe.py` embedded `BYTE-EXACT` and re-run verbatim; the Lean mirror `TfptCarrier/FlavorFeedback.lean`, 11 theorems); `v875_shell_global_lean.py` (15/15 + 4/4, `SHELL-GLOBAL-THEOREM`; probe `gaussian_shell_global_probe.py`; the Lean skeleton `TfptCarrier/GaussianShells.lean`, the `ShellSystem` hypothesis bundle + 7 consequence theorems); `v876_carleson_polygon_wave.py` (6/8 + 7/8 + 5/5 with the three frozen-honest FAILs pattern-gated at exit 1; probes `phase_polygon_probe.py`, `residual_quadrature_probe.py`, `pi_resonance_anatomy_probe.py`, ~9 min); `v877_complete_comb_reversal.py` (8/8 + 9/9; probes `deep_alpha_sign_probe.py` + `cotlar_v2_complete_comb_probe.py`, ~17 min; the v1/v2 spec-SHA chain machine-warded); `v878_ledger_integrity.py` (10/10, the CI guard). Suite 866 → 871; all five standalone green; run\_all ALL CHECKS PASSED.
- **The reversal (the round’s deliverable).** The discipline machinery worked end to end on its own freshest negative: the residual horizon flagged the deep stalls, `deep_alpha_sign` typed the silent comb truncation at certificate grade (`ATOM_MAX` =  $4 \times 10^5$ ; complete-comb floors strictly positive — NO negative canonical window), and the IDENTICALLY frozen round-35 bars re-ran on complete data and REVERSED the closure: `COTLAR-BOUNDED-V2`, run 3 unlocked and passed, the route reopened as `PRIME.PRUEFER.COMPENSATION.02`. The correction is booked DATED on every surface (ledger, papers, website) while v873 and the round-35 records stand as what was measured; the remaining gap is typed, not hidden ( $B_C \sim 4$  vs 1; local defects 7–10). NO RH claim.
- **Ledger.** 13 new rows (`FLAVOR.FEEDBACK.NORMALFORM.01`, `E8.WINDING.DECODER.01`, `FORM.FLAVOR.FEEDBACK.01` [F], `E8.SHELL.GLOBAL.01`, `FORM.E8.SHELL.GLOBAL.01` [F], `PRIME.PHASE.POLYGON.01`, `PRIME.RESIDUAL.QUADRATURE.01`, `PRIME.PI.RESONANCE.ANATOMY.01`, `PRIME.CARLESON.PRIME.01` [O], `PRIME.DEEPALPHA.SIGN.01`, `PRIME.PRUEFER.COMPENSATION.02`, `PRIME.TRUNCATION.AUDIT.01`, `GATE.LEDGER.01`) + the dated round-36 CORRECTION notes on `PRIME.KREIN.CONTRACTOR.01` (the stop-list entry amended, the route reopened, the  $\alpha$ -not- $h$  law retired) and `PRIME.PRUEFER.COMPENSATION.01` (the v1 record stands as-measured) + dated notes on `E8.SIMPLEX.FOURIER.01` (census → theorem), `E8.G31.CLOCK_ALPHABET.01` (the decoder) and `PRIME.FLOOR.SKELETON.01` (the ladder end was a data boundary) + THE QA FIX: `FORM.QGEO.03` was double-booked since 2026-06-18 (a primary-key violation); the `SeamDeckClosure` flat-closure row is RENAMED `FORM.QGEO.04` with a dated QA note, all references updated (papers, scripts, registry, website, and the two 2026-06-18 entries of this changelog corrected in place — documented here); 958 → 971 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round-36 section “The complete-comb reversal” (`\veri{v876/v877/v878}`; the artifact, the reversal, the Carleson chain, the theorems crossref) + the DATED CORRECTION paragraph appended to the round-35 section + the open-fronts chain extended with round 36 and its round-35 clause corrected; `tfpt_1`: two new Gaussian-code keybox entries (the global shell theorem `\veri{v875}`; the winding decoder `\veri{v874}`) + the census sentence upgraded; `tfpt_2`: the feedback normal form added to the winding keybox (`\veri{v874}`); `tfpt_3`: five new audit cards + dated corrections on the v857/v873 cards; `tfpt_research_contracts`: the `PRIME.CARLESON.PRIME.01` [O] registration, three round-36 recorded blocks, the round-36 correction notes on the contractor and paircorr blocks, and the `GATE.LEDGER.01` QA record.
- **Website.** Prime-front feed round 36 (`primeFront.ts` + the verdict enum); counters 871/v878 (`HonestyBanner`, prime-front page metadata and hero); DAG +v874–v878; `papers.ts` (the

research-contracts round-36 correction, the architecture  $-1/15$  upgrade); `ResidualChain.tsx` carries the renamed `FORM.QGEO.04`; generated surfaces (`ScriptIndex`, `changelog.ts`, `suite.ts`, `discipline.ts`, `maps`) regenerated; `README` + Zenodo synced to 871/v878; manifests regenerated last (Lean manifest 88  $\rightarrow$  90 files).

- **Infrastructure note (typed as a bug, follow-up registered).** `core.build_window` silently truncates the canonical comb at its atom-table cap; until the loud guard lands, every deep-window readout must check the demand  $e^{2\alpha} \leq \text{ATOM\_MAX}$  explicitly (the round-36 modules do).

**2026-08-08 (LXXVIII)** (reflux round thirty-five — the Cotlar decision: the first fully preregistered uniformity test of the program, **EXECUTED: ONE** module (v873, the executed round-34 preregistration `PRIME.PRUEFER.COMPENSATION.01`; suite 865  $\rightarrow$  866) — the composite verdict cotlar-growing / route dead **BY THE CONTRACT'S OWN FROZEN RULES**, booked exactly as frozen: (1) **THE PROVENANCE CHAIN** (machine-warded in the module's gate) — contract frozen + hashed (SPEC SHA-256 4621b899...7440811; 8/8 synthetic freeze wards; the decisive analysis locked behind the operational flag `pruefer_unfreeze.flag`, **ABSENT** at freeze)  $\rightarrow$  committed byte-exact **PRE-EXECUTION** in the round-34 commit 526ca3eb (FILE SHA-256 93208a1b...4825d9fd — the proof of precedence)  $\rightarrow$  the unfreeze flag created **AFTER** that commit with its audit line committed  $\rightarrow$  executed with exactly **TWO** documented **MECHANICAL** fixes, both typed verbatim in the probe (the  $h \leq 900$  battery filter that the frozen spec's own battery definition mandates — the committed code omitted the spec's own text; the per-step joint positive renormalization against float overflow at near-breakdown chain tails — invariant for `atan2` and the wrapped increments in exact arithmetic, validated at  $\leq 6.6 \times 10^{-4}$  against a 60-digit `mpmath` reference), the **FROZEN\_SPEC** literal byte-identical throughout (SHA re-verified at every stage; executed FILE SHA-256 15d35b26...5ffb4cc)  $\rightarrow$  re-executed verbatim at promotion (embedded **BYTE-EXACT**, **RUN 1** + **RUN 2**,  $\sim 22$  min); (2) **RUN 1** (the anatomy, anchors kz 9/12/13/26/40) — the predeclared danger geometry is **REAL**: the frozen cells  $\{0, 15\} \cup \{7, 8\}$  capture 0.98/0.61/0.94/0.85/0.81 of  $\lambda_{\min}(T_1)$  — source-only Prüfer phase cells localize the negativity **WITHOUT** the computed soft eigenvector (kill 3 silent); but the **LOCAL** compensation fails ( $\sum_r \varepsilon_r = 6.88\text{--}10.08$ ) and the cross-block envelope does **NOT** decay (Spearman  $-0.17\text{--}0.00$  vs the frozen  $\leq -0.8$  bar, **BIMODAL** with the second peak exactly at the predeclared  $\pi$ -resonance cells  $d = 7\text{--}8$ ); (3) **RUN 2** (the Cotlar decision, the 42-rung battery + the frozen deep holdouts kz 90/116/142/177/243) — the battery is **FLAT** ( $S_U$  5.55–6.30,  $S_C$  3.97–4.78) and the deep holdouts **EXPLODE** (kz 142:  $S_U$  32.07/ $S_C$  58.78; kz 177: 14.29/30.76; kz 243: 13.94/44.85), both channels **GROWING** per the frozen rule (log–log slopes  $0.209/0.408 \geq 0.20$ ), the contrast bars **PASS** (Epstein max rel 3927% / local-defect ratio 224 $\times$ ; scramble 314462% /  $1.26 \times 10^6$  — kill 4 silent), and the measured carry-forward law is typed: **THE GROWTH TRACKS** kz-DEPTH ( $\alpha$ ), **NOT WINDOW SIZE**  $h$  (kz 90/116 at  $h = 1430/1433$  stay **INSIDE** the battery band while kz 177 at **SMALLER**  $h = 1219$  explodes) — any uniform certificate must be  $\alpha$ -aware; **RUN 3** correctly **NOT** executed (gated on **BOUNDED**); (4) **THE KILL TABLE** — kills 2 (power-law Cotlar growth) and 5 (the compensation stays global) **FIRE**, kills 1/3/4 do **NOT**; the blockwise Cotlar–Stein / phase-cell class on Prüfer phases is **CLOSED** and stop-listed; the surviving demand narrowed: a **NON-STATIONARY** phase bound on the explicit kernel with the  $\pi$ -resonance as the known critical structure; the honest **FAILs** pattern-gated, nothing refit — preregistration + frozen bars +



**committed hashes is the standard the program now holds itself to; NO marker moves; NO RH claim anywhere)**

- **New module (1).** `v873_pruefer_cotlar_decision.py` (the executed preregistration: the probe `pruefer_compensation_probe.py` embedded BYTE-EXACT in its EXECUTED form and re-run verbatim in an isolated namespace — RUN 1 + RUN 2, ~22 min; the pattern gate encodes the frozen census INCLUDING the honest FAILs: the provenance wards P1–P3 (byte-equality vs the experiments source, the executed file SHA, the committed unfreeze-flag audit line), the spec-SHA ward G0, the danger-geometry capture bands, the local-dominance and envelope-decay FAILs, the battery/holdout Cotlar censuses, the frozen GROWING decision with slope bands, the  $\alpha$ -not- $h$  law, the contrast bars, the kill table, and the RUN-3-withheld ward). Suite 865  $\rightarrow$  866; standalone green; the two mechanical fixes documented verbatim in the probe’s own docstrings with the spec literal SHA unchanged.
- **The verdict (the round’s deliverable).** The program’s first fully preregistered uniformity test returns its own frozen verdict, and the negative is booked with the same machinery as a positive would have been: the blockwise Cotlar–Stein route on source-only Prüfer phase cells is DEAD — the Cotlar sums are  $h$ -stable on the battery but explode on the frozen deep holdouts as a power law in depth. What survives, typed: the danger-geometry positive (source-only cells find the dangerous blocks without the soft eigenvector), the truth/Epstein/scramble discrimination, the  $\pi$ -resonance as the named critical structure, and the  $\alpha$ -not- $h$  growth law as a constraint on every future candidate. NO RH claim.
- **Ledger.** 1 new row (`PRIME.PRUEFER.COMPENSATION.01`, the executed contract with the full provenance chain and the frozen census) + the dated round-35 closing note on `PRIME.KREIN.CONTRACTOR.01` (the blockwise Cotlar/phase-cell class CLOSED by the preregistered test’s own rules; the STOP-LIST entry: blockwise Cotlar–Stein on Prüfer phase cells; the demand narrowed to a non-stationary phase bound with the  $\pi$ -resonance critical structure and the  $\alpha$ -not- $h$  constraint; the danger-geometry positive and the discrimination recorded as carry-forward assets); 957  $\rightarrow$  958 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The Cotlar decision: the first fully preregistered uniformity test, executed” (`\veri{v873}`; the contract, the execution discipline, the verdict, what survives, and the honest note that preregistration + frozen bars + committed hashes is the program’s standard) + the round-34 section’s “not run” prose given its forward pointer + the open-fronts chain extended with round 35; `tfpt_3`: the audit card; `tfpt_research_contracts`: the round-35 dated notes on the `PRIME.KREIN.CONTRACTOR.01` block and the contractor registration block (the execution record with the executed file SHA verbatim).
- **Website.** Prime-front feed: the round entry “reflux round thirty-five”; counters 865  $\rightarrow$  866 modules / `v535–v873`; DAG extended with `v873`; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `maps`); README and Zenodo updated to 866 modules / `v873`.
- **Scope.** One module, one executed contract: the verdict is a route decision under the strictest protocol the program has used — every bar, kill and holdout frozen and hashed BEFORE execution, the frozen file committed pre-execution, the two mechanical fixes documented verbatim with the spec literal SHA unchanged, the honest FAILs pattern-gated and NOT refit, RUN 3 correctly withheld by the contract’s own gating; the unfreeze flag with its audit trail is committed with this round; NO marker moves; NO RH claim anywhere.

**2026-08-08 (LXXVII) (reflux round thirty-four — the Christoffel–Darboux wave: the kernel NAMED, the compensation identity EXACT, the measure-relation closure, and the preregistered next test: FOUR modules (`v869–v872`, from seven frozen probes; suite 861  $\rightarrow$  865) plus the FROZEN preregistra-**

tion **PRIME.PRUEFER.COMPENSATION.01** committed byte-exact and **NOT** run — the consolidation of the **2026-08-08** evening **CD/Uvarov** campaign in three waves plus one commit-time contract: (1) **THE CD WAVE** (the kernel identification) — the reproducing kernel  $K_+$  of the closed-form SourceContractor (v866) is **EXACTLY** the Christoffel–Darboux kernel of an explicit source measure  $\mu = \sum [2 \sin^2(\varphi/2) d_+(\varphi)/L] \delta_{2 \cos(D\tau)}$ : the displacement rank collapses from the growing  $\tau$ -coordinate budget (12–14, the  $\sqrt{L}$  law) to **EXACTLY** 2 in the multiplication coordinate  $x = 2 \cos(D\tau)$  with a hard  $\sigma_2/\sigma_3$  gap of  $6.6 \times 10^{12}$ – $4.8 \times 10^{13}$  on **ALL** 7 rungs including the blind holdouts **kz** 40/49/60, and the **FIREWALLED** chain-only reconstruction (Stieltjes–Lanczos source Jacobi chain + node/geometry data **ONLY**, enforced by the reconstruction signature) rebuilds  $C$  entrywise at  $8.1 \times 10^{-14}$ – $5.3 \times 10^{-13}$  — the whole contractor is three named objects: the arm weights, the phase/geometry factors, and **ONE** source Jacobi chain, with the honest boundary typed **PRE-RUN** and confirmed (the rank-2 collapse is **WINDOW GEOMETRY** — the controls collapse too, exactly as the derivation forces for any even density; the arithmetic lives in the **MEASURE**: control chains  $O(1)$  off at 0.038/0.062 vs the  $10^{-2}$  bar); the  $\mu_4$  quarter-turn Clark-phase hypothesis is falsified **SHARPLY** (the coupled pairs obey the **EQUAL-PHASE** law: offset  $-0.5..+0.6^\circ$  at  $\bar{R}$  0.97–0.99, **NOT**  $90^\circ$ ; the Clark rank leg dies at  $E$ -rank 85–99; the phase coordinate itself source-only clean, circularity fence exact); the factorization  $C = D_- U D_+$  assembles **EXACTLY** ( $1.3 \times 10^{-14}$ ) with  $\|D_+\| \leq 0.9654$  by the Christoffel variational theorem,  $\|D_-\| \leq 0.8655$  measured, and  $U$  non-unitary **BY THEOREM** (the deployed grid  $2\times$ -oversamples the folded pairs: rows  $\geq 1.158$  — the  $\sqrt{2}$  over-normalization is structural); and the Gauss-node frame **DECIDES** the repair: exact where algebra promises ( $D_+^G = I$  at  $4.7 \times 10^{-11}$ , lossless bridge, frame-invariant port share — theorem P5) yet the unitary core does **NOT** exist (co-isometry defect  $O(1)$ , 0.93–1.30, cross-measure node misalignment at near-pole pairs), the surviving asset being the **LADDER-WIDE** damping bound  $D_-^G \leq 0.997041$  with  $34\times/2900\times$  discrimination and the named hypotheses (H-D)/(H-U)/(H-PORT) registered; (2) **THE UVAROV WAVE** (the measure-relation decision, two honest kills with typed residues) — **NONE** of the three classical spectral transformations (Uvarov/Christoffel/Geronimus) relates the two channel measures exactly or at fixed small rank at the source pole  $x_{\text{pole}} = 2 \cosh(D/2)$  (machinery warded exact on synthetics at  $9.2 \times 10^{-16}$ ; typed residue: Geronimus with **SMALL NEGATIVE MASS** uniformly closest at 3.5–8.1% Weyl residual, non-growing, discriminating — misses the 5% bar, no  $\lambda(D)$  law), and the co-isometry defect is **DIFFUSE**: full-rank, growing **LINEARLY** with  $h$  (Pearson +0.996,  $\times 3.39$  first-to-last third), pole removal changes nothing, the closest-co-isometry bracket  $O(1)$  at near-full rank — **NO** finite-matrix compensation exists at this ingredient list (both closures stop-listed); (3) **THE HEADLINE** (the compensation identity) — the defect splits **EXACTLY** into two source-built terms,  $I - C^*C = T_1 + T_2$  with  $T_1 = I - U^*U$  (the **INDEFINITE** cross-measure Jacobi geometry) and  $T_2 = U^*(I - D_-^2)U$  (the **PSD** arithmetic Christoffel damping), warded at  $10^{-12}$  on all 42 rungs and transported losslessly into the Gauss frame; the **EXACT** global minimizer of the directional compensation ratio **IS** the soft direction (overlap  $\geq 0.9991$ ; min ratio =  $1 + \tau/|v^*T_1v|$  to all printed digits), the rank-one pole deflation gives  $\Delta = \Delta_{\text{bulk}} + s pp^*$  with the certified bulk floor at 98–100% of  $\lambda_2$  and  $s = \kappa\tau$  reproducing the v864  $\kappa$  law — **THE WALL AS TWO EXPLICIT SOURCE NUMBERS** (kz 9 at the soft direction:  $-0.0328 + 0.0330 = 1.68 \times 10^{-4} = \tau$ ), with the honesty line typed (the truth-side ratio  $\geq 1$  is implied by upstream certified positivity on these finite rungs — the

identity buys the FORM: one two-term inequality at one named direction) and the discrimination anatomy measured (Epstein/scramble BREAK the damping's sign,  $\max D_- = 34.2/2917 > 1$ , and under-compensate on their OWN soft directions at worst ratios  $-175/-446513$ ); (4) THE PREREGISTRATION (committed, NOT run) — PRIME.PRUEFER.COMPENSATION.01: source-only Prüfer phases of the two arm chains, EXACTLY 16 fixed phase cells (boundaries  $k\pi/8$ , entrywise pairing primary, danger families predeclared), Cotlar–Stein decision rules and all success/kill criteria frozen BEFORE any deployed-data run, deep holdouts kz 90/116/142/177/243 never used by the CD wave, the decisive analysis locked behind an operational unfreeze flag, 8/8 synthetic freeze wards green; SPEC SHA-256 4621b899...7440811, FILE SHA-256 93208a1b...4825d9fd, committed byte-exact (any modification invalidates the preregistration); NO marker moves; NO RH claim anywhere)

- **New modules (4).** v869\_cdcore\_clarkphase.py (10 + 8 checks, zero fails, CDCORE-IDENTIFIED + CLARKPHASE-DEAD; probes cdcore\_probe.py + clarkphase\_probe.py: the frame factorization  $F = \text{diag}(\gamma) \text{diag}(s) V_{\text{poly}}$  machine-exact ( $3.0 \times 10^{-13}$ , Chebyshev 4th kind), the reproducing/projection wards ( $P^2 = P$ ,  $PF = F$  at  $10^{-14}$ ), the decisive displacement test (x-rank 2 with the  $10^{13}$  gap vs  $\tau$ -rank 12–14, Spearman +0.96, slope +0.17 vs  $\sqrt{L}$ 's +0.50), the firewalled CD reconstruction within certified budgets on construction AND holdouts, the entrywise weighted transport ( $6.8 \times 10^{-13}$ ), the leverage trace ward ( $\sum l_j = h$  at  $9.3 \times 10^{-16}$ ), the structural control collapse typed pre-run with the  $O(1)$  chain discrimination; the Clark falsification: passivity and spread wards, the E-rank leg (85–99 vs the  $\leq 8$  bar), the equal-phase offset law ( $\bar{R} \geq 0.97$ ), the exact circularity fence ( $\max |dz| = 0.0$  under comb perturbation), Epstein/scramble discriminate in E-rank/offset); v870\_christoffel\_gauss\_frame.py (17/17 + 9/9, CHRISTOFFEL-PARTIAL + GAUSS-STILL-DEFECTIVE; probes christoffel\_transition\_probe.py + gauss\_node\_unitary\_probe.py: the classical wards W1–W6 (trace, variational, Golub–Welsch, bridge, assembly, frame transfer), the row theorem P3, the payoff census (product bound  $\leq 1$  on 0/42 rungs — honest), the softport regression  $\beta(kz9) = 0.6057$  / ladder max 0.8591 with the sharpening question answered (pole share  $\sim 0.86$  in all three coordinate systems); the Gauss frame: tightness/Gauss–Christoffel/unit-rows exact,  $\sigma$ -spectra match the grid contractor,  $\tau$  lossless, port share frame-invariant, the co-isometry certificate 0.93–1.30 BREACHED (the measured content),  $D_-^G \leq 0.997041$  ladder-wide, the named hypotheses registered); v871\_pole\_uvarov\_crossdefect.py (7 + 11, POLE-TRANSFORM-DEAD + CROSSDEFECT-DIFFUSE; probes jacobi\_pole\_uvarov\_probe.py + gauss\_crossdefect\_probe.py: the machinery ward on synthetics ( $9.2 \times 10^{-16}$ ), the two-measure wards with the cdcore regression, the three-transformation census (all FULL/PARTIAL False), the Geronimus residue with the blind holdout transfer (ratio 0.97) and the  $2.58 \times / 2.14$  discrimination; the defect anatomy: traceless ward ( $10^{-13}$ ), the rank-growth law ( $\times 3.39$ , Pearson +0.996), mode stability ( $0.123 \leq 0.15$ ), the  $U_0 + R$  split with the PSD base theorem and the  $\tau$  ward ( $\lambda_{\min}(\Delta_{\text{pos}}) = \tau$  to all digits)); v872\_damping\_compensation.py (10, COMPENSATION-EXACT; probe cd\_damping\_compensation\_probe.py, spec v2 with the basis-free exact-minimizer amendment: the split ward ( $10^{-12}$ ), the unitary transport ( $7.7 \times 10^{-11}$ ), the exact Feshbach deflation ( $\lambda_{\min}(\Delta_{\text{bulk}}) = 0$  at  $2.6 \times 10^{-15}$ ), the  $\kappa$  regressions, the T2 sign theorem (zero negative entries), the exact global compensation census (min ratio 1.000008, implied by upstream positivity — the anatomy, not new evidence), the bulk floor ( $c/\lambda_2 \geq 0.98$ ), the four-way alignment table, the isolated scalar, and the discrimination anatomy). Suite 861  $\rightarrow$  865; all four standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all seven probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).

- **The preregistration (committed, NOT run).** experiments/

`tfpt-discovery/pruefer_compensation_probe.py` — the FROZEN contract `PRIME.PRUEFER.COMPENSATION.01`: source-only Prüfer phases (structural AST firewall on the construction functions), EXACTLY 16 fixed phase cells with the entrywise pairing primary and the danger families `DGR1/DGR2` predeclared, `RUN 1/2/3` protocols (anatomy / Cotlar–Stein decision / uniformity candidate) with all bars, verdict rules and kills frozen BEFORE any deployed-data run, deep holdouts `kz 90/116/142/177/243` (never used by the CD wave), the decisive analysis locked behind the operational flag file `pruefer_unfreeze.flag`, and 8/8 synthetic freeze wards green at freeze time. Committed `BYTE-EXACT` with this round: `SPEC SHA-256 4621b89958531d6d3baae9fe762dbd10d23dc22eed77a57aa1c6b179a7440811`, `FILE SHA-256 93208a1b2edebde31388a87192055c9bf3e5a3de29af9fb3e34ddc904825d9fd`; any modification invalidates the preregistration. Booked as a dated `PREREGISTRATION` note on `PRIME.KREIN.CONTRACTOR.01` (the contract’s next test); NO vN module, NO analysis run, NO result claimed.

- **The compensation statement (the round’s deliverable).** The wall’s operator coordinates gain their classical names: the contractor’s kernel IS the Christoffel–Darboux kernel of one explicit source measure (the “non-monomial band-spreading letter” of rounds 32–33 is NAMED at kernel level — with the honest boundary that the naming is geometry, the arithmetic is the chain and the weights), the defect is the EXACT sum of an indefinite Jacobi cross-geometry and a PSD Christoffel damping, and the floor demand is ONE two-term inequality at ONE named direction with both terms explicit source objects ( $s = \kappa\tau > 0$ ). The classical escapes are closed with certificates and stop-listed: classical pole transformations at  $x_{\text{pole}}$  (typed Geronimus residue), finite-rank cross-defect corrections (linear rank growth),  $\mu_4$ -offset Clark kernels (the equal-phase law). The next test is `PREREGISTERED`, not run. NO RH claim.
- **Ledger.** 4 new rows (`PRIME.CONTRACTOR.CDCORE.01`, `PRIME.CONTRACTOR.CHRISTOFFEL.01`, `PRIME.JACOBI.POLE.UVAROV.01`, `PRIME.CD.DAMPING.COMPENSATION.01`) + the dated round-34 note on `PRIME.KREIN.CONTRACTOR.01` (the kernel NAMED; the compensation identity; the Uvarov/finite-rank/Clark closures with the stop-list entries; the Prüfer preregistration with BOTH SHAs recorded verbatim) and the dated round-34 note on `PRIME.FLOOR.PAIRCORR.01` (the wall as the two-term inequality at the soft direction, both terms source-built); 953 → 957 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The Christoffel–Darboux wave: the kernel named, the compensation exact, the anatomy of discrimination, and the preregistered next test” (`\veri{v869..v872}`) + the part-intro chain extended; `tfpt_3`: four new audit cards; `tfpt_research_contracts`: the round-34 dated notes on the `PRIME.KREIN.CONTRACTOR.01` and `PRIME.FLOOR.PAIRCORR.01` blocks including the preregistration record with both SHAs.
- **Website.** Prime-front feed: the round entry “reflux round thirty-four”; counters 861 → 865 modules / `v535–v872`; DAG nodes extended with `v869–v872`; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `maps`); `README` and `Zenodo` updated to 865 modules / `v872`.
- **Scope.** The CD-wave modules are exact structural namings and typed route decisions under frozen preregistered protocols (the derivations typed BEFORE running; the structural non-discrimination of the rank collapse typed PRE-RUN — window geometry, not a kill; the non-unitarity of  $U$  a THEOREM, not a miss; the compensation census implied by upstream positivity and typed as anatomy); the Uvarov/crossdefect modules are honest kills with typed residues (machinery warded on synthetics first); the preregistration is a COMMIT-TIME contract — no analysis mode was executed, the file hash is the proof of freezing; NO marker moves; NO RH claim anywhere.

**2026-08-08 (LXXVI) (reflux round thirty-three — the wall’s operator form: the Krein coordinates, the SourceContractor formula, the Perron closure, and the quadrature reach: EIGHT modules (`v861–v868`, from sixteen frozen probes**

plus one Lean module; suite 853  $\rightarrow$  861) — the consolidation of the 2026-08-08 Krein/port/formula campaign in five strands: (1) THE KREIN CHAIN — the deployed window form gets an EXACT Krein normal form  $Q_h(t) = \|B_+t\|^2 - \|B_-t\|^2$  (entrywise ward  $\leq 10^{-15}$ , both canonical cuts;  $B$  carries  $\sqrt{|\Lambda|}$  — linear in  $\Lambda$ , the round-32 grade barrier passed by construction) with the Douglas pencil identity  $1 - \|C_2\|^2 =$  the  $\tau$ -margin in the positive-side metric (rel  $\leq 3.9 \times 10^{-9}$ ): truth contracts on every rung, Epstein  $h=2$  explodes to  $\|C_E\| = 46.8$ , the scramble to 3693 — the discriminator IS the Douglas equivalence; the register wiring typed (the  $\mu$ -sign and the deck are TWO separate  $C_2$ s, the identification one faithful- $\mu_4$  character away with three candidate constructions machine-verified; the Redheffer/Mertens echo exact:  $\det R_n = M(n)$  with the vacuum column's ENTIRE Smith deviation in the single invariant factor  $|M(n)|$ ; no resolvable floor–Mertens correlation, min  $4p = 0.378$ ); the defect theorem kernel-checked (TfptCarrier/KreinDefect.lean: contraction  $\Leftrightarrow$  PSD with the full-rank Douglas converse, SourceContractor the NAMED hypothesis with the circularity warning typed, krein\_cofinal\_weil composing contractors on a cofinal ladder to Weil nonnegativity — FORM.PRIME.KREIN.DEFECT.01 [F]); the defect has rank ONE ( $\lambda_1 =$  the  $\tau$ -margin exactly, separation  $3.9\text{--}14.1\times$  — the Redheffer Smith invariant, the vacuum code's one-qubit price and the Krein soft port are ONE rank-one structure), the MONOMIAL CLOSURE THEOREM extends the word-census emptiness to EVERY word of ANY length (all source letters monomial, contractor monomial mass  $0.23\text{--}0.24 \Rightarrow$  miss  $\geq 0.874$ ; the 408 sign flips kill the corridor form, typed v2), and the 2-torsion weld law supplies the missing operator's TYPE (the deck's anticommutant is entirely faithful- $\mu_4$  — the deck welds only as  $J = MD$ ; all four candidates weld exactly, the control refuses exactly, the contractor connection null at  $\sqrt{2}$ ); (2) THE MACHINE — the 16-slot colligation architecture makes contraction a CONSEQUENCE (exactly unitary,  $\|D\| = 1/4$ ,  $\sigma_{\max} \leq 1 + 10^{-9}$  automatic, provably comb-blind) with the compiler-side discovery booked in Paper 1:  $\mathbb{F}_2^4$  IS the divisor lattice of 210 with Walsh–Hadamard carrying the lattice Möbius function and the scalar-limit determinant  $\prod_p(1 - 1/p) = 8/35$  EXACT; the identification gate fails honestly and localizes the gap — the contractor couples DIFFERENT  $|\tau|$  bands, the  $J$  flip couples equal  $|\tau|$ : the missing object is the grid-side band intertwiner, the explicit-formula kernel itself; (3) THE SCALAR REDUCTION — the exact Feshbach reduction turns the floor into ONE port impedance ( $\tau > 0 \Leftrightarrow s > 0$ ; four source ports pass; the pole port carries 84% of the soft mode with the exact skeleton  $\kappa^{-1} = \beta^2 + \lambda_1\rho$ , the closed-form  $a$ -term — the Poisson/Cauchy average of the signed density at  $s = 1/2$  — and  $\kappa - 1 \approx 6e^{-\alpha/2} \rightarrow 1$  at Pearson  $-0.99$ ; the backflow  $\leq 17$  bulk modes cancelling 99.9%: the classical shape EXISTS, the boundable term is exactly the arithmetic); the divisor-tower Weyl load repairs the hull's comb-blindness at LOAD level ( $O(1)$  response) and the repaired phase readout discriminates at 1–2 orders (Epstein  $\kappa = 42.7$  / scramble 77.6 outside the  $1.5\times$  truth window; the typed caveat carried: Spearman( $s, \tau$ ) =  $-0.8$ , the discrimination is the load-bearing content); (4) THE FORMULA (the campaign's headline) — the Douglas contractor factors EXACTLY as the target-free closed-form source expression  $C = W_-FG_+^{-1}F^HW_+$  ( $1.4 \times 10^{-14}$  on construction rungs AND blind holdouts; no zeros, no  $\tau$ , no fit) with CERTIFIED defect transfer on all seven rungs (the frozen Sylvester-12 kill fires as designed, correcting rank-12 to  $\sqrt{L}$  growth,  $k^* = 24.48$ ); the norm statement is the exact reformulation  $\|C\| \leq 1 \Leftrightarrow G_- \preceq G_+ \Leftrightarrow K \succeq 0$  — and the PERRON CLOSURE is proven:

$\rho(|C|) = 2.4\text{--}3.1 > 1$  everywhere, so NO absolute Schur/Hilbert-type inequality can ever certify — cancellation provably carries the arithmetic, the certificate must be PHASE-AWARE (the sharpest demand registered: PRIME.KREIN.CONTRACTOR.01 [O], one oscillatory-integral-grade bound beating 1 by the measured  $e^{\text{slope } \alpha}$  margin — it would discharge SourceContractor and with it ( $H_{\text{cof}}$ )); quadrature certificates for  $s > 0$  EXIST on every rung ( $m^* \leq 426$ , the Neumann frontier killed) but the depth law  $m^* \sim \sqrt{\text{cond}(G)}$  (Spearman +0.99) is INTRINSIC — all three source preconditioners fail informatively, the  $m \approx 17$  prediction honestly failed,  $J_\infty$  named the free Chebyshev chain (Rakhmanov-generic typed): the certificate depth is the price of the cancellation; (5) THE AUDITS — the 210 register is canonical up to PROVEN gauge (Boolean layer generic 210/210 as warned; the Euler determinants pin  $\{2, 3, 5, 7\}$  uniquely; the anchor prime 2 forced by ramification; two gauge classes = the chirality pair, then proven gauge by two exact no-go wards — the orientation functional vanishes identically via the Möbius complement, quadratic readouts provably blind by transposition; the 6-vs-7 note stands), and the  $17 = 12+3+1+1$  moving-sector identification is honestly buried at space level (principal angles  $\sim 90^\circ$ ); NO marker moves; NO RH claim anywhere)

- **New modules (8).** `v861_krein_normalform_lean.py` (37/37 + 8/8, zero fails, DECK-MU-PARALLEL + REDHEFFER-ECHO-EXACT + KREIN-CONTRACTOR-STABLE-NO-WORD, +4 Lean-mirror checks; probes `mobius_deck_redheffer_probe.py` + `krein_normalform_probe.py`: the two- $C_2$  wiring census with the three candidate constructions (diagonal characters  $j \in \{1, 3\}$ ; Galois  $\mu_K((p)) = \chi_4(p)$  exact on 609 + 619 primes; Hall chain parity row  $1 = \mu$ ),  $\det R_n = M(n)$  on 62 sizes with SNF  $(1, \dots, 1, |M(n)|)$  and the mod-2 shadow to  $2 \times 10^6$ , the Krein Gram ward entrywise on both cuts, the pencil identity, the Epstein/scramble discriminators, the 155-word census); `v862_defect_polar_weld.py` (8/8 + 22/22, DEFECT-ONE + MONOMIAL-CLOSURE-THEOREM + WELD-WITHOUT-CONTRACTOR; probes `krein_defect_polar_probe.py` + `mu4_clifford_weld_probe.py`); `v863_redheffer_colligation.py` (7/7, COLLIGATION-CONTRACTS-NOT-IDENTIFIES; probe `redheffer_colligation_probe.py`: unitarity  $10^{-12}$ , automatic contraction, the exact 8/35 scalar limit, the band-transfer census); `v864_softport_kappa_law.py` (5/5 + 6/6, SOFTPORT-FOUND + CAUCHY-RANK-SMALL + KAPPA-LAW-PARTIAL; probes `softport_cauchy_probe.py` + `pole_port_kappa_probe.py`); `v865_weyl_port_readout.py` (9 + 5/5 with the TWO frozen-honest FAILs S1.SPR + S4.RSP kept and pattern-gated, NOT refit, WEYL-PORT-BLIND + READOUT-REPAIRED; probes `divisor_weyl_port_probe.py` + `weyl_readout_repair_probe.py`); `v866_source_contractor_formula.py` (10 + 8/8 with the ONE frozen-honest FAIL S2.2 at exit code 1 per the frozen kill design, SYLVESTER12-ANGLES-FAIL + NORM-REFORMULATION; probes `displacement_sylvester12_probe.py` + `source_contractor_norm_probe.py`); `v867_radau_conditioning.py` (7 + 8 with the TWO frozen-honest FAILs S2.G2 + S3.G, RADAU-DEPTH-GROWS + CONDITIONING-INTRINSIC + J-INFINITY-NAMED; probes `softport_radau17_probe.py` + `preconditioned_port_probe.py`); `v868_divisor210_audits.py` (36/36 + 12/12 + 15/15, DIVISOR210-GAUGE-FAMILY(2) + MOVING12-DIMENSION-ONLY + CHIRALITY-DEGENERATE; probes `divisor210_canonicity_probe.py` + `moving12_soft17_probe.py` + `chiral_phase_probe.py`). Suite 853  $\rightarrow$  861; all eight standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all sixteen probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).
- **Lean (1 module promoted).** `TfptCarrier/KreinDefect.lean` — the Krein-defect theorem frame (14 declarations, 10 theorems; lake build green, 3416 jobs, no sorry, no `native_decide`,

axioms `propext/Classical.choice/Quot.sound` only, verified by `#print axioms` on all ten): the defect representation (`defect / format_defect`), the main theorem `defect_psd_of_contraction` (via the exact factorization  $B_+^H(1 - C^H C)B_+$ ), the full-rank Douglas converse (the singular case a typed classical citation), the named-hypothesis structure `SourceContractor` (contractor + factorization + INDEPENDENT certificate; the circularity warning typed), and the cofinal composition `krein_cofinal / krein_cofinal_weil`. Imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round,  $87 \rightarrow 88$  files. Mirrored in v861 part C (numeric witnesses, v856/v849 precedent).

- The operator form (the round’s deliverable).** The wall now has exact operator coordinates: the Krein normal form with the Douglas pencil identity; the rank-one defect; the closed-form target-free contractor  $C = W_- F G_+^{-1} F^H W_+$  with certified defect transfer; the scalar pole-port shadow with the closed-form leading term. And three PROOFS about the tool space: no absolute-value bound can ever certify (the Perron closure  $\rho(|C|) > 1$  everywhere); no source word of any length reaches the contractor (the monomial closure theorem); no fixed-depth quadrature family certifies uniformly (the intrinsic conditioning law). The sharpest demand is registered: ONE phase-aware bound on the closed-form kernel (`PRIME.KREIN.CONTRACTOR.01 [O]`) — it would discharge `SourceContractor` and, through `krein_cofinal_weil`, hypothesis ( $H_{\text{cof}}$ ). Stop list extended (source-word searches at all lengths; absolute-value kernel bounds; fixed-depth quadrature without conditioning control; chirality deciders via quadratic readouts). NO RH claim.
- Ledger.** 10 new rows (`PRIME.KREIN.NORMALFORM.01`, `FORM.PRIME.KREIN.DEFECT.01 [F]`, `PRIME.KREIN.DEFECT_ONE.01`, `PRIME.REDHEFFER.COLLIGATION.01`, `PRIME.SOFTPORT.FESHBACH.01`, `PRIME.WEYL.PORT.01`, `PRIME.SOURCECONTRACTOR.NORM.01`, `PRIME.KREIN.CONTRACTOR.01 [O]`, `PRIME.SOFTPORT.RADAU.01`, `E8.DIVISOR210.CANONICITY.01`) + the dated round-33 operator-form note on `PRIME.RELATION.SKELETON.01` (the Krein coordinates; the three tool-space proofs; the extended stop list; the sharpened demand) and the dated round-33 note on `PRIME.FLOOR.PAIRCORR.01` (the floor demand in Krein coordinates: the phases of the pair kernel, not its magnitudes);  $943 \rightarrow 953$  rows; NO marker moves.
- Papers.** `tfpt_prime_front`: the new round section “The wall’s operator form: the Krein coordinates, the `SourceContractor` formula, the Perron closure, and the quadrature reach” (`\veri{v861..v867}`) + the part-intro chain extended; `tfpt_1`: the divisor-lattice-of-210 entry with its canonicity guard and the Redheffer/Mertens Smith-echo entry in the Gaussian code keybox (`\veri{v861/v862/v863/v868}`); `tfpt_3`: eight new audit cards; `tfpt_research_contracts`: the `FORM.PRIME.KREIN.DEFECT.01` and `PRIME.KREIN.CONTRACTOR.01` registration blocks, the round-33 canonicity record, and the dated round-33 notes on the `PRIME.RELATION.SKELETON.01` and `PRIME.FLOOR.PAIRCORR.01` blocks.
- Website.** Prime-front feed: the round entry “reflux round thirty-three”; counters  $853 \rightarrow 861$  modules / v535–v868; DAG nodes extended with v861–v868; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 861 modules / v868.
- Scope.** The Krein/port/formula modules are typed route decisions and reformulations under frozen preregistered protocols (the five frozen-honest FAILs are preregistered adjudications on record, bars NOT refit; the Perron closure and the monomial theorem are proofs about the TOOL SPACE, not bounds on the target; the formula is a REFORMULATION — the Loewner note typed); the canonicity module audits a deployed discovery (selection census, burial, proven-gauge wards — no new physical claim); the Lean module consumes and formalizes, never evaluates, positivity — `SourceContractor` stays a hypothesis; NO marker moves; NO RH claim anywhere.



**2026-08-08 (LXXV)** (reflux round thirty-two — the grade law, the connected current, and the E8 audits: FIVE modules (v856–v860, from seven frozen probes plus two Lean modules; suite 848 → 853) — the consolidation of the 2026-08-08 morning campaign in three waves: (1) **THE CONNECTED-CURRENT ARCHITECTURE** — the global-centering KMS covariance of the Möbius commutator current  $\mathcal{L} = -[D, \log Z] = T(\Lambda)$  (exact at  $7.2 \times 10^{-15}$ ; Epstein  $h=2$  leaks 76% off the prime powers at **RELATION** level, before any state) is the **FIRST** design that **EVADES** the nuclear tax (the relapse kill does not fire: ratio  $\approx 0.3$  against the frozen band, geometry similarity 0.3–0.4 — genuinely different bookkeeping from the event-local Schur completions) and dies on the typed **GRADE GAP** instead: the covariance is the pair-correlation form (pair weights  $\lambda_m^2 = \Lambda(m)^2/m$ , **QUADRATIC** in the comb, sim 0.56–0.63; the centered main term **IS** the deployed smooth half-line subtraction at corr 0.99), the deployed window form is **LINEAR**, and routing the linear read through the correlator costs the variance itself:  $2.5\text{--}4.1 \times 10^5 \tau$  — an instrument, not a floor certificate; the phase-cell census is **EMPTY** on the entire deployed torus (0 positive cells of 220–1760 per anchor incl.  $\{2, 3, 5\}$  and  $\{2, 3, 5, 7\}$ ; grid  $\lambda_{\min}$  down to  $-3.48$ ; the phases control  $\sim 1\%$  of the demand; the independent tail kill:  $\lambda_{\min}(\text{tail})/\tau = -3.4 \times 10^3 \dots -4.8 \times 10^3$  — the split moves the demand onto an object **FARTHER** from PSD); and the conditional frame is kernel-checked: `TfptCarrier/CofinalCurrent.lean` — `ConnectedTail + PositiveHeadCell + PhaseRecurrence`  $\Rightarrow (H_{\text{cof}})$ , composed through `CofinalWeil` to Weil nonnegativity on the whole dense family, with the  $\delta = 0$  relaxation typed and the counterexample `connectedTail_not_sufficient` proven — hypothesis 2 measured **EMPTY** at the deployed frame: a named continuation contract, not a discharge (`FORM.PRIME.COFINAL_CURRENT.01 [F]`); (2) **THE E8 AUDITS** — the simplex-Fourier character theorem: the Gaussian census  $240 = 15 \times 16$  **IS** the spectral statement  $\hat{r} = (240, -16^{15})$  (the uniform-nonzero channel  $P = (J - I)/15$  with spectrum  $\{1, (-1/15)^{15}\}$  and two-step contraction  $1/225$ , exact in Fractions), the v817 packet numerator **EQUALS** the code Walsh transform at every level  $\leq 16$  (Eisenstein normal form  $15\Theta_{D8} = 7\Theta_L + 8\Theta_0$ ), and  $\hat{m}_2(n) = -1/15$  holds as an **INTEGER IDENTITY** at every odd  $n \leq 16500$  (all 1911 odd primes) — v817's per-prime measurement is the prime restriction of a **CHARACTER THEOREM** of the code, typed honestly spectral-only (the gradings do not refine each other); the G31 clock alphabet:  $\text{Deg}(G_{31})/4 = \{2, 3, 5, 6\} = \{|Z_2|, N_{\text{fam}}, g_{\text{car}}, |R^+(A_3)|\}$  as a **NORMAL FORM** ( $\gcd 4 = |\mu_4|$ ,  $\text{lcm } 120 = |R^+(E_8)|$ ,  $\text{lcm}/\gcd 30 = h(E_8)$ , audit identity  $16 \times 15 = 240$  **REALIZED** on the roots), (8, 12, 20, 24) the unique product-46080/sum-64 quadruple, the 607-group rank-4 kill scan ( $G_{31}$  the **SOLE** full-battery passer; impostors typed by name; honesty line: degree-only audits cannot see the reducible torus), the  $W(D_5) \times W(A_3)$  fence re-killed by **COMPUTED** centers 4 vs 1; the winding quadratic: the whole `tfpt_2` winding line is **ONE** adjugate quadratic  $q_{\text{wind}} = e_1^T \text{adj}(tI - R)1 = t^2 - g_{\text{car}}t + |Z_2|$  (irreducible, disc 17) with the triple lock  $s = 6$  (trace/det/Coxeter each alone),  $\Delta(6) = 39200 = 2 \cdot 140^2$  **NOT** a collision point (the sole  $2 \times \text{square}$  hit in the window) and the reality threshold  $s^* \approx 2.8250$  the unique  $\Delta$ -root in (2, 3) (deployed  $\approx 2.83$ ); (3) **ROUND 33, THE GRADE THEORY** (the day's conceptual core) — `TfptCarrier/GradeNoGo.lean`: **THE GRADE LAW** kernel-checked — the entire direct-positivity graveyard reduces to **ONE** homogeneity no-go (a 1-homogeneous target can never be matched exactly by a 2-homogeneous Gram square: evaluate on a scaling ray at  $t = 1, 2$  — two lines, **CLASS** closure), the affine trilemma with the exact PSD tax (error = constant-Price + quadraticPrice) and the sharpened `affine_grade_no_go` (the background killed too), and the elevator lemma `tangent_psd_on_kernel_null` (derivatives

of PSD families at kernel-null directions are positively signed — RATES, not amplitudes: the ONLY licensed grade-1 mechanism) — measured exchange rate  $10^4$ – $10^5$  across the four graveyard instances (FORM.PRIME.GRADE.NO\_GO.01 [F]); the instantiated elevator is REAL (PSD compound-Poisson Euler factors with no eigenvalue input, exact tangent within the certified Taylor bound, free null-sum positivity  $x^\top \Psi x = w|X(\theta_q)|^2$ , rates Euler-indexed at rel dev 0.0) and dies honestly on SIGNED spectral densities on BOTH deployed sides: the comb Lévy density at exactly 50% negative mass (first flip  $\theta^* = \pi D/(2u_q)$ , prediction matched to  $\sim 1\%$ ) and the arch source on the DIGAMMA BAND ending at  $\tau^* = 6.27$  where  $\operatorname{Re} \psi(\frac{1}{4} + i\tau/2) = \log \pi$  (the closure’s sharpest new number) — the Lévy/conditional-positivity elevator class CLOSED on the deployed sides (Epstein  $h=2$  refuses the Euler indexing at construction grade, 82% off-pp); and the phase-lever theorem closes the finite-head phase-rescue FAMILY with certificates: granting every window atom a PRIVATE best phase, the budget never pays the arch deficit on any of the 67 rungs ( $d_0 - \text{budget}(\{2, 3\}) \geq 1.3894$ ;  $d_0 - \text{budget}(\{2, 3, 5, 7\}) \geq 0.6296$  — uniformly negative at every phase, no Kronecker rescue), three certified scope tiers (exact census / budget inequality / greedy sharp control), the D-frontier saturating at  $0.73 < 1$  (structural, not a boundary effect), the anti-vacuity ward firing; NO marker moves; NO RH claim anywhere)

- **New modules (5).** v856\_connected\_current\_lean.py (14/14 + 5/5, zero fails, CONNECTED-COVARIANCE-PARTIAL + PHASE-CELLS-EMPTY, +4 Lean-mirror checks; probes connected\_covariance\_probe.py + cofinal\_phase\_cell\_probe.py: the current exact  $(-[D, \log Z] = T(\Lambda)$  at  $7.2 \times 10^{-15}$ ;  $\chi_4/\zeta_{\mathbb{Q}(i)}$  pass in their own completions; Epstein  $h=2$  leaks 75.8% off-pp at relation level), states PSD with exact centering bookkeeping, the main subtraction = the deployed smooth subtraction (corr 0.991–0.995; the wrong half-weight loses the pair-weight carrier), the relapse kill does NOT fire (ratio2 0.29–0.31 outside  $[0.5, 20]$ ), the grade gap carries the identity residual (16–45 Ah units; variance price  $2.49$ – $4.06 \times 10^5 \tau$ ); the torus census 0 positive cells with per-cell affineness exact, the Kronecker UFD lemma warded, the tail kill  $-3.4/-4.8/-3.7 \times 10^3 \tau$ , scramble 0/200); v857\_simplex\_fourier\_winding.py (26/26 + 20/20, SIMPLEX-FOURIER-EXACT + INTERTWINER-IDENTIFIED + WINDING-QUADRATIC-EXACT; probes simplex\_fourier\_probe.py + flavor\_winding\_quadratic\_probe.py: the Walsh law with ramification/equidistribution at every level, the packet-code intertwiner identity (both ladders  $[-16, 112, -448, 1136, \dots]$ ), the value theorem to 16500; the adjugate quadratic with the triple lock, the locus census and four firing controls); v858\_g31\_clock\_alphabet.py (19/19, CLOCK-ALPHABET-SELECTIVE; probe g31\_clock\_alphabet\_probe.py: BFS closure 46080 exact, the 60-reflection census with zero quasi-reflections (v634 reproduced), the unique quadruple, the seven-gate battery over the 607-entry catalog, the family boundary to  $m = 120$ , X1/X2 controls, the computed-center fence kill); v859\_grade\_no\_go\_elevator.py (11/11, LOCAL-SIGN-FAILS, +4 Lean-mirror checks; probe euler\_schur\_semigroup\_probe.py: the mechanism real (Schoenberg PSD at all tested  $t$ , exact tangent, free null-sum positivity at  $10^{-10}$ , rates warded exact, constructor fence passed), the lag-form kill (cos-sim 0.004–0.099), the signed-density measurements (comb 0.500 exactly; arch flip bands ending 6.26–6.27 with the analytic  $\tau^* = 6.27$ ), scramble and Epstein controls); v860\_phase\_lever\_closure.py (7/7, PHASE-LEVER-THEOREM; probe phase\_lever\_theorem\_probe.py: the warded fast route, the census regression, the analytic lever bound dominating the exact sup ( $10.3\times$ ), the budget margins on all 67 rungs, the break-even census, the greedy exact-landscape control (S5b: still zero cells), the D-frontier saturation and the recalibrated anti-vacuity ward (v1.1, declared in the frozen spec)). Suite 848  $\rightarrow$  853; all five standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all seven probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).

- **Lean (2 modules promoted).** `TfptCarrier/GradeNoGo.lean` — the grade barrier as a class-closure theorem (13 declarations, 10 theorems; lake build green, 3415 jobs, no `sorry`, no `native_decide`, axioms `propext/Classical.choice/Quot.sound` only, verified by `#print axioms` on all ten): `grade_no_go` (the two-scalar core), `gram_two_homogeneous / grade_no_go_gram`, the affine trilemma (`affine_gram_expansion / affine_gram_tax / affine_grade_no_go`) and the elevator lemma `tangent_psd_on_kernel_null`. `TfptCarrier/CofinalCurrent.lean` — the conditional route to  $(H_{\text{cof}})$  (13 declarations, 7 theorems, axioms clean): `HeadTailSplit`, the three NAMED hypotheses with provenance typed, `cofinal_current` (extraction from the recurrence), `cofinal_current_weil` (composition through the minimal H theorem), the  $\delta = 0$  relaxation, the Rayleigh reading, and `connectedTail_not_sufficient`. Both imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round, 85  $\rightarrow$  87 files. Mirrored in v859 part A and v856 part C (numeric witnesses, v849/v843 precedent).
- **The grade law (the round’s deliverable).** Free positivity lives at QUADRATIC grade, the deployed target at LINEAR grade, and no exact match exists on any scaling ray — the four graveyard instances (Schur–Gram tax, covariance grade gap, wedge uniformity failure, phase-lever closure) are instances of ONE kernel-checked homogeneity law with measured exchange rate  $10^4$ – $10^5$ ; the affine escape is priced exactly; the only licensed grade-1 mechanism (rates) is proven abstractly, instantiated, and its deployed instantiation honestly killed by signed densities on both sides ( $\tau^* = 6.27$  typed analytically); the phase-rescue family is closed by theorem; and the two typed positive cores enter the record: the tax-evading zero-free pair-correlation instrument and the conditional kernel-checked frame three hypotheses  $\Rightarrow (H_{\text{cof}})$ . Stop list extended (finite-head phase rescues; amplitude-Gram constructions without grade accounting; Lévy/conditional-positivity elevators on the deployed sides). NO RH claim.
- **Ledger.** 7 new rows (`PRIME.RELATION.CONNECTED_COVARIANCE.01`, `FORM.PRIME.COFINAL_CURRENT.01` [F], `E8.SIMPLEX.FOURIER.01`, `E8.G31.CLOCK_ALPHABET.01`, `FORM.PRIME.GRADE.NO_GO.01` [F], `PRIME.EULER.SCHUR.SEMIGROUP.01`, `PRIME.PHASE.LEVER.01`) + the dated round-32 GRADE-LAW note on `PRIME.RELATION.SKELETON.01` (the unifying diagnosis, the elevator closure, the extended stop list) and the dated round-32 instrument note on `PRIME.FLOOR.PAIRCORR.01` (the pair-type object constructed zero-free; the round-31 transport note also restored to the contract block, mirroring the ledger); 936  $\rightarrow$  943 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The grade law: the graveyard unified by one homogeneity theorem; the licensed elevator and its honest death; the phase-rescue family closed” (`\veri{v856/v859/v860}`) + the part-intro chain extended + the Lean jobs count refreshed; `tfpt_1`: the simplex-Fourier character theorem and G31 clock alphabet entries in the Gaussian code keybox (`\veri{v857/v858}`); `tfpt_2`: the winding-quadratic addendum in the Winding Line block (`\veri{v857}`); `tfpt_3`: five new audit cards + the round-32 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the `FORM.PRIME.GRADE.NO_GO.01` and `FORM.PRIME.COFINAL_CURRENT.01` registration blocks, the round-32 E8 audit compressions record, and the dated round-32 notes on the `PRIME.RELATION.SKELETON.01` and `PRIME.FLOOR.PAIRCORR.01` blocks (Lean counts refreshed to 3415 jobs / 87 files).
- **Website.** Prime-front feed: the round entry “reflux round thirty-two”; counters 848  $\rightarrow$  853 modules / v535–v860 (the prime-front page metadata refreshed from its stale 841/v848); DAG nodes extended with v856–v860; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `maps`); README and Zenodo updated to 853 modules / v860.
- **Scope.** The E8 audit modules (v857, v858) are exact-arithmetic corpus compressions and audits of deployed claims — normal forms, character identities and armed kill rules, no new physical claim; the architecture modules (v856, v859, v860) are typed route decisions under frozen preregistered protocols (the relapse expectation and the sign expectation both REFUTED BY MEASUREMENT and typed either way in the frozen specs — the kills moved to where the

numbers put them); the Lean modules consume and formalize, never evaluate, positivity — all named hypotheses stay hypotheses; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXIV)** (reflux round thirty-one — the invariance atlas, the graded kernel, the code complex, and the minimal hypothesis kernel-checked: SEVEN modules (v849–v855, from fifteen frozen probes plus one Lean module; suite 841 → 848) — the consolidation of the 2026-08-07 evening harvest in three waves: (1) **THE F-PLAN** — the compiler object is **UNIQUE** (all 14400 admissible tuples **ONE**  $Sp(4, 2) \times S_5$  orbit,  $\text{Aut}(C_{\text{fin}}) \cong C_6$ , unique up to **NON-unique** isomorphism) and the minimal H theorem is kernel-checked (TfptCarrier/CofinalWeil.lean: the  $\varepsilon/2$  implication proven with **NO** diagonal argument, the strict hierarchy  $\text{uniform} \subsetneq \text{pointwise} \subsetneq \text{cofinal}$  — **UniformMarginBound** formally **DEMOTED** to an over-strong sufficient lemma; the load-bearing chain consumes **EXACTLY** ( $H_{\text{cof}}$ )); the **TAX THEOREM** closes the positive-dilation completion class at **CLASS** level ( $\text{tr } P + \text{tr } Q \geq 2\|R\|_*$  exact with zero-gap dual certificates, **BASIS-INVARIANT** — the v846 geometry-independence now a corollary; the exact optimum already  $\geq 4057.6 \times \tau$  on every rung, diverging  $\sim h^{0.24}$  vs  $h^{-3/2}$ ; the measured Schur designs only  $13.7\text{--}20.4\times$  above it — optimality could not have saved them); the index route closes by measurement (the density endpoint **NOT** PSD,  $n_- = 5.12$  — premise corrected; all 634 crossings upward, endpoint-determined; the velocity test at ratio 1.00: distance-to-crossing = metric margin / flow velocity); the corner cluster expansion is exact but its weights are **POSITION-GEOMETRIC** (route closed, two frozen-honest **FAILS** kept); the cone violations resolve into the **GRADED KERNEL LAW**: the strict census wanders ( $P_0 \sim X^{1.03}$ , max 244333 vs cap 100) but the depth- $10^{-2}$  **G1** kernel is  $n \leq 73$  on **ALL 67** rungs and the skin depth obeys  $D(X) = 0.754 X^{-0.548}$  ( $R^2$  0.99) — vanishing-depth skin, not kernel growth (**PRIME.CELLCONE.GRADEDKERNEL.01** [O] registered; the tube field transports clean 0/355199 but is self-calibrating, four must-break controls kept as **FAILS**); (2) **THE CODE COMPLEX** — the Arf-vacuum syndrome theorem ( $a(t+U) = \hbar(u, v)$  on all 140 flats; the two-bit table 15/20/45/60; the **LOCAL** Hecke theorem  $12/16 \rightarrow 28/-4$  at all 15 points; the code pencil  $[15, 11, 3] \supset \{[15, 10, 4], 2 \times [15, 10, 3]\} \supset [15, 9, 4]$  with the canonicity lemma; the vacuum bit = the **RM(2, 4)** parity bit), the ovoid decoder ( $\text{SNF}(N) = \text{diag}(1^{10}, 0^5)$  torsion-free;  $N^+ = N^T/4 - J/36$  exact;  $P_5 = I/2 - B/4 + J/12$ ;  $\ker =$  the  $S_6$  standard rep 11/11; three rings:  $\mathbb{R}$  ovoid span /  $\mathbb{Z}$  torsion-free /  $\mathbb{F}_2$  code  $A_q = [15, 5, 6]$ ), the bent CSS pair ( $\hat{s}_q = -4s_q$  on all 16; the (16, 6, 2) difference set;  $[[15, 5, 3]] \rightarrow [[16, 4, 4]]$  with the vacuum transformation typed; **THE -4 TRIPLE POINT**: Hecke = Gauss = Walsh =  $a_3$  as **ONE** machine-checked integer; the 16 bent translates the MUB partner of the v800 rays, no upgrade), and the von Mangoldt commutator theorem ( $\mathcal{L} = -M[D, Z] = -[D, \log Z] = T(\Lambda)$  **EXACT** three ways;  $\log Z$  **RATIONAL**, **BCH** truncates; the four-comb discipline at operator level with the Epstein leak  $[6, 14, 21, \dots]$  exact and repaired by  $a_A + a_B = 1 * \chi_{-20}$ ; the deployed carrier **IS** the first row — typed honestly as classical incidence algebra); the factorization complex (multiplicativity **IS** flatness at machine zero; Epstein curvature localized at class-group products with the  $\mathbb{F}_2$  Arf-lift obstruction **PROVEN**; gate 4 fails typed — the form is 0-chain-supported; the Pfaffian has no anchor-independent normalizer); (3) **THE STRATEGIC ATTACKS** — **THE INVARIANCE ATLAS**: the demand rewritten **EXACTLY** in Selberg-hierarchy coordinates and audited against on-range-verified elementary constants (the sieve reaches the  $\alpha \in 1..2$  band in **SCOPE** but **ZERO** coverage in **GRADE**, best factor  $205\times$ ; the demand at  $0.99\text{--}1.00 \tau_{\text{pnt}}$ , the factor-2 gate unclosable at truth — **INPUT-CLASS-INVARIANT**; the named sieve object  $d\text{Chain}_2$ ); the criteria align

(the Li coefficients  $\lambda_n > 0$  to  $n = 20$  computed with NO zeros;  $d_N^2$  exact to  $N = 64$ ; BD-constant =  $2\lambda_1$  at  $4.9 \times 10^{-17}$ ; the NB span IS the spectral-mother geometry term by term; the Li transfer fails by CONE GEOMETRY) — no coordinate system holds hidden supply; the minimizer census (no frozen closed-form limit; the internal direction exists at 0.0217 rad and IS the comb block's soft direction at 0.0056 rad; the drift real, no identification claimed,  $-4/\pi$  a coincidence) — **THE WALL IS INVARIANT ACROSS COMPRESSIONS, GEOMETRIES, CRITERIA AND INPUT CLASSES**; the one object in four languages:  $e_1$  bound = form-factor band =  $d\text{Chain}_2$  = comb-soft direction; NO marker moves; NO RH claim anywhere)

- **New modules (7).** `v849_cfin_unique_cofinal_lean.py` (22/22, CFIN-UNIQUE, +4 Lean-mirror checks; probe `cfin_uniqueness_probe.py`: the WLOG ladder exact (28 alternating forms one  $GL(4, 2)$  orbit;  $Sp(4, 2)$  transitive on the 6 Arf-1 refinements; 20 admissible  $\sigma$  per refinement, one orbit; 120 admissible equivariant  $\iota$ , one orbit); the census 14400, ONE orbit (orbit-stabilizer  $14400 \times 6 = 86400$  exact);  $\text{Aut}(C_{\text{fin}}) \cong C_6$  (orders  $[1, 2, 3, 3, 6, 6]$ , faithful into  $Sp(4, 2)$ , slot permutation determined); the Gaussian diagram typed (18-element centralizer torsor; the strict terminal reading FAILS honestly); controls fire (dot form 0 refinements; wrong Arf class: empty  $\iota$  set; no- $\sigma$ : canonicity dies  $16 \rightarrow 8 \rightarrow 4 \neq 1$ )); `v850_completion_tax_flow.py` (15 + 11, zero fails, TAX-GAP + FLOW-CROSSINGS; probes `psd_completion_tax_probe.py` + `spectral_flow_pivot_probe.py`: the nuclear-norm identity with zero-duality-gap certificate pairs on every anchor ( $\lambda_{\min} \geq -8.9 \times 10^{-15}$ ,  $\|W\|_{\text{op}} = 1.000000000000$ ,  $\langle W, R \rangle \text{ rel} \leq 1.6 \times 10^{-16}$ ; symbolic  $2 \times 2$  at 50 dps, gap  $2.7 \times 10^{-51}$ ), basis-invariant ( $\leq 1.1 \times 10^{-15}$ ); tax/ $\tau \geq 4057.6$  on all 67 rungs (max  $7.5 \times 10^5$ , slopes 1.99/1.86/0.46); scramble/Epstein fingerprints move the norm (0.54–0.96 / 0.12–0.22); the flow: 634 upward crossings,  $SF = n_-(0) - n_-(1)$  verified 67/67, last crossing hugs the deployed point (median gap  $1.6 \times 10^{-5}$ , slope  $-2.11$  vs  $\tau$ 's  $-1.65$ ), velocity ratio 1.00 (IQR 0.99..1.00); Jacobi pivot certification eigenproblem-free; scramble/Epstein/density controls typed); `v851_cluster_kernel_field.py` (15 + 10 + 7 with the SIX frozen-honest FAILS S3.2/S4.1/S2.C1/S2.C2/S4.C1/S4.C2 kept and pattern-gated, NOT refit; CLUSTERS-DIFFUSE + BOTH-PARTIAL + KERNEL-GRADED; probes `multiplicative_cluster_probe.py` + `cone_kernel_field_probe.py` + `graded_kernel_field_probe.py` over the read-only `cell_cone_transport_probe.py` parent: the expansion exact (sympy telescoping; Möbius elimination;  $2^{16}$ -subset ward 0.0; weights =  $-\text{tr}(B^{-1}E_i B^{-1}E_j)$ ) but  $E_2 = 0.883$  vs bar 3.0 and the scramble does not collapse it — position-geometric; strict censuses wander ( $P_0 \sim X^{1.03}/X^{1.02}$  at  $R^2$  1.00, max 244333/283607) while the G1 depth- $10^{-2}$  kernel is  $n \leq 73$  on all rungs; the causal tube field clean on all four rule $\times$ grouping combos (0/355199, cond  $\leq 57.3$ ) but self-calibrating; the skin law  $D(X) = 0.754 X^{-0.548}$  with verified  $X_0(\varepsilon)$ ; the G2 mechanism typed (1653 $\times$  / 39 $\times$ ); scramble depth control fires, Epstein anchor holds); `v852_arf_vacuum_ovoid.py` (24/24 + 23/23, ARF-VACUUM-SYNDROME-EXACT + OVOID-DECODER-EXACT; probes `affine_arf_vacuum_probe.py` + `doily_ovoid_decoder_probe.py`: the syndrome table, the local Hecke theorem, the code pencil with the canonicity lemma,  $\text{RM}(2, 4)$  restored with the vacuum bit as parity bit;  $N^T v_a = 0$ , Gram  $36I - 6J$ , SNF torsion-free with a det  $\pm 1$  minor, Penrose exact in  $\mathbb{Q}$ , the decoder exact on the predeclared set, the  $S_6$  standard rep, the three-ring correspondence; five controls fire); `v853_bent_css_mangoldt.py` (22/22 + 30/30, ARF-BENT-CSS-EXACT + MANGOLDT-COMMUTATOR-EXACT; probes `arf_bent_css_probe.py` + `mangoldt_incidence_probe.py`: the Walsh eigenvector at  $-4$ , the (16, 6, 2) set, the 31-word census, the two CSS codes with the 60 minimal words = the 60 isotropic planes, the  $-4$  triple point, the MUB partner (no upgrade);  $\mathcal{L} = -M[D, Z] = T(\Lambda)$  exact at  $N = 60$  (sympy) /  $N = 360$  (integer) /  $N = 10^4$  (float ward  $5.3 \times 10^{-15}$ ), the one-line proof termwise, powers = chains with nilpotency  $\lfloor \log_2 N \rfloor + 1$ ,  $\log Z$  rational with BCH truncation, the four-comb operator table with the Epstein leak exact and class-average repaired, the ideal-level Dedekind  $\Lambda_K$  with norm aggregation, the carrier bridge per-divisor

exact,  $L^2 - [D, L] = T(\mu * \log^2)$ ); `v854_relation_hodge_pfaffian.py` (18/18, CURVATURE-DISCRIMINATES-ONLY; probe `relation_hodge_pfaffian_probe.py`:  $d_1 d_2 = 0$  exact,  $b_0 = 1$ ,  $b_1 = 16/21/22$  on the same-prime chords; the true comb exactly flat on ALL squares ( $\leq 1.8 \times 10^{-15}$ ); Epstein curvature 321/756/905 squares at the class-group products; the Arf lift PROVEN obstructed (GF(2) rank vs augmented); gate 4 fails typed ( $S_{\text{cob}} \sim 2.9\text{--}3.9 \times 10^4$ ); the Pfaffian log-spread infinite); `v855_invariance_atlas.py` (19 + 8 + 7 + 3, zero fails, SIEVE-SAME-GAP (spec v2, the S5.1 bar recalibration 5.0  $\rightarrow$  2.0 declared, measured 3.9) + ATLAS-SAME-WALL + PROFILE-DIVERGES + DIRECTION-TRANSIENT; probes `selberg_supply_probe.py` + `criteria_atlas_probe.py` + `minimizer_profile_probe.py` + `minimizer_direction_probe.py` over the read-only `spectral_flow_pivot_probe.py` frame: the hierarchy decomposition exact ( $D_0 = d\text{Head}_2 - d\text{Chain}_2$  at  $10^{-9}$ ; levels 2/3 exact in Fractions); the envelopes verified on range and dominating (Chebyshev  $0.111x$  measured 0.0956); the ledger (best factor  $2.05 \times 10^2$ , route 2 up to  $1.1 \times 10^4$ ;  $|D_0|/\tau_{\text{pnt}} = 0.99..1.00$ ;  $\min |dS|_F/\text{GATE}_W = 2.28 > 1$ ); Brun–Titchmarsh typed UNCOVERED, the large sieve NOT-APPLICABLE per rung;  $\lambda_1 = 1 + \gamma/2 - \log(4\pi)/2$  at dev 0.0, literature wards  $3 \times 10^{-10}$ ;  $d_N^2$  exact to  $N = 64$ ;  $\text{BD} = 2\lambda_1$ ; the NB span = the spectral-mother geometry (the exact  $\mu$  floor identity; the  $1/x$  mirror = J); the Li cone residual 60.6% vs 5%; the minimizer census (RAY 0.4530 / ARCH 1.0976 rad; dispersion 0.0217; C03 tracking 0.0056; the Bonferroni demonstration 8 of 15 with the golden-ratio control rejected)). Suite 841  $\rightarrow$  848; all seven standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all fifteen probes re-run identically at promotion (2026-08-07); promoted VERBATIM with no downscoping (the round-31 embedding convention: the frozen probe sources embedded byte-exact and executed verbatim in isolated namespaces — every printed spec SHA-256 reproduces).

- **Lean (1 module promoted; wording corrected by 2026-08-13 XCII).** `TfptCarrier/CofinalWeil.lean` — the minimal H theorem, kernel-checked (11 declarations; lake build green, 3411 jobs, no sorry, no `native_decide`, axioms `propext/Classical.choice/Quot.sound` only, verified by `#print axioms` on all eight theorems; imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round, 84  $\rightarrow$  85 files): `CofinalHypothesis` ( $H_{\text{cof}}$ ) stores the MATHEMATICAL CORE — a strictly monotone selected ladder with PSD rungs; it does not by field order enforce that the ladder was predefined or sign-independent. `limit_nonneg_of_cofinal_seq` — THE MINIMAL IMPLICATION, the  $\varepsilon/2$  argument proven; `ladderForm_nonneg / weil_nonneg_of_cofinal` — the dense-family form with NO diagonal argument (one PSD certificate per rung covers every element simultaneously); the STRICT HIERARCHY `uniformMarginBound_to_cofinal / pointwise_to_cofinal / cofinal_not_uniform` (witness  $1/(m+1)$ , reusing the v843 kernel gap) / `cofinal_not_pointwise` (witness  $\pm 1$  on the even ladder); the assembly `cofinal_weil`. THE FORMAL DEMOTION: the load-bearing chain of `PRIME.EXTRACTION.CHAIN.01` consumes EXACTLY the mathematical ( $H_{\text{cof}}$ ) payload; `UniformMarginBound` implies it and the converse fails kernel-checked — from this round an over-strong SUFFICIENT lemma, no longer the named wall. PREDEFINED/noninterference is the separate explicit external contract added by `CofinalPredefinition.lean` in XCII. ( $H_{\text{cof}}$ ) stays a hypothesis everywhere. Cited in v849's part B (numeric witness mirror, v843/v837/v817 precedent).
- **The wall statement (the round's deliverable).** THE INVARIANCE ATLAS: the wall is proven invariant across COMPRESSIONS (the diverging nuclear-norm tax closes the positive-dilation class — in any basis; the index is endpoint-determined; the corner cluster expansion is position-geometric), GEOMETRIES (basis invariance — the v846 TV-price geometry-independence now a corollary), CRITERIA (Weil = Li = Nyman–Beurling;  $\text{BD-constant} = 2\lambda_1$  at  $4.9 \times 10^{-17}$ ; the NB span IS the spectral-mother geometry; the Li transfer fails by cone geometry — certified partial data everywhere, one uniform statement missing everywhere) and INPUT CLASSES (statistics = sieve: the demand at truth exceeds the factor-2 gate on every

rung — unclosable by ANY upper-bound class). The one object in four languages: the uniform  $e_1$  bound = the form-factor band  $\alpha \in 1..2$  = the bilinear chain fluctuation  $d\text{Chain}_2$  = the comb block's soft direction (0.0056 rad). The positive cores: the graded kernel law (depth, not sign) and the minimal kernel-checked hypothesis ( $H_{\text{cof}}$ ). Stop list extended (positive-dilation completions with fixed comb coupling in any basis; finite-dimensional index reformulations; corner cluster expansions; elementary upper-bound supply rows against the factor-2 gate). NO RH claim.

- **Ledger.** 9 new rows (CFIN.UNIQUE.01, FORM.PRIME.COFINAL.WEIL.01 [F], PRIME.PSD.COMPLETION.TAX.01, PRIME.CELLCONE.KERNELFIELD.01, PRIME.CELLCONE.GRADEDKERNEL.01 [O], ARF.VACUUM.SYNDROME.01, ARF.BENT.CSS.01, PRIME.RELATION.HODGE.01, PRIME.CRITERIA.ATLAS.01) + four dated round-31 notes: the invariance-atlas closing note on PRIME.RELATION.SKELETON.01 (the four-axis invariance; the one object in four languages; the extended stop list), the transport note on PRIME.FLOOR.PAIRCORR.01 (the pair-type object =  $d\text{Chain}_2$  = the comb-soft direction), the demotion note on FORM.PRIME.EXCESS.SKELETON.01 (UniformMarginBound  $\rightarrow$  over-strong sufficient lemma), and the kernel-check note on PRIME.EXTRACTION.CHAIN.01 (step 2 kernel-checked; (H) formalized as ( $H_{\text{cof}}$ )); 927  $\rightarrow$  936 rows; NO marker moves.
- **Papers.** `tfpt_1`: the Arf-vacuum syndrome + code pencil, the bent CSS pair + the  $-4$  triple point, and the ovoid decoder entries in the Gaussian code keybox + the new  $C_{\text{fin}}$  uniqueness keybox (Aut  $\cong C_6$ ) after the normal-form keybox (`\veri{v849/v852/v853}`); `tfpt_prime_front`: the new round section “The invariance atlas: the wall is the same across compressions, geometries, criteria and input classes; the minimal hypothesis kernel-checked” (`\veri{v849/v850/v851/v853/v854/v855}`) + the part-intro chain extended; `tfpt_3`: seven new audit cards + the round-31 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the FORM.PRIME.COFINAL.WEIL.01 and PRIME.CELLCONE.GRADEDKERNEL.01 registration blocks, the round-31 code-layer compressions record, and the dated round-31 notes on the PRIME.RELATION.SKELETON.01, PRIME.MARGIN.LAW.01/FORM.PRIME.EXCESS.SKELETON.01 and PRIME.EXTRACTION.CHAIN.01 blocks.
- **Website.** Prime-front feed: the round entry “reflux round thirty-one”; counters 841  $\rightarrow$  848 modules / v535–v855; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 848 modules / v855.
- **Scope.** The code-layer modules (v849 part A, v852, v853 part A) are exact-arithmetic compressions and a uniqueness theorem — normal forms and set equalities, not new claims (the P2 residual  $R1'$  and the MUB reading stay typed, no upgrades); the no-go modules (v850, v851, v854) are typed route decisions under frozen preregistered protocols (exact identities [E]-grade; the six control FAILS preregistered-honest adjudications on record, bars NOT refit); the atlas module (v855) is an honest invariance mapping (the one bar recalibration declared in the frozen spec; no number invented for inapplicable supply rows); the Lean module consumes and formalizes, never evaluates, positivity — ( $H_{\text{cof}}$ ) stays a hypothesis; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXIII)** (reflux round thirty — the three-architecture decision, the extraction chain complete, and the finite-compiler compressions: FIVE modules (v844–v848, from eight frozen probes; suite 836  $\rightarrow$  841) — the v5.4 strategy campaign consolidates: the three candidate proof architectures are DECIDED, all honest negatives with typed positive cores — (i) the relational Schur–Gram route dies STRUCTURALLY (three of four gates green incl. the exact  $\mu \log$  descent with the  $h=2$  Epstein refusal at construction grade; the corner identity fails in trace AND form by exactly the forced Cauchy–Schwarz diagonal



price,  $\text{price}/\tau \sim 3.1 \times 10^4 - 4.5 \times 10^4$  at the density-anchored floor) and the price is GEOMETRY-INDEPENDENT: the unitary spectral mother ( $J^2 = I$  exact; the deployed form IS the  $J$ -odd Toeplitz compression; the Euler product the exact operator algebra  $U_d U_m = U_{dm}$ ) gains 20–33× at symbol grade but never at Gram grade — harvesting interference is Fejér–Riesz of the total symbol = the positivity itself; (ii) the sign-register wedge lift EXISTS as exact algebra (nonneg wedge weights per frame where the naive signed Lagrange provably cannot) but the frame-uniform law fails at  $\sim 15\%$  vs the  $4.4 \times 10^{-5}$  floor need — the commutant uniformity wall transported; (iii) the completed-cell Lorentz transport BREAKS the cone at the  $n=2$  cell on 67/67 rungs under both predeclared groupings (endpoint back in-cone = the certified  $\tau > 0$ ; ray cascade to  $(5, -3, 4)$  CONFIRMED, Kendall  $\pm 1.000$ ; INVERTED Epstein signature: the fake stays in-cone, the arithmetic sits IN the violations) — all three die on the SAME finer-than-statistical substance; AND the continuum extraction is separated from all RH semantics and CLOSED: the implication chain cofinal finite positivity (H)  $\Rightarrow$  Weil positivity  $\Rightarrow$  (Weil’s criterion) the target is theorem-grade modulo named citations with a QUANTIFIER REDUCTION (no Mosco compactness, no uniform  $\delta$ , no diagonal argument; the pure-box class carries the EXACT Weil value at every finite level) — the wall is EXACTLY hypothesis (H) and nothing else (PRIME.EXTRACTION.CHAIN.01 [O] registered); plus two exact finite-compiler compressions: the message ladder  $M_n = 15 \cdot 2^n$  with  $h(E_8) = 30$  derived from the rebuilt root system, the Doily–Pascal rank theorem (the P2 integers as the singular-value data of the Cremona–Richmond incidence), and the finite compiler normal form  $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$  as ONE object with the code-to-matter kill built in; NO marker moves; NO RH claim anywhere)

- **New modules (5).** `v844_message_doily_rank.py` (23/23 + 16/16, MESSAGE-LADDER-EXACT + DOILY-PASCAL-RANK-EXACT; probes `binary_message_ladder_probe.py` + `doily_pascal_rank_probe.py`, promoted verbatim:  $M_n = 15 \cdot \Delta p_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$  against the ACTUAL rebuilt objects, one named structure per rung —  $M_1 = 30 = h(E_8)$  derived here from the rebuilt root system (8 simple roots, Cartan det 1, E8 degree sequence, highest root unique at height 29 with marks  $\{2, 2, 3, 3, 4, 4, 5, 6\}$ ),  $M_2 = 60$  Gaussian lines,  $M_3 = 120 = |R^+|$ ,  $M_4 = 240 = 15 \times 16$  census, coda  $248 = 15 \cdot 16 + 8$ ; wrong anchor  $(1, 2, 2)$  and wrong multipliers 14/16 miss all targets; AND the rank theorem:  $NN^T = B + 2I$  entrywise under the  $K_6$  duad model (all 225 cells),  $\text{charpoly}(NN^T) = (x - 9)(x - 4)^9 x^5$  exact — rank  $N = 10 = A_\Lambda$ ,  $\dim \ker N = 5 = g_{\text{car}}$  (the six ovoid indicators SPAN it), multiplicity  $9 = N_{\text{fam}}^2$ , top  $3 = N_{\text{fam}}$ , recovery  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}}$ , six-step  $(2/3)^6 = 64/729$  exact; NO  $(1/3)^6$  doily mode (v814 typed, no upgrade); wrong-pairing controls fire); `v845_cfin_normal_form.py` (28/28, SPEC v2 declared, NORMAL-FORM-ASSEMBLED; probe `finite_compiler_normal_form_probe.py`:  $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$  READ OFF the Gaussian quotient — the unique selector-picked refinement (16 refinements, Arf 6 + 10), the order-3 family cycle, the parity lift onto  $C_{\text{even}}(5)$  with  $\beta \equiv \hbar$  and  $q_{\text{wt}} \equiv q^*$ ; derived:  $|Sp(4, 2)| = 720$  full census,  $\text{Stab}(q^*) = 120$  faithful  $S_5$ , orbits  $[1, 5, 10]$ , Pascal  $(1, 10, 5) = \left(\binom{5}{0}, \binom{5}{2}, \binom{5}{4}\right)$ , the  $\bar{5}$  an OVOID of the doily, budget  $240 = 16 \cdot 15 / 248 = 16 \cdot 15 + 8$  against the actual census; the kill built in: the 128-spinor counting bound ( $\geq 7$  of 15 classes always adjoint-side; saturated  $8 + 7$ , zero mixed) + the weight-level kill (8 pairs, never 1) — ROOTCLASS-MIXED re-verified, NOT overturned; P2 narrowed to the v799 residual  $R1'$  ([C]), not eliminated; controls fire); `v846_schur_spectral_mother.py` (19 + 16 checks, zero fails, SCHUR-INDEFINITE + SPECTRAL-TV-UNIVERSAL; probes `relational_schur_gram_probe.py` + `spectral_mother_probe.py`: the Schur program closed as a typed structural negative — G1/G3/G4 green (manifest positivity; exact  $\mu$  log routing, Epstein bookkeeping defect 0.73–0.83 vs TRUE 0; Selberg safety), the G2 form gate fails structurally (the P0 deviation IS  $\text{diag}(\text{price})$  at  $\leq 10^{-9}$ ;  $\text{price}/\tau \sim 3.1 - 4.5 \times 10^4$ ;

trace fails by the same object); the spectral mother built as typed ( $J^2 = I \text{ dev } 0$ ;  $K$  the  $J$ -odd Toeplitz compression rel 0;  $U_d U_m = U_{dm}$  exact at integer commensurability): symbol-grade gain 20–33 $\times$ , Gram-grade best legal overhead  $34\text{--}55 = 5.7 \times 10^4\text{--}1.2 \times 10^5 \times \tau$  — THE TV PRICE IS GEOMETRY-INDEPENDENT); `v847_wedge_cellcone_transport.py` (13 + 9 checks with the TWO frozen-honest FAILS S5.SCR + S5.C3 kept and pattern-gated, NOT refit; WEDGE-PARTIAL + CONE-BROKEN + RAY-EDGE-CONFIRMED; probes `relational_lagrange_probe.py` + `cell_cone_transport_probe.py`: the naive signed Lagrange no-go EXACT (forced weights;  $P^-/\det \sim 2.0\text{--}3.2 \times 10^4$ ; the arch block NEGATIVE definite, eigen  $-2.285/-1.139$ ); the  $C_2$  register lift feasible on every anchor (LP resid  $\leq 8.9 \times 10^{-16}$ ); level C fails (design 0.170/0.151, held-out 0.153 vs floor  $4.4 \times 10^{-5}$ ; scramble  $\times 7.5$  vs  $\times 10$  bar — typed, not excused); the completed cells telescope exactly ( $3.4 \times 10^{-15}$ ;  $\Phi^\top J \Phi = cJ$  at  $2.2 \times 10^{-16}$ ; min  $e_1 = 4.8546$  reproduces the envelope); both groupings exit on 67/67 rungs at  $n = 2$  (first-exit median cell 1–2; endpoint back in-cone 67/67); ray cascade  $r_{\text{end}} 0.5700 \rightarrow 0.8018$  / angle  $46.26^\circ \rightarrow 34.39^\circ$  (Kendall  $\pm 1.000$ ); incomplete contrast 67/67; equal-weight and wrong-norm controls fire 14/14; the Epstein comb does NOT fire (0/14) — the inverted signature kept); `v848_extraction_chain.py` (19/19, EXTRACTION-CHAIN-COMPLETE; probe `continuum_extraction_probe.py`, run-1 $\rightarrow$ run-2 declared calibration (P2.0 exact-class certificate, v762 D2.0 precedent): the  $(D, X)$  ladder  $j = 4..11$  ( $D_j = 2^{-j}$ , 68–8704 lags,  $ka = 28$  pinned); the pure-box class EXACT at every level (max err  $5.3 \times 10^{-14}$ ); identification rates  $-1.58..-1.84$  (median  $-1.818$ , dyadic prediction  $-2$ ); ledger geometric (med3 0.302, law  $-1.752$ ); X-direction EXACTLY stable (44 cutoff pairs at float zero); recovery constructive + Cauchy (exact strong- $L^2$  law in  $\mathbb{Q}$ ); F2/F3 band margins TYPED never gated; W1 regression wards ( $5.2 \times 10^{-13}$  /  $2.1 \times 10^{-6}$  /  $8.0 \times 10^{-8}$ ); controls fire (wrong-limit stall  $1.6 \times 10^6$ ; scrambled tower  $5.5 \times 10^5$ )). Suite 836  $\rightarrow$  841; all five run standalone green with pattern gates encoding the frozen verdicts (incl. the two expected-FAIL signatures); all eight probes re-run identically at promotion (2026-08-07); promoted VERBATIM with no downscoping (spec/rule SHA-256 values reproduce byte-exactly).

- **The wall statement (the round’s deliverable).** The three v5.4 architectures die on the SAME substance — the total-variation price (geometry-independent by the unitary mother), the frame-uniformity failure (the commutant wall transported) and the unrenormalized  $n = 2$  cell (the arithmetic IN the violations) are three coordinate presentations of the one finer-than-statistical datum the GUE-side rounds typed; the stop list extends (diagonal relational mothers; constant wedge laws; unrenormalized small- $n$  cell transports). On the other side the extraction is COMPLETE: cofinal finite positivity (H)  $\Rightarrow$  Weil positivity  $\Rightarrow$  the target, theorem-grade modulo named citations (Weil 1952, Bombieri 2000, Suzuki arXiv:2606.09096, Iwaniec–Kowalski Thm 5.12), with the quantifier reduction — the arithmetic wall is EXACTLY hypothesis (H), equivalently the uniform demand of PRIME.RELATION.SKELETON.01 restricted to the cofinal ladder, AND NOTHING ELSE. NO RH claim.
- **Ledger.** 5 new rows (MESSAGE.LADDER.01, NORMALFORM.CFIN.01, PRIME.SCHUR.GRAM.01, PRIME.WEDGE.LAGRANGE.01, PRIME.EXTRACTION.CHAIN.01 [O]) + the dated round-30 closing note on PRIME.RELATION.SKELETON.01 (the three-architecture decision; the extended stop list; the extraction separation) with its dependencies/evidence extended to v846–v848; 922  $\rightarrow$  927 rows; NO marker moves (the compressions are normal forms; the architecture modules are typed route decisions; the extraction registration is OPEN [O] with hypothesis (H) the isolated wall).
- **Papers.** `tfpt_1`: the message-ladder and Doily–Pascal rank entries in the Gaussian code keybox + the new  $C_{\text{fin}}$  normal-form keybox + the dated addendum in the P2 reduction-state keybox (`\veri{v844}`, `\veri{v845}`); `tfpt_prime_front`: the new round section “The three-architecture decision, and the extraction chain complete” (`\veri{v846/v847/v848}`) + the part-intro chain extended; `tfpt_3`: five new audit cards + the round-30 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the PRIME.EXTRACTION.CHAIN.01 registration block, the finite-compiler compressions record, and the dated round-30 closing note on the PRIME.RELATION.SKELETON.01 block.

- **Website.** Prime-front feed: the round entry “reflux round thirty”; counters 836  $\rightarrow$  841 modules / v535–v848; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 841 modules / v848.
- **Scope.** The compressions (v844/v845) are exact-arithmetic normal forms of existing corpus surfaces — compressions, not new claims (the  $(1/3)^6$  gap and the P2 residual stay typed, no upgrades); the architecture modules (v846/v847) are typed route decisions under frozen preregistered protocols (exact wards [E]-grade; the two control FAILS preregistered-honest adjudications on record, bars NOT refit); the extraction module (v848) never evaluates hypothesis (H) — the implication is measured/cited, the classical legs carry named citations, and the wall stays OPEN [O]; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXII) (reflux round twenty-nine — the margin as a doubly-derived law, and the wall kernel-checked: ONE module (v843, from one frozen probe plus one Lean module; suite 835  $\rightarrow$  836) — the certified margin series becomes a typed mathematical object:  $\tau = e_1(\alpha) h^{-3/2} \tau_{\text{pnt}}(\alpha)$  with the certified envelope constant reproduced EXACTLY across two independent coordinate systems ( $e_1 \in [4.855, 24.209]$  on all 67 rungs,  $\min \geq 4.335$  — the double-derivation ward;  $\tau$  vs  $\lambda_{\min}(A_h)$  at  $2.1 \times 10^{-12}$ ), the ONE preregistered-honest FAIL kept and pattern-gated (fit-free  $h$ -exponent median  $-1.341$  vs the frozen bar edge  $-1.35$ , NOT refit), the tower gives SCALE not recursion (ratio defects non-decaying), the excess is a GROWING cancellation (CI median 15.6, rising  $5.65 \rightarrow 29.46$  — the wall self-similar at cell level; no per-prime positive control: 22 287 of 351 122 aggregates nonpositive), the compiler connection is NULL (0/96, Bonferroni-honest) — and the finite-table limit becomes a KERNEL-CHECKED THEOREM: TftptCarrier/ExcessSkeleton.lean proves pointwise\_pos\_not\_uniform (per-rung positivity at every rung does not yield a uniform bound) and the bridge theorem excess\_floor consumes UniformMarginBound as the SINGLE NAMED HYPOTHESIS; the quantifier now sits on the ONE scalar series  $e_1 \geq 4.335$ ; same-class margin-law re-attacks stop-listed; NO marker moves; NO RH claim anywhere)**

- **New module (1).** v843\_margin\_law\_excess\_lean.py (10 checks with the ONE preregistered-honest S2.2 FAIL pattern-gated + 4 Lean-mirror checks, verdict MARGIN-RECURSES-THE-WALL; discovery probe margin\_law\_probe.py, 2026-08-07, re-run identically at promotion, promoted VERBATIM with no downscoping,  $\sim 6$  s): Q1 THE LAW — the margin series recomputed in identified-corner coordinates ( $\tau = \lambda_{\min}(S + C) = \lambda_{\min}(A_h)$  at  $2.1 \times 10^{-12}$ ; cell ward  $\sum_j \text{cell}_j = v^T(A - B)v$  at  $10^{-8}$  on all rungs) reproduces the floor-strand envelope EXACTLY:  $e_1 = (\tau/\tau_{\text{pnt}})h^{3/2} \in [4.855, 24.209]$ ,  $\min \geq 4.335$  on all 67 rungs — two independently derived laws, one object;  $\tau_{\text{pnt}}$  carries the entire  $\alpha$ -decay (segment slopes  $-0.23..-0.30$ ),  $e_1$  the arithmetic; the fit-free  $h$ -exponent (21 near-equal-alpha rung pairs) measures median  $-1.341$ , 0.009 outside the frozen bar  $[-1.65, -1.35]$  — the honest miss kept, NOT refit (v829/v831 precedent); Q2 THE TOWER — the induced recursion on the dyadic mass subsequence  $4^m$  ( $m = 4..9$ ): ratio defects (0.6749, 0.5159, 0.0305, 0.3635) non-decaying — scale, not recursion; no quantifier reduction (the  $m = 12$  extrapolation typed HEURISTIC); Q3 THE CELLS — CI median 15.6 rising  $5.65 \rightarrow 29.46$ ; 22 287/351 122 per-prime aggregates nonpositive (deep  $C_2 = -0.623$ ) — the excess is NOT a positive sum over primes: the wall RECURSES into the cells; Q4 — compiler connection NULL (0/96 at  $10^{-3}$ ); CONTROLS — the scramble turns the margin negative on every rung; the  $h = 2$  Epstein routed margin negative AND below its own unrouted baseline at all anchors ( $-0.786/-1.100/-1.228$ , exact rational  $\Lambda_F$  routing). Suite 835  $\rightarrow$  836; standalone green with the pattern gate encoding the expected-FAIL signature.
- **Lean (1 module promoted).** TftptCarrier/ExcessSkeleton.lean — the excess-skeleton logical geography, kernel-checked (18 declarations; lake build green, 3410 jobs,

no sorry, no native\_decide; imported in TfptCarrier.lean; lean\_manifest.sha256 regenerated with this round, 84 files): the two-giants decomposition (formAt\_add / excessAt\_eq\_comb — the excess carried EXACTLY by the comb block; the deployed structural block NOT assumed PSD), the interval-certificate lemma (pos\_of\_certified — the exact formal shape of every v842 certificate), the certified-negative discriminator (certified\_negative\_excludes — the Epstein rejection logic), the finite-ladder quantifier (ladder\_pos\_on\_finset), the bridge theorem excess\_floor with UniformMarginBound the SINGLE NAMED HYPOTHESIS (implying the per-rung IdentificationPositivity instances of SectorPositiveDescent; uniform\_to\_identification proven), and THE KERNEL-CHECKED GAP pointwise\_pos\_not\_uniform: margin sequences strictly positive at EVERY rung with NO uniform lower bound exist (witness  $1/(m+1)$ ) — no finite certificate table, certified or not, discharges the uniform statement: the finite-table limit, until now a typing discipline, is a theorem. Cited in v843's part B (numeric witness mirror, v837/v817 precedent).

- **The wall statement (the round's deliverable).** The infinite quantifier sits on ONE scalar series:  $e_1(m)$ , certified  $\geq 4.335$  on every reachable rung, DOUBLY DERIVED (the consistency ward), self-similar at cell level, with no tower recursion and no compiler-constant shortcut; what any future route must supply is a UNIFORM lower bound on  $e_1$  — by the GUE-side findings a finer-than-statistical datum — with its arithmetic entry point declared. The bridge contract PRIME.FLOOR.PAIRCORR.01 [O] unchanged; same-class margin-law re-attacks (tower-recursion / cell-factorization of the excess) stop-listed per the measured self-similarity. NO RH claim.
- **Ledger.** 2 new rows (PRIME.MARGIN.LAW.01, FORM.PRIME.EXCESS.SKELETON.01 [F]) + the dated round-29 closing note on PRIME.RELATION.SKELETON.01 (the quantifier in its final coordinates; the self-similarity; the kernel-checked finite-table limit; the stop-list) + the dated round-29 margin-law note on PRIME.FLOOR.RATIO.01 (the envelope constant doubly derived — the consistency ward); 920  $\rightarrow$  922 rows; NO marker moves.
- **Papers.** tfpt\_prime\_front: the new round section “The margin as a doubly-derived law, and the wall kernel-checked” (\veri{v843}) + the part-intro chain extended; tfpt\_3: one new audit card + the round-29 follow-up notes on the floor-contract prose; tfpt\_research\_contracts: the PRIME.MARGIN.LAW.01/FORM.PRIME.EXCESS.SKELETON.01 registration block + the dated round-29 notes on the PRIME.FLOOR.RATIO.01 and PRIME.RELATION.SKELETON.01 blocks.
- **Website.** Prime-front feed: the round entry “reflux round twenty-nine”; counters 835  $\rightarrow$  836 modules / v535–v843; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 836 modules / v843.
- **Scope.** The envelope ward, the regressions and the cell ward are [E]-grade machine checks on the frozen protocol; the law, the tower defects, the cell census and the compiler null are measured finite-level evidence; the one FAIL is a preregistered-honest adjudication on record (the bar NOT refit); the Lean statements are kernel-checked with UniformMarginBound deliberately NOT a theorem — the wall itself; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXI) (reflux round twenty-eight — the wall gets coordinates: the corner-closure theorems, the GUE saturation, the relation carrier, and the certified skeleton: SIX modules (v837–v842, from eleven frozen probes plus one Lean module; suite 829  $\rightarrow$  835) — the corner route's perimeter CLOSES as measured compression-class theorems (the 90-cell class map has no identity $\wedge$ visibility $\wedge$ placement cell; the quantifier extends to tower levels  $m \leq 3$ ; all 5276 subgroup expectations, the full pinching and Stinespring compressions obey the dichotomy; position-dependent carriers are EXTREMALLY PINNED —  $\hat{c}_{GL1} = -1$  locks the entire  $C_2$  mass, so identity-true carriers read zero on EVERY self-consistent comb), the zero side is QUANTIFIED (the bridge demand in Montgomery  $F(\alpha)$ -form with the named minimal missing**

input; the demand/GUE ratio saturates **STRUCTURALLY** at  $R_\infty = 1.11$  with the tight band pinned at unfolded  $\alpha \in 1..2$ ; the bootstrap loop is **SHORT** —  $g = 1/(k^2 R^2 c_{\text{sup}})$ , even ideal supply gives  $0.655 < 1$ : the wall's self-conservation **IS** the saturation), and the route **REOPENS** with the relational input: the identification carrier **EXISTS** (`PRIME.RELATION.MULT.01 [O]`; all four gates incl. the self-consistency null; the  $h = 2$  Epstein  $x^2 + 5y^2$  separates at 200.0, Selberg-class-correct blindness), the identified corner's excess is **POSITIVE** on all 67 rungs ( $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$ ), and the certified skeleton (`PRIME.RELATION.SKELETON.01 [O]`) encloses  $\tau_X$  **STRICTLY POSITIVELY** on all 67 rungs — the wall = the infinite quantifier over strictly-positive certified enclosures, sharpened demand: a **UNIFORM** lower bound on the excess margin; the corner/descent core kernel-checked in Lean (`SectorPositiveDescent.lean`, `IdentificationPositivity` the single named hypothesis); **NO** marker moves; **NO** RH claim anywhere)

- **New modules (6).** `v837_corner_closure_quantifier.py` (19/19 + 17/17 + 3 Lean-mirror checks, `DOORS-CLOSED` + `LEVEL2-CLOSED`; probes `hjelmslev_open_doors_probe.py` + `hjelmslev_level2_corner_probe.py`, promoted verbatim: Door 1 — the GL1 corner reads the full lock-pair **COMPRESSION**, not a trace, so factor dressings are generically visible, but every visible dressing breaks the deployed-form identity by exactly its action on the lock pair (mass multiplier  $1.215 \times 10^{-1}$ , unital phase  $5.849 \times 10^{-2}$ ; the commutant invisible at  $2.7 \times 10^{-14}$ ; sweep dichotomy on all 24 draws: visibility  $\Leftrightarrow$  identity defect); Door 2 — the product law  $\hat{c}(\chi) = (-1)^e V_{\text{fac}}(w) M_{\text{fac}}(j; n)$  verified on the full 128-dim carrier Fourier for ALL  $128 \times 136$  character–event pairs (exact Fractions; 18-class collapse exact; the mass ward **GENERALIZED**: one position-blind V-mode + one  $\mu_4$ -mode per event, bit-identical under position scramble); THE CLASS MAP (90 cells): identity  $\Rightarrow$  NOT visible everywhere, carrier cells **NONE**; the quantifier extends to  $m \leq 3$  (rings  $\mathbb{Z}[i]/(1+i)^m$ , complete duals 16/256/4096; identity characters exactly  $\text{Ann}(\langle S_m \rangle)$ ,  $|S_m| = 7 \cdot 8^{m-1}$ ; Vfac spectra  $\{1, 0, -1/7\}$  — the jet refines position-blind data only); PART C: the Lean mirror of `TfptCarrier/SectorPositiveDescent.lean` (corner CP + trace split, descent transport, PSD limits, `sector_floor` with `IdentificationPositivity` the single named hypothesis; lake build green, no sorry, imported in `TfptCarrier.lean`, v817 precedent)); `v838_corner_expectation_position.py` (13/13 + 13/13, `EXPECTATION-CLOSED` + `POSITION-CARRIER-TRADEOFF`; probes `conditional_expectation_probe.py` + `position_carrier_probe.py`: ALL 5276 subgroups (74259 components) — identity( $E_H$ ) for EXACTLY 2 subgroups = the predicted criterion  $\text{Ann}(H) \subseteq \{1, \psi_0\}$ , 0 dichotomy violations, the full average reads **ZERO**; the pinching complete with the two pinned sectors as the **ONLY** PSD summands (the joint-use kill); Stinespring: the position data **IS** in the dilation yet the identity pins the read to the constant 1; the four position-carrier designs all fail the **DECISIVE NULL** (the self-consistent Epstein comb reads excess 0.0 identical to truth) — the fourth wall coordinate named **EXTREMAL PINNING**); `v839_paircorr_bridge_saturation.py` (9/9 + 7/7, `BRIDGE-FULLY-BEYOND` + `GUE-SATURATING`; probes `paircorr_bridge_map_probe.py` + `gue_saturation_probe.py`, promoted verbatim with no downscoping ( $\sim 0.8$  min per part, v830/v831 precedent; import remap to v823/v829/v830/v831 per the v831 precedent, shim symbols declared): the demand in Montgomery  $F(\alpha)$ -form on all 73 frames with the **SCHOLARSHIP CORRECTION** (Montgomery RH-conditional) carried in every supply row; the two-tier gap (unconditional misses by 52 $\times$ ; the conditional window covers 43–86%; full-GUE rms closes 39/73); the NAMED OBJECT: unconditional  $F(\alpha, T) \leq C_F$  (0.33–1.7) out to  $\alpha \approx 3$ ; the saturation:  $R$  rises  $0.61 \rightarrow R_\infty = 1.114$  (deepest 20) inside  $[0.5, 1.5]$ ,  $\text{rms}(z) = 0.72$ , tight band unfolded  $\alpha \in 1..2$  (64% of the tower budget); the F-INSTRUMENT:  $\hat{F}(\text{win}) = 0.44 \pm 0.14$ , Poisson control  $0.94 \pm 0.46$  calibrates to 1, scramble fires); `v840_gue_ablation_loopgain.py` (10/10 + 8/8, `SATURATION-STRUCTURAL` + `LOOP-SHORT`; probes `gue_ablation_probe.py` + `bootstrap_loop_gain_probe.py`: all

6 source-native Guinand-admissible variants LAND at the plateau (1.058–1.199 vs 1.154, tol 0.35; 0 broke positivity), the tight band pinned in the UNFOLDED coordinate under  $D = 1/48, 1/64, 1/96$  and geometric- $h$  — a property of the tower class and the unfolded zeros, not the grid; the thinned comb breaks positivity ( $-0.105$ ), the scramble explodes ( $\times -954770$ ), Epstein positive-trivial ( $\times 9461$ ); the loop: supply  $c_{\text{hat}} = 0.88$  GUE-consistent, demand  $R$  climbing to 1.24, headline  $g(k=1) = 0.529$ ,  $g(k=2) = 0.132$ ,  $N_{\text{max}} = 0$  induction steps, IDEAL supply gives  $1/R^2 = 0.655 < 1$ ; the Poisson circularity ward passes ( $g_P = 0.296$ ; base case breaks 17.6%)); `v841_relation_carrier_ladder.py` (18/18 + 13/13, RELATION-CARRIER-EXISTS + EXCESS-NONNEGATIVE; probes `multiplicative_relation_probe.py` + `relation_corner_ladder_probe.py`: THE BREAKTHROUGH — the identification carrier EXISTS with the relational input:  $\mu * 1 = \delta$  and  $\Lambda = \mu * \log$  EXACT in the truncated convolution frame; the four-comb discriminator exact (TRUE 0;  $s_2s/\chi_4$  0 — Selberg-class-correct blindness; the  $h = 2$  Epstein  $x^2 + 5y^2$  reads 200.0 on 24 class-group-obstruction sites,  $21 = 3 \times 7$  first); ALL FOUR GATES pass incl. the self-consistency null (14/70 missing product relations, excess  $-0.297$  vs 0 at truth); the amplitude wiring fixes L-safety AUTOMATICALLY (H2 reads 196/392/496 exact on the anchors); the EXCESS positive on all 67 rungs (min  $+2.285$ , max  $+3.704$ ; scramble  $-1.224$ ) while  $\tau_X$  decays  $5.98 \times 10^{-4} \rightarrow 1.71 \times 10^{-5}$ ); `v842_excess_certified_skeleton.py` (9/9, SKELETON-CERTIFIED; probe `excess_certified_skeleton_probe.py`: strictly positive certified interval enclosures of  $\tau_X$  on ALL 67/67 rungs (widths  $5.4 \times 10^{-11}$ – $6.6 \times 10^{-9}$ , three to five orders below the margin; `chol_cert` discipline; midpoints reproduce the float ladder at  $4.9 \times 10^{-13}$ ); the  $h = 2$  fake certified-NEGATIVE on every anchor ( $-0.786/-1.100/-1.228$ , exact rational  $\Lambda_F$  routing); the scaling law typed: width slope  $+0.62$  vs margin slope  $-0.52$  per  $\alpha$ , horizon  $\alpha^* \approx 9.1$  (mass  $\sim 7.5 \times 10^7$ ) far beyond the ladder end 6.304 — certifiability not binding). Suite 829  $\rightarrow$  835; all six run standalone green with v757-precedent pattern gates; all eleven probes re-run identically at promotion (2026-08-07); the two zero-side modules promoted VERBATIM with no downscoping (measured  $\sim 1.5$  min each, v830/v831 runtime precedent).

- **Lean (1 module promoted).** `TfptCarrier/SectorPositiveDescent.lean` — the corner/descent core of the two hardest fronts, kernel-checked (lake build green, 3409 jobs, no `sorry`, no `native_decide`; imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round): corner compression by a hermitian idempotent is linear/idempotent/CP with the corner-trace split; descent towers with the measured-zero compatibility defect transport compatible positive families; entrywise limits of corner states are PSD (via `GramCompactness`); and the final theorem `sector_floor` passes `IdentificationPositivity` through as the SINGLE NAMED HYPOTHESIS — the kernel-checked logical geography of the wall: structural positivity proven, the identification datum deliberately NOT a theorem. Cited in v837's part C (numeric witness mirror, v817 precedent).
- **The wall statement (the round's deliverable).** The corner route's perimeter is fully surveyed and CLOSED (register, tower levels, compressions, position carriers — four stop-list entries), the zero side is quantified (the demand SATURATES the GUE boundary structurally; the loop is SHORT: the wall conserves itself through the saturation), and the route REOPENS with the relational input: the identification carrier exists, the identified corner's excess is positive on all 67 rungs, and the certified skeleton is strictly positive — the wall takes its sharpest form,  $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$ , and what remains is the INFINITE QUANTIFIER over strictly-positive certified enclosures: any future route must supply a UNIFORM lower bound on the excess margin, by the GUE-side findings a finer-than-statistical datum that no finite table settles. The bridge contract `PRIME.FLOOR.PAIRCORR.01` [O] unchanged. NO RH claim.
- **Ledger.** 6 new rows (`PRIME.CORNER.OPENDOORS.01`, `PRIME.CORNER.EXPECTATION.01`, `PRIME.FLOOR.BRIDGEMAP.01`, `PRIME.FLOOR.GUEABLATION.01`, `PRIME.RELATION.MULT.01` [O], `PRIME.RELATION.SKELETON.01` [O]) + the dated round-28 note on `PRIME.FLOOR.PAIRCORR.01` (the four closure coordinates complete; the saturation structural; the loop short; the relation route registered with the sharpened demand); 914  $\rightarrow$  920 rows; NO marker moves (the closures

are typed route decisions, the registrations OPEN [O], the zero-side modules necessary-side consistency).

- **Papers.** `tfpt_prime_front`: the new round section “The wall gets coordinates: the corner closure theorems, the GUE saturation, the relation carrier, and the certified skeleton” (all six `\veri`) + the part-intro chain extended; `tfpt_3`: six new audit cards + the round-28 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the round-28 note on the PRIME.FLOOR.PAIRCORR.01 block + four new blocks (the corner-closure quantifier closed at registration, the bridge/saturation/loop evidence block, and the two OPEN registrations PRIME.RELATION.MULT.01 / PRIME.RELATION.SKELETON.01).
- **Website.** Prime-front feed: the round entry “reflux round twenty-eight”; counters 829 → 835 modules / v535–v842; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 835 modules / v842.
- **Scope.** The closure modules (v837/v838) are typed route decisions under frozen preregistered protocols (exact wards [E]-grade, float window legs typed); the zero-side modules (v839/v840) are necessary-side demand/supply mappings and instrument calibrations with every supply row carrying its conditionality tag (Montgomery RH-conditional — the scholarship correction is part of the frozen protocol); the relation modules (v841/v842) register an OPEN route — the carrier exists and the skeleton is certified, the positivity along the ladder remains UNPROVEN and is exactly the wall; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXX)** (reflux round twenty-seven — the two-route decision at the wall, and three exact corpus compressions: FIVE modules (v832–v836, from seven frozen probes; suite 824 → 829) — the corner route’s STRUCTURAL half closes weight-generically (the character-corner identity  $T_{\text{GL1},X} = V^* e_{\text{GL1}} \pi(P_X) e_{\text{GL1}} V$  holds as polynomial algebra in free event weights with  $\hat{c}_j = -1$  exact for all events; the Gaussian Hjelmslev CP tower  $\mathbb{Z}[i]/(1+i)^m$ ,  $m = 1..5$ , is strictly projective at certified cb defect ZERO and carries the identity verbatim) — BUT identity and state corner are COMB-BLIND and the level- $m$  motion is the pure register-dilution law  $16^{1-m}$ : the arithmetic wall RELOCATES into the identification step and stays PRIME.FLOOR.PAIRCORR.01; the commutant SOS route is CLOSED DEFINITELY by exact rational certificates (unique trivial feasible point; forced  $G = \text{diag}(1, -1, -1)$  with dual point  $q(0, 1, 0) = -1$ ) — both routes stop-listed; plus the anchor affine normal form ( $T(x) = 2x - 2$ , the whole budget from ONE recursion), the flavor bidirectional checksum ( $Ra = (4, 10, 13) / R^T a = (6, 18, 8)$ , 1 of 12), the  $(1+i)$  ramification ladder (four corpus roles = ONE object) and the PG(3, 2) flag normal form ( $105 = 45 + 60$  as the  $\text{Sp}(4, 2)$ -orbit split, the doily obstruction typed); NO marker moves; NO RH claim anywhere)

- **New modules (5).** `v832_anchor_flavor_checksum.py` (18/18 + 15/15, ANCHOR-AFFINE-EXACT + FLAVOR-BIDIR-CHECKSUM (selective); probes `anchor_affine_probe.py` + `flavor_bidir_probe.py`, promoted verbatim: the anchor power sums  $p_n = 2 + 2^n$  obey  $p_{n+1} = 2p_n - 2$  identically (sympy);  $T(x) = 2x - 2$  has the UNIQUE fixed point  $2 = |\mathbb{Z}_2|$  and the  $T$ -orbit of 4 is the compiler quintet (4, 6, 10, 18, 34),  $p_0 = 3 = N_{\text{fam}}$  prepended — the whole budget from one recursion:  $|R(E_8)| = p_1 p_2 p_3 = 240$ ,  $\dim E_8 = 240 + q_3 = 248$  (ladder identity  $p_4 - p_3 = p_3 - 2 = 8$ ),  $h(E_8) = p_2 p_3 / 2 = 30$ ,  $|R(D_5)| = 40$ ,  $\Omega_{\text{adm}} = 48$ ,  $10b_1 = 41$ ; the flavor matrix decodes the SAME anchor bidirectionally —  $Ra = (4, 10, 13) = (p_1, p_3, N(3+2i))$ ,  $R^T a = (6, 18, 8) = (p_2, p_4, p_3 - 2)$ ; of the 12 down-row candidates exactly ONE (the accepted (1, 5, 2)) passes the full identity set, the new one-number kill being the anchor contraction  $(Ra)_2 = 10 = p_3$  (siblings: 11, 12); controls fire incl. the sibling reproducing `tfpt_2`’s failure numbers verbatim); `v833_gaussian_ramification_ladder.py` (33/33, RAMIFICATION-LADDER-EXACT; probe `gaussian_ramification_probe.py`: the four documented roles of



$\pi_2 = 1+i$  re-certified from scratch as ONE ladder — norm doubling  $((1+J)^\top(1+J) = 2I_8$  exact, zero class empty  $0/240$ ,  $240 = 15 \times 16$ , elementary divisors  $(1^4, 2^4)$ ), the 4-bit address (deck trivial), the non-split jet  $(\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$  unique among 24 bijections, deck  $= 1+\epsilon$ , exactly  $0/65536$  deck-equivariant sections, the 120 root pairs the  $q=1$  shell), and the metaplectic lift ( $\zeta_8 = (1+i)/\sqrt{2}$ ,  $(SH)^3 = \zeta_8 I$ ,  $(T_3 K_3)^3 = iI$  = the deck; exact  $\mathbb{Z}[\zeta_8]$  BFS census  $|C_2/\mu_8| = 11520$ , ZERO Galois-mixed classes,  $C_2 = K \sqcup \zeta_8 K$ ,  $\chi(H \otimes I) = 1$  the RM-CSS phase bit); controls: the split ring admits 0 isos, the split extension splits  $65536/65536$ ); `v834_pg32_flag_completion.py` ( $26/26$ , PG32-FLAGS-EXACT + INCIDENCE-SPLIT; probe `pg32_flag_completion_probe.py`:  $\binom{4}{2}_2 = 35$ ,  $140 = 35 + 105$ ,  $105 = 35 \times 3$  flags  $\cong$  point pairs; the RM layer rebuilt as set equalities ( $[15, 10, 4]$  enumerator exact, Grams  $22I+6J / 6I+J / 28I+7J$ ); the vacuum coordinate is ORIGIN RESTORATION; the kill typed as a split:  $105 = 45 + 60$  is the  $\mathrm{Sp}(4, 2)$ -ORBIT decomposition on both sides (720 brute force), the polarity dictionary on the 60 non-isotropic rungs is bijective/equivariant/incidence-preserving, the 45 doily rungs are obstructed  $((t+U) \cap U = \emptyset$  45/45, flag Gram  $18I+3J \neq 22I+6J$ ) — “105 checks = 105 flags” is a deck-equivariant bijection, NOT an incidence isomorphism (the typed reason the v822 dilation died)); `v835_corner_hjlemslev_tower.py` ( $21/21 + 20/20$ , CORNER-IDENTITY-SYMBOLIC + HJELMSLEV-STRUCTURE-ONLY; probes `character_corner_probe.py` + `hjlemslev_cp_tower_probe.py`: the corner identity holds as POLYNOMIAL ALGEBRA in free event weights (symbolic toy leg exact; all 3 deployed windows on two independent code paths); the entire equality step consumes ONE piece of arithmetic — the glue identity  $\sum_m \Theta_m = 240\sigma_3$  + the uniform  $H^*$  mass — forcing  $\hat{c}_j = -1$  EXACTLY for all 136 events (exact group-algebra Fourier, 128-dim register); honest scope: the identity-carrying object is NOT positive (1 of 12 sectors PSD), the positive GNS corner is NOT  $\tau_X$ , and the scramble proves identity + state corner COMB-BLIND while the lock data moves ( $\tau = +5.98 \times 10^{-4} \rightarrow -3.51$ ); the Hjlemslev ladder  $m = 1..5$  carries exact counts  $15 \cdot 8^{m-1}/35 \cdot 16^{m-1}/105 \cdot 32^{m-1}$ , strictly projective CP channels (certified cb defect identically ZERO), level-exact  $\hat{c}^{(m)} = -1$ , fence silent — but H5 decides AGAINST identification: the level- $m$  corner is comb-blind with pure dilution  $16^{1-m}$  (ratio 0.0625 exactly); the one live door: position-dependent Kraus data on the level- $m$  flags — register-side dressings provably cannot carry it (`hjlemslev_position_kraus_probe.py`, POSITION-KRAUS-TRADEOFF, 13/13, probe-level, cited in the provenance)); `v836_commutant_sos_closure.py` ( $29/29$ , COMMUTANT-SOS-INFEASIBLE; probe `commutant_sos_probe.py`, the declared SPEC  $v1 \rightarrow v2$  witness-triple repair documented: the degree-2 SOS ansatz over the canonical five-dim abelian subalgebra  $\{\Pi_A(0), \Pi_A(1), \Pi_A(4), \Pi_A(7), \Pi_A(9)\}$  of the 39-dim commutant is closed DEFINITELY — (a) the exact reduction lemma collapses the SDP (4 blocks  $S_+^5$ , 60 unknowns, 85 constraints) to a rational LP with the TRIVIAL diagonal rewriting as unique feasible point (Vandermonde kernel  $\{0\}$ ): no cross-sector positive transfer; (b) the rank-3 det form has signature (1, 2) ( $\det S = t^2 - x^2 - y^2$ , compiler null vector  $q(5, -3, 4) = 0$ ), coefficient matching FORCES  $G = \mathrm{diag}(1, -1, -1)$  (not PSD) with the rational dual certificate  $q(0, 1, 0) = -1 < 0$ , and  $q$ -negative directions are achievable even on the nonneg cone (832/2799 truncation cuts); FENCE-CLEAN on the algebra layer, FENCE-HIT on the positivity branch — the nontrivial version is pair-correlation substance; controls fire (comb-specific identification, the v815 sector pattern, wrong-ratio inconsistency)). Suite  $824 \rightarrow 829$ ; all five run standalone green and encode their frozen patterns per the v757 precedent; all seven probes (plus the position-Kraus companion) re-run identically at promotion (2026-08-07).

- **The wall statement (the round’s deliverable).** Both corner-era routes are DECIDED: the corner route delivers everything structural (identity, projector, strictly projective CP tower, level-exactness) and shows the structure is placement-blind — the arithmetic lives entirely in the IDENTIFICATION step (a positive  $P_{\mathrm{lock}}$  whose corner IS the deployed window form); the SOS route shows no positive algebraic rearrangement can manufacture the floor. The wall stays exactly PRIME.FLOOR.PAIRCORR.01 [O], relocated into identification, with both routes stop-listed per house discipline. NO RH claim.

- **Ledger.** 5 new rows (ANCHOR.AFFINE.01, GAUSSIAN.RAMLADDER.01, PG32.FLAGS.01, PRIME.CORNER.CHARACTER.01, PRIME.COMMUTANT.SOS.01) + the dated round-27 two-route note on PRIME.FLOOR.PAIRCORR.01 (corner structural half closed / wall relocated; SOS closed with the rational dual certificate; both stop-list entries; the position-Kraus trade-off finding); 909 → 914 rows; NO marker moves (the compressions are normal forms of existing corpus surfaces; the corner/SOS results are typed route decisions).
- **Papers.** `tfpt_1`: the anchor affine normal form keybox (Theorem 8) + the affine pointer in the anchor-first refinement + the  $(1+i)$  ramification-ladder entry in the Gaussian code keybox (`\veri{v832}`, `\veri{v833}`); `tfpt_2`: the bidirectional anchor checksum keybox after the spectral-selector block (`\veri{v832}`); `tfpt_prime_front`: the new round section “The two-route decision at the wall” (`\veri{v834/v835/v836}`) + the part-intro chain extended; `tfpt_3`: five new audit blocks + the round-27 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the PRIME.CORNER.CHARACTER.01/PRIME.HJELMSLEV.CPTOWER.01 and PRIME.COMMUTANT.SOS.01 registration blocks + the dated round-27 note on the PRIME.FLOOR.PAIRCORR.01 block; `note_e8_gaussian_code`: a dated pointer to the ramification ladder in the machine-verification section.
- **Website.** Prime-front feed: the round entry “reflux round twenty-seven”; counters 824 → 829 modules / `v535–v836`; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 829 modules / `v836`.
- **Scope.** The three compressions (`v832–v834`) are exact-arithmetic normal forms of existing corpus tables — compressions, not new claims; the corner/Hjelslev/SOS results are typed route decisions under frozen preregistered protocols (exact wards [E]-grade, float window legs typed, the honest scope measured); the queued Lean follow-up (`SectorPositiveDescent.lean`) stays queued; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXIX) (reflux round twenty-six — the typed circularity boundary: ONE module (`v831`, from one frozen probe; suite 823 → 824) — the correlation-corrected alias second moment is EXACT (sympy-proven bilinear form  $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$ , identity-warded on all 73 battery frames), the round-25 premise “the phases carry the gap” is OVERTURNED (the AMPLITUDES carry it: the tower law decomposes exactly as  $-2.505 = -3.172$  amplitude law + 0.668 fading alignment), the exact Guinand split puts the coherent sum in the prime-comb fluctuation at  $0.01\times$  the Poisson square-root scale (smooth  $-994$  / fluctuation  $+995$ , each  $\sim 1000\times$  the residual, cancelling; the comb scramble explodes it  $10^6\times$  — THE CANCELLATION IS THE ARITHMETIC) — and the analytic-envelope route of PRIME.FLOOR.RATIO.01 is CLOSED HONESTLY at its typed circularity boundary: by Guinand the comb’s self-cancellation IS the zero-side floor statement itself, the required control is even deeper than Montgomery-type pair-correlation input, the envelope’s constant is arithmetic substance equivalent to the target — stop-list entry, no further analytic-envelope variants; the demand moves to the NEW bridge contract PRIME.FLOOR.PAIRCORR.01 [O]; NO marker moves; NO RH claim anywhere)**

- **New module (1).** `v831_prime_alias_second_moment.py` (15 checks with the SEVEN preregistered-honest premise-overturning FAILs `S1.SIGN/W2B/W2C/W2D/S4.GEOM/C1/C2` pattern-gated — the FAIL pattern IS the finding; verdict ALIAS-CORRELATION-PAIRCORR; discovery probe `prime_alias_second_moment_probe.py`, 2026-08-07, re-run identically at promotion; promoted VERBATIM with NO downscoping — measured runtime  $\sim 0.1$  min in-tree, the zero list is the `v823` cache + vectorized RS scan): the EXACT BILINEAR FORM (sympy-proven,  $S_0\ 3/3$ ): in the pole frame the zeros+pole gram is exactly  $M_2 =$

$[[P_0 + (S_2 + S_c)/2, S_s/2], [S_s/2, (S_2 - S_c)/2]]$  with  $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$  — the correlation combination at the leading term is the coherent phase sum  $S_c$  (rotation-invariance ward at the conditioning-aware machine bar on all 73 frames, truncated-series dev  $3.8 \times 10^{-7}$ ); THE PREMISE OVERTURNED: the diagonal (random-phase) part  $\tau_{\text{diag}} = S_2/2$  slopes  $-3.172$  not  $-1$  (W2B), phase scramble  $-3.054$  (W2C), Poissonized zeros  $-3.654$  with  $q_{\text{poiss}}$  median  $0.34 > 0.5 q_{\text{true}}$  (W2D), and  $q = S_c/S_2$  changes sign across the battery  $[-0.324, 0.552]$ , median  $0.19$ ; deep  $M = 1632$  suppression only  $\times 1.1$  — the full tower  $-2.505$  ( $R^2$  0.98, W2A PASS) decomposes EXACTLY as amplitude law  $-3.172$  + fading alignment  $+0.668$ ; THE GUINAND SPLIT (exact, identity-route residue 0.019, soft bar met):  $G_Z = (B - S_p - P) + (S_p - S)$  gives smooth share  $-994.15$  / prime-fluctuation share  $+995.15$ , cancelling to  $|S_c^{\text{fluc}}|/\sigma_{\text{sqr}} = 0.01$  (SQRT-CANCELLATION REGIME); the smooth surrogate does NOT carry it (S4.GEOM 0.868, deficit-slope dev 2.274); comb scramble explodes the coherent sum  $\times 2.0 \times 10^6 - 1.7 \times 10^7$ , the Epstein  $x^2 + 5y^2$  comb  $\times 5.1 \times 10^3 - 1.3 \times 10^4$  — comb-specific arithmetic; THE TYPED CIRCULARITY BOUNDARY: proving the tower  $\tau$ -law requires bounding an oscillatory prime sum AT its square-root scale = variance-type, pair-correlation-equivalent substance (Montgomery-type on the zero side by the explicit formula), and by Guinand the comb's self-cancellation IS the zero-side floor statement itself. Suite 823  $\rightarrow$  824; standalone green with the v757-precedent pattern gate.

- **Contracts.** NEW: **PRIME.FLOOR.PAIRCORR.01** [O] — the floor  $\leftrightarrow$  pair-correlation bridge: the floor's envelope constant is Guinand-EQUIVALENT to the zero-side floor statement; demands typed in both directions (FORWARD: an unconditional variance-type bound  $|\text{aniso}(S_p - S)| \leq C\sigma_{\text{sqr}}$  uniform in  $h$  with the pair-correlation-grade entry point DECLARED — no hidden circularity; BACKWARD: instrument calibration of floor margins into effective pair-correlation statements, never a theorem about zeros); KILL: fluctuation share collapsing below 0.6 while the suppression persists, or the scale ratio rising systematically above 3 (one-point regime — the boundary mis-typed, the route reopens on PNT-grade input); CLOSE: the forward demand delivered with its entry point named; explicitly does NOT promise an unconditional proof (the proving direction contains pair-correlation-grade arithmetic, fenced). **PRIME.FLOOR.RATIO.01** [O] **ANALYTIC-ENVELOPE ROUTE CLOSED** (dated closure note): the round-25 alias-phase blocker resolves INTO the typed boundary; NO further analytic-envelope variants (STOP-LIST entry per house discipline); kill criteria unchanged, still [O], promotion fenced.
- **Ledger.** 2 new rows (PRIME.FLOOR.ALIASMOMENT.01 + the contract row PRIME.FLOOR.PAIRCORR.01) + the dated round-26 closure note on PRIME.FLOOR.RATIO.01; 907  $\rightarrow$  909 rows; NO marker moves (the bilinear form and its expansion sympy-proven [E]-grade wards; the overturned premise and the sqrt-scale cancellation measured; the contracts [O]).
- **Papers.** `tfpt_prime_front`: the new round section “The typed circularity boundary: the analytic-envelope route closes, and the floor–pair-correlation bridge is named” (`\veri{v831}`); `tfpt_3`: one new audit block + the round-26 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the PRIME.FLOOR.PAIRCORR.01 registration block, the dated closure note on PRIME.FLOOR.RATIO.01, and the typed-blocker note on the depth-kill/budget block.
- **Website.** Prime-front feed: the round entry “reflux round twenty-six”; counters 823  $\rightarrow$  824 modules / v535–v831; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 824 modules / v831.
- **Scope.** The bilinear form, its  $1/P_0$  expansion and the det identity are sympy-proven [E]-grade wards; the overturned premise, the exact Guinand split shares, the sqrt-scale ratio and the comb-scramble/Epstein fingerprints are measured finite-level evidence on the frozen battery; the seven FAILs are preregistered-honest adjudications on record (the FAIL pattern IS the finding, pattern-gated); the boundary is a TYPED BOUNDARY, not a theorem — the route closure is a discipline statement, not a positivity result; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXVIII)** (reflux round twenty-five — the exclusion detector and the depth kill: SIX modules (v825–v830, from eleven frozen probes; suite 817 → 823) — the verified PSD rungs of the GL1 prime-comb tower become a hash-preregistered *window detector*: certified exclusion regions with rigorous Cholesky/Higham + witness-Rayleigh certificates to  $X = 24.81$ , a zero locator validated OUT-OF-SAMPLE on a disjoint untouched window (83% detection, 0% false positives, precision 0.086), and the capstone comb-native window verification whose census reconciles against the Riemann–von Mangoldt count (21+4 typed misses = 25 = cache = RvM; new deepest certified rung  $X = 25.5$ , comb cap  $1.2 \times 10^{11}$ ) — registered as the new contract PRIME.DETECTOR.WINDOW.01 [O]; and the PRIME.FLOOR.RATIO.01 kill gates, played at full sieve depth, SURVIVE with growing margins while the  $\alpha^* \approx 11$  horizon falls as a float-convention artifact (citation grade to  $\alpha = 12.75$ ; envelope certified-explicit  $\rho \geq 4.335 h^{-3/2}$  on 73/73; the alias-phase proof blocker named) — every statement typed strictly weaker than classical verification; NO marker moves; NO RH claim anywhere)

- **New modules (6).** v825\_prime\_exclusion\_ladder.py (26/26 + 21/21, LADDER-EXTENDED + SATURATION-TYPED: every verified PSD rung of the GL1 tower ( $D = 1/64$ ) carries a rigorous  $\lambda_{\min} > 0$  certificate — Cholesky PD verification with the Higham backward-error bound plus a Rayleigh-residual enclosure, e.g.  $\lambda_{\min}(X=24.81) = 1.078 \times 10^{-6} \geq 9.70 \times 10^{-7}$  CERTIFIED — and is inverted through the tent-read Guinand identity into quadruple-exclusion regions  $\{1/2 \pm \delta \pm i\gamma\}$  on the frozen rank-4 battery, every published boundary point re-proved by a witness-Rayleigh certificate; the census threshold  $\Xi_{\text{eff}} = X \delta_{\text{mb}}$  falls  $0.639 \rightarrow 0.457 \rightarrow 0.274 \rightarrow 0.218$  and PLATEAUS at  $0.218 \pm 3\%$  (gates recomputed from the frozen deep series; mechanism SUPPORT-ACTIVE-BUT-X-STALLED: support slope  $-0.68$ , rank-12 gain growing  $19.3 \rightarrow 25.0\%$  below the 30% bar); multi-zero double injection breaks without interference rescue; honest reach typed: Nyquist  $\pi/D = 201$ , region INSIDE the Platt–Trudgian strip — consistency re-derivation, not new territory); v826\_prime\_exclusion\_battery2.py (16/16 + 13/13, FLOOR-LOWERED + RANGE-DECIDES-DECLINE: battery v2 designed a priori from instrument diagnostics only and SHA-256-hashed fd39fb42... BEFORE any evaluation (no zero datum;  $v1 \subset v2$ , pointwise dominance live-verified); old-grid floor  $\leq 0.0979$  with the grid-censoring caveat typed; the measurement-range extension (exact-subset grid, bit-for-bit ward) UNCENSORS the series 0.1066/0.0982/0.0960/0.0913/0.0816 at slope  $-1.39$  — the v825 plateau was the instrument, not the tower; benchmark  $\delta \geq 10^{-3}$  at  $X^* = 81.6$  (extrapolation, typed); side-finding: rank correlation  $-0.872$  of the uncensored boundaries with the distance to the nearest true ordinate, hash-verified non-circular); v827\_prime\_zero\_locator.py (the ONE preregistered-honest C3 FAIL + 13/13, LOCATOR-NULL + LOCATOR-V2-RESOLVES: the width profile  $W(\gamma)$  PEAKS at true ordinates (explicit-formula redundancy: a real zero makes the injected quadruple redundant, the boundary rises); v1 honestly failed its frozen gates (12/13 peaks but 61% false peaks at  $\ln 1.5$ ) — on record; the post-hoc separation (matched  $\geq 0.82$ , false  $\leq 0.73$ ) froze  $\ln 2.2$  for v2, preregistered f57a2e7f... BEFORE any test evaluation; out-of-sample on the disjoint window [60, 120]:  $20/24 = 83\%$  detection, 0% false positives, precision 0.086, all four controls (scramble/Epstein no peaks, shifted null below the structural re-lock ceiling, grid refinement 7/7 stable); depth law  $54 \rightarrow 62 \rightarrow 83\%$ ); v828\_prime\_comb\_window.py (12/12, COMB-VERIFIES-WINDOW, the capstone: on  $\gamma \in (60, 120]$  at  $X = 24.8125$  ONE certified positivity object yields location (21/25 ordinates to  $\pm 0.25$ , max error 0.242, ZERO unmatched peaks), exclusion ( $\delta \geq \delta_{\text{mb}}(\gamma)$  pointwise, uniform bound 0.0496, witness-certified at six frozen samples) and the census (21+4 typed misses = 25 = cached verified count = rounded RvM main term  $25.237 \rightarrow 25$ ); miss autopsy complete (1 pair-merge below the grid resolution +3 prominence-limited; the depth-recovery prediction at the NEW deepest certified rung  $X = 25.5$ ,  $\lambda_{\min} = 8.07 \times 10^{-7} \geq 7.26 \times 10^{-7}$  CERTIFIED, verified as typed: 0/3 + 0/1); the scramble fails ALL THREE axes; WEAKER-THAN-CLASSICAL typing mandatory and carried verbatim: Turing proves  $N(T)$  and  $\delta = 0$  exactly, the comb

leaves a  $\delta_{\text{mb}}$ -strip unresolved and can never close it at exponential comb cost — a necessary-side consistency demonstration; does NOT bear on RH); `v829_prime_floor_depth.py` (16/17 with the ONE preregistered-honest S5.HLAW FAIL + 12/12, ENVELOPE-HOLDS-DEEP + FAMILY-SCALES: the `PRIME.FLOOR.RATIO.01` kill gates K1 ( $\rho h^{3/2} \geq c = 4.85$ , NO refit) and K2 ( $\cos^2 \theta \geq 1/2$ ) played at  $X = 18.375\text{--}25.5$  on uniform-grid tower frames whose  $h = 588\text{--}816$  sit INSIDE the deployed  $h$ -range while the depth doubles — the test decouples depth from dimension — and SURVIVED: margins  $\times 5.76/6.24/6.53/7.16/8.17/8.92$  growing monotonically,  $\min \cos^2 \theta = 0.849$ , non-decay  $+0.060$  per log-unit (the  $h$ -law wins); the typed miss: the single pair collapses (gap  $40\times \rightarrow 673\times$ ) and the certified top-100 family takes over (0.977–0.981 of the floor,  $\rho_{\text{certfam}}$  within 2%);  $\alpha$ -coverage typed: citation grade only  $\alpha \in \{9.19, 10.36, 11.05\}$  — exactly the v830 blocker); `v830_prime_float_budget.py` (10/10 + 19 checks with the FIVE preregistered-honest asymptotic FAILs A7/A8/A9/A10/S2.EXP, BUDGET-LINEAR-CLOSES-ALL + ENVELOPE-ASYMPTOTIC-PARTIAL: the deployed quadratic float convention  $100 \varepsilon \|A_2\|_F^2$  was four orders above the true tail term at depth; the explicit Higham-linear chain (compensated banded accumulation with TwoSum,  $N_{\text{SUB}} = 2^{16}$ ; the irreducible atom-position term  $\varepsilon X/D$ ; exact fsum dots; dps-40 spot anchors with  $\times 10$  caps) is  $7\times\text{--}171\times$  tighter and closes the citation-grade family gate 6/6 (old: 3/6) — citation grade extends to  $\alpha = 12.75$ , FULL SIEVE DEPTH; CONVENTION NOTE: the deployed v818 convention stays FROZEN in the existing suite, the linear budget is registered as convention for NEW modules only; the analytic part runs near-verbatim: family closed form exact on ALL 73 battery frames ( $3.6 \times 10^{-11}$ ), sympy-proven limit algebra, phase conspiracy excluded (dip floor 0.170), and THE EXPLICIT CERTIFIED ENVELOPE  $\rho \geq \rho_{\text{certfam}} \geq 4.335 h^{-3/2}$  on 73/73, non-decaying; the proof blocker typed: alias phase correlations — random-phase predicts  $\tau \sim h^{-1}$ , the tower measures  $h^{-2.5}$  — the envelope stands certified-explicit, not proven). Suite 817  $\rightarrow$  823; all six run standalone green and encode their frozen patterns (incl. the expected-FAIL patterns) per the v757 precedent; all eleven probes re-run identically at promotion (2026-08-07); the deep sieves (to  $1.2 \times 10^{11}$ ,  $\sim 17\text{--}25$  min per probe) are downscoped in the suite modules to fast live sub-ladders at  $X \leq 18.375$  with the full-depth numbers recorded as frozen reference constants and every decision gate RECOMPUTED from them — predeclared in each module's PROVENANCE.

- **Contracts.** NEW: `PRIME.DETECTOR.WINDOW.01` [O] — the comb-native window detector (location + exclusion + census, hash-warded end to end, validated out-of-sample); OPEN: the depth-to-width law as theorem, the completeness mechanism, window extension beyond Nyquist (explicitly NOT promised); KILL: out-of-sample detection  $< 40\%$  on a deeper certified rung, a census contradiction on any PD-certified window, or a false positive surviving the grid-refinement ward; mandatory weaker-than-classical typing carried by every module of the strand. **`PRIME.FLOOR.RATIO.01` [O] DEPTH-HARDENED** (dated contract-text update): kill gates played at the contract's own preregistration and survived at  $X = 25.5$ ; the round-24  $\alpha^* \approx 11$  horizon REMOVED as a float-convention artifact (citation grade to  $\alpha = 12.75$ ; deployed v818 budget convention unchanged, the Higham-linear budget is convention for NEW modules only); the envelope now certified-explicit ( $c_{\text{cert}} = 4.335 h^{-3/2}$  on 73/73); the remaining proof blocker NAMED (alias phase correlations, random-phase  $h^{-1}$  vs tower  $h^{-2.5}$ ); kill criteria unchanged, still [O], promotion fenced.
- **Ledger.** 7 new rows (`PRIME.EXCLUSION.LADDER.01`, `PRIME.EXCLUSION.BATTERY2.01`, `PRIME.EXCLUSION.LOCATOR.01`, `PRIME.EXCLUSION.WINDOW.01`, `PRIME.FLOOR.DEPTHKILL.01`, `PRIME.FLOOR.BUDGET.01` + the contract row `PRIME.DETECTOR.WINDOW.01`) + the dated depth-hardening update on `PRIME.FLOOR.RATIO.01`; 900  $\rightarrow$  907 rows; NO marker moves anywhere (certificates and preregistration hashes [E]-grade wards; exclusion maps, locator scores, envelope constants measured; the contracts [O]).
- **Papers.** `tfpt_prime_front`: the new round section “The exclusion detector: the certified ladder becomes a window-verification instrument, and the floor survives its kill gates at full sieve depth” (all six `\veri` citations; the weaker-than-classical typing carried in the

body); **tfpt\_3**: six new audit blocks + the two round-25 follow-up notes on the floor-contract prose; **tfpt\_research\_contracts**: the PRIME.DETECTOR.WINDOW.01 registration, the depth-kill/budget registration block, and the dated PRIME.FLOOR.RATIO.01 note.

- **Website.** Prime-front feed: the round entry “reflux round twenty-five”; counters 817  $\rightarrow$  823 modules / v535–v830; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 823 modules / v830.
- **Scope.** The certificates (Cholesky/Higham, witness-Rayleigh, preregistration hashes, sympy algebra, dps-40 anchors) are [E]-grade machine wards at float precision; the exclusion maps, locator scores, envelope margins and closure constants are measured finite-level evidence on frozen instruments; the honest negatives (LOCATOR-NUL, the S5.HLAW single-pair collapse, the five asymptotic FAILs of the analytic envelope) are preregistered adjudications on record; the mandatory typing is everywhere: locations to  $\pm 0.25$  and exclusions to  $\delta \geq \delta_{\text{mb}}$  are strictly weaker than classical verification, every verified window lies inside the Platt–Trudgian strip, and the strand is necessary-side evidence — NO marker moves; NO RH claim anywhere.

**2026-08-06 (LXVII) (reflux round twenty-four — the floor-theorem skeleton: TWO modules (v823–v824, from four frozen probes; suite 815  $\rightarrow$  817) — the sector floor  $\det \hat{A}_2 = \lambda \tau$  becomes an EXACT machine-verified sum of squares over zero+pole rank-one carriers (Lagrange identity), the fixed pair (pole  $\times \gamma_1$ ) is certified strictly positive on all 14 rungs with the budget tightened  $\sim 10$  orders by the psd-remainder monotonicity chain, and the three-piece uniform certified lower-bound skeleton closes at CITATION grade for all  $h$  at the deployed  $\alpha$  up to the validity horizon  $\alpha^* \approx 11$  — PRIME.FLOOR.RATIO.01 stays [O], NARROWED: the floor bound is battery-relative and  $\alpha$ -bounded, necessary-side only, NO RH claim)**

- **New modules (2).** v823\_prime\_lagrange\_floor.py (7/8 with the ONE preregistered-honest old-budget FAIL S3.CERT + 9/9, LAGRANGE-CONCENTRATED + PAIR-CERTIFIED: every per-zero layer of the v692 master identity is EXACTLY rank-one — the odd-extension symmetry factors the common phase,  $v_\gamma = 2\sqrt{w(\gamma)}(S_1, S_2)$  with the real closed form  $S_i(\phi) = \sum_j t_{ij} \sin((h-j-\frac{1}{2})\phi)$  and the alias weight  $w(\gamma) = D \text{csinc}^2(\gamma D/2) \geq 0$  (layer dev  $\leq 1.2 \times 10^{-16}$ , pole rank-one  $\leq 4.8 \times 10^{-17}$ ); the LAGRANGE IDENTITY  $\det(G_Z + P) = \sum_{\text{pairs}} w_i w_j (a_i b_j - a_j b_i)^2$  holds at machine precision on all 14 rungs (ward  $\leq 1.4 \times 10^{-9}$ ); the POLE is the universal non-collinear leg (the pole family carries 1.0000 of the pair total, every top-100 pair is pole  $\times$  zero) and the moving zero leg obeys the ALIAS-COMB LAW (median comb distance 0.0007–0.0061 vs uniform 0.25); A0 typed honestly: the atom-side rank-one framing FAILS as frozen ( $0.10-0.23 < 0.5$ ) — the exact SOS realization lives on the zero+pole side; THE ROUTE CHANGE:  $R = \hat{A}_2 - G_Z - P$  is an exact float  $2 \times 2$  with  $\lambda_{\min}(R) = 1.6 \times 10^{-7} - 2.8 \times 10^{-6} \geq$  margin verified per rung, and the  $2 \times 2$  psd fact  $\det(B+R) \geq \det B$  ELIMINATES the tail estimate from the chain — BUD 0.14–7.1  $\rightarrow 2.4 \times 10^{-12} - 2.7 \times 10^{-9}$ , median 9.8 ORDERS tightened; the certified interval  $X^2(\text{pole}, \gamma_1)$  – BUD is strictly positive on 14/14 ( $\gamma_1$  pinned to **zetazero**(1) within  $10^{-11}$ ; dps-40 mpmath ward inside the interval); the fixed pair carries 0.0063–0.5832 of the floor with slope  $h^{+1.06}$  — a GROWING brick; controls fire (scramble  $\times 45.8$ –49.0, Epstein  $\times 1784$ –50029 at det scale, collinear  $\det = 9.2 \times 10^{-16}$  with interval containing 0)); v824\_prime\_floor\_skeleton.py (12/12 + 9/9, FLOOR-PARTIAL + TAIL-CLOSED-ALL-H: PIECE 2 UNIFORM-VERIFIED — the ANALYTIC  $h \rightarrow \infty$  limit  $X_\infty(\alpha) = 16\alpha\pi^2 \sin(\alpha\gamma_1) \sinh(\alpha/2)$  [bracket] (Dirichlet closed form = direct sum at  $5.7 \times 10^{-13}$ , pipeline  $8.5 \times 10^{-13}$ , ZERO sign flips over 14  $\alpha \times 10$  synthetic  $h$ ; the explicit bound  $L(h) = X_\infty^2(\alpha)(1 - C(\alpha)/h)$  for  $h > h_0(\alpha) = C(\alpha) \leq 499.6$ ; honest limitation: 25  $\alpha$ -NODES of  $\sin(\alpha\gamma_1)$  on  $[2, 12]$  —  $h$ -uniform per  $\alpha$ , NOT ladder-uniform in  $\alpha$ ); PIECE 3 UNIFORM-VERIFIED — FAMILY EXHAUSTION: the certified top-100 (pole  $\times$  zero) family carries median 0.952 (min 0.931, max 0.974) of the floor, all carriers on-line by verified

computation ( $\leq 2 \times 10^4$ ); the residue 0.015–0.049 with trend slope +0.010 (truncation depth, not a structural carrier); the analytic family limit stays positive across ALL fixed-pair nodes (min  $1.62 \times 10^{-4}$  at  $\alpha = 2.68$ ) — the family never dies simultaneously; PIECE 1 CLOSED BY THE TAIL REPAIR: the crude coherence-discarding envelope closes only 4/14 ( $h \leq 434$ ; the autopsy names the coherence discard — measured band slack  $\times 2.7 \times 10^3$ – $2.4 \times 10^5$ ); the PRODUCT-SUP envelope (exact Dirichlet closed form, grid-converged with doubling ward; the rigorous Lipschitz certificate of the sup is a NAMED elementary step, typed not claimed) + Abel summation against RvM counting with EXPLICIT unconditional Trudgian-grade constants (factor 2.13; the Vinogradov–Korobov region typed honestly USELESS, 0.985) closes ALL 14 rungs at the FIXED verified horizon  $T_{\text{ver}} = 3 \times 10^{12}$  with margins  $10^3$ – $10^6$  (the secondary top-100 family gate closes 14/14 too); THE LAW CHANGE:  $\log h$  coefficient  $+1.98 \rightarrow +0.13$  — the  $h e^\alpha$  growth was an ARTIFACT of the coherence discard, the sharpened law is  $h$ -FREE at fixed  $\alpha$ ; remaining growth  $\alpha$ -only ( $q = +1.79$ , the off-line allowance  $e^{2\alpha\delta}$  of the critical strip), re-crossing  $X_\infty^2$  near  $\alpha^* \approx 11.2$  — far beyond the deployed battery ( $\alpha \leq 6.146$ ) but finite and honestly typed; controls fire (synthetic off-line pairs break the psd chain 4/4 exactly at the off-line locus AND stay inside both envelopes; the scramble kills the family structure; the regression ward reproduces the old {184, 210, 218, 434} closure boundary)). Suite 815  $\rightarrow$  817; both run standalone green and encode their frozen patterns per the v757 precedent; all four probes re-run identically at promotion.

- **The net floor statement (the round’s deliverable, verbatim honesty).** On every rung of the deployed 14-rung battery,  $\lambda\tau = \det \hat{A}_2 \geq X^2(\text{pole}, \gamma_1) - \text{pert}(h) > 0$  at CITATION grade — no per-rung eigencheck, no zero-location input beyond the critical strip; inputs: the strip, verified on-line zeros to  $T_{\text{ver}} = 3 \times 10^{12}$  (Platt–Trudgian, cited), RvM counting with explicit unconditional constants (cited), the analytic pair limit, the product-sup per-zero envelope. The complete three-piece skeleton: exact SOS identity [E-grade wards] + analytic fixed-pair bound ( $h$ -uniform per deployed  $\alpha$ ) + certified 0.93–0.97 family exhaustion + psd remainder closed for all  $h$  up to  $\alpha^* \approx 11$ . This is NECESSARY-side evidence on a frozen battery — NOT RH; the floor bound is battery-relative and  $\alpha$ -bounded.
- **Contracts.** NEW: PRIME.FLOOR.LAGRANGE.01 (the exact Lagrange SOS identity + the pair certification) and PRIME.FLOOR.SKELETON.01 (the three-piece skeleton + the all- $h$  tail closure). PRIME.FLOOR.RATIO.01 [O] NARROWED (dated contract-text update): WHAT IS NOW CLOSED — the deployed-battery skeleton at citation grade (exact SOS, analytic fixed-pair bound with explicit  $h_0(\alpha) \leq 500$ , certified family exhaustion,  $h$ -free psd closure at  $T_{\text{ver}}$ ); WHAT REMAINS — (i)  $\alpha > \alpha^* \approx 11.2$  needs verified on-line DEPTH beyond  $T_{\text{ver}}$  (never location), (ii) the family version of the analytic bound at the  $\sin(\alpha\gamma_1)$  nodes, (iii) the product-sup Lipschitz certificate (named elementary step), (iv) the  $V_\infty$  positivity itself — the  $O(1)$  capture and the  $\rho > 0$  demand at infinity remain the contract; kill criteria unchanged; promotion of any  $\rho$  bound stays FENCED.
- **Ledger.** 2 new rows (PRIME.FLOOR.LAGRANGE.01, PRIME.FLOOR.SKELETON.01) + the dated narrowing update on PRIME.FLOOR.RATIO.01; 898  $\rightarrow$  900 rows; NO marker moves ([O] stays [O]; exact identities [E]-grade wards, certified intervals and shares measured/[C] per the ledger fine-typing).
- **Papers.** tfpt\_prime\_front: the new round section “The floor-theorem skeleton: the floor as an exact sum of squares, and the tail closed for all  $h$ ” (both \veri citations); tfpt\_research\_contracts: the two registrations + the PRIME.FLOOR.RATIO.01 narrowing note.
- **Website.** Prime-front feed: the round entry “reflux round twenty-four”; counters 815  $\rightarrow$  817 modules / v535–v824; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 817 modules / v824.
- **Scope.** The exact identity layer (rank-one factorization, Lagrange SOS) carries identity-grade machine wards at float precision; the certified intervals, shares and closure margins are measured



finite-level evidence; the citation-grade chain uses ONLY the critical strip + cited verified depth + cited explicit RvM constants — no zero-location input, no RH-strength assumption; the honest boundaries (battery-relative,  $\alpha$ -bounded, the  $\alpha$ -nodes, the named sup certificate) are typed in the modules and the contract; NO marker moves; NO RH claim anywhere.

**2026-08-06 (LXVI) (reflux round twenty-three — one inequality: the floor reduces to the prime comb against its own density: NINE modules (v814–v822, from thirteen frozen probes; suite 806  $\rightarrow$  815) — the selection problem of the unified compactness contract is finite-level SOLVED by Mosco form convergence + Friedrichs minimality, the positive-descent master theorem applies to BOTH instances with the new prime summability number (kernel-checked Lean core), and the sector floor reduces to the SINGLE ratio inequality  $\rho = \tau/\tau_{\text{pnt}} > 0$  with a measured  $h^{-3/2}$  envelope (the new contract PRIME.FLOOR.RATIO.01 [O]); the K5 follow-up locates the deployed transport bit-exactly ( $T_{v221} = B^6$ , the dated SIXTHROOT correction) and types the six-step composition clock-only with the canonical  $T_{10}$  proposal; the Kraus tower gains the spread/commutant deciders and its 39-dim nonabelian commutant closure; the RM cascade lands (the 15-label channel = punctured  $\text{RM}(1, 4)^*$  with the  $[[15, 1, 3]]$  CSS code and the closing self-reproduction cycle, the rule-A protocol set = the 105 minimal dual checks, the  $E_7$  vacuum completion  $112 + 21 = 133$  as a matrix identity); and the vacuum-completion reading of the continuum dies honestly in both transcriptions with the  $c^*$  pencil bound as the frozen fence)**

- **New modules (9).** `v814_k5_sixstep_transport.py` (23/23, SIXSTEP-CLOCK-ONLY + TRANSPORT10-PROPOSED: the rate  $(2/3)^6$  arises canonically on THREE spatial routes (doily alternation, hexagon compression, hexagon circulation), each blocked by a typed obstruction (no unit mode — the norm obstruction [E neu]; the bipartite sign  $-2/3$ ; the wrong space  $\times 9$ ) — the deployed six is the CLOCK exponent ( $v124 \lambda_n = (1 - n/3)^6$ , v486 hand, fixed eigenbasis); sixth-power blindness [E neu]:  $C_6$  and  $2K_3$  have identical sixth powers, only per-step data distinguish walk from clock; THE CORRECTION (dated note on the v808 row):  $T_{v221}$  (the exact spectral sum of v221's transfer  $(1 - w)^6 = B^6$  BIT-EXACTLY — the sixth-root verdict upgrades from SPECTRAL-ONLY to BITEXACT-vs-v221; the canonical proposal  $T_{10} = P_{10}^6$  in the  $\Lambda^2$  basis passes ALL gates (spectrum  $\{1, (1/3)^6 \times 5, (2/3)^6 \times 4\}$  with multiplicities  $(5, 4) = (g_{\text{car}}, |\mu_4|)$ , equitable v774 quotient =  $(Q_{\text{Pet}}/3)^6$  bit-exact) — a promotion-ready proposal, NO deploy claim); `v815_kraus_spread_commutant.py` (21/21 + 24/24, SPREAD-SIGMA-BROKEN + COMMUTANT-EXTENDS + AORB-NOT-MAXIMAL:  $\sigma$  acts on the 6 spreads as two 3-cycles, ZERO fixed (outer-class forced), no invariant coarsening — but the direct-sum protocol is  $\sigma$ -stable as a FAMILY with invariant witnesses  $\sum B_{45}(s) = 4I + 2B$  and per-orbit  $A_{\text{orb}}$  (integer spectrum  $\{0^5, 1^2, 4^5, 7^2, 9^1\}$ );  $[E, K] = 0$  lifts EXACTLY to channel / KMS tower / odd places (the  $C_2$ -character expectation unique per place) and the 600-dim register; the  $\{K, \sigma\}$  commutant is 39-dim nonabelian  $\mathbb{R} \oplus M_3(\mathbb{R}) \oplus M_3(\mathbb{C}) \oplus M_3(\mathbb{R}) \oplus M_1(\mathbb{C})$  with  $A_{\text{orb}}$  the canonical abelian 5-dim subalgebra, window-silent beyond the  $K$ -spectrum); `v816_prime_mosco_selection.py` (23/23 guards+controls, 6/6 legs, MOSCO-SELECTS: the decider table on identical rungs — moment oscillation 22.0829 (the failed selector, reproduced exactly) vs Friedrichs resolvents 0.0004 / ledger-adjusted Weyl 0.0843 / raw  $\mu$ -face 0.0000; cofinal-unique  $2.5 \times 10^{-5}$ ; M2 recovery exact  $1.2 \times 10^{-12}$ ; M1 margins +2.63 with the EDGE family typed a genuine boundary channel; the import diagnosis completed at form level — nothing whitened at the core, the whitened control reproduces 5.8772 exactly); `v817_positive_descent_master.py` (14/14, MASTER-BOTH-INSTANCES: the 7-hypothesis master theorem with the kernel-checked Lean core (`PositiveDescentMaster.lean`: telescoping + Cauchy tail, positivity closure, sector commutation);  $G_{\text{net}}$  re-verified (Choi PSD, rate 0.267/doubling); THE NEW NUMBER: the prime packet state defects are summable at 0.701/doubling  $\approx 2^{-1/2}$  with  $\hat{m}_2(p) = -1/15$  exactly for every odd prime; control K4: positivity

is a SEPARATE hypothesis); `v818_sector_floor_attack.py` (9/13 + 13/15 with exactly the preregistered FAILs, FLOOR-MECHANISM-PARTIAL + CAPTURE-BOUNDED / THEOREM-SHAPE / AMPLIFIER-MECHANISM:  $\text{onem} = F(r)\varepsilon K$  symbolically exact with the sign entirely in the vanishing weight; interlacing capture  $\tau/\lambda_{\min}(A_h)$  median 1.53 (range 1.04–4.36) — the 2-mode lock block is an  $O(1)$ -faithful witness of the full floor; capture angle  $\cos\theta$  median 0.990, scramble-robust (frame content); the rotation law closes symbolically with 100% staircase sign-match —  $r$ -monotonicity is now a calculus statement; the amplifier is carried by the smallest prime powers, envelope  $\rho h^{3/2} \in [4.85, 24.2]$  non-decaying; the contraction constant is density-level (Epstein  $s_E = 0.241$ )); `v819_prime_packet_rm14.py` (21/21, RM14-EXACT, Lean companion `PacketRM14.lean`: the 15-label channel row hull = punctured  $\text{RM}(1, 4)^* = [15, 5, 7]$ , enumerator exact;  $\chi_{\text{NSR}}$  is ONE codeword (7+8 its two sides,  $C_2$  the affine bit); the cycle  $\text{RM}(1, 3) \rightarrow E_8 \rightarrow V \rightarrow \text{RM}(1, 4)^* \rightarrow \text{RM}(1, 3)$  closes with exactly 1344 equivalences; CSS  $[[15, 1, 3]]$  with dual  $[15, 10, 4]$ , logical distances (7, 3), triorthogonality 8/4/2;  $\zeta_8$  typed [H]); `v820_prime_kraus_rm24.py` (16/16 + 14/14, KRAUS-RM24-RULE-CANONICAL + PLANEFRAMES-EXACT: the literal leg-support claim dead twice (weights  $1/2/3 < \text{distance } 4$ ; Sp orbits 15+90 vs 45+60); the canonical  $\omega$ -built rule identifies the rule-A protocol set with the 105 weight-4 checks of  $[15, 10, 4]$  exactly, Sp/ $\sigma$ /deck/KMS-covariant;  $A_3 = 12 / B_3 = 16$  certified as PLANE-COUNT statements ( $\sigma_3(3) = 28$ ,  $a_3 = -4$ ); the graded frame intertwiner  $V^*V = \sigma_3(n)I$ ,  $V^*TV = a_nI$  exact for  $n = 3, 5, 7$ ); `v821_prime_vacuum35.py` (22/22, VACUUM35-EXACT:  $140 = 105 + 35$ ; Grams  $22I+6J / 6I+J / 28I+7J$  entrywise; the  $E_7$  completion  $112 + 21 = 133$  as a matrix identity (v170 anchor [C]); Weitzenböck  $H_{105}^T H_{105} = 2B^2 + 14I$  as the canonical label-side left factor with the honest v758 scope; the  $\pm B$  cancellation inside one Gram; tight frame constant 28 on mean-zero;  $\text{RM}(2, 4)$  dual-containing); `v822_prime_vacuum_dilation.py` (17/19 + 12/14 with the four FAILs being the preregistered PSD deciders, UNSHORTEN-DEAD + DILATION-DEAD: the 35-row Stinespring form fixes the shape exactly ( $6I+J$  ward, Weitzenböck bookkeeping integer) and buys  $\times 13.1$  — and both sectors STILL fail under both normalizations; THE GENERAL OBSTRUCTION QUANTIFIED:  $\text{PSD} \iff \text{coupling} \leq c^* = (2/21)\lambda_{\text{pencil}}(T_{\text{dep}}, T_{\text{cont}}^2)$ , measured deficits  $\times 3 \times 10^7$  (GL1) /  $\times 2 \times 10^2$  (f8) — the reading dies with the calibration, not the geometry; the continuum's completion role is already exhausted inside  $T_{\text{dep}}$ ;  $c^*$  is the frozen fence; module 7 / path-state NOT buildable).

- **Contracts.** NEW: PRIME.MOSCO.SELECTION.01 (selection solved finite-level; the three infinite-level ingredients typed), TFPT.POSITIVE\_DESCENT.MASTER.01 (both instances; the Lean core; positivity a separate hypothesis), PRIME.FLOOR.RATIO.01 [O] — **the new main contract candidate**: the sector floor reduces to the single ratio inequality  $\rho(X) = \tau/\tau_{\text{pnt}} > 0$  with the measured  $h^{-3/2}$  envelope (constant  $\approx 4.85$ , non-decaying), direction owned by the density rotation law (calculus statement, hypotheses verified), capture angle-certified frame geometry ( $\cos\theta$  0.990, scramble-robust), the amplifier carried by the smallest prime powers; KILL = the envelope failing at depth or the capture angle collapsing; NO RH claim, PROMOTION FENCED. Also registered: K5.SIXSTEP.TRANSPORT.01, PRIME.AORB.REFINEMENT.01, PRIME.PACKET.RM14.01, PRIME.KRAUS.RM24.01 (carrying PRIME.C2.PLANEFRAMES.01), PRIME.VACUUM35.01, and PRIME.CONTINUUM.UNSHORTEN.01 as the vacuum-route CLOSURE row (both transcriptions dead, the  $c^*$  pencil bound the frozen fence, the exhaustion insight). Dated notes on CARRIER.PETERSEN.RADIAL.01 (the SIXTHROOT BITEXACT upgrade with the v221 evidence — the frozen SPECTRAL-ONLY stands as that probe's adjudication, ABEL-DEAD precedent — and the  $T_{10}$  proposal), PRIME.KRAUS.DOILY.01 (spread/commutant/aorb ad-denda), PRIME.Z1.COMPACTNESS.01 and PRIME.POSITIVE\_DESCENT.01 (selection solved via Mosco, unique limit via the master theorem, the floor the single remaining object  $\rightarrow$  cross-ref PRIME.FLOOR.RATIO.01). NO marker moves; the honest deaths recorded as preregistered adjudications; all declared repairs carried verbatim in PROVENANCE.
- **Ledger.** 10 new rows (the nine module rows K5.SIXSTEP.TRANSPORT.01, PRIME.AORB.REFINEMENT.01, PRIME.MOSCO.SELECTION.01, TFPT.POSITIVE\_DESCENT.MASTER.01,

SECTOR.FLOORATTACK.01, PRIME.PACKET.RM14.01, PRIME.KRAUS.RM24.01, PRIME.VACUUM35.01, PRIME.CONTINUUM.UNSHORTEN.01 + the contract row PRIME.FLOOR.RATIO.01) + 4 dated updates; 888  $\rightarrow$  898 rows; NO marker moves anywhere.

- **Papers.** `tfpt_prime_front`: the new round section “One inequality: the floor reduces to the prime comb against its own density” (the Mosco selection, the master theorem, the floor mechanism and the new contract, the Kraus commutant tower, the vacuum-route closure); `tfpt_1`: the round-23 keybox entries (the clock-only six-step with the  $T_{10}$  proposal and the SIXTHROOT correction, the RM cascade with the  $[[15, 1, 3]]$  CSS code, the  $E_7$  vacuum completion); `tfpt_2`: the plane-frame Hecke match ( $\sigma_3(3) = 28$  and  $a_3 = -4$  as plane counts) and the RM-code reading of the 15-label channel join their sections; `tfpt_3`: nine new audit blocks; `tfpt_research_contracts`: the registrations and dated updates above.
- **Website.** Prime-front feed: the round entry “reflux round twenty-three”; counters 806  $\rightarrow$  815 modules / v535–v822; papers.ts and VerificationDag updated; generated surfaces regenerated (ScriptIndex, changelog.ts, version.ts, maps); README and Zenodo updated to 815 modules / v822.
- **Scope.** The exact results (the bit-exact transport location, the commutant tower, the RM enumerators and Gram identities, the  $E_7$  completion) sit behind the standing fences (ROOTCLASS-MIXED intact; no matter semantics; the  $E_7$ /fingerprint readings [C]; the  $T_{10}$  and vacuum readings proposals/[H], not deployments); the analytic strand is measured finite-level evidence only — PRIME.FLOOR.RATIO.01 carries its kill criteria and its promotion fence, the honest kills (the vacuum route in both transcriptions, the four preregistered floor FAILs) are on record as adjudications; no marker moves; NO RH claim anywhere.

**2026-08-06 (LXV)** (reflux round twenty-two — the K5 edge machine: five exact structures, two precise middles, and the honest kills: ELEVEN modules (v803–v813, from fifteen frozen probes; suite 795  $\rightarrow$  806) — the jet-code wave gives the Gaussian compiler its ring-theoretic normal form ( $W = L/2L$  free over the dual numbers, non-splitness = the no-origin obstruction), reconstructs the Hamming code from the NS/R bit alone ( $AGL(3, 2) = \text{Aut}(H_8)$ ), extends the CP intertwiner by the affine Gray register chart, and records two preregistered honest kills (the syndrome algebra and the analytic Clifford-activation ordering); the K5 wave certifies the Petersen carrier geometry whose radial sixth power IS the deployed transport spectrum, the doily incidence identity behind the recovery base rate, the cut-code reading of the Arf split, the Kraus context protocol with its exact commutant surprise, the flavor-filtration fingerprint — and closes the round with the P1 index-KMS identity  $c_3 = 1/(I \cdot \beta)$  and its convention audit)

- **New modules (11).** `v803_e8_jetcode_affine_nsr.py` (18/18 + 15/15, JETCODE-EXACT+AFFINE-NSR-EXACT:  $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$  with  $\epsilon = 1+i$  (2 ramifies; unique iso; the split ring  $\mathbb{F}_2 \times \mathbb{F}_2$  fails all 24 bijections);  $W = L/2L$  FREE of rank 4 over the dual numbers (Hermite-certified  $\mathbb{Z}[i]$ -basis); the exact self-extension  $0 \rightarrow \epsilon W \rightarrow W \rightarrow V \rightarrow 0$  with  $\ker \epsilon = \text{im } \epsilon$ ; the deck IS the jet unit  $J = 1+\epsilon$ , re-deriving the v782 transition-bus torsor on all 240 roots; ZERO  $\mu_4$ -equivariant splittings among all 65536 sections — non-splitness IS the no-canonical-origin obstruction; the 120 root pairs = the  $q=1$  unit shell ( $240 = 15 \times 8 \times 2$ : root = (address, error jet, orientation));  $\text{Stab}(\chi_{\text{NSR}}) = AGL(3, 2)$  of order 1344 =  $\text{Aut}(H_8)$  with the SAME natural 8-point action, and affine functions on the NS/R torsor reconstruct  $RM(1, 3) = H_8$  with exactly 1344 equivalences — typed internal reconstruction, NOT a P2 removal); `v804_prime_carrier_gray.py` (23/23, GRAY-INTERTWINER-EXISTS: the affine Gray map encodes  $C_2 \times \mu_4$  into the R-class torsor respecting the v786 kill (a group iso is impossible — exponent 4 vs 2 — and not demanded); the fiber family deck-covariant EXACTLY with TRIV-

IAL measured  $\mu_4$ -cocycle (residual = a pure  $\mathbb{Z}_3$  line twist on the  $\sigma$ -fixed classes); the extended Stinespring map's GL1 channel re-lands on the deployed Weil window at  $6.0 \times 10^{-16}$ ; one declared control repair); `v805_e8_syndrome_algebra.py` (13/13, SYNDROME-DEAD, the preregistered kill fires: max rank 222/256 over all four frozen phase conventions, the  $\pm r$  proportionality obstruction structural for the whole diagonal-phase family, both post-hoc escape directions score worse (215/160); survives:  $16 + 240 = 256$ , the 16 `v112` neutral kernels = the diagonal MASA, same-class products  $A_0$ -valued with  $\mu_4$  coefficients exactly); `v806_hecke_local_clifford.py` (16/18 with the two FAILs the preregistered decider gates themselves, controls 4/4, LOCAL-CLIFFORD-BREAKS: all four  $p \bmod 8$  class sectors fail PSD — signed character combinations carry no completed  $L$ -function, the trivial mod-8 component is imprimitive; class 1 is the UNIQUE positivity-completable class (Euler repair exact,  $+1.4 \times 10^{-5}$ ); the margin ordering does NOT track the proven activation depths (Spearman +0.40) — the Clifford activation reading stays decorative analytically, the arithmetic multirate theorem untouched; POSITIVE side-result:  $\chi_8/\chi_{-8}$  are two NEW PSD instances of the  $\Gamma_{\mathbb{R}}$  conductor rule (exact parities:  $\chi_8$  even,  $\chi_{-8}$  odd) — the rule map now covers the full mod-8 dual group; one declared bar re-freeze); `v807_lorentz_nullselector.py` (22 checks with the three honest  $h$ -order FAILs encoded +14/14, LORENTZ-COORDS-REVEAL+NULLSELECTOR-ALGEBRAIC:  $M(5, -3, 4) = 2(1, 2)^T(1, 2)$  exact — the compiler triple is a rank-one spinor square, Euclid  $(2, -1) \mapsto (5, -3, 4)$ ; THE REVEAL: the spinor slope is strictly monotone in the window width  $\alpha$  (66/66, Kendall 1.000, cross-ratio contraction CV 0.004) — the  $h$ -oscillation was  $\alpha$ -jitter; the approach  $2 - r \sim \alpha^{-0.23}$  stays glacial; the null selector  $|v_d|^2 = \det M_d = d(4 - d)$  forces  $d = 4$  on the RR/Serre functor vector with the anchor identities  $i+j = N_{\text{fam}}$ ,  $ij = |\mathbb{Z}_2|$ ,  $i^2+j^2 = g_{\text{car}}$ ,  $2ij = |\mu_4|$ ; THE BRIDGE:  $g^2 - N^2 = 4d = |H|^2$ , so null  $\iff$  glue index =  $d$  — independent of the `v781` census, same output {4}; honest downgrade: the seam family has no  $d$ -dependence to vary, the missing theorem typed); `v808_petersen_sixthroot.py` (19/19+16/16, PETERSEN-RADIAL-EXACT+SIXTHROOT-SPECTRAL-ONLY: the K5 edge machine — charges  $(-4, +1, +6)$  = the  $SU(5)$  decuple against the certified `v774` table; Petersen  $SRG(10, 3, 0, 1)$  with  $|\text{Aut}| = 120$  census; distance shells  $(1, 3, 6)$  charge-pure; the flavor hexagon = the distance-2 shell edge-by-edge; equitable quotient with hypercharge eigenvector  $(6, -4, 1) \rightarrow -2$ ;  $\text{spec}(P^6) = \{1, 64/729, 1/729\}$  = the deployed transport spectrum EXACTLY; the 10-dim multiplicity prediction ( $\times 5 = g_{\text{car}}$  fast,  $\times 4 = |\mu_4|$  slow) typed [H neu];  $B = \frac{1}{18}[[13, 1, 4], [1, 13, 4], [4, 4, 10]]$  is the unique symmetric doubly stochastic PSD sixth root in the corpus basis,  $B^6 GL(3, \mathbb{Q})$ -conjugate to `v486`'s  $M^6$  and the cusp-diagonal form but bit-equal to NEITHER — the corpus freezes the spectrum, not the basis); `v809_curve_code_doily.py` (21/21, DOILY-EXACT:  $NN^T = B + 2I$  entrywise,  $\text{spec} = \{9, 4^9, 0^5\}$ ;  $N/3$  has singular values  $\{1, 2/3, 0\}$  — the recovery base rate  $2/3$  as the unique nontrivial singular value of the canonical duad-to-syntheme incidence, the  $(2/3)^6$  identification typed pending the six-step composition; Petersen edges = synthemes 15/15; every census bridge factors exactly  $N_{\text{code}} = P_{\Delta}^T N_{\text{curve}}^T P_{\Gamma}$  — a duality, not an isomorphism; the Arf switch set  $T_6$  = the six within-block duads); `v810_cutcode_moment.py` (17/17 + 16/16, CUTCODE-EXACT+MOMENT-INCIDENCE-PARTIAL: the cut map is injective-isometric on all 256 cells and the Arf split  $16 = 1 + \bar{5} + 10$  IS the cut classification  $(0|5) + (1|4) + (2|3)$ ; Clebsch  $40 = 5 + 20 + 15$  over the Arf shells with within-10 = Petersen (census 120); roots/lines grade  $80 + 160 / 20 + 40$  — Arf/cut grading, NOT matter; the master moment  $\text{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$  from one symbolic identity; the  $210 = \binom{10}{4}$  map DECORATIVE per the frozen rule — the equivariant bijection killed by the exhaustive fixed-point obstruction, the partial  $80 \rightarrow 5 \times 16$  miss-slot map and the Newton tie survive as audit); `v811_prime_kraus_doily.py` (21/21, KRAUS-DOILY-PROTOCOL: the 105 Kraus legs = 15 diagonal +90 off-diagonal; the canonical  $45 + 60 =$  reversal-quotient (the 90 legs double-cover the 45 flags 2:1) + fiber expansion ( $15 \times 4 = 60 =$  the `v783` stabilizer rays); the  $Sp(4, 2)$  obstruction certifies no leg-subset partition — the literal partition is spread-relative = the ledger T3 census; the 45 transports quadratic  $(P_x P_y = \pm P_{x+y})$  spanning  $\text{End}(\mathbb{C}^4)$  at rank 16; THE SURPRISE:  $[B_{45}, B_{60}] = 0$  exactly —  $K = \frac{3}{7}E + \frac{4}{7}R$  is a direct-sum protocol with  $E$  a conditional expectation; measured: the certified spread is

NOT  $\sigma$ -invariant — the named next falsifier); `v812_flavor_graph_filtration.py` (16/16, FILTRATION-FINGERPRINT-ONLY: the  $Q/K/L$  row budgets are exact, order-exact counts of the  $K_{3,2}/\text{prism}/K_5/H_8$  incidence chain with level-wise additivity mirroring  $L = K + Q$ ; NO entry-level reconstruction from any frozen candidate; the exhaustive false-positive rates 0.005–0.75% make the misses informative — typed audit fingerprint); `v813_p1_index_kms.py` (13/13, P1-IDENTITY-CLOSED:  $I \cdot \beta \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$  EXACT in formal arithmetic; the index is an independent  $c_3$ -free theorem (zero occurrences in all four sources, gated),  $\beta = 2\pi$  independently measured (v526; the radian pin is the one shared convention and cancels),  $c_3$  stays the P1 axiom the identity EXPLAINS; the finite evenness precursor measured ( $w_1 = w_3 = \frac{1}{4}$  exactly at every level, symmetric zero-sum offset  $\pm 2^{-(l+1)}$  — the no-anomaly shape); the sea state does NOT equidistribute (dev 0.188) — the bridge must normalize the modular response; controls: index  $2 \rightarrow 1/(4\pi)$ , booked elsewhere by the corpus). Suite 795  $\rightarrow$  806; all eleven run standalone green and encode their frozen patterns per the v757 precedent (the two honest kills encode their expected-FAIL patterns explicitly); all fifteen probes re-run identically at promotion.

- **Contracts.** NEW: `E8.RAMIFIED.JETCODE.01` [E] (carrying `E8.AFFINE.NSR.01`), `PRIME.CARRIER.GRAY.01` (demands D1–D3: odd-place fiber charts, the  $\mathbb{Z}_3$  line twist as crossed-product structure, the infinite-level CP extension), `E8.SYNDROME.ALGEBRA.01` (DEAD as frozen, successor criterion named: root-individual operator data), `HECKE.LOCAL.CLIFFORD.01` (analytic ordering DEAD; next falsifier: primitive sector data, not class indicators), `PRIME.LORENTZ.SPINOR.01` (the reveal + the named 1D monotonicity candidate), `CARRIER.PETERSEN.RADIAL.01` [E/H] (carrying `TRANSPORT.SIXTHROOT.01`, the basis-freeze question named), `CURVE.CODE.DOILY.01` (the  $(2/3)^6$  six-step target named), `CARRIER.CUTCODE.01` [E] (carrying `CARRIER.MOMENT.INCIDENT.01`, the 210 decorative), `PRIME.KRAUS.DOILY.01` (protocol-grade; the commutant/spread falsifier named), `FLAVOR.GRAPH.FILTRATION.01` (fingerprint), `P1.INDEX.KMS.01` registered OPEN [O] (the one open bridge: even modular-response normalization over the  $I$  deck sectors; kill = any asymmetric sector offset). Dated notes on `ARF.BOUNDARY.CODE.01` (the affine NS/R reconstruction sharpens the attack surface;  $R1'$  unchanged), `PRIME.CP.INTERWINER.01` (the Gray extension; D1–D3 unchanged), `PRIME.SECTOR.CONTINUA.01` (the rule map covers the full mod-8 dual group) and `E8.ONEOBJECT.01` (the null-selector addendum: a second independent  $d = 4$  mechanism on the same invariant  $4d$ ). NO marker moves; both kills recorded as preregistered adjudications; the declared repairs (Gray C2 scramble, local-Clifford C1b bar, Kraus report text) carried verbatim in `PROVENANCE`.
- **Ledger.** 11 new rows (`E8.RAMIFIED.JETCODE.01`, `PRIME.CARRIER.GRAY.01`, `E8.SYNDROME.ALGEBRA.01`, `HECKE.LOCAL.CLIFFORD.01`, `PRIME.LORENTZ.SPINOR.01`, `CARRIER.PETERSEN.RADIAL.01`, `CURVE.CODE.DOILY.01`, `CARRIER.CUTCODE.01`, `PRIME.KRAUS.DOILY.01`, `FLAVOR.GRAPH.FILTRATION.01`, `P1.INDEX.KMS.01`) + 4 dated updates; 877  $\rightarrow$  888 rows; measured/typed throughout; NO marker moves anywhere.
- **Papers.** `tfpt_prime_front`: the new round section “The K5 edge machine: five exact structures, two precise middles, and the honest kills” (the Kraus protocol, the Gray intertwiner, the local-Clifford negative, the doily and Lorentz strands, the P1 index identity); `tfpt_1`: the jet-code normal form and the affine NS/R reconstruction join the architecture narrative (keybox entries; the cut-code grading of Theorem 2’s Clebsch geometry); `tfpt_2`: the Petersen carrier geometry of the transport spectrum and the flavor-filtration fingerprint join their sections; `tfpt_3`: eleven new audit blocks; `tfpt_research_contracts`: the registrations and dated updates above.
- **Website.** Prime-front feed: the round entry “reflux round twenty-two”; counters 795  $\rightarrow$  806 modules / v535–v813; papers.ts and VerificationDag updated; generated surfaces regenerated (ScriptIndex, changelog.ts, version.ts, maps); README and Zenodo updated to 806 modules / v813.

- **Scope.** The exact results (the jet-code normal form, the affine NS/R reconstruction, the Petersen/doily/cut-code incidence identities, the Kraus protocol identities, the index-KMS identity) sit behind explicit fences (ROOTCLASS-MIXED intact everywhere; no matter semantics; the Hamming reconstruction is internal, not a P2 removal;  $c_3$  stays the P1 axiom — the identity explains, it does not derive); the honest kills (syndrome algebra, analytic Clifford ordering) and the precise middles (the 210 decorative, the filtration fingerprint, the spectral-only sixth root, the glacial  $\alpha$ -approach) are preregistered adjudications on record; no marker moves; no RH claim anywhere.

**2026-08-06 (LXIV) (reflux round twenty-one — the descent functor completes at finite level and the boundary code becomes three physical axioms: ELEVEN modules (v792–v802, from thirteen frozen probes; suite 784 → 795) — the prime channel delivers all three demands of PRIME.POSITIVE\_DESCENT.01 at finite level (sector continua as one  $\Gamma_{\mathbb{R}}$  functor, the exact limit object with the first measured sector floor, the Stinespring carrier intertwiner) plus the halving-tails/ $k=5$  tower mechanism and the  $R$ -microstate honest negative; the seam/code strand forces Type II from three physical axioms, identifies the missing Clifford bit as the metaplectic  $S$ -lift at  $\tau = i$ , decides the canonical outer bridge and reads the carrier Pascal at character level; the clock strand turns the coset assignment into a theorem; and the  $G_{\text{net}}$  strand measures the martingale limit hypotheses as met)**

- **New modules (11).** `v792_ftransfer_kms_coset.py` (18 checks, KMS-COSET-THEOREM: the coset carried by a clock chart is a FUNCTION of its KMS/modular typing datum —  $S = -(\ln \lambda)^2/2$  with  $\lambda$  the time-1  $\text{PGL}_2$  holonomy multiplier per clock e-fold; Schwarzian-free, derivative-free classifier (non-circularity gated on its own source), blind assignment ( $F_{\text{pole}}/F_{\text{Boltzmann}}$  modular with  $|\ln \lambda| = \Delta = 6 \log(3/2)$ ,  $F_{\text{QCD}}$  autonomous,  $F_{\text{relic}}$  degenerate) and sympy-exact unblinding; controls incl. the v578 Möbius clock as the positive case — the executor  $\rightarrow$  groupoid  $\rightarrow$  coset chain of the CONTRACT.F.01 refinement is whole); `v793_f8sector_conductor.py` (15/15+16/16, F8SECTOR-PSD+CONDUCTOR-FUNCTORIAL: the  $f_8$  sector, broken at  $-132..-151$  under the register-trivial GL1 continuum, is PSD on the entire frozen ladder with its OWN weight-4/conductor-8/no-pole continuum ( $+1.52 \times 10^{-4} \rightarrow +3.20 \times 10^{-5}$ ); the continuum assignment is ONE closed rule  $\chi \mapsto q^{s/2} \prod_j \Gamma_{\mathbb{R}}(s + \mu_j)$  (duplication ward  $2.7 \times 10^{-15}$ ); the twist  $f_8 \otimes \chi_{-4}$  has conductor 16 by Atkin–Li + the Fricke ward ( $6.7 \times 10^{-16}$ ) and lands PSD with the rule continuum; four-sector coherence; the conductor datum load-bearing at exactly  $\ln 2$ ); `v794_cp_extension_gate.py` (17/17, CPGATE-UNDECIDED-FALLING: restriction exactness/interlacing/uniform bounds measured (the frozen  $e^{X/2}$  pole prediction recalibrated, declared); the limit object = the per-sector positivity floor on  $V_{\infty} = \text{Z1-COMPACTNESS}$  sector-decorated (anchor  $3.882 \times 10^{-6}$  reproduced); deep trend: no crossing, GL1 power-law to  $3.9 \times 10^{-6}$  at  $X = 18.375$ ,  $\chi_4$  SATURATES at  $\sim 1.5 \times 10^{-6}$  — the first measured sector floor); `v795_halving_tails_k5.py` (20/20+17/17, TAILS-IDENTIFIED+K5-MECHANISM-FOUND: exact identities  $v_2(D_p) = 6 + v_2(X_7) / 8 + v_2(X_1)$  with termwise-integral divisor sums; class-7 base  $\iff R_3(p) \equiv 16 \pmod{32}$ ;  $k$  deterministic on the 2-power residue-tower cells with zero exceptions at  $10^6$ ,  $k = 5$  only in cell (5, 3) of density  $1/64 \Rightarrow P = 1/128$  = the observed half mass; the class-7 mirror law  $v_2(D_p) = 3 + v_2(p+1)$ ; the governing-field derivation typed open); `v796_duad_syntheme_bridge.py` (24/24, BRIDGE-CANONICAL: the Sylvester outer bridge  $\beta$  with  $\beta\sigma = \tau\beta$ ,  $\beta(q^*) = S^*$  unique up to the order-6 joint stabilizer;  $q_{\Theta}$  pulls back constantly to  $q_{\text{even}}^* = q^* + \bar{h}(\cdot, A)$  — the anchor shift resolves the v784 Arf mismatch;  $t_A \mapsto \rho = J$ ;  $\chi_{\text{NSR}}$  an exact two-variant torsor); `v797_r_microstate_identification.py` (14/14, R-NO-ARROW-REALIZATION: the honest negative — no arrow-level realization of the  $R$ -grading; the count identity  $16B_n = 256R(n)$  exact but count-level only); `v798_seam_clifford_modular_s.py` (27/27, MODULAR-S-PARTIAL: the per-factor Hadamard IS the metaplectic  $S$ -lift at  $\tau = i$ ,

landing exactly in the missing coset  $\zeta_8 G_{31}$ ; the deck  $J$  = the metaplectic anomaly of the total Fourier (R2 placed); the strict  $q^*$  census EMPTY (0/46080), one  $G_{31}$ -conjugacy class of survivors, the  $\sigma$ -orbit Arf defects XOR to the anchor  $A$  (R3 named)); `v799_seam_code_typeii.py` (19/19, TYPEII-FORCED: exhaustively over all 308,993 subspaces of  $\mathbb{F}_2^8$ , locality  $\iff$  self-orthogonal, integer spin  $\iff$  doubly even,  $S$ -closure/ $|\text{disc}| = 1 \iff$  self-dual = the 30  $\hat{e}_8$  copies; residual  $R1 \rightarrow R1'$  = the three physical axioms A1–A3); `v800_e8_torsor_fourier.py` (30/30, TORSOR-FOURIER-PARTIAL: 120 canonical projective Fourier rays; the carrier block =  $(1+5+10) \boxtimes (1+3)$  under  $W(D_5) \times W(A_3)$  at character level — impossible at point level; the weight/character bases exactly mutually unbiased, ROOTCLASS-MIXED intact; the 45 deck-even modes = the 45 context Pauli observables); `v801_prime_cp_intertwiner.py` (29/29, CP-INTERTWINER-EXISTS:  $\Phi = V^* \pi V$  from the 105 Kraus legs, arrow algebra = the full  $M_{15}$ , Choi exact-rational ( $105 \times 1/7$ ); the four automorphic channels of ONE map land on the deployed GL1 Weil window ( $6.0 \times 10^{-16}$ ) and the  $\Gamma_{\mathbb{R}}$ -rule windows, all PSD; F1 retyped as a negative Choi eigenvalue); `v802_gnet_martingale.py` (15/15, MARTINGALE-HYPOTHESES-MET: exact Haar filtration, defects summable at  $\sim 0.25$  per doubling, unique limit by telescoping, determinant witness  $\geq 0.99$ ; decimation converges to the WRONG degenerate limit — the coherent bond load-bearing). Suite 784  $\rightarrow$  795; all eleven run standalone green and encode their frozen patterns per the `v757` precedent; all thirteen probes re-run identically at promotion.

- **Contracts.** NEW: `SEAM.CODE.TYPEII.01` ( $R1 \rightarrow R1'$ : the seam satisfies A1 mutual locality, A2 integer conformal spin, A3 holomorphy/index one — the three-axiom equivalence certified exhaustively), `SEAM.CLIFFORD.MODULAR_S.01` (partial: coset positive, R2 solved as the metaplectic anomaly, R3 named through the Arf defects), `CURVE.CODE.OUTERSPIN.01` (decided CANONICAL; the `v784` inner kill stands), `PRIME.CP.INTERTWINER.01` (D1–D3 registered: dilation family, normality on the Z1-COMPACTNESS object, the limit identification — contains RH, fenced), `GNET.MARTINGALE.LIMIT.01` (hypotheses measured; steps 5–7 the typed analytic remainder). UPDATED with dated notes: `PRIME.POSITIVE_DESCENT.01` (all three demands now finite-level; the remaining analytic core restated as the per-sector positivity floor + D1–D3), `PRIME.Z1.COMPACTNESS.01` (the sector-decoration cross-reference), `CONTRACT.F.01/FTRANSFER.CLOCKS.01` (the coset theorem completes the fibered refinement), `HECKE.MULTIRATE2ADIC.01` (the tails identities, the residue tower, the mirror law; governing field typed open), `ARF.BOUNDARY.CODE.01` (all three residuals moved), `CURVE.CODE.2TORSION.01`, `E8.TRANSITIONBUS.01`, `E8.CLIFFORD2Q.01`, `HECKE.ARROW_MESSAGE.01`.
- **Ledger.** 11 new rows (`FTRANSFER.KMSCOSET.01`, `PRIME.SECTOR_CONTINUA.01`, `PRIME.CP.EXTENSION.01`, `HECKE.HALVINGTAILS.01`, `CURVE.CODE.OUTERSPIN.01`, `HECKE.R.MICROSTATE.01`, `SEAM.CLIFFORD.MODULAR_S.01`, `SEAM.CODE.TYPEII.01`, `E8.TORSOR.FOURIER.01`, `PRIME.CP.INTERTWINER.01`, `GNET.MARTINGALE.LIMIT.01`) + 10 dated notes; 866  $\rightarrow$  877 rows; honest deaths and declared repairs carried in the module provenance blocks ( $[C]/[E]$ /measured typing per the probes); NO marker moves anywhere.
- **Papers.** `tfpt_prime_front`: the new round section “The descent functor completes at finite level, and the boundary code becomes three physical axioms” (demands (1)–(3), the depth-class mechanization, the two honest adjudications) + the positive-protocol close retyped; `tfpt_1`: five round-21 keybox entries (outer bridge, Type II, metaplectic  $S$ -lift, torsor Fourier; the  $R1$ – $R3$  residual prose retyped); `tfpt_2`: the Type-II forcing, the martingale follow-up and the Clifford completion join their sections; `tfpt_3`: eleven new audit blocks + the stale next-step/residual prose retyped; `tfpt_4_frontier`: the coset-theorem note; `tfpt_research_contracts`: the five registrations and the dated updates above.
- **Website.** Prime-front feed: the round entry “reflux round twenty-one”; counters 784  $\rightarrow$  795 modules / `v535`–`v802`; `papers.ts`, `VerificationDag`, `OpenGates`, `HonestyBanner` and prime-front counters updated; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `version.ts`, `maps`); `README` and `Zenodo` updated to 795 modules / `v802`.



- **Scope.** The exact results (the coset theorem, the  $\Gamma_{\mathbb{R}}$  rule instances, the tails/tower identities, the outer-bridge census, the Type-II equivalence chain, the metaplectic coset identities, the character-level carrier block, the Stinespring/Choi certificates) sit behind explicit fences; the honest caps and negatives (the empty strict census, the mutually-unbiased cap, the no-arrow-realization, the undecided-falling trend) are preregistered adjudications on record; sector PSD stays finite-level evidence — GRH-type statements unproven; the limit identification contains RH and is NOT claimed; no marker moves; no RH claim anywhere.

**2026-08-05 (LXIII) (reflux round twenty — the largest promotion of the day: ELEVEN modules (v781–v791, from sixteen frozen probes; suite 773 → 784) — the strategic round lands the  $d = 4$  forcing census with its honest clock kill, the transition-bus torsor, the two-qubit Clifford identification and the curve/code 2-torsion partial; the prime channel lands the check32 theorem, the packet-480 kill and the arrow-ledger/broadcast pair; and the positive protocol lands the positive  $C_2$  lift, the multirate ladder with its constant-depth proof, the SNF/ $\mu_4$  adjudication and the positive-descent decider with the new open contract PRIME.POSITIVE\_DESCENT.01)**

- **New modules (11).** v781\_one\_object\_clock\_census.py (21/21 + 23/23, ONE-OBJECT-PARTIAL+CLOCK-FIBER-DEAD: the TFPT glue pattern admits an even unimodular DIAGONAL glue of  $A_{d-1} \oplus D_{d+1}$  iff  $d = 4$  (exact census  $d = 2..12$ ; the two v92 Lagrangians reproduced; explicit  $E_8$  certificate;  $d = 9$  dies at isotropy; controls fire) with the corollaries  $g_{\text{car}} = d+1 = 5$ ,  $N_{\text{fam}} = d-1 = 3$ ,  $|\mu_4| = 4 = d$ ; honesty gate:  $c_3 = 1/(2\pi d)|_{d=4} = 1/(8\pi)$  is a corpus-legitimate REWRITING, NOT a new derivation of P1; the  $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$  clock action on the 60 Gaussian lines is DEAD at three levels — full line-compatible group order 23040, max element order 12); v782\_e8\_transition\_bus.py (35/35, TRANSITION-BUS-TORSOR: the 240 roots are 15 coordinate 8-frames whose pair fibers are canonical  $v^\perp$ -torsors; NO deck-equivariant  $V \times (V \setminus 0)$  labeling exists (proof-grade obstruction); the frozen symplectic predicate reproduces all 57600 inner products;  $240 = 5 \times 3 \times 16$  spread layer; the  $A_{15}$  comparison quantified); v783\_two\_qubit\_clifford.py (32/32, CLIFFORD-PARTIAL(H4): the 60 Gaussian lines ARE the 60 two-qubit stabilizer rays entry-exactly over  $\mathbb{Z}[i]$  (60/60, support census  $4 + 24 + 32$ );  $G_{31} = \text{index-2 Clifford subgroup with } 46080 = \mu_4 \times \text{Pauli}_{16} \times Sp(4, 2)$ , classes = Pauli contexts, spreads = MUB pentads; the index bit is the Galois  $\sqrt{2}$ /Hadamard bit, not  $J \leftrightarrow -J$ ); v784\_curve\_code\_2torsion.py (33/33, CURVE-CODE-PARTIAL: every CM generator period certified as an integral  $\mathbb{Z}[\zeta_{12}]$  recognition; the correspondence dies at two exact layers — deck syntheme-side  $[3, 3]$ -free vs code duad-side, and curve theta even/Arf-0 ( $1+6+9$ ) vs  $q^*$  odd/Arf-1 ( $1+5+10$ ); joint census 0 with positive control 6); v785\_hecke\_check32.py (20/20, CHECK32-THEOREM:  $f_8 \equiv E_{\text{odd}} \pmod{32}$  to  $q^{100000} = 25000 \times$  the Sturm bound 4 (one cited classical ingredient);  $a_p \equiv 1 + p^3 \pmod{32}$  on 9591/9591 odd primes with the cube-map decoder; mod-64 fails at  $q^3$ ; controls 91.7% and  $\sim 31/32$ ); v786\_prime\_packet480.py (28/28, PACKET480-ADDRESS-ONLY: the header ( $p \bmod 30$ ,  $a_p \bmod 32$ ) is exact 7-bit address arithmetic but the canonical intertwiner to the 128 spinor roots is IMPOSSIBLE — exponent 8 vs 2; 3 of 7 bits realizable; the 128 stays a fingerprint; kill fired as preregistered); v787\_hecke\_arrow\_broadcast.py (48/48 + 21/21, ARROW-LEDGER-STRUCTURED+BROADCAST-EXACT: the 15 ramified arrows ARE the 15 polar labels (7 NS + 8 R), spread census (3, 1, 1, 1, 1),  $T/196 = (4/49)I + (45/49)\Pi_0$  exact with the 105 leg classes = the v756 Kraus set,  $a_7 = +24$  sign-derived from the full 137600-line census;  $\bar{T}_n = M_n \otimes I_{16}$  on all odd shells 3..13 with  $B_n = 16R(n)$  arrow-exactly, new identities  $b = 2A_p$  and  $n_B = A_p B_p/2$ ; honest negative at  $p = 7$ ); v788\_positive\_c2\_lift.py (42/42, C2LIFT-THEOREM:  $E_{\text{odd}} \pm f_8 = 2A/2B$  are POSITIVE theta series with  $B = 16R$  and  $a_n = \sigma_3(n) - 32R(n)$  — check32 becomes a kernel corollary; microstates 1, 4, 10, 24, 43; automaton censused on 41053 pairs + 70 prime powers; five-condition validator with joint detection 1.000000 and the tamper blind

spot exactly the  $16\mathbb{Z}$  lattice); `v789_multirate_constdepth.py` ( $23/23 + 24/24$ , MULTIRATE-THEOREM/SPINLIFT-VERIFIED+CONSTDEPTH-THEOREM:  $v_2(1+p^3-a_p) \geq 5 + 2[\chi_{-4}] + [\chi_8]$  on all 78497 primes  $< 10^6$  with sharp witnesses  $3/7/5/17$ ; sifts Sturm-certified at level  $\leq 512$ ; mod-512 breaks at  $n = 9$ ;  $\text{Sh}(g) = -8E_{\text{odd}} + 256H$  exact with Kronecker factor  $\chi_8$ ; the constant depths PROVEN via the two named  $h = 1$  ingredients); `v790_snf_mu4_theta.py` ( $18/18 + 17/17$ , SNF-FINGERPRINT+MU4-PARTIAL: the  $256R(n)$  representation identity and the SNF  $(1^4, 2^4)$  match are exact but the isometry census is EMPTY at every admissible level (kissing 240 vs 8;  $0/20160$ ; the naive  $4+4$  split inadmissible by theorem); the NEW exact identity  $c_{\text{sig}} = \frac{1}{15}E_4(q^2) - \frac{6}{5}E_4(q^4) + \frac{32}{15}E_4(q^8) - 8f_8$  — the cusp part of the quartic  $\mu_4$  character is exactly  $-8f_8$  with  $-8 = 52 - 60 = -\text{rank}(E_8)$ ; the  $C_2$  channel is pure Eisenstein; the all-shell quartic vanishing is FALSE); `v791_positive_descent.py` ( $30/30$ , DESCENT-PARTIAL: the packet GNS state on  $\mathbb{N}[C_2] \otimes \mathbb{N}[\mathbb{F}_2^4] \otimes \mathbb{N}[\mu_4]$  is manifestly positive in exact rationals; the naive linear pushforward is NOT a state (F1); the GL1 descent is bit-identical to the deployed Weil window and its UNIQUE PSD sector of 24 (F2: the breaking tensor factor is the continuum/pole leg)). Suite 773  $\rightarrow$  784; all eleven run standalone green and encode their frozen patterns per the v757 precedent; all sixteen probes re-run identically at promotion.

- **Lean.** `TfptCarrier/Check32.lean` (13 kernel theorems: the cube bijection on the 16 odd residues, the first-coefficient congruences as exact integer statements, the mod-64 failure witness) and `TfptCarrier/PositiveC2Lift.lean` (the positive-lift kernel with check32 as a corollary), both kernel-checked, lake build green (3406 jobs), Lean tree 80 modules; dated Lean notes carried in the `HECKE.CARRIER_CHECK32.01` and `HECKE.POSITIVE_C2_LIFT.01` ledger rows.
- **Contracts.** `PRIME.POSITIVE_DESCENT.01` registered OPEN [O] with the three demanded objects — (1) sector-adapted continua (the  $f_8$ -sector continuum as the finite-level falsifier), (2) GL1-sector PSD persistence in the Z1-COMPACTNESS frame (Weil positivity retyped as sector compression of a manifestly positive object), (3) the carrier intertwiner — plus stop conditions and inherited falsifiers; cross-referenced both ways with `PRIME.Z1.COMPACTNESS.01` (dated add-on note). Dated notes on `HECKE.GEOM.01` (the 2-adic skeleton now load-bearing), `HECKE.GEOM.HALFINT.01` (the ladder form of the Shimura lift; Kronecker factor  $= \chi_8$ ), `RTF.GNS.LEDGER.01` (the four-character envelope closed multiplicatively), `E8.GAUSSIAN.CODE.01` (the Clifford identification and the bus torsor), `PRIME.Z1LOOKAHEAD.01` (the packet-480 impossibility), `ARF.SPINORCOMPILER.01` (the  $q^*$  spread deployment and its Arf-parity boundary) and `ARF.ROOTCLASS.01` (the fence held through the round).
- **Ledger.** 11 new rows (`E8.ONEOBJECT.01`, `E8.TRANSITIONBUS.01`, `E8.CLIFFORD2Q.01`, `CURVE.CODE.2TORSION.01`, `HECKE.CARRIER_CHECK32.01`, `PRIME.PACKET480.01`, `HECKE.ARROW_MESSAGE.01`, `HECKE.POSITIVE_C2_LIFT.01`, `HECKE.MULTIRATE2ADIC.01`, `HECKE.SNF_THETA.01`, `PRIME.POSITIVE_DESCENT.01`) + 8 dated updates; 855  $\rightarrow$  866 rows; measured/typed throughout, honest kills recorded as preregistered adjudications, the [C neu]/[E neu] typing carried, NO marker moves anywhere.
- **Papers.** `tfpt_1`: the four strategic entries join the code keybox (the  $d = 4$  forcing census with the P1 honesty gate, the transition-bus torsor, the Clifford identification, the curve/code partial); `tfpt_2`: the stabilizer-identification follow-up joins the  $G_{\text{net}}$  page; `tfpt_3`: eleven new audit blocks; `tfpt_prime_front`: the new section “The positive protocol: the prime channel as a positive routing code, and what the descent still needs” (check32 theorem, positive lift, multirate ladder + proof, arrow ledger + broadcast, the two preregistered adjudications, the descent and its three-object contract); `tfpt_research_contracts`: the `PRIME.POSITIVE_DESCENT.01` registration and the Z1-COMPACTNESS add-on note.
- **Website.** Prime-front feed: the round entry “reflux round twenty”; counters 773  $\rightarrow$  784 modules / v535–v791; papers.ts narrative mirrors updated; README and Zenodo updated to 784 modules / v791.
- **Scope.** The exact results (the forcing census, the torsor structure theorem, the stabilizer

identification, the check32/positive-lift/multirate theorems modulo their named classical ingredients, the broadcast factorization, the  $c_{\text{sig}}$  identity) sit behind explicit fences; the three fired kills (clock fiber, packet-480, all-shell quartic vanishing) are preregistered adjudications firing inside green machinery; the  $c_3$  link stays typed INTERPRETATION (P1 is not derived); ROOTCLASS-MIXED stays binding (no code→matter reading anywhere); the descent contract names what is still missing; no marker moves; no RH claim anywhere.

**2026-08-05 (LXII) (reflux round nineteen: SEVEN promotions (v774–v780, from twelve frozen preregistered probes, all first-run adjudications reproduced identically at promotion; suite 766 → 773 modules) — the Arf compiler lands with its fired matter fence and the boundary-code uniqueness chain, the external-clock contract is EXECUTED on its expected K1 kill, remainder (a) of the (L1) majorant closes in the corrected  $(1 + \ln M)$  form, the  $G_{\text{net}}$  GNS precursors cohere, and the sharpened-Z1 trilogy measures the z1-compactness unification)**

- **New modules (7).** `v774_arf_spinor_compiler.py` (46/46, ARF-SPINOR-EXACT: the Arf/spinor compiler on  $V = L/(1+i)L$  — parity lift  $\iota \cong C_{\text{even}}(5)$  with  $\beta = \hbar$  exact, unique  $\sigma$ -selected  $q^*$  (census  $4 \rightarrow 2 \rightarrow 1$ ; the weight form IS  $q^*$ ), Arf census  $6+10$  with  $Sp(4, 2) \cong S_6$ ,  $16 = 1+\bar{5}+10$  as  $\text{Stab}(q^*) \cong S_5$  orbits,  $K_6$  duads,  $B = I + A_{KG(6,2)}$  with exact ovoid eigenvectors, hypercharge table  $X = (-2, -2, -2, 3, 3) \rightarrow$  the v310/v14 dictionary, moments  $(16, 0, 120, 0)$  with the chain  $40/41/240/248$ , Pati–Salam  $8+8$  with  $B-L$  exact,  $\chi_{\text{NSR}} =$  anchor bit;  $A_3^{\text{Fam}} \neq A_3^{\text{PS}}$  naming discipline); `v775_gaussian_class_d5_purity.py` (18/18, ROOTCLASS-MIXED — THE FIRED PHYSICAL KILL: no  $D_5 \oplus A_3/SU(5)$  convention makes even one class set pure,  $\geq 7$  adjoint-side classes for EVERY embedding by the 128-spinor-root bound,  $0/56$  splits; the code→matter reading incl. the “empty mailbox =  $\nu_R$ ” reading is fenced out of all main claims); `v776_boundary_hamming_uniqueness.py` (31/31, BOUNDARY-CODE-UNIQUE: census  $135 \rightarrow 30 \rightarrow 14 \rightarrow 2 \rightarrow 1$  without Hamming/ $E_8$  input, ONE deck-invariant  $\mathbb{Z}_2$  flag bit, Construction A  $\rightarrow E_8$  exact,  $g_{\text{car}} = 1+4 = 5$  as OUTPUT; residuals R1–R3 typed, NO P2-removal claim); `v777_ftransfer_clock_jets.py` (24+14 checks, 0 failures: FTRANSFER.CLOCKS.01 EXECUTED — the K1 kill fired exactly as the frozen prior expected, NO-COMMON-CONNECTION, transported Schwarzians sympy-exactly  $\{-\Delta^2/2, -\Delta^2/2, 0\}$ ; + the fibered CONTRACT.F.01 refinement FTRANSFER-FIBERED-CARRIES: discrete  $\text{PGL}_2$  base with  $\text{CR} = 4/3$  exact, constant seam cocycle  $\{0, \pm\Delta^2/2\}$ , thermal-vs-RG coset split); `v778_qgeo_gridsup_logsum.py` (19/19+19/19: remainder (a) of the v745 (L1) majorant FULLY closed in the corrected  $(1 + \ln M)$  form; the literal v715 all-refinements clause is FALSE at  $r_1 = 1$  and repaired; certified grid Fourier log-sum theorem with explicit constants; GATE.QGEO does not move); `v779_gnet_gns_arf.py` (27/27+27/27: GNS precursors COHERENT — exact Haar tower, commuting index-4 expectations, duality defect 0 with ratio 4; the exact dim law  $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1} \cos(\pi c/2)$  makes the  $\mathbb{C}[\mathbb{Z}_4]$  obstruction exactly  $2^{-\ell}$  and explains the  $\ell = 1$  anomaly; the honest negative: the  $1+5+10$  echo is non-geometric,  $K_{\text{matter}}$  does not govern transport); `v780_z1_compactness_trilogy.py` (gates  $3/4+1/4+$  legs  $3/6$  with exactly the preregistered fails: the N2 family carries contact/collision/coupling natively but not the widening cadence; the finite boundary triple is exactly Herglotz and dies as a whitened  $K=M/2$  import, cell-mass exactly 0.500 beyond the half-window; the raw variable-edge family is bounded (0.936) + equicontinuous (0.807) with summable  $\sim$ rank-one entry poles,  $c_{\text{mass}} 0.032 \ll 1$ ). Suite 766 → 773; all seven run standalone green and encode their preregistered patterns per the v757 precedent.
- **Lean.** `TfptCarrier/ArfSpinorCompiler.lean` (23 kernel theorems: parity lift,  $\beta =$  symplectic form, selector uniqueness, Arf census, orbit partition, traceless/primitive  $X$ , moments  $16/0/120/0$  with the chain,  $\chi_{\text{NSR}} =$  anchor bit, PS split, must-fail dot-form control) and `TfptCarrier/CellCocycle.lean` (18 declarations: exact Banachiewicz/bordered-Schur update, PD cell  $\Rightarrow$  PSD correction  $\Rightarrow$  monotone Loewner flow, bounded-monotone entrywise convergence naming the missing uniform Loewner bound, Redheffer/linear-fractional composition

over arbitrary rings), both kernel-checked, `lake build green`; dated Lean notes added to the `PRIME.QFCOCYCLE.01` and `PRIME.GRAM.DIAGONAL.01` ledger rows.

- **Contracts.** `FTRANSFER.CLOCKS.01` moves `PENDING`  $\rightarrow$  `EXECUTED` (K1 kill as the frozen prior expected; freeze guards byte-verified; the four dictionary legs pass at  $571 \times / 1.15 \times 10^{-4} / 0.83\% / 0.0602$ ); `CONTRACT.F.01` carries the dated jet-route-closure + fibered-refinement note (next named step: coset assignment from KMS/modular clock typing); `ARF.BOUNDARY.CODE.01` registered `OPEN [O]` (derive evenness/self-duality from the seam axioms; residuals R1–R3; kills named); `GATE.QGEO` dated note (residue (a) closed in  $(1 + \ln M)$  form, the literal v715 clause corrected, (b)/(c) open, gate does not move); `GATE.METRIC.08/.11` dated notes (the  $2^{-\ell}$  obstruction law as the finite precursor of the identification); `PRIME.Z1.OPERATOR.01` and `PRIME.KMS.INDUCTIVE_STATE.02` carry the measured unification (compactness carried, selection = import-faithful coupling as the single shared open object; merged target Z1-COMPACTNESS, cross-referenced, no row deletion).
- **Ledger.** 6 new rows (`ARF.SPINORCOMPILER.01`, `ARF.ROOTCLASS.01`, `ARF.BOUNDARY.CODE.01`, `QGEO.GRIDSUP.01`, `GNET.GNSLIMIT.01`, `PRIME.Z1.COMPACTNESS.01`) + dated updates on `FTRANSFER.CLOCKS.01`, `CONTRACT.F.01`, `QGEO.CARL1.01`, `GNET.LOCALNET.01`, `GNET.PPINDEX.01`, `GATE.METRIC.08/.11`, `PRIME.Z1.OPERATOR.01`, `PRIME.KMS.INDUCTIVE_STATE.02`, `PRIME.QFCOCYCLE.01`, `PRIME.GRAM.DIAGONAL.01`; 849  $\rightarrow$  855 rows; all measured/typed, NO marker moves anywhere.
- **Papers.** `tfpt_1/tfpt_2`: the Arf compiler, the fired matter fence and the boundary-code uniqueness join the code sections (after the symplectic code lemma), and the  $G_{\text{net}}$  follow-ups gain the v779 continuum-strand paragraphs; `tfpt_3`: seven new audit blocks + dated notes on the v715/v745/v746/v773 blocks; `tfpt_prime_front`: the Arf-compiler round in the incidence programme, the QGEO  $(1 + \ln M)$  note, and the new section “The sharpened-Z1 trilogy: compactness carried, selection open”; `tfpt_research_contracts`: the executed clock contract, the fibered `CONTRACT.F.01` note, the `ARF.BOUNDARY.CODE.01` registration, the `GATE.QGEO/GATE.METRIC` notes, the Z1/KMS unification notes and the `CellCocycle.lean` legacy; `tfpt_4_frontier`: the jet-route-closure note.
- **Website.** Prime-front feed: the round entry “reflux round nineteen”; counters 766  $\rightarrow$  773 modules / v535–v780; research-contracts and F\_transfer mirrors updated (`papers.ts`, glossary, OpenGates, DAG nodes); README and Zenodo updated to 773 modules / v780.
- **Scope.** The exact results (Arf finite algebra + Lean, boundary-code census, sympy-exact clock kill and cocycle, the grid Fourier log-sum theorem, the dim law) sit behind explicit fences; the fired matter kill and the preregistered partials are criteria firing inside green machinery and stay on record; the honest residuals (R1–R3, residues (b)/(c), the import-faithful coupling) are named, not claimed; no marker moves; no RH claim anywhere.

**2026-08-05 (LXI)** (the qf offensive: FIVE promotions (v769–v773, from eight frozen preregistered probes, all first-run adjudications; suite 761  $\rightarrow$  766 modules) — the diagonal Gram route CLOSES at a moving spectral edge, `PRIME.GRAM.DIAGONAL.01` is closed per its own frozen stakes, and the sharpened `PRIME.Z1.OPERATOR.01` takes over. THE GATE-3 AUDIT PAIR: v769 (merged, 5/5 gates + 34/168 cells, 14/14 + 14/14 guards+controls, qf-gauge/gate3prime-dead) — Gate 3’s full-ladder Cauchy demand on  $q_f$  was provably STRONGER than the diagonal contract needs: a deterministic  $\varepsilon$ -net subsequence (density 0.144) makes all 14  $q_f$  profiles Cauchy at the exact parent bars the full ladder failed, and the contract’s own  $E_Y$  corner objects are BLIND to  $q_f$  motion (corr +0.038; the 0.34 mode-entry jump at  $M = 888$  leaves the corners at routine level; 370 cross-cluster pairs, median  $\rho = 1.13$ ) —  $q_f$  is REPRESENTATION GAUGE, owning the  $\varepsilon$ -scale distribution of the Yosida deficit, not the value (`VAR_RATIO` 0.021);

the double limit is non-uniform (deficit  $X$ -slope grows  $88\times$  over four decades) but not certified singular; yet the corner-native weakened Gate 3' DIES on its own predeclared  $1.6 \times 10^7$  comb ( $X = 16.5625$ ): the corner increments fall to  $X \sim 12.9$  and then RISE sustained ( $+0.10..+0.34$  per  $X$  unit at  $\varepsilon = 10^{-3}$ , 0/56 cells, broad-spectrum) — the v764 precondition is REFUTED at the corner level. THE TRANSPORT PAIR: v770 (merged, gates 4/5 + 4/5 with exactly the preregistered decider/Z fails, 19/19 + 14/14, qf-bundle-partial/qf-settles-positive) — the Kato/polar transport of the rank-6 near-kernel is well-defined with wide margins (rank stable on 888..972, gap 6/7 = 0.34..0.56, angles  $\leq 2.72^\circ$ , TV flattening) and repairs the rigid frame for 14/14 functions (median med5 ratio  $26.7 \rightarrow 1.004$ ); the residual non-Cauchy part is the frame-independent band-weight drainage (radial share  $\sim 0.64$ ) — and the drainage was a TRANSIENT: on the  $10^8$  comb ( $X = 18.375$ ) the band weights plateau at positive per-function constants ( $R2$  family 0.2249–0.3583,  $R1$  family 0.0082–0.0793; zero-bar depth  $X^* \sim 227$ , i.e. never), the 6/7 gap bottoms at 0.1008 then recovers to 0.1397, the 7th/8th threshold crossings (992/ $\sim 1108$ ) are typed, the holonomy exponent is  $\sim 1$  (step refinement closed) — the Feshbach/boundary-triple treatment becomes MANDATORY. THE CENSUS: v771 (gates 6/12 with exactly the preregistered fails, 26/26 guards+controls, qf-representation-open) — the  $6 = 1+5$  hypothesis  $E_{qf} \cong E_7 \oplus E_{-2}$  FAILS the frozen census (the deployed scalar comb realizes only two characters of  $V$ ; no rank-6 lift exists; the near-kernel is a continuum-vs-odd-places balance, ro-dominant count 0/6 vs predicted 5) while the label side is exact and uniquely canonical ( $BB^T = 4I + 3J$ , spectrum (1,9,5); 27/27 wrong forms + 5/5 wrong labelings die on  $\sigma$ -sharpness); the honest residue: ONE rung-stable ram-odd contact direction ( $\cos = 0.997$ ) plus a second contact at  $\cos \sim 0.64$ ; the dimension match stays typed DECORATIVE. THE FIXED- $d$  PAIR: v772 (merged, gates 9/12 with exactly cauchy-6/gap-7/gap-8 failing + edge-a fail/edge-b pass/KM2 quiet, 28/28 + 20/20,  $\sim 180$  s at full frozen ladders per the v677 suite precedent, feshbach-partial/edge-keeps-moving; one declared guard-normalization correction of the G1.9  $z$ -independence Ward, no gate bar touched, repeat run identical) — the  $d = 6$  reduction is EXACT (reconstruction  $8.8 \times 10^{-13}$  against independent complex dense solves, 5 rungs  $\times$  5  $z$ ) and matrix-Herglotz certified, but its Cauchy tail flattens where mode 7 grazes the band;  $d = 7$  converges strongly ( $b_2 = +0.502$ ) but loses separation at the top rung;  $d = 8$  has an internal 8/9 crossing ( $87^\circ$ ); on the  $10^9$  comb ( $X = 20.6875$ , 160 s) the 7/8 edge closes into an AVOIDED CROSSING (gap 0.0039 at  $M = 1240$ , tail 0.0074 vs bar 0.10) while the transported  $d = 7$  block stays Cauchy straight through; the 9th mode enters at  $M = 1276$ ; the entry cadence WIDENS ( $\Delta X = 1.625/1.8125/2.625$ , mean 2.02, spread 0.381 — not regular at the frozen 0.25 bar) — fixed- $d$  Feshbach is closed PERMANENTLY: convergence was never the problem, band ownership is. THE CLOSURE DECIDER: v773 (gates 2/8 with exactly the six convergence-cell fails, 14/14 guards+controls, cocycle-domain-only; one declared implementation correction: Hotelling refinement of the carried cocycle state after a numerically destroyed first execution, coefficients/algebra/gates untouched, the eigh-based layer reproduced identically) — STRUCTURE WITHOUT LIMIT: the exact Möbius/Redheffer cell identity holds at  $9.7 \times 10^{-12}$  over 73 rungs  $\times$  5  $z$  (coefficients = the added source rows/columns; independent dense-solve Ward  $1.3 \times 10^{-13}$ ), every cell Schur block is PD (min eig  $D_k \in [0.0605, 0.0668]$  — a monotone Loewner flow), and the Nevanlinna/Stieltjes domain is preserved with ZERO breaches in all 730 rung $\times$ cell tests INCLUDING both typed mode-entry transitions ( $M = 992$ ,  $M = 1108$ ; transition increments unremarkable) — but no renormalized

object converges at reachable depth (0/6 cells: the corner absorbs a flat  $\sim 0.1\%$  per rung, the band increments rise with the moving edge). THE CONTRACT MOVES: PRIME.GRAM.DIAGONAL.01 CLOSED per its own frozen stakes (dated closure note with the full decision cascade: Gates 1/2/4 positive from round LX, Gate 3 dead at the corner level via v769, Gate 5/v764 never executed, cocycle domain-only; survivors typed: the exact anti-alias theorem v760, the unconditional paired Abel bound v761, the canonical dense family v762, the Kato bundle frame and settled levels v770, the Herglotz-certified exact Feshbach reduction v772, the domain-preserving cell cocycle v773; NO new variants, the ten-item stop-list stays binding); PRIME.Z1.OPERATOR.01 stays OPEN [O] and is SHARPENED (dated note: a self-adjoint geometric bulk operator with a canonical finite-dimensional boundary/threshold structure whose Weyl  $M$ -function delivers the qf block and whose relative trace formula reproduces the full Weil formula; mandatory measured constraints: the Hecke/E8 bulk + archimedean lift, the moving-edge cadence (entries 888/992/1108/1276,  $\Delta X$  widening  $1.625 \rightarrow 2.625$ , avoided crossing 0.0039), the ram-odd contact direction ( $\cos = 0.997$ ), the settled coupling levels ( $R_2$  0.225–0.358,  $R_1$  0.008–0.079), the cell-ordered domain-preserving prime transport; additional kills: cadence/level contradictions = dead on arrival); the PRIME.KMS.INDUCTIVE\_STATE.02 compactness note carries the dated follow-up (the  $q_f$  pointer is adjudicated; handover to the sharpened Z1). Papers: `tfpt_prime_front` gains the closure section “The qf offensive: the route closes at a moving spectral edge, and what survives”; `tfpt_research_contracts` carries the closure cascade + the Z1 sharpening; `tfpt_3` gains the five audit blocks. Both declared implementation corrections are carried verbatim in the module PROVENANCE blocks. No marker moves on evidence lines; NO RH claim anywhere)

2026-08-05 (Two working notes join the published document set)

- **note\_e8\_gaussian\_code (working note N1) published.** The Gaussian code bridge:  $E_8$  over  $\mathbb{Z}[i]$  via Construction A over the extended Hamming code  $[8, 4, 4]$  placed  $\mu_4$ -equivariantly, the canonical four-bit quotient  $L/(1+i)L \cong \mathbb{F}_2^4$  (the 240 roots distribute exactly  $15 \times 16$ ; the code returns as the message space), and the  $G_{31}$  quartic companion — exact-arithmetic probes  $26/26 + 22/22$  (promoted as v689/v690) and 65 kernel-checked Lean 4 theorems. No claim beyond the stated algebra.
- **note\_hilbert\_polya\_truncations (working note N2) published.** A computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert–Pólya candidate (v714, v716–v721, v727–v734); SHA-256-frozen truncation spectra hit 100% of the first 377 zeta zeros at tolerance 0.25 with no prime table and no zero data in the construction path; the remaining statement (Weil positivity in the limit) localized in four machine-verified equivalent languages. Explicit no-proof fence: no theorem about  $\zeta$ , no RH claim.
- **Structure sync.** Both notes registered in `build.sh` / `make_docs_map` / `make_manifest` / the document-set box, mirrored to the website (`papers.ts/release.ts`, new “working note” class) and added to the Zenodo deposit (description + upload list). The German-language big-picture overview (`big_picture_2026-08-02_de`) stays repo-only: the published set and website are English by house rule and no supplement pattern exists.

2026-08-04 (LX) (the diagonal gram round: NINE promotions (v759–v763, v765–v768; no v764 — the named unexecuted gate; suite  $752 \rightarrow 761$  modules) — the frequency front closes exactly, the paired tail is unconditional, and the wall moves into  $q_f$ . THE OFFENSIVE FOUNDATION: v766 (17/17, handoff-bulk-converges)

— the  $\varepsilon$ -regulated resolvent  $G^\varepsilon = (A_X + \varepsilon I)^{-1}$  is the ADMISSIBLE operator-system evaluation; local Cauchy defects fall at  $b = 0.174\text{--}0.310$  per  $X$  unit,  $\varepsilon$ -robust over three decades; the bare  $A^{-1}$  does NOT stabilize (+45%, named negative block); the layer decomposition is a CANCELLATION BALANCE (leading channel up to  $1.9 \times 10^{12} \times$  the total defect); v767 (6/6, handoff-gram-converges) — the growing-frequency PSD source Gram: Candidate A missed honestly (slope +0.471 territory), the predeclared Candidate B (anti-alias  $N_f = 2M+1$ ) converges (errors  $0.0855 \rightarrow 0.0377$  over  $\dim Q$   $738 \rightarrow 5734$ , slope  $-0.369$ ; PSD by construction, min symbol  $+7.4 \times 10^{-3}$ ); v768 (merged from two probes, gates  $11/13 + 4/4$ , tail-weil-partial+compat-eps3-converges) — the quadrature tail TERMINATES ALGEBRAICALLY ( $2.4 \times 10^{-13}$  uniform: no quadrature tail exists, every layer error is structural), Weil identification on the frozen battery  $\kappa = 0.97768 \rightarrow 0.99028$  (agreement 0.0485), all layer slopes  $-0.365$  with atom+pole a  $10.5 \times$  cancellation pair, and the two frozen  $\varepsilon = 10^{-3}$  endpoint fails of part 1 are DECIDED POSITIVELY by the part-2 preregistration on the deepest 36-rung ladder ( $b = 0.178/0.186$ , med5  $0.343/0.467$ ,  $b_2 = 0.069/0.096$ ; first wall sign at  $\varepsilon = 3 \times 10^{-4}$ :  $b_2 = -0.011$ , reported never gated). THE SIX-STRAND DECISION LINE: v759 (8/8, resolution-boundary) —  $E_{\text{total}}$  is  $q$ -invariant under  $N_f = q(2M+1)$  to  $\sim 10^{-13}$  on all three fixed windows: the remaining handoff error is boundary/cutoff/object-gluing, NOT quadrature; v760 (14/14, antialias-corrected) — the anti-alias claim becomes an EXACT theorem of the  $M$ -cell space (Parseval  $2 \times 10^{-14}$ , moments  $3.5 \times 10^{-13}$ , Gram  $5.9 \times 10^{-13}$ ; exact alias formula matched at every count; must-fail witness error exactly 1 at  $2M-2$ ); the sharp minimal count is CORRECTED to  $N_{f,\min} = 2M-1$  (spread- $S$ :  $M+S$ ), the deployed  $2M+1$  stays exact, nothing patched; v761 (merged,  $24 + 10$  checks, combined abel-paired-bound) — the Stieltjes/Abel split is exact (6/6 sympy certificates), the matrix identity holds to  $1.45 \times 10^{-11}$ ,  $E_{\text{origin}}$  closes via the W1/Suzuki bookkeeping (comb-independent, verified on the controls), and the UNCONDITIONAL bound  $\eta_K = |\det_K| + |\Theta_K(0)E(1)| + B_1 I_K$  with  $B_1 = 3.734$  frozen on the finite range holds for all 24 kernels on all 5 rungs with pure  $1/\log X$  decay — NO RH-strength input; the companion DERIVES the correct breaker standard (symmetric zero-violation gate + envelope hypothesis): Epstein breaks at  $B_1^E = 2.95 \times (7/24)$ , scramble  $262 \times$ , truth passes  $24/24$ ; the parent's frozen abel-dead stays on record as the calibration lesson; honest typing: identity and slope-1 normalization are measure-generic, the certified  $\zeta$ -specificity is envelope quality + measure positivity; v762 (26/26, dense-core-canonical) — the canonical countable dense test family of the  $(D, X)$  tower: admissibility LOCATED per class (hats direct; the bare box fails the  $r^{-2}$  criterion and enters only correlated/tent-read — named caveat), density rates hat  $-1.566$  / box  $-0.503$  per level, exact  $\mathbb{Q}$  correlation closure on 8/8 type pairs, explicit stage map with  $I_{64}$  exact, and the 24 battery typed an initial diagnostic package (hat 8 IS family member hat(2,3), rank 35); v763 (merged, gates  $9/10 + 7/12$  preregistered, yosida-partial/qf-dead) — the bounded Yosida transform carries the handoff rates with NO  $1/\varepsilon$  bound and NO PD-margin assumption anywhere ( $|b_Y - b_G| \leq 10^{-13}$ , sandwich verified; the circularity kill did NOT fire), the near-kernel converges as an identified stable OBJECT, every fixed  $f$  has a monotone contraction-certified  $\varepsilon$ -limit — but  $q_f$  (the battery near-kernel pairing) is NON-CAUCHY at honest maximal depth ( $X = 15.1875$ , 279849 atoms,  $4 \times$  rung density) and SURVIVES orthogonalization by structural subordination (89–99.9% of every battery function lies in the 6-mode near-kernel span): Q3 closes NEGATIVELY — THE  $q_f$  WALL is the round's named wall object (density and near-kernel avoidance collide on the same function class); v764 NOT EXECUTED — its precondition (theorem-capable bounds



from v761 AND v763) failed via Gate 3; documented as the named open gate inside the new contract, no module fabricated; v765 (merged, 16 + 8 checks, redteam-green-combined) — firewall CLEAN (zero blacklist hits across all 21 source-path functions), baseline reproduced exactly, 8/8 controls break; RT6 decided: the symbol gate is the ONLY detector for excess-mode deck corruption (min symbol  $-0.45..-1.72$ , all other gates measurably blind), the deficit mode is caught by rate/ratio/ $\kappa$  — BINDING: certificates must always carry BOTH gate families (symbol/PSD and rate/ratio/identification). THE NEW CONTRACT: PRIME.GRAM.DIAGONAL.01 registered OPEN — PSD source matrices  $G_n$  (no target decomposition, no zeros, no positivity tests) with corner error  $\leq 2^{-n}$  against  $W$  on the canonical dense family  $\Rightarrow$  Weil positivity on the rational span by PSD limits + density; decision state recorded (Gates 1/2/4 positive, Gate 3 negative on the canonical class = the  $q_f$  wall, Gate 5 not executed, redteam green; kills that did NOT fire: no RH-strength input, no spectral gap; the 10-item stop-list binding). LEAN ROUND 7: GramCompactness.lean (8 theorems), AntiAliasExact.lean (19 general- $N$  theorems), DenseWeilCore.lean (42 theorems); lake build green (3402 jobs), manifest at 76 files. No marker moves; NO RH claim anywhere)

- **New modules (9).** v759\_handoff\_fixed\_window\_resolution.py (8), v760\_antialias\_exact.py (14), v761\_atom\_pole\_abel.py (24+10, one module from two probes per the v518/v668 precedent, the preregistered C.EPSTEIN miss encoded per the v757 pattern), v762\_dense\_weil\_core.py (26), v763\_yosida\_handoff.py (12+14 guards/controls, gates 9/10 + 7/12 preregistered), v765\_handoff\_redteam.py (16+8), v766\_handoff\_bulk.py (17), v767\_handoff\_frequency\_gram.py (6), v768\_handoff\_tail\_weil.py (12+8 guards/controls, gates 11/13 + 4/4) — all wired into run\_all.py (suite 752  $\rightarrow$  761, ALL CHECKS PASSED); discovery probes remain in experiments/tfpt-discovery/ with frozen results.
- **Lean.** TfptCarrier/GramCompactness.lean (8 theorems — the abstract finite-stage Gram compactness/diagonal-extraction closure for the contract's compactness step), TfptCarrier/AntiAliasExact.lean (19 general- $N$  theorems: discrete orthogonality, offset-invariant exact quadrature, Parseval, moment reconstruction, threshold witness, midpoint alias sign; axioms only propext/choice/Quot.sound), TfptCarrier/DenseWeilCore.lean (42 theorems: enumeration bijection, stage bounds, grid module closure; honest analytic remainder in the header); imports added to TfptCarrier.lean; lake build green (3402 jobs); lean\_manifest.sha256 at 76 files.
- **Contracts.** PRIME.GRAM.DIAGONAL.01 registered OPEN [O] in tfpt\_research\_contracts + ledger (target form, the round's measured decision state, the not-fired kill criteria, the 10-item stop-list, and the unexecuted Gate 5 = the planned v764 documented as the named open gate); PRIME.KMS.INDUCTIVE\_STATE.02 gets the dated note: the compactness step now has MEASURED handoff evidence (v766/v767/v768) and the abstract Lean closure (GramCompactness.lean); the surviving open object is the  $q_f$  wall of v763.
- **Ledger.** 9 new module rows (PRIME.HANDOFFRES.01, PRIME.ANTIALIAS.01, PRIME.ABELPAIR.01, PRIME.DENSECORE.01, PRIME.YOSIDAQF.01, PRIME.HANDOFFREDTEAM.01, PRIME.HANDOFFBULK.01, PRIME.HANDOFFGRAM.01, PRIME.HANDOFFTAIL.01) + the contract row + the dated KMS note; finite convergences typed measured, exact results typed [E]; the honest deaths/PARTIALs (tail-weil  $\varepsilon = 10^{-3}$  first pass, QF-DEAD, parent ABEL-DEAD) documented as such; NO marker moves.
- **Papers.** tfpt\_prime\_front: the new round section “The diagonal gram round: the frequency front closes exactly, the paired tail is unconditional, and the wall moves into  $q_f$ ” (all nine \veri citations); tfpt\_research\_contracts: the PRIME.GRAM.DIAGONAL.01 contract block + the KMS dated note; tfpt\_3: the round's audit block.

- **Website.** Prime-front feed: the round entry “the diagonal gram round”; counters  $752 \rightarrow 761$  modules / v535–v768; README and Zenodo updated to 761 modules / v768.
- **Scope.** The exact results (anti-alias identities, Abel identity + unconditional  $\eta_K$ , dense-core algebra, three Lean modules) sit behind explicit fences; the PARTIALs and deaths are preregistered criteria firing inside green machinery and stay on record; no RH claim, no marker moves; the  $\varepsilon \rightarrow 0$  wall (PD persistence) and the battery-limited identification remain open exactly as quantified.

**2026-08-04 (LIX)** (the incidence programme: SEVEN promotions (v752–v758, 134 new checks, all green; suite  $745 \rightarrow 752$  modules) — the second external review, fully adjudicated: five exact wins, one decorative seven, and the keystone negative that names the wall as the missing factor. THE CENTERPIECE: v752 (36/36, proj-hamming-exact) — the projective Hamming incidence lemma on  $V = L/(1+i)L$ :  $h = H/4$  is  $\mathbb{Z}[i]$ -valued with hermitian Gram determinant 1 (the UNIMODULAR hermitian E8 lattice over  $\mathbb{Z}[i]$ ),  $\bar{h}$  alternating and nondegenerate, so  $(V, \bar{h})$  is symplectic over  $\mathbb{F}_2$ ; on the 15 points  $BB^T = 4I + 3J$  EXACT with the design parameter set  $2/3/4/7/15/16$  and eigenvalues  $7^1(+2)^9(-2)^5$  (the multiplicities  $(1, 9, 5)$  are the symplectic signature);  $K^2 = (4/49)I + (45/49)\Pi_0$ ; NS/R is a POINT of the geometry ( $\chi_{\text{NSR}} = \bar{h}(\cdot, \text{FSUM})$ , the anchor bit;  $248 = 120 + 128$ ); THE v737 RESOLUTION: the seam-48 distribution  $0:96/1:64/2:64/3:16$  derived exactly ( $16 \times$  the  $3 \times 5$  label table) AND the  $5 \times 48$  coordinatization exists via exactly ONE  $\bar{u}$ -invariant spread (kill explained, not revised); two honest corrections to the review derivation documented; companion Lean module ProjectiveHamming.lean (32 theorems, kernel-checked, lake build green 3391 jobs). FOUR MORE WINS: v753 (25/25, polarity-match) — the 15 ramified Hecke hyperplanes ARE the  $\bar{h}$ -orthogonal spaces, canonical bijection  $y(\varphi) = \bar{G}^{-1}\varphi$ ,  $\sigma$ -functorial, deck and parity commute; canonicity census: exactly 1 of 28 nondegenerate alternating forms is  $\sigma$ -invariant  $= \bar{h}$ ; v754 (19/19, twostep-exact) — the real  $n \rightarrow 4n$  down-up pass IS exactly  $K^2$ , degree factor  $196 = (2 \cdot 7)^2$  closed, KMS half-weight exponent-exact, identical integer matrix on all 9 contexts,  $T = 4B^2$ , cone-preserving; honest boundary: naive label chains do not carry  $K^2$  (correspondence composition mandatory); v755 (13/13, schur-recursion-alive) —  $S_k \succeq 0$  on all 8 ram-odd stages of the tower nesting ( $\lambda_{\min}$  flat, long-double-hardened), Douglas trivial, controls break immediately; the Albert/Haynsworth induction frame stands; honestly: for  $\zeta$  data equivalent to tower PD — the remainder is the structural Gram certificate; v756 (7/7, kms-incidence-cp-covariant) — 105 explicit Kraus terms, all four compatibilities exact ( $\sigma$  at 0, half-weight  $10^{-16}$ ,  $\beta=1$  balance  $7 \times 10^{-18}$ , Gate-0) — step 3 of the master contract SOLVED on the ramified layer; local candidate state  $\tau_{\text{inc}} \otimes \tau_{\text{Toeplitz}}$ . THE DECORATIVE SEVEN: v757 (26 checks, one preregistered break, fano-disc7-decorative) — the preregistered four-condition Fano  $\rightarrow$  discriminant(−7) test breaks exactly at condition 2: 2 splits in  $\mathbb{Q}(\sqrt{-7})$  so the 2-rank is  $\leq 2 < 3$ , the code  $\mathbb{F}_2$ -structures are arithmetically unrealizable; the three-cycle half is real (Fano  $\bar{\sigma}$  = the  $\times 2$  multiplier, C corner = fixed point, norm  $14 = \dim G_2$ ); the seven-correlation is permanently typed decorative. THE KEYSTONE: v758 (8/8, cert-continuum-root-dead) — the review certificate [continuum root  $\oplus$  Kraus] is NOT mountable, the wall IS the missing left factor (arch+pol PSD at no stage, Levinson break exactly between the  $n=2/n=3$  cells ladder-wide, already  $W_0$  indefinite); real remain the [E] building blocks + the corrected target shape: cell-wise cascade factorization, rescue strictly cell-ordered. CONTRACTS: PRIME.PD.PERSISTENCE.01 gets the big dated update (induction

form registered / assembled certificate closed / cell-wise factorization the named open object); PRIME.KMS.INDUCTIVE\_STATE.02 carries the step-3-solved note. No marker moves; no RH statement anywhere)

- **New modules** (7). v752\_projective\_hamming\_incidence.py (36), v753\_ramified\_polarity.py (25), v754\_ramodd\_twostep.py (19), v755\_simpler\_schur\_recursion.py (13), v756\_kms\_incidence\_stinespring.py (7), v757\_fano\_disc7\_bridge.py (26, one preregistered break), v758\_simpler\_certificate.py (8). Suite 745 → 752 modules, ALL CHECKS PASSED; the one [FAIL] line in the green run (v757 C2.d) is exactly the preregistered honest break. All seven probes re-run before promotion (outputs identical modulo runtimes).
- **Lean.** TfptCarrier/ProjectiveHamming.lean committed (32 theorems, kernel decide/norm\_num only, no sorry/native\_decide; lake build green, 3391 jobs); import added to TfptCarrier.lean; lean\_manifest.sha256 regenerated (73 files).
- **Contracts.** PRIME.PD.PERSISTENCE.01: the big dated update — (i) the induction form registered (the Schur ladder on the canonical tower, v755; open need = the structural Gram certificate per step, measured boundary data recorded), (ii) the assembled certificate a CLOSED route (the v758 keystone negative: the wall is the missing left factor), (iii) the corrected target shape (cell-wise cascade factorization) the named open object. PRIME.KMS.INDUCTIVE\_STATE.02: dated note — step 3 (complete positivity) solved on the factorized ramified 15-label Toeplitz layer (v756); the remainder stays exactly the compactness step.
- **Ledger.** 7 new module rows (E8.PROJHAMMING.01, PRIME.HECKEPOLARITY.01, PRIME.HECKETWOSTEP.01, PRIME.SCHURREC.01, PRIME.KMSSTINESPRING.01, QGEO.FANODISC7.01, PRIME.CERTROOT.01) + dated notes on PRIME.PD.PERSISTENCE.01, PRIME.KMS.INDUCTIVE\_STATE.02 and E8.SEAM48.01 (the v737 resolution); 828 → 835 rows.
- **Papers.** tfpt\_prime\_front: the new section “The incidence programme: five wins, one decorative seven, and the wall as the missing factor” (the  $BB^T = 4I + 3J$  formula prominent with the 2/3/4/7/15/16 design set, the v737 resolution, the keystone negative). tfpt\_1/tfpt\_2: the symplectic code lemma added to the code sections. tfpt\_3: seven audit blocks.
- **Website.** Prime-front feed: the round entry “the incidence programme”; counters 745 → 752 modules / v535–v758; README and Zenodo updated to 752 modules / v758.
- **Scope.** The exact lemmas are finite census/incidence statements behind explicit fences; the two honest negatives (v757, v758) are preregistered criteria firing inside green machinery; excluded on purpose: big\_picture\_\*, note\_\*, articles/, the news-checker note and the other worker files; no marker moves; no RH statement anywhere.

**2026-08-04 (LVIII)** (the news round: a pure data-refresh against the August-2026 literature — no new analysis of our own, honest in both directions. **CONFIRMING: JUNO’s Neutrino-2026** update (207 days) lands at  $\sin^2 \theta_{12} = 0.3036 \pm 0.0064$  — the sharpest single measurement of the solar angle,  $+0.49\sigma$  from the frozen prediction 0.306747, central value moving *toward* it from the 59.1-day run; ICARUS’s first standalone result (no  $\nu_\mu$  disappearance, new 3+1 exclusions at 90% CL) adds another sterile null exactly as  $N_{\text{fam}} = 3$  requires; the sequentially-valid DESI reanalysis localises the evolving-dark-energy signal almost entirely in the LRG2 bin (without it the evidence collapses, mildly pro- $\Lambda$ CDM — the dissolution trend the  $w = -1$  watchdog predicts); two new independent birefringence analyses ( $\beta = 0.30^\circ \pm 0.05^\circ$  constant-model-preferred; PR4 field-level  $0.32^\circ \pm 0.12^\circ$ ) stay consistent with the  $0.2424^\circ$  seed; CMB-SPA  $H_0 = 67.19 \pm 0.38$  lands  $+0.11\sigma$  from the parameter-free budget 67.15 (correlated legs, never a new independent

hit). **WEAKENING**, equally prominent: CMB-SPA (SPT-3G D1+ACT+Planck) gives  $n_s = 0.9679 \pm 0.0033$  — the central value sits *above* the TFPT band edge 0.967 for the first time (edge at  $-0.27\sigma$ , preferred branch 0.9611 at  $-2.06\sigma$ ; not yet the dated kill “robustly  $\geq 0.967$ ”, typed an honest mild weakening, scorecard row tension + watch-flag); the DESI DR2 effective-mass posterior keeps the  $\Sigma m_\nu = 0.0588$  eV floor under  $+3.1\sigma$  model-dependent pressure (unchanged, on record). **ONE BRIDGE DIES**: the muon  $g-2$  seam-vertex bridge’s dated kill condition (2026-07-02) has FIRED — WP25 (Phys. Rep. 2025) adopts the lattice HVP as the SM baseline and the BMW hybrid evaluation (Nature 2026) confirms it, experiment–SM =  $0.5\sigma$ , so the converged  $\Delta a_\mu$  lies outside  $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ : the vertex identification is excluded in its present form, the scorecard row is typed kill\_channel, ledger FR.MUONG2.01 carries the dated kill note, and tfpt\_4\_frontier gets the dated update paragraph + summary-table move [C]→[X] on the bridge — the exact compiler number  $\delta_2/(2\pi)$  and the compiler core are untouched (no core marker moves): the watchdog delivered exactly what it was built for. **OPERATIONAL**: the Vela GO-programme preregistration is FROZEN as experiments/p9-forecast/hypotheses/p9\_go\_programmes\_v1.yaml ahead of the expected 2026/27 giant-glitch window (no glitch since 2024-04-29; long-component decision rule and the 8-TOAs/day cadence requirement inside; draft SHA recorded as provenance); CHIME/FRB Catalog-2 + RepCat4 data-availability notes recorded in repeater-cascade and frb-ontology. Scorecard 120 → 121 typed rows (consistent 49 → 47, null 27, parked 4, kill\_channel 1); predictions.ts mirrors the JUNO/ $n_s/g-2$  status; no verification module touched, no core marker moves)

- **Scorecard (regenerated)**.  $\theta_{12}$  row: JUNO 207-day value  $0.3036 \pm 0.0064$  ( $+0.49\sigma$ ; supersedes the 59.1-day 0.3092).  $n_s$  row: CMB-SPA  $0.9679 \pm 0.0033$  added, status CONSISTENT→TENSION + watch-flag (central value above the 0.967 band edge). Flat-budget row: CMB-SPA  $H_0$  leg  $+0.11\sigma$  (correlated). Sterile watchdog: ICARUS null leg.  $w$  watchdog: LRG2-localisation note. Birefringence row: two new supporting legs.  $g-2$ : WP2025-lattice row TENSION→KILL\_CHANNEL (kill condition fired), WP2020-dispersive row PARKED (baseline withdrawn by WP25).
- **Ledger**. FR.MUONG2.01: dated kill note “(6) KILL FIRED 2026-08-04” — bridge dissolved, exact value stays [E], core untouched. No other row touched.
- **Papers**. tfpt\_4\_frontier: dated update paragraph in the muon section + summary-table bridge move [C]→[X] (core untouched); no other paper touched.
- **Experiments**. Vela/Auger GO-prereg frozen (p9\_go\_programmes\_v1.yaml); CHIME Cat2/RepCat4 data notes; news-check note experiments/news\_check\_2026-08-04.md on record.
- **Website**. predictions.ts: JUNO 207-day confrontation,  $n_s$  watch-flag (+ CMB-SPA numbers), the dated news-round entry (verdict MIXED), audit counters 121 rows; papers.ts: muon section mirror updated (bridge dissolved).

**2026-08-04 (LVII)** (the full day wave: TWELVE promotions (v740–v751, 157 new checks, all green; suite 733 → 745 modules), the frozen external-clock preregistration, Lean round 6 (47 new theorems), and the retroactive record of the editorial commit. **THE KMS END-FORM**: v740 (8/8, kms-extension-dead) — the strong master contract form is algebraically dead twice:  $\beta = 1$  KMS is impossible on Gate-0’s nonzero-grade Laurent unitaries ( $T(u_p u_p^*) = 1$  vs  $T(u_p^* \sigma_i(u_p)) = 1/p$ , exact defects  $1/2, 2/3, 4/5$ ), and exact covariant \*-restrictions do not exist; local extended Gram positivity survives (55,296 monomials/window); v741 (9/9, kms-

toeplitz-dead) — the Bost–Connes semigroup Toeplitz correction repairs the algebraic KMS step COMPLETELY (boundary error  $10^{-2} \rightarrow 10^{-9}$ , directed system), but the covariant inductive step stays dead (UCP covariance nonmonotone, preregistered); the master contract is refined to PRIME.KMS.INDUCTIVE\_STATE.02 (operator-system compactness; keep the proven local blocks, ask only for a weak-\* cluster state). THE CHANNEL ANATOMY: v742 (14/14, interference-collective) — the W3 margin is ONE collective scalar balance (arch + ram-odd vs split + inert), strictly monotone over 35 windows, with TWO exact Ward identities ( $\sum c = 0$  to  $1.8 \times 10^{-15}$ ; rigid block =  $\dim \cdot \dim \cdot m_0$ ); no pair reduction; scramble  $\times 2 \times 10^3$ – $3.9 \times 10^6$ ; v743 (7/7, schur-cone-dead) — exact Moebius update ( $2.9 \times 10^{-41}$ ) + exact  $T^*JT$  identity, then the one-liner kill: cone-preserving iff layer-positive, and no prime layer is ( $c_0 = 0$ ) — layer-wise recursion is the wrong granularity level; v744 (15 checks, hecke-sos-channels-partial) — no canonical column choice factorizes (cos-half indefinite; PSD- $B^*B$  consumes 5–6 orders beyond the margin); the Ihara channel route fails coherently (channels grade positions, positivity lives in frequencies — structural, not  $\zeta$ -specific); finiteness measured:  $\dim 16$  vs 168–862 required frequencies. THE PROOF BLOCK: v745 (11/11, l1-sector-proven-modulo-elementary) — (L1) of the CAR majorant lemma proven modulo elementary steps for all 6 sectors (exact geometric sums instead of Euler–Maclaurin, leading term at  $6 \times 10^{-4}$ ,  $\gamma = 2$  derived, 39744+108 inequalities 0 violations,  $r_1 = 1$  sharp); GATE.QGEO does not move (L2 hard-technical, L3 medium remain). THE NET THEOREM: v746 (16/16, gnet-local-functor-alive) — net axioms exact (isotony/graded locality/clock covariance), Watatani index LOCALLY exactly 4 for every interval  $\ell \geq 2$ , Ramond = half-line/bond defect (solitonic; leakage  $\sqrt{2/N}$ ); 4 must-fails fire. THE ZUSE/WOLFRAM ADJUDICATION: v747 (24/24, mixed) — the tower as a confluent multiway system: diamond exact (CRT theorem), branchial = Grassmann  $[4, 2]_q$  at the prime places (the one new fingerprint),  $\Lambda$  recursion = causal path statistics; A2/A4 renamings, A1/A3 useful typings; v748 (15/15, equivalently-useful) — the  $\sigma$ -quotient is confluent everywhere (two-line lemma:  $\sigma$  is a multiway automorphism); only the half-orbit WEIGHTING breaks, exactly at the ramified place, same location as  $[D, A]$ , not the same object; rests as exploration + the one reference sentence in the introduction’s Zuse positioning. THE SIMPLIFICATION TRILOGY: v749 (8/8, tower-nested-reduces) — THE structural upgrade: the window quantifier reduces exactly to the nested  $(D, X)$  tower (X-nesting  $6.7 \times 10^{-18}$ , dyadic  $D$ -refinement exact — PD fine forces PD coarse; every deployed window a tower member at dev 0.0); the canonical truncation replaces the historical window family in PRIME.PD.PERSISTENCE.01; honest limits: incommensurable chains, falling margins; v750 (5/5, gluing-detector) — the gluing identity is an off-line detector at identity level (sees  $\delta = 0.01$ , sharper than the PD break  $\delta^* = 0.02$ ; envelope slope measures  $\delta$ ); the rigidity thesis honestly false (the identity holds unconditionally); v751 (15/15, ward-monotone-mixed) — running deterministic in direction (100%, parameter-free  $\zeta/L$  constants,  $\mp u/2$  drifts from  $p^2 \equiv 1 \pmod{4}$ , slope-Ward cancellation  $4 \times 10^{-4}$  — the margin inherits ZERO deterministic drift), but the balance sign is zero content (budget  $2.9 \times 10^4 \times$  margin) — the induction route is RH-circular. THE CLOCK PREREGISTRATION: experiments/ftransfer-clocks/committed (design YAML byte-frozen + contract note) and FTRANSFER.CLOCKS.01 registered in the contracts paper — four frozen external clocks (proper time / inverse temperature / RG time / e-folds), the one-class Riccati/Schwarzian prediction (the seam channels transport to  $-\Delta^2/2$  exactly), three kills (K1/K2/K3), honest

prior “expected: K1 kill”; executor PENDING (the FTC.01 data interfaces need curated  $g_{*s}$ /PDG overlay tables first). **LEAN ROUND 6: TraceLedger.lean** (28 theorems — the finite Gauss-quadrature trace formula as an exact ledger kernel, monic-Jacobi intertwining certificate, sharp must-fail at  $m_6$ ) + **PinningLemma.lean** (19 theorems — L-PIN-1 Wilkinson + L-PIN-2 Kato–Temple/Temple positivity in sqrt-free rational form); lake build green (3390 jobs), no sorry/native\_decide. **RETROACTIVE RECORD: commit 414471b2** (2026-08-04, between LVI and this round) was the pure editorial restructuring of **tfpt\_prime\_front** — no content changes, the  $\sim 248$ -page paper reorganized into parts with part intros; recorded here for the dated trail. **NO RH claim, NO marker moves**)

- **New modules (12).** v740\_kms\_extension\_switch.py (8), v741\_kms\_toeplitz\_semigroup.py (9), v742\_channel\_interference.py (14), v743\_schur\_cone\_recursion.py (7), v744\_hecke\_channel\_columns.py (15), v745\_qgeo\_car\_l1\_sector\_lemma.py (11), v746\_phys\_gnet\_local\_functor.py (16), v747\_automaton\_multiway\_tower.py (24), v748\_automaton\_sigma\_orbifold\_confluence.py (15), v749\_simpler\_tower.py (8), v750\_simpler\_gluing.py (5), v751\_ward\_monotone.py (15) — 157 checks, each promoted verbatim from its discovery probe with numbers frozen from the promotion run (all twelve probes re-run before promotion; outputs identical modulo runtimes; sibling imports resolve against the promoted v696/v716/v718/v735/v740/v747 modules). The three [FAIL] lines in the green run (v740 K1.ii/K1.iii, v741 T2.ii) are exactly the preregistered honest kills.
- **Lean.** TfptCarrier/TraceLedger.lean (28 theorems) + TfptCarrier/PinningLemma.lean (19 theorems) committed (kernel-checked, no sorry/native\_decide; lake build green, 3390 jobs); imports added to TfptCarrier.lean; lean\_manifest.sha256 regenerated.
- **Contracts.** PRIME.KMS.INDUCTIVE\_STATE.02 registered (operator-system compactness end-form; .01 annotated as settled at the finite level); PRIME.PD.PERSISTENCE.01 restated on the canonical nested tower (PRIME.TOWERNEST.01 paragraph); FTRANSFER.CLOCKS.01 registered (executor pending); dated GATE.QGEO note (the (L1) proof block).
- **Ledger.** 12 new module rows + 2 new contract rows (PRIME.KMS.INDUCTIVE\_STATE.02, FTRANSFER.CLOCKS.01) + dated notes on PRIME.KMS.INDUCTIVE\_STATE.01, PRIME.PD.PERSISTENCE.01, GATE.QGEO.01 and Lean notes on PRIME.MOONSHOT.05/PRIME.K1CAPTURE.01; 813  $\rightarrow$  827 rows.
- **Papers.** tfpt\_prime\_front: the new section “The simplification wave” (the tower upgrade, the two closed hopes, the KMS end-form, the channel anatomy). tfpt\_2: the  $G_{\text{net}}$  local-net paragraph (the algebraic-QFT counterpart of the OS package). tfpt\_3: 12 new audit blocks. introduction: the one Zuse-positioning reference sentence (the multiway reading measured out, equivalently useful).
- **Website.** Prime-front feed: the round entry “the full day wave”; counters 733  $\rightarrow$  745 modules / v535–v751; README and Zenodo updated to 745 modules / v751.
- **Scope.** Five honest kills promoted as encoded expected outcomes (preregistered criteria fire inside green machinery); the exact lemmas are finite/census statements behind explicit fences; the clock contract is a preregistered search-target design typed [X] with the honest prior “expected: K1 kill”; no marker moves; no RH statement anywhere.

**2026-08-04 (LVI)** (the morning after: **FOUR** review strikes (v736–v739, 104 new checks, all green) — two exact structural lemmas, two honest kills, and two new research contracts. **STRIKE 1, EXACT** (v736, 26/26, orbit-packet-exact): the Gaussian ORBIT PACKET lemma — Burnside  $7 = (15 + 3 + 3)/3$  on the 15 nonzero classes; the canonical packet split  $240 = 5 \times 48$  (one fixed block + four moved blocks

of 12 Gaussian lines each, label-free canonical; packets =  $1 + 4 = 5 = g_{\text{car}}$ ); the  $(1, 1, 2)$  anchor from the fixed-syndrome leader weights  $(0, 1, 1, 2)$  with leader pair  $e_6/e_7$  = the v638 anchor pair; the  $q = 16$  normal form  $240 = 16 \cdot 15$ ,  $248 = 240 + 8$ ; the CIRCULARITY FENCE is part of the lemma — P1/P2 come back as EXACT-INTERNAL-RECONSTRUCTION (reconstruction, NOT derivation), and the packet law is  $\sigma$ -CLASS-specific (160 free order-3 elements of G31 give Burnside 5); companion Lean module `OrbitPacket.lean` (34 theorems, kernel-checked, lake build green at 3388 jobs). STRIKE 2, DEAD (v737, 24/24 machinery, seam48-dead): the seam-48 intertwiner — the sharpest structural sharpening — dies on its preregistered criterion: the  $48 \times 5$  incidence matrix is NOT identically 1 (entry distribution 0:96/1:64/2:64/3:16), representative-independent (0 of 2304 order-5 elements) with the must-fail control firing (0 of 25 random operators); valuable residues: the four moved blocks carry the v623 character table EXACTLY, the fixed block deviates exactly by the 12 lifted marks,  $u^5 = \pm J$  exact —  $240 = 48 \cdot 5$  remains a factorization WITHOUT coordinatization. STRIKE 3, EXACT (v738, 34/34, hecke-mod-ramified-channels): the Gaussian-E8 Hecke tower projects onto  $V = L/(1+i)L$  exactly and  $\sigma$ -functorially over ALL 5907 submodules (no kill); odd layers act as degree  $\cdot \text{id}$  (the fiber is Hecke-rigid); the ramified layer IS the 15 hyperplanes with the 2:1 deck; channels = the 7  $\sigma$ -orbit channels +1 trivial (resolving 7-vs-8); NS/R = the  $\sigma$ -fixed parity character (v722 re-derived); ram-odd ( $n = 2, 8, 32, \dots$ ) is the unique negative-pressure channel; honest boundary: the W3 margin is CROSS-CHANNEL CANCELLATION — no transfer-matrix reduction from the projection alone. STRIKE 4, DEAD (v739, 20/20, rank3-jet-dead): the review's first-jet reading of the rank-3 wall is killed by its own  $\varepsilon^0$  test — the jet read itself is EXACT (the lock block as the first jet of ONE function to  $4.6 \times 10^{-16}$ , Loewner coordinates, odd-sector Schwarzian form  $[1 - r_{12}^2]_{\text{odd}} = (3/8) S[\tilde{g}](1) \varepsilon^4$ ), but the  $\varepsilon^0$  coefficient is identically 1 (not 0), the even sector exceeds the net entry ( $1.15\text{--}6\times$ ), and the pure jet block carries no collinearity (factor  $\geq 1.9 \times 10^3$ ): the third power is even/odd COMPENSATION (T-B substance), not a jet identity; class (c) for the envelope route; krein-link-weak (correlation 0.678 vs control 0.946). TWO NEW CONTRACTS: PRIME.KMS.INDUCTIVE\_STATE.01 (the review master contract — extend the v733 Gate-0 functor to a positive global KMS state at  $\beta = 1$  with  $T_h = \omega \circ \iota_h$ ; five separation steps individually named; kill = complete positivity failing already on a finite extension) and PRIME.PD.PERSISTENCE.01 (the universal-theorem form of the wall with the important correction  $\lambda_{\min} \geq 0$  instead of  $\geq \delta$  — the Ihara lesson: exact zero modes after rank saturation are admissible; target = cofinally infinitely many PD windows via state/Gram/cone-preserving recursion, NOT eigenvalue bounds; the review blocklist confirmed as permanent kills). NO RH claim, NO marker moves)

- **New modules (4).** `v736_orbit_packet.py` (26), `v737_seam48_intertwiner.py` (24 machinery), `v738_hecke_mod_ramified.py` (34), `v739_rank3_jet.py` (20) — 104 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (all four probes re-run before promotion; outputs identical modulo runtimes; sibling imports resolve against the promoted v563/v683/v696 modules).
- **Lean.** `TfptCarrier/OrbitPacket.lean` committed (34 theorems, kernel `decide/norm_num` only, no `sorry/native_decide`; lake build green, 3388 jobs); import added to `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with the new file.
- **Ledger.** 4 new module rows (`E8.ORBITLET.01`, `E8.SEAM48.01`, `PRIME.HECKEMODRAM.01`, `PRIME.RANK3JET.01`) + 2 new contract rows (`PRIME.KMS.INDUCTIVE_STATE.01`, `PRIME.PD.PERSISTENCE.01`) + dated notes on `E8.GAUSSIAN.CODE.01` (orbit packet



closed, seam-48 dead), PRIME.Z1.MOONSHOT.01 (the ramified fiber typed exactly) and PRIME.RANK3FUNC.01 (the jet simplification dead); 807 → 813 rows.

- **Papers.** `tfpt_prime_front`: the new section “The morning after: four review strikes” (the two exact structures, the two graves, the 7-channel structure, the documented exact jet/Loewner/Schwarzian forms, the two new contracts). `tfpt_research_contracts`: the two new contracts + the review blocklist confirmed as permanent kills. `tfpt_1/tfpt_2`: the orbit packet lemma in the code sections (the compact number-family table with the circularity fence; the seam-48 kill recorded). `tfpt_3`: 4 new audit blocks.
- **Website.** Prime-front feed: the round entry “the morning after: four review strikes”; counters 729 → 733 modules / v535–v739; README and Zenodo updated to 733 modules / v739.
- **Scope.** Two honest kills promoted as encoded expected outcomes (preregistered criteria fire inside green machinery); the exact lemmas are census/reconstruction statements behind explicit fences (no P1/P2 derivation claim); no marker moves; no RH statement anywhere.

2026-08-03 (LV) (the closing slice: ONE promotion (v735, 10 construction checks, green) — the L1 route portfolio completed honestly, plus one number-fidelity fix on the website. THE STATE-TRANSPORT ROUTE STAGNATES (v735, 10/10, ucp-stagnates): for each window the simple-spectrum GNS shift generates the commutative spectral algebra  $C^*(J_h, 1) = \mathbb{C}^h$ , so a UCP map is exactly a row-stochastic transport kernel and the Choi SDP reduces to a finite transport LP — all eight adjacent maps along the nine-window ladder exist CANONICALLY as monotone quantile couplings (unitality/state preservation to  $3.1 \times 10^{-12}/1.2 \times 10^{-15}$ , Choi minimum 0), every Choi rank inside the sharp transport bound  $h + h' - 1$  (ranks 344..2688, log-log slope 0.991 vs bar 1.25: LINEARLY controlled sparse-transport ranks) — but the preregistered covariance gate S2.3 FAILS HONESTLY (aggregate shift/grade/phase defects 0.797..1.227..0.058: slope  $-0.487$ , tail not monotonically decreasing): mere state transport is NOT the missing L1 mechanism; the optimal maps are quantile Markov/conditional-expectation couplings, NOT \*-homomorphisms (multiplicativity defects  $\sim 5-8 \times 10^{-3}$  quantify the operator-system-vs- $C^*$  obstruction); the full noncommutative groupoid algebra was not tested. THE PORTFOLIO, complete: Sonin transfer dead (v732), UCP stagnates (v735), Gate 0 weak functor (v733), Suzuki bridge a measurement (v734) — what remains for L1 is atomicity/identification substance, not transport machinery. QUALITY FIX (website only): the prime-front feed entry for v727 misquoted the Carleman evidence as “band 0.34–0.41 vs the 0.5 threshold”; corrected to the docstring values — the measured Gauss/Freud growth band  $[0.638, 1.117]$  makes the Carleman sum diverge (all repository surfaces — ledger, registry, run\_all, papers — already carried the correct band; the remaining v727–v734 feed numbers spot-checked against the docstrings, no further discrepancy). NO RH claim, NO marker moves)

- **New module (1).** `v735_strat3_ucp_inductive.py` (10 construction checks) — promoted verbatim from `strat3_ucp_inductive_probe.py` (probe re-run before promotion; output identical modulo runtime; the sibling import rewired to the promoted v718 module).
- **Ledger.** 1 new row (PRIME.UCPLIMIT.01) + the dated LV closing-slice note on PRIME.Z1.MOONSHOT.01 (the L1 route portfolio complete; stays OPEN); 806 → 807 rows.
- **Papers.** `tfpt_prime_front`: one closing-slice passage in the keystone section (the portfolio sentence); `tfpt_research_contracts`: PRIME.Z1.MOONSHOT.01 update V; `tfpt_3`: 1 new audit block.

- **Website.** Counters  $728 \rightarrow 729$  modules/v535–v735; the Carleman number-fidelity fix in the keystone feed entry; README + Zenodo 729/v735; generated surfaces regenerated.

2026-08-03 (LIV) (the keystone round: EIGHT promotions (v727–v734, 90 new checks, all green) — THE WALL IN FOUR EQUIVALENT LANGUAGES, the K1 diagram closed classically, and honest graves for two strategy routes. THE WALL, MAXIMALLY PRECISE (v727, 11/11, l1-determined-wall-typed): determinacy of the limit moment problem is CLASSICAL (the raw  $\tau$ -measure does not carry the limit; Gauss-damped Weil moments identify against the closed geometric side at  $1.4 \times 10^{-3}$  over  $m_0..m_{16}$ ; the measured Gauss/Freud band  $[0.638, 1.117]$  makes Carleman diverge) and the identification chain (a)–(d) contains RH NOWHERE (Weil 1952 unconditional); the seeming paradox resolves honestly — “ $T_h$  is PD” is a MEASUREMENT per  $h$ , not a theorem (scramble: Levinson death at depth 61); the one remaining statement is PD PERSISTENCE: infinitely many PD windows suffice for RH, each window a FINITE Levinson decision, measured 9/9 (min headroom 0.718); the converse is NOT shown; equivalently (Riesz–Haviland): the geometric side defines a positive measure on  $\mathbb{R}$ . THE FOURTH LANGUAGE (v734, 10/10, s1-canonical-suzuki-bridged): the Jacobi/Wheeler data define via an EXACT Krein correspondence (Weyl identity  $1.6 \times 10^{-15}$ , nilpotency  $3.1 \times 10^{-16}$ ) a canonical-system Hamiltonian per window with  $H_h \geq 0$  MEASURED on all nine (min  $l_k = 0.033..0.377$ ); the boundary phase carries the  $\gamma$ -side at scale EXACTLY 1; scramble breaks the Wheeler/Krein chain at depth 31; Suzuki (JFA 2021, GRH  $\equiv$  positivity of the canonical family, unconditional only  $\omega > 1$ ) becomes the fourth anchor — so the wall now reads: moments (Hankel-PD) / state (Levinson-PD) / symbol (Fejér  $\geq 0$ ) / Hamiltonian (Krein  $l_k > 0$ ), each finitely decidable per window, the  $h$ -quantifier IS the RH substance. THE K1 DIAGRAM CLOSES CLASSICALLY: the memo’s locality/symmetry mechanism for the  $\sim 900\times$  super-resolution is DEAD on two preregistered bars that die honestly (v728, 11/13, k1b-atom-pinning) — what carries the scale is ATOM PINNING of the raw measure (Nevai-type; lab rate  $K^{-1.84}$ , weight ratio  $\rightarrow 1.03$ , real correlate 0.64; off-line pairs break POSITIVITY instead of mislocating nodes); behind it the explicit PINNING LEMMA is certified (v730, 12/12: residual bound + Kato–Temple, one-line proofs, 0 violations at 598 peaks/535 matched zeros incl. ALL 377 of the largest window, SHA256-frozen; the certificate SATURATES at window width — sub-width precision needs atomicity beyond every window depth = L1, the sharpest localization); the gap layer: strong measured universality KILLED on the preregistered bar (v731, 5/5, gap-univ-killed: deviation plateaus 0.33; the uniform-in- $h$  Szegő condition breaks like  $h^{-0.64}$ ; a fixed-window CONDITIONAL candidate remains via Lubinsky 2009/Levin–Lubinsky 2008/Avila–Last–Simon 2010/Totik 2009) — so: capture theorem (v721) + pinning lemma (v730) + conditional gap rigidity (v731) + atom structure (= L1) = K1. TWO HONEST GRAVES: naive reflection positivity is DEAD on the truncations (v729, 5/5, rp-dead-on-truncations: the total window fails plain AND KMS-twisted RP at full scale; the FE symmetry buys exactly the atom-free Connes–Consani 2021 regime  $|u| < \log 2$  and no more; bulk universality EMERGING at gap points,  $0.304 \rightarrow 0.150$ ; the pinning lab reproduces the super-resolution scale, spacing/error up to  $211\times$ ); and the DIRECT transfer of the Connes–Consani Sonin compression is DEAD (v732, 21/21 construction checks, sonin-rank-grows: bad rank  $19 \rightarrow 86$ ; fenced — not the CC mechanism itself, retuning forbidden). PLUS THE FIRST EXACT BRIDGE INTO CONNES’ FRAMEWORK (v733, 13/13, gate0-weak-

**functor:** the  $\sigma$ -descended groupoid maps by an exact coefficient-one graded  $*$ -algebra functor candidate into the  $S$ -local periodic-orbit subalgebra,  $S = \{\infty, 2, 3, 5\}$ , 6859 convolutions at deviation 0,  $\kappa \equiv 1$  at all three places; full-algebra extension typed OPEN). THE LITERATURE ANCHOR, as the dual council reads it: the wall has three names — Weil 1952 (explicit-formula positivity), Connes 1999/Connes–Consani 2021 (trace formula/atom-free regime), Suzuki 2021 (canonical-system positivity) — with Meyer 2005 the proven boundary line (unconditional spectral realization WITHOUT positivity); TFPT adds the computable zeta-free truncation framework with emergent prime places and per-window decidability. NO RH claim, NO marker moves)

- **New modules (8).** v727\_l1\_identification.py (11), v728\_k1b\_superresolution.py (13; the two preregistered memo-bar deaths B1.1/B2.2 encoded as the expected promoted outcome), v729\_strat2\_rp\_universality.py (5), v730\_strat2\_pinning\_lemma.py (12), v731\_strat2\_gap\_universality.py (5), v732\_strat3\_sonin\_prolate.py (21), v733\_strat3\_gate0\_census.py (13), v734\_s1\_canonical.py (10) — 90 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (all eight probes re-run before promotion; outputs identical modulo runtimes; sibling imports rewired to the promoted v696/v714/v716–v721 modules).
- **Ledger.** 8 new rows (PRIME.L1IDENT.01, PRIME.K1BPIN.01, PRIME.RPUNIV.01, PRIME.PINLEMMA.01, PRIME.GAPUNIV.01, PRIME.SONIN.01, PRIME.GATE0.01, PRIME.S1CANON.01) + the dated LIV keystone note on PRIME.Z1.MOONSHOT.01 (L2 resolved classically as atom pinning + pinning lemma; L1 typed in four equivalent languages; the two strategy kills and the weak functor recorded; the literature anchor; stays OPEN); 798  $\rightarrow$  806 rows.
- **Papers.** tfpt\_prime\_front: the new section “The keystone round: the wall in four equivalent languages” (the maximally precise wall statement, the four languages, the literature-anchor paragraph with Weil 1952/Connes 1999/Meyer 2005/Connes–Consani 2021/Suzuki 2021, the closed K1 diagram, the two graves, the weak functor); the two stale “open K1b question” spots in the moonshot-arc section now point forward. tfpt\_research\_contracts: PRIME.Z1.MOONSHOT.01 update IV — the contract at its final state of the day. tfpt\_3: 8 new audit blocks.
- **Website.** Prime-front feed: the round entry “the wall in four languages” (with the explicit no-proof fence); counters 720  $\rightarrow$  728 modules / v535–v734; README and Zenodo updated to 728 modules / v734.
- **Scope.** No marker moves anywhere: PD persistence remains a MEASUREMENT per window (9/9 is not a universal quantifier); the two kills are honest deaths of strategy routes, cleanly fenced (the CC mechanism itself untouched); the weak functor is a finite-basis statement, the full-algebra extension open; no RH statement anywhere.

**2026-08-03 (LIII)** (the evening bundling: ELEVEN promotions (v716–v726, 211 new checks, all green) — THE MOONSHOT ARC COMPLETES AT MEASUREMENT LEVEL (“the Hilbert–Pólya candidate, measured” — with the explicit no-proof fence), and the three physics interfaces move: the  $v_{\text{geo}}$  torsor closes in calibration form,  $G_{\text{net}}$  gets its first exact witnesses, and both thermal-time routes for  $F_{\text{transfer}}$  die. THE MOONSHOT (v716–v721): stage 2 GLUES (v716, 18/18, moonshot-stage2-glued: one object, ONE normalization — the free 3-scalar LSQ returns  $\kappa = (1, 1, 1)$  to  $< 10^{-12}$  on all five windows; the  $1/4$  in  $\text{Re} \psi(1/4 + i\tau/2)$  DERIVED as the  $\mu_4$  fixed-sector offset;  $\log \pi$  only through the declared self-dual UV cell; E8 forced AT THE GLUING —  $\mu_2$  tower 0.326 / wrong  $\mu_4$  class 1.000 / wrong twists 0.474 all rip); stage 3 is a STATE (v717, 21/21,

state-on-truncations: constructive GNS vector state on every truncation via Levinson PD; the naive tower reading misses by  $|\Delta|/W3\text{-margin} = 104\text{--}1563$  — the trace-formula boundary IS the localized RH substance; KMS  $\beta = 1$  detailed balance measured; Ihara an exact state, Epstein breaks); stage 4 CONVERGES MEASURED (v718, 10/10: 100% of 377 frozen zero targets at tol 0.25 at  $h = 1433$ , rate  $-1.61$ , monotone GNS  $K$ -ladder  $0.4678 \rightarrow 0.0003$ , SHA256 prediction fire-wall, GUE light  $\langle r \rangle = 0.6178$  vs  $0.6189$ , scramble dies as a state at depth 61); the Satz-1 slice is EXACT (v719, 24/24, satz1-ledger-exact: the finite trace formula is exact Gauss quadrature at  $1.7 \times 10^{-11}$  and the term dictionary to the classical Weil formula CLOSES — pole  $1.7 \times 10^{-13}$  / atoms  $1.8 \times 10^{-15}$  / arch  $3.4 \times 10^{-16}$ ;  $\Delta \rightarrow \Gamma'/\Gamma$  term + prime term classically,  $\sigma = 1.000161$ ; E8 census over  $\{\mathbb{Z}^8, E8\}$  (Mordell 1938): ONLY the Gaussian E8 glues); the K collapse (v720, 14/14, k2k3-collapsed-classical:  $K3 \rightarrow$  measured tightness (sigma-finite, vague topology forced),  $K2 \rightarrow$  classical analysis on  $s > 1/2$  (unconditional Chebyshev majorant, exact slowdown law), the boundary EXACTLY  $s = 1/2$  — Mertens slope 1.9994); and the K1 capture lemma (v721, 14/14, k1-lemma-capture-only: capture proof-near from Markov–Stieltjes + symmetrized Christoffel + a closed capture condition; the certified radii sit  $\sim 900\times$  above the measured errors — the super-resolution is trace-formula content, the open K1b question). NET: everything a Hilbert–Pólya candidate needs exists at MEASUREMENT level; the remaining hard substance is (L1) identification at the line  $s = 1/2$  (no mass loss) + (L2) node convergence as a theorem; the contracts stay OPEN; NO RH claim. THE PHYSICS INTERFACES:  $G_{\text{net}}$  (v722, 9/9, ramified-sees-nsr: the NS/R grading IS the parity character of  $E8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$  — the Ramond projection is  $(1+i)$ -ADIC at the one ramified edge of the Hecke tower;  $15 \times 16$  classes parity-PURE, both controls fire; and v726, 57/57, t2-slice-go: optimal Pimsner–Popa constant EXACTLY  $1/4$  and Watatani index EXACTLY 4 on the CAR ladder via an explicit quasi-basis with FORCED weights,  $\mu_4$  DERIVED from the clock, the Ramond sector healed STATE-PRESERVINGLY by the Klein dressing — GNET.RAMIFIED.01 registered with kills; GATE.METRIC.08/10 typing unchanged);  $F_{\text{transfer}}$  (v723, 11/11, stt-killed + internal-pair-confirmed: the QCD 1-loop clock is PARABOLIC and not diagonalizable — no constant-rescaled modular flow reaches it; the pole multiplier exactly  $(2/3)^6$ ; and v724, 12/12, t3b-dead: conditioned modular flows of one internal mother state come out loxodromic/complex, the frozen solver classes do NOT arise — BOTH thermal-time routes closed, the external clock contract CONFIRMED and sharpened to  $\{\text{one unit}, \theta_i, C_p\}$ );  $v_{\text{geo}}$  (v725, 21/21, torsor closure pass: the interface is structurally CLOSED as an  $\mathbb{R}_+$  scale torsor — complete export table, dimension-matrix rank 1, all conditions  $\lambda$ -homogeneous, two-anchor consistency 0.11%, the calibration theorem stated; FENCE: no scale derivation (No-Unit); the one remaining dimensionless ratio NAMED:  $H_{\text{EW}} = \ln(\bar{M}_{\text{Pl}}/v_{\text{EW}}) = 37.1776$ ). PLUS LEAN ROUND 5: SineGramKeystone.lean + WatataniIndexFour.lean committed (lake build green, 3387 jobs, kernel-checked, no sorry/native\_decide) — the v701 sine-Gram keystone kernel and the  $\mu_4$  index-4 certificate in exact arithmetic)

- New modules (11). v716\_moonshot\_arch\_glue.py (18), v717\_moonshot\_state.py (21), v718\_moonshot\_spectral.py (10), v719\_moonshot\_traceformula.py (24), v720\_moonshot\_k2k3.py (14), v721\_k1\_node\_capture.py (14), v722\_phys\_ramified\_ns\_r.py (9), v723\_phys\_modular\_clock.py (11), v724\_phys\_t3b\_modular\_flows.py (12), v725\_phys\_vgeo\_torsor\_audit.py (21), v726\_phys\_car\_pp\_index.py (57) — 211 checks, each module promoted verbatim from its

discovery probe with numbers frozen from the promotion run (all eleven probes re-run before promotion; outputs byte-identical modulo runtimes).

- **Ledger.** 11 new rows (PRIME.MOONSHOT.02–.06, PRIME.K1CAPTURE.01, GNET.RAMIFIED.01, FTRANSFER.STT.01, FTRANSFER.COCYCLE.01, VGeo.TORSOR.01, GNET.PPINDEX.01) + dated LIII notes on PRIME.Z1.MOONSHOT.01 (the big update: stages 2–4, Satz-1 slice and K collapse done; kill audit K1/K3/K4 do not fire, K2 half-standing; remaining substance (L1)+(L2)+(L3); stays OPEN), ANCHOR.VGeo.02 and METROLOGY.CLOSURE.01 (torsor closure in calibration form; typing unchanged), FIRSTPRINCIPLES.BOUNDARY.01 (both thermal-time kills; anchor census  $\{\text{unit}, \theta_i, C_p\}$ ), GATE.METRIC.08 (index measured on the ladder; the identification half registered under GNET.RAMIFIED.01) and GATE.METRIC.10 (the NS/R grading’s arithmetic home); 787 → 798 rows.
- **Papers.** `tfpt_prime_front`: the new section “The moonshot arc: the Hilbert–Pólya candidate, measured” (the four-stage narrative + the end ledger, no-RH fence stated twice) and the stage-2 fence in the L1/moonshot section updated. `tfpt_research_contracts`: PRIME.Z1.MOONSHOT.01 rewritten to the post-run state (stages done, kill audit, remainders), PRIME.Z1.OPERATOR.01 update III, the GATE.METRIC.08/10 follow-ups (v726/v722), the No-Unit keybox torsor paragraph (v725), and the F\_transfer thermal-time kill paragraph (v723/v724). `tfpt_3`: 11 new audit blocks + the stage-1/montage fences updated. `tfpt_1/tfpt_2`: the code-section follow-ups extended (E8 forced at the gluing; census) and the  $G_{\text{net}}/v_{\text{geo}}/F_{\text{transfer}}$  interface pointers added. `tfpt_4_frontier`: the first-principles boundary follow-up (clock kills + torsor).
- **Website.** Prime-front feed: the round entry + a separate moonshot highlight (“the Hilbert–Pólya candidate, measured” with the explicit no-proof fence); counters 709 → 720 modules / v535–v726; the open-gates, status and glossary surfaces absorb the torsor/ $G_{\text{net}}$ /thermal-time state; README and Zenodo updated to 720 modules / v726.
- **Lean round 5.** `SineGramKeystone.lean` + `WatataniIndexFour.lean` committed (lake build green, 3387 jobs, kernel-checked, no sorry/native\_decide): the finite algebraic kernel of the v701 keystone (the product-to-sum identity  $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$  over an arbitrary commutative ring via Chebyshev recursion — the half-integer sine-moment form IS a Gram matrix,  $S = \sum_i 2w_i s_i s_i^T$ , hence PSD for nonnegative weights, with explicit Pythagorean-triple certificates; the FFT limit and the deployed window wiring honestly NOT formalized) and the  $\mu_4$  index-4 certificate of the v726 slice in its smallest faithful matrix model ( $M_4(\mathbb{Z}[i])$ ): the clock-order shadow  $S^{12} = -1 \Rightarrow H^4 = 1$  forced by the NS spin structure, the explicit 13-element quasi-basis with  $\sum vv^* = 4 \cdot 1$  EXACTLY  $= N(1+i)^2$ , the Pimsner–Popa constant  $\lambda^* = 1/4$  sharp via an SOS certificate at the mixing minimizer,  $\mu_2$ /three-sector controls give index 2/3; the CAR/Fock layer and the Ramond dressing honestly NOT formalized); `TfptCarrier.lean` imports added; `lean_manifest.sha256` regenerated with the new files.
- **Exploration (firewalled).** `experiments/p9-forecast/` committed as a standalone preregistered forecast workspace (FRB/GW/PSR/UHECR detector kernels + hypothesis YAMLS; no ledger row, no paper claim, no website surface — experiments firewall). The K1b super-resolution probe and the L1 determinization probe are NOT in this round (workers still running; next round).
- **Scope.** No marker moves anywhere: the moonshot results are MEASUREMENTS (diagnostic SHA256 typing; the operator family stays zeta-free); PRIME.Z1.OPERATOR.01 and PRIME.Z1.MOONSHOT.01 stay OPEN; the  $v_{\text{geo}}$  closure is the calibration form, not a scale derivation (No-Unit stands); GATE.METRIC.08/10 typing unchanged (the Q-system identification is the registered open half); the external clock contract of F\_transfer is CONFIRMED, not internalized; no RH statement anywhere.

**2026-08-03 (LII) (the third daily bundling: FIFTEEN promotions (v701–v715,**

158 new checks, all green) — the T-B census reaches 69/70, the Z1 continuum candidate is ASSEMBLED, and the moonshot’s atom layer is groupoid-internal. **HIGHLIGHT 1** (v712, 5/5 — the P4 strike): the v692 margin identity SCALARIZED along the transverse direction (the penalty pays only the transverse zero mass; median gain  $\times 4.23$  on the open ten) closes nine of the ten remaining windows: 69/70 COMPLETE FAMILY WINDOWS CLOSE at the cited  $3 \times 10^{12}$  (unconditional-modulo-citations); the single remainder is  $h = 5690$  with  $T^* = 1.6 \times 10^{14}$  (down from  $8.5 \times 10^{14}$ ); the deck lever honestly dead (triangle inequality). **HIGHLIGHT 2** (v713+v714): the L1 MONTAGE (v713, 16/16, 11-assembled-measured) assembles the Z1 continuum candidate explicitly — cover limit  $6.2 \times 10^{-8}$ , point masses at the  $D^2$  rate, one UV slot, the 5b negative exposed as a normalization artifact, divergence exactly at  $s \leq 1/2$  (the critical line as the boundary of the construction, a consistency signature, not positivity), the missing theorems NAMED; and the MOONSHOT stage 1 (v714, 19/19, moonshot-go) shows the atom layer is GROUPOID-INTERNAL from the  $\mathbb{Z}[i]$ -E8 Hecke tower — primitive degrees = Gaussian prime norms exactly (no prime projector anywhere), de-divisorization at  $4 \times 10^{-13}$ ,  $\sigma$  descent 100 positions/100 masses/368 moments at 0.00; honestly fenced: the finite counting facts are E8-UNSPECIFIC, specificity is stage 2 of the newly registered contract PRIME.Z1.MOONSHOT.01. **HIGHLIGHT 3** (the chain arc S-A-S-G, v702–v711): the lookahead question DECIDED (v702, 13/13, flow-verifier-not-generator: full comb verified  $913 \geq 898$ , autonomous reach 2–4 slots,  $g \sim 5$ –12 bits/slot information-theoretic); tolerance induction DEAD at RH grade (v703, 10/10); G1 open as a closed form (v704, 5/5: best law C2 median 0.9947, max 0.3130); the arch IS a deck-sector trace (v705, 7/7: exact tower traces  $3.4 \times 10^{-16}$ , scalar forced 1, must-fail  $\times 82$  — the geometry anchor); WEYL-NUL with the Schur-parameter identity ( $10^{-40}$ ) and the log 2 find (v706, 10/10); the section edge is an exact resolvent identity ( $3 \times 10^{-13}$ ) with the corridor position  $\sim 0.53$  open (v707, 9/9); the best position functional transports scalar-free (Levinson energy extremum, median 0.9986; outliers exactly on 16/64/81; v708, 7/7); the residual carries NO zero signature behind a temporal SHA256 firewall (v709, 10/10, layering-rejected); the two-stage law dies by its own bar but its Hecke stage heals the prime powers 11.7%  $\rightarrow$  0.53% circle-free from E8 counting (v710, 7/7); the first-birth error CONVERGES (promotion run  $\theta = 1.851$ ,  $R^2 = 0.692$ , bars 0.5/0.6; v711, 7/7) — net:  $G1 = [\text{E-edge}] + [\text{E-Hecke}] + [\text{M-rate}]$ . **THE REST:** the keystone identity (v701, 11/11:  $B = \text{sine-moment form}(c + \text{pole}) + \text{rank-1 pole square exact}$ ; the three gaps collapse onto ONE; triplication  $\mu_3$ -unique; V5 numerology dead) and the QGEO CAR route (v715, 22/22, qgeo-car-rates-summable: all 6+1 channels Schatten-Cauchy with summable rates, FH renormalization lattice-derived via Barnes  $G$  at  $8.8 \times 10^{-7}$ , RP cone stable; the majorant lemma (L1)–(L3) named — GATE.QGEO does not move). Between the bundlings, two website-side rounds landed separately: the visualization round (commit 52bcf00c: four animated schematics of the August-3 offensives on /prime-front, promoted and sandbox clearly separated, no new claims) and the prime-front explainer film (commit 76b092a0: a 2:39 German faceless-explainer video with burned-in captions, embedded on /prime-front; states “kein RH-Beweis” explicitly))

- New modules (15). v701\_big\_picture\_hunt.py (11), v702\_z1\_lookahead.py (13), v703\_chain\_tolerance\_scaling.py (10), v704\_chain\_mass\_law.py (5), v705\_chain\_deck\_sector.py (7), v706\_chain\_weyl\_mass.py (10), v707\_chain\_section\_edge.py (9), v708\_chain\_position\_functional.py (7), v709\_chain\_zero\_layer.py (10), v710\_chain\_two\_stage\_hecke.py (7),

v711\_chain\_firstbirth\_scaling.py (7), v712\_rank3\_transverse\_deck.py (5), v713\_l1\_montage.py (16), v714\_moonshot\_hecke\_groupoid.py (19), v715\_qgeo\_car\_continuum.py (22) — 158 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (v711: the worker-run point estimate  $\theta \sim 1.25/R^2 \sim 0.90$  vs the promotion run 1.851/0.692 documented; both far inside the preregistered bars).

- **Ledger.** 16 new rows (PRIME.KEYSTONE.01, PRIME.Z1LOOKAHEAD.01, PRIME.CHAINTOL.01, PRIME.CHAINMASS.01, QGEO.DECKSECTOR.01, PRIME.WEYLMASS.01, PRIME.SECTIONEDGE.01, PRIME.POSFUNC.01, PRIME.ZEROLAYER.01, PRIME.TWOSTAGE.01, PRIME.FIRSTBIRTH.01, PRIME.RANK3TRANSV.01, PRIME.L1MONTAGE.01, PRIME.MOONSHOT.01, QGEO.CARLIMIT.01, plus the new contract PRIME.Z1.MOONSHOT.01); dated LII notes on PRIME.Z1.OPERATOR.01 (montage + moonshot; stays OPEN — the continuum theorems are missing), PRIME.RANK3DENSITY.01 (69/70,  $T^* = 1.6 \times 10^{14}$ ) and GATE.QGEO.01 (rates summable, lemma named — the gate does not move); 772  $\rightarrow$  788 rows.
- **Papers.** tfpt\_3: 15 new audit blocks. tfpt\_research\_contracts: the PRIME.Z1.OPERATOR.01 update II (montage, moonshot, chain arc; formally OPEN), the new contract PRIME.Z1.MOONSHOT.01 (groupoid program; stage-2 gluing as the next gate; kill criteria K1–K4), the prob:R1 update II (69/70), and the GATE.QGEO note. tfpt\_prime\_front: three new subsections (the chain arc S–A–S–G as an honest section narrative; the L1 montage and the moonshot; the T–B census at 69/70). parity\_toeplitz\_classification: the transverse-strike paragraph (69/70). tfpt\_1/tfpt\_2: the Hecke-tower follow-up in the code sections (E8-unspecificity fence stated).
- **Website.** Prime-front feed entry (the third daily bundling); suite counters 694  $\rightarrow$  709; README and Zenodo description updated to 709 modules / v535–v715. (The two separate website rounds referenced above: 52bcf00c visualization, 76b092a0 explainer film.)
- **Scope.** No marker moves; the 69/70 closure is unconditional-MODULO-CITATIONS on the declared finite family (T–B itself stays the conjecture); PRIME.Z1.OPERATOR.01 stays OPEN (continuum theorems missing); v714 is honestly fenced (E8-unspecific finite counting facts — specificity is stage 2); GATE.QGEO does not move; no RH statement anywhere.

**2026-08-03 (LI)** (the afternoon bundling: **NINE** promotions (v692–v700, 140 new checks, all green) plus the **TWO** Lean modules committed — the T–B chain closes 60/70 windows and the Z1 series records its ground. **HIGHLIGHT 1** (the T–B chain, v692+v693): T–B(block) is **TYPED** (v692, 7/7:  $\hat{A}_2 =$  positive zero-Gram + psd rank-1 pole layer, exact per entry; the margin identity  $\det(G_Z + P) = \det G_Z + c_P(s_\perp^\top G_Z s_\perp)$  — the razor-thin margin **IS** the transverse zero mass, a **SUM OF SQUARES**; on-line zeros only help by  $2 \times 2$  psd superadditivity; “*E kippt nicht*” = a **QUANTIFIED PARTIAL-RH TAIL** statement, neither a 2c-envelope question nor a W3 renaming) and then **LARGELY CLOSED** (v693, 5/5: the explicit Ingham-form zero density arXiv:2507.15184 + Hasanalizade–Shen–Wong + halving + Platt–Trudgian built into a sharpened safe-side envelope — sinh difference pairing,  $V(0) = 0$ , on-line subtraction, PEN down  $\times 6.5$ –14.4: 60/70 complete windows close **UNCONDITIONALLY-MODULO-CITATIONS**; exact remainder  $9 \times T^* \approx 1$ – $3 \times 10^{13} + h = 5690$  at  $8.5 \times 10^{14}$ ; extension grows  $\sim e^{3.74a}$ ; Simonič honestly listed trivial-in-range). **HIGHLIGHT 2** (the Z1 series 5–5d, v695–v698, contract stays OPEN): the **MEASURE** comes from counting (v695, 25/25: seam+orbifold dead null-calibrated; the E8 counting route delivers the Weil measure **EXACTLY** — atoms 0.0, arch/pole via  $\Gamma$  duplication, 7/8 target spikes  $q = 0.000$ ; comb+pole positive-feasible, signedness = pole subtraction); the **CANONICAL OPERATOR** exists (v696, 18/18, z1-jacobi-opaque with the



renaming clause **REFUTED**: CMV/Jacobi moments  $2 \times 10^{-13}$ ; features on prime-power slots  $q = 0.000$ , amplitude–mass link  $r = +0.76$ , positivity load-bearing 170/171 vs 2865); the **MASSES** transfer exactly (v697, 14/14, z1-uvarov-sequential-closed: atoms are lag insertions, duality proven;  $\Delta\alpha = w_1/E$  at  $5.6 \times 10^{-17}$ ; the **INVERTED** stabilization law — the  $\Gamma$  flow predicts the counting masses to  $\sim 10\%$ ); the **POSITIONS** are forced (v698, 15/15, z1-recursion-semi: windows 0.5–2 cells, jitter null 30/30, adversary  $-516$ ; shooting recovers masses to 0.11% median; honest negatives: greedy saturation needs lookahead, residual noise-like, E-transport recursive — the continuum reading is the remainder). **THE REST**: v694 (20/20, closure-both-near-proof: retention closed with an  $O(k^2)$  certificate + straddle boundary case, separation law **FORM-CORRECTED** additive  $\alpha^* = C_{\text{cell}}/\delta + C''/\Delta\gamma$ ,  $C'' \leq 0.59$ , 0 exceptions in 97 tests; the two open parent configs detect at  $\alpha^* = 7.90$ ), v699 (18/18, cone-dynamics-dead: prime updates are translations,  $\sim 99\%$  increments spacelike, no semigroup — the review-7.3 dynamic-cone route killed honestly; the leaf functional separates scramble/Epstein where det is blind), v700 (18/18, orbit60-parked: all cascade routes dead, no involution of type  $2^{27}1^6$  in the line group of order 11520, Coxeter does not descend, no  $A_5$  signature). **PLUS LEAN ROUND 4**: GaussianCodeBridge.lean + QuarticHalf.lean committed (lake build green, 3385 jobs, kernel-checked, no sorry/native\_decide): the Gaussian-code-bridge algebraic core (SNF certificate,  $15 \times 16$  census with the zero class provably empty,  $\sigma$  semantics) and the quartic-half mechanism ( $\mu_4$ -orbit vanishing,  $\sum \langle \alpha, x \rangle^4 = 576q^2$  over the explicit 240 roots); Chevalley stays cited-not-formalized)

- **New modules (9)**. v692\_rank3\_lockgram.py (7), v693\_rank3\_density\_close.py (5), v694\_interpolation\_lemma\_closure.py (20), v695\_z1\_trace\_operator.py (25), v696\_z1\_jacobi.py (18), v697\_z1\_uvarov.py (14), v698\_z1\_flow\_recursion.py (15), v699\_cone\_dynamics.py (18), v700\_orbit60\_census.py (18). All nine discovery probes committed alongside; the zero-comb caches reused repo-relative with local fallbacks.
- **Lean round 4**. Two new committed proof modules (imports added to TfptCarrier.lean; lake build green, 3385 jobs, no sorry, no native\_decide; lean\_manifest.sha256 regenerated WITH the new files): GaussianCodeBridge.lean (the v689 algebraic core:  $\mathbb{Z}[i]$ -module and SNF certificate with unimodular  $P, Q$  in both directions, the 240-root census  $15 \times 16$  with the zero class provably empty,  $\mu_4$  stability,  $\sigma$  as the order-3 family action with coordinate block  $= F_1 + F_2 + F_3$ ) and QuarticHalf.lean (the v690 provable part: the  $\mu_4$ -orbit factor  $1 + (-i)^d + (-1)^d + i^d = 0 \iff d \not\equiv 0 \pmod{4}$ , the per-pair weight law, the full vanishing half incl. the trivial extras, and the quartic power sum  $\sum_{\alpha} \langle \alpha, x \rangle^4 = 576 (\sum_i x_i^2)^2$  over the explicit 240 roots with holomorphic isotropy; the Chevalley basic-invariant step cited, not formalized).
- **Ledger**. 9 new rows (PRIME.RANK3LOCKGRAM.01, PRIME.RANK3DENSITY.01, PRIME.INTERPCLOSURE.01, PRIME.Z1MEASURE.01, PRIME.Z1JACOBI.01, PRIME.Z1UVAROV.01, PRIME.Z1FLOWREC.01, QGEO.CONEDYN.01, E8.ORB60.01); dated notes on PRIME.RANK3UNIF.01 (T-B typed and largely closed), PRIME.INTERP.01 (both blocks near-proof), PRIME.Z1.OPERATOR.01 (the 5–5d series status; stays OPEN), E8.GAUSSCODE.01/E8.QUARTICHALF.01 (formally verified in Lean; Chevalley cited-not-formalized).
- **Papers**. tfpt\_3: 9 new audit blocks. tfpt\_research\_contracts: the prob:R1 update inside the tenth slice (T-B typed + 60/70 closed with the exact  $T^*$  remainder list) and the PRIME.Z1.OPERATOR.01 series-5–5d update (measure from counting [E], canonical operator, masses+positions forced (measured); remaining: lookahead autonomy + continuum reading); Lean addenda on the two review contracts. tfpt\_prime\_front: two new subsections (the T-B chain; the Z1 series). parity\_toeplitz\_classification: the T-B closure paragraph

(v692–v693).

- **Website.** Prime-front feed entry (the afternoon bundling); suite counters  $685 \rightarrow 694$ ; README and Zenodo description updated to 694 modules / v700.
- **Scope.** No marker moves; the T-B closure is unconditional-MODULO-CITATIONS on 60/70 windows of the declared finite family (the remainder carries exact  $T^*$ ; extension beyond the family typed); PRIME.Z1.OPERATOR.01 stays OPEN (lookahead autonomy + continuum reading); the two kills (CONE-DYNAMICS-DEAD, ORBIT60-PARKED) are honest closures of review routes; no RH claim anywhere.

**2026-08-03 (L)** (the great bundling: TEN promotions (v682–v691, 182 new checks, all green) — the five proof offensives of the day land in one round, and the master contract PRIME.UNIFPOS.01 is registered. **HIGHLIGHT 1** (v685, surface-closed, 8/8 — the theorem-level result):  $\det S > 0$  UNCONDITIONALLY on the entire declared window surface — the symbolic envelope  $K_{\text{env}}(a, h) \leq 0.9798 < 1$  for all 62 complete windows  $a \geq 3.434$ , the five below by finite certificates  $\kappa' \leq 0.108$ ; exact  $U_0$  rebooking  $M' = M + \Delta$ , inputs typed [E]/a-cited (Gram/Hutchinson  $\gamma_1$ ; Platt–Trudgian  $3 \times 10^{12}$ ); a SURFACE theorem on the declared family, NOT uniform in all  $a$  — T-B (the absorption margin) stays open, no RH statement. The chain below it: v683 (19/19, three functionals exact  $4.9 \times 10^{-16}$ , all entry-wise routes typed circular,  $\kappa_{\text{max}} = 0.0895$ ) and v684 (8/8, the zero side by the explicit formula — residuum  $\leq 6.9 \times 10^{-6}$ , 22 491 budget-certified zeros, unconditional  $\kappa_{\text{unc}} = 0.039\text{--}0.190 < 1$  class (a), the pretentious escape blocked  $\times 634$  by the classical zero-free strip). **HIGHLIGHT 2** (v691, hecke-sos-ihara-mechanism-extracted, 27/27): the target factorisation  $A = B^*B + P$  EXISTS EXACTLY in the Ihara lab ( $B$  by Chebyshev recursion,  $P \succeq 0 \iff$  Ramanujan), the deployed  $\zeta$  form is exactly the sine/defect HALF of the canonical split (index lemma), the Euler mechanism is measurable (resonance lattice exact 12/12, depth  $\times 1.98$ ), and the missing piece is named exactly: Z1, a Hilbert–Pólya-type operator (PRIME.Z1.OPERATOR.01, registered OPEN — offensive 5 running). **THE REST OF THE BUNDLE:** v682 (lk-split-dies, 20/20: the naive smooth  $L + K$  split is structurally dead — the pencil maximizer sits at  $\gamma_1 = 14.13$  found from primes alone, spike law  $\mu_{\text{max}} \approx 1 + 2a/\Omega$ ,  $\theta \sim 2.47a$ : the windows resolve individual zeros; one-sided/multi-resolution survive), v686 (geomsos-trace-exact-transport-open, 19/19:  $\Lambda(n)$  circle-free from  $E_8$  shell counting at  $7.1 \times 10^{-31}$ , cover-SOS canonical on dim 3, transport open at 91.4% top-3 mass; contract PRIME.GEOMSOS.01 with K1–K5), v687 (kernel-class-too-narrow, 18/18: the band-limited positive kernel class collapses exactly onto  $\{\lambda \text{Fejér}\}$ , class supremum  $0.0901684 < 0.10076$  — W2 stays closed via the v680/v681 route), v688 (interp-falsifier-constructive, 15/15: the in-kernel matched filter detects every off-line quadruple from  $2\alpha\delta \geq 1.974$ ,  $\times 2$  over the v677 mode map, masking adjudicated 48/48), v689 (gaussian-code-bridge-exact, 26/26:  $E_8(\mathbb{Z}[i])/(1+i) \cong \mathbb{F}_2^4$  by exact SNF, the  $240 = 15 \times 16$  census,  $\sigma$  as the RM(1,3) family permutation, must-fails fire), v690 (quartic-half-alive, 22/22: the vanishing half  $\{2, 14, 18, 30\}$  proven but honestly typed trivial ( $\mu_4$ -orbit factor); the substance:  $F_8/F_{12}/F_{20}/F_{24}$  algebraically independent G31 BASIC invariants, Chevalley cited, Jacobi carries 60/60 mirrors). **THE MASTER CONTRACT:** PRIME.UNIFPOS.01 (the Uniform Positivity Theorem — Suzuki’s localized Weil operator positive semidefinite for every  $a > 0$  from an operator/geometric representation defined without zeros and without RH-equivalent input; the machine-checkable intermediate form  $B_a = L_a + K_a$ ,  $|\langle K_a v, v \rangle| \leq \theta \langle L_a v, v \rangle$ , annotated with the v682 verdict; kill criteria: zero-defined  $C_a$

or RH-equivalent contractivity = renaming) as the tenth slice, plus the review contracts E8.GAUSSIAN.CODE.01/E8.QUARTIC.HALF.01/PRIME.W3.INTERPOLATION.01 and the OPEN PRIME.Z1.OPERATOR.01)

- **New modules (10).** v682\_lk\_split\_theta.py (20), v683\_rank3\_functionals.py (19), v684\_rank3\_zeroside.py (8), v685\_rank3\_uniformity.py (8), v686\_geometric\_sos.py (19), v687\_extremal\_kernel.py (18), v688\_interpolation\_detector.py (15), v689\_gaussian\_code\_bridge.py (26), v690\_quartic\_half.py (22), v691\_hecke\_sos.py (27). All ten discovery probes committed alongside; the zero-comb caches (zero\_comb\_cache\_n2000.json, c1\_zero\_ext\_n2500.json, coverage\_hole\_zscan\_cache.npz) were already committed and are reused repo-relative with local fallbacks.
- **Ledger.** 15 new rows (PRIME.LKSPLIT.01, PRIME.RANK3FUNC.01, PRIME.RANK3ZERO.01, PRIME.RANK3UNIF.01, PRIME.GEOMSOS.01, PRIME.KERNELCLASS.01, PRIME.INTERP.01, E8.GAUSSCODE.01, E8.QUARTICHALF.01, PRIME.HECKESOS.01; contracts PRIME.UNIFPOS.01, E8.GAUSSIAN.CODE.01, E8.QUARTIC.HALF.01, PRIME.W3.INTERPOLATION.01, PRIME.Z1.OPERATOR.01); dated notes on PRIME.WEIL.OPERATOR.01, PRIME.W3STRUCT.01, PRIME.ZEROGAP.01, E8.CODESEM.01, E8.ST31.01.
- **Papers.** tfpt\_3: 10 new audit blocks. tfpt\_research\_contracts: the tenth slice (the master contract PRIME.UNIFPOS.01 with target statement, intermediate form, v682 annotation and kill criteria; the four review contracts registered). tfpt\_prime\_front: the great-bundling paragraph (rank-3 surface theorem, kernel collapse, falsifier, Hecke-SOS mechanism). parity\_toeplitz\_classification: the rank-3 chain paragraph (v683–v685). tfpt\_1/tfpt\_2: the Gaussian code bridge and the quartic half in the code sections.
- **Website.** Prime-front feed entry (the great bundling); suite counters 675 → 685; README and Zenodo description updated to 685 modules / v691.
- **Scope.** No marker moves; v685 is a SURFACE theorem on the declared window family, NOT a uniform-in- $a$  statement; T-B (the absorption margin, prob:R1) stays open; PRIME.Z1.OPERATOR.01 is registered OPEN (offensive 5 running); no RH claim anywhere.

**2026-08-03 (XLIX) (the Monday-morning promotion round: TWO promotions (v680–v681, 24 new checks, all green) — THE PINCH BREAKS AND THE HOLE CLOSES: the pointwise W2 density map at the anchor window is now GAPLESS. HIGHLIGHT 1 (v680, pinch-broken-split, 13/13): the v678 11.7% pinch was a BOOKKEEPING ARTIFACT of one-sided capture — centered capture (the same zero-gap theorem spent two-sided,  $\text{dist}(t, Z) \leq H_{\min}(t - 26)/2$ , machine-checked on 6595 comb grid points with 0 violations) DOUBLES the threshold to  $A_1 < 1/(2a_0) = 0.18034$ : the existing Bellotti–Wong constant 0.10076 passes with 79% headroom; and the Beurling–Selberg minorant ELIMINATES  $A_1$  from the counting chain entirely (mass loss exactly  $\pi/a_0$ , extremal per Logan/Littmann; explicit prime term  $P = 2.534$ , positive from  $t^* = 1.11 \times 10^7$ ; no Platt cap, every window incl.  $a \rightarrow \infty$ ; Weil identity on the comb at  $9.0 \times 10^{-5}$ ); the family reach  $[10, 870]$  is closed verification-backed (min  $s_{\text{tot}} = 0.0063$  over all 5 windows  $\times$  lattice modes); the honest residue at that point: the  $O(1)$  coverage hole (2500,  $7.27 \times 10^6$ ) = 3.46 decades. HIGHLIGHT 2 (v681, hole-closed-split-type, 11/11): the hole attacked on the three named paths and CLOSED with split typing — H3 as the main lever (the exact almost-periodic prime term with a hierarchical Lipschitz certificate moves the abstract entry from  $t^* = 1.11 \times 10^7$  down to  $t_x = 1.53 \times 10^5$ , a factor 72.5); the vectorized Riemann–Siegel scan with 223 949 budget-certified zeros (Gabcke remainder, correctness 603/603 against the certified strip, mpmath**

12/12, found/expected 0.99774) carries  $(2515, 1.56 \times 10^5]$  with floor 0.01664; H1 honestly adjudicated (the Platt constant reproduces the hole boundary exactly — it cannot enter); THE GAPLESS MAP:  $s_{\text{tot}}(t; a_0) \geq 0.02259 \log(2 + t) - 0.5185$  for ALL  $t \geq 10$ , every region with named support and type (comb [E] / RS scan [E]-verification-consistent / exact-prime Weil chain abstract / Selberg uniform / Trudgian-2S unconditional from  $1.74 \times 10^{25}$ ); THE W2 CHAIN AT THE ANCHOR COMPLETE: density (v669) + frame-Gårding (v674/v678) + pointwise (v681); the calibration history documented with the pinned stage-1 duplicate bug)

- **New modules (2).** v680\_pinch\_attack.py (13), v681\_coverage\_hole.py (11). Both discovery probes committed alongside (pinch\_attack\_probe.py, coverage\_hole\_probe.py) together with the committed stage-1 scan cache coverage\_hole\_zscan\_cache.npz ( $\sim 1.2$  MB; regenerated from scratch on a cache miss — a speed-up, not an input).
- **Ledger.** 2 new rows (PRIME.PINCHBREAK.01, PRIME.HOLECLOSED.01); dated notes on PRIME.ZEROGAP.01 (the pinch is broken — the quantified target was a one-sided bookkeeping artifact; the residue moved to the coverage hole and was closed the same day), PRIME.FEJERDENS.01 (the pointwise map at the anchor is gapless: the F3 obstruction resolved by two-sided bookkeeping + minorant replacement) and PRIME.PACKETGARD.01 (the W2 chain at the anchor is complete: density + frame + pointwise, each with split typing); the ninth consolidated slice on PRIME.WEIL.OPERATOR.01.
- **Papers.** tfpt\_3: 2 new audit blocks. tfpt\_research\_contracts: the ninth slice of PRIME.WEIL.OPERATOR.01 (the W2 end-state after the pinch break and the hole closure: the pointwise density floor at the anchor window is a gapless split-type map — comb-certified / scan-certified / abstract-Weil / unconditional-asymptotic; remaining: deep family windows  $a > 4.43$  (Selberg-only entry), the formal Mosco writeup, the projection-norm form,  $a \rightarrow \infty$  (= W3/W4 territory); kill criteria unchanged; no RH statement).
- **Website.** Prime-front feed entry (the pinch-break round); suite counters 673  $\rightarrow$  675; sandbox counters 267  $\rightarrow$  269 probes and 5923  $\rightarrow$  5947 checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the pointwise map at the anchor is closed with MIXED typing — the abstract region is modulo the off-line window above  $3 \times 10^{12}$  until the unconditional Trudgian-2S entry; the deep family windows  $a > 4.43$ ,  $A_5(a)$ , the formal Mosco writeup and  $a \rightarrow \infty$  stay open; uniform W3 remains the conjecture); both verdicts are split-type by construction, the typing ledger is explicit; no RH statement anywhere.

**2026-08-02 (XLVIII)** (the sixth and final promotion round of the day: TWO promotions (v678–v679, 29 new checks, all green) — THE DAY ENDS AT THE WALL WITH A DOOR HANDLE: the last W2 gap now carries a concrete number (11.7% in one explicit constant), and the orbifold front reaches the continuum-convergence level. **HIGHLIGHT 1** (v678, zerogap-floor-only): the best explicit UNCONDITIONAL zero-gap theorem, documented and machine-verified — the  $S$ -difference route with  $S_{\text{bound}} = \min\{\text{Platt } |S| \leq 2.5167 \text{ to } 3.06 \times 10^{10}; \text{Trudgian 2014; Bellotti 2025 Cor. 1.5, asymptotic floor } 4\pi 0.10076 = 1.26619\}$ , verified on ALL 1999 Turing-certified comb gaps (0 violations on each chain separately, min air  $2.05\times$ , median  $5.24\times$ ); adaptive bands  $\delta_p = \kappa H_{\min}$  unconditionally hold  $\geq \kappa$  ordinates (0 misses)  $\Rightarrow$  the v674 FRAME packet floor is now UNCONDITIONALLY positive ( $0 < B_p \leq V_p$  everywhere, median discount 0.037) — the frame-Gårding chain is fully grounded in cited unconditional inputs; the projection form does NOT stabilize adaptively either (typed residual with inverted expectation, v642/v662/v674 pattern: 6/6 ( $\kappa, C$ ) cells miss, 1/log competitive, minimizer single-mode 0.898 — the pointwise obstruction is re-partition-invariant);

**THE QUANTIFIED PINCH:** pointwise capture of a guaranteed zero needs the explicit  $S(t)$  constant  $A_1 < 1/(4a_0) = 0.09017$  — the best cited constant 0.10076 (Bellotti 2025) misses by 11.7% at every height: the last W2 gap as a concrete named target inequality at the frontier of explicit analytic number theory, independent of RH. **HIGHLIGHT 2** (v679, orb-os-continuum-slice, 19/19): the first continuum-OS slice of the seam orbifold B — correlator convergence with UNDERSTOOD rates (all six  $\sigma$ -channel observables at the uniform raw rate 0.659 = the Fisher–Hartwig 2/3 branch;  $\tau$ -channel contrast 1.298 vs 4/3, the channel law  $\rho = 2(1 - \beta)^2 - 2\beta^2$ ; after one FH term the limits sit  $< 0.05\%$  at the CFT values; the  $\varepsilon$  channel exact); **RP SURVIVES THE LIMIT** ( $\lambda_{\min}$  extrapolations bounded away from zero:  $\sigma$   $1.36 \times 10^{-3}$  margin  $7\times$ ,  $\tau$   $2.80 \times 10^{-4}$  margin  $358\times$ , mixed  $1.34 \times 10^{-3}$  margin  $31\times$ ; must-fail  $\eta = -1$  flips); cluster exact in space AND Euclidean time ( $2\Delta = 0.222220$  vs  $2/9$  at  $8 \times 10^{-6}$ ; time gap = the  $\varepsilon$  level, cosh form, new time machinery); characters  $1/36, 1/9, 5/18, 4/9$  at  $10^{-12}$  with the measured  $N^{-2}$  rate — E0/E2/E3/E4 measured at the convergence level, E1 discrete, the formal-limit remainder named (GNS, uniform bounds); **GATE.QGEO** does not move)

- **New modules (2).** v678\_zero\_gap\_theorem.py (10; the probe’s declared FAIL at the literal Z3.3 adaptive stabilization bar promoted as a typed-residual check with inverted expectation, v642/v662/v674 pattern, numbers unchanged), v679\_orbifold\_continuum\_os.py (19). Both discovery probes committed alongside (zero\_gap\_theorem\_probe.py, orbifold\_continuum\_os\_probe.py).
- **Ledger.** 2 new rows (PRIME.ZEROGAP.01, QGEO.ORBOS.01); dated notes on PRIME.PACKETGARD.01 (the typed OPEN residue supplied: the frame floor is now unconditional), PRIME.FEJERDENS.01 (the theorem family extended by the sharper Bellotti floor 1.26619), PRIME.TURINGCERT.01 (all 1999 certified gaps verified against the explicit theorem) and GATE.QGEO.01 (the continuum-OS data measured at the convergence level; the gate residual is now the formal limit construction + the v622/v623 seam identification — NO marker move); the eighth consolidated slice on PRIME.WEIL.OPERATOR.01.
- **Papers.** tfpt\_3: 2 new audit blocks. tfpt\_research\_contracts: the eighth slice of PRIME.WEIL.OPERATOR.01 (the W2 end-state: frame-Gårding fully unconditionally grounded — zero-gap theorem + Fejér chain; the last gap carries a concrete number: the pointwise/projection form closes iff the explicit  $S(t)$  constant  $A_1$  improves from 0.10076 to  $< 0.09017$  (11.7%) — a named target at the frontier of explicit analytic number theory, independent of RH; kill criteria unchanged; no RH statement).
- **Companion note (no module).** The second citable short note note\_w3\_detector\_structure.pdf (8 pages) stands ready uncommitted for the user: the W3 structure theorem (v677) + the two-lab validation (Epstein census, Ihara threshold map) + the  $C = 1$  quadrature mechanism (v676) as paper #2 of the prime-front line.
- **Website.** Prime-front feed entry (the zero-gap round); suite counters  $671 \rightarrow 673$ ; sandbox counters  $265 \rightarrow 267$  probes and  $5894 \rightarrow 5923$  checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the frame form is now unconditionally grounded, the pointwise A5(a) statement stays open with a quantified external target; uniform W3 remains the conjecture); **GATE.QGEO** stays open (convergence-level data, the formal limit named); the honest typed-residual FAIL (ZEROGAP-FLOOR-ONLY) is promoted as the finding, not hidden; no RH statement anywhere.

**2026-08-02 (XLVII)** (the fifth and final promotion round of the day: **FOUR** promotions (v674–v677, 76 new checks, all green) — **THE WALL IS SURVEYED:** two theorems, one frame-Gårding, and the honest typing “uniform W3 = the

conjecture”. **HIGHLIGHT 1 (v676, c1-quadrature-mechanism):**  $C = 1$  is a **DISCRETIZATION THEOREM** on the declared surface —  $q_r$  is the exact zero-side reading of the lock profile (Weil explicit formula on the complete comb, median residual  $1.1 \times 10^{-2}$  on 67 windows,  $q_{\text{pred}} > 0$  on 67/67), the  $h^{-1}$  is the DST normalization ( $q_r(N/2)/F_\alpha$   $h$ -flat at slope  $+0.04$ ), the constant is the zero-free RvM density integral (ratio 1.005; tertiles 0.66/0.43/0.37 predicted vs 0.61/0.45/0.39 measured), the lock sign via the v591 pole killer; honest: the v596 exponent  $-1.01$  was flip-contaminated (clean  $-1.28$ ; dated note on PRIME.LOCKPROJ.01), “norm  $\leq 1/h$ ” is a direction statement, and  $\sup \leq 1$  does NOT follow from the mean (the  $\sin^2$  sampling budget is the precise remainder, factor-2 headroom). **HIGHLIGHT 2 (v677, w3st-structure-theorem):** the equivalence chain as a **THEOREM** — odd-Toeplitz = Cantoni–Butler compression ( $1.5 \times 10^{-15}$ ), the sandwich theorem instead of naive DST diagonality (parity defect measured 0.92..1.55), the per-lag Weil dictionary unconditional with the master identity  $x^\top Ax = \sum T_x(\gamma) + P(x)$ ,  $T_x \geq 0$  on-line (alias comb) and cosh-amplified off-line terms ( $5.8 \times 10^{-16}$ ); the own Epstein winding census (exactly 12 off-line zeros,  $\zeta_K$  control 0) predicts the form break quantitatively ( $\lambda_{\min}$  ratio 0.803, factor-2 gate); the threshold map  $2\alpha s_{\min} \approx 1.4..2.6$  over 67 windows (the Ihara echo) — W3-on-the-family = theorem + detector certificate, UNIFORM W3 = the conjecture (Weil/Yoshida), typed as such. **PLUS** the frame-Gårding (v674, packet-average-only: the projection packet norm does NOT escape the  $1/\log$  drift — the central honest negative promoted as a typed-residual check with inverted expectation, v642/v662 pattern, minimizer single-mode tight 0.90 — BUT the frame inequality  $Q + C_0G \succeq c_0Y$  holds at every  $M$  with the v669 theorem constants,  $c_0 = 0.3058$ , margins  $+1.52.. + 2.29$ ,  $a$ -uniform, and the Mosco mechanism is numerically complete,  $8/8$ ; named remainder: within-packet equidistribution,  $\theta \rightarrow 0.20$ ) and the needle saturation (v675, needle-emergent-bulk: all four assembly candidates miss — jump set too dense  $p = 0.007$ , phase gradient wrong sign, lock rotation cleanly killed, pole weight significant but **INVERTED** — the needles are an emergent bulk property; W3 recommendation: MARGIN-regime form + Ihara calibration instead of a needle predicate))

- **New modules (4).** v674\_packet\_garding.py (21; the probe’s declared FAIL at the literal P2.1 stabilization bar promoted as a typed-residual check with inverted expectation, v642/v662 pattern, numbers unchanged), v675\_needle\_mechanism.py (21), v676\_c1\_mechanism.py (12), v677\_w3\_structure\_theorem.py (22). All four discovery probes committed alongside (packet\_garding\_probe.py, needle\_mechanism\_probe.py, c1\_mechanism\_probe.py with the committed c1\_zero\_ext\_n2500.json extension cache, w3\_structure\_theorem\_probe.py).
- **Ledger.** 4 new rows (PRIME.PACKETGARD.01, PRIME.NEEDLEMECH.01, PRIME.C1MECH.01, PRIME.W3STRUCT.01); dated notes on PRIME.LOCKPROJ.01 (the exponent correction:  $-1.01$  was flip-contaminated, clean  $-1.28$ ), PRIME.UNIFC.01 (the  $C = 1$  mechanism grounded), PRIME.GROUNDTRUTH.01 (the Ihara detection law echoed in the threshold map), PRIME.FEJERDENS.01 (the frame-Gårding connection) and PRIME.W3LAND.01 (the mechanism search saturates); the seventh consolidated slice on PRIME.WEIL.OPERATOR.01.
- **Papers.** tfpt\_3: 4 new audit blocks. tfpt\_research\_contracts: the seventh slice of PRIME.WEIL.OPERATOR.01 (W2 end-state: density plane closed, Gårding in frame form with theorem constants, Mosco complete, remainder = within-packet equidistribution; W3 contracted to “MARGIN-regime generic bound + needle risk map”,  $C = 1$  demystified-and-grounded as a quadrature theorem, the structure theorem types W3-on-the-family as theorem + certificate and uniform W3 as the conjecture; kill criteria unchanged).
- **Website.** Prime-front feed entry (the structure-theorem round); suite counters 667  $\rightarrow$  671;

sandbox counters 261  $\rightarrow$  265 probes and 5818  $\rightarrow$  5894 checks; README and Zenodo description updated.

- **Scope.** No marker moves; W2/W3 stay open (the frame form holds with theorem constants, the pointwise A5(a) statement stays open; W3-on-the-family is a theorem + certificate, uniform W3 IS the conjecture — typed, no RH statement); the honest typed-residual FAIL (PACKET-AVERAGE-ONLY) and the fourfold needle miss (NEEDLE-EMERGENT-BULK) are promoted as findings, not hidden; the  $C = 1$  statement carries no RH lever beyond the density and is typed as such.

**2026-08-02 (XLVI)** (the fourth promotion round of the day: **FIVE** promotions (v669–v673, 65 new checks, all green) — **THE W2 DENSITY PLANE CLOSES** (v669, the highlight of the round: the exact identity  $s_{\text{tot}} = 2\pi(F_a \star dN)$  via the Weil explicit formula on the tent test pair, verified  $2.9 \times 10^{-6}$  pointwise /  $7.2 \times 10^{-8}$   $t$ -averaged against the Turing-certified comb — the envelope peaks ARE the zeros ( $\gamma_{1..4}$  to 0.0002), the dips ARE the zero gaps — plus the UNCONDITIONAL chain RvM + Trudgian  $\Rightarrow \rho_{a,\delta} \geq 0.306 \log(2+t) - C_0$   $a$ -uniformly with 10/10 finite certificates on the zero-free grid, margins  $\geq 1.36$ ; the honest floor: the literal  $1/a$  and the plane-wave  $\pi/a$  widths sit BELOW the RvM threshold  $4\pi A_1 = 1.4074$ , the structural explanation of the Gårding drift — the remaining W2 piece is the packet-to-point translation below the floor); **THE NEEDLE SEARCH NARROWS THREEFOLD** (v670 block deflation honestly FAILED: fidelity  $p_{90}$  only  $50.53 \rightarrow 49.35\%$  for  $K = 1 \rightarrow 8$ , leak median 1.000 everywhere — the coupling is bulk-spectral, not low-rank; v671 the Lehmer link is a clean NULL:  $h^3$ -partials all  $p > 0.1$ , placement control unremarkable — the needles are NOT zero-comb-driven; together with v667 not-in-the-BD-frame: the mechanism search moves frame-/assembly-side); and **THE LI DOUBLE** (v672 the W1  $\Rightarrow$  Li corollary:  $g_n = \frac{1}{2}e^{-|u|/2}L_{n-1}^{(1)}(|u|)$  derived exactly (Laguerre), the  $h = 1433$  window contains 30/32 finite Li coefficients at  $< 10\%$  form error ( $n = 1$  honestly 24%, a jump artifact;  $n = 5$  hair-thin 10.03%), pole cancellation to six digits, odd sector certified by  $\lambda_{\min} = +1.516 \times 10^{-3} > 0$ ; v673 the  $E_8 \leftrightarrow$  comb Li lock:  $\Lambda(E_4, s) = (2\pi)^{-s}\Gamma(s)\zeta(s)\zeta(s-3)$  completed with residues  $\pm 1/240$  — the  $E_8$  240 IS the residue normalizer —  $\lambda_n^L = 2\lambda_n^\zeta$  exact,  $E_8$ -arithmetic vs comb at budget usage 0.124, the naive single-map Li refused as non-tempered))

- **New modules (5).** v669\_fejer\_density.py (14), v670\_w3\_block\_deflation.py (18; verdict BLOCK-DEFL-FAIL promoted as the honest preregistered result), v671\_lehmer\_resonance.py (10; verdict LEHMER-NULL), v672\_li\_corollary.py (11), v673\_li\_e4.py (12). All five discovery probes committed alongside (fejer\_density\_bound\_probe.py, w3\_block\_deflation\_probe.py, lehmer\_resonance\_probe.py, li\_corollary\_probe.py, li\_e4\_probe.py).
- **Ledger.** 5 new rows (PRIME.FEJERDENS.01, PRIME.BLOCKDEFL.01, PRIME.LEHMERNULL.01, PRIME.LICOROLLARY.01, PRIME.LIE4.01); dated notes on PRIME.GARDING.01/PRIME.GARDENV.01 (v669 explains the drift: per-mode control below the RvM floor is not certifiable at any height), PRIME.W3LAND.01/PRIME.THETAPRED.01 (the threefold narrowing v670/v671), PRIME.KEIPERLI.01 (the Li strand connects, v672/v673) and PRIME.SHADOW.01 (the 240-residue identity + the Li lock); the sixth consolidated slice on PRIME.WEIL.OPERATOR.01.
- **Papers.** tfpt\_3: 5 new audit blocks. tfpt\_research\_contracts: the sixth slice of PRIME.WEIL.OPERATOR.01 (W2: the density plane closed, remainder = the packet-to-point translation below the RvM floor; W3: narrowed threefold — not low-rank, not zero-comb-driven,



not in the BD frame  $\Rightarrow$  frame-/assembly-side; new: the Li corollary and the  $E_4$  Li lock; kill criteria unchanged).

- **Website.** Prime-front feed entry (the Fejér-density round); suite counters 662  $\rightarrow$  667; sandbox counters 256  $\rightarrow$  261 probes and 5753  $\rightarrow$  5818 checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the density plane closes at the wave-packet level only — the pointwise A5(a) Gårding inequality stays open); the honest FAIL verdict (BLOCK-DEFL-FAIL) and the clean null (LEHMER-NULL) are promoted as findings, not hidden; positivity statements remain finite (30/32 Li coefficients,  $n \leq 32$ ), NOT Li’s criterion; no RH statement.

**2026-08-02 (XLV)** (the closing bundle of the day: **NINETEEN** promotions in one pass (v650–v668, 301 new checks, all green) — the **ORBIFOLD FRONT COMPLETES AT THE LATTICE LEVEL** (modular data v650: nine  $Z[g, h]$  match  $|\theta/\eta|^2$ , S-covariance measured at a clean  $N^{-4}$  rate, T exact as the  $\mathbb{Z}_6$  Dehn shift, deck<sup>3</sup> =  $(-1)^F$  as exact  $\mathbb{Z}_6$  trace arithmetic; assembly v651:  $H^2(\mathbb{Z}_3, U(1)) = 0$  enumerated, of six candidates exactly  $B$  survives, T kills the naive  $\mathbb{Z}_3$  exactly; Arf hardening v652: the  $6^6$  classification leaves  $\{B, C3\}$ , the one-Fock-space pin forces  $B$ , the  $\mu_6$  ladder measures the C3-excised states directly; bond-defect theorem v653: twist = deck = bond defect as a lattice theorem in  $\mathbb{Q}(\zeta_6)$  with  $D_{\text{can}} = L^4$ ,  $D_{\text{can}}^3 = -1$ ); the ST31  $d/4$ -THEOREM (v654: every Springer-regular  $d$ -clock has  $x^{d/4} = \pm i \text{Id}$ , converse exact at  $d \in \{8, 20, 24\}$ , the unique  $d = 12$  exception is the compiler-clock class; “free  $\Rightarrow$  regular” killed); the W2/W3 SERIES (v655 Mosco preparation: resolvent Cauchy + uniform  $H_{\log}$ ; v656 margin bridge typed as lock identification; v657 the exact 2D formula  $r_{\text{id}} = (1 - q \tan^2 \theta)/(1 - \tan^2 \theta)$ , the deficit angle-driven; v658 the uniform bound typed **FRAGILE** ( $\alpha$ -fragility  $\pm 60\%$ ); v659 the landscape:  $\lambda_{\min} > 0$  on ALL 635 points, every peak a rotation artifact, MARGIN-regime share 0.0%; v660 the rotation predicate honestly **FAILED** (necessary, not sharp; 47/52 needles are diagonal crossings; successor block deflation); v661–v663 the Gårding arc: drift undecidable on 4 stages  $\rightarrow$  both remedies refuted, the drift lives in the total symbol  $\rightarrow$  **THE ENVELOPE HOLDS** measured  $a$ -uniformly,  $(c_0, C_0) \approx (0.021, 0.055)$ , “flat” was a  $t$ -range artifact, remaining task = a Fejér spectral-density bound); **DISCIPLINE AND CONTROLS** (v664 look-elsewhere: 42 slots / 19 hits / 6 kills, global  $p = 5.1 \times 10^{-3}$  conservative, NO single observation significant — re-types executed; v665 Keiper–Li on two independent routes, consistent at budget usage 0.124, injection detected from  $n = 1$ ; v666 Turing certificate: the comb complete below  $\gamma_{\max}$ , integrals at 5% of the Lehman bound; v667 Baez–Duarte control frame: Vasyunin  $1.7 \times 10^{-31}$ ,  $C_{\text{BD}}$  cross-anchor  $6.4 \times 10^{-8}$ , drift **SHARED** = intrinsic, needles **NOT** shared = frame-side; v668 ground truth + firewall, one module from both probes: true positivity has NO uniform margin,  $\delta(K) \rightarrow 0$  exactly, detection reach  $K^* \times s \approx 2\text{--}3$ , Fejér reads blind — and without the Euler product the machinery breaks by  $\sim 13$  orders of magnitude: **THE** central W3 recalibration); plus **LEAN ROUND 3** (three new certificate files, lake build green, 3383 jobs, no sorry) and the Nature-Comms primon-gas reference typed honestly in the prime-front paper)

- **New modules (19).** v650\_orbifold\_modular.py (17), v651\_orbifold\_assembly.py (17), v652\_orbifold\_arf.py (15), v653\_bond\_defect.py (18), v654\_st31\_degree8.py (30), v655\_w2\_mosco.py (7), v656\_margin\_link.py (11), v657\_rid\_alignment.py (15), v658\_w3\_uniform\_bound.py (11), v659\_w3\_landscape.py (14), v660\_theta\_predicate.py (20), v661\_garding.py (12), v662\_garding\_edgeband.py (16; the probe’s declared FAIL at the

literal symbol-stability bar promoted as a typed-residual check with inverted expectation, v642 pattern, numbers unchanged), v663\_garding\_envelope.py (18), v664\_look\_elsewhere.py (14), v665\_keiper\_li.py (13; shared zero cache zero\_comb\_cache\_n2000.json committed), v666\_turing\_cert.py (6), v667\_baez\_duarte.py (12), v668\_ground\_truth.py (35 = 22 + 13; ONE module merged verbatim from the two probes ihara\_ground\_truth\_probe.py + epstein\_firewall\_probe.py). All discovery probes committed alongside.

- **Ledger.** 19 new rows (QGEO.ORBMOD.01, QGEO.ORBASM.01, QGEO.ORBASF.01, QGEO.BONDDEF.01, E8.ST31D8.01, PRIME.WEIL.MOSCO.01, PRIME.MARGINLINK.01, PRIME.RIDGEOM.01, PRIME.W3BOUND.01, PRIME.W3LAND.01, PRIME.THETAPRED.01, PRIME.GARDING.01, PRIME.GARDEDGE.01, PRIME.GARDENV.01, META.LOOKELSEWHERE.01, PRIME.KEIPERLI.01, PRIME.TURINGCERT.01, PRIME.BAEZDUARTE.01, PRIME.GROUNDTRUTH.01); dated notes on QGEO.TWISTOPE.01, QGEO.KLEINRP.01 and GATE.QGEO.01 (the orbifold front is complete at the lattice level; the [C] remainder is the continuum CFT statement + the seam identification — *no marker moves*), on E8.ST31.01/E8.ST31DEG.01 (v654 + Lean round 3), on EXTREV.LATTICE.01 ( $6 = p_2(a)$ : observation without statistical weight,  $p = 0.169$ ), E8.CODESEM.01 (number battery  $p = 0.593$  — the v646 finding quantified) and E8.CODE.01 (14/16 typed FORCED + the Lean code certificate), and the consolidated W2/W3 note on PRIME.WEIL.OPERATOR.01.
- **Lean round 3.** Three new committed proof modules, lake build green (3383 jobs, no sorry): G31WordOrders.lean ( $\chi_w \mid X^{24} - 1$  and  $\chi_u \mid X^{20} - 1$  via conjugate-quartic witnesses  $\chi\bar{\chi} = \Phi_{24}/\Phi_{20}$ , no proper divisor by companion-matrix kernel decide — all eigenvalues primitive), HammingCode.lean (self-duality, weight distribution  $\{0:1, 4:14, 8:1\}$ , coset-leader distribution  $\{0:1, 1:8, 2:7\}$ , covering radius 2), SquareParity.lean (the combinatorial kernel of the  $c = w^2$  parity kill at the multiset level, the permutation-level bridge honestly typed as not formalized).
- **Papers.** tfpt\_3: 19 new audit blocks. tfpt\_research\_contracts: the fifth slice of PRIME.WEIL.OPERATOR.01 — the consolidated W2/W3 state (W2: envelope measured valid, Mosco prepared, drift intrinsic, remainder = Fejér spectral-density bound + symbol proof; W3: reduced to lock sign +  $q \tan^2 \theta \leq 1 - \delta$ , landscape mapped without positivity loss, ground-truth recalibrated — shrinking margins are the expected behavior of true positivity, the alarm signal is only  $\lambda_{\min} < -\text{floor}$ ; kill criteria unchanged). tfpt\_prime\_front: the closing-bundle recalibration paragraph + the external-context note on Wei/Zhai/Lu et al., “The Riemann Hypothesis manifested in dynamical quantum phase transitions” (Nat. Commun. 2026, DOI 10.1038/s41467-026-74935-8) — their primon-gas energies  $\log n$  with weights  $\Lambda(n)/\sqrt{n}$  are literally the TFPT atom table and the DQPT times are the zero comb; typed honestly as complementary verification, not RH progress.
- **Website.** Prime-front feed entry (the closing bundle); suite counters 643 → 662; sandbox counters 236 → 256 probes and 5452 → 5753 checks; README and Zenodo description updated.
- **Scope.** No marker moves anywhere; GATE.QGEO does not move (the continuum statement stays open); W2/W3 stay open (measured/typed progress only); the two honest FAIL verdicts (THETA-PRED-FAIL, EDGEBAND-UNHEALED) are promoted as findings, not hidden; no positivity claim beyond the measured surfaces, no RH statement.

**2026-08-02 (XLIV)** (the discipline audit: the suite measures its OWN scientific process discipline and freezes it as checks (v649, META.DISCIPLINE.01) — ledger census 715 rows / 710 active with 214 documented kills / honest negatives / retypes, must-fail census 643 modules / 6040 static check() sites / 1469 negative-control keyword occurrences across 233 modules, preregistration census 47 contract rows (29 naming kill criteria before execution; executed chains FTR.PGL2.01 v533→v632 and PRIME.WEIL.OPERATOR.01 K1–K3→v643 verified in the ledger), deterministic

replay 12/12 on a declared sample spanning v55–v648, five classical anchors recomputed against the literature (Jacobi  $r_4$ , Construction A  $\rightarrow E_8$ , Shephard–Todd G31 degrees, Suzuki archimedean constants, completed-zeta functional equation), 54 committed Lean proof modules — and the honest limit typed [C]: process discipline, not physics truth, the open gates stay open; PLUS the public mirror: a new /method page (“the verification framework”) built from a new generated data surface website/lib/discipline.ts)

- **New module.** v649\_discipline\_audit.py (18 checks, ~200s; discovery probe discipline\_audit\_probe.py 17/17, verdict DISCIPLINE-MEASURED): a read-only meta-module that parses the suite’s own files — ledger, module sources, contracts paper, Lean tree — and certifies the measured discipline metrics as frozen minimum bars. D1 ledger census (715 rows, 710 active, display classes E/C/O/X/Axiom/other = 169/222/282/37/2/3 at freeze; 214 rows with negative-evidence keywords: kill, ill-posed, honest negative, must-fail, retyped, superseded, dead, bingo, numerology); D2 must-fail census (643 modules, every one with checks; 6040 static check() call sites, clearing the published 5452 counter; 1469 must-fail/kill/scramble occurrences in 233 modules; typing marks [E]  $\times$  3579 vs [C]  $\times$  1323); D3 preregistration census (47 contract rows, 29 naming kill criteria; executed chains machine-verified: FTR.PGL2.01 preregistered v533  $\rightarrow$  executed v632, PRIME.WEIL.OPERATOR.01 kill tests K1–K3  $\rightarrow$  W1 theorem-closed v643, REG.FREEZE.01 frozen 2026-06-09); D4 reproducibility (deterministic replay 12/12 on the declared sample v55, v91, v108, v205, v305, v405, v505, v555, v600, v634, v643, v648 — two fresh subprocess runs each, identical up to timing lines; 396 sympy-exact modules, 77 with explicit seeds); D5 external anchors (five classical recomputes against the literature: Jacobi 1834 four-square counts exact  $n = 1.40$ , Construction A  $[8, 4, 4] \rightarrow E_8$  with 240 minimal vectors, Shephard–Todd G31 degrees (8, 12, 20, 24), Suzuki  $\psi(\frac{1}{4})/L/A$  to 25+ digits, completed-zeta functional equation at  $3 \times 10^{-31}$ ); D6 Lean layer (54 committed TfptCarrier modules via git ls-files; lake build honestly *not* re-run in-suite); D7 the limit, typed [C] (process discipline, not physics truth; 278 class-O rows stay open incl. SEAM.THEOREM.01 and PRIME.WEIL.OPERATOR.01 W2–W4). Freeze discipline: every bar is the value measured at freeze time; a future census regression fails the module and forces an honest dated re-freeze.
- **Ledger.** New row META.DISCIPLINE.01 (canonical status DISCIPLINE-CERTIFIED); no other row changes, no marker moves anywhere.
- **Website.** New generated data surface website/lib/discipline.ts written by the new verification/make\_discipline\_stats.py (stdlib-only, hooked into bash build.sh gen; freshness enforced by audit\_sync.py section A.generated). New page /method (“The verification framework”): the sandbox  $\rightarrow$  probe-with-kill-criteria  $\rightarrow$  suite  $\rightarrow$  ledger  $\rightarrow$  generated-mirrors pipeline as a static SVG; the measured D1–D6 numbers rendered from discipline.ts; the “buried coincidences” graveyard ( $|G31| = |W(D_5)| \cdot |W(A_3)|$  killed three ways v634; the bytecode  $p_n$  reading 0/11 v646;  $c = w^2$  dead by parity theorem v647; the fine-invariant  $\leftrightarrow C = 1$  bridge dead after  $h$ -control v637; the naive  $48 \times 5$  incidence v629; the ill-posed B-test v624; the same-day Lerch erratum v643); “what would falsify this” (the open contracts’ kill criteria); the E/C/O/X status typology with the measured distribution; and a sober AI-transparency paragraph (agents execute, the deterministic suite decides — python3 verification/run\_all.py replays everything). Navigation entry, a compact “how we keep ourselves honest” band on the home page, and the /method link added to the prime-front honesty banner; suite counters 642  $\rightarrow$  643 on the affected pages, README and Zenodo description.
- **Scope.** The meta-module measures *process* discipline, not the truth of any physics claim; it is read-only and moves no marker; the /method page renders only generated or machine-checked numbers — no new claims anywhere.

**2026-08-02 (XLIII)** (the second round of the day: **THE LERCH ERRATUM** (honest, corrected the same day) plus six promotions (v643–v648) — the W1 chain (v631, v640–v642) had read Suzuki’s eq. (1.3) with Lerch coefficient  $-1$ ; the paper’s own §2.2 data lock  $+1/4$  (machine evidence v643 C0.1), every measured number transfers VERBATIM via the exact identity  $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4c_{\text{gal}}^{\text{sm}}(g)$ , and only the labels change: Suzuki’s smooth layer is  $+\rho$ , the true dictionary is the SINGLE scalar  $+1/D$  on both layers (sign-compatible with positivity), and the origin constant vanishes,  $\kappa = 0$  exactly; **ON THE CORRECTED READING:** W1 becomes a MEASURE-LEVEL THEOREM (the projection lemma closes the  $L_0^2$  remainder — Suzuki’s mean-zero condition is automatic on the  $u$ -side;  $A_{\text{arch}} = -g''_{\text{smooth}}$  exactly at every lag,  $3.4 \times 10^{-52}$ ; form equality  $1.28 \times 10^{-10}$ ), W2 STARTS in honest slices (classical FEM density at rate; Rayleigh–Ritz monotone from above on all six nested sequences;  $\lambda_a = 0^+$  within  $\sim 10^{-9}$ , consistent with Thm 1.3, no sign statement; Mosco remainder typed), the  $\mathbb{Z}_3$  TWIST SECTOR IS REFLECTION POSITIVE under the parafermionic Klein-twist pairing (mirror  $\times$  charge conjugation,  $\omega^{kq}$  phases,  $\eta = (1, 1)$  unique among 36;  $N$ -stable 48..384; full mixed OS Gram PSD — the named v639 open item resolved at the lattice OS level), the RM(1,3) REVERSE READING is honestly two-part (anchor decomposition  $(0, 1, 1, 2)/\text{norm } 6$  exact with the  $\sigma$  action selecting the anchor sector; the bytecode  $p$ -sequence has NO code reading — the bingo is buried), the ST31 DEGREE ROUND anchors the  $\mu_4$  center as a POWER of both regular clocks ( $w^6 = \pm J$  with  $\chi_w = x^4 + ix^2 - 1$  exactly primitive  $\zeta_{24}$ ;  $u^5 = \pm J$  central;  $u^4$  free with census  $\{5:48\}$ ,  $48 = 240/5$ ) and KILLS  $c = w^2$  by a parity theorem, and the W3 SIGN-UNCERTAINTY TOOLBOX is typed: a real 25-digit dictionary to the critical strip (phase slope  $= L = \log \pi - \psi(\frac{1}{4})$ , the W1  $\delta_0$  weight) whose mass lever dies at  $d = 1$  ( $e^{-E^*} = 0.967$  where  $< \frac{1}{2}$  is needed;  $d \geq 21$  required) — while the W3 surface itself is empirically positive on all 67 complete windows ( $\min \lambda_{\min} = +8.26 \times 10^{-4}$ ; the incomplete combs are exactly the v618 flip set  $\{1219, 1292, 1445\}$ ))

- **The erratum (first, honest, dated).** The v631/v640–v642 chain read the Hurwitz–Lerch block of Suzuki’s screw function (eq. 1.3) with coefficient  $-1$ ; the correct coefficient is  $+1/4$ , locked by the paper’s own §2.2 data ( $A = \frac{1}{2}(\log 2\pi - \psi(2)) = 0.70754637$  reproduced only with  $+1/4$ ;  $g(0) = 0$ ; machine evidence v643 C0.1). All chain identities are correct identities of the kernel  $\tilde{g} = g - \frac{5}{4}$  Lerch, and all measured numbers transfer verbatim via  $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4c_{\text{gal}}^{\text{sm}}(g)$  — no check flips. Corrected labels: Suzuki’s smooth layer  $= +\rho$ ; the two-layer dictionary ( $-4D$  smooth,  $D^2$  atomic) collapses to the *one* scalar  $+1/D$ ;  $\kappa = 0$  exactly (the  $-\frac{5}{4}L\delta_0$  term of v640 was an artifact of the misreading). Documented as dated CONVENTION ERRATUM blocks in the four modules (all re-run green, numbers unchanged), erratum notes on the five PRIME.WEIL.\* ledger rows (CONTACT/DICT/BOUNDARY/PORTABLE/MATRIX), erratum sentences in the tfpt\_3 audit blocks, the contracts paragraph and the prime-front theory-open section, and the website mirrors (feed, W1 dictionary map, honesty banner).
- **New modules.** v643\_w1\_theorem.py (11 checks: convention lock/erratum evidence, projection lemma,  $A_{\text{arch}} = -g''_{\text{smooth}}$  exact,  $\kappa = 0$ , derived  $D^8$  moment brackets, form equality  $1.28 \times 10^{-10}$ , pole rank 1); v644\_w2\_form\_density.py (7 checks: classical FEM density at rate, Rayleigh–Ritz monotone on nested spaces,  $\lambda_a$  scan, typed Mosco/norm-resolvent remainder); v645\_klein\_rp.py (17 checks: dressed-string reality, charged Klein blocks PSD  $N$ -stable,  $\eta = (1, 1)$  unique of 36, must-fail controls, full mixed OS Gram); v646\_rm13\_reverse.py (20 checks: exact anchor identity, generic/specific split, bytecode negative, selective kill control); v647\_st31\_degree24.py (25 checks: regular 24/20 clocks with central powers, the parity-theorem kill of  $c = w^2$ ,  $u^4$  census  $\{5:48\}$ ); v648\_sign\_uncertainty.py (12 checks: 25-digit

dictionary anchors, the  $d = 1$  mass-budget death, surface positivity 67/67, flip-set reproduction, typed verdict).

- **Ledger.** New rows PRIME.WEIL.THEOREM.01, PRIME.WEIL.W2.01, QGEO.KLEINRP.01, E8.RM13REV.01, E8.ST31DEG.01, PRIME.SIGNUNC.01; dated erratum notes on PRIME.WEIL.CONTACT.01, PRIME.WEIL.DICT.01, PRIME.WEIL.BOUNDARY.01, PRIME.WEIL.PORTABLE.01, PRIME.WEIL.MATRIX.01; dated notes on PRIME.WEIL.OPERATOR.01 (contract state: W1 theorem-closed, W2 started, W3 open with tool diagnosis, W4 unchanged), QGEO.TWISTOPE.01 (open item resolved), E8.CODESEM.01, E8.ST31.01, PRIME.UNIFC.01 (independent flip-set confirmation,  $\text{corr} = +0.704$ ).
- **Scope.** No marker moves; the erratum changes labels, not numbers (the transfer identity is exact and machine-checked); W1 closes as a theorem at the window level, W2 is started (not closed), W3 stays open (the sign-uncertainty route is closed as a typed negative — the toolbox needs a large parameter the  $\zeta$  strip does not have), W4 unchanged; GATE.QGEO does not move; no positivity claim beyond the measured surface, no RH statement.

**2026-08-02 (XLII)** (website: the home page catches up with the 2026-08 arcs — a new “geometric realization” section (the seam axioms force the  $\mu_3$ -cover  $y^3 = x^4 - 1$ ,  $N = 3$  pinned, the physical seam is the conformal seam, the covered seam and its  $\mathbb{Z}_3$  orbifold; static SVG map seam  $\rightarrow$  curve  $\rightarrow$  E8 with the 60-line/ST31 quotient) and a new “code and primes” section (E8 as an error-correcting code with the RM(1,3) bit dictionary; primes as the shadow of the finished geometry; the W1 Weil-operator contact with its honest W2/W3 scope — no RH claim); claim-stack honesty updates)

- **Website only, no theory change.** New home sections GeometryArc (mirrors v617, v620, v622, v623, v628, v629, v633, v634, v639) and CodePrimesBand (mirrors v625, v626, v638 and v630, v631, v640–v642), placed between the claim stack and the honesty band; every card carries its display marker and the open items are stated in place.
- **Claim stack.** The declared-inputs layer records the conditional v624 reduction (three selectors force  $d = 4$ , so the two inputs reduce to  $d = 4$  plus  $\pi$  under the named architecture); the numerical layer now names the honest  $\sin^2 \theta_{13}$  stress ( $\approx 2.0\sigma$ , the largest single pull) instead of leaving it to the predictions page.
- **No marker moves; no new claims.** All content mirrors existing ledger rows; the bedrock (GATE.QGEO) and the RH-hard steps W2/W3 are stated open where the new sections mention them.

**2026-08-02 (XLI)** (the ten-front batch lands (v632–v642 promoted) — **THE PGL<sub>2</sub> CONTRACT EXECUTED:** the four transfer solvers share ONE Möbius action on the disc  $-7$  norm line where they act at all (Koide reading exactly intertwined with the base translation,  $g_{\text{QCD}}$  exactly conjugate to it, Boltzmann = pole by the single-flow theorem; kill criterion NOT met); **THE  $\mathbb{Z}[i]$ -E8 QUOTIENT AND ITS STABILIZER:** the  $\mu_4$ -quotient of the 240 roots IS the hermitian-unimodular  $\mathbb{Z}[i]$ -E8 with the 60-hyperplane system of ST31,  $G_{31}$  is the FULL unitary stabilizer with Molien-unique degrees (8, 12, 20, 24) and the exact clock identities  $\sigma = c^4$ ,  $J = c^9$  — while the numerology  $|G_{31}| = |W(D_5)| \times |W(A_3)|$  is **KILLED** three ways (real structure:  $(\mathbb{Z}_4 \circ 2^{1+4}).\text{Sp}_4(2)$ ; the compiler clock is NOT the regular 12-element); **P CANONICAL AND CONSTRUCTED:** the Lorentz congruence matrix is the unique minimal-Frobenius operator-compatible class in the full census AND, mod sign, the Frobenius-minimal integer congruence transporting the  $C_V$  null frame onto the canonical rank-one rays (finite 16-family); an **HONEST**

**NEGATIVE:** after  $h$ -control the fine Hodge invariants carry **NO** independent information about the  $C = 1$  margin ( $\rho = -0.04$ ,  $p = 0.75$  — the direct window-level bridge route closes); **THE CODE DICTIONARY:** the unique equivariant Hamming placement is Reed–Muller  $\text{RM}(1, 3)$  on  $\text{AG}(3, 2)$  with information bits one per  $\mu_4$  pair, the  $\sigma$  3-cycle on the syndrome flag, and decode = projection verbatim (3840/3840); **THE TWIST OPE:** the interacting  $\mathbb{Z}_3$  orbifold slice stands at the abelian vertex level ( $h_\sigma = 1/36$  three ways, exponent  $2\Delta = 2/9$ , crossing exactly symmetry-protected, closed four-point form; RP validated in the real class, the  $\mathbb{Z}_3$  Klein-twist pairing typed open); and **THE W1 CLOSURE TRIO:** the boundary equation  $A_{\text{arch}} = \frac{1}{4}g''_{\text{smooth}} - \frac{5}{4}(\log \pi - \psi(\frac{1}{4}))\delta_0$  **EXACT** with the derived  $1 + 1/(6d^2)$  moment law, the frozen dictionary **PORTABLE** on three fresh windows (atom constant  $D^2$  at machine precision), and the full-matrix operator level **CLOSED** (block norm  $2.7 \times 10^{-3}$  derived,  $6.3 \times 10^{-6}$  after measure-level re-binning; pole block rank 1; two typed lattice residuals) — plus Lean certificates for the W1 dictionary identities, the Lorentz congruence, and the  $G_{31}$  order identities)

- **New modules.** `v632_ftransfer_pgl2.py` (25 checks: substrate, discrete intertwining, native  $\text{PGL}_2$  classification, five must-fail controls); `v633_orbit60_quotient.py` (17 checks: hermitian-unimodular  $\mathbb{Z}[i]$ -E8,  $\sigma$  census  $19 \times 3 + 3$ , line group 11520, cascade killed); `v634_st31_structure.py` (56 checks: full stabilizer count, Molien uniqueness, Springer regulars, the numerology kill,  $\sigma = c^4/J = c^9$ , honest clock typing); `v635_p_canonicity.py` (18 checks: the 40-solution census, minimal-Frobenius class, fine-invariant separation); `v636_p_construction.py` (25 checks: null-cone census, boost normal form, the finite 16-family, must-fail  $\text{gl}_2$  ansatz); `v637_fine_c1_bridge.py` (8 checks: the preregistered honest negative); `v638_code_semantics.py` (27 checks:  $\text{RM}(1, 3)$  dictionary, syndrome flag, decode = projection, the 30-census kill control); `v639_twist_ope.py` (25 checks: three weight routes, two-point, crossing, closed four-point, OPE data, differentiated RP); `v640_w1_boundary.py` (11 checks: the exact boundary equation, three boundary constants, the derived moment law); `v641_w1_portability.py` (10 checks: the frozen-dictionary kill test on  $h = 285/540/997$ ); `v642_w1_matrix.py` (8 checks: operator-level closure, pole-block rank 1, the two typed lattice residuals as inverted-expectation checks, numbers unchanged from the probe).
- **Lean.** Three new certificate files in `experiments/lean4-carrier-rigidity` (lake build green, no sorry): `WeilDictionary.lean` (the exact v631 W1-dictionary identities: Lerch collapse, structure theorem), `LorentzCongruence.lean` ( $P^\top J_{\text{det}} P = J_{\text{fix}}$  by kernel decide), `G31Orders.lean` (the  $G_{31}$ /glue order identities  $8 \cdot 12 \cdot 20 \cdot 24 = 46080 = 1920 \cdot 24$ ,  $60 = 7 + 11 + 19 + 23$ ).
- **Ledger.** New rows `FTR.PGL2.01`, `E8.ZIQUOTIENT.01`, `E8.ST31.01`, `PRIME.PCANON.01`, `PRIME.PCONSTRUCT.01`, `PRIME.FINEC1.01`, `E8.CODESEM.01`, `QGEO.TWISTOPE.01`, `PRIME.WEIL.BOUNDARY.01`, `PRIME.WEIL.PORTABLE.01`, `PRIME.WEIL.MATRIX.01`; dated notes on `PRIME.WEIL.DICT.01`, `PRIME.WEIL.OPERATOR.01`, `E8.INCIDENT.01`, `E8.CODE.01`, `QGEO.ORBCAS.01`, `PRIME.HODGECON.01`, `FTR.DISC7.NORM.01`, `EXTREV.LATTICE.01`.
- **Scope.** No marker moves; the honest negatives (cascade route, fine-invariant bridge,  $\mathbb{Z}_3$  mirror pairing,  $\text{gl}_2$  ansatz) are typed results that close routes, not gaps; W1 is theorem-capable at the window level with the  $L_0^2$  projection the last remainder; W2/W3 stay open and unmoved; no positivity claim, no RH statement; `GATE.QGEO` does not move.

**2026-08-02 (XL)** (the W1 dictionary lands (v631 promoted) — the v630 non-scalar residual **IS** the zeta pole term: the Lerch collapse  $[e^{-t/2}\Phi(e^{-2t}, 2, \frac{1}{4})]'' = 4e^{-t/2}/(1 - e^{-2t})$  (exact) yields the structure theorem  $g'' = -2 \cosh(t/2) - 4 \times$  (the TFPT arch density) — Suzuki's smooth layer is  $-4 \times$  the window density **MINUS** the pole block (TFPT's separate v591 rank-one); the closed per-lag ratio law

$-D[4+(e^t+1)(1-e^{-2t})]$  verifies, after pole subtraction the smooth conversion is the SCALAR  $-4D$  (monotone to 1.0006), and the atom-layer constant is  $D^2$  exactly: W1 of PRIME.WEIL.OPERATOR.01 closes at the measured level on both halves —  $\hat{A}_{\text{arch}} = -(1/4D)$  Galerkin<sub>smooth</sub> + pole rank-one; the theorem-level remainder and the RH-hard steps W2/W3 stay typed open)

- **New module.** v631\_w1\_dictionary.py (7 checks: the exact Lerch collapse, the structure theorem, the route certificate, the closed ratio law, the scalar  $-4D$  closure, the  $D^2$  atom constant, the typed remainder).
- **Ledger.** New row PRIME.WEIL.DICT.01; resolution note on PRIME.WEIL.CONTACT.01.
- **Scope.** The implication map stays explicit: W1 measured-closed  $\rightarrow$  W2 (open)  $\rightarrow$  W3 (open, RH-hard)  $\rightarrow$  W4 (classical given W2+W3); closing W1 does not move W3; no positivity claim, no RH statement.

2026-08-02 (XXXIX) (the six-front batch lands (v626–v630 promoted) — THE E8 CODE: Construction A on the self-dual extended Hamming code  $[8, 4, 4]$  yields E8 exactly (shells 240/2160; every single-bit error exhaustively correctable: “E8 is an error-correcting code” is a THEOREM); THE HODGE CHAMBER: all 67 complete windows land in the POSITIVE cone of the cover polarization lattice on ONE sheet (the measured half of the geometric positivity route; honest typing: chamber membership is density-layer, the fine  $C = 1$  arithmetic lives inside); THE ORBIFOLD CASIMIR: the covered seam’s deck classes  $\{1/6, 1/2, 5/6\}$  carry exactly the  $\mathbb{Z}_3$ -twist Casimir data (exact closed form, sector table  $(1/72, -1/24, 1/72)$ , twist gap  $1/18$ ); THE INCIDENCE CENSUS: the naive  $48 \times 5$  root incidence is KILLED ( $19 \times 12 + 3 \times 4$ ) with the sharp positive residue  $60 = D_{\text{start}}$  as the free clock-orbit count on the roots; and THE SUZUKI FIRST CONTACT: the TFPT atom table IS Suzuki’s prime measure literally (positions  $\log n$ , weights  $\Lambda(n)/\sqrt{n}$ , exact) — W1’s atomic half CLOSED, the smooth conversion measured non-scalar (the preregistered residual, now with data))

- **New modules.** v626\_e8\_code.py (7 checks: Hamming  $[8, 4, 4]$  self-dual, Construction A even unimodular = E8, shells, exhaustive syndrome decoding, typed compiler ties  $8/4/16/14$ ); v627\_hodge\_chamber.py (5 checks: exact transport  $z = P^{-1}y$ ,  $\det S > 0$  on  $67/67$ , one sheet, honest scramble typing); v628\_orbifold\_casimir.py (5 checks: exact closed form, exact  $1/N$  expansion  $\pi(6\nu^2-6\nu+1)/12$ , sector table, twist gap  $1/18$ ); v629\_root\_incidence.py (6 checks: 240-root model, the kill,  $60 = D_{\text{start}}$  free clock orbits, 12  $\sigma$ -fixed roots, controls); v630\_suzuki\_contact.py (4 checks: the literal atom identity, tent structure, the measured non-scalar Galerkin profile, contract typing).
- **Ledger.** New rows E8.CODE.01, PRIME.HODGECONE.01, QGEO.ORBCAS.01, E8.INCIDENCE.01, PRIME.WEIL.CONTACT.01.
- **Scope.** No marker moves; no positivity claim beyond the measured surface; no RH statement; the interacting orbifold statement and the W1 normalization dictionary stay the named residues.

2026-08-02 (XXXVIII) (the prime-shadow audit lands (v625 promoted): the checkable core of “primes are the shadow of the finished geometry” is EXACT —  $\Theta_{E_8} = E_4$  with shell counts  $240 \sigma_3(n)$  (multiplicative over coprime addresses: the ‘address space’ reading IS unique factorization), the primes act as COMMUTING Hecke channels ( $T_p \Theta = (1+p^3)\Theta$  for  $p = 2, 3, 5, 7$ ;  $[T_2, T_3] = 0$ ) with the compiler’s theta as SIMULTANEOUS eigenvector, and  $L(E_4, s) = \zeta(s)\zeta(s-3)$ : the zeta function is the factorized shadow of the  $E_8$  counting function — primes



enter AFTER the geometry; the speculative framings (eigenfrequencies, RH as coherence) stay typed hypotheses)

- **v625\_prime\_shadow.py** (PRIME.SHADOW.01, 6 checks; probe 6/6, PRIME-SHADOW-EXACT). Theta from the glue decomposition  $(\theta_2^8 + \theta_3^8 + \theta_4^8)/2$ ; shells  $240\sigma_3(n)$  exact for  $n = 1..12$ ; multiplicativity census with the non-coprime must-fail ( $\sigma_3(4) = 73 \neq 81$ ); Hecke eigenvalues  $1 + p^3$  and commuting channels;  $L(E_4, 6) = \zeta(6)\zeta(3)$  to  $5 \cdot 10^{-8}$ . Honest scope: classical Jacobi/Hecke facts realized verbatim by the compiler's own objects — the direction of explanation (geometry first, primes as readout) fixed within the narrative.
- **Sync.** Ledger row + note on EXTREV.LATTICE.01; tfpt\_3 (audit block); diary; suite 618  $\rightarrow$  619 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-02 (XXXVII)** (the external-review lattice audit lands (v624 promoted) — **THE FOUR-MARK CLOSURE**: under the  $D_{d+1} \oplus A_{d-1}$  glue architecture three independent selectors force  $d = 4$  (disc-group census, the norm equation  $(d+1)/4 + (d-1)/d = 2$ , the Pythagorean  $(3, 4, 5)$  triple of cycles/marks/carrier) — the two-input presentation reduces to ONE boundary degree  $d = 4$  plus  $\pi$ , conditional on the architecture; **THE ANCHOR BYTECODE** ( $\chi = (t-1)^2(t-2)$ , two-state recursion,  $240+8 = 248$ ); **THE LORENTZ CONGRUENCE**  $P^\top J_{\det} P = J_{\text{fix}}$  EXACTLY (index 6) — the prime-front determinant form and the cover polarization lattice are the SAME rational quadratic form; the review's  $B$ -test retyped ill-posed ( $B$  complex symmetric); and the RH architecture preregistered as PRIME.WEIL.OPERATOR.01 with verified Suzuki citations)

- **v624\_external\_lattice\_audit.py** (EXTREV.LATTICE.01, 14 checks; probe 14/14, LATTICE-AUDIT-PASSED). Disc-group census with negative controls  $d = 3, 5, 8$ ; unique norm-equation solution (no other even target  $2k$ ,  $k \leq 10$ ); Pythagoras at  $d = 4$ ;  $(1, 1, 2)$  unique partition;  $c_3 = 1/(8\pi)$ ; bytecode identities exact; the congruence with  $\det P = -6$ ,  $36 = 6^2$ ,  $6 = p_2(a)$ ;  $B = M^\dagger J$  complex symmetric ( $\det 8$ ) — the literal integer- $R$  test cannot be posed; third  $\mathbb{Q}(\omega)$  ratio an exact square. AX.P2.01 keeps its typing.
- **PRIME.WEIL.OPERATOR.01 (new research contract)**. The W1–W4 chain (exact Galerkin identification with Suzuki's localized Weil operator  $A_a$ ; form density; uniform discrete positivity — the RH-hard step, with the  $C = 1$  contraction reading as candidate mechanism; continuum passage via Yoshida's odd criterion), kill tests K1–K3, citations verified by web lookup (arXiv:2606.09096, arXiv:2607.24830). No RH statement.
- **Sync.** Ledger rows EXTREV.LATTICE.01 + PRIME.WEIL.OPERATOR.01 + notes on AX.P2.01/QGEO.EQORDER.01; tfpt\_research\_contracts (contract block); tfpt\_3 (audit block); diary; suite 617  $\rightarrow$  618 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXVI)** (the covered seam lands (v623 promoted) — the lattice skeleton of the covering-level identification: the seam lifted to the  $\mu_3$ -cover is ONE 48-site NS circle for the uniform weights (the alternating pair DISCONNECTS it — a new independent kill), the lifted clock has EXACT ORDER 12 with  $L^4 = \text{deck}$  ( $r^4 = \omega$  as lattice combinatorics), the v622 kernel identity generalizes verbatim to  $N = 48$ , and the deck-character mode grid is the 12-grid ( $\{1/6, 1/2, 5/6\}$ ) with the untwisted sector equal to the base seam VERBATIM and the twisted pair carrying  $\omega$  on bilinears (the  $t = \omega/\bar{\omega}$  sheets); the fermionic deck lift satisfies  $\text{deck}^3 = -1$  (the spin double))

- **v623\_covered\_seam.py** (QGEO.COVERSEAM.01, 9 checks; probe 9/9, COVERED-SEAM-SKELETON). Connectivity census ( $j = 1, 2$ : one circle; trivial and alternating: 3 components);

the explicit 48-walk (three base turns, 12 crossings);  $L^4 = \text{deck}$  and  $L^{12} = \text{id}$  on all 48 points with proper order 12; the  $N = 48$  mode-sum identity; three 16-mode deck classes with the offset-1/2 class = the base NS frequencies exactly;  $\text{deck}^3 = -1$  on modes, bilinear eigenvalue  $= \omega$ ; the 12 lifted marks in a single 12-cycle. Residue named: the interacting/CFT orbifold statement (twist-field OPE).

- **Sync.** Ledger row + note on QGEO.SEAMID.01; tfpt\_3 (audit block); diary; suite 616  $\rightarrow$  617 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXV) (the seam identification lands (v622 promoted) — THE PHYSICAL SEAM IS THE CONFORMAL SEAM: the v519 chiral seam kernel is EXACTLY the discrete NS (antiperiodic) chiral mode sum on the 16-site circle, including the even-distance zeros (the flat band IS the NS spectrum; the Ramond sum fails as must-fail control); the four mark-bond midpoints sit EXACTLY at  $\mu_4$  under the site map  $\theta_a = (2a + 1)\pi/16$  (no site touches a mark — the v519 “marks at bond midpoints” becomes literal geometry); the group dictionary is exact (clock  $= z \mapsto iz$ , RP reflections = diameter reflections); and the v534 STRADDLE LAW becomes geometry — straddled cuts = exactly the through-mark axes; the second v617 bedrock residue closes at the kernel + dictionary level, leaving the covering-level identification)**

- **v622\_seam\_identification.py** (QGEO.SEAMID.01, 10 checks; probe 10/10, SEAM-IDENTIFIED). Mode-sum identity for all  $d = 1..15$  with the closed form  $\sum_{j<8} \sin((2j+1)x) = \sin^2(8x)/\sin(x)$ ; Ramond must-fail; mark midpoints  $= \mu_4$  exact; reflection axes  $(k+1)\pi/16$  with cuts  $\mapsto \{\pi/4, \pi/2, 3\pi/4, \pi\}$ ; through-mark axes = the straddled cuts  $\{7, 15\}$ ; v617’s bond reflection = cut  $k = 11$  with the v599  $(k, 5-k)$  pattern.
- **Sync.** Ledger row + notes on QGEO.SEAMFORCE.01 and GATE.QGEO.01 (bedrock state after v617/v620/v622; no status move); tfpt\_3 (audit block); diary; suite 615  $\rightarrow$  616 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXIV) (the follow-through round (v619–v621 promoted): the two  $C = 1$  violators are TRUNCATION ARTIFACTS of the prime-data cap (flip set = truncation set exactly, mechanism reproduced by injection at both scales,  $C = 1$  EXCEPTION-FREE on the complete-comb surface); the cyclic- $N$  census makes the bedrock residue CONDITIONAL (seam-admissible primitive-Lorentz orders  $= \{3, 5\}$  exactly,  $N = 6$  reduces to  $N = 3$ , and the compiler’s machine-checked 27/81 pin  $N = 3$  uniquely); and the interacting Klein–Landau semigroup CONTRACTS to the  $\mu_4$ -commensurate steps with the clock surviving as a positive transfer over the whole coupling grid)**

- **v619\_flip\_mechanics.py** (PRIME.FLIPMECH.01, 6 checks; probe 6/6, FLIPS-ARE-TRUNCATION). Atom demand  $u \leq 2\alpha$  vs data cap  $U_{\max} = 12.899$ : exactly the two v618 flip windows are truncated ( $\Delta u = 0.677/0.089$ ); no zero resonance (collective spectrum); injection reproduces both scales ( $-5e-5/ -1e-4$  vs observed  $-1.0e-4$ ;  $-0.18/-0.22$  vs  $-0.29$ );  $C = 1$  with ZERO exceptions on the 67 complete windows (max 0.982). The flip windows are retired as data-boundary artifacts.
- **v620\_cyclic\_n\_census.py** (QGEO.NCENSUS.01, 9 checks; probe 9/9, N3-PINNED). Full  $\infty$ -ramification iff  $\gcd(N, 4) = 1$ ; genus table 1, 3, 3, 6, 7; primitive sheet dims; Chevalley–Weil signatures ( $N = 3$ : both sheets Lorentz;  $N = 6$ : no primitive Lorentz sheet – reduces to  $N = 3$ ); the pinning  $\det_{\mathbb{Z}}(1 - \text{deck}) = N^3$ : 27 (the v597 constant) vs 125; saturation 81 vs 625:  $N = 3$  unique.

- **v621\_interacting\_semigroup.py** (WOIT.GAMMA.SEMI.01, 6 checks; probe 6/6, **CLOCK-SURVIVES-INTERACTING**). Free v524 pattern reproduced at  $N = 16$  (even steps PD, odd steps zero one-particle diagonal); at  $g > 0$  Hermiticity survives exactly on  $\{4, 6, 7\}$  (the  $\mu_4$ -commensurate window; mechanism: the survivor is  $\alpha_4$ -invariant exactly, breaking translation invariance to the quartet stabilizer); the clock step stays PD over  $g \in \{1/32..8\}$ : the reconstructed rotation group of the interacting toy is the clock tower.
- **Sync.** Three ledger rows + notes on PRIME.UNIFC.01, QGEO.SEAMFORCE.01, WOIT.GAMMA.TOY.01; tfpt\_3 (three audit blocks); tfpt\_prime\_front (flip-mechanism sentence); diary; suite 612  $\rightarrow$  615 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXIII)** (the three-front round (v616–v618 promoted): the internal **SU(2) SEPARATES** kinematically (kill test (6)’s first slice —  $\rho_{\text{tw}} = \text{right-}j$  verbatim, the clock factorizes as  $\text{left-}u \circ \text{right-}u$  exactly, the deck is spacetime-side, and the complex choice breaks the internal factor to  $U(1)$  with the mark torsor as its  $\mu_4$ ); the conformal seam axioms **FORCE** the  $\mu_3$ -cover ( $\mathbb{Z}_4$ -Möbius rigidity + uniform weights + genus 3 + the seam reflections matching v599’s pattern — the bedrock residue narrows to the cyclic-3 choice and the physical  $\leftrightarrow$  conformal identification); and the equidistribution constant  $C = 1$  is **FROZEN** on the declared surface with the sharpened sign-flip dichotomy (violator set = flip set exactly; honest correction of the XXXIII note))

- **v616\_su2\_internal\_kinematic.py** (WOIT.SU2.KIN.01, 17 checks; probe 17/17, **SU2-KINEMATIC-SEPARATED**). Right-sp(1) fiber-preserving, commutes exactly with the spacetime quaternion action, intersection  $\{0\}$ ;  $\text{RHO4} = \text{left-}e^{i\pi/4} \circ \text{right-}e^{i\pi/4}$  exact;  $\text{DECK4} = \text{left-}i$ ; internal  $SU(2) \rightarrow U(1)$  by the complex choice;  $\sigma_{\text{std}}$  fixes right- $j$  and inverts the internal charge. Kill test (6)’s kinematic branch does not fire; the dynamical half stays live on  $\mathcal{A}_{\text{hol}}$ .
- **v617\_seam\_cover\_forcing.py** (QGEO.SEAMFORCE.01, 14 checks; probe 14/14, **SEAM-FORCES-COVER**). Order-4 Möbius multiplier forced primitive; harmonic cross-ratio orbit  $\{2, 1/2, -1\}$  (must-fail:  $4/3$  not in it); weight census  $81 \rightarrow 4 \rightarrow 2$  (mark equivalence kills the alternating pair);  $y^3 = x^4 - 1$  exact; Riemann–Hurwitz genus 3; bond reflection = the v599  $(k, 5-k)$  pattern with the two v519 cut points.
- **v618\_uniform\_constant.py** (PRIME.UNIFC.01, 6 checks; probe 6/6, **UNIFORM-C1**). 69 windows,  $q_{\text{model}}$  one-signed on the whole ladder;  $\varepsilon h \leq 0.982$  on all 67 lock-sign windows ( $C = 1$ ); tertiles 0.61/0.45/0.39; exactly two  $q_{\text{real}}$  sign-flip windows (1219, 1445) = the violator set exactly; scramble breaks by  $> 10^4$ . Honest correction: the XXXIII note’s mechanism (“ $q_{\text{model}}$  crossing”) was wrong and the edge flip 1445 was missed.
- **Sync.** Three ledger rows + notes on WOIT.GAMMA.TOY.01, QGEO.COVER.01, PRIME.LOCKPROJ.01; tfpt\_3 (three audit blocks); tfpt\_research\_contracts (kill-test-(6) slice note); tfpt\_prime\_front ( $C=1$  frozen-module note); diary; suite 609  $\rightarrow$  612 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXII)** (the interacting gamma slice lands at toy level (v615 promoted): the gamma battery executed ON the RP-surviving interaction class (the alignment bit  $\delta = \pi/2$ , positive coupling, v534) — **THE INTERACTING MIRROR FLIP**: at the forced twist  $\eta = +i$  the mirror state’s odd sector is **STRICTLY NEGATIVE DEFINITE** on every cut and every coupling  $g \in \{1/32..8\}$  (bounded away from zero: kill test (3) cannot fire on the surviving class — v608 upgraded from  $g = 0$  to the full grid); the mark transport exists interacting (kill test (7) does not fire); the odd transfer spectrum stays multiplicity-free; and the dead

member has no RP home — the dynamical bit selection and the chirality data CO-LOCATE on the same member)

- **v615\_gamma\_toy\_interacting.py** (WOIT.GAMMA.TOY.01, 11 checks; probe 11/11, GAMMA-TOY-LANDED). Chiral/mirror parents ( $I \pm iC$ , gap 1); v534 survivor anchor reproduced; sector-exact mirror flip (even (29,0,0) PD, odd (0,8,0) ND, max odd eigenvalue  $\leq -1.5 \cdot 10^{-3}$ );  $\eta = -i$  convention control; mark transport  $\alpha_4$  exact 16/16 with Gram and transport form Hermitian PD at every  $g$ ; odd transfer spectrum simple at every  $g$  (honest finding: strong-coupling near-doubling at  $g = 8$  without closing;  $g = 0$  top 1.641438 = numerically the v524 clock top); dead-member control ( $m=2, s=+$ ) indefinite. Kill tests (3)/(6)/(7) stay formally live on  $\mathcal{A}_{\text{hol}}$ ; kill test (6) untouched.
- **Sync.** Ledger row + note on WOIT.GAMMA.FREE.01; tfpt\_research\_contracts ( $\gamma$ -stage note); tfpt\_3 (audit block); diary; suite 608  $\rightarrow$  609 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXI) (the Gauss–Manin transport round lands (v614 promoted) — THE DECK STEP IS THE BOUNDARY MONODROMY: transporting the segment cycles along the rigid quarter-rotation loop, the discriminant  $\lambda(\tau) = e^{2\pi i \tau}$  winds ONCE around 0 and the transported periods factorize EXACTLY,  $I_m(\tau) = e^{i\pi\tau(m+1)/2} e^{-2\pi i \tau j/3} I_m(0)$ ; at  $\tau = 1$  the deck factor is  $\omega = t^4$  on EVERY sheet (the exponent  $4 = |\mu_4|$  forced by the puncture count); the canonical transport matrix in the segment basis is EXACTLY  $\omega M$  — the v613 dictionary  $r \sim \omega M$  is CANONICAL, no conjugator freedom left, and  $(\omega M)^4 = \omega$  re-derives  $r^4 = \omega$ ; v613’s residue F1 closed at the transport level)**

- **v614\_transport\_deck.py** (QGEO.TRANSPORT.01, 12 checks; probe 12/12, TRANSPORT-CANONICAL). Exact rotation substitution identity; uniform deck  $\omega$  across all three omega-sheet rows (must-fail:  $t^3 \neq \omega$ ,  $t^5 \neq \omega$ ); char(transport) = char( $r$ ) exact; 25-digit branch-tracked certificates at  $\tau = 1/3, 1/2, 2/3, 1$  (dps 80, max  $5 \cdot 10^{-28}$ ;  $\tau = 1$  at  $2.7 \cdot 10^{-81}$ ). The reading: the braid fixes the boundary, the rigid rotation does not — their homological difference is one boundary loop of local-system monodromy.
- **Sync.** Ledger row + note on QGEO.PERNORM.01; tfpt\_3 (audit block); diary; suite 607  $\rightarrow$  608 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXX) (the canonical period normalization lands (v613 promoted) — THE ANALYTIC BRIDGE CLOSES: the quarter rotation acts DECK-FREE on the four branch segments (all 12 wrap factors exactly 1), and the exact dictionary is  $r = \text{deck} \circ \text{rotation}$  — char( $r$ ) = char( $\omega M$ ) EXACTLY, explaining  $r^4 = \omega$  structurally, with the deck power unique; the period rows (hol  $dx/y^2$ , hol  $x dx/y^2$ , antihol  $\overline{dx/y}$ ) realize the rotation eigenframe; the Hodge weights are explicit Gamma monomials ( $h_1 = (3\sqrt{3}/16\pi)\Gamma(1/3)^2\Gamma(1/6)^2$ , complex Beta formula); and  $J = -h$  PER CHARACTER: the compiler polarization is MINUS the analytic Hodge form — Riemann–Hodge positivity realized, v612’s named step B5 executed)**

- **v613\_canonical\_periods.py** (QGEO.PERNORM.01, 19 checks; probe 19/19, PERIOD-NORMALIZED). Deck-free rotation on segments (40-digit certificates) with the must-fail control (raw period rows are NOT  $r$ -eigenvectors, residual 0.42 — the naive ansatz falsified before the honest construction); char( $r$ ) = char( $\omega M$ ) exact with explicit conjugator  $G$ ; character match antihol  $\leftrightarrow \zeta_{12}$ ; omega-sheet signature (2, 1) with the unique negative on the antiholomorphic line =  $J$ ’s unique positive line (v612): all three character-matched sign products negative. Residue named: the canonical fork-basis identification (basepoint bookkeeping for  $G$ ).

- **Sync.** Ledger row + notes on QGEO.POLFRAME.01 and QGEO.PERIODS.01; tfpt\_3 (audit block); diary; suite 606 → 607 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIX) (the polarization frame lands (v612 promoted): the braid rotation has three simple eigenvalues  $\zeta_{12} \cdot \{1, -1, -i\}$ , the polarization  $J$  diagonalizes EXACTLY in the rotation eigenbasis (values  $(1 + \sqrt{3})/2$ ,  $(1 - \sqrt{3})/2$ ,  $-1$ ), and THE UNIQUE POSITIVE (HODGE) DIRECTION IS THE PRIMITIVE  $\zeta_{12}$  EIGENSPACE — the analytic Hodge structure and the finite rotation agree on where positivity lives; the canonical period normalization (Beta weights) is the named next step)**

- **v612\_polarization\_frame.py** (QGEO.POLFRAME.01, 9 checks; probe 9/9, POLARIZATION-FRAMED). Simple spectrum,  $\det r = -1$ ; exact diagonalization (honest typing: diagonality follows from J-unitarity, the content is the values and the signature distribution); product of diagonal values =  $\det J / |\det W|^2 = 1/2$  exactly.
- **Sync.** Ledger row + note on QGEO.PERIODS.01; tfpt\_3 (audit block); diary; suite 605 → 606 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVIII) (the periods round lands (v611 promoted): the analytic layer of the cover is CLASSICAL — all five basic period integrals are exact Beta values with Gamma denominators on the  $\zeta_{12}$  grid ( $\Gamma(11/12), \Gamma(7/12), \Gamma(5/12), \Gamma(1/12)$ ); the  $\mu_4$  rotation acts on periods with the exact characters  $i^{k+1}$ ;  $\Omega_E = B(1/3, 1/6)/2$  exactly; and the cross-factor period ratio admits no low-degree integer relation (PSLQ null) — the two CM factors are period-independent, as the v610 splitting demands)**

- **v611\_periods.py** (QGEO.PERIODS.01, 6 checks; probe 6/6, PERIODS-CLASSICAL). Five symbolic Beta identities; three 40-digit rotation-character certificates; the classical  $\omega$ -CM period; the PSLQ-null cross-factor consistency; the period dictionary named as the Hodge/RP resource.
- **Sync.** Ledger row + note on QGEO.LANDSCAPE.01; tfpt\_3 (audit block); diary; suite 604 → 605 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVII) (the curve landscape lands (v610 promoted): the Jacobian SPLITS into a  $\mathbb{Q}(\zeta_{12})$ -CM abelian surface — the field of deck AND marks together — times the  $\mathbb{Q}(\omega)$ -CM elliptic quotient  $E : y^3 = u^2 - 1$ , point-count-confirmed ( $a_p(C) = a_p(E)$  exactly for all ten tested  $p \not\equiv 1 \pmod{12}$ ); the integer-spectrum census honestly relativizes single-char hits (7894 integer entries — single spectra are cheap; the anchor class is the most frequent nontrivial family at 286; the compiler's distinction is the joint structure); two new canonical period resources named)**

- **v610\_curve\_landscape.py** (QGEO.LANDSCAPE.01, 9 checks; probe 9/9, LANDSCAPE-MAPPED). Differential characters (12, 12, 6); splitting confirmed at 14 primes (ten exact zero-differences; surface traces  $(-8, 4, -20, -32)$  at  $p \equiv 1 \pmod{12}$ );  $a_p = 0$  for  $p \equiv 2 \pmod{3}$ ; census: anchor class 286 = max nontrivial,  $\{1, 1, 3\}$  at 254; v598/v599 char hits retroactively confirmed as correctly typed candidates.
- **Sync.** Ledger row + notes on QGEO.COVER.01/QGEO.DICT.01; tfpt\_3 (audit block); diary; suite 603 → 604 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVI) (the translator factorization lands (v609 promoted): the geometric identity of  $c$  is CLOSED — the mirror intertwiner  $d$  ( $dA = \Gamma(A)d$ ) is unique, unimodular ( $\det d = 1$ ),  $\mathbb{Z}[\omega]$ -integral and  $\Gamma$ -involutive ( $\Gamma(d)d = 1$  exactly), and the v605 element factorizes exactly as  $c = \lambda K_1 d$  with  $K_1 = J^{-1} \overline{F_2}^T M^{-1}$ :  $\theta_{\text{corpus}} = J \text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$  — all three v605 residuals are now resolved (seam reversal v603, zero-mode rule v607,  $c$ -identity v609))**

- **v609\_translator\_factorization.py** (QGEO.CFACTOR.01, 8 checks; probe 8/8, **C-FACTORIZATION-LANDED**). Intertwiner space 1-dim;  $\det d = 1$ ;  $d \in M_3(\mathbb{Z}[\omega])$ ;  $\Gamma(d)d = 1$ ;  $c = \lambda K_1 d$  exact ( $\lambda = 1/7 - 3\sqrt{3}i/7$ );  $\text{inner}(K_1 d)$  reproduces  $\varphi$  on the pair; honest scope:  $F_2$  carries the corpus datum by definition; the v606 arithmetic negative explains why the cyclotomic route had to fail.
- **Sync.** Ledger row + notes on QGEO.TRANSLATOR.01/QGEO.MODESEP.01; tfpt\_3 (audit block); diary; suite 602  $\rightarrow$  603 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXV) (the free gamma slice lands (v608 promoted) — the first executed step of the WOIT  $\gamma$  milestone: mirror modes are STRICTLY OS-negative sector-exactly (every mirror fermion vector has negative OS norm — no mirror survives reconstruction, kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly  $\{1, \sqrt{2} - 1\}$ ), and the mark transport exists ( $\tau_2$  is the mark-A $\rightarrow$ B transport, PD — kill test (7) does not fire free);  $\gamma$  itself stays open on  $\mathcal{A}_{\text{hol}}$ ; side finding:  $14c$  is unipotent mod  $(1 - \omega)$ )**

- **v608\_gamma\_free\_slice.py** (WOIT.GAMMA.FREE.01, 9 checks; probe 9/9, **GAMMA-FREE-SLICE-LANDED**; the v524 machinery imported read-only). Chiral  $(16, 0, 0)$ ; anti-chiral  $(8, 8, 0)$  with even  $(8, 0, 0)$  PD and odd  $(0, 8, 0)$  strictly ND; clock spectrum  $\{1, \sqrt{2} - 1\}$  multiplicity-free on the fermion domain; mark algebras PD  $(4, 0, 0)$  with  $\dim 4 = |\mu_4|$ , products generate the full 16-dim basis,  $\alpha_2$  transports mark A to mark B.
- **Sync.** Ledger row + notes on WOIT.OS.TWISTOR.01/WOIT.BETA2.OS.01; tfpt\_research\_contracts ( $\gamma$ -stage free-slice note); tfpt\_3 (audit block); diary (incl. the  $c$ -unipotency side finding); suite 601  $\rightarrow$  602 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIV) (the selection rule lands (v607 promoted): the infinity torsion is a genuine  $\mathbb{F}_3$  space, the deck rotation acts with char  $(x + 1)(x^2 + 1) \bmod 3$  — EXACTLY ONE projective eigenline, and it IS the top mode; the three modes carry mutually exclusive canonical labels (seam =  $R$ -real, top = deck-eigen, zero = the only unlabeled) and  $S_0$  flips exactly the labeled pair: BOTH v590 demands expressed through cover structure (typed reading; the gate residual is the continuum derivation))**

- **v607\_selection\_rule.py** (QGEO.SELRULE.01, 10 checks; probe 10/10, **SELECTION-RULE-LANDED**).  $\omega$ -multiplication acts as the identity on the torsion; the induced  $r$ -action is  $[[0, 1, 0], [1, 0, 2], [2, 2, 2]] \bmod 3$  with only fixed class 0;  $x^2 + 1$  irreducible over  $\mathbb{F}_3$  gives structural uniqueness of the eigenline ( $\lambda = -1$ ), census-confirmed; the top-mode class is that eigenline, the zero mode and the seam are not projectively fixed (nor under  $r^2$ ).
- **Sync.** Ledger row + notes on QGEO.MODESEP.01/QGEO.INVOL.01; tfpt\_3 (audit block); diary; suite 600  $\rightarrow$  601 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIII) (the mode-separation round lands (v606 promoted — the suite reaches 600 scripts): the infinity torsion SEPARATES the three  $V$ -eigenlines**

(all pure 3-torsion at infinity, classes pairwise distinct — the selection carrier the real structure could not provide; the canonical “+1” rule stays open); the arithmetic-identity route for  $c$  is closed (cubic field disc  $-783$ , Galois  $S_3$ , not abelian); the equidistribution conjecture is posed with the explicit constant  $C = 1$  in the prime-front paper; and the contract survey executes its bookkeeping (the disc-7 preregistration’s discrete half marked executed via v571/v575/v578; the WOIT  $\gamma$  milestone documented as the designed next step))

- **v606\_mode\_separation.py** (QGEO.MODESEP.01, 7 checks; probe 7/7, MODES-SEPARATED). Unique intertwiner (rank 3, 1-param); all three eigenline lifts pure torsion; fixed-frame signatures  $(2, 0, 2)/(2, 2, 0)/(1, 2, 0)$ , pairwise distinct, none trivial;  $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$  irreducible with discriminant  $-783 = -3^3 \cdot 29$  non-square.
- **Sync.** Ledger row + notes on QGEO.TRANSLATOR.01/QGEO.SEAMMARKS.01 and the cross-reference on FTR.DISC7.NORM.01; tfpt\_prime\_front (the constant-1 statement in sec:theory-open); tfpt\_3 (audit block); diary; suite 599  $\rightarrow$  600 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXII)** (the audit-and-measurement close of the day: the gate audit documents the cover program as MODEL-COMPLETE on the operator half (ledger note on GATE.QGEO.01, no status move — the residual is the conformal seam identification, the zero-mode selection, and the identity of  $c$ ); honest negatives quantified ( $c$  without independent geometric identity; the zero-mode line is not  $R$ -real); tfpt\_1 gains the consolidated “cover program at a glance” paragraph; and the prime-front uniformity sharpens:  $\varepsilon \cdot h \leq 0.98$  on all 68 floor-passed windows —  $|q_{\text{real}}/q_{\text{model}}| \leq h^{-1}$  uniformly on the declared surface (note on PRIME.LOCKPROJ.01))

- **No new modules.** Sandbox measurements and typed notes only:  $c^2$ ,  $c\Gamma(c)$ ,  $\Gamma(c)c$  outside the word algebra (curiosities documented:  $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$  integer monic;  $|\det(14c)|^2 = 28^3$ );  $\ker V$  not  $R$ -real and no  $0 \leftrightarrow 2$  swap; tercile medians of  $\varepsilon \cdot h$ : 0.61/0.44/0.39 (stable, slightly decreasing); the single blow-up ( $h = 1219$ ) is exactly the declared  $q_{\text{model}}$  zero-crossing exclusion.
- **Sync.** Ledger notes on GATE.QGEO.01 and PRIME.LOCKPROJ.01; tfpt\_1 consolidation paragraph; diary; suite unchanged (599, green); mirrors regenerated.

**2026-08-01 (XXI)** (the translator round lands (v605 promoted): the corpus RP anti-automorphism IS the J-adjoint twisted by an explicit inner element  $c$  ( $\varphi = J_{\text{adj}} \circ \theta_{\text{pull}}$  is inner,  $c$  unique up to scale,  $14c$  integral over  $\mathbb{Z}[\omega]$  — the single-sheet negatives resolved, the  $\bar{\omega}$  detour unnecessary); the seam is a NONZERO pure 3-torsion class at infinity (torsion coordinates  $(2, 2, 0)$ ); the 81 survives frame-joining as the saturation defect (chain indices  $3^5, 3^4$ ); the  $J_{\text{fix}}$  duality certified with a Lean addendum)

- **v605\_translator.py** (QGEO.TRANSLATOR.01, 11 checks; probe 11/11, TRANSLATOR-LANDED).  $\varphi(A) = cAc^{-1}$  exact (1-dim intertwiner space,  $c$  invertible);  $14c$   $\mathbb{Z}[\omega]$ -integral (denominator  $14 = 2 \cdot 7$ ); seam lift zero in the free part,  $(2, 2, 0)$  in the torsion of  $\text{coker}(1 - T_\infty)$ ; order chain  $O_1 \subset O_{\text{joint}} \cap \text{span}_1 \subset P_1$  with indices 243 and 81;  $C_V^T J_{\text{fix}} = J_{\text{fix}} C_{V_{\text{dual}}}$  integer;  $c$ ’s independent geometric identity typed open.
- **Lean.** CoverEmbedding.lean addendum: Jfix\_symmetric, Jfix\_det = 72, CV\_dual\_certificate, CVdual\_cubic, dual\_joint\_trace; lake build green (3375 jobs).



- **Sync.** Ledger row + notes on QGEO.EQORDER.01/QGEO.SEAMMARKS.01/QGEO.DUALFORMS.01; tfpt\_3 (audit block); diary; suite 598 → 599 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XX) (the equivariant-order round lands (v604 promoted):**  $R$  IS an anti-isometry of the polarization ( $M^\dagger JM = J^T$  exact — correcting a diary side remark),  $J$  restricts to the integer form  $J_{\text{fix}}$  with  $\det = 72 = 8 \cdot 9$ ; the four mark frames generate a FULL order of index  $27 = 3^3$  in  $M_3(\mathbb{Z}[\omega])$ ; and the 3-torsion localizes at infinity ( $\text{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$ , finite cusps torsion-free): 81 at the marks, 27 at infinity — the integral infinity/order-torsion bridge; translator negatives extended (the  $\bar{\omega}$ -sheet named))

- **v604\_equivariant\_order.py** (QGEO.EQORDER.01, 12 checks; probe 12/12, **EQUIVARIANT-ORDER-LANDED**). Anti-isometry exact;  $J_{\text{fix}} = [[16, 2, 4], [2, -2, 2], [4, 2, -2]]$ ; all 28 rotated words  $\mathbb{Z}[\omega]$ -integral, joint lattice full rank 18, Smith divisors  $(1^{15}, 3, 3, 3)$ ;  $\text{coker}(1 - T_\infty)$  divisors  $(1, 1, 1, 3, 3, 0, 0)$ ; each finite cusp twist torsion-free; three objects share  $(\mathbb{Z}/3)^3 = 27$  (infinity monodromy, equivariant order,  $\det(1 - \text{deck})$ );  $F_2$  vs  $J_{\text{fix}}$ -menu and algebra-twist negatives quantified.
- **Sync.** Ledger row + notes on QGEO.SEAMMARKS.01/QGEO.COVER.01; tfpt\_3 (audit block); diary (incl. the exploration round XX:  $\infty$ -embedding,  $R$ -orbit classes, translator hunt); suite 597 → 598 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XIX) (the seam-and-marks round lands (v603 promoted):** the seam line is a REAL line of the cover (unique common eigenline,  $R$ -fixed) and  $S_0$  reverses it — the v590 seam-reversal demand holds geometrically; the infinity cusp separates in the unreduced Burau module; the first per-mark  $\mathbb{Z}/3$  witness lands ( $E_1$ , class order 3 in the word-order cokernel); the corpus-transpose translator negatives are quantified)

- **v603\_seam\_marks.py** (QGEO.SEAMMARKS.01, 15 checks; probe 15/15, **SEAM-REAL-LANDED**). Seam:  $Uv = 0$ ,  $Vv = v$  unique,  $v = (0, (\sqrt{3} - i)/(\sqrt{3} + i), 1)$ ,  $Rv \sim v$ ,  $S_0v = -v$ ; infinity: unreduced Burau braid relations, invariant column  $(1, 1, 1, 1)$ , full twist eigenvalues  $\{1, \omega^{\times 3}\}$ ; marks:  $E_1$  integer restriction, in the word algebra with denominator lcm 3 (order-3 cokernel class);  $E_4$  trivial,  $E_2/E_3$  outside (naive global map fails, equivariant framing typed); translator:  $F_2 = g^T \Sigma g$  matches no canonical menu form (all typed open).
- **Sync.** Ledger row + notes on QGEO.DUALFORMS.01/QGEO.INVOL.01; tfpt\_3 (audit block); diary; suite 596 → 597 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVIII) (the duality-and-forms round lands (v602 promoted):** the J-adjoint exchanges the two parabolic types exactly ( $(U_1, V_1^J) \sim \text{compiler pair}$ ); the dictionary completes in the J-projector language (the corpus anchor formula  $H = 1 + V(V - 1)/2$  holds verbatim on the curve); Weil positivity holds exactly (the Riemann bilinear relations on the  $\omega$ -sheet); the canonical bilinear form  $B = M^\dagger J$  is symmetric with  $\det B = 8$ ; the v600 integer core is Lean-certified (FORM.QGEO.04); the contracts paper gains the  $\mu_3$ -cover model paragraph)

- **v602\_duality\_forms.py** (QGEO.DUALFORMS.01, 17 checks; probe 17/17, **DUALITY-FORMS-LANDED**).  $U_1$  J-self-adjoint,  $V_1^J$  the exact product reversal,  $(U_1, V_1^J)$  conjugate to the compiler pair;  $H$ ,  $Q$ ,  $Q_+$ , ladder all exact in the J-projector language; Weil-twisted form positive definite ( $\{2\sqrt{2} - 2, 2, 2 + 2\sqrt{2}\}$ ); honest negatives:  $g^T g \neq \lambda B$ , naive mark-local map fails, seam normal not a deck eigenvector (all typed open).

- **Lean (FORM.QGEO.04).** `TfptCarrier/CoverEmbedding.lean`: the v600 integer core via determinant-free Smith certificates ( $W = TS$ ,  $PT = \text{diag}(1, 1, 1, 3, 3, 3, 3)$  with unimodular  $P$ ,  $S$ -primitivity via unimodular minor); kernel `decide`; lake build green (3375 jobs).
- **Sync.** Ledger row + notes on QGEO.CANON.01/QGEO.EMBED.01; `tfpt_research_contracts` ( $\mu_3$ -cover model paragraph after the bedrock chain), `tfpt_3` (audit block); diary; suite 595  $\rightarrow$  596 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVII) (the canonicity round lands (v601 promoted): the J-adapted cusp projectors give a manifestly  $D_4$ -equivariant presentation of the pair (mark choice = winding choice), which realizes the DUAL parabolic exactly (conjugate to the transposed compiler pair; the direct intertwiner is singular); the v590 Klein four-group lives integrally on the curve; the Weil-twisted RP comparison and the seam identification stay open)**

- `v601_equivariant_dual.py` (QGEO.CANON.01, 14 checks; probe 14/14, **EQUIVARIANT-DUAL-LANDED**).  $GJ_k = (1 - T_k)J^{-1}(1 - T_k)^\dagger J$  idempotent rank-1 J-projectors rotating exactly under  $D_4$ ; pair family  $U_k = 3GJ_k$ ,  $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ :  $\mathbb{Z}[\omega]$ -integral, compiler-exact, equivariant by construction; duality: conjugate to  $(U_c^T, V_c^T)$  exactly (must-fail to  $(U_c, V_c)$  documented); Klein group: all eight spectral-sign involutions integral,  $S_0$  explicit,  $AB = S_0$ ; honest: naive J-word-Gram inertia  $(3, 2, 2) \neq (4, 0, 3)$  (Weil twist named open).
- **Sync.** Ledger row + note on QGEO.EMBED.01; `tfpt_1/tfpt_2` (canonicity clauses), `tfpt_3` (audit block); diary; suite 594  $\rightarrow$  595 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVI) (the joint embedding lands (v600 promoted): the compiler pair  $(U, V)$  is REALIZED on the  $\mu_3$ -cover — a  $\Gamma$ -even degree-2 pair,  $\mathbb{Z}[\omega]$ -integral, seven-word rank 7, simultaneous conjugation to the compiler pair solved exactly, integer restrictions on the saturated  $\text{Fix}(R)$  lattice, and saturation index  $81 = 3^4$  with the exact v566 divisor pattern; canonicity typed open, the gate does not move)**

- `v600_joint_embedding.py` (QGEO.EMBED.01, 15 checks; probe 15/15, **JOINT-EMBEDDING-LANDED**). The pair  $U^* = -2E_1 + (G_1\Gamma(G_1) + \Gamma(G_1)G_1) + 3$ ,  $V^* = -E_3 + \frac{1}{2}(G_2\Gamma(G_2) + \Gamma(G_2)G_2) + 1$ : chars and joint traces compiler-exact ( $\text{tr}(UV) = 3$ ,  $\text{tr}(UV^2) = 6$ ,  $\text{char}(U+V) = \text{char } Q$ ); seven words independent; explicit invertible  $g$  with  $gU^*g^{-1} = U_c$ ,  $gV^*g^{-1} = V_c$ ; the real structure integer on  $\mathbb{Z}^6$  with explicit saturated fixed-lattice basis (SNF certificate); integer restrictions; Smith divisors  $(1, 1, 1, 3, 3, 3, 3)$ , index 81 — the v566 self-code order structure on the curve. Must-break control passes; canonicity open (existence claim only).
- **Sync.** Ledger row + notes on QGEO.REAL.01, QGEO.DICT.01, QGEO.COVER.01 and DIAMOND.PARA.SELFCODE.01; `tfpt_1/tfpt_2` (realization sentences), `tfpt_3` (audit block); diary; suite 593  $\rightarrow$  594 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XV) (the real structure lands (v599 promoted): the anti-holomorphic  $D_4$  reflection on the  $\mu_3$ -cover exists and is unique up to scale with  $R^2 = +1$ ; its grading functor closes all three dictionary gaps at char level ( $V, H$  with the anchor spectrum  $(1, 1, 2)$ ,  $Q_+$ ); the  $\Gamma$ -decomposition carries 5, 9, 27; the joint algebra embedding stays open (quantified negative); and the invariant form has  $\det J = 8$  — a sheet-diamond determinant-list member — with Lorentz signature**

(1, 2))

- **v599\_real\_structure.py** (QGEO.REAL.01, 18 checks; probe 18/18, **REAL-STRUCTURE-LANDED**). Existence and uniqueness of  $R(v) = M \bar{v}$  (1-dimensional intertwiner space,  $R^2 = +1$  exact, four twist intertwinings  $RT_k R^{-1} = T_{5-k}^{-1}$ ); the grading functor  $\Gamma(A) = M \bar{A} M^{-1}$  locates  $V \sim G_2 + G_3 - \Gamma(G_3)$ ,  $H \sim 1 + G_2 - \Gamma(G_3)$  (the anchor spectrum) and  $Q_+ \sim 1 + G_2 + G_3 - \Gamma(G_3)$  exactly at char level; the  $\Gamma$ -even/odd cusp elements carry  $\{0, 1, 5\}$ ,  $\{0, 3, 9\}$ ,  $\pm\sqrt{5}$ ,  $\pm\sqrt{27}$  and one nilpotent; the joint negative quantified ( $4 \times 6$  candidates, zero pairs pass  $\text{tr}(UV) = 3$  and  $\text{char}(U + V) = \text{char } Q$ ; kill not triggered); the invariant hermitian form is unique with  $\det J = 8$  and signature (1, 2) — Strategy II's Hodge prediction at form level, pinned.
- **Sync.** Ledger row + notes on QGEO.DICT.01/QGEO.COVER.01; tfpt\_3 (audit block); diary; suite 592  $\rightarrow$  593 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XIII) (the cover dictionary reaches THREE exact entries (v598 promoted):  $U \sim G_2$ ,  $Q_- \sim G_1 - G_3$  — the  $\mu_4$  sheet coupling is the diamond-diagonal Gram difference — and the binary AnchorLadder  $(1, 2, 4) \sim 1 + G_1 + G_3$ ; the grading half is absent from 7296 canonical combinations, with the real structure as the named suspect)**

- **v598\_cover\_dictionary.py** (QGEO.DICT.01, 4 checks; probe 4/4, **DICT-THREE-ENTRIES**). All three entries exact (numeric prefilter, symbolic confirmation); supporting structure: opposite-twist products nilpotent,  $Q_-^2 \sim x(x-3)^2$ ; negatives quantified (7296 combinations, 0 matches for  $Q_+/V/H/Q$ ); the Strategy-I kill criterion not triggered; GATE.QGEO does not move.
- **Sync.** Ledger row + note on QGEO.COVER.01; tfpt\_1/tfpt\_2 (dictionary sentences updated), tfpt\_3 (audit block); diary; suite 591  $\rightarrow$  592 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XII) (dictionary round 2 honest negative (the  $Q = U + V$  double criterion over  $\sim 50$  canonical elements: no  $Q/V$  hit; kill not triggered, the flag-adapted search is the designed next session); the real-structure map documented as blocked on the dictionary; and the self-contained equidistribution problem statement added to tfpt\_prime\_front (sec:theory-open) — the v596 identity family posed TFPT-free for external consumption)**

- **No module.** Exploration and editorial only: ledger untouched except mirrors; the one dictionary entry stays  $U = G_2$  (2026-08-01 XI); the open problem now has a publication-grade statement (explicit couplings  $c_\gamma$ , closed constant  $K(\alpha, D)$ , the observed collective cancellation, and the uniformity question named as the open content).

**2026-08-01 (XI) (the dictionary's first entry: U IS LOCATED — the winding operator appears as a cusp Gram on the  $\mu_3$ -cover; exploration round, documented as a note on QGEO.COVER.01)**

- **The hit.** The cusp Gram  $(1 - T_2)(1 - T_2)^\dagger$  has characteristic polynomial  $x^2(x-3) = \text{char}(U)$  exactly — the rank-one winding operator (trace 3 =  $N_{\text{fam}}$ ) as a Gram of a  $D_4$ -conjugate cusp twist: the fifth fingerprint. Additional exact structure: opposite-twist products  $(1 - T_1)(1 - T_3)$  are nilpotent ( $x^3$ ),  $G_1 + G_3$  has spectrum  $\{0, 1, 3\}$ . Honest negative:  $V/H/Q$  not located in the scanned canonical ranges (products/sums/ $\omega$ -twists of length  $\leq 2$ ) — the  $V$ -dictionary likely requires the parabolic flag (the seam-cusp filtration); named next step; kill criterion unchanged.
- **Sync.** Ledger note on QGEO.COVER.01; diary; mirrors regenerated.

**2026-08-01 (X) (Strategy I/II executed: the  $\mu_3$ -cover  $y^3 = x^4 - 1$  hits FOUR compiler fingerprints (v597 promoted) — rank-3 Eisenstein  $H_1$ , projective  $D_4$  via the deck, cusp-class group  $(\mathbb{Z}/3)^4$  of order 81, and the unique invariant hermitian form of LORENTZ signature (1, 2) matching Paper II’s polarization: the first geometric candidate that could close GATE.QGEO and the RP slot with one construction)**

- **v597\_mu4\_cover\_model.py** (QGEO.COVER.01, 6 checks; probe 6/6, FINGERPRINTS-HIT). Genus 3, five cusps ( $\mu_4 + \infty$ ), Chevalley–Weil  $H^{1,0} = (1, 2)$ ; Burau at  $t = \omega$  with verified braid relations,  $r^4 = \omega \cdot 1$ ,  $r^{12} = 1$ ; cusp classes  $(\mathbb{Z}/3)^5$ /one relation  $= (\mathbb{Z}/3)^4$ , order  $81 = 3^4$  with the per-mark  $\mathbb{Z}/3$  mechanism ( $\det_{\mathbb{Z}}(1 - \text{deck}) = 27$  complementarily); invariant form unique up to scale, eigenvalues  $\{-2, 2-2\sqrt{2}, 2+2\sqrt{2}\}$ , signature (1, 2); single-cusp Gram spectrum  $\{0, 0, 2\}$  all-integer. OPEN and named: the  $Q/V$  operator dictionary, the real-structure/RP map to the v572 Gram data; kill criteria live, not triggered; GATE.QGEO does not move.
- **Sync.** Ledger row + notes on QGEO.INVOL.01 and DIAMOND.PARA.SELFCODE.01; tfpt\_1/tfpt\_2 (candidate paragraphs), tfpt\_3 (audit block); diary; suite 590  $\rightarrow$  591 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (IX) (the C-D-E round: the 1D lock projection (v596 promoted, T199 — the near-identity along the closed direction, median  $7.7 \times 10^{-4}$ , coupling spectrum collective); the v590 involution group formalized in Lean with its Klein-four closure  $A \cdot B = S_0$ ; and the big-picture strategy survey (the open core funnels into GATE.QGEO + the RP slot; two new geometric strategies preregistered))**

- **v596\_lock\_projection.py** (PRIME.LOCKPROJ.01, 6 checks; T199 6/6, NEAR-IDENTITY-1D). Along the closed v591 direction (parameter-free):  $|q_{\text{real}}/q_{\text{model}}|$  median  $7.7 \times 10^{-4}$ , decay  $h^{-1.01}$ ; the coupling spectrum is collective (max single-zero share 5.9%,  $N_{50} = 56$ , heavy zeros 14.1/25.0/30.4/40.9); 400-zero truncation fails (v589 plateau); scramble breaks by six orders — the arithmetic content of Problem 7.1 is ONE explicit identity family, equidistribution-flavored.
- **Lean.** A, B, klein\_closure ( $A \cdot B = S_0$ ), involutions and  $V$ -commutation added to ParityWeightLaws.lean; lake build green (3374 jobs).
- **The big-picture survey (E).** The gate/contract census shows the open core is SMALL: GATE.METRIC.19 funnels the whole  $G_{\text{net}}$  line into GATE.QGEO + standard AQFT assembly; GATE.QGEO itself is narrowed (v574/v590) to the parabolic  $\mu_4$  realization; the RP slot is the one dynamical open; absolute scales and external interfaces are typed contracts. Two new strategies preregistered in the diary: (i) the  $\mu_4$ -cover homology model ( $H_1$  of the cyclic covers  $y^4=x^4-1$  /  $y^3=x^4-1$  as  $\mathbb{Z}[i]$ - resp.  $\mathbb{Z}[\omega]$ -modules of rank  $3/2$  — the compiler  $Q$  as a canonical homology operator; testable by finite computation); (ii) RP positivity as HODGE positivity (the seam kernel as the polarization form of the cover’s Hodge structure — the Riemann bilinear relations would make positivity a theorem; testable against the v572 Gram data via exact Beta-function periods).
- **Sync.** Ledger row + notes; tfpt\_prime\_front (48  $\rightarrow$  49 modules, sec:v596/sec:t199, 199 probes / 4944 checks); Paper II (“thirty modules”); tfpt\_3; diary; suite 589  $\rightarrow$  590 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VIII) (the 0-1-2-3-4-5 round: repository pushed (v576–v594, 111 files); the residual bookkeeping executed (v595 promoted, T198: the v593 residual was the  $\sigma$  MAPPING — identity-level closed forms 1.00000–1.00017, honest amendment of the v593 attribution); derivation sketches added to the theorem**

section; the determinant law formalized in Lean (`det_S_closed`, build green))

- **v595\_mapping\_completion.py** (`PRIME.MAPCLOSE.01`, 4 checks; **T198** 4/4, **MAPPING-CLOSES**). The exact dictionary is  $\sigma = d/N$ , hence  $a_{\text{eff}} = ND/2$  with limits  $[u_0/(ND), 2\alpha/(ND)]$  ( $ND = 2\alpha + O(D)$  — why the  $\alpha$ -form carried a first-order residual); entries AND det match at 1.00000–1.00017 (1.00000 at  $h = 1445$ ); kernel piece  $9 \times 10^{-6}$  (= the genuine  $O(1/N^2)$ ); the v593 numbers stand, its residual attribution is superseded (note on `PRIME.CUTOFF.01`).
- **Manuscript-grade theory section.** `sec:theory-proofs` added to `tfpt_prime_front`: derivation sketches for all seven theorems (Dirichlet sums and branch merge; geometric ladders; factorization/separability; the direction and determinant laws; the exact dictionary; partial summation + Bütthe).
- **Lean.** `det_S_closed` and `det_denominators_factor` added to `ParityWeightLaws.lean` (the closed determinant law as a field identity; the  $E^2$  cancellation structural); full lake build green (3374 jobs).
- **Repository.** Commit 0701e1e pushed (v576–v594, 111 files, +19133 lines); the remaining hard opens (determinant-level zero theorem; OS/RP; the  $\mu_4$  geometric model) stay documented.
- **Sync.** Ledger row + amendment note; `tfpt_prime_front` (47  $\rightarrow$  48 modules, `sec:v595/sec:t198`, 198 probes / 4938 checks); Paper II (“twenty-nine modules”); `tfpt_3`; diary; suite 588  $\rightarrow$  589 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VII)** (the full follow-up slate lands: the cutoff completes the density layer **END-TO-END** (v593, T196), the **FIRST UNCONDITIONAL STATEMENT** via Bütthe 2018 (v594, T197), the consolidated theorem section in `tfpt_prime_front`, and the Lean-4 formalization of the round’s algebraic cores (`ParityWeightLaws.lean`, builds green against Mathlib))

- **v593\_cutoff\_completion.py** (`PRIME.CUTOFF.01`, 4 checks; **T196** 4/4, **CUTOFF-CLOSES**). The v592 finite- $N$  det gap (0.19–0.50) was the integration cutoff:  $\sigma_0$ -corrected closed integrals match entries to 0.7–1%, det to 0.2–1%; residual  $O(1/N^2)$ , shrinking — the density layer of Problem 7.1 is analytically closed end-to-end.
- **v594\_unconditional\_cert.py** (`PRIME.UNCONDCERT.01`, 5 checks; **T197** 5/5, **verdict UNCONDITIONAL ENTRY CERT**). Partial summation (machine-verified) +  $|\psi(x) - x| < 0.94\sqrt{x}$  (Bütthe 2018, unconditional to  $10^{19}$ ; external data)  $\Rightarrow \sup |\text{Osc}| \leq 17.4$  unconditional (measured 1.474); certificate margin  $565\times$  — the first unconditional statement of the program; the determinant level stays with the zero-oscillation theorem.
- **Theory section + Lean.** `tfpt_prime_front` gains `sec:theory` (“The closed-form theory: seven statements”, Theorems 1–7 with module and Lean citations); new Lean-4 module `ParityWeightLaws.lean` (branch merge, rank-one determinant, null vector, direction law + approach, sheet involution  $S_0^2 = 1/[S_0, V] = 0/\text{tr} = -1$ ) — full lake build green (3374 jobs).
- **Sync.** Ledger rows + notes on `PRIME.DETLAW.01` and `PRIME.ZEROCOMB.01`; `tfpt_prime_front` (45  $\rightarrow$  47 modules, `sec:v593/sec:v594/sec:t196`, 197 probes / 4934 checks); Paper II (“twenty-eight modules”); `tfpt_3`; diary; suite 586  $\rightarrow$  588 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VI)** (the continuum determinant law (v592 promoted, T195):  $\det S(a)$  closed with the  $e^{2a}$  coefficient identically zero and exact leading law  $640\pi^2 e^a/a^3$ ; the real dominance tracks the closed formula at correlation 0.9985 (the v570 growth derived); entry-level arithmetic certificate with margin  $47.6\times$ ; **PLUS** the

## Paper-II consolidation section (the deterministic–arithmetic dissection as one narrative))

- **v592\_continuum\_det\_law.py** (PRIME.DETLAW.01, 6 checks; probe T195 6/6, DET-LAW-DERIVED).  $e^{2a}$  coefficient  $\equiv 0$  (rank-one);  $e^a$  term with exact closed coefficient carries the determinant at window scales (0.995–1.0002); leading law converges monotonically; dominance correlation 0.9985; finite- $N$  gap typed (0.19  $\rightarrow$  0.50); entry certificate  $\sup |\text{Osc}| \times \text{TV}$ , margin  $47.6\times$ , conditional on the measured sup (1.47) — the unconditional sup IS the zero-oscillation theorem.
- **Paper-II consolidation (follow-up 5)**. New subsection “The deterministic–arithmetic dissection (machine survey)” before the cross-reference table: the v582–v592 arc as one narrative (two layers; the density layer closed-form with the rank-one pole term and the two laws; the arithmetic layer identified as the zero comb with a structural recovery plateau; the two named remaining opens). Follow-ups 2/3/4 stay open with reasons documented (no forced modules).
- **Sync**. Ledger row + notes on PRIME.PENCIL.SEPFLOOR.01 and PRIME.POLERANKONE.01; **tfpt\_prime\_front** (44  $\rightarrow$  45 modules, **sec:v592/sec:t195**, 195 probes / 4925 checks); Paper II (“twenty-six modules” + dissection section); **tfpt\_3**; diary; suite 585  $\rightarrow$  586 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (V) (the pole term is EXACTLY rank-one and the locking-direction law is derived (v591 promoted, T194):  $S^{\text{pole}} = -32\pi^2 a e^a g g^T$  with  $\det \equiv 0$  symbolically;  $v_2/v_1 = -(a^2 + 16\pi^2)/(2(a^2 + 4\pi^2))$  with limit exactly  $-1/2$  — the v577 null-ray conjecture DERIVED, the v586 drift explained; plus the honest B/C rounds: deeper zero sums plateau (structural, note on v589) and the grading shortcut for the involution demands is falsified (note on v590))**

- **v591\_pole\_rank\_one.py** (PRIME.POLERANKONE.01, 5 checks; probe T194 5/5, POLE-RANK-ONE). Denominators factor completely ( $D_{11}D_{22} = D_{12}^2$ ); pole part exactly separable, residuals exponential-free; direction law with limit  $-1/2$  and approach  $-6\pi^2/(a^2 + 4\pi^2)$ ; ladder match 0.04–1.4% ( $h = 1445$ :  $-1.2248$  vs  $-1.2253$ ); via v586 the law describes the real corpus locking direction; the density layer = the correction theory of a rank-one matrix.
- **The B and C rounds, honest**. Deeper zero sums (500/1000 zeros,  $\gamma \leq 1419$ ) do NOT improve the depth recovery (plateau  $\sim 31$ – $37$  at  $h = 184$ ; entries degrade): the bottleneck is sub-cell placement reconstruction, exponentially beyond finite sums — note on PRIME.ZEROCOMB.01. And NO  $V$ -commuting involution implements the corpus  $\mu_4$  grading (all six fail;  $\Sigma S_0$  is an infinite-order shear): the two v590 demands stay independent — note on QGEO.INVOL.01. The OS/RP positivity theorem (D) remains open (no quick construction; the v572/v584 design space is exhausted).
- **Sync**. Ledger row + notes on PRIME.NULLRAY.01 and PRIME.LOCKDIR.01; **tfpt\_prime\_front** (43  $\rightarrow$  44 modules, **sec:v591/sec:t194**, 194 probes / 4919 checks); Paper II (“twenty-five modules”); **tfpt\_3**; diary; suite 584  $\rightarrow$  585 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (IV) (the sheet involution exists unconditionally (v590 promoted): the integral  $V$ -commuting involutions are exactly  $(\mathbb{Z}/2)^3$  (rigidity, both twins); the v574 type theorem + seam reversal + zero-mode fixing single out the explicit  $S_0$  with exact  $\Sigma$ -signature uniquely — GATE.QGEO.01’s involution-existence input relocates to DERIVED)**

- **v590\_involution\_existence.py** (QGEO.INVOL.01, 7 checks; probe 7/7, EXISTENCE-UNCONDITIONAL). All eight spectral sign choices on  $V$ ’s eigenspaces are integer, involutive

and order-stabilizing on both twins; narrowing  $6 \rightarrow 3 \rightarrow 2 \rightarrow 1$ : the trace  $+1$  half dies by the v574 rank theorem,  $Se_3 = -e_3$  and  $+1$  on  $\ker V$  single out  $S_0 = [[1, -1, 0], [0, -1, 0], [0, 0, -1]]$ ; honest scope: the corpus RP reflection does not commute with  $V$  (different structure), the alt twin has the same classification (not a bit selector); the two demands' continuum derivation ( $\mu_4/D_4/H^1$ ) stays the named open.

- **Sync.** Ledger row + note on QGEO.WINDING.SELF.01; `tfpt_1` (Stage-B passage), `tfpt_2` (new paragraph), `tfpt_3` (audit block); diary; suite 583  $\rightarrow$  584 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (III) (the zeta zeros enter the instrument (v589 promoted, T193): the arithmetic depth layer is IDENTIFIED as the explicit-formula zero comb — correlation 0.82 with the first 200 damped zeros, a factor 5.7 of the lock depth recovered at the smallest window; the zero-oscillation bound becomes a quantified target)**

- **v589\_zero\_comb.py** (PRIME.ZEROCOMB.01, 4 checks; probe T193 4/4, **ZERO-COMB-IDENTIFIED**). The residual mass oscillation of the atom table is the explicit-formula comb  $-4 \sum_{\gamma>0} \sin(\gamma u)/\gamma$  (dense-grid correlation 0.82, slope 1.28 = truncation deficit; pointwise 1.00 at  $u = 3$ ); adding the comb to the density model improves the  $h = 184$  lock ratio  $176 \rightarrow 31$  (factor 5.7; non-monotone in truncation — honest), deep windows out of reach ( $h = 540$ :  $2646\times$ ); the zero-oscillation bound is now the quantified target “control  $\sum \sin(\gamma u)/\gamma$  at resolution  $\pi/D$  with amplitude accuracy = lock depth”. No bound claimed, NO RH statement.
- **Sync.** Ledger row + notes on PRIME.CLOSEDELTA.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (42  $\rightarrow$  43 modules, `sec:v589/sec:t193`, 193 probes / 4914 checks); Paper II (“twenty-four modules”); `tfpt_3`; diary; suite 582  $\rightarrow$  583 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (II) (the deterministic layer of Problem 7.1 closes in one function (v588 promoted, T192):  $G_{ij}(\beta)$  carries the comb side at the pole exponent and the archimedean side as the TRIVIAL-ZERO LADDER; closed  $B$  to  $2 \times 10^{-5}$ , the closed defect to 0.2% on all 70 windows, census decay  $h^{-1.40}$  — the explicit formula's skeleton, pole vs  $\Gamma$ -factor, in closed form)**

- **v588\_closed\_delta.py** (PRIME.CLOSEDELTA.01, 6 checks; probe T192 6/6, **CLOSED-DELTA-EXACT**).  $G_{ij}(\beta) = \sum_d e^{-\beta d} W_{ij}(d)$  machine-exact (finite geometric-trig series over the v587 formulas);  $c_{\text{ar}}(d) = -Df(dD)$  with  $f(w) = \sum_n e^{-(2n+1/2)w}$  verified 1.000002 (near field  $d \leq 32$  as explicit quadrature constants); closed  $B$ : entries  $2 \times 10^{-5}$ , det  $6 \times 10^{-5}$ , all 70 windows; closed defect vs the v585 deterministic layer: 0.9999–1.0023 on all 70 windows incl. deep/anomalous; census decay  $h^{-1.40}$  (= v585's  $-1.43$ ). Remaining for a theorem: elementary asymptotics of  $G$  and the ladder; the arithmetic layer stays with the primes.
- **Sync.** Ledger row + notes on PRIME.WCLOSED.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (41  $\rightarrow$  42 modules, `sec:v588/sec:t192`, 192 probes / 4910 checks); Paper II (“twenty-three modules”); `tfpt_3`; diary; suite 581  $\rightarrow$  582 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (I) (a NEW EXACT FORMULA closes the deterministic layer's entries (v587 promoted, T191): the diagonal lag weight is the single closed expression  $W_{kk}(d) = (2/N)[(N-1-d) \cos(\omega_k d) + \sin(\omega_k(d+1))]/\sin \omega_k$ , machine-exact for all lags; the exact cell-sum reproduces the prime-free model's entries and**



determinant to 0.03%)

- **v587\_w\_closed\_form.py** (PRIME.WCLOSED.01, 6 checks; probe T191 6/6, **CLOSED-FORM-EXACT**). The diagonal companion of the v576 cross formula, machine-exact ( $1.9 \times 10^{-15}$ ) for  $h = 7\text{--}1000$ ,  $k = 1\text{--}3$ , all lags; the building blocks are closed Dirichlet sums, the apparent piecewise break at  $d = m$  cancels on assembly, the reflection-edge suppression is an exact two-term cancellation; the exact cell-sum reproduces the v583 entries (1.0011–1.0036) and det (0.9999–1.0003); macro limit  $g_k(\sigma) = 2(1-\sigma)\cos(2\pi k\sigma) + \sin(2\pi k\sigma)/(\pi k)$  with the v579 kernel its edge-anchored version identically. The first executed step of the v585/v586 classical program; named next: the arch/ $B$ -side closed form, then the  $\delta_{\text{PNT}}$  expansion.
- **Sync.** Ledger row + notes on PRIME.CHEBLOEWNER.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (40  $\rightarrow$  41 modules, `sec:v587/sec:t191`, 191 probes / 4904 checks); Paper II (“twenty-two modules”); `tfpt_3`; diary; suite 580  $\rightarrow$  581 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIX) (the locking DIRECTION is density-fixed (v586 promoted, T190): real vs prime-free locking eigenvector  $0.030^\circ$  median on all 69 windows while the depth stays  $1350\times$  apart — THE PRIMES ARE A PURE DEPTH AMPLIFIER ALONG A DENSITY-FIXED DIRECTION; the v577 slow-decay puzzle dissolves as a comparator artifact)**

- **v586\_pnt\_lock\_direction.py** (PRIME.LOCKDIR.01, 7 checks; probe T190 7/7, **DIRECTION-DENSITY-FIXED**). Census on all 69 regular windows: angle max  $0.331^\circ$ , median  $0.030^\circ$ , decay  $h^{-1.16}$ ; witness certificate — the quadratic form of the real  $B - S$  along the deterministic direction reproduces the minimal eigenvalue within  $[0.98, 2.9]$  (median 1.42): the arithmetic content of Problem 7.1 is ONE quadratic form along ONE deterministic vector; the deterministic direction drifts ( $-1.73 \rightarrow -1.19$ , limit  $-0.551$  under  $1/\log h$ , consistent with the null-ray  $-0.5$ , not settled) — resolving v577’s comparator artifact; scramble must-break  $6\text{--}25^\circ$  vs  $0.026^\circ$ .
- **Sync.** Ledger row + notes on PRIME.NULLRAY.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (39  $\rightarrow$  40 modules, `sec:v586/sec:t190`, 190 probes / 4898 checks); Paper II (“twenty-one modules”); `tfpt_3`; diary; suite 579  $\rightarrow$  580 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVIII) (the two-layer split of the locking cancellation (v585 promoted, T189): THE DOMINANCE IS DENSITY, THE LOCKING IS ARITHMETIC — the prime-free model misses the Paper-II defect by 2–4 orders (median  $1350\times$ ) while reproducing  $\det S$ ; layers  $h^{-1.43} \times h^{-1.08} = h^{-2.51}$ ; the Problem-7.1 budget splits into a classical density part and a zero-oscillation part)**

- **v585\_pnt\_locking\_split.py** (PRIME.LOCKSPLIT.01, 6 checks; probe T189 6/6, **TWO-LAYER-SPLIT**). The v583 model through the normalised defect on all 70 windows: median ratio 1350, growing  $h^{1.08}$  — the depth of the lock is the primes; the one-mode geometry is density-level (coarse lock  $10^{-2}\text{--}10^{-4}$  without primes); density layer  $h^{-1.43}$  always positive (scatter 0.70), real  $h^{-2.51}$ ; real sign overshoots exactly at  $h = 1219, 1445$  (+ the typed 1292 anomaly), deterministically never. Budget: density  $\sim -1.4$  classical, primes  $\sim -1.1$  zero-oscillation.
- **Sync.** Ledger row + notes on PRIME.PNTMODEL.01 and PRIME.PENCIL.ONEMODE.01; `tfpt_prime_front` (38  $\rightarrow$  39 modules, `sec:v585/sec:t189`, 189 probes / 4891 checks); Paper II (“twenty modules”); `tfpt_3`; diary; suite 578  $\rightarrow$  579 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVII) (the interacting RP route of the alignment bit, executed: EXT-SOURCE COUPLING SEPARATES THE 81-TWINS DIRECTIONALLY IN THE UNIT BOX (v584 promoted) — exact rational flip thresholds with the v572 pencil factors as numerators; positive coupling  $J \leq 15/16 < 1$  heals exactly the compiler twin, the alt twin is rigid; response pole on the golden polynomial)**

- **v584\_ext\_source\_response.py** (DIAMOND.BIT.EXTSOURCE.01, 8 checks; probe 8/8, **UNIT-BOX-SEPARATED**). The interacting kernel  $K(s, J) = K_0 + sK_\Pi + Jvv^T$  ( $v$  from the shared Ext arrows  $E_{31}/E_{32}$ ) has exact thresholds  $J_c(s) = -1/(v^T K(s)^{-1}v)$ : compiler  $(15s^2 - 16s - 16)/(16(s^2 - s - 1))$  and  $(15s^2 - 16s - 16)/16$ , alt  $(s^2 + 16s + 16)/(16(s + 1))$  and  $-(s^2 + 16s + 16)/16$  — numerators exactly the v572 pencil factors; verified numerically (16 flip events). In the crossed phase the compiler is healed by positive  $J \leq 15/16 < 1$  (either arrow); the alt twin cannot be flipped by any positive coupling in  $|J| < 1$  (E31 threshold  $\geq 1$ , equality exactly at  $s = 0$ ; E32 negative). Negative coupling never heals (direction consistent with v534's toy RP selection). Exact anchors  $15/16, 1, -1$ ;  $J_c(s^*) = 0$  via Ext-visibility of the critical mode; response pole on  $s^2 - s - 1$  ( $-1/\varphi$ , next to  $s^*$ ) — observed. Honest: response-level, no PSD anywhere, the alignment-bit contract stays open.
- **Sync.** Ledger row + note on DIAMOND.BIT.RPFORM.01; **tfpt\_1** (follow-up passage extended), **tfpt\_2** (new interacting-route paragraph), **tfpt\_3** (audit block); diary; suite 577  $\rightarrow$  578 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVI) (the named next step executed the same day: THE SMOOTH FUNCTIONAL IS IN CLOSED, PRIME-FREE FORM (v583 promoted, T188) — the two-term classical law  $4e^{u/2} - 2(\zeta'/\zeta)(1/2)$  (no zeros) reproduces  $\det S$  on all 69 regular windows parameter-free to 1.016–1.143; the  $h = 1292$  sign flip is genuine prime-fluctuation content)**

- **v583\_pnt\_model.py** (PRIME.PNTMODEL.01, 8 checks; probe T188 8/8, **PRIME-FREE-CLOSED-FORM**). The two-term law holds on the atom table (oscillation band  $< 1.5$  around the parameter-free constant  $-2(\zeta'/\zeta)(1/2) \approx -5.372$ ); the prime-free grid model (density  $2e^{u/2}$ , hard cutoff, theoretical constant) reproduces  $\det S$  on all 69 regular windows to 1.016–1.143 (mean 1.039), the  $S$  entries to 0.995–1.001, the dominance growth with correlation 0.998; honesty: constant sensitivity  $\sim 0.38/\text{unit}$  — the residual band is the zero-oscillation; the anomalous  $h = 1292$  window typed (prime-free +1844 vs real -1465: the sieve-horizon sign flip IS prime-fluctuation content). FINAL RELOCATION:  $h$ -uniform dominance = prime-free two-integral inequality + zero-oscillation bound.
- **Sync.** Ledger row + notes on PRIME.DENSITYDOM.01 and PRIME.PENCIL.SEPFLOOR.01; **tfpt\_prime\_front** (37  $\rightarrow$  38 modules, **sec:v583/sec:t188**, 188 probes / 4885 checks); Paper II (“nineteen modules”); **tfpt\_3**; diary; suite 576  $\rightarrow$  577 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XV) (the quantitative occupation question answered at its first level: THE DOMINANCE IS DENSITY-DRIVEN (v582 promoted, T187) — the smoothed comb carries 99.8–99.9% of  $\det S$  on every regular window, the scrambled control only 70%: the real comb is anomalously smooth, and the  $h$ -uniform dominance question relocates to PNT-level regularity of the smooth density functional)**

- **v582\_density\_dominance.py** (PRIME.DENSITYDOM.01, 5 checks; probe T187 5/5, **DENSITY-DRIVEN**). Replacing the atom lag coefficients by their moving average (declared windows 2–20%) and recomputing  $\det S$  through the exact lag-side identity: 0.998–0.999 at 2%, 0.987–0.993 at 5%, degrading smoothly to 0.81–0.89 at 20% on every regular window —

the dominance is carried by the deterministic density profile, fluctuations contribute  $\sim 1\%$ ; the scrambled control captures only 0.70 (random placement has real fluctuation content — the actual comb is anomalously smooth at these scales); the anomalous  $h = 1292$  window deviates (1.24), boundary named. THE RELOCATION: the v570  $h$ -uniform dominance reduces — up to a measured  $\sim 1\%$  correction needing only a crude bound — to a positivity/growth statement about the smooth density functional (PNT-level regularity); the named next step is a closed-form bound on that functional.

- **Sync.** Ledger row + notes on PRIME.PENCIL.SEPFLOOR.01 and PRIME.OCCUPATION.01; `tfpt_prime_front` (36  $\rightarrow$  37 modules, `sec:v582/sec:t187`, 187 probes / 4877 checks); Paper II (“eighteen modules”); `tfpt_3`; diary; suite 575  $\rightarrow$  576 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIV) (Regime B’s arithmetic half gets its measured object (v580: the lag-side identity is exact and the primes POPULATE the geometric sign map — every significant occupation cell matches  $K_\infty$ ), and the review’s multilevel transport lemma is measured and killed (v581: drift small but growing  $h^{+0.36}$  against the  $h^{-1-\delta}$  bar) + backflow parts T185/T186 + full sync)**

- **v580\_occupation\_map.py** (PRIME.OCCUPATION.01, 5 checks; probe T185 5/5, **OCCUPATION-FOLLOWS-KERNEL**). The lag-side polarization identity —  $\det S$  as the atom-lag double sum  $\frac{1}{2} \sum c_{\text{at}} c_{\text{at}} D(W, W)$  — is exact (the T163 correlation theorem,  $10^{-9}$  on four windows); every significant cell of the binned occupation map sign-matches  $K_\infty(\sigma_1, \sigma_2)$  on every tested window (50/50, 67/67, 65/65, 65/65): the arithmetic does not fight the kernel, it populates it — the open Regime-B content is the AMOUNTS; mid-diagonal negative, cross-scale positive; scramble control clean.
- **v581\_locking\_transport.py** (PRIME.TRANSPORT.01, 4 checks; probe T186 4/4, **TRANSPORT-KILLED**). The Priority-4 lemma measured on 40 doubling pairs: the locking-direction drift is small ( $0.3\text{--}6.7^\circ$  per octave, median  $2.35^\circ$ ) but GROWS ( $h^{+0.36}$ ) against the proposed summability bar — the review’s own kill criterion fires; no fixed-direction scalar transport; future transports must carry the window-dependent macro direction.
- **Sync.** Two ledger rows + notes on PRIME.MACROKERNEL.01 and PRIME.NULLRAY.01; `tfpt_prime_front` (34  $\rightarrow$  36 modules, `sec:v580/sec:v581/sec:t185/sec:t186`, 186 probes / 4872 checks); Paper II (“seventeen modules”); `tfpt_3`; diary; suite 573  $\rightarrow$  575 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIII) (the review’s remaining strands executed: the jet contract natively (v578, MOEBIUS-IN-OWN-CLOCK — no shared wall-time Schwarzian, each flow projective in its own clock), the two-scale kernel signs verified globally (v579, SIGNS-MATCH 332/332 — the geometric half of Regime B closes), and the Ext-resolvent marking honestly killed (sandbox:  $\chi_{\text{Ext}} \equiv 0$  for both twins, the radical sublattices are identical))**

- **v578\_native\_jets.py** (FTR.JETS.01, 6 checks; probe 6/6, **MOEBIUS-IN-OWN-CLOCK**). The exact v99 trajectory has  $\{q, t\} = -\Delta^2/2$  in wall time and Schwarzian 0 in its own exponential clock; the 1-loop QCD flow has Schwarzian 0 in its own RG time; the native quadruple shares no wall-time jet invariant — each instance is projective in its own clock, the obstruction is the clock (anchored by the external data); a future joint jet test needs external anchor-level jets, as v575 fenced.
- **v579\_macro\_kernel\_signs.py** (PRIME.MACROKERNEL.01, 6 checks; probe T184 6/6, **SIGNS-MATCH**).  $\text{sign } K_\infty(\sigma, \tau)$  matches the real two-lag kernel on every resolved cell (332/332 at  $h = 300$ ); the diagonal  $\det W^\infty(\sigma)$  is negative on  $\sim 72\%$  (same-scale coupling

negative, analytically); the geometric half of Regime B closes, the  $\Lambda$ -weighted arithmetic occupation stays open.

- **Sandbox honest negative.** The review’s Ext-sensitive marked resolvent  $\chi_{\text{Ext}}(z)$  vanishes identically for both twins with the natural markings, and the radical sublattices of the two integral orders are IDENTICAL ( $\text{span}\{3E_{31}, 3E_{32}\}$ ): the review’s own kill criteria fire; the Gibbs pencil (v572) remains the one working marked object. Diary record, no module.
- **Sync.** Two ledger rows + notes; `tfpt_research_contracts` (jet block), `tfpt_prime_front` (33  $\rightarrow$  34 modules, `sec:v579/sec:t184`, 184 probes / 4863 checks), Paper II (“fifteen modules”), `tfpt_3`; diary; suite 571  $\rightarrow$  573 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XII) (the second review’s Toeplitz strand audited – everything EXACT (v576: the cross-weight formula, the Chebyshev–Loewner edge theorem with the rank-one first read, the Pythagorean null bridge (5, −3, 4) typed compression, the exact  $N^{-4}$  defect coefficient, the pair-kernel sign polynomial) and the null-ray census executed (v577: correlated on all 70 windows, honestly partial) + backflow parts T182/T183 + full sync)**

- **v576\_cheb\_loewner\_edge.py** (PRIME.CHEBLOEWNER.01, 9 checks; probe T182 9/9, **FORMULAS-EXACT**). The polarized cross-weight formula  $w_d^{(ij)} = (2/N)[\sin \omega_j \sin(dw_i) - \sin \omega_i \sin(dw_j)]/(\cos \omega_i - \cos \omega_j)$  is exact against the corpus definition (symbolic  $N \leq 11$ ,  $10^{-11}$  surface) — the  $K = 2$  read block closes analytically; the reflection edge is a scaled Loewner matrix of  $U_{s-1}$  (exact; rank  $\leq s-1$ , the first read rank ONE); the leading profile’s Lorentz image is the null triple (5, −3, 4) — the same as  $(g_{\text{car}}, N_{\text{fam}}, |\mu_4|)$  via Euclid, typed COMPRESSION; the edge defect is the exact coefficient  $-\frac{3\pi^4}{175}(s^2 - 4)(11s^2 + 6)N^{-4}$  (locally better than the  $h^{-3}$  demand); the pair polynomial  $P(s, z)$  ( $P(2, 3) = 0$  exact, transition  $s/z \approx 0.503/1.987$ ) is the analytic prototype of the v573 pattern; macro kernels verified. Three-regime split: edge CLOSED at  $N^{-4}$ , macro and transition OPEN; Problem 7.1’s quantifier untouched.
- **v577\_nullray\_census.py** (PRIME.NULLRAY.01, 5 checks; probe T183 5/5, **NULL-RAY-CORRELATED**). The locking mode aligns with the exact null ray (2, −1) at 23.5–33.8° on all 70 windows (below random; scramble destroys it, +21°), declining slowly ( $h^{-0.10}$ ): the clean success bar is honestly NOT met — a real partial carrier; the macro regime bends the direction; follow-up:  $K_\infty(\sigma, \tau)$ .
- **Sync.** Two ledger rows + notes on PRIME.PAIRBAND.01 and PRIME.PENCIL.SEPFLOOR.01; `tfpt_prime_front` (verified layer 31  $\rightarrow$  33, subsections `sec:v576/sec:v577/sec:t182/sec:t183`, counters 183 probes / 4857 checks); Paper II (cross-reference rows, “fourteen modules”); `tfpt_3` audit blocks; diary; suite 569  $\rightarrow$  571 scripts; README/zenodo/website counters; mirrors regenerated.

**2026-07-31 (XI) (the continuous half of the cross-ratio check executed and closed: UNIFLOW-DEGENERATE (v575 promoted) — the native exports are forced degenerate by the single-flow theorem, so the continuous half closes onto v425; both halves of the preregistered check are now executed)**

- **v575\_cr\_continuous\_uniflow.py** (FTR.CRCRONT.01, 7 checks; probe 7/7, **UNIFLOW-DEGENERATE**). The frozen protocol exports, recomputed from primitives ( $F_{\text{pole}} = e^{-\Delta} = (2/3)^6$  via the v99 generator;  $F_{\text{Boltzmann}}$  the SAME contraction by the v425 single-flow theorem;  $F_{\text{QCD } b_3} = -7$ ;  $F_{\text{relic adiabatic } 1}$ ), form the quadruple  $\{(2/3)^6, (2/3)^6, 1, -7\}$  with a forced repetition — and a repeated quadruple gives  $\text{CR} \in \{0, 1, \infty\}$  in every corner assignment (theorem + all 24 permutations): the continuous half can never pass natively. It closes onto the

single-flow reduction; the anchors ( $v_{\text{geo}}$ ,  $M_R$ ,  $C_p$ ,  $\theta_i$ ) stay [O] walls; the meaningful cross-ratio content remains the discrete half (v571). Contract markers unchanged.

- **Sync.** New ledger row FTR.CRCONT.01; update notes on FTR.CRDISC.01 and CONTRACT.F.01; contracts-paper protocol block extended with the execution result; suite 568 → 569 scripts; diary; README/zenodo/website counters; mirrors regenerated.

**2026-07-31 (X) (closing the round: the winding form's content located exactly (v574 → 11 checks — without it a four-column family survives, exactly one uniform: the compiler), and the half-sided RP restrictions shown blind (v572 → 11 — the interacting construction is the one remaining RP route))**

- **v574 extended (probe 12/12).** Dropping the winding demand entirely (arbitrary first column), the self-code package leaves the four-column family  $\{(0, 1, 1), (0, 3, 3), (3, 3, 3), (5, 1, 5)\}$  — exactly one uniform: the compiler. The winding demand is the uniformity choice (1 of 4). Final declared list of the reconstruction: involution existence, column uniformity,  $N_{\text{fam}} = 3$  (E8-glue).
- **v572 extended (probe 11/11).** The Gibbs kernels restricted to the Levi and radical sublattices have identical inertia on both twins across the sampled range: the RP-positivity selection resists every kinematic and half-sided design tried — the genuinely interacting construction (the v534 FK-quartet analogue) is the one remaining route, typed open.
- **Sync.** Ledger/registry/run\_all counts; tfpt\_1/tfpt\_2 passages; diary; mirrors regenerated.

**2026-07-31 (IX) (three final deepenings: the maximal ablation drops ( $E$ ) too (v574 → 10 checks — the reconstruction needs only involution existence, the winding column with  $N_{\text{fam}} = 3$  and the self-code), the Gibbs-critical mode is identified as the seam-row Ext-carrier direction (v572 → 10), and the per-atom profile of the dominance oscillates standing-wave-like (v573 → 10))**

- **v574 extended (probe 11/11).** The maximal ablation: dropping the exponent demand ( $E$ ) entirely, the winding column  $3(1, 1, 1)$  plus integrality plus the self-code package (parabolic + ladder + saturation 81) still pin the compiler  $Q$  uniquely (29 ladder candidates, one hit). ( $E$ ), ( $M$ ), ( $L$ ) are ALL redundant at the declared bound — what stays declared is the involution existence, the winding form and  $N_{\text{fam}} = 3$  (independently anchored in the  $E_8$  glue,  $240 = 16 \times 5 \times 3$ ); GATE.QGEO.01 narrowed by five items, not closed.
- **v572 extended (probe 10/10).** The critical mode at the compiler twin's Gibbs temperature is identified: rank one, supported entirely on the seam row (the radical /  $A_2$ -Ext-carrier row), fixes the seam line, is a fixed point of the sheet flow ( $VX = X$ ), annihilated by the dominant projector — the Gibbs criticality lives exactly in the Ext-carrier slot: the typed resonance with the alignment bit deepens to a shared carrier.
- **v573 extended (probe 10/10).** The per-atom share of  $\det S$  oscillates across the window with a fixed sign pattern (positive at  $x \sim 0.3/0.8$ , negative past the midpoint), amplitudes growing with  $h$  while the net stays 1 — a standing-wave-like anatomy, measured with its ladder, no mechanism claim.
- **Sync.** Ledger/registry/run\_all; tfpt\_1/tfpt\_2 passages; tfpt\_prime\_front counters (4843 checks, T181 10/10); diary; mirrors regenerated.

**2026-07-31 (VIII) (two decisive deepenings: the anchor-Hamiltonian Gibbs pencil delivers the first DYNAMICAL distinction of the reconstruction bit (v572 → 9 checks: the compiler twin is Gibbs-critical at  $s^* = (8 - 4\sqrt{19})/15$ , the alternative**

never), and the redundancy ablation shrinks the reconstruction inputs to three (v574  $\rightarrow$  9 checks:  $(M)$  and  $(L)$  fall relative to the self-code demands))

- **v572 extended (probe 9/9).** Along the Gibbs family  $\omega_g \propto \text{tr}(e^{-gH})$  of the v566 anchor operator  $H = I + \Pi$ , the pencil determinant factors exactly differently on the twins — compiler  $\frac{6561}{4096}(s^2+16s+16)^2(15s^2-16s-16)$ , alternative  $-\frac{6561}{4096}(s^2+16s+16)^3$  ( $6561 = 81^2$ ) — and only the compiler's extra factor has a root inside the physical Gibbs range: the compiler twin's RP kernel degenerates at a finite critical temperature, the alternative twin's never does. Honest typing: a rank-drop marker, not a positivity selection; the interacting RP question stays open.
- **v574 extended (probe 10/10).** Redundancy ablation: relative to the self-code demands (parabolic + ladder + saturation 81) the monodromy budget  $(M)$  and the seam-line demand  $(L)$  are redundant at  $B = 6$  — dropping both widens the class to 1274 members and the self-code still picks exactly the compiler  $Q$ : the declared geometric input list shrinks to  $(S)$ -existence,  $(W)$ -form and  $(E)$ ; GATE.QGEO.01 narrowed by four items, not closed.
- **Exploratory (diary only).** The pair-mechanism hunt on the reference window: no dominant pair exists ( $\max |\text{contribution}| \sim 0.006 \times \det S$  — consistent with the dead two-atom witness), and the fine structure by smaller atom shows the smallest primes  $(2, 3)$  dragging negative while medium primes  $(11..41)$  carry positive net.
- **Sync.** Ledger/registry/run\_all counts; tfpt\_1/tfpt\_2 passages; diary; mirrors regenerated.

**2026-07-31 (VII) (the four lines deepened in the same round: the dominance anatomy sharpened to heavy-pair window-scale coherence (v573  $\rightarrow$  9 checks), the RP blindness shown robust under the compiler flow (v572  $\rightarrow$  7), the Stage-B forcing decomposed by the spectrum ablation (v574  $\rightarrow$  8), and the continuous-half export protocol frozen on the contract surface)**

- **v573 extended (probe T181 9/9).** The excess is *scale-locked* (pairs  $\geq h/2$  carry it; bands 16..256 stay negative) and *weight-concentrated*: the top-10%-weight pairs alone net  $0.85\text{--}1.00 \times \det S$  on every selected window — essentially  $\det S$  itself; the lag-grid phase is flat (mechanism killed). The certificate target is typed precisely: heavy-pair coherence at window-scale separation.
- **v572 extended (probe 7/7).** The spectral blindness is robust under the compiler's own flow: the  $V$ -semigroup-deformed kernels keep identical inertia on both twins for every declared state and every  $n = 1..3$  — a dynamical selection needs genuine interaction.
- **v574 extended (probe 9/9, full 35-spectrum scan).** The honest decomposition: freeing the exponent spectrum, the one surviving double-selfconsistent family is exactly  $\{1, 2, N\}$  with the compiler shape — the self-consistency forces the *sheet* spectrum  $\{1, 2\}$  and the shape, while the  $A_3$  exponent demand contributes exactly one datum,  $N = 3$ ; both factors load-bearing.
- **Contract surface.** The frozen export protocol for the continuous cross-ratio half preregistered (each instance exports the fixed-point multiplier of its native flow, cf. v425, before evaluation); design [X], markers unchanged.
- **Sync.** Ledger/registry/run\_all counts updated; papers (tfpt\_1, tfpt\_2, tfpt\_3, tfpt\_prime\_front counters 4842 checks, Paper II cross-reference row, tfpt\_research\_contracts); diary; website mirrors regenerated.

**2026-07-31 (VI) (all four review follow-ups executed: the cross-ratio check passes on its discrete half (v571,  $y_F = 53$  Möbius-forced), the RP form is integrally anchor-visible on the twins (v572), the dominance is typed long-range (v573, T181), and Stage B forces the winding normalisation (v574,  $N_{\text{fam}} = 3$  by double**

## self-consistency) + ledger/paper/diary/website sync)

- **v571\_cr\_discrete\_corners.py** (FTR.CRDISC.01, 9 checks; probe 9/9, **CR-PASSES**). The preregistered cross-ratio check (v566 S9), executed on its discrete half: the four solver corners are the in-repo kernels  $J, K, C, F = M(1, t)$ ,  $t = -2.1$  (v224), and the corpus' one canonical declared solver reading — the Koide anchor response  $\ell(M) = a^T M \mathbf{1}$  (v183) — reads (26, 35, 44, 53): CR = 4/3 exactly, and the fit-free reconstruction *forces*  $y_F = 53$  from the three upstream corners (a new exact cross-corner identity). Honesty theorem: every linear functional passes (the check separates linear from nonlinear exports); negative controls fail as preregistered (determinants 28/25, dominant eigenvalue 1.3222). The continuous-solver half stays open per CONTRACT.F.01.
- **v572\_rp\_bit\_form.py** (DIAMOND.BIT.RPFORM.01, 6 checks; probe 6/6, **ANCHOR-VISIBLE**). The kinematic RP form on the 81-twins: spectrally blind (identical inertia for every declared state; never PSD — no v534-style kinematic positivity selection exists, honest negative), but the *integral* Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor  $9 = N_{\text{fam}}^2$ ) — the bit is visible to the RP form only through anchor-coupled states; trace discriminant  $-81^2$  on both twins; the dynamical RP question stays the open RP-slot.
- **v573\_pair\_band\_structure.py** (PRIME.PAIRBAND.01, 6 checks; probe T181 6/6, **LONG-RANGE**). Where the v570 dominance lives: every near band ( $< 16$  lag cells) and the diagonal drag negative on the declared subset; the far band carries the whole excess with stable net  $([1.55, 1.91] \times \det S, \text{gross growing to } 13\times)$ ; the scramble explodes  $\det S \times 49$  and collapses the far share — long-range coherence, arithmetic placement; a certificate must control far-pair coherence (near-diagonal budgets aim at the wrong sign).
- **v574\_stageb\_winding.py** (QGEO.WINDING.SELF.01, 7 checks; probe 7/7, **FORCED**). Stage B, first narrowing: the involution *type* is killed by a rank theorem (a two-dim-positive involution cannot carry the binary trace code),  $(E)$  bounds  $N \leq 3$ , the ladder self-consistency selects one member per  $N$  ( $28/16/14 \rightarrow 1/1/1$ ), and  $\text{sat} = N^4$  holds *only* at  $N = 3$  ( $81 = 3^4$ ;  $32 \neq 1$ ,  $512 \neq 16$ ): the double self-consistency pins  $(N, Q) = (3, \text{compiler } Q)$  —  $N_{\text{fam}} = 3$  relocates from declared input to consistency-forced; GATE.QGEO.01 narrowed by two items, not closed; no P2 reduction, markers untouched.
- **Sync.** Four ledger rows + update notes on CONTRACT.F.01, DIAMOND.BIT.SELECT.01, PRIME.PENCIL.SEPFLOOR.01, QGEO.QRECON.01; tfpt\_research\_contracts (CR block “executed, discrete half”), tfpt\_1/ tfpt\_2 (RP-form and Stage-B passages), tfpt\_3 (pair-band audit block), tfpt\_prime\_front (verified layer  $30 \rightarrow 31$ , subsections `sec:v573/sec:t181`, counters 181 probes / 4839 checks), Paper II (cross-reference rows, “twelve modules”); diary entries; suite 564  $\rightarrow$  568 scripts; README/zenodo/website counters; mirrors regenerated.

## 2026-07-31 (V) (the T179 open question closed on the surface: the spectator separation certified from closed scalars (FLOOR-CERTIFIED) + v570 promoted + backflow part T180 + ledger/paper/diary/website sync)

- **Probe separation\_floor\_probe.py** (T180, 9/9, verdict **FLOOR-CERTIFIED**). The open question v569 named — does the spectator eigenvalue  $\lambda_1$  of the relative pencil stay separated from 1? — answered on the reachable surface by *reduction*: two one-line theorems ( $P = \det S / \det B > 0$ ,  $T = \text{tr } Z > 0 \Rightarrow \lambda_1 \geq \sqrt{P}$ ;  $P \leq 0 \Rightarrow \lambda_1 \geq T$ ) turn the separation into a closed per-window floor, and on all 70 declared T170 frame-A windows the floor clears the margin,  $\min F = 2.130 \geq 1.05$  (measured  $\min \lambda_1 = 4.534$ ): the separation follows from  $\det S / \det B$  and  $\text{tr } Z$  alone. The dominance is large and grows with depth ( $P$  median 30.1, slope  $h^{0.90}$  on the surface ladder), and  $1 - T + P = \det(B - S) / \det B$  links trace and determinant dominance exactly. *Honest negative*: the cheap two-atom witness route is dead —  $\det S$  is itself



a pair-level cancellation object (Cauchy–Binet exact; negative pair mass up to  $13.7 \times \det S$ ; best single pair  $0.13\text{--}0.34 \times \det B$  against the  $\geq 1.10$  needed) — so the  $h$ -uniform dominance stays open, typed as pair-level size-vs-cancellation. Must-break: the position scramble flips  $\text{tr } Z$  to  $-78.28$ .

- **Promotion** `v570_separation_floor.py` (PRIME.PENCIL.SEPFLOOR.01, 8 checks,  $\sim 13$  s). New ledger row; update note on PRIME.PENCIL.ONEMODE.01; suite 563  $\rightarrow$  564 scripts. **Named limits:** declared finite surface only — the  $h$ -uniform dominance  $\det S \geq (1+c)^2 \det B$  is the same open question `v569` named, in sharper closed-scalar form; Paper-II Problem 7.1 untouched; no rate, no bound, no RH statement.
- **Sync.** `tfpt_prime_front` (verified layer 29  $\rightarrow$  30 modules, subsections `sec:v570` and `sec:t180`, index T11–T180, counters 180 probes / 4833 checks); Paper II (cross-reference row, reproducibility “eleven modules”); `tfpt_3` audit block; diary entry in `experiments/next.txt`; README/zenodo/website counters 564; mirrors regenerated.

**2026-07-31 (IV) (the relative pencil on the declared Paper-II surface: the one-mode question of the self-code review answered (ONE-MODE, labels corrected) + `v569` promoted + backflow part T179 + ledger/paper/diary/website sync)**

- **Probe** `lambda_pencil_probe.py` (T179, 4/4, verdict ONE-MODE). Priority 5 of the self-code review, executed on the real surface: the `v566` S8 pencil identities  $\det(B-S) = \det B (1-\lambda_1)(1-\lambda_2)$ ,  $D(B, S) = \det B (\lambda_1+\lambda_2)$ ,  $\det S = \det B \lambda_1 \lambda_2$  are machine-tight on all 70 declared T170 frame-A windows (worst residual  $2 \times 10^{-14}$ ; the `v563` surface imported read-only, bit for bit). *The labels are the reverse of the review guess:* the locking mode is  $\lambda_2$ , hugging 1 (median 1.000021), while the spectator  $\lambda_1$  is large and uniformly separated —  $\min |1-\lambda_1| = 3.53$ ,  $\lambda_1 \in [4.53, 180.64]$ , every window *above* 1. *The one-mode answer:* exactly one eigenvalue can approach 1, so the two-dimensional cancellation of Paper II reduces to the single scalar  $1-\lambda_2$  on the whole reachable surface; the  $h = 540$  reference window reads  $1-\lambda_2 = -2.27 \times 10^{-5}$  (reproducing  $\det \hat{A} = 4.22 \times 10^{-3}$ ), and the position scramble explodes  $|1-\lambda_2|$  by  $\times 1.9 \times 10^6$  while the identities keep holding: the locking is arithmetic placement.
- **Promotion** `v569_lambda_pencil_onemode.py` (PRIME.PENCIL.ONEMODE.01, 6 checks,  $\sim 2$  s). New ledger row; update notes on PRIME.PAPER2.READOUT.01 and DIAMOND.PARA.SELFCODE.01; suite 562  $\rightarrow$  563 scripts. **Named limits:** declared finite surface only — Paper-II Problem 7.1 (the uniform quantifier) is untouched, typed not bounded; whether  $\lambda_1$  stays separated as  $h$  grows is a new named open question, strictly weaker than Problem 7.1; no rate, no bound, no RH statement.
- **Sync.** `tfpt_prime_front` (verified layer 28  $\rightarrow$  29 modules, subsections `sec:v569` and `sec:t179`, index T11–T179, counters 179 probes / 4824 checks); Paper II (cross-reference row for Problem `prob:R1`, reproducibility “ten modules”); diary entry in `experiments/next.txt`; README/zenodo/website counters 563; mirrors regenerated.

**2026-07-31 (III) (the residual reconstruction bit located and given a canonical witness: the ladder self-consistency selects the compiler  $Q$  uniquely (SELECTOR-FOUND); the split geometry excluded; Priority 2 sharpened + `v568` promoted + ledger/paper/diary/website sync)**

- **Probe** `bit_selector_probe.py` (10/10, verdict SELECTOR-FOUND). Priorities 2+3 of the self-code review, fused with the `v567` residual bit. *The bit is integrally visible:* the two 81-twins ( $q_{32} = 2=|\mathbb{Z}_2|$  vs 0) generate *different* integral orders (equal index, equal Smith divisors, different lattices). *The selector:* the dominant mode of the sheet direction is  $v(c) = (1, 2, 2c)$  identically, and the *ladder self-consistency* — the dominant mode reproduces its own eigenvalue as the geometric ladder  $(2^0, 2^1, 2^2) = (1, 2, 4)$  — selects  $c = 2$ , the compiler  $Q$ , uniquely on all 28

members of the carrier-free class (no 81-pin, no column budget); the witness is prior structure (binary AnchorLadder steps  $2^n$ ,  $2 = |\mathbb{Z}_2|$ , the v566 trace code). *The split geometry located and excluded*: a module complement exists iff  $q_{32} = 1$ , and that member's algebra collapses to dimension 5 — the  $A_2$  Ext arrow is nonzero on the whole reconstruction class; neither twin splits: the bit toggles *which* nonsplit integral order is realised, not split-vs-nonsplit. *Priority-2 candidates, honest both ways*: the  $\mu_4$ -grading fails (the order is not graded; the 81 stays a fingerprint, the kill condition does not fire), while left  $V$ -multiplication induces a canonical  $\mathbb{F}_3$ -action on the cokernel  $(\mathbb{Z}/3)^4$  — *computed*: its characteristic polynomial is  $x(x-1)^2(x+1) \bmod 3$  (eigenvalues  $\{0, 1, 1, -1\}$ , NOT the regular  $\mu_4$  pattern  $x^4 - 1$ ;  $U$  acts as zero; both twins identical) — after two candidates the 81 stays a fingerprint.

- **Promotion v568\_bit\_selector\_ladder.py** (DIAMOND.BIT.SELECT.01, 11 checks,  $\sim 1$  s). New ledger row; update notes on QGEO.QRECON.01 and DIAMOND.PARA.SELFCODE.01; suite 561  $\rightarrow$  562 scripts. **The physical selection is not derived**: why the dominant mode should satisfy the ladder self-consistency is the relocated open question — the same structural slot as the RP-selection of the alignment bit (v528/v534); no marker moves, no P2 reduction.
- **Sync.** tfpt\_1 (selector appended to the Stage-A passage), tfpt\_2 (a dedicated bit paragraph after the audit paragraph); diary entry in experiments/next.txt; README/zenodo/website counters 562; mirrors regenerated.

**2026-07-31 (II) (the No-Carrier-Reconstruction audit, Stage A: the anchor self-code is carrier-free, and  $Q$  reconstructs up to ONE binary sheet-coupling choice (RECONSTRUCTS-TO-FAMILY) + v567 promoted + ledger/paper/diary/web-site sync)**

- **Audit probe carrier\_free\_q\_reconstruction\_probe.py** (10/10, verdict RECONSTRUCTS-TO-FAMILY). Priority 1 of the self-code review executed. *The circularity located*: v11's uniqueness certificate for  $Q$  uses row sums (4, 5, 6), column sums (9, 5, 1) and  $\chi_Q = (t-1)(t^2-5t+3)$  — the 5 =  $g_{\text{car}}$  in the budgets and in  $\chi_Q$ ; all of it *forbidden* in the audit. Under the corpus' own carrier-free data (sheet involution, v10 winding normalisation,  $A_3$  exponent spectrum, monodromy budget, integrality, seam-line demand): exactly 52  $\rightarrow$  28 (parabolic) admissible reconstructions — and **all 28 code the same anchor** ( $\chi_{H'} = x^3 - 4x^2 + 5x - 2$ , spectrum (1, 1, 2) =  $a$ ): the anchor self-code sits *upstream* of the carrier budgets. The single family invariant  $81 = N_{\text{fam}}^4$  (saturation index) pins  $Q$  to a *two-element* residual family ( $q_{32} \in \{2=|\mathbb{Z}_2|, 0\}$ , difference  $2E_{32}$ ) on which every classical carrier-free certificate coincides (anticommutator spectrum  $\{6, -4, -2\}$ , commutator charpoly  $t(t^2+12)$ , determinant, saturation) — the one separating certificate is the carrier column budget (9, 5, 1): exactly the forbidden input. Ablations: dropping the winding input widens the bound-3 set 28  $\rightarrow$  1376 (real load); bound-stable at  $B = 9$ .
- **Promotion v567\_carrier\_free\_q\_reconstruction.py** (QGEO.QRECON.01, 9 checks,  $\sim 1$  s; integer prefilters take the probe's 555 s to 1.2 s). New ledger row; update notes on FLAV.QGEO.01 and DIAMOND.PARA.SELFCODE.01; suite 560  $\rightarrow$  561 scripts. **Stage A only, no marker moves**: whether the geometric inputs themselves derive from  $\mu_4/D_4/H^1$  parabolic geometry is exactly GATE.QGEO.01 (the audit is relocated, not closed); the carrier-free selector of the residual bit is open, its resonance with the split/nonsplit alignment bit a typed observation; no P2 reduction, AX.P2.01 untouched.
- **Sync.** tfpt\_1 (Stage-A result appended to the self-code paragraph), tfpt\_2 (a dedicated audit paragraph after the self-code block); diary entry in experiments/next.txt; README/zenodo/website counters 561; mirrors regenerated.

**2026-07-31 (I) (the parabolic anchor self-code: an external structural review audited (SELFCODE-VERIFIED) — the anchor  $a = (1, 1, 2)$  as the spectrum**

of a sheet-direction operator, double-coded by flag/Levi/radical, unique in its parabolic family; the 81 as a saturated lattice index; the  $A_2$  bit carrier + v566 promoted + ledger/paper/diary/website sync)

- **Audit probe** `parabolic_selfcode_audit_probe.py` (23/23, verdict **SELF-CODE-VERIFIED**). An external review proposed exact statements about the v218 sheet-diamond direction operators  $U = Q \operatorname{diag}(1, 0, 0)$ ,  $V = Q \operatorname{diag}(0, 1, 1)$ ; every claim holds exactly against the corpus objects, *including the saturation check the review had not run* (the integral order stabilises at word length 4; the word-basis Smith form survives). *The parabolic theorem*:  $\mathcal{A} = \mathbb{Q}\langle I, U, V \rangle$  is the full  $2|1$  line-stabiliser algebra (dim 7, word determinant  $-81$ , multiplicatively closed, mutation-fragile). *The double self-code*:  $H = I + \frac{1}{2}V(V - I)$  has spectrum exactly  $\{1, 1, 2\} = a$ , and  $\chi_H = x^3 - 4x^2 + 5x - 2$  carries  $(4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$  and  $= (\text{flag}, \text{Levi}, \text{radical})$  of  $\mathcal{A}$  simultaneously. *The uniqueness theorem*:  $(x - m)(x^2 - mx + 1)$  with all-positive-integer roots forces  $m = 2 - a = (1, 1, 2)$  is the only anchor this family can code. *The 81*: the saturated order has elementary divisors  $(1, 1, 1, 3, 3, 3, 3)$  — index  $3^4 = 81$ , cokernel  $(\mathbb{Z}/3)^4$  — equal to the v218 determinant-staircase sum  $(N_{\text{fam}}^4)$ ; the mark-local mechanism stays open. *The  $A_2$  bit carrier*: the basic algebra is the  $A_2$  path algebra with  $\dim \operatorname{Ext}^1 = 1$ : exactly two module classes (split / one nonsplit) — a canonical carrier for the alignment bit; the selection stays with RP/geometry. *Re-encoding typed as such*:  $p_n(a) = 1 + \operatorname{tr}(V^n)$  (AnchorLadder through  $V$ ; 240 and rank 8 as trace readouts). *Test design*: the cross-ratio check  $\text{CR} = 4/3$  for the four CONTRACT.F.01 solvers (universal-progression honesty wired; negative control 28/25). *Reformulation*: the Paper-II rank-3 form as  $(1, 2)$  Lorentz geometry with the relative-pencil identities (eigenvalue locking as the named prime-front instrument).
- **Promotion** `v566_parabolic_anchor_selfcode.py` (DIAMOND.PARA.SELFCODE.01, 14 checks,  $\sim 1$  s, exact). New ledger row; update notes on DIAMOND.AXIS.01 (self-code + 81-index cross-link) and CONTRACT.F.01 (the preregistered cross-ratio check added to the contract surface); suite 559  $\rightarrow$  560 scripts. **No P2 reduction and no marker moves**: the “ $g_{\text{car}} = \dim(\mathcal{A}/J)$  derived” route is conditional on the open *No Carrier Reconstruction* audit (rebuild  $Q/V$  without carrier-aware inputs) — the review’s Priority-1 workstream, recorded as the named open door; the 2..9 dimension table is typed as compression (no-free-pattern).
- **Sync**. `tfpt_1` (self-code paragraph after v23), `tfpt_2` (self-code + 81-index +  $A_2$ -carrier paragraph after the v218 block), `tfpt_research_contracts` (cross-ratio check in the CONTRACT.F.01 section); diary entry in `experiments/next.txt`; website counters and mirrors regenerated; README/zenodo 560.

**2026-07-30 (II) (WOIT  $\beta_3$  executed: the  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality typed against  $\sigma_{\text{std}}$  — the OS cut and the  $(2, 2)$  signature forced, the kill branch empty by algebra, the induced member = the  $\Theta_{\text{phys}}$  carrier (SUCCESS) + v565 promoted + contract/paper/diary/website sync)**

- **Discovery probe** `woit_os_beta3_pt_dual_probe.py` (25/25, 2.4 s, verdict **SUCCESS** per the frozen preregistration; contract wording verbatim: **success = Woit’s Minkowski conjugation identified with the family-D conjugation on the quotient, kill = the duality forces a clock-centralising (family A) structure**). *The class (block theorem)*: a  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality is a nondegenerate Hermitian form  $\Phi$  on  $\mathbb{C}^4$ ; clock compatibility has entrywise residual  $(\lambda_j \lambda_k - 1)\Phi_{jk}$  and forces the four cross-eigenspace entries to zero: two  $2 \times 2$  Hermitian blocks,  $16 \rightarrow 8$  real parameters (the deck adds nothing; the wrong clock lift breaks the split). *The dichotomy and the forced  $(2, 2)$* : the  $\rho_{\text{tw}}$  pairing  $M^\dagger \Phi M = \varepsilon \Phi$  splits the class exactly in two — the Euclidean branch ( $\sigma_{\text{std}}$ ’s home) has seam-line norm  $a(|z|^2 + 1)$ , no null point, no OS cut ever; the Minkowski branch has norm  $a(|z|^2 - 1)$ : *the null locus meets the seam line exactly in the seam circle* for every member (the OS cut is forced, not imposed), and  $\Phi_1 = -\bar{\Phi}_i$  exactly (charpoly mirrored), so the total signature is  $(2, 2)$  for every nondegenerate

member — Woit’s Minkowski signature is a theorem of clock +  $\rho_{\text{tw}}$  pairing, not an input. *The kill is empty by algebra*: every compatible duality inverts clock and deck (family D); family A would need  $\rho_4^2 = \text{scalar}$ , but  $\rho_4^2 = \text{diag}(-1, 1, -1, 1)$  — no solution, decided not scanned;  $s_{\Phi}^2 > 0$ : the Kramers-free class. *The identification*: every seam-preserving Minkowski member induces the same conjugation  $z \rightarrow -\bar{z}$  ( $\mu = -1$ , fixed points  $\pm i$ ), perpendicular to  $\sigma_{\text{std}}$ ’s  $z \rightarrow +\bar{z}$ ; under the declared RP-side placement (exact rational- $\pi$  arithmetic, offsets  $\mp\pi/2$ ) it is  $\theta_{\perp}$  — the mirror whose descent v524 certified:  $\Theta_{\text{phys}}$  is the quotient shadow of the duality (regressions re-verified: Gram PD (16, 0, 0), anti-unitary with  $\eta_{\perp} = +i$ ,  $\Theta^2 = +1$  per sector, time reversal, torsor closure). Role separation complete:  $\sigma_{\text{std}} \sim$  the OS cut (the metric), the  $\Phi$ -twisted duality  $\sim$  the descending mirror. Free/quotient level only;  $\mathcal{A}_{\text{hol}}$  untouched; all seven kill tests stay live; no marker moves.

- **Promotion v565\_woit\_beta3\_pt\_duality.py** (WOIT.BETA3.DUALITY.01, 22 checks,  $\sim 2$  s, sympy-exact + two 40-digit inertia certificates). New ledger row; WOIT.OS.TWISTOR.01 updated ( $\beta_3$  executed,  $\gamma$  next); suite 558  $\rightarrow$  559 scripts.
- **Sync. tfpt\_research\_contracts**: title line ( $\alpha/\beta_1/\beta_2/\beta_3$  executed,  $\gamma$  next), the  $\beta_3$  executed block, the roadmap item, the status paragraph, the straddle-hurdle wording ( $\gamma$  and beyond); **tfpt\_3**: new Step 32, story counters thirty-one  $\rightarrow$  thirty-two (title, intro, preamble), the Woit-bridge paragraph gains the  $\beta_3$ -stage status; diary entry in **experiments/next.txt**; website: **papers.ts** WOIT milestone list + v565, **suite.ts** 558  $\rightarrow$  559, **ScriptIndex/changelog** mirrors regenerated; **README/zenodo** counters 559.

**2026-07-30 (I) (T178 backflow: the frame door of the CP mirror decided — R1 by theorem, R2 by anchor, the degeneracy explained (FIBER-PINNED) + v564 promoted + v563 Paper-II readouts registered + ledger/paper/diary/website sync)**

- **Prime-front diary T178** (**cp\_frame\_carrier\_probe.py**, contract **FRAME.CARRIER**, 31/31, 2.4 s, verdict **FIBER-PINNED**, sandbox; **fifth reverse-flow probe — T177’s two open items are pure finite algebra and are *decided*, not measured**). *R1, natural half, negative by theorem*: the fiber acts on the frame by a scalar on the  $n$ -row, so every homogeneous frame block factorises exactly ( $P(\rho n) = \rho^m P(n)$ , checked on fully generic degree-1..4 polynomials with symbolic coefficients), and the metric-free invariants of  $(\mathbf{1}, d, n)$  have *one* generator (bracket enumeration exact: the six permutations  $= \pm\Delta$ , the other 21 vanish identically in a generic  $n$ ; Weyl 1946 FFT as the cited class boundary) — every scale-free ratio is constant, so no frame-coupled scale-free  $\pi/3$ -carrier exists without a metric: T177’s obstruction sentence is now a theorem. *R1, metric half, non-canonical*: declared metrics admit frame-coupled Tier-2 carriers, but they split into two character classes  $((-1, -1, +1)$  vs  $(-1, -1, -1))$  and the admissible normalisers contain the continuum  $q_t = t n_1^2 + n_2^2 + n_3^2$  (values  $441 \sin(\pi/3)/(25t + 117)$ , strictly monotone) — no canonical frame factor; the factors 21/8, 21/2, 441/142 are normaliser-relative frame data with the anti-numerology fence wired as a check. *The single-sine lemma (why T177 ended DEGENERATE)*:  $\{|\text{Im } \rho_n^k|\}$  has exactly one nonzero element iff  $n$  is crystallographic ( $n \in \{1, 2, 3, 4, 6\}$ ,  $\varphi(n) \leq 2$ ), by  $\sin(4\pi/n) - \sin(2\pi/n) = 2 \sin(\pi/n) \cos(3\pi/n)$  (only zero at  $n = 6$  on  $n \geq 5$ ) — the compiler’s two clocks  $\mu_4$  ( $j = 1728$ ) and  $\mu_6$  ( $j = 0$ ) are exactly the two maximal single-sine fibers, so value-based pinning of functionals is impossible there by theorem: T177’s verdict was *forced*. *R2 selected by the anchor*: the Jarlskog invariant carries the full character tuple  $(-1, -1, -1)$  under (conjugation, sheet flip, deck reflection), each certified at 40 dps — matching the sheet-odd orientation class only; the unique anchor-matching *canonical* carrier is the fiber-pure  $\text{Im } \rho = \sin(\pi/3)$  (selected, conditional on the anchor — not derived). No marker moves; **REG.FREEZE.01** untouched; the CP residual is relocated with a proof, not closed.
- **Promotion v564\_cp\_fiber\_pin.py** (CP.FIBER.PIN.01, 22 checks,  $\sim 2$  s, sympy-exact).

The exact cores as one load-bearing module: the generic Euler factorisation, the one-generator bracket enumeration, the character split and the  $q_t$  continuum (anti-numerology as a check), the single-sine lemma (identity + enumeration to  $n = 24 + \text{totient criterion}$ ), and the anchor tuple with the selection and the CP.MU6.01 consistency control. New ledger row CP.FIBER.PIN.01; update note on CP.CHANNEL.IDENT.01 (R1 decided / R2 selected, markers unchanged).

- **Housekeeping:** `v563_paper2_readouts.py` (PRIME.PAPER2.READOUT.01, 24 checks) **registered**. The Paper-II readout closure was written in the previous session but never entered `run_all/registry/ledger`; registered now with its own ledger row. Paper II (parity-toeplitz\_classification): the cross-reference list of non-recomputed readings narrows from three to one (the fitted exponents of the rate measurement), four new cross-reference rows, reproducibility section now “nine modules v555–v563”.
- **Sync.** Diary entry Teil 178 in `experiments/next.txt` (counters 178 probes / 4820/4820 checks); `tfpt_prime_front`: verified layer 27  $\rightarrow$  28 modules (+ v563 subsection), new T178 subsection (`sec:t178`), T177 rest-list pointer updated, probe index T11–T178 + T178 row, counters and phase-2 range labels; `tfpt_5_redteam`: Target-D keybox (the fiber location now a theorem) and the CP-location winbox (frame door decided, single-sine lemma, anchor selection) citing `v564_cp_fiber_pin.py`; suite 556  $\rightarrow$  558 scripts; website mirrors regenerated.

**2026-07-29 (XXII) (prime-front phase 2, the endgame: T176 — the density curve sits at zero and the measurement programme closes (SITS-AT-ZERO) + v562 dense-limit identities promoted + paper/diary/website sync)**

- **Prime-front diary T176** (`dense_limit_probe.py`, contract DENSE.LIMIT, 24/24, 118.7 s, verdict SITS-AT-ZERO, sandbox; lands *after* T177 — part numbering by contract, series counters raised by T176’s checks). The big sieve:  $\text{ATOM\_MAX } 1.2 \times 10^6 \rightarrow 2.5 \times 10^7$  (factor 20.8; cost 0.08 s / 59.7 MB peak — the sieve was never the binding constraint), 1 566 727 prime-power atoms, density ceiling 361  $\rightarrow$  6120 (6118 reached), 14 532 cells on 692 anchors, 0 indefinite. *The extended curve* (eight ratio-4 bins, cluster-robust):  $+0.8886 \pm 0.1027 \rightarrow +0.5148 \rightarrow +0.1948 \rightarrow +0.3922$  (declared monotonicity violation, explained by curvature)  $\rightarrow +0.1520 \rightarrow +0.0579 \rightarrow +0.0376 \pm 0.0410$  (dens  $\sim 779$ ,  $0.9\sigma$ )  $\rightarrow +0.1046 \pm 0.0881$  (dens  $\sim 2512$ ,  $1.2\sigma$ ); pooled over the new bins  $+0.0496 \pm 0.0372$  ( $1.3\sigma$ ) — no sign change, no stabilisation above zero; density slope  $-0.0725 \pm 0.0090$  ( $8.0\sigma$  over 4.0 decades); *the plateau window narrows*  $0.264 \rightarrow 0.074$  ( $2\sigma$ ), a factor 3.6; all six old bins reproduce at  $0.0\sigma$  against T175. Both curvature channels survive ( $\log h +0.1529 \pm 0.0258$ ,  $5.9\sigma$ ;  $\log \text{dens} -0.0434 \pm 0.0060$ ,  $7.3\sigma$ ); none of the six preregistered reparametrisations makes it one exponent. *R1 (T175) closed*: the Feynman–Hellmann form  $d\lambda_{\min} = v^\top d\hat{A} v$  reproduces the rebuild to  $3.37 \times 10^{-6}$  at the minimum of a declared step ladder with the U-shape exhibited;  $d\text{DEL}/\text{DEL}$  carries 84%. *The  $300\times$  harmonic anomaly explained as a pole*: PD survives 2% of the phase period ( $\delta_{\text{PD}} = 5.9 \times 10^{-3}$ , product  $5.8 = O(1)$ ), and the two lowest modes are already 13.4% apart at  $\delta = 0$  — the intervention runs along a near-degeneracy, it does not create one. *Anti-promotion*: the estimator is LADDER-DEPENDENT at the 0.20 level — every quoted bin deficit must cite its rung ladder. True zero / power-law approach / low plateau stay indistinguishable (one more decade of window would cost  $\text{ATOM\_MAX} \sim 2.5 \times 10^9$ ); **the phase-2 measurement programme closes as planned** — no follow-up probe; the work continues in the classification papers and the backflow lines.
- **Promotion v562\_dense\_limit\_identities.py** (PRIME.DENSE.LIMIT.01, 16 checks,  $\sim 0.5$  s, deterministic, small sieve). The exact cores per instance: the Feynman–Hellmann identity with the step-ladder U-shape check and must-break mutations (R1.CLOSED); the phase-pole mechanism (PD-survival window, near-degeneracy at  $\delta = 0$ , sign change bracketed); sieve consistency at construction level (complete-comb lag vectors bit for bit under the bigger table; the truncated must-break moves); and the curve estimator as a deterministic procedure *with the ladder caveat wired as a check and into the ledger row*. SITS-AT-ZERO typed as a

MEASUREMENT with the ladder caveat; the plateau/zero distinction explicitly OPEN; no RH statement. Suite 555 → 556 scripts.

- **Sync.** Diary entry Teil 176 in `experiments/next.txt` (counters 177 probes / 4789/4789 checks; phase-2 closure noted); prime-front paper T176 subsection (the extended curve as a table, the sieve result, the pole finding, the ladder caveat; keybox “After T176: the measurement programme closes”) + probe-index row + abstract counters + the final “Where phase 2 now stands” paragraph; website: prime-front feed entry T176 + v562 promotion entry, status counters, `ScriptIndex` / changelog mirrors regenerated.

**2026-07-29 (XXI) (T177 backflow: the CP mirror — two invariants carry  $\pi/3$ , four CP ledger rows narrowed, none upgraded (DEGENERATE) + v561 CP channel identities promoted + ledger/paper/diary/website sync)**

- **Prime-front diary T177** (`cp_invariant_probe.py`, contract `CP.INVARIANT`, 49/49, 0.2 s, verdict **DEGENERATE**, **sandbox**; the reverse flow resumed while T176 runs). The gauge-degree toolkit of T173/T174 turned on the four CP ledger rows (`REDTEAM.D.01`, `E8.CHAN.01`, `CP.MOD.01`, `DUAL.FRAME.01`). All four constructions rebuilt exactly from the compiler primitives ( $\delta = \pi/3 + 3\lambda_C^2 = 1.198231638$  rad certified against v88/v220;  $\det(\mathbf{1}, d, n) = 21$ ;  $J = 3.327 \times 10^{-5}$  with the CKM unitary to  $10^{-41}$ ). *New exact identity*: on the hexagonal unit  $\rho = e^{i\pi/3}$  the phase channel carries  $\sum_k \rho^{d_k} = 4\rho$  (arg =  $\pi/3$  exactly, sheet-even — the  $E_8$  degrees are even) against the magnitude channel  $\sum_k \rho^{m_k} = 4$  (arg = 0 exactly, sheet-odd — the exponents are odd). *Tier-1 invariance is empty by algebra* (both kernels commute with conjugation, so every  $\pi/3$ -carrier is conjugation-odd); *two* inequivalent Jarlskog-class invariants carry  $\pi/3$  (sheet-even channel angle vs sheet-odd orientation), neither coupled to the frame — what is frame data is the oriented volume 21 and the sign, not the angle. The Jarlskog benchmark passes the invariance criteria and fails only the  $\pi/3$ -carrier test (its content is  $\delta$ ), validating the criterion instead of the value.
- **Four ledger rows narrowed, NO marker moves.** `REDTEAM.D.01`:  $\arg \prod c = 0$  exactly on both sectors; the residual is a *fiber* datum, not a magnitude datum (branch blindness = order divides  $N_{\text{fam}} = 3$ ). `E8.CHAN.01`: the exact hexagonal realisation + parity typing recorded; the phase-channel angle is sheet-blind (cannot distinguish quark from lepton). `CP.MOD.01`: scope narrowed — the invariant part of the frozen reading is the leading term  $\arg \rho = \pi/3$ , the full  $\delta$  is branch/frame-relative (no character; `REG.FREEZE.01` untouched). `DUAL.FRAME.01`: the orientation *reading* is degree +1 in the frame normalisation ( $21 \sin(\pi/3)$  is frame data); the invariant core is the fiber datum  $\text{Im } \rho = \sin(\pi/3)$ , the sign a declared orientation. Prose narrowed in `tfpt_5_redteam` (Target D keybox + the located-residual box).
- **Promotion v561\_cp\_channel\_identities.py** (`CP.CHANNEL.IDENT.01`, 25 checks,  $\sim 1$  s, **sympy-exact**). The exact cores as one load-bearing module: the hexagonal channel identity with parity typing and mutation controls ( $e^{i\pi/5}$ , mutated lists) plus the honest caveat wired as a check; Tier-1 emptiness as algebra; the frame removal ( $\text{Im}(\rho \det)/\det = \text{Im } \rho$  exactly, degree +1 account); the two Tier-2 characters as verified homomorphisms; the Jarlskog anchor ( $J$  passes rephasing/CP-oddness, fails only the carrier test). Narrowing statements only — no CP derivation claimed, R1/R2 explicitly OPEN. Suite 554 → 555 scripts.
- **Sync.** Diary entry Teil 177 in `experiments/next.txt` (counters 176 probes / 4765/4765 checks; T176 and Paper II run in parallel); prime-front paper T177 reverse-flow subsection + probe-index row + counters; website: prime-front feed entry T177 + v561 promotion entry, status counters, DAG node, `ScriptIndex` / changelog mirrors regenerated.

**2026-07-29 (XX) (prime-front diary T175 (the placement phases are real and causal but non-smooth, the heterogeneity dissolves into the error bar, and the**

**densest reachable bin is consistent with zero — PHASES-RESIST) + paper + website sync; NO promotion, NO marker moves)**

- Prime-front diary T175** (`phase_placement_probe.py`, contract `PHASE.PLACEMENT`, 38/38, 20.2s, 1504 cells on 188 anchors — every prime power in  $[40, 1090]$  is an anchor, no selection, no frame rule anywhere — crossed with the T174 ladder  $h = 160 \dots 1280$  plus the declared floor  $h = 128$ ; verdict **PHASES-RESIST, sandbox**). (1) *The phases*. The preregistered anchor invariant  $\theta_p = \text{frac}(\log p / \alpha)$  is unusable and the probe says so out loud (a smooth monotone function of  $\alpha$ , Rayleigh  $Z = 1296$  — a regression on it would be an  $\alpha$ -regression in disguise); only the cell phase  $\varphi_p(h) = \text{frac}(h\theta_p)$  is free and equidistributed (largest  $Z = 1.41$  on 1316 cells), with the generating identity  $\text{frac}(\log p / D) = \text{frac}(h \text{frac}(\log p / \alpha))$  certified at  $1.1 \times 10^{-13}$  (Weyl 1916). *The phases are real in log R*:  $F = 10.80$  against a within-rung-scrambled null with 95% point 1.69, +10.3% held-out residual variance under a rank-parity split, a composite placebo family at  $F = 0.81 / -0.0275$ , amplitudes monotone in  $p$  (0.018 at  $p = 5$  to 0.141 at  $p = 19$ ). *But they are not the heterogeneity channel*: residual  $\chi^2/\text{dof} = 2.52$  from 3.14 (target  $< 2$  missed), the held-out excess scatter gets worse (−15.4%), the deficit untouched ( $0.18\sigma$ ); the exploratory scan excludes every other anchor statistic (all improve in sample and worsen held out). *What resolves the heterogeneity is the error bar*: with jackknife slope errors the excess scatter goes  $0.1224 \rightarrow 0.0000$  and the robust pull spread  $1.41 \rightarrow 1.22$  (model-free interleaved half-split  $0.2722$  against  $\text{sd} = 0.2789$ ) — T174's  $\chi^2$  was substantially an OLS independence assumption, not a physical spread; *the 5 $\sigma$  deficit is unaffected* (its bar never came from  $\text{se}_n$ ). (2) *The dense limit*. The corner is exhausted ( $\text{dens} \leq 361$  forced by the sieve cap and  $h \geq 128$ ; 359 reached); the deficit( $\text{dens}$ ) curve over three decades (6 declared ratio-4 bins, cluster-robust) falls  $+0.982 \pm 0.213 \rightarrow +0.576 \rightarrow +0.207 \rightarrow +0.194 \rightarrow +0.088 \rightarrow -0.057 \pm 0.132$  — monotone within  $2\sigma$  at every step, density slope  $6.1\sigma$ , *the densest reachable bin consistent with zero*; the extrapolation fence is explicit: a true zero, a power-law approach and a plateau below 0.132 are all consistent — undecidable under this sieve. The sharpened confound test finds *both* channels significant ( $\log h + 0.2435 \pm 0.0670$ ,  $\log \text{dens} - 0.0631 \pm 0.0172$ ; curvature  $a_2 = -0.137 \pm 0.029$ ) — T174's density driver survives but is no longer unique; the deficit is not a single exponent. Unplanned load-bearing finding: mass-preserving alternative assemblies are *inadmissible* (lump 0%, wide 46%/0% PD survival); the admissible continuous deformation gives sensitivities +0.384 (sparse) vs −1.054 (dense), ratio 0.4 — the sparse corner is *not* an assembly artefact. (3) *Rule proposal + intervention*.  $\text{dens} \geq 1$  proposed for the chain convention ( $+0.722 \pm 0.094$  unresolved vs  $+0.099 \pm 0.019$  resolved,  $6.5\sigma$  separation, 6% of cells; explicitly not retroactive); comb scramble destroys 12/12; *the new phase intervention* (with an exact  $\delta = 0$  rebuild control): displacing the six family atoms inside their cells gives  $d \log R / d\delta = 879$ , linear to a factor 1.011 over four declared decades — *the placement phase is causal for R*, the first interventional statement of the series — and it overturns the *shape* of the D1 result: the fitted first harmonic predicts only 2.89 (a factor  $\sim 300$  too small), positive definiteness dies for  $\delta \geq 10^{-3}$ ,  $\text{GAP} = 1.6 \times 10^{-9} - 4.8 \times 10^{-6}$  — the Schur floor is a near-degeneracy and  $\log R$  is necessarily *non-smooth* in the phases. Candidates P175-1–P175-8 PENDING, no promotion. The rest: the closed-form  $d\lambda_{\min} = v^T d\hat{A} v$  response (R1); *the dense-corner zero needs a larger sieve, nothing else* (R2); the curvature vs the exponent reading (R3); an  $h$  floor for the positivity margin (R4); the last 22% of the pull spread (R5). T176 (`DENSE.LIMIT`, `dense_limit_probe.py`) is running — the larger sieve. Diary counters 175 probes / 4716 checks (series (CLXXXVI)).
- Paper + website sync.** `tfpt_prime_front`: new phase-2 subsection (T175: the cell-phase construction with the invariant trap said out loud, the causality finding as key result, the error-bar resolution of the heterogeneity as the honest correction, the density curve with the zero-consistent end bin and the extrapolation fence, keybox “After T175” with the larger sieve as the only road and T176 running), section title T126→T175, abstract counters 175/4716, probe index extended (T175 row, running totals, T176 running); `experiments/next.txt` T175 entry (counters 175 Proben / 4716 Checks, Serie (CLXXXVI)); website: `primeFront.ts` feed



entry T175 (verdict enum PHASES-RESIST added), status surfaces Teile 11–175 / 4716 checks (page.tsx, HonestyBanner.tsx, ModuleLadder.tsx). NO promotion; NO marker moves; not RH evidence.

**2026-07-29 (XIX) (v560 — the frame/deficit identities promoted; prime-front diary T174 (the gauge route is exhausted by theorems, and the deficit is measured frame-free at  $5.0\sigma$  — PARTIAL-CANCEL) + paper + website sync; NO marker moves)**

- New module v560 (PRIME.FRAME.DEFICIT.01).** Promotion of the theorem cores of discoveries T172 (contract FRAME.BEYOND), T173 (FRAME.RATE) and T174 (CANCEL.IDENTITY) as one module (v560\_frame\_deficit\_identities.py, 24 checks,  $\sim 0.5$  s) — *the gauge route of phase 2, closed as algebra*, the companion to v559's sixteen-link map, recomputed on a small declared surface (a  $4 \times 3$  (anchor,  $h$ ) rectangle of gap-blind frame-B cells, prime-power anchors  $n = 41 \dots 557$  crossed with the declared ladder  $h = (160, 226, 320)$ , comb COMPLETE on every cell, PD low block on 12/12; one non-prime-power instance  $n = 210$ ; one truncated-comb stress instance  $n = 1009$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$ ; the builder's signature is  $(n_{\text{zone}}, \alpha, M)$  — no frame label is an argument of any observable). (1) *The identity transfer + the sieve-horizon discriminator* (T172): the KMS/Schur pair ( $3.2 \times 10^{-16}$ ), the lag-sum/correlation identity ( $1.4 \times 10^{-16}$  of the larger half; Abel 1826), the R4-free det collapse (at 0.020 of the conditioned bar  $64\varepsilon/|1 - r_{12}^2|$ ) and the additive split ( $7.6 \times 10^{-16}$ ) re-derived EXACTLY on a frame-B and a non-prime-power instance; the truncated-comb instance turns INDEFINITE ( $\lambda_{\min} = -9.6 \times 10^4$ ) against 12/12 PD complete-comb cells, the identity links holding either way (none uses positivity) — indefiniteness sits at the declared sieve horizon, not at a frame or zone. (2) *The  $q = 1 - s$  identity + the calibration* (T173):  $hD = \alpha$  per cell forces the OLS slopes to obey  $q = 1 - s$  to  $2.9 \times 10^{-14}$  (the  $h^{-3}$  target was the  $q = 1$  idealisation of a  $\Theta(D^3)$  law, as an identity); the gauge-degree account comes out as integers —  $G2 = t/\lambda_{\max}(B_{LL})$  and the delivery  $1 - r_{12}^2$  both (0,0), so  $G2$  is the unique doubly gauge-invariant arithmetic member;  $G3 = \mu_1^P G2$  identically (the offset-2 explained);  $(D/\alpha)^3 h^3 = 1$  a tautology. (3) *The blindness theorems + the additive limit* (T174): the amplitude  $c \rightarrow \kappa c$  is a gauge of both sides and of  $R$  over twelve decades (worst 0.026 of the per-cell conditioned bar); the scalar  $\mu_1^P$  — *the entire  $h^{-2}$  channel of the demand* — cancels exactly on both sides; the delivery is invariant under the full positive diagonal group while a non-scalar diagonal moves the demand by 0.23–1.40; and neither the arch term nor the comb term alone has a positive Schur floor on a single cell ( $0/12 + 0/12$  against 12/12 for the sum): no factorisation exists, so T173's P6 is NOT promoted to a theorem — only its multiplicative half. (4) *The KMS shape band* (unconditional, declared  $\varepsilon = (K\pi/(2h+1))^2/6$ ): the  $k^2$ -swap moves log GAP by  $\leq 0.312$  of the certified band while the delivery does not move at all. (5) *The placement discriminator*: the value-multiset scramble destroys the floor on 24/24 draws. Each statement carries a mutation control that fails loudly. THE FRAME-FREE DEFICIT IS A MEASUREMENT, NOT A CLAIM OF THE MODULE ( $+0.1111 \pm 0.0222$  stays sandbox MEASURED); R1 typed OPEN (frame-free form); NO RH statement. Ledger row PRIME.FRAME.DEFICIT.01; registered in run\_all.py and script\_registry.csv; suite counter  $553 \rightarrow 554$  scripts.
- Prime-front diary T174 (cancellation\_identity\_probe.py, contract CANCEL.IDENTITY, 37/37, 6.5 s, 84 rectangle + 1020 power-check + 147 anchor-ladder + 178 leg windows; verdict PARTIAL-CANCEL, sandbox).** The cancellation question turned into blindness algebra: both sides of the T173 deficit are scale-free functionals of ONE raw Gram matrix, and the ledger reads — amplitude gauge THEOREM ( $\leq 0.028$  of the per-cell conditioned bar over twelve decades); *the complete  $\mu_1^P \approx h^{-2}$  channel THEOREM* ( $\leq 0.009$  — exactly absent from the deficit); full positive diagonal group THEOREM (delivery exact; the demand keeps the SHAPE — the asymmetry is the whole anatomy); KMS shape against  $k^2$  CERTIFIED (unconditional band,  $\leq 0.0075$  in deficit units = 4.8% of T173's number); arch/comb mixture

DOES NOT CANCEL (additive and exact; no factorisation; neither term alone has a positive Schur floor on one of 84 cells); frame rule THEOREM ( $R = R(\alpha, M)$  identically — a sampling map, two frames on one cell agree bit for bit). Explained along the way:  $G2$  is the unique doubly gauge-invariant arithmetic member of T173’s family, and the  $G3$ – $G2$  offset of exactly 2 is the  $\mu_1^P$  exponent (measured  $h^{-1.9975}$ ). The key result is a direct measurement: on the preregistered  $12 \times 7$  ( $\alpha, h$ ) rectangle the  $h$ -direction carries the deficit ( $a = -0.281 \pm 0.066$ ,  $4.3\sigma$ ; anchor direction flat); the exact between/within split puts it in the GRID response ( $3.3\times$  the anchor selection, whose sign is mostly wrong — the selection reading tested and rejected); the driver is the COMB DENSITY per lag cell (monotone  $+0.281 \rightarrow +0.062$  across the density cuts; on legs  $\log \nu$  and  $\log \text{dens}$  collinear at  $r = -0.921$  — T173’s  $2.8\sigma \log \nu$  driver named and discharged); and the dense corner, exhausted (1069 cells, 151 anchor clusters, cluster-robust), gives  $\text{DEFICIT}(\text{frame-free}, \text{dens} \geq 3) = +0.1111 \pm 0.0222$  —  $5.0\sigma$  from zero,  $0.4\sigma$  from T173’s  $+0.155 \pm 0.102$ . P6 does NOT become a theorem (only the multiplicative half) but is less load-bearing: the deficit is measured on a D-rule-free surface, and the fixed- $\nu$  invariance is explained (fixed  $\nu = \text{fixed density regime}$ ). Stress green (scramble 24/24 destroys the block and moves  $R$  by  $10^2$ – $10^5$ ; jackknives stable; horizons declared). Candidates P9–P15; the rest: the per-anchor scatter ( $\chi^2/\text{dof} = 3.02$ ; candidate: the placement phases  $\text{frac}(\log p/D)$ ), the admissibility question ( $0.11$  vs  $\geq 3$  atoms per cell: factor 2.5), the sparse corner. T175 (PHASE.PLACEMENT, `phase_placement_probe.py`) is running — the phases and the dense limit. Diary counters 174 probes / 4678 checks (series (CLXXXV)).

- **Paper + website sync.** `tfpt_prime_front`: new phase-2 subsection (T174: the cancellation ledger as a compact table, the  $\mu_1^P$ -channel theorem, the  $G2$ /offset-2 explanation, the sampling-map reading of the frame rule, the frame-free rectangle measurement as key result with the density ordering, keybox “After T174” with the phases / dense limit and T175 running), the verified layer extended to twenty-six modules (new subsection for v560), section title T126→T174, abstract counters 174/4678, probe index extended (T174 row  $\rightarrow$  v560, running totals, T175 running); companion blocks for v560 in `tfpt_1_architecture_e8`, `tfpt_3_e8_audit_bootstrap` and `tfpt_research_contracts`; `experiments/next.txt` T174 entry (counters 174 Proben / 4678 Checks, Serie (CLXXXV)); website: `primeFront.ts` feed entries T174 + v560 promotion (verdict enum PARTIAL-CANCEL added), status surfaces Teile 11–174 / 4678 checks / 26 modules v535–v560 (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`), suite counter 554 (`suite.ts`). NO marker moves; not RH evidence.

**2026-07-29 (XVIII) (prime-front diary T172 (the map is portable, the numbers are not — PARTIALLY-PORTABLE) + T173 (the demanded rate is itself a frame datum; the deficit  $+0.155 \pm 0.102$  is the number of the phase and no frame closes it — DEFICIT-VARIES) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T172** (`frame_beyond_probe.py`, contract FRAME.BEYOND, 44/44, 1.3 s, 54 windows on four surfaces plus 3 truncated-comb stress windows,  $h = 128$ – $1444$ ; verdict PARTIALLY-PORTABLE, sandbox). T171’s inherited question — how far does the certified map carry beyond frame A? — answered by a recipe inventory (four degrees of freedom: the frame rule  $D$ , the subdivision  $\nu$ , the zone family; the atom set is NOT free —  $\Lambda$  lives on prime powers, “beyond” means different *anchor* arithmetic) and a transfer table: *13 of the 16 links transfer unchanged — the whole identity tower plus both direction lemmas — and none breaks*; the three that shift are exactly the number-carrying ones (the two window certificates and the rate of link 16). Indefiniteness is *localised at the declared sieve horizon* ( $\lambda_{\min} = -3.7 \times 10^5$  –  $-6.2 \times 10^6$  on 3/3 cut-comb stress windows against 0/54 complete-comb; the identity links keep holding there), not at a frame or zone. Constants map: 8/11 stay in frame A’s band, 3 wander on the congruence surface — a certificate on a new surface is a NEW certificate; the procedure and the directions transfer (54/54). R1’s near-degeneracy persists on every

surface (angle  $< 10^{-2}$  on 54/54, decay on 9/9 legs) while the rate is frame-bound ( $h^{-1.83}h^{-2.87}$ ; zone arithmetic secondary, spread 0.40 vs 1.03); sharpest new result, the placement finding: scrambling atom positions at a fixed  $\Lambda$ -value multiset raises the angle by a median  $6.6 \times 10^4$  and removes the decay — the collapse belongs to the actual prime-power placement. Candidates P1–P4 PENDING (no promotion — bundling with T174 planned). Diary counters 172 probes / 4601 checks (series (CLXXXIII)).

- **Prime-front diary T173** (`frame_rate_probe.py`, contract `FRAME.RATE`, 40/40, 2.2 s, 284 windows plus 109 scramble twins over eleven preregistered frame data (**A** at  $\nu \in \{3, \dots, 16\}$ , **gap-blind B**, **smoothed C**),  $h = 128\text{--}1447$ ; **verdict DEFICIT-VARIES, sandbox**). The demand side itself put on the measurement table:  $q = 1 - s$  is an *exact identity* ( $1.9 \times 10^{-15}$ , from  $hD = \alpha$ ) and  $q = 0.667\text{--}0.777 < 1$  on every frame datum — *T171's  $h^{-3}$  target was the  $q = 1$  idealisation of a  $\Theta(D^3)$  law*. The gap functional is calibrated instead of guessed (six preregistered candidates; the relative I5 gap  $t/\lambda_{\max}(B_{LL})$  hits the chain's 3 on frame A, median 3.033, and is the only dimensionless one;  $(D/\alpha)^3$  exposed as a tautology). The deficit, read as one slope of the per-window ratio  $R = \text{GAP}/(1 - r_{12}^2)$ : pooled  $+0.155 \pm 0.102$  ( $\times 1.54$  over-scatter inflation,  $3.0\sigma$  above zero),  $1.8\times$  flatter than either side ( $\chi^2/\text{dof}$  REQ 4.87 / DEL 4.17 / difference  $2.38 = 2.4\sigma$  — constancy rejected), split-stable to 0.214 where both single sides drift  $\sim 0.5$ , *invariant under the anchor-to-grid rule at fixed  $\nu$*  ( $\chi^2/\text{dof} \leq 0.64$  — theorem candidate for that half only), not invariant under  $\nu$  ( $2.8\sigma$ , confounded with arithmetic content under the cap;  $q$  the weakest driver at  $0.2\sigma$ ). Honest correction to T172 (P7): in the fixed- $h^{-3}$  convention the wider surface gives  $+0.015 \pm 0.084$  — consistent with zero — so T172's  $C$ -growth was neither sign- nor split-stable (its verdict is untouched). *R1 now stands frame-free*: the delivered near-degeneracy closes slower than the relative I5 gap it must pay, by  $h^{+0.155 \pm 0.102}$ ; no frame closes it (best 2.s.e. upper edge  $+0.170$  at B; 8/11 legs individually consistent with zero — pooled with the penalty, not claimed per leg) — frame shopping is over, by numbers. Stress: scramble removes the decay on 4/4 declared legs; jackknife  $\leq 0.026$ ; shuffle null  $3.7\sigma$  away. Candidates P5–P8 PENDING (no promotion — bundling with T174 planned). T174 (`CANCEL.IDENTITY`, `cancellation_identity_probe.py`) is running — the cancellation identity. Diary counters 173 probes / 4641 checks (series (CLXXXIV)).
- **Paper + website sync.** `tfpt_prime_front`: two new phase-2 subsections (T172: the four-degrees-of-freedom inventory, the 13/16 transfer, the sieve-horizon localisation, the placement finding as key result; T173: the  $q = 1 - s$  identity as display, the calibration, the compact deficit table, the honest T172 correction, the frame-free R1 statement as key result, keybox “After T173” with the cancellation identity and T174 running), section title T126→T173, abstract counters 173/4641, probe index extended (T172 + T173 rows, running totals, T174 running); `experiments/next.txt` T172 + T173 entries (counters 172/4601 (CLXXXIII) and 173/4641 (CLXXXIV)); website: `primeFront.ts` feed entries T172 + T173 (verdict enums PARTIALLY-PORTABLE and DEFICIT-VARIES added), status surfaces Teile 11–173 / 4641 checks (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`). NO promotion (T172s P1–P4 and T173s P5–P8 PENDING — bundling with T174 planned); NO marker moves; not RH evidence.

**2026-07-29 (XVII) (v559 — the phase-2 capstone promoted; prime-front diary T171 (the map is complete: all sixteen links in one connected run, eight no-gos as classified, the precision ledger closed — MAP-COMLETE) + paper + website sync; NO marker moves)**

- **New module v559 (PRIME.PHASE2.CAPSTONE.01)**. Promotion of discovery T171 (contract `FINAL.MAP`, verdict `MAP-COMLETE`) as the phase-2 capstone (`v559_phase2_capstone.py`, 35 checks,  $\sim 0.3$  s) — *the capstone verifies the map and claims nothing new*: T126–T170 reduced the I5 uniformity question through sixteen exact reformulations to ONE object (R1), and this module is the load-bearing version of T171's connected run, everything recomputed on

v551–v558’s declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$ ; the  $16 \times 16$  low block positive definite on 12/12 — a numerical fact, never routed through the Weil criterion). *The chain, sixteen links in one pass* (13 THEOREM / 3 CERT): T152 Schur two-block (KMS  $1.3 \times 10^{-15}$ ; holds at  $0.5\lambda_{\min}$ , fails at  $1.02\lambda_{\min}$  — sharp; Haynsworth  $3.8 \times 10^{-15}$ ); T151 Sobolev  $b_k \leq C_S k$  (worst ratio 0.4816); T153  $\Psi$ -collapse with the Z-law ( $6.3 \times 10^{-16}$  /  $9.0 \times 10^{-17}$ ); T154 Ritz ceiling from below ( $g_{32}/s = 0.9520$ – $0.9849$ ); T155  $12 \times 12$  floor, sharp ( $t = 0.2295$ – $0.3719$ ); T156  $F(P, r)$  ( $L(16) = 6.6$ – $15.6$ ); T158 Thomson/ladder ( $3.1 \times 10^{-12}$ ; 36/36 trials below); T160 sampling (closed Dirichlet weights = correlation weights  $1.1 \times 10^{-15}$ ; sampled  $\Lambda$ -mass  $9.9 \times 10^{-15}$ ); T163 Abel swap with the TV floor (overshoot  $\times 5.2 \times 10^4$ – $1.1 \times 10^6$ ); T164 gauge/quantifier ( $\Psi(x^*) = 1/g_{16}$  to  $7.0 \times 10^{-12}$ ; 48/48 perturbed vectors larger); T165  $P_{\text{pr}} = 1$  at  $x^*$ ; T166 cascade  $1/(1 - R_{16}^2)$ , diagonal-invariant; T167  $K = 2$  exactness (milder  $\times 1.62$ – $5.20$ ); T169 det collapse R4-free ( $\leq 4.2 \times 10^{-12}$ , no positivity of  $A_h$  consumed); T170 rank-3 ( $\{-2, -1, +1\}$  exact;  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ ); T170 R1 shape via the  $2 \times 2$  Lagrange identity ( $\sin \angle = 1.169 \times 10^{-4}$ – $2.467 \times 10^{-3} \leq 10^{-2}$  per window; *rate and quantifier typed OPEN*). *The no-go battery*, 8/8 failing as classified (size budget  $\times 2.7 \times 10^3$ – $1.6 \times 10^4$ ; symbol infimum  $< 0$  on 12/12; sector invariance; perturbation scale; atom band dominates; sieve operator-norm  $\times 8.8 \times 10^3$ – $7.5 \times 10^4$ ; scramble  $\times 679$  median with rank-3 untouched on 60 draws;  $L_P$  gain = 1 exactly). *The precision ledger*: needed  $2.752 \times 10^{-4}$  at the deepest window — finer than the RH yardstick  $h^{-0.5}$  by  $215\times$  and the best unconditional  $h^{-0.996}$  by  $13\times$  on this surface; both self-similarity identities  $\leq 6.8 \times 10^{-13}$ ; mutation controls fail loudly. ZERO new uniform-in- $m$  statements; the one open phase-2 object R1 explicitly typed OPEN; NO RH statement (RH\_DELTA = 0.5 a yardstick in exactly one role); Weil fence documented (link 14 R4-free by construction). Ledger row PRIME.PHASE2.CAPSTONE.01; registered in `run_all.py` and `script_registry.csv`; suite counter 552  $\rightarrow$  553 scripts.

- **Prime-front diary T171** (`final_map_probe.py`, contract FINAL.MAP, 43/43, 0.6 s, 12 windows over two frames  $\nu = 4/6$ ,  $h = 142$ – $1445$ , lever arm  $10.2\times$ ; verdict MAP-COMplete, sandbox). The capstone stands — the map is complete: all sixteen links of the reduction chain, from the I5 floor to the final R1 shape, reproduce in ONE connected run (13 THEOREM / 3 CERT; honest exclusion said out loud: on  $\nu = 4$ ,  $h = 1445$  the low block is indefinite,  $\lambda_{\min} = -343$ , so the ladder links carry 11/12 while the identity links carry 12/12); all eight classified no-go routes fail on instances exactly as classified (size budget  $h^{+1.32}$ ; symbol infimum negative on 12/12; sector change an invariance  $\leq 2.1 \times 10^{-9}$ ; Kato  $2.0 \times 10^4$ – $1.9 \times 10^7$  off scale; atom band  $h^{+0.79}$  faster; sieve  $\delta = +1.044$  against 3.0; scramble type-flip  $h^{-2.57} \rightarrow h^{+0.21}$ ,  $\times 11421$ ;  $L_P$  gain = 1 exactly); and the precision ledger closes: needed  $2.284 \times 10^{-7}$  at  $h = 1444$  (rate  $h^{-3.0}$ ) against the RH yardstick  $h^{-0.5} = 2.63 \times 10^{-2}$  — short by  $1.2 \times 10^5\times$  — and the best unconditional  $h^{-0.996} = 7.13 \times 10^{-4}$  — short by  $3.1 \times 10^3\times$ ; both self-similarity identities check ( $4.2 \times 10^{-10}$  /  $9.9 \times 10^{-10}$ , sign-agnostic incl.  $\nu_2 < 0$  on 1/12). At link 16 the RATE is frame-dependent ( $\nu = 4$ :  $h^{-1.86}$ ;  $\nu = 6$ :  $h^{-2.87}$ ) while the COLLAPSE is not. Balance of the quotable parts: 40 THEOREM / 5 CERT-UNIF / 9 CERT-WINDOW / 16 MEASURED / 9 NO-GO (T126–T161 honestly unaggregated). The file adds ZERO new uniform-in- $m$  statements — exactly why it is promotable: phase 2 is now a certified map with ONE open object (R1, classified as a near-degeneracy, not a size); R2/R3 booked, R4 removed. T172 (FRAME.BEYOND, `frame_beyond_probe.py`) is running — how far the map carries beyond frame A. Diary counters 171 probes / 4557 checks (series (CLXXXII)).
- **Paper + website sync.** `tfpt_prime_front`: new capstone subsection (T171) with the sixteen-link table, the no-go list and the precision-ledger display, a closing paragraph of the phase-2 section, the verified layer extended to twenty-five modules (new subsection for v559), section title T126 $\rightarrow$ T171, abstract counters 171/4557, probe index extended (T171 row, running totals, T172 running); companion blocks for v559 in `tfpt_1_architecture_e8`, `tfpt_3_e8_audit_bootstrap` and `tfpt_research_contracts`; `experiments/next.txt` T171

capstone entry (counters 171 Proben / 4557 Checks); website: `primeFront.ts` feed entries T171 + v559 promotion (verdict enum MAP-COMplete added), status surfaces Teile 11–171 / 4557 checks / 25 modules v535–v559 (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`), suite counter 553 (`suite.ts`). NO marker moves; not RH evidence.

**2026-07-29 (XVI) (v558 — the bilinear/rank identities of phase 2 promoted; prime-front diary T170 (the bilinear form collapses by a rank-3 theorem, R1 classified as a near-degeneracy — SIEVE-RESISTS) + paper + website sync; NO marker moves)**

- **New module v558 (PRIME.BILINEAR.RANK.01).** Promotion of the closed theorem cores of T168 (contract LAGRANGE.MINORS), T169 (contract TSTAR.RATIO) and T170 (contract BILINEAR.SIEVE) as one deliberately narrow module (`v558_bilinear_rank_identities.py`, 27 checks,  $\sim 0.2$ s), everything recomputed on v551–v557’s declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table): (1) the Lagrange/Wronskian anatomy — the lag-sum identity  $\hat{a}_{ij} = t_i^T A_h t_j = \sum_d c_d W_d^{ij}$  with the closed correlation weights ( $1.6 \times 10^{-15}$ ; the  $1/\sqrt{\mu}$  factors cancel, T169-TH3), Euclidean orthogonality  $t_1 \cdot t_2 = 0$  ( $4.4 \times 10^{-16}$  — the near-parallelism is created *entirely* by the arithmetic metric), the Wronskian telescope  $(\mu_1 - \mu_2)u_r v_r = W_{r-1} - W_r$  with  $W_{-1} = W_{h-1} = 0$  ( $1.9 \times 10^{-17}$ ) and the maximal-minor identity  $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$  ( $4.6 \times 10^{-16}$ ; Lagrange 1773); (2) the exponent account  $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$  identically ( $2.5 \times 10^{-13}$ ) with the unconditional Gershgorin certificate  $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$  (12/12, loss 1.1885–1.1947; the  $|\hat{a}_{12}|$ -free mutation fails on every window); (3) the T169-TH7 det-collapse in closed form,  $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$  ( $8.7 \times 10^{-13}$ ) — the loop T167 scalar  $\rightarrow$  T168 factor  $\rightarrow$  T169 candidate closes as an identity, with the usable bound  $\nu_2 \leq 2 \det \hat{A} / (\hat{a}_{11} + \hat{a}_{22})$  tight to  $< 2$  exactly; (4) the exact bilinear von Mangoldt form (T170 TH1–TH3) —  $\hat{A} = B - S$  against the direct block ( $1.7 \times 10^{-14}$ ), the polarisation  $\det \hat{A} = \det B - D(B, S) + \det S$  ( $1.0 \times 10^{-10}$ ) and  $\det S = \sum_{n,m} \Lambda(n)\Lambda(m)K(n,m)/\sqrt{nm}$  with  $K = \frac{1}{2}D(X_n, X_m)$  through the *explicit* kernel matrix ( $7.5 \times 10^{-16}$ ; Cauchy–Binet 1812/1815), the wedge reading to the measured rank-one defect  $4.6 \times 10^{-3}$ – $3.1 \times 10^{-2}$  (honest CERT); (5) the rank-3 theorem (T170-TH4: polarisation matrix eigenvalues  $\{-2, -1, +1\}$  exactly,  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ , nonzero spectrum =  $\text{spec}(M P^T P)$  — the bilinear form *is* the rank-3 polynomial  $S_{11}S_{22} - S_{12}^2$  in three linear  $\Lambda$ -sums) and the Type II collapse (T170-TH5: Vaughan’s identity coefficient-exact for every  $U = V \in \{3, 4, 5, 6\}$ , Type II kernel 99.9%-energy rank  $[2, 2, 2, 2]$  against generic  $[20, 15, 11, 9]$ ); (6) the R4-free chain identity  $1 - r_{12}^2 = \det \hat{A} / (\hat{a}_{11} \hat{a}_{22})$  (T170-TH6 — the Weil fence never approached), the  $L_P$  no-go ( $\hat{a}_{12} = 1.1 \times 10^{-17}$  identically, no decay) and the scramble discriminator (rank-3 *untouched* on all 60 draws while the value of  $1 - r_{12}^2$  moves by a median factor 834.8: the arithmetic sits in the *joint values* of  $(S_{11}, S_{22}, S_{12})$  against  $B$ ) as must-breaks; every statement with a mutation control; the one open object R1 (the collinearity/near-degeneracy rate) explicitly typed OPEN; NO RH statement. Ledger row PRIME.BILINEAR.RANK.01; registered in `run_all.py` and `script_registry.csv`; suite counter 551  $\rightarrow$  552 scripts.
- **Prime-front diary T170 (`bilinear_sieve_probe.py`, contract BILINEAR.SIEVE, 40/40, 1.8s, 70 frame-A windows  $h = 142$ – $1445$  on the T163–T169 union recipe, 67 arithmetically complete; verdict SIEVE-RESISTS, sandbox).** T169’s standard shape cashed in: the bilinear form is written down *exactly* —  $\hat{A} = B - S$  (TH1,  $3.5 \times 10^{-15}$ ),  $\det \hat{A} = \det B - D(B, S) + \det S$  (TH2,  $1.3 \times 10^{-10}$ ),  $\det S$  a genuine double von Mangoldt sum against the closed wedge kernel  $K = \frac{1}{2}D(X_n, X_m)$  (TH3,  $1.5 \times 10^{-14}$ ) — and on the reference window ( $h = 540$ ) the three pieces read 6.18,  $-198.8$ ,  $192.6$  and sum to  $4.22 \times 10^{-3}$ : a *five-order cancellation* between three  $\sim 200$ -sized pieces, with absolute target  $1.10 \times 10^{-5}$ . The route table (target  $\delta \geq 3.0$ ): truth  $+2.747$  MEASURED; Cauchy–Schwarz  $-0.000$ ; large sieve/duality  $+0.669$ ; Montgomery–Vaughan 1973  $-0.384$ ; the exact split  $+0.996$  (best); Vaughan Type I+II

+0.669 — all THEOREM. The collapse is structural, by two theorems: the **rank-3 theorem** (TH4:  $D$  is the polarisation of  $\det$  on  $2 \times 2$  symmetric matrices, rank 3, signature  $(1, 2)$ , so  $\text{rank } K \leq 3$  for every  $h$  and  $X = \sigma_4/\sigma_1 = 2.0 \times 10^{-15}$ , effective rank 1.38–2.71 against  $N \sim 10^3$ : the form is  $S_{11}S_{22} - S_{12}^2$  in three linear  $\Lambda$ -sums) and the **Type II collapse** (TH5: the closed weights see  $n$  only through  $\log n$ , so Vaughan Type II blocks have effective rank  $O(1)$  for every  $U, V$  — 2 singular values carry 99.9% against 153 generic,  $U = 12$ –70 scanned). Honest T161 comparison: GAINED — 32 frequencies reduce to THREE linear functionals, by theorem; NOT GAINED — the precision, which binds  $(5.3 \times 10^{-5})$  per sum at  $h = 285$ , sharpening  $h^{-3}$ , the RH yardstick  $1.1 \times 10^3$  too coarse;  $\ell^1$  mass  $41\times$  over the signed value). The chain becomes R4-free for free (TH6:  $1 - r_{12}^2 = \det \hat{A}/(\hat{a}_{11}\hat{a}_{22})$  an identity, sharper than T169-TH9 at  $-2.747$  vs  $-2.771$ ; the Weil fence never approached); end to end the best unconditional bound carries  $h^{-0.996}$  against the required  $\leq h^{-2.50}$  — missing by 1.50 in the exponent, and the shortfall is a theorem. Stress: the scramble leaves rank-3 and the unconditional bounds unchanged while the truth moves 2.76 in the exponent — **the arithmetic sits entirely in the joint values of  $(S_{11}, S_{22}, S_{12})$  against  $B$** ; flat-weight breaks;  $L_P$  exact; the  $\nu$ -decoupling is reported unsmoothed (frames  $\nu = 5/8/11$  fall at  $-1.70/-1.27/-1.74$  — T169's  $h^{-2.95}$  is frame-A, the shortfall is not). Balance 6 THEOREM / 1 CERT-UNIF / 5 NO-GO. **R1 finally classified**:  $\det \hat{A}/(|\text{row}_1||\text{row}_2|)$  is the sine of the angle between the two rows of  $\hat{A}$ , falling  $h^{-2.711}$  — R1 asks for an unconditional certificate that two explicit finite  $\Lambda$ -sum vectors become collinear at rate  $h^{-3}$ : a *near-degeneracy*, not a size, beyond any size-bounding tool; needs beyond-RH joint precision. R2/R3 booked in T169; R4 off the path. T171 (`final_map_probe.py`, contract FINAL.MAP) is running — the capstone map of the whole phase.

- **Docs + website sync.** `tfpt_prime_front` gains the T170 subsection (the exact bilinear decomposition as a display, the rank-3 theorem and the Type II collapse as key results, the route table with its  $\delta$  numbers, the five-order cancellation, the final R1 classification as a row-angle near-degeneracy, the honest T161 comparison, the “After T170” keybox with R1-final and T171 running) and the v558 subsection in the verified layer (now *twenty-four* modules); the phase-2 section title moves to T126–T170, the probe index gains the T170 row and the running totals move to 170 probes / 4514 checks; companion v558 blocks in `tfpt_1`, `tfpt_3` and `tfpt_research_contracts`; `experiments/next.txt` gains the entry (series CLXXXI); website: `primeFront.ts` feed entries T170 + v558 promotion, status surfaces Teile 11–170 / 4514 checks / 24 modules v535–v558, T171 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XV) (prime-front diary T168+T169 (the self-similarity of the hardness is *proved*, the one open object becomes a bilinear von Mangoldt sum — MINORS-RESIST / RATIO-RESISTS) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T168 (`lagrange_minors_probe.py`, contract LAGRANGE.MINORS, 39/39, 2.3 s, the same 63-window union as T163–T167, recipe untouched; verdict MINORS-RESIST, sandbox).** The determinant attacked as a sum of squares via the Lagrange identity — and the sum of squares is *real*: the  $h \times h$  arithmetic kernel  $A$  is **positive definite on all 63 windows** ( $\lambda_{\min} = 5.0 \times 10^{-6}$ – $8.5 \times 10^{-4}$ , 8–12 orders above the solver floor), so the Lagrange identity is a genuine sum of strictly positive squares ( $\rho_{\text{LAG}} = 1.000$  exactly, no sign cancellation) — hard-fenced as a statement about one finite matrix per window; the Weil criterion is never tested, assumed or reverse-inferred. The anatomy is exact:  $Q = X^T A X$  ( $3.9 \times 10^{-15}$ ), the mode vectors are **Euclidean-orthogonal** ( $r_{12} = 0$  to  $6.2 \times 10^{-17}$  — the near-parallelism is created *entirely* by the arithmetic metric), the Wronskian telescopes exactly ( $W_r - W_{r-1} = (\mu_1 - \mu_2)u_r v_r$ ,  $1.7 \times 10^{-13}$ ), the minor sup is closed and  $m$ -free ( $\sim N/\pi^2$ , attained to 0.770), Cauchy–Binet verified as algebra. The key measurement splits by basis: in the closed basis the minors

are *maximal* ( $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$  exactly  $\sim h^4$  — the Wronskians carry zero arithmetic, all smallness sits in the PSD kernel at one closed vector); in the eigenbasis the sum is *thin* (90% from 4–101 pairs, top pair always  $(\lambda_{\min}, \lambda_{\max})$  at 6.8–41.7% alone, raw  $\kappa_w = 0.30$ –0.94). The exponent ledger closes arithmetic-free:  $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$  with  $\nu_1 \sim h^{+0.81}$ ,  $\hat{a}_{11} \sim h^{+0.76}$ ,  $\hat{a}_{22} \sim h^{+0.91}$  — every factor but one at the right power; best  $m$ -free-shaped bound  $h^{-0.548}$  union /  $h^{-0.880}$  frame A, short by  $h^{2.45}/h^{2.04}$ ; the open factor is identified exactly:  $\nu_2$  with minimiser  $t^* = Q_{12}/Q_{11}$ , needed to  $6.05 \times 10^{-3}$  median /  $2.82 \times 10^{-4}$  worst ( $h^{-1.146}$ ) — by T168-TH7 that factor *is* the target rewritten (ONE-TERM-MISSING deliberately not rendered). Scramble breaks harder than T167 (8/8 windows lose  $A \succ 0$  *and* the Gram property,  $n_{\text{neg}}$  up to 73);  $L_P$  control exact. **The hardness is self-similar under exact reformulation** — determinant  $\rightarrow$  PSD compound at a closed vector  $\rightarrow$  near-null Rayleigh value  $\rightarrow$  one ratio of two closed lag sums, same accuracy price. Balance 8 THEOREM / 0 CERT-UNIF / 3 CERT-WINDOW / 11 MEASURED / 7 NO-GO.

- **Prime-front diary T169** (`tstar_ratio_probe.py`, contract `TSTAR.RATIO`, 41/41, 3.0 s, same union surface; verdict **RATIO-RESISTS** — the self-similarity *proved*, sandbox). T168’s one open factor attacked as a preregistered candidate table: every genuinely closed candidate misses by  $O(1)$  and *diverges* from the threshold at  $h^{+1.99}$  (the block is atom-dominated,  $|\text{atom/arch}| = 3.5$ –43 resp. 14–461, and  $\hat{a}_{11}^{\text{arch}} < 0$  on 63/63 — the archimedean half is not even PSD); **the only candidate that meets the threshold**,  $K7 = \sqrt{\hat{a}_{22}/\hat{a}_{11}}$  (63/63), **does so by a new theorem (T169-TH7) because it puts  $\det \hat{A}$  back into the numerator**:  $R(K7)$  collapses in closed form to  $2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$  ( $1.1 \times 10^{-9}$ ) — **the  $t^*$ -language is provably exactly as hard as T167’s determinant: the loop closes as an identity** (T167 scalar  $\rightarrow$  T168 factor  $\rightarrow$  T169 candidate = the same object). The real gains: **R3 is booked CERT-UNIF** —  $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$  (**Gershgorin, unconditional, uniform in  $h$** ), 63/63, losing only 1.10–1.19 — **the CERT-UNIF column moves off zero**; R2 is honestly downgraded from cheap to OPEN (the absolute-value budget cannot even certify positivity); the chain is **rebuilt R4-free** ( $1 - r_{12}^2 \leq [\max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|] \cdot R(\hat{t}) / (\hat{a}_{11}\hat{a}_{22})$ , valid 63/63, frame-A trend  $h^{-2.948}$ , i.e.  $\varepsilon = 0.052$  inside the 0.5 carry window) so the Weil-shaped positivity of  $A_h$  never enters it — the Weil fence is respected by construction (uniform-in- $h$  positivity of  $A_h$  is RH-equivalent-shaped and is never routed or claimed); and by T169-TH3 the  $1/\sqrt{\mu}$  factors cancel, so the cond  $\sim 10^8$  ladder leaves the file. Stress:  $L_P$  exact (the  $t^*$ -recipe is a near-degeneracy artefact — its tightness *measures*  $\det \hat{A}$ ), scramble 8/8, flat-weight null 8/8, parity-pair control only  $6.7\times$  (honestly PARTIAL). Balance 10 **THEOREM** / 2 **CERT-UNIF** / 1 **CERT-WINDOW** / 4 **NO-GO**. The rest: **R1, for the first time in standard analytic shape — an  $m$ -free bound on  $\det \hat{A}$  at  $h^{-3+\varepsilon}$  as a bilinear von Mangoldt sum against closed Dirichlet weights** (equivalently the diagonal ratio to  $1.2 \times 10^{-2}/5.6 \times 10^{-4}$ ); R2 = R1; R3 closed; R4 not needed. T170 (`bilinear_sieve_probe.py`, contract `BILINEAR.SIEVE`) is running the unconditional sieve toolbox (Montgomery–Vaughan, Vaughan decomposition) on the bilinear form — T161’s beyond-RH triage applied to *linear* sums; for *bilinear* forms the large sieve gives unconditional square-root cancellation.
- **Docs + website sync.** `tfpt_prime_front` gains the T168 subsection (the Lagrange identity and the PSD finding with the hard Weil fence, the basis split of the key measurement, the closed exponent ledger, the self-similarity statement) and the T169 subsection (the candidate table, the collapse theorem TH7 with the loop closure as the key result, the CERT-UNIF gain, the R4-free chain with  $\varepsilon = 0.052$ , the standard-shape restatement, the “After T169” keybox with R1 in bilinear shape and T170 running); the phase-2 section title moves to T126–T169, the probe index gains the T168+T169 rows and the running totals move to 169 probes / 4474 checks; `experiments/next.txt` gains both entries (series (CLXXIX), (CLXXX)); website: `primeFront.ts` feed entries T168+T169, status surfaces Teile 11–169 / 4474 checks, T170 running. NO promotion (T168/T169 candidates PENDING — bundling with T170 planned). Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed



## THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XIV) (v557 — the cascade/vector identities of phase 2 promoted; prime-front diary T167 (three dresses are one inequality, the closed vector exact at  $K = 2$  — VECTOR-RESISTS) + paper + website sync; NO marker moves)**

- **New module v557 (PRIME.CASCADE.VECT.01).** Promotion of the closed theorem cores of T166 (contract SCHUR.CASCADE) and T167 (contract NULL.VECTOR) as one deliberately narrow module (v557\_cascade\_vector\_identities.py, 23 checks,  $\sim 0.1$  s), everything recomputed on v551–v556’s declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table): (1) the cascade identities — prefix (one Cholesky, all sixteen rungs,  $1.1 \times 10^{-12}$  on 192 comparisons), minor form  $1/g_K = \det G_{[1..K]} / (\mu_1^P \det G_{[2..K]})$  on the raw arithmetic Gram block ( $2.7 \times 10^{-12}$ ; Schur 1917) and regression form  $g_K/g_1 = 1/(1 - R_K^2)$  ( $8.8 \times 10^{-13}$ ); (2) the diagonal invariance of the gain under  $B \rightarrow DBD$  ( $2.7 \times 10^{-12}$  on 144 random diagonals — the gain belongs to the arithmetic Gram block alone); (3) the exact  $K = 2$  identity — the closed vector  $(1, -Q_{21}/Q_{22})$  attains  $1/g_2 = Q_{11}(1 - r_{12}^2)$  exactly and  $B_{11}g_2 = 1/(1 - r_{12}^2)$  identically ( $1.2 \times 10^{-12}$ ): the fast-null-vector is FREE at  $K = 2$ , dress (c) collapses onto dress (b); (4) the pivot identity  $u^\top Q_K u = 1/g_K + \delta^\top Q_K \delta$  for  $\delta_1 = 0$  ( $2.0 \times 10^{-12}$  on 252 evaluations; the  $\delta_1 = 0.1$  mutation breaks by exactly the closed cross term  $2\delta_1$ ) with the exact entry threshold and the unification  $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$  — ONE inequality, not three; (5) the trial theorem (288 trials, every candidate a certified lower bound), the  $L_P$  no-go (cascade gain exactly 1, deviation 0.0) and the scramble type-change ( $B_{11}$  itself loses positivity on 49/60 declared draws, a majority on 12/12 windows) as must-breaks; every statement with a mutation control; the one open scalar R1 (an  $m$ -free upper bound on  $1 - r_{12}^2$ ) explicitly typed OPEN; NO RH statement. Ledger row PRIME.CASCADE.VECT.01; registered in run\_all.py and script\_registry.csv; suite counter 550  $\rightarrow$  551 scripts.
- **Prime-front diary T167 (null\_vector\_probe.py, contract NULL.VECTOR, 39/39, 1.4 s, the 63-window union of T166 rebuilt bit for bit; T163–T166 reproduced as fits; verdict VECTOR-RESISTS with a strong reduction, sandbox).** The most constructive dress attacked and closed *as a construction*: 22 families preregistered, all evaluated through the pivot identity  $u^\top Q u = 1/g_K + \delta^\top Q \delta$  (10206 evaluations,  $3.2 \times 10^{-9}$ ) — **gram2 is EXACT at  $K = 2$**  ( $\rho = 0$ , a theorem: the closed vector attains  $1/g_2$  on 63/63,  $B_{11}g_2 = 1/(1 - r_{12}^2)$  an identity, certified gain exponent  $h^{+2.921}$  on frame  $A = h^{3-0.079}$ ,  $h^{+2.246}$  union). Perturbation theory dies structurally: Jacobi diverges (radius 1.0–12.7) but **the Kato series CONVERGES (radius 0.067) — to the WRONG object** (the exact bottom eigenvector is itself a useless trial,  $\rho = 6.04$ , overlap 0.083, relative gap  $5.4 \times 10^{-7}$ ) — closed off in every order. Two thresholds separated (envelopes on 189 perturbations each); **monotonically sharper in  $K$  — mildest at  $K = 2$**  ( $1.90\times$ ,  $h^{+0.353}$ ), **against the hypothesis and exactly where the vector is free**; at  $K = 6$  no closed family hits (separation  $h^{+1.138}$ , ratio 264 $\times$ ); anchor  $\text{eps}_{\text{ent}}(2) \rightarrow (1 - r_{12}^2)/4$  — T166’s  $3.57 \times 10^{-4}$  is literally the quarter. End-to-end gain 2207– $7.84 \times 10^6$  certified; the deficit is the higher rungs, not the vector; **the scramble breaks harder than T166 saw: 8/8 windows lose positivity of the  $2 \times 2$  diagonal itself** ( $Q_{11} = -1.35 \times 10^5 \dots -1.02 \times 10^4$ ) — **a change of TYPE, not a factor**; anti-fitting cheat only  $h^{-0.823}$ ; off-recipe  $1 - r_{12}^2 \sim h^{-2.740}$  vs  $h^{-2.921}$ . **The unification as an identity:**  $\text{eps}_{\text{ent}}(K) = \rho(1/g_K)/S_K$  to machine precision — the determinant ratio (dress a),  $1 - r_{12}^2$  (dress b) and the entry threshold (dress c) are ONE inequality, not three. Balance 6 THEOREM / 0 CERT-UNIF / 2 CERT-WINDOW / 7 MEASURED; ONE-TERM-MISSING deliberately NOT rendered (the one missing term IS dress (b)); rest: R1, the only item — an  $m$ -free upper bound on  $1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C h^{-3+\varepsilon}$  (required  $1.25 \times 10^{-5}$  median,  $3.19 \times 10^{-8}$  worst); redirected: work at  $K = 2$ . T168 (lagrange\_minors\_probe.py, contract LAGRANGE.MINORS) is running — the Lagrange identity, the determinant as a sum of squares.

- **Docs + website sync.** `tfpt_prime_front` gains the T167 subsection (the pivot identity and the gram2 exactness as displays, the Kato-converges-to-the-wrong-object lesson, the threshold monotonicity against the hypothesis, the unification identity as the key result, the type-change scramble, the “After T167” keybox with R1 as the only item and T168 running), the v557 subsection in the verified layer, the module table row, the phase-2 section title moves to T126–T167, the probe index gains the T167 row and the running totals move to 167 probes / 4394 checks; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v557 companion blocks; `experiments/next.txt` gains the T167 entry (series (CLXXVIII)); website: `primeFront.ts` gains the T167 and v557 feed entries, status surfaces Teile 11–167 / 4394 checks, 23 modules v535–v557, T168 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT; ledger row PRIME.CASCADE.VECT.01.

**2026-07-29 (XIII) (prime-front diary T166 (the cascade gain is a Gram-determinant cancellation, one inequality in three dresses — CASCADE-RESISTS) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T166** (`schur_cascade_probe.py`, contract SCHUR.CASCADE, 30/30, 2.1 s, the two surfaces rebuilt as a *union*: 27 frame-A windows  $h = 142\text{--}1320$  plus 36 decoupled- $\nu$  windows over 6 zones,  $\nu \in \{4, 5, 6, 8, 11, 16\}$ ,  $h = 96\text{--}1371$  — 63 windows with a  $14\times$  lever arm on which zone depth and resolution are independent knobs ( $\log h$  vs  $\log \nu$  correlated 0.33); **T165’s three numbers reproduced on frame A, the union reading**  $h^{+2.759}/h^{+0.193}/h^{+2.566}$ ;  $\kappa = 0.038821$  verified at every jump point in  $[100, 4 \times 10^5]$ ; **verdict CASCADE-RESISTS, sandbox**). The one open object of T165 — the cascade lower bound  $g_{16}/g_1 \geq ch^{3-\varepsilon}$  uniform in  $m$  — attacked head-on and dissected completely. *The anatomy*: four identities, all theorems re-verified on 63/63 windows — the prefix property (one Cholesky yields all sixteen rungs), the minor form (Schur 1917), the **regression form**  $g_K/g_1 = 1/(1 - R_K^2)$  (the open object *is* a near-collinearity: mode 1 explained by modes 2...16 up to a residual  $h^{-3+\varepsilon}$ ), and the **diagonal invariance** of the gain under  $B \rightarrow DBD$  — it does not see the KMS  $1/\sqrt{\mu}$  normalisation at all: the  $h^3$  is a property of the arithmetic Gram block  $G = TAT^\top$  alone. The rung profile is extreme: rung 2 alone carries a median 59% of the gain, rungs 2–5 72%,  $K_{\text{half}} \in \{2..5\}$  on 63/63; rung 2 in closed form has  $1 - r_{12}^2 = 1.3 \times 10^{-7}\text{--}4.5 \times 10^{-4} \sim h^{-2.92}$  (frame A) — one rung multiplies  $g$  by  $2207\text{--}7.8 \times 10^6$  where the  $m$ -free budgets are sharp only to a factor 21–26.
- **The cancellation finding — a third mechanism.** No entry of the  $2 \times 2$  bottom block cancels (each keeps 60–99% of its larger half) and neither half alone is collinear (arch  $1 - r^2 = 0.98$ , atoms 0.29) — yet the sum is degenerate to seven digits: **the cancellation lives in the  $2 \times 2$  Gram determinant** — the polarisation split  $\det(G_{\text{ar}} + G_{\text{at}}) = \det G_{\text{ar}} + \det G_{\text{at}} + \Pi$  (exact to  $6.7 \times 10^{-14}$ ) has pieces 2.6–630 cancelling to  $1.0 \times 10^{-3}\text{--}1.7 \times 10^{-2}$ , a ratio  $2.6 \times 10^{-6}\text{--}5.1 \times 10^{-4} \sim h^{-1.45}$  — the same arch-against-atom mechanism as T159/T160, one level up, in a quadratic functional of the lag sums. Pinned to the arithmetic twice: the  $L_P$  no-go breaks completely (gain  $\equiv 1.0000000000$  exactly), and the anti-fitting scramble (uniform random atom positions, same total mass) destroys the effect by a factor 4569.
- **The best lower bound, the verdict, and two honest contract corrections.** Every route is an instance of the trial-vector theorem ( $1/g_K = \mu_1^{P-1} \min\{u^\top Gu : u_1 = 1\}$ ); the best closed route (two-rung,  $\beta$  from the atom block) certifies gain  $h^{+1.319}$  against the target  $h^{+3.110}$  — **an exponent gap of  $h^{+1.791}$ , not a constant: the one missing inequality *is* the cancellation** (required  $\beta$  precision  $3.57 \times 10^{-4}$ , all closed candidates miss by 12.4–2540); sharpest restatement  $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ , first flat at  $K = 6$  — exactly one uniform inequality missing, in three dresses (the det-ratio bound, the closed  $\beta$ , a closed near-null vector of  $A$  in the low modes). Corrections recorded honestly: form F2 (gain  $\geq ch^{3-\varepsilon}$ ) is strictly *weaker* than  $\inf_m g_{16} > 0$  ( $B_{11} \sim h^{+3.110}$ , F1 needs  $\varepsilon \leq -0.110$ ); the confinement route was a sub-surface

artefact ( $h^{+2.714}$  on frame A  $h \leq 780$  vs  $h^{+1.963}$  on the full union — exposed by raising the eigen-horizon to the cap). Errand closed:  $U_{\text{ref}}(\text{union}) = 1.15 \times 6.4781 = 7.45$  supersedes T164's 4.90 and T165's 5.7327 on the union (off-recipe rule declared, CERT-WINDOW, never CERT-UNIF); the union supremum is zone-depth-driven (multi-covariate gain  $\sim h^{+2.92}\nu^{-1.53}$ ). Balance +5 THEOREM / +3 CERT-WINDOW / +4 MEASURED, deliberately no new CERT-UNIF; candidates P166.1...P166.9 all PENDING (no promotion — bundling with T167 planned). T167 (`null_vector_probe.py`, contract `NULL.VECTOR`) is running the most constructive dress.

- **Docs + website sync.** `tfpt_prime_front` gains the T166 subsection (regression form, diagonal invariance and the Gram-minor ratio as displays, the Gram-determinant key result with the T159/T160 connection, the scramble factor, the two contract corrections, the “After T166” keybox), the phase-2 section title moves to T126–T166, the probe index gains the T166 row and the running totals move to 166 probes / 4355 checks; `experiments/next.txt` gains the T166 entry (series (CLXXVII)); website: `primeFront.ts` feed entry T166, status surfaces Teile 11–166 / 4355 checks, T167 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XII) (v556 — the gauge/ $P_{\text{pr}}$  identities of phase 2 promoted; prime-front diary T165 (the successor closes by certificate, one quantifier remains — ETA-RESISTS) + paper + website sync; NO marker moves)**

- **New module v556 (PRIME.GAUGE.PPR.01).** Promotion of the closed theorem cores of the discovery parts T164 (`sector_change_probe.py`) and T165 (`alignment_eta_probe.py`) — 23 checks,  $\sim 0.3$ s, everything recomputed on v551–v555's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (Chebyshev 1852 / Rosser–Schoenfeld 1962), nothing cited from the sandbox; the companion to v555, which certified the *price structure* of the pairing — this module certifies its *gauge structure*. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; the T164/T165  $h$ -exponents ( $1/g_{16} \sim h^{+0.061}$ , the ascent price  $\sim h^{+3.3}$ , the split  $+3.110 - 3.049 = +0.061$ ) are sandbox fits and are *not* consumed; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made; the ONE open object after T165 ( $\inf_m g_{16}(m) > 0$ , the cascade-gain quantifier) stays open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The gauge theorem (T164, S2.1) [E].**  $x_1 = 1$  fixes  $a_1 = 1/\sqrt{\mu_1}$ , i.e. only the *scale* of  $a$ ;  $Q$  and TV are homogeneous of degree two in  $a$ ; hence the demand ratio  $Q/\text{TV}$  of the same  $a$ -direction — and with it  $\delta_{\text{bnd}}$  — is the *same number in every sector*: unchanged to  $2.9 \times 10^{-11}$  on 2520 sector-by-trial-vector combinations (5 weight families  $\times$  7 shifts  $\times$  6 taper widths on 12 windows), with  $\text{TV} \geq |w_0|$  in every sector and the pure shift scaling the certified value by exactly  $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$  ( $2.2 \times 10^{-11}$  on 432 points); candidate (i) settled in closed form — full-space  $\mu_0 = 0$  exactly, strictly worse. Mutation: the wrong transport map (the  $x$ -coordinates instead of the  $a$ -direction) moves the ratio by  $2.0 \times 10^2$ – $5.4 \times 10^3$ .
- **(2) The  $h^2$  conservation law (T164, S2.2) [E].** floor  $\times$  transfer  $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$  *identically* at every shift on the declared grid (72 window-by-shift points,  $2.2 \times 10^{-16}$ ): a sector shift can flatten the floor but pays the identical factor back as a transfer — the  $h^2$  is conserved, never removed; it lives in the *ratio*, not in the floor. Mutation: the wrong transfer misses by 0.9995 at the largest shift on every window.
- **(3) The power-one spend as an identity (T164, S1) [E].** The  $r$ -ceiling  $r \leq U/L$  with  $L = \lambda_1(A)/\mu_1^P = 1.1019$ – $1.5845$  (computed per instance from the full window) is *linear* in the entry ceiling, so  $d \log r / d \log U = +1$  exactly (central difference at  $U = 1/g_{16}$ , within  $1.1 \times 10^{-14}$  on 12/12): an  $h^\varepsilon$  ceiling costs  $h^{-\varepsilon}$  of  $r$ -margin for every  $\varepsilon > 0$  — no free tolerance in the

consumption step (the measured power of the full T156 loss margin stays sandbox). Mutation: the quadratic consumption reads slope 2.0000 and is recognised.

- **(4) The  $P_{\text{pr}}$  identity and the predicate equivalence (T165, T1) [E].** The gauge-invariant dashboard satisfies  $P_{\text{pr}} = g_{16} \cdot R \cdot (\text{TV}/(t_1 v)^2)/\mu_1^P$  with no inequality anywhere — four factors, each a named object of the chain: the R-E-A quantifier  $g_{16}$ , the demand  $R = Q/\text{TV}$ , the T163 floor  $\text{tvf} \geq 1$ , and the Kac–Murdock–Szegő 1953 scale  $1/\mu_1^P = 943.6\text{--}8609.6 \sim h^2$  — exact to  $4.3 \times 10^{-16}$  on all 192 battery vectors (knob settings, random sixteen-vectors *and* random full-window vectors), gauge-invariant under  $v \rightarrow tv$  to  $1.5 \times 10^{-11}$ ; the predicates  $R > 2\kappa$  and  $P_{\text{pr}} > 2\kappa g_{16} \text{tvf}/\mu_1^P$  agree as booleans on all 192 — the demand half of R2'' and the crossing condition are *one* predicate, so R-F was never independent and its  $m$ -uniform form *is* R-E-A — and every crossing vector pays  $P_{\text{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4\text{--}142.8$ , above the preregistered tolerance  $\Theta_{\text{TOL}} = 10$  on 12/12: **the R-F conjunction is empty by an identity plus a floor, a certificate and not a failed search** (T165's ascent stays sandbox, no search is run). Mutation:  $(t_1 v)$  unsquared scales like  $t$  and moves by  $1.0\text{--}1.0 \times 10^4$ .
- **(5) The Schur-cascade direction and the exponent closure (T164/T165) [E].**  $s = \mu_1^P t_1^T A^{-1} t_1 = 0.1696\text{--}0.2472 \geq g_{16} \geq g_1$  window by window ( $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  — the T158 ladder is a chain of upper bounds with the route-(0) rung the loosest),  $P_K = \hat{a} s = 254.8\text{--}7.7 \times 10^3 \geq 1$  (Kantorovich 1948); the exponent closure is an *identity of the lists* — per instance  $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$  to  $1.6 \times 10^{-15}$ , the least-squares fits closing to  $1.3 \times 10^{-15}$  by linearity (T165's  $+3.110 - 3.049 = +0.061$  is this identity; the exponent values are sandbox fits, not consumed — what is promoted is that the certification lives in the cascade gain  $g_{16}/g_1$ , the object of the one open quantifier). Mutations: the  $g_8$  closure breaks by  $0.10\text{--}0.24$ ; a corrupted sixteen-block makes the Cholesky *refuse* on 12/12. Plus the monotone no-go staircase (damage  $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$  over  $\nu = 4/2/1/0.5$ , exact null control) and exact Dirichlet/parity controls.
- **Prime-front diary T165 (alignment\_eta\_probe.py, contract ALIGNMENT.ETA, 30/30, 15.1 s, the T163/T164 surface rebuilt and re-verified, window ladder up to  $h = 1445$ , verdict ETA-RESISTS in the strong form, sandbox).** The successor question R-F closes *by certificate, not by failed search*: the new gauge-invariant identity  $P_{\text{pr}} = g_{16} \cdot R \cdot (\text{TV}/(t_1 v)^2)/\mu_1^P$  (machine-checked to  $3.3 \times 10^{-16}$ ) makes demand and price *one* equation, so with T163's floor every crossing vector pays  $\geq 2\kappa g_{16}/\mu_1^P$  — the old KMS  $h^2$  — and the two R-F clauses cannot be chosen independently. The anatomy: the free optimum is a *low-harmonic* object (93.7–96.8% of its energy in the first 32 parity modes against a 2.4–22.5% baseline), essentially orthogonal to the PNT main term (cosine  $-0.070\ldots - 0.031$ ), with heavy edges on prime-power cells (lift  $1.36\text{--}2.12$ ); the price decomposes *exactly* (to 0.0):  $h^{+3.261} = +1.997$  (KMS — the same  $h^2$  since T152)  $+1.279$  (demand overshoot the chain never asked for)  $+0.093$  (floor)  $-0.108$  ( $g_{16}$ ); the bar-tight counterfactual stays  $h^{+1.982}$ .  $\eta(\text{Cap})$  is two-sided: the ceiling is a theorem, and at  $\text{Cap} = 10$  it falls short of the requirement by  $5.5\text{--}392\times$  on every window, 0/27 ladder points reach the bar. The equivalence:  $R > 2\kappa$  and T163's crossing predicate are the *same* predicate (105/105 vectors)  $\Rightarrow$  R-F is strictly *stronger* than the R2'' demand, and the clauses collide — **R-F is empty; its  $m$ -uniform form is R-E-A**. T2 localises the quantifier further: the  $m$ -free budgets bound  $B_{11}$  sharply (factor 21–26) — but the wrong object:  $1/g_1$  grows  $h^{+3.110}$ ,  $1/g_{16}$  is flat  $h^{+0.061}$ , exponents close exactly; the whole certification lives in the 15 ladder increments — “bound  $g_{16}/g_1$  from below by  $h^{3-\varepsilon}$ ”, one scalar per window, a cancellation and hence provably beyond absolute-value budgets. T3, the decoupled surface ( $\nu$  as a knob: 25 admissible windows over 6 zones,  $\nu \in \{4, 5, 6, 8, 11, 16\}$ , collinearity  $0.87 \rightarrow 0.68$ ): **the quarter-bar drift is zone depth, not window length (factor 19)** — and two honest costs reported openly:  $\sup 1/g_{16}$  rises to 5.7327, exceeding T164's  $U_{\text{ref}} = 4.90$  by 17% (the constant was frame-A-specific and is retired as a surface constant), and the  $h$ -coefficient of  $1/g_{16}$  is no longer negligible at free  $\nu$ . End-to-end on the union (52 windows,  $h = 142\text{--}1320$ ): every crossing vector has  $P_{\text{pr}} \geq 54.6\text{--}5865 > \Theta_{\text{TOL}}$ , closed for every  $h > 84.9$ . Stress: T145 coarsening monotone

(1.0/2.8/12.1/27.5, up to  $102\times$ ); surrogate  $8.6 \times 10^3$ – $6.9 \times 10^4$ ; Dirichlet/KMS exact. Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED; 6 new PENDING candidates. **The honest bottom line: exactly ONE genuine open object remains** —  $\inf_m g_{16}(m) > 0$ , **a quantifier over a certified flat list; no missing term, no missing device, no sector.** T166 (`schur_cascade_probe.py`, contract SCHUR.CASCADE) is running at exactly that quantifier. NOT RH evidence; no marker of any pre-existing contract moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T165 subsection (the  $P_{\text{pr}}$  identity as the key display, the exact price decomposition, the R-F  $\iff$  R-E-A equivalence, the decoupled surface with the zone-depth answer and the retired  $U_{\text{ref}}$  reported honestly) and the After-T165 keybox (the one cascade-gain quantifier, T166 running), the v556 subsection in the verified layer (now twenty-two modules), the phase-2 section retitled T126–T165, the T165 row in the probe index and the updated counters (165 probes, 4325 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v556 companion blocks; `experiments/next.txt` gains the Teil-165 entry; `website/lib/primeFront.ts` gains the T165 and v556 feed entries; the prime-front status surfaces move to Teile 11–165 / 4325 checks / 22 modules (v535–v556); suite counter 549  $\rightarrow$  550 scripts. Ledger row PRIME.GAUGE.PPR.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (XI) (prime-front diary T164 (both arms of the successor decided: the  $O(1)$  gate discharges to a single quantifier, sectors are a gauge — TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B)) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T164** (`sector_change_probe.py`, contract SECTOR.CHANGE, 28/28, 31.7 s, the T163 surface rebuilt and re-verified: 28 log-spaced prime-power zones  $h = 142 \dots 1445$  with declared caps, 27 windows inside the cond horizon,  $|\psi(x) - x| \leq \kappa x$  with  $\kappa = 0.038821$  verified at every jump point in  $[100, 4 \times 10^5]$ ; verdict TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B), sandbox). Both arms of T163's successor R-E are decided. *Arm A (the tolerance analysis)*: the station chain is instrumented with a power at every station — the T158 ladder really is a chain of upper bounds on the entry ceiling ( $s \geq g_{16} \geq g_1$  window by window), and at the heart the T156 kernel spends the ceiling at power *exactly one*:  $d \log(1 - F)/d \log U = -1.0005 \dots -1.0000$  with  $d \log r/d \log U = +1.0000$  exactly, so a ceiling growing like  $h^\varepsilon$  costs a margin decaying like  $h^{-\varepsilon}$  for every  $\varepsilon > 0$  — the declared one-sided rule returns  $\varepsilon^* = 0.50$  and the probe *disowns* it at once as the  $\varepsilon = 0$  tailwind ( $h^{+0.391}$ , bought from the measured drift of  $L$ ) divided by power one: no structural slack. But the sharp question is answered the other way: the single constant  $U_{\text{ref}} = 4.9008 = \max_h 1/g_{16}$  carries all 9 sub-surface windows out of sample,  $1/g_{16} = 1.7527$ – $5.3286$  is flat ( $h^{+0.061}$  over a  $9\times$  lever arm, split halves  $-0.010/+0.103$ ), and every  $g_K$  is strictly increasing (Schur 1917) — **the  $O(1)$  gate is not open as a quantity: it is discharged window by window by a Cholesky identity**, and what remains is exactly one quantifier,  $\sup_m 1/g_{16}(m) < \infty$  — the  $m$ -freedom of a certified flat list, the same grammatical form as every other open uniformity item and no longer an existence question about trial vectors. *Arm B (the sector change)*, *negative by a theorem*:  $x_1 = 1$  fixes only the *scale* of  $a$ ;  $Q$  and TV are homogeneous of degree two, hence  $Q/\text{TV}$  and  $\delta_{\text{bnd}}$  are the *same numbers in every sector* — the entry normalisation is a **gauge** (machine-checked to  $8.8 \times 10^{-10}$  over 5670 sector-by-vector combinations, declared horizon  $2.6 \times 10^{-8}$ ); the full space is strictly worse ( $\mu_0 = 0$  exactly, no finite floor); a shifted or weighted sector does flatten the floor ( $h^{+0.000}$  at  $\kappa_s = 4$ ) but pays the transfer factor  $1/\theta \sim h^{+1.997}$ , and floor  $\times$  transfer  $= 1/\mu_1^P$  *identically* — the two exponents sum to  $+1.997$  at every shift on the declared grid to  $10^{-9}$ : the  $h^2$  lives in the ratio, never in the floor. The genuine surprise, stated and immediately priced: a normalisation-free first-order ascent on  $Q/\text{TV}$  over the full window reaches 2.38–74.57 against the crossing bar  $2\kappa = 0.0776$  —  $31$ – $960\times$  over on 9/9 windows ( $\delta_{\text{bnd}} = -0.124 \dots -0.090$ , alignment efficiency  $\eta = 7.9$ – $11.1\%$  of the Abel ceiling

$\|C\|_\infty = 30.3\text{--}930.6 \sim h^{+1.185}$ ) — and T163's TV floor *holds on the unconstrained optimiser* (TV  $\mu_1^P = 8.30\text{--}11.72 \geq 1$ , a family T163 never searched), at a price  $h^{+3.299}$ , worse than T163's crossing price  $h^{+1.91}$ : dropping the normalisation relocated the difficulty without reducing it.

- **R-B''' narrowed, R-D settled in type, R-F named.** R-B''' is a *naming* problem — two different quantities carried one name: the T162  $\alpha$ -weighted quarter-bar object is  $0.1001\text{--}0.1623 \sim h^{-0.172}$  (T163's  $-0.172$  reproduced to  $0.0004$ , under the  $0.25$  bar on all 27 windows with 35% margin), the raw arch/atom cancellation is a different number ( $h^{-2.04}$ ); a two-variable regression puts  $-0.046$  of the drift on  $\log h$  against  $-0.477$  on  $\log \alpha$  ( $10\times$  more on the zone scale), and on this recipe  $\alpha$  and  $h$  are coupled by construction — narrowed, not closed: a surface with independent  $\alpha$  and  $h$  is needed. R-D is settled in *type*: under the shift  $s \rightarrow s/\theta$ ,  $\hat{a} \rightarrow \theta\hat{a}$ ,  $L \rightarrow \theta L$ , so  $P_K$ ,  $r$  and  $F(P_K, r)$  are unchanged to  $3.5 \times 10^{-16}$  on 54 combinations — every object the chain consumes downstream is a ratio of equally-homogeneous forms, hence no fifth device can come from a sector change ( $\rho = 1.0036\text{--}1.0140 > 1$  flat, T161, quoted and untouched). The one genuinely new item is **R-F**: is there a  $v$  with  $Q(v)/\text{TV}(v) > 2\kappa$  and  $Q(v)$  comparable to  $1/s$  — equivalently, can the alignment efficiency  $\eta$  stay above  $2\kappa/\|C\|_\infty$  at bounded price? No sector, no normalisation, no prime. Stress: the uniform surrogate breaks by  $1.5 \times 10^3\text{--}4.9 \times 10^7$ , the T145 no-go staircase is monotone ( $1.0 \rightarrow 9.1 \rightarrow 25.4 \rightarrow 129 \rightarrow 434$ , exact null control at  $\nu = 4$ , damage  $\geq 26.1$  at the coarsest setting on every window), the gauge break is  $3.4 \times 10^{-2}$ , Dirichlet/parity/orthonormality controls  $\leq 2 \times 10^{-14}$ , and the random- $v$  pairing identity holds to  $3.8 \times 10^{-16}$ . Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED, zero load-bearing fit rows; candidates T164-Q1...Q9 all PENDING (no promotion — to be bundled with T165). NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T164 subsection (the power-one measurement, the quantifier collapse as the key result, the gauge theorem with the product identity, the unconstrained ascent with the TV-floor hold, the R-B''' narrowing with the  $\alpha/h$ -coupling lesson) and the After-T164 keybox (R-E-A/R-F/R-B'''/R-D, T165 running), the phase-2 section retitled T126–T164, the T164 row in the probe index and the updated counters (164 probes, 4295 checks); `experiments/next.txt` gains the Teil-164 entry; `website/lib/primeFront.ts` gains the T164 feed entry; the prime-front status surfaces move to Teile 11–164 / 4295 checks. T165 (`alignment_eta_probe.py`, contract ALIGNMENT.ETA) is running at the alignment successor R-F. No ledger row (sandbox); **no marker of any pre-existing contract moves.**

**2026-07-29 (X) (v555 — the Pareto/total-variation identities of phase 2 promoted; prime-front diary T163 (the front resists as a theorem — the  $h^2$  is the parity spectral gap, not the primes, FRONT-RESISTS) + paper + website sync; NO marker moves)**

- **New module v555 (PRIME.PARETO.TV.01).** Promotion of the closed theorem cores of the discovery parts T162 (`third_split_probe.py`) and T163 (`pareto_front_probe.py`) — 23 checks,  $\sim 0.4$ s, everything recomputed on v551–v554's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (Chebyshev 1852 / Rosser–Schoenfeld 1962, reproduced to  $4.2 \times 10^{-7}$ ), nothing cited from the sandbox; the companion to v554, which certified what the sixteen-form *pairs against* — this module certifies the *price structure* of that pairing. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; the T163  $h$ -exponents ( $P_{\text{aff}} \sim h^{3.05}$ ,  $P_{\text{cross}} \sim h^{1.91}$ , the TV exponent  $\sim h^{2.00}$ ) are sandbox fits and are *not* consumed; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 (R-E the successor with both arms prime-free; R-B''' the  $h$ -uniform positivity margin; R-D a fifth R1'' device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.



- **(1) The exchange law and the crossing criterion (T163, R1.2) [E].**  $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}(x)/P(x))/\log X$  is an *identity* at all 612 grid points of the declared Fejér knob (50 geometric  $\sigma$  on  $[1, 4096]$  plus  $\infty$ , 12 windows;  $5.1 \times 10^{-16}$  relative, admissibility  $Q(x) > 0$  everywhere), and the crossing criterion  $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16} \text{TV}$  holds as an exact equivalence at every point — price and demand are two coordinates of *one* object. Mutation: the wrong constant moves the law by  $6.5 \times 10^{-2}$ –3.4.
- **(2) The total-variation floor (T163, R2.2) [E].** (T1)  $w_0(x) = \|a(x)\|^2$  ( $4.9 \times 10^{-16}$ ); (T2)  $\text{TV}(x) \geq |w_0(x)|$  by telescoping with  $w_M = 0$ ; (T3)  $x_1 = 1$  forces  $\|a\|^2 \geq 1/\mu_1^P$ , hence  $\mu_1^P \text{TV}(x) = 5.00$ –28.84  $\geq 1$  on all 708 trial vectors built, against the closed floor  $1/\mu_1^P = 1/(4 \sin^2(\pi/N)) = 943.6$ –8609.6  $\sim h^2$ ; (T4) every sub- $\frac{1}{2}$  grid point pays  $P > 2\kappa g_{16}/\mu_1^P = 15.4$ –142.8 (all 54 respect it) — a flat-price sub- $\frac{1}{2}$  demand is impossible in the parity sector on this surface: **R-C''' is closed negatively, by an inequality.** Negative control: the *inadmissible* vector  $0.01x^*$  undercuts the floor ( $2.5$ – $2.9 \times 10^{-3}$ ) and is recognised — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953).
- **(3) The chain-derived price axis (T163, R1.0) [E].**  $\sigma = 1$  gives  $x = e_1$  exactly with  $P = g_{16}/g_1$  (the  $K = 1$  rung of the T158 Cholesky ladder, the T157 route-(0) certificate;  $4.1 \times 10^{-16}$ ),  $\sigma = \infty$  gives  $g_{16}Q(x^*) = 1$  (the  $K = 16$  top rung;  $1.2 \times 10^{-12}$ ); the ladder carries all 16 terms strictly positive with monotone partial sums, the knob is monotone in both coordinates on every window (the knob curve *is* the Pareto front). Mutation: a corrupted sixteen-block makes the Cholesky *refuse* on 12/12.
- **(4) The Mellin cell moments and the Abel step (T162, C2/C5) [E].** The  $k$ -th Mellin rung  $\sum_d w_d C_d(-(2k + \frac{1}{2}))$  with  $C_d(s)$  the closed tent integral (Mellin 1896) agrees with an independent per-cell Gauss–Legendre quadrature to  $2.8 \times 10^{-13}$  on every rung  $k = 1 \dots 4$  (shifted tent centres break by  $\geq 1.3 \times 10^{-2}$ ); the boundary-free Abel identities hold at levels  $k = 1 \dots 3$  ( $8.3 \times 10^{-17}$ ; Abel 1826), the dropped-boundary level-1 form agrees because the gauge identity kills the boundary term exactly, the optimal level is *one* for T162's closed reason ( $32\pi/\alpha = 40.6$ –59.4  $> 1$ , measured ratio 30.8–69.1), and a gauge violation breaks by its closed prediction  $v_0 c_0 M$  to  $1.2 \times 10^{-15}$  (the positive control).
- **(5) The Lerch/Frullani log-moment (T162, R-A') [E].**  $L = \sum_d \Psi_d w_d$  agrees on three routes per window: direct; double-Abel *with* the Wronskian boundary term ( $2.8 \times 10^{-14}$ ; dropping it breaks by  $2.2 \times 10^{-4}$ – $1.3 \times 10^{-2}$ ); the Lerch/Frullani integral with the  $d = 1$  term peeled closed ( $1.8 \times 10^{-15}$ ) — the peel constant  $2 \log 2$  reproduced to  $2.2 \times 10^{-16}$  and  $\Psi_1 = -\log 2$  exactly (Frullani 1828 / Lerch 1887 / Clausen 1832). Plus the monotone no-go staircase (damage  $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$  over  $\nu = 4/2/1/0.5$ , exact null control) and exact Dirichlet/parity controls.
- **Prime-front diary T163 (pareto\_front\_probe.py, contract PARETO.FRONT, 32/32, 1.9 s, 28 log-spaced prime-power zones  $h = 142 \dots 1445$  with declared caps, 27 windows inside the cond horizon, verdict FRONT-RESISTS, sandbox).** The survey of the Pareto front, and the front resists *as a theorem*: the price axis is chain-derived ( $P_{\text{aff}} = g_{16} B_{11} \sim h^{+3.049}$ , the T157 route-(0) certificate), the Fejér knob has both endpoints in the chain and is measured monotone in both coordinates (the knob curve *is* the front), no flat price crosses the yardstick (0/27 at caps 1.25/2/10 and every flat rung tier; only the full  $P_{\text{aff}}$  reaches 27/27), the crossing price  $P_{\text{cross}} \sim h^{+1.906}$  sits at 0.0066–0.2047 of  $P_{\text{aff}}$  — 5–152 $\times$  strictly inside the already-accepted certificate, margin widening — and the price anatomy is an *identity*:  $\delta_{\text{bnd}} = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}/P)/\log X$  exactly at all  $27 \times 50$  grid points, the crossing exponent decomposing additively ( $\text{TV} + 2.013$ ,  $g_{16} - 0.061$ ). The four-line theorem (T1–T4) gives every admissible trial vector the total-variation floor  $\text{TV} \geq 1/\mu_1^P \sim h^2$  (verified on all 1674 built, slack 5.0–23.2), so the crossing price is bounded below by a quantity growing like  $h^2$ : the  $h^{2.86}$  of T162, the  $h^{1.91}$  of the crossing and T161's granularity are *all* the reciprocal smallest parity eigenvalue — **the spectral gap of the parity Laplacian meeting the entry normalisation**



$x_1 = 1$ : **it was never in the primes**. The mode sweep  $K = 16 \rightarrow 64$  confirms it from the other side (price dividend, worse demand, TV exponent unmoved at  $h^{+1.81}$ ); R-B'' reduces to an  $h$ -uniformity question (the chain never needed the sign: duality plus an evaluated number, headroom  $\geq 10^{14.9}$ ); R-D is a type mismatch (operator smoothing degrades the floor by the same  $h^2$ ). Balance 11/1/4/10, zero fit rows. T164 (`sector_change_probe.py`), running, tests the successor R-E. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T163 subsection with the exchange-law display, the T1–T4 floor theorem as the key result, the chain-derived price axis, the monotone no-go staircase and the After-T163 keybox (R-E/R-B'''/R-D, T164 running), the v555 subsection in the verified layer (now twenty-one modules), the phase-2 section retitled T126–T163, the T163 row in the probe index and the updated counters (163 probes, 4267 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v555 companion blocks; `experiments/next.txt` gains the Teil-163 entry; `website/lib/primeFront.ts` gains the T163 and v555 feed entries; the prime-front status surfaces move to Teile 11–163 / 4267 checks / 21 modules (v535–v555); suite counter  $548 \rightarrow 549$  scripts. T164 (`sector_change_probe.py`) is running at the sector-change successor. Ledger row PRIME.PARETO.TV.01 [E]; **no marker of any pre-existing contract moves**.

**2026-07-29 (IX) (v554 — the sampling/harmonics identities of phase 2 promoted; prime-front diary T161 (the circularity triage: the hardness sits in the split, not in the primes, CLOSURE-RESISTS) + paper + website sync; NO marker moves)**

- **New module v554 (PRIME.SAMPLING.HARM.01).** Promotion of the closed theorem cores of the discovery parts T160 (`pairing_probe.py`) and T161 (`classical_closure_probe.py`) — 21 checks,  $\sim 0.5$  s, everything recomputed on v551–v553's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ) plus a declared  $\Lambda$ -battery of six log-spaced cut-offs  $X = 10^3 \dots 3.2 \times 10^5$  for the PNT item (finite von-Mangoldt sums only, no matrix anywhere), nothing cited from the sandbox; the companion to v553, which certified the algebra of the sixteen-form — this module certifies *what that form pairs against*, and it closes no open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made, and the  $\delta$  triage of T161 (required cancellation depth against RH strength) is a documentation item, *not* a module claim; the open terms after T161 (R-A' the prime-free log-moment; R-B' the  $m$ -free  $16 \times 16$  Gram fraction bound; R-C' one split of the pairing with  $\delta_{\text{eff}} < \frac{1}{2}$ ; R-D a fifth R1'' device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The sampling identity (T160, P2) [E].** The atom half of the pairing equals  $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n}) \hat{w}(\log n)$  (plus one reflected term below the grid;  $\hat{w}$  the linear interpolant of the lag weights), exact to  $5.6 \times 10^{-15}$  of the absolute atom scale on 12/12 windows: identically a finite combination of sums  $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$  at 32 explicit frequencies. Mutation: a one-index grid shift breaks by  $5.1 \times 10^{-5} - 3.1 \times 10^{-3}$ .
- **(2) The moment laws and the TV theorem (T160, P1/P4) [E].**  $m_0 = 0$  ( $1.0 \times 10^{-16}$  of  $\|w\|_1$ ),  $m_1 = -[S_0^2 + 2 \sum_j P_j^2]$  ( $9.2 \times 10^{-16}$ ) and  $m_2 = -[2S_1 - (M-1)S_0]^2$  ( $3.4 \times 10^{-14}$ , a perfect square), both strictly negative on every window; U3 as a theorem with *checked* hypotheses ( $c^{\text{arch}} \leq 0$  and monotone on  $d \geq 1$ , 12/12):  $\text{TV}(c^{\text{arch}}) = 3.9817 - 4.8991 \leq |c_0| + 2 \sup < 6$ , closed and window-independent. Mutations: a one-index prefix shift breaks  $m_1$  by  $\geq 1.25 \times 10^{-2}$ , an  $(M-2)$  mis-normalisation breaks  $m_2$  by  $\geq 6.1 \times 10^{-2}$ , the full atom-including kernel violates the monotone hypothesis on 12/12.
- **(3) The harmonic-frequency theorem and the PNT main term (T161, P3) [E].**  $t_j(2\alpha) = 2\pi j$  *exactly* ( $2.8 \times 10^{-14}$ ) on every window and every battery cut-off — the 32

frequencies of the sampling identity are precisely the Fourier harmonics of the log-window  $[0, 2\alpha]$ ; at those frequencies  $\theta = 2\pi j$  collapses the partial-summation main term to the closed, parameter-free prediction  $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$  (de la Vallée Poussin 1896; the Mellin factor of  $x^{-1/2}$ ), matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12, the subtraction reducing the aggregate residual to  $0.008\text{--}0.466 \leq 0.5$  on every cut-off. Mutations: a wrong Mellin pole worsens the aggregate residual by  $\times 1.8\text{--}\times 95.8$ ; a mass-matched *uniform* fake measure misses by 0.23–0.51 — the agreement is a property of the von-Mangoldt measure, not of any measure with the same mass.

- **(4) The head split and the scale-free Bernstein rate (T161, P1) [E].**  $c_d^{\text{arch}} = \Psi_d + D\hat{G}(dD, D)$  with  $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$  closed,  $D$ -free and  $m$ -free (the triangle-smeared simple pole; the  $\log D$  cancels identically), exact to  $3.2 \times 10^{-16}$  of  $\sup |c^{\text{arch}}|$  at *every* lag of every window, the two  $\Psi$  representations agreeing to  $2.9 \times 10^{-14}$ ; and  $\rho^* = (3 + \sqrt{5})/2 = 2.618034$ : the root of  $x^2 - 3x + 1$ , returned exactly by the  $D \rightarrow 0$  limit of the peeled interval  $[0.4\alpha, 2\alpha]$  and invariant under  $\alpha \times 10$  — scale-free, every window's  $\rho(D) = 2.588\text{--}2.608$  below it (Bernstein 1912). Mutations: dropping  $\Psi$  misses  $c^{\text{arch}}$  by 0.99–1.00; the wrong interval ratio moves the limit to 3.0.
- **(5) The off-diagonal fraction bound and the sign refutation (T161, P2) [E].** The gauge identity licenses one boundary-free Abel step,  $Q^{\text{arch}} = \sum_{k,l} a_k a_l \sum_{d \geq 1} (\Delta c^{\text{arch}})_d R_{kl}^1(d)$  exactly ( $2.0 \times 10^{-13}$ ), and the certified per-window inequality is  $|\text{off-diagonal block}| \leq \frac{1}{4}|Q^{\text{arch}}|$  (measured 0.171–0.212 on 12/12) — with the *refuted* sign law wired as the negative control: the arch half is negative while its off-diagonal block is positive on 12/12, and 176/240 off-diagonal pairs violate the per-pair form on every window — nothing single-signed is promoted. Mutation: a one-index  $R$ -kernel shift breaks by  $\geq 6.8 \times 10^{-5}$ . Stress: the head split misses the T145 no-go reconstruction by 2.39; the Dirichlet cosine-sum identity (including the degenerate branch), the KMS eigenpairs and  $\text{odd\_toeplitz}(c^L) = L_P$  (zero tolerance) are exact.
- **Prime-front diary T161 (classical\_closure\_probe.py, contract CLASSICAL.CLOSURE, 35/35, 1.2s, 19 log-spaced prime-power zones  $h = 50 \dots 1445$ , verdict CLOSURE-RESISTS, triage verdict NEEDS-BEYOND-RH-STRENGTH, sandbox).** The two classical rests plus the circularity triage, and the triage returns its most consequential answer: *the chain is not circular*. R-A closes  $m$ -free —  $A = D\hat{A}$  exactly, only the  $1/s$  head binds, the scale-free Bernstein rate  $\rho^* = (3 + \sqrt{5})/2$ , the closed head split (no peeling), and an explicit degree schedule  $K(h) = O(\log h)$  (a *fixed* degree is refuted with a number; the residual is the prime-free log-moment, outside the polynomial ladder). R-B is refuted in all four readings (3168/4320 pairs; the aggregate by *sign*) and replaced by the certified fraction bound  $0.1035\text{--}0.1713 \leq \frac{1}{4}$ , flat. The triage: the 32 frequencies are exactly the Fourier harmonics of the log-window (theorem), the measured cancellation is fully the PNT main term, and the required depth is  $\delta = 1.1482\text{--}1.8809$  against RH strength  $\frac{1}{2}$  — in absolute terms 0.012–0.31 of the last term of the sum, below the boundary term of every partial summation, so no strengthening of the  $\psi(x) - x$  input reaches it: the chain needs *more* than RH-strength input would supply, which localises the  $h^2$  in the *split* rather than in the primes; a re-split moves the demand to  $\delta_{\text{eff}} = 0.9839\text{--}1.3767$  — it moves, but stays above  $\frac{1}{2}$  on 18/18. Balance 20/10/4/9/3. T162 (third\_split\_probe.py), running, searches for one split with  $\delta_{\text{eff}} < \frac{1}{2}$ . NOT RH evidence; no marker moves.
- **Docs + website sync.** tfpt\_prime\_front gains the T161 subsection with the  $\delta$  table, the head-split display and the After-T161 keybox, the v554 subsection in the verified layer (now twenty modules), the phase-2 section retitled T126–T161, the T161 row in the probe index and the updated counters (161 probes, 4205 checks); the big-picture section states the split-design reading soberly; tfpt\_1, tfpt\_3 and tfpt\_research\_contracts gain the v554 companion blocks; experiments/next.txt gains the Teil-161 entry; website/lib/primeFront.ts gains the T161 and v554 feed entries; the prime-front status surfaces move to Teile 11–161 / 4205 checks

/ 20 modules (v535–v554); suite counter 547 → 548 scripts. T162 (third\_split\_probe.py) is running at the third-split search. Ledger row PRIME.SAMPLING.HARM.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (VIII) (v553 — the exact-form identities of phase 2 promoted; prime-front diary T159 (the gauge identity and the Z-matrix law, cancellation is the weight, FORM-RESISTS) + paper + website sync; NO marker moves)**

- **New module v553 (PRIME.EXACT.FORM.IDENT.01).** Promotion of the theorem cores of the discovery parts T158 (schur\_entry\_probe.py) and T159 (exact\_form\_probe.py) — 25 checks, ~0.2s, everything recomputed on v551/v552's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; no-go ladders  $m = 48 \dots 288$  on both T145 forms; parity and Dirichlet controls at  $m = 64, 128, 256$ ), nothing cited from the sandbox; the companion to v552, which certified the four angle instruments — this module certifies the *algebra of the sixteen-form itself*, and it closes neither open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the two open terms after T159 — the one explicit pairing inequality for  $R2''$ , needing a correlation bound instead of sizes; a third device for the atom off-band block for  $R1''$  — stay open and typed open); each statement carries at least one mutation control that must fail loudly.
- **(1) The Thomson dual form and the positive Cholesky ladder (T158, P1) [E].**  $s = (B^{-1})_{11} = \max_x (2x_1 - x^T Bx)$  (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted: 8 random trial vectors per window never exceed  $s$ , so every trial is an upper bound on  $1/s$  — exactly the variational structure Cauchy–Schwarz lacks (v552 wired that refutation as a control; here the direction itself is the check). The ladder  $g_K = \sum_{j \leq K} y_j^2$ ,  $y = L^{-1}e_1$  from *one* Cholesky of the leading  $16 \times 16$  block: all terms strictly positive, partial sums strictly monotone,  $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  literally on 12/12, tight to 1.0642–1.1522 against the declared cap 1.5 (T158 measured  $\leq 1.2738$  at  $h = 254 \dots 1393$ , sandbox); mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12.
- **(2) The two-dimensionality (T158, P2) [E].**  $\text{span}\{t_1, A^{-1}Lpt_1\}$  attains  $s$  exactly ( $g/s = 1$  to  $9.0 \times 10^{-13}$  on 12/12) for the theorem reason  $Lpt_1 = \mu_1^P t_1$  (KMS 1953, eigenpair residual  $8.1 \times 10^{-13}$ ): the entry is a two-dimensional quantity the moment one Green column is granted. Negative controls: the two-sine span (no Green column) misses by 1.42–4.8, the corrupted Green column (no solve) by  $255\text{--}7.7 \times 10^3$ .
- **(3) The gauge identity and the TmH signature (T159, P4/P7) [E].** The reflection-odd section  $A_{rs} = c_{|r-s|} - c_{M-1-r-s}$  annihilates constant lag vectors *entrywise with zero tolerance* ( $\text{odd\_toeplitz}(1) = 0$  exactly — the mechanism in one line), hence the gauge identity  $\sum_d w_d = 0$  holds to  $1.1 \times 10^{-14}$  of  $\sup |w|$  for the per-window optimiser and 3 random sixteen-vectors: the form is *blind* to the lag mass of  $c$ , and the lag mass is exactly where the archimedean and atom halves of the sixteen-form are  $h^2$ -sized and cancel. The premise — the Toeplitz-minus-Hankel signature  $A_{r,s} - A_{r+1,s+1} = f(r+s)$  — holds to  $1.4 \times 10^{-16}$ . Mutations: a one-index Hankel shift breaks the identity by  $2.7 \times 10^{-7}\text{--}4.2 \times 10^{-5}$ ; the T145 form  $R = aa^T + \varepsilon I$  has signature defect  $7.4 \times 10^{-2}$  — not a parity section, the gauge identity is not even *defined* for it.
- **(4) The exact lag sum and the two closed scalars (T159, P2/P5) [E].**  $x^T B_{LL} x = \sum_d c_d w_d$  with  $w_d = T_d - H_d$  against the direct assembly to  $2.7 \times 10^{-16}$  of the absolute scale  $\sum_d |c_d w_d|$  — and the honest price *measured*: relative to the cancelled total the same identity holds only to  $1.4 \times 10^{-11}$  on this small surface (the sandbox at  $h \leq 1293$  loses 4.0–8.1 digits; that depth *is* the  $h^2$  obstruction). The two closed scalars:  $w_0 = \sum_k x_k^2 / \mu_k^P$  (the  $N^2/4\pi^2$  factor itself) to  $4.2 \times 10^{-16}$  and  $2w_0 - w_1 = \|x\|^2$  to  $7.4 \times 10^{-16}$  (KMS 1953). Mutations: a 1% perturbation of the largest lag breaks the lag sum by  $2.2 \times 10^{-3}$ ; a one-index  $\mu$  shift breaks  $w_0$  by 0.60.

- **(5) The Z-matrix law and the Collatz–Wielandt floor (T159, U1/U2) [E].**  $B_{HH}^{\text{arch}}$  is a symmetric Z-matrix *raw* — every off-diagonal entry  $\leq 0$  with no tolerance on 12/12; the closed sign-based floor  $\min_k (B_{HH}^{\text{arch}} \mathbf{1})_k = 0.9093\text{--}1.1284 \geq t = 0.25$  with the theorem direction verified per instance against the exact  $\lambda_{\min}(B_{HH}^{\text{arch}}) = 0.9138\text{--}1.1432$  (Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8 on every window. The honest limit as content: the atom block obeys *no* sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is *negative* ( $-48.3\ldots - 0.79$ ) against a positive exact bottom — admissible and vacuous, R1'' stays open. Mutation: the checkerboard similarity  $S = \text{diag}((-1)^k)$  — an isometry,  $\lambda_{\min}$  moves by 0 exactly — destroys the raw law (fraction  $1.0000 \rightarrow 0.494\text{--}0.498$ ): the Z law is an entrywise fact, not a spectral one, and T159's checkerboard reading (S2) is refuted. Stress: the T145 no-go breaks on three axes of this module's structure; on the  $c(l) = 1/(1+l)$  reconstruction the ladder theorem survives while  $1/g_{16}$  explodes ( $1187 \rightarrow 7.11 \times 10^4$ ) — the instrument reports the collapse instead of hiding it; parity: for  $A = L_P$  the machinery returns  $s = 1$  with every ladder partial sum constant in  $K$ ; Dirichlet controls exact.
- **Prime-front diary T159 (exact\_form\_probe.py, contract EXACT.FORM, 41/41, 2.9 s, 23 of 24 prime-power zones  $h = 142\ldots 1293$ , the sign surface on 18 windows  $h = 142\ldots 963$ , verdict FORM-RESISTS, sandbox).** The cancellation executed algebraically: seven machine-checked identities to  $10^{-12}$  against the absolute scale (the cancellation eats 4.0–8.1 decimal digits, reported as a measurement), among them the closed 256-term Dirichlet-kernel form of the geometric weights and the *gauge identity*  $\sum_d w_d = 0$  exactly — the form is blind to the lag mass, exactly where the  $h^2$  halves live. The honest core answer is negative: the  $h^2$  does *not* telescope away, it is the common *weight* of both halves (both grow  $h^{+2.104}$ ), and four routes are closed with exponents — T158's own fixed sixteen-vector refuted at  $h^{+3.510}$ , one sine at  $h^{+3.066}$ , the preregistered ansatz family at  $h^{+3.001}$ , Abel- $\ell^1 \times \sup$  at every rung (best  $h^{+3.604}$ ). The structural win:  $B_{HH}^{\text{arch}}$  is exactly a symmetric Z-matrix with a closed sign-based Collatz–Wielandt floor 0.819–1.165 comfortably above  $t = 0.25$  — but the atom block obeys no sign law, and the full-block criterion is vacuous. Balance  $6/3/4/1 \rightarrow 13$  theorem / 6 cert-unif / 4 cert-window / 1 measured; a numerical horizon declared as its own measurement ( $\text{cond}(B_{LL}) > 10^{12}$  from  $h = 1292$ ); one fully explicit pairing inequality remains, under attack via correlation structure in T160 (pairing\_probe.py), running. NOT RH evidence; no marker moves.
- **Docs + website sync.** tfpt\_prime\_front gains the T159 subsection with the After-T159 keybox, the v553 subsection in the verified layer (now nineteen modules), the phase-2 section retitled T126–T159, the T159 row in the probe index and the updated counters (159 probes, 4124 checks); tfpt\_1, tfpt\_3 and tfpt\_research\_contracts gain the v553 companion blocks; experiments/next.txt gains the Teil-159 entry; website/lib/primeFront.ts gains the T159 and v553 feed entries; the prime-front status surfaces move to Teile 11–159 / 4124 checks / 19 modules (v535–v553); suite counter  $546 \rightarrow 547$  scripts. T160 (pairing\_probe.py) is running at the pairing bound. Ledger row PRIME.EXACT.FORM.IDENT.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (VII) (v552 — the four fixed-size angle instruments of phase 2 promoted; prime-front diary T157 (the confinement theorem and the one-entry reduction, ANGLES-RESIST) + paper + website sync; NO marker moves)**

- **New module v552 (PRIME.ANGLE.INSTR.01).** Promotion of the instrument cores of the discovery parts T155 (bottom\_floor\_probe.py) and T157 (angle\_floor\_probe.py) — 18 checks,  $\sim 0.4$ s, everything recomputed on v551's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29\ldots 139$ ,  $m = 96\ldots 291 \leq 300$ ; a no-go size ladder  $m = 48\ldots 288$ ; parity and Dirichlet controls at  $m = 64, 128, 256$ ), nothing cited from the sandbox; the companion to v551, which certified the fixed-size ceiling step — this module



certifies the four fixed-size *instruments* the angle campaign produced, and none of them closes an angle. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all*  $D$  (the two open terms after T157 — an  $m$ -free lower bound on  $\hat{g}_1^2$ , equivalently an  $m$ -free upper bound on the one Schur diagonal entry  $1/s$ ; and an  $m$ -free  $R1''$  domination with a non-shrinking margin — stay open and typed open); each statement carries at least one mutation control that must fail loudly.

- **(1) The complement-floor certificate (T155) [E].**  $\min_{v \perp W} v^\top L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^\top)M^{1/2})$  with  $M = \text{diag}(\mu_{13}^P - \mu_k^P)$  and  $G = T_{12}Q_W$  — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every  $m$  and every subspace, direction checked: the certificate never exceeds the exact size- $m$  complement bottom (12/12) and agrees to  $1 - \text{ratio} \leq 1.01 \times 10^{-3}$  against the declared cap  $2 \times 10^{-3}$  on this small surface (T155 measured  $\leq 2.2 \times 10^{-7}$  at  $h = 142 \dots 1293$ , sandbox). Exactness controls both ways:  $W = \text{span}\{t_1..t_8\}$  returns  $\mu_9^P$  and the *worthless* configuration  $W = \text{span}\{t_2..t_{13}\}$  returns  $\mu_1^P$ , both to  $2.0 \times 10^{-11}$  — the instrument reports its own worthlessness; mutation: one corrupted overlap row moves the certificate by  $\geq 0.549$ , only downward.
- **(2) The sine-block confinement (T157) [E].** From the certified pencil floor  $A \succeq tL_P$  and the certified ladder ceiling  $\lambda_1 \leq S\mu_1^P$  *alone*,  $\|\gamma_H\|^2 \leq \lambda_1/(t\mu_{17}^P) \leq (S/t)/\rho_{17} = 0.0153\text{--}0.0245 < 1$  on 12/12: the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a *per-instance theorem* (measured tail  $6.7 \times 10^{-6}\text{--}3.0 \times 10^{-5}$ ; T146’s measured “98 per cent” replaced by one line). Negative control: the pencil ceiling  $\kappa$  in place of the ladder  $S$  makes the bound vacuous ( $17.6\text{--}477 > 1$  on 12/12); mutation: the inflated floor  $\kappa L_P$  is refuted at  $e_1(A)$  by  $3.3 \times 10^3\text{--}1.0 \times 10^5$ .
- **(3) The Schur-entry identity (T157) [E].**  $s = \mu_1^P t_1^\top A^{-1} t_1 = (B^{-1})_{11}$  exactly ( $1.1 \times 10^{-11}$ ) and  $(B^{-1})_{LL} S_L = I$  to  $5.8 \times 10^{-11}$  with  $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$  the *same* fixed-size  $16 \times 16$  Schur complement the chain already forms:  $1/s = 4.0460\text{--}5.8962$  is an  $O(1)$  number carried by *one* entry, so T156’s  $R2''$  loss ceiling  $r \leq 1/(Ls)$  is a statement about one number. Negative control (T157’s refutation wired as a check): the inversion-free Cauchy–Schwarz ceiling is valid and useless, off by  $250\text{--}7.27 \times 10^3$  — the cancellation in  $b^\top B_{HH}^{-1} b$  is nearly complete and Cauchy–Schwarz cannot see it; mutation: a one-index shift breaks the identity by  $1.55\text{--}2.15$ .
- **(4) The adaptive Lipschitz ceiling (T157) [E].** Interval bisection with the *local* Lipschitz constant certifies  $\inf f^{\text{arch}}/(4 \sin^2(\theta/2)) \geq 0.25$  on 12/12 windows in 519–1201 symbol evaluations (certified floor  $0.2505\text{--}0.3062$ , never above the fine-grid minimum); the global constant is documented insufficient (a uniform grid would need up to  $61.4\times$  more points); mutation: fed a target above the true grid minimum the certificate *refuses* on 12/12. Stress: on the T145 no-go family the Schur floor survives ( $t = 0.05$ ) while the confinement goes vacuous ( $\times 52.0$ ) and  $(S_L)_{11}$  explodes ( $\times 60.0$ ) against the flat real band; parity control exact to  $4.9 \times 10^{-14}$ ; the Dirichlet residual hits the predicted  $2/\sqrt{N}$  to  $7.4 \times 10^{-5}$ .
- **Prime-front diary T157 (angle\_floor\_probe.py, contract ANGLE.FLOOR, 32/32, 3.9 s, 16 prime-power windows  $h = 142 \dots 1293$ , the alignment surface on 12 ( $h = 50 \dots 1077$ ), verdict ANGLES-RESIST, sandbox).** Neither angle falls, but both change shape: the  $p_1$  floor becomes a theorem times one measured fixed-size number — the sine-block confinement  $\|\gamma_H\|^2 \leq (S/t)/\rho_{17} = 0.0165\text{--}0.0293$  proves  $e_1(A)$  lives 97.1–98.4% inside the first sixteen parity sines, and the resolvent route pins  $p_1 \geq \hat{g}_1^2(1 - \text{tail}) = 0.1968\text{--}0.3228$ , 97.9% of the measurement (the classical Rayleigh angle bound is empty on 16/16, its block version empty at every  $J$ , the angle-free Cauchy–Schwarz route loses exactly the Kantorovich  $P = 5.6 \times 10^2\text{--}1.0 \times 10^6$ ); the structural gem  $1/s = (S_L)_{11} = 2.3359\text{--}6.2049$  flat makes the whole first term a bound on *one* diagonal entry of the  $16 \times 16$  Schur complement the chain already forms (Cauchy–Schwarz off by  $5.4 \times 10^2\text{--}5.2 \times 10^5$ ); the arch half is uniformly certified for the first time (executed adaptive Lipschitz ceiling, 12/12, cost  $h^{0.85}$ ) with the two extremals at opposite ends of the band (the

pointer: use the  $\theta$ -growth, not the infimum at  $\pi$ ); the alignment term stays a per-window domination (1.0003–1.0907, margin  $7.3 \times 10^{-4}$ , shrinking trend); balance 9 theorem / 2 cert-unif / 2 cert-window / 3 measured (all three now fixed-size in the statement); the no-go breaks on *five* axes. T158 (`schur_entry_probe.py`) is running at the cancellation-seeing bound. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T157 subsection with the After-T157 keybox, the v552 subsection in the verified layer (now eighteen modules), the phase-2 section retitled T126–T157, the T157 row in the probe index and the updated counters (157 probes, 4047 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v552 companion blocks; `experiments/next.txt` gains the Teil-157 entry; `website/lib/primeFront.ts` gains the T157 and v552 feed entries; the prime-front status surfaces move to Teile 11–157 / 4047 checks / 18 modules; suite counter 545  $\rightarrow$  546 scripts. T158 (`schur_entry_probe.py`) is running at the cancellation-seeing bound. Ledger row PRIME.ANGLE.INSTR.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (VI) (v551 — the fixed-size Ritz ceiling certificate of phase 2 promoted; prime-front diary T154 (the ceiling closes exactly at fixed size, ALIGN-RESISTS) + paper + website sync; NO marker moves)**

- **New module v551 (PRIME.RITZ.CEIL.01).** Promotion of the certificate core of the discovery part T154 (`green_align_probe.py`) — 16 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v550, which certified the bottom ladder  $\lambda_k(A) \leq S\mu_k^P$  by per-window  $\text{LDL}^T$  counts of *size*  $m$  — this module certifies the *fixed-size* replacement T154 found for the ceiling step. Scope fence: per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all*  $D$  (the two open inputs after T154 —  $R1'$ , an  $m$ -free floor for  $\lambda_{\min}(B_{HH})$  on the bulk parity block, and  $R2'$ , an  $m$ -free lower bound for the  $L_P$  floor on the complement of the eight bottom Ritz directions — stay open and typed open; every floor consumed is a per-window certificate of size  $m$ ). Battery: the 12 *deepest* prime-power zones admitting a frame-A window inside the cap ( $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ,  $D = 7.248 \times 10^{-3} \dots 1.815 \times 10^{-2}$  — mirroring T154's own surface selection, declared rather than tuned: on the shallowest zones  $n \leq 11$  more than one bottom direction leaves the sine block and the fixed-depth certificate is *not* claimed there), a no-go size ladder  $m = 48 \dots 288$ , parity controls at  $m = 64, 128, 256$ , and 3 Dirichlet control windows; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The direction lemma (T154) [E].** For orthonormal  $Q$  and  $W = Q^T A Q$ ,  $\lambda_k(A) \leq \theta_k$  for *every*  $k \leq \dim$  — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument at all; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module. Negative control: the *reversed* reading (Ritz values as lower bounds) is recognised as false on 12/12 windows,  $\theta_k$  overshooting  $\lambda_k$  by a factor  $\geq 18.6$  somewhere on every window.
- **(2) The sixteen-column certificate (T154) [E].**  $\text{span}\{t_1..t_8\} + A^{-1}L_P \text{span}\{t_1..t_8\}$  has *fixed* dimension 16, independent of  $m$  (one Green step by Cholesky back-solves, no inverse formed), and its ceiling  $K^F = \max_k \theta_k / \mu_k^P$  (8 completed Choleskys of matrices  $\leq 8 \times 8$ ) equals 1.1019–1.5845 with  $K^F \geq \max_k \lambda_k / \mu_k^P$  on every window *and*  $|K^F / K_{\text{bot}} - 1| \leq 9.9 \times 10^{-7}$  against the declared cap  $10^{-4}$  on 12/12, where  $K_{\text{bot}}$  is the inertia-certified true ladder constant (8 completed  $\text{LDL}^T$  counts of size  $m$  per window; count = direct spectral count on all 96 pairs): the fixed-size ceiling *attains* the truth, and the size- $m$  counts are retired from the ceiling step (controls: the parity sines alone overshoot by 41.6–739 — T152/T153's  $O(m^2)$  overshoot reproduced at fixed

size — and corrupting the Green columns breaks the attainment on 12/12).

- **(3) The one-direction anatomy (T154) [E] (angles measured).** The median of the seven aligned principal angles between the bottom eight of  $A$  and the bottom eight of  $L_P$  is  $0.03\text{--}0.14^\circ$  while the largest is  $83.79\text{--}89.70^\circ$  on every window (Björck–Golub 1973) — the misalignment is carried by *one* nearly orthogonal direction, and that single direction *is* the collapse price  $\lambda_1(A)/(t\mu_1^P) = 4.408\text{--}7.042$ , recovered in full per window by one completed Cholesky of  $A - \gamma I$  (size  $m$ , certified, not fixed size, labelled as such); control:  $A = L_P$  through the same instrument returns  $K_{\text{bot}} = K^F = 1$  and zero misalignment (deviations  $< 10^{-5}$ , the inertia ladder’s backoff step) with the price exactly  $1/t$ .
- **(4) The no-go discriminator (T154) [E].** On the T145 no-go reconstruction  $c(l) = 1/(1+l)$  the odd section is positive definite and the Schur floor survives ( $t = 0.05$  on all 4 sizes) but  $K_{\text{bot}}$  explodes ( $\times 35.4$ ) and the sixteen-column ceiling explodes *with* it ( $\times 57.5$ ) over  $m = 48 \dots 288$  — the instrument does not report flatness where flatness is false and manufactures no closure; the Courant–Fischer direction is violated nowhere (16/16 positive-definite cases); the Dirichlet control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite —  $\lambda_1 = -5.60 \dots -3.55$  certified on 3/3 — so no ceiling exists there: the reflection half is load-bearing for positivity.
- **Prime-front diary T154 (green\_align\_probe.py, contract GREEN.ALIGN, 29/29, 5.0 s, 12 prime-power windows  $h = 50 \dots 1077$ , verdict ALIGN-RESISTS, sandbox).** The ceiling half of T153’s R2 closes exactly at fixed size ( $K^F = 1.1019\text{--}1.9964$  flat, agreement  $5.17 \times 10^{-7}$  on 12/12); the floor half is refuted at fixed size ( $\|R\|^2/(\beta\lambda_1) = 6.3 \times 10^3\text{--}5.6 \times 10^9$ ,  $x^{4.50}$ ) and recovered at full size (one Cholesky per window recovers the whole collapse price  $4.408\text{--}7.985$ ; per-window end to end  $4.45 \times 10^{-2}\text{--}3.13 \times 10^{-1}$ , inside the target band, while the  $m$ -free-in-shape number stays  $1.01 \times 10^{-2}\text{--}3.92 \times 10^{-2}$ ); the obstruction is named (seven bottom directions agree to  $0.15\text{--}1.35^\circ$  median, *one* sits at  $82.93\text{--}89.79^\circ$ ) and the missing fixed-size ingredient is priced (the  $L_P$  floor on the Ritz complement, arithmetic-free, flat  $5.93\text{--}8.25\mu_1^P$ , worth 91–100% of the price); the Kato route to R1 fails by the same geometry (loss  $1.97\text{--}121$  attained  $24.9\text{--}89.8^\circ$  apart; the Hankel-reflection culprit refuted twice). Two open terms remain (R1’/R2’, both uniformity terms); T155 (bottom\_floor\_probe.py) is running at exactly those two floors. NOT RH evidence; no marker moves.
- **Docs + website sync.** tfpt\_prime\_front gains the T154 subsection with the After-T154 keybox, the v551 entry in the verified layer (now seventeen modules), the phase-2 section retitled T126–T154, the T154 row in the probe index and the updated counters (154 probes, 3951 checks); tfpt\_1, tfpt\_3 and tfpt\_research\_contracts gain the v551 companion blocks; experiments/next.txt gains the Teil-154 entry; website/lib/primeFront.ts gains the T154 and v551 feed entries; the prime-front status surfaces move to Teile 11–154 / 3951 checks / 17 modules; suite counter  $544 \rightarrow 545$  scripts. T155 (bottom\_floor\_probe.py) is running at the two uniformity floors. Ledger row PRIME.RITZ.CEIL.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (V) (v550 — the odd-sector identities of phase 2 promoted; prime-front diary T151 (the one inequality closes by rerouting off the ladder, ODD-CARRIES) + paper + website sync; NO marker moves)**

- **New module v550 (PRIME.ODD.SECTOR.IDENT.01).** Promotion of the identity/certificate core of the discovery part T151 (odd\_ladder\_probe.py) — 19 checks,  $\sim 0.2$  s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v549, which certified the parity compression and the quadratic ladder certificate  $\nu_k^P \leq Ck^2$  — this module certifies the *reroute* T151 found: off the  $\nu$  ladder, through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in  $k$ . Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit,



no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open input after T151 — the  $m$ -freeness of the bottom pencil ratio  $R = K_{\text{bot}}/\kappa$  for the odd parity sector of a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), parity control models at  $m = 241$  (the certified (1, 1) pencil) and  $m = 2001$  (the  $\pi$  price), and a no-go size ladder  $m = 64 \dots 256$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .

- **(1) The grid step-over and the honest negative (T151) [E].** The parity sines live on the *shifted* grid  $\theta_k = 2\pi k/N$ ,  $N = 2m+1$ , which never contains  $\theta = 0$ , with  $\mu_k^P = 2 - 2\cos\theta_k$  *exactly* ( $2.2 \times 10^{-16}$ ; eigen-residual  $2.3 \times 10^{-13}$ ); the symbol has  $f(0) = -47.6 \dots -6.3 < 0$  (T148's honest negative, re-read and never re-derived) with a nonempty negative window  $[0, \theta_c)$ , and  $\theta_c/\theta_1 = 0.371\text{--}0.485 \leq 0.9$  on *every* window: the odd grid *steps over* the negative window — odd-sector positivity is a grid fact, stepped over rather than cancelled. The honest negative beside it, promoted as content (T150's rest item 3 answered *negatively*): the symbol-only pencil floor  $\kappa_{\text{sym}}(1 - \rho)$  is vacuous ( $\rho \geq 1$ ) on 12/12 windows and the symbol moves by 0.45–3.27 relative across *one* mode spacing — no pointwise symbol statement is admissible at the bottom mode's resolution: the local model is a *matrix* statement, not a symbol statement (controls: the Dirichlet corner breaks the eigenpairs at 1.0, a one-index shift misses by  $6.5 \times 10^{-3}$ ).
- **(2) The certified bottom ladder (T151) [E].**  $\lambda_k(A) \leq S\mu_k^P$  for  $k \leq 8$  is certified per window by 8 completed LDL<sup>T</sup> inertia counts (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list),  $S = 1.1019\text{--}6.4725$ : the odd sector's bottom spectrum *is* the parity-Laplacian ladder up to a printed  $O(1)$  factor — order 2 in  $k$ , with no trace of  $f(0) < 0$  in it (the Rayleigh floor  $L_P \geq \mu_1^P I$  absorbs it); the count equals the direct spectral count on all 96 (window,  $k$ ) pairs (control:  $S \times 10^{-3}$  certifies no eigenvalue below it anywhere).
- **(3) The discrete Sobolev step and the  $\pi$  price (T151) [E].**  $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\text{last}}^2$  exactly ( $7.6 \times 10^{-15}$ ),  $\|\nabla\psi\|_2^2 \leq \langle\psi, L_P\psi\rangle$ , and  $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\nabla\psi\|_2$  on the true window modes, the parity sines and random unit vectors — elementary, licensed by the odd sector's virtual node  $\psi_{-1} = 0$  (Hardy–Littlewood–Pólya 1934 / Agmon 1965 the continuum address); calibration: on the exact parity model the pencil is (1, 1) (certified  $|\kappa - 1| = 1.1 \times 10^{-12}$  at  $m = 241$ ) and the Sobolev per-mode price is  $\pi$  to  $5.1 \times 10^{-8}$  at  $m = 2001$  — the honest,  $m$ -free cost of leaving the  $\ell^2$  world (controls: dropping the boundary term breaks the identity at  $4.8 \times 10^{-2}$ , the Neumann energy undercounts a first-node vector by 0.50).
- **(4) The closed archimedean minimum (T151; T150's rest item 2 closed and attained) [E].** The archimedean lag kernel has two shape properties, checked per window and never assumed —  $c_i^{\text{arch}} < 0$  for  $i \geq 1$  and  $c_i^{\text{arch}}$  non-decreasing on  $i \geq 1$  — and given those,  $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$  *exactly* (max rel 0.0; value 2.1719–3.4674), attained at  $r = 0$ : a closed two-term functional of the smooth kernel, no matrix, no eigenvector, no loss (control: the full atom-including kernel violates the monotone shape on every window and the closed formula misses its minimum by  $\geq 8.7 \times 10^{-2}$ ).
- **(5) The linear per-mode bound and the no-go discriminator (T151) [E].** With  $\kappa = 0.1939\text{--}0.3599$  the certified pencil floor (*one* completed Cholesky of  $A - \kappa L_P$  per window, the floating-point floor carried as  $\text{fl}/\mu_1^P$ ; the direction re-read on the actual bottom eigenvalue) and  $K_{\text{bot}} = S$  from item (2), the chain  $b_k := 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \leq C_S k$  holds with  $C_S = 12.081\text{--}26.953 \leq 2\pi\sqrt{K_{\text{bot}}/\kappa}$  on the bottom 8 modes of every window — *linear* in  $k$  where the quadratic  $\nu$  ladder grew, every  $m$ -power cancelled, nothing additive anywhere (the  $\Theta(D^3)$  bottom that killed the Weyl / Bauer–Fike / Davis–Kahan route never enters) — and the bound dominates the truth  $m\|\psi_k\|_\infty^2$  and the parity Dirichlet energy on every read mode; the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.7362\text{--}18.6607$  is the *one* scalar the chain still owes, printed per window and typed open in  $m$ . The discriminator: on the T145 no-go form the ratio *explodes* on the size ladder ( $1.686 \times 10^3 \rightarrow 2.667 \times 10^4$ , factor 15.8 over  $m = 64 \dots 256$ ) while the exact parity model

holds  $R = 1$  to  $1.1 \times 10^{-12}$  — a route that did not fail there would prove a false statement (controls: the ladder  $\times 10^{-3}$  fails against the true sups, the inflated floor  $1.5\kappa$  makes the Cholesky *refuse* on 12/12).

- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the symbol-only route stays certified vacuous; TV(log  $\Lambda$ ) stays withdrawn as a hypothesis (T149),  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T151 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the address of the symbol question — the computed answer is that the pointwise symbol language does *not* extend to the bottom mode), Weyl 1912, Parlett, Hardy–Littlewood–Pólya 1934 / Agmon 1965, Sylvester 1852 / Bunch–Kaufman 1977, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Wilkinson 1968 / Higham 2002, Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.ODD.SECTOR.IDENT.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T151, odd-carries).** The T151 probe (`odd_ladder_probe.py`, contract ODD.LADDER, 27/27, 43.4s, 72 prime-power windows  $m = 96 \dots 1491$ ) closes the one inequality by rerouting off the ladder: the odd grid steps over the symbol’s negative window ( $\theta_c/\theta_1 = 0.328\text{--}0.407$  on 72/72), the bottom spectrum is certified against the parity Laplacian ( $\lambda_k \leq S\mu_k^P$ ,  $S = 1.1019\text{--}2.3870$ ), and the discrete Sobolev step yields the *linear* per-mode bound  $b_k \leq C_S k$  with a *non-growing* constant ( $C_S = 11.5137\text{--}19.5731$ , trend  $x^{0.020 \pm 0.007}$ ) where the quadratic ladder grew ( $13.81\pi$  at  $x^{0.258}$ ); end to end moves to  $2.01 \times 10^{-2}\text{--}3.52 \times 10^{-2}$  of the true gap (gain up to  $3.18\times$ ), the bottleneck relocating from  $Q^*$  to  $\Psi$ . Two of T150’s three rest items settled: the minimum of the archimedean diagonal is closed *and attained exactly* ( $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}} = 2.5928\text{--}4.8981$ , deviation 0.0), and the symbol route is a computed dead end (the local model is a matrix statement — the pointwise symbol language fails at the bottom mode’s resolution, margin vacuous on 72/72). One scalar remains a fit: the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.3634\text{--}9.7108$  at  $x^{0.037 \pm 0.015}$  — and the T145 no-go breaks exactly there ( $x^{1.986}$ ). Verdict ODD-CARRIES: all six preregistered gates met, every layer typed (theorem / certified / elementary-verified), the one fit named. `experiments/next.txt` part T151 (series totals 151 probes / 3852 checks); `tfpt_prime_front.tex` gains the T151 subsection with the “After T151” keybox, the v550 entry in the verified layer (now sixteen modules), the phase-2 section retitled T126–T151 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v550 companion blocks; `website/lib/primeFront.ts` gains the T151 and v550 feed entries; suite counter 543  $\rightarrow$  544 scripts. T152 (`pencil_ratio_probe.py`) is running at exactly the remaining scalar — the  $m$ -freeness of the bottom pencil ratio — and is *not* part of this change.

**2026-07-29 (IV) (v549 — the gauge/parity identities of phase 2 promoted; prime-front diary T150 (the parity mechanism named, ONE-TERM-MISSING) + paper + website sync; NO marker moves)**

- **New module v549 (PRIME.GAUGE.PARITY.IDENT.01).** Promotion of the identity/certificate core of the discovery parts T149 (`weight_smoothing_probe.py`) and T150 (`mode_ladder_probe.py`) — 21 checks,  $\sim 0.3\text{s}$ , everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v548, which certified the factorisation  $\Gamma = \sqrt{Q^*}S_w$  and the Liouville machinery in the Jacobi gauge — this module certifies the gauge *freedom* around that machinery and the parity *structure* underneath it. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open input after T150 — an  $m$ -free constant  $C$  in the odd-parity-sector ladder  $\nu_k \leq Ck^2$  for a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 48

gauge assemblies (Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal), the full-section parity items on the 6 windows with  $M \leq 300$ , and a parity control model at  $m = 2001$  (vector operations only); each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .

- **(1) The gauge freedom (T149) [E].** For *any* positive diagonal  $\tilde{\Lambda}$  the whitening  $\tilde{E} = \tilde{\Lambda}^{-1/2} A \tilde{\Lambda}^{-1/2}$  is an exact congruence, so  $\lambda_{\min}(A) \geq \text{cert } \lambda_{\min}(\tilde{E}) \cdot \min_j \tilde{\Lambda}_j$  is a valid certified lower bound for every gauge — an identity plus one Rayleigh step — and the family maximum is valid too (verified on all 48 assemblies); the factorisation identity  $\Gamma = \sqrt{Q^*} S_w$  holds *gauge-invariantly* (max rel  $2.4 \times 10^{-16}$ ); the const gauge (the geometric mean of the Jacobi diagonal) has  $\text{TV}(\log \tilde{\Lambda}) = 0$  *exactly* on every window — T149’s elimination, not reduction, of T148’s blocking scalar — with the certified entrywise sandwich  $\sigma = c_2/c_1 = 1.3868\text{--}2.5805$  (controls: a 1%  $Q^*$  perturbation breaks the identity at  $4.99 \times 10^{-3}$ , and on the shifted indefinite form  $A - 1.05\lambda_{\min} I$  the completed-Cholesky floor *refuses* on 12/12 windows).
- **(2) The two-regime discriminator (T149, the refutation as a check) [E].** The flutter-smoothing gauge (moving geometric mean, preregistered half-width  $m/16$ ) removes  $\text{TV}(\log \Lambda)$  by  $\geq 2.88\times$  on every window while moving the Liouville-*transported* bottom-block smoothness functional  $\nu_{\tilde{L}}$  by at most 0.0088 — and even the matched-amplitude *rough* re-gauge (lattice-scale oscillation) moves it by at most 0.0046 (bar 0.01 for both): the roughness  $\nu_{\tilde{L}}$  responds to is *not* in the multiplier — T149’s withdrawal of the TV hypothesis, wired as a per-instance check (control: the same functional read *without* the transport, on the whitened modes  $\psi$  which carry the multiplier, jumps by 0.41–9.76 under the same rough re-gauge — the invariance is carried by the transport, not by numbness of the functional).
- **(3) The explicit zone diagonal (T150) [E].**  $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$ , assembled lagwise with no matrix anywhere, equals the lumped-pair diagonal on every window (max rel  $5.9 \times 10^{-16}$ ) — the whitening diagonal is an *explicit* functional of the zone’s lag arithmetic and of nothing else; the lag vector splits *exactly* as  $c = c^{\text{arch}} + c^{\text{atom}}$  with  $c^{\text{atom}} \leq 0$  entrywise, and the arithmetic mass carries the closed budget  $\|c^{\text{atom}}\|_1 \leq \sum_j \mu_j \leq 4B\sqrt{N}$  with  $B = 1.038821$  the *verified* maximum of  $\psi(n)/n$  over the whole atom table (Chebyshev 1852, computed and never assumed; Abel summation carries the closed form). The honest negative as content: the archimedean section alone is *indefinite* on 12/12 windows (completed  $\text{LDL}^T$ , up to 5 negative eigenvalues) while the full section has zero — the atoms are *co-responsible* for the positivity, so the additive perturbation reading (Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970, cited, computed dead by T150 at 2.4–5.8 orders, *not* used) is structurally empty and only the multiplicative / Loewner gauge step transfers (control: dropping the positive part in the zone formula breaks the identity at  $4.25 \times 10^{-1}$ ).
- **(4) The parity compression (T150, the named mechanism) [E].** With  $U_-$ ,  $U_+$  the antisymmetric / symmetric parity isometries of the full window,  $U_-^T T_M(c) U_- = T - H$  and  $U_+^T T_M(c) U_+ = T + H$  *exactly* (max rel  $3.9 \times 10^{-16}$ ) with cross block  $U_+^T T_M U_- = 0$  ( $1.4 \times 10^{-16}$ ) on all 6 windows with  $M \leq 300$ : the reflection-odd section used since T106 *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector (Basor–Ehrhardt 2009: the parity sectors are the natural objects of the Toeplitz+Hankel algebra, cited as the address and never as an authority); the completed  $\text{LDL}^T$  counts *localise* the negative inertia — the full section’s negative eigenvalues (forced by the sign-changing symbol,  $f(0) = -29.2 \dots - 6.3 < 0$ ; T148’s honest negative *re-read*, never re-derived) sit entirely in the *even* sector while the odd sector counts zero:  $f(0) < 0$  is explained rather than worked around (control: the sign-flipped basis does not reproduce  $T - H$ ,  $3.99 \times 10^{-1}$ ).
- **(5) The parity-sine calibration and the ladder certificate (T150) [E].** The parity sines  $t_k(r) = \frac{2}{\sqrt{N}} \sin(2\pi k(r+1)/N)$ ,  $N = 2m+1$ , are the exact eigenpairs of the parity Laplacian (reflecting corner 3, forced by antisymmetry) at  $\mu_k^P = 4 \sin^2(\pi k/N)$  — eigen-residual  $2.3 \times 10^{-13}$  — and the smoothness ladder *calibrates* on the exact model:  $\nu_k^P = \pi k^2$  for  $k = 1 \dots 12$  at  $m = 2001$

(max deviation  $3.0 \times 10^{-5}$ ), so the ladder form  $\nu_k \leq Ck^2$  is the right question and  $\pi$  its exact floor; on every window, in the const gauge ( $\varphi = \psi$  exactly — the ladder of the *pure* Toeplitz-minus-Hankel section), the parity coefficient licence holds,  $\nu_k^P \leq Ck^2$  with the per-window certified constant  $C = 11.70\text{--}17.28$ , and the derived per-mode ceiling  $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k + 1)^2$  is a certified instance inequality — an  $m$ -free  $C$  is exactly the one open input and is *not* claimed (controls: the ladder  $\times 10^{-3}$  fails everywhere, a one-index calibration shift misses by 0.75).

- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ;  $\text{TV}(\log \Lambda)$  stays withdrawn as a hypothesis (T149),  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T149/T150 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953 (in the parity sector), Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the parity sectors as the algebra’s natural objects — the address, never an authority; the sign-changing symbol is exactly what Basor–Ehrhardt does *not* cover, and that gap is the open input), Sylvester 1852 / Bunch–Kaufman 1977, Chebyshev 1852 (verified on the table), Abel, Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970 (cited, computed dead, *not* used), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.GAUGE.PARITY.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T150, one-term-missing).** The T150 probe (`mode_ladder_probe.py`, contract `MODE.LADDER`, 36/36, 48.0s, 72 prime-power windows  $m = 50 \dots 1393$ ) names the mechanism: the form is exactly the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the sign-changing symbol’s entire negative inertia sits in the *even* sector (certified 72/72) — T148’s honest negative is explained rather than worked around. The flutter amplitude becomes a certified form functional ( $0.0606\text{--}0.1993$ , trend  $x^{-0.165 \pm 0.015}$ , T149’s fit reproduced fit-free), the whitening diagonal an explicit zone functional, and the arithmetic atoms turn out to be co-responsible for the positivity (the archimedean section alone has 2–7 negative eigenvalues); the additive perturbation route is dead by 2.4–5.8 orders while the multiplicative gauge step closes. One gate remains: the ladder constant still grows —  $C \leq 43.391 = 13.81\pi$  certified per stratum, trend  $x^{0.258 \pm 0.009}$  against the flatness bar 0.25 — and the gap from  $\pi$  is exactly the arithmetic leakage of the bottom mode ( $\ell^1$  mass 1.41–1.79). Verdict ONE-TERM-MISSING: one named inequality left —  $\nu_k \leq Ck^2$  with  $m$ -free  $C$  for the odd parity sector at a sign-changing symbol. `experiments/next.txt` part T150 (series totals 150 probes / 3825 checks); `tfpt_prime_front.tex` gains the T150 subsection with the “After T150” keybox, the v549 entry in the verified layer (now fifteen modules), the phase-2 section retitled T126–T150 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v549 companion blocks; `website/lib/primeFront.ts` gains the T150 and v549 feed entries; suite counter  $542 \rightarrow 543$  scripts. T151 (`odd_ladder_probe.py`) is running at exactly the remaining inequality — the odd-sector ladder — and is *not* part of this change.

**2026-07-29 (III) (v548 — the Green/Szegő identities of phase 2 promoted; prime-front diary T148 (lifting statement, ONE-INPUT-MISSING) + paper + website sync; NO marker moves)**

- **New module v548 (PRIME.GREEN.SZEGO.IDENT.01).** Promotion of the identity/certificate core of the discovery parts T147 (`green_asymptotic_probe.py`) and T148 (`szego_bottom_probe.py`) — 21 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v547, which certified the a-priori level constant  $c_0^{\text{ap}}$  — this module certifies the two scalars that constant is made of. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open lifting input —  $\nu_L \leq F(\text{form})$  with  $F$   $m$ -free, equivalently bounded  $\text{TV}(\log \Lambda)$  or any weaker smoothness of the multiplier — stays open and typed open). Battery:



12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 7 random positive definite forms for the form-independent items, and a weight-class model ladder  $m = 64 \dots 288$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .

- **(1) The factorisation identity (T147) [E].**  $\Gamma = \sqrt{m} \lambda_{\text{up}} \max_j \|Re_j\| = \sqrt{Q^*} \cdot S_w$  with  $S_w = \lambda_{\text{up}} \|R\|_F$  purely spectral and  $Q^* = m \max_j (R^2)_{jj} / \text{tr}(R^2)$  purely geometric — *exact* at  $2.2 \times 10^{-16}$  on all 12 windows and 7 random forms: the all- $D$  question splits into two named scalars and into nothing else; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true  $\Gamma$  everywhere ( $S_w = 1.1186\text{--}1.5777$ ,  $Q^* = 2.0879\text{--}2.8634$ ; controls: a 1%  $Q^*$  perturbation breaks the identity at  $5.0 \times 10^{-3}$ , perturbing the largest Green column breaks the factorisation).
- **(2) The bottom-block equidistribution constant (T147) [E].** With  $\Pi_B$  the spectral projector onto  $B = \{k : \lambda_k \leq 10\hat{\lambda}\}$  (preregistered rule,  $|B| = 3 \dots 5$ ),  $\sum_j (\Pi_B)_{jj} = |B|$  *exactly* ( $3.3 \times 10^{-16}$ ), hence  $Q_B \geq |B|$  is a theorem ( $\max \geq \text{mean}$ ); the *measured*  $Q_B/|B| = 1.5091\text{--}1.7980$  lies in the preregistered band  $[1, 2.2]$  against the Szegő/Widom sharp reference  $Q_B \leq 2|B|$  — a measurement on a finite list, typed as such (controls: the rank- $|B|$  coordinate projector gives  $Q_B = m$  and overshoots by  $\geq 3.9$ , a 1% basis scaling breaks the trace identity).
- **(3) The second-difference  $\ell^1$  ceiling (T148, the lever) [E].** From the exact Dirichlet eigenpair  $L_0 s_k = \mu_k s_k$  and  $\mu_k \geq 4k^2/(m+1)^2$ : the coefficient bound  $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^p \psi\|_1 / \mu_k^p)$  for  $p = 1, 2$  against the *true* coefficients on every window bottom mode and on random unit vectors (worst excess  $\leq 0$ ), the *m-free* closed form  $\|a\|_1 \leq 2\sqrt{\nu} + 1$  with  $\nu = (m+1)^{3/2} \|L_0 \psi\|_1 / (2\sqrt{2})$  — every power of  $m$  cancelled — and the chain  $m\|\psi\|_\infty^2 \leq 2\|a\|_1^2$ ; calibration: for the Dirichlet bottom mode  $\|a\|_1 = 1$  *exactly* and  $\nu = 3.1415 \rightarrow \pi$  at every size (controls: the ceiling  $\times 10^{-3}$  fails on every window, a one-index  $\mu$  shift breaks the coefficient identity at  $1.5 \times 10^{-1}$ ).
- **(4) The layer-cake  $S_w$  certificate (T148) [E].** The completed LDL<sup>T</sup> inertia count at every rung of the preregistered ladder  $\tau_i = \rho_i \lambda_{\text{lo}}$  ( $\rho = 1.5 \dots 1024$ ,  $\lambda_{\text{lo}}$  a certified positive floor) equals the direct spectral count *exactly* on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), and the layer cake certifies  $S_w \leq 1.3288\text{--}1.8220$  per window (true  $1.1186\text{--}1.5777$ , slack  $\leq 1.216$ ); nothing here says *m-free* (control: dropping the bottom layer breaks the bound on every window — the bottom eigenvalue is the weight).
- **(5) The Liouville transform and the weight-class discriminator (T148) [E].** The chain  $m\|\psi\|_\infty^2 \leq \kappa_\Lambda m\|\hat{\varphi}\|_\infty^2 \leq 2\kappa_\Lambda \|a(\hat{\varphi})\|_1^2 \leq 2\kappa_\Lambda \text{ceil}(\hat{\varphi})^2$  with  $\varphi = \Lambda^{-1/2}\psi$  (the exact pencil equivalence, a change of variables and not a perturbation) holds stepwise on every bottom-block mode;  $\kappa_\Lambda = \max \Lambda / \min \Lambda = 1.3868\text{--}2.5805$  is a priori (a ratio of two diagonal entries of the form). The discriminator: on a model section with a genuine order-2 zero ( $f(0) = 0$ ,  $f''(0)/2 = 1$  exact), over the ladder  $m = 64 \dots 288$  at matched  $\kappa_\Lambda$  (mismatch 0.0013), the bounded-variation multiplier keeps  $\nu$  flat ( $\max/\min$  1.003) while the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning (control: the first Liouville step  $\times 10^{-3}$  fails on every mode, provably — the step slack is bounded by  $\kappa_\Lambda^2$ ).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the symbol-level order-2 hypothesis is *refuted* (T148:  $f(0) < 0$  on every surface window) and enters nowhere as a theorem;  $\sigma_{\text{tot}}$  stays retired as a route; every T147/T148 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the mechanism’s address, never an authority), Sylvester 1852 / Bunch–Kaufman 1977, Euler 1735, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Davis–Kahan 1970 (cited, computed vacuous, *not* used), Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.GREEN.SZEGO.IDENT.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T148, one-input-missing).** The T148 probe (szego\_bottom\_

`probe.py`, contract `SZEGO.BOTTOM`, 31/31, 13.0s, 48 windows  $m = 96 \dots 1077$  in four  $D$ -strata, stress ladders to  $m = 1500$ ) dissects the lifting statement: the second factor  $S_w$  *closes* on the surface ( $\text{LDL}^\top$  layer cake,  $S_w \leq 1.9587$  certified against true 1.282–1.683, trend flat, per  $D$ -stratum), at the price of an honest negative — the arithmetic Toeplitz symbol has  $f(0) < 0$  on 48/48 windows, so positive definiteness is a finite-section effect of the minus-Hankel part and the lumping and the KMS order-2 hypothesis survives only as a measurement ( $\alpha = 1.643$ – $1.989$ ) — while the lifting statement resists with its hypothesis isolated to one named scalar ( $\text{TV}(\log \Lambda)$ , proved to be the roughness and not the conditioning by the weight-class experiment); the first end-to-end number arrives: the certified chain delivers 8.36–15.86% of the true gap on all 48 windows (median 11.74%), and the a-priori-shaped chain is still a lower bound at a factor 10.5. Verdict ONE-INPUT-MISSING: exactly one scalar side ( $Q^*$ ) lacks an  $m$ -free certified statement, target  $\nu_L \lesssim 34$  against 282 measured. `experiments/next.txt` part T148 (series totals 148 probes / 3759 checks); `tfpt_prime_front.tex` gains the T148 subsection with the “After T148” keybox, the `v548` entry in the verified layer (now fourteen modules), the phase-2 section retitled T126–T148 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v548` companion blocks; `website/lib/primeFront.ts` gains the T148 and `v548` feed entries; suite counter  $541 \rightarrow 542$  scripts. T149 (`weight_smoothing_probe.py`) is running at exactly the remaining input — whether a smoothed whitening removes the TV roughness — and is *not* part of this change.

## 2026-07-29 (II) (`v547` — the level-lemma identities of phase 2 promoted; prime-front diary T146 (level lemma, LEVEL-CARRIES) + website sync; NO marker moves)

- **New module `v547` (PRIME.LEVEL.LEMMA.01).** Promotion of the a-priori core of the discovery part T146 (`level_lemma_probe.py`) — 20 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v546`, whose item (5) certified  $S_1'$  with a  $c_0$  *measured* at the minimiser’s own layer cake — this module supplies the a-priori side, every factor a functional of the form alone. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the asymptotic delocalisation of the Green columns, T146’s remainder, stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 9 random positive definite forms for the form-independent items, and a stress ladder  $m = 64 \dots 288$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The union identity for nested cakes (T146) [E].** For a nested decreasing chain with positive coefficients,  $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T \|\psi_t\|_1 - \|\psi_t\|^2$  with  $T = \sum_j c_j$  and  $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$  — form-independent and  $O(m)$ , the reason the cake base is movable at all; exact (0.0) against T145’s  $K \times K$  double sum on all 12 window minimisers and 9 random vectors, *and* against the union sizes computed from the sets themselves with no nestedness used (controls: one shuffled proper level set  $3.2 \times 10^{-2}$ , a 1% perturbation of the heaviest coefficient  $4.8 \times 10^{-3}$ ).
- **(2) The layer-cake counting lemma at a general base (T146) [E].**  $G(b) \leq 2b^2 \|v\|_\infty \|v\|_1 / \|v\|^2 + \varepsilon(b)$  for *any* vector at all 6 preregistered bases (margin  $\geq 1.08$  on the battery and on every window minimiser); at  $b = 2$  the leading constant is the classical dyadic 8 *exactly*, and the base is *free* because the chain consumes only the lower domination  $|v| \leq \psi_t$  — the a-priori constant gains  $4/b_*^2 = 3.7612$  over base 2 on every window (controls: the bound  $\times 10^{-3}$  fails on every draw and base, and the one-level-down cake breaks the domination on every draw — the +1 in the cake exponent is load-bearing).
- **(3) The resolvent delocalisation bound (T146) [E].**  $\psi = \lambda R\psi$  to  $9.0 \times 10^{-14}$  — an *identity*, so no Davis–Kahan transfer step appears anywhere — with  $\lambda_{\text{up}} = \min_j \text{Rayleigh}(Re_j) \geq \lambda_{\text{min}}$  (Rayleigh–Ritz; window ratio 1.0592–1.2883), the coordinate bound  $|\psi_j| \leq \lambda \|Re_j\|$  (Cauchy–

Schwarz) and the column norms certified from above with the linear-solve residual paid for, giving  $\|\psi\|_\infty \leq \lambda_{\text{up}} \max_j \|Re_j\|$  on every window (within 1.061–1.337 of the truth) and  $\|\psi\|_1 \leq \sqrt{m} \min(1, \Gamma_1)$ . The lever: the trivial ceiling  $\max_j \|Re_j\| \leq 1/\lambda_{\text{min}}$  would give  $\Gamma \leq \sqrt{m} = 5.1$ –16.9 (useless); the certified columns sit a factor 2.85–10.17 *below* it (preregistered bar 2.0), giving  $\Gamma = 1.8356$ –2.2797 — read off the form, never off the minimiser (controls: a 1%  $\lambda$  perturbation breaks the identity at  $1.0 \times 10^{-2}$ , the sup bound  $\times 10^{-3}$  fails on every window).

- **(4) The composite level lemma (T146) [E].**  $c_0^{\text{ap}} = 2b^2 \Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042$ –4.8488 dominates the measured level constant on all 12 windows (slack 1.986–2.418) and stays under the preregistered ceiling  $16 = 4 \times \text{Maz'ya's Dirichlet value } 4$ ; the certified window inequality  $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}} \Psi_{\text{abs}}$  holds on 12/12 (both sides by completed Cholesky, the density the  $|R|$  one with Charikar's cited constant), closing the chain  $\hat{\lambda} \geq 1/(c_0^{\text{ap}} \Psi_{\text{abs}})$ ,  $\text{gap} \geq \hat{\lambda}/\kappa_{\text{up}}$  at loss factor 0.0423–0.1272 of the exact gap. The built-in discriminator: on T145's no-go form  $R = aa^T + \varepsilon I$ ,  $a_i = i^{-1/2}$ ,  $\Gamma$  grows  $3.676 \rightarrow 6.798$  over  $m = 64 \dots 288$  (ratio  $1.85 \geq$  the preregistered 1.5) and  $c_0^{\text{ap}} = 11.10$  *exceeds* every window value — the bound must be worse there, since a candidate that did not fail on the no-go would prove a false statement — while the theorem half survives on every no-go size and the Dirichlet path-Laplacian control stays flat (max/min 1.0063; control:  $c_0^{\text{ap}} \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; only the sign-free / Green constant is bounded a priori, the energy-route  $\sigma_{\text{tot}}$  stays *measured*, and every T146 exponent is a fit that stays in the sandbox. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Miclo 1999, Davis–Kahan 1970 (cited, deliberately *not* used), Perron–Frobenius (not applicable to  $E$ , not used), Fukushima–Ōshima–Takeda 1994, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.LEVEL.LEMMA.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T146, level-carries).** The T146 probe (`level_lemma_probe.py`, contract LEVEL.LEMMA, 30/30, 7.6 s, 64 windows  $m = 26 \dots 1173$ , 9/9 preregistered verdict conditions) closes L1 on the measurement surface as a chain of theorems and certified window inequalities with no step reading the minimiser: four hand-written profiles lose by 5–9 orders of magnitude, the Green column survives ( $\lambda_{\text{up}}/\hat{\lambda} = 1.0497$ –1.3764), Davis–Kahan is instrumented and discarded (informative on 0/64; the spectrum bottom is a near-degenerate block,  $\lambda_2/\hat{\lambda} = 1.2542$ –3.4471), the cake base proves free (the dyadic 8 falls to  $2b^2 \approx 2$ , gain 3.76 at zero cost) and  $c_0^{\text{ap}} = 3.9042$ –4.8488 on 64/64 windows lands at the size of Maz'ya's classical Dirichlet value 4, closing the chain at loss factor 0.0422–0.1586; the one genuine remainder is  $D$ -uniformity for *all*  $D$  — the asymptotic delocalisation of the Green columns. `experiments/next.txt` part T146 (series totals 146 probes / 3689 checks); `tfpt_prime_front.tex` gains the T146 subsection with the “After T146” keybox, the v547 entry in the verified layer (now thirteen modules), the phase-2 section retitled T126–T146 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v547 companion blocks; `website/lib/primeFront.ts` gains the T146 and v547 feed entries; suite counter  $540 \rightarrow 541$  scripts. T147 (`green_asymptotic_probe.py`) is running at exactly the all- $D$  asymptotics and is *not* part of this change.

**2026-07-29 (I) (v546 — the capacity-chain identities of phase 2 promoted; prime-front diary T145 (Maz'ya proof, ONE-STEP-MISSING) + website sync; NO marker moves)**

- **New module v546 (PRIME.CAPCHAIN.IDENT.01).** Promotion of the identity-shaped core of the capacity cycle T142 (`conductance_profile_probe.py`), T143 (`sharp_capacity_probe.py`), T144 (`capacity_inequality_probe.py`) and T145 (`mazya_proof_probe.py`) — 26 checks,



$\sim 0.5$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and *no level lemma* (T145's L1 stays open, unclaimed). Battery: 12 windows ( $n = 4 \dots 139$ ,  $h = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ , 32–5102 positive off-diagonal edges); the covering kernel is positive definite on 11, where items (1)–(2) run; each identity carries a mutation control that must fail by  $\geq 10^{-3}$ .

- **(1) The capacity decomposition (T142) [E].**  $K^{-1} = D^T J^{-1} D + x x^T / \text{cap}$  with  $J = D K D^T$  (assembled entrywise,  $3.8 \times 10^{-15}$ ),  $x = K^{-1} \mathbf{1}$  and  $\text{cap} = \mathbf{1}^T K^{-1} \mathbf{1}$ , as a matrix identity ( $1.4 \times 10^{-13}$ ); under the congruence  $u = K^{1/2} v$  the Dirichlet half is an *orthogonal projection* ( $P^2 = P$  at  $1.9 \times 10^{-12}$ ,  $\Omega = \lambda_{\max}(P) = 1$  to  $5.1 \times 10^{-12}$  — the T142 normalisation, where T141 had guessed 20.7–2724), its complement *is* the capacity rank-one and it annihilates  $K^{-1/2} \mathbf{1}$  (controls: dropping the rank-one  $3.4 \times 10^{-1}$ , a 1% cap perturbation  $3.4 \times 10^{-3}$ ).
- **(2) The exact capacity–Rayleigh identity (T143) [E].** With  $H = \text{diag}(s) + L_N$  ( $4.7 \times 10^{-16}$ ),  $L_N = D^T B D$  for the signed crossing kernel ( $4.9 \times 10^{-16}$ ) and  $\rho(W) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ), the assembled quotient  $[d^T (J^{-1} - B) d + (x^T v)^2 / \text{cap} - \sum_k s_k v_k^2] / [d^T J^{-1} d + (x^T v)^2 / \text{cap}]$  equals  $v^T (K^{-1} - H) v / v^T K^{-1} v$  for *every*  $v$  ( $9.5 \times 10^{-14}$  on 24 random vectors per window), reproduces  $1 - \rho(W)$  at the exact top eigenvector ( $1.7 \times 10^{-9}$  under the *declared* 1/gap conditioning bar  $5.4 \times 10^{-8}$ ), and every random quotient stays  $\geq 1 - \rho(W)$ ; on the constant vector the whole denominator *is* the capacity term (controls: dropping it,  $\text{rel} = 1$ ; the  $|N|$  crossing kernel,  $2.6 \times 10^{-1}$ ).
- **(3) Cholesky nestedness (T144) [E].**  $\text{cap}_E([a, a+j])$  equals the prefix sum of squared forward-substitution entries of *one* triangular solve: 3404 interval capacities from 36 solves at  $2.0 \times 10^{-14}$  against independent Schur solves, exhaustive on the median window (every prefix,  $2.8 \times 10^{-14}$ ), monotone in  $b$  by construction (control: the reversed ordering computes *suffix* capacities and deviates by  $\geq 4.6 \times 10^{-1}$ ).
- **(4) The whitening certificate (T144) [E].**  $\lambda_{\max}(W) \leq \kappa_{\text{up}}$  (completed Cholesky)  $\leq$  the Gershgorin ceiling on every window ( $\kappa_{\text{up}} = 1.2245\text{--}1.4236$  against  $\leq 2.1213$ ); the licensed chain direction  $\lambda_{\min}(E) / \kappa_{\text{up}} \leq \lambda_{\min}(A, A_B)$  holds on every window — a denominator comparison, untouched by T142's numerator obstruction — and the Cholesky refuses at the shrunk shift  $0.995 \lambda_{\max}(W)$ .
- **(5) The M4 split and the certified chain (T145) [E].** The mass half of Maz'ya's step M4:  $\sum_k f_k^2 \leq \psi^2$  *pointwise* (a theorem for nonnegative truncations, exact slack  $\geq 0$ ), the row-wise implication on rows with  $(E\mathbf{1})_i \geq 0$ , measured mass ratio  $\leq 0.2995$  and the full step  $\sigma_{\text{tot}} = 0.2197\text{--}0.4425 < 1$  on every window although the Markov hypothesis is absent. Licence 4 is stated with  $|R|$  and *not*  $R^+$ , with the witness  $R = [[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$ ,  $x = (1, -1)$  ( $3.0 > 2.0$ ) built in as a counterexample check. The sign-free chain is ordered on every window, and S1' is *certified per window on both routes*:  $\text{cert\_lam\_max}(R) \leq 12 \sigma_{\text{tot}} \Psi_{\text{pos}}$  (energy,  $c_0 = 2.6369\text{--}5.3099$ ) and  $\leq G_{\text{dy}} \Psi_{\text{abs}}$  (sign-free,  $c_0 = 2.5154\text{--}6.1449$ ); the better route gives  $c_0 = 2.5154\text{--}4.2265$  against the classical Dirichlet value 4. The all-sets density bound carries Charikar's cited constant, bracketed by exhaustive enumeration of all 4095 subsets of a small block (control:  $c_0 \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in  $D$ ; the  $c_0$  of item (5) is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere — it is the thing the cycle transcribes), Muckenhoupt 1972, Fukushima–Ōshima–Takeda 1994, Miclo 1999, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.CAPCHAIN.IDENT.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.

- **Diary, paper and website (T145, one-step-missing).** The T145 probe (`mazyia_proof_probe.py`, contract MAZYIA.PROOF, 33/33, 8.0s, 64 windows  $m = 26 \dots 1173$ ) transcribes Maz'ya's classical proof step by step onto the non-Markovian form: M1, M2, M3 are theorems, the Markov property sits in exactly one line (M4), that line splits, its mass half is a theorem and dominates ( $\sigma_{\text{tot}} = 0.215\text{--}0.443 < 1$ ),  $c_0$  becomes explicit (2.248–4.227, fit  $D^{0.028 \pm 0.017}$ , zone-uniform) and S1' is certified on 64/64 windows; an explicit no-go proves a level-profile hypothesis is *necessary*; S2' retired, R5 downgraded, R4 closed — the mathematical rest is *one* named lemma, L1. `experiments/next.txt` part T145 (series totals 145 probes / 3659 checks); `tfpt_prime_front.tex` gains the T145 subsection, the v546 entry in the verified layer (now twelve modules) and the updated probe index; `website/lib/primeFront.ts` gains the T145 feed entry; suite counter 539 → 540 scripts. T146 (`level_lemma_probe.py`) is running at exactly L1 and is *not* part of this change.

### 2026-07-28 (III) (v545 — the Hardy-core identities of phase 2 promoted; prime-front diary T140/T141 + website sync; NO marker moves)

- **New module v545 (PRIME.HARDY.IDENT.01).** Promotion of the identity blocks of the discovery parts T140 (`signed_band_probe.py`) and T141 (`discrete_hardy_probe.py`) — 36 checks,  $\sim 1$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: *identities*, closed-form quantities and the *validity* — never the *size* — of two certified upper-bound shapes, plus one negative typing. No fit is promoted, no  $D$ -exponent, no zone-uniform statement, and no bound that beats the target. Battery: 11 windows ( $n = 8 \dots 139$ ,  $h = 58 \dots 285 \leq 300$ , core  $m = 57 \dots 284$ , 193–5102 positive off-diagonal edges, 10–87 loaded stripes),  $D = 7.578 \times 10^{-3} \dots 2.388 \times 10^{-2}$  (a factor 3.15, so item (8) is a comparison and not a single point), exact  $\rho(W) = 0.997759\text{--}0.999981$ ; the explicit  $n_e \times n_e$  signed Gram is formed only on the 5 windows with  $n_e \leq 900$ .
- **(1)–(3) The form lifting, the exact finite core and the energy reordering [E].** With  $M$  the edge–interval incidence matrix and  $C = \text{diag}(\sqrt{\Delta})M$ , the T139 telescope lifts from the *entries* of the signed Gram to its *form*:  $\text{Gram} = CHC^T$  at  $3.0 \times 10^{-15}$ , hence  $\text{rank}(\text{Gram}) \leq h-1$  (measured 57–183 against 193–882 edges; control: unweighted incidence 1.6). The covering kernel in *closed geometric form*,  $K_{rs} = W([r \wedge s, r \vee s])$ , equals  $M^T \Delta M$  ( $1.3 \times 10^{-15}$ ) and the spectral radius is an *exact* eigenvalue of the finite core,  $\rho(W) = \lambda_{\max}(K^{1/2} H K^{1/2}) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ) =  $\lambda_{\max}(\text{Gram})$  ( $7.8 \times 10^{-16}$ ) — three independent routes, no truncation (control: the factorwise product  $\lambda_{\max}(K)\lambda_{\max}(H)$  misses by  $7.8 \times 10^2$ ). The energy reordering  $H = \text{diag}(s) + L_N$  holds at  $4.7 \times 10^{-16}$ : a mass term plus a long-range Dirichlet form, its weights positive on 0.676–0.872 of the off-diagonal area (a measurement, no law).
- **(4) The checkerboard split [E].** At group length exactly  $b$ ,  $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$  *entrywise* (0.0) on all 9 (window,  $b$ ) pairs over  $b = 4, 8, 16$ , and the resulting three-step Weyl sequence dominates  $\lambda_{\max}(\text{Band}_b)$  with 0 exceptions (worst ratio 1.6940) at a step count independent of the up to 9 groups — three steps instead of  $O(nb)$  (control:  $L = b/2$  loses entries,  $3.5 \times 10^{-1}$ ).
- **(5)–(6) The four Hardy identities and the closed Muckenhoupt quantity [E].** (i) the  $K^+$  Rayleigh form  $\rho(W) = u^T H u / u^T K^+ u$  at  $u = K^{1/2} v$  ( $6.6 \times 10^{-13}$ ); (ii) the signed crossing kernel  $L_N = D^T B D$  with  $B_{kl} = \sum_{r \leq k \wedge l, k \vee l \leq s} N_{rs}$  — the same closed covering formula as  $K$ , on the increment grid ( $4.9 \times 10^{-16}$ ); (iii)  $Q = B_+ \mathbf{1}$  ( $1.4 \times 10^{-15}$ ): the Cauchy–Schwarz weight T140 introduced by hand *is* the row sum of  $B_+$ , so that step is the Gershgorin diagonal step on  $B_+$ ; (iv)  $DKD^T = L_\Delta$  on the interior nodes ( $2.7 \times 10^{-14}$ ) with diagonal  $(\Delta \mathbf{1})_{k+1} = 0.4886\text{--}15.15$ , the endpoint edge mass — a *local* quantity and not  $\lambda_{\max}(K)$  (control: the crossing kernel of  $|N|$  does not reproduce  $L_N$ ,  $2.6 \times 10^{-1}$ ; the signs are load-bearing). The two-weight quantity  $A_{M0} = \max_k Q_k (\Delta \mathbf{1})_{k+1}$  is computed by two closed routes agreeing to  $8.3 \times 10^{-15}$ ; its value (65.5–4244) is *reported*, its size and zone behaviour are not promoted (control:  $\lambda_{\max}(K)$  in place of the local mass inflates it by  $\geq 358$ ).

- **(7) Two certified shapes, promoted as *valid* bounds only [E].** Every Loewner step is certified by a completed Cholesky ( $-1.5 \times 10^{-14}$ ,  $-2.3 \times 10^{-10}$ ). The additive Hardy chain dominates  $\rho(W)$  on 11/11 windows (factor 3.074–10.758); the exact two-term Weyl split dominates too (1.691–2.503) and lies *below* the chain everywhere, so it is the floor of the whole additive family; the genuinely joint shape  $\Lambda(H, Y) \cdot \Omega(Y)$  with a positive definite Jacobi Hardy form  $Y$  dominates on 11/11 with the closed-form geometric profile (8.815–74.872 at  $\Omega = 8.33$ –36.2) and on 11/11 with the per-zone tridiagonal read-off of  $K^+$ . Promoted is the *dominance* on these instances and nothing about the factors. Controls: the same  $Y$  without its mass term is singular — the constant vector lies in its kernel ( $5.7 \times 10^{-17}$ ) while  $H$  does not vanish there — so the certificate *refuses* on 11/11; and dropping the *positive* part instead of the negative one falls below  $\rho(W)$  on 11/11.
- **(8) The sign-versus-absolute separation — a negative typing [X].** Over the measured  $D$  range the exactly bounded object (the signed Dirichlet share) spreads by  $1.36\times$ , while every absolute variant spreads by at least  $54.94\times$  (Loewner drop 62.03, Cauchy–Schwarz 54.94, product 56.92), and dropping the positive part of  $H$  alone costs  $9.96$ – $464.63\times$ . The corresponding power laws are *fits* and labelled as fits:  $D^{-0.138 \pm 0.094}$  (signed) against  $D^{-0.776 \pm 0.352}$  (certified Muckenhoupt) and  $D^{-2.748}$ ,  $D^{-2.820}$ ,  $D^{-2.918}$  (absolute). Promoted is the instance comparison of the measured spreads, not the exponents.
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in  $D$ ; item (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*; item (7) promotes validity and not size — neither shape beats the target. Abel summation, Cauchy–Schwarz, Weyl 1912, Gershgorin 1931, Haynsworth 1968, Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979, Muckenhoupt 1972, Bradley 1978, Opic–Kufner 1990, Bennett 1987/1991, Gantmacher–Krein 1950/1960, Karlin 1968, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.HARDY.IDENT.01 [E] + [X]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website.** experiments/next.txt part T141 (HARDY-RESISTS, 22/22; series totals 141 probes / 3547 checks); tfpt\_prime\_front.tex gains the T141 subsection, the v545 entry in the verified layer and the updated probe index; website/lib/primeFront.ts gains the T141 feed entry and the v545 promotion entry. T142 (conductance\_profile\_probe.py) is running at the one open ingredient and is *not* part of this change.

**2026-07-28 (II) (v543 + v544 — the lumped  $M$ -matrix pair and the long-lag support structure promoted; prime-front diary T136/T137 + website sync; NO marker moves)**

- **New module v543 (PRIME.MMATRIX.IDENT.01).** Promotion of the identity/theorem block of the discovery part T136 (35 checks,  $\sim 1$  s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* per-instance result; no fit, no  $D$ -exponent, no uniform-in-zone claim, no floor value. Battery: 20 seam windows ( $n = 4 \dots 59$ ,  $\rho = 1.25 \dots 4.00$ ,  $h = 25 \dots 299$ ; every factorised or diagonalised matrix  $\leq 300$ ) giving 40 symmetric matrices (pole-free odd sections + border Schur blocks), 39 with positive off-diagonals, plus 312 shifted ladder rungs and 8 whitened two-grid rungs.
- **(1)–(2) The lumped pair and the congruence [E].** With  $\Delta$  the positive off-diagonal part and  $L_\Delta = \text{diag}(\Delta \mathbf{1}) - \Delta$ , the pair  $A_B = A + L_\Delta$  is Stieltjes (off-diagonals exactly  $\leq 0$ ),  $L_\Delta$  is a graph Laplacian hence PSD with zero row sums, so  $A_B \mathbf{1} = A \mathbf{1}$  ( $1.6 \times 10^{-15}$ ) and  $A_B \succeq A$  — an *upper* bound on  $\lambda_{\min}(A)$  and a floor *never* (control: lumping the negative off-diagonals destroys Stieltjes,  $2.0 \times 10^{-1}$ ). The congruence  $A = A_B^{1/2} (I - W) A_B^{1/2}$ ,  $W = A_B^{-1/2} L_\Delta A_B^{-1/2} \succeq 0$ , holds as a *matrix* identity ( $4.4 \times 10^{-15}$ ) with  $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$  ( $4.4 \times 10^{-15}$ ), and the *declared*

*circularity*  $\lambda_{\max}(W) < 1 \iff A \succ 0$  is checked in *both* directions (0 exceptions forward; 39/39 shifted controls in the indefinite range give  $\lambda_{\max}(W) \geq 1$ ) — the Loewner sandwich is vacuous by itself, and that is the content.

- **(3)–(4) The anchor and the empty shift ladder [E].** With  $x = A_B^{-1}\mathbf{1} \geq 0$  and  $A_B^{-1} \geq 0$  entrywise (Ostrowski 1937, *checked*),  $\lambda_{\min}(A_B) \geq 1/\max_r x_r$  on all 39 rows (min slack  $3.0 \times 10^{-3}$ , sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue (control: a positive off-diagonal makes it false,  $1.9000 > 0.1000$ ). Lumping commutes with diagonal shifts,  $(A - sI)_B = A_B - sI$  ( $2.5 \times 10^{-16}$  on 312 rungs), and  $(A_B - sI)^{-1} \geq A_B^{-1} \geq 0$  entrywise (Berman–Plemmons 1979, violation 0), so  $\max_r x_r(s)$  and the a-priori margin  $\|L_\Delta\|_\infty \max_r x_r(s)$  are non-decreasing along the whole ladder: the shifted ladder is *structurally empty* — a theorem, not a measurement.
- **(5)–(6) Varga’s identity and the a-priori Jacobi radius [E].** For the regular splitting  $A_B = D_0 - N_0$  (all three hypotheses checked on 39/39),  $\rho(J) = \tau/(1 + \tau)$  with  $\tau = \rho(A_B^{-1}N_0)$  at  $4.8 \times 10^{-15}$  ( $\rho(J) = 0.7347\text{--}0.9860$ ); the Collatz–Wielandt bracket at the anchor vector gives  $\min_r (Tx)_r/x_r \leq \tau \leq \max_r (Tx)_r/x_r$  (0 exceptions) and hence an *a-priori*  $\rho(J) \leq 0.9861 < 1$  with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. Controls: a non-regular splitting breaks the identity ( $1.1 \times 10^{-1}$ ); at a sign-changing test vector the quotient bound is false on 39/39.
- **(7) The  $k$ -scanned M18 majorant — a negative result [X].** The whitened mass identity  $\|L^{-1}\hat{s}\|^2 = \hat{s}^\top U^{-1}\hat{s}$  holds at  $7.2 \times 10^{-16}$  and the triangle chain is a *valid* majorant at every  $k$  separately (39 (rung,  $k$ ) pairs over  $k = 2 \dots 64$ , 0 violations), so the minimum over the ladder is legitimate — and it does *not* reach the preregistered T134 bar 1/2 on any rung (best 3.080; best over the three values T134 tried, also 3.080) while the *measured* bad mass is  $\leq 0.0120$ . The localisation term alone already exceeds the bar (0.542): the loss is the bound, and what is missing is a localisation statement. The bar is *not* moved and nothing is upgraded.
- **New module v544 (PRIME.LONGLAG.SUPP.01).** Promotion of the structure block of the discovery part T137 (24 checks,  $\sim 0.2$  s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons).
- **(1)–(4) The support is a comb stripe set, and its amplitude is certified [E].** The kernel splits exactly as  $c = c_{\text{arch}} - c_{\text{comb}}$  with  $c_{\text{comb}} \geq 0$ , and the archimedean section alone has *all* off-diagonals  $\leq 0$  on 20/20 windows: every positive off-diagonal of  $A$  is comb-generated (control: the full section carries  $+2.0 \times 10^{-1}$ ). Every one of the 20555 positive edges has its Hankel index  $M - 1 - r - t$  inside the atom-hat index set — per-window fraction exactly 1.000000 — so  $\text{supp}(\Delta)$  is a union of anti-diagonal stripes at prime-power atom positions (the converse is reported, *not* claimed). Each stripe is a perfect matching (0 endpoint collisions), so its edge vectors  $b_e = e_r - e_t$  are *exactly* orthogonal and  $L_\Delta$  restricted to a stripe is a direct sum of  $2 \times 2$  blocks (control: edges across two stripes overlap). And  $A_{rs} = A_{rs}^{\text{arch}} - c_{\text{comb}}[|r - s|] + c_{\text{comb}}[\text{anti}]$  gives the amplitude certificate  $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$  on all 20555 edges (excess  $-1.9 \times 10^{-6}$ , sharpest ratio 0.99998532) — no eigenvalue, no norm; control: the Toeplitz lag in place of the Hankel index breaks it on 20415/20555. Honestly undecided and *not* promoted: at  $h \leq 300$  the hats never overlap, so the tighter one-atom refinement coincides with the certificate here.
- **(5) The Green function, bracketed two-sidedly [E].** For the lumped comparison  $B = D_0 - N_0$ ,  $J = D_0^{-1}N_0 \geq 0$ , the anchor identity  $q_J = \max_r (Jx)_r/x_r = 1 - \min_r 1/(d_r x_r) < 1$  ( $8.2 \times 10^{-16}$ ) needs no measurement; then the Neumann partial sum to  $K = 8$  is an *entrywise* lower bound for  $B^{-1}$  ( $-2.1 \times 10^{-4}$ ) *because the discarded terms are nonnegative*, and partial sum  $+(x_r/x_s)q_J^{K+1}/((1 - q_J)d_s)$  is an entrywise upper bound ( $-1.4 \times 10^{-1}$ ), on 39 rows (20 pole-free sections + 19 border Schur blocks). Control: dropping the tail makes the upper bound false. This delivers item (2) of the T136 rest list.
- **(6) The absolute-value envelope family is certified dead [X].** With  $E = B^{-1}L_\Delta$ , Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight



are all  $\geq \rho(|E|)$ , so one rigorous *lower* bound decides the whole family at once: a Collatz–Wielandt quotient from below at a power-iterated weight gives  $\rho(|E|) \geq 1.3141 \geq 1$  on 20/20 rows (the cheapest four members all  $\geq 1.5937$ ). No member of the family can ever certify  $\lambda_{\max}(W) < 1$  — a certificate of *death*, not a measurement. The *signed* radius  $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$  on the same rows: the absolute value alone destroys the cancellation, and what survives is a sign-preserving bound that does not exist yet.

- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing uniform in the zone index, in  $D$  or in the gap. The T136  $D$ -exponents (anchor  $D^{-0.565}$ , margin gap  $D^{+2.72}$ , slack  $D^{0.116}$ ) are fits and stay in the sandbox; the structure bound assembled from the support and amplitude facts does *not* beat the margin (T137: 0 of 35 windows) and is *not* promoted; entrywise positivity of the *target* inverse at depth stays open. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Frobenius 1912–Perron 1907 / Collatz 1942–Wielandt 1950 / Gershgorin 1931 / Neumann series / Haynsworth 1968 / Wilkinson 1968–Higham 2002 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Diary Teile 136/137 (sandbox).** `m_matrix_pair_probe.py` (contract M.MATRIX.PAIR, 30/30, verdict ONE-CARRIES): the  $\rho(K)$  item closes with Varga’s regular splitting and the Collatz–Wielandt bracket (sharp 1.00–1.03 on 900/900), the exact three-way bookkeeping  $\lambda_{\min}(A) \sim D^{2.27} = \text{anchor } D^{-0.56} \times \text{gap } D^{2.72} \times \text{slack } D^{0.12}$  attributes the whole degradation to the margin (which is only  $2.5 \times 10^{-6}$ – $2.5 \times 10^{-2}$  wide while every cheap bound on  $\rho(E)$  sits at 1.82–3.11), and M17 closes *negatively* (the bad pencil subspace is delocalised, participation ratio 0.54–0.98, its dimension tracking the atom count at  $n_{\text{bad}}/n_{\text{at}} = 1.64$  rather than the grid, so  $\sigma_b(k) \geq 0.646$  everywhere); promotion list 15  $\rightarrow$  22 ready. `long_lag_probe.py` (contract LONG.LAG, 22/22, verdict BOTH-RESIST):  $L_{\Delta}$  is fully comb-generated (the archimedean half has a positive off-diagonal on 0 of 40 windows), the structure bound does *not* beat the margin (0/35, overshoot  $24$ – $7.1 \times 10^5$  gaps, second differences decaying only algebraically as  $\text{dist}^{-0.63}$  where Demko–Moss–Smith would need a geometric rate), the envelope family is certified dead from below ( $\rho(|E|) \geq 1.3218$  on 35/35), and the Green bracket closes two-sidedly (lower recovers 0.011–0.406 of the minimum entry, upper overshoots by 2.49–12.24). Series counter: 137 probes, 3438/3438 sandbox checks. Sandbox only for the probes: no ledger row, no marker move, not RH evidence.
- **Ledger/papers/website.** Two new ledger rows: PRIME.MMATRIX.IDENT.01 and PRIME.LONGLAG.SUPP.01; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *ten* modules; phase 2 extended to T126–T137); website mirrors (/prime-front feed entries T136/T137, status surfaces Teile 11–137 / 3438 checks, verification DAG, papers data) updated in the same change.

**2026-07-28 (I) (v542 — the margin-chain identities of phase 2 promoted; v379 hardened to an exact positive-mixture Gram certificate; prime-front diary T132/T133 + website sync; NO marker moves)**

- **New module v542 (PRIME.MARGIN.IDENT.01).** Promotion of the *identity/theorem block* of the discovery parts T128–T131 into one narrow load-bearing module (44 checks,  $\sim 0.2$  s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: identities and *per-instance* theorems only; no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ( $n = 3 \dots 97$ ,  $\rho = 1.25 \dots 4.00$ ,  $h_c = 24 \dots 214$ ,  $h_f = 36 \dots 299$ ; every inverted or diagonalised matrix  $\leq 300$ ), 43 graded/fine grid pairs ( $m \leq 172$ ) and 400 randomised profiles; each identity is a numerical residual against a *preregistered* tolerance plus at least one mutation control that must fail by  $\geq 10^{-3}$  relative.
- **(1)–(2) Border geometry and the (T) identity [E].**  $\sum_i (1 - g_i) = t_l + t_r$  from the exact

interval geometry (max rel  $5.5 \times 10^{-13}$ ), so the  $\kappa$  weights are a probability vector and the flat border vector gives  $\kappa_{\text{flat}} = 1$  *exactly* (control: a 1% perturbation of  $g_i$  breaks it by  $7.9 \times 10^{-2}$ );  $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$  at  $2.2 \times 10^{-16}$ , with  $\tau$  independently recomputed from the *full* odd overlap matrix ( $1.1 \times 10^{-13}$ ); dropping any of the three correction terms breaks the identity.

- **(3)–(4) The profile identity and the curvature chain [E].**  $p_{N-1} = 2 - p_0 + 2E$  at  $2.3 \times 10^{-16}$  (instances) /  $1.4 \times 10^{-15}$  (400 random profiles), hence the equivalence  $2E > p_0 \iff p_{N-1} > 2$  with 0 exceptions and both branches realised (60/400); the Abel/Hölder chain  $\kappa \leq 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\text{max}} - p_{N-1})$  holds per instance with 0 violations (min slack  $1.0 \times 10^{-1} / 3.7 \times 10^{-2}$ ), while the *same* chain without the curvature term is violated on 88/400 profiles — the curvature term is load-bearing, not decorative.
- **(5)–(6) Assembly and the C  a/Strang defect [E].** The matrix-free two-scale assembly equals  $J^T Q J$  ( $5.2 \times 10^{-16}$ ) and the graded overlap operator equals  $J$  ( $5.2 \times 10^{-14}$ ), reducing to the uniform odd overlap when both partitions are uniform;  $S_{\text{graded}} - S_{\text{uniform}} = R^T X^{-1} R$  as a *matrix* identity on all 43 grids ( $2.1 \times 10^{-14}$ ) with a PSD defect on every grid, so the compression error is one-sided *by* the identity (controls: unnormalised merge weights  $8.4 \times 10^{-1}$ ; dropping the  $-B$  term in  $R$   $4.0 \times 10^{-1}$ ).
- **(7)–(9) Sandwich, Perron implication, counting chain [E].** The secular sandwich  $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon/\|u\|^2$  holds on all 48 pole-free sections at which  $A > 0$  and  $\varepsilon > 0$  are *checked* (relative width  $\leq 1.4448$ ), with Albert’s sign equivalence verified in *both* directions (6 rescaled controls with  $\varepsilon < 0$  all give  $\lambda_{\min}(Q) < 0$ );  $S^{-1} > 0$  entrywise (measured 48/48)  $\Rightarrow$  sign-constant *and* simple ground vector as an *implication* (tridiagonal control: hypothesis fails, conclusion fails with it);  $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$  and  $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$  with 0 exceptions on a non-vacuous battery ( $s_2$  up to 31).
- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing is uniform in the zone index; the  $\kappa$  law is *not* claimed (T129 found it false in general), only the chain underneath it; (7) assumes nothing — positivity of the pole-free section at depth (the open M25d) is not extended and no floor value is carried; (8) has a *measured* hypothesis; (6) says nothing about the *size* of the defect (the open T130 exponent). Abel / H  lder / Schur–Haynsworth 1968 / Albert 1969 / C  a 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Hardening of v379 (SEAM.S3.RP.01).** T133 (CERT.FLOOR) certified that the  $10 \times 10$  Hankel matrix v379 assembles has, *as a double-precision object*, a negative direction at  $-7.75 \times 10^{-18}$  (60 digits) — inside the module’s old `min_eig > -1e-10` tolerance. The mathematical matrix is fine and the exact rescue is now in the module: the RP Gram is built explicitly as a *positive mixture*  $M = V^T \text{diag}(1/1600)V$ ,  $V_{ki} = e^{-\varepsilon_k \tau_i}$ , so  $x^T M x = \sum_k w_k (v_k \cdot x)^2$  is a *sum of squares* and  $M \succeq 0$  holds exactly; the mixture identity is zero at the rounding level ( $1.1 \times 10^{-15} \leq 64u|M|$ ) and the sum-of-squares route is nonnegative on 510/510 test directions where the assembled float form is negative on 2. The eigenvalue check is *retained and relabelled a diagnostic measurement*. Ledger row SEAM.S3.RP.01 reworded accordingly (method text: exact positive-mixture Gram identity, eigenvalue check as diagnostic); **marker stays [E] — no upgrade**. 7 checks (was 5), `run_all` green.
- **Diary Teile 132/133 (sandbox).** `bd_seam_probe.py` (contract BD.SEAM, 21/21, verdict SPECTRUM-ONLY): the model boundary-Dirac and the marks kernel share their spectrum to  $7.5 \times 10^{-13}$  but are *not* the same operator — the gap sits in the killing mass 0.1746 ( $N$ -stable), the killing measure equals the marks profile pointwise, the Markov remainder is exactly zero with a closed jump-weight form (odd lags positive, even lags zero — which explains T126’s 86.1%), and the  $D_4$  lattice symmetries leave the triad exactly invariant; ceiling: the DtN is a model DtN, the gate is *not* closed. `cert_floor_probe.py` (contract CERT.FLOOR, 23/23, verdict

MIXED, 840 control validations): the `v379` finding above plus a recommendation table — the hardening is implemented in this same change. Series counter: 133 probes, 3352/3352 sandbox checks. Sandbox only: no ledger row for the probes, no marker move, not RH evidence.

- **Ledger/papers/website.** New ledger row `PRIME.MARGIN.IDENT.01`; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *eight* modules; phase 2 extended to T126–T133); website mirrors (`/prime-front` feed entries T132/T133, status surfaces Teile 11–133 / 3352 checks, verification DAG, papers data) updated in the same change.

**2026-07-27 (IV) (prime front: T111 deep ladder maps three walls, crossing measured at  $n \approx 462$  — CROSSING-CONFIRMED; paper + diary + website sync; NO marker moves)**

- **Diary Teil 111 (sandbox).** `experiments/tfpt-discovery/deep_zone_stress_probe.py` (contract `DEEP.ZONE.STRESS`, 23/23 checks, 140 s, verdict `CROSSING-CONFIRMED`): the probe stops extrapolating and measures — the T110 grid is reproduced bit-exactly, then the ladder is extended to every prime-power zone  $n \leq 1000$  plus a thin tail to 3001 (199 zones; chain ladder  $n = 2 \dots 521$  with 117 handovers). The ratio  $m_k/\text{need109}$  falls  $179.59 \rightarrow 0.0174$ ; the crossing is *measured*: last zone  $\geq 1$  at  $n = 461$ , first below at  $n = 463$  (0.935) — T110’s  $n \approx 170$  was a fit artefact, but the wall is real, and a depth-preserving resolution study makes  $n^* \approx 462$  an *upper* bound. The failure follows the prime branch (prime powers 512/529/625/729 survive longer). New and harder: the *ladder wall* — the nested step needs a lag gap  $> 2D$ , and the twin-prime pair  $521 \rightarrow 523$  (gap  $0.003831 < 2D = 0.005617$ ) kills the ladder arithmetically (arithmetic scan to 40 000: 4052/4284 pairs endangered; co-moving  $D \sim 1/n$  forces  $h \sim n \log n/4$ ). The no-reserve finding softens ( $f_{\text{crit}} \sim n^{-0.39}$ , the reserve opens with depth); the circle breaks first at  $k = 109$ ,  $n = 463$  — it tears at the ratio, not at a step: all 117 handovers certify at retention 1.000000 (largest single loss  $5.96 \times 10^{-8}$ ), atom entry free 117/117. Three separate walls: margin  $n \approx 462$  (measured), ladder  $521 \rightarrow 523$  (twin primes), requirement  $n = 727$  ( $\omega_{\text{cert}} \geq 1$ , need109 vacuous on 46 zones). Reinterpretation: the operating variable is the window *depth*, not  $n$  (cheek doubling moves the wall  $463 \rightarrow 47$ ); a proof must beat the exponent gap  $-0.974 = -0.681 - 0.293$  and couple  $\delta_0$  to the local prime gap. Follow-up probe T112 (`adaptive_scaling_probe.py`, contract `ADAPTIVE.SCALING`, the  $n$ -adaptively scaled frame) is running. Series counter: 111 probes, 2781/2781 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T111 (new subsection “The three walls (T111)”); the T110 three-gaps box and the  $n \approx 170$  mentions corrected to the measured  $n^* \approx 462$  (fit artefact noted); reduction-cascade caption, two-doors figure chain, three-way status split and asymptotic-mountain section updated (the three walls as the measured formulation; depth-not- $n$  reinterpretation; T112 running); abstract and probe-index appendix updated (T111 row; series counter 111/2781; T112 recorded as running and deliberately absent).
- **Website.** `/prime-front`: T111 feed entry (`lib/primeFront.ts`, new verdict enum `CROSSING-CONFIRMED`), status line in the two-doors convergence panel (T111 + T112 running), hero counter Teile 11–111 / 2781 sandbox checks, program-status callout sentence for T111.

**2026-07-27 (III) (prime front: T110 margin propagation closes the measured-zone circle — MARGIN-PROPAGATES; paper + diary + website sync; NO marker moves)**

- **Diary Teil 110 (sandbox).** `experiments/tfpt-discovery/margin_propagation_probe.py` (contract `MARGIN.PROPAGATION`, 28/28 checks, 7.6 s, verdict `MARGIN-PROPAGATES` with a precision statement): the induction circle closes end to end on the measured zones — the base case



$n = 2$  is certified by an explicit Cholesky at three resolutions ( $\lambda_{\min} \geq 6.93 \times 10^{-2}$ , factor 88–180 over need), the graded Loewner minorant holds all 15 handovers with retention 1.0000 (the strictly scalar variant closes 0/15), and the atom entry costs nothing — it *raises*  $\lambda_{\min}$  on 15/15 (first-order Kato term with helping sign; at  $n = 29$  the atom-free window is not even positive,  $-9.2 \times 10^{-3}$ ). Margin map:  $m_k \geq \text{need109}$  on 16/16 zones (ratio 2.09–179.6); the growth step splits exactly (embedding error  $\leq 2.6 \times 10^{-14}$ ;  $\max |\mu N| = 0.0$  on 15/15); scalar routes killed (bordered Weyl 0/15; Schur/Friedrichs cap vacuous 15/15). Three sharp gaps remain: no reserve ( $f_{\text{crit}} = 1.00$  at the first handover; retention  $1 - 2 \times 10^{-7}$ ), no scalar step law (structurally excluded by the boundary layer), no  $k$ -uniformity (trend  $m_k \sim n^{-1.93}$  against  $\text{need109} \sim n^{-0.98}$ , extrapolated crossing  $n \approx 170$ ). Follow-up probe T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, the deep ladder to  $n \sim 200$ ) is running. Series counter: 110 probes, 2758/2758 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.

- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T110 (new subsection “The margin propagates: the circle closes on the measured zones” with the three gaps as a key box); reduction-cascade caption and two-doors figure chain extended to T110; kill list extended by the T110 scalar-step-law kill; three-way status split and asymptotic-mountain section sharpened (the three gaps as the precise formulation; the  $n \approx 170$  warning; T111 running); abstract and probe-index appendix updated (T110 row; T111 recorded as running and deliberately absent); symbol-table row for  $m_k/\text{need109}$ .
- **Website.** `/prime-front`: T110 feed entry (`lib/primeFront.ts`, new verdict enum `MARGIN-PROPAGATES`), status line in the two-doors convergence panel (T110 + T111 running), hero counter Teile 11–110 / 2758 sandbox checks, program-status callout sentence for T110.

**2026-07-27 (II) (prime front: T109 boundary certificates close (R) conditionally — BOUNDARY-CERTIFIED; paper certification arc T102–T109 + diary + website sync; NO marker moves)**

- **Diary Teil 109 (sandbox).** `experiments/tfpt-discovery/boundary_decay_probe.py` (contract `BOUNDARY.DECAY`, 29/29 checks, 47.2 s, verdict `BOUNDARY-CERTIFIED`): both remaining scalars of (R) are certified —  $\omega$  *unconditionally* via a graded matrix cap in PSD order (one level per soft direction, Cholesky-verified,  $\omega_{\text{cert}} = 0.2561\text{--}0.7569$  against measured  $0.2366\text{--}0.7531$ , factor 1.004–1.124, no hypothesis input; the T108 compression Schur distance shrinks  $0.18\text{--}0.52 \rightarrow 0.0024\text{--}0.039$ ), the boundary value via a residual / goal-oriented certificate (Prager–Synge) that *carries* the cancellation (sharpness 1.0000–2.4937, 16/16 in budget). Mechanism finding: no decay — *cancellation* ( $44\text{--}4.6 \times 10^4$  on 15/16 zones); the Combes–Thomas route is refuted with the exact Green row (1/16). The full chain closes 16/16 (margin 1.3–10.4,  $M$ -stable to 3000) conditional on exactly *one* input: the strict positivity margin  $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}\text{--}7.5 \times 10^{-3}$ , i.e.  $10^2\text{--}10^6$  below  $\mu/2$ . Follow-up probe T110 (`margin_propagation_probe.py`, contract `MARGIN.PROPAGATION`) is running. Series counter: 109 probes, 2730/2730 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T109 (new subsection “Boundary certificates: cancellation, not decay”); the reduction cascade figure now ends at the one strict-positivity margin input; two-doors figure chain extended to T109; kill list K8 extended by the T109 Combes–Thomas kill; three-way status split (certified / measured / classical) updated; abstract and probe-index appendix updated (T109 row; T110 recorded as running and deliberately absent); symbol table rows for  $x/h/x_0$  and  $\omega/\omega_{\text{cert}}$ .
- **Website.** `/prime-front`: T109 feed entry (`lib/primeFront.ts`, new verdict enum `BOUNDARY-CERTIFIED`), status line in the two-doors convergence panel, hero counter Teile 11–109 / 2730 sandbox checks, program-status callout sentence for T109.

**2026-07-27 (I) (new companion `tfpt_prime_front` — the prime / zeta line documented end to end as RESEARCH DOCUMENTATION; NO new claims, NO marker moves)**

- **New companion paper `tfpt_prime_front` (P).** 22 pages, 9 sections plus two appendices, 8 TikZ figures. It documents the complete prime / zeta line of the project from the first  $\mu_4 \Rightarrow \chi_4$  observation to the current residual — explicitly a *documentation of a research route*, not a results paper. Registered in `tex-artefacts/tfpt_docset.tex` as the fifth companion.
- **Framework interpretation stated once, up front.** Section 3, “The big picture”, collects the reading that motivated the line and labels it as interpretation: the  $E_8$  shell count  $N(2n) = 240\sigma_3(n)$  is the Dirichlet series  $240\zeta(s)\zeta(s-3)$ ; the  $\mu_4$  glue carries  $\chi_{-4}$ , so the  $4k+1 / 4k+3$  split is a fibre property of the compiler and the signed channel is the only pole-free one; Hecke structure is emitted by lattice adjacency (`v535`) rather than put in;  $E_8$  is unimodular, so Poisson summation gives the theta sum an exact functional equation whose fixed axis is the critical line — the same reflection that reappears as  $J_1 \circ J_{1/2} = e^{\pm u}$  (the transport operator, T83) and as the parity split of the Weil pole (T106). Explicitly typed as a picture, not a reformulation, and not used as a premise anywhere.
- **Verified layer restated, not extended.** Sections 4.1–4.7 restate the seven load-bearing modules `v535–v541` with their ledger status markers quoted verbatim (Hecke from geometry, Eichler trace layer, half-integral bridge, relative-trace identity, Weil structure family, amplitude linear carrier, matching-lemma / transport ledger). The `\veri` citations in this paper are the only citations it makes; no ledger row is added, moved or upgraded.
- **Sandbox layer marked as sandbox throughout.** The localization program (I5, the Connes–Consani dictionary, the tightness map, the Sonin crossing, the blind prime demo), the handover induction and the certification arc are carried as clearly tagged *sandbox* content with a per-section provenance marker; every quantitative statement carries its probe label (Tn) and is reproduced verbatim from the research diary. All measured quantities are declared as measured, all window results as window results, all hypothesis inputs as hypothesis inputs.
- **Kill list as a first-class section.** Section 8 records the refuted readings and dead routes of the program (essential-singularity reading, the  $C/g$  proxy law, the naive margin route, the density chain, the Landau–Widom anchor, the accumulated invariant with self-propagation, the symbol route for the odd channel, three certificate candidates for the avoidance norm), each with the probe that killed it.
- **Explicit RH fence.** The paper states in the abstract, in the scope-and-status statement, in the reduction figure and in the status section that *no progress on the Riemann Hypothesis is claimed*; the  $I5 \Leftrightarrow$  Weil positivity  $\Leftrightarrow$  RH relation is carried, as in the ledger, as an equivalence *typing* only. ZETA.HP.CARRIER untouched.
- **Appendices.** Probe index T11–T108 (98 parts, 99 runs, 2509 green checks; the diary’s own series counter at T108 reads 108 probes / 2701 checks including the ten parts that predate the current diary file) with contract, verdict and check count per probe, plus a symbol table for the sandbox objects used in the figures. T109 (`boundary_decay_probe.py`) was running when the document was written and is deliberately absent.

**2026-07-25 (XIX) (`v541` — matching lemma and transport ledger package: two named limits as content; NO marker moves)**

- **New module `v541` (RTF.GNS.LEDGER.01).** Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 into one load-bearing module (33 checks,  $\sim 10$ s); every core check is *recomputed* from scratch, not cited. Companion to `v538/HECKE.GEOM.RTF.01`, `v539/RTF.GNS.WEIL.01` and `v540/RTF.GNS.AMP.01`. Classical Gronwall 1913 / Robin 1983

*unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet  $L(1, \chi)$  / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri / Alaoglu–Erdős named classical — NEW is the compiler-native consolidation and the machine-checked window proofs.

- **(A) Window proof [E].** The matching lemma is *proved exact-integer* on  $[4, 10^6]$ :  $7S(j) < 40A(j)$  at every atom (0 violations over 939 870 clash atoms; full enumeration, no-overflow proven, big-integer rechecks); exact margin  $X = 22242575693/270725400000 = 0.082159$  at  $j^* = 554400$ ;  $\rho_{\text{crit}} = 1.1440 > 21/20$ ; maximal-greedy identity  $\Rightarrow$  uniform in the target set and the greedy weights. Structure laws at 0 tolerance: 8-law ( $n \leq 250000$ ), seed-tower +  $\psi$   $w$ -table ( $n \leq 50000$ ), Cohen seeds  $\{-48, -80, -8, -8\}$  ( $s \leq 2000$ ), bracket  $\Theta \leq 2|\psi| \leq 3\Theta$  on  $10^6$ .
- **(B) Transport ledger [E].**  $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$  closes with  $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$  *proven* (sympy kernel collapse; quadrature  $2.8 \times 10^{-16}$ ; exact scale invariance); ledger residual  $4.4 \times 10^{-16}$  on 24 battery rows; the odd-prime side of the Weil functional equals the certified plus combination  $P_{\zeta}(g_-) + P_{\zeta}(g_+)$  exactly (rel  $3.9 \times 10^{-16}$ ).
- **(C) Character-exact signed envelope [E].**  $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$  coefficient-exact on all odd  $n \leq 50000$  (both builds); signed certificate 0 violations, margin factor  $8.86\times$ ; confinement *set equality* on  $10^6$ : zero-credit clash atoms =  $\chi_{-4}$ -coherent odd composites.
- **(D) Archimedean term internal [E].** Legendre duplication bridge  $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$  (sympy +  $1.8 \times 10^{-41}$ ); kernel identity  $k_{\zeta} = K_{\text{fam}} - K_{\text{shift}}$  pointwise  $7.9 \times 10^{-31}$  on 51 points;  $\Delta_{\text{arch}}(h) = A_{\text{fam}}(h) - A_{\text{shift}}(h)$  on a 10-row battery subset at rel  $9.9 \times 10^{-16}$ .
- **(E)  $\lambda$ -equivariant channel [E].** CM carrier  $g_{\lambda} = \sum_{\mathbf{a}} \lambda_1(\mathbf{a})q^{N_{\mathbf{a}}}$  exact in two independent routes ( $\mathbb{Z}[i]$  lattice vs Cornacchia reconstruction, 0 mismatches  $n \leq 10^4$ ); CM laws exact;  $\text{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$  as a set equality; canonical phase mean  $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$  exact;  $\lambda$ -window certificate 0.0653 vs 0.2365 unlifted ( $3.6\times$ ), exact Fraction rechecks.
- **(F) Frontier facts [C].** Unlifted coherent chain crossing  $k^* = 14$  (exact Fractions;  $\log_{10} N^* = 23.1$ ); absolute Robin tail factor 6.16 at  $J = 10^6$  and divergent — the constant route stays open (window constants with declared classical all- $n$  extension).
- **Named limits as load-bearing content.** (i) *One* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped  $\neq$  formal proof. (ii) I5 in *one-family* form ( $Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \geq 0$ ) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity  $\iff$  RH (an equivalence *typing*, no progress claim). NOT “almost RH”; ZETA.HP.CARRIER untouched; no marker moves.
- **Ledger/papers/website.** New ledger row RTF.GNS.LEDGER.01; \veri citations in tfpt\_1, tfpt\_3, tfpt\_research\_contracts; Wolfram extension mirrors the exact algebraic identities; website mirrors (/prime-front feed, verification DAG, papers data) updated in the same change.

## 2026-07-25 (XVIII) (v540 — amplitude route and positive linear carrier: open boundary $\lambda^*$ as content; NO marker moves)

- **New module v540 (RTF.GNS.AMP.01).** Promotion of the discovery amplitude chain T67/T68/T69/T70/T71/T72 into one load-bearing module (34 checks,  $\sim 3$  s). Companion to v538/HECKE.GEOM.RTF.01 and v539/RTF.GNS.WEIL.01: the machine-checked route out of the square plane behind the two v539 obstructions. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Möbius square sieve / Shimura / Shintani / Weil 1952 / Farkas named classical — NEW is the compiler-native consolidation.

- **(A) Amplitude Dirac [E].**  $D = \begin{pmatrix} 0 & V \\ V^\top & 0 \end{pmatrix}$  with  $V[d, m] = \sqrt{w_d} b(d) \chi_d(m)$ :  $D = D^\top$ ;  $D^2 = \text{diag}(VV^\top, K)$ ,  $V^\top V = K$  (rel  $2.7 \times 10^{-16}$ ); spectrum =  $\pm$  singular values; Hecke intertwining exact ( $p = 3, 5, 7$ ); Shimura  $b(dp^2) = b(d)(a_p - \chi_d(p)p)$  integer-exact on 644 pairs.
- **(B) Geometric polarisation + Cohen seed [E].**  $b = N_+ - N_-$ ,  $\Theta = N_+ + N_-$  lattice-exact;  $N_\pm \geq 0$  and  $b^2 = \Theta^2 - 4N_+N_-$  for all  $n \leq 50000$ ;  $\Theta$  is a *pure* Siegel–Weil Eisenstein  $\sigma_3$ -eigenform ( $p \leq 13$ , cusp component zero); Cohen seed  $\Theta(d) = -48 L(-1, \chi_d)$  exact-rational on 159 live  $d$ .
- **(C) Deletion universal = square-class double counting [E].** Cauchy–Littlewood window lemma (free Satake symbols,  $k \leq 6$ ): numerator  $1 - stuv X^2$ , hence  $1 - p^6 X^2$  pair-independent at  $st = uv = p^3$  (five channels); and fam  $[1, 0, 2, 0] + 2b[0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$  generating-function exact (Möbius square sieve exact  $n \leq 2000$ ).
- **(D) Positive linear carrier + plus balance [E].**  $\ell^2(d, \mu)$  with  $\mu = 48 |L(-1, \chi_d)| |d|^{-a}$ : full weights  $\lambda_k = 1 + p^{3k} - (\chi p)^k$  (layer  $[1, 1, 1, 1]$ );  $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$  (rel  $2 \times 10^{-15}$ ); the  $b$ /doubling kernel enters *with plus* (sign inverted vs the v539 square-plane minus).
- **(E) Functional equation exact [E].**  $\Theta(i/(8y)) = 8y^{5/2} \Theta^\dagger(iy)$  (rel  $6 \times 10^{-41}$ , Fricke closed);  $\Lambda_\Theta(s) = 8^{1-s} \Lambda_{\Theta^\dagger}(5/2 - s)$  in the strip (rel  $1.4 \times 10^{-40}$ , independent split points); poles  $5/2 / 0$  swapped; centre  $5/4$ , plus-locus offset exactly  $1/4$ .
- **(F) Cone facts + named open boundary [C].** Guaranteed cone and library hull FE-self-dual, gap functional  $\lambda^*$  FE-covariant (36/36 sampled); mirror sign law  $\text{sign } \psi(n) = (-1)^{\lfloor n/2 \rfloor + 1}$  exact ( $n \leq 50000$ ); constraint class  $C_{L3} = \{n \equiv 6 \pmod{8}\}$ ;  $\lambda^*$  exists with one exact Farkas witness ( $\lambda_{L3}^* = 0.0511$  at  $n^* = 6$ ); coverage saturates — the residual distance to the Weil cone *is*  $\lambda^*$  on  $n \equiv 6 \pmod{8}$ .
- **Fences.** Euler-region positivity only (edge  $L$ -values, not the central line); not “almost RH”;  $\lambda^*$  is the named open boundary *inside* the claim; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger RTF.GNS.AMP.01; `\veri in tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (613  $\rightarrow$  618); website `papers.ts` + VerificationDag E8 + `primeFront.ts` + ScriptIndex via `bash build.sh gen`.

## 2026-07-25 (XVII) (v539 — Weil structure of the compiler family: two isolated obstructions; NO marker moves)

- **New module v539 (RTF.GNS.WEIL.01).** Promotion of discovery probes T55/T61/T62/T63 (Route-B chain) into one load-bearing module (25 checks,  $\sim 6$  s). Companion to v538/HECKE.GEOM.RTF.01: the compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* (obstructions are prominent verified content, not a footnote). Classical Weil 1952 / Gelfand / direct integrals / SU(2)–Chebyshev / Sato–Tate /  $2^\omega$ -series / GNS named classical — NEW is the compiler-native identification and the machine-checked isolation of the two obstructions.
- **(A) GNS fibre structure [E].** Metric  $\ell^2(d, b^2/|d|)$  diagonal; fibres  $\sigma(d) = (\chi_d(3), \dots, \chi_d(13))$  partition the live family (1191 fund.  $d \leq 12000$ );  $\Lambda_{\text{fam}}$  exact constant on each fibre; twist-mix per fibre  $\leq 5.037 \times 10^{-16} < 10^{-12}$ .
- **(B) Isotype / GL(1) core [E].** Character expansions  $\Phi_1 = \chi_0 + \chi_2$ ,  $\Phi_2 = \chi_4 - \chi_2$ ,  $\Phi_3 = 2\chi_0 - \chi_4 + \chi_6$ ,  $\Phi_4 = \chi_8 - \chi_6$ ; core series  $c_{k,0} = [1, 0, 2, 0, 2, 0]$ ;  $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$ ; global  $\zeta(u)^2/\zeta(2u) = \sum 2^{\omega(n)} n^{-u}$  (Dirichlet exact through  $n \leq 2000$ ).
- **(C) Weil linear relation [E].** Prime<sub>F</sub> =  $2P_\zeta(g) - 2P_\zeta(g_b)$  (max rel  $9 \times 10^{-16}$ );  $Q_{\text{fam}} = Q_F + \text{Corr}$  (rel  $2 \times 10^{-16}$ ).
- **Obstructions as verified content [E].** (1) Doubling enters with a *minus* (family positivity does *not* imply  $Q_\zeta \geq 0$ ). (2) Plancherel Corr non-automorphic: polar/arch absorption FAIL; finite Euler factor FAIL;  $e^{-\sum p^{-u}}$  PASS.

- **Finite-class positivity** [C].  $Q_{\text{fam}} \in [4.369, 11.486]$  on the finite test class; dense-class positivity open / RH-adjacent — *not* claimed.
- **Fences.** Not “almost RH” — the claim structurally shows why family positivity does not imply RH; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger RTF.GNS.WEIL.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (608 → 613); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-25 (XVI) (v538 — compiler relative-trace identity: three projections of one finite RTF; NO marker moves)

- **New module v538 (HECKE.GEOM.RTF.01).** Promotion of discovery probe T53 (relative\_trace\_compiler\_probe.py) into one load-bearing module (18 checks,  $\sim 14$ s). Companion synthesis of v535/HECKE.GEOM.01, v536/HECKE.GEOM.EICHLER.01 and v537/HECKE.GEOM.HALFINT.01: these are three *projections* of one finite relative-trace identity  $[E_8/\text{glue orbital counting}] = [\text{Newform/Waldspurger spectral sum}]$ , not three separate results. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical — NEW is the compiler-native realisation and the machine-checked bilateral independence.
- **(R1) Trace identity, both sides independent** [E].  $\text{Tr}_V(\nu_p \circ \pi)$  geometric ( $a_p = -c(p)/8$  from the glue census, no modular input) equals spectral ( $a_p$  from  $\eta(2\tau)^4\eta(4\tau)^4$ , no geometry) for all  $(p, \pi) \in \{3, 5, 7\} \times \{\text{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ . Anchors (Id, Eis, cusp,  $E_4$ ,  $f_8$ ):  $p=3$ : (6304, 5600, 704, 1120, 352);  $p=5$ : (105848, 98280, 7568, 19656, 3784);  $p=7$ : (727680, 688000, 39680, 137600, 19840).
- **(R2) Orbit dictionary** [E]. The Eichler split  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is the RTF orbit decomposition (principal orbit  $a \cdot \text{Id} \rightarrow \lambda_{\text{Eis}} = L - \sigma_3^2$ ; elliptic  $b \cdot T_p \rightarrow a_p^2$ ; cross terms cancel). Anchors  $\lambda_{\text{Eis}} = 336/3780/19264$ ,  $a_p^2 = 16/4/576$ .
- **(R3) Period side is lattice counting** [C].  $g$  is the quaternary form  $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$  ( $x, y$  odd; sign  $(-1)^{|u|+|w|}$ );  $b_{\text{rep}} = b$  exactly through  $n \leq 200$ ; then  $[\text{signed lattice count}]^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$  with  $R = 23.1873585645$  (rel  $\sim 10^{-12}$ ). Verdict ONE-FORMULA.
- **Fences.** Identity is *finite* ( $p \in \{3, 5, 7\}$ , finite  $d$ -windows); the *infinite* RTF (sum over all  $p$  and  $d$  + archimedean term) is the named open step; no RH; not ZETA.HP.CARRIER-relevant; no marker moves.
- **Surfaces.** Ledger HECKE.GEOM.RTF.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (603 → 608); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-24 (XV) (v537 — half-integral bridge: unique Shimura preimage + $T(p^2)$ Hecke equivariance + Kohnen scope fence + Waldspurger/Baruch–Mao constancy; NO marker moves)

- **New module v537 (HECKE.GEOM.HALFINT.01).** Consolidated promotion of discovery probes T38/T41/T44/T45 into one load-bearing module (20 checks,  $\sim 90$ s). Companion to v535/HECKE.GEOM.01 and v536/HECKE.GEOM.EICHLER.01 without duplicating weight-4 Kneser/Eichler checks. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical — the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant  $R$ .
- **(A) Unique Shimura preimage at scale  $-8$**  [E]. In the 70-element compiler weight-5/2 theta-monoid exactly one object satisfies  $\text{Sh}_{t=2, \psi=1}(g) = -8 f_8$  (witness  $g = \vartheta_2(q^2)^2 \vartheta_3(q^2) \vartheta_4 \vartheta_4(q^2)$ , verified through  $n \leq 120$ ); three further monoid elements lift to scales  $+8, -16, +16$ .



- **(B) Hecke equivariance [E]**. Exact  $T(p^2)$ -eigenform, trivial nebentypus, with  $\lambda = a_p(f_8)$  for  $p = 3, 5, 7, 11, 13$   $(-4, -2, 24, -44, 22)$ ; trial FE  $N = 512$ ,  $\varepsilon = +1$  (centre  $5/4$ ).
- **(C) Kohnen scope fence [E]**.  $U_4(g) = 0$  (mass mod 4 on  $\{1, 2\}$ ); level  $32 = 4 \cdot 8$  outside Kohnen 1982 — plus-route structurally closed (negative result part of the claim).
- **(D) Waldspurger/Baruch–Mao constancy [C]**. AFE twist  $L$ -values ( $\varepsilon_d = \chi_d(8)$ );  $R(d) = 23.1873585645\dots$  constant on ten fundamental  $d \equiv 1 \pmod{8}$  (spread  $\leq 10^{-12}$ , including  $d = 89$ );  $d \equiv 5 \pmod{8}$  vanishes.
- **Fences**. GL(2) centre  $s = 2$ , not the  $\xi$ -line; no RH; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces**. Ledger HECKE.GEOM.HALFINT.01; `\veri` in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension  $+5$  ( $598 \rightarrow 603$ ); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

**2026-07-24 (XIV) (v536 — Eichler trace layer at the  $E_8$  lattice: Witt  $\lambda_{\text{Eis}}$  + two-sided Eichler identity + Type-A/B local densities + signed  $a_p$ ; NO marker moves)**

- **New module v536 (HECKE.GEOM.EICHLER.01)**. Consolidated promotion of discovery probes T33/T36/T37/T42 into one load-bearing module (23 checks,  $\sim 30$  s). Builds on v535/HECKE.GEOM.01 (Kneser counts,  $\nu_p$  law, oldforms) without duplicating those checks. The claim is *not* RH-related: the Eichler/Witt package of the  $\mu_4$ -glue census is lattice-native and two-sided in-suite.
- **(A) Witt layer [E]**.  $\lambda_{\text{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$  identically (sympy) and for all 24 odd primes  $p \leq 100$ ; anchors  $\lambda_{\text{Eis}}(3, 5, 7) = (336, 3780, 19264)$ .
- **(B) Eichler identity [E]/[C]**.  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ ; type-(i)  $N_{\perp} = (p^6 - 1)/(p - 1)$  live constant on all 240 roots at  $p = 3, 5$ ; type-(ii) live at  $p = 3$  ( $352 = 364 - 12$ ) with Kneser filter = identity;  $p = 5$  T36-validated freeze ( $3784 = 3906 - 122$ ,  $R = 4$ ; Shell(25) re-enumeration budget-gated); assembler  $R(p) = \sigma_3 - 1 - c(p^2)/8 = a_p^2$ ;  $R(7) = 576$  prognosis (Shell(49) out of scope).
- **(C) Local densities [E]**. Type-A = isotropic cosets representing the integer  $p$ ;  $N_A = \min(240(1 + p^3), p^7 + p^4 - p^3 - 1)$ ,  $N_B = (\#\text{iso} - 1) - N_A$ ; B empty iff  $p^4 < 241$  (only  $p = 3$ );  $p = 7$  live  $82560/743040$ ; injectivity at  $p = 11, 13$ ; fibre refinement scoped to  $\{5, 7, 11\}$ .
- **(D) Signed extraction [E]**.  $a_p = -c(p)/8$  from geometric Shell( $p$ ):  $(-4, -2, +24)$  at  $p = 3, 5, 7$ ;  $b = \sigma_3 + a_p \in \{24, 124, 368\}$  (hooks the  $\nu_p$  law of v535).
- **(E)  $O(1)$  assembler [E]**.  $\lambda_{\text{geom}}(p)$  and  $R(p) = a_p^2$  for all odd  $p \leq 31$  against the eta product  $f_8$  (Ramanujan bound included).
- **Fences**. Classical Eichler/Siegel/Witt named classical — the claim is in-suite two-sidedness (mod- $p$  geometry left, eta product right) and the closed compiler densities; no RH statement; Shell( $p^2$ ) for  $p \geq 7$  out of scope; weight-4  $\rightarrow \text{GL}(1)$  stays **[O]**; no marker moves of any pre-existing contract.
- **Surfaces**. Ledger HECKE.GEOM.EICHLER.01; `\veri` in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension mirrors (exact Witt/density/assembler/ $a_p$  identities); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

**2026-07-24 (XIII) (v535 — Hecke from geometry: Kneser correspondence + affine neighbour-sum + 2-adic oldform recovery on the  $E_8$  census channels; NO**

## marker moves)

- **New module v535 (HECKE.GEOM.01).** Consolidated promotion of discovery probes T27/T31/T32 into one load-bearing module (25 checks,  $\sim 11$  s). The claim is *not* RH-related: the Hecke structure of the  $\mu_4$ -glue census is lattice-native.
- **(A) Kneser carries Hecke [E].**  $\#iso\_pts = p^7 + p^4 - p^3$  and  $\#iso\_lines = \sigma_3(p) \# \mathbb{P}^3(\mathbb{F}_p)$  identically and by full enumeration at  $p = 2, 3, 5, 7$  (135/1120/19656/137600 lines) — the Eisenstein eigenvalue  $\sigma_3(p) = 1 + p^3$  is an exact factor of a lattice-native correspondence count.
- **(B) Neighbour-sum = affine Hecke element [E].**  $\nu_p = a \text{Id} + b T_p$  exactly (overdetermined, residual 0) with  $b = \sigma_3 + a_p$ ,  $a = \#lines - b\sigma_3$ ; cusp rule  $a_p = b - \sigma_3$  recovers  $a_3 = -4$ ,  $a_5 = -2$  ( $p = 3$  live orbit-reduced;  $p = 5$  T31-validated freeze of  $S[1]$  with  $\lambda_{\text{odd}} = 3784$ );  $p = 7$  consistency  $\lambda_{\text{odd}} = \#lines - \sigma_3^2 + a_p^2 = 19840$ ; placebo fails.
- **(C) Redundancy = 2-adic oldform structure [E].**  $\dim_{\mathbb{Q}} V = 7 = 5$  ( $E_4$ -copies, levels 1, 2, 4, 8, 16) + 2 ( $f_8$ -copies, levels 8, 16); every  $\Theta_j$  an explicit  $\mathbb{Q}$ -linear combination; recovery  $\pi_{\text{cusp}} = (28 - T_3)/32$ ,  $\pi_{\text{Eis}} = (T_3 + 4)/32$ , newform via  $(1 - V_2 U_2)$ ; new quotients each dimension 1; level census purely 2-adic.
- **Fences.** Classical theorems (Kneser, Hecke, Atkin–Lehner, multiplicity one) named classical — the TFPT content is the in-suite mechanics. No RH statement. Weight-4 restriction named:  $GL(1)/\text{weight-drop transport}$  stays **[O]**. No marker moves.
- **Surfaces.** Cited in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; ledger row HECKE.GEOM.01; Wolfram extension +5 exact mirrors ( $588 \rightarrow 593$ ); website sync.

**2026-07-24 (I) (v534 — the straddle cone executed: THE FIRST DYNAMICAL SELECTION OF THE ALIGNMENT BIT — reflection positivity keeps exactly ONE member of the interacting mark family alive,  $\delta = \pi/2$  with positive coupling; the literal leading-order cone formalization dies; NO marker moves)**

- **New module v534 (SEAM.STRADDLE.CONE.01).** The 2026-07-23 review-round contract (recorded “not promoted” in entry (X)) is *executed*, double-verdicted. **V2 = ERFOLG — the selection law (the central datum):** with independent couplings  $\lambda_q$  on the four mark quartets of  $M(\delta = m\pi/8)$  and the equivariant (uniform) direction, reflection positivity holds on *every* admissible cut over the whole grid  $g \in \{1/32..8\}$  for *exactly one* member  $\times$  sign:  $\delta = \pi/2$  ( $m = 4$ ) with *positive* coupling — all four Grams PD (37, 0, 0) *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ( $1.78 \cdot 10^{-6} \rightarrow 5.5 \cdot 10^{-5}$  near  $g = 1/4$ : the  $\pi/2$  interaction pushes *away* from the boundary), Gibbs  $\beta = 2$  identical [E]. The asymmetric members die:  $m = 2/6$  with  $s = +$  already at  $g = 1/32$ , with  $s = -$  from  $g = 1/8$ . The v529 straddle law *refines*: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it.
- **V1 = KILL (the literal cone formalization).** The leading-order soft-block cone  $K_{\text{RP}} = \text{Fix}(G_\delta) \cap \bigcap_c \{\lambda : G_c^{\text{lead}} \succeq 0\}$  is *empty* for every member and both signs (first-order PT via the exact reduced resolvent, Hermitian to  $10^{-15}$ , central-difference regression  $2 \cdot 10^{-8}$ ) — the linear pencil cannot carry the selection; the protection at  $\pi/2$  is *nonperturbative*. Recoverable leading-order content: the straddled-cut drift signs are exact ( $\pi/2$ :  $+1.4 \cdot 10^{-4}$  for  $s = +$ ,  $-3.7 \cdot 10^{-2}$  for  $s = -$ ; asymmetric members reversed) — necessary, not sufficient.
- **The exact geometry behind it [E].**  $\delta = \pi/2$  is the *unique symmetric-straddle member*: stabilizer = the full clock+mirror group (rotations  $\{0, 4, 8, 12\}$ , reflection axes  $\{3, 7, 11, 15\} =$  four admissible bond cuts, straddled  $\{7, 15\}$ ), the four quartets *partition* the 16-circle, and every straddling quartet is reflection-*closed* across its cut with exact sign +1;  $m = 2/6$  straddle asymmetrically; odd  $m$  carry no bond axis (the typed v521/v525 placement artifact).  $\text{Fix}(G_\delta)$  is 1-dimensional throughout: uniform on  $m = 2..6$ , NS-wrap-twisted  $(1, -1, 1, 1)/(-1, -1, -1, 1)$



on the tight members  $m = 1/7$ . Typed cross-link: the RP-selected member  $m = 4$  is exactly the carrier of the [v528](#) twist-class order parameter  $O = 1/2$  — the dynamical selector and the kinematic definition point at the *same* member.

- **Honest fences [C]**. One toy, one interaction class (FK quartets), flat-band parent,  $\deg \leq 2$  OS bases, finite coupling grid — toy-level evidence *for* the bit's dynamical origin, *not* a derivation; nothing about the continuum  $\mathcal{A}_{\text{hol}}$  is claimed; `WOIT.OS.TWISTOR.01` stays Open, NO marker moves. Transparency: the run-1 preregistered verdict logic conflated formalization and substance (UNENTSCHIEDEN, mismatch typed); run 2 split V1/V2 with no tolerance change and no number changed — note kept verbatim in the probe. Discovery provenance: `experiments/tfpt-discovery/seam_straddle_cone_probe.py` (15/15, 2026-07-24).
- **Wolfram mirror**. Four exact mirrors (stabilizer census,  $\pi/2$  partition + straddle incidence, reflection-closure signs, equivariant Fix rays incl. the wrap-twisted  $m = 1/7$  rays) in `wolfram/tfpt_readouts_extension.wl`, verified engine run 2026-07-24,  $584 \rightarrow 588$  (the 256-dim Fock spectra, PT corrections and finite- $g$  scans stay Python-only, flagged); suite  $527 \rightarrow 528$  modules.

**2026-07-23 (X) (v530–v533 — the external review round promoted: center quotient compiler, Coxeter Stage A tensor clock, dual-degree modular checksum 744, sheet-transfer disc  $-7$  norm line; every identity machine-confirmed exactly before promotion (probe 76/76); NO marker moves)**

- **New module v530 (DIAMOND.CENTER.QUOTIENT.01)**. The Sheet-Diamond center  $C = M(1, 0)$  carries the fixed primitive line  $Cv = v$  ( $v = (-1, 1, 0)$ ), and the induced operator on  $\mathbb{Z}^3/\mathbb{Z}v$  is *exactly* the atom matrix  $A = \begin{pmatrix} 8 & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} \text{rank } E_8 & |\mathbb{Z}_2| \\ g_{\text{car}} & N_{\text{fam}} \end{pmatrix}$  [E] with  $\text{tr } A = 11$ ,  $\det A = 14$ ,  $\text{disc} = 65 = 5 \cdot 13$ ,  $\text{perm } A = 34 = p_5(a)$ ,  $\prod A_{ij} = 240$ ,  $\det(A - I) = 4$ . The center *self-codes*: row sums  $C\mathbf{1} = (7, 11, 13) = (\det A/|\mathbb{Z}_2|, \text{tr } A, \text{disc}/g_{\text{car}})$ ; with the fixed eigenvalue 1 this is  $(1, 7, 11, 13)$  = the CP-halved  $E_8$  Coxeter exponents. The [v414](#) resolvent ladder factorises:  $\det(C + kI) = (k+1)\det(A + kI)$  identically, quotient ladder  $(14, 26, 40, 56)$ , differences  $(12, 14, 16)$ , endpoint  $224 = 248 - 24$ . *Honest correction of the review*: eigenvalue 1 is NOT unique to  $C$  ( $K$ ,  $Q$  share it) — uniqueness holds for the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays [O]. New paragraph in `tfpt_2` (Sheet Diamond section).
- **New module v531 (COX.STAGEA.TENSOR.01)**. The `tfpt_1` two-stage Coxeter audit (Stage A: a natural order-30 operator with  $\chi = \Phi_{30}$  on the even-flip atlas *or its 8-dim primitive-character space*) is *closed on the character-space alternative*:  $T_{30} = \text{Comp}(\Phi_5) \otimes \text{Comp}(\Phi_6)$  has exact order 30 and  $\chi_{T_{30}} = \Phi_{30}$  [E] (CRT,  $\gcd(5, 6) = 1$ : carrier clock  $\times$  family clock = Coxeter clock as *one integer operator*); Ramanujan trace tomography  $\text{tr}(T_{30}^k) = c_{30}(k) = c_5(k)c_6(k)$  for all  $k$ , signed table  $(-1, 1, 2, 4, -2, -4, -8, 8)$ , atom readout  $(2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \text{rank } E_8)$ ; honest kill scope ( $\Phi_{15}$  passes the unsigned triple — the *signed* table discriminates;  $\Phi_{16}/\Phi_{20}/\Phi_{24}$  die on the triple). The flip-atlas intertwiner (COX.FLIP.INTERWINER) and Stage B stay [O]; `tfpt_1` Stage A prose updated.
- **New module v532 (E8.DEGREE.MODULAR.01, audit-only)**. The dual products of the  $E_8$  invariant degrees,  $(60, 192, 240, 252)$ , satisfy simultaneously  $\sum = 744 = 3 \cdot 248$  (= the  $j$ -function constant, recomputed from the  $q$ -series  $E_4^3/\Delta$ ),  $\prod = |W(E_8)|$ ,  $\gcd = 12$ ,  $\frac{1}{12}(\cdot) = (g_{\text{car}}, \dim S^+, \det L, 7N_{\text{fam}})$  [E]. *Mandatory fence*:  $D_{16}$  passes the same checksum ( $1488 = 3 \cdot 496 = 2 \cdot 744$  — heterotic, NOT  $E_8$ -exclusive); six non-heterotic controls fail. *Prime-totative fingerprint (audit)*:  $\{2, 3, 5\} \cup (\text{Exp}(E_8) \setminus \{1\})$  = all primes below 30, and 30 is the *largest* integer whose totatives are all 1 or prime (scan to  $10^4$ ) —  $h_{E_8}$  as the maximal prime-totative Coxeter number; per the no-free-pattern rule nothing here is load-bearing. New keybox in `tfpt_3`, cross-note in the `tfpt_1` 496 box.

- **New module v533 (FTR.DISC7.NORM.01).** The branch-preserving transfer path  $J, K, C, F = M(1, -2..1)$  is a *norm line*:  $\det M(1, t) = 4t^2 + 14t + 14 = ((4t+7)^2 + 7)/4 = N(\alpha_t)$  with  $\alpha_t = (4t+7+\sqrt{-7})/2$  integral in  $\mathbb{Z}[(1+\sqrt{-7})/2]$ , the path one translation  $\alpha_{t+1} = \alpha_t + 2$ , norms  $(2, 4, 14, 32)$ , constant discriminant  $-7$ , vertex  $7/4 > 0$  [E]; 2 splits in  $\mathbb{Q}(\sqrt{-7})$  ( $-7 \equiv 1 \pmod{8}$ , class number 1); controls: winding axis linear, anchor block disc +105. [C]/[O] preregistered next test: do the four external transfer solvers share *one* fractional-linear action on  $\alpha_t$  (kill: four incompatible actions)? GATE.FTRANSFER unchanged. New paragraph in `tfpt_2` + frame note in `tfpt_research_contracts`.
- **Wolfram mirror.** All four modules mirrored exactly in `wolfram/tfpt_readouts_extension.wf` (20 new checks, verified engine run 2026-07-23, 564  $\rightarrow$  584); suite 523  $\rightarrow$  527 modules.
- **Not promoted (honest fences).** The review’s remaining proposals stay research contracts, not verifications: the straddle-cone selector on the interacting FK quartets (the highest-leverage attack on the alignment bit after v529) and the metrology-torsor formalisation of  $v_{\text{geo}}$  are recorded as next steps in `experiments/next.txt`; the primes-below-30 identity stays an audit fingerprint exactly as the no-free-pattern rule demands.

## 2026-07-23 (IX) (website entry redesign: narrative home, honesty band, proof strip, guided-tour route — editorial only, NO new modules, NO marker moves)

- **Website home.** Slim one-viewport hero (proof strip: 523 machine-checked modules · Wolfram mirror · Lean carrier · 0 external replications yet; copyable `python run_all.py`  $\rightarrow$  ALL CHECKS PASSED; CTA trio film / browser check / kill board). Narrative order: Hero  $\rightarrow$  IntroVideo (`#intro-video` kept)  $\rightarrow$  In one breath  $\rightarrow$  WhyThisMatters (plain-English column)  $\rightarrow$  Trust-Contract  $\rightarrow$  ClaimStack  $\rightarrow$  HonestyBand (honest negatives: RXTE/NICER nulls, 2DCS/v529, ten bit tests)  $\rightarrow$  Safeguards (four core stats)  $\rightarrow$  Overview (TheoryUnpacking/ThreeDecoderMap behind progressive disclosure)  $\rightarrow$  PathChooser  $\rightarrow$  Downloads teaser.
- **Archive routes.** Full PapersSection  $\rightarrow$  /papers; PredictionsSection  $\rightarrow$  /predictions; ReconstructionChain + HorizonStory + Gravity  $\rightarrow$  /architecture; RosettaLexicon  $\rightarrow$  /orientation; OpenGates  $\rightarrow$  /verification; EcosystemSection off home (footer adjacent link). Legacy hash redirects on home for old anchors.
- **Nav / design.** Primary nav: Start · Tour · Verification · Kill board · Replication · Papers · FAQ (More dropdown for secondary). SuiteBadge (static per release from `suite.ts/version.ts`). Design pass: drop purple-gradient pills/glow in hero and accent tokens; lab-notebook proof strip as above-the-fold visual.

**2026-07-23 (VIII) (prominence pass — the v519–v529 breakthroughs lifted into the FRAME texts of the papers and the website data layer; purely editorial, NO new modules, NO marker moves — (1) `tfpt_5_redteam` gains the seam round Target G: the ten-test side-blind scoreboard on the alignment bit (v512/v521/v525/v529) and the first *firing* kill test of the programme — the interacting straddle law (v529), documented as a named threat AND the first hard selection principle for  $\mathcal{A}_{\text{hol}}$ , fenced at toy level; (2) the bit-reduction statements in `tfpt_1`, `tfpt_2` and `origin_theory` extended from the v512 state to the current state (ten side-blind tests, the twist-class definition with gauge-robust order parameter, “in principle interferometrically readable”, v528 — the bit stays formal input); (3) the thermal seam lifted into the `origin_theory` horizon chapter (new keybox:  $T_{\text{seam}} = 4c_3$  measured, temperature the third leg of  $c_3$ , v526) and into the `tfpt_horizon_readouts` head keybox; (4) `tfpt_3` title/keybox now carry the thirty-step story and “the measure chain is derived” (v520/v523) prominently; (5) `tfpt_research_contracts` title/intro keybox now show the executed stages of**

**CELEST.SEAM.01** (M1–M3,  $\delta$ -chains decided,  $w_m$  derived) and **WOIT.OS.TWISTOR.01** ( $\alpha/\beta_1/\beta_2$  executed,  $\beta_3$  next under the straddle constraint); both contracts stay Open)

- **Papers.** `tfpt_5_redteam` (Target G + method/outcome/title honesty pass); `tfpt_1`, `tfpt_2` and `origin_theory` (bit-reduction statements  $\rightarrow$  ten tests + twist-class definition); `origin_theory` (thermal-seam keybox, horizon chapter); `tfpt_horizon_readouts` (head keybox third leg); `tfpt_3` (title + keybox: thirty-step story, measure chain derived); `tfpt_research_contracts` (title + intro keybox: executed contract stages).
- **Website.** `papers.ts` (redteam Target G, horizon third leg, audit/contracts abstracts, architecture reduction state); `orientation/StatusMatrix.tsx` (v512  $\rightarrow$  current state); `glossary.ts` (new terms: twist-class choice, thermal seam, straddle law, side-blind scoreboard); new visual components `ThermalSeamLegs` (three legs of  $c_3$ : geometry v58 + anomaly/modular v208 + temperature v526) and `WoitBridgeProgress` ( $\alpha/\beta_1/\beta_2$  done,  $\beta_3$  open, straddle warning) on the home and verification pages.
- **Counters.** Predictions counter made consistent with `predictions.ts` (27 falsifiable test surfaces): `README.md` badge/hero/prose and the intro-film description (23  $\rightarrow$  27; the film narration already says twenty-seven).
- **Honesty.** No marker moves anywhere; **SEAM.EQUIV.01** and **WOIT.OS.TWISTOR.01** stay Open; the bit stays input; all fences (“derived at probe level”, “at the free level”, “toy model”) preserved verbatim.

**2026-07-23 (VII)** (v529 — the interacting Fidkowski–Kitaev seam toy promoted as the ledger row **SEAM.INT.FKTOY.01**: the **FIRST** firing **Kill-Test-2** shadow — triple verdict **F1 = KILL / F2 = KILL / F3 = ERFOLG**; the celestial/twistor story in `tfpt_3` gains Step 30 and the header goes twenty-six  $\rightarrow$  thirty steps — (1) **F1 = KILL [E]**:  $\Theta$  exists exactly on the interacting model ( $U_r^2 = 2^8$ ,  $\Theta^2 = +1$ , clock-tower inversion,  $[\Theta, H_g] = 0$ ) so **Kill-Test 1** does *not* fire; but **Kill-Test 2** *fires at toy level*: RP breaks for every  $g > 0$  — inertia ladder  $(37, 0, 0) \rightarrow (33, 4, 0) \rightarrow (31, 6, 0) \rightarrow (29, 8, 0)$ ; Gibbs  $\beta = 2$  same ladder; mechanism = interference; **STRADDLE LAW** (24/24); **[C]** fence one toy / one interaction class / flat-band parent — narrows the Woit route, does *not* close it; **WOIT.OS.TWISTOR.01** stays Open with the toy shadow documented as a named **Kill-Test-2** threat; (2) **F2 = KILL [E]**: the **TENTH** side-blind test (9  $\rightarrow$  10) —  $\Theta_{15}$  maps  $H(m) \rightarrow H(8 - m)$  **EXACTLY** despite  $\delta$ -seeing spectra; (3) **F3 = ERFOLG [E]**: grading +  $\mathbb{Z}_8$  tower + **FK** anchor + constructive OS quotient at  $\pi/2$ ; Wolfram extension 552  $\rightarrow$  564 (verified engine run); suite 519  $\rightarrow$  523; **NO marker moves**)

- **Suite.** New module `v529_seam_interacting_toy_fk.py` (24 checks,  $\sim 97$  s); registered in `run_all.py` + `script_registry.csv`; suite 523 modules green.
- **Ledger.** New row **SEAM.INT.FKTOY.01**; **WOIT.OS.TWISTOR.01** text-precised (Kill-Test-2 toy shadow + straddle law); scoreboard rows text-precised (9  $\rightarrow$  10); no status moves.
- **Papers.** `tfpt_3` Step 30; `tfpt_research_contracts` (Kill-Test-2 toy threat + bit scoreboard 10); introduction (tenth side-blind test).
- **Wolfram.** Three exact mirrors (Z8 identities; straddle combinatorics; FK binomial ladder).
- **Website.** `papers.ts` (story 30 steps + scoreboard 10), generated indices via build; `README.md`, `zenodo_description.html` (523 modules, highest ID v529, Wolfram 564/564), `next.txt` notes.

**2026-07-23 (VI) (v528 — the alignment bit as an exact measurable TWIST-CLASS choice promoted as the ledger row SEAM.BIT.TWISTCLASS.01: the physical definition — verdict SUCCESS; the celestial/twistor story in tfpt\_3 gains Step 29 — (1) Facets #14 ( $\eta$ -flip  $\Leftrightarrow \delta = \pi/2$ ) and #15 (twist-frustration  $\Leftrightarrow \delta = \pi/2$ ) both directions exact [E](cardinality lemma  $|A_+| = |A_-| \Rightarrow m = 4$ ;  $g$ -class census 224 axes: flip class exactly at  $\{15, 31\}$ ); (2) order parameter  $O = 1/2$  at  $m = 4$  else 0 [E], gauge-robust; (3) unification #14=#15=flag-transitivity = ONE  $\mathbb{Z}_2$  invariant [E](v512 web 13  $\rightarrow$  15); Kramers = POSITION half (derived), twist class = MODULUS half (the input); (4) [C] measurement sketch (interferometric in principle); the bit remains formal input; *NO marker moves*)**

- **Suite.** New module v528\_seam\_bit\_twist\_class\_definition.py (18 checks,  $\sim 46$  s).
- **Ledger.** New row SEAM.BIT.TWISTCLASS.01; SEAM.TAU.FLAG.01 text-precised (web 13  $\rightarrow$  15); no status moves.
- **Papers.** tfpt\_3 Step 29; contracts + introduction (“symmetry-lift bit” now also “the twist-class choice, in principle interferometrically readable”).
- **Wolfram.** Three exact mirrors (cardinality lemma; 224-axis census;  $O$ -values).

**2026-07-23 (V) (v527 — the silver clock eigenvalue EXPLAINED and DEMYSTIFIED, promoted as the ledger row SEAM.CLOCK.SILVER.01: double verdict STRUCTURE = SUCCESS / CONNECTION = KILL (anti-numerology style); the celestial/twistor story in tfpt\_3 gains Step 28 — (1) STRUCTURE SUCCESS [E]:  $\{1, \sqrt{2} - 1\}$  closed via diagonal Hankel pencil at  $N = 8$ ,  $\tan(\pi/8) = C(3)/C(1)$ , doubling = exact second quantisation, carrier = silver-midpoint axes, gap =  $\operatorname{arcsinh}(1) = \operatorname{gd}^{-1}(\pi/4)$ ;  $N$ -scaling = pure  $N = 8/16$  lattice datum (not continuum); positive clock only for  $N \equiv 0 \pmod{8}$ ; (2) CONNECTION KILL [E]: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable; (3) cross-link v526:  $\sqrt{2} - 1 =$  thermal kernel ratio; *NO marker moves*)**

- **Suite.** New module v527\_seam\_clock\_silver\_spectrum.py (22 checks,  $\sim 10$  s).
- **Ledger.** New row SEAM.CLOCK.SILVER.01; no status moves.
- **Papers.** tfpt\_3 Step 28; contracts thermal/silver follow-on.
- **Wolfram.** Three exact mirrors ( $\tan(\pi/8)$  pencil;  $N$ -scaling witness; gap =  $\operatorname{arcsinh}(1)$ ).

**2026-07-23 (IV) (v526 — the thermal seam promoted as the ledger row SEAM.THERMAL.KMS.01: KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain — the THIRD LEG of  $c_3$ ; verdict SUCCESS; the celestial/twistor story in tfpt\_3 gains Step 27 — (1)  $\beta$  MEASURED =  $N$  at both levels [E]( $N = 8$ :  $\omega = \operatorname{arcsinh}(2^{-3/4})$ ,  $\kappa = q^{-8}$ ;  $N = 16$ : pairing exponent  $(N/2) + 1 = 9$ ; wrong  $\beta$  excluded); (2)  $\beta_{\text{angle}} = 2\pi$  EXACT =  $1/(4c_3)$ ,  $T_{\text{seam}} = 4c_3 =$  Bisognano–Wichmann/Hawking normalisation (v208) [E]; (3) modular Hamiltonian closed [E]( $\nu_1 = \cos(\pi/24)$ ,  $\nu_2 = \cos(7\pi/24)$ ; Tomita KMS- $\beta = 1$  on all 256 pairs); (4)  $c_3$  third-leg class-1 selection [E]/[C](Hawking/Nariai chain; v104 cross-link); (5) entropy  $S(8/16/32) = 0.52187/0.63510/0.75006$ ,  $\Delta S/\ln 2 \rightarrow 1/6$ , horizon fractions do NOT transfer [E]; (6) [C]-5 silver =  $C(3)/C(1)$  thermal kernel ratio; non-compact  $\Rightarrow T = 0$ ; *NO marker moves*)**

- **Suite.** New module v526\_seam\_thermal\_kms\_nariai\_bridge.py (22 checks,  $\sim 8.4$  s).



- **Ledger.** New row SEAM.THERMAL.KMS.01; no status moves.
- **Papers.** tfpt\_3 Step 27; tfpt\_horizon\_readouts (third leg of  $c_3$ ); tfpt\_research\_contracts (thermal follow-on); introduction.
- **Wolfram.** Three exact mirrors (Q-polynomial/ $\beta$  measurement;  $2\pi = 1/(4c_3)$  chain;  $\nu$  charpoly).

2026-07-23 (III) (v525 — the twist-state attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.TWISTBLIND.01: “mark-decorated states do not see  $\delta$ ” — an honest, decided NEGATIVE result (v508 style), the NINTH side-blind test on the **v512** scoreboard ( $8 \rightarrow 9$ ); the free-plus-twist class (everything Wick-computable) is EXHAUSTED; the celestial/twistor story in tfpt\_3 gains Step 26 and the header goes twenty-three  $\rightarrow$  twenty-six steps — (1) ALL LIFTINGS WELL-DEFINED AND BLIND [E]: (a) the  $\sigma$ -gauge arc decoration ( $\mathbb{Z}_2$ ) satisfies  $\sigma \circ r = \sigma$  on the swap axes  $\Rightarrow$  the twisted Gram is a diagonal unitary congruence  $DMD$ , spectra IDENTICAL to untwisted (dev 0.0); (b) the Kadanoff–Ceva 4-twist insertion  $\omega(D \cdot)/\omega(D)$  is well-defined on the  $N = 32$  ladder ( $\omega(D)$  real nonzero; the  $N = 16$  odd- $m$  degeneration is the  $C(\text{even}) = 0$  checkerboard artifact) and its RP SPECTRA are the FIRST free-class data of the programme that depend on  $\delta$  (pairwise up to 1.05) — yet the INERTIA is  $(16, 0, 0)$  PD with the same  $\eta = +i$  for EVERY  $m = 1..7$ : the decoration sees  $\delta$ , positivity does not; (c) the  $\mu_4$  twist at  $\beta = \pi/2$  is pure plane gauge; (c') the genuine Bogoliubov defect at  $\beta = \pi/4$ :  $\Theta$ -compatibility selects exactly 2 of 16 sign patterns (NS-wrap-forced), the winner genuinely non-monomial — still PD, spectra identical; STRUCTURE THEOREM (mechanism exact): the defect planes never straddle a cut bond, so every  $\Theta$ -compatible quasi-free mark decoration is RP-spectrally INVISIBLE; (2) SUB-RESULTS [E]: the defect energy  $E_{\text{def}} = 1.2752872$  for EVERY  $m$  (spread  $4 \times 10^{-40}$ ) —  $\delta$ -blind; the 4-twist Casimir  $\ln |\omega(D)|$  measures  $\delta$  ( $-0.890.. - 1.348$ ) but exactly mirror-symmetric under  $m \leftrightarrow 8-m$  — a family gauge, never a side selector; Kramers  $\langle \Gamma \rangle_{\text{tw}} = +1$  for every  $m$ ; (3) the HARVEST [E]: two bit-presupposing  $\pi/2$  structures (the  $\eta$  flip  $+i \rightarrow -i$  on the mark-fixing axes via  $\sigma \circ r = -\sigma$ ; the twist frustration of the mark-fixing mirror,  $(4, 4, 0)$  indefinite for both  $\eta$  — the lattice shadow of the Ising twist field’s mirror-oddness) — both formulable only on the fixing axes existing iff  $\delta = \pi/2$  (the **v512** facet class); (4) NEGATIVE CONTROLS [E]: the twist-free limit = v519/v521 exactly; the site cut keeps failing; the **v512** counterwitness is arc-equivalent at  $N = 16$  to the blind member  $\pi/4$  — no state selector can exclude it; a single twist has  $\omega(D) = 0$  identically; [O]: the door narrows to a genuinely INTERACTING  $\mathcal{A}_{\text{hol}}$  whose OS data are NOT Pfaffian-reducible to the chiral vacuum; Wolfram extension (with v523/v524)  $540 \rightarrow 552$  (verified engine run 2026-07-23,  $552/552 + \text{base } 116/116$ ); suite 516  $\rightarrow$  519 modules; *NO marker moves — the alignment bit remains genuine discrete input*)

- **Suite.** New module v525\_seam\_bit\_twist\_blind.py (20 checks,  $\sim 80$  s; Pfaffian-exact insertion machinery cross-validated three ways, 40-digit certificates); registered in run\_all.py + script\_registry.csv; suite 519 modules green.
- **Ledger.** New row SEAM.BIT.TWISTBLIND.01 (KILL, honest negative, preregistered); SEAM.BIT.RPBLIND.01 and SEAM.TAU.FLAG.01 text-precised (scoreboard  $8 \rightarrow 9$ , “the free-plus-twist class is exhausted”); no status moves.
- **Papers.** tfpt\_3 Step 26 + the story header (twenty-six steps) + the Step-22 gap sentence; tfpt\_research\_contracts (the QGEO.REALIZE.01 reduction paragraph  $8 \rightarrow 9$  and the WP2

relocation note extended by **v525**); **introduction** (the ninth side-blind test added to the reduction-state sentence).

- **Wolfram.** Three exact mirrors (the  $N = 32$  integer frame + the  $\sigma \circ r = \pm\sigma$  congruence mechanism; the  $\pi/2$  existence solveset + the  $(3 + 4i)/5$  root-of-unity census; the  $\delta \leftrightarrow \pi - \delta$  mirror equivariance); the Pfaffian spectra, the 2-of-16 pattern census and  $E_{\text{def}}$  stay Python-only.
- **Website.** `papers.ts` (tfpt\_3 story mirror 26 steps + scoreboard 9), generated indices (`ScriptIndex.tsx`, `changelog.ts`, `maps`) via `build`; `README.md`, `zenodo_description.html` (519 modules, highest ID **v525**, Wolfram 552/552), `next.txt` notes.

**2026-07-23 (II) (v524 — the WOIT- $\beta_2$  milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.BETA2.OS.01: “the OS quotient of the free system made explicit” — verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the celestial/twistor story in tfpt\_3 gains Step 25 — (1)  $\mathcal{H}_{\text{phys}}$  EXPLICIT [E]:  $N = 16$  ( $\deg \leq 2$ ): the  $37 \times 37$  bond-cut OS Gram exactly Hermitian, parity-block-diagonal, inertia  $(37, 0, 0)$  PD (min eigenvalue  $1.7801 \times 10^{-6}$  at 40 digits), null space  $\{0\}$ ,  $\dim 37 = 29 \oplus 8$ ;  $N = 8$  (complete half algebra):  $(16, 0, 0)$  PD (min  $3.3471 \times 10^{-3}$ ),  $\dim 16 = 8 \oplus 8 = 4^2$  — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate  $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$ ; (2) the TRANSFER SEMI-GROUP (Klein–Landau) [E]:  $\tau_k$  exactly Hermitian on all domains ( $k = 1..7$ , dims  $29, 22, 16, 11, 7, 4, 2$ ), chain identities exact, vacuum fixed; the positivity pattern = the site/bond dichotomy: even  $k$  PSD via the exact square identity  $T(2j) = A_j^* A_j$  ( $((22, 0, 0)/(11, 0, 0)/(4, 0, 0))$ , odd  $k$  indefinite ( $((10, 12, 7)/(5, 6, 5)/(2, 2, 3))$ ) — the one-step transfer is NOT positive (the chirality datum, kill-3 shadow sharpened), its even powers are; (3) THE CLOCK =  $T^{N/4}$  [E]:  $N = 16$ :  $\tau_4$  PD  $(11, 0, 0)$  (min  $6.6821 \times 10^{-5}$ ), compressed spectrum  $[0.0194517, 1.64144]$ , vacuum eigenvalue exactly 1, spectral projections  $\sim 10^{-40}$ ;  $N = 8$ : spectrum EXACTLY  $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_{\text{Silber}}\}$  in both parity sectors — the silver axes of the **v519** torsor return as the clock eigenvalue; the reconstructed rotation group  $U(s) = e^{isH}$  unitary with the group law; (4) the **v522** RESOLUTION [E]: the census  $(745, 96, 0)$  reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus; pre-registered KMS deviation: no contraction on the compact circle ( $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$  exactly,  $\det(G - \tau_4) < 0$ ); (5) GSO/ $\Theta$  [E]: the grading survives; the perpendicular torsor mirror descends anti-unitarily ( $\eta_{\perp} \in \{\pm i\}$ ),  $\Theta_{\text{phys}}^2 = +1$  on every sector (Kramers-free),  $\theta_{\text{cut}} \circ \theta_{\perp} = \alpha_{N/2}$  exactly; (6) CONTROLS [E]: site cut  $(3, 3, 1)/(2, 2, 4)$  — the kill branch fires there; family A  $(4, 4, 0)$ ; anti-chiral  $(8, 8, 0)$  no quotient; wrong direction non-Hermitian (all mismatches wrap-overlap); KILL TESTS: (1)/(2) strengthened, (3) shadow sharpened, none fires, all seven stay live on  $\mathcal{A}_{\text{hol}}$ ; TRANSPARENCY: run 1 = 18/20 (a sympy simplify failure on a TRUE  $\pi/16$  product formula, fixed by a Laurent-polynomial certificate; no criterion changed); *NO marker moves — WOIT.OS.TWISTOR.01 stays Open,  $\beta_3$  is next*)**

- **Suite.** New module `v524_woit_beta2_os_quotient.py` (20 checks,  $\sim 5$  s; exact sympy + Pfaffian-Wick + 40-digit inertia certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row WOIT.BETA2.OS.01 (SUCCESS, [C]-typed per precision (iii)); WOIT.OS.TWISTOR.01 claim/canonical status extended ( $\beta_2$  executed,  $\beta_3$  next); no status moves.
- **Papers.** `tfpt_3`: Step 25 plus the Woit-bridge paragraph with the  $\beta_2$ -stage status.

**tfpt\_research\_contracts**: a full “ $\beta_2$  stage — executed” block with `\veri`; the  $\beta/\gamma$  roadmap entry marked executed; the scope-fence status sentence extended.

- **Wolfram**. Four exact mirrors (the  $\alpha_1^N = (-1)^F$  wrap arithmetic + the certificate identity; the Klein–Landau one-particle transfer forms with odd- $k$  zero diagonal and the even- $k$  square identity; the exact  $N = 8$  clock spectrum  $\{1, \sqrt{2} - 1\}$  + the transparency  $\pi/16$  product formula certified symbolically; the KMS silver witnesses); the  $\mathcal{H}_{\text{phys}}$  Grams, inertia spectra and the rotation group stay Python-only.
- **Website**. `papers.ts` (`tfpt_3` + `contracts mirrors`), generated indices via `build`.

**2026-07-23 (I) (v523 — the constructive  $w_m$  derivation executed and promoted as the ledger row CELEST.WP5E.WM.01: “ $1/\det_j$  is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources” — verdict ERFOLG per the frozen preregistration; the residual [O] of v516/v520 is closed at the fibre-zero-mode level and narrows to the GLOBAL BCOV integral on  $\mathbb{P}T/\mathbb{Z}_4$  beyond the fibre zero-mode factor; the celestial/twistor story in `tfpt_3` gains Step 24 — (1) ATIYAH–BOTT (route i) [E]: the equivariant mode ledger of the fibre  $\mathbb{C}^2$  equals the closed form  $1/((1 - q i^j)(1 - q i^{-j}))$  exactly in  $\mathbb{Q}(i)$  for all  $n \leq 120$ ; REGULAR at  $q = 1$  with Abel value  $(1/2, 1/4, 1/2) = 1/\det_j$ ,  $\det_j = \det(1 - g^j) = (2, 4, 2)$ ;  $(C, 2)$  Cesàro convergence in exact arithmetic; fixed-point splitting at EVERY truncation level  $d \in \{7, 12, 25\}$  for the phased (skeleton) AND the v520-forced unphased (completion) trace — the Abel-limit  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  is the v516 contact vector COMPUTED, not declared; (2) ZETA/QUILLEN (routes ii/iii) [E]:  $\det_\zeta$  of the  $g^j$ -twisted circle Laplacian  $= 4 \sin^2(\pi j/4) = \det_j$  EXACTLY (Lerch + reflection formula symbolically, Hurwitz certificates  $\sim 10^{-41}$ ); Quillen split  $\det \Delta = \det_j^2$ ,  $|\det \bar{\partial}| = \det_j$ , the real positive holomorphic section UNIQUE via the  $SU(2)$  conjugation pairing; the  $\delta_{1f} f_1 f_3$  block at the  $(0, b)$  nodes carries the exact constant term  $1/\det_b$  (deviations  $\leq 5.2 \times 10^{-28}$ ); (3) the CONSISTENCY CHAIN [E]:  $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$ ;  $T_5 = 0$  at every scale;  $h_2 : h_1 = 4 : 3$ ; the  $T_3$  budget  $32 - 8c$  forces  $c = 4 = |\mu_4|$  uniquely (leftovers  $24/16/8/0/-8$ ); per-sector Okubo squares  $(18, 9, 18)\langle x, x \rangle^2$ ; total  $45\langle x, x \rangle^2$ ;  $\Phi_{T_3} : 32 \rightarrow 0$ ,  $\Phi_P : 72 \rightarrow 0$ ;  $\psi(A_{\text{fix}}) = +64$ ,  $\psi(\text{contact}) = -64$ ; (4) NEGATIVE CONTROLS [E]:  $1/\det^2$  gives  $w' = (0, 5/8, 1, 5/8)$ , leftovers  $(2, 12)$ , Quillen contradiction;  $1/|1 - i^j|$  gives  $w''_1 = 1 + \sqrt{2} \notin \mathbb{Q}$ ; the  $\mathbb{Z}_2/\text{Eguchi–Hanson}$  anchor PASSES with the same derivation ( $\det = 4$ ,  $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ,  $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $c = 2 = |\mathbb{Z}_2|$ ); the  $SO(16)/D_8$  KILL fires;  $\text{diag}(i, i)$  breaks the zeta = AB identity itself;  $k = 3, 5$ :  $w_m = m(k - m)/2 = k h_m$  — the weights wander correctly, only  $k = 4$  carries the  $E_8$  chain; TYPED PREMISES [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation = the definition of the equivariant character), TP-Q (Quillen structure of BCOV  $F_1 = \text{v520 TP-4}$ ), TP-NUM (unphased numerator = v520 ERFOLG-A), TP-CH (insertion channel  $(0, j)$ ); [O]: the global BCOV integral beyond the fibre zero-mode factor, the resolved-ALE route not built, SEAM.EQUIV.TWISTOR.01 stays Open; *NO marker moves*)**

- **Suite**. New module `v523_celestial_wp5e_wm_derived.py` (26 checks,  $\sim 1$  s; exact  $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{2})/\text{cyclotomic}$  arithmetic + 40-digit Hurwitz/block certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger**. New row CELEST.WP5E.WM.01 (ERFOLG); the v516 row CELEST.WP5E.M2.01 and the v520 row CELEST.WP5E.MEASURE.01 text-precised (“the  $w_m$  normalisation is now DERIVED at probe level; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor”); no status moves.



- **Papers.** `tfpt_3` Step 24 + the “What remains, honestly” narrowing; `tfpt_research_contracts` (the WP5e measure-question block extended by the executed derivation with `\veri`, the M2 fence, the celestial summary blocks and the section status list); `SEAM.EQUIV.TWISTOR.01` stays Open.
- **Wolfram.** Five exact mirrors (the mode ledger = closed form with Abel value; the zeta/Quillen route with the symbolic reflection formula and the  $\text{diag}(i, i)$  control; the derived-weight contact vectors + success table +  $\mathbb{Z}_2/\text{EH}$  anchor on the 240 glue roots; the consistency chain + wrong-weight controls; the  $k = 3, 5$  cyclotomic wandering controls); the Cesàro/Hurwitz/stripped-block numerics stay Python-only.
- **Website.** `papers.ts` (`tfpt_3` + contracts mirrors), generated indices via build.

**2026-07-22 (XIII) (HFQPO scorecard extension — the NICER/MAXI J1820+070 third-tooth null and the AstroSat/LAXPC block integrated as a new evidence-scorecard row (search target, never a claim):** the preregistered single-QPO extension of the H3 ladder discriminator (frozen `hypotheses/hfqpo_ext_v1.yaml` BEFORE any photon download) ran on 6 NICER ObsIDs (13.22 GB,  $2013 \times 16 \text{ s} = 32.2 \text{ ks}$  stack at 19.05 kc/s) — sanity gate PASS (the published 55.12 Hz anchor reproduced at 55.03 Hz, 0.94% rms,  $3.8\sigma$ ), injection calibration PASS ( $\geq 90\%$  recovery at 0.93% rms), the geometric tooth at 82.68 Hz NOT detected ( $0.69\sigma$  single-trial;  $3\sigma$  upper limit 0.75% rms BELOW the detected anchor strength  $\Rightarrow$  `null_with_sensitivity`), the integer line at  $\sim 110.6 \text{ Hz}$  at  $3.82\sigma$  after trials — BELOW the preregistered  $4\sigma$  threshold, documented as a sub-threshold excess consistent with the weak second harmonic of ATel #11951, NOT a hit; AstroSat/LAXPC (GRS 1915+105, scientifically the strongest channel) `infrastructure_blocked` (ISSDC login-gated); the RXTE + NICER nulls now cover five sources on two instruments — the tooth channel is recommended DORMANT; anti-double-counting: the parent RXTE row and the new extension row share `independence_group hfqpo_ladder_tooth` (one hypothesis, two instruments, never two independent nulls); scorecard regenerated (121 rows), `predictions.txt` HFQPO row extended (“Kanal ruhend”); experiment-only integration — no verification claim, no marker moves)

- **Scorecard.** New row “BH HFQPO third-tooth extension on non-RXTE archives” (NICER/MAXI J1820+070 plus AstroSat/LAXPC; `hfqpo-ladder` extension) in the generator `experiments/build_evidence_scorecard.py` (status `null`, stage `search_target`, signature `S1e`); OVERRIDES give both HFQPO rows the shared `independence_group hfqpo_ladder_tooth`; generator re-run: `evidence_scorecard.json` 121 rows, README stats block refreshed.
- **Experiment surfaces.** `experiments/hfqpo-ladder/extension-nicer-laxpc/` (preregistration, frozen manifest, download log, per-ObsID caches, stack/sanity/injection/scan reports, `results_ext.json`, README) — executed 2026-07-22; `predictions.txt` HFQPO row updated with the NICER finding and the dormant-channel recommendation; `next.txt` round notes.

**2026-07-22 (XII) (v522 — the  $\text{WOIT-}\beta_1$  milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.BETA1.GSO.01`: “the clock is time-like, GSO is the gauge datum” — a TYPED result (verdict UNDECIDED per the frozen preregistration, the mechanism identified exactly, gauge-fixed RP established under the corrected typing; the celestial/twistor story in `tfpt_3` gains Step 23) — (1) the HERMITICITY WITNESS**

[E]: the  $\mu_4$  clock average violates Hermiticity EXACTLY (the one-step insertion  $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$  has entry  $-i/(8 \sin(5\pi/16))$  where the conjugate transpose demands  $+i/(8 \sin(5\pi/16))$ ); entrywise census 745 Hermitian-matching, 96 ANTI-matching, 0 violations — the pairing is complex-SYMMETRIC, strictly weaker than Hermitian: “positivity after gauge fixing” is not well-posed for the clock reading); (2) TIME-LIKE TYPING [E]: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ( $RSR^{-1} = S^{-1}$ ), NONE commutes — the  $\mu_4$  clock IS Woit’s euclidean rotation; the gaugeable ( $\theta$ -compatible) part of the  $\mathbb{Z}_8$  Fock tower is EXACTLY its 2-torsion  $\{1, (-1)^F\}$  = the GSO/fermion-parity  $\mathbb{Z}_2$  (Hermiticity census over all powers;  $\alpha^4$  = the full wrap = the  $2\pi$  rotation of the seam = fermion parity); (3) the POSITIVE PART [E]: GSO-fixed RP holds EXACTLY —  $N = 16$  even  $\deg \leq 2$  parity-projected bond-cut Gram  $(29, 0, 0)$  PD (min eigenvalue  $1.78 \times 10^{-6}$  at 40 digits),  $N = 8$  complete half algebra: all odd rows/columns killed exactly, even block  $(8, 0, 0)$  PD ( $3.35 \times 10^{-3}$ ), no degree truncation; the site-cut defect SURVIVES gauge fixing  $((7, 9, 6)$  — the bond placement is gauge-invariant information); family A keeps failing  $((17, 12, 0)$  indefinite vs family D’s  $(29, 0, 0)$ ); Kramers:  $\Theta_t^2 = \text{id}$  on  $\mathcal{A}^G$ ,  $\Theta_{\text{Fock}}$  restricts with  $\Theta^2 = +1$ ; Ramond control: the wrap +1 “true  $\mathbb{Z}_4$ ” lift fails state invariance — the NS/ $\mathbb{Z}_8$  tower is forced; (4) KILL-TEST BOOKKEEPING: kill test (2)’s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction), kill test (1) stays discharged also gauge-invariantly; both stay formally live on  $\mathcal{A}_{\text{hol}}$ ; (5) TRANSPARENCY: the first frozen run scored 6/13 (the clock-reading checks failed exactly as the mechanism predicts) and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census  $28 + 4 = 32$ ) — documented in full, the verdict by the frozen [C-3] logic (Hermiticity failure  $\Rightarrow$  UNDECIDED); CONTRACT PRECISION (iii) (prose, no marker): “gauge-invariant subalgebra” at the free level means the GSO/fermion-parity  $\mathbb{Z}_2$  — the only  $\theta$ -compatible part of the  $\mu_4$  tower; the clock is the euclidean rotation itself (timelike), its equivariant statement moves to  $\beta_2$  (OS quotient first, then the clock as reconstructed transfer operator); Wolfram extension (with v520/v521)  $522 \rightarrow 540$  (verified engine run); *NO marker moves* — *WOIT.OS.TWISTOR.01 stays Open*,  $\beta_1$  *proper re-routed via  $\beta_2$* )

- **v522\_woit\_beta1\_gso\_gauge.py** (WOIT.BETA1.GSO.01, 16 checks,  $\sim 5$  s; exact sympy Pfaffian–Wick + exact Cl(16) Fractions; definiteness via 40-digit spectra, tolerance  $10^{-25}$ ; deterministic). G1 the gauge datum (Z8/NS tower, Ramond control, invariant dimensions  $120 > 64 > 32$ ); G2  $\Theta$  normalises the tower and restricts as an involution; G3 the  $(-1)^F$  class does not penetrate  $\mathcal{A}^G$ ; R1 the tower average is OS-symmetric, the odd sector dies; M1 the time-like census; M2 the exact Hermiticity witness; M3 the GSO 2-torsion census; R2 GSO-fixed RP ( $N = 16/N = 8$ ); C1–C3 site-cut/wrong-projection/family-A controls; V1 the frozen verdict. Registered in `run_all.py` (suite 516 modules, green), `script_registry.csv` (cluster *frontier*), ledger row WOIT.BETA1.GSO.01 (new).
- **Contract updated (markers unchanged).** WOIT.OS.TWISTOR.01: the  $\beta_1$  stage moved to *done* (typed UNDECIDED) with its own block, precision (iii) added, the  $\beta_1$  roadmap bullet annotated *executed*, the closing status line carries the  $\beta_1$  state; ledger WOIT.OS.TWISTOR.01 text extended (beta-1 executed via v522).
- **Paper integration.** `tfpt_3` (new Step 23 “the clock is time-like, GSO is the gauge datum”; the Woit-bridge paragraph extended by the  $\beta_1$  status); `veri` citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the Z8/NS tower  $S^4 = -1 +$  the Ramond witness  $1/(4 \sin(5\pi/16))$ ); the invariant-dimension chain by exact ranks; the time-like reflection census; the exact Hermitic-

ity witness with the full 745/96/0 census via exact Pfaffian–Wick; the GSO 2-torsion census; the family-A tower-normalisation failure) added to `tfpt_readouts_extension.wl`; the GSO-fixed RP inertia certificates stay Python-only, flagged in the `.wl` comment.

- **Provenance.** Discovery probe `woit_os_beta1_gauge_invariant_theta_probe.py` (16/16,  $\sim 3$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

**2026-07-22 (XI) (v521 — the RP/ $\Theta$  attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.RPBLIND.01: “free OS positivity does NOT see  $\delta$ ” — an honest, decided NEGATIVE result in the v508 style (the celestial/twistor story in `tfpt_3` gains Step 22):** does free reflection positivity (v519) or anti-linear  $\Theta$  existence break for  $\delta \neq \pi/2$  on the deformed mark family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$  — which would DERIVE the v512 alignment bit from OS positivity? NO — (1) the CENSUS [E] (exact solvesets, symbolic  $\delta$ ): for EVERY  $\delta$  exactly the two mark-SWAPPING reflections  $\Theta_{\pm e^{i\delta}}$  exist (axes  $\delta/2, \delta/2 + \pi/2$ ; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at  $\delta = \pi/2$  (solveset  $\cos \delta = 0$ , lost immediately at  $\pi/2 - \varepsilon$ ); the v510/v512 counterwitness  $\{\pm 1, \pm(3+4i)/5\}$  carries the swap pair exactly; (2) RP IS  $\delta$ -BLIND [E]: bond-cut Gram inertias  $(8, 0, 0)$  PD for all tested  $\delta$  with the full sorted spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO  $\delta$  information; the  $N = 16$  odd- $m$  failure (site axes,  $(3, 3, 1)$ ,  $\det = 0$ ) is a lattice-PLACEMENT artifact: at  $N = 32$  ALL  $m = 1..7$  have mark-compatible BOND axes with PD inertia  $(16, 0, 0)$ ; continuum: the OS kernel in axis-adapted coordinates is EXACTLY  $1/\sin((s+t)/2)$  — the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram  $(6, 0, 0)$ ); (3)  $\Theta$  EXISTENCE IS  $\delta$ -BLIND [E]:  $\Theta^2 = +1$ , deck normalisation and Fock implementability ( $U^2 = +2^7/2^8 > 0$ , exact  $\text{Cl}(16)$ ) for every  $\delta$ ; the ONLY  $\delta$ -sensitive v519 clause ( $\Theta_\rho \Theta = \rho^{-1}$ ) is for  $\delta \neq \pi/2$  not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff  $\delta = \pi/2$ ) — it PRESUPPOSES the bit; same for the clock-tower clause; (4) the battery HAS TEETH at every  $\delta$  [E]: family A/antipode fails RP structurally (zero diagonal,  $(4, 4, 0)$ ), the chirality- $\eta$  pinning persists ( $(0, 8, 0)$  ND) — RP separates the two real-structure FAMILIES, never the SIDES; (5) precondition controls: hexagonal (marks not concyclic — no seam circle), silver (deck not free), lifting audit (B = index relabelling,  $\delta$ -blind by construction; B' fails the  $\pi/2$  negative control — discarded); [C] FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is  $V_4$  for every  $\delta < \pi/2$  and closes to  $D_4$  exactly at  $\delta = \pi/2$  ( $\Theta_i \circ \Theta_1 =$  the clock) —  $12 \rightarrow 13$  faces, a typed REFORMULATION of the bit, not a derivation; [O] the honest gap: marks decorating the STATE (defect/twist insertions, interacting  $\mathcal{A}_{\text{hol}}$ ) = exactly v519's [O] scope; VERDICT: KILL exactly as preregistered — the free RP/ $\Theta$  battery is the EIGHTH side-blind test on the v512 scoreboard ( $7 \rightarrow 8$ ); the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; *NO marker moves — the bit stays the one [C] carrier input*)

- `v521_seam_bit_rp_blind.py` (SEAM.BIT.RPBLIND.01, 25 checks,  $\sim 25$  s; exact sympy solvesets + exact Gram entries + exact  $\text{Cl}(16)$  Fractions; inertia via 40-digit spectra, tolerance  $10^{-25}$ ; deterministic). S the anti-linear census (family D/A, counterwitness, face #13 closure, cut incidence, side conditions); L the lattice  $\delta$  ladder (frame,  $\pi/2$  reproduction,  $\pi/4$  PD, odd- $m$  artifact,  $N = 32$  refinement, site-axis control, spectra blindness, family-A rejection, chirality pinning); C the continuum axis elimination + counterwitness; LB the lifting audit;

T Fock implementability + the clock-tower clause; W scoreboard + hex/silver controls + the preregistered KILL verdict. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.BIT.RPBLIND.01` (new).

- **Ledger/paper text precisions (markers unchanged).** `SEAM.TAU.FLAG.01`: scoreboard text extended (7  $\rightarrow$  8 side-blind tests, face #13, web 12  $\rightarrow$  13); `tfpt_research_contracts` (the v512 reduction paragraph now carries the v521 kill and the honest [O] gap; the WP2 residual sentence extended), `introduction` and `tfpt_3` Step 2 (12-fold  $\rightarrow$  13-fold, the eighth side-blind test); `veri` citations in `tfpt_research_contracts`, `introduction` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the census solvesets incl. the exact-rational counterwitness; the  $V_4 \rightarrow D_4$  closure / face #13 as symbolic composition identities; the 20 incidence solvesets + the eps jump + the integer lattice frame; the continuum axis elimination + Cauchy–Stieltjes minors + the counterwitness cut ordering; the clock-tower clause + hex/silver controls; the  $\delta$ -blind side conditions) added to `tfpt_readouts_extension.wl`; the 40-digit lattice inertia spectra stay Python-only.
- **Provenance.** Discovery probe `seam_alignment_bit_rp_selector_probe.py` (25/25,  $\sim 24$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

**2026-07-22 (X)** (v520 — the WP5e BCOV measure question DECIDED at probe level and promoted as the ledger row `CELEST.WP5E.MEASURE.01`: “single-valuedness is derived, the completion reading wins” — the  $\delta_{1e} + \delta_{1f}$  consolidation in the v518 style, ERFOLG-A on the preregistered decision layer (the celestial/twistor story in `tfpt_3` gains Step 21): which of the two exact, contradictory measures on  $\mathbb{PT}/\mathbb{Z}_4$  is the true BCOV measure — (A) the DECLARED completion reading (v516) or (B) the DERIVED chiral measure (v518)? — (1) the INVARIANT SUBSPACE [E]:  $\dim_{\mathbb{Q}}\{v : \rho(T)v = v, \rho(S)v = v\} = 8 = 4 \times 2$  in the dressed 16-component Weil system (exact rational nullspace); EVERY kernel vector lies in the  $\mathbb{Q}(\zeta_8)$ -span of  $\{e_H, e_{H'}\}$  — the two Lagrangian gluings, both  $\theta = E_4$ , level-1 quartic  $= Q^{(0)}$  = the UNPHASED trace; strict trivial-character solutions at physical weight  $(m, k) = (0, 0)$ : ZERO for every dressing (11 combinations); (2) SOURCE 1 — the F-LEMMA [E]: the only strict chiral families are  $\chi_4$  (dims (3, 3)) and  $\chi_{10}$  (dims (1, 1)), both with nontrivial  $T$ -character EXACTLY; the canonical column-matched member obeys  $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$  pointwise (certificate  $3.9 \times 10^{-16}$ ,  $|G| \geq 59.6$ ); EXACT change of variables  $\Rightarrow$  fundamental-domain independence forces  $f = 0$  for EVERY chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is SHARPENED — its unfolded values are convention bookkeeping); (3) SOURCE 2 — the QUILLEN PAIRING [E]:  $|t\bar{t} - 1| < 8.3 \times 10^{-40}$  on all 15 pairs and both generators;  $|\chi_4|^2 = 1$  exactly vs one-sided  $\chi_4^2 = e(2/3) \neq 1$  ( $T$ -fixed:  $640 \neq 0 \bmod 960$ , 15/15 defective); the doubled 256-dim hol $\times$ antihol system closes STRICTLY at trivial character and physical weight (nullspaces  $(28, 17) \geq 11$ ), contains the unphased diagonal ( $\sim 2 \times 10^{-15}$ ) AND the physical columns  $w_b \otimes \bar{w}_b$  ( $\sigma$   $4.9/8.3 \times 10^{-16}$ ); negative control hol $\otimes$ hol: EMPTY nullspace (0, 0); (4) the CONSISTENCY CLOSURE [E]: the forced unphased numerator in the v518 scaffold reproduces  $\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x \rangle^2$  EXACTLY on every level  $n \leq 8$ ;  $J$ -weighted spreads  $\sim 5 \times 10^{-41}$ ,  $T_5/T_3 \sim 7 \times 10^{-42}$ , zero negative cells;  $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$  — the  $J$ -weighted mirror image of  $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$ ; linearity  $4.6 \times 10^{-41}$ : the circle to v516 is CLOSED; (5) D3 — the  $\mathbb{Z}_2/\text{EH}$  ANCHOR [E]: only the declared reading hits  $(9\langle x, x \rangle^2 = \frac{1}{4} \times 36, \psi = -8, \text{scale tooth } c = 1/2 = |\mathbb{Z}_2|h^{A_1})$ ; all three derived instantiations fail (spreads 1.12/0.895/0.516,  $\psi$  sign wrong); (6) D1 — COLUMN CANONICALISATION [E]:



the physical AB column  $\sum_m i^{bm} e_{(m,m)}$  is contained EXACTLY in the  $\chi_4$  family ( $\sigma$   $1.8/1.7 \times 10^{-16}$ , one scalar per orbit — canonicalising  $(N_1, N_2)$ ); the canonical member STILL fails every tester; **NEGATIVE CONTROLS**:  $SO(16)/D_8$  ( $T_5 = 20/T_3 = -40$  exact,  $\mu_2$ -only), wrong form ( $|\text{Gauss}|^2 = 64$ ), wrong weight (dev 0.78), shuffle ( $T_5 = -14/T_3 = 36$ ), planted family detected ( $1.5 \times 10^{-16}$ ),  $\text{hol} \otimes \text{hol}$  empty; **[C] TYPED PREMISES** (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source  $\psi = 64$ ), TP-3 (v518 kernel convention), TP-4 (Quillen structure of BCOV  $F_1$ ); **[O]** the constructive BCOV derivation of the  $w_m$  normalisation itself; **WIRKUNG** (text only, no marker moves): the v516 fence and the v518 tension now read “decided at probe level for the declared reading under TP-1..TP-4 (v520)”; **SEAM.EQUIV.TWISTOR.01** stays Open; **VERDICT ERFOLG-A** per the preregistered criteria — both constructive sources fire independently, the consistency closure carries)

- **v520\_celestial\_wp5e\_measure\_decision.py** (CELEST.WP5E.MEASURE.01, 39 checks, ~20 s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/\text{sympy}$  rational nullspaces + 1/960-grid phase arithmetic + **mpmath** 30-digit kernels + **numpy** double SVD, deterministic). S0 exact foundations (roots, ledgers, A-core +  $\mathbb{Z}_2$ -anchor targets, Weil relations, characters); S1 transports + derived families + the  $k = 0$  slice; S2 D1 column canonicalisation + the sharpened kill; S3 D2 the exact invariant subspace; S4 D3 the  $\mathbb{Z}_2/\text{EH}$  anchor at the integral level; S5 source-1 (the F-lemma); S6 source-2 (the Quillen pairing, doubled 256-dim system); S7 the point-4 consistency closure; S8 negative controls; S9 the ERFOLG-A verdict + typed premises. Registered in **run\_all.py**, **script\_registry.csv** (cluster *frontier*), ledger row CELEST.WP5E.MEASURE.01 (new).
- **Ledger/paper text precisions (markers unchanged)**. CELEST.WP5E.M2.01 (the v516 fence: the completion reading is now DERIVED at probe level from two independent constructive sources under TP-1..TP-4; residual **[O]** = the constructive  $w_m$  derivation), CELEST.WP5E.DELTA1.01 (the v518 kill sharpened, the tension decided); **tfpt\_research\_contracts** (the WP5e measure-question prose updated throughout: object diagram, WP4 arc,  $\varepsilon_1/\text{M2}$  fences, the measure-question target block now carries the executed decision, the closing status lines); **tfpt\_3** (Step 17/19 fences updated, new Step 21, the “what remains” block narrowed to the constructive  $w_m$  derivation); **tfpt\_4\_frontier** (celestial keybox updated). **veri** citations in **tfpt\_research\_contracts** and **tfpt\_3**.
- **Wolfram**. Six exact mirrors (the 16-component Weil data with both Lagrangian gluings; the invariant-subspace forcing  $\dim = 2 = \text{span}\{e_H, e_{H'}\}$  by exact NullSpace + rank; the character arithmetic of the two sources  $|\chi_4|^2 = 1$  vs  $\chi_4^2 = e(2/3)$  with the exact grid defects; the A-core +  $\mathbb{Z}_2/\text{EH}$ -anchor targets with the scale teeth; the level-1 Okubo identity of BOTH gluings (240 norm-2 vectors each, summed quartic  $36\langle x, x \rangle^2$ ); the  $\text{sol6}/\text{shuffle}/\text{diag}(i, i)$  controls) added to **tfpt\_readouts\_extension.wl**; the Harvey–Moore integrals, SVD memberships and  $J$ -weighted closures stay Python-only, flagged in the **.wl** comment; combined engine run with the v521/v522 mirrors: extension 522  $\rightarrow$  540 (verified, Wolfram Engine 14.3; **GATE.WOLFRAM.02**, Wolfram README, **tfpt\_safeguards**, website counters and **zenodo\_description.html** synced 116/116 + 540/540).
- **Provenance**. Discovery probes in **experiments/tfpt-discovery/** (2026-07-22): **celestial\_seam\_wp5e\_delta1e\_bcov\_measure\_decision\_probe.py** (28/28, ~15 s) and **celestial\_seam\_wp5e\_delta1f\_single\_valuedness\_probe.py** (23/23, ~17 s); suite 513  $\rightarrow$  516 modules (green); website mirrors regenerated (**ScriptIndex**, DAG *qft/P2* nodes, **papers.ts** story counters, glossary, changelog mirror); **next.txt** round notes.

**2026-07-22 (IX)** (v519 — the  $\text{WOIT-}\alpha$  milestone of the OS twistor bridge **WOIT.OS.TWISTOR.01** executed and promoted as the ledger row **WOIT.THETA.FREE.01**: “the real structure exists — and free reflection positivity picks the same family”;

the celestial/twistor story in `tfpt_3` is now TWENTY steps — (1) THETA CLASSIFICATION COMPLETE [E]: exactly TWO families of anti-linear structures on  $\mathbb{C}^2$  normalise the clock  $\rho = \text{diag}(i, 1)$  — family D ( $z \mapsto \mu\bar{z}$ ,  $|\mu| = 1$ ):  $\Theta\rho\Theta = \rho^{-1}$  EXACTLY at matrix level,  $\Theta^2 = +1$  (−1 IMPOSSIBLE:  $MM = \text{diag}(|\mu|^2, 1)$  — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ( $z \mapsto \mu/\bar{z}$ ): centralises the clock projectively (NEVER inverts it),  $\Theta^2 = \pm 1$  per  $\mu$ , fixed set  $|z| = 1$  resp. empty; ROLE SEPARATION: family A DEFINES the euclidean section (Woit’s  $\rho_{\text{tw}}$  side:  $\rho_{\text{tw}}^2 = -1$ , no real points — replicated exactly on  $\mathbb{C}^4$ ), family D REFLECTS it (the OS conjugation,  $\sigma_{\text{std}} =$  conjugation with real points  $\mathbb{RP}^3$ ); (2) MARK PINNING [E]:  $z \mapsto \mu\bar{z}$  preserves the  $\mu_4$  marks iff  $\mu \in \mu_4$  — a 4-element  $\mu_4$ -TORSOR (2 mark axes, 2 silver-midpoint axes), clock conjugation  $\rho\Theta_\mu\rho^{-1} = \Theta_{-\mu} \Rightarrow 2$  clock orbits; (3) SPIN/FOCK [E]: the  $\mathbb{Z}_8$  plane has no phase leaks ( $\zeta_8^k = \zeta_8^{-k}$ ); exact  $\text{Cl}(16)$ :  $\Theta_{\text{Fock}} = U_r \circ K$  with  $\Theta_{\text{Fock}}^2 = 2^7 > 0$  (normalised  $+1$ ), inverts the clock tower ( $V \mapsto 4096 V^{-1}$ ) and normalises the deck; the KRAMERS DICHOTOMY: the deck-induced candidate  $\Theta_t = \tilde{U} \circ K$  has  $\Theta_t^2 = (-1)^F$  — *the v510 split/nonsplit dichotomy IS the  $\Theta^2 = +1$  vs  $(-1)^F$  dichotomy*; the deck does NOT furnish  $\Theta$ , the seam-circle REFLECTION does; (4) KILL TEST 1 (free level) [E]: does NOT fire —  $\Theta$  exists with  $\Theta^2 = +1$  AND  $\Theta\rho\Theta = \rho^{-1}$  and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock); (5) FREE RP [E]: the NS vacuum  $C(d) = (2/N)/\sin(\pi d/N)$  has  $C^2 = -1$  exactly; preconditions exact ( $RCR^T = -C$ ,  $SCS^T = C$ ,  $RSR^{-1} = S^{-1}$ ; the deck FAILS anti-invariance); BOND CUT: one-particle Gram  $(8, 0, 0)$  PD (min eigenvalue  $1.888 \times 10^{-3}$ , 40 digits), even  $\deg \leq 2$  sector  $(29, 0, 0)$  PD,  $N = 8$  complete half-sided algebra PD (even + odd each  $(8, 0, 0)$ ) — RP with NO degree truncation; twist  $\eta = +i$  FORCED ( $\eta = 1$  non-Hermitian); CONTROLS: the site cut fails exactly ( $\det = 0$ , inertia  $(3, 3, 1)$ ); continuum control: Cauchy–Stieltjes strictly positive — a lattice-PLACEMENT artifact, RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally ( $((4, 4, 0))$ ) — *RP and  $\Theta\rho\Theta = \rho^{-1}$  select the SAME family*; bonus: the anti-chiral state flips the odd sector to  $(0, 8, 0)$  ND (the free shadow of kill test 3); [C]/[O] NOT shown: the interacting algebra  $\mathcal{A}_{\text{hol}}$  (kill test 1 stays live there), gauge-fixed RP (kill test 2), OS reconstruction, chirality without mirrors (kill tests 3/6),  $\mu_4$  incidence (kill test 7) — the WOIT- $\beta/\gamma$  roadmap named ( $\beta_1$   $\Theta$  on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant  $\text{SDYM}(E_8)$  sector;  $\beta_2$  the OS quotient of the free system explicit;  $\beta_3$  the  $\text{PT} \leftrightarrow \text{PT}^*$  duality vs  $\sigma_{\text{std}}$  typed;  $\gamma$  the chirality theorem + the mark incidence); TWO CONTRACT PRECISIONS (prose, no marker): (i) “ $\Theta$  induced by the seam reflection” means the seam-circle REFLECTION, not the deck — the deck furnishes the  $(-1)^F$  Kramers class (the v510 bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle; Wolfram extension  $516 \rightarrow 522$  (verified engine run); *NO marker moves — WOIT.OS.TWISTOR.01 stays Open: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra*)

- `v519_woit_theta_rp_free.py` (WOIT.THETA.FREE.01, 34 checks,  $\sim 20$  s; exact sympy + exact  $\text{Cl}(16)$  Fractions; RP definiteness via 40-digit spectra of exact Hermitian matrices, tolerance  $10^{-25}$ ; deterministic). Part W: W-S1 the two-family classification (symbolic solve); W-S2 family D (relations, Kramers-free, mark pinning, seam action, torsor); W-S3 family A (equatorial/ $\rho_{\text{tw}}$ ); W-S4 the  $\mathbb{C}^4$  twistor level ( $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ); W-S5 the  $\mathbb{Z}_8$  spin level; W-S6 the  $\text{Cl}(16)$  Fock level ( $\Theta_{\text{Fock}}^2 = 2^7$ ,  $V \mapsto 4096 V^{-1}$  via the implementer/centre route — a consolidation of the 105-s discovery probe to  $\sim 20$  s with identical claims and tolerances —

deck normalisation, Kramers dichotomy,  $\tilde{U}V = V\tilde{U}$  control); W-S7 verdict. Part R: R1 the exact preconditions; R2 the site cut fails; R3 the bond cut passes (incl. the complete  $N = 8$  half-algebra); R4 the continuum control; R5 controls (twist forced, antipode fails, chirality flip); R6 verdict. Registered in `run_all.py` (suite 513 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `WOIT.THETA.FREE.01` (new).

- **Contract updated (markers unchanged).** `WOIT.OS.TWISTOR.01`: the  $\alpha$  stage moved to *done* with the core numbers and `veri{v519}`; the two precisions added as their own sentences (the input definition of  $\Theta$  now names the seam-circle reflection, with the deck carrying the  $(-1)^F$  Kramers class; the marks at the bond midpoints of the 16-Majorana circle); the  $\beta/\gamma$  roadmap bullets added with success and kill criteria; the object-diagram OS row and the closing status line carry the  $\alpha$ -stage state. Ledger: `WOIT.THETA.FREE.01` new; `WOIT.OS.TWISTOR.01` text extended (“alpha stage executed via v519 ... kill test 1 does NOT fire at the free level and stays live on the interacting algebra; beta/gamma milestones named”). NO marker moves anywhere.
- **Paper integration.** `tfpt_3` (new Step 20 “the real structure exists — and free reflection positivity picks the same family” — the celestial/twistor story is now twenty steps; the Woit-bridge paragraph extended by the  $\alpha$  status: “the missing global object now exists at the free level; the interacting statement stays [O]”). `v519_woit_theta_rp_free.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the two-family classification; mark pinning + the  $\mu_4$  torsor with the clock-orbit map and the cut-point census; the  $\mathbb{C}^4$  level  $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ; the  $\mathbb{Z}_8$  tower conjugation; the  $\text{Cl}(16)$  dichotomy  $2^7$  vs  $(-1)^F$  on explicit Jordan–Wigner gammas incl.  $U_r V U_r V = 2^{19}$  and  $\tilde{U}V = V\tilde{U}$ ; the free-RP preconditions + site-cut degeneration + the Cauchy–Stieltjes continuum control) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $516 \rightarrow 522$ ; the 40-digit PD inertia certificates of the bond-cut Grams stay Python-only, flagged in the `.wl` comment; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ( $116/116 + 522/522$ ).
- **Provenance.** Discovery probes `woit_os_theta_realstructure_probe.py` (21/21,  $\sim 105$  s) and `woit_os_theta_rp_freelevel_probe.py` (13/13,  $\sim 5$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v519, `papers.ts` “20 steps” + the WOIT-contract mirror, glossary  $\Theta$ /Woit-bridge entries, changelog mirror); `next.txt` round notes.

**2026-07-22 (VIII) (v518 — the WP5e  $\delta_1$  chain ( $\delta_{1b} + \delta_{1c} + \delta_{1d}$ ) DECIDED and promoted as the ledger row `CELEST.WP5E.DELTA1.01`: “the derived chiral measure kills the  $\delta_1$  route” — an honest, decided NEGATIVE result in the v508 style, with the v516 tension stated: does the Harvey–Moore/BCOV fundamental-domain integral on  $\mathbb{PT}/\mathbb{Z}_4$ , evaluated under a measure DERIVED from blockwise  $SL(2, \mathbb{Z})$  covariance of the dressed 16-component Weil system (solved for — not declared, not scanned), deliver the twisted contact vector  $-A_{\text{fix}} = -(9, -30, -15, 0, 32)$  or the  $\psi = 64$  slice? NO — (1) the 16-COMPONENT WEIL COMPLETION ( $\delta_{1c}$ ): the discriminant module of  $D_5 \oplus A_3$  is  $\mathbb{Z}_4 \times \mathbb{Z}_4$  with  $q = (5x^2 + 3y^2)/8$ , Gauss sums  $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$  (signature  $0 \bmod 8 = \text{rank } E_8$ ), the Weil relations hold EXACTLY in  $\mathbb{Q}(\zeta_8)$  ( $S^2 = C$ ,  $(ST)^3 = S^2$ ,  $S^4 = 1$ ,  $T^8 = 1$ , tensor split  $S = S_{D_5} \otimes S_{A_3}$ ), the diagonal  $H$  and anti-diagonal  $H'$  are the two invariant Lagrangians (both counts =  $E_4$ : two  $E_8$  gluings),  $P_H S P_H = \frac{1}{4}J$  with  $3/4$  of every  $S$ -image weight on the 12 shifted classes, and the 16-vector theta  $S$ -covariance CLOSES: the  $\delta_{1b}$  E1.5 residual  $2.91 = O(1)$  (naive 4-character rule) drops to  $\sim 10^{-39}$ ; (2) the OBSTRUCTION IS A FINITE  $\mu_4$  CHARACTER, IDENTIFIED ( $\delta_{1d}$ ): the dressed gauge blocks  $M = G[a, b] \eta^{-8}$  carry constant transport  $(t_p, s_p)$  (28+ digits,**



recognised as exact  $1/960$ -grid phases at  $10^{-25}$ ), the  $T$ -fixed pairs  $(0, b)$  carry  $t = e(-1/2) = -1$  EXACTLY (the naked level-matching defect), the transport-groupoid loops land in  $\Gamma_1(4)$  resp.  $\Gamma_0(2)^\pm$  with joint holonomy eigenspaces  $\dim 8/16$  resp.  $6/16$  and every eigen-phase of order  $\leq 4$ ; the KOBOUNDARY TEST: the  $SL(2, \mathbb{Z})$  relation defects  $c_{S^4}$ ,  $c_{(ST)^3S^{-2}}$ ,  $c_{[S^2, T]}$  are  $(1, 1, 1)$  EXACTLY on all 15 pairs — the multiplier system is a CHARACTER of the orbit stabilisers, NOT a genuine 2-cocycle, and on  $\Gamma_1(4)$  it is  $\lambda(\gamma) = i^{2B+C/4}$ ; (3) the CANCELLATION EXISTS AND IS THE TWISTED FIBRE BLOCK  $f_1f_3$ , NOT THE THREE SPHERE AXIONS:  $G[a, b] = f_1f_3$  is an exact identity (30+ digits; the axion  $T$ -phases are v502 vacuum-energy data), the T-fix mechanism is one line of exact phase arithmetic —  $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$  at every  $T$ -fixed node — and the exact CANCELLATION TABLE reads: none  $\rightarrow$  residual order 4,  $f_1f_2f_3$  (the three sphere axions)  $\rightarrow 4$ ,  $f_2 \rightarrow 6$ ,  $f_1f_3 \rightarrow 1$  (hol AND anti), wrong weights  $(1, 1, 2)/(2, 2, 2) \rightarrow 4$ ; the strict solutions are exactly  $\chi_4$  (dims  $(3, 3)$ ) and  $\chi_{10}$  (dims  $(1, 1)$ ) at  $\eta^0$ , certified on the dressed functions at a fresh  $\tau$  ( $10^{-15}$ , NO residual multiplier); (4) the DECISION — ALL THREE PREREGISTERED TESTERS FAIL: under both derived solutions the integral (30-digit kernels, negative-exponent cell audit, cut-6 vs cut-8 scan) fails  $T_5(V) = 0$  (fractions  $0.5/0.83$ ), fails  $W_2:W_{13} = 4:3$  ( $W_{13} = 0$ ), fails  $V = -A_{\text{fix}}N$  (spreads  $1.05/1.47$ ) and the  $\psi = -64N$  slice, and the honest  $(N_1, N_2)$  orbit rebalance rescues nothing in the positive cone ( $\chi_4$ :  $T_5 = 0$  only at  $\psi > 0$ ;  $\chi_{10}$ : only at  $N_2/N_1 = -4 < 0$ ) — a GENUINE KILL on the derived surface; (5) NEGATIVE CONTROLS:  $\mathbb{Z}_2/\text{Eguchi-Hanson}$  (gauge-only residual order 2 — the  $\mu_4$  defect is  $\mathbb{Z}_4$ -specific — and the SAME fibre dressing cancels it to 1),  $SO(16)/D_8$  (only even sectors, all axion  $b$ -characters in  $\mu_2$ : the order-4 supply structurally absent), wrong form  $(x^2+y^2)/4$  ( $|G|^2 = 64 \neq 16$ , its  $S$ -rule fails at  $O(1)$ ), wrong signature ( $S^2 = e(-1/2)C$  exactly); [C] the  $f_1f_3 = \text{KS-weight identification}$  (deck weights  $(1, 3) = \text{the twistor fibre weights forced by the v514 incidence ledger}$  — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete  $\chi_k$  choice (both evaluated, both fail); the MANDATORY TENSION FENCE: the DECLARED completion reading (v516/M2) delivers the  $\psi = 64$  slice and cancels  $32T_3$ ; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure — neither result is hidden behind the other; [O] the BCOV-integral derivation on  $\mathbb{PT}/\mathbb{Z}_4$  that decides between the two measures; Wolfram extension  $510 \rightarrow 516$  (verified engine run); NO marker moves — the  $\delta_1$  route is decided negative under the derived measure, the cubic-GS question stays dissolved, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched)

- `v518_celestial_wp5e_delta1_derived_kill.py` (CELEST.WP5E.DELTA1.01, 30 checks,  $\sim 20$  s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/1\text{-}960\text{-grid}$  phase arithmetic + `mpmath` 30-digit kernels + `numpy` double SVD, deterministic). S0 conventions + 16-class ledgers + targets; S1 the 16-component Weil system (exact) + the E1.5 closure; S2 gauge transport + the  $\mu_4$  multiplier system (koboundary test, character identified); S3 the cancellation ( $G = f_1f_3$ , T-fix mechanism, exact table); S4 the strict solutions (constructive scan,  $m \in \{0, 8, 16\}$ ); S5 the decision (function certificates, integrals, three testers,  $(N_1, N_2)$  scan); S6 negative controls; S7 verdict + the tension fence. Registered in `run_all.py` (suite 512 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.DELTA1.01 (new — “DECIDED: kill under the derived measure, tension with v516 stated”).
- **Contract updated (markers unchanged).** CELEST.SEAM.01: the  $\delta_1$  strand moves from “UNDECIDED at exploration level” to DECIDED by v518; the named remaining target is now

the BCOV MEASURE QUESTION (the declared v516 completion reading vs the derived v518 chiral measure — both exact, which is the true BCOV measure is the named [O]); the cubic-GS question stays dissolved (v518 does not revive the  $d$ -channel). SEAM.EQUIV.TWISTOR.01 dependencies extended by v518 (preparation range now v492–v518; the route stays Open). CELEST.WP5E.M2.01 cross-referenced (its [O] pointer “ $\delta_{1b/1c/1d}$  territory” is executed and decided). NO marker moves anywhere.

- **Paper integration.** `tfpt_research_contracts` (the  $\delta_1$  stage moved to *done*, *DECIDED* with the four exact results and the mandatory tension fence as its own paragraph; “WP5e proper” renarrowed from the  $\delta_{1d}$  derivation to the declared-vs-derived measure question; the object-diagram row, the honest status line and the per-ID status list extended by CELEST.WP5E.DELTA1.01); `tfpt_3` (new Step 19 “the derived measure decides — and disagrees with the declared one” — the celestial story is now nineteen steps; the Step-16/17 fences and “what remains” updated to the measure question); `tfpt_4_frontier` (keybox synced). `v518_celestial_wp5e_delta1_derived_kill.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil system with Gauss sums and tensor split; the two invariant Lagrangians  $+ P_H S P_H = \frac{1}{4}J +$  the  $3/4$  shifted-class weight; the koboundary test  $(1, 1, 1)$  on all 15 pairs from the recorded exact transport phases; the  $\Gamma_1(4)$  character formula  $\lambda(\gamma) = i^{2B+C/4}$  on the recorded loops; the T-fix mechanism  $(-1)e(-1/6) = e(1/3) = \chi_4(T)$  uniform for  $f_1 f_3$  vs inconsistent  $\{\chi_3, \chi_4\}/\{\chi_5, \chi_6\}$  for  $f_1 f_2 f_3/f_2$ ; the wrong-form/wrong-signature controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $510 \rightarrow 516$ ; the Harvey–Moore integrals, SVD contraction solves, theta covariance certificates and block transport measurements are numerical and stay Python-only, flagged in the `.wl` comment; GATE.WOLFRAM.02, Wolfram README and the website counters synced  $(116/116 + 516/516)$ .
- **Provenance.** Discovery probes `celestial_seam_wp5e_delta1b/1c/1d_*.py` — the Harvey–Moore unfolding  $(30/30)$ , the Weil-16 completion  $(31/31)$  and the order-4 multiplier cancellation  $(36/36)$  — in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v518, `papers.ts` “19 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (VII) (v517 — WP5e back-reaction milestone M3 “the  $a_0$  uplift”** executed (SUCCESS on the preregistered v514 S8.3 criterion), promoted as the ledger row CELEST.WP5E.M3.01: the  $(4, \pm 4)$  GH multipole (= the  $a_0$  direction) is uplifted to the twistor-potential kernel  $\chi = \log P_4$  — the LOG of the v515 family polynomial — and the induced Kähler-potential correction is a log-type correction COUPLED TO THE CENTRE COUNT ON FOUR SCALES — (1) the KERNEL BRIDGE: the  $O(2)$  section is a NULL coordinate ( $\Delta_{3d}G(\eta_p) = 0$  identically for an ARBITRARY kernel  $G$ : the uplift is well-typed), the RESIDUE IDENTITY  $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$  holds EXACTLY (residue  $1/(2r)$ ), so the GH potential  $V = \sum_p 1/r_p$  IS the  $x$ -derivative of the contour transform of  $\chi$ ; flux  $-4\pi$  per centre, source charge  $4 = |\mu_4|$  (the  $V$ -ledger matched); (2) the ASYMPTOTIC KERNEL LOG:  $\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$  with LOG COEFFICIENT  $4 = |\mu_4|$  = centre count (the CPS “ $N \log$ ” analogue at kernel level); at the clock-fixed seam fibre  $\zeta = 0$  the  $a_2$  slot closes and the first correction is EXACTLY  $a_0/\eta^4$  = the  $(4, \pm 4)$  multipole in kernel clothes; exact  $m$ -grading ( $m = \pm 4$  pieces PURE  $a_0$ ,  $m = 0$  piece  $-6t_0^4 \varphi_0$ -free: the  $U(1) \rightarrow \mathbb{Z}_4$  breaking sits exclusively in the  $a_0$  pieces); NO  $\log \times \text{power}$  terms (the v514 honest zero preserved — the  $a_0$  correction is power-law at infinity); the once-integrated GLT tower  $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + p_8/(56\eta^7) + \dots$  with  $p_n = 0$  unless  $n \equiv 0 \pmod 4$  and  $p_{4k} = 4(-a_0)^k$ : EVERY correction carries the prefactor  $4$  = centre count; (3) the EXCEPTIONAL-LOCUS LOG:  $\chi(\eta = 0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$  ( $t_0$ -coefficient  $4 = |\mu_4|$ , clock phase  $4i$ ); the modulus response  $a_0 \partial_{a_0} \chi = a_0/(\eta^4 + a_0)$  equals 1 AT the exceptional locus and falls off as  $a_0/\eta^4$  at infinity (no constant, no log): the  $a_0$ -log lives at the exceptional locus, exactly as v514 S6.3 demanded; (4) the PERIOD RESPONSE:  $\Pi_j(\lambda)$  is LINEAR in  $t_0$  (uniform first-order shift: lockstep + phase ratios  $(1, i, i^2)$  preserved exactly),  $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$  uniformly, and the integrated shift around the  $a_0$  circle is the monodromy  $e^{2\pi i/4} = i$  = ONE Coxeter clock step (the v493 monodromy REPRODUCED from perturbation theory; the finite monodromy is the cyclic centre permutation  $q_p \rightarrow q_{p+1}$ ); the node  $\lambda$ -support is  $t_0$ -FREE (the  $\mathbb{Z}_8$  seam bridge is  $a_0$ -rigid), all 6 pair discriminants  $\sim t_0^2 \neq 0$  ( $K_4$  connectivity: the forced uniform flux vector persists), the BM kernel is modulus-free with  $\int_{S^3} K = (2\pi i)^2$  (the quantised fluxes cannot move); (5) NEGATIVE CONTROLS: the  $(4, 0)$  multipole is  $\varphi_0$ -independent in  $V$  AND in the kernel (clock-invariant, breaks nothing); the  $\mathbb{Z}_2$ /Eguchi–Hanson analogue reads  $2 = |\mathbb{Z}_2|$  on EVERY dial ( $(2, \pm 2)$  amplitude  $\frac{3}{2} \sin^2 \theta$ , kernel log 2, exceptional log  $2 \log t_0$ , response  $1/2$ , monodromy  $-1$ ); wrong centre counts  $k = 3, 5$ : kernel log coefficient  $k$ , modulus slot  $O(6)/O(10) \neq O(8)$  — the observable MOVES with the centre count; forbidden family  $Z^4 - Z$ :  $p_3 = 3 \neq 0$  (selection rule broken),  $e_4 = 0$  (no  $a_0$ -log) — fails BOTH uplift dials; [C] the GLT dictionary ( $\eta_p \log \eta_p$  = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O] the full nonlinear Kähler potential of the resolved  $A_3$  ALE; with v516 the v514 fence M1–M3 is FULLY WORKED OFF; Wolfram extension  $504 \rightarrow 510$  (verified engine run); NO marker moves — the KILL (decoupling from the centre count) does NOT fire, SEAM.EQUIV.TWISTOR.01 stays Open (preparation, not construction), SEAM.EQUIV.01 untouched)

- v517\_celestial\_wp5e\_m3\_a0\_uplift.py (CELEST.WP5E.M3.01, 23 checks,  $\sim 7$  s; exact sympy, no floats; deterministic). S0 replication (GH multipole ledger + the twistor family at general orbit phase, v515 slice); S1 the kernel bridge (null coordinate, residue  $V$ -ledger identity, source flux); S2 the uplift (asymptotic log 4, seam-fibre correction  $a_0/\eta^4$ , exact  $m$ -grading, GLT tower + selection rule); S3 the exceptional-locus log; S4  $\Omega_N$  consistency (period response

1/4, monodromy  $i$ ,  $a_0$ -rigid nodes and fluxes); S5 negative controls; S6 verdict. Registered in `run_all.py` (suite 511 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M3.01 (new, SUCCESS).

- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M2+M3 executed via v516/v517 — the v514 fence M1–M3 is fully worked off: the twisted KS measure (declared completion reading,  $\text{ch}_2$ -weighted) cancels the  $32 T_3$  residual and delivers the  $\psi = 64$  slice without the cubic  $d$ -channel, and the  $a_0$  uplift couples to the centre count on four scales; what stays **[O]**: the BCOV-integral derivation of the completion reading ( $\delta_{1d}$ ) and SEAM.EQUIV.TWISTOR.01. SEAM.EQUIV.TWISTOR.01 dependencies extended by v516/v517 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M3 bullet moved to *done* with the four coupled scales and the GLT fence; “what remains” narrowed to  $\delta_{1d}$  + the BCOV derivation); `tfpt_3` (new Step 18 “the  $a_0$  uplift: four coupled centre-count scales” — the celestial story is now eighteen steps; “what remains” updated). `v517_celestial_wp5e_m3_a0_uplift.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (multipole ledger + family at general phase; kernel bridge (null coordinate, residue  $1/(2r)$ , flux  $-4\pi$ ); asymptotic log +  $m$ -grading + GLT tower; exceptional-locus log; period response  $1/4$  + monodromy  $i$  +  $a_0$ -rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $504 \rightarrow 510$ ; GATE.WOLFRAM.02, Wolfram README and the website counters synced ( $116/116 + 510/510$ ).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_a0_uplift_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + v517, `papers.ts` “18 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (VI) (v516 — WP5e back-reaction milestone M2 “the twisted Kodaira–Spencer measure” executed (SUCCESS on the preregistered v514 S8.2 criterion, ON THE DECLARED COMPLETION MEASURE — verdict B), promoted as the ledger row CELEST.WP5E.M2.01: the one-loop box coefficient per sector on  $\mathbb{PT}/\mathbb{Z}_4$  under the completion contact term  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  — the twisted-sector LOOP contributes the UNPHASED sector trace with the  $g^j$  zero-mode normalisation  $1/\det_j$ , of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part — (1) the COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class  $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$  EXACTLY ( $h_m = m(4-m)/8$  the v502 sector Casimir energies,  $\text{ch}_2(T_m)$  the McKay/Kronheimer charges, v505 S4.3 index bridge): THE THREE SPHERE AXIONS PAIR THROUGH THEIR OWN  $\text{ch}_2$  CHARGES — an exact identity, NO free scale, no fit; (2) PARAMETER-FREE LOCKS:  $T_5 = 0$  for ANY scale  $c$  (the  $\text{ch}_2$  pattern alone kills  $T_5$ ); ratio  $h_2 : h_1 = 4 : 3 =$  the  $\delta_1$ -forced leading ratio (REPRODUCED, not fitted); the  $T_3$  budget fixes  $c = 4 = |\mu_4|$  uniquely (leading system  $\det 256$ , unique weights  $(\frac{3}{2}, 2) = (4h_1, 4h_2)$ ); (3) the SUCCESS TABLE:  $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x \rangle^2/\det_j$  — a PERFECT Okubo square ( $T_5 = T_3 = 0$ ,  $\tilde{\lambda}^2 = 36$  in EVERY twisted channel:  $(18, 9, 18) \times \langle x, x \rangle^2$ ); total  $A_{\text{fix}} + \text{contact} = 45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo} =$  the UNIQUE quartic-free weighting of v505 S3.8 (the rigidity theorem is exactly what the measure lands on); both v508 certificates KILLED ( $\Phi_{T_3}: 32 \rightarrow 0$ ,  $\Phi_P: 72 \rightarrow 0$ ); the v511 slice  $\psi = 64$  SUPPLIED EXACTLY ( $\psi(\text{contact}) = -64$ ) — NO cubic  $d$ -channel needed ( $c_d$  free = 0); remainder inside the exchange span ( $45\langle x, x \rangle^2 = \frac{45}{16}E_{22} = \frac{1}{80}E_{00}$ ,  $\Phi_P = 0$ ): the BULK axion cancels every channel with the SAME  $\lambda^2 = 36$ , the twisted propagator carrying the AB zero-mode factor  $1/\det_j$ ; the v514 measure chain stays  $\mu$ -blind; (4) NEGATIVE CONTROLS: wrong scale  $c \in \{1, 2, 3, 5\}$  leaves  $T_3 = (24, 16, 8, -8)$  — only  $c = 4 = |\mu_4|$  clears the budget; the  $h_1 \leftrightarrow h_2$  shuffle breaks  $T_5 = -14$  AND  $T_3 = 36$ ;  $SO(16)$  glue keeps  $T_5 = 20 / T_3 = -40$  — the KILL branch FIRES there (Okubo failure inherited:  $E_8$  doubly special);  $\text{diag}(i, i)$  degenerates to the  $\mathbb{Z}_2$  target and breaks the conjugation pairing + the index bridge; the  $\mathbb{Z}_2/\text{Eguchi–Hanson}$  anchor PASSES with the same mechanism ( $A_2 + \text{contact} = 9\langle x, x \rangle^2 = \frac{1}{4} \times 36\langle x, x \rangle^2$ , coupling scale  $2 = |\mathbb{Z}_2| =$  centre count); (5) the MULTIPLIER NOTE (cross-reference to the parallel  $\delta_{1d}$  strand): the contact weights  $4h$  are REAL RATIONALS with no multiplier system of their own; the  $\delta_{1c}$  residual phases are the SQUARES of the  $A_3$  discriminant  $T$ -phases  $e(3a^2/8)$  — orders  $(1, 8, 2, 8)$  squaring to  $(1, 4, 1, 4)$ , a finite  $\mu_4$  system; Gauss sums  $2e(3/8)$ ,  $2e(5/8)$ , full Weil sum 4; [C] the MANDATORY FENCE: the completion reading (the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation) is DECLARED, supported by the  $\delta_1$  modular-completion finding, NOT derived from the BCOV integral; [O] the BCOV-integral derivation ( $\delta_{1b}/\delta_{1c}/\delta_{1d}$  territory); Wolfram extension  $498 \rightarrow 504$  (verified engine run); NO marker moves — the KILL (leftover independent quartic in ANY sector) does NOT fire, the completion reading stays [C]-declared, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched)**

- `v516_celestial_wp5e_m2_twisted_ks_measure.py` (CELEST.WP5E.M2.01, 23 checks,  $\sim 1$  s; exact sympy/Fraction, no floats; deterministic). S0 replication (roots, class quartics,  $A_{\text{fix}}$  two routes, bulk Okubo); S1 the index bridge + the completion weights (no free scale, parameter-free locks); S2 the contact term (two forms, genuine-contact certificate); S3 the success table (per-sector Okubo squares, Dedekind  $\times$  Okubo, certificate kills, exchange span,  $\mu$ -blind chain); S4 negative controls; S5 the multiplier note (cross-reference to  $\delta_{1d}$ ); S6 verdict. Registered



in `run_all.py` (suite 511 modules together with v517, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M2.01 (new, SUCCESS on the declared measure).

- **Contract updated (markers unchanged).** CELEST.SEAM.01: the M2 bullet moves to *done*; the v511 burden paragraph is closed out — the  $\psi = 64$  slice is DELIVERED by the declared KS measure (v516); the cubic  $d$ -channel is not needed — its physical-justification question dissolves unless  $\delta_{1d}$  revives it. The completion reading itself stays [C]-declared; its BCOV-integral derivation stays [O] ( $\delta_{1d}$ ).
- **Paper integration.** `tfpt_research_contracts` (the M2 bullet moved to *done* with the weight identity and the success table; the v511 fence paragraph extended); `tfpt_3` (new Step 17 “the twisted KS measure: every sector the same Okubo square” — on the way to the eighteen-step story). `v516_celestial_wp5e_m2_twisted_ks_measure.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (completion-weight identity + index bridge; root/quartic replication; parameter-free locks; contact term + success table + certificates +  $\mu$ -blind chain; negative controls; multiplier note) added to `tfpt_readouts_extension.wl` and verified with the active engine, 498  $\rightarrow$  504; GATE.WOLFRAM.02 and the Wolfram README synced.
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_twisted_ks_measure_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v516, changelog mirror); `next.txt` round notes.

**2026-07-22 (V) (v515 — WP5e back-reaction milestone M1 “the  $A_3 \Omega_N$ ” executed (SUCCESS on the preregistered v514 S8.1 criterion), promoted as the ledger row CELEST.WP5E.M1.01: the back-reacted holomorphic 3-form  $\Omega_N$  is CLOSED FORM on the  $A_3$  twistor family, every  $S^3/\mathbb{Z}_4$  period is  $(2\pi i)^2$ -integral with the lockstep flux vector, and the lens geometry FORCES the source charge  $4 = |\mu_4|$  — (1) the RESIDUE FORM DERIVED: on  $F = XY - P(Z)$  all three ambient representatives satisfy  $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$  exactly and pull back to  $\omega_2 = dX \wedge dZ / X$  (Poincaré residue, derived not assumed); orbifold-cover pullback at  $a_0 = 0$ :  $\omega_2 = 4 dz_1 \wedge dz_2 = |\mu_4| \times (\text{flat form})$  — the residue normalisation CARRIES the source charge 4; clock  $\omega_2 \rightarrow i\omega_2$  (v492 S5 replicated); weight ledger  $4+4+2-8 = 2$ ,  $\Omega \in O(4) = -\deg K_{\mathbb{PT}}$ ; (2) the TWISTOR FAMILY CLOSED: from the four  $O(2)$  centre sections  $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$  exactly  $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$  ( $e_1 = e_3 = 0$  identically;  $\lambda = 0$  gives exactly the v493 family;  $a_2 \in O(4)$  opens only off-seam), CY-compatible clock lift  $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$ ,  $\gamma^4 = 1$ ,  $\Omega \rightarrow +\Omega$ ; (3)  $\Omega_N$  CLOSED + PERIOD TABLE:  $\Omega_N = \Omega_0 + \sum_p N_p K_p$  with CPS/Bochner–Martinelli kernels on the centre twistor lines,  $dK = 0$  off-source; boundary  $S^3/\mathbb{Z}_4$  at the orbifold point  $(2\pi i)^2 \times 1$  (minimal invariant charge  $4 = |\mu_4|$  — the lens fundamental domain FORCES  $N \in 4\mathbb{Z}$ ), deformed phase  $(2\pi i)^2 \times 4$ , linking  $S^3$  uniformly  $(2\pi i)^2 \times N$ , per-sphere flux vector  $(2\pi i)^2 \times N(1, 1, 1)$  (forced uniform: clock orbit +  $K_4$  connectivity), seam-fibre phase vector  $2\pi i t_0 (i - 1)(1, i, i^2)$ ;  $\Omega_0$  carries ZERO 3-flux (all flux sourced);  $(2\pi i)^2 = \text{fibre residue} \times \text{transversal residue}$  (density  $-\sin 2\theta$ ); (4) CLOCK COVARIANCE:  $\Pi_{j+1}(-i\lambda) = i\Pi_j(\lambda)$  for ALL  $\lambda$ ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ( $8 = 2|\mu_4|$ , the  $\mathbb{Z}_8$  bridge), clock orbits  $4+4+4$ , honest conifold points (Hessian  $\det 512t_0^6$ , quadric rank 4); (5) the HONEST KILL PROBE: forbidden family  $Z^4 - Z$ : 0/24 lockstep (vs 8/24),  $\lambda$ -support 12 twelfth roots  $\neq \mathbb{Z}_8$ , no antipodal pairing — BUT the  $(2\pi i)^2$  quantisation holds there TOO: integrality alone is NOT the discriminator, the discriminator is the lockstep phase structure + the clock forcing; the v509 dial survives; (6) ANCHORS + CONTROLS: EH/ $\mathbb{Z}_2$  (charge  $2 = |\mathbb{Z}_2|$  forced), CPS original (every integer  $N$  allowed),  $\text{diag}(i, i)$  ( $\Omega_0 \rightarrow -\Omega_0$ , collapse at step zero), fractional charges  $N = 1, 2, 3$  excluded; [C] global kernel patching / Hitchin small resolution / CPS brane dictionary; [O] M2 (twisted KS measure), M3 ( $a_0$  uplift), the quantised BCOV coefficient; Wolfram extension  $490 \rightarrow 498$  (verified engine run); NO marker moves — M2/M3 and BCOV stay [O], SEAM.EQUIV.TWISTOR.01 stays Open (v515 is preparation, not construction), SEAM.EQUIV.01 untouched)**

- **v515\_celestial\_wp5e\_m1\_omega\_n.py** (CELEST.WP5E.M1.01, 30 checks,  $\sim 1$  s; exact sympy, no floats; deterministic). S1 the residue form (three routes, cover factor 4, weight ledger, EH anchor, wrong-weight control); S2 the twistor family (closed form,  $O(2)$  reality, clock lift, section-level dichotomy); S3 periods (generic reduction, lockstep, covariance, collision ledger on  $\mathbb{Z}_8$ ); S4  $\Omega_N$  (BM anchor, lens descent, source curve, conifold nodes, M1 criterion); S5 the KILL probe (honest); S6 negative controls; S7 fences + verdict. Registered in `run_all.py` (suite 509 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M1.01 (new, SUCCESS).
- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M1 executed via v515 — the back-reacted  $\Omega_N$  is closed-form on the  $A_3$  twistor family, all  $S^3/\mathbb{Z}_4$  periods are  $(2\pi i)^2$ -integral with the lockstep flux vector and clock covariance, and the lens geometry FORCES the source charge  $4 = |\mu_4|$ ; honest fence: integrality alone does not discriminate — the discriminator is the lockstep phase structure (0/24 on the forbidden family); M2/M3 stay [O]. SEAM.EQUIV.TWISTOR.01 dependencies extended by v515 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M1 bullet moved to *done* with the core



numbers; the object-diagram row  $\mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{PT}/\Gamma$  extended by v515 with the M1 kill retired to  $\delta_1/\text{M2-M3}$ ; WP4 handover, WP5 narrative, honest status line and status inventory synced); `tfpt_3` (new Step 16 “the back-reacted  $\Omega_N$ : closed form, integral periods, forced charge 4” — the celestial story is now sixteen steps; “what remains” updated: M2/M3 are the remaining preregistered milestones); `tfpt_4_frontier` stage list synced. `v515_celestial_wp5e_m1_omega_n.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (residue-form identities; twistor family closed; period reduction + lockstep + covariance; collision ledger on  $\mathbb{Z}_8$ ; BM anchor + lens forcing  $N \in 4\mathbb{Z}$ ; source curve  $K_4$ ; conifold nodes; KILL probe +  $\text{EH}/\text{diag}(i, i)$  controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $490 \rightarrow 498$ ; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 498/498).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m1_a3_omega_n_backreaction_probe.py` (30/30) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + v515, `papers.ts` “16 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (IV) (external review integrated: route renamed, object diagram, group-extension definition, SEAM.EQUIV route split (MMST/TWISTOR ids, no status change),  $A_3$  role table, WOIT.OS.TWISTOR.01 contract — a pure structure/text integration, NO new physics claims, NO marker moves, suite unchanged at 508)**

An external expert review of the celestial/twistor branch confirmed the substance and asked for *semantic* precision — role separation, status clarity, and the exact naming of what is missing. Six changes plus one new central research contract, all editorial/structural:

- **(Ä1) Route renamed + the Woit bridge.** The `tfpt_3` celestial section and the contract heading are now “**The celestial and twistor continuum route**” (the old “from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow” survives as the narrative-arc subtitle). A new paragraph block in `tfpt_3`, “*The Woit bridge: chiral Wick rotation and the real structure*” [O], introduces Woit’s Euclidean Twistor Unification ([arXiv:2104.05099](#)) as a *named external programme* (chiral Wick rotation problematic for purely chiral theories; Euclidean-holomorphic theory on  $\mathbb{PT}$ , Lorentzian theory via conjugation structure/boundary values), lists TFPT’s existing ingredients (RP v379, OS reconstruction,  $\rho$  = clock v492,  $\sigma$  = deck/seam reflection v510), and names exactly what is missing: a global anti-linear  $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \rightarrow \mathcal{A}(\mathbb{PT}/\Gamma)$  with  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , plus RP of the interacting BCOV+SDYM functional — “until that exists, the Woit connection is a strong analogy, not a mathematical integration”. No namedropping as confirmation.
- **(Ä2) The object diagram.** `tfpt_research_contracts` (CELEST section, up front) now opens with the chain  $(\Sigma, \mu_4, \rho, \Theta) \rightarrow \mathbb{P}^1 \setminus \mu_4 \rightarrow \mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{PT}/\Gamma \rightarrow \mathcal{A}_{\text{hol}} \rightarrow_{\text{OS}} \mathcal{A}_{\text{Mink}}$  as a table with one row per arrow (claim id, status, load-bearing modules v492/v493/v497–v514, kill test); the last two arrows are honestly [O] with WOIT.OS.TWISTOR.01 as the target — where mathematics ends and physics begins is visible at a glance.
- **(Ä3) The group extension, exactly.** A definition block:  $\Gamma_{\text{ALE}} = \langle \text{Deck} \rangle \cong \mathbb{Z}_4 \subset SU(2)$  (triholomorphic, preserves  $\Omega$ ; the quotiented  $\mathbb{Z}_4$ );  $\rho$  = Clock =  $\text{diag}(i, 1) \in U(2)$  (Kähler, *not* triholomorphic,  $\det = i$  rotates  $\Omega$  by  $i$  — v492 S5; normalises, is not quotiented); the global object is cyclic  $\langle \rho \rangle$  with  $\rho^2 = \text{Deck}$  (projectively  $\mathbb{Z}_4$ , spin lift  $\mathbb{Z}_8$ ; v492/v506/v507), plus the action table on  $K_{\mathbb{PT}}$ ,  $\Omega$ , BCOV fields, open-string fields and boundary states, each entry [E] (arithmetic) or [O] (quantisation question); “ $\mathbb{PT}/\mathbb{Z}_4$ ” is defined once for the whole contract (quotient by  $\Gamma_{\text{ALE}}$ , normalised by  $\langle \rho \rangle$ ) with a cross-reference sentence in `tfpt_3`.
- **(Ä4) SEAM.EQUIV route split (ledger restructuring, no status change in substance).** New ledger rows SEAM.EQUIV.MMST.01 (the conditional scaling-limit route — closed

modulo cited theorems, typing taken verbatim from the pre-split rows/prose; residual **[O]** = the cited continuum existence, v336) and SEAM.EQUIV.TWISTOR.01 (the open Costello–Li construction, **[O]**; dependencies on CELEST.SEAM.01/WP modules and WOIT.OS.TWISTOR.01). SEAM.EQUIV.01 *remains* as the parent target with the OR logic as claim text: *closed if MMST.01 OR TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open  $\Rightarrow$  parent stays **[O]** as an unconditional claim*. Every surface that said “SEAM.EQUIV.01 closed modulo cited theorem” (status cards in introduction, tfpt\_1/3/4, the central status card, README, origin\_theory, tfpt\_2, tfpt\_5, website StatusPyramid/Open-Gates/StatusMatrix/papers.ts/glossary/VerificationDag) now attaches the statement to the MMST route id — the assertions are unchanged, only the ids are clean.

- **(Ä5) The  $A_3$  role table.** In the contract:  $A_3^{\text{family}}$  (lattice family factor,  $\mathfrak{su}(4)$  flavour),  $A_3^{\text{ALE}}$  (ADE type of  $\mathbb{C}^2/\mathbb{Z}_4$ ), the three exceptional spheres (Coxeter/PL structure),  $D_5^{\text{carrier}}$ , and — as external reference rows — Woit’s  $SU(3)_{\text{color}}$  and  $SU(2)_{\text{weak}}$ : per pair either the connecting functor with its vN witness (McKay type-level; Kronheimer–Nakajima quiver v479) or an explicit “*not identical — no identification claimed*”; in particular  $A_3^{\text{family}} \not\cong SU(3)_{\text{color}}$ , and the Woit rows are pure programme references.
- **(Ä6) Anti-number-hunting programme note.** experiments/next.txt (dated 2026-07-22 (IV)): the next real gain lies at positivity, incidence, locality and the  $\Theta$  real structure — *not* at further factorisations; recorded as working discipline per the review.
- **New central contract WOIT.OS.TWISTOR.01 (Open, research contract; new ledger row).** In tfpt\_research\_contracts after CELEST.SEAM.01: input  $X_+ = (\mathbb{PT}/\Gamma)_+$ ,  $\mathcal{A}_{\text{hol}}$  (interacting open+closed twistorial algebra),  $\rho$  (order-4 clock),  $\Theta$  (seam-reflection-induced anti-linear real structure),  $\mu_{\text{BCOV}+\text{SDYM}}$ ; target theorem  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , RP of the gauge-invariant algebra, OS quotient  $\Rightarrow (\mathcal{H}, \Omega, U(\mathcal{P}_+^\uparrow), \mathcal{A}_{\text{Mink}})$  with positive Hilbert metric, positive energy, local causality, *one chiral generation without mirror doubling*, the TFPT charge lattice, internal  $SU(3) \times SU(2) \times U(1)$  action, and the  $(E_8)_1$  seam net as an *actual* boundary net; *seven* preregistered kill tests ( $\Theta$ –clock incompatibility; RP failure after gauge fixing; vector-like mirror generation; Penrose transform only free/self-dual; right character but wrong OPE/net equivalence;  $SU(2)$  stays a spacetime factor; the four  $\mu_4$  marks not incidence-compatibly extendable); typing “this is the actual bridge from compiler to physics — everything in CELEST.SEAM.01 is preparation for it”; honest scope fence (self-dual  $\neq$  full 4D physics: both helicities, generic amplitudes, local matter, full Einstein dynamics, EWSB, confinement are explicit non-claims of the celestial branch). Mirrored on the website (papers.ts section, glossary entries “Woit bridge” and “Real structure  $\Theta$ ”).
- **No marker moves; no new vN modules; suite unchanged (508 modules, green).**

**2026-07-22 (III) (v514 — WP5e- $\varepsilon_1$  “the  $O(-2)$  bulk-axion slot” executed (verdict B), promoted as the ledger row CELEST.WP5E.EPS1.01: the  $s = 0$  tower field of the celestial route is now a CONSTRUCTION,  $\tilde{\lambda} = 6$  is pinned three ways, and the first honest back-reaction step is taken in Gibbons–Hawking form — (1) the EQUIVARIANT  $O(-2)$  SLOT: the pushforward ledger  $\pi_* O(-2)$  on  $\mathbb{P}\mathbb{T}' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)$  gives per degree  $d$  exactly  $(d+1)^2$  classes = the exact wave-operator nullspaces (1, 4, 9, 16, 25, 36 for  $d = 0..5$ ), the v492 action on the fibre coordinates  $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$  with incidence-forced column weights  $(+1, +1, -1, -1)$  closes the equivariant bookkeeping BLOCK BY BLOCK (all  $d \leq 6$ , all 4 characters); character series  $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots$ ,  $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots$ ,  $P_2 = 6t^2 + 10t^4 + 28t^6 + \dots$ ; the  $d = 0$  slot has multiplicity  $(1, 0, 0, 0)$  — THE BULK AXION SURVIVES THE PROJECTION; Molien invariant ring  $= (1 - t^8)/((1 - t^4)^2(1 - t^2))$  —  $Z \in O(2)$ ,  $X, Y \in O(4)$ , relation degree 8 = the  $a_0$  weight (v492/v493 bridge); E3 bijection sharpened (twisted minimal content  $(2t, 6t^2, 2t) =$  the Coxeter eigenvalues,  $\det(A - 1) = -4$ ); graviton control:  $O(+2)$  invariant only from fibre degree 4 with multiplicity  $3 = \{X, Z^2, Y\}$ ; (2)  $\tilde{\lambda} = 6$  TRIPLY PINNED: Okubo  $\text{Tr}_{\text{adj}} X^4 = 36\langle x, x \rangle^2 = (6\langle x, x \rangle)^2$  on the 240 glue roots ( $36 = h^\vee + 6$ ,  $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\text{fam}}$ ); the measure chain cancels  $\mu = 1/4$  EXACTLY (vertex<sup>2</sup>  $\mu^2$ , propagator  $1/\mu$ , anomaly  $\mu$ ;  $\tilde{\lambda}_{\text{eff}}^2 = 36$  for arbitrary  $\mu$ ; wrong bookings EXCLUDED: anomaly-only  $\Rightarrow 3$ , missing propagator renormalisation  $\Rightarrow 12$ ); flux (minimal CPS quantum  $N = 1$ , single channel iff  $k = 1$ ,  $(\kappa/c_3)^2 = 12$  exact); (3) the GH/ $A_3$  BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches, the free orbit re-derives the v493 family ( $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$ ,  $a_0 = -z_0^4$ , monodromy = one clock step), the charge- $k$  GH point is exactly  $\mathbb{R}^4/\mathbb{Z}_k$  ( $k = 4$ : seam boundary  $S^3/\mathbb{Z}_4$ ), periods  $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$  LOCKSTEP (v509 counter-check), Coxeter clock from geometry ( $\Pi \cdot A = i\Pi$ ), source charge  $4 = |\mu_4|$ , EH asymptotic log coefficient EXACTLY 0 (the CPS log is an exceptional-locus statement; Burns contrast  $\det g \neq 1$ ), multipole rule  $m \equiv 0 \pmod{4}$  with  $(4, \pm 4)$  carrying  $-a_0$ ; controls  $\mathfrak{so}_8$  ( $\tilde{\lambda}^2 = 12$  irrational, squares only  $\{9, 36\}$ ),  $\text{diag}(i, i)$  (Veronese),  $k = 2$  (three channels); [C] conditional on Costello’s flat- $\mathbb{P}\mathbb{T}$  matching; [O] the quantised BCOV coefficient and the twisted channels ( $32 T_3$ ) — M1–M3 preregistered (the  $A_3$   $\Omega_N$  / the twisted KS measure / the  $a_0$  uplift, each with success + kill); Wolfram extension  $482 \rightarrow 490$  (verified engine run); *NO marker moves* — WP5e stays [O], SEAM.EQUIV.01 untouched, CELEST.SEAM.01 stays Open)**

- **v514\_celestial\_wp5e\_eps1\_axion\_slot.py** (CELEST.WP5E.EPS1.01, 34 checks,  $\sim 4$  s; exact sympy/Fraction, no floats; deterministic). S1 Penrose accounting; S2 equivariant ledger + character series + Molien + E3 sharpening + graviton control; S3 the  $\tilde{\lambda} = 6$  residue (Okubo / measure / flux) with honest gaps; S4–S5 GH centre dichotomy, periods, Coxeter clock; S6 the CPS “ $N \log$ ” ledger (kernel charge, multipoles, EH log); S7 negative controls; S8 the M1–M3 roadmap; S9 verdict. Registered in `run_all.py` (suite 508 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.EPS1.01 (new, verdict B).
- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads:  $\varepsilon_1$  executed via v514 — the slot is a construction,  $\tilde{\lambda} = 6$  pinned by three exact ledgers, the GH/ $A_3$  step re-derives the v493 family and the Coxeter clock from geometry; the quantised BCOV coefficient and the twisted channels stay [O] (M1–M3 preregistered).
- **Paper integration.** `tfpt_research_contracts` (the  $\varepsilon_1$  stage moved to *done* with its own stage block; the named remaining targets now carry the M1–M3 milestones verbatim; WP4 handover and honest status line synced); `tfpt_3` (new Step 15 “the bulk axion is a construction;  $\tilde{\lambda} = 6$  is pinned three ways” — the celestial story is now fifteen steps; “what remains” updated).

v514\_celestial\_wp5e\_eps1\_axion\_slot.py is cited via veri in both.

- **Wolfram.** Eight exact mirrors (equivariant Penrose ledger; character series + bulk-axion survival; Molien hypersurface + Veronese control; Okubo perfect square +  $\mathfrak{so}_8$  control; measure cancellation excluding 3 and 12 + flux dial; GH centre dichotomy; period lockstep + Coxeter clock; multipole rule + EH log ledger) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $482 \rightarrow 490$ ; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 490/490).
- **Provenance.** Discovery probes `celestial_seam_wp5e_eps1_o2_bulk_axion_probe.py` (20/20) and `celestial_seam_wp5e_eps1_gh_a3_backreaction_probe.py` (14/14) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node, `papers.ts` “15 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (II) (v513 — the  $c_d$  negative certificate, promoted as the ledger row CELEST.DTERM.NONDERIV.01 (anti-numerology certificate in the v508 style, honest rigid NEGATIVE result): can geometry derive  $c_d = 1920$ ? NO — the v511 [C] fence “ $1920 = |W(D_5)|$ ” does NOT survive the honest four-route battery with its PRE-DECLARED look-elsewhere ledger: (1) the CONVENTION TABLE gives  $c_d \in \{1920, 120, 256, 32, 69120\}$  across five normalisations (probe  $\text{tr-fund}/e_8$ -Killing; Gell-Mann;  $\mathfrak{su}(4)$ -Killing; plain trace;  $\text{ch}_3$ -normalised  $d/6$ ) — only 1 of 5 hits  $|W(D_5)|$ , while the convention-STABLE content is the factorisation  $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$ , both factors [E] (Killing propagator 60 from  $\sum_\alpha \langle \alpha, e \rangle^2 = 120$  over the actual roots;  $T_3$ -slot fixing via  $\det(\mathbf{P}\text{-block}) = -64 \neq 0$ ); (2) K1 DEAD: the charge-legal flux pairings  $(1, 3)/(2, 2)$  give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit  $64 \cdot 60/2$  is the FORBIDDEN  $(1, 2)$  pairing (charge 3 mod 4, v511 innerness); look-elsewhere: 924 expressions from 21 [E] numbers, 11 hits on 1920, controls  $1800 \rightarrow 8$ ,  $2048 \rightarrow 5$ ,  $2016 \rightarrow 0$ ; (3) K2: quantisation delivers only lattices ( $60\mathbb{Z}$  with 1920 the 32nd multiple, no minimality; strict  $\text{ch}_3$  units  $2160\mathbb{Z}$  EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots,  $3/8$  on weights), the integer 32 has 12 factorisations; (4) K3 PROVENANCE CLASH (the death blow): cubic indices  $A_m = (0, 16, 0, -16)$  with  $\sum_m i^m A_m = 32i$  numerically  $= \Phi_{T_3}(A_{\text{fix}})$  — but the quartic class decomposition  $(10, -4, 30, -4)$  comes from classes  $0/2$  vs the cubic  $16+16$  from classes  $1/3$ : same number, disjoint mechanism = coincidence; the Weyl side is WRONG ( $d$ -channel is  $A_3$ ,  $|W(A_3)| = 24$ ;  $D_5$  has no cubic;  $|W(E_8)|/|W(A_8)| = 1920$  a second accidental 1920); (5) K4 controls ( $SO(16)$ ,  $k = 2$  doubling 7680, false  $\mathfrak{g}_0$  with six cubics,  $\mathfrak{so}(10)$ ) separate consistently; VERDICT: NUMEROLOGY as a derivation claim — the v511 [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is  $c_d = 32 \times 60$ ; Wolfram extension  $476 \rightarrow 482$  (verified engine run); *NO marker moves — the fence stays [C], the physical generation of the 32 stays [O], CELEST.SEAM.01 stays Open*)**

- `v513_celestial_dterm_nonderivation.py` (CELEST.DTERM.NONDERIV.01, 24 checks,  $\sim 2$  s; exact sympy/Fraction, no floats; deterministic). N0 baseline (240 roots (52, 64, 60, 64),  $A_{\text{fix}} = (9, -30, -15, 0, 32)$ ,  $Q_{dd}$  two routes,  $c_d$  pinned by the  $T_3$  slot alone); N1 convention table + the stable factorisation  $32 \times 60$ ; K1 flux products + look-elsewhere ledger; K2 quantisation lattices; K3 cubic/quartic provenance clash + Weyl data; K4 negative controls. Registered in `run_all.py` (suite 507 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.DTERM.NONDERIV.01 (new).
- **Fence sharpened (text only, marker unchanged).** CELEST.WP5E.DELTA2.01 and the CELEST.SEAM.01 contract prose now carry: “the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded (11/924 vs. 8/924 for the control target 1800) and convention-contingent — v513”; the



honest remaining target of the cubic-GS item is the physical generation of the factor 32, not the Weyl fingerprint.

- **Paper integration.** `tfpt_research_contracts` ( $\delta_2$  fence environment, named remaining targets, status list — new CELEST.DTERM.NONDERIV.01 status line); `tfpt_3` (Step-14 paragraph + the “what remains” sentence); `tfpt_4_frontier` (celestial-route clause). `v513_celestial_dterm_nonderivation.py` is cited via `veri` in the  $\delta_2$ /fence environments.
- **Wolfram.** Six exact mirrors (propagator anchor +  $T_3$ -slot pinning; five-way convention table; flux enumeration + look-elsewhere counts; K2 lattices; K3 provenance clash + Weyl data; K4 controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 476  $\rightarrow$  482; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 482/482).
- **Provenance.** Discovery probe `celestial_seam_wp5e_dterm_coefficient_derivation_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 24/24); website mirrors regenerated (ScriptIndex, DAG *qft* node, papers.ts, changelog mirror); `next.txt` round notes.

2026-07-22 (I) (v512 — the flag-transitivity equivalence web promoted as SEAM.TAU.FLAG.01: the  $\tau = i$  attack executed on the free RP seam circle — BARE MARK-TRANSITIVITY IS FALSIFIED as a selector (on the test family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ ,  $\delta \in (0, \pi/2]$ , the symmetry group is the pair-exchanging  $V_4 = \{z, -z, b/z, -b/z\}$  for EVERY  $\delta$ , simply transitive on the four marks; complete census: deck-commuting Möbius maps are diagonal/antidiagonal; hardening: ALL undirected per-mark data are identical for every  $\delta$  — no local jet sees the bit), and THE SHARP SELECTOR is FLAG transitivity (marks AND their two sides indistinguishable)  $\iff \delta = \pi/2 \iff \tau = i \iff$  the clock — stabiliser jump  $1 \rightarrow 2$ , flag/arc orbits  $2 \rightarrow 1$ , order-4 rotation ( $b^2 = -1$  solveset  $\{\pi/2\}$ ), mark mirror  $z \mapsto 1/z \iff \cos \delta = 0$ ; welded into a 12-FOLD EQUIVALENCE WEB (every link exact): clock rotation,  $D_4$ , mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split  $= (2/\pi) \ln \cot(\delta/2)$ , zero only at  $\pi/2$  — counterwitness exactly  $(2/\pi) \ln 2$ ; arc-Laplacian degeneracy  $4/\pi$ ; the v215-K3 indicator in CLOSED FORM  $(2\sqrt{2}/\pi) |\ln \tan(\delta/2)|$  — the v503 lattice curve analytically),  $\rho$ -twist existence  $((m^2 - n^2)(t - 1/t) = 0 \iff t = 1$ ; Atiyah–Bott rationality  $\delta$ -BLIND —  $\text{Tr}(\sigma e^{t\Delta}) = 1$  for all  $\delta$ ,  $j_{\text{cw}} = 148176/25$  rational non-CM: rationality selects NOTHING), cusp-4-cycle implementability (H4 rigidity:  $V_4$  with orders  $\{1, 2, 2, 2\}$  cannot induce it; centraliser scalar,  $U^2$  nonscalar  $\Rightarrow$  no projective involution — projective order exactly 4,  $\mathbb{Z}_8$ -shadow consistent v510), harmonic,  $j = 1728$ ; NEGATIVE CONTROLS: hexagonal (not concyclic, fails preconditions), silver (harmonic but topologically dead), 3 marks (impossible on the free circle: 0 vs 28 invariant subsets; bare  $n = 3$  transitivity already forces equal spacing — “4 counts”); [C] THE REST: the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ( $V_4 \rightarrow D_4$ : side-indistinguishability of the four equally-decorated marks), v216 local equality does NOT deliver it, its negation is now concretely measurable; scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones; Wolfram extension 470  $\rightarrow$  476 (verified engine run); *NO marker moves — the bit stays [C]*, QGEO.REALIZE.01 and P2.TYPING.01/AX.P2.01 text-precised only, SEAM.BIT.FREEDOM.01 cross-referenced)

- `v512_seam_tau_flag.py` (SEAM.TAU.FLAG.01, 29 checks,  $\sim 6$  s; sympy-exact + mpmath 30 digits + one numpy lattice commutator; deterministic). S0 family frame (counterwitness inside the family:  $\lambda = 5/4$ ,  $j = 148176/25$ , cross-ratio  $-1/4$ ); H1 complete symmetry census

(census basis, rotation solvesets, stabiliser jump, flag/arc orbits, v216 separation, five-member table); H2 spectral criteria in closed form; H3 heat-trace rationality ( $\delta$ -blind) + twist existence; H4 monodromy rigidity (Schur); NC negative controls; SYN the 12-fold web + counterwitness scoreboard + the typed sentence. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.TAU.FLAG.01` (new).

- **Reduction state re-stated (text only, markers unchanged).** `AX.P2.01` and `P2.TYPING.01` synthesis notes now read: “four marks + one discrete symmetry-lift bit (flag transitivity  $\iff \tau = i) + \pi$ ”; `QGEO.REALIZE.01` precision: “the remaining input can equivalently be stated as the flag-transitivity postulate — v512; the search space for a future [E] closure is narrowed to the RP positivity of the odd sector”; `SEAM.BIT.FREEDOM.01` cross-referenced (the modulus half gains the discrete form).
- **Paper integration.** `tfpt_1` (the P2 reduction keybox re-dated 2026-07-22 + Theorem-7 skeleton keybox), `introduction` (narrative chain, consolidated reduction state, status-card addendum), `tfpt_3` (celestial Step 2 + the honest-status reduction bullet), `tfpt_research_contracts` (QGEO kill-test and marks-derived blocks, WP2 note); secondary mirrors `tfpt_2/origin_theory` aligned. `v512_seam_tau_flag.py` is cited via `veri` in all four primary documents.
- **Wolfram.** Six exact mirrors ( $V_4/D_4$  census table incl. flag/arc orbit counts; solveset equivalences + family identities; spectral faces in closed form; twist mode identity + AB  $\delta$ -blindness +  $j$ -rationality control; H4 Schur rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 470  $\rightarrow$  476; the `mpmath` heat-trace numerics and the `numpy` lattice commutator stay Python-only (flagged).
- **Provenance.** Discovery probe `seam_tau_i_established_selector_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 29/29); website mirrors regenerated (`ScriptIndex`, DAG *P2/masses* nodes, `papers.ts` reduction-state mirrors, `StatusMatrix`); `next.txt` round notes.

**2026-07-21 (XXIII) (v511 — WP5e- $\delta_2$  of CELEST.SEAM.01 “the full-tensor ledger” promoted as CELEST.WP5E.DELTA2.01 (intermediate verdict, exact):** the named escape route of the v508 exchange no-go executed on the FULL adjoint tensor structure of  $\mathfrak{e}_8$  — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  semisimple with NO  $\mathfrak{u}(1)$  (class-0 roots 40+12, root span rank 8),  $h$  in the Cartan OF  $\mathfrak{g}_0 \Rightarrow$  every invariant tensor carries total charge 0 mod 4: the bilinear Hom table is nonzero ONLY at  $j' = -j$  (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have Hom = 0), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ( $\{0, 0, 0\}: 3 = 1$  sym + 2 antisym,  $\{0, 1, 3\}: 2$ ,  $\{0, 2, 2\}: 2$ ,  $\{1, 1, 2\}: 1$ ,  $\{2, 3, 3\}: 1$ ; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir,  $\text{Inv}(S^3 45) = 0$  protects  $T_5$ ), whose exchange quartic  $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$  (TWO independent routes, Killing propagator  $2h^\vee = 60$  from the roots) has  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$  — the v508 master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$  ( $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  rank 3, augmented 4), and becomes EXACTLY solvable relaxed:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$  (residual 0); [C] fences  $64 = \dim \mathfrak{g}_1 = 2^6$ ,  $1920 = |W(D_5)| = 8 \cdot 240$ ; controls:  $SO(16)/D_8$  has NO symmetric cubic and no  $A_3$  block ( $E_8/\mu_4$  doubly special), false  $\mathfrak{g}_0$ /sector assignments separate; the  $\delta_1$  burden SHARPENED to the  $\psi = 64$  slice or the selection-rule relaxation — and the  $\delta_1$  exploration run itself stands UNDECIDED at exploration level (strict holomorphic  $q^0$  reading REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore  $\tau$ -integral with forced leading  $(T_5, T_3)$  ratio 4:3; massive coset levels  $n \geq 2$  mandatory); Wolfram extension  $463 \rightarrow 470$  (verified engine run); *NO marker moves — WP5e proper stays [O], SEAM.EQUIV.01 stays [O]*

- **v511\_celestial\_wp5e\_delta2\_tensor\_ledger.py** (CELEST.WP5E.DELTA2.01, 41 checks,  $\sim 4$  s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic). D0/D1 structure: 240 roots, class split (52, 64, 60, 64);  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  SEMISIMPLE without  $\mathfrak{u}(1)$  (40+12 class-0 roots, no mixed root, root span rank 8); sectors  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$  as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ( $h = (2, 2, 2, 2, 2; 0^4)$  in the  $\mathfrak{g}_0$  Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at  $j' = -j$  (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims (3, 2, 2, 1, 1) with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\text{Inv}(S^3 45) = 0$ ,  $\text{Inv}(S^3 \mathfrak{a}_3) = 1$ ). D5 the new channel:  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  by Killing-propagator contraction AND the hand route  $|x^2 - (\text{tr } x^2/4)\mathbf{1}|^2/60$ ;  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ ; brackets/tadpoles/ $d \times f$  die exactly; no  $T_5$  analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate  $\psi(A_{\text{fix}}) = 64$ ; relaxed reading EXACTLY solvable with  $c_d = 1920 = 32 \times 60$ . D7 controls: Killing control = v508 replication ( $A_{\text{fix}}, E_{13}, E_{22}, E_{00} = 225E_{22}$ );  $SO(16)/D_8$  bilinears (1, 0, 1), trilinears (1, 0, 1, 0),  $\text{Inv}(S^3 \text{adj } D_8) = 0$ ; false  $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$ : (5, 2, 2)  $\neq$  (2, 1, 1),  $\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$ ; false sector content (16<sub>s</sub>, 4) kills every pairing. Discovery provenance: experiments/tfpt-discovery/celestial\_seam\_wp5e\_delta2\_full\_tensor\_ledger\_probe.py (41/41), same day.
- **The  $\delta_1$  fence (exploration-level, NOT promoted — verdict UNDECIDED).** The twisted BCOV contact-term probe (celestial\_seam\_wp5e\_delta1\_bcov\_contact\_probe.py, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence



text: (i) the strict holomorphic  $q^0$  reading is REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading  $(T_5, T_3)$  ratio is 4:3 (sector 2 : sectors 1+3); (iii) the  $SL(2, \mathbb{Z})$  orbit decomposition  $15 = 12 + 3$  identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type  $\tau$ -integral), not a free number; (iv) massive coset levels  $n \geq 2$  are mandatory. The remaining  $\delta_1$  target is the  $\tau$ -integral, evaluated against the  $\delta_2$ -sharpened criterion ( $\psi = 64$  slice or selection-rule relaxation).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.DELTA2.01 ([E] Hom tables/innerness/ $Q_{dd}$ /certificate; [C] the number fences; [O] arity  $\geq 4$ , cubic GS term,  $\delta_1$ ); CELEST.SEAM.01 updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic  $d$ -symbol channel opens  $T_3$  with the weaker certificate  $\psi = 64$ ; delta-1 stands UNDECIDED at exploration level). Papers: tfpt\_research\_contracts WP5e bullet gains the executed  $\delta_2$  stage + the  $\delta_1$  exploration fence, WP5e-proper targets renamed ( $\tau$ -integral, cubic GS term,  $\varepsilon_1$ , back-reaction); tfpt\_3 celestial section extended to the FOURTEEN-step story (new Step 14 “the full-tensor ledger: the collapse is real, and one cubic door opens”) with the “What remains” block updated. Wolfram: seven mirrors (innerness count +  $\mathfrak{g}_0$  structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the  $SO(16)$ /false- $\mathfrak{g}_0$ / false-sector controls;  $Q_{dd}$  by two routes; the  $\psi$  certificate + the relaxed solution  $(-1, 8, 2; 1920)$ ) verified with the active engine the same day ( $463 \rightarrow 470$ , ALL WOLFRAM EXTENSION CHECKS PASSED). Registry + run\_all (505 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves):** the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ( $\tau = i$  — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v491/v499/v506/v507/v510) and  $\pi$ , plus the [C] identifications R1/R2 of v499; AX.P2.01 stays a DECLARED axiom and QGEO.REALIZE.01 stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (v491:  $4 = 2+1+1$  unique  $\Rightarrow g_{\text{car}} = e_2 = 5$ ), the weight typing (v499: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , residuals R1/R2 [C]), the residual convergence (v503: QGEO-R1  $\equiv$  P2-R1), the clock rigidity and bit origin (v506/v507: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (v510: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler’s dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ( $\tau = i$ , equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and  $\pi$  — plus the [C] identifications R1/R2 of v499.* No new claims; conservative wording; no “axiom abolished” language anywhere.
- **Papers + ledger + website.** tfpt\_1\_architecture\_e8: new keybox “The current reduction state of P2” next to the foundational-postulates keybox, with v491/v499/v506/v507/v510 citations. tfpt\_3: mirror bullet “The current reduction state” in the honest-status section. introduction: the anchor paragraph gains the consolidated one-sentence state and the status card row  $a = (1, 1, 2)$  now names the reduction state. Ledger: SYNTHESIS NOTE appended to AX.P2.01 (“reduction state 2026-07-21: four marks + one square-modulus datum ( $\tau = i$ ) +  $\pi$ ”), marker *Axiom unchanged*. Website: papers.ts mirrors of the changed sections. No gate closes, no marker moves.

**2026-07-21 (XXI) (v510 — the freedom attack on the v506/v507 alignment bit promoted as SEAM.BIT.FREEDOM.01: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the  $\mathbb{Z}_2$  seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in  $D_{16}$ , exactly 1 free;  $\text{Aut}(C_{16}) = D_{16}$  complete census, brute force  $8!$  at  $N=8$ ; quotient topology circle/antipode = CIRCLE  $\chi=0$  vs circle/reflection = INTERVAL  $\chi=1$  — the  $6\pi$  Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT  $\iff$  FREE over the complete 17-involution census in exact  $\text{Cl}(16)$  (free  $\Rightarrow \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F$  nonsplit with 2 roots = the  $\mathbb{Z}_8$  tower; ALL 16 reflections  $\Rightarrow U^2 = \text{scalar } \{2^7, 2^8\}$ , split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two;  $\mu_4 \rightarrow$  equatorial, deck FREE; silver  $\rightarrow$  real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal  $\rightarrow$  NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE  $\Rightarrow \mu_4$  exactly ( $b = \pm i$ ); honest counterwitnesses (“nonsplit  $\Rightarrow$  clock” is FALSE:  $\{0, 1, 8, 9\}$  and  $\{\pm 1, \pm(3+4i)/5\}$  with  $j = 148176/25 \neq 1728$ ); the alignment bit REDUCES from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; Wolfram extension 457  $\rightarrow$  463 (verified engine run); *NO marker moves — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays an axiom, P2-R1 halved)***

- **v510\_seam\_bit\_freedom.py (SEAM.BIT.FREEDOM.01, 30 checks,  $< 5$  s; sympy + exact  $\text{Cl}(16)$  Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in  $D_N$  on the refined  $2N$ -point seam circle ( $N = 16$ : 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness  $N = 8, 12$ ; odd control  $N = 9$ : ZERO free — an even carrier count is needed, consistent with  $16 = 2^{g_{\text{car}}-1}$ ); the census is COMPLETE because  $\text{Aut}(C_{16}) = D_{16}$  exactly (brute force over all  $8! = 40320$  permutations at  $N = 8$ , propagation at  $N = 16$ ); quotient topology: circle/antipode = CIRCLE ( $\chi = 0$ , closed), circle/reflection = INTERVAL ( $\chi = 1$ , two boundary corners) — a fixed-point deck would break the  $6\pi = 3 \times 2\pi$  closed-circle count, so “deck free on the seam circle” is exactly the established “the seam is a closed circle”, not a new premise; the complete Fock census: ALL 17 involutions implemented in exact  $\text{Cl}(16)$  — NONSPLIT  $\iff$  FREE with no exception, root dichotomy 2 vs 0; the chain: free  $\Rightarrow$  shift 8  $\Rightarrow$  crossing (all 28 invariant 4-sets)  $\Rightarrow U^2 = (-1)^F \Rightarrow$  central class  $\Rightarrow$  the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch  $\mathbb{Z}_2$  for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the  $\mathbb{Z}_2$  seam fibre; one-sidedness TYPES the  $\mathbb{Z}_2$  as the free antipode (quotient  $\mathbb{RP}^2 =$  the  $|\mathbb{Z}_2|$  of  $c_3 = 1/(|\mathbb{Z}_2|2\pi\chi)$ ); the deck-commuting real structures are EXACTLY two families ( $\mu\bar{z}$ : fix line through the poles, NOT free;  $\mu/\bar{z}$ ,  $\mu > 0$ : fix circle  $|z| = \sqrt{\mu}$ , FREE;  $\mu < 0$ : empty); the per-configuration table ( $\mu_4$  equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to  $b = \pm i$  (on free circles the silver family “harmonic AND edge” is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete  $\{0, 1, 8, 9\}$  and the continuous  $\{\pm 1, \pm(3+4i)/5\}$  (free, central,  $j = 148176/25$ , no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck  $\Rightarrow$  free [standard topology; the one-sidedness strand *origin\_theory/v456*]; (P-ii) marks on the  $\Theta$ -fix seam circle = existing REALIZE [C] content (QGEO.REALIZE.02 hypotheses, v264) — NO new input. Audit finding documented: “circle-freeness” was nowhere typed before (only mark-freeness, v216); v507 F1.1 explicitly admitted the reflection reading of the edge class — v510 closes that door (a

reflection is not a covering deck). The [C] rest, verbatim: *the alignment bit reduces from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.FREEDOM.01 ([E] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [C] the square-modulus datum); TEXT precisions (markers unchanged) in SEAM.CLOCK.RIGIDITY.01, SEAM.BIT.ORIGIN.01, QGEO.REALIZE.01 (bit  $\rightarrow$  square-modulus datum; edge class topologically excluded) and P2.TYPING.01 (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended (“the position half is topology”), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where v506/v507 are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on  $256 \times 256$  Jordan–Wigner gammas; the real-structure table; harmonic + free  $\Rightarrow \pm i$ ; the counterwitnesses) verified with the active engine the same day (457  $\rightarrow$  463, ALL WOLFRAM EXTENSION CHECKS PASSED; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves.

**2026-07-21 (XX) (v509 — WP5e- $\varepsilon_2$  of CELEST.SEAM.01 “the CPS level-from-flux dial” promoted as CELEST.WP5E.EPS2.01, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism (“level = flux quantum  $\times$  Dynkin index”) executed on the lockstep spheres — the CPS SKELETON exact ( $S^3$  period  $(2\pi i)^2 N$  with pullback density  $-\sin 2\theta$ , exceptional-sphere Kähler flux  $2\pi N$ , boundary level magnitude  $2N$  by the symplectic-boson contraction,  $T(\text{vector}) = 1$ ); the PAIRING MATRIX pinned, not postulated (ch<sub>2</sub> ledger + integrality + unimodularity + effectivity  $\Rightarrow P_{mi} = \delta_{mi}$  by complete enumeration: rows 8/6/8, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  with uniformly ONE quantum per charged current; “level = total flux” KILLED by the lockstep test itself ( $F_i$  unequal = flavour multiplicity), the per-current reading (1, 1, 1) anchored by the lattice current count (240, 0, 0, 0) and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors:  $\mu = 16 = \text{ord}^2$ , Lagrangian  $\mu_4$  glue, KLM  $16 \rightarrow 1$ ) with the NEW sector-counter dial  $\#\text{prim}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$  — exactly ONE sector iff  $k = 1$ ; the LOCKSTEP THEOREM lifts the v505 “one scale  $\Rightarrow$  one level” to a THEOREM of clock invariance ( $|\Pi_j| = \sqrt{2}t$  equal, monodromy = pure phase  $i$ ,  $\det(A-1) = -4$ : no invariant flux vector; NEW falsifier  $Z^4-Z$ : 0/24 orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls  $k=2/SO(16)/A_2$  separate honestly; Wolfram extension 450  $\rightarrow$  457 (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on  $\mathbb{P}^T/\mathbb{Z}_4$  typed [C], the type-I B-model back-reaction [O]), SEAM.EQUIV.01 stays [O])***

- `v509_celestial_wp5e_eps2_level_from_flux.py` (CELEST.WP5E.EPS2.01, 28 checks,  $< 1$  s; sympy/Fraction exact, no floats; deterministic). Target: derive the boundary level  $k = 1$  from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL 130, 061602; arXiv:2306.00940). *CPS skeleton replicated exactly*: the  $S^3$  period of the brane back-reaction 3-form is  $(2\pi i)^2 N$  (their eq. 3.33; large-gauge invariance quantises  $N$ ), the exceptional-sphere Kähler flux is  $2\pi N$  exactly ( $K = |u|^2 + N \log |u|^2$ , the flat part dies at  $t \rightarrow 0$ ), and the boundary level magnitude is  $2N$  via the symplectic-boson double contraction ( $N = 1, 2, 3$ ; their eq. 9.4), Dynkin normalisation  $T(\text{so}_8 \text{ vector}) = 1$ ,  $T(\text{adj}) = 6$ . *Pairing matrix pinned*: the v505 ch<sub>2</sub> ledger ( $f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$ )

- + integrality + unimodularity + effectivity force  $P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of  $PC^{-1}P^T = C^{-1}$ ; with  $P \geq 0$  exactly 2: identity +  $A_3$  diagram flip); fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ , total 188 = 248 – 60, flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: “level = total open-string flux” is KILLED by the lockstep test itself ( $k_i = (64, 60, 64)$  unequal —  $F_i$  is the FLAVOUR multiplicity, the brane matter content); what works:  $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$ , anchored non-circularly by the lattice current count (240, 0, 0, 0) for  $k = 1.4$  (v505 S4.8) and the embedding index  $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$  ( $60 = 2h^\vee$ ). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ( $\mu = \det(D_5)\det(A_3) = 16 = \text{ord}^2$ ; the  $\mu_4$  glue diagonal isotropic + Lagrangian; KLM  $16 \rightarrow 16/4^2 = 1$ , zero residual fractional flux); NEW DIAL:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1.6$  from the affine marks — exactly ONE sector iff  $k = 1$  (independent of the CPS transplantation). *The lockstep theorem* (the lifting of v505 S4.10): clock invariance  $P(iZ) = P(Z)$  forces  $a_3 = a_2 = a_1 = 0$  sharply, the four branch points are ONE  $\mu_4$  orbit,  $|\Pi_j| = \sqrt{2}t$  EQUAL on all three spheres, the monodromy is the pure phase  $i$  and  $\det(A-1) = -4 \neq 0$  (no invariant flux vector) —  $k_1 = k_2 = k_3$  IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family  $Z^4 - Z$  (smooth, disc  $-27$ , branch points  $\{0, 1, \omega, \omega^2\}$  not a  $\mu_4$  orbit): over ALL 24 orderings ZERO chains with three equal period moduli (1 vs  $\sqrt{3}$ ); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value (1 from the counters and the CPS minimal quantum  $N = 1$ ). *Negative controls*:  $k = 2$  (current count 0, residual 360, prefactor  $-31/48$ ,  $\#\text{prim} 3$  — while the lockstep dial honestly does NOT fire);  $SO(16)$  (fluxes (0, 60, 0): spheres 1/3 FLUX-DARK, defect  $-30 \neq -78$ , condensation stuck at  $4 = \det \text{Cartan}(D_8)$ );  $A_2$  ( $\det 3 \neq 4$ , glue diagonal  $23/24 \notin \mathbb{Z}$ ,  $\mu = 12$  no perfect square: NO Lagrangian subgroup, Coxeter  $3 \neq 4$ ). Typed honestly: the CPS dictionary itself on  $\mathbb{PT}/\mathbb{Z}_4$  stays [C] (their geometry is the  $\mathbb{C}^2$  blow-up with one  $(-1)$ -sphere and  $\mathfrak{so}(8)$  at negative level — not the  $A_3$  ALE); the type-I B-model back-reaction stays [O] (the same fence as v502/v505).
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.EPS2.01 ([E] CPS skeleton/pairing/lockstep/sector arithmetic + [C] CPS dictionary + [O] back-reaction); CELEST.SEAM.01 updated (“ $\varepsilon_2$  executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier  $Z^4 - Z$ ), level = 1 carries the new sector-counter dial  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ; the  $\mathbb{PT}/\mathbb{Z}_4$  back-reaction stays [O]”); the v505 “one scale  $\Rightarrow$  one level” sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: tfpt\_research\_contracts WP5e section extended by the  $\varepsilon_2$  block (done, verdict B; roadmap now  $\delta_1/\delta_2/\varepsilon_1$  + back-reaction); tfpt\_3 celestial section becomes a THIRTEEN-step story with Step 13 “one level is a theorem; the sector counter pins  $k = 1$ ” and the “What remains” paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the  $2N$  contraction + Dynkin normalisation; the pairing enumeration  $48 \rightarrow 2$ ; the fluxes (64, 60, 64); the  $\#\text{prim}$  table (1, 3, 5, 10, 15, 27); the lockstep periods + the 0/24 negative probe; the  $SO(16)/A_2$  controls) verified with the active engine the same day (450  $\rightarrow$  457, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag: v509 on the QFT/celestial node; registry + run\_all (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XIX) (e8-ladder-bed v2 — the 2DCS extension executed with an honest preregistered kill\_2d verdict at the DETECTOR level: the known-answer map recovers 8/8 rungs incl.  $m_7/m_8$  via F3 difference coordinates (MC  $p = 2 \times 10^{-4}$ ), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO 8/8, CoNb<sub>2</sub>O<sub>6</sub> 6/8) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)**

- experiments/e8-ladder-bed v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at  $B_y = 0\text{--}3\text{ T}$ , not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of CoNb<sub>2</sub>O<sub>6</sub> exists anywhere* (web-checked). amendment\_v2 was frozen in hypotheses/e8\_ladder\_bed\_v1.yaml BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); amendment\_v2\_1 discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (results/results2d.json, deterministic): known-answer KA2D 8/8 rungs with  $m_7/m_8$  recovered ONLY via F3 difference coordinates ( $\chi^2/\text{dof} = 1.30$ , MC  $p = 2.0 \times 10^{-4}$ , 0/5000) — the mechanism that pulls  $m_7/m_8$  inside a low-pass window works; the NEG2D localized-limit map (the paper's own non-E8 closed forms) is correctly rejected ( $p = 0.029$ ); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at  $54/200 = 27\%$  and `jitter2d` shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level)  $\Rightarrow$  preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The CoNb<sub>2</sub>O<sub>6</sub> leg stays 6/8 ( $p = 0.0038$ ) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row “E8 mass-ladder blind recovery bed”: `data_value` extended (“CoNb<sub>2</sub>O<sub>6</sub> 6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl.  $m_7/m_8$  via F3,  $p=2e-4$ ; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)”) and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, `experiments/next.txt` note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at  $B \parallel b$ , 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.



**2026-07-21 (XVIII) (v508 — WP5e- $\gamma$  of CELEST.SEAM.01 “the sphere-axion pairing check” promoted as CELEST.WP5E.GAMMA.01, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the  $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are EXACTLY  $\text{span}\{s_5, s_3\}$  (dim 2) and the quartics EXACTLY  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$  (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into  $\text{span}\{P_1, P_2, P_3\}$  — zero  $T_5/T_3$  content, so  $\Phi_{T_3}$  annihilates EVERY exchange image while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ ); the strict two-index  $\mathbb{Z}_4$  rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and the annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ ; side discovery  $K^{(0)} = -15 K^{(2)}$ ; naturalness DISSOLVES (ch<sub>2</sub>-natural and AB-weight couplings both certify (0,0) vs required (32,72) — scale-independent); controls:  $SO(16)$  has no sphere partners AND uncanceled  $T_5, D_8$  no  $T_3$  structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the 32  $T_3$  residual MUST be carried by twisted BCOV contact terms ( $\delta_1$  binary: coefficient exactly (9, -30, -15, 0, 32), computed not fitted; escape  $\delta_2$ : the full-tensor ledger); Wolfram extension 444  $\rightarrow$  450 (verified engine run); *NO marker moves — CELEST.SEAM.01 stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, SEAM.EQUIV.01 stays [O]*)**

- **v508\_celestial\_wp5e\_gamma\_pairing.py** (CELEST.WP5E.GAMMA.01, 27 checks,  $\sim 4$  s; sympy/Fraction exact, no floats; deterministic). Question: can the three twisted sphere axions  $\eta_m$  (Coxeter eigencharacters  $\{i, -1, -i\}$  on  $H^2(\text{ALE})$ , v505 S5.8) cancel the rigid 32  $T_3$  residual of the Atiyah–Bott ledger ( $A_{\text{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32T_3$ , v505 S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem:* exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees (1,0,1)) and quartics dim 5 (bidegrees (2,0,1,0,2)) — the vertex basis is COMPLETE; *product theorem* (the master kill):  $(as_5 + bs_3)(cs_5 + ds_3) = acP_1 + (ad + bc)P_2 + bdP_3$  identically — every exchange image (any axion count, any rule, any couplings) is annihilated by  $\Phi_{T_3}$ , and  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . *Selection enumeration:* the strict two-index reading collapses ( $C_{j,j'} \neq 0$  only for  $j' = -j$ , forcing  $m = 0$ : 16 raw triples, 4 nonzero, all  $m = 0$ ); twist insertion:  $\eta_1 \leftrightarrow K^{(3)}$ ,  $\eta_2 \leftrightarrow K^{(2)}$ ,  $\eta_3 \leftrightarrow K^{(1)}$ ,  $\eta_0 \leftrightarrow K^{(0)}$ ; channels  $E_{13} = (16, -96, 144, 0, 0)$ ,  $E_{22} = (16, 32, 16, 0, 0)$ ,  $E_{00} = 225 E_{22}$ ; SIDE DISCOVERY:  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are PARALLEL — this collapse drives the  $\Phi_P$  obstruction). *Unsolvability with certificate (the core):*  $M$  is  $5 \times 2$ , rank 2, kernel 0;  $\text{rank}([M | A_{\text{fix}}]) = 3$ ; annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions (32  $T_3$  structural, rule-independent;  $\Phi_P = 72$  selection-rule-driven);  $\ker M = 0$  forces the unique non-disturbing coupling  $c = 0$  (the sphere axions are RIGIDLY SILENT);  $\Phi_P(Q^{(0)}) = 0$  (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the  $P$ -block ( $\det = -64$ ) but  $T_5 = T_3 = 0$  persists. *Naturalness resolution:* ch<sub>2</sub>-natural couplings (9/64, 1/4) give (25/4, -11/2, 97/4, 0, 0) and AB-weight couplings (1/2, 1/4) give (12, -40, 76, 0, 0) — BOTH with certificate (0,0) vs required (32,72): the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”). *Negative controls:* invariant-only = v505 S3.9 replicated + sharpened; flipped rule  $m+j-j'$ :  $\Phi_{T_3}$  kill flip-invariant,  $\Phi_P$  rule-dependent;  $SO(16)$ : odd classes EMPTY and AB sum (15, -18, -15, -4, 40) —  $T_5$  does not even cancel ( $E_8$  doubly special);  $D_8$ : 16  $T_8 + 12 s_8^2$ , no  $T_3$  structure — the question cannot even be posed. Number fences [C] only:  $32 = (h, h) + (h', h') = 20 + 12$ ,  $72 = 2\lambda_{e_8}^2$ .
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.GAMMA.01 ([E] vertex space/selection/certificate + [C] fences + [O] contact terms); CELEST.SEAM.01 stays Open with the WP5e targets UPDATED (“the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms ( $\delta_1$ , binary test) and the full-tensor ledger ( $\delta_2$ )). Papers: tfpt\_research\_contracts WP5e section extended by the  $\gamma$  block ( $\gamma$

done — negative result with certificate; the  $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$  roadmap with success/kill criteria); **tfpt\_3** celestial section becomes a TWELVE-step story with Step 12 “the exchange no-go and the contact-term criterion” and the “What remains” paragraph updated. Wolfram: six mirrors (the vertex-space dims  $2/5$  by generator-wise nullspace arithmetic; the twisted quadratics with  $K^{(0)} = -15 K^{(2)}$ ; the product theorem, symbolic; the rank-3 certificate  $(0, 32, 72)$  with channels and naturalness images; the  $SO(16)$  AB sum  $(15, -18, -15, -4, 40)$ ; the  $D_8$  quartic  $16 T_8 + 12 s_8^2$ ) verified with the active engine the same day ( $444 \rightarrow 450$ , ALL WOLFRAM EXTENSION CHECKS PASSED). **VerificationDag**: v508 on the QFT/celestial node; registry + **run\_all** (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

**2026-07-21 (XVII) (v507 — the tautology attack on the v506 alignment bit, executed and refuted, promoted as SEAM.BIT.ORIGIN.01: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods  $\Lambda/2\Lambda$  of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  the free involutions are exactly the three  $\sigma_c$ : the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic  $\lambda$ ): a marks-preserving root of  $\sigma_c$  exists  $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$  (one per  $c$ ; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit  $\iff j = 1728 \iff \tau = i$ ; the EQUIVALENCE TETRAD “Uhr existiert  $\iff (\tau=i$  AND deck = CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift NONSPLIT”; the FERMION DICHOTOMY (the new physical face): central  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$  nonsplit vs edge  $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$  SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension  $438 \rightarrow 444$  (verified engine run); *NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01 text precised only***

- **v507\_seam\_bit\_origin.py** (SEAM.BIT.ORIGIN.01, 26 checks,  $\sim 2.6$  s; Part A sympy-exact torus/Möbius arithmetic + Part B exact  $Cl(16)$  Fractions; deterministic). Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double  $T_\tau^2 \rightarrow S^2$ , v214/v260), so the common origin would force the central alignment automatically and the v506 bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of  $w \mapsto -w$  on  $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =$  the half-periods  $\Lambda/2\Lambda$  (a  $\frac{1}{3}$ -coefficient never qualifies);  $t_c$  descends iff  $2c \in \Lambda$  — exactly the three half-period translations (control  $c = \frac{1}{3}$  fails); the descended maps are the EXPLICIT rational involutions  $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$  with the exact curve-lift identity  $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$ ,  $\psi_i$  RATIONAL (the  $\wp$ -addition formula as pure algebra), closing to Klein  $V_4$ ; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic  $V_4$   $(4, 3, 3)$ , harmonic  $D_4$   $(8, 5, 3)$ , hexagonal  $A_4$   $(12, 3, 3)$ ) are EXACTLY the three  $\sigma_c \Rightarrow$  every mark-free collar deck (v216) IS a half-period translation. *Core solve (symbolic  $\lambda$ )*:  $\sigma_c$  admits a marks-preserving  $PGL_2$  root  $\iff \lambda = -1$  ( $\sigma_1$ ),  $2$  ( $\sigma_2$ ),  $\frac{1}{2}$  ( $\sigma_3$ ) — per half-period exactly ONE solution, intrinsically “the deck pairs separate harmonically” (pair cross-ratio  $-1$ ); generic modulus:  $V_4$  abelian, ALL  $\sigma_c$  central yet clockless (“central alone is empty” — the bit is the CONJUNCTION  $D_4 \wedge$  central); at  $\tau = i$  the stabiliser jumps to  $D_4$  with  $\sigma_1$  central (2 roots = the clock pair),  $\sigma_2/\sigma_3$  edge-type (0 roots); the v506 seam deck lands on the central  $\sigma_1 = -1/x$  under all 8 mark bijections; CM dictionary: mult-by- $i$  fixes exactly  $c^* = (1+\tau)/2$  (mult-by- $\rho$  3-cycles all three), the clock lift needs the CM unit  $i$  ( $\psi = 2i/(x+1)^2$ ,  $t^2+1$  irreducible over  $\mathbb{Q}$ ) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists  $\iff (\tau=i$  AND deck = the CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift nonsplit; the v493 converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by



8):  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ , nonsplit  $\mathbb{Z}_4$ , 2 roots in the 32-element dihedral lift (the  $\mathbb{Z}_8$  tower); edge arrangement (reflection  $j \mapsto 4-j$ , fixed sites = non-marks = the silver  $\{0, \infty\}$  between neighbour marks): the implementer EXISTS but  $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$ , the extension SPLITS, 0 roots, no  $\mathbb{Z}_8$ ; one-particle  $(\lambda^2+1)^8$  vs  $(\lambda^2-1)^8$ ; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*:  $\{\pm 1, \pm(3+2\sqrt{2})\} = “\tau = i$  with deck = a non-CM-fixed half-period translation” (all 8 bijections  $\rightarrow \sigma_2/\sigma_3$ , never  $\sigma_1$ ) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; “no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION”. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.ORIGIN.01 ([E] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: SEAM.CLOCK.RIGIDITY.01 (“the bit is now fully characterised as a tetrad ... and carries a physical face: the non-split NS Fock extension class — v507”), QGEO.REALIZE.01 (“the class of the deck IS derived (half-period translation, v507); only its position stays the input”, stays [C]/[O]), P2.TYPING.01 (origin cross-reference on the common R1 bit). Papers: tfpt\_3 celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, introduction/tfpt\_1/tfpt\_2/origin\_theory one-sentence updates where v506 is cited. Wolfram: six mirrors (half-period descent  $2c \in \Lambda$  with the rational  $\sigma_i$  lift identity; the core solve  $\lambda \in \{-1, 2, \frac{1}{2}\}$ ; the stabiliser counts  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  via the complete 24-triple enumeration; the CM fixed point  $c^*$  + the lemniscatic  $\psi = 2i/(x+1)^2$ ; the Cl(16) dichotomy  $256 \gamma$ -monomial vs scalar  $2^7$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ( $438 \rightarrow 444$ , ALL WOLFRAM EXTENSION CHECKS PASSED); the v493 audit direction and the 28/21 arrangement census stay Python-side (flagged). VerificationDag: v507 on the P2/carrier and QGEO nodes; registry + run\_all (501 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XVI) (v506 — seam clock rigidity, Route B of the v503 classification, promoted as SEAM.CLOCK.RIGIDITY.01: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck  $z \mapsto -z$  has EXACTLY TWO square roots  $z \mapsto \pm iz$  in  $\text{PGL}_2(\mathbb{C})$ , both *automatically* order 4, and a marks-preserving root exists  $\iff$  the deck is the CENTRAL involution of the mark- $D_4$  (counterexample  $\{\pm 1, \pm(3+2\sqrt{2})\}$ : harmonic with full  $D_4$  but deck non-central  $\Rightarrow$  0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 —  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$ , i.e.  $U^2 = (-1)^F$  EXACTLY, a nonsplit  $\mathbb{Z}_4$ , with the  $\mathbb{Z}_8$  tower  $V^2 \propto U$ ,  $V^4 \propto (-1)^F$  ( $8 = 2|\mu_4|$  fermionically explained) and the R-sector control split ( $U_R^2 = +1$ ); the one [C] rest: “the collar deck is the central involution of the mark- $D_4$ ” — the fermionic  $\mathbb{Z}_4$  forces the ORDER, not the spatial alignment; Wolfram extension  $434 \rightarrow 438$  (verified engine run); *NO marker moves — QGEO.REALIZE.01 stays [C]/[O] (text precised), P2.TYPING.01 R1 mapped onto the same bit)***

- **v506\_seam\_clock\_rigidity.py** (SEAM.CLOCK.RIGIDITY.01, 34 checks,  $\sim 3.5$  s; **Part A** sympy-exact Möbius/divisor arithmetic + **Part B** exact Cl(16) Fractions; **deterministic**). v503 localised the one remaining identification input (“why does the raw seam carry the order-4 clock”, QGEO-R1  $\equiv$  P2-R1); v506 reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give  $g^2 = \text{deck} \Rightarrow \text{tr } g \neq 0$  forces  $g$  diagonal with  $(a/d)^2 = -1$ , and  $\text{tr } g = 0$  forces  $\det g = 0$  (impossible) — exactly two roots  $z \mapsto \pm iz$ , both projective order 4, mutually inverse ( $\text{Aut}(\mathbb{Z}_4)$ ), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set  $\{u, -u, w, -w\}$  admits a marks-preserving root  $\iff$

$w = \pm iu \iff \text{marks} = \text{one } \mu_4 \text{ orbit} \Rightarrow \text{cross-ratio class } \{2, \frac{1}{2}, -1\}, j = 1728, \tau = i, \text{ free 4-cycle, deck} = \text{clock}^2, \text{ only the scale free; group form: } |\text{Stab}(\mu_4)| = 8 = 2|\mu_4| \text{ with order profile } \{1:1, 2:5, 4:2\} = D_4, \text{ center} = \{\text{id}, \text{deck}\}, \text{ exactly 2 order-4 elements} = \text{the clocks} \text{ — “deck central in the mark-}D_4\text{”} \iff \text{“clock exists”}; \text{ the counterexample } \{\pm 1, \pm(3+2\sqrt{2})\} \text{ is deck-free AND harmonic } (j = 1728, \text{ cross-ratio } \frac{1}{2}) \text{ with full } D_4, \text{ but the deck sits non-centrally (the central involution crosses the } \pm \text{ pairs)} \Rightarrow 0 \text{ roots; the RP filter passes BOTH roots (the } \text{Aut}(\mathbb{Z}_4) \text{ ambiguity is irreducible, v492 NC3). } B3/B4 \text{ (} n\text{-arithmetic + controls): free orbit } \Rightarrow n \mid 4; \text{ transitivity } \Rightarrow n = 4; \text{ faithfulness kills } n \geq 5 \text{ (kernel argument through } S_4); n = 3 \text{ fixes a mark; } n = 6 \text{ dies independently at the cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); } n = 8 \text{ impossible on 4 marks; hexagonal stabiliser } A_4 \text{ (0 order-4 elements, squares only orders } \{1, 3\}); \text{ generic stabiliser } V_4 \text{ (exponent 2); stabiliser trichotomy } V_4/D_4/A_4 \text{ with order-4 counts } (0, 2, 0). \text{ Theorem B (Fock side, the core result): } S^2 \text{ has } b_1 = 0 \Rightarrow \text{unique spin structure } \Rightarrow \text{NS/bounding on every circle (established P1 datum); one-particle: } D^2 = -1 \text{ (charpoly } (\lambda^2+1)^8), \text{ quarter-shift } C^2 = D, C^8 = 1 \text{ (charpoly } (\lambda^4+1)^4, \zeta_8 \text{ spectrum); Fock (exact Cl(16)): } \tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8}) \text{ implements the deck on all 16 generators and } \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F \text{ EXACTLY; } (cU)^2 = c^2(-1)^F \text{ is never 1 (center of Cl(16) = scalars): the } \mathbb{Z}_4 \text{ lift is FORCED, } 1 \rightarrow \mathbb{Z}_2^{(-1)^F} \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2^{\text{deck}} \rightarrow 1 \text{ nonsplit; the quarter-shift lift } V \text{ (unique even block implementer) has } V^2 \propto \tilde{U}, V^4 \propto (-1)^F, \text{ projective order } 8 = 2|\mu_4| \text{ — the v492 } \mathbb{Z}_8 \text{ spin bridge realised on the 16-Majorana Fock space; R control } U_R^2 = +1 \text{ (split — the forcing rests on NS = established geometry); 16-fold way } \theta_\nu = 1 \text{ at } \nu = 16; \text{ robustness: the forcing already holds at 8 sites (16 = the carrier count } 2^{g_{\text{car}}-1}, \text{ not a tuning). The one [C] rest (verbatim): “the collar deck is the central involution of the mark-}D_4\text{” (} \iff \text{ square-root existence on the mark data) — ONE alignment bit; the fermionic } \mathbb{Z}_4 \text{ forces the ORDER, not the spatial alignment. No overclaim.}$

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.CLOCK.RIGIDITY.01 ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: QGEO.REALIZE.01 (“the rest is now ONE bit: the central position of the deck in the mark- $D_4$ ; order/uniqueness/orbit/ $\tau=i$  follow — v506”, stays [C]/[O]), P2.TYPING.01 (R1 mapped onto the same bit, fermionic halving), QGEO.EMERGE.LIGHT.01 context sharpened. Papers: tfpt\_3 celestial Step 1 (fermionic  $U^2 = (-1)^F$  sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, introduction/tfpt\_1/tfpt\_2/origin\_theory one-sentence updates at the v499/v503 citation sites. Wolfram: four mirrors (the complete square-root solve;  $D_4$  center + order-4 counts + the silver counterexample; the Cl(16) identity  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the split R control; the  $n$ -arithmetic + generic kill) verified with the active engine the same day (434  $\rightarrow$  438, ALL WOLFRAM EXTENSION CHECKS PASSED); the 24-triple stabiliser enumeration and the  $A_4$  profile stay Python-side (flagged). VerificationDag: v506 on the P2/carrier node; registry + run\_all (500 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XV) (v505 — WP5e- $\beta$  of CELEST.SEAM.01 “the equivariant anomaly ledger on twistor space” promoted as CELEST.WP5E.BETA.01: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators  $(2, 4, 2)$  with Dedekind sum  $\frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ , equivariant characters  $(248, 0, -8, 0)$  by two routes, invariant Okubo  $36\langle x, x \rangle^2$  with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual  $32T_3$ , the index bridge  $f(m) = \text{ch}_2(T_m)$  exact with glue defect  $-78$  both routes, the level dials  $k = 1$  geometric (current count  $240/0/0/0$ ), and the honest E3 REFUTATION:  $a_0$  fills the GRAVITON slot  $O(2)$ , NOT the axion slot  $O(-2)$  — the three sphere classes fill the three twisted axion slots instead; the preregistered kill (“inflow demands a level  $\neq 1$ ”) does NOT fire on the equivariant skeleton; WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed, the global BCOV quantisation stays [O]; Wolfram extension  $429 \rightarrow 434$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py** (CELEST.WP5E.BETA.01, 47 checks,  $\sim 1.3$  s; sympy/Fraction exact, no floats; deterministic). The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $E_8$   $\mu_4$ -glue monodromy. *AB ledger*:  $\det(1 - g_j^{-1}) = (2, 4, 2)$ ;  $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ ;  $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$  two routes; invariant average  $60 = \text{carrier}$ ; Frobenius  $61568$ ;  $\text{tr}_{g_2} = -8 = 120 - 128$ . *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ( $Q^{(0)} = 36\langle x, x \rangle^2$ ,  $36 = \tilde{\chi}_{e_8}^2$  — v495 re-derived through the glue grading); the twisted sectors do NOT reduce ( $T_5$ :  $(-8, 32, -8)$ ,  $T_3$ :  $(48, -64, 48)$ ); the AB sum cancels the  $D_5$  quartic exactly and leaves the rigid  $32T_3$  (the only quartic-free weighting is the  $\mathbb{Z}_4$  projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities ( $\text{Tr}_{120}, \text{Tr}_{128}, \text{Tr}_{E_8} = 9(\text{tr}^2)^2$ ) exact. *Index bridge*:  $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$  EXACT (E1 = E2 arithmetic); glue  $\text{ch}_2$  defect  $-78$  both routes (classes  $(-24, -30, -24)$ ); Killing sums  $(320, 320, 240, 320)$ , total  $1200 = 2h^\vee(h, h)$ .  $k = 1$  *geometric*: lattice current count  $240$  at  $k=1$ , exactly  $0$  at  $k = 2, 3, 4$ ; embedding residual  $47k^2 + 219k - 266 = (0, 360, 814, 1362)$ ; integrality alone fixes nothing (honest, as v502); one scale  $\Rightarrow$  one level (v493 periods). *E3 refutation (prominent)*:  $a_0 \in O(8)$  fills the GRAVITON slot  $O(2)$ , NOT the BSS axion slot  $O(-2)$  ( $X = -2 \neq +2 = Y$ ; mismatch  $4 = |\mu_4|$  as a [C] coincidence) — “the theory brings its GS axion as  $a_0$ ” is REFUTED; instead the three  $H^2(\text{ALE})$  classes carry exactly the Coxeter eigencharacters  $\{i, -1, -i\} =$  the three twisted slots (bijection); slot count  $1+3 = 4 = |\mu_4|$  soft. *Negative controls*:  $\text{diag}(i, i)$  breaks the ledger four ways ( $\sum 1/\det = \frac{1}{4}$ ,  $\omega$ -weight  $-1$ ,  $5$  generators);  $SO(16)$  glue: characters  $(120, 0, +120, 0)$ , defect  $-30 \neq -78$ , bulk Okubo fails (independent 4th Casimir);  $k = 2$ : current count  $0$ ,  $c$ -prefactor  $-31/48 \neq -\frac{1}{3}$ . [C] dictionaries ( $\text{CS}(\rho) \bmod 1 = \text{discriminant weight}$ ;  $32 = (h, h) + (h', h')$ ;  $-30 = -h^\vee$ ;  $\Omega$ -contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the  $O(-2)$  bulk axion, CPS level-from-flux, the pairing  $32T_3 \leftrightarrow$  sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.BETA.01; CELEST.SEAM.01 updated to “WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed; the global BCOV quantisation stays [O] — with the rigid  $32T_3$  twisted residual and the sphere-axion pairing as the named remaining targets” at *unchanged* Open status. Papers: contracts WP5e bullet  $\rightarrow$  “ $\alpha + \beta$  done” with the  $a_0$  refutation as an honest finding (kill-test and honest-success paragraphs consistent); **tfpt\_3** celestial section becomes the *eleven-step* story (new Step 11 “the equivariant anomaly ledger on twistor space”, “what remains” = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge +  $-78$  defect; the per-sector Okubo sums over the  $240$  glue roots with the  $32T_3$  residual; the lattice current counts + embedding residuals; the

axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429 → 434, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag qft node gains v505; registry + run\_all (499 modules at this step; 500 with v506, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIV) (v504 — WP5d- $\beta$  of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ( $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ ), GF2 spans full, rank 32/32; disjoint exactly index 2 = the v501  $\ln 2$  bit), the entropic touching defect BOUNDED  $< \ln 2$  with the Ising  $\frac{1}{4}$ -exponent approach ( $p = 0.2444$ ) while the preregistered  $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate  $\pi K(1-x)/K(x)$  (1.3–2.0%) with EXACT orbifold inheritance ( $C \rightarrow -C$  + the  $\Lambda^2 C$  compound — Longo heredity), and Pimsner–Popa  $E(a) - a/2 = PaP/2$  identically ( $\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$ ,  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ,  $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$  — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension 424 → 429 (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)**

- **v504\_celestial\_wp5d\_klm\_completeness.py** (CELEST.WP5DB.01, 37 checks, ~12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval  $\mu$ -index; v501 (WP5d- $\alpha$ ) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the  $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd  $\otimes$  odd is the v501  $\ln 2$  bit, localised at the split point); the neutral  $U(1)$  algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law  $((c/3) \ln 2$  increments, 0.115525, dev  $1.2 \times 10^{-6}$  — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the  $\mathbb{Z}_2$ /orbifold deficit rises  $0.4045 \rightarrow 0.5899$  ( $N = 64 \rightarrow 4096$ ), stays  $< \ln 2$ , and  $(\ln 2 - \Delta_2) \sim N^{-p}$  with  $p = 0.2444$  vs  $2\Delta_\mu = \frac{1}{4}$  (Ising disorder exponent); split-point scan reflection-symmetric to  $3.9 \times 10^{-14}$ ; ED vN same direction; gap continuity to the v501  $x \rightarrow 1$  endpoint; honest note: “defect  $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded  $\Leftrightarrow$  finite index (Longo–Xu) is the correct statement; the  $U(1)$  control bursts  $\ln 2$  from  $L = 128$  and grows as  $\frac{1}{2} \ln \text{Var } Q_A$  (Klich–Levitov slope 0.10134 vs  $1/\pi^2 = 0.10132$ ). *Split/nuclearity*: coupling-block  $\sigma_k$  fall exponentially at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at  $d = 16/64/256/512$ );  $\sigma_1 \sim x^{0.5044}$  (vs  $\frac{1}{2}$ ); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*:  $P_A$  flips  $C \rightarrow -C$  ( $\sigma_k$  identical,  $8.3 \times 10^{-17}$ ) and the even-bilinear coupling Gram is the second compound  $\Lambda^2 C$  (singular values  $\{\sigma_a \sigma_b\}$ ,  $2.8 \times 10^{-14}$ ) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*:  $E(a) - a/2 = PaP/2$  identically; attainment in integers (all 16384 (even, odd) monomial pairs at  $N = 8$ : 8192 extremal with  $M^2 = 4$ ,  $\text{Tr } M = 0$ ,  $PMP = -M$ , 0 violations; 2048/2048 at  $N = 12$ ; pencil min  $\lambda = 0.5000000000$ );  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ; index consistency  $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$  and  $e^{2\Delta_\infty} = 4 = \mu$ ;  $U(1)$ :  $\lambda = 1/(m+1) \rightarrow 0$ . *Controls + honesty*: dimerised phase trivial (drift  $3.6 \times 10^{-15}$ ); wrong prescription  $\{1, P_{AB}\}$  deficit exactly 0;  $\mathbb{Z}_2$  finite-size honesty (14.9% below



$\ln 2$  at  $N = 4096$  — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is Xu’s (cited, not claimed), the concrete seam quotient net and the condensed  $(E_8)_1$  net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d:  $\alpha$  ( $\mu$ -index) +  $\beta$  (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (Xu’s theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet  $\rightarrow$  “both stages done” with the  $\beta$ -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group  $\lambda_{E_4} = \frac{1}{4}$  + index consistency; the  $\Lambda^2/\text{Minors}$  compound with Cauchy–Binet and  $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$ ; the  $U(1)$  dephasing  $\lambda = 1/(m+1)$  family) verified with the active engine the same day (424  $\rightarrow$  429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field’s global  $\det'$  winner is the *hexagonal* torus,  $\tau=i$  is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff*  $\tau=i$ , an isolated zero); and the residual *converges*: QGEO-R1  $\equiv$  P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420  $\rightarrow$  424 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays a declared axiom)**

- `v503_qgeo_emergence_light.py` (QGEO.EMERGE.LIGHT.01, 17 checks; `mpmath` 40 digits + v280-style lattice DtN + `sympy` exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*:  $\log F = \log(\tau_2|\eta|^4)$  (Osgood–Phillips–Sarnak) is exactly  $S$ -invariant ( $\text{dev} < 10^{-35}$ ), so  $t=1$  is symmetry-forced critical on the rectangular family  $\tau = it$ ; there  $g''(1) = -1.298 < 0$  (strict maximum) but the  $\tau_1$ -direction has  $h''(0) = +0.298 > 0$ :  $\tau=i$  is a *saddle* of the full moduli space; the only critical points are the CM points ( $E_2^*(i) = E_2^*(\rho) = 0$  at 40 digits, generic  $0.163 \neq 0$ ); the *global*  $\det'$  winner at fixed area is the *hexagonal* torus ( $\log F(\rho) = -1.03352 > \log F(i) = -1.05469$ , diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to  $8 \times 10^{-15}$  exactly at  $\tau=i$  and rises strictly monotonically both ways (min  $8.4 \times 10^{-2}$  away) — an isolated zero;  $R^2 = \mathbb{Z}_2$  exact as permutation matrices (the v492  $\mathbb{Z}_8$  bridge on the lattice);  $j(it) \geq 1728$  with equality only at  $t=1$ . *The  $\mathbb{Z}_2$  chain is modulus-blind*: 4 marks +  $[\mathbb{Z}_2, \Delta_t] = 0$  on *every* torus (v216 confirmed without mark input); pillowcase  $\det' = \frac{1}{2}$  torus  $\det'$  exactly ( $\pm$ -pair structure) — only the order-4 clock distinguishes  $\tau=i$ ;  $\text{Aut}(E_\rho) = \mathbb{Z}_6$  has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ( $j = 1728$  identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on  $H^1$  is a Coxeter element of  $W(A_3)$  (char poly  $\lambda^3 + \lambda^2 + \lambda + 1$ , order 4 =  $|\mu_4| = N_{\text{fam}} + 1$ ) = the P2 cusp-class carrier datum of v499 R1  $\Rightarrow$  QGEO-R1  $\equiv$  P2-R1, *one common residual*; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input [[C]], rest = its dynamical realisation [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data  $\Rightarrow$  order-4 clock”

is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker **[C]**/**[O]**; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), **tfpt\_3** Step 2 (light-converse sentence), **origin\_theory** necessity paragraph, **introduction/tfpt\_1/tfpt\_2** one-residual cross-references. Wolfram: four mirrors (CM-point identities  $E_2(i) = 3/\pi$ ,  $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$  exact,  $j/\text{DedekindEta}$  special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock  $\Rightarrow$  square + Coxeter) verified with the active engine the same day (420  $\rightarrow$  424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). **VerificationDag** masses node gains v503; registry + **run\_all** (497 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XII) (v502 — WP5e- $\alpha$  of CELEST.SEAM.01 “prefactor and level bookkeeping” promoted as CELEST.WP5E.ALPHA.01: the  $q^{-1/3}$  prefactor of  $E_4/\eta^8$  IS exact  $\mu_4$  vacuum-energy bookkeeping (the clock is inner  $\Rightarrow$  shift orbifold  $\theta = 0^8$ , common  $-c/24 = -\frac{1}{3}$ ; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and  $k = 1$  is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial  $31124 - 27000 = 4124$ ); honest sharpening: glue- $h$  integrality holds for ALL  $k$  and fixes nothing; the BCOV/Costello–Li twistor inflow stays **[O]** — WP5e is subdivided ( $\alpha$  done, WP5e proper open); Wolfram extension  $416 \rightarrow 420$  (verified engine run); SEAM.EQUIV.01 stays **[O]**, no marker moves)**

- **v502\_celestial\_wp5e\_prefactor\_level.py** (CELEST.WP5E.ALPHA.01, 33 checks, exact **sympy/Fraction**, no floats). The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor:* Hurwitz- $\zeta$  exact ( $\zeta(-1, \theta) = -B_2(\theta)/2$ ;  $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ ; Ising shift  $\frac{1}{16}$  per Majorana); the clock is *inner* ( $h = (2, 2, 2, 2; 0^4)$  and  $h' = (0^5; 1, 1, 1, -3)$  both read the class mod 4 on all 240 roots;  $(h, h) = 20$ ,  $(h', h') = 12$ , sum 32)  $\Rightarrow$  twist  $\theta = 0^8$  in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so  $E_{\text{osc}} = -\frac{1}{3} = -c/24$  is sector-independent ( $c = 8 = \text{rank} = \text{Sugawara } 248/31$ ; one common  $\eta^8$ ); sector ground exponents  $-\frac{1}{3} + (0, 1, 1, 1)$ ; *discriminant form = Casimir energy* two ways (spectral flow  $j^2(h, h)/32 \bmod 1$ ; free fermions exact:  $D_5 = 10$  Majorana  $\rightarrow \frac{5}{8}$ ,  $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$ , glue diagonal  $= 16 = 2^{g_{\text{car}}-1} = \text{the WP5d seam carrier} \rightarrow 1$ ); the rotation reading *fails* (deck  $\frac{3}{16} \neq \frac{3}{8}$  — “clock  $\neq$  deck” at the Casimir level); the four sector characters sum exactly to  $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$ . *Part B — the level:*  $k = 1$  forced three independent ways — (i) the current condition  $h(J) = k = 1$  (adjoint closure  $60+64+60+64 = 248$  at spin 1); (ii) the conformal embedding ( $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$ ;  $D_8: 128k(1-k) = 0$ ); (iii)  $c = 8 \Leftrightarrow 240(k-1) = 0$  (= the prefactor); plus the v498 singular-vector dial ( $\|(E_{-1}^\theta)^m 0\|^2 = m! k(k-1) \cdots (k-m+1)$ , singular vector at grade  $k+1$ ;  $31124 - 27000 = 4124$  at  $k=1$  only,  $\langle s|s \rangle = +4$  at  $k=2$ ). *Honest sharpening (mandatory):* glue- $h$  integrality  $h(J^j; k) = k(0, 1, 1, 1)$  is integer for ALL  $k = 1..8$  — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls:*  $D_8/SO(16)_1$  (same  $c = 8$ , SAME  $q^{-1/3}$  prefactor, but  $h = (0, \frac{1}{2}, 1, 1)$ : the discriminator is glue- $h$  integrality, not the prefactor;  $\Theta_{D_8} + \Theta_s = E_4$ );  $\mathbb{Z}_2$  rotation ( $E = +\frac{1}{6}$ ,  $h = \frac{1}{2}$ );  $\mu_4$  rotation (sector-dependent  $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$  — the common prefactor breaks); wrong  $k \in \{2, 3, 4\}$  fails all five dials coherently (residuals  $360/814/1362$ ;  $\langle s|s \rangle = 4/12/24$ ;  $31124 \neq 4124$ ). **[C]** Kac generation for  $k \geq 3$  ( $k = 1, 2$  machine-anchored by WP5b/WP5c). **[O]** the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion  $\lambda_g/(8\pi\sqrt{3})$  on  $\mathbb{P}T$ , the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level  $\neq 1$ ”) is not triggered

on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- $\alpha$  (done) [E]/[C]: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper [O]: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- $\alpha$  landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status  $\rightarrow$  “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]”. Wolfram: four exact mirrors (Hurwitz- $\zeta$  vacuum energies; discriminant = Casimir with the deck failure; the  $k$ -fixing equations incl. the “fixes nothing” integrality table; the character sums +  $D_8$  control) verified with the active engine the same day (416  $\rightarrow$  420, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

**2026-07-21 (XI) (v501 — WP5d- $\alpha$  of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the  $\ln 2$  offset at machine precision,  $\mu_{\text{gauged}} = 4 =$  the v490 census, and the condensation arithmetic  $16 \rightarrow 4 \rightarrow 1$  closes with  $\mu = 1$ ; the preregistered KILL (“ $\mu$ -offset  $\neq 0$  after condensation”) does NOT fire; contract now “WP1–WP5d executed; WP5e the single remaining milestone”; Wolfram extension 412  $\rightarrow$  416 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator —  $(E_8)_1 \Rightarrow \mu=1$  (Haag-dual two-interval net, one sector) vs  $SO(16)_1 \Rightarrow \mu=4$  — executed at the  $\alpha$  (lattice-index) stage on the v367 seam edge (16 critical Majorana layers,  $c_- = 8$ , v450-style ring; Peschel-exact, every Gaussian formula ED-validated to  $\leq 1.5 \times 10^{-15}$ ; the superselection identity  $[\rho, P_{AB}] = 0$  machine-exact). *The  $\mu = 1$  reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* —  $c$  fit 0.5000,  $\max |F(x)| = 5.97 \times 10^{-5}$  at  $N=512$ , falling monotonically to  $1.3 \times 10^{-6}$  at  $N=1024$  (Casini–Huerta extensivity as a lattice witness). *The  $\mu$ -offset*: Rényi-2 exactly  $\Delta_2 = -\ln[(1+R)/2]$ ; the 16-layer seam carrier saturates  $\ln 2$  at machine precision ( $|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$  at  $N=512$ ;  $R \sim N^{-0.500}$ ); index reading  $[F : F_{\text{even}}] = 2.000000$ ,  $\mu_{\text{gauged}} = 4.000000 =$  the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs  $16 \ln 2$  (the offset measures  $\ln[\text{index}]$ , not a blob). *Haag-duality indicator*: fermionic  $S(E) = S(E')$  to  $8.2 \times 10^{-12}$ ; the orbifold *breaks* complementarity ( $D(E) - D(E') = 0.427$  at  $a=64$ ); the complementary-pair sum is exchange-symmetric and  $\leq \ln 4 = \ln \mu(SO(16)_1)$  with max 1.2637 at the self-dual  $x = \frac{1}{2}$  — a *double-limit* statement (the first naive fixed- $x$  check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*:  $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$ ; KLM/Longo–Rehren  $16/4^2 = 1$  and  $16/2^2 = 4 \rightarrow 4/2^2 = 1$  (both routes exact);  $\Sigma d_i^2 = (4, 4, 1)$ ;  $\theta_v = 1$  exactly at  $\nu = 2c_- = 16$ , rivals  $\neq 1$  — so  $\mu = 1$  after condensation ([C] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*:  $\nu=1$  single Majorana (offset non-removable,  $\theta_v = e^{i\pi/8} \neq 1$  — the discriminator has teeth); trivial phase (all  $\leq 10^{-7}$ ); wrong sector sum  $\{1, P_{AB}\}$  (offset exactly 0). [O] WP5d- $\beta$  (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet  $\rightarrow$  “ $\alpha$  stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor



uplift) is the remaining open piece”. **tfpt\_3** celestial section: new Step 8 (one paragraph, conservative); **tfpt\_4** keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + **[C]** dictionary); CELEST.SEAM.01 status → “WP1–WP5d executed; WP5e **[O]** (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients;  $\Sigma d^2$  + index-reading arithmetic; 16-fold-way  $\theta_v$ ) verified with the active engine the same day (412 → 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). wolfram/README.md, GATE.WOLFRAM.02, tfpt\_safeguards Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + run\_all (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family  $\omega_w$  exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 =  $V(2\theta)$  weight by weight;  $|s\rangle$  IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); SEAM.EQUIV.01 stays **[O]**, no marker moves)**

- **v500\_celestial\_wp5c\_gns\_limit\_state.py** (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*.  $\omega_w$  is quasi-free: loop sector = the affine  $k=1$  vacuum  $n$ -point functions via the *machine-determined* compact anti-involution  $\theta(e_\alpha) = -e_{-\alpha}$  (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings  $x^{wr}$  (the exact Gibbs regulator,  $x$ -adic). *Convergence sharp*:  $\omega_w = \omega_\infty \bmod x^{N+1}$  for all  $w \geq N+1$  ( $N=8$ ), not yet at  $w = N$  — the WP5a threshold  $w = n+1$  holds *identically at the state level*; positivity of every  $\omega_w$  checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the  $E_8$  Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel =  $V(2\theta)$  *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at  $(0, \omega, \theta, 2\theta)$ ; direct ideal checks  $F_i|s\rangle$ ,  $F_j F_i|s\rangle$  norm 0.  $\mu_4$ -equivariance:  $\omega_w \circ \text{clock} = \omega_w$  for all  $w$ ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) =  $\Theta_{C_j}/\eta^8$  at  $q^2$  — the two-routes identity at the *state level*. *Resolution of WP5b S4.10*:  $|s\rangle$  is the zero vector of  $\text{GNS}(\omega_\infty)$  — the twisted two- $C_1$ -mode floor (6 quarters =  $q^{3/2}$ ) concerns only pre-limit objects (GNS has no state there,  $6 \bmod 4 \neq 0$ ). *Negative controls*:  $k=2$  ( $\langle s|s\rangle = +4$ , dominant blocks full rank — no ideal in the kernel);  $k=0$  (level-1 Gram  $\equiv 0$  — no 248 layer without the central extension);  $D_8$  (rank 2076 = 7380 – 5304, kernel =  $V(2\theta_{D_8})$  PSD, but  $h = (0, \frac{1}{2}, 1, 1)$ : one block of four — one-block completeness stays  $E_8/\mu_4$ -specific); wrong family  $E'_w = mw + r$  (the 248 layer erased, 710955  $\neq$  248); no damping (897266  $\neq$  248); *CCR obstruction*: at fixed radial pairing  $c > 0$  positivity forces  $\omega(bb^*) \geq c$  — no  $w$ -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. **[C]** quasi-free extension to the full  $*$ -algebra; Kac positivity beyond level 2. **[O]** local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website**. **tfpt\_research\_contracts**: the WP5c bullet → done with the core numbers (state-level threshold  $w = N+1$ , the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing v500\_celestial\_wp5c\_gns\_limit\_state.py; the WP5b handover marked answered. **tfpt\_3** celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity **[E]** + **[C]** identifications + **[O]** net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with  $\dim V(\omega) = 3875$  by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial +  $D_8$  census) verified with the active engine the same day (408  $\rightarrow$  412); the exact Gram/PSD computation stays Python-side (flagged). VerificationDag qft node gains v500; registry + run\_all (494 modules at this step, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor  $a = (1, 1, 2)$  typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the tfpt\_2 “balanced” selection typed as stability; Wolfram extension 403  $\rightarrow$  408 (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])**

- **v499\_p2\_weights\_deligne\_bg.py** (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank  $3 = \# \text{marks} - 1$ , v137/v228; cusp-class local monodromy  $\lambda^3 - 1$  and exact irreducibility  $|\langle U, M_0 \rangle| = 24 = |S_4|$ ,  $\sum_g |\chi|^2 = 24$ , v117; unitarity (U)), the Deligne canonical extension  $E$  of the flavor connection satisfies:  $\deg E = -4$  as a theorem (residue exponents  $\{0, \frac{1}{3}, \frac{2}{3}\}$  the unique lift to  $[0, 1)$ , trace 1 per mark; par-deg 0;  $\infty$  apparent, monodromy product = 1 exactly);  $h^0(E) = 0$  as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary  $h^0=0 \Leftrightarrow a_i \geq 1$ ); integrality = Birkhoff–Grothendieck. Hence  $\{a\} = \{1, 1, 2\}$  unique and  $g_{\text{car}} = e_2 = 5$ . *Selector typed*: the Schur–Horn permutohedron has integer points exactly  $\{(2, 2, 0), (2, 1, 1)\}$  (the two options of the tfpt\_2 “Global computation” theorem) and  $h^0=0$  kills  $\{0, 2, 2\}$  — the “balanced/minimal-variance” heuristic is now stability/positivity. *The  $h^1$  discrepancy, real and resolved*:  $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ , so “generations =  $H^1$  of the rank-3 bundle” is FALSE (negative control: forcing it demands  $\deg -6$  and loses uniqueness,  $e_2 \in \{9, 11, 12\}$  never 5); correct two-object typing:  $L = \mathcal{O}(-D)$  carries the generations ( $h^1(L) = 3 = N_{\text{fam}}$ , the v137 basis =  $H^0(\Omega^1(\log D))$ ),  $E$  carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator  $3 = N_{\text{fam}}$  forces the  $\mathbb{Z}_3$  cusp cover  $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$  (genus 2; NOT the pillowcase double, genus 1,  $j = 1728$  — the deck side);  $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ ; the regular-representation model lands on  $\{0, 2, 2\} =$  the *unstable* Schur–Horn companion ( $h^0 = 1$ ): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity  $\rightarrow$  4/31 solutions;  $\deg -3/ -5$  breaks;  $n$ -mark scan  $e_2 = (n-2)(n+1)/2 = 5$  ONLY at  $n = 4$  (the value, not the uniqueness, is the  $n=4$  content); MS-rational  $e_2 = \frac{16}{3}, \frac{44}{9}$ . *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s  $\text{diag } A_0 = a/4$  input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499” (v491 stays active, no supersede). Registry + run\_all (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control +  $h^1$  discrepancy; Deligne residue-trace degree with the  $A_0^*$  witness char poly; Schur–Horn lattice points; Biswas  $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double  $j = 1728$ ; the  $n$ -mark formula), verified with the active engine the same day (403  $\rightarrow$  408, ALL WOLFRAM EXTENSION CHECKS PASSED); wolfram/README.md + GATE.WOLFRAM.02 + tfpt\_safeguards Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): introduction anchor paragraph, tfpt\_1 (5, 3) keybox, tfpt\_2 H2 winbox and the “Global computation” theorem (the balanced selection typed as stability), origin\_theory (5, 3) keybox — all citing v499\_p2\_weights\_deligne\_bg.py; no status card changes. Website: VerificationDag P2

node gains **v499**; generated indices via build; no marker moved anywhere.

**2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **tfpt\_3\_e8\_audit\_bootstrap:** new section “The celestial route: from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow”. The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible  $\pi$ ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the  $E_8$   $\mu_4$ -glue as a flat  $\mathbb{Z}_4$  monodromy on the  $A_3$  ALE space,  $\text{clock}^2 = \text{deck}$  ( $\text{spin } \mathbb{Z}_8, 8 = 2|\mu_4|$ ) [E] (v492); (2) deformation —  $XY = Z^4 + a_0$  pure scale,  $j = 1728$  frozen,  $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$  [E] (v493); (3) consistency honestly demoted —  $(\kappa/c_3)^2 = 12$  exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the  $(E_8)_1$  character is not a jet-grading conformal block (growth  $n^{2/3}$  vs  $n^{1/2}$ ) but the  $\mu_4$  period sum 248 holds exactly: a boundary-limit shadow of SEAM.EQUIV.01/MMST shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold  $w = n+1$ ,  $27000 = (h^\vee)^3$  derived, quotient = sector sum = 4124,  $SO(16)_1$  contrast [E] (v497); (6) the deleting operator explicit —  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ ,  $J_1^a|s\rangle = 0$  on all 248 (tally 190/57/1), level dial  $2(1-k)$ , one-block closure  $E_8/\mu_4$ -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close SEAM.EQUIV.01; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. **tfpt\_research\_contracts:** a reader’s note at the head of the CELEST.SEAM.01 contract cross-references the new Paper-3 section (the contract stays the full technical reference). **tfpt\_4\_frontier:** one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

**2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v498\_celestial\_wp5b\_singular\_vector.py** (CELEST.WP5B.01, **53 checks, exact integer/Fraction, deterministic, no floats**). Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction:*  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$  (weight  $2\theta$ , level  $2 = 8$  quarter units =  $q^2$ , an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the  $[e_\alpha, e_{-\alpha}]$  sign *forced* by the Jacobi identity ( $\text{SGN} = -1$ , not chosen), the invariant form  $\kappa$  derived ( $\kappa(\theta^\vee, \theta^\vee) = 2$ ), Jacobi exact on all 57600  $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496  $\theta$ -adjacent, all  $N_{\alpha\beta} = \pm 1$ . *Singularity (machine):*  $J_1^a|s\rangle = 0$  for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ( $a = F^\theta$  via the central-term cancellation — the *only* case that sees the level  $k$ ); plus  $J_2^a|s\rangle = 0$ ,  $E_0^a|s\rangle = 0$ , exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial:* the  $F_1^\theta$  coefficient

is  $2(1-k) - 2/0/-2$  at  $k = 0/1/2$ : without the central extension the deletion operator does not exist.  $\mu_4$  data: glue class  $j(\theta) = 1$  (machine-built  $h$ -adapted chamber,  $\langle \theta, h \rangle = 5$ , height  $29 = h^\vee - 1$ ), clock phase  $i^{2j} = -1$ , class( $2\theta$ ) = 2 sheet-even; the deleting  $\theta$ - $\mathfrak{sl}_2$  crosses the sheet-odd classes (1, 3) while its square is sheet-even; cross-table  $\theta$ -grading  $\times$  glue class consistent (rows (1, 56, 126, 56, 1), columns (52, 64, 60, 64)); 8 quarters =  $q^2$  via the per-period dictionary. *Level-2 match*: weight  $2\theta$  has multiplicity 1 in the level-2 Fock, so  $U(\mathfrak{g})|s\rangle$  is the  $27000 = (h^\vee)^3$ ; the direct  $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of  $V(2\theta)$  exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient  $31124 - 27000 = 4124$ ; lattice reading  $(2\theta, 2\theta)/2 = 4 > 2$ :  $|s\rangle$  is the kernel generator of  $\text{Fock} \rightarrow E_4/\eta^8$ . *Negative controls (honest separation)*: the level-1 current state  $E_{-1}^\theta|0\rangle$  (one witness,  $\rightarrow -k|0\rangle$  — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at  $k=2$  the *cube* is (generic Kac  $(E^\theta)^{k+1}$  mechanics, honestly typed);  $SO(16)_1$  has the same singular vector but keeps three extra level-1 primaries ( $h = \frac{1}{2}$  breaks one-block fusion;  $5304 \neq 14^3$ ) vs  $E_8$  (comarks all  $\geq 2$ , glue weights (0, 1, 1, 1) integer) — the *one-block closure* is the  $E_8/\mu_4$ -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- $C_1$  modes never sum to 8 quarters (minimum  $6 = q^{3/2}$ ) — the per-period dictionary (v496), not the per-slot identification, carries  $|s\rangle$  to  $q^2$ : exactly the WP5c GNS/limit-state question; WP5c-e stay open; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial  $2(1-k)$ ;  $j(\theta) = 1$ , clock phase  $-1$ ; orbit BFS through depth 4; the  $D_8$  contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O]”). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O] (roadmap)”). Wolfram: 5 exact v498 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (398  $\rightarrow$  403, ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c-e roadmap”), DAG `qft` node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b-e [O] (roadmap); Wolfram extension 394  $\rightarrow$  398 (verified engine run); no gate closes, no marker moves, SEAM.EQUIV.01 untouched)**

- **v497\_celestial\_wp5a\_null\_ideal.py** (CELEST.WP5A.01, 34 checks, exact integer/Fraction, no floats). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family  $\chi_w$  of graded Fock characters on the chiral jet generators (loop energy  $m$ , radial number  $r$ ;  $E_w = m + wr$  in quarter units  $u = q^{1/4}$ ) contains the chiral jet grading as its  $w=2$  member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its  $u^n$  coefficient equals the quarter-moded loop Fock for *all*  $w \geq n+1$ , strictly larger for every  $w \leq n$  (verified  $n \leq 8$ ,  $w \leq 10$ ) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit  $\varepsilon = 1/w \rightarrow 0$  with explicit threshold  $w = n+1$ , not a slice (convention clarified: 897266 = the quarter-moded loop Fock at  $u^4 = \text{integer level 1}$ ). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give  $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$  with residual exactly zero (independent Weyl-orbit cross-check);  $27000 = 30^3 = (h^\vee)^3$  exactly;  $C_2 = (124, 96)$ , Dynkin indices 6750/750; level-1 integrability ( $\langle 2\theta, \theta^\vee \rangle = 4$ ,  $\langle \omega, \theta^\vee \rangle = 2$ ) leaves the vacuum as



the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient  $31124 - 27000 = 4124 = 1 + 248 + 3875$  exactly. *Two routes, one number*: the  $\mu_4$  theta-split sector sum at  $q^2 = (1036, 1024, 1040, 1024)$  gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space ( $q^3$ :  $34752 = \chi_3$ ). Negative controls:  $SO(16)_1$  through the same pipeline gives *four* components ( $5304 + 1820 + 135 + 1$ ),  $5304 \neq 14^3$  (the  $h^{\vee 3}$  identity is not generic), quotient  $2076 = \Theta_{D_8}/\eta^8$  at  $q^2$  (the recipe validated on a second algebra), block weights  $(0, \frac{1}{2}, 1, 1)$  break one-block fusion (vs  $E_8$ 's integer  $(0, 1, 1, 1)$ ); only  $P = 4 = |\mu_4|$  periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ( $64 \neq 60$ ). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold  $w = n+1$ ;  $27000 = (h^{\vee})^3$  derived; quotient = sector sum = 4124;  $SO(16)_1$  contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel  $\supseteq$  ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function =  $E_4/\eta^8$  incl. the  $q^{-1/3}$  prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $394 \rightarrow 398$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors  $116/116 + 398/398$ . Website: `papers.ts` contract mirror (“WP5a done, WP5b–e roadmap”), DAG `qft` node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak  $F_0 = 11.19275$  Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute  $\varepsilon$ ), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)**

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs  $\rightarrow$  491 usable  $\rightarrow$  119 segment-coherent  $\nu(t)$  points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection”  $F_0 = 11.19275$  Hz is corrected on record (true Vela  $F_0 = 11.1861692$  Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear*  $\delta\nu(\tau)$  residual ( $\mu$ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches  $\varepsilon_{90}$  below  $\varepsilon = 1$  ( $\varepsilon_{50} = 0.55$  in 2019 only)  $\Rightarrow$  no absolute per-channel  $\varepsilon$ -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude  $\sqrt{2g} = 0.40$  (leg gains 0.046/0.114, min  $p = 0.014$  — the 2019 audit-level chance feature: Bonferroni  $\times 10 = 0.14$ , shuffle  $p = 0.40$ ).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute  $\varepsilon$ ); all-channel universal-DSI 95% UL  $\varepsilon < 0.11954$  ( $\approx 12.0\%$ ; HKSJ  $t_2 = 2.92$ ,  $\rho = 0.3$ -inflated 0.137;  $6.91\times$  the predicted  $\varepsilon = 0.0173$ ), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5%  $\rightarrow$  recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list,

PG.06b FULL bullet), `results/results.json`, scorecard row `data_value/source` via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

**2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)**

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; `tfpt_readouts_extension.wl` → 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); v479 registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

**2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v495\_celestial\_wp3\_gs\_alignment.py (CELEST.WP3.01, 25 checks, exact Fraction/sympy).** The pre-registered WP3 “alignment test only”: the Okubo coefficient  $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$  is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ( $\text{sl}_2, \text{sl}_3, \text{g}_2, \text{so}_8, \text{f}_4, \text{e}_6, \text{e}_7, \text{e}_8$ ;  $\text{e}_6/\text{e}_7$  carved out of the v492  $E_8$  root set; negative control:  $\text{sl}_4$  fails,  $5/32$  vs  $3/32$ ); the closed form  $\tilde{\lambda}^2 = 10h^{\vee}/(\dim + 2) = h^{\vee} + 6$  holds exactly across the Deligne series;  $E_8$ :  $\tilde{\lambda} = 6$  (unit-trace) resp.  $\lambda_{\text{fund}} = 1/10$  (adjoint-trace) exact, so  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$  resp.  $1/300$  — with  $\kappa/c_3$  itself *irrational* ( $2\sqrt{3}$ ). Byproduct: the printed  $\lambda_{\text{so}_8}^2 = 3/2$  in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity),  $\tilde{\lambda}$ -integrality  $2/8$  (shared with  $\text{sl}_3$ ), only  $g_{\text{car}}=5 \mid h^{\vee}$  is  $\text{e}_8$ -selective ( $1/8$ ), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not  $\text{e}_8$ -selective evidence. Verdict B.
- **v496\_celestial\_wp4\_character\_shadow.py (CELEST.WP4.01, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the  $(E_8)_1$  character  $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$  is *not* a conformal block of the glue-equivariant  $E_8[\mathbb{C}^2]$  S-algebra in its own jet grading — localised three ways: (a) spin  $1 - d/2$  unbounded below and 248 never a tower dimension ( $d \leq 100$ ); (b) growth  $n^{2/3}$  vs  $n^{1/2}$  (quadratic cumulative count  $31s^2 + 92s + 60$ ,  $31 = k + h^{\vee}$ ;  $f_n/\chi_n$  strictly increasing  $n = 1..12$ ); (c) the level-2 null ideal deletes exactly  $27000 = 30^3 = (h^{\vee})^3$  out of  $\text{Sym}^2(248) = 1 + 3875 + 27000$  — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes =  $60 = C_0$  currents, jet slice cycles  $(60, 64, 60, 64)$ , one  $\mu_4$  period sums to 248, glue weights  $(0, 1, 1, 1)$  integer; free loop Fock  $897266 \gg 248$ , so the truncation must come from a limit) — the same MMST

scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray  $\nu(t)$  leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) → 491 usable → 240 per-obs detections → **119 segment-coherent  $\nu(t)$  points** (108 fully coherent), median  $\sigma_\nu \approx 0.6 \mu\text{Hz}$ ; barycentring validated against PINT to  $3.4 \mu\text{s}$  — the *first X-ray  $\nu(t)$  probe* of the search program. Sanity gate PASS:  $\Delta\nu/\nu$  2019 =  $2.39 \times 10^{-6}$  vs  $2.5012 \times 10^{-6}$  published, 2021 =  $1.272 \times 10^{-6}$  vs  $1.26 \times 10^{-6}$ ; the 2024 glitch is not reducible (only  $\sim 150$ -s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ( $\omega = 2.5827$ ,  $\varepsilon_{\text{pred}} = 0.0173$ ): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods ( $<$  the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw  $p = 0.014$  has the kernel as smallest- $p$  battery member BUT Bonferroni  $\times 10 = 0.14$  and shuffle  $p = 0.395 \Rightarrow$  chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021  $p = 0.731$ ;  $\ln \tau$  stack  $p = 0.049$ /shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise:  $\varepsilon_{50} \approx 0.55$  (2019 only),  $\varepsilon_{90}$  never reached below  $\varepsilon = 1$ , 0% power at the predicted  $\varepsilon = 0.0173$  — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag:  $\lambda = e$  in the 2021 leg (raw  $p = 0.002$ , Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider:  $<3$ -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs “pipeline proven” detection at  $F_0 = 11.19275 \text{ Hz}$  ( $H = 18.4$ ) was a noise peak; the true Vela frequency at that epoch is  $F_0 = 11.1861692 \text{ Hz}$  ( $H = 99.7$ , consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray  $\nu(t)$  (PG.06b FULL)” (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.



**2026-07-21** (the two 2026-07-20 prediction-of-record rows promoted to the suite: **v494** —  $\alpha_s$  running +  $\mu$ -distortion branch discriminator machine-checked, ledger rows **COSMO.RUNNING.01/COSMO.MUDIST.01**, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494\_cosmo\_running\_mu\_record.py** (23 checks, all green). The  $\alpha_s$ -running and  $\mu$ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact):  $\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6$ ,  $\beta_s = -4/N_*^3$  (NOT  $-8/N_*^3$ ); LO band  $N_* \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ . [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check < 1%):  $\alpha_s(51.4) = -7.09 \times 10^{-4}$ ,  $\beta_s(51.4) = -2.70 \times 10^{-5}$ . [C] data: Planck 2018 ( $-0.0045 \pm 0.0067$ ) +  $0.57\sigma$ ; record leg P-ACT-LB ( $+0.0062 \pm 0.0052$ )  $-1.33\sigma$  (watch channel: positive central value; kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch). [C]  $\mu$ -distortion (Chluba window, +21% calibrated on the  $\Lambda$ CDM anchor): sharp branch  $N_*=51.4$   $\mu = 1.6 \times 10^{-8}$ , profiled record  $N_*=56.1$   $\mu = 1.94 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; branch split  $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$   $A_s/N_*$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; PIXIE baseline  $9 \times 10^{-8}$  does not reach; kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ ). [E] comb blindness (honest S15): the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is structurally invisible in the  $\mu$  window ( $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage-2050). Fences:  $N_*$  stays an external reheating input (GATE.NSTAR.01); the frozen band  $[50, 60]$  stays the surface of record (v84/v86). Ledger rows **COSMO.RUNNING.01** and **COSMO.MUDIST.01** added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. **predictions.txt** gains the two rows ( $\alpha_s$  [C] with the  $-1.33\sigma$  P-ACT-LB pull and the positive-running kill;  $\mu$ -distortion [C] data-limited with the Voyage-2050 branch split); website **lib/predictions.ts** gains the matching cards **alpha-s-running** and **mu-distortion** (confrontation + experiment blocks pointing at **experiments/tfpt-discovery**); the cosmology DAG node lists **v494** and states the running/ $\mu$  readouts.
- **Paper bodies**. **tfpt\_1** (Inflation interface) gains the running consistency-relation paragraph ( $\alpha_s = -2/N_*^2 = -r/6$ , exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; **tfpt\_2** (inflation-pivot keybox) gains the  $\mu$ -distortion branch-decider sentence (sharp  $1.6 \times 10^{-8}$  vs profiled  $2.0 \times 10^{-8}$ ,  $3\sigma$  at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

**2026-07-20 (VIII)** (E8 mass-ladder blind-recovery bed EXECUTED: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov–E8 spectra of  $\text{BaCo}_2\text{V}_2\text{O}_8$  (8/8) and  $\text{CoNb}_2\text{O}_6$  (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8\_ladder\_bed\_v1.yaml)**. The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. **hfqpo-ladder**) on the only public *known-answer* mass-ladder dataset:  $\text{BaCo}_2\text{V}_2\text{O}_8$  (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**,  $\chi^2/\text{dof} = 0.35$ , MC  $p = 0.0056$  (singles-only run  $p = 0.0004$ ;  $\sigma$ -variant robust);  $\text{CoNb}_2\text{O}_6$  (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**,  $\chi^2/\text{dof} = 0.52$ ,  $p = 0.0038$  ( $m_7/m_8$  lie outside the 0.6-THz low-pass window; the broad 0.43-THz ( $m_1+m_2$ )-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of  $2I - A(E_8)$  equals the Zamolodchikov ladder to  $1.5 \times 10^{-15}$  ( $m_2/m_1 = \varphi$  to  $2.2 \times 10^{-16}$ ; lowest Cartan

eigenvalue  $2 - 2\cos(\pi/30)$  to  $4.5 \times 10^{-14}$ ).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset  $\pm\{0.3, 0.5\}m_1$ : 0/8 detections; jitter dose-response  $\leq 5\%$  at the destructive 30% level; wrong ladders on the real data (harmonic 1..8,  $A_3$ -PF,  $D_8$ -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates,  $\chi^2$ -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked  $\rightarrow$  consistent (stage not\_applicable, tier analog\_positive\_control, leakage detector\_control — the executed-analog basket of qc-recovery-kernel and qgeo-eit-soff); counters 120 rows: 49 consistent / 3 parked; experiments/README.md catalog row + stats (generator); website predictions.ts EXPERIMENTS\_AUDIT counters; experiments/next.txt. This validates the ladder-detector pipeline — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [E], no \veri, no ledger row.

**2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation  $XY=Z^4+a_0$  is a PURE SCALE with the  $\tau=i$  pillowcase frozen, the  $\mu_4$  clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$  with no sector leak; verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O])**

- **v493\_celestial\_wp2\_clock\_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the experiments/tfpt-discovery/probe celestial\_seam\_wp2\_clock\_deformation\_probe.py. *Geometry:*  $P(iZ)=P(Z)$  forces  $a_3=a_2=a_1=0$  sharply and two-sided ( $\prod_k (Z - i^k w) = Z^4 - w^4$ );  $XY=Z^4+a_0$  is smooth iff  $a_0 \neq 0$  (disc =  $256 a_0^3$ ); the branch points  $Z^4 = -a_0$  are one free  $\mu_4$  orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of  $w^2=Z^4+a_0$  are  $I=12a_0$ ,  $J=0$  *identically*, so  $j=1728$  and the  $\tau=i$  pillowcase shape (v168/v214) is *frozen* for all  $a_0$  —  $a_0$  is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:*  $\Pi = t(i-1)(1, i, i^2)$ , all three vanishing spheres carry the same volume  $\sqrt{2}t$  (lockstep); the clock acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3+x^2+x+1$ , eigenvalues  $\{i, -1, -i\}$  = the  $A_3$  exponents,  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ), fixes no cycle ( $\det(A-1)=-4$ ), and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$ , *not* the naive invariant diagonal; the versal weights  $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$  are the regular  $\mu_4$  representation. *Algebra (BHS  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ , arXiv:2305.09451):* mode dictionary  $\{w[m, n] : m \equiv n \bmod 4\}$ ; the fibre bracket  $-4\text{Nambu}(XY-Z^4-a_0)$  anchored at the  $a_0=0$  orbifold ( $\{X, Y\}=16Z^3$ , Jacobi + Casimir exact, uniform clock-degree  $-1$ ); S-algebra corrections exactly *linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade — no sector leak, the v492 equivariant sector deforms consistently;  $a_0 \leftrightarrow$  BHS’s weight-0  $c^2$  slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/ $\mathbb{Z}_2$  middle node,  $4=2 \times 2$ ). *Negative controls:*  $a_1 Z \Rightarrow j=0$  (hexagonal); pure  $a_2 Z^2$  stays singular ( $A_1$ );  $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$ ; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications ( $-4\text{Nambu}$  = the  $k=4$  CCA bracket; period = root difference; seam-scale reading via  $\text{clock}^2=\text{deck}$ ); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** tfpt\_research\_contracts CELEST.SEAM.01 section: WP2 bullet “finite, next”  $\rightarrow$  *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction =  $\chi_1$ -diagonal,  $a_0$  pure scale); kill-test paragraph K1

“survived by WP1 and WP2”; status line CELEST.WP2.01 [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02 (four exact v493 mirrors added to tfpt\_readouts\_extension.wl, counted with the next verified engine run — engine unactivated, same convention as v491); registry + run\_all (487 modules, suite green); website mirrors (papers.ts contract section, generated script index). SEAM.EQUIV.01 stays [O]; no marker moves anywhere. Exact (sympy).

**2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data\_limited→null, no marker moves)**

- **H3 stage 2 executed (experiments/hfqpo-ladder/results/archival/).** The preregistered archival scan (hypotheses/hfqpo\_v1.yaml, archival\_design) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms  $4.4\sigma$ , 280.1 Hz at 3.71%  $9.9\sigma$ ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth  $\nu_3 = 1.5\nu_u$  (661.5/414/252/363 Hz) nor the integer control line  $4\nu_0$  (588/368/224/322.7 Hz) is detected in any source ( $\sim 0\sigma$  single-trial; trials correction  $N=8$  as preregistered).  $3\sigma$  rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the harmonic\_hit branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) data\_limited→null (120 rows: 26 null / 33 data\_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in tfpt\_horizon\_readouts (+ the papers.ts mirror); website predictions.ts (hfqpo-tooth entry + EXPERIMENTS\_AUDIT counters); predictions.txt; experiments/README.md; experiments/next.txt. No verification/ module, no ledger row, no marker move (the search target stays [O]).

**2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant celestial chiral algebra on the  $A_3$  ALE space, with the exact correction clock<sup>2</sup>=deck (spin bridge  $\mathbb{Z}_8$ ,  $8=2|\mu_4|$ ); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])**

- **v492\_celestial\_z4\_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128  $\mu_4$ -glue grading of  $E_8$  ( $240 = 52+64+60+64$ , additive on all 6720 root pairs) is *inner* —  $h = (2, 2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$ , an order-4 inner torus automorphism = a flat  $\mathbb{Z}_4$  monodromy in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the same class = the diagonal (1, 1) of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  (the v92/v125 Lagrangian glue).  $\mathbb{C}^2/\mathbb{Z}_4$  is verified as the  $A_3$  singularity  $XY=Z^4$  (Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ , Molien exact); the glue-equivariant SDYM( $E_8$ ) sector *closes* with graded dimensions  $60(d+1)/64(d+1)$  — possible only because  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$  — with zero modes = the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$  and density  $1/4 = 1/|\mathbb{Z}_4|$ ; the four glue-sector characters sum exactly to  $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$  (v377), with sector weights = the discriminant form  $(5x^2+3y^2)/8$  and integer glue-diagonal  $h = (0, 1, 1, 1)$  (= locality of the  $(E_8)_1$  extension). *Critical correction*

(*verdict B*): the  $A_3$  deck acts on the celestial sphere as  $z \mapsto -z$  (order 2), *not* as the order-4 clock; the clock is the  $U(2)$  phase  $\text{diag}(i, 1)$  (normalising the deck,  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator), with the exact spin bridge (spin clock)<sup>2</sup> = deck ( $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$ , the  $c_3 = 1/(8\pi)$  winding integer). Clock-invariance selects the 1-parameter  $A_3$  deformation  $XY = Z^4 + a_0$  whose four branch points are one  $\mu_4$  orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action  $\text{diag}(i, i)$  killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity =  $\text{Aut}(\mathbb{Z}_4)$ .

- **New research contract CELEST.SEAM.01** in `tfpt_research_contracts` (own section, the fourth contract alongside  $U_{\text{wall}}/G_{\text{metric}}/\text{CONTRACT.F.01}$ ): hypothesis (seam = glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the CCA on the  $A_3$  ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum  $(E_8)_1$  existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (`papers.ts` section, script index). SEAM.EQUIV.01 stays **[O]**; no marker moves. Exact (sympy).

**2026-07-20 (IV) (evidence scorecard: five new typed rows — the  $\alpha_s$  running prediction of record, the  $\mu$ -distortion  $A_s/N_*$  branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)**

- **$\alpha_s$  running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index:  $\alpha_s = -2/N_*^2 = -7.6 \times 10^{-4}$  at  $N_*=51.4$  (exact slow roll  $-7.09 \times 10^{-4}$ ; band  $N_* \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ ) and  $\beta_s = -4/N_*^3$  (leading order; NOT  $-8/N_*^3$  — the identity  $\alpha_s = -r/6$  holds only at leading order ( $\sim 7\%$  off exact)). Data: Planck 2018  $-0.0045 \pm 0.0067$  ( $+0.57\sigma$ ), P-ACT  $+0.0060 \pm 0.0055$  ( $-1.22\sigma$ ), record leg P-ACT-LB  $+0.0062 \pm 0.0052$  ( $-1.33\sigma$ , Calabrese et al. 2025 JCAP);  $\beta_s$  (Planck run+runrun)  $0.010 \pm 0.013$  ( $-0.77\sigma$ ). Forecast: LiteBIRD+CMB-S4  $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$  ( $0.14\sigma$ ); CMB-S4+DESI+Euclid statistical  $\sim 3.3 \times 10^{-4}$  ( $2.2\sigma$ , systematics-limited). Same `N_star_reheating` independence group as the  $n_s/r/A_s$  rows (anti double-counting: never an independent hit); kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **$\mu$ -distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch  $N_*=51.4$  ( $A_s=1.76 \times 10^{-9}$ )  $\mu \approx 1.6 \times 10^{-8}$ , profiled record  $N_*=56.1$  ( $A_s=2.10 \times 10^{-9}$ )  $\mu \approx 2.0 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; the 23% branch split ( $\Delta\mu = 3.0 \times 10^{-9}$ ) gives a  $3\sigma$   $A_s/N_*$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; the PIXIE baseline  $|\mu| < 9 \times 10^{-8}$  does NOT reach — factor 4–7 above the band; FIRAS  $|\mu| < 9 \times 10^{-5}$ ). Honest S15 typing: the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is STRUCTURALLY BLIND in the  $\mu$  window — window attenuation  $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two  $A_s$  rows (one branch question, several readings). Kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ . Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (parked\_analog; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall:  $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2 / 3h)$  with no charge Hall (seam edge  $c_-=8$ , v450/v451/v489); no engineered material yet ( $\alpha\text{-RuCl}_3$   $c_-=1/2$  contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated  $\kappa_{xy}$ , PRX 12, 021025). (ii) E8 mass-ladder validation bed ( $\text{CoNb}_2\text{O}_6/\text{BaCo}_2\text{V}_2\text{O}_8$ ): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201,



NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz  $m_1$ – $m_6$ ; Coldea et al. 2010 Science  $m_2/m_1=\varphi$ ) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning:  $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$  at  $M/m = 8.67$  (2 resonant pairs); best candidate  $^{52}\text{Cr}_2$ – $^6\text{Li}$  ( $M/m=8.635$ ,  $s_0=1.2890$ ,  $\lambda=11.44$ ,  $+0.44\%$  vs  $(3/2)^6$ , gap deviation  $+0.18\%$ ); alternatives  $^{39}\text{K}_2$ / $^{41}\text{K}_2$ – $^6\text{Li}$  ( $\lambda = 11.59/11.20$ , needs double Feshbach control),  $^{176}\text{Yb}_2$ – $^{23}\text{Na}$  ( $+5.9\%$ ); solver validated on the Naidon–Endo anchors (homonuclear 22.69,  $\text{Cs}_2\text{Li}$  4.877); first end-to-end HARDWARE injection of the comb detector at tunable  $\lambda \approx 11.39$ ; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

**2026-07-20 (III) (v491 —  $g_{\text{car}}=5$  as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely  $\{1, 1, 2\}$ , sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)**

- `v491_p2_partition_corollary.py` (P2.PARTITION.01). Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE —  $\{1, 1, 2\}$  — so  $e = (e_1, e_2, e_3) = (4, 5, 2)$  and hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = e_2 - e_3 = 3$  are *corollaries*, not inputs (consistency fixed point  $(2^4 - 1)/5 = 3 = \text{rank } H^1$ ). [E] negative controls, each assumption load-bearing: drop positivity  $\rightarrow$  4 solutions,  $e_2$  ambiguous  $\{0, 3, 4, 5\}$ ; sum 3  $\rightarrow e_2=3$ ; sum 5  $\rightarrow$  not unique; 2/4 parts  $\rightarrow e_2 \in \{3, 4\}/\{6\}$ ; Mehta–Seshadri RATIONAL weights  $\rightarrow e_2 = \frac{16}{3}$  resp.  $\frac{44}{9} \neq 5$  (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins  $(1, 1, 2)$  with ALL THREE targets  $(4, 5, 2)$ ; the partition needs only  $e_1=4$ . [E] sum rule:  $4 \times \frac{N-1}{2} = 4$  iff  $N=3$ ; the v115 route  $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$  agrees. Independence chain: #marks = 4 from v216 (P1-side, no  $g_{\text{car}}$  input); rank  $H^1 = 3$  pure topology (v137/v228);  $|a|_1 = 4$   $g_{\text{car}}$ -free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity  $h^0=0/\text{sum}$  “given (U)”); AX.P2.01 stays a declared axiom, and the  $N_{\text{fam}}$  direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in introduction (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

**2026-07-20 (II) (v490 — the  $T^2$  spin-structure parity census:  $\mu_{\text{pre}}=4$  witnessed on the lattice, the 16-fold-way bridge  $\theta_v=1$  on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)**

- `v490_seam_parity_census.py` (SEAM.MU.01). The  $T^2$  ground-state fermion parity of the v367  $p+ip$  collar, resolved per spin structure ( $AA, AP, PA, PP$ ) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial  $M=3$  phase, with the twisted real-space spectra matching the Bloch bands to  $9 \times 10^{-15}$ . [E] per layer  $(+, +, +, -)$  (the  $\nu=1$  Ising signature, TRIM  $\equiv$  Pfaffian in every sector); [E] the 16-layer carrier has parity<sup>16</sup> = + in ALL FOUR sectors  $\Rightarrow$  gauged torus GSD = 4 =  $\mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$  — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$  before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers:  $\theta_v = e^{2\pi i \nu/16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489), rivals  $\nu = 1, 2, 8$  not holomorphically condensable; [E]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH  $M=1$  and  $M=3$  ( $\gamma_f=0.000001/0.000000$ ) — it does NOT discriminate  $(E_8)_1$  vs  $SO(16)_1$ . [C] the KLM step  $\mu: 4 \rightarrow 4/2^2 = 1$  stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$ . [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact  $\theta_v$ ).

**2026-07-20 (v489 — the chiral central charge  $c_-=8$  measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); no gate closes, no marker moves)**

- `v489_seam_modular_commutator.py` (SEAM.CCC.01). The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367  $p+ip$  seam collar model, entirely from the Peschel Majorana covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ,  $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$ ), on a disk about the torus centre cut into three  $120^\circ$  wedges. [E] ground-state purity  $\text{eig}(i\Gamma) = \pm 1$  to  $6 \times 10^{-15}$ ; [E] per-layer  $c_- = 0.499998$  at  $L=32$  (convergence  $0.499793$  at  $L=16 \rightarrow 0.500000$  at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ); [E] controls: trivial  $M=3 \rightarrow 0.000000$ , orientation flip  $M=-1 \rightarrow -0.499986$ ; [E] layer additivity exact ( $1.4 \times 10^{-14}$ ); [E] the 16-layer carrier lands on  $c_-=8=g_{\text{car}}+N_{\text{fam}}$  (direct  $7.9967$  at  $L=16$ , additivity route  $8.0000$ ); [E] clip invariance  $5 \times 10^{-8}$ . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ( $c_-=8$ ) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

**2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; no result, no status change)**

- `introduction.tex` — new section “TFPT in the light of algorithmic physics”. Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.

**2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$  Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)**

- **v488\_majorana\_clebsch\_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung  $M_R = c_3^3 \bar{M}_{PI}$ : a third-generation Majorana factor  $r \approx 1.6696$ , 0.18% from  $g_{car}/N_{fam} = 5/3$  — declined for lack of mechanism. The mechanism search is now EXECUTED: **[E]** NO RENORMALISABLE CHANNEL —  $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$  and the  $\mathbf{126}$  is NOT in the  $E_8$  hull  $\{1, 10, 16, 45\}$  (v247), so  $M_R$  only via the  $d=5$  operator  $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$ ; channels  $\mathbf{16} \times \mathbf{16} = \mathbf{1} + \mathbf{45} + \mathbf{210}$  with only singlet +  $\mathbf{45}$  admissible. **[E]** CHARGE TABLE:  $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$ ,  $\text{Tr}_{16} Y^2 = \frac{10}{3}$  (v485). **[E]** THE  $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero  $\nu^c$  channel weight  $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$  is  $\{2, 3\}$ -smooth, and the multiplicative closure of  $\{2, 3\}$ -smooth rationals never contains  $\frac{5}{3}$  (no factor 5); the UNIQUE natural 5 of the embedding,  $k_Y = \frac{5}{3}$  (Ginsparg, v470), is a full-multiplet trace whose direction has  $Y(\nu^c) = 0$  EXACTLY — structurally decoupled from the Majorana operator. **[E]** GENERATION-BLINDNESS: Clebsches are family-space scalars;  $\text{diag}(1, 1, \frac{3}{5})$  cannot arise from group theory by construction. **[O]** compiler-flavor inventory typed and declined: exact  $\frac{3}{5}$ ’s only in wrong slots ( $K$  lepton gen2/gen1;  $R + L$  lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate ( $M_R$  free inside the PS window, the  $\Sigma m_\nu = 0.059$  eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk’s one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)**

- **v487\_transfer\_clock\_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$  vs uniform  $(\frac{1}{3}, \frac{1}{3})$  is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: **[E]** RUNG READING — the lazy rule’s one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124:  $\lambda_n^{1/p^2} = 1 - n/N_{fam}$ , pole at  $n=N_{fam}$ ):  $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$ , while v327’s uniform rule puts its second pair mode ON the wall ( $0 = 1 - 3/3$ ) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. **[E]** DECK PARITY = RUNG INDEX:  $[B, \sigma] = 0$ ; the  $\sigma$ -even pair mode carries rung  $n=1$  (survival  $\frac{2}{3}$ ), the  $\sigma$ -odd mode rung  $n=2$  (survival  $\frac{1}{3}$ ) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment ( $\omega_1$  even-channel,  $\omega_2$  odd-channel) that `experiments/frb-parity-comb` tests. **[E]** FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) +  $\mathbb{Z}_2$  equivariance + non-negative rates forces UNIQUELY both the split  $(\frac{1}{2}, \frac{1}{6})$  AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs  $h = -\frac{1}{6} < 0$ ); the “sheet-symmetric choice” candidate is superseded by this proof. **[E]** BEND COROLLARY:  $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$  — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. **[O]** what remains open is exactly what was open before v486: the arity  $\{2, 3\}$  (anchor input) and the clock’s semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.



**2026-07-15 (v486 — generator hunt: the FULL transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  from ONE forced local rule — the unique lazy  $\mathbb{Z}_2$ -pair walk (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$ ; the previously generator-less  $(1/3)^6$  mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)**

- **v486\_transfer\_full\_rule.py (HYP.REWRITE.02).** The verified seam transfer spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  (v54/v221), but the only local generator in the suite (v327) produces  $\{1, \frac{2}{3}, 0\}$  — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUE-NESS: for the symmetric 3-channel rule  $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$  (one absorbing family channel +  $\mathbb{Z}_2$  pair) the survival spectrum is  $\{s+h, s-h\}$ ; demanding the PHYSICAL pair  $\{\frac{2}{3}, \frac{1}{3}\}$  forces UNIQUELY (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$  — the lazy  $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum  $\rightarrow$  rates). [E] SIX-HAND GENERATION: over the order-6 =  $2N_{\text{fam}} = |R^+(A_3)|$  hand,  $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$  exactly — both decay gaps now have one recursive generator:  $6 \ln \frac{3}{2}$  (recovery, v76) and  $6 \ln 3$  (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its  $\lambda_2$  arity story ( $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ) is unchanged. [O] the arity  $\{2, 3\}$  stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay =  $1/|\mathbb{Z}_2|$  as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits  $a^T K a=41$ , Koide 53, and the recorded-only  $1^T Q a=16$  — and the lepton-exponent observation  $(5, 3, 2) = K\text{-row} = \text{atom triple}$ ; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)**

- **v485\_contact\_diagonal\_closed.py (SEAM.CONTACT.UNIT.02).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of  $|D|$  on circumference  $\ell$  is  $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$  EXACTLY (symbolic limit + 30-digit checks); it vanishes iff  $\ell = 2\pi = 1/(4c_3)$ , the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure:  $F_{U(1)}$  carries exactly ONE log term, the transport  $8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$ , while the  $\varphi_0^{\text{ret}}$  split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly —  $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ ,  $u = \varepsilon c_3 \ln 2$ , all orders, the linear term ABSENT precisely because the diagonal vanishes ( $\text{Tr } C = 0$ ); the series matches v484's  $\text{Tr}(C^k) = 4^k + 2(-2)^k$  term by term. [E] WEIGHT DICHOTOMY: the SAME 48 =  $3 \times 16$  admissible states carry BOTH multiplicities — counted flat they give the  $\varphi_0^{\text{ret}}$  puncture count  $48 = \Omega_{\text{adm}}$ ;  $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget  $41 = 10b_1 = 40 + 1$  ( $\text{Tr}_{16} Y^2 = \frac{10}{3}$  exact,  $\nu^c = 0$ ; Ginsparg  $(3/5)(41/6) = 41/10$ , v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHI0.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam  $\zeta$ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update

notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

**2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ $c_3$  per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHI0.PUNCTURE.01 is the KMS seam unit  $2\pi = 1/(4c_3)$  with  $\frac{1}{4} = 1/|\mu_4|$ , derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)**

- **v484\_seam\_contact\_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight  $c_3$  per boundary contact insertion”: the  $F_{U(1)}$  ladder  $\{0, 3, 6\}$ , the  $\varphi_0^{\text{ret}}$  puncture  $\{1, 4\}$ ). This module shows the shared rule is NOT a new unknown: [E] the unit chain  $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$  — one *bare* boundary insertion divided by the orbit order  $|\mu_4|=4$  is exactly  $c_3$ ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function  $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$  (polylog identity, 40 digits) takes INTEGER multiples of  $c_3 \ln 2$  at the  $\mu_4$  separations ( $G(\pi/2) = -4c_3 \ln 2$ ,  $G(\pi) = -8c_3 \ln 2$ ; mark matrix  $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$ , spectrum  $\{4, -2, 0, -2\}$ ), so the  $\mu_4$ -invariant (orbit-averaged) insertion carries  $-(c_3 \ln 2)C$  exactly and the log-det contact expansion is exactly graded in  $c_3$  per insertion ( $\text{Tr } C^k = 4^k + 2(-2)^k$ ). [E] the grammar is suite-wide: the  $\Lambda$  prefactor  $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$  (v60) carries the SAME multiplicity  $\Omega_{\text{adm}}=48$  as the  $\varphi_0^{\text{ret}}$  puncture at TWO insertions; the  $g-2$  seam loop divides by the same  $2\pi = 1/(4c_3)$ . [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ( $48 = \Omega_{\text{adm}}$ ,  $41 = 10b_1$ ); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

**2026-07-14 (II) (v482–v483 — the deep-dive round: the  $\nu$ -scale decision computation EXECUTED (the integer  $c_3^3$ -rung excluded unstructured, the  $5/3$ -proximate structure escape recorded and declined) and the  $\varphi_0^{\text{ret}}$  puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are  $t$ -independent rationals, so the  $\pi^{-4}$  must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)**

- **v482\_seesaw\_rung\_decision.py (FLAV.NUSCALE.03).** The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung  $M_R = c_3^3 M_{\text{Pl}}$  needs a rescue factor  $\times 1.670$  in  $m_3$  (0.0301 vs 0.0503 eV), while  $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$  ( $> 3\sigma$ ),  $\alpha_3^{-1} \mp 0.10$  ( $> 3\sigma$ ), a  $\pm 10\%$  band on the ADKLR  $\kappa$  run-down and a PS-leg Yukawa-beta envelope  $\times [0.5, 1.5]$  move it by at most  $\times 1.10$  singly and  $\times 1.165$  ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise  $y_\nu = y_t$  (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor  $r \sim 1.67$ ; its post-hoc 0.18% proximity to  $g_{\text{car}}/N_{\text{fam}} = 5/3$  is recorded and DECLINED per v354/v355 (no forcing mechanism; the  $t_0 \sim 30$  discipline of v478). [O] the v481 one-parameter window candidate (NO floor  $\Sigma m_\nu = 0.059 \text{ eV}$ ) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` ( $\nu$ -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483\_pillowcase\_exact\_traces.py (HYP.PHI0.GEOM.01).** The GEOMETRIC side of the  $\varphi_0^{\text{ret}}$  puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat  $\tau=i$  pillowcase is a  $t$ -INDEPENDENT RATIONAL —  $\text{Tr}(\gamma e^{t\Delta}) = 1$  for

$\gamma = \sigma, \rho, \rho^3$  at every tested  $t$  (position-space Gaussian image sums  $< 10^{-9}$ ; mode-space proof: only  $n=0$  is  $\gamma$ -diagonal), matching the Atiyah–Bott fixed-point counts  $4 \times \frac{1}{4}$  ( $\det = 4 = |\mu_4|$ ) and  $2 \times \frac{1}{2}$  ( $\det = 2 = |\mathbb{Z}_2|$ ); the pillowcase contact term is EXACTLY  $\frac{1}{2} = 4 \times \frac{1}{8}$  per mark and the clock-equivariant trace exactly 1, for all  $t$ . THEREFORE the  $\pi^{-4}$  in  $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$  cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ( $4 = |\mu_4|$  marks of weight  $c_3$  give  $c_3^4$ , multiplicity  $\Omega_{\text{adm}}=48$ ; the bare  $1/(2\pi)$ -propagator 4-cycle differs by the exact rational  $\frac{3}{16}$ ). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight =  $c_3$ ” (the same residual class as the EM-Ward  $c_3$ -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows HYP.PHIO.GEOM.01 + HYP.PHIO.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

**2026-07-14 (website + figure refresh — the /verification journey views become live HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)**

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“~295-script journey”, “260 scripts” → the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ~13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (`AuditRound`: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long ALPHA.QUILLEN residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the SEAM.EQUIV.01 bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\ctthree`, ...) rendered as red KaTeX errors on the site (e.g. the  $\varphi_0^{\text{tree}}$  icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + “Show more” toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

**2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four  $\mu_4$  marks realised as the exactly solvable *four-interval* multilocal modular geometry with  $\omega \circ \rho = \omega$  manifest; and the absolute  $\nu$ -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation  $y_\nu = y_t$  — *no gate closes, no marker moves*)**

- **v479\_kronheimer\_quiver\_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the  $\det K=1$  keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  are the null vector of the affine- $E_8$  Cartan matrix AND the Perron vector  $A\delta=2\delta$  of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the  $\mathbb{C}^2/\Gamma$  geometry;  $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$  forced by the null identity); the gauge datum is  $|2I|=120=|\mu_4|h(E_8)$ ,  $h(E_8)=30=2 \cdot 3 \cdot 5$ ; the resolution carries the unimodular  $-E_8$  form ( $\det=1$ , SNF = identity — the Poincaré link) with  $3 \cdot 8=24$  deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the  $(2, 3, 5)$  cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over  $\mathbb{Z}$ ,  $\sum \delta_i^2 = 120 = 4 \cdot 30$ ,  $30 = 2 \cdot 3 \cdot 5$ , dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).



- **v480\_multilocal\_four\_interval.py** (QGE0.MULTILOCAL.01). The seam circle minus its four marks IS the  $n=4$  multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic  $\Leftrightarrow$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  EXACTLY (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$ , each sector a single-interval quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1}=C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at  $10^{-2}$  (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in `tfpt_research_contracts` (marks keybox + rigidity contract + v478 leg A), `origin_theory` (BW paragraph), `tfpt_4_frontier` (v358 companion note); Python-only by nature.
- **v481\_seesaw\_carrier\_ladder.py** (FLAV.NUSCALE.02, **CANDIDATE** class, **genre v467/v468**). FLAV.NUSCALE.01's  $y_\nu=1$  probe was not the carrier normalisation: one  $SO(10)$  16 per family + the minimal Yukawa sector ( $\mathbf{10}_H$  / PS  $(1, 2, 2)$ ) forces  $y_\nu=y_t$  at the matching scale (named premise). With explicit 1-loop RG (gauge/ $y_t/\lambda$  up; ADKLR Weinberg-operator running down, hep-ph/0108005) the  $(y_\nu, M_R)$  trade-off collapses to  $M_R$  alone: the observed  $m_3=0.0503$  eV demands  $M_R=9.3 \times 10^{13}$  GeV — INSIDE the PS window  $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$  (v249) at  $\log_{c_3}(\bar{M}_{Pl}/M_R)=3.15$  (the  $y_\nu=1$  inversion gives the structureless 2.58). [X] honesty gate: the integer rung  $M_R=c_3^3 \bar{M}_{Pl}$  predicts  $m_3=0.030$  eV — 40% low at 1-loop ( $\kappa$  run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band  $m_3 \in [0.002, 0.115]$  eV and the NO floor  $\Sigma m_\nu=0.059$  eV; frozen record untouched. Integrated in `tfpt_2_standard_model` (new paragraph after v272), `introduction` status card (third named candidate), `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the  $\phi_0$  puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights  $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$  verified) but does NOT force the value  $48c_3^4$  (pure rationals vs  $\pi^{-4}$ ) — recorded in `experiments/theory-contracts/phi0_donnelly_equivariant_probe.py`; the target stays [O].

**2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy  $c_{\text{est}} \rightarrow 1$ , CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$  with the  $h(E_8)=30$  near-miss explicitly declined; both legs stay [O], no status moves)**

- **v478\_entropic\_continuum\_legs.py** (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has  $S(L)=(c/3) \ln L+k$  with  $c_{\text{est}}=1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1=2 \times \frac{1}{2}$ , the v450 reading; gapped control  $\sim 6 \times 10^{-8}$ , area law); the Peschel modular Hamiltonian  $K=\ln((1-C_A)/C_A)$  organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr  $0.87 \rightarrow 0.99$ , shrinking plateau drift; constants deliberately not asserted), the even-range bands

vanish identically (particle-hole,  $\sim 10^{-36}$ ) and the odd tail is 1.8% with parabolic envelope — the state-side  $\Delta_\Sigma$  (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary —  $d\mu=\delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2=e^{t_0}$  exactly, so the v477 one-moment condition forces  $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$  (one dimensionless KMS time, closed form); the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

**2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving  $R^2$  route typed as *one moment condition*  $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$  with the exact closure  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$  — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)**

- `v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`). [E] The Frullani identity upgrades to a scale-flow representation:  $-\ln a = \int \frac{dt}{t}(e^{-ta}-e^{-t}) = 2 \int \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$  (symbolic +  $10^{-20}$  quadrature) — Bianconi’s log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT’s  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight  $w(t) dt/t$  gives  $3\beta R\mu_1 + \frac{17}{24}\beta^2 R^2\mu_2$ ; the flat weight has  $\mu_1=\mu_2=1$  and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding  $m^2=c_3^7\bar{M}_{P1}^2$  forces exactly  $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$  — the v475 kill-test number IS a moment ratio — and the closure identity  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$  holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency:  $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{P1}^2=c_3^7$  — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein +  $\Lambda$  level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the  $R^2$  gate and the v359 typing stay [O], no marker moves.

**2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)**

- `v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`). The last open work package of the v473 round, exercised on a gapped staggered-mass chain ( $N=16$ ). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to  $10^{-15}$ ; thermal boundary:  $\text{spec}(PCP) \subset (0,1)$ , Peschel  $K=\ln((1-C_A)/C_A)$  finite,  $S>0$  — v155 in the bridge context). **The reading is decided:** on the pure bulk  $f(x)=x/(1-x)$  is singular on  $\text{spec}(C)=\{0,1\}$ , so the literal operator-side reading  $Pf(C)P$  of the v473 conjecture is *ill-posed* while the state-side  $f(PCP)$  is finite — the compression identity can only mean “build  $\Delta_\Sigma$  from the *compressed* relative metric”, which matches Bianconi’s own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic):  $[C^2]_{AA}-(C_{AA})^2=BB^t$ ,  $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$  — first order in the cross block vanishes identically,



so for any analytic  $f$  the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity  $f(C_T)=e^{-h/T}$  holds to  $10^{-11}$  relative; the mismatch obeys the quadratic law  $d \leq x^2$  ( $d/x^2 \in [0.4, 1.0]$ ) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

**2026-07-13 (v475 — the  $R^2$  kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron  $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$  ( $\approx 51.7 \bar{M}_{\text{Pl}}$ , trans-Planckian)  $\Rightarrow$  the light-trace-mode reading is dead, KMS renormalisation the only surviving  $R^2$  route; plus the explicit Lorentzian-positivity witness; *no status moves*)**

- **v475\_entropic\_scalaron\_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. **[E]** On a maximally symmetric background the relative curvature operator  $\tilde{\mathcal{R}}\tilde{g}^{-1}$  has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (Bianconi's Appendix-B flattening verified symbolically, incl. the  $6 \times 6$  two-form block), giving the exact vacuum action  $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) (= the v473 pin),  $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$  is the exact tensorial  $R^2$  factor. **[E]** With the pinned  $\beta' = c_3/6$  both mass routes (Starobinsky matching and  $f'/3f''$ ) agree: the raw entropic scalaron is  $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ , i.e.  $m_{\text{raw}} \approx 51.7 \bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$  GeV — TRANS-PLANCKIAN, not light; v473's interpretation 2 (the TFPT scalaron as the light  $\mathcal{G}$ -trace mode) is *dead in naive form*; a viable mechanism must supply exactly  $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup> (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at  $R=1/\beta=384\pi^2\bar{M}_{\text{Pl}}^2$  (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), spacelike/Euclidean stay positive — the v473 **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein +  $\Lambda$  level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving  $R^2$  route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

**2026-07-13 (v474 — the operator level of the entropic-action bridge: the  $D_5$  Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the  $5 \rightarrow \bar{5}$  conjugation, and the v473  $Q$ -target *decided* — supports exactly  $\{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , the naive pair-block reading killed; *no status moves*)**

- **v474\_entropic\_hodge\_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4-algebra of the v473 bridge, machine-checked on the finite carrier Fock space  $\Lambda^\bullet \mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}}=32$ ). **[E]** CAR $\Rightarrow$ Clifford: the ten  $\Gamma$ 's from  $(a_k^\dagger, a_k)$  satisfy  $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$  exactly (all 100 pairs, integer matrices); the 45 bilinears  $[\Gamma_a, \Gamma_b]/4$  commute with  $\gamma = (-1)^N$ , span exactly  $45 = \dim \mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). **[E]** the Hodge–Dirac symbol:  $c(\xi) = a^\dagger(\xi) - a(\xi)$  satisfies  $c(\xi)^2 = -|\xi|^2 \text{Id}$  for symbolic  $\xi$  — Bianconi's  $D = d + \delta$  is Clifford multiplication at symbol level. **[E]** the fold is the  $5 \rightarrow \bar{5}$  conjugation: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16=1+\bar{5}+10$  (the SM carrier decomposition). **[E]** honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant ( $\Lambda^2 \rightarrow \Lambda^3$  leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the  $Q$ -target enumeration (the v473 **[C]** conjecture decided at the algebraic level):  $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$  with integer  $k$  has exactly the supports

$(d, k) = (2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8, 2^{g_{\text{car}}} = \dim \Lambda^\bullet\}$  — minimal uniform solution  $q = c_3^2 = 1/(64\pi^2)$  per Fock mode, bonus identity  $\text{Tr } Q^2 / \delta_{\text{top}} = 32/48 = 2/3 = \dim \Lambda^\bullet / \Omega_{\text{adm}}$ ; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers  $\sqrt{16/5}, \sqrt{2}$  — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the  $R^2$  mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

**2026-07-13 (v473 — the entropic-action bridge: Bianconi’s *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned  $\beta'_B = c_3/6 = 1/(48\pi)$ , the  $R^2$  gap  $3(8\pi)^9$  pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])**

- **v473\_entropic\_action\_bridge.py** (GRAV.ENTROPIC.ACTION.01). TFPT’s classical field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$  (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via  $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$ ) is a local covariant action whose variation gives Einstein + an emergent nonnegative  $\Lambda$  at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her  $\Omega^{0,1,2}$  fiber on the five-slot carrier  $\mathbb{C}^5$  counts  $1+5+10=16=2^{g_{\text{car}}-1}=\dim S_{D_5}^+$ , while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling  $3\beta R$  to  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,  $\beta'_B = c_3/6 = 1/(48\pi)$ ; [E] her  $\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point, and with  $\varepsilon_\star = e^{-\alpha^{-1}} Q$  it reproduces the closed v60 branch  $(3/4\pi^2)e^{-2\alpha^{-1}}$  with the exact target  $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  ([C] until  $Q$  is constructed); [E] the  $R^2$  kill test — the raw log expansion sits *exactly*  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$  below the TFPT Starobinsky coefficient  $1/(12c_3^7)$ , so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT’s OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

**2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient  $g_{a\gamma\gamma} = -4c_3$ , the free-seam Higgs band  $m_H \approx 129\text{--}134$  GeV, and the never-run BH HFQPO  $\times 1.5$  ladder-tooth search; no new vN, no status moves)**

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ,  $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$  (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$  — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction  $\lambda(\bar{M}_{\text{Pl}}) = \beta_\lambda(\bar{M}_{\text{Pl}}) = 0 \Rightarrow m_H = 133.5$  GeV (band 129–134,  $m_t/\alpha_s$ -sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO  $\lambda(M_{\text{Pl}}) = -0.0143 \pm 0.0057, 2.5\sigma$ ) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- $(m_t, \alpha_s)$  RGE pull off the surface at  $> 5\sigma$ ). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at  $\nu_3 = (3/2)\nu_u$  (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (experiments/hfqpo-ladder, preregistered): added as an [O] search-target bullet to the tfpt\_horizon\_readouts “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226  $+9.2\sigma$ , GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (predictions.ts 23  $\rightarrow$  25 cards, papers.ts frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

**2026-07-03 (v472 — the determinant line over the  $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whiteness probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)**

- **v472\_quillen\_detline\_moduli.py** (ALPHA.QUILLEN.DETLINE.01). ALPHA.QUILLEN.EXACT.01 names the  $\alpha^3$  level as the curvature of the *determinant line over the  $U(1)$  seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ( $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$ , v367/v470) in real space on an  $L \times L$  torus with *twisted* boundary conditions — the twist torus  $(\theta_x, \theta_y)$  is the moduli space of flat  $U(1)$  connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus:  $\oint = 2\pi \cdot 1$  exactly (integer to  $10^{-9}$ ) at  $M=1$  for  $L=4$  AND  $L=6$  (size-independent); controls: trivial collar  $M=3 \rightarrow 0$ , orientation flip  $M=-1 \rightarrow -1$ ; the twist-moduli integer equals the Bloch-BZ integer for all three  $M$  (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap ( $= 2.0$ ) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]**: det-line holonomy over the  $U(1)$  moduli = inflow level  $k_0 = |C| = 1$  — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” holds *verbatim at the finite/model level*, where  $\log \det$  of the occupied frame replaces  $\log \det_\zeta$ . **Not closed [O]**: the *continuum*  $\zeta$ -determinant identification on the abstract seam (= the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: tfpt\_1 (sixth honest step in the EM closure), tfpt\_research\_contracts (residual certification), tfpt\_safeguards ( $\alpha$ -form lever), root README, zenodo\_description.html. Exploration original: experiments/tfpt-discovery/quillen\_detline\_twist\_moduli.py (same day).
- **Bird’s-eye whiteness rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted)**. (i) **electron\_sector\_cabibbo\_probe.py**: the  $\alpha$  metrology triangle (TFPT  $\alpha^{-1} = 137.0359992168$  vs Rb-2020  $+0.99\sigma$  / CODATA22  $+1.90\sigma$  /  $a_e$ -route  $+3.39\sigma$  / Cs-2018  $+6.33\sigma$  — TFPT takes the Rb side of the  $5.5\sigma$  metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of  $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$  is excluded by the electron at  $\sim 2 \times 10^4\sigma$ ; linear scaling at  $\sim 10^2\sigma$ ; quadratic survives — the [C] bridge must scale at least quadratically in  $m_\ell$ ); the Cabibbo-angle anomaly as a third dissolution watchdog ( $\lambda_C = 0.2243762$  exact sits  $+0.08\sigma$  on the PDG-2026 rescaled kaon average, between  $K_{\ell 3}$  and  $K_{\mu 2}$ ; exact unitarity puts  $V_{ud} = 0.97450$  — superallowed is the outlier at  $+2.6\sigma$ , all neutron/pion routes agree). (ii) **kernel\_calderon\_uv\_universality.py**: the raw-collar

Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol  $mI'_\nu(mR)/I_\nu(mR)$ , Richardson residual  $3 \times 10^{-4}$ ;  $N_\theta$ -independent;  $|k|$  principal symbol;  $\mu_4$  block structure at machine precision; both spec- $T$  masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays **[O]**). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure  $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$  [frozen],  $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$  [the 2026-07-02 numerology-flagged Nebenbefund],  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  [frozen],  $\Sigma m_\nu = 0.0588$  eV [v468 route]] + spatial flatness determines the whole flat  $\Lambda$ CDM budget:  $H_0 = 67.15$  km/s/Mpc ( $-0.39\sigma$  Planck18,  $-5.67\sigma$  SH0ES22 — the parameter-free Planck side of the Hubble tension),  $\Omega_m = 0.3133$  ( $-0.27\sigma$ ),  $t_0 = 13.86$  Gyr; honest stresses recorded:  $\omega_b -2.02\sigma$  (BBN-only  $-0.73\sigma$ ),  $\omega_c -1.19\sigma$ , DESI-DR2  $H_0 -2.16\sigma$  (one correlated low- $h$  direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the  $\rho_\Lambda$  leg the  $\alpha$  engine),  $\Omega_c$  leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

**2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays **[O]**. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)**

- **v471\_seam\_horizon\_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope  $2C(\gamma)$  on deficit cones (0.01–0.05%) and replica sheets  $n=2$  (0.8%),  $n=3$  (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the  $\zeta$ -scale  $\mu_{\text{lat}}$  drifts — the v152 anchor exhibited on data, with *no* canonical  $\ln(m/\mu)=3/4$  emerging (recorded honestly); the BFK/Schur split is exact ( $\sim 10^{-16}$ ) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying  $2C_{\text{cone}}$  via Kac doubling; the v302 transfer masses  $6\ln(3/2)$ ,  $6\ln 3$  sit on the calibrated EH line ( $\leq 0.05\%$ ); and the Perron/attractor mode ( $m=0$ ) is IR-divergent — **the recovery gap  $\Delta=6\ln(3/2)$  is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes  $1/G$  finite). SEAM.THEOREM.01 stays **[O]**; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the 10 loudest ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at  $f_0/(1+z)$  — all templates moved to detector-frame  $M(1+z)$  (before the fix GW190521,  $z=0.56$ , was filtered  $\sim 1.6\times$  too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked  $p=0.262$ , kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni  $\times 12$ ): ratio semantics



(amplitude  $(2/3)^6$  vs *energy reading*  $(2/3)^3$  vs step  $2/3 \times \mu_4$  per-bounce phases  $\{0, \pi/2, \pi, 3\pi/2\}$  (the boundary-birefringence analogue — path birefringence from parity violation  $c_- = 8$  cancels in the inter-echo ratio; only a per-reflection phase does not)  $\times$  lags 0.5–350 ms (ECO through Planckian window), with joint  $(2, 2, 0) + (2, 2, 1)$  subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

## 2026-07-02 (v295 robustness patch — external-review follow-up: the numeric $a_2$ -kernel tolerance must dominate its own finite-difference floor)

- `v295_flataway_a2_exact.py` (FLATAWAY.A2.01). Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference  $a_2$  kernel is compared against an  $\varepsilon = 10^{-3}$  second finite difference whose *own* truncation error is  $O(\varepsilon^2) \sim 10^{-6}$ , yet the tolerance was  $10^{-6}$  — on his machine  $k=3$  landed at  $1.1 \times 10^{-6}$  (ours:  $2.9 \times 10^{-7}$ ). Fix: tolerance widened to  $5 \times 10^{-6}$  so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

## 2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the $\alpha^3$ level = the *computed* bulk Chern invariant + the seam $F$ -normalisation = the embedding index $k_Y = 5/3$ ; both targets stay [O])

- `v469_seam_crossedproduct_route.py` (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the  $SO(16)_1$  spinor has  $h_s = N/16 = 16/16 = 1 \in \mathbb{Z}$ , so its statistics phase is  $+1$  — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1 \rtimes \{1, s\}$  is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B) = 4/2^2 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4  $\mu_4$  glue runs on the same integer:  $h(J^k) = \{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1)) = k^2$  at net level),  $\mu = 16/4^2 = 1$ , same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_- = 8$  [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern  $|C| = 1$  vs 0 (control),  $\nu = 16 \times |C| = 16$ ,  $c_- = \nu/2 = 8$  — and  $\nu \equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of  $c_-$  is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed'`: the invariant-level axiom `collar_invariants` (R1’) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` ( $h_s = 16/16 \in \mathbb{Z}$ ), `glue_locality_integers` ( $\mu_4 h(J^k) \in \mathbb{Z}$ ), `klm_holomorphic` ( $4/2^2 = 1, 16/4^2 = 1$ ), `sixteenfold_pin` ( $\nu = 16 \equiv 0 \pmod{16}, c_- = 8$ ), bundled as `crossedproduct_arithmetic` ([propext] only); `#print axioms seamResidualClosed'` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).

- **v470\_alpha\_inflow\_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ( $\alpha^{-1}=137.0359992168$ ; Quillen split  $<10^{-35}$ ). [C] *the level is computed, not minimal*: the  $a^3$  coefficient  $k_0=1$  equals the FHS bulk Chern invariant  $|C|=1$  of the same p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez–Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016)  $\eta$ =CS part of  $\delta \log \det$ ; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), replacing v435’s minimality assumption. [E]  $k_Y = \text{tr}(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$  (Ginsparg, Phys. Lett. B197 (1987)):  $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$  — the “GUT 3/5” is the inverse embedding index;  $\frac{41}{6} = \frac{20}{3} + \frac{1}{6}$  (carrier decomposition). [C] the seam  $F$ -normalisation (the one [C] of v434) is *retyped*: once the seam net is  $(E_8)_1$  level 1,  $\langle JJ \rangle = k/(z-w)^2$  is fixed by  $k_Y$  — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses —  $c_- = 8$  (gravitational, feeds SEAM.EQUIV.01/S3) and  $C=1$  ( $U(1)$ , feeds EM1) — so closing  $R1'$  feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E].
- **Integration.** Both registered in run\_all.py + script\_registry.csv (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed’ route); the exact joints Wolfram-mirrored (365  $\rightarrow$  368: the LR/glue locality integers, the KLM  $\mu$ -arithmetic, the sixteen-fold pin, the  $k_Y=5/3/b_1$  identities); cited in tfpt\_1 ( $\alpha$  fixed point + EM-Ward residual prose), tfpt\_research\_contracts (seam residual + Lean section), introduction (closure ledger), tfpt\_5\_redteam (Target A residual wording) and the README/website status surfaces; exploration provenance experiments/tfpt-discovery/closure\_route\_\*.py (2026-07-02 note in experiments/next.txt). Suite ALL CHECKS PASSED through v470. Curated version bumped to v5.4 (Zenodo deposit updated).

**2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three  $\sim 2\sigma$  mixing tensions as *one* third-generation  $-\varphi_0$  pattern, and the splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$ )**

- **v467\_thirdgen\_mixing\_pattern.py** (FLAV.THIRDGEN.PATTERN.01). The suite’s three  $\sim 2\sigma$  mixing tensions ( $\sin^2 \theta_{13} + 2.0\sigma$ ,  $\sin^2 \theta_{23} + 1.76\sigma$ ,  $|V_{cb}| + 1.79\sigma$ ; v307) are exactly the three *third-generation* channels and share *one* relative deviation  $-5.36\% \pm 1.67\%$  — non-zero at  $3.2\sigma$ ,  $-0.03\sigma$  from  $-\varphi_0$ . [E] vocabulary identities:  $\lambda_C^2 = \varphi_0(1-\varphi_0)$  (the frozen Cabibbo);  $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$ ; the suppression size  $\varphi_0 = 5.3\%$  sits inside the seam-gapped band  $(2/3)^6 = 8.8\%$  (v393). [N] the ladder-base forms  $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2/(1+\lambda_C) = 0.041119$  drop the joint four-observable  $\chi^2$  from 10.4 to 0.11 with *zero* new parameters;  $|V_{ub}|$  (no bare  $\varphi_0$ ) and the 1–2 channels stay untouched and correct (dressing  $\lambda_C$  would fail at  $-24\sigma$  — scope checked, not chosen); Daya Bay alone already prefers the pattern form ( $\Delta\chi^2 = 4.3$ ). [C] **honest**: post-hoc (5/16 equally-simple dressings fix the two PMNS tensions;  $(1-\varphi_0)/(1-c_3)/\text{resolvent degenerate} < 1.4\%$ ). [O] no seam-transport derivation (the Bernoulli-variance reading  $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$  is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric:  $\theta_{13}$  stable at 0.0231 kills the pattern, at  $\sim 0.0219$  (sub-1%) kills the record’s bare- $\varphi_0$  reading, exclusive  $|V_{cb}| \sim 0.0395$  kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468\_dm2\_ratio\_jarlskog.py** (FLAV.DM2RATIO.01). The splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2$  was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270,  $\delta = 4\pi/3$ ) has one canonical derived CP invariant, and [N]  $|J_{\text{PMNS}}| = 0.029653$  meets the NuFIT 6.0 measured ratio



$0.029805 \pm 0.000792$  at  $-0.19\sigma$  (equivalently  $m_2/m_3 = \sqrt{|J|} = 0.17220$  vs  $0.17264(229)$ ) — zero dials. [N] the informal `tfpt_2` table heuristic  $m_2/m_3 \sim \pi\varphi_0 = 0.16704$  (a “ $\sim$ ” entry, never frozen) is now  $-2.44\sigma$  and is *superseded as the comparator*. [C] spectrum consistency:  $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$  eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the  $\mu\tau$  texture gives  $\theta_{13} = 0$  at leading order). [X] JUNO shrinks the window  $\sim 4\times$ ; adopting the v467 pattern would shift  $|J| \rightarrow 0.028849$  ( $-1.21\sigma$  today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the  $m_2/m_3$  table row retyped to the  $|J|$  comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328’s “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

**2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  back-solves the same axiom seed to  $-0.11\%$ , extending the empirical over-determination to *five* sectors)**

- **v466\_seed\_leptonmass\_channel.py (SEED.LEPTONMASS.01).** The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  is a sixth independent readout of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$  to  $10^{-9}$ . [N] **new sector (headline):** back-solving  $\varphi_0$  from the measured *pole* ratio  $m_e/m_\mu = 0.004836$  (PDG 2024) gives  $\varphi_0 = 0.053113$ , only  $-0.11\%$  from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound  $(2/3)^6 = 8.78\%$ , v393) but largely cancels for  $m_e/m_\mu$ , so the  $0.11\%$  residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306’s five + this) has  $\chi^2/\text{dof} < 1$  and the error-weighted mean stays at the axiom to  $< 0.1\%$ ; with the corroborating heavy-quark  $m_t/m_b$  ( $-0.23\%$ ) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster  $\chi^2$  by  $> 100\times$ , and the scheme-ambiguous  $m_e/m_s = (34/47)/\varphi_0$  (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence;  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

**2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within  $0.01\sigma$ , the  $\theta_{13}$ -independent externally-runnable slice of v306)**

- **v465\_seed\_crosssector\_joint.py (SEED.CROSSSECTOR.01).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation  $\beta$  (a CMB *EB/TB* polarisation observable) and the Cabibbo angle  $\lambda_C = |V_{us}|$  (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$

reproduces the frozen  $\lambda_C$ ,  $\sin^2 \theta_{12}$ ,  $\beta$  to  $10^{-9}$ . [N] **cross-sector (headline)**: back-solving  $\varphi_0$  *independently* from  $\beta$  (CMB; ACT DR6 + Planck PR4 combined) and from  $\lambda_C$  (quark; PDG 2024) gives the same seed within  $0.01\sigma_{\text{comb}}$ . [N] **one-parameter fit**: the error-weighted mean of the three back-solved  $\varphi_0$  reproduces the axiom to 0.06%; the joint  $\chi^2$  at the axiom (zero free parameters, 3 dof) is 0.007,  $\chi^2/\text{dof} < 1$ , combined pull  $0.08\sigma$ . [N]  $\theta_{13}$  **exclusion declared**: adding  $\sin^2 \theta_{13}$  raises  $\chi^2$  by  $\sim 4$  (the documented  $\sim 2\sigma$  pressure point, v328/v338), so this slice is the honest  $\theta_{13}$ -independent statement, not a hidden dilution. [N] **power**: a data $\leftrightarrow$ formula shuffle explodes the cluster  $\chi^2$  by  $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306);  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row SEED.CROSSECTOR.01; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

## 2026-06-30 (v463/v464 + Lean + Wolfram — the $c=8$ holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a  $c=8$  holomorphic limit must be  $(E_8)_1$ , and (A) the *realisation* input  $R_1$  that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463\_seam\_c8\_holomorphic\_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The  $c=8$  holomorphic-uniqueness pin. [E]  $c=8$  is NOT unique at level 1:  $c = \dim \mathfrak{g}/(1+h^\vee) = 8$  has THREE simple solutions  $A_8$  ( $\dim 80$ ),  $D_8 = \mathfrak{so}(16)$  ( $\dim 120$ ),  $E_8$  ( $\dim 248$ ), so the  $\det K=4$   $SO(16)$  rival is a genuine same- $c$  competitor. [E] HOLOMORPHY FORCES  $\dim V_1=248$ : the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3} = E_4/\eta^8$ ,  $q^1$  coefficient 248 (the 1-dim candidate space, vacuum  $a_0=1$ ). [E]  $\dim 248 + \text{level-1 } c=8 \Rightarrow E_8$  uniquely ( $248/(1+30)=8$ ); tower  $\{1, 248, 4124, 34752, 213126\} = E_4/\eta^8 = j^{1/3}$ . [C] Dong–Mason/Schellekens: the holomorphic  $c=8$  VOA is unique  $= V_{E_8}$  (even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt) — so the identification half is classification-forced.
- **v464\_seam\_oneparticle\_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking  $R_1$ . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol  $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with  $P =$  projection onto  $K < 0$  ( $P^2 = P$  to  $< 10^{-12}$ ),  $K$  fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ( $\max |P_{2N} - P_N| \sim 1/N \rightarrow 0$ ). [E] the limit is  $c=1$  Dirac: entanglement  $S(L) = (c/3) \ln L$  gives  $c \rightarrow 1$ , 16 copies  $\Rightarrow c_- = 8$  (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ( $A_8/D_8/E_8$  each  $\dim = 8(1+h^\vee)$ ) and `holomorphy_selects_e8` (forced  $\dim V_1=248=E_8 \neq 80, 120$ ) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms` `seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the  $E_4/\eta^8$  tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the

SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

**2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)**

- **v461\_seam\_strict\_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367  $p+ip$  collar. [E] CHERN INTEGER: the FHS Chern is  $C=+1$  (chiral  $M=1$ ), 0 (trivial  $M=3$ ),  $-1$  (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding =  $|C|=1 \neq 0$  and  $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$  (the same 8 as  $c_3$ ’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0,  $C=0$ ; since  $C=1 \neq 0$  NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **v462\_seam\_spinor\_continuum.py (SEAM.EQUIV.SPINOR.01).** The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  exhibited three ways. [E]  $(E_8)_1$  TOWER FROM  $SO(16)_1$  SECTORS: the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  gives  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  — the holomorphic  $(E_8)_1$  character IS the  $SO(16)_1$  vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL  $120 \rightarrow 248$ :  $\chi_o$  gives 120,  $\chi_s$  gives 128, summing to the exact tower  $\{1, 248, 4124, 34752\}$  ( $248=120+128$ ). [E] FINITE 16-MAJORANA FOCK:  $C(16, 2)=120$  NS bilinear currents and  $2^{16/2}=256$  Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$  CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives  $c_- \rightarrow 8$  and the lowest scaled NS gap  $x_1 \rightarrow 1$  with  $|x_1 - 1| \sim 1/L^2$  ( $L=64..1024$ ). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact ( $\theta$  identity + tower +  $120/128/256$ , Wolfram-mirrored) + numerical (finite- $L$ , Python-only).
- **Lean (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01).** The two former cited axioms `mmst_existence`  $\circ$  `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding =  $|C|=1 \neq 0$ ,  $c_- \neq 0$ ) and `fock_sectors` (v462:  $C(16, 2)=120$ ,  $2^8=256$ , spinor 128,  $120+128=248$ ); `residual_arithmetic` (now including both) depends only on [propext]. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram. tfpt\_readouts\_extension.wl** adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462  $\theta$ -identity +  $120+128=248$  sector / Fock counts.

**2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the  $\mathbb{Z}_2$  reflection and the  $\mu_4$  clock)**

- **v460\_seam\_s3\_lto.py (SEAM.S3.LTO.01).** NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a  $\mathbb{Z}_2$  reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- $N$  tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367  $p+ip$  collar. [E] FLAT-BAND PARENT: the flattened Bloch  $Q=h/|h|$  has  $Q^2=I$  (flat  $\pm 1$  bands) and  $P=(I-Q)/2$  is an exact projector EQUAL to v367’s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a  $k$ -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel  $|P(n,0)|$  decays rapidly ( $\xi \approx 1.14$  at  $M=1$ ,  $\exp R^2=0.995$ ,  $|P(20,0)| < 10^{-6}$ ) and  $\xi$  grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law  $|R|^{-2.1}$  ( $R^2=0.999$ ) — locality needs the gap. [E]  $\mathbb{Z}_2$  REFLECTION +  $\mu_4$  INHERITED: the flattened modular Hamiltonian  $K_{\text{flat}} = \text{sign}(K)$  inherits  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  ( $u_\Theta = J$ ) and  $[\rho, K_{\text{flat}}] = 0$  EXACTLY (sign is odd;  $\rho$  commutes with any  $f(K)$ ), with side-even and off- $\mu_4$  neg controls firing. [C][O] the chiral  $c_- = 8 \neq 0$  (FHS Chern  $|C|=1$ ) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756;  $C \neq 0$  forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in run\_all.py, the registry, the ledger and cited in tfpt\_research\_contracts (route (xxvii) of the convergent attacks).

**2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, chirality forced by  $P_1$ , an  $(E_8)_1$  character kill test, the exact MMST citation audit and the complementary lattice-VOA route)**

- **v454\_seam\_edge\_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits  $c_{\text{Dirac}}=1.000$  ( $\frac{1}{2}$  per Majorana,  $L=64..1024$ ); [E] the lattice  $U(1)$  current closes its affine level exactly ( $\langle 0|J_n J_{-n}|0\rangle = n$ ,  $n=1..8$ ; a gapped control vanishes); [E] the level-1 Sugawara  $c = \dim G / (1+h^\vee) = 8$  for *both*  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31). [C]  $16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  is the level-1 current-algebra  $c_-$ ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455\_seam\_bw\_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- $N$  Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition  $u_\Theta = J$  — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E]  $\|\Theta K \Theta + K\| = 0$  exactly at every  $N$  ( $N=8..128$ ); [E] an  $N$ -independent gap  $6 \log \frac{3}{2}$ ; [E]  $[\rho, K] = 0$  exactly at every  $N$ ; [E] an  $N$ -independent neg-control margin  $2\|K\|$ . [C][O] the inductive limit inherits  $u_\Theta = J$ ; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456\_seam\_chirality\_from\_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ( $c_- \neq 0$ , the S3 phase) is *forced* by the one-sidedness that defines  $c_3 = 1/(8\pi)$  (P1) (G6). [E]  $c_3$ ’s “8” is the one-sided count  $8 = |\mathbb{Z}_2| \cdot (\int K/\pi) = 2 \cdot 4$ ; [E] a reflection sends the Chern integer  $C \rightarrow -C$  ( $+1 \rightarrow -1$  on the  $p+ip$  model); [E] a two-sided boundary forces  $C = -C = 0$ . [C] the one-sided seam (no reflection) forces  $C \neq 0$ , magnitude  $c_- = 8 = \text{rank } E_8$  — S3 is a consequence of  $P_1$ ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.



- **v457\_seam\_character\_killtest.py** (SEAM.EQUIV.KILLTEST.01). An explicit character/-sector kill test of  $(E_8)_1$  against  $SO(16)_1$  (G7). [E] the  $(E_8)_1$  character  $\chi=E_4/\eta^8$  with tower degeneracies  $\{1, 248, 4124, 34752, \dots\}$ ; [E] the weight-1 count  $248=120+128$  (not the free 120); [E] the sector count  $|\det \text{Cartan } E_8|=1$  (not  $|\det \text{Cartan } D_8|=4$ ). [X] a measured 120, four sectors, or a wrong coefficient would *falsify*  $(E_8)_1$ ; the data give  $248/1/\{1, 248, 4124\}$  (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **v458\_seam\_mmst\_citation\_audit.py** (SEAM.EQUIV.MMST.AUDIT.01). The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy  $c=\frac{1}{2}$  (Thm 4.11), [C] H4 decoupled additivity to  $c=8$ , H5 the WZW range  $8 \leq 8 \leq 16$  with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  ( $\det K 4 \rightarrow 1$ ), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459\_seam\_lattice\_voa\_route.py** (SEAM.EQUIV.LATTICEVOA.01). The second, independent route to  $(E_8)_1$  supplying exactly the residual of v458 (G5). [E]  $E_8$  even unimodular ( $\det=1$ ); [E] the weight-1 space  $248=8+240$ ; [E] the 240 roots split 112 integer + 128 half-integer (the spinor,  $248=120+128$ ). [C] the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar =  $A_{E_8}$ , tied to  $P_1$  by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **Lean** (FORM.SEAM.RESIDUAL.01, `TfptCarrier/SeamResidualAxiom.lean`). Kernel-checks the whole G-block arithmetic (the one-sided  $8=|\mathbb{Z}_2| \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the MMST range  $8 \leq 8 \leq 16$ , decoupled additivity,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K 1$  vs 4) by `decide`, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST v458, AGT v459); `lake build` OK.
- **Wolfram + integration**. Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara  $c=8$  for 120/15 and 248/31; the one-sided  $8=2 \cdot 4$  and reflection  $C=-C \Rightarrow 0$ ; the  $(E_8)_1$  tower  $\{1, 248, 4124, 34752\}$  with  $248=120+128$  and  $|\det \text{Cartan}| 1$  vs 4; the root split  $112+128$  and  $248=8+240$ ) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

**2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- $N$  convergence, three independent reads of  $c_-=8$ , the Ising operator tower, the  $(E_8)_1$  modular data, and  $\mu_4$ -symmetry derived from the four marks)**

- **v449\_seam\_mmst\_uniform.py** (SEAM.EQUIV.MMST.UNIFORM.01). Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A  $Q(1, 2, 4, 7)/12 \rightarrow 2+\sqrt{3}$ , the deviation shrinking monotonically; [E] an *independent* cross-ratio B  $Q(1, 3, 5, 9)/12 \rightarrow$  its own value 2, so convergence is not a tuned point; [E] a uniform  $1/N$  rate  $|Q(N)-Q_\infty| \cdot N \leq 4$  for *both* observables at every  $N_x \in \{36, 48, 60\}$  (one  $N$ -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450\_seam\_edge\_entanglement.py** (SEAM.EQUIV.EDGE.ENTANGLEMENT.01). A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives  $c_{\text{Dirac}}=1.000$  for the critical free-fermion chain; [E] so  $c$  per Majorana =  $1/2$ ; [E] a gapped control saturates ( $c=0$ ). [C] bulk–edge:  $c_-=N_{\text{Maj}} \cdot \frac{1}{2}=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ . SEAM.EQUIV.01 stays [O].

- **v451\_seam\_edge\_cardy\_tower.py** (SEAM.EQUIV.EDGE.CARDY.01). Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension  $h_\sigma=1/16$  (the periodic–antiperiodic ground-energy split) and [E] the energy dimension  $h_\epsilon=1/2$  (the lowest NS pair excitation), both [E]  $L$ -independent (conformal  $1/L$  scaling). [C]  $\{c, h_\sigma, h_\epsilon\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the chiral Ising (free-Majorana) minimal model  $M(3, 4)$ ; 16 copies give  $c_-=8$ . SEAM.EQUIV.01 stays [O].
- **v452\_seam\_e8\_modular.py** (SEAM.EQUIV.E8.MODULAR.01). The  $(E_8)_1$  torus modular signature, extending v156’s  $q$ -expansion to the transformation data. [E] the character  $\chi=E_4/\eta^8$  is  $S$ -invariant  $\chi(-1/\tau)=\chi(\tau)$  (one primary,  $S=[1]$ ); [E]  $T$ -phase  $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$  ( $c=8$ ); [E] leading power  $\chi q^{1/3} \rightarrow 1$  so  $q^{-c/24}=q^{-1/3} \Rightarrow c=8$ ; [E] holomorphic  $c=8 \equiv 0 \pmod{8}$  with  $T$ -phase order  $24/\gcd(8, 24)=3$ . [C] the  $(E_8)_1$  identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453\_seam\_mu4\_from\_marks.py** (SEAM.RIGIDITY.MU4FROMMARKS.01). Folds the last structural premise of the rigidity chain (v446’s residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are*  $\mu_4$ = roots of  $z^4-1$ ; [E] the order-4 clock  $z \rightarrow iz$  is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio  $\lambda=2$  (in the  $D_4$  stabiliser, v168); [E] the form basis  $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$  is  $\mu_4$ -graded (weights  $(1, 2, 3)=A_3$  exponents). [C] so QGEO.SYM.01 follows from marks= $\mu_4$  + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].
- **Lean: TfptCarrier/SeamEdgeChern.lean** (FORM.SEAM.EDGE.CHERN.01). Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers  $(+1/0/-1)$ , the bulk–edge channel count  $|C|=1$ ,  $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$ , the assembly  $2c_-=N_{\text{Maj}} \cdot |C|$  (so  $c_-=8$ ),  $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$ , holomorphic  $c \equiv 0 \pmod{8}$  and the order-3 modular  $T$ -phase — all by kernel *decide* (edge\_kernel\_facts, axioms propext only). lake build OK, #print axioms clean.
- **Wolfram mirror.** tfpt\_readouts\_extension.wl gains nine exact checks (v452 modular data  $\times 4$ , v450 entanglement assembly, v451 Ising Kac weights, v453 marks= $\mu_4 \times 3$ ): 345 → 354 checks, all passing (GATE.WOLFRAM.02).  $E_4$  is evaluated via its  $q$ -series since EisensteinE stays symbolic for complex  $\tau$ .
- **Papers.** The adversarial audit (tfpt\_5\_redteam, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge  $c_-=8$  from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the  $(E_8)_1$  torus modular data (v452), and the  $\mu_4$  premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

**2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from  $\mu_4$ -symmetry, an independent topological reading of the edge  $c_-=8$ , and pinning the one external MMST fact at four-point order)**

- **v446\_seam\_clock\_invariance.py** (SEAM.RIGIDITY.CLOCKINV.01). Closes v445’s single residual one notch further: the operator condition  $[\rho, \Lambda_\Sigma]=0$  (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer  $\Lambda=C(1)C(0)^{-1}=e^{-h}$  (fixed by RP+reflection, v54) of a  $\mu_4$ -invariant RP covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ); [E] a  $\mu_4$ -breaking covariance gives  $[\rho, \Lambda] \neq 0$  (iff-sharp neg control); [E]  $\Lambda$  is a positive contraction (spectrum in  $(0, 1]$ ); [E] the free modular Hamiltonian  $K=\log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki), broken control  $\neq 0$ . [C] so with v445 (commute  $\Leftrightarrow$  block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal  $\Lambda_\Sigma$ ; the residual reduces from an operator commutation to “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].



- **v447\_seam\_edge\_chern.py** (SEAM.EQUIV.EDGE.CHERN.01). An *independent* (topological) reading of the edge central charge, corroborating v444's correlator route from a different observable. [E] the  $M=1$  collar has integer Chern number  $C=+1$  (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E]  $C=0$  in the trivial phase and  $C=-1$  at  $M=-1$  (opposite chirality); [E] the open strip carries exactly  $|C|=1$  chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at  $1/2$  only in the topological phase. [C] bulk–edge:  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the same  $c_- = 8$  as v444. SEAM.EQUIV.01 stays [O].
- **v448\_seam\_edge\_fourpoint.py** (SEAM.EQUIV.EDGE.FOURPOINT.01). Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector  $G^2=G$  ( $\sim 10^{-13}$ ), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio  $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement; [E]  $Q$  converges to the free-fermion conformal cross-ratio  $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$ ; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (v444) and four-point order. SEAM.EQUIV.01 stays [O].
- **Lean: TftptCarrier/SeamRigidityForcing.lean** (FORM.SEAM.RIGIDITY.FORCING.01). Kernel-hardens v445 (and v442's uniformity): the entrywise forcing iff  $\text{clock4 } a = \text{clock4 } b \Leftrightarrow a=b$  (i.e.  $i^a=i^b \Leftrightarrow a \equiv b \pmod{4}$ , residue-only hence uniform in  $N$ ), the four distinct eigenphases, the exact commutant dimension  $\sum n_s^2=64<256$  and the order discriminator ( $128>64$ ) are all proved by kernel `decide` (`forcing_kernel_facts`, no domain axioms); only the operator-level lift to the full  $L^2$  is the named premise. `lake build OK`, `#print axioms clean`.
- **Wolfram mirror. tftpt\_readouts\_extension.wl** gains three exact v445 checks (entrywise forcing for all  $a, b \in 0..31$ ; commutant dimension  $64<256$ ; order-2 commutant  $128>64$  swept  $N=4..64$ ):  $342 \rightarrow 345$  checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (`tftpt_5_redteam`, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling v444, forcing converse v445 + Lean, derived clock-invariance v446, independent topological  $c_- = 8$  v447, four-point pinning v448) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (v336). The research-contracts note lists the same as “eleven convergent attacks”.

**2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, upgrading the rigidity step from “permit” to “force”)**

- **v445\_seam\_rigidity\_forcing.py** (SEAM.RIGIDITY.FORCING.01). The rigidity route (v398/v442) showed a  $\mu_4$ -block-diagonal  $K$  *commutes* with the order-4 clock  $\rho = \text{diag}(i^n)$ . This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff):  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \not\equiv j \pmod{4}$ , verified entrywise and swept  $N=4..128$  — commuting *forces*  $\mu_4$ -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$  (the  $\text{ad}_\rho$  nullspace), a proper subspace of the full  $N^2$  algebra ( $N=16$ :  $64<256$ ) — a genuine quantitative restriction. [E] OFF-BLOCK  $\Leftrightarrow$  NON-COMMUTING: every off-block matrix unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2} > 0$  (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ( $N=16$ :  $64$  vs  $128$ ) — the four Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ ; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues  $\{1, i, -1, -i\}$  for all  $N \geq 4$ . [C] SYNTHESIS: commuting with the order-4  $\mu_4$  clock is *equivalent* to  $\mu_4$ -block-diagonality, so RP-definability (v54/v398)

+ clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$  — v442’s “permit” upgraded to “force”. [O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

**2026-06-30 (Lean kernel-hardening of two of the seven SEAM.EQUIV.01 routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)**

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build OK`; `#print axioms clean`. Tracked as two new FORM.\* ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over  $\mathbb{Z}$  are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom):  $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$  (and  $\text{tr } K=0$ ). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ( $8 \leq 8 \leq 16$ ) and `npw_outside_bucket` ( $\det K=1 \neq 4$ ) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

**2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral  $c=1/2$  Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])**

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368  $p+ip$  collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), `\veri`-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444\_seam\_edge\_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral  $c=1/2$  Majorana CFT:

[E] a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2 = 0.9999$ , with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$ , versus the *exponential* bulk row and the exponential trivial-phase ( $M=3$ , no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ( $\eta = 1.001 \pm 0.007$ , successive profile spread  $0.028 \rightarrow 0.013$  across  $N_x = 36..60$ ) — the numerical hallmark of an existing scaling limit. [C]  $\eta = 1 \Leftrightarrow$  one chiral Majorana ( $c = 1/2$ ) per edge, so the 16-copy carrier has a chiral  $c_- = 8$   $(E_8)_1$  edge CFT ( $c_- = g_{\text{car}} + N_{\text{fam}}$ ). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.

- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

**2026-06-30 (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the  $\beta$ -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)**

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends `ALL CHECKS PASSED`), and are \veri-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438\_seam\_hsmi\_borchers.py (SEAM.EQUIV.BW.HSMI.01) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation  $P \geq 0$ , boost  $K$  with  $[K, P] = P$ , reflection  $J$  with  $JKJ = -K$ ,  $JPJ = P_-$ ), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact,  $sl(2, \mathbb{R})$  positivity ( $\min \text{eig} > 0$ ); [C] synthesis; [O] the continuum standard pair.
- **v439\_seam\_lieb\_robinson.py (SEAM.EQUIV.CONTINUUM.LR.01) — continuum motor.** The explicit gapped  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440\_seam\_lto\_rp\_beta.py (SEAM.EQUIV.LTORP.BETA.01) —  $\beta$ -homotopy.** The (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K] = 0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta = 1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441\_seam\_applicability\_ledger.py (SEAM.EQUIV.APPLICABILITY.01) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.

- **v442\_seam\_rigidity\_uniform.py** (SEAM.RIGIDITY.UNIFORM.01) — **uniform rigidity.** The band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$ : exact 4-sector commutation to  $N=256$  (not merely the  $N \leq 64$  sweep), a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]).
- **v443\_seam\_e8\_discriminator.py** (SEAM.E8.DISCIMINATOR.01) — **kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$ : torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots;  $c=8$  alone cannot decide (v277), only  $\det K=1$  (the genus-1 statement, v90) selects  $E_8$ . A seam with  $GSD>1$  would falsify  $(E_8)_1$ . [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

**2026-06-30 (shadow-export coherence — the tfpt5-shadow mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)**

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh` notes  $\rightarrow$  11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh` audit crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **make\_docs\_map.py is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh` notes/gen leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **audit\_sync.py is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh` audit therefore runs to a clean AUDIT OK on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` on the *filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says ALL CHECKS PASSED while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or vN module changed — this is an export/infrastructure fix only.

**2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated /1lms.txt, explicit AI-crawler robots.txt, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)**

- **New /1lms.txt context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.
- **robots.txt** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root README.md and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (E8.DEGREE.JOINT.01) — a consolidation of v6/v66/v355/v431 that derives more from the  $E_8$  Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of  $E_8$  is fixed by its degree multiset.** From  $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$  alone:  $\text{rank} = \# \deg = 8$ ,  $h = \max \deg = 30$ ,  $\# \text{positive roots} = \sum \exp = 120$ ,  $\# \text{roots} = 240$ ,  $\dim E_8 = 248$ ,  $|W(E_8)| = \prod \deg = 696,729,600$ ,  $\sum \deg = 128 = \dim S^+$ .
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers ( $g_{\text{car}}, N_{\text{fam}}$ ) are the *unique* root pair of  $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ : sum of roots =  $\text{rank } E_8 = 8$  (the rank-fill, v6), product =  $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$  ( $2g_{\text{car}}N_{\text{fam}} = h$ , v431);  $\{3, 5\}$  is the unique positive-integer factor pair of 15 summing to 8. So  $g_{\text{car}} = 5$  and  $N_{\text{fam}} = 3$  are forced *together* by two  $E_8$  degree-invariants, not chosen one at a time. The three primary-readout degrees  $\{2, 8, 30\}$  are exactly  $\{\min, \text{rank}, \max\}$ .
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (E8.DEGREE.JOINT.01); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339  $\rightarrow$  342). Exact integer/lattice; no new physical number, no status change.

**2026-06-29 (v436 — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the  $\alpha^3$  Chern level)**

- **New module v436 (OVERDET.FLOOR.02) — it *hardens* OVERDET.FLOOR.01 (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432’s nominal  $\sim 10^{-10}$  multiplies the counting census by hand-chosen plausibilities (input-“8” independence  $0.1^3$ , seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.



- **[E] assumption-minimal floor.** The  $\alpha$  cubic-class census (v100 layer D) is *pure counting*: of  $N = 94500$  complexity-matched  $F_{U(1)}$  variants only *one* lands in the  $4 \times 10^{-8}$  CODATA window — the TFPT instance, reproducing  $\alpha^{-1} = 137.0359992$  — so  $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$  with *no* subjective probability, prior or multi-observable grammar.
- **[E] monotone concession ladder.**  $-\log_{10} p$  from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$  scorecard-only  $\sim 25.8 \rightarrow \alpha$ -census-only  $\sim 4.98$ ; each rung uses strictly less evidence and the “not chance ( $> 4\sigma$ )” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ( $\sim 1.8 \times 10^{-6}$ ) leaves the floor at  $4.40\sigma$  — they only sharpen it.
- **[C] honest  $5\sigma$  gap.** The conventional  $5\sigma$  line ( $p = 5.7 \times 10^{-7}$ ) is crossed unconditionally by the census times exactly *one* further independent factor  $\leq 0.054$  (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the  $\alpha^3$  Chern-level reading of v435 in named theorems — Dai–Freed ( $\eta$ -invariants and determinant lines, the exponentiated  $\eta$ -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the  $\alpha^2$  Calderón side) and Redlich/Stora–Zumino (the cubic  $\text{Tr } F^3$  anomaly descends to a Chern–Simons level, quantised to  $\mathbb{Z}$ ). It records that the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains **[O]** external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

**2026-06-29 (v435 — a fourth honest step on the  $\alpha^{-1}$  Quillen target: a  $\pi$ -power test isolates the cubic  $\alpha^3$  as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)**

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining **[O]** left after v434 (residual (2), the cubic  $\alpha^3$  Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays **[O]**;  $\alpha^{-1} = 137.0359992$  stays **[E]**.
  - **[E]  $\pi$ -power / metric-independence test.** The three EM-closure coefficients carry  $\pi$ -powers exactly  $\{0, 3, 6\}$ , equal to their  $c_3$ -orders: Maxwell  $k_0 = 1 \rightarrow \pi^0$ , Calderón  $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$ , transport  $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$ . So the Maxwell  $a^3$  is the *unique*  $\pi^0/c_3^0$  (metric-independent) term. A heat-kernel boundary coefficient always carries the  $(4\pi)^{-d/2}$  measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level ( $a^3$ ) + *two* heat-kernel boundary terms, confirming v342/v434’s “ $a^3$  is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
  - **[C] conditional level quantisation.** If the  $a^3$  term is the seam  $U(1)$  Chern–Simons/ $\eta$  boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to  $\mathbb{Z}$ , so  $k_0 = 1$  is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
  - **[O] not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that  $\partial_\tau \log Z_{\text{Maxwell}} = a^3$  — stays external math (type A, v384); the seam  $F$ -normalisation stays the one **[C]** (residuals 1 & 3, v434);  $\alpha^{-1}$  stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv`



(ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside `v433/v434`; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-28 (v434 — a third honest step on the  $\alpha^{-1}$  Quillen target:  $b_1$  is the  $a_4$  heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])**

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03) — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O];  $\alpha^{-1} = 137.0359992$  stays [E].
  - [E]  $b_1$  *is* the  $U(1)$   $a_4$  **heat-kernel coefficient (the  $\beta = a_4$  theorem).** The one-loop  $\beta$  function of a gauge coupling *is* the  $a_4$  (Seeley–DeWitt  $a_2$  in  $d=4$ ) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ( $\frac{2}{3}$  per Weyl fermion,  $\frac{1}{3}$  per complex scalar),  $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$  (SM) and  $(3/5) \cdot \text{that} = 41/10 = b_1$  (GUT). So the  $b_1$  in the counterterm  $8b_1c_3^6$  is the  $a_4$  coefficient of the carrier  $U(1)_Y$  content (the same number as the  $\beta$ , v159/v246), the *same kind* of heat-kernel object as the  $\alpha^2$  Calderón  $a_4$  (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off:**  $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$ , with  $\text{rank } E_8 = 8$  the  $\chi=2$  seam boundary count (v216) and  $c_3^6$  the six boundary insertions (the  $\pi$ -power test, v342).
  - [C] **residual (1)  $\equiv$  residual (3).** With  $b_1$  and the  $c_3$ -powers heat-kernel-forced, the only freedom left in the exact seam  $a_4$  value is how the gauge curvature couples to the boundary measure — the seam  $F$ -normalisation. So “the actual seam  $a_4 = 8b_1c_3^6$ ” and “the seam  $F$ -normalisation” are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed):** the determinant-line variation factors  $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$ ;  $a^2$  is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading  $a^3 = a \cdot a^2$  is the *first moment* of the Chern density — a topological level at boundary order  $c_3^0$ , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
  - [E] **net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the  $\alpha^3$  Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside `v433`; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-27 (v433 — a second honest step on the  $\alpha^{-1}$  Quillen target: a solvable 4D model reaches the  $a_4$  order, connecting v391 and v342)**

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02) — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (v382); it *grounds and connects*, it does *not* close it.** The target stays [O] (external math, type A, v384); the value  $\alpha^{-1} = 137.0359992$  stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose  $\zeta$ -determinant variation reached only the low Seeley–DeWitt orders  $a_0, a_1$ ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient  $a_4$  ( $30/360 = \frac{1}{12}, \Omega^2/12$ ). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the  $a_4$  order:** the flat operator  $\Delta = -\partial^2 + m^2$  on a 4-torus of volume  $V$  has heat trace  $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$ , with Gilkey coefficients  $a_0 = V/(4\pi)^2$ ,  $a_2 = -Vm^2/(4\pi)^2$  and  $a_4 = Vm^4/(2(4\pi)^2)$ ; the  $t^0$  (conformal-anomaly) order  $a_4$  is *non-zero* — exactly the order v391’s 1D circle toy could not reach.

- **[E] it is the order v342 uses:** so v391 (“a  $\zeta$ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey  $a_4$ ”) are unified — the variation is heat-kernel data *at the  $a_4$  order*, not only the low orders v391 reached.
- **[E] the measure is  $c_3$ -built:** the universal 4D heat-kernel measure  $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$  in the seam’s one-sided 2D-boundary normalisation  $c_3 = \frac{1}{2}(4\pi)^{-1}$ ; and the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ .
- **[O] not closed:** computing the *actual* seam  $U(1)$   $\zeta$ -determinant  $a_4$  coefficient ( $= 8b_1c_3^6$ ) on the  $\mu_4$  seam moduli, and the origin of the cubic  $\alpha^3$  Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam  $F$ -normalisation fixing the coefficient *value* stays **[C]** (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.02); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

## 2026-06-27 (v431, v432 — the “unmapped” $E_8$ degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped”  $E_8$  Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430):  $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \text{det-ladder}\{8, 14, 20\})$ . **[E]** the two-family split is *forced* by  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  — the exponents are the  $\varphi(30) = 8$  totatives of 30, hence coprime to 6, hence  $\equiv \pm 1 \pmod 6$ , so the degrees occupy only the residue classes  $\{0, 2\} \pmod 6$ . **[E]** the  $6k$  family  $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$  is the v91 spine; the  $6k+2$  family  $\{8, 14, 20\} = (\det R, \det C, \det L)$  is the winding line  $\det M(s, 0) = 6s + 8$  of the flavor diamond (v135). **[E]** “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to  $E_8$  among the simply-laced exceptionals ( $E_6/E_7/D_5$  fail). **[C] honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$  Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays **[C]** (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the  $L_1 \leftrightarrow L_2$  bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness  $\alpha^{-1} \sim 137$ . **[C]** multiplying only across disjoint pieces (input over-determination  $\times$  the  $\alpha$  census  $1/94500 \times$  the  $\varphi_0^{\text{ret}}$ -seed cluster counted once  $\times$  one downstream witness) gives a floor  $\leq 10^{-6}$  across a grid of generous assignments, nominal  $\sim 10^{-10}$ . **[E]** that floor sits  $\sim 20$  orders *above* the v100 conditional  $10^{-30.7}$ ; the gap is exactly the evidential value of proving the forms forced, so the  $\alpha$  form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. **[C] honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

## 2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from $E_8$ ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five  $E_8$  Casimir degrees  $\{12, 14, 18, 20, 24\}$  with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided  $\mathbb{Z}_2$  collar / conjugate half-spinor  $S^-$ ), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes  $|\mathbb{Z}_2| = 2$  (the  $\frac{1}{2}$  in  $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$ , v58/v216), and  $2 = \min \deg(E_8)$  is the quadratic/metric — one of the *matched* primary readouts  $\{2, 8, 30\}$ , never one of the unmapped degrees. [E] the two sheets are the 128-spinor:  $\dim S^+ = \dim S^- = 16$ , the spinor block  $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$ , the spinor half of  $248 = 120 + 128$ ;  $S^-$  enters 248 as the conjugate  $(\overline{16}, \overline{4})$  ( $128 = 2 \cdot 64$ ), not a spare singlet. [E] forced-disjoint: the sheet/deck set  $\{2, 16, 32, 128\}$  has empty intersection with  $\{12, 14, 18, 20, 24\}$ , its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits  $\geq 2$  readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the  $S^-$ -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier  $E_8$  scan (tfpt\_4\_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run\_all.py, script\_registry.csv, status\_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt\_5\_redteam, the magnitude/phase keybox of tfpt\_3\_e8\_audit\_bootstrap, the WIMP-no-go passage of tfpt\_4\_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

## 2026-06-26 (v429 — the ‘unmapped’ golden $E_8$ structure is the geometry of the one external input $\theta_i$ )

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle  $E_8$  structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio  $\varphi$  is *unmapped* (it appears only internally — the affine- $E_8$  attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude  $\varphi$  cannot be numerology. [E] this sharpens that statement: because  $N_{\text{fam}} = g_{\text{car}} - 2$ , the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is  $\theta_i = \pi N_{\text{fam}} / g_{\text{car}} = (g_{\text{car}} - 2)\pi / g_{\text{car}}$ , which is *exactly* the interior angle of the regular  $g_{\text{car}}$ -gon — for  $g_{\text{car}} = 5$  the pentagon,  $\theta_i = 3\pi/5 = 108^\circ$ . [E] its cosine is golden,  $\cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$  (partner  $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$ ,  $\varphi = 2\cos(\pi/5)$ ). [E] the golden character is *unique* to  $g_{\text{car}} = 5$ : among the small regular  $n$ -gons only  $n = 5$  has an irrational interior-angle cosine ( $n = 3, 4, 6$  give  $1/2, 0, -1/2$ ), so  $\theta_i$  is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts —  $\varphi$  is the unmapped  $g_{\text{car}} = 5$  signature (v354/v313) and  $\theta_i = 3\pi/5$  is the one external cosmological input (v211) — and *refines* v354 from “ $\varphi$  in no physical readout” (it is the unmapped  $g_{\text{car}} = 5$  signature, v313) to “ $\varphi$  touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop  $\theta_i \approx 170^\circ$  (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite- $T$  solver that decides the branch (v185/v211). It does *not* derive  $\theta_i$ , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

## 2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** (“External works built upon”), \input into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

## 2026-06-25 (External AQFT citations + the four-fold $\mu_4$ identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold  $\mu_4$  identity made explicit (v422).** v422 already proved that the seam deck/divisor  $\mu_4$ , the carrier Galois group  $\text{Gal}(\mathbb{Q}(\zeta_5))$ , the discriminant  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}/4$  and the  $(E_8)_1$  simple-current glue are ONE cyclic  $\mathbb{Z}/4$  (with the cyclic-vs-Klein negative control). It is now named the *four-fold  $\mu_4$  identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

## 2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the  $(2, 3, 5)$  singularity is the ONE generator behind the order-30 clock (Milnor  $\mu = (2-1)(3-1)(5-1) = 8 = \text{rank } E_8$ ,  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ ,  $\varphi(30) = 8$ ), so each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or  $E_8$ ): all



seven collapse to one generator  $\Rightarrow$  *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in  $c_3=1/(8\pi)$  from rank  $E_8$ ,  $h(D_5)=2(5-1)$ ,  $\varphi(30)$  and the Milnor number — plus the foreign witness  $\alpha^{-1}\approx 137$ . [C] refines and partly walks back v427, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), tfpt\_safeguards Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (papers.ts, Safeguards.tsx, release.ts, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

## 2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (make\_figures.py, in manifest.sha256): the layered defense, the  $\alpha^{-1}$  uniqueness scan (one  $F_{U(1)}$  variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs  $\leq 5/13$ ).
- **tfpt\_safeguards expanded.** Layer 5 now states the null model explicitly ( $\prod p_i=10^{-25.8}$ ,  $2\times 200k$  pseudo-theories  $\leq 5/13$ ,  $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$ , shuffle 1.2) and the  $\alpha$  uniqueness ( $\leq 10^{-30.7}$ ,  $\sim 102$  bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading  $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$  MeV, a Belle II kill test. No new module.

## 2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of  $F_{\text{transfer}}$ .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap  $6\ln(3/2)$ ) restricted to each sector. [E] the v99 Koide generator linearised at  $q=2$  has rate  $-\Delta$ , time-1 contraction  $(2/3)^6$ ; Boltzmann washout is that contraction (CP phase  $4\pi/3$  native, v320); QCD rate  $b_3=-7$ . [C] only the anchors ( $v_{\text{geo}}$ ,  $C_p$ ,  $M_R$ ) stay external;  $F_{\text{relic}}$ ’s cosmological IC  $\theta_i$  is the one wall. A reduction (extends v383 static $\rightarrow$ dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible  $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds by direct Tomita–Takesaki;  $|\det \text{Cartan}(E_8)|=1$  (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss  $N(3+2i)=13$ , Eisenstein  $N(3+2\omega)=7$ , cyclotomy  $N(3+2\zeta_5)=55$ , Galois  $|(\mathbb{Z}/5)^\times|=4$ ,  $|\det \text{Cartan}(E_8)|=1$ , Pascal  $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=11$  ( $2^4=16$ ), Coxeter  $h(E_8)=30=2\cdot 3\cdot 5$  with  $\varphi(30)=8$  — and the anchor  $a=(1,1,2)$  is the compression generator. Extends v305.

- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the  $F_{\text{transfer}}$  and keystone sections), `tfpt_safeguards`; manifests refreshed.

## 2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01: v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`’s intrinsic Bisognano–Wichmann lemma (the `v308/v309` residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [E] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K]=0$ , `v309/v199`), not the flow —  $\sigma^\psi=\rho$  is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26’s axiom (LTO-RP) means  $u_\Theta=J$  (reflection = modular conjugation) = the intrinsic BW condition  $\theta K \theta = -K$ , *distinct* from  $\omega \circ \rho = \omega$ : a positive character-block-diagonal  $K$  has  $[\rho, K]=0$  yet never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code  $|\det K|=4$  via a trace; Levin–Wen anyonic), while the seam is invertible ( $|\det \text{Cartan}(E_8)|=1$ , no anyons) and a  $\beta=1$  KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via `v89/v281`).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

## 2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01: v423`)

- **New module `v423` (`QG.AMB.NODEP.01`).** Sharpens the `v369/v384` prose redundancy (“`QG.AMB.01` is a [C] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin  $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$  ( $31 = 2^{g_{\text{car}}} - 1$ ), susceptibility  $\chi = 729/665$  finite (negative control gap  $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$ ); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while  $\alpha^{-1}$ , the mixing angles and cosmology are each *computed* by a compiler script, `v3/v9/v268/v7`); [E] the one transitive link is architectural — of the frozen readouts only  $\theta_{13}$  references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — `v369` as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).



- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic*  $|\det K|=1$  point that selects  $(E_8)_1$  — the central charge  $c=8$  alone does *not*, being shared by the non-holomorphic  $SO(16)_1$  counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

## 2026-06-25 (The Galois $\leftrightarrow$ Net bridge: v422)

- **New module v422 (SEAM.GALOIS.NET.01).** The seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$  is the SAME cyclic  $\mathbb{Z}/4$  as the  $(E_8)_1$  simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to  $G_{\text{net}}/\text{SEAM.EQUIV.01}$ . [E]  $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$  (one Smith invariant factor  $4 = \text{cyclic, det } 4$ );  $D_n$  disc is  $\mathbb{Z}/4$  for  $n$  odd,  $\mathbb{Z}_2 \times \mathbb{Z}_2$  for  $n$  even, so the carrier  $D_5$  (rank  $5 = g_{\text{car}}$ , odd) is cyclic while  $D_8$  (rank 8, even) is Klein; [E]  $(\mathbb{Z}/5)^\times = \langle 2 \rangle$  is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in  $\mathbb{Z}_4 \times \mathbb{Z}_4$  ( $16 = \dim S^+$ ) the glue  $(1, 1)$  is order 4 with Lagrangian quotient  $16/4^2 = 1 = (E_8)_1$  (v89/v154); [E] negative control: the Klein / order-2  $\langle (2, 2) \rangle$  gives  $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16) (\text{det } 4)$ , NOT  $E_8 (\text{det } 1)$  — cyclic  $\mathbb{Z}/4 \rightarrow E_8$ , Klein  $\rightarrow D_8$ . [C] the bridge: one cyclic  $\mathbb{Z}/4$  threads the seam deck  $\mu_4$  (v419),  $\text{Gal}(\mathbb{Q}(\zeta_5))$  and the  $(E_8)_1$  glue — the Galois gearbox  $(\mathbb{Z}/30)^\times$  IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ( $324 \rightarrow 327$ ).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts ( $327/327$ ) and the website verification count; manifests refreshed.

## 2026-06-25 (F\_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide  $Q$  is rescaling-invariant ( $Q(cm) = Q(m)$ ), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit  $\frac{2}{3} - Q_{\text{src}} = 0.00220$  is one-loop-EM-sized ( $= \varphi_0/24$  to 0.6%,  $= \alpha/\pi$  to  $\sim 5\%$ ), the PDG pole sits at  $2/3$  (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$  per-lepton-log estimate shifts  $Q$  by EM-loop order toward  $2/3$  (the v82 direction); [O] the exact flow time  $t \approx 2.84$  (v93) is scheme/scale-dependent external QED, only sized. [C] verdict:  $F_{\text{pole}}$  is a one-loop EM effect — the lens forces the rate  $(2/3)^6$  (v327), external EM supplies the flow; `F_pole` stays [C], firewalled (v187), so  $F_{\text{transfer}}$  is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

## 2026-06-25 (The seam $\mu_4$ is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, `RES.COXETER.SYMMETRY.01`): why the seam  $\mu_4$  sits on the Galois side, not on the cyclic  $\zeta_{30}$  dial. [E]  $h(E_8) = 30$  is squarefree, so  $\mathbb{Z}/30$  has element orders  $\{1, 2, 3, 5, 6, 10, 15, 30\}$  — *no* order 4, so the seam  $\mu_4$  cannot be a rotation hand and is forced onto the Galois side; [E] by CRT  $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$  (order  $\varphi(30) = 8 = \text{rank } E_8$ ), and the  $\mu_4 (\mathbb{Z}/4)$  factor IS  $(\mathbb{Z}/5)^\times$  (the carrier prime 5), the  $\mathbb{Z}_2$  IS  $(\mathbb{Z}/3)^\times$  (family 3 = CP conjugation, v316); [E] so the seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ , realised on the carrier clock  $C_5$

(v418) by the Frobenius  $\zeta_5 \mapsto \zeta_5^2$  — the explicit order-4 operator  $G$  with  $GC_5G^{-1} = C_5^2$ ,  $G^4 = I$ ,  $G^2C_5G^{-2} = C_5^{-1}$ . [C] this is the positive content of v409: the  $2^2=4$  is the order of  $(\mathbb{Z}/5)^\times$ , and “no contraction rate” is because the seam is an *automorphism*, not a hand; so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants (cyclic  $h=30$ , Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321  $\rightarrow$  324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ( $\mu_4 = \text{Aut}(\zeta_5)$  the seam,  $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$  the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

## 2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split (3,2) read in the three atom-rings gives (7,13,55)=(scalaron,  $\Delta_Q$ , quark numerator), and the carrier-5 clock missing from v417 is found as the  $4 \times 4$   $\Phi_5$ -companion  $C_5$ . [E]  $C_5^5 = I$ ,  $\text{tr } C_5 = -1$ ,  $\det C_5 = 1$  (the four primitive 5th roots), with the golden ratio as its real-part data ( $2 \cos 72^\circ = 1/\varphi$ ,  $2 \cos 144^\circ = -\varphi$ ; output-side, v313/v349; dim 4, not 3 — this refines v417’s gap from “no operator” to “no  $3 \times 3$  operator”); [E]  $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5) = \det(3I+2C_5) = 55 = 5 \cdot 11 = g_{\text{car}} \| \text{Pl}(K) \|_1$ , the  $c_u/c_d$  numerator (v410/v411); [E] the triple  $\det(3I+2 \text{Comp}(\Phi_n)) = (7,13,55)$  for  $n=3,4,5$  over  $\mathbb{Z}[\omega]$ ,  $\mathbb{Z}[i]$ ,  $\mathbb{Z}[\zeta_5]$  (atoms 3,2,5, v416); [E] NEG control: the anchor (5,4)  $\rightarrow$  (21,41,461) reaches named values only in the  $\omega$ - and  $i$ -rings ( $21=N_\omega(5,4)$ ,  $41=10b_1$ , v222) but 461 (prime) in the carrier ring, so the carrier ring separates (3,2)  $\rightarrow$  55 from (5,4)  $\rightarrow$  461. [C] honest: (3,2) is the *forced* carrier split (v14), not the unique clean one — a scan finds  $\{(2,1) \rightarrow (3,5,11), (3,2) \rightarrow (7,13,55)\}$ , a small ladder. The sharper discriminator is *cross-sector*: only (3,2) spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while (2,1) is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318  $\rightarrow$  321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

## 2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the  $\tau=\omega$  flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation  $P = (1\ 2\ 3)$  is the order-3 Eisenstein deck ( $P^2+P+I = \text{ONES}$ , so  $\omega^2+\omega+1=0$  on the plane  $\perp (1,1,1)$ ), the order-3 dual of  $J^2=-I$ ; [E] it realises the hex CM norm as  $(3I+2P)(3I+2P^2) = 7I+6 \text{ONES}$ ,  $\text{Spec}\{7,7,25\}$  ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ = scalaron,  $25=g_{\text{car}}^2$ ) — the dual of v415’s {13,13,9}; [E] the CP clock  $W=-P^2$  ( $\rho$  of v233) has  $W^2=P$ ,  $W^3=-I$ ,  $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , giving  $\delta_{\text{CKM,lead}} = \arg \zeta_6 = \pi/3$  and  $\delta_{\text{PMNS}} = \arg \zeta_6^4 = 4\pi/3$  (v316/v320); [E] all flavor clocks are powers of  $\zeta_{30}$  ( $\zeta_{30}^{15}=-1$ ,  $\zeta_{30}^{10}=\omega$ ,  $\zeta_{30}^6=\zeta_5$ ,  $\zeta_{30}^5=\zeta_6$ ) while the seam  $\mu_4=i$  is the Galois side  $(\mathbb{Z}/30)^\times$  of order  $\varphi(30)=8=\text{rank } E_8$  — so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants ( $h=30$  and rank=8). [C] honest gap: the carrier  $\mu_5$  has no  $3 \times 3$  operator (the golden  $\varphi$  is an output-side trace, v313/v349); negatives: no  $U, V$ -diamond operator carries  $\zeta_6$ ,  $[D, P] \neq 0$  (no  $\mu_{12}$ ),  $N(3+2\zeta_6)=19 \neq 7$ . Cited in `tfpt_2`; Wolfram-mirrored (315  $\rightarrow$  318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

## 2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the  $\mu_4$  deck becomes an intrinsic diamond object. [E] the integer commutator  $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$  has  $\text{Spec}(J) = \{i, -i, 0\}$ ,  $J^2 = -I$  on the rank-2 image and  $J^3 + J = 0$  — a  $\mu_4$  quarter-turn born from the binary  $V$  and ternary  $U$  compilers (v410/v95); [E]  $3I + 2J$  and  $5I + 4J$  have spectra  $\{3+2i, 3-2i, 3\}$ ,  $\{5+4i, 5-4i, 5\}$ , so the Gaussian integers  $3+2i$  (norm  $13=\Delta_Q$ ) and  $5+4i$  (norm  $41=10b_1$ ) are *literal eigenvalues*; [E]  $(aI+bJ)(aI-bJ)$  reads  $(a^2+b^2)$  on the  $\mu_4$ -plane and the real part<sup>2</sup> on the kernel ( $\{13, 13, 9\}$ ,  $\{41, 41, 25\}$ ); [E] the intrinsic order-4 deck  $D = -I + J - J^2$  ( $D^4 = I$ ,  $\chi_D = (x+1)(x^2+1)$ ) has  $\chi_2$ -line  $\ker[U, V] = a - 1$ . [C] the geometric-deck identification sharpens (does not close) the v140  $\mathbb{Z}_3$  residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms  $\{2, 3, 5\}$  are ramified/inert/split in the two exceptional CM rings. [E] in  $\mathbb{Z}[i]$  (square/seam)  $2=|\mathbb{Z}_2|$  ramifies,  $3=N_{\text{fam}}$  is inert,  $5=g_{\text{car}}$  splits as  $(2+i)(2-i)$ ; [E] in  $\mathbb{Z}[\omega]$  (hex/flavor)  $3=N_{\text{fam}}$  ramifies,  $2, 5$  are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets  $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$  of  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$  (v403) ramify over exactly one atom each, product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ , two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension 308  $\rightarrow$  315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns  $-\chi$  for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

## 2026-06-24 (The sheet generator $V$ as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis  $V = Q \text{diag}(0, 1, 1)$  of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form  $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$  (proved by the inductive step  $F(n)V=F(n+1)$ ), so  $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; [E] the four bilinear families  $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$ ,  $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$ ,  $a^T V^n a = 11 \cdot 2^{n-1}$ ,  $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$  give  $(6, 7, 9, 11)$  at  $n=1$  and  $\mathbf{1}^T V^2 \mathbf{1}=13=\Delta_Q$ ,  $\mathbf{1}^T V^3 \mathbf{1}=27=\mathbf{1}^T R a$ ,  $\mathbf{1}^T V^4 \mathbf{1}=55$ ,  $\mathbf{1}^T V^4 a=56=\dim \mathbf{56}_{E_7}$ ; [C] the binary spine is FORCED by  $\text{Spec}(V)=\{0, 1, 2\}$ , so the carrier/Lie matches (rank  $E_8=8$ ,  $\dim S^+=16$ ,  $248/8=31$ ) and the theta cross-link ( $\sigma_3(2)=9=a^T V \mathbf{1}$ ,  $\sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R)$ ,  $\sigma_3(5)=126$ ) are audit-typed (the v95  $G_2$ -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure  $V$ -power readout,  $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1})/((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$ , identical to the established  $g_{\text{car}} \|\text{Pl}(K)\|_1/(N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$  (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the  $u/d$  identification stay conditional, coupled to the ( $U_{\text{wall}}$ ) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the  $\mathbb{Z}_2$  wall ( $t=-2$ ,  $\det=2$ , v135),  $J=M(1, -2)=C-2V$ , completing the sheet axis  $J, K, C, F=M(1, -2..1)$ . [E]  $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)$ ;  $a^T J a=30=h(E_8)$ ,  $\det(I+J)=12=\dim \mathbf{g}_{\text{SM}}$  and  $\det(2I+J)=40=|R(D_5)|$ ; [C] the Lie readings audit-typed like the  $G_2/F_4$  shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On  $M_t=C+tV$  the elementary-symmetric coefficients encode the atoms as difference orders:  $e_1=3t+12$ ,  $e_2=2t^2+15t+25$ ,

$e_3=4t^2+14t+14$  give  $\Delta e_1=3=N_{\text{fam}}$ ,  $\Delta^2 e_2=4=|\mu_4|$ ,  $\Delta^2 e_3=8=\text{rank } E_8$  (the  $e_3$  curvature is the  $\sqrt{218}/\sqrt{224}$  sheet quadratic; the full triple is new); [E] the anchor energy  $a^T M_t a = 52 + 11t$  ( $30 \rightarrow 41 \rightarrow 52 \rightarrow 63$ , step  $11 = a^T V a$ ); [C] atom/Lie readings audit-typed.

- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center  $C=M(1,0)$  is a resolvent portal: [E]  $\det C=14=\dim G_2$ ,  $\det(I+C)=52=\dim F_4$ ,  $\det(2I+C)=120=|R^+(E_8)|$ , the exceptional shell ladder  $G_2 \rightarrow F_4 \rightarrow E_8^+$  as  $\det(C+kI)$ ,  $k=0,1,2$  (with  $\text{tr } C=12=\chi_{E_8}(i)$ ,  $\sum C=31=2^{g_{\text{car}}}-1$ , v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \veri citations) and the  $E_8$  theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror  $303 \rightarrow 308$ .

## 2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHI0.PUNCTURE.01).** Names the  $\varphi_0$  puncture term  $48c_3^4$  as the order- $|\mu_4|=4$  local boundary heat-kernel (Seeley–DeWitt) contact term of the  $\mu_4$  puncture divisor, summed over  $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$  admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split  $\varphi_0=1/(6\pi)+48c_3^4$ , the puncture count, the closed form  $3/(256\pi^4)$  and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the  $c_3$  Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*:  $\sigma=\text{diag}(1, -1, -1)$  has spectrum  $\{\pm 1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0, 1\}$ , none in  $(0, 1)$ , so prime-2 carries no contraction rate, whereas the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0, 1)$  and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ( $|\mu_4|=2^2$  the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O]  $\rightarrow$  [E]/[C].
- **Axion branch consistency (tfpt\_4\_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- $T$  result already stated later in the same section: the robust dominant-DM branch is the *spine* angle  $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$  (untuned, in band), while the seam hilltop  $\theta_i \approx 170^\circ$  over-produces ( $\sim 5.4\times$ ) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ( $r^2=1$ ) is  $\pm 1$ , a projector eigenvalue ( $r^2=r$ ) is 0 or 1, neither lies in the open interval  $(0, 1)$ , while  $2/3$  does — so *no prime-2-only attractor exists*. Built clean (lake build); the new theorems are axiom-pure (`#print axioms = propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant  $11=\|Pl(K)\|_1=\det(1|a|\sigma)$  is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

## 2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain



v114–v117 keeps .06–.09 (as asserted by v123\_inventory\_update.py and run\_all.py); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal depends\_on carried along). No claim content changed.

- **Website.** papers.ts document count  $8 \rightarrow 9$  (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E*  $\rightarrow$  *A–F* with a new Target F section + contribution bullet (the abstract already carried F); StatusMatrix.tsx Doc 7  $\rightarrow$  Doc 8 for the research contracts and the residual interfaces re-typed  $v_{\text{geo}}$  [O],  $G_{\text{net}}$  [C],  $F_{\text{transfer}}$  [C]; objections Paper 1  $\rightarrow$  Doc 1.
- **Papers (marker/wording alignment, no canonical status change).** tfpt\_1 Strong CP [E]  $\rightarrow$  [E]/[C] (given RP) and  $\theta_{12}$  “genuinely open”  $\rightarrow$  “conditionally derived”; tfpt\_2 Starobinsky  $A_s$  [E]  $\rightarrow$  [E]/[C] ( $M$  exact,  $A_s$  a band over  $N_\star \in [50, 60]$ ) and the Part-II abstract “opens the one remaining gap”  $\rightarrow$  “closes the last structural gap”; tfpt\_research\_contracts the QFT4D “nonperturbative measure stays open” line and the “only  $U_{\text{point}}$  stays open” headline reconciled with the [C] redundancy /  $v_{\text{geo}}$  reduction stated alongside; origin\_theory class C re-typed to the ambient-measure [C] redundancy; introduction ( $U_{\text{wall}}$ )  $\rightarrow$   $v_{\text{geo}}$ . Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the tfpt\_3 cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ( $v_{\text{geo}}$  [O];  $G_{\text{net}}$ /SEAM.EQUIV.01,  $F_{\text{transfer}}$ ,  $N_\star$ , QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated status\_heatmap (ledger-driven) and extended the narrative figures script\_timeline/residual\_chain/qft\_skeleton through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the qft\_skeleton “Stelle ghost” open box is retyped (resolved truncation artefact). Added qft\_skeleton/qft\_unification to the manifest figure list.

## 2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] v405\_seam\_equiv\_omega.py (SEAM.EQUIV.02): the *dual keystone* at  $\tau=\omega$  – the family/flavor sector as the order-3 Eisenstein/ $A_2$  face of  $E_8$ , dual to the seam  $= (E_8)_1$  at  $\tau=i$  (v404). A named contract (genre of v286/v398), NOT a closure. [E]  $\chi_{E_8}(\omega) = 0$  (the family CM point degenerates  $E_8$ ) vs  $\chi_{E_8}(i) = 12$ ; [E]  $A_2$  is the  $\tau=\omega$  Eisenstein lattice (Cartan  $\det = 3 = N_{\text{fam}}$ ,  $h = 3 = N_{\text{fam}}$ ); [E]  $R$  lives in  $E_6 \times A_2$  ( $248 = 78 + 8 + 2 \cdot 27 \cdot 3$ ,  $\det R = 8 = \dim A_2 = \text{rank } E_8$ ); [E]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  CM phase; [E] the cusp weights  $\{0, 1/3, 2/3\}$  are the order-3 deck  $= N_{\text{fam}} = \det Q$ . [C] the family sector is the order-3 Eisenstein/ $A_2$  deck; the two keystones sit at the only two elliptic points of  $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical  $\mathbb{Z}_3$  deck map (v140) + the  $v_{\text{geo}}$ -class scale. Cited in origin\_theory; Python-only.
- [E]/[C]/[O] v406\_pmns\_phase\_origin.py (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the  $\tau=\omega$  keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one  $v_{\text{geo}}$ -class unit, so PMNS has *zero free flavor dials beyond*  $v_{\text{geo}}$ . [E] the three angles (v9/v268)  $\rightarrow$  unitary  $U_{\text{PMNS}}$  (v270); [C]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C]  $J_{\text{PMNS}} = -0.0297$  derived; [E] the absolute scale = one seesaw ratio =  $v_{\text{geo}}$ -class (v272/v153). [O] residual = the seesaw operator realisation ( $M_R$ , v184) + the  $\Sigma m_\nu$  floor kill test. Cited in tfpt\_2\_standard\_model; Python-only.
- [E]/[C]/[O] v407\_dn\_pairings\_omega.py (FLAV.SELECTOR.CLOSE.01): the  $R4'$  selector residue folds into the  $\tau=\omega$  keystone – the three  $(d, n)$  pairings ARE the  $\tau=\omega$  family-slice atoms. [E] the frame  $(\mathbf{1}, a, \sigma)$  has  $\det = 11 = \|Pl(K)\|_1$  and  $n = (5, -9, 6)$  is the unique covector with  $n \cdot \mathbf{1} = 2$ ,  $n \cdot a = 8$ ,  $n \cdot \sigma = 121 = 11^2$  ( $\Rightarrow (d, n) \Rightarrow R$ , v136/v139);

[E]  $n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$  (the  $A_2$  block reading  $R$  in  $E_6 \times A_2$ , v405); [E] the pairings  $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$ . [C] deriving  $(d, n)$  folds into SEAM.EQUIV.02 ( $d$  anchor-derived, v139); [O] the lone residue “why  $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (The arithmetic hull + the $E_8$ -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ . [E] the squarefree parts  $-1, -3, 5$  are independent in  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so  $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$  with Galois group  $(\mathbb{Z}/2)^3$  (order 8); [E] exactly 7 quadratic subfields  $\{-1, -3, 5, 3, -5, -15, 15\}$ , splitting 4 imaginary  $= |\mu_4|$  (CM/dynamic) + 3 real  $= N_{\text{fam}}$  (RM/static); [E] ramified *only* over  $\{2, 3, 5\}$  = the atoms = the primes of  $h(E_8) = 30$ ; [E]  $2^{\omega(h)} = \text{rank}$  is  $E_8$ -special (the  $E_7$  control  $h = 18$  gives  $2^2 = 4 \neq 7$ ). [C]  $K$  is the abelian  $(\mathbb{Z}/2)^3$  *arithmetic* hull dual to the  $E_8$  *lattice* hull; the 7 quadratic subfields are a candidate for the 7  $E_8$  audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] `v404_e8_char_cm_duality.py` (E8.CMDUAL.01): the  $(E_8)_1$  vacuum character  $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$  read at *both* CM points, extending v282 ( $\chi_{E_8}(i) = 12$ ) to the second elliptic point. [E]  $\chi_{E_8}^3 = j$  (q-series); [E] SEAM  $\tau=i$  ( $\mathbb{Q}(i)$ , order 4): cross-ratio  $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$  ( $E_8$  present, v214/v282); [E] FAMILY  $\tau=\omega$  ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic  $\lambda$  ( $\lambda^2 - \lambda + 1 = 0$ )  $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$  ( $E_4$  vanishes, v220/v267); [E]  $\tau=i, \tau=\omega$  are the *only* elliptic points of  $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where  $E_8$  is present ( $\chi = 12$ , the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where  $E_8$  degenerates ( $\chi = 0$ :  $U_{\text{wall}}, v_{\text{geo}}$ , the CP texture), a candidate dual keystone at  $\tau=\omega$ . Cited in `origin_theory`; core values  $j = 1728/j = 0$  already mirrored (v214/v220/v267/v282), Python-only.

## 2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] `v398_seam_state_rigidity.py` (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer  $T=e^{-H}$  from RP + reflection alone, v54, quasi-free v155, no  $\mu_4$  normal form); [E] band-limited character-block-diagonal core  $[\rho, |k|]=0$  swept  $N=4.64$  (beyond the 3-dim  $H^1$ ); [E] holomorphy  $\Rightarrow (E_8)_1$  ( $c=8, |\det \text{Cartan}|=1$  vs 4, index  $4=|\mu_4|$ ). [O] the one residual:  $[\rho, \Lambda_\Sigma]=0$  on the full  $L^2$  (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- [E]/[C]/[O] `v399_gravity_complete.py` (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton  $8\pi$ ,  $\Lambda$ , scalaron  $c_3^7$ , horizon  $1/|\mu_4|$ , recovery  $(2/3)^6$ ) are all boundary/seam/gap, so  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ ; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- [E]/[O] `v400_grav_nonlocal_action.py` (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local  $R+R^2$  as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire  $a=e^u \Rightarrow$  one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces  $R+R^2$  with the atom-fixed scale  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$  (v36/v253). [O] the entire- $a$  assumption + perturbative-only.



- **[E]/[O] v401\_metrology\_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set  $\{a, \pi, v_{\text{geo}}\}$  certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping  $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$ ; **[E]** the  $G$ -vs- $\Lambda$  over-determination 0.11% (v274); **[O]**  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ ).
- **[E]/[C] v402\_ftransfer\_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor  $F_{\text{transfer}}$ , the composition  $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$  (consolidating the FR.\*.SOLVE solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

## 2026-06-24 ( $\varphi_0$ leading term from icosahedral combinatorics: v396)

- **[E]/[C] v396\_phi0\_icosahedral.py** (HYP.PHI0.01): the tree-level seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi) = F/(|\text{Aut}|\pi) = (F/(g_{\text{car}}N_{\text{fam}}))c_3$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ );  $F/h = |Z_2|/N_{\text{fam}} = 2/3$  (the same survival ratio as v327). **[E]** Gauss–Bonnet consistent with  $c_3 = 1/(|Z_2|4\pi)$  (v216/v342). **[E]** honest negative: the puncture  $48c_3^4$  is still analytic, not a hypergraph fraction. **[C]**  $\varphi_0^{\text{tree}}$  is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

## 2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- **[E] v395\_hypergraph\_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) unifies v299 (carrier growth), v327 (branching rule  $M$ ) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column +  $M$  on each node’s family vector ( $T_{\text{micro}} = T_{\text{net}} \otimes M$ ); joint attractor marks  $\otimes e_0$ ; one clock hand (6 family steps) gives recovery  $(2/3)^6$ ; v324’s  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis. **[O]** the 3-arm seed ( $P_2$ ) and analytic  $\varphi_0$  remain irreducible (v312).

## 2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — **do the “external” frontier scheme-numbers land on the  $\{2, 3, 5\}$  clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). **[E]** the  $\eta_B$  decuple  $A_\Lambda = 10 = |Z_2|g_{\text{car}}$  is ON the clock (v212); **[E]** the Koide rate  $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$  is ON the clock (v303); **[E]** the axion spine angle  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  is ON the clock (v211/v373); **[E]** the proton  $C_p \sim 2.83 \pm 0.15$  is *not* a discriminating clock value — within its  $\pm 5.3\%$  band several simple values fit  $(2^{3/2}, 17/6, 14/5)$ , so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number  $C_p$ . Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

## 2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)

- **v394 (COXETER.ATOMS.01) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye [E] re-reading tying v390 (the prime facets) to the anchor  $a = (1, 1, 2)$  and v319 (the  $5 \times 6$  clock). [E]  $a = (1, 1, 2)$  has  $e_3=2=|Z_2|$ ,  $p_0=3=N_{\text{fam}}$ ,  $e_2=5=g_{\text{car}}$ , product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  — the clock's primes *are* the anchor's atoms. [E] atoms = facet fields (v390):  $2 \rightarrow \mathbb{Q}(i)$ ,  $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ ,  $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] the QG susceptibility is pure  $\{2, 3\}$ :  $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$ ,  $(2/3)^6 = 2^6/3^6$ , exponent  $6 = 2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2:  $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$  with  $|\mu_4|=4=2^2$ . [E]  $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$  (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01 — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383's "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

## 2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value  $\pm$  computed first correction". [E] FIXED-POINT band (the  $\varphi_0$ -derived  $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$ ) = the explicit  $\varphi_0$  puncture term  $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$  (exact;  $= \theta_{12}$ 's  $\varepsilon$  first-correction,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ); [E] SEAM-GAPPED band (Koide  $F_{\text{pole}}$ , recovery)  $= (2/3)^6 \approx 8.78\%$ , the QG-decoupling bound capped by  $\chi - 1 = 64/665 \approx 9.62\%$  (v337); [E] EXACT-IDENTITY band  $= 0$  structurally ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ); [E] EXTERNAL-RATE band = external physics, quoted  $(m_p/m_e \pm 7.5\%$  v374;  $n_s/r$  the  $N_\star \in [50, 60]$  band;  $\eta_B 1.07 \times$  v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

## 2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01's one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope  $c_1(L)$  at  $L = 66, 130, 258, 386, 514$  is a monotone Cauchy-like sequence approaching 1 ( $|c_1(L) - 1|$  strictly decreasing,  $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$ ); [E] a  $1/L$  Richardson fit extrapolates to  $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar  $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$  is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin  $(E_8)_1$  over the same- $c$   $SO(16)_1$  (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01** — the multi-dimensional case, reduced to the scalar core. The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): [\[E\]](#) `component_tendsto` (a non-dominant eigencomponent  $|r| < 1$  decays,  $r^n c \rightarrow 0$ ); [\[E\]](#) `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [\[E\]](#) `deviation_unique` (two starts give deviation vectors both  $\rightarrow 0$ , so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [\[E\]](#) `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , [v56/v82/v383](#)). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

## 2026-06-24 (Website: an animated “Universal Gap Lab” visualising [v383–v391](#))

- **Website** — the `UniversalGapLab` animation on /verification. A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction  $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$  racing to its attractor at  $r = (2/3)^6$  (seam-gapped, [v303/v387/v388](#)) or the golden  $(\varphi + 2)/4$  ([v312](#)), with a gapless  $r = 1$  track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size ([v388](#)); (B) the entire-form-factor graviton ratio  $e^{-u}$  falling faster than any power, ghost-free and UV-soft ([v386/v389](#)); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms ([v384](#)). Mirror-only: every number is grounded in a module; no theory change.

## 2026-06-24 (Honest attempt at an external target: [v391](#) reduces it, not a closure)

- [v391](#) (`ALPHA.QUILLEN.PROGRESS.01`) — an honest attempt at the external target `ALPHA.QUILLEN.EXACT.01` ([v382](#), door 5). It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays [\[O\]](#) (external math, type A, [v384](#)). [\[E\]](#) a solvable model  $\det_\zeta(-\partial_\tau^2 + m^2)$  on a circle of circumference  $\beta$  equals  $(2 \sinh(\beta m/2))^2$ , whose  $m$ -variation  $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$  is a closed heat-kernel/Laurent series (the  $m^3$  coefficient  $-1/360$  verified). [\[E\]](#) so a  $\zeta$ -determinant’s coupling-variation is structured heat-kernel data, and [v382](#)’s counterterm  $8b_1c_3^6$  (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the [v382](#) split is re-verified at  $\alpha^{-1} = 137.0359992$ . [\[E\]](#) the reduction: the open step now reduces to the named sub-target “the seam  $U(1)$  operator’s Seeley–DeWitt coefficient  $= 8b_1c_3^6$ ”. [\[O\]](#) *not closed*: the actual seam-operator  $\zeta$ -determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, [v336](#)) stays the other external [\[O\]](#); the value  $\alpha^{-1}$  stays [\[E\]](#). An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01** — a Lean 4 proof of the gap  $\Rightarrow$  unique-attractor  $\Rightarrow$  parameter-free core (door 4). New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of [v303/v383/v387](#) at the one-dimensional reduction — the affine gap contraction  $x \mapsto x_\star + r(x - x_\star)$  with multiplier  $r = \lambda_2/\lambda_1$ . [\[E\]](#) `iter_sub`: the closed form  $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$  (general  $n$ ); [\[E\]](#) `iter_tendsto`:  $0 \leq r < 1$  (the gap)  $\Rightarrow \text{iter}_n \rightarrow x_\star$  (the unique attractor); [\[E\]](#)

**starts\_agree**: two starts differ by  $r^n(a-b) \rightarrow 0$ , so the attractor is independent of the initial state (parameter-freeness); **[E] no\_gap\_no\_forget**:  $r = 1$  (no gap) leaves  $\text{iter}_n - x_\star = x_0 - x_\star$  constant — the initial condition is then a free parameter, so the gap is exactly what removes it; **[E]** the seam rate  $(2/3)^6$  and the golden  $(\varphi + 2)/4 = (5 + \sqrt{5})/8$  both lie in  $[0, 1)$ . Honest scope: the scalar reduction (the contraction core **v303/v383** iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in **origin\_theory**; ledger row **FORM.SPECTRALGAP.01**.

## 2026-06-24 (Completing the Coxeter clock: **v390 the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam)** facet of $30 = 2 \cdot 3 \cdot 5$ )

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** **v383/v387** named two number-field facets of the clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  — the golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (prime-5, carrier) and  $(2/3)^6 \in \mathbb{Q}$  (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field  $\mathbb{Q}(i)$ , i.e. the CM point  $\tau = i$  and the order-4  $\mu_4 = \mathbb{Z}/4$  *square* seam deck (**v282/v288/v333**). **[E]**  $h(E_8) = 30$  factors into exactly three primes; one quadratic facet per prime, each ramified at its prime:  $\text{disc } \mathbb{Q}(i) = -4$  (2 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$  (3 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$  (5 ramifies); the ramified primes are  $\{2, 3, 5\}$  with product  $30 = h(E_8)$ . **[E]** the prime-2 (Gaussian) facet is the  $\mathbb{Z}/4$  seam deck: min. poly  $x^2 + 1$ ,  $j(i) = 1728 = 12^3$ ,  $\tau = i$  fixed by the modular  $S$  of order  $4 = 2^2$ ,  $|\mu_4| = 4 = 2^2$  — the static, even, seam-orientation facet. **[C]** so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set  $\{2, 3, 5\}$  is forced by  $h(E_8) = 30$ , not chosen. Cited in **origin\_theory**; a synthesis, no new number; Python-only.

## 2026-06-24 (Graviton loop finiteness: **v389 the entire form factor is UV-finite at the loop level by power counting**)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the loop level.** The honest next step past **v386** (which established the tree amplitude), extending **v304/v370/v380/v386** from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. **[E]** super-polynomial falloff:  $p^k e^{-p^2/M^2} \rightarrow 0$  for every degree  $k$ . **[E]** one dressed line turns a quadratic divergence into the finite  $(M^2/2)e^{-p_0^2/M^2}$ , while the GR  $\int p^3/p^2 dp$  diverges quadratically. **[E]** the superficial degree  $\rightarrow -\infty$  at every loop order  $L$  (the form factor multiplies the integrand by  $e^{-(\#\text{internal})p^2/M^2}$ ; dressed cutoff-ratio  $\rightarrow 1$  vs GR  $\rightarrow 4$ ). **[E]** the regulator is the seam scale  $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  (**v253/v259**), not a dial. **[C][O]** loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is **v386**; not claimed solved. Cited in **tfpt\_4\_frontier**; Python-only.

## 2026-06-24 (Typed correction budget: **v388 the gap correction is keyed to the mechanism class, not a uniform band**)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (**v387**) is a *typed* budget.** A machine-checked answer to “does the  $(\lambda_2/\lambda_1)^n$  correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading  $\lambda_2$  exists, so the prediction surface splits into four classes. **[E]** *seam-gapped* (Koide  $F_{\text{pole}}$  **v303**, recovery **v221**, the QG bound capped by  $\chi = 729/665$  **v337**): rate  $(2/3)^6$ ; the discrete compiler: golden  $(\varphi + 2)/4$  (**v312**). **[E]** *exact identity* ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$ ): no  $\lambda_2$ , so the correction is 0 structurally and a band would contradict the **[E]** status. **[E]** *external-rate*



( $\eta_B$  washout, axion freeze,  $m_p/m_e$  RG): the gapped *shape* with an external rate — only  $F_{\text{pole}}$  has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g.  $\sin^2 \theta_{12}$ ,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ). [E] the Koide first correction (door 2):  $F_{\text{pole}}$  is the one seam-gapped prediction, rate  $(2/3)^6 \approx 0.0878$  (v371), the source quotient 0.33% below  $2/3$  — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

## 2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ( $\sin^2 \theta_W = 3/8$  vs 0.231); [C] the optional carrier-PS branch sits at  $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$  GeV ( $\kappa \in [1.0, 1.2]$ , v253); [E] minimal PS gauge bosons are  $SU(4)$  leptoquarks that mediate rare LFV ( $K_L \rightarrow \mu e$ ), *not*  $p \rightarrow e^+ \pi^0$ , so the branch clears the binding bound by  $M_{PS}/M_{\text{bound}} \sim 10^7$  – proton-safe, with *no* fake  $\tau_p$  window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2$  has its only pole at  $p^2 = 0$  (residue  $1 > 0$ , ghost-free), is exponentially softened in the UV (the ratio  $e^{-u}$  falls faster than any power) and reduces to GR + a calculable  $-p^2/M^2$  correction at low energy ( $M = M_{\text{scal}}$ , v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering  $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$  is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] flavor, horizon recovery and the QG bound all share the  $(2/3)^6$  rate (the family-3 facet; QG susceptibility  $\chi = 729/665$ ), while the discrete compiler transient decays at the golden  $(\varphi + 2)/4$  (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

## 2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for  $F_{\text{transfer}}$ ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer  $\{1, (2/3)^6, (1/3)^6\}$  with gap  $6 \ln(3/2)$  (v56/v82); the affine- $E_8$  adjacency with Perron eigenvalue 2 (Kac marks) and subleading  $2 \cos(\pi/5) = \varphi$ , gap  $2 - \varphi = 1/\varphi^2$  (v312/v313); the decoupling margin  $\Delta_{\text{eff}} \approx 1.648$  (v76/v337). [E] the two number-field facets (v314): golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (carrier-5, static) and  $(2/3)^6 \in \mathbb{Q}$  (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ . [C] the other sectors are the same structure — gravity ( $\delta S=0$  equilibrium, v358/v359), the QG measure (Gaussian saddle, v365),

the boundary QFT (holomorphic  $c=8$  RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221),  $\tau=i$  (v333). [C] the meta-theorem: gapped  $\Rightarrow$  unique attractor  $\Rightarrow$  parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden ( $v_{\text{geo}}$ , the No-Unit theorem v153/v364), [E] (C) external physics ( $F_{\text{transfer}}$ , v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

**2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for  $S_{\text{pert}}$ , and v382 the Quillen determinant-line variation as a tracked target)**

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix  $S_{\text{pert}}$  (the previous documents had the  $S_{\text{pert}}$  skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order  $\omega \leq 4$  and the counterterm space is finite at every order; [E] BRST nilpotency  $s^2=0$  verified exactly via the Jacobi identity for  $su(2)$  and  $su(3)$  (the carrier weak+colour content); [E] the seam gap  $\Delta = 6 \log \frac{3}{2} > 0$  (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order  $T_n$  existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the  $R^2/\text{Weyl}^2$  gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and  $S_{\text{pert}}$  is not the ambient measure QG.AMB.01 (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this*  $F_{U(1)}$  functional” obligation (previously tracked only as EM.WARD.01’s residual, v341/v342) is elevated to its own citable target:  $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0$  on the determinant line over the  $\mu_4$  seam moduli. [E] the Quillen split holds at the  $\alpha^{-1}=137.0359992$  root with every coefficient a named atom (index  $8b_1$ , heat-kernel  $c_3^{3,6}$ , discriminant  $q(D_5)+q(A_3)=2$ ); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of SEAM.EQUIV.01 (v336) — the seam  $U(1)$  determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.



## 2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **v369 (QG.AMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy  $\Rightarrow$  DHR( $(E_8)_1$ ) = Vec: #primaries =  $|\det \text{Cartan}| = 1$  for  $E_8$  vs 4 for  $SO(16)$   $\Rightarrow$  no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ( $|\det K| = 1$ ); [E] finite Petz recovery rate  $(2/3)^6$  (v221); [E] the gap-decoupling margin  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$  is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, introduction) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root README.md, the Zenodo description and `website/lib/papers.ts`: SEAM.EQUIV.01 is *closed modulo a cited theorem* (lattice v367/v368 + the  $S_3$  stack v376–v379 + Lean FORM.SEAM.MMST.01; residual [O] = the cited MMST/OS existence only); QG.AMB.01 is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; AUDIT OK.

## 2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. [E] the seam KMS cutoff  $f(u)=e^{-u}$  (v259) gives the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$  with  $a(u)=e^u$  entire and nowhere zero (only the  $p^2=0$  pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of  $a$  and carries a Stelle-ghost zero ( $T_1=1+u \rightarrow u=-1$ ;  $T_2 \rightarrow -1 \pm i$ ); the entire  $a$  has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order  $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$  — the ghost decouples, “entire” = “do not truncate”. [C]  $\Rightarrow$  perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

## 2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — FORM.SEAM.MMST.01)

- **FORM.SEAM.MMST.01 — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms):  $D = 16$ , rank =  $c = 8$ ,

the range  $8 \leq c \leq 16$ ,  $c_- = 8 \neq 0$ , and  $\det K = 1$  vs  $SO(16)$ 's 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion  $(E_8)_1$  follows by a well-typed composition whose residual (**#print axioms**) is exactly those two published theorems plus the carrier gap — no hidden assumption, no sorry. Mirrors the S3 closure stack v376–v379; so SEAM.EQUIV.01 is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

**2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge  $c=8$  from the lattice (v376), the  $(E_8)_1$  character (v377), the genus-1 torus count = 1 (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)**

- **v376 (SEAM.S3.CENTRALCHARGE.01).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives  $c = 1.0000$  per complex mode (Peschel correlation-matrix EE; the  $L=4m+2$  sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , chiral ( $c_- = 8$ , v367/v368). Numerical  $c=8$  from the lattice. Python (numpy).
- **v377 (SEAM.S3.E8CHARACTER.01).** The exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary); the  $\mu_4/\text{GSO}$  promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, 4 primaries). Python (sympy q-series).
- **v378 (SEAM.S3.MODULAR.01).** The genus-1 discriminator: KLM condensation  $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$ ,  $|\det \text{Cartan}(E_8)|=1$  vs  $D_8=4$ , and the single character's modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 (holomorphic =  $(E_8)_1$ ) — the “hard part” v344 isolated. Python (sympy).
- **v379 (SEAM.S3.RP.01).** Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With  $c=8 + (E_8)_1$  in hand, the OS inputs are complete and C7/QG.AMB.01 is then discharged by the redundancy theorem v369. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net  $(E_8)_1$  is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem v336), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

**2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (v375))**

- **v375 (OBSERVATORY.REGISTRY.01).** A CI over `freeze_file.csv`: **[E]** every prediction carries a kill criterion (falsifiability complete, no orphan claims); **[E]** the headline values re-derive from  $\{\varphi_0^{\text{ret}}\}$  ( $\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$ ,  $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}$ ,  $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$ ); **[C]** a live scorecard records  $\theta_{12}$  vs the JUNO first run ( $0.28\sigma$ , compatible),  $\theta_{13}$  vs NuFIT 6.0 NO ( $2.0\sigma$ , the flagged most-tensioned core prediction),  $\beta_{\text{rad}}$  vs ACT DR6 ( $0.37\sigma$ , a hint),  $r < 0.036$  (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

**2026-06-23 (Track 3 — the  $F_{\text{transfer}}$  functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371),  $\eta_B$  Boltzmann (v372), axion finite- $T$  relic (v373),  $m_p/m_e$  uncertainty budget (v374))**

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes  $Q$  above  $2/3$ ; the required  $\epsilon$  is negative while QED's is positive), so the transfer is the non-QED  $53/54$  operator /  $(2/3)^6$  Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen  $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$  (v212) gives  $\eta_B$  within a factor  $\sim 1.1$  of  $6.1 \times 10^{-10}$  with no  $M_R$  dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- $T$  axion misalignment ODE decides the branch: the frozen spine angle  $3\pi/5$  gives  $\Omega_a h^2 \sim 0.12$  (in  $[0.08, 0.16]$ , lands on  $\Omega_{\text{DM}}$  untuned) while the hilltop  $170^\circ$  over-produces ( $\sim 5.4\times$ ); replaces the v326 toy. [C]/[X] (lattice  $\chi(T)$  input).
- **v374 (FR.QCD.BUDGET.01).** The  $m_p/m_e$  uncertainty budget: with  $b_3 = -7$  [E] and the two declared externals ( $\alpha_s(M_Z)$ ,  $C_p$ ,  $\sim 5.3\%$  each), the central  $\sim 1824$  (v262) carries a propagated  $\pm 7.5\%$  band [ $\sim 1690, \sim 1960$ ] CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

**2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ( $p+ip$ ) lattice model with numerical Chern  $|C|=1$  realises the  $c_-=8$  edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))**

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the  $p+ip$  Bloch model is gapped at  $M=1$  with a numerically computed Fukui–Hatsugai–Suzuki Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ); [E]  $N_{\text{Maj}} = \dim S^+ = 16$  copies give  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). [C] the  $\mu_4$  condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ( $\min_{k_x} |E| \sim 0$ ) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ( $c_-=8$ ). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

**2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))**

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar  $1/(p^2(p^2+M^2))$  pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ( $\text{tr } P_2 = (d-2)(d+1)/2$ ,  $\text{tr } P_1 = d-1$ ,  $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$  sum to  $d(d+1)/2$ ;  $d=4$  split  $5+3+1+1=10$ ), the EH propagator is  $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$ , and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor  $e^{p^2/M^2}$  (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour

(v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304’s entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE:  $\text{tr } P_2=5$  is dimension counting, not  $g_{\text{car}}$  (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

**2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST’s free-fermion class so the scaling limit is  $(E_8)_1$  (v366))**

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is  $1/c_3 = 8\pi$  (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped  $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337)  $\Rightarrow$  mode-by-mode convergence, finite model  $\log\text{-det } \text{Tr}' \log M = 6 \log \frac{9}{2}$ ; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ( $\chi = 729/665$ , v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST’s free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E]  $D = \dim S^+ = 16$  Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8$ ,  $\text{rank } E_8 = 8$  so  $\text{rank} \leq c \leq D$  is  $8 \leq 8 \leq 16$  (in range); [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ( $c_- = D/2 = 8 \neq 0$ ). [C] the MMST scaling-limit theorem then applies, and [E] the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K = 1$  vs  $SO(16)$   $\det K = 4$ , v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)**

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the  $R + R^2$  regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit  $v_{\text{geo}}$ . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume  $\Rightarrow$  Einstein tensor, Lovelock  $\Rightarrow$  conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 +  $v_{\text{geo}}$  residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of  $c_3$  converging to  $1/(8\pi)$ ; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.



**2026-06-23 (Direction 6,  $v_{\text{geo}}$  sharpened: after the parameter-free gravity, the gravity sector adds NO new dimensionful input either, so  $v_{\text{geo}}$  is the SINGLE dimensionful input of the complete theory — final tally  $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials: v364)**

- **v364 (VGE0.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made  $v_{\text{geo}}$  a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  reduce to  $v_{\text{geo}}$  times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)  $\times$  (a power of  $v_{\text{geo}}$ ); TFPT's free content is  $\{v_{\text{geo}}, \pi\}$  with ZERO dimensionless dials, a  $\sim 26 \rightarrow 1$  reduction vs the SM. [O]  $v_{\text{geo}}$  stays irreducibly primitive (No-Unit;  $1/G$  UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced  $c_- = 8$ , and the matter vacuum-energy backreaction is finite ( $\rightarrow$  the forced  $\Lambda$  from  $\alpha$ , not  $M_{\text{Pl}}^4$ ) — the loop closes; no new SM-quantum-number read-out (declined): v361)**

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced:  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$  (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE  $\rightarrow \Lambda$  from  $\alpha$ : the matter vacuum energy through  $G_{ab} = c_3^{-1} T_{ab}$  is gap-suppressed (Decoupling Thm v337) to  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$  (v60), the 123 orders =  $119.03 + 3.92$  — *not* the  $M_{\text{Pl}}^4$  catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit:  $\delta\langle K_B \rangle = (8\pi^2 R^4/15)\delta\langle T_{00} \rangle$ , carrier  $T_{ab}$  = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond  $c_- = 8$  and  $\Lambda$  — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

**2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known  $R^2$  scalaron — already falsifiable via  $n_s/r$ ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)**

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap  $\{1, (2/3)^6, (1/3)^6\}$  yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known  $R^2$  scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the  $R^2$  scalaron (spectral-action  $a_4 = R^2/72$ , v36;  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ , exponent  $7 = g_{\text{car}} + N_{\text{fam}} - 1$ ) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law  $\rightarrow f(R) = R + R^2$ , v28/v359). [E] it is ALREADY falsifiable via  $n_s = 1 - 2/N_*$ ,  $r = 12/N_*^2$  (live CMB-S4 kill test). [E] the gap  $\Delta = 6 \ln \frac{3}{2}$  is a dimensionless rate, so a distinct gap-induced curvature<sup>2</sup> term sits at  $\xi v_{\text{geo}} \sim$  the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent  $7 = g + N - 1$  and the gap exponent  $6 = 2N_{\text{fam}}$  are different;  $c_3^7$  and  $(2/3)^6$  unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

**2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with BOTH coefficients TFPT-fixed ( $8\pi = 1/c_3$  from v358,  $\Lambda$  from  $\alpha$  via v60); B6 closed at the local level: v359)**

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR  $G_{ab}$  (not just  $R_{ab}$ ) — trace  $g^{ab}G_{ab} = (1 - d/2)R = -R$  in  $d=4$ , the structure that carries  $\Lambda$ . [E] Lovelock:  $G_{ab}$  is the unique symmetric divergence-free second-order metric tensor, so  $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$  forces  $\nabla^a T_{ab} = 0$  — conservation is an output. [E] coefficient 1:  $8\pi G \rightarrow 8\pi = 1/c_3$  fixed (v358). [E] coefficient 2: Jacobson leaves  $\Lambda$  free, TFPT fixes it —  $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions  $\rightarrow$  the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate, gap-decoupled);  $v_{\text{geo}}$  the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the  $\Lambda$  here matches the  $\alpha$ -readout.)

**2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law  $\delta S = \delta \langle K \rangle$  with TFPT’s atoms gives the LINEARISED Einstein equation with  $G$  from  $c_3$ ; the thermodynamic and geometric origins of  $c_3$  coincide via  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ ; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)**

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement  $\Rightarrow$  Einstein equation” with TFPT’s atoms. [E] PARAMETER-FREE COEFFICIENT:  $\delta S = \delta \langle K \rangle$  gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  fixed; all four  $c_3$ -roles consistent (Unruh  $1/(2\pi) = 4c_3$ , EH  $1/(16\pi) = c_3/2$ , Einstein  $8\pi = 1/c_3$ , Bekenstein  $1/4 = 1/|\mu_4|$ ) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law’s coefficient  $2\pi/\eta$  equals the geometric  $|\mathbb{Z}_2|2\pi\chi$  iff  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$  (v216) — the thermodynamic and geometric origins of  $c_3$  are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux  $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$ . [E] ENTROPY DENSITY ATOM-FIXED:  $1/4 = 1/|\mu_4|$  (v57) + central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  (form v59). [O] residual: linear  $\rightarrow$  full non-linear (the original B6) + the  $v_{\text{geo}}$  unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

**2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)**

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ( $\Delta = 6 \ln(3/2) > 0$ , v302) quasi-free 16-Majorana collar (v155/v160)  $\rightarrow$  **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly



reduces the v336/v351 “does the collar have a chiral edge?” residual  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range rank  $\leq c \leq D$ ,  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3=1/(8\pi)$  a one-sided Gauss–Bonnet  $(|\mathbb{Z}_2| \nmid K, \text{v58})$ . [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs  $\Phi=0$ ) and  $\det R=8$  is read off — not a PDE. Only the absolute amplitude  $U_{\text{point}}$  (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

**2026-06-22 (bandwidth search of the unmapped  $E_8$  region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ( $\sum \deg = 128 = \dim S^+$ ,  $\prod \deg = |W(E_8)|$ , exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)**

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like  $E_8$  returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about  $E_8$ ), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number):  $\sum \deg = 128 = \dim S^+$ , the spinor half of  $248 = 120 + 128$  (theorem:  $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$ ) – the invariant budget of  $E_8$  equals the fermion content the carrier reads;  $\sum \exp = 120 = \# \text{pos roots}$ ;  $\prod \deg = |W(E_8)| = 696,729,600$ ; exponents = the  $\varphi(30)=8$  totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set;  $\max \deg = 30 = h(E_8)$ . [C] SUGGESTIVE, not a theorem:  $12=h(E_6)$ ,  $18=h(E_7)$ ,  $30=h(E_8)$  are all degrees of  $E_8$  (the  $2T/2O/2I$  McKay ladder, v219), but “ $\deg(E_8)$  contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches  $14=\dim G_2$ ,  $20=\det L$ ,  $24=|W(A_3)|$ ,  $12=|R(A_3)|$ ,  $18=p_4(a)$  (v66 already flagged 12, 18 as fittable) each admit  $\geq 2$  readings, and the lone extra prime in  $|W(E_8)| = 2^{14}3^55^27$  “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

**2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of  $E_8$ ’s 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)**

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an  $E_8$  structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much  $E_8$  structure

maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of  $E_8$ 's eight Casimir invariant degrees  $\{2, 8, 12, 14, 18, 20, 24, 30\}$  (exponents  $+1$ ), exactly THREE feed a readout — 2 (metric), 8 (rank  $\rightarrow c_3=1/(8\pi)$ ), 30 (Coxeter  $\rightarrow g_{\text{car}}=5$  and the order-30 cycle) — and FIVE carry no primary readout  $\{12, 14, 18, 20, 24\}$  (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8,  $h=30$ ,  $\det K=1$ ,  $D_5 \oplus A_3$ , the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of  $E_8$  untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED  $\Rightarrow$  NOT NUMEROLOGY:  $\varphi = 2 \cos(\pi/5)$  appears only in internal structure (the affine- $E_8$  attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation,  $\varphi$  is fitted to nothing — it is the algebraic signature of the 5-fold ( $h=30=2 \cdot 3 \cdot 5$ ), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD ( $E_8$  is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

**2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system;  $\pi$  is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)**

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms  $\rightarrow$  theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop:  $g_{\text{car}}=5 = \max$  prime of the output  $h(E_8)=30$ , and the 8 in  $c_3 = \text{rank } E_8$  – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E]  $\pi$  is a FIXED geometric constant (the boundary 2-sphere,  $\oint K=4\pi$ ), not a dial. [C] this refines v352's “framework +  $\pi$ ”:  $\pi$  is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

**2026-06-22 (two consequences: the  $c=8$  which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework +  $\pi$ : v351, v352)**

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$  ambiguity” ( $E_8 \det 1$  vs  $SO(16)=D_8 \det 4$ , both  $c=8$ , flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4  $\mu_4$  clock: the index-4 simple-current extension of  $D_5 \oplus A_3$  is  $E_8$  (v154), the index-2 would be  $SO(16)$ , so the order-4 clock (the four Gauss–Bonnet marks, v216) selects  $E_8$  (full  $\mu_4$  condensation  $16 \rightarrow 4 \rightarrow 1$ ) not  $SO(16)$  (partial  $\mathbb{Z}_2$   $16 \rightarrow 4$ ). The bootstrap agrees ( $h(E_8)=30$  max prime  $5=g_{\text{car}}$  vs  $h(D_8)=14$  max prime 7). [E] so  $\det K=1$  is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E]  $g_{\text{car}}=5$  is forced three ways (v6); the 8 in  $c_3$  four ways (rank  $E_8 = h(D_5) = \phi(30) = \det R$ , v54); the  $E_8$  hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles

are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental  $\pi$ . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework +  $\pi$ . Wolfram mirror +2 (297 → 299).

**2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms –  $g_{\text{car}}=5$  IS the icosahedral 5 of  $h(E_8)=30$ , and the golden ratio is emergent from the  $\mu_4$ -glue, not external: v350)**

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6):  $g_{\text{car}}=5$  is over-determined three ways – rank-fill ( $g_{\text{car}}+N_{\text{fam}}=8$ ), Coxeter-match ( $g_{\text{car}}=\max$  prime of  $h(E_8)=30$ ), Pascal ( $2^4=C(5,0)+C(5,1)+C(5,2)$ ). [E] so  $g_{\text{car}}=5$  IS the icosahedral 5-fold (the 5 in  $h(E_8)=30=2\cdot3\cdot5$ ), not the  $D_5$  rank; the atoms (2,3,5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue:  $D_5$  ( $h=8$ )/ $A_3$  ( $h=4$ ) are non-golden alone, but  $D_5\oplus A_3 \rightarrow E_8$  ( $\mu_4$  index-4, v154) gives  $h(E_8)=30$  (golden) – the seam’s own  $\mu_4$  produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

**2026-06-22 (the honest test – does the raw seam carry  $\varphi$ ? – answer NO: the bare carrier gives  $\sqrt{2}$ , golden is the icosahedral input; the keystone reduces to “is  $g_{\text{car}}=5$  a pentagon or a count?”: v349)**

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio  $2\cos(\pi/5)$  is the signature of a 5-fold rotation ( $5 \mid h$ ); the carrier  $D_5$  has  $h=8$  (eigenvalue  $\sqrt{2}$ ) and  $A_3$  has  $h=4$  – no 5-fold; the dynamical data (cusp  $\{0, 1/3, 2/3\}$ , recovery  $(2/3)^6$ , seed  $\varphi_0$ ) are rational/transcendental. [E]  $5 \mid h$  holds only for  $A_4=SU(5)$  ( $h=5$ ),  $H_3$  ( $h=10$ ),  $E_8$  ( $h=30$ ) – the output/icosahedral side, so  $\varphi$  is the icosahedral INPUT. [E]  $\varphi$  alone would not suffice ( $2I \neq C_5 \times \mu_4$ ,  $20 \neq 120$ ). [O] the resolution: the keystone reduces to one question – is  $g_{\text{car}}=5$  a PENTAGON (a 5-fold rotation  $\Rightarrow$  golden  $\Rightarrow$  icosahedral  $2I = E_8$ ) or just a COUNT (the rank of  $D_5 \Rightarrow \sqrt{2}$ )? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of  $g_{\text{car}}=5$ , not derivable from  $g_{\text{car}}=5$ -as-a-count); there is no hidden  $\varphi$ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

**2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)**

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form  $z \mapsto iz$  (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians =  $2I$ , golden-ratio quaternion norm) IS the rank-8  $E_8$  lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders  $\{1, 2, 3, 4, 6\}$  (the 5-fold absent), so  $\varphi = 2\cos(\pi/5)$  is exactly what extends the crystallographic  $\mu_4$  to the non-crystallographic icosahedral  $2I$  (the quasicrystal). [O] so

the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes,  $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$  and rigidity fixes the rest. [O] honest subtlety: TFPT’s  $\varphi$  is currently  $E_8$ -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries  $\varphi$  non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294  $\rightarrow$  295). An analysis/reduction; closes nothing.

**2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom  $P_3$ : v347)**

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ , v216) while the  $2I/E_8$  structure is the Seifert base over the *spherical*  $S^2(2, 3, 5)$  ( $\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$ ), so L2 bridges two geometric types – why it needs the carrier  $(2, 3, 5)$  input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom  $P_3$  (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293  $\rightarrow$  294). An analysis/roadmap; closes nothing.

**2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam  $\Rightarrow \det K=1$  is a six-link chain, five theorems + one open orbifold realisation, the group forced by the  $(2, 3, 5)$  atoms: v346)**

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam  $\Rightarrow \det K=1$  as six links and shows exactly one is open. [E]/[C] L1 raw seam  $\rightarrow \mu_4$  + four marks (Gauss–Bonnet, v216); [O] L2  $\mu_4$  + the  $(2, 3, 5)$  atoms  $\rightarrow$  the  $2I$  orbifold  $\mathbb{C}^2/\Gamma$  (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3  $2I \rightarrow$  affine  $E_8$  McKay (v219); [E] L4  $2I \rightarrow E_8$  du Val (v232); [E] L5  $\mathbb{C}^2/2I \rightarrow$  Poincaré link  $H_1=0$  (v345); [E] L6  $H_1=0 \rightarrow \det K=1$ . [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional  $SU(2)$  axis signatures  $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$ ;  $(2, 3, 5)$  unique to  $2I$ , and  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$  ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point  $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

**2026-06-22 (executing route R3: the  $(2, 3, 5)$ -rewrite attractor link computed to be the Poincaré homology sphere ( $\det K=1$ ), the combinatorial core complete, only the geometric bridge open: v345)**

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has  $H_1 = \text{coker}(\text{Cartan})$ . [E] the Smith normal form of the  $E_8$  Cartan matrix is the identity, so  $\text{coker}(E_8) = 0$  – the  $E_8$ -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of  $\det E_8=1$ ); among ADE only  $E_8$  does this ( $A_n \rightarrow \mathbb{Z}/(n+1)$ ,  $D_n \rightarrow \mathbb{Z}/4$ ,  $E_6 \rightarrow \mathbb{Z}/3$ ,  $E_7 \rightarrow \mathbb{Z}/2$ ). [E]  $\pi_1 = 2I = SL(2, 5)$  is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so  $H_1 = 0$  and the link is THE Poincaré sphere,  $|2I| = 120 = |\mu_4| h(E_8)$ . So R3’s combinatorial half is



[E]-complete: the  $E_8$  attractor link is forced to be the Poincaré sphere ( $\det K=1$ ), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this  $E_8$  du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292  $\rightarrow$  293).

**2026-06-22 (the one keystone bit  $\det K=1$  mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)**

- **v344 (SEAM.DETK.01).** A bird's-eye, full-spectrum map of SEAM.EQUIV.01's only open residual, the holomorphy bit  $\det K=1$ . [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link  $H_1=0$  / one 1-dim irrep /  $\det \text{Cartan} = 1$  / full  $\mu_4$  condensation /  $\tau=i$  CM), unified by  $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$  across ADE ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ) – only  $E_8$  gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87);  $\det K$  is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the (2,3,5) rewrite attractor is the affine- $E_8$  graph (v312) and the  $E_8$  du Val link is the Poincaré sphere (v232), so  $\det K=1$  becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

**2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)**

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter  $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$  finite,  $\chi = 729/665$ ) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ $c$  ( $c=8$  consistent, but the Carlip ambiguity and no holomorphy  $\det K=1$ ) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

**2026-06-22 (the heat-kernel origin of the  $\alpha$  terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)**

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the  $F_{U(1)}$  determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E]  $c_3 = 1/(8\pi)$  *is* the boundary Gauss–Bonnet coefficient ( $\oint_{S^2} K = 2\pi\chi = 4\pi$ ,  $\chi=2$ , v216). [E] the  $\alpha^2$  (Calderón) term *is* the Gilkey gauge-curvature coefficient ( $30\Omega^2/360 = \frac{1}{12}\Omega^2$ ,  $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$ ), so it is forced, not an ansatz. [E] the  $c_3$ -orders  $\{0, 3, 6\}$  are an arithmetic ladder and  $2c_3^3 = 1/(256\pi^3)$  carries  $\pi^3$  (three boundary insertions). [C] the exact coefficient needs the seam  $F$ -normalisation. [O] the cubic  $\alpha^3$

Maxwell moment (a topological/Chern level) + the full “this is the seam  $U(1)$   $\zeta$ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01.  $\alpha^{-1}$  stays [E] (v3). Wolfram mirror +1 (290  $\rightarrow$  291).

**2026-06-22 ( $\alpha$  as the stationarity of a  $U(1)$  determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)**

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the form of  $\alpha$ , not more decimals”. The EM closure  $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \log(1/\varphi_{\text{seam}}) = 0$  is rewritten as a *stationarity* of the boundary  $U(1)$  determinant line:  $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$ . [E] every ingredient is a NAMED object, not a knob:  $M=41$  is a  $U(1)$  index ( $\frac{4}{5}M = 8b_1$ ,  $b_1=41/10$  the SM  $U(1)_Y$  beta, EM Ward v48);  $c_3=1/(8\pi)$  the Gauss–Bonnet boundary coefficient ( $8=\text{rank } E_8$ );  $-5/4 = -q(D_5)$  a discriminant form and  $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$  carries the other,  $c_3/\varphi_{\text{base}}=3/4=q(A_3)$ ;  $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$ . The Quillen split is verified at the  $\alpha^{-1}=137.0359992$  root. **No new gate:**  $\alpha^{-1}$  stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that  $F_{U(1)}$  is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289  $\rightarrow$  290).

**2026-06-22 ( $A_s$  reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)**

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have  $A_s$  / the old versions derived it” and “can a ratio define mass itself”. [E]  $A_s$  *is* predicted and *dimensionless*:  $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2c_3^7/(24\pi^2)$  with  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$  a pure ratio, so  $A_s$  pins the e-fold count  $N_\star$ , NOT  $\bar{M}_{\text{Pl}}$  — it cannot define an absolute mass (No-Unit, v153, stands). [C] the  $-11.4\sigma$  at the slow Higgs channel ( $N_\star=51.4$ ) is the slow-vs-fast reheating dichotomy; the  $A_s$ -preferred  $N_\star=56.2 \sim$  the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean  $A_s = 2.124 \times 10^{-9}$  using the algebraic  $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$ ; at that  $N_\star$  the current normalisation gives  $2.047 \times 10^{-9}$  ( $-1.9\sigma$ ) — the old clean  $A_s$  is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic  $N_\star$  with one derived from reheating physics and exposed honestly that  $A_s$  requires fast (pre)heating. Record band  $N_\star \in [50, 60]$  (v84) untouched.

**2026-06-22 (the first-principles boundary, mapped: can  $M_{\text{Planck}}$  be computed? can  $F_{\text{transfer}}$  be first-principles? both NO, with the exact reason: v339)**

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* —  $v_{\text{geo}}/M_{\text{Planck}}$  is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E]  $F_{\text{transfer}}$  *cannot be made first-principles* either: for  $m_p/m_e$  the structure is in place (v262, carrier  $b_3=-7$  + closed electron) but two inputs stay external. [X]  $\alpha_s(M_Z)$  is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). The lattice  $C_p \sim 2.83$  is parameter-free but lattice-only (v35);  $\eta_B$ /axion ride on cosmological initial conditions; Koide  $Q=2/3$  is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.



**2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the  $\theta_{13}$  budget: v336, v337, v338, FORM.CARTAN.DET.01)**

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur  $\rightarrow$  Virasoro, correct  $c$ ); WZW range  $\text{rank} \leq c \leq D$ , and  $(E_8)_1$  ( $c=8$ , rank 8,  $D=16$  Majoranas,  $SO(16)_1 \rightarrow (E_8)_1$ ) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by  $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ . [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic  $(E_8)_1$  net” ( $\det K=1$ ), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ; finite susceptibility  $\chi = 1/(1 - (2/3)^6) = 729/665$  gap-suppresses the ambient contribution (margin  $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ , with  $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$  exactly), so no readout depends on the ambient measure  $\Rightarrow$  QG.AMB.01 decoupled; negative control (gap closed  $\Rightarrow \chi$  diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the  $\theta_{13}$  budget, fenced.** The  $+2\sigma$  pressure point (v328) quantified against *both* NuFIT 6.0 NO variants:  $+2.0\sigma$  vs the no-SK central (0.02195) but  $+1.7\sigma$  vs the with-SK best fit ( $0.02215 \pm 0.00057$ ), a  $\sim 0.3\sigma$  systematic spread; [E] 0.023108 lies inside the with-SK  $3\sigma$  band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a  $\sim 6\sigma$  kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit  $D_8 = SO(16)$  Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator:  $D_8$  is even and symmetric just like  $E_8$  (same  $c=8$ ) but  $4 = \det \text{Cartan}(D_8)$  is genuinely not a unit while  $\det \text{Cartan}(E_8)$  is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

**2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)**

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a **COROLLARY** of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations  $U(1)$  are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the  $\mu_4$  clock = rotation by  $\pi/2$  ( $(e^{i\pi/2})^4=1$ ) fixes it  $\Rightarrow \omega \circ \rho = \omega$ . So SEAM.EQUIV.01  $\Rightarrow$  QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ( $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ , gap  $6 \ln \frac{3}{2} > 0$  derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] QG.AMB.01 decoupled, not a gate: margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76/v311/v330/v332)  $\Rightarrow$  no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items

$2 \rightarrow 1$ ; live residual =  $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance (=QGEO.SYM.01)` follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). `lake build green` (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (OpenGates, FAQ, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

**2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)**

- **v333\_cm\_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual:  $\tau=i$  (the square) is the *unique* order-4 modulus —  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $S^4=I$  and fixes  $i$ , and an order-4 deck selects  $j=1728$  (vs the order-3 hexagon  $j=0$ ) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal  $\pi/2$  spacing). So given the  $\mu_4$  deck,  $\tau=i$  is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGEO.SYM.01). NOT a closure.
- **v334\_conformal\_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation  $\phi \rightarrow i\phi$  flips the wrong-sign kinetic term, making the divergent conformal Gaussian  $\int e^{+|c|\phi^2}$  into the convergent  $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$ ; the entire IDG form factor  $e^{p^2/M^2}$  dresses the propagator pole-free (the polynomial  $R+R^2$  ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended (FORM.CARTAN.DET.01, [E]).** The Conway–Sloane lattice side: the  $E_8$  Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ( $|\det|=1$ ) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness:  $E_8$  is the only even unimodular lattice in  $\dim < 16$ ) is the cited classification residual.

**2026-06-22 (endgame round: the necessity of  $H$  reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem,  $|\det E_8|=1$  and the  $\mu_4$  commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)**

- **v331\_necessity\_of\_H.py (QGEO.NECESS.01, [E]/[O]).** The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam  $\Lambda=|k|+M_f$  commutes with the  $\mu_4$  clock *iff* the four marks sit at the square ( $\tau=i$ ) — moving a mark off the square breaks  $\mathbb{Z}_4$  and  $[\rho, \Lambda] \neq 0$ . The square has cross-ratio 2 ( $j=1728$ ), the unique 4-config with an order-4 automorphism (v214/v267), so  $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$  and “necessity of  $H$ ” sharpens to: the raw

dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **v332\_qgamb\_metric\_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient  $-(D-1)(D-2)/4$  ( $= -\frac{3}{2}$  at  $D=4$ ), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial  $R+R^2$  ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator  $|\det \text{Cartan}(E_8)|=1$  ( $E_8$  has one anyon, vs  $D_8$ ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the  $8 \times 8$  PRODUCT gives `IsUnit (det E8)` via  $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$  — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers  $\mathbb{Z}[i]$ :  $i^4=1$ ,  $i^2=-1$ , and (on concrete  $6 \times 6$  matrices) an offset-4 coupling commutes with the  $\mu_4$  clock while an offset-2 coupling does not — all by kernel `decide`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)**

- **v329\_os\_gap\_reduction.py** (QGE0.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization  $d\Gamma(-\log T)$ , whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy  $= 6 \ln \frac{3}{2}$ , *independent of system size*. Together with  $H \Rightarrow \omega \circ \rho = \omega$  and  $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ , this exhibits the rotation-invariant flat-pillowcase premise  $H$  as the *single sufficient premise* of both bedrock IDs; residual = the necessity of  $H$  on the raw collar.
- **v330\_qgamb\_admissible\_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ( $C(n)/C(n-1) \rightarrow (2/3)^6$ ), with finite susceptibility  $\chi = 729/665$  so  $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$  converges  $\Rightarrow$  uniform tightness (Prokhorov)  $\Rightarrow$  the projective limit exists; OS reconstruction gives  $H_{\text{OS}} \geq 0$  with gap  $6 \ln \frac{3}{2}$ . The full  $\text{QG}_{\text{amb}}$  (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom):  $|\det \text{Cartan}(A_3)|=4=|\mu_4|$  ( $3 \times 3$ , `det_fin_three`) and  $|\det \text{Cartan}(D_5)|=4$  ( $5 \times 5$ , `decide`), giving the carrier  $|\det \text{Gram}|=4 \cdot 4=16$  (the v281 anyon count). The rank-8  $E_8/D_8$  case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the  $F_{\text{transfer}}$  solver suite, the cusp weight derived from a minimal rewrite, and the  $\theta_{13}$  pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)**

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum`  $\rightarrow$  `GeometricModularFlow`  $\rightarrow$  `Mu4ModularSymmetry`  $\rightarrow$  `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator  $(1 \mid 4) \wedge \neg(4 \mid 1)$  (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325\_pillowcase\_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat  $\tau=i$  pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks,  $\mu_4$  deck,  $\omega \circ \rho = \omega$ , holomorphic  $c=8 \Rightarrow (E_8)_1$ , with the recovery gap as input); all pieces machine-checked (incl. the pillowcase  $\chi_{\text{orb}}=0$ ), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326\_ftransfer\_suite.py** (FR.SUITE.01, [E]/[C]). The four  $F_{\text{transfer}}$  readouts productised into one runnable harness, each a gapped relaxation to a unique attractor:  $F_{\text{pole}}$  (Koide, seam rate  $(2/3)^6$ ),  $F_{\text{Boltzmann}}$  ( $\eta_B$ , H-theorem),  $F_{\text{relic}}$  (axion adiabatic invariant),  $F_{\text{QCD}}$  ( $m_p/m_e$ , RG to the UV fixed point), plus the  $F_{\text{RareKaon}}$  bridge. Only  $F_{\text{pole}}$  carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327\_hypergraph\_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight  $2/3$  is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum  $\{1, 2/3, 0\}$ , so  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$  emerges from the rule arity and  $(2/3)^6 = 64/729$  is the recovery gap; with a PROOF that  $2/3$  is not a root of the affine- $E_8$  charpoly ( $p(2/3) = -3520/19683 \neq 0$ , sharpening v312). v324's injected datum is reduced to the anchor arity  $\{2, 3\}$ .
- **v328\_theta13\_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified:  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$  is  $+2.0\sigma$  above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG < 1%, exponent  $5/6$  exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon  $F_{\text{transfer}}$  bridge row.
- **Wolfram mirror** 285  $\rightarrow$  288. v325 ( $\chi_{\text{orb}}=0$ ) and v327 (rewrite spectrum  $\{0, 2/3, 1\}$ ,  $(2/3)^6 = 64/729$ , and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

**2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)**

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ” framing was replaced by the current  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  with the two structural



items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, tfpt\_4\_frontier, origin\_theory, tfpt\_5\_redteam, tfpt\_research\_contracts, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/212→v324/322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

**2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)**

- **Bug fix —  $\delta_{\text{CKM}}$  display value.** The predictions table (`tex-artefacts/predictions.tex`) showed  $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ , but that formula evaluates to  $68.65^\circ$  (the canonical frozen `DELTA_CKM_RAD` = 1.198232 rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to  $68.65^\circ$ . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202\_rare\_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  ( $\sim 1.2\sigma$  above) to the full 2016–2024 combination  $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$  (La Thuile 2026), on which the prediction  $9.45 \times 10^{-11}$  lands at  $+0.08\sigma$ . Framed honestly: a strong *consistency* hit for the closed CKM point, but an  $F_{\text{transfer}}$  bridge — *not* a unique TFPT-vs-SM discriminator (the SM  $(8.6 \pm 0.4) \times 10^{-11}$  is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new  $F_{\text{transfer}}$  bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ( $\pi/3$ ) component of the quark phase”: the lock is to the structural  $\pi/3$ , not the full measured  $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$ , so  $\delta_{\text{PMNS}} = 240^\circ$  (not  $248.7^\circ$ ). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305\_witness\_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule:  $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$  is the *static* carrier ( $\mathbb{Z}/5$ ) field, every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6$  hand), so a dynamic rate explained by  $\phi$  is a cross-field false friend (v314/v315).
- **v203\_eht\_achromatic.py — the  $\mu_4$  Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at  $m \equiv 0 \pmod{4}$  (the  $\mu_4$  fingerprint of the four marks, v201); a  $\mu_4$ -orbit profile has  $\sim 0$  off-mode power while a  $\mathbb{Z}_2$  control leaks at  $m = 2$ .

- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed  $\varphi_0$ , the error-weighted mean matching the axiom to 0.04%, with  $\sin^2 \theta_{13}$  the  $\sim 2\sigma$  pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

## 2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks,  $275 \rightarrow 285$ , *all passing*): v313 (the affine- $E_8$  characteristic polynomial with the golden factor  $x^2-x-1$ ; the  $(2, 3, 5)$  atoms as  $2 \cos(\pi/k)$ ; the 31/30 selection), v314 (the rational kernel vs  $\mathbb{Q}(\sqrt{5})$  dynamic-rate split), v315 ( $30=5 \cdot 6$ ,  $\varphi(30)=8$ , the Galois group  $\mu_4 \times \mathbb{Z}_2$ , the  $\sqrt{5}$  Gauss sum), v316 ( $\rho=\zeta_6=\zeta_{30}^5$ ,  $\rho^4=-\rho$ ,  $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$ ), v317 ( $N_{\text{fam}}=3$  as the  $\mu_3$  orbit, the  $1+2$  Galois split), v318 ( $\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$  a pure function of  $\pi$ ) and v320 (the CP lock  $\delta_{\text{PMNS}}=240^\circ$ ). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322\_cp\_lock\_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes  $\arg \rho^k = k \cdot 60^\circ$ , and *which* one is selected is the deck order:  $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$  (node 4); [C] the sub-leading correction is bounded by the quark analogue  $3\lambda^2 \approx 8.7^\circ$ , so the prediction is the band  $240^\circ \pm \sim 9^\circ$ ; [C] against NuFIT 6.0 (NO,  $\delta_{\text{CP}}=212^{+26}_{-41}^\circ$ ) the leading value is  $+1.08\sigma$ , the band edge  $+0.74\sigma$  – compatible today; [X] the nearest wrong node is  $60^\circ$  away, far beyond the budget, so DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323\_bw\_geometric\_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the  $\mu_4$  clock is a geometric rotation  $\rho=\text{diag}(i^n)=\exp(i\frac{\pi}{2}L)$ ,  $\mu_4 \subset U(1)_{\text{rot}}$ ; [E] for a rotation-covariant covariance  $C=f(L)$  the modular Hamiltonian  $K=\log((1-C)/C)=g(L)$ , so  $[K, L]=0$  (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric,  $[K, L] \neq 0$ ); [C] given the seam is the  $(E_8)_1$  chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ( $\omega \circ \rho = \omega$ ) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **v324\_hypergraph\_fiber.py (HYP.FIBER.01, [E]/[C]).** The constructive v312 follow-up: the *minimal* substrate that carries both the  $E_8$  skeleton and the recovery gap is a *product*. [E] the carrier network  $T_{\text{net}}=(A+2I)/4$  (attractor = Kac marks = skeleton) fibred by the 3-node family cusp  $T_{\text{cusp}}=\text{diag}((1-w)^{2N_{\text{fam}}})$ ,  $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$  (spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ); the fibred  $T_{\text{net}} \otimes T_{\text{cusp}}$  has top eigenvalue 1 (eigenvector marks  $\otimes (w=0)$ ) and  $(2/3)^6$  as a genuine eigenvalue (marks  $\otimes (w=\frac{1}{3})$ ) – the  $N_{\text{fam}}=3$  cusp is the minimal fibre and the substrate = carrier  $\times$  family = the  $30=g_{\text{car}}(2N_{\text{fam}})=5 \times 6$  of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of  $2/3$  from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

## 2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321\_killtest\_board.py (KILL.BOARD.01, [E]).** The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO  $\sin^2 \theta_{12}=0.306747$  (fastest), #2 CMB-S4/LiteBIRD  $r$



(kill  $r > 0.01$ ), #3 DESI/Euclid  $w$  (kill  $w \neq -1$ ), #4 DUNE/Hyper-K  $\delta_{\text{PMNS}} = 240^\circ$  (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering +  $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  now), #8 LiteBIRD  $\beta$ . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

## 2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes  $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$  assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Mger / Kawahigashi–Longo–Mger holomorphy; Conway–Sloane  $E_8$  uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a decide-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the  $|\det \text{Cartan}|$  that separates the holomorphic  $E_8$  from the same- $c$  rival  $D_8 = SO(16)$ ). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

## 2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ : v320)

- **v320\_galois\_cp\_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit  $\rho = \zeta_6$  of the family Galois factor:  $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$  ( $60^\circ$ ),  $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$  ( $240^\circ$ ), and  $\rho^4 = -\rho$  locks them as  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  (the  $\mathbb{Z}_2$  sheet flip); [C] this upgrades the previously assigned  $\delta_{\text{PMNS}} = 240^\circ$  to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a  $\delta_{\text{PMNS}}$  robustly incompatible with  $240^\circ$  ( $> 3\sigma$  at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation;  $240^\circ$  is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

## 2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319\_translation\_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314);  $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  and  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (two coprime hands, v315). [E] the *dynamic* hand is  $\mathbb{Z}/6 = 2N_{\text{fam}}$  (ticks 0, ..., 5): the recovery exponent is exactly  $6 = 2N_{\text{fam}}$ , rate  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ , order-6 generator  $c^5$  ( $\zeta_6 = \zeta_{30}^5$ , v316). [E] position 0 is the *conserved law* — the resummed clock rate( $n$ ) =  $-p_2 \ln(1 - n/N_{\text{fam}})$  (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from  $n=1$ , and the  $E_8$  live phases (totatives of 30) start at 1 ( $\{1, 7, \dots, 29\}$ ), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is  $\mathbb{Z}/5 = g_{\text{car}}$  (ticks 1, ..., 5): the golden field  $\mathbb{Q}(\sqrt{5})$  ( $\varphi = 2 \cos(\pi/5)$ ), field-obstructed (no rational dynamic image, v314) — static-only;  $\zeta_5 = \zeta_{30}^6$  carries no phase. [E] one full turn

$= \text{lcm}(5, 6) = 30 = h(E_8)$ , rank  $E_8 = 8 = \varphi(30)$  live phases pairing into  $|\mu_4| = 4$  invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring)  $\times$  (dynamic 6-ring), the dynamic hand carrying the rate (exponent  $6 = 2N_{\text{fam}}$ ), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation\\_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin\\_theory](#); ledger row TRANSLATE.CLOCK.01.

**2026-06-21 (capstone: the SM structural sector is  $\mathbb{Q}(\zeta_{30})$ + Galois  $\mu_4 \times \mathbb{Z}_2$ , and the complete input is  $\{a, \pi, v_{\text{geo}}\}$  — zero dimensionless free parameters: v318)**

- **v318\_arithmetic\_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field  $\mathbb{Q}(\zeta_{30})$  with Galois group  $\mu_4 \times \mathbb{Z}_2$  (degree  $\varphi(30) = 8 = \text{rank } E_8$ ,  $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$ ), forced by the anchor atoms  $\{2, 3, 5\} = e(a)$ ; [E] the magnitude seed  $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$  ( $c_3 = 1/(8\pi)$ ) is a pure function of  $\pi$  (no free parameters, anchor-derived coefficients), so the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ ) are  $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters,  $\pi$  the unique transcendental primitive (ARCH.CORE.01),  $v_{\text{geo}}$  the unique dimensionful input (No-Unit Theorem) — complete input  $\{a, \pi, v_{\text{geo}}\}$ . [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin\\_theory](#); ledger row ARITH.CAPSTONE.01.

**2026-06-21 (are the 3 generations a  $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)**

- **v317\_galois\_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$  (with  $\text{Spec}(Q_+) = \{1, 2, 3\}$ , the  $A_3$  exponents) are the cube roots of unity  $\{1, \zeta_3, \zeta_3^2\}$  under  $w \mapsto e^{2\pi i w}$ , and the cyclic family symmetry  $w \mapsto w + \frac{1}{3}$  is a *transitive*  $\mu_3$  orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues  $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$  attach to the weights (attractor = eigenvalue 1 at  $w=0$ ); [E] the family Galois  $\mathbb{Z}_2$  ( $\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$ , = the v316 CP conjugation) refines the orbit to 1+2 — fixing  $w=0$  (rational) and swapping  $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$ ; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a  $\mu_3$  orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in [origin\\_theory](#); ledger row GALOIS.FAMILY.01.

**2026-06-21 (does the Galois  $\mu_4 \times \mathbb{Z}_2$  organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois  $\mathbb{Z}_2$  = CP conjugation: v316)**

- **v316\_galois\_readout.py** (GALOIS.READOUT.01, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling  $\mu_4 \times \mathbb{Z}_2$  reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit  $\rho = \zeta_6 = \zeta_{30}^5$  is exactly the order-6 =  $2N_{\text{fam}}$  factor  $c^5$  of v315 (the carrier being  $\zeta_5 = \zeta_{30}^6$ ); [C] the leading CP phases are its cyclotomic arguments  $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$  and  $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$  (power 4 =  $|\mu_4|$ ), with  $\zeta_6^4 = -\zeta_6$  the  $\mathbb{Z}_2$  sheet flip and  $\mu_6 = \mu_3 \times \mu_2$  (inherits v231/v233); [E] the family Galois  $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$  ( $\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$ ) *is* CP conjugation  $\delta \mapsto -\delta$ ; [E] the carrier factor  $\zeta_5$  (golden  $\sqrt{5}$ ) carries no phase ( $\mu_4$  static), and the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ , transcendental  $\varphi_0$ ) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in [origin\\_theory](#); ledger row GALOIS.READOUT.01.

## 2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$ : v315)

- **v315\_coxeter\_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  couple the carrier (5-fold, golden,  $\mathbb{Q}(\sqrt{5})$ ) and family ( $6 = 2 \cdot 3$ , rational) sectors, or just factorize? [E] the cyclic group *factorizes*:  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (direct product, group-decoupled, in line with the v314 field obstruction); [E] the  $E_8$  exponents = totatives(30) split as totatives(5)  $\times$  totatives(6),  $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$ ; [E] but the *field* couples:  $\mathbb{Q}(\zeta_{30})$  is the compositum of the carrier  $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$  and the family  $\mathbb{Q}(\zeta_3)$ , degree  $8 = \text{rank } E_8$ ; [E] the Galois group is exactly  $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$  (order 8, max element order 4), so the  $\mu_4$  seam clock *is* the Galois group of the carrier's golden field and  $\mathbb{Z}_2$  that of the family field (refining v223); [E]  $\sqrt{5} = \phi + 1/\phi$  is the quadratic Gauss sum in  $\mathbb{Q}(\zeta_5)$ . [C] verdict:  $30 = 5 \cdot 6$  does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in *origin\_theory*; ledger row COX.COUPLE.01.

## 2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$ : v314)

- **v314\_rate\_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- $E_8$  network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate  $(2/3)^6$ ) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part  $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$  plus the golden part  $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$ , while the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles  $2\cos(\pi/2)=0 = |\mathbb{Z}_2|$  and  $2\cos(\pi/3)=1 = N_{\text{fam}}$  build  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$ , same field, same atoms), but [E] the carrier 5-fold (golden,  $\mathbb{Q}(\sqrt{5})$ ) has NO rational dynamic image – a field obstruction ( $2/3$  is neither an adjacency nor a lazy-update eigenvalue; heat-kernel  $t = \Delta/(2 - \phi) \approx 6.37$  not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , and the  $\phi$ -quasicrystal  $E_8 = H_4 + \phi H_4$  ( $240 = 2 \cdot 120$ ) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in *origin\_theory*; ledger row RATE.TRANSLATE.01.

## 2026-06-21 (the golden ratio made precise: the $(2, 3, 5)$ atoms are the affine- $E_8$ network spectral angles, $\phi = 2\cos(\pi/5)$ the $g_{\text{car}}=5$ signature: v313)

- **v313\_golden\_atoms\_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- $E_8$  adjacency characteristic polynomial factors as  $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , so the golden minimal polynomial  $x^2-x-1$  divides it and  $\phi = 2\cos(\pi/5) = (1+\sqrt{5})/2$  is an *exact* eigenvalue; [E] the non-trivial eigenvalues are  $2\cos(\pi/k)$  for exactly  $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$  ( $2\cos(\pi/2)=0$ ,  $2\cos(\pi/3)=1$ ,  $2\cos(\pi/5)=\phi$ ,  $2\cos(2\pi/5)=1/\phi$ ), so the golden ratio is *specifically* the  $g_{\text{car}}=5$  (5-fold) signature and  $g_{\text{car}}=5$  gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the  $(2, 3, 5)$  spherical excess  $1/2+1/3+1/5 = 31/30 > 1$  (the condition selecting the finite  $2I \Rightarrow E_8$ ) ties the v63/v76 gap-margin numerator  $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$  to the Coxeter  $30 = h(E_8)$  — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null:  $\phi$  is a structural lattice/network quantity, not a phenomenological readout. [C] direction:  $\phi$  is the  $E_8$ -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in *origin\_theory*; ledger row GOLD.ATOMS.01.

## 2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **v308\_seam\_equiv\_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic  $c=8$   $(E_8)_1$  net” assembled into ONE well-typed logical chain  $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$ , with the exact discriminators  $|\det \text{Cartan}(E_8)|=1$  vs  $|\det \text{Cartan}(D_8=SO(16))|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$  and  $c=g_{\text{car}}+N_{\text{fam}}=8$  all machine-checked; the input gap is the derived recovery gap  $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309\_modular\_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock  $\omega \circ \rho = \omega$  via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so  $\mu_4$  is derived; the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  round-trips to the seam operator and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310\_carrier\_sm\_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  is exactly one anomaly-free Standard-Model generation ( $SU(5) \subset SO(10)$ :  $10+\bar{5}+1$ ), with  $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$  (exact) and the one-loop  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts  $b_1$ , so  $N_{\text{fam}}=3$  pins the RG. Closes the spectrum/anomaly/ $\beta$  sub-question; the interacting Epstein–Glaser  $S$ -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311\_gap\_decoupled\_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  gives geometric clustering at rate  $(2/3)^6$  (correlation length  $1/(6 \ln(3/2))$ ), finite susceptibility  $729/665$ , and the v76 margin  $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$  decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312\_hypergraph\_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- $E_8$  Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5) =$  the golden ratio, but the recovery rate  $(2/3)^6$  is *not* any eigenvalue and  $\varphi_0$  is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and the analytic seed  $\varphi_0$ . The open question is precisely delimited, not closed. Cited in introduction; ledger row HYP.INJECT.01.

## 2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305\_witness\_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h(E_8)=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the single anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive (13 atoms from 2 free primitives,  $6.5 \times$



compression made explicit), so the recurring  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio; the  $E_8$  spine is overdetermined ( $\geq 3$  structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by **v100**, never multiplied); the negative control shows  $\alpha^{-1}$ 's “137” is foreign to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, **v3**). Cited in the **tfpt\_5\_redteam** body; ledger row **WITNESS.ECON.01**. Python-only.

- **v306\_seed\_crossval.py** (**SEED.CROSSVAL.01**, [E]). A leave-one-out cross-validation turning the “many observables, one seed  $\varphi_0$ ” worry into a falsifiable out-of-sample test: back-solving  $\varphi_0$  from each of the five seed-grammar observables ( $\sin^2 \theta_{12}$ ,  $\sin^2 \theta_{13}$ ,  $\beta$ ,  $\Omega_b$ ,  $\lambda_C$ ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  out of sample (most-tensioned =  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the  $\chi^2$  by  $> 100\times$ . Cited in **tfpt\_5\_redteam**; ledger row **SEED.CROSSVAL.01**. Python-only.
- **v307\_data\_watchdog.py** (**DATA.WATCHDOG.01**, [E]). The operational decision pipeline over the frozen registry (**v84**): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live  $n_\sigma$  against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are  $\alpha^{-1}$  ( $1.9\sigma$ ),  $s_{23}$  ( $1.8\sigma$ ) and the most-tensioned core prediction  $\sin^2 \theta_{13}$  ( $2.0\sigma$ , the honest current pressure point);  $\sin^2 \theta_{12} = 0.306747$  is compatible ( $0.02\sigma$ , the JUNO falsifier), the  $r \in [0.0033, 0.0048]$  band is below the BK18 limit and the kill, and  $w=-1$  is flagged as the DESI dark-energy watchdog. Cited in **origin\_theory**; ledger row **DATA.WATCHDOG.01**. Python-only.
- **Target-A holomorphy kill test (freeze\_file.csv)**. A new **seam\_holomorphy** freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ( $|\det K| > 1$ , the same- $c$  rival  $SO(16)_1$ ) would break holomorphy and kill “seam net =  $(E_8)_1$ ” — since holomorphic  $c=8 \Leftrightarrow E_8$  is machine-proven (**v83/v143**) and  $\det K=1 \Leftrightarrow$  invertible/SRE (**v237/v301/v302**).

## 2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of $c$ ” framing of **SEAM.EQUIV.01**: **v304**)

- **v304\_idg\_nonlocal\_ghost.py** (**QGAMB.IDG.01**, [C]/[O]). A conditional, parameter-free narrowing of the red-team **rt\_F/v278** Stelle-ghost attack on the perturbative gravity sector. [E] the local  $R+R^2/\text{Weyl}^2$  four-derivative propagator  $1/(p^2(p^2+M^2))$  carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight  $f(u) = e^{-u}$  (**v259**), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close **QG.AMB.01**: the trace regulator is not yet shown to resum to an entire  $a(\square)$ , and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row **QGAMB.IDG.01**.
- **Double-pushout figure (tfpt\_1\_architecture\_e8)**. A new shared TikZ macro (**TFPTdoublepushout** in **tex-artefacts/tfpt\_figures.tex**) draws the commuting square showing that the order- $|\mu_4|=4$  glue is *one* operation in two categories: the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  (**v1**) and the conformal-net / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$  (**v92/v281/v234/v277/v260**). Expository only; no new claim.
- **Framing (introduction, tfpt\_research\_contracts)**. **SEAM.EQUIV.01** is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of  $c$  plays in relativity (cf. **v267**). Prose sharpening; no status change.

## 2026-06-19 (the discrete→dynamic lens on **F\_transfer**: one shape, one seam-rate, v303)

- **v303\_ftransfer\_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the  $E_8$  marks, gap  $6 \ln(3/2)$ ) — is also the shape of the four **F\_transfer** instances, or an unrelated bolt-on. It is the *same* shape: **[E]** **F\_pole** (Koide) *is* the seam dynamics (Möbius multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock eigenvalue  $\Rightarrow$  a unique attractor at the seam rate, **v82/v54**); **[C]** **F\_Boltzmann** ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0,1)$ , H-theorem); **[C]** **F\_relic** (axion) freezes at an adiabatic fixed point; **[E]** **F\_QCD** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3 = -7$  (asymptotic freedom, Gaussian UV attractor). **[O]** Verdict: one shape underlies all four; **F\_pole** has the seam rate  $(2/3)^6$  *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F\_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (**v187**), not reduced to the seam. **[E]** suite v303 5/5, AUDIT OK.

## 2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$ , v302)

- **v302\_seam\_gap.py** closes the chain on the single statement **v301** had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. **[E]** the frozen transfer spectrum is  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (**v160/v162**), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  (multiplicative gap  $1 - (2/3)^6 = 665/729$ ;  $(2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (**v76**) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian. **[O]** **Bottom line:** with **v300/v301**, **SEAM.EQUIV.01**'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the **v297** AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ) — *no undischarged TFPT-internal assumption remains*. It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite v302 5/5, AUDIT OK.

## 2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301\_route\_a\_invertible.py** attacks the single remaining *unconditional* gap of **SEAM.EQUIV.01**. Since **v300** collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (**v297** LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (**v148**); **[E]**  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, **v154**); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) vs 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. **[O]** LIT-A's ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (**v148/v160**+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O]** **Bottom line:** with **v300**, the whole open content of **SEAM.EQUIV.01** reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not



manufactured. [E] suite v301 6/6, AUDIT OK.

## 2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$ , v300)

- **v300\_flataway\_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden*: the flat seam was only a *soft* minimum (“the spectral mismatch grows with  $\varepsilon$ ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint —  $\text{spec}(|D_\theta|)$  is the integer ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (sympy  $2 \times 2$   $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , + an  $O(g^2)$  common shift), validated against the full Toeplitz operator; [E] any  $g_k \neq 0$  lifts the resonant degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every  $g_k = 0 \Rightarrow f = 0$  — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin*: [C] the  $(E_8)_1$  character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients (248 at level 1;  $E_8$  even unimodular  $\Rightarrow$  integer weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam, i.e. the external pin *is* the  $(E_8)_1$  rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

## 2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches $E_8$ as the last finite point, v299)

- **v299\_hypergraph\_0d\_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ( $|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$ ; the McKay group is the  $SU(2)$  binary lift  $2I$ , a *different* order-120 group whose McKay graph is affine  $E_8$ ; 30 edges  $= h(E_8)$ ); [E] the  $1 \rightarrow 4$  subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the  $2I$  fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine  $E_8$  McKay graph; [E] Smith ( $\rho \leq 2 \Leftrightarrow$  affine ADE) + Collatz–Wielandt ( $\rho \leq 2 \Leftrightarrow$  a local witness  $Am \leq 2m$ ); [E]\* the autonomous rule (diffusion-generated witness + *local* gate  $Am \leq 2m$ , Perron only as audit) runs  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ ; [E] the negative controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$  (closed form), so neither family ever reaches  $E_8$ . Honest type: [O] the  $(2, 3, \cdot)$  seed is the single irreducible input = P2 — *the rule reaches  $E_8$  as the last finite point on this seed; it is not a derivation of P2*. [E] build green, AUDIT OK.

## 2026-06-18 ( $E_8$ as a network attractor — the Wolfram-program reading, v298)

- **v298\_e8\_network\_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- $E_8$  Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4m$  on the  $(2, 3, 5)$  McKay graph (converges from any positive start;  $Am = 2m$ , top eigenvalue 2); the spherical tower terminates at  $E_8$  and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ $E_8$  is the universal attractor of a minimal  $(2, 3, 5)$  network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives:  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

## 2026-06-18 ( $a_2$ closed form + Route A literature stack, v296–v297)

- **v296\_flataway\_a2\_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact  $a_2$  coefficient in *closed form*: the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ) to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle; e.g.  $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ . Manifestly positive, small- $t$  law  $W_k = t/2 + O(t^3)$  (the  $t^2$  term cancels). Reproduces v295 to machine precision.

- **v297\_route\_a\_literature\_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta \text{Tr} \geq 0$  for all deformations,  $= 0$  iff  $f = 0$ . The exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (Duhamel), validated to rel. err  $< 10^{-6}$ .
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal  $\Delta=0 \Leftrightarrow$  flat step — a positive-weighted sum of squares is positive-definite (heatDeviation\_eq\_zero\_iff); built and axiom-checked (only propext, Classical.choice, Quot.sound). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes  $a_2$ ) remains. [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292\_flataway\_heat\_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation  $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a *positive-definite* quadratic form in the off-mark curvature ( $c_k(t) > 0$  for every  $\mathbb{Z}_4$  mode, positive on all combinations), leading invariant  $\|f\|_{L^2}$  — so Flat-Away reduces to the single fact “the seam fixes the  $a_2$  heat coefficient”.
- **v293\_flataway\_spectral\_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$  space. Mode  $4k$  splits the degenerate Steklov level  $|2k|$  at first order; the Hessian of the spectral mismatch over modes  $\{4, 8, 12\}$  is positive-definite (eigenvalues  $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294\_flataway\_rp\_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles  $\pi$ , Gauss–Bonnet  $4\pi$ ), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy  $\int f^2$  is strictly convex (Hessian  $2\pi I$ ) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290\_z4\_smooth\_curvature\_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that  $\mathbb{Z}_4$  block-diagonality is *not* mark-locality: a smooth  $\mathbb{Z}_4$  off-mark curvature  $f = \varepsilon \cos(4\theta)$  [E] passes the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , same as mark-local) yet [E] shifts the DtN spectrum ( $\max |\Delta\lambda| = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not  $[\rho, \Lambda]=0$  — which is exactly why Flat-Away cannot be read off the commutator.

- **v291\_flataway\_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family — deviation zero only at  $\varepsilon=0$ ) and [C] RP-energy/Trojanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289\_marklocal\_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks  $\Rightarrow$  curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on  $4\mathbb{Z}$  (v264/v288), L5 full-operator  $[\rho, \Lambda]=0$  (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- $L^2$  QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk  $\Rightarrow$  single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

## 2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286\_seam\_equivalence\_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ $E_8$  into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible  $\Rightarrow \mu=1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary* (the AQFT form of  $\det K=1$ ). 4/5 discharged.
- **v288\_full\_l2\_subprincipal\_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ , so  $\Lambda = |n| + M_f$  commutes with  $\rho = \text{diag}(i^n)$  on the full operator,  $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$ ; generic curvature  $\rightarrow 4.44$ ), lifting v201/v284 to  $L^2$ . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

## 2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284\_route\_i\_rp\_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP $\rightarrow$ DtN (v194), L2 marks $\rightarrow$ pillowcase (v195/v216/v214), L3 flat Steklov =  $\sqrt{-\Delta_{\text{flat}}}$  (verified here), L4 covariance from DtN (v258), L5  $\mu_4$ -invariance (v276/Lean/v280); [C] Trojanov ( $\chi_{\text{orb}}=0 \Rightarrow$  unique flat

$\tau=i$ ) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.

- **v285\_route\_ii\_seam\_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E]  $3/4$  discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$ ; the tower  $16 \rightarrow 4 \rightarrow 1$ ), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the  $(E_8)_1$ -at- $\tau=i$  object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$ . Build green, AUDIT OK.

## 2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280\_pillowcase\_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat  $\tau=i$  pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281\_holomorphy\_modular\_data.py** (QGAMB.MODULAR.01):  $\# \text{anyons} = |\det \text{Gram}|$  ( $E_8 \rightarrow 1$ ,  $SO(16) \rightarrow 4$ ), the anyon-condensation tower  $16 \rightarrow 4 \rightarrow 1$ , Gauss–Milgram  $\rightarrow c=8$ ; holomorphy ( $\det K=1$ ) is the single Tier-B discriminator.
- **v282\_e8\_tau\_i\_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* —  $\chi_{E_8}=j^{1/3}$  gives  $\chi_{E_8}(i)=1728^{1/3}=12$ , so  $\tau=i$  is both the order-4 CM point (QGEO) and the  $(E_8)_1$  modulus (QG.AMB): the two obligations are two faces of one object (the seam is  $(E_8)_1$  at  $\tau=i$ ), **obligation count**  $2 \rightarrow 1$ .
- **v283\_live\_killtest\_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$  floor 0.0586 eV; proton decay  $1.55 \times 10^{35}$  yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

## 2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.04)

- **v279\_qgeo\_obligation\_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat  $\tau=i$  pillowcase metric* (Trojanov-unique,  $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$ ), then  $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow \text{mark-locality} \rightarrow E_8$ . [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal,  $\rho$  commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.04: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

## 2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278\_lszy\_bridge\_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG  $T$ -products of  $S_{\text{pert}}$  (v269) are the OS Wick-rotated correlators

(v240), so amputate+on-shell-residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ; [C] the gap  $\Delta=6\log(3/2)>0$  gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity  $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$  *exactly* = the two-body phase space, so  $2\text{Im } M = \sum \int d\Pi |M|^2$  (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt\_F) is non-unitary  $\rightarrow$  QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

## 2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277\_seam\_calderon\_e8\_match.py** (QGAMB.TIERB.01) **reduces Tier-B to ONE bit**. [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$ ,  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248=\dim E_8$  weight-1 currents (240 roots). [E] the discriminator is holomorphy — the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (non-holomorphic), so the one condition is  $|\det K|=1$ . [C] the seam side matches via v276 (flat  $\tau=i$ ) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

## 2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276\_qgeo\_flat\_closes\_commutator.py** (QGEO.SYM.03) **collapses the bedrock to ONE geometric postulate**. The state-level residual the whole bedrock rests on is the operator equation  $[\rho, H]=0$ . [E] if  $H=f(\Delta_{\text{flat}})$  and the order-4 deck  $\rho(z \mapsto iz)$  is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (verified on the discretised square torus:  $\rho^4=I$ ,  $[\rho, \Delta]=0$ ,  $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full  $H$ . [C]  $\tau=i$  is forced by order-4 ( $j=1728$ ; hexagonal  $j=0$  is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat  $\tau=i$  pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

## 2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt\_F)

- **redteam/rt\_F\_qft4d.py** (REDTEAM.F.01) **stress-tests the just-published round** (the perturbative  $S_{\text{pert}}$  layer, the PMNS Jarlskog, the  $\nu$ -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity  $R^2/\text{Weyl}^2$  sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator  $1/(p^2(p^2+M^2))$  has a negative-residue massive spin-2 mode), so  $S_{\text{pert}}$  is unitary for the matter+gauge sector only — which is exactly why the nonperturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the  $\Lambda$ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

## 2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo**. Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser  $S_{\text{pert}}$  skeleton v269, the



concrete one-loop quartic **v271** and gauge  $\beta$ -coefficients **v273**), the complete complex PMNS matrix and leptonic CP strength **v270**, the honest absolute  $\nu$ -mass-scale typing **v272**, the over-determination of the single mass anchor **v274**, and the **QG.AMB.01** obligation roadmap **v275**. Suite 273 modules (highest ID **v275**; **v186/v226** skipped), Wolfram 271/271, **run\_all.py** green, **AUDIT OK**, **npm** build clean. The Zenodo description (**zenodo\_description.html**) and the public website were synced in the same release.

## 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, **v272–v275**)

- **v272\_nu\_mass\_scale.py** (**FLAV.NUSCALE.01**) **types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the **v184** firewall). **[E]** the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{EW})^2/M_R$  (one UV input, like  $v_{geo}$ , **v153**); **[C]** the observed  $m_3=0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (**v249**) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; **[O]**  $M_R$  is not a clean  $\bar{M}_{Pl}$  power, so the absolute value stays open; **[X]** the NO floor  $\Sigma m_\nu=0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio).
- **v273\_eg\_gauge\_running.py** (**QFT4D.SPERT.03**) **derives the SM one-loop  $b_i$  via the EG gauge self-energy**. **[E]** the gauge 2-point is the same scaling-degree-4 extension as the quartic (**v271**), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, **v159**); **[O]** two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- **v274\_scale\_overdetermination.py** (**SCALE.OVERDET.01**) **shows the single mass anchor is over-determined**. **[E]/[C]** two independent routes to  $\bar{M}_{Pl}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda/\bar{M}_{Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV) =  $2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens **v153/v78**.
- **v275\_qgamb\_roadmap.py** (**QGAMB.ROADMAP.01**) **gives the open gate QG.AMB.01 a module**. **[E]** Tier A gap decoupling ( $\Delta_{eff} = 1.648 > 0$ , **v36/v76**); **[E]/[C]** Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (**v77**); **[C]** the perturbative layer in hand (**v269/v271/v273**); **[O]** the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, **run\_all** green, **AUDIT OK**, build green.

## 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, **v271**)

- **v271\_eg\_oneloop\_quartic.py** (**QFT4D.SPERT.02**) **takes the **v269**  $S_{pert}$  skeleton from “exists” to an actual EG number** (no path integral). **[E]** the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $sd=2$  in  $d=4$ , the one-loop bubble has  $sd=4=d$ , so  $\omega=sd-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. **[C]** the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(UV) = 1/(16\pi^2)$ . **[O]** order-2/one-loop only; the all-order EG construction and the nonperturbative measure **QG.AMB.01** stay open. Exact core mirrored in Wolfram (270/270). **[E]** build green, **AUDIT OK**.

## 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, **v270**)

- **v270\_pmns\_jarlskog\_assembly.py** (**FLAV.PMNS.03**) **assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{ret}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{ret} e^{-5/6}$ ,  $\delta = 4\pi/3$ ) **into one exactly-unitary**



**complex PMNS matrix and derives the leptonic CP strength.** [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a *derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\text{max}}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6$ /triality phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\dim \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py (FLAV.TH13.01) closes the  $\theta_{13}$  exponent origin.** The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py generates verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (status\_ledger.csv for claim→status→dependencies→supersedes, script\_registry.csv/script\_clusters.csv for the theme grouping, docs\_map.csv for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into bash build.sh gen. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”

( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio  $2 \Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

## 2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266\_ps\_threshold\_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the **v265 fork**. The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr  $<$  Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed  $(15, 1, 1)$  [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be*  $+(15, 1, 1)$ ), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

## 2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265\_qft4d\_fork\_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

## 2026-06-18 (Three open-residual advances: the $F_{QCD}$ solver, $M_\nu$ from $D_F$ , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262\_fqcd\_mp\_me.py** (FR.MPME.02) implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z)=0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py** (FLAV.PMNS.02) makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric;

[C] it reproduces  $\theta_{23}=45^\circ$  and  $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13}=0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264\_raw\_seam\_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] AUDIT OK, build green.

**2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)**

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

**2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)**

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

**2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)**

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian

$H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum); [E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .

- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).
- **v260\_k3\_kummer\_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes =  $\dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.



## 2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257\_quasiorient\_modpd.py** (PS.NCG.ORIENT.02) **corrects the slightly too-generous v256 typing**. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit **v252** triple: **[E]** strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); **[E]** *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); **[E]** strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); **[C]** modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

## 2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254\_inner\_fluctuations.py** (PS.NCG.FLUCT.01) **closes v250 #5 + the no-junk obligation**. The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: **[E]** they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)$ /**126**. **[E]** the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and **[E]**  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . **[C]**  $\sigma$  = scalaron.
- **v255\_spectral\_action\_expansion.py** (PS.SPECACT.01) **closes the spectral-action expansion (v252 #11)**. The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : **[E]** structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); **[E]**  $b/a^2 = 1/3$  (top-dominated); **[E]**  $\sin^2 \theta_W = 3/8$ ; **[E]** the  $R^2$  spin-0 mode = the Starobinsky scalaron; **[C]** Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; **[C]**  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; **[O]** exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py** (PS.NCG.ORIENT.01) — **honest status, no faked proof**. KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$  to be *antisymmetric*: **[E]** skew form (verified), **[E]** rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; **[E]** the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . **[C]** the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); **[O]** a self-contained machine proof needs the complete canonical bimodule.

## 2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and $\kappa$ pinned to $O(1)$ , v252–v253)

- **v252\_full\_finite\_triple.py** (PS.DIRAC.02) **advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]**. The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: **[E]** reality ( $J^2 = +1$ ,  $JD = DJ$ ), grading ( $\gamma^2 = 1$ ,  $[\gamma, a] = 0$ ,  $J\gamma = -\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$ . **[C]** Poincaré duality; **[O]** orientability, no-junk, spectral-action expansion.

- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) discharges residual 6(b) of **v249**.  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the **126**-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

**2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$  — the CCvS Pati–Salam mechanism, v251)**

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) advances the Dirac contract **v250** from a contract to an exhibited mechanism. On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2=+1$ ,  $\gamma^2=1$ ,  $D\gamma=-\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

**2026-06-17 (Carrier-native Pati–Salam program:  $E_8$  forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)**

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247\_e8\_branching\_no126.py** (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one  $45 \Rightarrow$  proton-marginal). **v248\_ps\_algebra.py** (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). **v249\_ps\_unification.py** (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . **v250\_dirac\_triple\_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

**2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)**

- **v246\_unification\_data\_crosscheck.py** (QFT4D.RGTEST.01) — the first hard data confrontation of the spectral-action prediction. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft\_skeleton.pdf**, **figures/qft\_unification.pdf**) added via **make\_figures.py** (PDF + website PNG) and embedded in **tfpt\_research\_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.



## 2026-06-17 (Spectral-action matter content: the $SO(10)$ **16** = one anomaly-free SM generation, v245)

- **v245\_spectral\_matter\_content.py** (QFT4D.MATTER.01) discharges the computable core of the 4d contract **v244**. The carrier half-spinor (the  $SO(10)$  **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: **16** =  $SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of **CONTRACT.QFT4D.01** is now **[E]**; only the seam→Dirac realisation and the run-down couplings stay **[O]**. Wired into **run\_all.py**, the ledger, the registry, cited in **tfpt\_research\_contracts.tex**, mirrored to the website.

## 2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238)**. **v239\_kms\_thermal\_time.py** (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. **v240\_gns\_os\_reconstruction.py** (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the **v64** mass gap). **v241\_dhr\_sectors.py** (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (**v235/v237**). **v242\_spin\_statistics\_modular.py** (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . **v243\_haag\_ruelle\_braiding.py** (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the **v238** mass gap supplying Haag–Ruelle asymptotic states. **v244\_spectral\_action\_contract.py** (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor **16** is one generation, the SM gauge group sits in the carrier, gravity is **v36**; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into **run\_all.py**, the ledger and registry, cited in **tfpt\_research\_contracts.tex**, mirrored to the website.

## 2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238\_modular\_lindblad\_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (**v196/v198**) and the recovery code (**v221/v64**) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the **v198** state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap =  $-\log((2/3)^6) = 6 \log(3/2)$  = the OS mass gap  $\Delta$  (**v64**) — the static recovery bound (**v221**) and the dynamical mass gap (**v64**) are one number, one exponentiated. **[E]** the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (**v198**) — a target, not a closure. Wired into **run\_all.py**, **status\_ledger.csv**, **script\_registry.csv** and cited in **tfpt\_research\_contracts.tex**; mirrored to **website/public/verification**.

## 2026-06-17 (Experiments: black-hole-cosmology signatures, the $\eta_B$ Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the problem\_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ( $w_{\text{in}}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{\text{in}} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (**alternative\_group w\_de\_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai  $Q_{\text{geom}}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a **[C]** structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES  $\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale (experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent **[C]** readout (the full *flavored* density-matrix solve is the next refinement); tfpt\_4\_frontier Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (**internal\_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** experiments/evidence\_scorecard.json now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website EXPERIMENTS\_AUDIT mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone experiments/ surfaces (not in the verification suite or ledger); the tfpt\_4\_frontier/tfpt\_horizon\_readouts and papers.ts/predictions.ts edits are the in-same-change sync.

## 2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A** re-synced to `tfpt_5_redteam (rt_A_e8net.py, redteam_table.txt, redteam/README.md, infrastructure)`. The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level (v143) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , v148), while the  $q(A_3)$  normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays [C]/[O] (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/** (`scripts/export-shadow.sh, verification/make_script_index.py, infrastructure`). `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet** (v231\_cp\_mu6\_phases, `tfpt_5_redteam`, [E]/[C]). Sharper than v220/v225 (which only located the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}}$  leading  $= \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the v88 misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays [C] (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity** (v232\_e8\_kleinian\_seam, `origin_theory`, [E]/[C]). The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise QGEO.REALIZE.01 a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves = rank  $E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; TOP0.E8.01, bedrock stays [O]).
- **CP is the universal family/triality phase, only sheet-split** (v233\_cp\_triality\_phase, `tfpt_5_redteam`, [E]/[C]). One step past v231 in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate** (v234\_seam\_holomorphy\_selection, `tfpt_research_contracts`, [E]/[O]). The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the

seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; v232), (iii) exactly one 1-dim irrep (v219). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]):  $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$  (v160)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (v235\_seam\_chern\_simons, tfpt\_research\_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\#\text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic =  $E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton** (v236\_brieskorn\_capstone, origin\_theory, [E]). Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms ( $|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE** (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2,3,5\}$**  (v219\_icosahedral\_mckay, origin\_theory, [E]).  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1,2,3,4,5,6,10\}$  — it contains the 2,3,5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1,2,2,3,3,4,4,5,6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep *is* the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222\_cm\_norm\_duality, origin\_theory, [E]). The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index *is* the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3,2) split generates (5,6,7,13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).



- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle (v223\_coxeter\_totative\_clock, origin\_theory, [E]).**  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h) = \text{rank}$  and  $h = 2 \cdot 3 \cdot 5$  (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120 = |R^+(E_8)|$  (magnitude channel), degrees sum  $128 = \text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D = 4 = |\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0 = 5 = g_{\text{car}}$ ,  $\text{rank } H_1 = 3 = N_{\text{fam}}$ ,  $h^0 + H_1 = 8 = \text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5) = 16 = \dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGE0.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+ = 16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]).** The seam deck is the square modulus  $j = 1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j = 0$  ( $\mathbb{Z}/6$ ,  $\arg \rho = \pi/3$ ). The frozen  $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$  has leading term  $\pi/3 = \arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame  $(d, n)$  with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).**  $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$  Nariai;  $n = (5, -9, 6)$  reads  $\det$  on the first-generation column;  $\det(1, d, n) = 21 = 3 \cdot 7$ ;  $\text{Im } \det(1, d, \rho n) = 21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- **$F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).**  $M(s, t) = R + Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source  $\rightarrow$  pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).**  $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$ , product  $\frac{32}{9}$ ;  $A = \mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).**  $C = R + Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$  (hex),  $13 = N_{\mathbb{Z}[i]}(3 + 2i)$  (square),  $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C + xU) = 14 + 6x$  (slope  $6 = |R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C + yV) = 14 + 14y + 4y^2$  (2nd difference  $8 = \text{rank } E_8$ ); the anchor block has  $\det B(C + xU) = 2 \det(C + xU)$  and sheet curvature  $6 = |R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the

decouple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28 = 2 \dim G_2$ ,  $52 = \dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (v216\_marks\_gauss\_bonnet, tfpt\_research\_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6)$ ,  $(2, 4, 4)$ ,  $(3, 3, 3)$ ,  $(2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling v217\_marks\_emergence\_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension → 248/248); v217 numerical (Python-only). Suite → 217 modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (v215\_seam\_deck\_killtest, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ , Aut =  $\mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via freeze\_file.csv (seam\_deck\_symmetry), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via v214/v54/v56). Suite → 215 modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (v214\_seam\_pillowcase, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$



( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (v168)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2$ ); all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values mpmath-numerical. Suite  $\rightarrow$  214 modules.

## 2026-06-15 (the $F_{\text{transfer}}$ transfer functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (v213\_ftransfer\_functor, tfpt\_research\_contracts, **[C]**). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F) = (1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1 = 11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3 = -7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

## 2026-06-15 (frontier: two sharper **[C]** scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, **[C]**). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]**  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch** (v212\_leptogenesis\_decuple, tfpt\_4\_frontier, **[C]**). Both Boltzmann inputs share the decuple  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184’s  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, **[O]**). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho: z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents =  $\text{Spec}(Q_+)$ , rank 3 =  $N_{\text{fam}}$ . The MODULE *parity* is also

proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^*\omega_1 = \omega_3$ ,  $\sigma^*\omega_2 = \omega_2$ ,  $\sigma^*\omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`Möbius Uniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the clock  $\rho : z \mapsto iz$  has  $\rho^4 = \text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma\rho\sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1,  $-1$ , swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext`, `Classical.choice`, `Quot.sound` only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]). A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of v175). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]). The carrier’s second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs

lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.

- **An independent gravitational 3/4 (v205\_xi\_threequarter, tfpt\_4\_frontier, [E]/[C]).**  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$ ) — the same  $3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative  $H_0$  from the  $\Lambda$  branch (v206\_h0\_lambda\_branch, tfpt\_4\_frontier, [C]).**  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207\_asymptotic\_safety, tfpt\_4\_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208\_bh\_thermodynamics, tfpt\_horizon\_readouts, [E]/[C]).** Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]).** No singularity ( $\varphi \rightarrow \varphi_*$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

## 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt\_3, [O]).** Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

## 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to  $c_3$  (tfpt\_4\_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

## 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a

horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_{\pm}$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62  $\text{BR}(K^+)$  outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II  $\text{BR}(K_L)$  off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, v198); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \bmod 4]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.



- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py).** A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py [E]/[O] (claim QGEO.STATE.01, work package A).** Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200\_seam\_variational\_scan.py [C]/[O] (claim QGEO.VARI.02, work package B).** The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i =$  the  $(1, 2, 3)$  grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198\_modular\_commutator\_reduction.py [O] (claim QGEO.MODULAR.01).** The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- **v195\_marks\_lefschetz\_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **v196\_seam\_energy\_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional  $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197\_rr\_carrier\_clifford\_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^* w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the  $(1, 2, 3)$  grading on  $H^1$  (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by  $5/3$ ) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.



- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

## 2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side rank  $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by  $\text{rank}(D_5) = 5$ . The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known  $2/3$  ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192\_qgeo\_energy\_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_{\Sigma} \rho = \Lambda_{\Sigma}$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

## 2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F\_transfer-firewall round (v184–v188): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

## 2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)

- **v184\_etaB\_anchored\_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_{\Lambda}$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_{\Lambda} = 10$  atom, in the strong-washout window) [C]; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{\text{P1}}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so

$M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.

- **v185\_axion\_relic\_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **v187\_ftransfer\_laws.py** [E] (claim FR.TRANSFER.01). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be [E], the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188\_frontier\_wording\_guard.py** [E] (claim AUDIT.WORDING.01). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  v184 [C]; the old “open relic scale” dark-matter line  $\rightarrow$  v185; the old “near, not exact” Koide line  $\rightarrow$  v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183\_koide\_f\_corner\_transfer.py** [E]/[C] (claim FR.KOIDE.07). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T R a = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (v80). Hard negative control:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182\_reviewer\_residual\_map.py** [E]/[C] (claim REVIEW.MAP.01). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ( $\equiv \text{QGEO.SYM.01}$ ), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator =  $F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , JUNO;  $r \approx 0.004$ , CMB-S4) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

## 2026-06-14 (external-review response: CSV fix, G\_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox**. “no free fundamental number” softened to “no free *dimensionless* compiler dial beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.

- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

## 2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual\_chain, script\_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make\_figures.py; PDF + website PNG). (i) **residual\_chain** — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in tfpt\_research\_contracts. (ii) **script\_timeline** — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in introduction ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py** [E]/[O] (claim QGEO.SYM.01). The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain REALIZE(v175)→...→SYM.01(v181) ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- `v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`). Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$



giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee-Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` [E] — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168's cross-ratio to the full uniformisation); `COHOM` [E] —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE` [E] — a *unique*  $\mu_4$ -equivariant iso (multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); `NET` [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone}$  RP all  $m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.



## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175\_net\_existence\_full\_cone.py [E]/[O] (claim QFT.NETRP.01).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$* : the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every  $m$*  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2)  $A_2$  *assembled + verified*: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 v156 check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174\_seam\_fock\_readings.py [E] (claim QFT.FOCK.01).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170\_diamond\_slice\_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **v171\_os\_moment\_cluster.py** [E]/[C] (claim QFT.OSMOMENT.01). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in `tfpt_research_contracts` (G5).
- **v172\_trace\_anomaly\_seed.py** [E] (claim SEED.ANOMALY.01). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2 / \dim(D_5)$  factorisation. Cited in `tfpt_2` (seed).
- **v173\_pfaffian\_cp\_car.py** [E] (claim FLAV.STRONGCP.02). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, `origin_theory`, `tfpt_1–5`, `horizon`, `research_contracts`, `changelog`) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron

28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- `v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- `v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- `v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ 's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py`** [E]/[C] (claim `EM.BUDGET.02`). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,

$\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5}Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

## 2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + $F_{\text{transfer}}$ + irreducible core, **v167**)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1=N_{\text{fam}}=3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (**v152**) and the ambient measure into  $G_{\text{net}}$  (**v76**). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit **v153**); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

## 2026-06-13 (premise A maximal honest closure: $A_2$ by assembly, $R_1 = \text{GATE.QGEO}$ , zero independent open gates, **v165**)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the `v160-v162` (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo  $Q$ -system of index 4 (**v125**),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (**v156**); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $\text{DtN} = |k|$  (**v156/v157**).  $R_1$  (**seam clock**) **reduces to GATE.QGEO** [C]: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , **v115/v116**), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary =  $\mathbb{P} \setminus \mu_4$ ” = `GATE.QGEO` (cohomology established, **v137**,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (**v153**) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.



## 2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, **v162**)

- **New module `v162_seam_transport_identification.py`** [E]/[O]. Pushes the **v161** residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced**: a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,  $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (**v115**) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (**v137**). ( $\alpha$ ) is the rank-8 Calderón polarisation of  $\omega_k=|k|$  (**v113/v156**). **Closure**: with **v160+v161**, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping**. Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, **v161**)

- **New module `v161_one_particle_reduction.py`** [E]/[O]. *Discharges* the scope premise of **v160** (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by **v113** (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and **v158** makes every non-Gaussian deformation irrelevant. **Consequence**: **v160**’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).
- **Bookkeeping**. Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py`** [E]/[O]. Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment  $\leftrightarrow$  cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim

needs only  $\text{gap} > 0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual: v56's** gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

### 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

### 2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral  $c=8$  fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (v157, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee-Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The



Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual*: exact given (i) the free-bulk premise (DtN =  $|k|$ , **v155**) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger **GATE.METRIC.13**).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN =  $|k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, **v155**)

- $G_{\text{net}}$ ’s last premise reduces to a single shared physical premise (**v155**, Python-only). The seam-side identification of the Simple-Current Extension Theorem (**v154**) splits (**v110**) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (**v59**) and the EH mechanism (**v150/v151**). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger **GATE.METRIC.12**).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (**v155** is numerical/numpy, Python-only); contracts / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, **v153–v154**)

- $v_{\text{geo}}$  **re-typed: the No-Unit Theorem (**v153**)**. Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger **ANCHOR.VGEO.02**).
- $G_{\text{net}}$  **stated as the central closing theorem (**v154**)**. The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (**v89/v92/v125/v143/v148**); the only residue is the seam-side identification. Pulled to the front of **tfpt\_research\_contracts**, **tfpt\_5\_redteam** (Target A final form) and **origin\_theory** (ledger **GATE.METRIC.11**).
- **Status surfaces re-typed** (review): the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  (**[E]/[C]**), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in **origin\_theory**, the introduction, the README and **next.txt**.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/**origin\_theory**/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, **v152**)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (**v152**).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (**v68**). So  $v_{\text{geo}}$  (**v78**),  $1/G$  (**v68**) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (`rev {git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`).  $\sim 270$  files /  $\sim 3$  MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a **keyeq**. The exact result keyboxes (**v53–v56**, **v101**, **v105**, **v113**, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, **v151**)

- **R3: the Calderón/DtN kernel is conically clean (**v151**).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the **v150** EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by **v150** needs no separate derivation. `SEAM.THEOREM.01` narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays **[O]** (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, **v149–v150**)

- **R4': both dual normals are anchor data (**v149**).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact.  $R4'$  residue = *one* discrete assignment (ledger `GATE.UWALL.14`).

- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2-1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m/12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0$  = the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$  ( $4+5=9$ ,  $\sigma \perp w_0(a)$ ) and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line (=  $T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p^2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day)*: the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue,  $2/3$  motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{net}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  error was in fact a correct-but-treacherous notation switch (see above).

## 2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+ = 1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q_+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading =  $Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion =  $\mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_b r_c = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays `[C]` with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its `[O]` typing untouched.



- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

## 2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: `suite` ↔ `run_all.py` ↔ `registry` ↔ `ledger`; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.



- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$  rows are made consistent with `v86/COSMO.NSTAR.01`:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_\star \in [50, 60]$  with the scalaron-reheating conditional point  $N_\star=51.4$  ( $n_s=0.9611, r=0.0045$ ), replacing the stale point values  $N_\star=55$  (predictions table) and  $N_\star \simeq 57$  (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger `[P]` typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.4 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.

- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record  $0.306747$  with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with  $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt}_N$ ). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tftpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tftpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

## 2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue  $Q$ -system (v124/v125).** R1's bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue  $Q$ -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\det'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+) =$  the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.

- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the  $R4'$  residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the `GATE.QGEO` residue collapses to one  $\mathbb{Z}_3$  choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

## 2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank =  $c$ ) — the QBL input merges with the  $R2$ /holomorphy premise; communicated across all status surfaces.
- **$U_{\text{wall}}$  made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  (diag =  $a/|\mu_4|$ , off-weights  $(8, 0, 5)/144$ ); resonance theorem ( $M_\infty = \mathbf{1} \Leftrightarrow a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad  $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ ; the 121 audit lemma; the lepton words are the compiler atoms  $(8, 5, 3)$  with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and  $\det 8$  — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.

- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).

## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.